$\mathbb{Z}$ is Diophantine in $\mathbb{Q}$

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Abstract

We construct an integer-coefficient polynomial in 292 890 231 905 variables whose rational zero projection is exactly $\mathbb{Z}$. Its degree is less than $2 \uparrow\uparrow 7$, and the absolute value of every coefficient is less than $2 \uparrow\uparrow 8$.

We show the minimum degree of a Diophantine definition for $\mathbb{Z}$ in $\mathbb{Q}$ is 4 and describe two such quartics. The first has 314 variables, 6 672 nonzero monomials, and absolute coefficients below 2^{6500}. Its complete decimal coefficient table is 264 430 bytes. The second has 233 variables, 2 514 nonzero monomials; we do not have an effective bound on its coefficients.

Well-known consequences of $\mathbb{Z}$ being Diophantine in $\mathbb{Q}$ include a refutation of Mazur's conjecture that the topological closure of the rational points of a variety over $\mathbb{Q}$ has finitely many components, and existence of a universal rational Diophantine polynomial. Moreover, the set $\mathrm{HTP}(\mathbb{Q})$ of integer polynomials with a rational zero is many-one equivalent to the halting set $0'$.

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1 The main theorem and proof structure

This construction builds on OpenAI's preprint *Hilbert's tenth problem over the rational numbers* [94].

The strategy of using rank-one elliptic curves and existential local-divisibility conditions to approach undecidability over $\mathbb{Q}$ goes back to Pheidas [110]. His work also developed partial valuation predicates anticipating the pole conditions used here.

Earlier definability results provide the logical context. Robinson gave a first-order definition of $\mathbb{Z}$ in $\mathbb{Q}$ [87, Theorem 3.1]. Poonen gave a definition with universal quantifiers followed by existential quantifiers [88, Theorems 3.1 and 4.1]. Koenigsmann gave a universal definition and proved that $\mathbb{Q} \setminus \mathbb{Z}$ is Diophantine in $\mathbb{Q}$ [99, Theorem 1 and Corollary 2]. These are historical context, not additional premises of our construction.

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$. For a reduced rational number a/b, with $b > 0$, put $h(a/b) = \log \max(|a|, b)$. Define $2 \uparrow\uparrow 1 = 2$ and $2 \uparrow\uparrow (r+1) = 2^{2 \uparrow\uparrow r}$; every tower is associated to the right.

Theorem 1.1 (The fully decimal 314-variable quartic)**.** *There is an explicitly specified polynomial $P_{314} \in \mathbb{Z}[x_1, \ldots, x_{314}]$ of degree four and content one such that, for every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (x_2, \ldots, x_{314}) \in \mathbb{Q}^{313} : P_{314}(x, x_2, \ldots, x_{314}) = 0.$$

It has 6672 nonzero monomials and coefficients of magnitude below 2^{6496}. The complete ordinary decimal coefficient table has 264430 bytes. The construction and its proof are given in Section A12, with the final constant-coordinate lift in Section A12.11.

Theorem 1.2 (The 233-variable family)**.** *For $N \in \mathbb{N}$, let $A_N = 149^{2N+1}$, and let $Q_{1,N}, \ldots, Q_{207,N}$ be the integer quadratic expressions printed in Section A46, in the 233 coordinates $z_1, \ldots, z_{233}$. Put $P_N = \sum_{i=1}^{207} Q_{i,N}^2$. There exists an integer N_0 such that for every $N \geq N_0$ and $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (z_2, \ldots, z_{233}) \in \mathbb{Q}^{232} : P_N(x, z_2, \ldots, z_{233}) = 0.$$

Every P_N has degree four and content one. For $N \geq 53$ it has 2514 coordinate monomials and largest absolute coefficient $A_N^2 + 1 = 149^{4N+2} + 1$. The threshold N_0 is ineffective; no particular numerical N is certified as defining $\mathbb{Z}$ by the coefficient calculation.

The proof is given in the internal contact-height, aggregate-height, scalar, prime-selection and transformation sections culminating in Sections A45–A46.2. The arithmetic foundation developed throughout this manuscript is reused there with its exact domains. The 248-variable starting construction is due to José Brox; its new height argument, aggregate argument, and local construction are included here, rather than used as external black boxes. The 314-variable quartic is the other featured quartic. The large polynomial construction below supplies the original quantitative definition. Earlier quartic systems are retained as proof intermediates for the two featured quartics.

Theorem 1.3 (A polynomial definition of the integers)**.** *There exists one polynomial*

$$P \in \mathbb{Z}[X_0, X_1, \ldots, X_{292890231904}]$$

such that

$$\begin{aligned}
&\deg P < 2 \uparrow\uparrow 7, \\
&|[X^\alpha]P| < 2 \uparrow\uparrow 8 \qquad \textit{for every multi-index } \alpha, \\
&q \in \mathbb{Z} \iff \exists (w_1, \ldots, w_{292890231904}) \in \mathbb{Q}^{292890231904}, \quad P(q, w_1, \ldots, w_{292890231904}) = 0 \quad (q \in \mathbb{Q}).
\end{aligned}$$

All conclusions concern the same polynomial. Here $q \in \mathbb{Z}$ means the ordinary embedded integers. The ambient variable count does not assert that every variable occurs essentially.

The file `ordinary-integer-definition.json` specifies the polynomial P_*. The argument first represents integer indices by points on an elliptic curve over $\mathbb{Q}(\sqrt{2})$. A separate value relation identifies the ordinary rational value of an index. The point coordinates and the integer index have distinct roles throughout.

Theorem 1.3 refutes Mazur's conjecture that the closure of the rational points of a variety in its real locus has finitely many connected components. Mazur proposed this in 1992 [86]; his observation that it obstructs a Diophantine definition of $\mathbb{Z}$ is recalled in [92, p. 166]. It also gives a universal rational Diophantine polynomial, with $\mathrm{HTP}(\mathbb{Q})$ many-one equivalent to $0'$. Section 13.1 proves these three consequences.

The earlier construction gives a quartic in 299 total variables and an ordinary decimal-coefficient quartic in 314 total variables. Section A12 proves the reductions and specifies their exact coefficient tables. The Version 13 constructions and their arithmetic inputs remain as proof baselines. Version 11's index-set and retained-class identification in the published Kato application is unchanged.

Version 10 clarified determinant specialization, the published prime-two Euler-system input, the local valuation deduction, and the identification of the saved templates, following the review in repository Issue 2. It also corrects and expands the historical attribution. The polynomial specifications and numerical bounds are unchanged. We thank `manvirjaswal` for the detailed review [1].

Version 9 incorporated Gabriel Conant's odd-index product-height argument and asymmetric formal-depth refinement, developed with ChatGPT. Section A9 specifies the resulting 523- and 540-variable quartics and proves their connection to the arithmetic inputs established in this manuscript. The earlier quartics below are retained as proof and source baselines; they are not the current variable records.

Section 15 gives the explicit 4 973-variable quartic. Its arithmetic source revises the base residual compiler of Section 14, with a separate proof of soundness and completeness. The complete coefficient table and exact algebraic conversion certificate are included with this manuscript. A third polynomial, in 962 variables, is constructed in Section 17. Its complete monomial table records the large integer coefficients using exact powers. The 4 973-variable polynomial is retained with its ordinary, fully expanded integer coefficients. Section 18 proves that a single polynomial defining $\mathbb{Z}$ in $\mathbb{Q}$ must have total degree at least four, following the agentic notes cited there. Thus the degree of a quartic definition is optimal.

The final composition uses the following precise statements.

G Canonical point codes form exactly the infinite cyclic group $2E(\mathbb{Q}(\sqrt{2})) = \mathbb{Z}P_0$. The membership polynomial types arbitrary inputs; raw addition describes the group law on canonical codes.

[1] `https://github.com/elliotglazer/diophantine-z-q/issues/2`

M	The actual guarded multiplication template has rational witnesses exactly for triples with indices a, b, ab.
H	The six-contact height theorem holds with the constants used by the precision computation and, separately, by multiplication.
PC	The explicit pole polynomial is soluble for every input in Theorem 5.2.
C	The actual five-input natural polynomial and its rational translation have the exact computation and substitution semantics of Section 10, including the power template.
B	The templates, fixed constructors and private witness substitutions specify the same saved expression P_*. The quantitative syntax calculation applies to that expression.

Lemma 1.4 (Assembly)**.** *Assume* G, M, H, PC, C, B*. Then P_* satisfies Theorem 1.3.*

Proof. Section 11 proves the exact ordinary-value relation using one shared shift and disjoint private witness blocks. The literal substitutions in B identify its existential polynomial with P_*. Section 12 proves the stated bounds for that same expression. □

Proof of Theorem 1.3. Section 3, including Proposition 3.1, proves G with its canonical membership and addition formulas. Theorem 4.2 proves H for each of the two sets of constants used below. Sections 7–9, with Appendices A5–A7, prove the restricted converse in Theorem 5.3. The simultaneous descent in Section 5 then proves PC. Using these results, Section 6 proves M on all rational input codes, including the finite false-triple guard.

Section 10 proves C for the fixed five-input natural polynomial and its rational model translation. Its witness variables remain natural before that translation. Sections 2 and 12 and Appendix A3 prove B, including the fixed scalars and private witness substitutions. All six statements therefore hold, and Lemma 1.4 applies to the specified P_*. The versioned supporting arguments, their source identities and their stated hypotheses are indexed in `research/version14/SUPPORTING-NOTES.md`. □

2 A finite specification of the polynomial

The delivered specification is `construction/ordinary-integer-definition.json`, a 23 179 783-byte JSON expression with SHA256

`4d639d4c44e69ea360d36d4591b8fdc82269d68b9363af0157d1e5c1174c7af6.`

It contains 23 library templates and seven named recipes. The root recipe is `integers`; its only parameter is the rational variable at index zero. Every other variable is existentially quantified over $\mathbb{Q}$. Sources S01–S03 fix the construction and its semantics. Their historical specification had 23 180 023 bytes and SHA256

`a86790bbf1095d6b1cea9c66067ad8432dd35adc320254eecde21f9f98574714.`

The delivered expression replaces two diagnostic `sourceFile` strings and the path key in `sourceHashes` by portable relative paths. Direct comparison shows that every polynomial-defining field is unchanged. Since these three provenance fields contribute no equations, both byte representations denote exactly the same integer polynomial, with the same variables and bounds. The shipped `construction/SOURCE_MANIFEST.json` records both identities and the normalization; the same manifest retains the historical source references.

2.1 Literal operations and witness scope

A scalar node is a variable, integer constant, sum, product, or a call to an earlier template. A call substitutes its parameter expressions and a private consecutive witness block into the callee's polynomial. Shared nodes represent repeated subexpressions, not new equations. A recipe sums squares of constraint polynomials or multiplies branch polynomials. Over $\mathbb{Q}$,

$$\sum_i f_i^2 = 0 \iff \bigwedge_i f_i = 0, \qquad \prod_i f_i = 0 \iff \bigvee_i f_i = 0.$$

Private witness blocks can be moved to the front because the rational domain is nonempty. A whole inactive product-call block may receive arbitrary values. This does not discard fixed equations inside an active callee. A rational constant p/q, $q \neq 0$, is represented by a fresh c and the globally conjoined equation $qc = p$. In particular the cached coordinates of P_0 satisfy

$$8g_0 = 1641, \qquad 4g_1 = -15465, \qquad 32g_2 = 87483.$$

These equations remain active outside internal disjunctions in the multiplication, signed-encoding and exception templates. Inverses are equations $uv = 1$, never partial evaluation operations.

Strict positivity has the polynomial expression

$$s > 0 \iff \exists z_1, z_2, z_3, z_4 \in \mathbb{Q}, \quad s \sum_i z_i^2 = 1.$$

Indeed, if $1/s = a/b > 0$, express ab as four integer squares and divide their coordinates by b. The reverse direction follows from the order on $\mathbb{Q}$. Absolute value has the exact expression

$$a = |x| \quad \iff \quad \exists z_1, \ldots, z_4 \in \mathbb{Q}, \quad a = \sum_i z_i^2, \quad a^2 = x^2.$$

2.2 Fixed finite constructors

Three node types abbreviate finite polynomials. A `fixed_integer` inserts a fixed integer expression; `psi_component` inserts one coordinate of a division polynomial; `finite_guard_product` inserts a product over a fixed box of integers. Their bounds never depend on the free rational input.

For example, on $y^2 = x^3 + Ax$, write $\psi_n = f_n$ for odd n and $\psi_n = 2yf_n$ for even n. The bases are

$$f_0 = 0, \qquad f_1 = f_2 = 1,$$
$$f_3 = 3x^4 + 6Ax^2 - A^2, \qquad f_4 = 2(x^6 + 5Ax^4 - 5A^2x^2 - A^3).$$

For $m \geq 3$, $f_{2m} = f_m(f_{m+2}f_{m-1}^2 - f_{m-2}f_{m+1}^2)$. For $m \geq 2$, put $L = f_{m+2}f_m^3$, $R = f_{m-1}f_{m+1}^3$, and $D_0 = 16(x^3 + Ax)^2$. Then $f_{2m+1} = D_0L - R$ for even m and $L - D_0R$ for odd m. Negative indices satisfy $\psi_{-n} = -\psi_n$. This is a terminating recursion. These are the short-Weierstrass normalizations of [59, §§5.5–5.6, Lemmas 5.20 and 5.24, Theorem 5.21]. In its Jacobian-coordinate convention, $x([n]P) = \phi_n(P)/\psi_n(P)^2$ and $y([n]P) = \omega_n(P)/\psi_n(P)^3$. The numerator formula has a minus sign:

$$\omega_n = \frac{\psi_{n+2}\psi_{n-1}^2 - \psi_{n-2}\psi_{n+1}^2}{4y}, \qquad \psi_{2n} = 2\psi_n\omega_n.$$

Thus the fixed-scalar ordinate equation is $2y([n]P)\psi_n(P)^4 = \psi_{2n}(P)$. For $n \neq 0$ and a nonidentity point in the torsion-free carrier, $\psi_n(P) \neq 0$, so clearing these denominators is reversible.

The fixed prime catalogue consists of the first 22 primes above 324 in each unit residue class modulo $W = 4\prod_{p\le 324} p$. Dirichlet's theorem ensures termination. Put $N_s = (\prod_{p\le 324} p)H_{\mathrm{cat}}$, where H_{cat} is the catalogue product. Assign inverse abscissa zero at the identity. For the normalized representation $1/x([m]P_0) = (a+bs)/c$ with $a, b, c \in \mathbb{Z}$, $c > 0$, and $\gcd(a, b, c) = 1$, define M_0 to be the least $m \ge 1$ such that $N_s^4 \mid a, b$ and $\gcd(c, N_s) = 1$ (S31). This is a fixed integer, with a successful index bounded in Section 12. The fixed point coordinates in the guard have nonzero denominators by the infinite order of P_0. Thus finite expansion gives one integer polynomial without requiring a monomial expansion. Its arithmetic zero set remains a separate issue. In particular, bounding the integer T does not prove that it excludes all false multiplication triples.

3 The elliptic model of integer indices

The rank-one model and its generator calculation adapt [94, Proposition 2.1 and its proof].

Let

$$K_0 = \mathbb{Q}(s), \quad s^2 = 2, \quad d = 75 - 53s, \quad E : y^2 = x^3 - d^2x,$$

and fix

$$P_0 = \left(\frac{1641 - 1160s}{8}, \frac{-123720 + 87483s}{32}\right).$$

The identity has code $(1, 0, 0, 0, 0)$. An affine point has code $(0, x_0, x_1, y_0, y_1)$ with $x = x_0 + x_1 s$, $y = y_0 + y_1 s$. Linear independence of $1, s$ makes this unique. Write $\iota(n)$ for the code of $[n]P_0$.

Proposition 3.1 (Generator certificate)**.** *The point P_0 has infinite order and $2E(K_0) = \mathbb{Z}P_0$.*

Proof. We give the argument of S05–S07, making its finite certificate explicit. The ring of integers $\mathbb{Z}[s]$ is norm-Euclidean, hence has class number one. Its units are $\{\pm(1+s)^k : k \in \mathbb{Z}\}$, so it has four unit squareclasses. The sign vectors of $-1, 1+s$ are independent, so every totally positive unit is a square.

The x-squareclass map in the two-isogeny descent on E has image contained in the eight classes generated by $-1, 1+s, d$. Away from (d), odd $v(x)$ makes the minimum of $3v(x)$ and $v(x) + 2v(d)$ odd, impossible for y^2. The torsion points give $1, -1, d, -d$. Put $\epsilon = 1 + s$, $\delta = 3 - s$. The change $(x, y) \mapsto (\epsilon^4 x, \epsilon^6 y)$ takes E to $C : y^2 = x^3 - \delta^2 x$, and $R = (1 + s, 2s)$ provides the remaining unit class. Thus the first image has order eight.

On $E' : y^2 = x^3 + 4d^2x$, a nonzero x is positive in both real embeddings. Its valuations are even. At (s) the two terms have valuations $3m$ and $m + 4$, with equality only at $m = 2$. At (d) the only odd case would require $u^2 = -4$ in $\mathbb{F}_7$. Thus the second image is trivial. The exact two-isogeny sequence gives $[E(K_0) : 2E(K_0)] = 8$; its possible extra kernel term vanishes since a nonkernel rational two-torsion point maps to the dual kernel.

The good reductions at $(17, s - 6)$ and $(23, s - 5)$ have 20 and 24 points. Prime-to-residue-characteristic torsion injects, so the torsion group has order at most four, hence is exactly the displayed full two-torsion. Mordell–Weil now gives rank one. The x-class of R has opposite real signs, whereas all torsion classes have equal signs. Its free index is therefore odd.

The duplication forms on C are

$$F_2(a, b) = (a^2 + \delta^2 b^2)^2, \qquad G_2(a, b) = 4ab(a^2 - \delta^2 b^2).$$

At either real embedding their maximum lies between $\max(|a|, |b|)^4$ and $676 \max(|a|, |b|)^4$. For primitive a, b their common ideal has valuations at most six at (s) and four at (δ), and none elsewhere. For the latter bound use that -1 is nonsquare modulo seven. For the former, a local

unit is $A + Bs$ with A odd, and $v_s((A + Bs)^2 + 1)$ is two or three according to the parity of B. Hence

$$4h_x(T) - \log 392 \le h_x(2T) \le 4h_x(T) + \log 676.$$

With $2\widehat{h}(T) = \lim_m 4^{-m} h_x(2^m T)$ this gives $h_x(T) \le 2\widehat{h}(T) + \log 392/3$. Direct doubling gives $x(2R) = (57 - 28s)/8$, with height $\log 41$. If $R = mS + T$ for torsion T and $|m| \ge 3$, it follows that

$$h_x(S) \le \frac{\log(41^3 \cdot 676)}{108} + \frac{\log 392}{3} < \log 9.$$

The strict inequality is the exact integer comparison $41^3 \cdot 676 \cdot 392^{36} < 9^{108}$.

A nonrational $x \in K_0$ in this height range has primitive polynomial aX^2+bX+c with $1 \le a \le 81$, $|b| \le 162$, $|c| \le 81$, and discriminant $2k^2$, $1 \le k \le 162$. Both roots are included. Rational possibilities have coprime numerator of absolute value at most nine and denominator from one to nine. The exact finite square test of $x^3 - \delta^2 x$ gives precisely

$$0, \quad \delta, \quad -\delta, \quad 1 + s, \quad 23 - 17s, \quad 2 + 4s, \quad (5 - 4s)/2.$$

The certificate covers 7,004 primitive quadratics, their 14,008 roots, and 111 rational candidates. To test whether $u + vs$ is a square, take both signs of a rational square root t of $u^2 - 2v^2$, test $y_0^2 = (u + t)/2$, then set $y_1 = v/(2y_0)$ if $y_0 \ne 0$. The $y_0 = 0$ case requires $v = 0$ and $y_1^2 = u/2$. Substitution verifies each result. This is a necessary and sufficient test, including zero. The last four listed coordinates are those of $R + T$ for $T \in C[2]$. Every surviving nontorsion point therefore has height $\widehat{h}(R)$, contradicting $\widehat{h}(R) = m^2\widehat{h}(S)$. This finite computation is an explicit part of the proof, with a recorded independent replay; it is not replaced by a rank oracle.

Thus $2C(K_0) = \mathbb{Z}(2R)$. The inverse coordinate change sends $2R$ to P_0, as asserted. □

3.1 Polynomial membership and addition

Apart from the identity branch, membership in $2E(K_0)$ is expressed by choosing $X, Y, H \in K_0$, $u \in \mathbb{Q}$, and imposing

$$Y^2 = X^3 - d^2X, \quad u\,\mathrm{N}(Y) = 1, \quad 2YH = 3X^2 - d^2, \quad x = H^2 - 2X, \quad y = H(X - x) - Y.$$

Every nonidentity double has a preimage with nonzero ordinate. Each field equation is two rational equations. Membership is an all-input statement. Raw addition has the narrower exact contract

$$\mathrm{Add}_{\mathrm{raw}}(\iota(a), \iota(b), R) \quad \Longleftrightarrow \quad R = \iota(a + b).$$

Its branches cover both identities, opposite points, distinct abscissas and doubling with nonzero ordinate. Every later use supplies canonical input codes. Multiplication first imposes membership of all three roots and is sound on arbitrary rational tuples. In the natural predicate, four initially untyped tuples enter multiplication calls, so are typed before their squared values are added. Lagrange's theorem then gives exactly $\iota(\mathbb{N})$. This uses multiplication completeness; the generator proposition alone does not supply the natural predicate.

4 The six-contact height estimate

The contact and odd-support framework follows [94, Section 4]. The effective geometric estimates below give the height input used in this manuscript; the preprint's height assertion is not taken as an additional premise.

Set

$$\mathcal{C} = \{0, -15, -20, -24, -30, -60\}, \qquad F(u) = u(u+15)(u+20)(u+24)(u+30)(u+60),$$

and $G(u) = 7u^2 + 300u + 3600$. For a finite prime set S containing all primes at most 324 and $u \notin \mathcal{C}$, define

$$M_S(u) = \prod_{\substack{p \notin S:\ v_p(u-c)>0 \text{ odd} \\ \text{for some } c \in \mathcal{C}}} p.$$

Negative valuations of u do not enter the product. Put

$$\begin{aligned} F_0 &= 324!, \qquad D = F_0\, 3^{256F_0^4}, \qquad G_0 = 116064000000, \\ E_0 &= 10(DG_0)^2, \qquad Q_R = R^{RE_0}, \\ B_{\mathrm{cm}} &= D^2\, 3^{21000D^2} + 1, \\ C_{\mathrm{cm}} &= 2B_{\mathrm{cm}} + 4096(1028 + 5D + 84000D^2), \\ H_6(S;R) &= 2 \cdot 10^{13} \max(1, C_{\mathrm{cm}}, Q_R^{17}), \\ R &\geq \max(D, \max S, 10^{12}). \end{aligned}$$

When the second argument is omitted, we use $H_6(S) = H_6(S; \max(D, \max S, 10^{12}))$. The geometric construction is followed by the published prime-five modularity theorem and a separate effective CM argument. The constants in the displayed definition of $H_6(S;R)$ are unchanged.

Lemma 4.1 (Modularity after rational scalar descent)**.** *Let A be an abelian surface over a finite Galois extension of $\mathbb{Q}$, with full geometric endomorphism algebra a quaternion division algebra B over $\mathbb{Q}$. Suppose that B-compatible quasi-isogenies $\mu_\sigma : {}^\sigma A \to A$, with $\mu_1 = 1$, form a finite locally constant choice on $G_\mathbb{Q}$, and put*

$$c(\sigma, \tau) = \mu_\sigma {}^\sigma\mu_\tau \mu_{\sigma\tau}^{-1} \in \mathbb{Q}^\times.$$

Suppose that a locally constant cochain $t : G_\mathbb{Q} \to C^\times$ in a number field C satisfies

$$t_\sigma t_\tau c(\sigma, \tau) = t_{\sigma\tau}.$$

The added coefficients have trivial Galois action. Enlarge C to split B, and fix its embedding into $\overline{\mathbb{Q}}_5$. If $\rho_5(\sigma) = t_\sigma \mu_\sigma \circ \sigma$ is written as $r_5 \oplus r_5$ using the matrix units of B, then r_5 is the Tate realization of a primitive weight-two cuspidal newform, after a finite coefficient extension. No bound on an auxiliary field or abelian variety is part of this assertion.

Proof. We apply the published prime-five modularity statement reported in [106, p. 624, with the definition on pp. 611–612] to this same representation. We check its hypotheses and the finite twist needed for the conclusion.

The cocycle equation gives $t_1 = 1$. Choose a finite Galois extension $F/\mathbb{Q}$ over which A and all its geometric endomorphisms are defined and on whose Galois subgroup both locally constant choices are the identity. Then

$$V_5(A) \otimes_{\mathbb{Q}_5} E \simeq r_5|_{G_F} \oplus r_5|_{G_F}$$

for a finite coefficient extension $E/\mathbb{Q}_5$. The continuous Tate action on this open subgroup and the finite coset decomposition prove continuity of r_5 on $G_\mathbb{Q}$. Faltings' semisimplicity and endomorphism theorem [117, Theorem 1(a)–(b), p. 204] identifies the commutant on the left with $B \otimes E = M_2(E)$.

The multiplicity commutant on either copy is therefore E. The same calculation after coefficient extension, together with semisimplicity, shows that $r_5|_{G_F}$ is absolutely irreducible. Thus r_5 is absolutely irreducible. Good reduction of A outside a finite set, enlarged by the primes ramified in $F/\mathbb{Q}$, proves that r_5 is unramified outside a finite set.

At a place of F above five, proper-smooth comparison [17, Corollary 1.8 and Theorems 9.1, 9.3], with [18], makes $H^1_{\mathrm{et}}(A, \mathbb{Q}_5)$ de Rham. The canonical dual is $V_5(A)$ [116, Theorem 15.1(a), p. 129]. In the convention that $\mathbb{Q}_5(1)$ has weight -1, its weights are $0, 0, -1, -1$. De Rham representations are closed under duals, tensor products and subquotients [19, Theorem 3.8, p. 142]. The two identical summands therefore each have weights $0, -1$. This conclusion descends to $\mathbb{Q}_5$: after a finite Galois local extension $K'/\mathbb{Q}_5$, the full-rank filtered space $(B_{\mathrm{dR}} \otimes r_5)^{G_{K'}}$ has its natural semilinear Galois action. Descent of that space and each filtration step gives the full-rank filtered space over $\mathbb{Q}_5$ with the same weights. No good-reduction or crystalline hypothesis at five is used.

For oddness, fix an embedding into $\mathbb{C}$ and its complex conjugation c. On Betti homology, c interchanges the two Hodge summands, whereas the holomorphic quasi-isogeny μ_c preserves their types. Thus $t_c \mu_c c$ has trace zero. The complex lattice description [21, Theorem 6.8 and the preceding paragraph, p. 319] identifies its 5-power torsion functorially with the homology lattice modulo 5^n. This also respects conjugation; clearing isogeny denominators and taking the inverse limit transfers the trace identity to the Tate module under the chosen coefficient embedding. Hence $2\,\mathrm{tr}(r_5(c)) = 0$. Since $r_5(c)^2 = 1$, its eigenvalues are $1, -1$, so it is odd.

The cited published statement gives, after a finite coefficient extension, $r_5 \simeq U_g \otimes \delta$ for a nonzero classical cuspidal eigenform g of positive integral weight k. We use arithmetic Frobenius for the associated semisimple representation. We first choose a normalized primitive representative without assuming that $k = 2$. The decomposition in [68, §1.9, p. 510] writes g as a sum of primitive oldforms $f_j(d_j z)$; its Eisenstein part is zero. For a prime ℓ not dividing the original level, the coefficient formula in [68, §1.6, p. 509] shows that T_ℓ commutes with each such degeneracy map. Uniqueness of the decomposition implies that every nonzero constituent has the same good-prime eigenvalues as g. Choose one primitive constituent f, normalized by $a_1(f) = 1$ as in [68, §1.7, p. 509]. Its induced nebentypus and good-prime characteristic polynomials agree with those of g. Semisimple uniqueness [68, Lemma 3.2, p. 513] therefore gives $r_5 \simeq U_f \otimes \delta$ with the same character. All coefficient embeddings are extended compatibly with the one already fixed.

If $k = 1$, the transformation under $-I$ makes the nebentypus odd. Theorem 4.1 of [68, pp. 513–514] supplies the associated irreducible finite-image representation. Finiteness here uses the discrete complex topology specified in its §3.1; its Frobenius is arithmetic. Moreover, §§8.2, 8.5–8.6, pp. 525–526, construct this representation over a number field, so it has a finite-image 5-adic realization with the same good-prime polynomials. Lemma 3.2 identifies it with U_f. After a finite local extension killing this finite image, r_5 is the direct sum of two copies of the same character. De Rham subobject closure and finite-extension compatibility then give two equal Hodge–Tate weights, contrary to $\{0, -1\}$. These closure and base-change properties are also stated in [69, §9.2, p. 187]. Thus $k \geq 2$.

For this range, let E_f be the coefficient field of f in Kato's construction, with the place λ above five fixed by the chosen embedding. Write $V = V_{E_{f,\lambda}}(f)$ for its realization. It has dimension two over $E_{f,\lambda}$ and $\dim_{E_f} S(f) = 1$ [69, §6.3, pp. 161–162]. Section 9.4, p. 188, states that V is de Rham and gives the decreasing filtration

$$\mathrm{Fil}^i D_{\mathrm{dR}}(V) = \begin{cases} D_{\mathrm{dR}}(V), & i \leq 0, \\ S(f) \otimes_{E_f} E_{f,\lambda}, & 1 \leq i \leq k-1, \\ 0, & i \geq k. \end{cases}$$

This holds at every level, including levels divisible by five; §§5.4 and 8.3, pp. 155 and 181, include the small-level objects through fine-level invariants. No non-CM or crystalline hypothesis is used. Writing ε_f for the nebentypus and ϵ_5 for the cyclotomic character, [69, (14.10.2)–(14.10.3), p. 240] gives

$$\det V = \epsilon_5^{1-k}\varepsilon_f^{-1}$$

and puts the good-prime Hecke polynomial on inverse arithmetic Frobenius in V. Hence arithmetic Frobenius on $V^\vee$ has the polynomial defining U_f. Their semisimplifications agree by the same uniqueness lemma. Since U_f is irreducible, being a character twist of r_5, the two-dimensional $V^\vee$ is itself irreducible and isomorphic to U_f. Duality therefore yields

$$\mathrm{HT}(U_f) = \{0, 1-k\}, \qquad \det U_f = \varepsilon_f \epsilon_5^{k-1}.$$

There is no further Tate twist in this identification.

The δ-line in $\mathrm{Hom}(U_f, r_5)$ is Galois-stable; tensor, dual and subobject closure make it de Rham. If its weight is m, then

$$\{m, m+1-k\} = \{0, -1\}.$$

Since $k \geq 2$, comparison of the larger entries and then the smaller entries gives $m = 0$ and $k = 2$. To prove that the twist is finite, put $\alpha = \det(r_5)/\epsilon_5$. The polarization pairing on $V_5(A)$ gives $\det V_5(A) = \epsilon_5^2$, so $\alpha^2 = 1$ on G_F. Its image on all of $G_\mathbb{Q}$ is consequently finite. The determinant formula just proved now gives $\delta^2 = \alpha/\varepsilon_f$. The right side has finite image, and each nonzero scalar has at most two square roots in characteristic zero. Thus δ has finite image; this is not inferred from weight zero alone.

The fixed field of $\ker\delta$ is a finite abelian extension of $\mathbb{Q}$. By [118, Theorem 14.1, p. 321], it lies in a cyclotomic field. Let χ be the corresponding primitive Dirichlet character, with $\chi(q) = \delta(\mathrm{Frob}_{q,\mathrm{arith}})$. If $M = \mathrm{level}(f)$ and $r = \mathrm{cond}(\chi)$, Shimura's translate formula [37, Section 4, equation (4.1), p. 531] gives a nonzero Gauss sum times $h = \sum_{n\geq 1}\chi(n)a_n(f)q^n$. For completeness, conjugating the translation matrices by an element $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) \in \Gamma_0(Mr^2)$ permutes their indices by $a \mapsto aD^2 \pmod r$. The weighted sum therefore transforms with character $\varepsilon_f\chi^2$ and remains cuspidal. The Hecke coefficient formula gives $T_\ell h = \chi(\ell)a_\ell(f)h$ for $\ell \nmid rM$. Li's old/new decomposition [36, p. 294 and Theorem 5, p. 298] supplies a primitive weight-two newform with these good-prime eigenvalues. Its Tate representation is $U_f \otimes \delta$ by Chebotarev and semisimplicity, hence is the original r_5. No conductor bound is inferred from this qualitative twisting step. □

Scope of the modularity input. The immediate supplier is the published statement in [106, p. 624]; Pan's work [32] is its proof source. This use imposes no residual restriction at five. It does not assert that every input to Pan's proof has been reproved here. In particular, his ordinary inputs and his local Hu–Tan ring calculations are distinct from the global Hu–Tan support argument. The separate GL2-type and Serre weight-transfer reconstructions are not inputs to the present lemma.

The exact source-to-object and historical-dependency map is given in [1].

Theorem 4.2 (Six-contact estimate)**.** *For every such S, R and every $u \in \mathbb{Q} \setminus \mathcal{C}$,*

$$h(u) \leq H_6(S;R)M_S(u)^{24}.$$

Proof. Take $B = (2, -3)$, the maximal order $\mathcal{O} = \mathbb{Z}[i, (1+j)/2]$, and $\mu = 2j + ij$. With its positive sign choice, the form $\mathrm{Trd}(\mu^{-1}x\overline{y})$ has matrix

$$\begin{pmatrix} 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -2 \\ 1 & -1 & 0 & 0 \\ -1 & 2 & 0 & 0 \end{pmatrix}$$

and determinant one, in the ordered basis $(1, i, (1+j)/2, (i+ij)/2)$. The Riemann form is the convention of [30, §2, pp. 779–780, Lemma 3]; its Rosati involution is $a \mapsto \mu^{-1}\overline{a}\mu$. The form printed in [23, §2.1], $\mathrm{Trd}(\overline{x}\mu y)$, is six times this form, since $\mu^2 = -6$. It has determinant 6^4 on this lattice. We use the determinant-one form above; the positive scalar does not change the Rosati involution or which isomorphisms preserve the form. The corrected form has the prescribed positive Rosati involution, and the principal polarization inducing it is unique [29, p. 592]. Thus it gives the abstract polarized datum in [23, §2.1, Theorem 2.1]. For this order the two twisting involutions are w_2, w_3, and they generate the full four-element Atkin–Lehner group W. The theorem identifies the polarized forgetful quotient with V/W, where $V = \mathfrak{H}/\mathcal{O}^1$. The canonical model and the involutions are defined over $\mathbb{Q}$. Lemma 3.1 and the fixed-point table of [23, pp. 739–740] give a genus-zero quotient with three distinct rational marked points. The two elliptic stabilizer orders on V are 2 and 3. A finite subgroup fixing a point of a smooth characteristic-zero curve acts faithfully on its tangent line, so it is cyclic. Hence a point fixed by a nontrivial element of W has W-stabilizer of order two, rather than four. The three resulting branch indices are therefore $4, 6, 2$. We take the unique coordinate j with values $0, \infty, -16/27$ at the CM points of discriminants $-4, -3, -24$, respectively. This is the coordinate defined in [30, Example 30]. The principal-form correction and the abstract coarse-moduli application are detailed in [24].

We also need its compatibility with the explicit Igusa invariants. Put $s_2 = \psi_4$, $s_3 = \psi_6$, $s_5 = 2^{12}3^5\chi_{10}$ and $s_6 = 2^{12}3^6\chi_{12}$. The divisor calculations in [30, Examples 29–30, pp. 795–797] give

$$\frac{s_3^2}{s_2^3} = c_1 j, \qquad \frac{s_5}{s_2 s_3} = c_2, \qquad \frac{s_6}{s_2^3} = c_3 j + c_4.$$

Its product-curve parameterization at elliptic invariant 1728 gives $c_4 = 2^{12}3^6/1728^2 = 1$. For the remaining constants use its Y_5' intersection at discriminant -24. The stated eta-coordinate $x = -6 \pm 6\sqrt{-3}$ satisfies $x^2 + 12x + 144 = 0$; substitution in the published parameterization gives

$$s_2 = -36x, \quad s_3 = 6912, \quad s_5 = -248832x, \quad s_6 = -32845824.$$

The three ratios are $-16/27, 1, 11/27$, and their denominators are nonzero. Since $j = -16/27$ at this point, $c_1 = c_2 = c_3 = 1$. On the finite nonzero j-locus, a common weighted rescaling gives $[s_2, s_3, s_5, s_6] = [j, j^2, j^3, j^3(j+1)]$ in weights $2, 3, 5, 6$. Equations (13)–(14) of [30, p. 793], followed by rescaling by -12 in weights $1, 2, 3, 5$, therefore give

$$[J_2, J_4, J_6, J_{10}] = [12(j+1), 6(j^2+j+1), 4(j^3-2j^2+1), j^3]. \tag{1}$$

This calculation uses the published intersection, divisor and eta-value inputs at their stated scope; its source comparisons and weighted conventions are in [25]. It does not use the product-isomorphism arguments of [23, Propositions 3.7–3.8], the subsequent numerical CM evaluation, or its explicit conic descent. Let Y_3 be the generic full-three-marked moduli curve, with all its determinant components. Its full adelic description and the same positive-normalizer forgetful quotient are proved below, after the integral construction. We first compute the arithmetic index which that identification will permit us to apply. Put $K_0 = \widehat{\mathcal{O}}^*$, $K_3 = \ker(K_0 \longrightarrow (\mathcal{O}/3\mathcal{O})^*)$, and let N be the finite-adelic normalizer of $\widehat{\mathcal{O}}$. It normalizes K_3, since it preserves the ideal $3\widehat{\mathcal{O}}$.

The completion $\mathcal{O}_p$ is maximal. Indeed a larger local order inside $p^{-k}\mathcal{O}_p$ would intersect $\mathcal{O}[1/p]$ in a larger global order inside $p^{-k}\mathcal{O}$; coordinate density in B_p makes that containment strict. At a split prime identify $\mathcal{O}_p$ with $\mathrm{End}_{\mathbb{Z}_p}(L)$ for a rank-two lattice L. If g normalizes this order, the matrix units show that $gL = Ie_1 \oplus Ie_2 = p^r L$. Thus the local normalizer is $\mathbb{Q}_p^*\mathcal{O}_p^*$. At each division prime the maximal order is the unique nonnegative valuation ring, with $w = v_p(\mathrm{nrd})/2$.

Its unit group has valuation zero, while central elements have integral valuation and a division uniformizer has half-integral valuation. The quotient by center and units consequently has order two. These local facts, including the prime two, are used with the precise normalization of [105, §13.3]. Since the division primes are exactly two and three, their independent valuation parities give $|N/(\mathbb{A}_f^* K_0)| = 4$.

A central finite idele acts trivially on this full level-three curve. Multiply it by a rational scalar to cancel all finite valuations and choose the sign so that its residue at three is one. The remaining idele is in K_3; the rational central scalar is absorbed on the left of the adelic quotient and acts trivially on the symmetric domain. The given normalizer action thus factors through $N/(\mathbb{A}_f^* K_3)$. Its quotient is the same $\mathbf{P}_j^1$, and a geometric generic fiber is an orbit. Therefore

$$\deg(Y_3 \longrightarrow \mathbf{P}_j^1) \leq 4[K_0 : K_3] \leq 4|(\mathcal{O}/3\mathcal{O})^*| \leq 4 \cdot 81 = 324.$$

This uses the kernel definition of K_3, so no surjectivity of reduction is needed. It also retains all components throughout.

We construct an integral model without changing this generic degree. Write $R_0 = \mathbb{Z}[1/30]$ in the following construction. The full proofs and the exact source hypotheses for its six phases are recorded in Section A33.

The initial model. Apply [29, Theorem 2.1 and Corollary 2.3] to $V_1(5)$. At five its determinant image is full because it contains the diagonal matrices $\mathrm{diag}(a, 1)$ for every unit a. At the other split primes, maximal-order units have arbitrary determinant. At a division prime, the maximal order contains the integers of the unramified quadratic field, whose unit norms are all of $\mathbb{Z}_p^*$; restriction of reduced norm gives the same surjectivity. Thus the entire adelic determinant condition holds. The level is maximal at the division primes, and $5 \geq 4$ is prime to six. The result is a smooth projective fine model X_{V_1}/R_0 with its universal object. Here Corollary 2.3 identifies the actual groupoid with the stack of a scheme, so its objects have trivial automorphisms over arbitrary base schemes, including nonreduced ones.

The full five-marking. The full-five marking sheaf and its H-orbit sheaf in [29, Definition 1.1, p. 593] apply at this prime-to-discriminant level. Pulling the full-marking torsor back along the existing $V_1(5)$-marking gives a finite étale surjective H-torsor of refinements. Its total space X_5 represents full five-markings and retains every determinant component. It is smooth projective over R_0. For a left-module marking m and a unit g, set $g \cdot m = m \circ r_g$, where $r_g(x) = xg$. Since $r_h \circ r_g = r_{gh}$, this is the left action used below. An automorphism preserving a full five-marking preserves its induced $V_1(5)$-marking, so the full-five groupoid is also rigid.

The full three-marking. For an abelian scheme and an invertible integer n, multiplication by n is finite étale of rank n^4 on its kernel in relative dimension two. The fiberwise multiplication theorem, flatness and étale-fiber criteria give the assertion over arbitrary bases; properness and quasi-finiteness give finiteness. Finite étale schemes are finite locally constant sheaves. Requiring a bijection to respect addition and a finite generating set of $\mathcal{O}$ imposes finitely many equations on a finite set. Hence

$$X_{35} = \mathrm{Isom}_{\mathcal{O}}(\mathcal{O}/3\mathcal{O}, A[3]) \longrightarrow X_5$$

is finite étale. Its surjectivity requires a separate calculation. On every geometric fiber, $T_3(A)$ is free of rank four over $\mathbb{Z}_3$. The division algebra B_3 acts unitally on its four-dimensional rational Tate module. Multiplication on any nonzero vector embeds B_3 into that module, so it is a rank-one left B_3-module. Its $\mathcal{O}_3$-stable lattice has a nonzero element x of least division valuation: the values are discrete and bounded below by the minimum of the valuations of a lattice basis. For every y in the lattice, $w(yx^{-1}) \geq 0$, so $y \in \mathcal{O}_3 x$; the reverse inclusion follows from stability. Reduction modulo the central element three gives $A[3] \simeq \mathcal{O}/3\mathcal{O}$. Thus $X_{35} \to X_5$ is surjective in every allowed

characteristic. It is smooth projective over the affine, hence quasi-compact and quasi-separated, base R_0.

Freeness and rigidity. Let $G_5 = (\mathcal{O}/5\mathcal{O})^*$ change only the five-marking. A geometric stabilizer gives a polarized automorphism α preserving the three-marking. Polarized automorphisms form a finite group, and the action on $T_3(A)$ is faithful, by the integral-lattice proof in Section A32. The matrix of α lies in $1 + 3M_4(\mathbb{Z}_3)$, which is torsion-free. Indeed, a nontrivial torsion element has a power of prime order ℓ; write it as $1 + 3^r M$ with $r \geq 1$ and M having a unit entry. If $\ell \neq 3$, the linear term in its ℓth power has entry valuation r, while higher terms have valuation at least $2r$. For $\ell = 3$, these bounds are $r + 1$ and at least $r + 2$. Neither power is the identity. Hence $\alpha = 1$, and the five-marking then forces the element of G_5 to be one. For each nonidentity element of this constant group, its fixed subscheme has no geometric points and is empty. This proves a free scheme action without assuming that $|G_5|$ is invertible on R_0.

The same argument gives geometric rigidity of every full-three object. To pass to an arbitrary base T, its automorphism acts identically on the finite étale $A[5]$, since morphisms of finite locally constant sheaves are detected on geometric fibers. After adding a five-marking it is the identity by the full-five rigidity above. The existence of that faithfully flat marking cover for every object is proved below; descent of equality of morphisms then gives rigidity over T.

The quotient and the exact descended object. The free finite-group quotient theorem gives a scheme $Y = X_{35}/G_5$ and a G_5-torsor $q : X_{35} \to Y$. Because G_5 is finite étale, so is q. Smoothness descends; the diagonal is closed after pullback, where it is the union of the action graphs, and properness descends along this finite surjection. An ample invertible sheaf descends in existence by its norm, making Y/R_0 projective. This norm is used only for the base quotient, not for its abelian-scheme polarization. The regular-base projectivity theorem for abelian schemes makes the universal abelian scheme projective over X_{35} before any use of duality. Projective schemes descend effectively along the finite locally free surjection q. The relabeling isomorphisms obey the cocycle identity, and faithful descent carries the group maps and their identities to an abelian scheme A/Y. Base-compatible duality identifies the pullback of $A^\vee$ with the original dual. The particular polarization morphism therefore descends; being an isomorphism is faithfully-flat local, and its geometric fibers are the original polarizations. It remains principal, of degree one. The $\mathcal{O}$-action and three-marking likewise descend as actual maps. The projectivity, Picard and descent inputs, in this order, are specified in Section A33.

The whole full-three functor. Let a full-three object be given over any R_0-scheme T. On a geometric fiber, $A[5]$ has dimension four over $\mathbf{F}_5$ and $\mathcal{O}/5\mathcal{O} \simeq M_2(\mathbf{F}_5)$ acts unitally. Its matrix units split $A[5] = e_{11}A[5] \oplus e_{22}A[5]$; the two summands are isomorphic through e_{21}, e_{12} and have dimension two. For a basis v_1, v_2 of the first, $E_{ij} \mapsto e_{i1}v_j$ is a left $M_2(\mathbf{F}_5)$-module isomorphism with the regular module. Thus the finite étale sheaf of full five-markings is surjective for this arbitrary object and is a G_5-torsor $P \to T$. The tautological extra marking gives an equivariant map $P \to X_{35}$, whose invariant composite descends to $T \to Y$. Conversely, a map to Y pulls back the descended universal object. These constructions are inverse after the faithfully flat cover P; the rigidity just proved makes their comparison isomorphisms unique, so they are inverse over T and compatible with base change. Consequently Y represents the entire full-three problem.

The generic adelic curve and its rational model. We now identify this constructed full-three scheme with the precise generic curve used in the degree calculation. Use $B = (2, -3)$, $\mathcal{O} = \mathbb{Z}[i, (1 + j)/2]$, $\mu = 2j + ij$ and

$$E_0(x, y) = \mathrm{Trd}(\mu^{-1} x \bar{y}), \qquad a^\dagger = \mu^{-1}\bar{a}\mu, \qquad \mu^2 = -6.$$

The principal-datum calculation above supplies the integral unimodular positive form and its prescribed Rosati involution. In particular we do not replace it by the six-times form in the printed

source. Put $K_n = \ker(K_0 \to (\mathcal{O}/n\mathcal{O})^*)$ and $K_{35} = K_3 \cap K_5$. Everything in this paragraph is in characteristic zero. The integral three-marking construction has already been proved separately.

Take $V = B$ as a left B-module. Its B-linear automorphisms are right multiplications, and R_b scales E_0 by $\mathrm{nrd}(b)$. For an ordinary B^* coordinate rather than its opposite group use $b \mapsto R_{b^{-1}}$, whose multiplier is $\mathrm{nrd}(b)^{-1}$. This inversion preserves every congruence subgroup K_n. The identity $E_0(ax, y) = E_0(x, a^\dagger y)$ verifies the adjoint condition of the PEL datum in [27, §4.9]. For $z = x + iy$ in either half-plane set

$$v_z = (z, 1)^t, \qquad J_z = y^{-1} \begin{pmatrix} x & -|z|^2 \\ 1 & -x \end{pmatrix}.$$

Then $J_z^2 = -1$ and $J_z v_z = \mathrm{i} v_z$. Right multiplication by J_z is the complex structure for the period map $u \mapsto u v_z$. Deligne's homological convention has characters $w^{-1}, \bar{w}^{-1}$, so set $h_z(a + b\mathrm{i}) = R_{(a+bJ_z)^{-1}}$. In particular $h_z(\mathrm{i}) = -R_{J_z}$. On the positive half-plane $E_0(u, h_z(\mathrm{i})u) > 0$; on the other half-plane the effective positive form is $-E_0$. Negative-norm elements exchange these choices, as allowed by the homogeneous polarization in [27, §§4.8–4.10]. The left B-action on the complex Lie space has its reduced characteristic polynomial. The associated conjugacy class of cocharacters has the two standard weights and is rational. Thus the reflex field is $\mathbb{Q}$.

Conversely, for a polarized $\mathcal{O}$-surface over $\mathbb{C}$ with this Rosati action, $H_1(A, \mathbb{Q})$ is a faithful four-dimensional left B-module and hence has rank one, since B is division. In a rational frame its alternating form is a rational scalar multiple of E_0. Indeed write $E'(x, y) = E_0(x, Ty)$. The adjoint identities force T to commute with left B, so $T = R_b$. Alternation makes it self-adjoint for E_0, hence $b = \bar{b} \in \mathbb{Q}$. The complex structure is likewise right multiplication by an element squaring to -1, and gives one of the two displayed half-planes. Its effective polarization chooses the positive sign. These checks give the actual PEL and Lie-trace conditions of [27, §4.10] on every object, without fixing a determinant component.

In such a frame let $I = H_1(A, \mathbb{Z}) \subset B$. At a division prime $p = 2, 3$, a least-valuation element x of its full $\mathcal{O}_p$-stable lattice gives $I_p = \mathcal{O}_p x$, exactly by $w(yx^{-1}) \geq 0$ as above. At a split prime, matrix idempotents give $I_p = e_{11} I_p \oplus e_{22} I_p$ and e_{21}, e_{12} identify the two rank-two $\mathbb{Z}_p$-summands. For a basis v_1, v_2 of the first, $E_{ab} \mapsto e_{a1} v_b$ is a left $M_2(\mathbb{Z}_p)$-module isomorphism. It extends to an invertible right multiplication on B_p. Consequently $\widehat{I} = \widehat{\mathcal{O}} g$ for $g \in B(\mathbb{A}_f)^*$. Conversely $B \cap \widehat{\mathcal{O}} g$ is a full lattice with these completions: clear the finitely many denominators and apply the additive Chinese remainder theorem. This uses no multiplicative strong approximation or ideal-class-number assertion.

Choose $c > 0$ rational with $v_p(c) = v_p(\mathrm{nrd}(g_p))$ at every prime. The form $c^{-1} E_0$ on I is integral and unimodular, since in its local frame it is E_0 times the unit $\mathrm{nrd}(g_p)/c$. Choose the effective sign with the half-plane. Riemann's equivalence [27, Theorem 4.7] then gives the principally polarized surface. The uniqueness of the principal polarization with this Rosati involution in [29, p. 592] identifies the same polarization in our functor.

The actual analytic family is $\mathbb{C}^2/(I v_z)$. On a simply connected piece of a half-plane its homology local system is the constant lattice I. Changing rational frame by $b \in B^*$ changes (I, z) to (Ib^{-1}, bz), with the actual torus isomorphism given by $b v_z = (c_b z + d_b) v_{bz}$. The exponential sequence identifies $A[n] = (n^{-1} I)/I$ with I/nI as left $\mathcal{O}$-modules. A generator of $\widehat{I}$ thus supplies the actual torsion marking; its stabilizer is K_n. Every unit modulo p^a lifts to a local unit: lift it and a modular inverse and invert their product $1 + p^a t$ by its convergent geometric series. This applies also at division primes and proves the marking assertion at three directly. All these frames are symplectic similitudes, since right multiplication scales E_0 by reduced norm. No root of unity is trivialized.

Deligne's frames point out of the Tate module, whereas our markings point into torsion, so invert the frames in [27, §§4.11–4.12]. Explicitly put $a = g^{-1}$ when $\widehat{I} = \widehat{\mathcal{O}} g$. Changing generator

and rational frame gives $g \mapsto ugb^{-1}$ and $(z, a) \mapsto (bz, bau^{-1})$. Thus left relabeling $u \cdot \alpha = \alpha \circ r_u$ is the canonical right action by u^{-1}. We obtain the whole double quotient

$$B^* \backslash (\mathbb{H}^{\pm} \times B(\mathbb{A}_f)^*)/K$$

for $K = K_{35}$ or K_3. This includes all rational frames, lattices and determinant components. The explicit homology transitions and [27, §4.15] identify the analytic families and their markings, not just their geometric points.

To identify the same model over $\mathbb{Q}$, embed this PEL group in $\mathrm{GSp}(V, E_0)$. Proposition 1.15 of [27] gives a compact open K' containing K_{35} for which the PEL-to-Siegel map is a closed immersion. Intersect K' with the integral stabilizer of $\mathcal{O}$ and principal Siegel level three, both containing K_{35}. This shrinking preserves that conclusion: the equivalence relations in the proof decrease and the maps at the sufficiently small levels are finite and unramified. This Siegel level is fine and has its $\mathbb{Q}$-model from full principal level $N \geq 3$ and a finite level quotient [27, §§4.16–4.17].

There is an actual $\mathbb{Q}$-morphism from the constructed X_{35} to this Siegel model: forget the $\mathcal{O}$-action and take the K'-orbit of its frame. Choose N divisible by fifteen with principal Siegel level N contained in K'. Refine the $\mathcal{O}$-marking to level N on its finite etale surjective marking cover. The lattice calculation supplies geometric nonemptiness. The resulting K'-orbit is independent of refinement, since two refinements agreeing at both three and five differ by $K_{35} \subset K'$. It therefore descends over $\mathbb{Q}$. The primitive integral lattice and the degree-one effective polarization in Deligne's construction are the same ones used here, rather than an isogenous or scaled replacement.

Over $\mathbb{C}$ this morphism is the closed immersion just described. Its PEL image is a canonical closed subscheme Z over the reflex field $\mathbb{Q}$ [27, Theorem 4.21, Corollary 5.7 and Example 5.8]; the homological convention agrees with [102, 2.2.5 and 2.3.1]. The pullback of the ideal of Z vanishes after extension to $\mathbb{C}$, hence already over $\mathbb{Q}$. The morphism factors through Z and becomes an isomorphism over $\mathbb{C}$, so faithful flatness makes it an isomorphism over $\mathbb{Q}$. This is the actual map to a fine moduli scheme, not descent of an arbitrary complex isomorphism.

The projection at five gives $K_3/K_{35} = (\mathcal{O}/5\mathcal{O})^*$, by the local unit-lifting calculation and independence of the factors at three and five. The relabeling action and the canonical finite-adelic action agree over $\mathbb{C}$ by the explicit u^{-1} conversion. Both are $\mathbb{Q}$-defined, so they agree over $\mathbb{Q}$. Their quotients agree, and the already constructed $Y = X_{35}/G_5$ is the canonical level-K_3 curve. The auxiliary five-marking contributes no factor to its degree. The central -1 changes a full-three marking to its negative, which is isomorphic by $[-1]$. On X_{35} it changes both markings; changing only the five-marking is not a stabilizer. This agrees with the freeness and center calculation above.

Finally forget the three-marking. The unlevelled polarized curve is connected: Buzzard's $V_1(5)$ curve is geometrically connected and surjects onto this moduli problem, since every geometric object admits a five-marking by the split matrix-module calculation. Its distinguished analytic component $\mathbb{H}/\Gamma$ is therefore the whole curve, with the same family $\mathbb{C}^2/(\mathcal{O}v_z)$ and principal form E_0. The fixed A001 application of [23, Theorem 2.1] identifies its polarized coarse forgetful relation with the full positive-normalizer group W. The four arithmetic parity classes above are exactly these classes: i, j normalize $\mathcal{O}$, $1 + i$ is a unit of norm -1, and $i(1 + i) = 2 + i$ and j have positive norms two and three. They give the independent local parity classes, with no omitted sign. The $\mathbb{Q}$-defined coarse model and its already fixed marked coordinate therefore give the same map $Y \to \mathbf{P}^1_j$. This is a composition of the actual $\mathbb{Q}$-defined forgetful maps, not an unspecified fractional-linear change of j. The normalizer index calculation now applies to this identified $Y_3 = Y_{\mathbb{Q}}$ and this same j, and gives the stated bound 324. The full generic connection and its source hypotheses are included in Section A33.

Each geometric component of Y_3 is a torsion-free cover of the norm-one orbifold of Euler characteristic $-1/3$, and hence has genus at least two. The normalizer action extends to its smooth proper

model by uniqueness of the stable model, including permutations of components. Fix $p > 324$ and work after strict henselization. The action is tame. On each component use its effective action. A finite-order automorphism of order prime to p cannot specialize to the identity, by considering its first nonzero term modulo a power of the uniformizer. At a fixed point the same power-series argument shows that the tangent character of the stabilizer is faithful. Thus the stabilizer is cyclic.

For a nonidentity generator the tangent eigenvalue minus one is a unit. Its fixed-point equation therefore has a unique lift by Hensel's lemma. All powers of the generator fix that lift. It follows that no new branch orbit appears and that distinct branch orbits cannot collide: their stabilizers at a purported common specialization would belong to one cyclic group, whose nontrivial elements have the same unique fixed lift. Averaging a parameter gives a local invariant ring $R[[z^e]]$. Hence the quotient is smooth and retains the three disjoint branch sections and their indices $4, 6, 2$.

Its genus-zero model with these three ordered sections is isomorphic to $\mathbf{P}^1_{\mathbb{Z}_p}$ carrying the ordered triple $(0, \infty, -16/27)$, which is disjoint since 2 and 3 are units. The isomorphism agrees with j on the generic fiber, by uniqueness of a projective coordinate on three points. Thus it is the integral model in the stated j-coordinate; no further denominator prime has been discarded.

Pull back by $J(u) = -57600F(u)^2/G(u)^6$. If

$$H(u) = (u^2 - 60u - 1800)(13u^2 + 480u + 3600)(11u^2 + 600u + 7200),$$

direct calculation gives $G^6 - 97200F^2 = H^2$. The pullback indices are $2, 6, 2$. Tame normalization replaces e by $e/\gcd(e, a)$ and leaves $2, 1, 1$. The discriminants and resultants have prime support at most 13, so this persists at $p > 324$. The normalized cover of $\mathbf{P}^1_u$ has degree at most 324, not $12 \cdot 324$, and is branched only at the six contacts. Locally it has equation $u - c = z^2$ after unramified extension. Positive even contact valuation is unramified; negative valuation specializes to the unbranched point at infinity. This is exactly the support specified by $M_S(u)$.

Let K be the splitting field of the fiber over u of the normalized cover. The fiber is finite étale of degree at most 324, so $K/\mathbb{Q}$ is normal with $[K : \mathbb{Q}] \le F_0 = 324!$. The preceding local calculation makes it unramified outside $T = S \cup \operatorname{supp} M_S(u)$. Choose a fiber A/K. The proper model and its universal abelian scheme give good reduction outside the places over S.

All conjugates of A have the same geometric polarized moduli point. Indeed the polarized equivalence relation in [23, Theorem 2.1], with the corrected principal datum above, gives $V_{\mathbb{Z}} = V/W$; its two twisting involutions generate all of W for discriminant six. On the finite nonzero j-locus, (1) specifies the same point by its weighted Igusa invariants. Here $j = J(u)$ is rational, finite and nonzero: $F(u) \neq 0$ and $\operatorname{disc}(G) = -10800 < 0$. The exact coordinate connection is [25].

To define the Hom field without choosing a Galois action on a labelled collection, form $W_0 = \operatorname{Res}_{K/\mathbb{Q}} A$, of dimension $g = 2[K : \mathbb{Q}]$. Write $W_0(\mathbb{C}) = \mathbb{C}^g/\Lambda$. By [21, Proposition 6.5, pp. 318–319], every geometric endomorphism lifts uniquely to a complex-linear map preserving Λ. Since Λ spans $\mathbb{C}^g$ over $\mathbb{R}$, its action on Λ is faithful. Thus the geometric endomorphism group is a subgroup of $\operatorname{End}_{\mathbb{Z}}(\Lambda)$, hence is free of rank

$$r \le (2g)^2 = 16[K : \mathbb{Q}]^2 \le 16F_0^2.$$

Each member of a finite integral basis is defined over a finite extension, so the Galois image on this lattice is finite. Every finite subgroup of $\mathrm{GL}_r(\mathbb{Z})$ injects modulo three. Indeed, a nontrivial kernel would contain an element M of prime order q. Write $M = I + 3^k B$, with $k \ge 1$ and $B \not\equiv 0 \pmod 3$. If $q \neq 3$, expansion modulo 3^{k+1} gives $M^q - I \equiv q3^k B \not\equiv 0$. If $q = 3$, expansion modulo 3^{k+2} gives $M^3 - I \equiv 3^{k+1}B \not\equiv 0$, again a contradiction. If E is the kernel field of the Galois action and $L = EK$, then $L/\mathbb{Q}$ is normal and

$$[L : \mathbb{Q}] \le F_0\, 3^{r^2} \le F_0\, 3^{256F_0^4} = D.$$

At a prime p outside T, choose $\ell \neq p$. The field K is unramified at p, and A has good reduction at every place above p. The residue fields are finite and hence perfect, so [119, Theorem 1, p. 493] makes inertia trivial on the Tate module at every such place. The decomposition $K \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} \simeq \prod_{\sigma : K \hookrightarrow \overline{\mathbb{Q}}} \overline{\mathbb{Q}}$ identifies $W_{0,\overline{\mathbb{Q}}}$ with the product of the conjugates of A, with Galois acting by transport of factors. Thus $V_\ell(W_0) = \operatorname{Ind}_{G_K}^{G_{\mathbb{Q}}} V_\ell(A)$. Unramifiedness of K makes inertia fix the coset indices, so it acts trivially on this entire Tate representation. The action of geometric endomorphisms on $V_\ell(W_0)$ is faithful. Indeed, after fixing $\overline{\mathbb{Q}} \hookrightarrow \mathbb{C}$, the finite torsion schemes have the same geometric points, and [21, equation (41), p. 334] gives $T_\ell(W_0) \simeq \Lambda \otimes \mathbb{Z}_\ell$ compatibly with endomorphisms. An endomorphism with zero induced map therefore has zero integral lattice matrix, and hence is zero. This realization is Galois equivariant because it is defined on torsion points. Inertia consequently fixes the full endomorphism lattice by conjugation. Thus E, and hence L, is unramified outside T. Over K the conjugate factors of W_0 are individually defined. Their inclusions and projections identify its geometric endomorphism lattice with the direct sum of all ordered Hom groups between these factors. The group G_E fixes every endomorphism and G_K fixes the factor labels. Their intersection G_L therefore fixes every such morphism. Since its source and target are defined over L, Galois descent for morphisms defines it over L. In particular L defines every geometric endomorphism and every ordered conjugacy homomorphism of A.

By [21, Theorems 6.7–6.8 and Corollary 6.9, p. 319], polarizable complex tori are abelian varieties and their analytic homomorphisms are algebraic. After clearing lattice denominators, the same assertion identifies rational Hodge endomorphisms with algebraic quasi-endomorphisms. These endomorphisms are defined over $\overline{\mathbb{Q}}$: every automorphism of $\mathbb{C}/\overline{\mathbb{Q}}$ fixes their action on all torsion points, and torsion is Zariski dense. On rational homology the complex structure is right multiplication by an element $J \in B \otimes \mathbb{R}$ with $J^2 = -1$. In $\operatorname{End}_{\mathbb{Q}}(B) = B \otimes B^{\mathrm{op}}$, commutation with J requires each right coefficient to commute with J. Its rational centralizer is either $\mathbb{Q}$ or an imaginary quadratic field k. Hence the geometric endomorphism algebra is either B or $M_2(k)$. We treat these two cases separately. Throughout, put $M = M_S(u)$, $d = [L : \mathbb{Q}] \leq D$, and $Q = Q_R$.

CM fibers. Suppose $\operatorname{End}^0(A_{\overline{\mathbb{Q}}}) = M_2(k)$. All its elements are defined over L. Multiples of matrix units therefore give an elliptic curve E/L with complex multiplication and an L-isogeny $E^2 \to A$. Let $\Delta < 0$ be the discriminant of the full geometric endomorphism order of E. The degree of $j(E)$ is its order class number, so $h(\Delta) \leq d \leq D$. The effective bound of [40, Proposition 2.7] gives

$$\sqrt{|\Delta|} \leq 2D^2 \exp(21000D^2).$$

This bound includes every imaginary quadratic order and has no exceptional-field exclusion.

Here is the height comparison with its normalization explicit. After a finite extension E has everywhere good reduction, since its j-invariant is integral. This extension does not change its stable Faltings height. For a reduced CM period $\tau = x + iy$ we have $\sqrt{3}/2 \leq y \leq \sqrt{|\Delta|}/2$. The archimedean summand in the original Faltings normalization is

$$-\log(2\pi) + \frac{\pi y}{6} - \frac{1}{2}\log y - 2\sum_{n \geq 1} \log|1 - e^{2\pi i n \tau}|,$$

by the discriminant formula [41, Eq. (6.2)]. For $\rho = e^{-2\pi y} < 1/100$, the final sum is at most $2\rho/(1-\rho)^2 < 200/9801$, and $-\frac{1}{2}\log y < 1/10$. Averaging and discarding the negative $-\log(2\pi)$ gives

$$h_F(E) \leq \frac{\pi\sqrt{|\Delta|}}{12} + 1 \leq D^2 3^{21000D^2} + 1 = B_{\mathrm{cm}}.$$

Apply the published isogeny estimate [39, Theorem 1.4] over the original field L, with source E^2. It gives an isogeny to A of degree at most

$$\bigl(28^{256} d \max(h_F(E^2), \log d, 1)^2\bigr)^{8192}.$$

The maximum is at most $2B_{\mathrm{cm}}$. Height variation under isogeny [44] and additivity give

$$\begin{aligned} h_F(A) &\le 2B_{\mathrm{cm}} + 4096\bigl(256 \log 28 + \log d + 2\log(2B_{\mathrm{cm}})\bigr) \\ &\le 2B_{\mathrm{cm}} + 4096(1028 + 5D + 84000D^2) = C_{\mathrm{cm}}. \end{aligned}$$

For the last line use $\log 28 < 4$, $\log d \le D$, and $\log(2B_{\mathrm{cm}}) \le 2 + 2D + 42000D^2$. No bound for the degree of the auxiliary good-reduction extension enters the isogeny estimate.

Non-CM fibers and scalar descent. Now suppose $\mathrm{End}^0(A_{\overline{\mathbb{Q}}}) = B$. In particular A is absolutely simple. For each $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ choose a geometric isomorphism $\psi_\sigma : {}^\sigma A \to A$. The automorphism $b \mapsto \psi_\sigma {}^\sigma b \psi_\sigma^{-1}$ of B is inner. Correcting it by Skolem–Noether and clearing denominators gives a B-linear isogeny $\mu_\sigma : {}^\sigma A \to A$, defined over L because all the relevant Hom groups are defined there. Take $\mu_1 = 1$ and inflate this finite choice to $G_{\mathbb{Q}}$. The discrepancy

$$c(\sigma, \tau) = \mu_\sigma {}^\sigma \mu_\tau \mu_{\sigma\tau}^{-1}$$

commutes with B, hence belongs to $\mathbb{Q}^\times$. Set $a_\sigma = (\deg \mu_\sigma)^{-1/4} > 0$ and $\nu_\sigma = a_\sigma \mu_\sigma$ in the category with algebraic coefficients. Throughout these coefficient extensions Galois acts on the varieties and morphisms and acts trivially on the added scalars; the induced homology maps are extended linearly. In particular, the a_σ are constant coefficients in the products below. Here degree extends to rational quasi-isogenies by $D(u) = \deg([m]u)/m^4$, for any positive integer m clearing denominators. This is independent of m: comparison after multiplication by a second denominator uses $\deg[n] = n^4$ and multiplicativity [116, Theorem 8.2, p. 115]. It is multiplicative and invariant under field conjugation. Writing $c = r/s \in \mathbf{Q}^\times$ with $s > 0$, the equality $[s](cu) = [r]u$ gives $D(cu) = c^4 D(u)$. Thus

$$D(\mu_\sigma) D(\mu_\tau) = c(\sigma, \tau)^4 D(\mu_{\sigma\tau}), \qquad \frac{a_\sigma a_\tau}{a_{\sigma\tau}} = \frac{1}{|c(\sigma, \tau)|}.$$

The positive fourth-root convention therefore gives $\varepsilon(\sigma, \tau) = c(\sigma, \tau)/|c(\sigma, \tau)|$. Consequently

$$\nu_\sigma {}^\sigma \nu_\tau = \varepsilon(\sigma, \tau) \nu_{\sigma\tau}, \qquad \varepsilon(\sigma, \tau) \in \{1, -1\}.$$

Thus ε is a normalized sign cocycle inflated from $\mathrm{Gal}(L/\mathbb{Q})$. Only finitely many algebraic coefficients a_σ occur.

We need a splitting with controlled ramification, not merely existence of a projective lift. Identify $H^2(G_{\mathbb{Q}}, \{1, -1\})$ with $\mathrm{Br}(\mathbb{Q})[2]$. At a finite prime outside T the class of ε is zero: its restriction is unramified and the unramified local quotient has cohomological dimension one. Let T_- be the finite primes where its invariant is $1/2$. For each odd $p \in T_-$ choose an odd character of $(\mathbb{Z}/p\mathbb{Z})^\times$; at $2 \in T_-$ use the nontrivial character modulo four. Their product is a Dirichlet character χ, unramified outside T, whose order m satisfies

$$m \le 2 \prod_{\substack{p \in T \\ p \ne 2}} (p - 1) \le 2R^R M.$$

Here the product over an empty set is one.

To check that χ has the required square-root obstruction, use the exact sequence of constant discrete Galois modules $1 \to \{\pm 1\} \to \mu_\infty \xrightarrow{z \mapsto z^2} \mu_\infty \to 1$. By local reciprocity, a finite-order character of $G_{\mathbb{Q}_p}$ has a character square root exactly when its value on $-1 \in \mathbb{Q}_p^\times$ is 1. Indeed this is the only obstruction on the torsion part of $\mathbb{Q}_p^\times$; the valuation factor and the procyclic principal-unit factors admit such roots. Hence the boundary of χ and $[\varepsilon]$ agree at every finite place: the local two-torsion Brauer group has order two, and the square-root criterion distinguishes its two classes. The global Brauer exact sequence determines the real invariant from the finite ones and then identifies the two global classes [54, I.1.6(b) and proof of II.2.1, p. 164]. Choosing a continuous pointwise square root of χ and correcting it by a sign cochain consequently gives a continuous cochain $\beta : G_{\mathbb{Q}} \to \mu_{2m}$ such that

$$\beta^2 = \chi, \qquad \beta(\sigma)\beta(\tau)\beta(\sigma\tau)^{-1} = \varepsilon(\sigma, \tau).$$

Its restriction to G_L is a character invariant under $G_{\mathbb{Q}}$. Outside T, its inertia restrictions are quadratic and nontrivial at only finitely many primes. Twisting by the quadratic character of the product of those odd primes kills all these restrictions: at an odd prime there is a unique nontrivial quadratic character on inertia. This introduces ramification only at those primes and possibly at $2 \in T$, and changes neither displayed identity. We henceforth use this choice of β.

Let L' be the kernel field of $\beta|_{G_L}$. Conjugacy invariance shows that $L'/\mathbb{Q}$ is Galois, and

$$[L' : \mathbb{Q}] \le 2md \le 4DR^R M \le QM.$$

The last inequality follows from $R \ge \max(D, 10^{12})$ and $E_0 = 10(DG_0)^2$.

Modularity and a uniform conductor bound. Choose one number field C containing all a_σ, the values of β, and a splitting field of B. For each coefficient prime ℓ choose an embedding of C into $\overline{\mathbb{Q}}_\ell$ and extend scalars on $V_\ell(A)$ to a finite extension containing its image. With the constant-coefficient convention above, define

$$\rho_\ell(\sigma) = \beta(\sigma)^{-1} a_\sigma \mu_\sigma \circ \sigma.$$

The cocycle identities make this a representation of $G_{\mathbb{Q}}$. It commutes with B, so matrix units give $\rho_\ell \simeq r_\ell \oplus r_\ell$ with $\dim r_\ell = 2$. On $G_{L'}$ it is the actual Tate action of A. Faltings' semisimplicity and endomorphism theorems [44] show that $r_\ell|_{G_{L'}}$ is absolutely irreducible: after splitting B, its commutant is the scalars. It is unramified outside $T \cup \{\ell\}$, since there L is unramified, A has good reduction, and β is unramified.

Apply Lemma 4.1 with $t_\sigma = a_\sigma/\beta(\sigma)$. It identifies this same r_5 with the Tate realization of a primitive weight-two newform f. The qualitative modularity and finite-twist argument does not replace A, L', β, or r_ℓ in the ramification and height bounds below. For this final newform, Li's Fricke formula [36, p. 296, before Lemma 6] makes the span of its coefficient conjugates Fricke-stable. Shimura's quotient construction and Euler-factor identity [37, Theorem 1] therefore show that r_5 is a constituent of the Tate module of the modular abelian variety A_f, or equivalently $r_5^\vee$ is its cohomological constituent. This duality preserves the conductor.

The representations at all coefficient primes belong to this same system. Enlarge C to contain the coefficients of f, extending the original embeddings and retaining the chosen embedding at five. Fix a coefficient prime ℓ. Work over a common finite extension of $\mathbb{Q}_\ell$ containing both coefficient images. For $p \notin T \cup \{5, \ell\}$ away also from the level of f, choose a place v of L and an arithmetic Frobenius lift $\sigma \in G_{\mathbb{Q}_p}$ whose residue action is $x \mapsto x^p$. Since $L_v/\mathbb{Q}_p$ is unramified and A has good reduction, relative p-Frobenius maps $\overline{A}_v$ to the reduction of ${}^\sigma A$ at v. The actual isogeny

μ_σ extends over the local abelian schemes. Composing its reduction with this Frobenius gives an endomorphism u_p of $\overline{A}_v$. Its rational Tate trace t_p is independent of the coefficient prime different from p. Functorial specialization and the two-copy decomposition give

$$\operatorname{tr} r_\ell(\sigma) = \iota_\ell(a_\sigma \beta(\sigma)^{-1} t_p/2).$$

At five the algebraic number in parentheses equals $a_p(f)$. Injectivity of the embedding at five makes this an equality in C, so transport to ℓ gives the trace of the corresponding classical Tate realization $U_{f,\ell}$. Chebotarev and the characteristic-zero semisimple trace criterion identify r_ℓ with $U_{f,\ell}^{\mathrm{ss}}$. Since r_ℓ is absolutely irreducible, this semisimplification has one simple composition factor; hence $U_{f,\ell}$ itself is irreducible and isomorphic to r_ℓ. For each residue prime p choose $\ell \neq p$. Restricting this isomorphism to the decomposition group at p identifies the local Galois representations and their Artin conductors. The local conductor comparison for the primitive newform then identifies this exponent with $v_p(N_f)$ [120, §§2.1, 2.4, 4.1].

Here are uniform local bounds, including the small primes. For a finite extension of $\mathbb{Q}_p$ of degree at most a, its normal closure has degree at most $a!$. For a finite Galois local extension with ramification index e and different exponent δ, let f be the Eisenstein polynomial of a uniformizer π over the maximal unramified subfield. Then $\delta \leq e - 1 + e v_p(e)$. The derivative terms have distinct valuations modulo e, so there is no cancellation at the smallest valuation. The identity $f'(\pi) = \prod_{\sigma \neq 1}(\pi - \sigma\pi)$ over the inertia conjugates gives $\delta = \sum_{i \geq 0}(|G_i| - 1)$. Changing variables by the Herbrand function yields $\delta/e = 1 - 1/e + \int_0^\infty (1 - |G^u|^{-1})\, du$. Before the last positive upper break the integrand is at least $1/2$, so that break is bounded by $2\delta/e \leq 2(1 + \log_2(a!)) \leq 4a^2$ [43, Sections 4.1–4.3].

At p, take $F = L_v$ and adjoin $A[3]$, using $A[5]$ if $p = 3$. This extension has degree at most DG_0 over $\mathbb{Q}_p$. Raynaud's criterion [42, Theorem 3.5] gives semistable reduction there. Its wild inertia acts trivially on each $V_\ell(A)$ with $\ell \neq p$. The preceding break bound and the subgroup rule for upper numbering therefore show that all absolute ramification groups above $U_0 = 4(DG_0)^2$ act trivially on that Tate module. On these groups $a_\sigma \mu_\sigma = 1$ because they lie in G_L. For odd p, the character $\beta|_{G_F}$ is trivial on wild inertia: its square is the tame character $\chi|_{G_F}$, and a pro-p group has no quotient of order two. At $p = 2$, local reciprocity and the norm map give $\beta^4 = 1$ on the units of F, since the 2-primary local factor of χ has conductor at most four. If $e = v_F(2) \leq D$, the logarithm for $\widehat{\mathbf{G}}_m$ identifies U_F^{e+1} with $\mathfrak{m}_F^{e+1}$ since $e + 1 > e$ [60, IV.6.4, p. 132]. Multiplication by four gives $(U_F^{e+1})^4 = U_F^{3e+1}$. Thus the relative upper break of β is at most $3e$. To make the change of numbering explicit, the same normal-closure estimate bounds the last upper break of $F/\mathbb{Q}_2$ by $4D^2$. Above it, the absolute upper group $G_{\mathbb{Q}_2}^u$ lies in G_F, and the subgroup formula is

$$G_{\mathbb{Q}_2}^u \cap G_F = G_F^{\psi_{F/\mathbb{Q}_2}(u)}, \qquad \psi_{F/\mathbb{Q}_2}(u) \geq u \quad (u \geq 0).$$

Consequently β is trivial on every absolute upper group with $u > \max(4D^2, 3e)$, which is at most U_0. This proves the asserted absolute bound without a factor involving $[L' : \mathbb{Q}]$. The Artin conductor exponent of r_ℓ is consequently at most $2 + 2U_0 \leq E_0$.

For $p > R$, the extension $L_v/\mathbb{Q}_p$ is tame, since its degree is at most $D < p$. The surface has good reduction, and the same argument kills the wild part of β. Hence the conductor exponent is at most two there. Outside T it is zero. Local–global compatibility for the weight-two modular representation therefore bounds the primitive level N of f by

$$N \leq \Big(\prod_{p \leq R} p^{E_0}\Big) M^2 \leq R^{RE_0} M^2 = Q M^2.$$

Transfer from a modular Jacobian. Over L' the Tate modules of A and $J = J_1(N)$ have a common constituent. Invariants in their finite-dimensional Hom space commute with extension of the coefficient field. Faltings' Hom theorem thus gives a nonzero L'-homomorphism $A \to J$. Let $B_0 \subseteq J$ be the image of this homomorphism. Absolute simplicity implies that $A \to B_0$ is an L'-isogeny. Apply [116, Proposition 12.1, p. 122] over L': for an ample line bundle $\mathcal{H}$ on J, the reduced identity component Z of the kernel of $J \xrightarrow{\phi_{\mathcal{H}}} J^\vee \to B_0^\vee$ is a complementary abelian subvariety over L'. Indeed, its intersection with B_0 is finite because the restriction is the polarization associated with $\mathcal{H}|_{B_0}$, and $\dim Z \geq \dim J - \dim B_0$. Thus $B_0 \times Z \to J$, and hence $A \times Z \to J$, are L'-isogenies. In particular $g = \dim J \geq 2$. The index of $\Gamma_1(N)$ is at most N^2, so both the genus and the degree of its standard Belyi map are at most N^2. By [38, Theorem 1.1.1 and Section 2.3],

$$h_F(J) \leq 13000000N^{12} \leq Q^{13}M^{24}, \qquad g \leq Q^3M^4, \qquad d' = [L' : \mathbb{Q}] \leq QM.$$

The normalization in that theorem is the original stable Faltings height of the Jacobian.

Apply [39, Theorem 1.4] with source J and target $A \times Z$, both over L'. There is such an isogeny φ with

$$\log \deg \varphi \leq 1024g^3\bigl(64g^2 \log(14g) + \log d' + 2 \log \max(h_F(J), \log d', 1)\bigr).$$

For $Q \geq 10^{12}$ and $M \geq 1$, the three logarithms in order are at most $5QM^4$, $2QM^4$, and $19QM^4$. These follow from $\log Q \leq Q$ and $\log M \leq M^4/4$. It follows that

$$\log \deg \varphi \leq 368640Q^{16}M^{24} \leq Q^{17}M^{24}.$$

The general lower bound $h_F(Z) \geq -\frac{3}{2} \dim Z \geq -2g$ is recorded in [39, Section 9]. Additivity and height variation now give

$$h_F(A) \leq Q^{13}M^{24} + \tfrac{1}{2}Q^{17}M^{24} + 2Q^3M^4 \leq Q^{17}M^{24}.$$

Thus both cases prove the exact estimate needed below:

$$\max(1, h_F(A)) \leq \max(1, C_{\mathrm{cm}}, Q_R^{17}) M_S(u)^{24}.$$

Use the same Hauptmodul, normalized by the values $0, \infty, -16/27$ at the CM points of discriminants $-4, -3, -24$. The calculation leading to (1), using [30, equations (13)–(14), Examples 29–30, pp. 793–797] and [25], gives in one common weighted-projective representative

$$J_2 = 12(j+1), \quad J_4 = 6(j^2+j+1), \quad J_{10} = j^3, \qquad \psi_4 = J_2^2 - 24J_4 = 144j, \quad \chi_{10} = -j^3/4.$$

Write $z_m = \theta_m(\tau, 0)$ for the same principally polarized surface, where $m = (a/2, b/2)$ with $a, b \in \{0,1\}^2$; the characteristic is even when $a^t b = 0 \pmod 2$. Igusa's theta series and identities [46, pp. 833–834 and 848] give $\psi_4 = 2^{-2} \sum_{m \text{ even}} z_m^8$ and $\chi_{10} = -2^{-14} \prod_{m \text{ even}} z_m^2$. These are the half-characteristic coordinates used at $r = 2$ in [48, §2.3.4, pp. 30–31]; the exact convention and substitution are recorded in [47]. The quotient χ_{10}^2/ψ_4^5 is unchanged by weighted rescaling: multiplying a weight-k coordinate by s^k multiplies both numerator and denominator by s^{20}. In particular, the weight-dependent normalization in [30, §5, p. 792] cancels in this quotient. Since our j is finite and nonzero, its denominator is nonzero, and substitution gives

$$j = 16 \cdot 144^5 \frac{\chi_{10}^2}{\psi_4^5} = 3779136 \frac{(\prod_{m \text{ even}} z_m)^4}{(\sum_{m \text{ even}} z_m^8)^5}.$$

The two homogeneous polynomials have degree 40 and coefficient ℓ^1 norms 3779136 and 10^5. Hence $h(j) \leq 40h([z_m]) + 22$. The product and sum use the ten even theta nulls of the same principally polarized surface. Adjoining the six zero odd coordinates does not change their projective height.

In [48], the level $r = 2$ theta coordinates are the sixteen half-characteristic values used here; the six odd values vanish. The permitted changes of theta structure permute coordinates and multiply them by roots of unity. His Euclidean projective height dominates our supremum projective height. Put $X = \max(1, h_F(A))$ and $X_D = \max(1, h_F(A) + \log \pi) \le 3X$. His Corollary 1.3(2), in the Deligne metric, gives

$$h_\Theta(A) \le \tfrac{1}{2} X_D + C_2 \log(X_D + 2), \qquad C_2 = 1000 \cdot 2^4 (\log(2^4))^5 \le 1000 \cdot 2^{24}.$$

Using $\log(X_D + 2) \le 2X_D$ proves $h([z_m]) \le 10000 \cdot 2^{24} X$.

Write $j = a/b$ in lowest terms. The nonzero integer polynomial $-57600bF(X)^2 - aG(X)^6$ has u as a root and coefficient norm at most $2 \cdot 4000^6 \max(|a|, |b|)$. Remove any initial power of X and use the actual highest nonzero coefficient. The rational-root theorem bounds both numerator and denominator of u, giving $h(u) \le h(j) + \log(2 \cdot 4000^6) < h(j) + 74$.

A field extension for the theta structure changes neither absolute height. Since $40 \cdot 10000 \cdot 2^{24} + 96 < 10^{13}$, the preceding inequalities give

$$h(u) \le 10^{13} \max(1, h_F(A)) \le 10^{13} \max(1, C_{\mathrm{cm}}, Q_R^{17}) M_S(u)^{24} = \tfrac{1}{2} H_6(S; R) M_S(u)^{24}.$$

Every estimate used only the displayed lower bound on R, so the conclusion holds for every permitted larger R as well. □

The final value argument uses $S_{\mathrm{ht}} = \{p : p \le 324\}$. Multiplication uses $S_{\mathrm{mul}} = S_{\mathrm{ht}} \cup \mathcal{H} \cup \mathrm{supp}(c_*)$, where $\mathcal{H}$ is the fixed catalogue. A common numerical R may bound both sets. It does not identify them.

5 The pole predicate and its converse input

The pole construction and simultaneous-descent strategy adapt [94, Section 3].

Let $S_0 = \{2, 3, 5, 7, 13\}$. With input B and eleven witnesses $b_0, \dots, b_5, e, z, r_1, r_2, w$, define

$$\begin{aligned} m &= (1 - 2b_0)(1 + b_1)(1 + 2b_2)(1 + 4b_3)(1 + 6b_4)(1 + 12b_5), \\ A &= mB, \qquad H = A(A - B^2), \\ T_1 &= A(A - e)(A + 3e), \qquad T_2 = A(Az^2 - eH)(Az^2 + 3eH), \\ Q_{\mathrm{pole}} &= B\Big(\sum_{i=0}^{5} (b_i^2 - b_i)^2 + (eHzw - 1)^2 \\ &\qquad + (r_1^2 - T_1)^2 + (r_2^2 - T_2)^2\Big). \end{aligned}$$

Write $\Phi(B)$ for rational solubility. For $B \ne 0$, m is a signed divisor of 2730, and the equations give points $(A, r_1) \in E_e$ and $(Az^2, zr_2) \in E_{eH}$ with $eHz \ne 0$, where $E_\ell : y^2 = x(x - \ell)(x + 3\ell)$. The polynomial has total degree 75.

Lemma 5.1 (One-sided obstruction)**.** *If $B \ne 0$ and $\Phi(B)$, then $v_p(B)$ is not negative and odd for any $p \notin S_0$.*

Proof. If $k = v_p(B) < 0$ were odd, then $v_p(A) = k$, $v_p(H) = 3k$. Exactly one of e, eH has even valuation. Square scaling makes its parameter a unit, preserving the parity of the abscissa's valuation. For a unit parameter the cubic has valuation $v_p(x)$ if $v_p(x) > 0$ and $3v_p(x)$ if $v_p(x) < 0$. Its being a square forces even valuation. But both points have abscissa in the squareclass A, a contradiction. □

Theorem 5.2 (Pole completeness, PC)**.** *For every rational B with square reduced denominator, $0 < B < 1/2$ and $v_p(B) \geq 4$ for every $p \in S_0$ imply $\Phi(B)$.*

The following statement is the $E = E_\ell$, $s_2(E) = 1$ case of [90, Theorem 1.1], and also follows as a special case of [91, Theorem 1.1]. We give the restricted argument here; these preprints are cited for attribution, not assumed as additional converse inputs.

Theorem 5.3 (Restricted converse, RC)**.** *For every positive squarefree $\ell \equiv 11 \pmod{12}$,*

$$\operatorname{corank}_{\mathbb{Z}_2} \operatorname{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1.$$

Here $c_4 = 208\ell^2$, $\Delta = 2304\ell^6$, and $j = 35152/9$. The calculation in (9) and the rational-integrality argument following it show that these curves are non-CM, using the monic integral CM class polynomial of [40, §2.1, pp. 612–613]. They are different from the CM generator curve. Theorem 5.3 is proved in Sections 7–9, with its coefficient and normalization arguments in Appendices A5–A7. We now prove Theorem 5.2 from this restricted converse.

5.1 The exact descent reduction

The signed divisors of 2730 realize all real, 2-adic and 3-adic squareclass signatures. After $-1, 2, 3$ fix sign and valuation parities, the remaining unit vectors of $5, 7, 13$ are $(0, 1, 1), (1, 0, 0), (0, 1, 0)$ and form a basis. Choose positive m making $A = mB$ square at two and a nonsquare unit squareclass at three. Let β, θ be the positive squarefree representatives of A, H. Then $\beta \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, with disjoint supports, $\beta > 1$, and $\theta = 1$ permitted. Each is a local square at the support of the other. At a prime of β, the relevant odd valuation of B is positive, so $1 - B/m$ is a square unit. At a prime of θ, $v(B)$ must be zero, since otherwise $v(H)$ is even; there $B/m \equiv 1$, so A is a square.

The simultaneous construction in S11–S12 gives $n, n\theta$ positive squarefree and congruent to eleven modulo twelve, both with two-Selmer dimension three and with extra Kummer class $(\beta, 1)$. Its matrix conditions can be stated explicitly. For support primes q_i, set $s_i = [-1/q_i]$, $\lambda_i = [3/q_i]$, and

$$(Kv)_i = \sum_{j \neq i} [q_j/q_i](v_j + v_i).$$

One arranges $\ker K = \langle \mathbf{1}, \mathbf{1}_{\operatorname{supp}\beta} \rangle$ and $\lambda \notin \operatorname{im} K$ on both supports. Add $d - 1$ prime slots of type $(s, \lambda) = (0, 0)$ and one slot q_0 of type $(1, 1)$, where $d = \omega(\beta)$. The cross-block has columns $e_i + e_d$. After the rank-one correction $K^\sharp = K + \lambda a^t$, deleting q_0 leaves

$$\begin{pmatrix} D_1 & C_1 \\ C_1^t & 0 \end{pmatrix}, \qquad \mathbf{1}^t D_1 \mathbf{1} = 1.$$

This is nonsingular: $C_1^t x = 0$ makes x constant, and the first equation forces that constant and then y to vanish. Adding the support of θ preserves this block. Free q_0-symbols toggle the new diagonal entries; the determinant's full-product coefficient is one. Thus a single Boolean assignment works on both supports. Reciprocity, CRT and Dirichlet realize it by distinct auxiliary primes.

In Kummer coordinates $(x + 3\ell, x - \ell)$, a positive-second-coordinate Selmer representative is $(3^\delta \prod q_i^{u_i}, \prod q_i^{v_i})$. The necessary local equations are

$$Ku + \operatorname{diag}(\lambda)v = \delta\lambda, \quad (K + \operatorname{diag}(s))v = 0, \quad (\lambda + s)^t v = 0.$$

The final equation is the condition at three. The matrix properties force $v = 0$, $\delta = 0$, $u \in \ker K$, so there are at most eight classes in total. The extra class is actually local at every place, including

two, and independent of the two torsion classes, giving dimension three. This sufficiency check for the particular class is essential; arbitrary parity conditions at two would not suffice.

Lemma 5.4 (Dimension three gives corank one)**.** *For an elliptic curve with full rational two-torsion,* $\dim_{\mathbb{F}_2} \mathrm{Sel}_2 = 3$ *implies* $\mathrm{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty} = 1$.

Proof. The finite Kummer sequence gives $r + \dim \mathrm{Sha}[2] = 1$. Write $\mathrm{Sha}[2^\infty] = (\mathbb{Q}_2/\mathbb{Z}_2)^t \oplus F$ with F finite. The displayed divisible summand is the maximal divisible subgroup: any divisible subgroup maps trivially to the finite quotient F. The Cassels–Tate pairing on that quotient is nondegenerate and alternating for an elliptic curve [49, Corollaries 4, 7 and Theorem 8]. Thus no identification of divisible elements with infinitely divisible elements in an arbitrary group is required. Its finite primary cyclic factors are paired, so $\dim F[2]$ is even. Thus $F = 0$ and $r + t = 1$. The restriction of the pairing to $F[2]$ need not be nondegenerate and is not used. □

Lemma 5.5 (Forward analytic rank zero and one)**.** *Let* $A/\mathbb{Q}$ *be an elliptic curve. If* $\mathrm{ord}_{s=1} L(A, s) = r \le 1$*, then* $\mathrm{rank}\, A(\mathbb{Q}) = r$ *and* $\mathrm{Sha}(A/\mathbb{Q})$ *is finite. Consequently* $\mathrm{corank}_{\mathbb{Z}_2} \mathrm{Sel}_{2^\infty}(A/\mathbb{Q}) = r$.

Proof. Use the modularity theorem of Breuil, Conrad, Diamond and Taylor [62] in the published formulation [60, Theorem C.13.6, p. 442]. Apply [67, Theorem 1, pp. 430–431], whose statement includes both $r = 0$ and $r = 1$. Its Theorems 3–5 provide the auxiliary imaginary quadratic field, the nonzero Heegner height, and finiteness over that field; the restriction kernel for Sha is contained in the finite group $\mathrm{Sha}(A/\mathbb{Q})[2]$. The ordinary Kummer sequence

$$0 \longrightarrow A(\mathbb{Q}) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \longrightarrow \mathrm{Sel}_{2^\infty}(A/\mathbb{Q}) \longrightarrow \mathrm{Sha}(A/\mathbb{Q})[2^\infty] \longrightarrow 0$$

gives the corank assertion. For the modular quotient used below, let N be the conductor of A. Choose the finite surjective $\mathbb{Q}$-morphism $\rho : X_0(N) \to A$ supplied by [60, Theorem C.13.6], and compose it with the finite surjective forgetful morphism $u : X_1(N) \to X_0(N)$. Put $f = \rho \circ u$. Pushforward of divisors preserves degree and sends principal divisors to principal divisors by the function-field norm [60, Proposition II.3.6(c),(d)]. It therefore induces

$$q = f_* : J_1(N) = \mathrm{Pic}^0(X_1(N)) \longrightarrow \mathrm{Pic}^0(A) \simeq A,$$

where the last identification uses the identity point of A [60, Proposition III.3.4]. This is a homomorphism of abelian varieties over $\mathbb{Q}$: for the finite flat map f, the norm on line bundles is $\mathcal{L} \mapsto \det(f_*\mathcal{L}) \otimes \det(f_*\mathcal{O}_{X_1(N)})^{-1}$; this construction is defined over $\mathbb{Q}$, commutes with base change, and induces the displayed map of Picard varieties. Section A34 proves finite flatness and the relative-Picard construction used here. To check surjectivity, work over $\overline{\mathbb{Q}}$ and fix any $P_0 \in X_1(N)(\overline{\mathbb{Q}})$. Given $a \in A(\overline{\mathbb{Q}})$, choose P with $f(P) = a + f(P_0)$. Then $q([P] - [P_0]) = a$. No rational point on $X_1(N)$ is required. Independently, the $r = 0$ finiteness assertion follows directly from [69, Corollary 14.3], applied to the modular quotient $J_1(N) \twoheadrightarrow A$ over $\mathbb{Q}$ and the trivial character. This result includes the prime two and imposes no residual irreducibility hypothesis. □

Applying Theorem 5.3 to both n and $n\theta$, Lemma 5.5 gives rank one and finite Sha, hence $\mathrm{Sha}[2] = 0$. The Kummer class is then represented by actual rational points. Their abscissas have squareclass β, by the cubic equation. Write $H = \theta c^2$ and rescale the second point to E_{nH}. If the first abscissa is Aq^2, divide both parameters by q^2 and both coordinates by q^2, q^3. The abscissas become A, Az^2, giving precisely the eleven witnesses, with $w = (eHz)^{-1}$. If $\theta = 1$, one can reuse a point on the same curve. This proves Theorem 5.2, with both restricted-converse applications in their stated domain.

6 Multiplication and the finite false-triple guard

The local-test construction and prime-pattern strategy adapt [94, Section 2 and Lemma 6.1].

The actual multiplication template first forces all three input codes to be members of $2E(K_0)$. These membership calls are conjuncts inside the height certificate; they are not assumed caller preconditions. Thus even untyped rational inputs, if accepted, have unique indices. The template has 35,134 rational witnesses. Its construction is S03 and its mathematical specification is S13.

We use the following local conventions for the displayed integral Weierstrass equation. Put $z = -x/y$, and put $w = 1/x$ with $w(O) = 0$. This w differs from the auxiliary variable $-1/y$ used in [60, IV.1]. The formal expansions give $w = z^2$ times an integral unit, so $v(w) = 2v(z)$. For a complete discrete valuation with $e = v(p)$, the formal logarithm converges on $v(z) > 0$ and is an isometry with inverse exponential on $v(z) > e/(p-1)$ [60, IV.5.5, IV.6.3–6.4, pp. 129–132]. Indeed its degree-r coefficient has denominator dividing r, so each relative error has valuation at least $(r-1)v(z) - ev_p(r) > 0$ in this range. Thus $\log([n]Q) = n\log(Q)$ and $v(z([n]Q)) = v(z(Q)) + v(n)$ for nonzero Q and n there. These statements concern the formal neighborhood of our integral equation. When identifying it with the full reduction kernel of a minimal model, we use [60, VII.2.1–2.2, pp. 188–191].

For this particular equation the invariant differential is

$$\frac{dx}{2y} = \frac{dz}{\sqrt{1 + 4d^2z^4}}, \qquad \log(z) = \sum_{j\ge 0} \frac{(-1)^j \binom{2j}{j} d^{2j}}{4j+1} z^{4j+1}.$$

This proves the displayed coefficients of L below. At a contact prime $p > 324$, with $v(z) \ge 1$ and $v(p) = 1$, every omitted term relative to z has valuation at least $r - 1 - v_p(r) \ge 28$ for $r \ge 29$. Consequently $L(z)/16575 = \log(z)(1 + \epsilon_z)$ with $v(\epsilon_z) \ge 28$.

For indices n_1, n_2, n_3, each test group chooses $H = [h]P_0$, $U_i = [z_i]P_0$, and sets $V_0 = H - [c_*]P_0$, $V_i = U_i - [c_* n_i]P_0$, where $c_* = 4M_0$ is fixed. Use

$$\begin{aligned} L(z) = {} & 16575z - 6630d^2z^5 + 11050d^4z^9 - 25500d^6z^{13} \\ & + 68250d^8z^{17} - 198900d^{10}z^{21} + 612612d^{12}z^{25}. \end{aligned}$$

This is 16575 times the degree-25 truncation of the normalized formal logarithm. The twelve tests are four inverse abscissas at H, U_i, their four conjugate counterparts at V_0, V_i, the three differences $A_i - \sigma(A_i')$, and $A_1A_2 - A_3$, where

$$A_i = \frac{L(z(U_i))}{L(z(H))}, \qquad A_i' = \frac{L(z(V_i))}{L(z(V_0))} \quad (1 \le i \le 3),$$

and σ is the nontrivial automorphism of $K/\mathbb{Q}$. In particular the primed ratio is conjugated before the difference is taken. Identity inverse abscissas are zero; all denominators are guarded.

For a slope $t \notin \mathcal{C} \cup \{1\}$, put

$$R_0 = \eta F(t)/(t-1)^8, \quad \eta^2 = 1, \quad a = 1 - R_0q^2 \ne 0, \quad R = \ell_0^2 a^2 R_0.$$

Denote the twelve unscaled test values in group g by D_{gj}. Their normalized values are

$$C_{gj} = \frac{D_{gj}}{R_0^{k_j}(t-1)^{k_j+1}}, \qquad k_j = \begin{cases} 1, & 1 \le j \le 8, \\ 25, & 9 \le j \le 12. \end{cases}$$

For six independent groups set $W_g = \prod_j (1 + RC_{gj}^2)$ and $\Theta = \operatorname{Tr} \sum_g W_g^{-1}$, require them nonzero, and impose $\Phi(R/\Theta^2)$. The pole input is nonzero, so the prefactor B in Q_{pole} cannot make these constraints vacuous.

Here is the valuation deduction at a positive odd contact prime $p \notin S_{\mathrm{mul}}$. Normalize every $v \mid p$ by $v(p) = 1$ and put $\delta = v_p(R_0) > 0$. The contact roots are distinct modulo p, and $t - 1$ is a unit, so δ is the positive odd contact order. The fixed ℓ_0 is supported on S_{mul}. Since $\delta + 2v_p(q)$ is odd, the two summands in $a = 1 - R_0 q^2$ have unequal valuations. Thus $v_p(a) \le 0$ and

$$\alpha = v_p(R) = \delta + 2v_p(a) \le \delta, \qquad \alpha \text{ is odd.}$$

Every factor has valuation $v(1 + RC_{gj}^2) = \min\{0, \alpha + 2v(C_{gj})\} \le 0$. If test j fails its threshold $v(D_{gj}) \ge k_j$ at v, then

$$v(1 + RC_{gj}^2) \le \alpha + 2(k_j - 1) - 2k_j\delta \le \alpha - 2\delta < 0.$$

Consequently $v(W_g^{-1}) \ge 2\delta - \alpha$ for that group and place. If every group fails at every place above p, the trace of their sum satisfies

$$v_p(\Theta) \ge 2\delta - \alpha, \qquad v_p(R/\Theta^2) \le 3\alpha - 4\delta \le -\delta < 0.$$

The latter valuation is odd, contradicting Lemma 5.1. Therefore one common group and place has its first eight tests of valuation at least one and its last four of valuation at least 25. This argument does not require all factors in that group to be units. The first tests put the points in the formal group. At least one of $h, h - c_*$ is a unit at the prime, so its full-log ratio is integral. The truncated logarithm is the full logarithm times a unit congruent to one in valuation at least 28. Its ratio is therefore integral. The difference test transfers integrality to the other truncated ratio, then to the other full-log ratio. Only now do the relative errors give absolute congruences modulo p^{28}. Write $A_i = L(z(U_i))/L(z(H))$. The difference test and the full-log ratios give

$$z_i - hA_i \equiv 0, \qquad z_i - c_* n_i - (h - c_*)A_i \equiv 0 \pmod{p^{25}}.$$

Their difference is $c_*(n_i - A_i) \equiv 0$. Since $p \notin S_{\mathrm{mul}}$ implies $p \nmid c_*$, this yields $n_i \equiv A_i \pmod{p^{25}}$. The product test then implies $p^{25} \mid n_1 n_2 - n_3$.

To specify the height certificate, write the abscissa of each input as $x_{i0} + x_{i1}s$, using $x_{i0} = x_{i1} = 0$ at the identity. Choose $a_{i0}, a_{i1}, b_i \in \mathbb{Q}$ with $b_i \ne 0$ and $b_i x_{ij} = a_{ij}$. Express each of these nine rational quantities as a sum of three slopes. Impose the preceding slope predicate on all 27 slopes, with private witnesses, and impose membership on the three input codes. This conjunction is the height certificate. The multiplication predicate conjoins it with the fixed scalar graphs and the inverse of the finite guard below.

Proposition 6.1 (Finite-box deduction)**.** *Assuming these local test identities and the height estimate for S_{mul} with constant A_{ht}, every false triple accepted by the unguarded certificate has $\max_i |n_i| < T = (48A_{\mathrm{ht}})^{13}$.*

Proof. Put $f = n_1 n_2 - n_3 \ne 0$. For each of the 27 slopes, $M_{S_{\mathrm{mul}}}(t)^{25} \mid f$, so $h(t) \le A_{\mathrm{ht}}|f|^{24/25}$. The nine rational homogeneous quantities for the three abscissas are sums of three slopes. The sum and quotient height bounds give $h(x_j) \le 6A_{\mathrm{ht}}|f|^{24/25} + 2\log 3$ for each rational component. The fixed coordinate change and the duplication bounds yield

$$2n_i^2 \widehat{h}(P_0) \le 12A_{\mathrm{ht}}|f|^{24/25} + B_0,$$

where $2\widehat{h}(P_0) > 1$ and $B_0 = 4\log 3 + \frac{3}{2}\log 2 + 2\log(1+\sqrt{2}) + \frac{1}{3}\log 676 < 10$. For $N = \max|n_i|$, $|f| \le 2N^2$, so $N^2 < 24A_{\mathrm{ht}}N^{48/25} + 10$. If $N \ge 5$, this implies $N^{2/25} < 48A_{\mathrm{ht}}$, hence $N < (48A_{\mathrm{ht}})^{25/2} \le T$. The case $N \le 4$ is immediate. □

For each false triple $(a,b,c) \in [-T,T]^3$, write its fifteen code coordinates as A_j/B_j with $B_j > 0$, put $D_{abc} = \prod B_j$, and form

$$g_{abc}(X) = \sum_{j=1}^{15} (D_{abc} X_j - (D_{abc}/B_j) A_j)^2.$$

The guard requires the product of all these factors to have an inverse. A false accepted triple makes its own factor zero. At a true triple every factor is positive by injectivity of codes, and the inverse exists. Thus the sound direction uses the generator and corrected all-S height theorem but does not use pole completeness.

Proposition 6.2 (The fixed twenty-two-prime pattern)**.** *Put*

$$\mathcal{B} = \{0, -15, -20, -24, -30, -60, 1\}, \qquad P_s = \prod_{p \le 324} p, \qquad W = 4P_s.$$

Let $\mathcal{H}$ consist of the first twenty-two primes greater than 324 in each unit residue class modulo W, as in the fixed constant prescription, and set

$$\mathcal{K} = \{dh : d \mid P_s^4,\ d > 0,\ h \in \mathcal{H}\}.$$

For every $A \in \mathbb{Z}$ and every real B_0 there are integers u_1, u_2, v, with $v \neq 0$, such that, on putting $u_3 = Av - u_1 - u_2$, each of the twenty-two values

$$v, \qquad u_j - bv \quad (1 \le j \le 3,\ b \in \mathcal{B})$$

equals ϵkp, where $\epsilon \in \{1, -1\}$, $k \in \mathcal{K}$, and $p > B_0$ is prime. Every such p splits in $\mathbb{Q}(\sqrt{2})$, the two reductions of

$$E : y^2 = x^3 - (75 - 53\sqrt{2})^2 x$$

are good, and their orders d_1, d_2 satisfy

$$\gcd(d_1, d_2) = 4, \qquad p \nmid d_1 d_2.$$

The prime values need not be distinct. The assertion concerns the two orders at the same prime only.

Proof. Write $T = (u_1, u_2, v)$ and denote the twenty-two homogeneous forms by F_ν. Their coefficient vectors are

$$(0,0,1), \quad (1,0,-b), \quad (0,1,-b), \quad (-1,-1,A-b).$$

Each form has a coefficient equal to 1 or -1. Within each of the three seven-form families, a nonzero two-by-two minor is a contact difference, of absolute value at most 61. Between two different families the first two coordinates give a minor 1 or -1; a pair involving v also has such a minor. Thus every pair is independent over $\mathbb{Q}$ and over $\mathbb{F}_q$ for every prime $q > 324$, for every integer A. No condition on three-form dependencies is required below.

Choose a bounded open rectangular box $\Omega \subset \mathbb{R}^3$ of positive volume whose closure avoids all the zero planes. For each ν choose ϵ_ν such that $\epsilon_\nu F_\nu > 0$ on $\overline{\Omega}$. These signs allow all integers A, including zero and negative integers.

The fixed multipliers and initial progression. For each prime $q \le 324$, the proportion of T modulo q^5 for which at least one $F_\nu(T)$ vanishes is at most $22/q^5 < 1$. Choose a residue outside this union and lift it to a residue modulo q^8. On this residue each F_ν has a fixed valuation $e_{\nu,q} \in \{0,1,2,3,4\}$. Set

$$d_\nu = \prod_{q \le 324} q^{e_{\nu,q}}.$$

The unit $\epsilon_\nu F_\nu/d_\nu$ is now fixed modulo 8 at 2 and modulo q at each odd $q \le 324$. Prescribe the prime residues

$$c_2 = 1 \pmod 8, \qquad c_7 = 3 \pmod 7, \qquad c_q = 1 \pmod q \quad (q \le 324,\ q \ne 2,7).$$

Choose $h_\nu \in \mathcal{H}$ in the unit class specified by $h_\nu \equiv (\epsilon_\nu F_\nu/d_\nu)c_q^{-1}$ at these moduli. There are at most twenty-two demands in any class, so the h_ν can be chosen pairwise distinct. Put $k_\nu = d_\nu h_\nu$. Dirichlet's theorem proves that this is the same finite catalogue as the prescribed one; no extra catalogue prime is introduced.

At $h = h_\nu$, impose $F_\nu(T) = 0$. This plane has h^2 points. For every $\mu \ne \nu$, the restriction of F_μ is nonconstant, so excluding

$$F_\mu(T) = 0, \qquad \epsilon_\mu F_\mu(T) = -k_\mu$$

removes at most $42h < h^2$ points. Choose a remaining point. Among its lifts modulo h^2, the unit $\epsilon_\nu F_\nu/(d_\nu h)$ runs through every residue modulo h: changing a coordinate with coefficient 1 or -1 by h proves surjectivity. Choose this residue different from 0 and -1. The restrictions for $\mu \ne \nu$ remain valid.

The Chinese remainder theorem now gives one vector residue r_0 modulo

$$W_0 = \prod_{q \le 324} q^8 \prod_\nu h_\nu^2.$$

On $r_0 + W_0\mathbb{Z}^3$ the functions $\Psi_\nu = \epsilon_\nu F_\nu/k_\nu$ are integral. At every prime dividing W_0 they are nonzero, and at every odd such prime they are different from -1. In the affine coordinates on this progression every variable coefficient is divisible by each prime dividing W_0. Indeed, q^8/d_ν is divisible by q even when $e_{\nu,q} = 4$, and h^2/k_ν is divisible by h.

All these choices depend on A and the fixed catalogue, but not on B_0 or the cutoff L used next. Fix also

$$Q_* = \max(324, \max \mathcal{H}).$$

In both upper sieves below the excluded fixed primes are exactly those at most Q_*; in particular this exclusion does not grow with L. There are constants $0 < \alpha_A < \beta_A$ such that

$$\alpha_A X \le \Psi_\nu(T) \le \beta_A X \quad (T \in X\Omega,\ 1 \le \nu \le 22).$$

The lower count with its cutoff dependence. Take $L > \max(Q_*, B_0, 3)$. At each prime $q \le L$ not dividing W_0, exclude the forty-four affine planes

$$F_\nu(T) = 0, \qquad \epsilon_\nu F_\nu(T) = -k_\nu.$$

Each has q^2 points and all these primes exceed 324. With

$$W_L = W_0 \prod_{\substack{q \le L \\ q \nmid W_0}} q,$$

let $\mathcal{R}_L$ be the resulting vector residues modulo W_L that reduce to r_0 modulo W_0. Then

$$\frac{|\mathcal{R}_L|}{W_L^3} \ge W_0^{-3} \prod_{\substack{q \le L \\ q \nmid W_0}} (1 - 44/q) \ge c_A (\log L)^{-44}. \tag{2}$$

Here and below a positive constant with subscript A may depend on the choices already made, including the fixed catalogue, but never on L, X, or a varying fiber prime. The last inequality follows from $\log(1 - 44/q) = -44/q + O(q^{-2})$ and Mertens' theorem [97, Theorem 1, with modulus 2].

Fix L. For $r \in \mathcal{R}_L$ apply the finite-complexity prime asymptotic of [50, Main Theorem, p. 1763], with its hypotheses supplied by [51, Theorem 1.1] and Section I, to

$$n \longmapsto \frac{\epsilon_\nu F_\nu(r + W_L n)}{k_\nu}, \qquad r + W_L n \in X\Omega.$$

These are integral affine forms with pairwise independent linear parts. Singleton classes in [50, Definition 1.5] give complexity at most 20, so the inputs GI(20) and MN(20) suffice. The local inverse-theorem proof supplies the normalized finite-family statement with linear nilsequences of step at most 20; its constructions and published inputs are given in the accompanying supplement. All coefficient bounds and moduli are fixed before X tends to infinity. At $q \mid W_L$ the local factor is $(1 - 1/q)^{-22}$. At $q \nmid W_L$, scaling and changing variables preserve the twenty-two homogeneous zero planes, so

$$1 - 22/q \le \mathbb{P}(F_\nu \ne 0 \text{ for all } \nu) \le 1 - 22/q + \binom{22}{2} q^{-2}.$$

Consequently the local factor is $1 + O(q^{-2})$, and

$$\log \beta_q \ge \log(1 - 22/q) - 22 \log(1 - 1/q) \ge -484/q^2.$$

The singular product is therefore bounded below by an absolute positive constant, uniformly in L and r. The real factor is $\mathrm{vol}(\Omega) X^3 / W_L^3$.

There are finitely many r for fixed L, so their asymptotic errors can be made smaller than half their main terms with one threshold for X, depending on A, L. This uses no uniform prime-asymptotic error as L varies. Summing over r and using (2) gives a weighted lower bound $c_A (\log L)^{-44} X^3$. Proper prime powers contribute at most $O_A(X^{5/2} (\log X)^{23})$: there are $O_A(X^{1/2} \log X)$ possible proper powers in each form's range, each equation fixing a form value has $O_A(X^2)$ integer solutions in the box, and each weight is $O_A((\log X)^{22})$. Thus, for all sufficiently large X depending on A, L,

$$\#\{T \in X\Omega : \ T \bmod W_L \in \mathcal{R}_L, \ \Psi_\nu(T) \text{ all prime}\} \ge a_A \frac{X^3}{(\log L)^{44} (\log X)^{22}}, \qquad a_A > 0. \tag{3}$$

An upper sieve with uniform constants. We give the estimates needed for both upper bounds. Consider an axis-parallel square of side $N \ge 1$ in $\mathbb{R}^2$. For each sieving prime q let $\Omega_q \subset \mathbb{F}_q^2$ have size $\rho(q)$, where $0 < \rho(q) \le 21q$ and $q > Q_*$. Put

$$g(q) = \rho(q)/q^2, \qquad h(q) = g(q)/(1 - g(q)),$$

and extend both multiplicatively on squarefree integers supported on the sieving primes. For such d, CRT and the elementary count of lattice points in each residue square give

$$A_d = N^2 g(d) + R_d, \qquad |R_d| \ll \rho(d)(N/d + 1) \le Nd + d^2. \tag{4}$$

The implied absolute constant is uniform in the square's location and in the sets Ω_q. The last inequality uses $\rho(q) \le q^2$.

For $Z \ge 2$ define

$$G(Z) = \sum_{d \le Z} h(d), \qquad \lambda_d = \frac{\mu(d)}{G(Z) \prod_{q \mid d}(1 - g(q))} \sum_{\substack{m \le Z/d \\ (m,d)=1}} h(m) \quad (d \le Z),$$

and set $\lambda_d = 0$ otherwise. All sums here are over the stated squarefree support, with $g(1) = h(1) = 1$. These are Selberg's weights. They satisfy $\lambda_1 = 1$, $|\lambda_d| \le d$, and

$$\sum_{e,f \le Z} \lambda_e \lambda_f g([e,f]) = G(Z)^{-1}.$$

For completeness, set $y_r = \sum_{r \mid d} \lambda_d g(d)$. The displayed quadratic form equals $\sum_r y_r^2 / h(r)$, because $1/g(m) = \sum_{r \mid m} 1/h(r)$; the stated weights give $y_r = \mu(r)h(r)/G(Z)$. The bound on the weights follows from the inner sum being at most $G(Z)$ and $(1 - g(q))^{-1} \le q$ for $q > 324$.

Squaring the divisor sum majorizes the indicator of avoiding all Ω_q for $q \le Z$. Inserting (4), and using $[e,f] \le Z^2$, there are at most Z^2 pairs with $|\lambda_e \lambda_f| \le Z^2$. It follows that

$$S(N, Z) \ll \frac{N^2}{G(Z)} + NZ^6 + Z^8. \tag{5}$$

Here are the two denominator estimates, including truncation. For $z = Z^{1/100}$ put

$$H(z) = \prod_{\substack{q \le z \\ q \text{ sieved}}} (1 + h(q)).$$

Give each squarefree divisor d of this prime product probability $h(d)/H(z)$. The events $q \mid d$ are independent and have probability $g(q)$. Hence

$$\mathbb{E} \log d = \sum_{\substack{q \le z \\ q \text{ sieved}}} g(q) \log q \le 21 \sum_{q \le z} \frac{\log q}{q} \le 42 \log z = 0.42 \log Z$$

for sufficiently large z, using $\sum_{q \le z} (\log q)/q = \log z + O(\log\log z)$. Indeed, [97, Theorem 1, with modulus 2] gives $S(z) := \sum_{q \le z} q^{-1} = \log\log z + C + O(1/\log z)$, including the prime 2. Partial summation in $S(z)\log z - \int_2^z S(t)\,dt/t$ gives the asserted bound. Markov's inequality gives probability at least 0.58, and thus at least $1/2$, that $d \le Z$. Every such divisor is included in $G(Z)$, so

$$G(Z) \ge \tfrac{1}{2} H(Z^{1/100}). \tag{6}$$

This proves a lower bound for the truncated sum, rather than merely for an untruncated Euler product.

In the first application the sieved primes are $q > Q_*$ with $q \equiv 1 \pmod 4$, with at most one prime omitted, and $g(q) = 2/q - 1/q^2$. Mertens' theorem in the fixed progression $1 \pmod 4$ [97, Theorem 1] gives

$$\log H(z) = \log\log z + O_{Q_*}(1).$$

Omitting a prime ℓ changes $H(z)$ by $1 - g(\ell) \ge 1/2$ when $\ell \le z$, and has no effect otherwise. Therefore $G(Z) \gg_{Q_*} \log Z$ uniformly in the omitted prime. In the second application all primes $q > Q_*$ are sieved and

$$\frac{21}{q} - \frac{210}{q^2} \le g(q) \le \frac{21}{q}.$$

Then $\log H(z) = 21 \log\log z + O_{Q_*}(1)$, with an error independent of the individual sets Ω_q, and hence $G(Z) \gg_{Q_*} (\log Z)^{21}$. These statements and (5) establish every uniform upper sieve estimate used below.

The conjugate orders. Every prime counted by (3) satisfies $p \equiv 1 \pmod 8$ and $p \equiv 3 \pmod 7$. Thus it splits in $\mathbb{Q}(\sqrt{2})$ and $(7/p) = -1$. The two reductions of $d = 75 - 53\sqrt{2}$ have product 7 and opposite square classes. The two reduced curves are the quadratic twists of $y^2 = x^3 - x$ by these two classes. Their traces are opposite. The discriminant $64d^6$ is a unit at these primes, so both reductions are good.

For $p \equiv 1 \pmod 4$, choose $i \in \mathbb{F}_p$ with $i^2 = -1$. The automorphism $\iota : (x, y) \mapsto (-x, iy)$ satisfies $\iota^2 = [-1]$ and commutes with the arithmetic p-power Frobenius φ, directly from its coordinate formula. The injection

$$\mathrm{End}(\overline{E}) \otimes \mathbb{Q}_3 \hookrightarrow \mathrm{End}_{\mathbb{Q}_3}(V_3\overline{E})$$

is [60, III.7.4, pp. 89–91]; here $p > 324$. Since $T^2 + 1$ is irreducible over $\mathbb{Q}_3$, the centralizer of ι on this two-dimensional Tate space is $\mathbb{Q}_3(i)$. The rational centralizer therefore has dimension at most two and already contains $\mathbb{Q}(i)$, so it is exactly $\mathbb{Q}(i)$. The Frobenius endomorphism is integral over $\mathbb{Z}$ by [60, V.2.3.1(b), p. 142]. Hence it lies in $\mathbb{Z}[i]$; write it as $a + bi$ with $a, b \in \mathbb{Z}$. The trace and determinant formulas [60, III.8.6, p. 99] give

$$a^2 + b^2 = p, \qquad d_1 = p + 1 - 2a, \qquad d_2 = p + 1 + 2a.$$

Here d_1, d_2 are the cardinalities of the two reduced elliptic curve groups; the opposite twist changes the Frobenius trace's sign. In particular $a \neq 0$, since p is not a square. Both orders are divisible by 4 because all the two-torsion is rational. Their sum is $2(p+1)$, of 2-adic valuation exactly 2, so the 2-part of their gcd is exactly 4. Hasse's bound [60, V.1.1, p. 138] gives $0 < d_i < 2p$ for $p > 324$, and $d_i \neq p$ since $4 \mid d_i$. Thus $p \nmid d_1 d_2$.

If an odd prime ℓ divides both orders, it divides a and $p + 1$, and $b^2 \equiv -1 \pmod \ell$. The progression conditions already rule out every odd $\ell \leq L$. It remains to discard prime values with such an $\ell > L$.

The number of bad prime values. For a fixed form its prime value lies between $\alpha_A X$ and $\beta_A X$. Representations $p = a^2 + b^2$ lie in $[-C_A\sqrt{X}, C_A\sqrt{X}]^2$. Overcount them to bound the number of bad primes. If $L < \ell \leq X^{1/8}$, there are at most two residues b_0 with $b_0^2 \equiv -1 \pmod \ell$. Write

$$a = \ell U, \qquad b = b_0 + \ell V.$$

The required (U, V) lie in a square of side $N \asymp_A \sqrt{X}/\ell \gg_A X^{3/8}$. For every $q > Q_*$ with $q \equiv 1 \pmod 4$ and $q \neq \ell$, the condition $a^2 + b^2 = 0 \pmod q$ is the union of two distinct lines through the origin, of size $2q - 1$. The affine substitution is bijective modulo q. Take $Z = X^{1/100}$. Since the prime norm is at least $\alpha_A X > Z$ for large X, every required pair avoids these residues. Equations (5) and (6) give

$$\#(U, V) \ll_A \frac{N^2}{\log X} + NZ^6 + Z^8 \ll_A \frac{X}{\ell^2 \log X}.$$

The absorption is uniform: $Z^6/N \ll_A X^{0.06-3/8}$ and $Z^8/N^2 \ll_A X^{0.08-3/4}$, both tending to zero faster than $1/\log X$. Summing over $\ell > L$ gives $O_A(X/(L \log X))$.

For $\ell > X^{1/8}$ use the lattice count directly. Since $a \neq 0$, there are $O_A(\sqrt{X}/\ell)$ possible multiples a, with no additive constant. For each of the at most two root classes there are $O_A(\sqrt{X}/\ell + 1)$

choices of b. Also $\ell \le |a| \le C_A\sqrt{X}$. Bounding the sum over primes by the corresponding sum over positive integers yields

$$\sum_{X^{1/8}<\ell\le C_A\sqrt{X}} O_A\left(\frac{X}{\ell^2}+\frac{\sqrt{X}}{\ell}\right) \ll_A X^{7/8}+\sqrt{X}\log X.$$

We have therefore proved the bound, uniformly in L,

$$\#\{\text{bad prime values for a fixed form}\} \ll_A \frac{X}{L\log X}+X^{7/8}+\sqrt{X}\log X. \tag{7}$$

The number of patterns on each nonzero fiber. Fix ν and $p \in [\alpha_A X, \beta_A X]$, and put $c = \epsilon_\nu k_\nu p$. The integral solutions of $F_\nu(T) = c$ are parametrized by two original coordinates: solve for a coordinate whose coefficient is 1 or -1. For example, for $F_\nu = -u_1 - u_2 + (A-b)v$, use (u_1, v) and set $u_2 = (A-b)v - c - u_1$. This parametrization is valid over every $\mathbb{F}_q$, and the points from $X\Omega$ lie in a square of side $O_A(X)$ in these coordinates.

For $Q_* < q \le Z = X^{1/100}$, one has $q \nmid k_\nu$ and $q < p$ when X is large. Thus $c \ne 0$ modulo q. Each remaining form restricts to a nonconstant affine form. Their twenty-one zero lines are distinct: if the restrictions of F_j, F_k had the same zero line, then for some $\lambda \ne 0, \mu$,

$$F_j - \lambda F_k = \mu(F_\nu - c)$$

as affine forms over $\mathbb{F}_q$. Comparing constants gives $\mu c = 0$, hence $\mu = 0$, contradicting pairwise independence of F_j, F_k. Parallel lines and multiple intersections cause no difficulty. The union consequently has between $21q - \binom{21}{2}$ and $21q$ points.

Every target pattern on the fiber avoids this union: its other prime values are also at least $\alpha_A X > Z$, and every k_j is a unit at $q > Q_*$. Discarding the CRT progression conditions only increases the count. The second denominator estimate in (6) gives

$$\#\{\text{target patterns on } F_\nu = c\} \ll_A \frac{X^2}{(\log X)^{21}} + X^{1.06} + X^{0.08} \ll_A \frac{X^2}{(\log X)^{21}}. \tag{8}$$

This is uniform in the fiber prime p and independent of L. In particular $q = p$ is outside the sieving range; no varying exceptional-prime product has been suppressed.

The order of choices. Multiply (7) by (8) and sum over the twenty-two forms. The number of bad patterns is at most

$$b_A \frac{X^3}{L(\log X)^{22}} + O_A\left(\frac{X^{23/8}}{(\log X)^{21}} + \frac{X^{5/2}}{(\log X)^{20}}\right),$$

where b_A is independent of L. First choose $L > \max(Q_*, B_0, 3)$ so large that

$$\frac{b_A}{L} < \frac{a_A}{2(\log L)^{44}}.$$

Such an L exists because $(\log L)^{44}/L \to 0$. Then keep A, L and every preceding choice fixed, and take X large enough for (3), the upper bounds, and the remaining error to be less than $a_A X^3/(2(\log L)^{44}(\log X)^{22})$. A pattern survives. Its signed prime values tend to infinity with X, $v \ne 0$, and the preceding order calculation proves all the assertions. □

Apply this construction to the nine integral homogeneous quantities of a true triple. Each slope has at most six odd-contact primes. At one such prime with reduced group cardinalities d_1, d_2, which annihilate the reduced point, solve

$$h \equiv 0 \pmod{\operatorname{lcm}(M_0, d_1)}, \qquad h \equiv c_* \pmod{d_2}.$$

Compatibility follows from

$$\gcd(\operatorname{lcm}(M_0, d_1), d_2) \mid M_0 \gcd(d_1, d_2) = 4M_0 = c_*.$$

Choose a solution larger than c_*. Each contact prime uses its own private group within its own slope. No common shift across different primes is imposed, and selected prime values need not be distinct. The saved library allocates six private groups per slope and twenty-seven private slope calls. Set $U_i = [n_i]H$ and assign one group to each prime. Fixed local clearing at the old primes can be checked as follows. Put $S_b = S_{\mathrm{ht}} \cup \mathcal{H}$, $B_p = 4$ on S_{ht} and $B_p = 1$ on the remaining catalogue primes, $C_p = 14B_p$, and $D_p = 51C_p$. The selected integral forms give $|v_p(R_0)|, |v_p(t-1)| \le C_p$, while the small formal subgroup in the definition of M_0 gives $v_p(D_{gj}) \ge 0$ at S_b. Consequently $v_p(C_{gj}) \ge -D_p$. The fixed normalization has $v_p(\ell_0) = E_p = 52C_p + v_p(12) + 4$.

Weak approximation chooses q small at S_b, integral at the selected primes, and arbitrarily close to a real root of $1 - R_0 q^2$, without equality; choose η to make $R_0 > 0$. Then a is a unit at the specified finite primes. At $p \in S_b$,

$$v_p(RC_{gj}^2) \ge 2E_p - C_p - 2D_p > v_p(12), \qquad v_p(\Theta - 12) > v_p(12),$$

so $v_p(R/\Theta^2) \ge 103C_p + 8 \ge 4$. At a selected odd-contact prime, $v_p(R) = 1$. Each of the twelve summands of the trace of the six W_g^{-1} has residue zero or one; the assigned group and place contributes one. Since $p > 324$, their sum is a unit and the pole input has valuation one. At every other prime outside S_b, its valuation is even. Thus its reduced denominator is square. In both real embeddings each W_g is positive; as $a \to 0$, $W_g \to 1$, $\Theta \to 12$, and $R/\Theta^2 \to 0$ through positive values. This verifies exactly PC's hypotheses, including all nonzero guards. Negative indices, zero indices and coincident points are handled by the group-law branches. Therefore Theorem 5.2 supplies the required pole witnesses. This proves the full multiplication statement M, including completeness for every integer triple $(n_1, n_2, n_1 n_2)$ and soundness on arbitrary rational input codes.

7 The restricted converse and its dependency order

The organization of the restricted-converse argument follows [90], particularly its ring-class interpolation, determinant transfer, coefficient systems and fixed-curve assembly. The broader converse is stated in [91]. Our specialization uses the same full Selmer-corank convention, including any divisible part of Sha, as those sources.

For a positive squarefree $\ell \equiv 11 \pmod{12}$ put

$$E_\ell : y^2 = x(x - \ell)(x + 3\ell).$$

The distinct roots $0, \ell, -3\ell$ give full rational two-torsion. Expanding the cubic gives $a_2 = 2\ell$, $a_4 = -3\ell^2$ and $a_1 = a_3 = a_6 = 0$. With the usual Weierstrass conventions, $b_2 = a_1^2 + 4a_2 = 8\ell$, $b_4 = 2a_4 + a_1 a_3 = -6\ell^2$, $b_6 = 0$, and $b_8 = -a_4^2 = -9\ell^4$. Hence

$$\begin{aligned} c_4 &= b_2^2 - 24b_4 = 208\ell^2, \\ \Delta &= -b_2^2 b_8 - 8b_4^3 = 2304\ell^6 \neq 0, \\ j(E_\ell) &= c_4^3/\Delta = 35152/9. \end{aligned} \tag{9}$$

For a CM elliptic curve, its j-invariant is a root of the monic integral order class polynomial, including for nonmaximal orders [40, §1, p. 607, and §2.1, pp. 612–613]. A rational algebraic integer a/b in lowest terms, $b > 0$, is an integer: a monic integral equation of degree n, multiplied by b^n, gives $b \mid a^n$, so $b = 1$. Since $35152/9$ is in lowest terms and has denominator 9, the curve E_ℓ is not CM. The curve in this section is different from the fixed CM curve used to represent integer indices.

The assertion needed in Theorem 5.3 is

$$\operatorname{corank}_{\mathbb{Z}_2} \operatorname{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1. \tag{RC}$$

We prove this assertion in this section and the next two sections. Appendices A5–A7 give the fixed-form, theta, nonvanishing and normalization constructions used in this deduction. The modularity theorem of Breuil, Conrad, Diamond and Taylor [62], in the published formulation [60, Theorems C.13.6 and C.16.4(a),(c), pp. 442, 451], makes $E = E_\ell$ modular and supplies a rational parametrization from $X_0(N)$, where N is the conductor of E. Choose the rational weight-two newform $f \in S_2(\Gamma_0(N))$ with integral Fourier coefficients supplied by [63, Theorem 3.1], using the normalized convention $a_1(f) = 1$. For every prime $q \nmid 2N$ its coefficient satisfies $a_q(f) = q+1-\#E(\mathbb{F}_q)$. Apply [37, Theorem 1, pp. 526–527] with $\Gamma = \Gamma_0(N)$. Since the coefficient field is $\mathbb{Q}$, this gives an elliptic curve $A_f/\mathbb{Q}$ and a quotient $q_f : J_0(N) \twoheadrightarrow A_f$ over $\mathbb{Q}$. Here T_q on the Jacobian is the Hecke correspondence whose pullback on differentials is the weight-two operator, as in [37, equations (1.3)–(1.4), p. 525]. The quotient satisfies $q_f T_q = [a_q(f)]q_f$ for $q \nmid N$. Shimura's same theorem identifies the local Euler factors of A_f and f outside finitely many primes. Together with the displayed coefficient equality, this makes the local Euler factors of A_f and E equal outside finitely many primes. Thus [44, §5, Corollary 2, p. 361, with the 1984 erratum] places A_f in the $\mathbb{Q}$-isogeny class of E.

For the present curve E_ℓ, with ℓ odd, the displayed discriminant is $2304\ell^6$, of valuation eight at two. A change of Weierstrass model changes this valuation by a multiple of twelve. Good reduction at two would give an integral model of valuation zero, which is impossible. Thus $2 \mid N$, and every prime outside N is already outside $2N$ in the trace comparison.

We spell out the operator normalization in this application. The newform is a common eigenform for the Hecke operators, including the level-prime operators. At $q \nmid N$ the determinant-q correspondence acts on $f(z)\,dz$ by

$$\left(q f(qz) + \frac{1}{q} \sum_{j=0}^{q-1} f((z+j)/q) \right) dz.$$

The coefficient of $e^{2\pi i n z}$ is $a_{qn}(f) + q a_{n/q}(f)$, with the second term zero unless $q \mid n$. At a level prime the translation branches give U_p, with coefficient $a_{pn}(f)$. Since $a_1(f) = 1$, the eigenvalues are the corresponding Fourier coefficients. The level-prime observation does not assert a bad-prime trace formula absent from the cited modularity statement. Shimura's equations (1.3)–(1.4) identify this action on differentials with the action of the correspondence on the Jacobian. The coefficient field is $\mathbb{Q}$, so its unital coefficient action on A_f sends $a_q(f) \in \mathbb{Z}$ to multiplication by that integer. Hence $q_f T_q = [a_q(f)]q_f$ is an equality of the actual homomorphisms. There is no torsion ambiguity: a homomorphism of abelian varieties killed by an integer has connected image in a finite group scheme and is zero in characteristic zero.

For clarity, the good-prime correspondence is also the unaveraged geometric neighbor correspondence used below. An order-q cyclic subgroup of an elliptic curve with level-N subgroup gives its quotient with induced level structure. The dual q-isogeny returns that level subgroup to itself, since

multiplication by q is bijective on it. Thus the correspondence is symmetric under exchanging its two projections. Its $q+1$ upper-triangular branches fix the cusp at infinity, so $T_q[\infty] = (q+1)[\infty]$. Consequently its action on $[x] - [\infty]$ is precisely the sum of the $q+1$ corresponding degree-zero divisor classes. Composing with a $\mathbb{Q}$-isogeny $A_f \to E$ preserves the Hecke identity and introduces neither an averaging factor nor a cusp correction. Choose a $\mathbb{Q}$-isogeny $\alpha : A_f \to E$ and put $\varphi = \alpha q_f$ and $\pi(x) = \varphi([x] - [\infty])$. Thus $\pi(\infty) = 0$ and, for $q \nmid N$, $\varphi T_q = [a_q(f)]\varphi$. Fix the integral Tate lattice and the local squareclasses allowed at the primes of $2N$. Constants depending on this data are chosen before any conductor, companion field or binary dimension varies. Quotienting by a rational point of order two gives a rational two-isogeny. For the parity step, apply [61, Theorem 1.4 and the definition preceding Conjecture 1.2, p. 568] with $E/\mathbb{Q}$ and $p = 2$. Its full 2^∞-Selmer rank is the $\mathbb{Z}_2$-corank in (RC), so it gives root number -1 under the left side of (RC). This application does not replace Selmer corank by Mordell–Weil rank and does not assume finite Sha.

Remark 7.1 (Why this family is the input to the converse). The congruence $\ell \equiv 11 \pmod{12}$ comes from the pole-completeness construction. In the proof of Theorem 5.2, the two parameters $n, n\theta$ are positive squarefree numbers in this congruence class; their two-Selmer groups have dimension three. Lemma 5.4 gives full two-primary Selmer corank one, so these are exactly the two applications required of Theorem 5.3.

In the converse argument we fix one such ℓ. In particular ℓ is odd, is prime to three, and is greater than one, hence is not a square. We use the displayed full rational two-torsion and non-CM properties, the resulting rational two-isogeny, and the explicit roots $0, \ell, -3\ell$. The last roots enter the K-split seed argument below. The constants may depend on this fixed curve; there is no bound on $\omega(\ell)$ and no assertion that they are uniform as ℓ varies. The statements about coefficient graphs below are the rational, full-two-torsion case used by E_ℓ and its quadratic twists.

Here is the dependency order of the proof. A fixed-form Euler-system calculation gives the even coefficient bound. A varying-companion arithmetic estimate, proved by the two limits below, then gives the genus-point lower bound. Only after that lower bound do we normalize the odd coefficients. Their finite-stage forms and the even forms give one common binary family. Finally a fixed-field determinant argument excludes a zero derivative at the base. The even estimate does not use that final determinant argument on Heegner classes.

7.1 The even coefficient calculation

Use normalized central values $L_*(h)$ with the real and imaginary periods fixed by π and the integral lattice. Only their valuations are used; every subsequent comparison uses this same normalization. For every twist whose two-primary Tate–Shafarevich group is finite, let $s(h) = \operatorname{length}_{\mathbb{Z}_2} \operatorname{Sha}(E^{(h)}/\mathbb{Q})[2^\infty]$.

Proposition 7.2 (The fixed-form even scalar)**.** *Use the fixed E, π, integral lattice and allowed old local types specified above. Put $\Lambda = \mathbb{Z}_2[[t]]$. For a finite elementary abelian two-group G, let $\chi_x = \chi_0(x \circ \varepsilon)$ $(x \in \widehat{G})$ be a family of quadratic characters with a common real sign and the same restrictions at every place dividing $2N$. There is one scalar $u^{\mathrm{ev}} \in \Lambda[1/2][G]$ satisfying (3.2). Its clearing exponent and the bounded error in that formula depend only on the fixed data, not on G or the good-prime conductor support. The central identity is used only at corank-zero constituents with nonzero central value; at a corank-zero constituent the same central coordinate is zero exactly when the central value is zero.*

Proof. The following construction is expanded in Propositions A5.1 and A5.2, with its central calculation in Proposition A5.3. The fixed-form construction uses one integral symbol combination

with nonzero projections to both real signs. Kato's integral maps, with j_* cohomology taken modulo each power of two before passage to the limit, give classes whose inertia restrictions vanish integrally. Indeed the low-degree Leray edge map gives this vanishing before rationalization. Restriction of transfer is the Mackey sum of the corresponding upper inertia restrictions. Hence unnormalized transfer preserves the vanishing even when the intervening extension is ramified. No character projector divided by a growing group order is used.

For a finite binary character group, let F be the least common multiple of the odd parts of all its character conductors, including the base character. Write Z_F for the compatible Iwasawa class obtained by pushing the retained-smoothed Kato classes at levels $F2^n$ through the integral coefficient map. Here and throughout the common-scalar construction, the binary family is required to have one real sign and fixed local squareclasses at $2N$. Writing its characters as $\chi_x = \chi_0(x \circ \varepsilon)$, this means $\varepsilon(c) = 1$ for complex conjugation c, and $\varepsilon|_{G_{\mathbb{Q}_v}} = 1$ at every $v \mid 2N$. Indeed every character of G evaluates these elements to one, and the characters separate G. The lower bound uses $G = 1$; the upper-transfer cubes lie in the fixed signed unit filter of their coefficient source. The final cubes have the same property. At a character χ its unused-prime factors are

$$E_q(\chi) = 1 - \frac{a_q \chi(q)}{q}\sigma_q^{-1} + \frac{1}{q}\sigma_q^{-2}. \tag{3.1}$$

They are the actual determinants of the unused good local terms. Active ramified good primes have rational inertia -1, so have no such rational local term. The fixed bad and dyadic corrections form a factor F_E with nonzero central value.

The rational divisibility input is Kato's Theorem 13.4(2), together with the nonzero-class input in 13.5(2) [69]. At two we justify the divisibility through [89, Theorem 0.8]; Appendix A5 checks its stronger abelian-extension hypothesis and compares the two cohomology conventions. We use the height-one primes not containing two. Rational irreducibility is preserved on a quadratic subgroup of the non-CM open image. That subgroup contains a nontrivial unipotent, which supplies the required one-dimensional fixed space. Oddness supplies the two real eigenspaces, and purity supplies the common Frobenius weight. This application does not use residual irreducibility at two or the stronger hypotheses of Theorem 13.4(3).

The smoothing constants c, d_s can be chosen to satisfy

$$c, d_s \equiv 1 \pmod{3N_{\mathrm{odd}}F}, \qquad c, d_s \equiv 1 + 2^e \pmod{2^{e+1}},$$

where $e \geq \max\{3, v_2(N)\}$ is fixed by the original dyadic data. In weight two and cohomological degree one the two smoothing exponents are one. Kato's explicit reciprocity formula therefore contributes valuation $2e$, independent of F. For the central normalization, first descend to the primitive character conductor and then use the invariant basis $\sqrt{h}\, e_{\chi_h}$ of the quadratic character line. The resulting varying scalar $1/h$ is a two-adic unit. The complete calculation in Appendix A5 includes discriminant $4h$; a Gauss basis there has the extra factor $\sqrt{\Delta_h/h}$, which must also be included in its change of basis. The numerical sizes of c, d_s may grow, while their relevant valuation remains fixed.

For completeness, the integral calculation at two uses the positive complex. Finite continuous cochains are filtered unions of finite free modules and hence flat. Splitting the residual complex into cohomology and disks, then using the nilpotent perturbation calculation of Section 9, gives finite free models with their maps. In the fixed real cyclotomic tower the ordinary two-primary class group vanishes. One proves this inductively: only the prime above two ramifies in each quadratic step; the abelianization of a possible Hilbert two-class tower is generated by that inertia of order two. A finite two-group with cyclic abelianization is cyclic, and inertia excludes a larger cyclic

extension. The unit-signature calculation is separate. A norm-minus-one unit in positive layers, and -1 at the base, has signature vector a unit in the characteristic-two cyclic group algebra. For a scalar coefficient the resulting residual dimensions are

$$\dim H^1 = 2^n + s_n - 1, \qquad \dim H^2 = s_n - 1.$$

For each fixed support s_n is bounded. Universal coefficients give generic residual rank one in degree one; the two trivial elliptic factors give rank two. Unit cancellations thus identify the positive complex over $\mathcal{O}[G]$ with $\mathcal{O}[G]^2[-1]$. This last rank is independent of support, although preliminary models need not have a uniform size.

A generator of the fixed real invariant lattice gives a connecting class a. Its central image need not be saturated. Twice an ordinary class lifts because the relevant real H^1 is killed by two. Wedge a with a lift of $2F_E Z_F$ in the original determinant line. This produces a common scalar u^{ev}. The first assertion below holds for the full family. The central valuation identity is asserted only at a constituent of Selmer corank zero with $L_*(h) \neq 0$:

$$2^{C_{\mathrm{ev}}} u^{\mathrm{ev}} \in \Lambda[G], \qquad v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + O_E(1). \tag{3.2}$$

Here $\Lambda = \mathbb{Z}_2[[t]]$ and $\mathcal{O} = \widehat{\Lambda_{(2)}}$. The error and C_{ev} are fixed before the binary group and support. To justify membership in $\Lambda[G]$, first use rational characterwise divisibility and Fourier inversion for membership in $\Lambda[1/2][G]$, without estimating the Fourier denominator. Intersect coefficientwise with the integral $\mathcal{O}[G]$ result. The explicit coefficient construction also covers field discriminant $4h$, with its dyadic character retained before the cofinal norm limit, as in Appendix A5. At a corank-zero constituent the same coordinate vanishes centrally if and only if the central value vanishes. At such a nonzero corank-zero constituent, keeping the real saturation and torsion terms gives the local formula

$$v_2 u(0) = v_2 \exp^* z - s - \sum_v g_v + 2g + 1 - r_1 - e_2.$$

At an active good odd prime $g_v = 2$, since $A^{I_v} = E[2]$. Unused terms cancel (3.1); all other terms belong to the fixed local list. This proves (3.2) with the indicated active-prime coefficient. □

7.2 Nonvanishing with all support restrictions

Proposition 7.3 (Nonvanishing in a prescribed unit filter)**.** *The curve E is the fixed curve of this section. The following assertion allows any nonempty signed unit filter and any additional finite avoidance set; its bound on the number of prime factors does not depend on either set. A signed unit filter prescribes a real sign and unit squareclasses at a finite set S containing $2, 3$ and the bad primes. Let B be any additional finite set of forbidden primes. If its root number is even, there is a squarefree parameter in this filter, coprime to B, with at most five prime factors and nonzero central value. In an odd filter there is a simple-zero parameter, with no asserted prime-count bound.*

Proof. Here is the reduction to the precise classical statements. Choose a signed prime q_0 in the prescribed unit classes by Dirichlet's theorem, avoiding $S \cup B$. Apply Hoffstein–Luo [71] in the even case, or Bump–Friedberg–Hoffstein [70] in the odd case, to the actual curve $E^{(q_0)}$. Require their positive discriminant d to split at $S \cup B \cup \{|q_0|\}$ and at the conductor primes. Then $q_0 d$ has the prescribed original classes. Hoffstein–Luo gives at most four prime factors of d, so at most five in total. The BFH clause used in the odd case is the simple-zero clause.

The root-number check includes ramified dyadic quadratic characters. At the primes in S equal squareclasses give isomorphic local curves. At a good odd prime newly ramified in the parameter,

the root factor is $(-1/p)$ [72, Proposition 3.3]. Its product outside S is fixed by the real sign and the dyadic unit class. Thus unit classes 3 or 7 modulo eight are allowed; the field discriminant may be four times the parameter. We do not invoke an odd-discriminant root formula on those parameters. See also Proposition A7.1 for the same five-prime bound and its subsequent use in the companion construction. □

7.3 Coefficient symbols and the finite address lemmas

We state the finite interface before using it for the even upper bound. The graph language records quadratic residue data of the coefficient indices. It does not replace a Selmer group by an abstract graph. Throughout this subsection all vertices are in the rational full-two-torsion case. Thus there is one independent residue bit per pair; the split/simple distinction needed for other two-torsion fields does not occur here.

Definition 7.4 (Compatible typed graphs)**.** Fix the finite support S of a coefficient source, containing two, and the finite quadratic characters used to read its coefficients. The *type* of a prime $p \notin S$ records its values under these characters and its unit squareclasses at S, including its class modulo eight. We write the resulting finite elementary two-group additively as $\mathcal{L}$. We retain all relations among these coordinates; a tuple of arbitrary character values need not be a type. Additional fixed splitting conditions restrict us to their actual realizable types. For an odd theta stage, its reduction prime is part of this fixed support during that stage. Let S_{lab} also contain the ramification of any additional stationary label tests. Enlarging S_{lab} records labels and prime exclusions; it does not change the coefficient source or add relaxed places to a Selmer complex.

A *compatible typed graph* consists of a finite ordered vertex set V, an individually realizable type $\ell_v \in \mathcal{L}$ for each vertex, and quadratic residue bits. For prime representatives, write $z_{vw} = 0$ or 1 according as $(p_v/p_w) = 1$ or -1. For each unordered pair just one direction is freely specified; the other satisfies

$$z_{vw} + z_{wv} = \epsilon_v \epsilon_w, \qquad \epsilon_v = (p_v - 1)/2 \pmod 2,$$

where ϵ_v is part of the type. There are no self-residue bits. Only these compatible, realizable data are used below. A fixed filter prescribes the total type $\sum_v \ell_v$ needed for the coefficient index, including the signed parameter convention of the source. If that convention is $h = \sigma D$, then $D = \prod_v p_v > 0$ is the Fourier index and σ is fixed by the source. The signed factors $p_v^* = (-1/p_v)p_v$ will be used when the same graph defines a family of quadratic characters.

Contracting two vertices replaces their type by its sum and every incidence to an external vertex by the sum of the two incidences. Equivalently its template is the product $p_v p_w$; it is represented by a fresh prime with those data. Reverse incidences are determined by reciprocity. No bit internal to the contracted cluster is retained. A vertex is *primary* if its generator is a square unit at the fixed support and its type is neutral in every retained quadratic and stationary coordinate. It is *neutral relative to a graph* if in addition all its incidences to the other vertices are zero.

Two symbols may read projections of one common type group. The projections retain the relevant coordinates, preserve primary types, and commute with the product and external-incidence rules just given. We require each indicated projection to be surjective onto its own realizable alphabet; we do not require the map to their product to be surjective.

Definition 7.5 (Coefficient symbols)**.** A *symbol* is the function A with values in $\mathbb{F}_2$ obtained from one fixed coefficient source and one fixed normalized depth δ. On an allowed positive squarefree index D it is

$$A(D) = \begin{cases} 1, & v_2(a(D)) = \delta + \omega(D), \\ 0, & v_2(a(D)) > \delta + \omega(D), \end{cases}$$

and it is zero off that source's filter. For the even source, exact phase removal makes the relevant coefficient rational. For the odd source, (3.8) gives the required phase removal modulo the retained power 2^{M_θ}; use its integer representative at that precision. A coefficient zero modulo 2^{M_θ} is invisible. Whenever a unit is tested, the precision exceeds $\delta + \omega(D)$, so the test is independent of the representative. No exact rationality of an arbitrary lifted odd theta coefficient is asserted. The lower bound making these alternatives exhaustive is part of the coefficient construction, not part of this definition. For the even source it follows from (3.2) and the coefficient-square formula of Proposition A5.4.

For an odd source this definition at one finite stage is used only through a specified finite squarefree-weight cutoff and at a stage where the lower bound holds through that cutoff. The chosen cutoff and all required divisions must be included in the operator budget. The fixed minimum and the compatible finite tables are constructed in Propositions A7.3 and A7.5. The resulting limit symbol is a function on finite compatible typed graphs. It is not a valuation rule at all weights of one finite form. The same source and depth are kept when a vertex is omitted.

Lemma 7.6 (The coefficient rules)**.** *After the stated coefficient lower bounds have been proved, the even symbol and the stabilized odd symbol satisfy the following rules on compatible typed graphs. At an odd stage a rule is asserted only when all its inputs, outputs and coefficient divisions lie in the chosen finite budget. Configurations in these rules, and outputs of contractions, are nonempty.*

1. Graph dependence and whole-prime omission. *For a new vertex q adjoining a list U, its reading factors through the elementary quadratic extension unramified outside $S \cup U$. It is determined by its type and its incidences with U. The reading after omission of $p \in U$ uses the same original form, phase characters and depth, with varying support $U \setminus \{p\}$. That extension is unramified at the whole rational prime p. Changing inertia at the omitted prime has no effect on this reading.*

2. Contraction. *For the independent bit of a pair v, w, write $\Delta_{vw}A = A|_{z_{vw}=0} + A|_{z_{vw}=1}$, changing the reverse bit as required by reciprocity. Then*

$$\Delta_{vw}A(\Gamma) = A(\Gamma/(v, w)).$$

   The graph on the right has the summed type and every *summed external incidence from Definition 7.4. Its fresh prime representative is read using the omitted-prime source in the first rule.*

3. Primary isolation. *If a configuration contains a primary vertex neutral relative to all its other vertices, its symbol is zero.*

These are identities of the actual coefficient sources, rather than axioms imposed on an arbitrary binary function.

Proof. Proposition A6.5 proves the rules from the integral shell recurrence, the unary subtraction and the weight-two trace and inertia identities. Divisibility by the integer two precedes reduction modulo the residue field. In the present full-two-torsion case every new vertex is split in $\mathbb{Q}(E[2]) = \mathbb{Q}$; the contraction coefficient is one. The omitted form is constructed from the same original source, so its unramifiedness removes the entire prime and does not choose a new minimum. The final fresh-copy comparison in that proof shows dependence on the whole typed graph, including overlapping prime realizations. The same argument proves the rules for the even source; no odd normalization is needed in that application. □

Lemma 7.7 (Finite realization)**.** *Keep finitely many old prime vertices and all their mutual data fixed. Given individually realizable types for finitely many new vertices, every prescription of the independent forward bits involving a new vertex can be realized by distinct fresh primes outside any prescribed finite avoidance set. This includes primary vertices and extra fixed splitting conditions. A product template can likewise be replaced by a fresh prime with the same type and all external incidences whenever that output type is one of the permitted realizable types.*

Proof. For rational quadratic characters the fixed type is a reduced residue condition at a modulus supported on S_{lab}, including eight. Choose the new primes successively. A desired residue bit at an old prime is a nonzero residue class there. The Chinese remainder theorem combines these classes with the fixed type, and Dirichlet's theorem supplies infinitely many primes in the resulting reduced residue class. At each step avoid all previously chosen primes and the finite avoidance set. Reciprocity gives the required reverse bits. Apply the same construction to the external residues and support type of a product template. The graph dependence in Lemma 7.6 makes replacement by this fresh representative legitimate for the symbol. □

Definition 7.8 (Unit seeds and partitionable lists)**.** A *unit seed* is a finite compatible typed graph with symbol one. A *minimum nonempty seed* has the least positive vertex count among such seeds. This terminology does not exclude the possibility that the empty configuration is also a unit. A nonempty prime list is *partitionable* if its vertices admit a partition into nonempty clusters, with a contraction tree in each cluster, whose contracted graph is a unit seed. These trees and the resulting types and incidences constitute a contraction certificate. This condition makes no assertion about a Selmer group.

Lemma 7.9 (Minimum seeds and contraction certificates)**.** *At the least positive vertex count of a unit seed, its value is independent of all internal pair bits and depends only on its vertex types. A contraction certificate for a nonempty list makes its polynomial in the independent pair bits nonzero. In particular, a list with the same total type as a minimum nonempty seed is partitionable if it contains anchors for the first $r-1$ seed types and a nonempty remainder, where r is the seed's size.*

Proof. If $r \geq 2$, a nonzero derivative in an internal bit would, by contraction, give a unit on $r-1$ nonempty vertices. This contradicts minimality. If $r = 1$ there is no internal bit. Iterating the contraction identity along a forest gives its contracted symbol; an output equal to one therefore proves that the original polynomial is nonzero. For the last assertion keep the $r-1$ anchors and merge the whole remainder into the last cluster. Equality of total types forces the last seed type. The internal bits of the minimum seed do not matter. All contractions have nonempty output. □

Definition 7.10 (Addresses and terminal polynomials)**.** For an integer $b \geq 1$, an *address* is $x \in \mathbb{F}_2^b$. A vertex with activation $\lambda \in (\mathbb{F}_2^b)^* \setminus \{0\}$ is present at x exactly when $\lambda(x) = 1$. Stationary vertices are present at every address. At a fixed address, retain all vertex types and any fixed external incidences, and regard a symbol as a function of the independent pair bits between the active terminals. Its unique multilinear polynomial over $\mathbb{F}_2$ is the *terminal polynomial*. Its variables describe possible changes of those bits; the initial prime realization already has fixed bit values. Finite differences in these variables are formal derivatives.

Lemma 7.11 (One symbol with a stationary base)**.** *Fix a coefficient symbol satisfying Lemma 7.6, a nonempty stationary terminal list, its types and any stationary external data. Suppose its terminal polynomial is nonzero and its total data lie in the source's filter. For every $b \geq 1$ and every fixed assignment of the list's pair bits, finitely many new primary vertices with nonzero linear activations*

can be adjoined, with only their new incidences chosen, so that the symbol is one at every $x \neq 0$. The original vertices and all their mutual data are unchanged. No new vertex is present at zero. Fixed extra splitting and neutral external-incidence requirements may be included in the primary types. For a fixed even source the original prime representatives can be kept throughout. For a stabilized odd symbol, the fixed data in this statement are the abstract types and bits; its actual theta stage is chosen before its prime representatives, as specified in the proof.

Lemma 7.12 (Two symbols on one address family)**.** *Fix one or two coefficient symbols satisfying Lemma 7.6, reading compatible projections of one common realizable alphabet. Fix $b \geq 1$ and terminals with nonzero linear activations, including at least one terminal for each nonzero linear form. Stationary external data may also be fixed. Assume that at every $x \neq 0$ the active configuration is nonempty, its total data satisfy every required filter, and each symbol's terminal polynomial is separately nonzero. Then, starting with any fixed assignment of these compatible types and bits, finitely many new primary vertices with nonzero linear activations can be adjoined so that all the symbols are one at every $x \neq 0$. Only incidences involving the new vertices are chosen. All original data and the zero address are preserved, and the same fixed splitting and neutral external-incidence conditions may be imposed as above. If an odd limit symbol is present, this is first a statement about the finite typed graphs. It does not fix rational prime representatives before the eventual theta stage.*

Proof of the two address lemmas. The forest and network calculation in Proposition A7.6 proves both statements. For the stationary case its grounded-network columns attach to a fixed terminal, so no terminal activation anchors are required. For variable terminals the stated anchors give the row-sum identity on every active restriction. Singular networks isolate prescribed derivatives at one nonzero address; varying networks supply the corresponding difference when two symbols share a variable. The two-symbol proof takes the product of the top forms in the ordinary polynomial ring before Boolean reduction. Its nonzero coefficient forces a single assignment that works at all nonzero addresses. It does not require the two symbols to have a common nonzero evaluation before the auxiliary vertices are added.

Each construction uses a finite number of vertices and coefficient readings once b and the terminal polynomials are fixed. For an odd symbol, include all of them in one finite weight and operator budget, then use Proposition A7.5 to choose one actual theta stage carrying those tables and divisions. Only afterward use Lemma 7.7 to choose its primes. Any old prime representatives already chosen at that stage may be retained when they have its full prescribed types and bits, by the same realization lemma. In the even stationary application all original primes can be retained from the outset. The number of auxiliaries and the required theta precision may depend on b and the terminal data. The source's depth and the valuation constants attached to a unit do not change. No common finite stage for all values of b is asserted. □

The extra upper-bound argument. Nonvanishing alone does not bound the valuation of L_*, and (3.2) by itself gives a lower bound. The needed upper bound uses a separate even-symbol selection and transfer argument, which we now give.

Proposition 7.13 (The partitionable even upper estimate)**.** *Fix one even coefficient source, its signed unit filter and its depth. If a parameter d in that filter has a nonempty partitionable prime list and $E^{(d)}$ has full two-primary Selmer corank zero, then (3.3b) holds. Its constant depends only on the fixed source, not on the number or sizes of the primes of d or on later finite exclusions.*

Proof. For the even coefficient system described below, a unit symbol at a parameter d gives

$$L_*(d) \neq 0, \qquad v_2 L_*(d) \leq 2\omega(d) + C_0. \tag{3.3a}$$

This follows from its same-filter coefficient-square formula and its fixed normalization. That even normalization uses the lower bound from (3.2), not the as-yet undefined odd minimum. Partitionability is the nonempty contraction-certificate condition in Definition 7.8. By Lemma 7.9, its iterated finite difference is nonzero and hence so is the terminal polynomial. An empty unit, when present, satisfies its coefficient bound directly and does not require the stationary address lemma.

For a partitionable parameter d already known to have Selmer corank zero, keep all its prime factors stationary. Lemma 7.11 gives arbitrarily large binary cubes based at d, with no old prime removed, whose other addresses satisfy (3.3a). The fixed-form common scalar in (3.2) has one uniformly cleared denominator on every such cube. Choose a two-adic precision larger than the nonzero-address scalar valuation bound and its fixed clearing exponent. Its sign-evaluation coefficient is a Boolean function with the bounded bit degrees proved in Section 9. A cube dimension larger than their sum therefore supplies a nonzero address congruent to the base at this precision. If the base scalar were zero or had larger valuation than the bound, so would that nonzero address, a contradiction. The corank-zero central comparison in (3.2) consequently gives

$$L_*(d) \neq 0, \qquad v_2 L_*(d) - 2\omega(d) - s(d) \leq C_1. \tag{3.3b}$$

The constants are fixed by the even system. The dimension needed may depend on d; the valuation bound does not. This is the cyclotomic missing-address argument on the fixed-form Kato scalar, not the later Heegner transfer. □

Proposition 7.14 (A bounded companion in the prepared filter)**.** *For (3.3), restrict h to positive squarefree parameters in the prepared odd-sign unit filter containing one, prime to its fixed support. Thus $h \equiv 1 \pmod 8$ and $w(E^{(h)}) = -1$. The simple-zero parameters used in (3.4)–(3.6) lie in this filter. There are constants A, C_E fixed before h varies such that, outside any prescribed additional finite avoidance set, one can choose a negative odd fundamental discriminant k' which is a square at every place dividing $2Nh$, avoids the extra-unit fields, and has $L_*(hk') \neq 0$. It satisfies (3.3), in particular $\omega(k') \leq A$. The valuation estimate uses the fixed even sources of the original E, although the analytic nonvanishing theorem will be applied to an auxiliary twist of E.*

Proof.

Here is why the actual rough companion is partitionable. Prepare an even symbol for every possible total type at the finite fixed support. There are finitely many. In our full rational two-torsion case the alphabet is all split. A least nonempty unit seed has no dependence on internal edge bits: an edge derivative would be a smaller nonempty unit configuration. Its individual labels determine it. The negative-discriminant parity at two makes the total label nonzero, so this seed is nonempty. A list with the same total label and enough copies of the seed labels contracts to it: keep the first $r-1$ seed anchors and merge all remaining vertices into one nonempty cluster. Its label is forced to be the last seed label. Thus the list is partitionable, regardless of the number or labels of the remaining vertices.

Before h varies, fix a bounded multiset of dummy labels supplying the required anchors and reserves for every prepared total type. For each h, realize these labels by distinct primes avoiding the fixed support, the support of h, and all already fixed exclusions. Put in B the entire fixed support, the support of h, the chosen dummy primes, and every other already fixed forbidden prime. Include any already fixed reduction prime in B as well. Prescribe the total k' to be a negative odd fundamental discriminant, square at every place of $2Nh$ and outside the extra-unit cases. Absorb h and the dummy product into the curve and use the preceding nonvanishing theorem in its even filter. It adds at most five new primes. For the actual resulting total type, the already

chosen dummy supply makes hk' partitionable. Its analytic nonvanishing gives Selmer corank zero by the forward rank-zero theorem. Apply (3.3b), obtaining

$$\omega(k') \le A, \qquad v_2 L_*(hk') \le 2\omega(h) + 2\omega(k') + s(hk') + C_E. \tag{3.3}$$

Here A is the fixed dummy count plus five. This is the argument of the partitionable-even and rough-companion lemmas, with their stationary primes and order of choices retained. Neither prime sizes nor a uniform analytic threshold are asserted. Excluding a later coefficient prime does not enlarge the old Selmer support. The finite source preparation and the distinction between fixed dummy labels and their later prime realizations are also detailed in Proposition A7.2. □

7.4 The genus lower bound precedes normalization

Fix a negative genus partner h_* with $L_*(h_*) \neq 0$. Here h_* is chosen odd fundamental with $|h_*| > 4$ and in the same squareclasses at $2N$ as the prepared positive filter containing one. Throughout (3.4)–(3.6), h belongs to that filter and is prime to the fixed support, including h_*; the companion k' is supplied by the preceding construction with the primes of h_* also in its avoidance set. For a known simple-zero h, let P_h^{gen} be the unaveraged conductor-one genus sum over $\mathbb{Q}(\sqrt{hh_*})$. Let $P_{h,k'}^{\mathrm{rc}}$ be the unaveraged conductor-$|h|$ ring-class sum over $\mathbb{Q}(\sqrt{k'})$, weighted by χ_h. They are different points in different native lattices. Write $j_{\mathrm{gen}}(h)$ and $j_{\mathrm{rc}}(h,k')$ for their index valuations.

The native genus lattice. For the genus point specifically, put $K_h = \mathbb{Q}(\sqrt{hh_*})$ and $F_h = \mathbb{Q}(\sqrt{h}, \sqrt{h_*})$. If $h > 1$ and τ generates $\mathrm{Gal}(F_h/K_h)$, set

$$\Gamma_h = \{P \in E(F_h) : \tau P = -P\}, \qquad \Lambda_h = \Gamma_h/\Gamma_{h,\mathrm{tors}}.$$

For $h = 1$, set $\Gamma_1 = E(K_1)$ and take its torsion-free quotient. The native genus index is the divisibility index of the class of P_h^{gen} in Λ_h, at the known simple-zero parameters where this lattice has rank one. The CM reciprocity action and reindexing the unaveraged genus-character sum give exact membership $P_h^{\mathrm{gen}} \in \Gamma_h$, before multiplication by the later fixed multiplier. The standard h-twist isomorphism identifies Γ_h with $E^{(h)}(K_h)$. The remaining involution of $K_h/\mathbb{Q}$ fixes its free quotient: the only other rational character component is the h_*-component, whose rank is zero by the supplied forward theorem. Thus $Q + \sigma Q \in E^{(h)}(\mathbb{Q})$ represents twice the free class of $Q \in E^{(h)}(K_h)$. The rational subgroup consequently contains $2\Lambda_h$ and has index at most two. For $h = 1$ this is ordinary quadratic conjugation and the same norm argument. This calculation uses the stated rank and CM inputs; it does not infer rank from nonvanishing height, nor apply to the whole group $E(F_h)$.

Proposition 7.15 (The genus bound before odd normalization)**.** *Use h_*, h, k' and the two native index valuations defined in the preceding paragraph, with h a known simple-zero parameter and k' as in Proposition 7.14. Then (3.4)–(3.6) hold. In particular their constant in (3.6) is independent of h. The arithmetic lower estimate (3.5) holds uniformly for the admissible companions with $\omega(k') \le A$; it is only asserted here at these known simple-zero parameters.*

Proof. The explicit height formula [73, Theorem 1.1] applies twice. All level primes split, conductors are prime to N, extra-unit fields are excluded, and the parametrization is the same. Its conductor factor is $c\sqrt{|\operatorname{disc} K|}$ and its sum is unaveraged. The two Rankin products have the common factor $L(E^{(h)}, s)$. Each rational free line has index at most two in its native rank-one lattice: conjugation fixes the free class, and the norm makes twice the class rational. Thus

$$2j_{\mathrm{rc}}(h,k') - 2j_{\mathrm{gen}}(h) = v_2 L_*(hk') - v_2 L_*(h_*) + \epsilon, \qquad |\epsilon| \le 2, \tag{3.4}$$

up to the one fixed initial period convention.

The arithmetic estimate of Section 9, for each fixed A, is

$$2j_{\mathrm{rc}}(h,k') \geq 4\omega(h) + s(h) + s(hk') - C_R(E,A), \qquad \omega(k') \leq A. \tag{3.5}$$

Combining (3.3)–(3.5) cancels $s(hk')$ and gives

$$2j_{\mathrm{gen}}(h) \geq 2\omega(h) + s(h) - C. \tag{3.6}$$

This calculation uses no odd normalization. For the forward reference in (3.5), combine Proposition 9.6 with the same-class central identity in Proposition 8.10. Those arguments use the known nonzero point in the present setup; they do not use an odd coefficient minimum or the final family. □

7.5 A fixed nonzero witness and actual odd forms

Choose one positive squarefree D_w in the odd filter containing one, after adjoining the primes of h_* and their compatible unit tests. Section 7.2 supplies $L'(E^{(D_w)},1) \neq 0$. The explicit height formula makes $P^{\mathrm{gen}}_{D_w}$ nontorsion. Its unwanted h_* component has rank zero. One fixed multiplier L kills the bounded-degree torsion and the projection denominator, so $Q = LP^{\mathrm{gen}}_{D_w}$ is a rational nontorsion point on $E^{(D_w)}$.

Let C be complex conjugation on its Tate module and choose a nonzero integral row b with $bC = b$. The Kummer translation image of Q is a full lattice. Indeed, if it vanished, its rational cocycle would descend to the Tate image. A nonidentity central scalar z gives

$$(z-1)c(g) = (g-1)c(z),$$

so that descended cocycle is a coboundary, contrary to Kummer injectivity. Irreducibility makes a nonzero translation span the Tate space, and intersection with any fixed finite kernel preserves full rank. Twisting preserves open image by restriction to an index-at-most-two subgroup.

Choose one such translation v with $2b(v) \neq 0$. Multiplying conjugation by this translation gives square translation $(1+C)v$, detected by $2b(v)$ with one fixed valuation c_w. The element is a single actual Galois element on the compositum. It is trivial on $\sqrt{D_w}$ and nontrivial on $\sqrt{h_*}$ because $D_w > 0 > h_*$. Its restrictions therefore already agree on intersections. Finite Chebotarev, with a place chosen above the resulting prime, gives arbitrarily advanced stages (M_θ, ℓ_0) such that

$$\rho(\mathrm{Fr}_{\ell_0}) \equiv C \pmod{2^{M_\theta}}, \quad (D_w/\ell_0) = 1, \quad (h_*/\ell_0) = -1, \quad v_2\lambda_{M_\theta}(\overline{Q}) = c_w < M_\theta. \tag{3.7}$$

Here $\lambda_{M_\theta} = b\kappa_{M_\theta}$ is the actual finite-field Kummer map on $E(\mathbb{F}_{\ell_0^2})$. It has Frobenius sign $+1$. All reductions use that same place. This is a fixed-witness statement, not a bound for arbitrary points or arbitrary reduction primes.

The weighted ternary theta construction gives actual holomorphic forms at these stages. At original level primes the root weight uses the summand selected by the cyclic level structure. At two, the integral spectral projectors of $(1+x)/2$ specify that summand; the operator itself is not in general idempotent. The nilpotent weight at h_* and the division-algebra residue-line weight at ℓ_0 have the prescribed conjugation and uniformizer signs. Their product is even. The root-character field is fixed before ℓ_0 and M_θ.

Appendix A6 gives these functions, their exact transformation rules, and the direct extraction using the form Q/m. Poisson summation for the resulting compactly supported local functions gives the scalar transformation law and holomorphy at every cusp. A fixed stabilizer multiple, for example

one divisible by $|\mathrm{GL}_4(\mathbb{F}_3)|$, clears the finite automorphism groups. For the Hecke recurrence, a split or irreducible nonzero reduction has $1 + (-n/p)$ invariant lines, a nonzero nilpotent has one, and zero has $p + 1$. The unique integral primitive nilpotent neighbor supplies the remaining square term. Tracking the scaling characters gives the recurrence with the same character convention as the coefficient comparison.

For an admissible D with $(D/\ell_0) = 1$, the oriented embedding count from the conductor-one argument Section A35.3, with the retained geometric and CM inputs stated there, gives

$$a_{M_\theta}(D) = c_0\, 2^{\omega(N)} \zeta_{M_\theta}(D) \lambda_{M_\theta}(\overline{LP_D^{\mathrm{gen}}}) \pmod{2^{M_\theta}}, \tag{3.8}$$

where $c_0 \neq 0$ is fixed and ζ is a permitted root-of-unity phase. For admissible D with $(D/\ell_0) \neq 1$, including $\ell_0 \mid D$, the local Schwartz support instead gives $a_{M_\theta}(D) = 0$. Formula (3.8) is asserted only on the first support. There is no division by a class number. Formula (3.7) makes D_w visible with fixed upper valuation in (3.8). It remains on every stage's support, including the extra test $(D_w/\ell_0) = 1$.

After phase removal, the coefficients used for valuation are integers. Equation (3.6) gives a uniform lower bound for $v_2 a_{M_\theta}(D) - \omega(D)$ on visible bounded-weight sequences. Take the finite integer values constant on a set belonging to the chosen nonprincipal ultrafilter. This set of values is nonempty by D_w, is bounded below, and hence has an attained minimum δ. If the resulting lower inequality failed for a fixed weight bound on an ultrafilter-large set, choosing a violating index and then one of finitely many smaller integer values would contradict minimality. For any fixed finite operator budget, the coarse lower bound handles the large weights and stabilization handles the bounded weights. Consequently every required division is valid at one sufficiently advanced actual stage. No infinite-precision modular form or arbitrary-ring base-change assertion is needed.

7.6 One common family and the final order of choices

The even source is a fixed algebraic vector in the full good-Hecke eigenspace. Waldspurger's Corollary 2, including its same-dyadic-squareclass removal of the auxiliary condition, compares the actual squarefree Fourier functionals [79]. Local upper-unipotent projections retain a nonzero initial functional; smoothness supplies lower-unipotent invariance. The algebraic linear conditions choose a vector over a number field. Galois conjugation followed by the matching finite character twist shows that each normalized coefficient ratio is a rational number times a two-power root of unity. This supplies the valuation and nonvanishing comparison on the whole filter. Only flat field extension of the algebraic model is used. The comparison is proved at the original support S_0 before any extra coprimality sieve B. The sieve is applied last, and every later good recurrence prime avoids the full support $S_0 \cup B$.

For both coefficient systems the integral operators have the required omission and contraction rules. At a new prime V_p raises the level exponent only to one, with ramified quadratic character. The recurrence makes the relevant differences divisible by the integer two before residue reduction. Multiplying by ordinary theta, then taking a fresh trace, removes the integral unary corrections. Two fresh traces vanish. The matrix trace identity gives $t(gu^2) = t(g)$, hence elementary quadratic labels. At a new odd level prime the weight-two local representation has one unramified and one ramified line. The normalized-induction and Whittaker calculation below identifies the Fourier U_p eigenvalue with arithmetic Frobenius on the unramified line. The diagonal-matrix calculation gives the two inertia differences. These identities prove omission and primary isolation for the same original form and depth, without choosing another minimum.

If the intrinsic odd limit symbol is a unit at the empty configuration, choose an actual stage carrying that value with $M_\theta > \delta_o$. The index $D = 1$ lies in every moving support and then satisfies

$v_2 a_{M_\theta}(1) = \delta_o < M_\theta$. The visibility implication (O.8) gives $\operatorname{ord}_{s=1} L(E,s) = 1$, which is exactly (RC). No final companion or determinant bound is needed in this case. For the rest of the final-family and missing-address argument, assume that the empty configuration is not a unit. The bounded-weight witness then supplies a least nonempty odd seed, as in Appendix A7.

Fixing the final companion. Use a separate even symbol on negative discriminants square at $2N$, excluding the fixed odd support, the ramification support of $E[8]$, and the exceptional imaginary fields. Choose a unit k of this symbol. Then $L(E^{(k)},1) \neq 0$. Fix k and $K = \mathbb{Q}(\sqrt{k})$ before choosing the binary dimension. At least one prime ramified in K is outside all the old fixed label extensions and torsion ramification. Choose every later moving theta prime ℓ_0 away from k. The intrinsic odd symbol continues to omit the primes of k.

At any such stage, the normal closure of the old odd label extension is unramified outside its old support and ℓ_0. The fresh ramified prime of K lies outside this set, so its intersection with K is $\mathbb{Q}$. Thus each old odd label still has a lift split in K. The added stationary tests at the primes of k may have relations; include those relations in the common alphabet, rather than requiring independent arbitrary prescriptions. Stabilize its separate projection maps on the same ultrafilter. This preserves the old odd minimum. Fresh ramification also makes K independent of $\mathbb{Q}(E[8])$ and of the old fields $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{\ell})$.

The forward theorem at this known nonzero central value gives $s_2(E^{(k)}/\mathbb{Q}) = 0$. The sum-and-difference isogeny over $\mathbb{Q}$ gives

$$s_2(E/K) = s_2(E/\mathbb{Q}) + s_2(E^{(k)}/\mathbb{Q}) = 1.$$

Indeed its kernel and cokernel on the relevant restriction and corestriction maps are killed by a fixed power of two. Tensoring the compact Selmer modules, equivalently the Pontryagin duals of the discrete Selmer groups, with $\mathbb{Q}_2$ gives the direct sum and hence this identity of coranks. No integral direct-sum assertion is used.

Lemma 7.16 (The nonempty seed for the present curve)**.** *For $E = E_\ell$ and the k, K just chosen, the positive even source for $E^{(k)}$, normalized at index one, has a nonempty unit seed consisting of K-split primes.*

Proof. This is the specialization to the roots $0, \ell, -3\ell$ of the coefficient calculation in Proposition A7.4. If no such nonempty unit existed, its integral p^3 coefficient and a fresh trace would force $a_p(E^{(k)}) \equiv p + 1 \pmod 8$ at every fresh good K-split prime outside the support, whether or not it lies in the coefficient filter. Indeed, off the filter $a(p) = 0$ in (G.4a); the second fresh prime p' matches the old labels so that pp' lies in the filter, as used in (G.4d). The selected p below need not lie in the filter. The cited calculation uses positive Fourier indices and the positive twist character convention; it derives this congruence inside the assumed absence of nonempty units.

Choose such a prime with $p \equiv 1 \pmod 4$, $(\ell/p) = -1$ and $(k/p) = 1$, avoiding the finite support. In fact this p is outside the square filter: quadratic reciprocity and $p \equiv 1 \pmod 4$ give $\prod_{r|\ell}(p/r) = (\ell/p) = -1$, so p is a nonsquare at some odd $r \mid \ell$, and $r \mid N$. This is why the off-filter scope of the preceding calculation is needed. The fresh ramification of K permits these simultaneous conditions. Because k is a square modulo p, the reductions of $E^{(k)}$ and E_ℓ are isomorphic over $\mathbb{F}_p$. None of the three nonzero rational points of order two can be doubled: their necessary square differences include, respectively, $-\ell, \ell, -4\ell$. Each is a nonsquare under the chosen conditions. The two-primary reduction group therefore has order exactly four, contradicting the asserted congruence. This proof uses the particular curve E_ℓ; no nonempty-seed theorem for an arbitrary elliptic curve is being asserted. □

We record the finite combinatorial realization used here. Each label space is a finite elementary two-group; the common alphabet projects onto each separately. Product surjectivity is unnecessary. Take sufficient repeated copies of every label, with total label zero. Keeping the seed anchors, contract the unused copies into the last anchor. Minimal positive seed size makes the internal bits irrelevant, so each of the two terminal polynomials is nonzero. Grounded Laplacian minors express the constant and pair-contraction operators, including singular matrices. A cyclic singular network isolates its target radical and a path network isolates its selected pair coefficient. In the product of the two top forms, taken in the ordinary polynomial ring before Boolean reduction, a variable of exponent two uses one singular program in each factor; exponent one uses one varying program. The total degree kills the other contractions and leaves coefficient one. Thus one choice works for both forms. CRT and Dirichlet realize the prescribed forward residue bits by distinct primes; quadratic reciprocity supplies the reverse bits.

Proposition 7.17 (One family for the two coefficient tests)**.** *Keep the fixed E, h_*, normalized odd symbol and final k, K of the preceding paragraphs. Assume, as in the remaining branch above, that the intrinsic odd empty configuration is not a unit. In this remaining branch, for every sufficiently large b there is one finite set Q of distinct good odd primes split in K and outside the fixed support, and nonzero linear forms $\lambda_q : \mathbb{F}_2^b \to \mathbb{F}_2$, such that the family*

$$h_x = \prod_{q \in Q} (q^*)^{\lambda_q(x)} \quad (x \in \mathbb{F}_2^b), \qquad q^* = (-1)^{(q-1)/2} q,$$

has $h_0 = 1$, all fixed local types unchanged, and, for $x \neq 0$,

$$\operatorname{ord}_{s=1} L(E^{(h_x)}, s) = 1, \quad L_*(h_x k) \neq 0, \quad j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + C_o, \quad v_2 L_*(h_x k) \leq 2\omega(h_x) + C_e. \tag{3.9}$$

The constants are independent of b. By (3.4), now with fixed k, $2j_{\mathrm{rc}}(h_x, k) \leq 4\omega(h_x) + C_J$.

Proof. The packet argument just given supplies separate nonzero terminal polynomials in one common alphabet. Apply Lemma 7.12. The full proof in Proposition A7.6 and Proposition A7.7 preserves the projected total labels, all stationary tests and the original two normalizations.

The order is fixed data and D_w, then the genus lower bound and odd minimum, then the final $K = \mathbb{Q}(\sqrt{k})$ and common alphabet, then determinant precision M, then b and its finite operator budget, then a sufficiently advanced actual theta stage M_θ and its prime realization. In particular M_θ is not M. The finite graph construction, its local conditions and the coefficient tests all occur at that same final stage. Appendix A7 spells out the absolute five-prime completion, normalization constants, and the final use of the one corrected coordinate. □

8 Actual classes, local conditions and central comparison

Let $K = \mathbb{Q}(\sqrt{k})$ be imaginary, split at $2N$, and different from $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$. Active primes are good odd primes split in K. Derivative primes are good inert primes outside all current supports. The actual finite coefficients are

$$A_i = (\mathbb{Z}/2^i\mathbb{Z})[t]/t^i, \quad M_i = (T_2 E \otimes A_i)(\chi_h \Psi_i).$$

For a fixed binary group G, use $A_i[G]$ and the universal quadratic character. The character Ψ_i is finite, trivial at $2N$, ramified at a moving conductor prime r_i, and has prescribed odd scalar exponents at the finite active list. In a binary family let Q be the union of its prime supports. Apply Proposition 8.5 to an orientation above every $q \in Q$, prescribing $\xi_i(\mathrm{Fr}_q) = 1$ at each finite

stage. This prescription includes the primes unused by an individual character, including every graph prime at augmentation. The same Ψ_i is used for all characters of the family. The finite set Q is fixed before choosing the sequence of moving primes r_i; the trace bound c_E does not depend on its size. The prime r_i varies with precision. There is no assertion that one fixed odd prime supports an infinitely ramified $\mathbb{Z}_2$ character.

Definition 8.1 (Coefficient and local-condition conventions)**.** In Sections 8 and 9, retain the fixed non-CM curve E, parametrization π, Tate lattice, and old local types of Section 7. Put $\Lambda = \mathbb{Z}_2[[t]]$ and $\mathcal{O} = \widehat{\Lambda_{(2)}}$. A binary family uses one finite elementary abelian two-group G, one real sign and common local squareclasses at $2N$. Its universal varying quadratic character is therefore trivial at the real place and at $2N$. The active primes are good odd primes split in the indicated imaginary quadratic field K; the moving character is the finite character specified above, trivial at $2N$. The primitive and relaxed complexes use the actual derived Kummer, unramified, full and orthogonal local conditions specified below, with their maps and homotopies. At derivative primes the four local conditions are those of Lemma 9.5. Every determinant coordinate uses the retained integral determinant basis.

Lemma 8.2 (Conjugate coefficients and local duality)**.** *For the finite coefficient module M_i and coefficient ring $R_i = A_i$ or $A_i[G]$ above, let τ be an actual complex conjugation. Pullback of the Galois action gives an R_i-linear identification $\tau^* M_i \simeq M_i^*(1)$ through $(v,w) \mapsto e_2(v, \rho_T(\tau)w)$; it fixes t and the group-ring basis. With the conjugate places, actual orthogonal local conditions and their isotropy homotopies, finite Selmer duality gives*

$$C_U(M_i) \simeq R\,\mathrm{Hom}_{R_i}(C_{U^\perp}(M_i^*(1)), R_i)[-3].$$

The local Kummer and derivative comparisons used here are proved below and in Lemma 9.5.

Proof.

The dual coefficient system. Fix an actual complex conjugation $\tau \in G_{\mathbf{Q}}$, so that $\tau^2 = 1$. The moving character is constructed on the ring-class group:

$$\xi_i : \mathrm{Gal}(H_{r_i}/K) \longrightarrow \mathbf{Z}/2^{n_i}\mathbf{Z}, \qquad \Psi_i = (1+t)^{\xi_i},$$

where n_i is large enough that $(1+t)^{2^{n_i}} = 1$ in A_i. Conjugation acts on this ring-class group by inversion. Consequently

$$\Psi_i(\tau g \tau^{-1}) = \Psi_i(g)^{-1} \qquad (g \in G_K).$$

Let $R_i = A_i$ in the scalar construction and $R_i = A_i[G]$ in the binary construction. Write $\alpha_i = \chi_h \Psi_i$, respectively $\alpha_i = \chi_0 \varepsilon \Psi_i$, where $\varepsilon : G_K \to G$ is the restriction of the universal rational quadratic character. Then $\varepsilon^\tau = \varepsilon = \varepsilon^{-1}$ and $\alpha_i^\tau = \alpha_i^{-1}$. Here $\tau^* M_i$ denotes pullback of the Galois action; it does not act on R_i.

The Weil pairing gives the following R_i-linear perfect pairing:

$$M_i \otimes_{R_i} \tau^* M_i \longrightarrow R_i(1), \qquad (v,w) \longmapsto e_2(v, \rho_T(\tau)w).$$

Indeed $\rho_T(\tau)^2 = 1$, the two scalar characters multiply to one, and e_2 transforms by the cyclotomic character. Thus $\tau^* M_i \simeq M_i^*(1)$. We do not identify M_i with its ordinary dual when Ψ_i is non-quadratic. In this convention t and every element of the group-ring basis are fixed. Inverting $1+t$ is already accounted for by pullback of the character.

At a split place v, the dual local condition is transported from the conjugate place τv. At a fixed place it is transported by the actual local representative of conjugation. Kummer conditions at $2N$ are preserved by this transport. A full condition is paired with the zero condition; the full local complex is replaced by zero only at the localizations where its displayed contraction exists. At a derivative prime the finite and singular planes are checked with this transported pairing. In the chosen coordinates this is the pairing $\beta(x,y)=\epsilon\langle x,Jy\rangle$. The J is therefore part of the duality convention, not an extra identification of an arbitrary twist with its dual.

Use the coefficient-valued pairing in the finite Selmer diagrams, with the conjugate local places and their isotropy homotopies. The resulting duality is

$$C_U(M_i)\simeq R\,\mathrm{Hom}_{R_i}(C_{U^\perp}(M_i^*(1)),R_i)[-3].$$

These are the maps and homotopies retained in both limits. □

8.1 Finite Heegner descent and the derivative identities at two

Fix the non-CM elliptic curve $E/\mathbf{Q}$, its conductor N, its Tate module $T=T_2E$, and a modular parametrization $\pi: X_0(N)\longrightarrow E$. Normalize $\pi(\infty)=0$. Let K be imaginary quadratic, split at every prime dividing $2N$, and different from $\mathbf{Q}(i)$ and $\mathbf{Q}(\sqrt{-3})$. The conductor f below is prime to $2ND_K$; it may contain all the active primes in a finite binary family and the moving prime. Write H_f for the ring-class field of the order $\mathcal{O}_f=\mathbf{Z}+f\mathcal{O}_K$.

Let I be any finite set of good inert primes outside $6fND_K$, and put $n_I=\prod_{\ell\in I}\ell$ and $M_I=H_{fn_I}$. At the end, a finite character θ of $\Delta_f=\mathrm{Gal}(H_f/K)$ will take values in $A^\times$, where $A=(\mathbf{Z}/2^m\mathbf{Z})[t]/t^b$, or in $A[G]^\times$ for a finite binary group G. It has the form $\chi_h\Psi$, or the corresponding universal quadratic character times Ψ. At $2N$, Ψ is trivial and the universal quadratic character has the fixed local type already specified in the manuscript. These are hypotheses on the actual finite character. Its construction is separate from the descent proved here.

Proposition 8.3 (Finite Heegner descent with one multiplier)**.** *With E,π,K,f,I,M_I,A and the finite character θ as just specified, there are integers $e\ge 0$ and $L>0$ depending only on the fixed curve, parametrization and allowed old local types, and on any other prescribed fixed multiplier requirements, with the following properties. Form the derivative over the complete extension H_{fn_I}/H_f and descend it before applying the character. This gives finite classes $P_{I,m}$ with the old Kummer conditions, the stated good unramified conditions and full active and moving conditions. For $I=\varnothing$ the class is the Kummer class of the L-multiplied unaveraged weighted point.*

The unaveraged norms are (4.1). If $a\ge m+e$ and every new derivative prime has $\rho_T(\mathrm{Fr}_\ell)\equiv\rho_T(\tau)\pmod{2^{a+1}}$, the classes projected to precision m satisfy (4.2). These assertions hold over $A[G]$ before character evaluation. They are realized by finite continuous cocycles, local-condition representatives and boundary witnesses, with the weighted coefficient map applied to all this data. There is no multiplier depending on K,f,I,m, the truncation length, the number of active primes or $|G|$.

Proof.

1. One uniform torsion annihilator. By the non-CM open-image theorem [85], choose $n\ge 2$ such that $1+2^nM_2(\mathbf{Z}_2)\subset\rho_T(G_{\mathbf{Q}})$, and choose $g\in G_{\mathbf{Q}}$ with $\rho_T(g)=uI$, $u=1+2^n$. For an actual complex conjugation τ set

$$z=g^2\tau g^2\tau^{-1},\qquad e=v_2(u^4-1)=n+2. \tag{H1}$$

The element g^2 fixes every quadratic field. Its image in the abelian group $\mathrm{Gal}(H_d/K)$ is inverted by τ, for every d and every allowed K. Thus z fixes every such H_d, while $\rho_T(z) = u^4 I$. If $P \in E(H_d)[2^\infty]$, then $(u^4-1)P = 0$, so

$$2^e E(H_d)[2^\infty] = 0. \tag{H2}$$

This bounds the whole invariant torsion group, with no restriction on the number of conductor primes. It also holds for a compositum of the ring-class fields in a finite construction, since z fixes each of them.

For the matrix-span statement used elsewhere, square preimages of $I + 2^n E_{11}$, $I + 2^n E_{12}$ and $I + 2^n E_{21}$. The squares belong to every quadratic subgroup, and their images, together with I, are

$$I, \quad I + (2^{n+1} + 2^{2n})E_{11}, \quad I + 2^{n+1}E_{12}, \quad I + 2^{n+1}E_{21}.$$

Their $\mathbf{Z}_2$-span contains $2^{n+1}M_2(\mathbf{Z}_2)$: subtract I, use that $1 + 2^{n-1}$ is odd, and obtain E_{22} from $I - E_{11}$. This bound also depends only on E.

2. Point relations with the full old conductor. Choose the CM points $x_d \in X_0(N)(H_d)$ by compatible proper ideal lattices and the fixed orientations at N, and put $y_d = \pi(x_d)$. The split Heegner hypothesis identifies the curve $X_K(N)$ in Cai–Chen–Liu, Theorem 2.15 [74], with $X_0(N)$: $N_1 = 1$, and the condition $(*)$ reduces to $(d, N) = 1$. Take their old conductor to be any d currently in use and the new prime outside $6dND_K$. Their factor u_m is one because $\mathcal{O}_K^\times = \{\pm 1\}$. Applying the elliptic quotient to that theorem gives

$$\mathrm{Tr}_{H_{d\ell}/H_d}\, y_{d\ell} = a_\ell y_d \quad (\ell \text{ inert}), \qquad \mathrm{Tr}_{H_{dq}/H_d}\, y_{dq} = (a_q - \sigma_q - \sigma_q^{-1})y_d \quad (q \text{ split}). \tag{4.1}$$

The inverse Frobenius pair at a split prime follows because their product is the Artin symbol of the rational principal ideal (q). These are unaveraged sums. On degree-zero divisor classes, $\varphi T_q = [a_q]\varphi$. Since $q \nmid N$, $T_q[\infty] = (q+1)[\infty]$, so the displayed norm relations for $y_d = \pi(x_d)$ have no cusp correction. For other changes of CM convention, let $\mathcal{C}_N$ be the subgroup of $J_0(N)(\overline{\mathbb{Q}})$ generated by $[c] - [\infty]$ for all cusps c. This is a finite group by Manin–Drinfeld [65, p. 59, Theorem]. Fix $C_\pi \geq 1$ annihilating $\varphi(\mathcal{C}_N)$. This single integer depends only on N, π. For an Atkin–Lehner involution W acting on f by w, the exact formula is $\pi(Wx) = [w]\pi(x) + \pi(W\infty)$; the final constant lies in $\varphi(\mathcal{C}_N)$ and vanishes after multiplication by C_π.

At an inert new prime ℓ, a compatible place gives

$$\overline{y}_{d\ell} = F_\ell \overline{y}_d. \tag{H4}$$

Here F_ℓ is rational arithmetic Frobenius on the reduced elliptic curve. This is the usual CM congruence, and it does not require the old conductor to be an inert squarefree product. Indeed $(d, \ell) = 1$ makes the local CM order the maximal unramified quadratic order. The CM elliptic curve has supersingular reduction at this inert prime. Each of its $\ell + 1$ cyclic raising isogenies specializes to the quotient by the unique subgroup of order ℓ, namely the relative Frobenius kernel. The prime-to-ℓ cyclic N-subgroup specializes by the same Frobenius map. The resulting equality of reduced modular points passes through π, which extends over the good integral models at ℓ. This also proves (H4) after every old ideal-class translate, with its place transported. Howard, Section 1.7, formula (13) [95], records this congruence in his conductor subfamily; his odd-prime descent and residual-surjectivity theorem are not used here.

3. The relative group and invariant derivative class. The idelic description $\mathrm{Pic}(\mathcal{O}_d) = K^\times\backslash\mathbf{A}_{K,f}^\times/\widehat{\mathcal{O}}_d^\times$ identifies the kernel of $\mathrm{Pic}(\mathcal{O}_{fn_I}) \to \mathrm{Pic}(\mathcal{O}_f)$ with

$$\prod_{\ell\in I}\frac{(\mathcal{O}_K/\ell\mathcal{O}_K)^\times}{(\mathbf{Z}/\ell\mathbf{Z})^\times} \simeq \prod_{\ell\in I} C_{\ell+1}. \tag{H5}$$

There is no additional quotient by units: the only global units are ±1 and already belong to the denominator at every factor. This proves (H5) without choosing a splitting of the full class group. Let σ_ℓ be a generator of its ℓ-factor, with the local Artin orientation fixed. Set

$$D_\ell = \sum_{j=1}^{\ell} j\sigma_\ell^j, \qquad D_I = \prod_{\ell\in I} D_\ell, \qquad (\sigma_\ell - 1)D_\ell = (\ell+1) - \mathrm{Tr}_\ell\,. \tag{H6}$$

If 2^a divides a_ℓ and $\ell+1$ for all $\ell \in I$, (4.1) and (H6) show that

$$c_{I,a} = \delta_a(C_\pi D_I y_{fn_I}) \in H^1(M_I, E[2^a])^{\mathrm{Gal}(M_I/H_f)}. \tag{H7}$$

For example its difference under σ_ℓ is the Kummer class of $C_\pi D_{I\setminus\{\ell\}}((\ell+1)y_{fn_I} - a_\ell y_{fn_I/\ell})$, an actual 2^a-multiple of an M_I-point. The factors in (H5) commute.

4. Descend once, before twisting. Put $B = H_f$, $M = M_I$, $\Gamma = \mathrm{Gal}(M/B)$ and $V_a = E[2^a]$. The five-term sequence for finite discrete coefficients [56, formula (4.1.4.1), p. 96] gives

$$0 \longrightarrow H^1(\Gamma, V_a^{G_M}) \longrightarrow H^1(B, V_a) \xrightarrow{\mathrm{res}} H^1(M, V_a)^\Gamma \longrightarrow H^2(\Gamma, V_a^{G_M}). \tag{H8}$$

By (H2), multiplication by 2^e is zero on the coefficient module $V_a^{G_M}$ and hence on every cochain and cohomology group of Γ with those coefficients. For an invariant class c, choose b with $\mathrm{res}(b) = 2^e c$ and define

$$d_a(c) = 2^e b. \tag{H9}$$

Any two choices of b differ by the first group in (H8), so (H9) is independent of the choice. It is additive, natural for coefficient reduction, and equivariant for transport of the whole field diagram. Its restriction is exactly $2^{2e}c$. This is one application to M_I/H_f, not one application for each factor of (H5). The coefficient module in (H8) is untwisted $E[2^a]$; the scalar and binary coefficient push comes only later. Thus no invariant module containing a varying nonquadratic scalar character has been silently substituted in this argument. For $I = \varnothing$, the relative group is trivial and this same definition gives $d_a(c) = 2^{2e}c$. In particular it uses the same normalizing multiplier for the primitive class and its derivatives.

5. Integral local descent at the old places. We record the local fact needed to put $d_a(c_{I,a})$ in the old Kummer conditions. If F'/F is a finite unramified extension of nonarchimedean local fields and B/F is an elliptic curve, then

$$H^1(\mathrm{Gal}(F'/F), B_0(F')) = 0, \qquad c_B H^1(\mathrm{Gal}(F'/F), B(F')) = 0, \tag{H10}$$

where $c_B = \#\Phi_B(\overline{k}_F)$ is the geometric component group order. To prove the first assertion, reduce a cocycle to the connected smooth special fiber. Lang's theorem [96, Theorem 1 and its corollary] makes its H^1 over the finite residue field zero; inflation–restriction then makes its H^1 for the finite residue extension zero. Lift the resulting coboundary through the smooth reduction map. The remaining cocycle takes values in the formal group. For a uniformizer of F, fixed by the unramified

Galois group, each successive formal quotient is an additive copy of the residue field of F' [60, IV.3.2(a), p. 124]. A normal basis makes this an induced module, so its positive-degree cohomology vanishes. Correct the coboundary successively in these quotients. The corrections converge in $B_1(F')$, and the group is finite, so their limit solves all the cocycle equations. The component exact sequence proves the second assertion of (H10). No division by $[F' : F]$ was used.

If a class $v \in H^1(F, B[2^a])$ becomes Kummer over F', its image in $H^1(F, B)[2^a]$ belongs to the kernel of restriction. That kernel is the image of $H^1(\mathrm{Gal}(F'/F), B(F'))$. Thus $c_B v$ is Kummer over F, by the Kummer exact sequence and (H10).

All ring-class extensions in use are unramified at $2N$, since the conductors are prime to $2N$. The base fields there are unramified extensions of $K_v = \mathbf{Q}_p$. Geometric component groups are unchanged under unramified extension. Let C_{bad} be a common multiple of their orders for the finitely many allowed local quadratic types at $2N$. The same integer works for every conductor, every place above $2N$ and every K. Choose a fixed positive integer m_0, allowing the other fixed multiplier requirements in the argument to be included, and define

$$L = m_0 C_\pi C_{\mathrm{bad}} 2^{2e}, \qquad b_{I,a} = m_0 C_{\mathrm{bad}} d_a(c_{I,a}) \in H^1(H_f, E[2^a]). \tag{H11}$$

Thus any common fixed multiple of this descent multiplier and the detection multiplier is permitted; every class is scaled by the same m_0. Then $b_{I,a}$ is Kummer at $2N$ and restricts to $\delta_a(LD_I y_{fn_I})$. At a good place away from $2fn_I$, the same assertion has component order one. It gives the unramified local condition, since at good odd places the finite Kummer image equals the unramified image. At active and moving primes the required condition is full. The integer L is now fixed, independently of K, f, I, m, b and G.

6. Exact finite precision for one derivative prime. Let the retained torsion precision be m, choose $a \geq m + e$, and require, for every retained derivative prime,

$$\rho_T(\mathrm{Fr}_\ell) \equiv J \pmod{2^{a+1}}, \qquad J = \rho_T(\tau), \quad J^2 = I. \tag{H12}$$

Then 2^{a+1} divides a_ℓ and $\ell + 1$, since $\operatorname{tr} J = 0$ and $\det J = -1$. Choose $\ell \notin I$, and perform the construction for $I \cup \{\ell\}$. Apply the other derivatives first, writing

$$X = D_I y_{fn_I}, \qquad Y = D_I y_{fn_I \ell}.$$

The K-prime (ℓ) splits completely in H_{fn_I}: it is prime to the conductor and is a rational principal ideal in the ring-class Artin map. Thus its residue field is $\mathbf{F}_{\ell^2}$, and

$$F^2 \overline{X} = \overline{X}, \qquad \operatorname{Tr} Y = a_\ell X, \qquad \overline{Y} = F\overline{X}, \qquad F^2 - a_\ell F + \ell = 0. \tag{H13}$$

The same splitting makes the old class $b_{I,a}$ at this place exactly the Kummer class of LX, by its restriction identity in (H11); there is no local descent ambiguity for this old class. For $2^a Q = LX$ its finite coordinate is

$$f_{\mathrm{old}} = (F^2 - 1)\overline{Q} = (a_\ell F - (\ell + 1))\overline{Q}. \tag{H14}$$

Although the two integer coefficients are highly divisible, Q is a division point, so (H14) does not assert $f_{\mathrm{old}} = 0$.

Choose a continuous cocycle representing $b_{I\cup\{\ell\},a}$ and choose Z with $2^a Z = LD_\ell Y$. Its restriction to $G_{H_{fn_I\ell}}$ is cohomologous to $g \mapsto (g - 1)Z$. Their difference is the coboundary of a single element of $E[2^a]$; subtract the same global coboundary to make the restriction exactly this Kummer

cocycle. Denote the adjusted representative by b. Restrict it to $G_{H_{fn_I}}$ and choose a local inertia lift of σ_ℓ. Put

$$U = (\sigma_\ell - 1)Z - b(\sigma_\ell).$$

For $g \in G_{H_{fn_I\ell}}$, compare $b(g\sigma_\ell)$ with $b(\sigma_\ell(\sigma_\ell^{-1}g\sigma_\ell))$. Using the exact restricted Kummer representative gives $gU = U$. Consequently U is an actual point of $E(H_{fn_I\ell})$, and

$$2^a U = L((\ell+1)Y - a_\ell X).$$

The point

$$U_0 = L\frac{\ell+1}{2^a}Y - L\frac{a_\ell}{2^a}X \tag{H15}$$

also belongs to this field, with the same 2^a-multiple. Its coefficients are integers. Hence $U - U_0$ is upper-field 2-primary torsion and is killed by 2^e, by (H2). This comparison with an integer quotient point is necessary; comparison with an arbitrary division point would give no bound.

Inertia acts trivially on the reduction of every algebraic point at a good prime. Reduction is injective on 2-primary torsion. Thus the singular coordinate $b(\sigma_\ell)$ is

$$s_{\text{new}} = (a_\ell - (\ell+1)F)\overline{Q} + \epsilon = (F - a_\ell)f_{\text{old}} + \epsilon, \qquad 2^e\epsilon = 0. \tag{H16}$$

The second equality uses the exact endomorphism identity

$$(F - a_\ell)(a_\ell F - (\ell+1)) = a_\ell - (\ell+1)F. \tag{H17}$$

Reduction $T/2^aT \to T/2^mT$ is multiplication by 2^{a-m} on geometric torsion points. It kills ϵ. Conditions (H12) then turn (H16) into $s_{\text{new}} = Jf_{\text{old}}$.

For the new finite coordinate, all inertia translates of Y have the same reduction, so

$$\overline{LD_\ell Y} = L\frac{\ell(\ell+1)}{2}F\overline{X}. \tag{H18}$$

This is 2^a times an F^2-fixed reduced point, because $2^{a+1} \mid \ell+1$. The local extension obtained by adjoining the conductor ℓ is totally ramified, of degree $\ell+1$. Its residue field is still $\mathbf{F}_{\ell^2}$. Lift the indicated 2^a-division point in the residue field; the discrepancy lies in the formal group, where multiplication by 2^a has an integral formal inverse because ℓ is odd [60, IV.2.3 and IV.3.2, pp. 122–124]. This produces a local 2^a-division point of $LD_\ell Y$ in the upper local field itself. The restricted Kummer class is therefore zero. Choose a Frobenius lift in that upper local Galois group. It acts trivially on $E[2^a]$ by (H12), so its cocycle value is zero, independently of a coboundary adjustment. Hence

$$f_{\text{new}} = 0, \qquad s_{\text{new}} = Jf_{\text{old}} \quad \text{at precision } 2^m. \tag{H19}$$

The generator σ_ℓ fixes the tame orientation. A different generator changes the corresponding singular coordinate by the matching unit. Frobenius lifts and places in (H19) are part of the retained local diagram. The extra bit in (H12) pays exactly for the division by two in (H18). Repeat this calculation with each prime of $I \cup \{\ell\}$ singled out: all other lower conductors split at that prime by the same principal-ideal argument. Thus the previously imposed singular conditions are retained. The maximum of the required finite precisions suffices for the whole finite set; neither (H11) nor its exponent is repeated.

7. The actual coefficient push and its local conditions. Reduce $b_{I,a}$ to $E[2^m]$, extend coefficients freely to A, and only now use the character θ of Δ_f. Under Shapiro, followed by the equivariant coefficient map,

$$E[2^m] \otimes A[\Delta_f] \longrightarrow (E[2^m] \otimes A)(\theta), \qquad v \otimes [\delta] \longmapsto \theta(\delta)v, \tag{H20}$$

obtain the class $P_{I,m}$ over K. Here the induced module is written with the diagonal Tate action and the left regular action on its class-group basis. Formula (H20) is equivariant since left translation multiplies the value by $\theta(g)$. It remains equivariant with values in $A[G]$, before evaluation at any sign character.

For an explicit degree-one cochain formula, choose representatives r_δ of G_K/G_{H_f} using the normal quotient Δ_f, and write $gr_\delta = r_{g\delta}h(g,\delta)$ with $h(g,\delta) \in G_{H_f}$. The transferred cocycle is

$$(\mathcal{T}_\theta b)(g) = \sum_{\delta \in \Delta_f} \theta(r_{g\delta})\rho_T(r_{g\delta})b(h(g,\delta)). \tag{H21}$$

The identity $h(gk,\delta) = h(g,k\delta)h(k,\delta)$ proves the cocycle identity with target action $\theta\rho_T$. In degree zero the formula is the weighted sum $\sum_\delta \theta(r_\delta)\rho_T(r_\delta)Q$; applying its boundary gives (H21) applied to the boundary of Q. More generally, put $R(g) = \theta(g)\rho_T(g)$ and, for $g_1, \dots, g_n$, set $\delta_i = g_{i+1}\cdots g_n\delta$. The degree-$n$ map is

$$(\mathcal{T}_\theta c)(g_1,\dots,g_n) = \sum_{\delta \in \Delta_f} R(r_{g_1\cdots g_n\delta})c\big(h(g_1,\delta_1),\dots,h(g_n,\delta_n)\big).$$

With the ordinary inhomogeneous differential, the relation for h above gives $d\mathcal{T}_\theta = \mathcal{T}_\theta d$. These are the integral Shapiro and transfer maps of [56, Sections 8.1.1–8.1.6, pp. 189–194], written with left cosets. Changing sections gives the integral homotopy in Section 8.1.4; compatible sections give strictly transitive transfer. No averaging factor occurs.

Localizing (H21) separates the cosets into decomposition-group orbits: this is the local Mackey formula, a sum of the local corestrictions from the transported places. At $2N$, the deformation character Ψ is trivial locally; the fixed quadratic character need not be trivial. Under the fixed quadratic-twist identification, the local map is the ordinary integral norm on Kummer classes. One can see this without a projector: the sum $\sum_\delta \chi(\delta)\delta Q$ is a point of the quadratic twist over the lower field, and Kummer of this sum equals the cohomological transfer. The extra A coefficients take finite linear combinations of such maps. Thus the old Kummer conditions proved in paragraph 5 survive (H20), also in the universal family with its common local quadratic type. Local Kummer corestriction is natural even for a ramified field extension; in the present old-place descent the extensions used in (H10) are unramified.

At a derivative prime, H_f/K splits completely and θ is trivial on the local Galois group. Each Mackey term uses a transported place and a transported Frobenius. Transporting the Tate basis transports F as well, so (H19) is equivariant, not an assertion that an arbitrary change-of-place matrix commutes with a fixed untransported J. The same weights occur on the upper and lower class. Summing gives, in the common target coordinates,

$$f(P_{I\cup\{\ell\},m}) = 0, \qquad s(P_{I\cup\{\ell\},m}) = Jf(P_{I,m}). \tag{4.2}$$

Active and moving conditions are full; all other good conditions are unramified and preserved by the same maps.

The split norm formula in (4.1) also survives this construction. For a new conductor prime the class-field squares obtained by retaining the derivative factors are Cartesian, by (H5) and the same

local-unit calculation at that prime. Restriction and corestriction commute in these squares. If b lifts $2^e c$ in (H9), its corestriction lifts 2^e times the corresponding corestricted class; uniqueness after the second multiplication by 2^e proves the required compatibility of d_a. The weighted sums are transitive finite sums. Thus the two horizontal Artin terms in (4.1) become $\theta(\sigma_q)$ and $\theta(\sigma_q)^{-1}$, giving $a_q - \theta(\sigma_q) - \theta(\sigma_q)^{-1}$ on the class. For $I = \varnothing$, the class is precisely the Kummer image of the L-multiplied weighted point, so the same statement gives the central split-norm relation with that exact L.

The local diagrams underlying the coefficient push. At a retained derivative place put $q = \ell^2$ and choose tame and unramified generators τ_v, ϕ_v with $\phi_v \tau_v \phi_v^{-1} = \tau_v^q$. The local coefficient character is trivial. The two-generator tame resolution is

$$[M \xrightarrow{u \mapsto ((\Phi_v - 1)u, 0)} M \oplus M \xrightarrow{(x,y) \mapsto (\Phi_v - q)y} M], \qquad \Phi_v = \rho(\phi_v).$$

At the retained exponent $m \le a$, (H12) makes this the zero-differential model $D^0 \oplus (F_v \oplus S_v)[-1] \oplus D^2[-2]$. Moreover $q \equiv 1 \pmod{2^{a+2}}$. Represent the singular generator by a uniformizer adjusted by a unit so its Kummer value on the chosen ϕ_v is zero. Residue-unit Kummer characters allow this adjustment. The latter congruence makes its self-cup zero: -1 is a 2^m-th power, so the adjusted uniformizer Kummer class has square zero. The finite self-cup is zero by unramified cohomological dimension one. Retain the transported mixed pairing and the D^0, D^2 pairing.

In the derivative diagram the old ring-class fields split locally, and the new conductor extension is totally ramified. For the latter choose the lower Frobenius in the upper subgroup and take $\tau_{\text{upper}} = \tau_{\text{lower}}^e$; here $f = 1$. For split terms $e = f = 1$. Transport these choices with the Tate basis. With these compatible generators the local matrices, in the order D^0, F_v, S_v, D^2, are

$$\text{res} = \text{diag}(1, f, e, ef), \qquad \text{cor} = \text{diag}(ef, e, f, 1).$$

The possible transfer shear is divisible by $(q-1)/2$ and vanishes at retained precision. These matrices preserve each of the four conditions retaining D^0, and satisfy the local cup-product adjunctions. Split Mackey terms have $e = f = 1$; coefficient weights and reduction commute with these matrices. An arbitrary new Frobenius lift is not used without transporting the resulting finite/singular decomposition.

Compare the small free resolution with the continuous bar resolution and choose augmentation-preserving comparison maps and their composite homotopies. Continuous Hom into M retains these identities. Since the small model has zero differential, the composite on it is the identity; normalize the other homotopy to obtain a contraction. Here are its compatibility maps. For this retained contraction $1 - i_v p_v = dh_v + h_v d$ and a full local cochain arrow F, take $F_L = p_w F i_v$ and restrict it to U_v. The inclusion square has homotopy $h_w F i_v|_{U_v}$. For composable arrows the transferred composition homotopy is $p_z G h_w F i_v$, together with the transferred original bar homotopy when present. Every such degree-minus-one map on U_v lands in D^0, which is retained in every condition; thus these homotopies restrict too. For three composable maps, write $K(G, F) = pGhFi$ and $L(H, G, F) = pHhGhFi$, with the intermediate subscripts understood. Direct expansion gives

$$K(H, GF) - K(HG, F) + H_L K(G, F) - K(H, G) F_L = Ld - dL.$$

The degree-minus-two map L vanishes on every derivative condition. Transfer the original bar comparison homotopies and cubes as well; unrelated classwise boundaries are not substituted for them. For the isotropy diagrams, $U_v \otimes U_v'$ is free in degrees $0, 1, 2$, whereas the trace target is $A[-2]$. Its negative Hom groups vanish, so the restricted zero pairing has compatible null-homotopies and the

requisite 2-homotopies. This constructs the local data required in [56, 8.6.2.1–8.6.2.4] before using 8.6.4. At old Kummer places use the natural restriction/norm diagrams of the multiplication-by-2^m exact sequences, including local torsion, and their derived scalar extensions; at good unramified, full and zero places use the corresponding standard subcomplexes. The witnesses in (H22) are chosen in these same diagrams.

8. Actual finite Selmer-cone witnesses. Represent $P_{I,m}$ by a continuous global cocycle p. At each constrained place, membership in the indicated local condition gives a local cycle u_v and a degree-zero cochain q_v with

$$\mathrm{loc}_v\, p - \iota_v u_v = dq_v. \tag{H22}$$

These entries give a cycle in $\mathrm{Cone}(C^\bullet(G_{K,S}, M) \oplus \bigoplus U_v \to \bigoplus C^\bullet(K_v, M))[-1]$. For (H19), choose in the same way a degree-zero cochain whose boundary is the difference of the two representatives being compared. At the derivative primes use the already specified D^0-retaining finite/singular local complexes, not full/zero conditions. Their coordinate equality is the finite local cohomology equality just proved. All coefficient groups and coset sets at this stage are finite; the cocycles and the finite list of witnesses factor through finite quotients.

Apply the coefficient push to these witnesses at the same time as to the cocycle, including the natural Kummer comparison homotopies. If (i, p, h) is a retained contraction to a finite target model, a relation $z - z' = dq$ becomes $pz - pz' = d(pq)$. For a comparison F to another model with projection p', the correction between $p'Fz$ and $p'Fipz$ is the boundary of $p'Fhz$ (with the consistent contraction sign). Thus the finite arithmetic provides actual cycles, boundaries and comparison entries before compression. Adding a derivative prime extends this same finite diagram. Existence and size bounds for those target contractions, and the subsequent two limits, are separate inputs; this proof makes no inference from abstract isomorphism of cohomology groups to those bounds.

The resulting finite arithmetic uses the single multiplier (H11) and the precision rule $a \geq m + e$, (H12). No factor depending on $[H_f : K]$, $|G|$, $|I|$ or the number of active primes occurs. □

8.2 The derived Kummer condition

Lemma 8.4 (The actual derived Kummer condition)**.** *For the actual local twist $B = E^{(h)}$ over $F = K_v$, the condition $U_F = B(F)^\wedge_2[-1] \to R\Gamma(F, T_2B)$ includes the finite two-primary torsion. Its reduction modulo 2^a has $B(F)[2^a]$ in degree zero and $B(F)/2^aB(F)$ in degree one. Its Kummer image is its exact local annihilator, and the comparison has integral isotropy homotopies. At $v \mid 2N$, triviality of the moving character and the common old quadratic type identify the scalar and binary versions by derived coefficient extension. These are the local maps used in the Selmer-cone duality above.*

Proof. For the actual local twist $B = E^{(h)}$ over $F = K_v$, put

$$U_F = B(F)^\wedge_2[-1] \longrightarrow R\Gamma(F, T_2B).$$

The completion includes finite two-primary torsion. Derived reduction modulo 2^a has $B(F)[2^a]$ in degree zero and $B(F)/2^aB(F)$ in degree one. At $v \mid 2N$, Ψ_i is trivial, so derived coefficient extension defines the scalar A_i condition for this actual twist. In the universal binary family every h_x has the same local squareclasses at $2N$ as $h_0 = 1$. Hence every sign character evaluates the universal character ε trivially on G_{K_v}, and therefore $\varepsilon|_{G_{K_v}} = 1$ in G, before reduction modulo two. The $A_i[G]$ Kummer condition and its null-homotopy are consequently derived scalar extensions of the fixed local data. Omitting the scalar twist, the universal local-squareclass condition, or degree-zero torsion changes this argument.

The local finite dualities [54, I.2.3, I.3.2–I.3.5] make the Kummer map adjoint to $H^1(F, B[2^a]) \to H^1(F, B)[2^a]$. The latter has Kummer image as kernel, so the Kummer image is its exact annihilator. To obtain the integral chain statement, write $M = B(F)^{\wedge}_2$. The restricted cup map $M[-1] \otimes^L M[-1] \to \mathbb{Z}_2[-2]$ has homotopy class determined by $\mathrm{Hom}(M \otimes M, \mathbb{Z}_2)$. Finite isotropy makes every value divisible by every power of two, hence zero. A projective resolution therefore supplies a null-homotopy. The possible ambiguity group in degree minus one vanishes. Reduction modulo two, including the torsion in degree zero, gives the perfect quotient-to-dual comparison; derived Nakayama lifts it. The global Selmer duality theorem [56, Sections 5.1, 5.2.4, 5.4.3 and 6.3.4] is applied to these actual local maps and homotopies.

For clarity the Selmer complex is the cone

$$C_U = \mathrm{Cone}\left(R\Gamma(G_{K,S}, M_i) \oplus \bigoplus_{v \in S} U_v \longrightarrow \bigoplus_{v \in S} R\Gamma(K_v, M_i) \right)[-1]. \tag{4.3}$$

The compatible trace in (4.3) is coefficient-valued. A top-coefficient functional used to test a finite Frobenius ring's perfection is not used as a coefficient-reduction-compatible arithmetic trace. □

8.3 Actual support comparisons

At a tame split prime the local free model has

$$d^0(v) = (Av, Cv), \quad d^1(v, w) = -Cv + Dw, \quad A = \Phi - 1,\ C = g - 1,\ D = \Phi - N_q(g),$$

$$N_q(g) = \frac{q+1}{2} + \frac{q-1}{2} g, \qquad DC = CA. \tag{4.4}$$

At augmentation $g = 1$, this is the direct sum of the unramified complex $[\Phi - 1]$ in degrees zero and one and the singular quotient $[\Phi - q]$ in degrees one and two.

The support-change theorem [56, Proposition 7.8.8(ii)] retains arbitrary old local conditions and puts the unramified condition at new places. Its finite-module proof applies at two. Use it only when the actual coefficient representation is unramified at the removed place. There are five uses here: universal augmentation at Q; an unused scalar prime $q \nmid h$; characteristic-two residual removal after $\chi_h = 1$; the moving prime after $t = 0$; and a new good inert derivative prime outside h, r_i and the old support. In the last case the coefficient module is unramified and the chosen finite local plane is the unramified condition. It is not applied to an active ramified scalar prime or to r_i before central specialization. Replacing the unramified condition by the full condition then gives the singular quotient by a cone of Selmer cones. In particular the old complex is an actual subcomplex of an upper triangular model, not just a complex with the same Euler characteristic.

Proposition 8.5 (One moving character for the complete active list)**.** *Fix an allowed K and any finite list of oriented active primes. For every $n, B \geq 1$ and finite avoidance set there are infinitely many split rational primes $r \equiv 1 \pmod{2^B}$ and characters $\xi : \mathrm{Gal}(H_r/K) \to \mathbb{Z}/2^n\mathbb{Z}$ that are surjective on inertia at r, trivial on the local decomposition groups at $2N$, and take value one at every prescribed active orientation. They can be chosen with*

$$v_2(2 \pm a_r) \leq c_E, \qquad v_2 \det(1 \pm \mathrm{Fr}_r) \leq c_E,$$

where c_E depends only on the fixed Tate image, not on K, n, B or the active list. One such character serves the entire binary family before evaluation.

Proof.

The simultaneous moving character. We give the construction for the non-CM curve used here. For every $n, B \geq 1$ and every finite avoidance set, there are infinitely many split rational primes $r \equiv 1 \pmod{2^B}$ and characters

$$\xi : \mathrm{Gal}(H_r/K) \longrightarrow \mathbb{Z}/2^n\mathbb{Z}$$

which are surjective on inertia at r, trivial at $2N$, and equal to one at each prescribed oriented active prime. The trace bounds below have one constant c_E, independent of n, B, K and the active list. The degrees of the finite fields and the sizes of the primes chosen in the construction need not be uniformly bounded.

Choose prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_s$ in distinct conjugate pairs, including the active orientations, one prime over each prime dividing $2N$, and enough additional split prime ideals to generate $\mathrm{Cl}(K)$. The latter exist outside any finite set by Chebotarev in the Hilbert class field. Set $w_j = 1$ at the active primes and $w_j = 0$ at the other chosen primes. Put

$$\mathcal{R} = \ker(\mathbb{Z}^s \longrightarrow \mathrm{Cl}(K)), \qquad (\beta_a) = \prod_j \mathfrak{p}_j^{m_j^{(a)}}, \qquad \alpha_a = \beta_a/\overline{\beta}_a,$$

where the $m^{(a)}$ form a $\mathbb{Z}$-basis of $\mathcal{R}$. The generators β_a may be fractional. Disjointness of the conjugate pairs gives $v_{\mathfrak{p}_j}(\alpha_a) = m_j^{(a)}$.

Let $F_\infty = K(\mu_{2^\infty})$ and adjoin the 2^n-th roots of all the α_a. Prescribe the multiplier

$$\zeta_{2^n}^{-\sum_j m_j^{(a)} w_j} \quad \text{on } \alpha_a^{1/2^n}. \tag{4.4a}$$

These prescriptions respect every Kummer relation. Indeed, if $\prod_a \alpha_a^{b_a}$ is a 2^n-th power in F_∞, its root lies in a finite cyclotomic layer. At every nonzero-weight prime that layer is unramified over K, since the prime is odd. Consequently 2^n divides $\sum_a b_a m_j^{(a)}$ at each such prime, and hence divides $\sum_{a,j} b_a m_j^{(a)} w_j$. The dyadic primes have weight zero. This tests all multiplicative relations, including those involving units; it does not assume independence of the radicands. Kummer theory therefore gives the specified automorphism of the finite abelian extension

$$L_\infty = F_\infty(\alpha_a^{1/2^n} : a)/F_\infty.$$

Here is why these are precisely the extension conditions for ξ. For a split r avoiding the chosen primes, the ring-class sequence is

$$1 \longrightarrow \mathbf{F}_r^\times \longrightarrow \mathrm{Pic}(\mathbb{Z} + r\mathcal{O}_K) \longrightarrow \mathrm{Cl}(K) \longrightarrow 1.$$

The kernel is the ratio of the two residue factors in $(\mathcal{O}_K/r\mathcal{O}_K)^\times$ modulo diagonal rational residues. There is no additional unit quotient because $\mathcal{O}_K^\times = \{\pm 1\}$. Write q_j for the ideal-class lift of $\mathfrak{p}_j$, and orient the connecting map δ so that its principal relations are

$$\sum_j m_j^{(a)} q_j + \delta(\mathrm{res}\, \alpha_a) = 0.$$

The kernel and the q_j generate the ring-class group. Any relation between them projects to $\mathcal{R}$; subtracting its basis relations leaves a relation within the kernel. Thus these are a complete presentation, not merely necessary relations. For $r \equiv 1 \pmod{2^n}$ the power-residue character $\kappa_r : \mathbf{F}_r^\times \to \mathbb{Z}/2^n\mathbb{Z}$ is surjective. The assignments $\xi(q_j) = w_j$ and $\xi\delta = \kappa_r$ extend exactly when

$$\kappa_r(\mathrm{res}\, \alpha_a) = -\sum_j m_j^{(a)} w_j. \tag{4.4b}$$

Below we choose r so that its radical Frobenius is (4.4a), which gives (4.4b). Reversing the connecting-map convention reverses the power-residue convention as well. No class number is divided out.

We next preserve all these conditions while controlling the Tate trace. Let $J = \rho_T(G_{\mathbb{Q}(\mu_{2^\infty})})$. The determinant is the cyclotomic character, so J is the determinant-one part of the fixed Tate image. Non-CM open image [85] gives a fixed $d \geq 2$ with

$$\Gamma_d := \ker\bigl(\mathrm{SL}_2(\mathbb{Z}_2) \to \mathrm{SL}_2(\mathbb{Z}/2^d\mathbb{Z})\bigr) \subset J.$$

The subgroup $J_K = \rho_T(G_{F_\infty})$ has index at most two in J, and therefore contains its squares. Write $U(a)$ and $V(b)$ for the upper and lower unipotent matrices. Then J_K contains $U(2^{d+1}\mathbb{Z}_2)$ and $V(2^{d+1}\mathbb{Z}_2)$. The identity

$$U(a)V(b)U\bigl(-a/(1+ab)\bigr)V\bigl(-b(1+ab)\bigr) = \mathrm{diag}\bigl(1+ab, (1+ab)^{-1}\bigr)$$

and Gaussian decomposition show that a subgroup containing $U(2^k\mathbb{Z}_2), V(2^k\mathbb{Z}_2)$ contains Γ_{2k}. Hence $\Gamma_m \subset J_K$ for $m = 2d+2$. Commuting $\mathrm{diag}(1+2^m, (1+2^m)^{-1})$ with the upper and lower unipotents of level m gives all upper and lower unipotents of level $2m+1$, since $v_2((1+2^m)^2-1) = m+1$. Applying the same decomposition gives

$$\Gamma_{4m+2} \subset \overline{[J_K, J_K]}.$$

Every matrix commutator lifts to a commutator in G_{F_∞}. Such lifts fix L_∞, whose Galois group over F_∞ is abelian. The image of the compact group G_{L_∞} is closed. Thus, for the fixed integer $\nu = 4m+2 = 8d+10$,

$$\Gamma_\nu \subset \rho_T(G_{L_\infty}). \tag{4.4c}$$

This is a subgroup in the image of the actual radical-fixing kernel. It requires no disjointness of the radical and Tate fields, and is independent of the number of radicands and of K.

For the trace estimate one can take $c_E = 2\nu + 1 = 16d + 21$. We verify the elementary bound used here. Put $\delta = 2^\nu$ and $D(v) = \mathrm{diag}(v, v^{-1})$. For any $A = \left(\begin{smallmatrix} a & b \\ c & f \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}_2)$, one of

$$A, \quad AU(\delta), \quad AD(1+\delta), \quad AD(1+2\delta) \tag{4.4d}$$

has both trace distances to 2 and -2 of valuation at most $2\nu+1$. If A itself does not, its trace is within $2^{2\nu+2}\mathbb{Z}_2$ of one of these two values, while its distance to the other has valuation two. Its diagonal entries have the same parity. If both are even, c is odd by $af - bc = 1$, and $AU(\delta)$ changes the trace by $c\delta$, of valuation ν. If both are odd, the two diagonal multipliers change the trace by

$$\delta\frac{(a-f)+a\delta}{1+\delta}, \qquad 2\delta\frac{(a-f)+2a\delta}{1+2\delta}.$$

If $v_2(a-f) < \nu$, the first change has valuation at most $2\nu-1$. Otherwise write $a - f = \delta z$. For even z the first change has valuation 2ν; for odd z the second has valuation $2\nu+1$. All the chosen changes have valuation at least three. They therefore leave the distance to the other forbidden value of valuation two, and give the claimed bound for the nearer one.

Lift the prescribed Kummer automorphism to $\sigma_0 \in G_{F_\infty}$. By (4.4c), every right multiplier in (4.4d) lifts to an element of G_{L_∞}. Choose one giving the trace bound and multiply σ_0 by that lift. The resulting actual Galois element σ fixes F_∞, preserves every prescribed radical multiplier, and satisfies $v_2(2 \pm \mathrm{tr}\,\rho_T(\sigma)) \leq c_E$.

Choose $B' \geq \max(B, n)$ with $B' > c_E$, and take a finite normal extension of $\mathbb{Q}$ containing $K, \mu_{2^{B'}}, E[2^{B'}]$ and all the chosen radical roots. Restrict σ to this extension. Its restrictions on all intersections agree because it is already an actual Galois element. Finite Chebotarev supplies infinitely many rational primes in its Frobenius class. Choose a place realizing this representative and avoid all the finitely many excluded primes. Then r splits in K and $r \equiv 1 \pmod{2^{B'}}$. The Frobenius formula on $\alpha_a^{1/2^n}$ gives (4.4b), and hence the character ξ. Each local unit factor at r maps onto the residue-ratio kernel, so local reciprocity proves that ξ is surjective on inertia there. At $2N$ it is unramified with Frobenius value zero, and is therefore trivial on the whole local decomposition group. Conjugation negates its values because a prime ideal times its conjugate is rational principal.

Finally $a_r \equiv \operatorname{tr} \rho_T(\sigma) \pmod{2^{B'}}$ and $\det(\mathrm{Fr}_r) = r$. Since $B' > c_E$, we obtain

$$v_2(2 \pm a_r) \leq c_E, \qquad v_2 \det(1 \pm \mathrm{Fr}_r) = v_2(1 \pm a_r + r) \leq c_E.$$

The same bound controls $a_r - 2\chi_h(r)$ for every constituent unramified at r. Repeat with increasing n, B and avoidance sets to obtain distinct r_i. Taking $n_i \geq 2i$ makes $(1+t)^{2^{n_i}} = 1$ in A_i: for $1 \leq j < i$, $v_2\binom{2^{n_i}}{j} \geq n_i - v_2(j) \geq i$. Thus $\Psi_i = (1+t)^{\xi_i}$ is the required finite character. Its inertia generator can be oriented to have value $1+t$, and its active oriented Frobenius values have odd exponent. One character works for the entire fixed active list before any binary evaluation. □

8.4 The same functional at every localization

Proposition 8.6 (The paired functional and its integral representatives)**.** *Use the retained inner-limit diagrams of Lemma 9.1, with the local conditions of Definition 8.1, and let z be the actual retained class. Transport z by conjugation, lift it through the orthogonal comparison at the generic point, and pair by the retained duality. This defines one generic closed degree-two functional λ_z. It has an integral representative after multiplication by 2^a over $\mathcal{O}[G]$, for a fixed old-support exponent a independent of the active list and G. For a scalar constituent with its primitive local conditions, its paired functional has an integral representative at every height-one prime $\mathfrak{p} \neq (2)$ of Λ, including (t) and $(t+2)$. These representatives have the same generic value for that primitive constituent. The universal relaxed functional is asserted integral after the fixed power of two over $\mathcal{O}[G]$; scalar regularity of its corrected coordinate is obtained instead from the actual primitive comparison of Lemma 8.11. All these assertions concern the inner limit and the indicated localizations, not inverses of t in a finite Artin ring.*

Proof. After the inner limit, put $\mathcal{O} = \widehat{\mathbb{Z}_2[[t]]_{(2)}}$. In (4.4), the residual active Frobenius is $(1+t)^a I$ with a odd. Thus A, D are invertible over $\mathcal{O}[G]$. The formulas

$$h^1(v, w) = A^{-1}v, \qquad h^2(u) = (0, D^{-1}u)$$

contract the complex; the identities follow from $DC = CA$.

Which functional is paired with the class. Let z denote the retained class and transport it by τ and the preceding Weil identification to the conjugate coefficient system. At a localization where the full local complexes at active and moving primes are acyclic, the comparison from the orthogonal condition to the transported condition is a quasi-isomorphism. Apply its inverse in the derived category to the transported class, and call the resulting class $z^\perp$. Define

$$\lambda_z(w) = \mathrm{PD}(w, z^\perp), \qquad w \in H^2(C_U).$$

Thus λ_z is a closed degree-two functional, with a cochain representative obtained using the displayed local contractions and the retained global duality map. The inverse is unique in the derived category,

so representatives built at distinct localizations define the same generic functional. Integrality still requires the stated integral contractions and fixed old-support correction; it does not follow from abstract duality alone. The coordinate definition below uses this same λ_z.

Here the representatives must be constructed separately. Over $\mathcal{O}[G]$, the active Frobenius operators A, D are units, and moving-prime inertia minus one is t times a unit; the displayed homotopies are integral. For the height-one assertion take the scalar primitive constituent. Its active primes are precisely the primes ramified in that constituent, so inertia minus one is -2, a unit at $\mathfrak{p} \neq (2)$. At the moving prime inertia minus one is invertible unless $\mathfrak{p} = (t)$; at (t) the nonzero central Frobenius determinants give integral adjugate contractions. In particular $(t+2)$ is treated by the inertia contraction. Unused good primes in the primitive constituent retain the unramified condition and its orthogonal comparison; no inverse of a full local Frobenius complex at such a prime is required. This proves the scalar assertion only for the primitive functional. In the universal relaxed family an unused prime can specialize to trivial inertia, so the -2 contraction is not available there. Its correction is the determinant identity of Lemma 8.11. The common generic value alone would not prove any of these integrality statements.

In the inertia contraction the maps are

$$h^1(v, w) = C^{-1}w, \qquad h^2(u) = (-C^{-1}u, 0).$$

The fixed old-support lattice and the exponent. Here is the integral content of the old-support comparison. For each $p \mid 2N$ and each of the fixed allowed local quadratic types δ, choose once a bounded free $\mathbf{Z}_2$-model of the diagram

$$U_{p,\delta} \longrightarrow L_{p,\delta}, \qquad U_{p,\delta} = (E^{(\delta)}(\mathbf{Q}_p)_2^{\wedge})[-1], \qquad L_{p,\delta} = R\Gamma(\mathbf{Q}_p, T_2E^{(\delta)}),$$

together with its Kummer map, cup pairing, isotropy homotopy, and quotient-to-dual comparison. Finite torsion is retained in $U_{p,\delta}$; it is represented by a free resolution, not removed. The local duality and Kummer exact sequence identify that comparison integrally:

$$\mathrm{Cone}(U_{p,\delta} \to L_{p,\delta}) \simeq R\,\mathrm{Hom}_{\mathbf{Z}_2}(U_{p,\delta}, \mathbf{Z}_2)[-2].$$

This is the chain-level comparison of Lemma 8.4, with its specified isotropy homotopy. In particular it is not a rational isomorphism of free parts of cohomology. A quasi-isomorphism between these bounded free models has an integral homotopy inverse: its bounded acyclic free cone is contractible, by splitting from its last nonzero term and proceeding downwards. Consequently neither torsion lengths nor inverse determinants of torsion differentials are denominators in this comparison.

Every allowed K splits at $2N$. Its two completions above p are therefore $\mathbf{Q}_p$, with precisely two copies of this same diagram, interchanged by conjugation. The moving character and the universal varying quadratic character are trivial on their decomposition groups. All old-place diagrams over $A_i[G]$ are thus derived scalar extensions of this finite fixed list. Tensoring the chosen free models and their homotopy identities preserves all these identities. The remaining good unramified conditions use the integral unramified comparison, and the derivative conditions use their integral orthogonal planes. Neither requires a new lattice comparison when a prime is added.

For completeness, there is also no hidden group-order denominator in duality. The coefficient pairing is the $A_i[G]$-valued Weil pairing. If one checks its perfection over A_i, the auxiliary pairing on the group algebra is

$$(a, b) \longmapsto [1](ab), \qquad ([1](gh))_{g,h\in G} = (\mathbf{1}_{h=g^{-1}})_{g,h\in G}.$$

Its matrix is a permutation matrix, with integral inverse. It is not the normalized trace $|G|^{-1}\,\mathrm{Tr}$. The global cup-product and trace maps are integral as well. The hypotheses of Selmer duality here are a perfect coefficient pairing and the actual orthogonal local diagrams with their isotropy homotopies [56, Sections 6.2.5–6.2.8 and Theorem 6.3.4]; the old Kummer diagrams just described supply those hypotheses. Since K is imaginary there is no real-place Tate correction in this argument.

Over $S = \mathcal{O}[G]$ the only remaining differences between the transported conditions and their orthogonal conditions occur at the active and moving places. Their full local complexes have the integral contractions displayed above. Their direct sum, of any finite size, is still an integral contraction. The resulting comparison $j : C_{U^\perp} \to \tau^* C_U$ is an integral homotopy equivalence over S. Write j^{-1} for such a homotopy inverse and c for conjugate transport. On the retained bounded free models,

$$z^\perp = j^{-1}c(z), \qquad \lambda_z = \mathrm{PD}(-, j^{-1}c(z))$$

are integral cochains, and the second is closed. Every map in this formula is integral; no inverse on a torsion cohomology quotient is taken. The finite-model contractions and their retained limits preserve these identities. In detail, represent the comparison on the bounded free models and contract its acyclic cone over S. The cone contraction yields a chain map j^{-1} and homotopies $jj^{-1} \simeq 1$ and $j^{-1}j \simeq 1$ over S. Thus $d(j^{-1}c(z)) = j^{-1}c(dz) = 0$ as a cochain identity, and its image differs from $c(z)$ by an integral boundary. Applying the integral duality chain map gives a closed element of $\mathrm{Hom}_S(C_U^2, S)$, with the claimed generic value. This proves the existence of an actual integral closed lift, rather than only a lift after rationalization.

With the conventions of this paper the actual class z already contains the fixed multiplier L. Thus the preceding construction permits $a = 0$. One may keep any larger fixed nonnegative a in the subsequent inequalities. This does not change z, divide out L, or change the determinant basis. It proves the existence and independence of the exponent from G, the active graph, and the allowed varying K; a numerical computation of a new denominator is unnecessary because no such denominator occurs in these integral diagrams. This conclusion concerns S. The separate primitive height-one argument above, and the corrected relaxed coordinate comparison, retain exactly their stated scopes. □

Definition 8.7 (The class-functional coordinate)**.** For a constituent whose generic cohomology has dimensions $(1, 1)$ in degrees one and two and vanishes otherwise, use the determinant line

$$\mathcal{D} = \det{}^{-1} C \simeq H^1(C) \otimes H^2(C)^*.$$

The class-functional coordinate u is the coordinate of $[z] \otimes [\lambda] \in \mathcal{D}$ in the retained integral determinant basis. Here z is the actual class and λ is the closed functional just constructed. For a generically acyclic constituent set $u = 0$. The primitive and relaxed subscripts specify the corresponding local conditions on C.

8.5 Central identity, torsion and the primitive coordinate

Lemma 8.8 (Central specialization preserves the same class)**.** *The finite classes of Proposition 8.3 satisfy the retained boundary (4.5), with the same fixed L. For the functional of Proposition 8.6, the actual central Kummer comparison i satisfies*

$$z_0 = i(LmP), \qquad (\lambda_z)_0(ix) = \mathrm{PD}_K(x, c_K(LmP)).$$

Here P is the unaveraged ring-class point and c_K is its conjugate-dual transport. On a rank-one Mordell–Weil free lattice, both free-line coefficients consequently have valuation $j_{\mathrm{rc}}(h, k) + v_2(Lm)$.

The statement identifies the same point and functional; their integral torsion factors are still computed in the integral Kummer complex and local quotient triangles.

Proof. The split norm relation gives an actual finite-stage boundary

$$z_i(0) - Lm_i\, j_i(b_h) = dv_i, \qquad m_i = a_{r_i} - 2\chi_h(r_i). \tag{4.5}$$

The equality in global cohomology lifts to the Selmer cone because the local degree-zero maps are onto and the degree-one maps inject. Retain v_i. The uniform trace bound bounds $v_2(m_i)$ and the local determinants by c_E; their adjugate identities multiplied by a fixed power of two survive the limit.

Specialization of the same paired functional. Put $A = \Lambda_{(t)}$, with residue field $\mathbf{Q}_2$. The representative of λ_z over A is the one obtained from the actual orthogonal comparison and the retained cup pairing. At an active prime its contraction uses $C = -2 \in A^\times$. At the moving prime the central Frobenius determinants are nonzero; their adjugate inverses therefore belong to A. Both contractions commute with $A \to \mathbf{Q}_2$: their defining identities are matrix identities with invertible denominators in A. In particular the central full local complexes are rationally acyclic. The Kummer local objects at these changed odd places are also rationally zero: local points have finite two-primary completion, and the retained central Frobenius matrices have nonzero determinants. Thus the full/Kummer quotients are rationally acyclic as well. At the fixed old places the Kummer maps and their duality homotopies are retained unchanged.

Let j denote the resulting orthogonal comparison over A and let $c(z)$ be the conjugate-dual transport of the actual class. The definition is $z^\perp = j^{-1}c(z)$ and $\lambda_z = \mathrm{PD}(-, z^\perp)$. Derived base change preserves the specified chain maps, cup products, and homotopies. It also preserves j^{-1}, since the base change of the two inverse identities is again the two inverse identities. Consequently

$$(\lambda_z)_0 = \mathrm{PD}_0\big(-, j_0^{-1}c(z_0)\big).$$

To make the central identification explicit, write $R_0 = C_U(M)_0$, $R_0^c = C_{cU}(M^*(1))_0$, and $S_0 = C_{U^\perp}(M^*(1))_0$. Let K_0 be the ordinary rational Kummer complex and K_0^c its conjugate-dual transport. The actual comparison factors as

$$i : K_0 \longrightarrow R_0, \qquad s : S_0 \longrightarrow K_0^c, \qquad q : K_0^c \longrightarrow R_0^c, \qquad j_0 = qs.$$

All three maps are quasi-isomorphisms, by the preceding local vanishing and the unchanged old conditions. The retained duality homotopies give

$$\mathrm{PD}_R(ix, y) = \mathrm{PD}_K(x, sy), \qquad c(iP) = qc_K(P).$$

The boundary (4.5) gives $z_0 = i(LmP)$. Therefore

$$(\lambda_z)_0(ix) = \mathrm{PD}_K\big(x, s(qs)^{-1}qc_K(LmP)\big) = \mathrm{PD}_K\big(x, c_K(LmP)\big).$$

Thus this is the Kummer-dual functional of the same transported point. On the integral rank-one Mordell–Weil free lattice, the central conjugation transport is an involutive automorphism (including the quadratic sign), hence acts by $+1$ or -1. It preserves the point index; it need not fix the point. The two free-line coefficients both have valuation $j_{\mathrm{rc}}(h,k) + v_2(Lm)$. An orientation sign is a fixed unit; there is no independently chosen scalar multiplying the functional. The integral torsion correction is still computed from the ordinary integral Kummer complex and the integral local quotient triangles, not from their rationally acyclic reductions. This is the compatibility needed to apply the arithmetic length calculation in (4.6). □

Lemma 8.9 (Determinant specialization with no dimension jump)**.** *Let A be a discrete valuation ring with uniformizer t, fraction field F, and residue field k. Let C be a bounded finite free complex over A. If*

$$\dim_F H^i(C \otimes_A F) = \dim_k H^i(C \otimes_A k) \quad \text{for every } i,$$

then its cohomology is free and commutes with reduction modulo t. In particular, when the only nonzero dimensions are $(1,1)$ in degrees one and two, the coordinate of an actual cycle and an actual closed functional in $\det^{-1} C$ is regular at $t = 0$ and specializes to the coordinate of their central tensor in the specialized retained determinant basis.

Proof. Smith reduction splits the bounded free complex into free summands with zero differential and two-term summands $[A \xrightarrow{t^r u} A]$, where u is a unit and $r \geq 0$. A summand with $r > 0$ contributes one extra residue-field cohomology dimension in each of two adjacent degrees and contributes none over F. The dimension equalities therefore exclude every such summand. The remaining two-term summands are contractible over A. On the zero-differential summands both cohomology and the class-functional tensor commute with base change. The determinant identifications include the same contractible summands before and after reduction, with their unit differentials; there is no independent rescaling of the retained basis. □

Proposition 8.10 (Arithmetic calibration at known simple zeros)**.** *Let $K = \mathbb{Q}(\sqrt{k})$ and h satisfy the ring-class hypotheses above, with every prime dividing h split in K. Suppose $L(E^{(h)}, s)$ has a simple zero at 1 and $L(E^{(hk)}, 1) \neq 0$. Then the specialization at $t = 0$ of the generic coordinate in Definition 8.7 is regular and satisfies*

$$v_2 u_{\mathrm{prim}}(0) = 2j_{\mathrm{rc}}(h,k) - 4\omega(h) - s(h) - s(hk) - \sum_{p \mid k} t_p + O_{E,\pi}(1).$$

The error depends on the fixed old local types and fixed multiplier, but not on h or k. Here $0 \leq t_p \leq 2$, and $t_p = 2$ in the present full rational two-torsion setting. Only after a bound on $\omega(k)$ is imposed may the companion sum be absorbed. The finite-Sha assertion in this calibration is supplied by the two stated analytic hypotheses.

Proof. First we justify the specialization of the generic coordinate. The forward theorem, Lemma 5.5, gives rank one and finite Sha for the first twist and rank zero and finite Sha for its companion. Hence the central rational primitive complex has cohomology dimensions $(1,1)$ in degrees one and two and zero elsewhere. Over $A = \Lambda_{(t)}$, cancel the unit disks outside these two degrees; Nakayama's lemma permits these cancellations because the other central cohomology groups vanish. The retained cycle has nonzero central class by (4.5), the height formula at the known simple zero, and its specified nonzero multipliers. In the remaining two-term free complex this forces a nonzero generic kernel. Semicontinuity bounds its dimension by one. The Euler characteristic is zero, so the generic dimensions are also $(1,1)$. Lemma 8.9 now identifies the specialization of the generic coordinate with the central tensor computed in Lemma 8.8, in the same retained determinant basis. This step uses neither the numerical calibration (4.6) nor the later uniform determinant bound.

For the ordinary compact Kummer complex C_K, universal coefficients and the local degree-zero identifications give

$$H^1(C_K)_{\mathrm{tors}} = E^{(h)}(K)[2^\infty], \qquad 0 \to H^1(C_K) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \to \mathrm{Sel}_{2^\infty} \to H^2(C_K)_{\mathrm{tors}} \to 0.$$

The last finite group is Selmer modulo its maximal divisible subgroup. Self-duality identifies the degree-three torsion with the degree-one torsion and gives a unimodular pairing of the free degree-one and degree-two parts. Thus their rational dimensions equal full Selmer corank, including a possible divisible part of Sha.

At known simple zeros, Lemma 5.5 gives rank one and finite Sha. For restriction of scalars and the product of the two twists, the sum/difference isogenies are dual and compose to [2]. Their free-lattice indices satisfy $IJ = 2$, with regulator ratio $2/I^2$. At odd primes the differential lattices agree; at two the pullback determinants have bounded total valuation. Milne's isogeny ratio [54, proof of I.7.3, (7.3.1)] follows from duality without assuming the BSD leading-term equality. It applies here with finite Sha, canonical heights and pullback measures.

At every active $p \mid h$, K splits, so the isogeny local factors cancel exactly. At either split place the full/Kummer quotient has length $\dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p} = 2$, since all of $E[2]$ is rational. The two places therefore contribute exactly -4. At a companion prime $p \mid k$, the rational component number of the ramified twist is $\#E[2]^{\mathrm{Fr}_p}$, whereas the restriction of scalars has connected special fiber. Thus $t_p = \dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p} \le 2$; in the present full rational two-torsion case it is exactly two. The result is

$$v_2 u_{\mathrm{prim}}(0) = 2j_{\mathrm{rc}}(h,k) - 4\omega(h) - s(h) - s(hk) - \sum_{p \mid k} t_p + O_{E,\pi}(1). \tag{4.6}$$

Only the companion sum can be absorbed when $\omega(k)$ is bounded. No class number or unbounded degree-two torsion length is absorbed.

Here is the determinant and isogeny bookkeeping in more detail. This calculation is used only at the known simple-zero constituents, where the forward theorems give finite Sha. Put $\mathcal{A} = \mathrm{Res}_{K/\mathbb{Q}}(E_K^{(h)})$ and $\mathcal{B} = E^{(h)} \times E^{(hk)}$, and let $g_{\mathcal{A}}$ and $s_{\mathcal{A}}$ be the lengths of their two-primary rational torsion and Tate–Shafarevich group, respectively, with analogous notation for $\mathcal{B}$. The ordinary Kummer complex has torsion lengths $g_{\mathcal{A}}, s_{\mathcal{A}}, g_{\mathcal{A}}$ in degrees one, two and three. A torsion disk with cokernel of length e in degree i contributes $(-1)^{i+1}e$ to the inverse determinant coordinate. The central class $Lm_i P^{\mathrm{rc}}_{h,k}$ and its paired functional each have free-coordinate valuation $j_{\mathrm{rc}}(h,k) + v_2(Lm_i)$. Thus their ordinary Kummer coordinate has valuation $2j_{\mathrm{rc}} + 2v_2(Lm_i) + 2g_{\mathcal{A}} - s_{\mathcal{A}}$.

At either place above the split moving prime r_i, the central full/Kummer quotient has degree-two length

$$v_2(N_i), \qquad N_i = r_i + 1 - \chi_h(r_i)a_{r_i} = (r_i - 1) - \chi_h(r_i)m_i.$$

Use the moving-character construction with $r_i \equiv 1 \pmod{2^{B_i}}$, $B_i > c_E$ tending to infinity. Since $v_2(m_i) \le c_E$, both local lengths equal $v_2(m_i)$. Their total negative contribution cancels the two occurrences of $v_2(m_i)$ in the class-functional tensor. This equality persists in the common limit. At each active split place the corresponding quotient has only degree-two cohomology $E[2]$, of length two. The two places above each active prime therefore contribute -4.

For clarity, orient the isogeny indices as

$$I = [\mathcal{B}(\mathbb{Q})_{\mathrm{free}} : \phi\mathcal{A}(\mathbb{Q})_{\mathrm{free}}], \qquad J = [\mathcal{A}(\mathbb{Q})_{\mathrm{free}} : \psi\mathcal{B}(\mathbb{Q})_{\mathrm{free}}].$$

Then $IJ = 2$ and $\mathrm{Reg}_{\mathcal{B}} / \mathrm{Reg}_{\mathcal{A}} = 2/I^2$. Choose global generators $\omega_{\mathcal{A}}$ and $\omega_{\mathcal{B}}$ of the top cotangent lattices of their Néron models, and use their period measures over $\mathbb{Q}$ throughout. The integral pullbacks $\phi^*\omega_{\mathcal{B}} = a\omega_{\mathcal{A}}$ and $\psi^*\omega_{\mathcal{A}} = b\omega_{\mathcal{B}}$ satisfy $ab = 4$. Thus $a, b \in \{\pm 1, \pm 2, \pm 4\}$. Real change of variables gives

$$|a|\Omega_{\mathcal{A}} = \frac{\#\ker \phi_{\mathbf{R}}}{\#\operatorname{coker} \phi_{\mathbf{R}}}\Omega_{\mathcal{B}},$$

where both finite groups have order at most four. Hence the period ratio has uniformly bounded two-adic valuation, without a separate discriminant normalization. With these differential choices

the finite measure factors are the products $C_{\mathcal{A}}, C_{\mathcal{B}}$ of component numbers. The arithmetic isogeny identity gives

$$s_{\mathcal{A}} - s_{\mathcal{B}} = 2g_{\mathcal{A}} - 2g_{\mathcal{B}} + v_2\frac{\mathrm{Reg}_{\mathcal{B}}}{\mathrm{Reg}_{\mathcal{A}}} + v_2\frac{\Omega_{\mathcal{B}}}{\Omega_{\mathcal{A}}} + v_2\frac{C_{\mathcal{B}}}{C_{\mathcal{A}}}.$$

The $2g_{\mathcal{A}}$ terms cancel on substitution into the determinant formula. The remaining rational torsion is uniformly bounded by the fixed local types at two. Component factors cancel at every split prime. At $p \mid k$, put $E_0 = E^{(h)}$; it has good reduction, and $K_p/\mathbb{Q}_p$ is a tame ramified quadratic extension. By [58, Theorem 5.3 and Remark 5.4.1, pp. 299–300], the component group of the ramified twist is the fixed subgroup of the good special fiber under its descent action $[-1]$:

$$\Phi(E_0^{(k)}) \simeq \widetilde{E}_0[2].$$

This is an isomorphism of finite étale residue-field group schemes, so its rational component number is $\#E[2]^{\mathrm{Fr}_p}$. The Weil restriction of the good model has special fiber an extension of $\widetilde{E}_0$ by a vector group, and is connected. Thus its component number is one. All other component factors cancel at split primes or equal one at the remaining good primes. Consequently

$$v_2(C_{\mathcal{B}}/C_{\mathcal{A}}) = \sum_{p|k} t_p, \qquad t_p = \dim_{\mathbb{F}_2} E[2]^{\mathrm{Fr}_p}.$$

Substitution into the coordinate containing $-s_{\mathcal{A}}$ gives the negative sign in (4.6). In the full rational two-torsion case each $t_p = 2$. □

Lemma 8.11 (Unused-prime comparison and one corrected coordinate)**.** *At an unused good split prime, use the actual primitive-to-relaxed maps and the retained class and functional above. Write z for the scalar Frobenius value. With $e_q = a_q - z - z^{-1}$ and l_q the two-place quotient factor, they satisfy (4.7) with $\varepsilon_q \in \Lambda^\times$. For a finite binary family the universal correction η lies in $\mathcal{O}[G]$, and its corrected coordinate $\widetilde{u} = \eta u_{\mathrm{rel}}$ satisfies $\widetilde{u}_x = \varepsilon_x u_{\mathrm{prim},x}$ at every character, with $\varepsilon_x \in \Lambda^\times$. At any character where the primitive coordinate is regular at $t = 0$, this identity preserves its central valuation and vanishing. Only the product is specialized; η_x itself may have a pole.*

Proof. At an unused prime the exact point factor is $e_q = a_q - z - z^{-1}$. The second factor comes from the same closed functional, as follows. Let $i : C_P \to C_R$ be the primitive-to-relaxed map, and let $j : C_S \to C_{P^\perp}$ be the paired strict-to-primitive map. All old orthogonal conditions and their homotopies are retained. At q these are the maps from the unramified condition to the full condition and from zero to the unramified condition. Naturality of the cup pairing [56, Sections 6.4.1–6.4.3] gives

$$\beta_R(v, iw) = \beta_P(jv, w).$$

The actual norm identity is $z_R = e_q i(z_P)$, with its retained boundary. The generic paired strict lift satisfies $jz_S = e_q^c z_{P^\perp}$. Conjugation sends z to z^{-1}, so $e_q^c = e_q$. Consequently

$$\lambda_R \circ i = e_q \lambda_P.$$

Thus the class and its paired functional each contribute one e_q; no integral self-duality of the primitive condition is assumed. The two quotient determinants satisfy

$$\delta_q = \det(z\,\mathrm{Fr}_q - q)\det(z^{-1}\,\mathrm{Fr}_q - q) = q^4 l_q.$$

In the original integral determinant bases, the support triangle therefore gives

$$l_q u_{\mathrm{rel}} = e_q^2 \varepsilon_q u_{\mathrm{prim}}, \qquad \varepsilon_q \in \Lambda^\times. \tag{4.7}$$

To see the unit assertion, retain the finite mapping-cone bases. Their comparison is a sign; replacing them by the retained homotopy equivalent free models contributes only determinants of unit disks. These are A_i-units. Their limits and inverse limits are Λ-units, and q^{-4} is one also. An arbitrary $\mathcal{O}$-unit would not suffice.

We check the integrality needed for the universal correction. The graph-wide prescription at the start of this section applies also to this unused prime. Modulo the maximal ideal $(2, I_G)$ of $\mathcal{O}[G]$, every group element and fixed quadratic sign becomes one, and the scalar z becomes $1 + t$. Since $E[2]$ is rational, a_q is even at every good odd prime. Thus

$$\overline{e}_q = (1 + t) + (1 + t)^{-1} = \frac{t^2}{1 + t} \neq 0 \quad \text{in } \mathbf{F}_2((t)).$$

Consequently e_q is a unit in $\mathcal{O}[G]$. This assertion uses the prescribed exponent at every $q \in Q$, not merely at the primes ramified in a particular constituent. With

$$l_q = \det(1 - q^{-1} z \operatorname{Fr}_q) \det(1 - q^{-1} z^{-1} \operatorname{Fr}_q),$$

the trace and determinant of Fr_q are a_q and q. Expanding these two degree-two polynomials gives the exact identity

$$q^2 l_q = e_q^2 + (q - 1)(q - 1 - (z + z^{-1}) e_q).$$

Because q is odd and e_q is a unit, this identity implies $l_q / e_q^2 - 1 \in 2\mathcal{O}[G]$. Set $v_q = (l_q / e_q^2 - 1)/2 \in \mathcal{O}[G]$. Then $1 + v_q(1 + g_q)$ selects l_q / e_q^2 at an inactive character and one at an active character. Their finite product is therefore one integral $\eta \in \mathcal{O}[G]$, without any division by the group order. Put $\widetilde{u} = \eta u_{\mathrm{rel}}$. It already is one common family; no interpolation of the individual ε_x is required. Formula (4.7) makes $\widetilde{u}_x = \varepsilon_x u_{\mathrm{prim},x}$. The central zero and valuations therefore transfer unchanged. The multiplier η_x may itself have a pole at $t = 0$. Only the product $\widetilde{u}_x$, identified with the regular primitive coordinate times a Λ-unit, is specialized there.

Simultaneous prescriptions and local lifts. We spell out why the preceding calculation applies to the whole graph. First choose its actual finite set Q by Proposition A7.7. Its primes are good, odd, split in K, and outside $2ND_K$. The theta prime has already been chosen and is an exclusion. Choose one prime of K above each $q \in Q$. In Proposition 8.5, "active list" can be this entire list of orientations: the construction uses only these splitting and avoidance properties, not ramification in any particular quadratic constituent.

Here is the compatibility check in its class-group presentation. Adjoin orientations above $2N$ and auxiliary split primes generating $\mathrm{Cl}(K)$, all in distinct conjugate pairs. Assign weight one to every member of Q and weight zero to the other primes. For the relation basis $m^{(a)}$ and $\alpha_a = \beta_a / \overline{\beta}_a$ used there, a relation $\prod_a \alpha_a^{b_a} \in K(\mu_{2^\infty})^{\times 2^n}$ implies

$$2^n \mid \sum_a b_a m_j^{(a)} \qquad (j \text{ corresponding to } Q).$$

Indeed, a root belongs to a finite cyclotomic layer, which is unramified at each of these odd primes, and its valuation is integral. Multiplying by the weights and summing proves that the prescribed Kummer multipliers respect every relation. At a resulting split prime r, their power-residue values therefore satisfy every relation in the presentation of $\mathrm{Pic}(\mathbf{Z} + r\mathcal{O}_K)$. The resulting character has value one at every chosen orientation in Q, value zero at $2N$, and is surjective on the residue-ratio kernel, hence on inertia at r. The conjugate orientation has the negative value because the product ideal is rational principal. The radical-fixing subgroup (4.4c) preserves all these prescriptions while

imposing the two trace bounds. Its level depends only on the Tate image. Enlarging Q consequently changes the finite Chebotarev field, but not c_E.

Perform this construction separately at each coefficient stage, excluding Q, the theta prime, the old support and the previously chosen moving primes. Taking $n_i \geq 2i$ gives $(1+t)^{2^{n_i}} = 1$ in A_i, so these are actual finite characters. They need not be restrictions of one character of G_K: the retained inner limit is the limit of the finite diagrams of Lemma 9.1. At each fixed graph orientation their values are already exactly $1+t$, so that identity survives every coefficient reduction and that limit. The norm identities (4.1), their coefficient maps, and their boundary witnesses hold before character evaluation. This is the same construction for all constituents. At $t = 0$ the moving character is trivial, so its added prescriptions do not alter the central quadratic family.

There is a local-lift issue only for the universal quadratic character. Write $g_q \in G$ for its inertia generator at q; thus $x(g_q) = (-1)^{\lambda_q(x)}$. Fix a lift of arithmetic Frobenius at the chosen orientation and define z_q using that lift. The moving character is unramified there, and its value is $1+t$. The fixed quadratic character is unramified there also. Every other graph prime contributes an unramified quadratic character at q. Changing the Frobenius lift can therefore only replace z_q by $g_q z_q$. For an inactive character x, $x(g_q) = 1$, and $x(z_q)$ is the actual unramified Frobenius scalar. For an active character its value is immaterial to the correction, which is one. Thus we never assign an unramified Frobenius eigenvalue to a ramified constituent.

This independence also holds inside $R = \mathcal{O}[G]$, before evaluation. Both choices of z_q have reduction $1+t$ modulo $(2, I_G)$, so both corresponding e_q are units. Put $r_q = l_q/e_q^2$ and $r'_q = l'_q/(e'_q)^2$. They agree in $R/(g_q - 1)$, hence $(1+g_q)(r'_q - r_q) = 0$. Since R is two-torsion-free and $r_q = 1 + 2v_q$, $r'_q = 1 + 2v'_q$, this gives

$$(1+g_q)(v'_q - v_q) = 0.$$

The correction $1 + (1+g_q)v_q$ is consequently independent of the lift. Reversing the orientation replaces z_q by z_q^{-1} and leaves both e_q and l_q unchanged.

For clarity, write $s_q = z_q + z_q^{-1}$. Direct expansion gives

$$q^2 l_q = (q-1)^2 + a_q^2 - a_q(q+1)s_q + q s_q^2, \qquad v_q = \frac{q-1}{2q^2 e_q^2}(-(q+1)e_q^2 + (q-1) - s_q e_q).$$

Every term in the second expression is integral in R because q is odd and e_q is a unit. These inverses are taken only after the inner limit and localization at (2), never in the finite ring $A_i[G]$. Reduction modulo $(2, I_G)$ gives $\bar{e}_q = t^2/(1+t)$ and $\bar{l}_q = t^4/(1+t)^2$. The latter identity also holds at each sign evaluation modulo two. At augmentation the local two-place determinant is $q^4 l_q$, so its residual t-order is exactly four. Multiplying the correction factors over the finite set Q proves the asserted universal integrality and all the characterwise identities simultaneously. □

Lemma 8.12 (The unknown base has zero coordinate if its derivative vanishes)**.** *Use the base $h_0 = 1$ in the final fixed-K construction, with $L(E^{(k)}, 1) \neq 0$ and full two-primary Selmer corank one for E/K. If $L'(E/K, 1) = 0$, then $u_{\mathrm{prim},0}(0) = \widetilde{u}_0(0) = 0$ for the same coordinates as above. The primitive coordinate is regular at (t) by the local argument below, and its corrected relaxed coordinate has the same property by Lemma 8.11. No finite* Sha *assertion at the base is required.*

Proof. Put $A = \Lambda_{(t)}$, with residue field $\mathbf{Q}_2$ and fraction field F. Use the actual finite free primitive complex and its retained class and functional over A. The latter is integral over A by Proposition 8.6. The conductor-one height formula makes the specified point torsion when $L'(E/K, 1) = 0$. Thus its rational Kummer class vanishes, and the retained boundary (4.5) makes the specialization of the actual cycle z zero. No new normalization of L is made here.

Full Selmer corank one gives central dimensions $(1,1)$ in degrees one and two and zero in every other degree. This uses the full Selmer group, including a possible divisible part of Sha. Over the DVR A, a bounded finite free complex decomposes, by successive Smith reductions, into free zero-differential summands and two-term disks $[A \xrightarrow{t^r} A]$, up to units. A disk with $r > 0$ contributes one additional residue-field cohomology class in each of its two adjacent degrees. Semicontinuity and Euler characteristic therefore leave only two generic possibilities: the complex is acyclic, or its dimensions are $(1,1)$ in degrees one and two.

In the acyclic case the specified coordinate is zero by definition. In the second case the generic and central dimensions agree in every degree. Lemma 8.9 applies: no disk with $r > 0$ occurs. All cancellable disks have unit differential, so integral determinant base change identifies the coordinate of $[z] \otimes [\lambda_z]$ with the coordinate of its central tensor, in the retained determinant basis. That tensor is zero because $[z]_0 = 0$. This proves regularity at (t) and $u_{\mathrm{prim},0}(0) = 0$ directly. Finally $\widetilde{u}_0 = \varepsilon_0 u_{\mathrm{prim},0}$ with $\varepsilon_0 \in \Lambda^\times$ by Lemma 8.11, so the corrected coordinate is also regular and zero at the center. This proof uses neither the numerical lower bound of Proposition 9.6 nor finite Sha at the unknown base. □

9 The two limits and determinant transfer

This section uses the actual diagrams of Section 8. Their cohomology ranks alone would not suffice: maps to original cochains, local comparisons, cup products, classes and boundary witnesses are retained together. The same retained data include the integral duality maps, their homotopy inverses, and the homotopies expressing compatibility of duality with localization.

9.1 Finite free models and the inner limit

Lemma 9.1 (Finite free models and retained inner-limit diagrams)**.** *Let A be an Artin local ring with nilpotent maximal ideal, and let a complex of flat A-modules have finite bounded residual cohomology. This includes the actual continuous cochains with finite free coefficient modules used here. It admits a bounded finite free model with actual inclusion, projection and contraction homotopy. Every fixed finite diagram of maps and homotopies, including the duality and local-condition data used here, transports to these models.*

For one fixed active support and binary family, retain a tuple whose model ranks are bounded over the finite coefficient stages. One fixed nonprincipal ultrafilter gives a diagram over Λ at all coefficient precisions, preserving its matrix identities and retained homotopy inverses. This assertion imposes no rank bound uniform over all active supports.

Proof. Let A be an Artin local ring with nilpotent maximal ideal $\mathfrak{m}$. A flat A-module is free, including in infinite rank: lift a basis modulo $\mathfrak{m}$, use nilpotence for surjectivity, and flatness and nilpotence for vanishing of the kernel. A continuous finite-coefficient cochain module is a filtered union of modules of functions on finite clopen partitions, hence is flat.

Suppose the residual cohomology of a complex of such modules is finite and bounded. Split the residual complex into cohomology and disks, and lift graded bases. Let d_0 be the lifted disk differential and $d = d_0 + \epsilon$, where ϵ has entries in $\mathfrak{m}$. Use the graded decomposition into cohomology, disk sources and disk targets. Let i_0, p_0 include and project the cohomology summand, and let h_0 invert d_0 from disk targets to disk sources, zero on the other summands. Then

$$p_0 i_0 = 1, \quad h_0^2 = h_0 i_0 = p_0 h_0 = 0, \quad 1 - i_0 p_0 = d_0 h_0 + h_0 d_0.$$

With this normalized contraction, the finite geometric series give

$$d_C = p_0\epsilon(1+h_0\epsilon)^{-1}i_0, \quad i = (1+h_0\epsilon)^{-1}i_0,$$
$$p = p_0(1+\epsilon h_0)^{-1}, \qquad h = h_0(1+\epsilon h_0)^{-1}.$$

Substitution using $d^2 = 0$ gives $pi = 1$ and $1-ip = dh+hd$. The resulting bounded finite free model carries actual maps and homotopies to the original cochains. Transport any fixed finite collection of arithmetic maps by $f \mapsto p'fi$. The identity $1 - ip = dh + hd$ transports their compatibility homotopies as well, including the second homotopies in the cup-product diagrams.

Fix one nonprincipal ultrafilter on coefficient stages. Bounded ranks can be fixed on a set in that ultrafilter. Each finite tuple of matrix entries then has a compatible limit at every coefficient precision, giving matrices over Λ. Every polynomial identity between the matrices survives. Retained homotopy inverses remain homotopy inverses. Mere stagewise quasi-isomorphisms without those maps would not justify a nonflat coefficient change. □

Lemma 9.2 (The reduced generic residual rank bound)**.** *Fix E, an integer $A \geq 0$, and a number j of derivative labels. For the ring instances with $\omega(k) \leq A$, reduction to characteristic two followed by the stated support comparison and inversion of t gives a generic residual model of rank at most*

$$N(E, A, j) \leq 56\omega(2N) + 8A + 56j + 70.$$

This bound is independent of the number and sizes of active primes and of $v_2(r_i - 1)$. It bounds the reduced generic model, not the whole retained support-comparison tuple before t is inverted.

Proof. All ranks in this proof are sums of the ranks in all degrees of one complex. They are not the size of the diagram retaining all comparison maps. Put $v = \omega(2N)$ and $m = v + 1 + j$. Let S_0 be the finite places of K above $2N$, the moving prime, and the derivative primes. Let T be the set of rational primes below S_0, together with the primes ramified in $K/\mathbb{Q}$, and put $s = |T|$. Since $2 \mid 2N$,

$$s \leq v + A + 1 + j. \tag{5.1a}$$

Primes added to T solely because they ramify in K are used for the global estimate. They do not impose additional Selmer conditions.

We first give the global estimate at the residue field $B_i/(t) = \mathbb{F}_2$, where $B_i = \mathbb{F}_2[t]/(t^i)$. The residual coefficient module is the trivial module $W = E[2] = \mathbb{F}_2^2$. Write S_T for all finite places of K above T, and let r be the number of finite rational primes ramified in K. Kummer theory gives the exact sequence

$$0 \longrightarrow \mathcal{O}_{K,S_T}^{\times}/(\mathcal{O}_{K,S_T}^{\times})^2 \longrightarrow H^1(G_{K,S_T}, \mathbb{F}_2) \longrightarrow \mathrm{Cl}(\mathcal{O}_{K,S_T})[2] \longrightarrow 0.$$

Here $2 \in T$ and $\mu_2 = \mathbb{F}_2$ as a Galois module. The unit term has dimension $1 + |S_T|$: the unit rank is $|S_T|$ and the roots of unity contribute one dimension. Moreover

$$|S_T| \leq 2s - r, \qquad \dim_{\mathbb{F}_2} \mathrm{Cl}(\mathcal{O}_{K,S_T})[2] \leq r - 1. \tag{5.1b}$$

For completeness, the class-group inequality follows directly from the imaginary quadratic genus argument. Conjugation acts by inversion on the ideal class group. If $[\mathfrak{a}]$ has order dividing two, write $\overline{\mathfrak{a}} = (\alpha)\mathfrak{a}$. Taking ideal norms gives $N_{K/\mathbb{Q}}\alpha = 1$, since this norm is positive. Hilbert 90 gives $\alpha = \beta/\bar{\beta}$ (take $\beta = 1 + \alpha$ unless $\alpha = -1$, and take $\beta = \sqrt{k}$ in that case), so $\beta\mathfrak{a}$ is a conjugation-invariant fractional ideal. Modulo rational principal ideals such ideals are generated by the ramified prime ideals. Their classes have exponent two, and the principal ideal of $\sqrt{d}$, where

d is the squarefree radicand of K, supplies a nonempty relation among them. Thus their span has dimension at most $r-1$. The excluded field $\mathbb{Q}(i)$ would also satisfy the inequality, since its class group is trivial. Finally an S-class group is a quotient of the ordinary class group, and the two-rank of a quotient of a finite abelian group cannot increase. This proves (5.1b), including the possible ramification at two. The bound on $|S_T|$ counts one prime over each ramified rational prime and at most two over each other prime.

Inflation from G_{K,S_0} into G_{K,S_T} is injective on H^1 with trivial coefficients. Consequently

$$h^0(G_{K,S_0}, W) = 2, \qquad h^1(G_{K,S_0}, W) \le 4s.$$

The global Euler characteristic and cohomological-dimension statements [54, I.4.10 and I.5.1], applied over the imaginary quadratic field K, give

$$h^0 - h^1 + h^2 = -2, \quad H^{>2} = 0, \quad h^2 = h^1 - 4.$$

Thus the total residual cohomology dimension of the global complex is at most $8s-2$, and in particular at most $8s+2$. The latter weaker bound can also be obtained without genus theory: the maximal pro-two group over $\mathbb{Q}$ unramified outside T and infinity has $s+1$ generators, by the squareclasses of -1 and the primes in T; its subgroup for K has at most $2s+1$ generators by the index-two Schreier bound. The maximal pro-two group for G_{K,S_0} is a quotient of this subgroup. Hence $h^1(G_{K,S_0}, W) \le 4s+2$, giving total dimension at most $8s+2$. In either computation the H^2 estimate is supplied by the global Euler characteristic, not by a generator estimate alone.

We next bound the local terms in the Selmer cone (4.3). For a finite place w, put $d_w = [K_w : \mathbb{Q}_2]$ if $w \mid 2$ and $d_w = 0$ otherwise. Local duality and the local Euler characteristic [54, I.2.3 and I.2.8] give, for the trivial module W,

$$(h^0, h^1, h^2)(K_w, W) = (2, 4 + 2d_w, 2).$$

At an old Kummer place, the derived condition has residual dimensions $(2, 2+d_w)$ in degrees zero and one. Indeed $E^{(h)}(K_w)^\wedge_2$ is the direct sum of a free $\mathbb{Z}_2$-module of rank d_w and finite two-primary torsion; that torsion has two generators because $E^{(h)}[2]$ is rational. Derived reduction retains its two-dimensional degree-zero torsion. For the other conditions at the odd moving or derivative places, the following table records the sum of the dimensions of the local condition and of the full local complex. Cone shifts do not change this sum.

condition U_w	$\sum h^a(U_w)$	$\sum h^a(U_w) + \sum h^a(K_w, W)$
old Kummer	$4 + d_w$	$12 + 3d_w$
unramified, $w \nmid 2$	4	12
0, $w \nmid 2$	0	8
full, $w \nmid 2$	8	16
derivative strict	2	10
derivative finite or singular	4	12
derivative relaxed	6	14

The last three rows use exactly the conditions retaining D^0 in Lemma 9.5; the relaxed condition does not include D^2. There are at most two places over each rational prime, and $\sum_{w|2} d_w = 2$. The old prime two therefore costs at most 30, any other old rational prime at most 24, the moving prime at most 32, and a derivative prime at most 28 (in fact it is inert and costs at most 14). Each is bounded by 48. Thus all finite local terms cost at most $48m$. No prime dividing k alone is charged a local condition.

There are no real places of K. Its one complex place has full cohomology W in degree zero only. With the usual modified archimedean convention its local complex is zero; even retaining both an ordinary full term and a full local condition costs at most four additional generators. Complex conjugation used to transport duality is not an additional real local condition over K. In particular no infinite two-periodic real resolution is being included in this count.

At each finite coefficient stage, apply Lemma 9.1 separately to these global and local complexes and transport the actual local maps. The mapping cone then has a finite free model whose total rank is bounded by the sum just computed. Further cancellation can only decrease it. The estimate applies before inverting t to the smaller-support model, uniformly in the moving prime and in the order of its finite character: reduction modulo t kills that character. Allowing the weaker global estimate and the archimedean allowance still gives

$$\begin{aligned} \operatorname{rank}_{\mathrm{tot}} C_{S_0} &\le (8s+2) + 48m + 4 \\ &\le 8(s+1) + 6 + 48(v+1+j) \\ &\le 56v + 8A + 56j + 70. \end{aligned} \tag{5.1}$$

It remains to transfer this estimate to the original support. First consider a scalar ring instance. Reduction modulo two kills χ_h, so the coefficient system is unramified at every active prime. The actual support comparison of [56, Proposition 7.8.8(ii)] therefore identifies the smaller-support complex with the complex having unramified conditions at those primes. At an oriented active place, reduction of (4.4) has

$$C = 0, \qquad A = D = \big((1+t)^e - 1\big) I_2,$$

where e is odd; at the conjugate place the exponent is $-e$. Both $((1+t)^e-1)/t$ and $((1+t)^{-e}-1)/t$ are units of $\mathbb{F}_2[[t]]$. Thus the full, unramified and singular comparison blocks all become contractible after inverting t. For each fixed finite active support retain the comparison maps and homotopies, take the inner limit, and only then invert t. Every active comparison cone is then contractible, so the original complex has the same generic residual cohomology as the bounded smaller-support model. No count of active primes remains.

For clarity, a universal binary character is not trivial merely after reduction modulo two. In the binary setting make the preceding calculation after augmentation $\mathbb{F}_2[G] \to \mathbb{F}_2$. After the inner limit and inversion of t, the ring $\mathbb{F}_2((t))[G]$ is local with nilpotent augmentation ideal. A finite free model over this ring has a minimal model obtained by cancelling unit entries; its ranks are the dimensions of the cohomology after augmentation, and hence obey (5.1). These are ranks over $\mathbb{F}_2((t))[G]$, not dimensions over $\mathbb{F}_2((t))$. For scalar instances they are precisely the generic residual ranks needed in the outer limit. The whole retained comparison diagram can still grow with G or with the active support; no bound for that diagram is asserted. Finally the minimum ranks over $\mathcal{O}$ are the dimensions after reduction to $\mathbb{F}_2((t))$, so the same estimate bounds the scalar $\mathcal{O}$-minimal models used below. □

9.2 Evaluation and the outer valuation ring

Lemma 9.3 (Evaluation, coefficient change and the outer DVR)**.** *For the retained global models and for the primitive and derivative Selmer models, evaluations at sequences of Galois elements satisfy (5.2). A cycle defines an abstract crossed homomorphism of the indicated product group. The induced map on H^1 is injective, and the finite detecting-list comparison remains injective after field extension. Direct evaluation of other Selmer models is asserted only when their local maps induce isomorphisms on residual H^0 and injections on residual H^1. For orthogonal zero/full comparisons,*

the assertion may instead be transported at a test ring where the actual local comparison complexes are contractible.

For a sequence of ring instances with $\omega(k_j) \le A$ and a fixed number of derivative labels as in Lemma 9.2, the quotient $\mathcal{O}_\infty$ in (5.3) is a DVR with uniformizer two. The bounded minimal-model matrices, evaluations, duality maps and their compatibility homotopies pass to it. Sequences whose differentials tend to zero give actual crossed homomorphisms there. No continuity assertion on a single original Galois group, or completeness of $\mathcal{O}_\infty$, is required.

Proof. For fixed K let $\Gamma_K = \prod_i G_K$ and include one simultaneous complex conjugation. Evaluation of retained cochain maps at arbitrary sequences $g = (g_i), h = (h_i)$ gives

$$e^1_g d = (\rho(g) - 1)e^0, \qquad e^2_{g,h} d = \rho(g)e^1_h - e^1_{gh} + e^1_g. \tag{5.2}$$

One fixed ultrafilter supplies these identities for all sequences. A cycle therefore defines an abstract crossed homomorphism of the product group. We do not assert that its restriction to one original Galois group is a continuous pointwise limit.

Evaluation remains injective after field extension. At each finite stage take representatives V for H^1 and coefficient space W. Evaluate (c, w) by $b_c(g) - (\rho(g) - 1)w$. The intersection of all kernels is $(0, W^\Gamma)$. A finite list of at most $\dim V + \dim W$ evaluations already has that intersection. Choose and pad these lists at the finite stages before taking their common limit. The comparison to $[W \to W^d]$ is an isomorphism on H^0 and injective on H^1. Its cone has no residual cohomology in degrees at most zero. Cancel unit disks there so that the free cone starts in degree one; the same is true after every coefficient extension. The long exact sequence gives the claimed injection. For the primitive and derivative Selmer complexes the required residual comparison with global cochains follows from their actual local conditions. At an old Kummer place, derived reduction retains $B(F)[2]$ in degree zero, isomorphically onto the local invariants, and has the Kummer injection in degree one. The unramified condition at a good place has the same degree-zero invariants as the full local complex and injects in degree one. A full condition has the identity map. Every derivative condition retains D^0 and includes its stated degree-one subspace. The Selmer-cone exact sequence therefore gives an isomorphism on global H^0 and an injection on global H^1.

This does not apply directly to the zero conditions in the retained orthogonal complexes: at $(2, t)$ their local degree-zero invariants need not vanish. Indeed, two zero local conditions with residual invariant space W can contribute $(W \oplus W)/\operatorname{diag}(W)$ to Selmer H^1, and global cochain evaluation kills this contribution. Such orthogonal complexes are retained for duality, without asserting residual evaluation injectivity. If an actual zero/full comparison is contractible at a chosen test ring, its retained contraction identifies the two Selmer models there and transports the evaluation injection. This use requires that contraction at the chosen ring; it is not an assertion for arbitrary unused relaxed primes at every scalar height-one localization.

Now allow K_j, h_j to vary with $\omega(k_j) \le A$. For copies $\mathcal{O}_j$ of $\mathcal{O}$, form

$$\mathcal{O}_\infty = \left(\prod_{\mathcal{U}} \mathcal{O}_j \right) / \{(x_j) : v_2(x_j) \to_{\mathcal{U}} \infty\}. \tag{5.3}$$

A nonzero element has valuation one fixed nonnegative integer on a set in $\mathcal{U}$ and is that power of two times a unit. Every nonzero ideal has an element of least valuation and is principal. Thus (5.3) is a DVR with uniformizer two and characteristic-zero fraction field. Completeness is unnecessary.

Integral minimal-model matrices and (5.2) pass to (5.3). So do the retained duality maps, inverse homotopies and duality-to-localization compatibility homotopies. After minimal cancellation an integral homotopy equivalence is degreewise invertible: its residual matrices are invertible,

and Nakayama applies. These inverse matrices and all compatibility identities pass to the same outer quotient. Thus perfect duality and exact orthogonality are preserved there; they are not inferred merely from isotropy. Approximate cycles suffice: if $v_2(dw_j) \to \infty$, the degree-two term in (5.2) tends to zero, so w_∞ is an actual crossed homomorphism. The finite detecting lists have bounded length by (5.1); retain their cone contractions to preserve injection. One must not replace approximate cycles by unexplained exact cycles. □

9.3 One new prime detects two classes

Lemma 9.4 (The common kernel and finite realization of a rank-two test)**.** *Use the exact kernel of sequences acting trivially on the full Tate module, the relevant ring-class fields and quadratic characters. On the characteristic-zero tests used below, restriction of the evaluated cohomology to this kernel is injective. When the moving scalar character is nonquadratic, two independent classes have a rank-two localization at a single actual kernel element. Finite Chebotarev realizes the required conjugation coset and basis-cochain evaluations, with exact inner-limit matrix identity (5.4) before any $\mathcal{O}$-minimal-model denominators are applied.*

At the residual test only scalar-inertia vanishing, bounded cohomology and finite detecting lists are asserted. The quadratic character exception at characteristic-zero height-one tests is treated in Proposition 9.6.

Proof. The exact kernel consists of sequences trivial on the full Tate module, all relevant ring-class fields and quadratic characters. The scalar element of Section 8 is central modulo this kernel, though not in the kernel itself. For a cocycle zero on the kernel,

$$(u^4 - 1)b(g) = (g - 1)b(z)$$

makes it a coboundary over characteristic zero. The same fixed scalar survives the outer quotient. To eliminate residual invariants, use moving-prime inertia: its scalar difference is t, invertible in $\mathbb{F}_2((t))$. These are two separate arguments.

On the characteristic-zero tests used for rank reduction, the fixed finite matrix-span witness gives absolute irreducibility of each rank-two summand. When the scalar character is not quadratic, moving inertia has distinct characters on the two summands and simultaneous conjugation exchanges them. The induced rank-four representation is therefore absolutely irreducible with scalar endomorphisms. The outer fraction field is one such characteristic-zero test; the quadratic character exception at an inner test is treated separately below.

For two independent cohomology classes on these tests, the span of their joint values on the exact kernel projects onto both copies of this absolutely irreducible representation. A proper submodule with both projections onto would be the graph of a scalar isomorphism, contradicting independence. Thus the span is the full sum. If the determinant of their two finite localizations vanished on every actual kernel element, additivity and polarization would make it vanish on the entire span, impossible. Hence one actual element supplies a rank-two localization on these characteristic-zero tests.

The polarization identity itself is also valid in characteristic two, but no residual irreducibility or residual rank-two localization is asserted. Here $E[2]$ is trivial of rank two, so its scalar twist is isotypic and the induced residual representation has multiplicity two. At the residual test we use only scalar-inertia vanishing of H^0, the bounded residual cohomology dimensions, and finite detecting lists from intersections of linear evaluation kernels. Those lists need not consist of one rank-two localization. The rank-two selection for the uniform valuation bound occurs only after passage to the characteristic-zero fraction field of the outer DVR.

Chebotarev is applied at finite stages, not to the product group. Each of the finitely many original basis cochains in degrees zero through two factors through a finite quotient. Include their quotient, the required Tate precision, ring-class and quadratic fields and old diagrams in one compositum. The element just found, multiplied by conjugation, is a single actual automorphism of that compositum. Its restrictions agree on intersections. Choose the new prime and a place above it to realize that representative.

This also resolves the precision issue caused by localization. Choose the new prime to match all original basis-cochain evaluations at a common retained precision $n(i) \to \infty$. Taking the same inner ultrafilter gives an exact equality of Λ-matrices

$$L_j = E_{\sigma_j^2}. \tag{5.4}$$

Only afterward compose with the inclusion I_j from an $\mathcal{O}$-minimal model: $L_j I_j = E_{\sigma_j^2} I_j$. The denominators in I_j may depend on j; equality (5.4) makes this irrelevant. Long t-jets are never treated as approximations in the two-adic valuation of $\mathcal{O}$, where t is a unit. □

9.4 Switches and the varying-companion estimate

Lemma 9.5 (Derivative local conditions and determinant-preserving switch)**.** *At a derivative prime use $L_v = D^0 \oplus (F_v \oplus S_v)[-1] \oplus D^2[-2]$ with the transported pairing, and retain D^0 in all four conditions. Take $U_{\mathrm{str}} = D^0$, $U_{\mathrm{rel}} = D^0 \oplus (F_v \oplus S_v)[-1]$, $U_F = D^0 \oplus F_v[-1]$, and $U_S = D^0 \oplus S_v[-1]$. Then strict and relaxed are orthogonal, the two plane conditions are self-orthogonal, and their cone triangles are the ones below.*

Over the stated fraction or test field, suppose the actual old class spans generic H^1 and has nonzero finite localization v. Its derivative from Proposition 8.3 has finite coordinate zero and singular coordinate Jv. The strict generic H^1 is zero, the new generic H^1 is spanned by this derivative, and the two class-functional determinant coordinates have equal valuations, even when v is not primitive in the local integral lattice.

Proof.

The strict and relaxed conditions at a derivative prime. The words strict and relaxed have a different local meaning at a derivative prime from zero and full at an active or unused split prime. At a derivative prime use the actual split local model

$$L_v = D^0 \oplus (F_v \oplus S_v)[-1] \oplus D^2[-2]$$

with zero differential at the retained precision, and retain its pairing and isotropy homotopies. Here D^0 and D^2 are dual, and F_v, S_v are the finite and singular isotropic planes. Put

$$\begin{aligned} U_{\mathrm{str}} &= D^0, & U_{\mathrm{rel}} &= D^0 \oplus (F_v \oplus S_v)[-1], \\ U_F &= D^0 \oplus F_v[-1], & U_S &= D^0 \oplus S_v[-1]. \end{aligned}$$

Thus $U_{\mathrm{str}}^{\perp} = U_{\mathrm{rel}}$ and the two plane conditions are self-orthogonal under the transported duality. All four maps retain local degree-zero cohomology. In particular, the relaxed condition here omits $D^2[-2]$ and is not the full local complex. The actual cone triangles are consequently

$$C_{\mathrm{str}} \longrightarrow C_F \longrightarrow F_v[-1] \longrightarrow C_{\mathrm{str}}[1], \qquad C_{\mathrm{str}} \longrightarrow C_S \longrightarrow S_v[-1] \longrightarrow C_{\mathrm{str}}[1].$$

These are the triangles used in the determinant switch. Zero/full conditions at the other primes retain their previously stated meaning. Exact annihilator and dimension assertions in the switch

argument are made after passage to the stated fraction or test field, using these integral diagrams and their retained dualities. At a new inert prime the finite and singular planes are free of rank two, with cross pairing $\beta(x,y) = \varepsilon\langle x, Jy\rangle$. Suppose the old actual class A spans generic H^1 and has $f(A) = v \neq 0$. Its derivative B satisfies $f(B) = 0$, $s(B) = Jv$. The strict H^1 vanishes. The transported duality and localization identities identify the relaxed localization image with the kernel of its dual localization map, hence with its exact annihilator [56, Sections 6.3.3–6.3.4]. This image contains $(v, 0)$ and $(0, Jv)$, so these span it. Hence the new generic H^1 is spanned by B.

The determinant comparison is exact even if v is divisible. In the fraction field choose x with $v \wedge x$ of unit volume, and similarly y with $Jv \wedge y$ of unit volume. The finite and singular determinant exact sequences have the same strict factor, so their class-functional coordinates have equal valuations. All strict torsion remains in that common factor. Requiring x integral would be wrong when v is divisible. □

Proposition 9.6 (Uniform ring lower bound and height-one regularity)**.** *Fix E, π, the old local types and a bound A. For every ring instance satisfying the hypotheses of Proposition 8.10 and $\omega(k) \leq A$, the actual primitive coordinate satisfies*

$$v_2 u_{\mathrm{prim}} \geq -C(E, A), \qquad u_{\mathrm{prim}} \in \Lambda[1/2],$$

where the first valuation is that of $\mathcal{O}$ and the constant is independent of the active support and companion field. Consequently its central specialization and (4.6) imply (3.5).

The regularity assertion also holds at the unknown full-corank-one base used in Lemma 8.12. The uniform numerical lower bound here is asserted only for the known simple-zero ring instances; it is not asserted at that unknown base.

Proof. The starting generic kernel has rank one. Indeed, at a known simple zero the ordinary class is nonzero. Equation (4.5), with its bounded nonzero multiplier, makes the retained central class nonzero. Over $\Lambda_{(t)}$ cancel the degree-zero and degree-three disks. If the generic kernel of the resulting two-term free complex vanished, the actual lifted cycle would vanish, contradicting its central value. Semicontinuity bounds that kernel by the central dimension one.

Suppose no uniform lower bound exists. Choose instances with $v_2 u_j \to -\infty$, $\omega(k_j) \leq A$, and write their minimal completed models as $[\mathcal{O}_j^n \xrightarrow{D_j} \mathcal{O}_j^n]$, fixing the bounded n on an outer-large set. In each generic one-dimensional kernel choose a primitive vector v_j. Its outer image is nonzero, so the outer kernel dimension s is at least one. If $s > 1$, one new derivative prime gives rank-two localization including that vector. The relaxed localization image is self-annihilating and contains the whole finite plane, hence equals that plane. The strict kernel has dimension $s - 2$ and the singular switch adds no localized vector, so the new outer dimension is $s - 2$.

The actual inner class is a nonzero scalar multiple of v_j. Even if this scalar tends to zero, its generic localization is nonzero on an outer-large set. Exact switch preservation therefore keeps each inner determinant valuation. The new inner kernels again have primitive vectors. Thus their outer dimension remains at least one; a proposed drop to zero is already a contradiction. There can be only finitely many drops. For each fixed number of them (5.1) bounds the models; no bound uniform in an unbounded number of switches is needed.

At terminal outer dimension one a maximal boundary minor Δ is nonzero. Its inner minors have one finite valuation b on an outer-large set. Determinant-one Schur operations in a square two-term rank-one-kernel complex give $u = \pm\alpha\beta/\Delta$, where α is an integral surviving cycle coordinate and β has valuation at least $-a$. Consequently $v_2 u_j \geq -a - b$, the required contradiction. We obtain

$$v_2 u_{\mathrm{prim}} \geq -C(E, A). \tag{5.5}$$

Here (5.5) is asserted for the known simple-zero ring instances used to obtain the varying-companion estimate. Their nonzero actual central class supplied the rank-one start. It is not a uniform valuation assertion at the unknown corank-one base. Bounded matrix size alone would not prove it: $[2^j]$ is a counterexample to that inference.

Height-one regularity, including quadratic specializations. Fix a height-one prime $\mathfrak{p} \neq (2)$, put $R = \Lambda_{\mathfrak{p}}$ and $k = \mathrm{Frac}(\Lambda/\mathfrak{p})$, and use the actual primitive scalar Selmer complex over R. Here R is a DVR and k has characteristic zero. All subsequent switches use the derivative conditions of Lemma 9.5, retaining D^0. The primitive old local conditions and these derivative conditions have the evaluation injection of Lemma 9.3. Cancel the degree-zero and degree-three disks and then the unit disks. Write $C = [R^s \xrightarrow{d} R^s]$ in degrees one and two, with $d \otimes k = 0$. Its generic kernel has dimension one. Choose a primitive vector $v \in \ker d$. Such a vector exists because $\ker d$ is saturated in R^s: divide a nonzero kernel vector by its largest common power of a uniformizer. Thus $\bar{v} \neq 0$ in $V = H^1(C \otimes_R k)$ and $\dim_k V = s \geq 1$. The actual class z is a nonzero fraction-field multiple of v. We will only make switches whose finite localization is nonzero on $\bar{v}$. Consequently their generic localization of z is nonzero, which is the hypothesis needed for the actual-class determinant switch.

If the scalar character specialized to k is nonquadratic, the rank-two detection of Lemma 9.4 applies to $\bar{v}$ and an independent vector whenever $s > 1$. The finite localization then has rank two. Exact duality makes the relaxed localization image equal to the finite plane, so switching to the singular plane lowers s by two. We give the quadratic case in detail.

Let α be the specialized character on the index-two group. Since the moving inertia generator has value $1 + t$, quadraticity implies $t(t + 2) = 0$ in k. The only possible primes are therefore (t) and $(t+2)$. The equality $\alpha^\tau = \alpha^{-1} = \alpha$ uses the actual conjugation τ; it involves no automorphism of R or substitution for t. The action

$$c^\tau(g) = Jc(\tau^{-1} g \tau), \qquad J = \rho_T(\tau),$$

on cocycles gives a k-linear involution on cohomology. Extend α to the index-two overgroup by choosing its value on τ to be $+1$. The other extension differs by the character of K. The induced rank-four representation is consequently $W_+ \oplus W_-$, with each $W_\pm$ of rank two. The matrix-span witness in (H1)–(H2) and the paragraph following them makes their restriction to the index-two group absolutely irreducible. Twisting its matrices by nonzero scalar character values does not change their linear span. The summands are nonisomorphic: an intertwiner restricts to a scalar, whereas their actions of τ are J and $-J$. A scalar intertwining these two actions is zero since 2 is invertible. We write $V = V_+ \oplus V_-$ for the resulting cohomology signs.

At a new inert derivative prime write F, S for the finite and singular planes. The action on F is J, and that on S is $-J$. Indeed, the local rational Frobenius acts on the Tate module by J; it fixes its square, the finite Frobenius generator, and conjugates tame inertia by its ℓ-th power. The retained congruences give $\ell = -1$ in the limit, so pullback on the tame coordinate contributes $\ell^{-1} = -1$. These assertions hold at every previously chosen derivative prime as well. A common scalar sign would simply interchange the names $+$ and $-$. The transported cross pairing is exactly

$$\beta(x, y) = \varepsilon e_2(x, Jy), \qquad \varepsilon \in k^\times,$$

from Lemma 9.5. The finite and singular planes are isotropic. These are the retained local cup-product identities, including the extra finite precision in (H12), not a residual alternatingness assertion. Since $\det J = -1$ and $J^2 = 1$, $e_2(x, Jy) = e_2(y, Jx)$. Choose nonzero J-eigenvectors e_+, e_-. They form a basis and $e_2(e_+, e_-) \neq 0$. The two local cohomology signs are therefore

$$H_+ = F(e_+) \oplus S(e_-), \qquad H_- = F(e_-) \oplus S(e_+).$$

They are orthogonal nondegenerate symmetric hyperbolic planes. In either plane the self-pairing of $aF(e_\pm) + bS(e_\mp)$ is $2ab$ times a nonzero constant. Its only isotropic lines are thus its finite and singular axes.

All old conditions are stable under this involution. Conjugate old Kummer conditions are transported into each other; primitive ramified full conditions occur at the conjugate pair; unused good conditions are unramified; and both F and S at a previous derivative prime are stable under the actions just computed. At the new prime let C_{str} and C_{rel} have the actual truncated conditions D^0 and $D^0 \oplus (F \oplus S)[-1]$. Put

$$U = \mathrm{im}\big(H^1(C_{\mathrm{rel}} \otimes k) \longrightarrow F \oplus S\big), \qquad K_0 = H^1(C_{\mathrm{str}} \otimes k).$$

The retained localization-duality diagrams give $U = U^\perp$. Specifically, localization's connecting map is its dual under the strict/relaxed duality, so its kernel is both U and $U^\perp$; this is the exactness in [56, 6.3.3–6.3.4] used in Lemma 9.5. Isotropy alone would not suffice. The cone triangles, with D^0 retained, give exact sequences

$$0 \longrightarrow K_0 \longrightarrow V \longrightarrow U \cap F \longrightarrow 0, \qquad 0 \longrightarrow K_0 \longrightarrow V' \longrightarrow U \cap S \longrightarrow 0, \tag{5.5a}$$

where V' is the switched fiber cohomology. There is no preceding local degree-zero term, since the quotient conditions are $F[-1]$ and $S[-1]$. The space U is stable under signs. Each $U_\pm \subset H_\pm$ is its own annihilator in $H_\pm$, hence is one of that plane's two isotropic axes.

We also need one prime detecting prescribed classes of opposite signs. Choose nonzero $c_+ \in V_+$ and $c_- \in V_-$. Their restrictions to the exact kernel H of Lemma 9.4 are nonzero additive equivariant maps. Their joint k-linear evaluation span in $W_+ \oplus W_-$ surjects to each summand. Simplicity and nonisomorphism force this span to be the full direct sum. At an element $h \in H$, evaluation at $(\tau h)^2$ on a sign-δ cocycle is

$$c_\delta((\tau h)^2) = (1 + \delta J)c_\delta(h) \qquad (\delta = +1, -1). \tag{5.5b}$$

This follows from additivity on H and the sign relation; a coboundary is zero on H. Thus the two finite-coordinate maps on the joint span surject respectively onto ke_+ and ke_-. There is an actual h for which both coordinates are nonzero. For completeness, if their product vanished at every actual h, additivity and the identity $Q(x+y) - Q(x) - Q(y)$ for that product would make its polar form zero on all pairs of actual evaluations. It would then vanish on their linear span, contradicting surjectivity onto the two axes. Finite Chebotarev and the exact matrix identity (5.4) realize this choice by a new inert prime, avoiding every previously used prime. All character, Tate, cochain and old-diagram constraints are included in the same finite quotient as in that lemma. No residual irreducibility or infinite-level Chebotarev assertion is used. The same argument for one nonzero sign class gives nonzero localization in its finite axis.

Suppose first that both V_+ and V_- are nonzero. Choose the class in one sign to be a nonzero component of $\bar{v}$ and choose any nonzero class in the other. Detection as above puts both finite axes in U, so $U = F$ and $U \cap S = 0$. Equation (5.5a) gives $\dim V' = s - 2$. The localization of $\bar{v}$ is nonzero: its two sign components lie on different finite axes, so cannot cancel.

Suppose instead that only one sign occurs, say $V = V_+$, with $s > 1$. Detect $\bar{v}$ in its finite axis. Then $U_+ = F(e_+)$. Since $V_- = 0$, (5.5a) gives $U_- \cap F(e_-) = 0$. The isotropic-axis calculation forces $U_- = S(e_+)$. Consequently K_0 has sign dimensions $(s-1, 0)$, while V' has sign dimensions $(s-1, 1)$. This proves that the first switch introduces the other sign and preserves total dimension. Since $s - 1 \geq 1$, the next switch uses the preceding mixed-sign case and lowers the total dimension by two. After the first switch choose the primitive vector in the new integral generic kernel; include one of its nonzero sign components in the second detection. Thus the actual-class nonvanishing requirement holds at both steps. The case $V = V_-$ is identical with signs interchanged.

At each step the new class is the actual derivative $P_{I\cup\{\ell\}}$ of Proposition 8.3, with the same fixed multiplier L. Equation (4.2) gives finite coordinate zero and singular coordinate $Jf(P_I)$, which is nonzero generically. The construction retains the old Kummer conditions and all earlier singular conditions, by the final assertion following (H19). Lemma 9.5 therefore makes this class span the new generic kernel and preserves the valuation of its class-functional determinant coordinate. In particular we do not replace the actual class by the chosen primitive vector. That vector is used only to select the prime. The paired functional is the one transported from the same actual class. The determinant comparison uses the common strict determinant factor; fractional complements to $f(P_I)$ and $Jf(P_I)$ have unit exterior volume. Consequently a divisible localization creates no extra valuation, and all strict torsion factors remain present on both sides.

Each new generic kernel has rank one and has a primitive integral vector, so its fiber dimension is at least one. A proposed drop from dimension two to zero is therefore a contradiction; that configuration cannot occur. Whenever the dimension exceeds one, one or two switches give a strict drop by two. Thus finitely many switches leave dimension one. After cancellation the terminal complex has one free generator in each of degrees one and two; its differential is zero because its generic kernel has rank one. The actual class is integral. To apply Proposition 8.6 after these switches, we check its orthogonal comparison on the added local conditions. In the integral derivative model of Lemma 9.5, D^0 pairs with D^2 and S pairs perfectly with F, while the pairing on S is zero with its retained isotropy homotopy. Hence $(D^0 \oplus S[-1])^{\perp} = D^0 \oplus S[-1]$ as the specified local condition, with the retained integral comparison maps. Conjugate transport preserves this condition as well. At a derivative prime the comparison from the orthogonal condition to the conjugate condition is therefore the identity under those integral maps. It requires no inverse of a localization vector or of a local Frobenius operator. Using this identity in the Selmer-cone comparison, together with the same old-place comparisons as before, constructs the integral orthogonal lift of the new transported class. Composing with the retained integral global duality gives a closed integral degree-two functional. Generically this is exactly the functional of Proposition 8.6: both use the inverse of the same orthogonal comparison, which is unique in the derived category. Thus its primitive integrality proof extends to every finite list of these derivative conditions. At the active ramified primes this uses the unit -2; at the moving prime it uses t except at (t), where the central Frobenius determinants are nonzero and hence units in R. Unused good primes retain their unramified condition. No inverse at an unused relaxed prime is used. The terminal coordinate is the product of the two integral coefficients, up to a determinant-basis unit, so belongs to R. Valuation preservation transports this conclusion through every switch. This proves regularity at every $\mathfrak{p} \neq (2)$, including both quadratic specializations. Finally the normal-domain intersection formula for $\Lambda[1/2]$ gives $u_{\mathrm{prim}} \in \Lambda[1/2]$.

The same proof applies at the unknown full-corank-one base. If its actual generic class is zero, its coordinate is zero. Otherwise its nonzero generic cycle and the central dimension bound give the rank-one start used above. This requires no uniform outer valuation bound at that base. Combining (5.5), for the known simple-zero instances, with (4.6) and $\sum_{p|k} t_p \leq 2A$ proves (3.5). No fixed-field class number is absorbed into the varying-field bound.

□

9.5 Actual augmentation and the fixed-field count

Lemma 9.7 (Augmentation and the fixed-field rank budget)**.** *Fix the final field K and the family of Section 7, in the branch where the odd empty configuration is not a unit. There are no derivative labels. At augmentation all graph primes are unramified. The actual old-support subcomplex gives (5.6). With $d_K = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/K)$, $t_K = \mathrm{length}\, E(K)[2^\infty]$ and $h_K = \mathrm{length}\, H^2(C_K)_{\mathrm{tors}}$, it has the*

rank, Smith-disk and torsion bounds (5.7). These are fixed-K bounds, independent of the binary dimension and graph size.

At every sign character, the inverse new local blocks and the resulting old-rank Schur complex have no Laurent terms below $-4M$ modulo 2^M, uniformly in the number of new blocks. This is a bound on sign evaluations, not on coefficients in the group basis. The finite number h_K does not assume finiteness of the whole Sha *at the base.*

Proof. We now use the final K in the remaining branch of Section 7, where the intrinsic odd empty configuration is not a unit. There are no derivative labels in this last binary family. At augmentation every graph prime is unramified, and the actual support triangle gives

$$C_{\mathrm{old}} \longrightarrow (C_b)_0 \longrightarrow \bigoplus_{q\in Q} Q_q \longrightarrow C_{\mathrm{old}}[1], \tag{5.6}$$

with C_{old} an actual subcomplex of an upper triangular model. The graph-wide prescription gives scalar $1+t$ at each chosen orientation, even though the quadratic character is trivial at augmentation. The Euler-factor calculation in Lemma 8.11 therefore gives

$$\overline{l_{q,0}} = \overline{e}_{q,0}^{\,2} = \frac{t^4}{(1+t)^2}.$$

The two-place local determinant differs only by the unit q^4. Its residual t-order is consequently four, uniformly in q and the number of graph primes.

Put $T = T_2E$ and let C_K be the ordinary compact Kummer Selmer complex over the fixed imaginary quadratic field K, with the derived Kummer condition, including its degree-zero torsion, at every finite place. Lengths in this paragraph are lengths over $\mathbb{Z}_2$. Write

$$d_K = \dim_{\mathbb{F}_2} \mathrm{Sel}_2(E/K), \qquad t_K = \operatorname{length} E(K)[2^\infty], \qquad h_K = \operatorname{length} H^2(C_K)_{\mathrm{tors}}.$$

The finite Selmer group defines d_K, and perfection of C_K makes its cohomology finitely generated, so h_K is finite. This does not assert that the full two-primary Tate–Shafarevich group is finite. The already fixed companion has $L(E^{(k)},1) \neq 0$, so the sum-and-difference isogeny and the forward corank-zero theorem give full Selmer corank one for E/K. This includes any divisible part of the Tate–Shafarevich group.

Here are the cohomology calculations underlying the budget. At finite coefficients, the local degree-zero maps are isomorphisms onto local invariants and the local degree-one maps are the Kummer injections. The Selmer cone therefore has $H^0 = E(K)[2]$ and $H^1 = \mathrm{Sel}_2(E/K)$. The Weil pairing and the exact orthogonality of the Kummer conditions give the finite self-duality with shift three [56, 5.1.1–5.1.6 and 6.3.4]. Since K has no real places and $E[2]$ is rational of rank two, this gives

$$\dim_{\mathbb{F}_2} H^i(C_K \otimes^{\mathbf{L}} \mathbb{F}_2) = (2, d_K, d_K, 2) \quad (i = 0,1,2,3), \tag{5.7a}$$

with zero cohomology in other degrees. In particular a minimal finite free $\mathbb{Z}_2$ model of C_K has total rank $2d_K + 4$.

For the integral torsion calculation, $H^0(C_K) = T^{G_K} = 0$: a compatible system of rational points of unbounded two-power order cannot lie in the finite group $E(K)_{\mathrm{tors}}$. The exact sequence for $0 \to T \to T \otimes \mathbb{Q}_2 \to E[2^\infty] \to 0$, and the local Kummer conditions, identify

$$H^1(C_K)_{\mathrm{tors}} = E(K)[2^\infty].$$

Indeed this finite group injects as torsion global cohomology, all of its elements satisfy the Kummer conditions, and a torsion class maps to zero with rational coefficients, so is in this image. Integral Kummer self-duality $C_K \simeq R\,\mathrm{Hom}_{\mathbb{Z}_2}(C_K, \mathbb{Z}_2)[-3]$ now gives

$$H^3(C_K) \simeq \mathrm{Ext}^1_{\mathbb{Z}_2}(H^1(C_K), \mathbb{Z}_2), \qquad \mathrm{length}\, H^3(C_K) = t_K. \tag{5.7b}$$

It also identifies the free ranks in degrees one and two. They both equal one: universal coefficients with $\mathbb{Q}_2/\mathbb{Z}_2$ give

$$0 \longrightarrow H^1(C_K) \otimes \mathbb{Q}_2/\mathbb{Z}_2 \longrightarrow \mathrm{Sel}_{2^\infty}(E/K) \longrightarrow H^2(C_K)_{\mathrm{tors}} \longrightarrow 0,$$

and the middle group has full corank one. Thus the total integral torsion length of C_K is $h_K + 2t_K$.

At $t = 0$ the moving character is trivial. At either place over the good split moving prime r_i, the unramified condition is the ordinary Kummer condition. Replacing it by the full condition adds the singular complex

$$Q_{r_i,w} = [\,\mathbb{Z}_2^2 \xrightarrow{\Phi_{r_i} - r_i} \mathbb{Z}_2^2\,] \quad \text{in degrees one and two.}$$

Here Φ_{r_i} is arithmetic Frobenius on T, with trace a_{r_i} and determinant r_i. In particular

$$\det(\Phi_{r_i} - r_i) = r_i(r_i + 1 - a_{r_i}) = r_i \det(1 - \Phi_{r_i}),$$

so this determinant is nonzero and has valuation at most c_E. The same bound holds at the conjugate place. The singular complexes have zero degree-one cohomology and finite degree-two cokernels of total length at most $2c_E$.

This comparison is retained before the inner limit. The field K and C_K are fixed, while the two moving Frobenius matrices and the comparison maps may vary with i and with the graph. Pass to their limit at every coefficient precision using the same retained finite diagram. A matrix whose determinant has valuation at most c_E cannot acquire a zero determinant in this limit: the property is determined modulo 2^{c_E+1}. Consequently, if $D_c = C_{\mathrm{old}} \otimes^{\mathbf{L}}_{\Lambda} \mathbb{Z}_2$, the resulting triangle is

$$C_K \longrightarrow D_c \longrightarrow Q_r \longrightarrow C_K[1], \tag{5.7c}$$

where Q_r is the sum of two rank-two singular disks and $H^2(Q_r) = J_r$ has length at most $2c_E$. The notation r in this limiting complex does not designate a fixed rational prime supporting characters of unbounded order.

Reducing (5.7c) modulo two, the two singular disks contribute at most eight to total cohomology dimension. Equation (5.7a) thus gives a finite free model of C_{old} over Λ of total rank at most $2d_K + 12$, by cancelling unit entries at its maximal ideal $(2, t)$. The integral long exact sequence of (5.7c) gives

$$H^1(D_c) \simeq H^1(C_K), \qquad 0 \to H^2(C_K) \to H^2(D_c) \to J_r \to H^3(C_K) \to H^3(D_c) \to 0. \tag{5.7d}$$

In particular $H^3(D_c)$ is a quotient of $H^3(C_K)$. The image of the torsion of $H^2(D_c)$ in J_r has length at most length J_r, and its kernel is $H^2(C_K)_{\mathrm{tors}}$. Hence, defining the symbol used below by

$$h_1 := \sum_i \mathrm{length}\, H^i(D_c)_{\mathrm{tors}},$$

we have $h_1 \le h_K + 2c_E + 2t_K$.

Finally, a finite free complex over $\mathbb{Z}_2$ decomposes into free cohomology generators with zero differential and Smith disks $[\mathbb{Z}_2 \xrightarrow{2^e u} \mathbb{Z}_2]$, $u \in \mathbb{Z}_2^\times$, $e \ge 0$. Its only free cohomology generators here

are one in degree one and one in degree two. If m counts all its Smith disks, including those with $e = 0$, then $\operatorname{rank}_{\mathrm{total}} C_{\mathrm{old}} = 2 + 2m$. Every disk contributes e to h_1, so the sum of the valuations of all its nonzero Smith entries equals h_1. We have proved

$$\operatorname{rank}_{\mathrm{total}} C_{\mathrm{old}} \le 2d_K + 12, \qquad m \le d_K + 5, \qquad h_1 \le h_K + 2c_E + 2t_K. \tag{5.7}$$

The model and its matrices may vary with the moving characters, but these bounds depend only on the final fixed K and fixed old data. They do not depend on the binary dimension or graph size. The varying-field estimate (5.1) is not used with $j = b$, and h_K is not treated as a varying-field constant.

The lift is made over $R = \Lambda[G]$ before passing to $\mathcal{O}[G]$. Let C be the retained finite free family model. Its augmentation C_0 is the actual scalar cochain model, by the retained coefficient-change maps. After augmentation the representation is unramified at Q, so the support-change triangle (5.6) gives a finite free model D of C_0 whose underlying graded module is $C_{\mathrm{old}} \oplus \bigoplus_q Q_q$ and whose differential is upper triangular. This follows by representing the connecting map between these bounded free complexes by a chain map and forming its cone.

Cancel unit disks in C_0. The same pivots and elementary operations lift to C over R, since the augmentation ideal is contained in the Jacobson radical of R. Let C' be the resulting family model, so that C'_0 is minimal. Decompose D over Λ into a minimal complex $D_{\min}$ and unit disks E. The actual comparison gives a homotopy equivalence $C'_0 \simeq D_{\min}$. A homotopy equivalence between minimal bounded free complexes over a local ring is degreewise invertible, by reduction modulo its maximal ideal and Nakayama. Add to C' the same unit disks E_R and lift the resulting graded isomorphism $C'_0 \oplus E \simeq D$ to R. Its matrices are invertible because their augmented determinants are units. Conjugating the family differential by these lifted matrices makes its augmentation exactly the differential of D. All matrices and their inverses at this step lie in $\Lambda[G]$; thus their sign evaluations are power series. The added disks are precisely those needed for D, so this procedure does not enlarge its stated old-support rank. The block containing the local quotients has

$$B_0 = A = \operatorname{diag}(A_q), \qquad T_0 = 0$$

in a differential $\left(\begin{smallmatrix} P & Q \\ T & B \end{smallmatrix}\right)$. Over $\mathcal{O}[G]$, cancel B. The surviving Schur differential $P - QB^{-1}T$ augments to P_0.

Write $\det A_q = t^{d_q} v_q + 2w_q$, $d_q \le 4$, with v_q a power-series unit, and put $C_q = \operatorname{adj}(A_q)/(t^{d_q} v_q)$. For a character x, $B_x C = 1 + 2F$ with both C and F having no Laurent terms below degree -4. Therefore

$$B_x^{-1} = C \sum_{j=0}^{M-1} (-2F)^j \pmod{2^M} \tag{5.8}$$

has pole at most $4M$, independent of the number of blocks. For completeness the adjacent-differential assertion is an exact block identity. In the two degrees containing the new local block, write the differential as $d = \left(\begin{smallmatrix} P & Q \\ T & B \end{smallmatrix}\right)$, with B invertible. Then

$$\begin{pmatrix} 1 & -QB^{-1} \\ 0 & 1 \end{pmatrix} d \begin{pmatrix} 1 & 0 \\ -B^{-1}T & 1 \end{pmatrix} = \begin{pmatrix} P - QB^{-1}T & 0 \\ 0 & B \end{pmatrix}.$$

If the preceding differential has components $(X, Y)^{\mathsf{T}}$, then $d^2 = 0$ gives $TX + BY = 0$. In the new source coordinates its components are $(X, B^{-1}TX + Y)^{\mathsf{T}} = (X, 0)^{\mathsf{T}}$. If the following differential has components (U, V), then $UQ + VB = 0$, and in the new target coordinates they

are $(U, UQB^{-1}+V) = (U, 0)$. Thus the surviving entries of both adjacent differentials are literally unchanged; no second inverse is introduced there.

Here A_q is the direct sum of the two rank-two singular differentials at the conjugate graph places. Modulo two their scalar entries are t and $-t/(1+t)$, since their Frobenius scalars are $1+t$ and $(1+t)^{-1}$ and $E[2]$ is trivial. Consequently

$$\det \overline{A}_q = \frac{t^4}{(1+t)^2}.$$

This independently gives the residual order four in (5.6). The factor q^4 in the Euler-factor convention is a unit and does not affect it. Moreover every sign evaluation of a matrix over $\Lambda[G]$ has the same reduction modulo two as its augmentation, because each sign is congruent to one modulo two. Thus $B_x = A + 2V_x$ with V_x a power-series matrix. For the block-diagonal matrix C in (5.8), the identities

$$A_q C_q = 1 + \frac{2w_q}{t^{d_q} v_q}, \qquad B_x C = 1 + 2F_x$$

show that every entry of C and F_x has Laurent order at least -4. Formula (5.8) therefore has Laurent order at least $-4M$, because its jth summand has order at least $-4(j+1)$. The estimate counts the largest pole of an entry, not the number of summands or blocks. The remaining Schur differential has the same bound since P_x, Q_x, T_x are power-series matrices, and the adjacent surviving entries have no poles at all. These arguments take place after localization where t is invertible; they do not invert t in a finite Artin coefficient ring. □

9.6 Determinant clearing and the missing-address contradiction

Proposition 9.8 (A common denominator for the corrected coordinate)**.** *Under Lemma 9.7, use the paired functional exponent a of Proposition 8.6 and the universal correction η of Lemma 8.11. There are common $\Delta, Q \in \mathcal{O}[G]$ such that, wherever $\Delta_x \neq 0$, $\widetilde{u}_x = 2^{-a}\eta_x Q_x/\Delta_x^3$. Putting $f = 2^a\Delta^4$ and $d = \eta\Delta Q$ gives $f_x\widetilde{u}_x = d_x$ at every character, including $\Delta_x = 0$, and $f_0 \in \Lambda$. Moreover*

$$v_2 f_0(0) \le a + 4(h_K + 2c_E + 2t_K) = c_{\det}, \qquad L_T(M) = 16(d_K+5)M$$

bounds the negative Laurent support of every f_x modulo 2^M, uniformly in the sign character x. The elements f, d themselves belong to $\mathcal{O}[G]$ in the original group basis. These bounds concern the same corrected coordinate throughout; no pole bound for d_x or for the group coefficients of f is required.

Proof. Put $S = \mathcal{O}[G]$ and first perform the simultaneous new-local cancellation of Lemma 9.7. Its determinant comparison factor κ is an S-unit: the canceled block is invertible over S, so its determinant and inverse are integral over S. Retain this factor, with the determinant-line sign prescribed by the original basis. Transport the actual cycle z and the closed functional $\mu = 2^a\lambda_z$ to the remaining complex D. Both transported cochains are integral over S. The augmentation of D is the old complex over Λ, and each sign-evaluated differential entry has pole at most $4M$ modulo 2^M. No bound on the entries of the transported z or μ is needed below.

Write $D_c = D_0 \otimes_\Lambda \mathbf{Z}_2$. Its integral Smith decomposition gives graded decompositions

$$D^i = B_i \oplus H_i \oplus E_i,$$

lifted by constant $\mathbf{Z}_2$ basis changes, where centrally the only nonzero differentials are the Smith disks $E_i \to B_{i+1}$. Here H_1, H_2 have rank one and all other H_i vanish. Let $m_i = \operatorname{rank} E_i$, $m = \sum_i m_i$, and let A_i be the actual differential submatrix from E_i to B_{i+1}. Set

$$\delta_i = \det A_i, \qquad \Delta = \prod_i \delta_i.$$

An empty determinant is one. The entries in this definition are the original entries of D, before any of these old-disk cancellations. Thus Δ has degree m in those entries, $\Delta_0 \in \Lambda$, and

$$v_2\Delta_0(0) = \sum_{\text{central Smith disks}} v_2(\text{disk}) \le h_1 \le h_K + 2c_E + 2t_K.$$

Here is the full denominator calculation. Work temporarily in $S[1/\Delta]$. Group all sources and targets for a given degree, and cancel the whole block A_i once. For that differential the two elementary matrices are

$$\begin{pmatrix} 1 & 0 \\ -CA_i^{-1} & 1 \end{pmatrix} \begin{pmatrix} A_i & B \\ C & D' \end{pmatrix} \begin{pmatrix} 1 & -A_i^{-1}B \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} A_i & 0 \\ 0 & D' - CA_i^{-1}B \end{pmatrix}.$$

Both elementary matrices have determinant one. In the adjacent differentials, the equation $d^2 = 0$ makes the components into and out of this canceled pair zero after these changes; the other entries of those adjacent differentials are unchanged. Hence a cancellation does not modify any other surviving A_j. Every inverse used is $\operatorname{adj}(A_i)/\delta_i$, and each grouped block is used only once.

More explicitly, in degree k the projection associated to the incoming cancellation is

$$(b, v) \longmapsto v - CA_{k-1}^{-1}b,$$

and the inclusion associated to the outgoing cancellation is

$$v \longmapsto (-A_k^{-1}Bv, v).$$

The other map in each of these two degrees is the corresponding coordinate inclusion or projection. Compose these maps in the chosen cancellation order. At each step each entry is linear in the one inverse A_i^{-1} at that step, and no A_i is encountered again. Products of denominators therefore divide $\prod_i \delta_i$, not a power depending on the number of individual Smith disks. In particular, if $p : D \to H$ and $\iota : H \to D$ are the final projection and inclusion, the two scalars

$$\alpha = (pz)^1 \in H_1, \qquad \beta = (\mu\iota)^2 \in H_2^*$$

satisfy $Z := \Delta\alpha \in S$ and $W := \Delta\beta \in S$. The final complex H has just the map $H_1 \to H_2$.

For the inverse determinant line, a disk in degrees $i, i+1$ contributes $\delta_i^{(-1)^i}$ to the coordinate of a cohomology tensor. This can be checked on the based disk $e \mapsto \delta_i b$: replace the boundary basis b by $\delta_i b$ in $\bigotimes_k \det(D^k)^{(-1)^{k+1}}$. Thus the remaining comparison factor is

$$T = \epsilon \prod_i \delta_i^{(-1)^i}, \qquad P := \Delta T = \epsilon \prod_{i \text{ even}} \delta_i^2 \in S,$$

where ϵ is the integral unit from the fixed basis ordering and the constant Smith basis changes. Including the previously retained new-local factor, put

$$Q = \kappa ZWP \in S.$$

This definition takes place over S, before sign evaluation. It uses no interpolation and no division by $|G|$.

If $\Delta_x \neq 0$ and the constituent has the rank-one cohomology pattern, its coordinate in the original determinant basis is exactly

$$\widetilde{u}_x = \eta_x\kappa_x\alpha_x(2^{-a}\beta_x)T_x = 2^{-a}\eta_x Q_x/\Delta_x^3.$$

This is the same z, functional, basis and multiplier η as in the statement. If the constituent is generically acyclic, the remaining map between the two lines is nonzero. Its cycle α_x is then zero over the fraction field, so $Q_x = 0$ and the formula agrees with the defined coordinate zero. These are the only generic patterns in the present family, by central specialization and the central corank-one bound.

Now define $f = 2^a\Delta^4$ and $d = \eta\Delta Q$ in S. The preceding formula proves $f_x\widetilde{u}_x = d_x$ whenever $\Delta_x \neq 0$. When $\Delta_x = 0$, both f_x and d_x are zero; the already defined coordinate $\widetilde{u}_x$ is finite in its fraction field, and the same equality follows directly. No rational expression with a zero denominator is specialized. In particular $f_0 \in \Lambda$ and

$$c = v_2 f_0(0) \le c_{\det} := a + 4(h_K + 2c_E + 2t_K), \qquad L_T(M) = 16(d_K + 5)M. \tag{5.9}$$

Indeed f is a polynomial of degree $4m$ in the differential entries of D, with constant integral coefficients. Each such entry, at a sign character, has pole at most $4M$ modulo 2^M. A product of $4m$ entries has pole at most $16mM$, and sums cannot increase this bound. Finally $m \le d_K + 5$.

Only the last pole estimate is characterwise. The actual elements f, d belong to the original group ring S. For $i < -L_T(M)$ all sign values of $[t^i]f$ are divisible by 2^M; the group-ring coefficient itself need not be divisible by 2^M. No Laurent support bound for d or for the group coefficients of f has been used. The constants d_K, t_K, h_K are fixed only after the final K is fixed; a and c_E precede that choice. None depends on the graph or binary dimension. A varying-K estimate cannot be deduced by treating h_K as an absolute constant. □

Proposition 9.9 (Finite coefficient transfer and the final contradiction)**.** *Use the fixed field, common family, regular coordinates and integral identity of Proposition 9.8. Let B be the fixed upper bound on $v_2\widetilde{u}_x(0)$ at every nonzero address supplied by (3.9), (3.4), (4.6) and Lemma 8.11. Choose M, b as in (5.10), before the finite graph and its actual theta stage are realized. For some nonzero address x,*

$$v_2(\widetilde{u}_x(0) - \widetilde{u}_0(0)) \ge M - 2c > B, \qquad c = v_2 f_0(0) \le c_{\det}.$$

The convolution includes every negative Laurent coefficient of f_x. If $L'(E/K, 1) = 0$, Lemma 8.12 makes $\widetilde{u}_0(0) = 0$, contradicting the upper bound. Since the fixed complementary value $L(E^{(k)}, 1)$ is nonzero, this proves $\operatorname{ord}_{s=1} L(E, s) = 1$.

Proof. For a coefficient of any element of $\mathcal{O}[G]$, its sign-value function on $\mathbb{F}_2^b$ expands as

$$A(x) = a_\varnothing + \sum_{I \neq \varnothing} 2^{|I|} a_I \prod_{i \in I} x_i.$$

Its low bit is constant and its bit of index $j \ge 1$ has Boolean degree at most 2^{j-1}. One can see this by extracting the Y^{2^j} coefficient of $(1+Y)^{A(x)}$ modulo two: the first possible contribution of x_I has degree at least $2^{|I|}$ in Y, and $|I| \le 2^{|I|-1}$. Thus q coefficient equalities modulo 2^M have sum of Boolean degrees at most $q(2^{M-1} - 1)$.

If that sum is less than b, multiply the complementary indicators of the equations. Its degree is less than b, so its sum over the Boolean cube is zero: every monomial misses a variable. The support of this indicator is the simultaneous solution set. It has even cardinality and contains zero, hence another address.

Apply this to the coefficients of f from $-L_T(M)$ through zero and the single constant coefficient of d. It suffices to choose

$$M > \max(c_{\det}, B + 2c_{\det}), \qquad b > (L_T(M) + 2)(2^M - 1), \tag{5.10}$$

where B bounds the nonzero-address central valuations. These choices precede the graph and its theta precision. At the resulting nonzero address, negative coefficients of f_x vanish modulo 2^M, and its constant coefficient agrees with $f_0(0)$ and has valuation c. Positive coefficients may have smaller valuation. Nevertheless $v_{\mathcal{O}}(f_x) \le c$, so integrality of $d_x = f_x\widetilde{u}_x$ gives $v_{\mathcal{O}}(\widetilde{u}_x) \ge -c$. This estimate comes from the displayed product at the selected nonzero address; it does not apply (5.5) to the unknown base. Both $\widetilde{u}_x$ and $\widetilde{u}_0$ lie in $\Lambda[1/2]$ by the height-one argument above, so neither has negative powers of t. In particular, since $f_0 \in \Lambda$, $[t^0]d_0 = f_0(0)\widetilde{u}_0(0)$. Put $F_0 = [t^0]f_0$ and $F_x = [t^0]f_x$. Since d_0 is integral and $[t^0]d_0 = F_0\widetilde{u}_0(0)$, we also have $v_2\widetilde{u}_0(0) \ge -c$, without using (5.5) at the base. The convergent convolution has the form

$$[t^0]d_x = F_x\widetilde{u}_x(0) + E_x, \qquad E_x = \sum_{i<0}[t^i]f_x[t^{-i}]\widetilde{u}_x, \qquad v_2(E_x) \ge M - c.$$

The untested coefficients below the Laurent cutoff have the same divisibility by the pole bound, so this sum includes every negative coefficient. Subtracting the two constant terms gives

$$F_x(\widetilde{u}_x(0) - \widetilde{u}_0(0)) = ([t^0]d_x - [t^0]d_0) - (F_x - F_0)\widetilde{u}_0(0) - E_x.$$

Every term on the right has valuation at least $M - c$ and $v_2(F_x) = c$. Consequently

$$v_2(\widetilde{u}_x(0) - \widetilde{u}_0(0)) \ge M - 2c > B. \tag{5.11}$$

By (3.9), (3.4), (4.6) and the unit comparison (4.7), every nonzero address has $v_2\widetilde{u}_x(0) \le B$ for one B fixed before b; the finite Sha terms have negative sign and may be discarded for this upper bound. If the base derivative vanished, Section 8 would give $\widetilde{u}_0(0) = 0$, contradicting (5.11). Hence $L'(E/K, 1) \ne 0$. The complementary value $L(E^{(k)}, 1)$ is nonzero, and $L(E/K, s) = L(E, s)L(E^{(k)}, s)$, so E has analytic rank one. This proves (RC), and hence Theorem 5.3, using the classical applications and finite constructions specified above. □

10 Exact precision computation and polynomial compilation

Encode a signed integer by

$$\mathrm{enc}(n) = \begin{cases} 2n, & n \ge 0, \\ -2n - 1, & n < 0. \end{cases}$$

The five natural inputs decode in the fixed order (n, κ, a, l, j). The decision predicate rejects unless $n \notin \mathcal{C}$, $\kappa \ne 0$, $a, l \ge 1$, and $j = na$. For nonzero multiples of P_0, $z = -x/y$ is well-defined. Set

$$z([l]P_0) = u_l + v_l s, \qquad \epsilon_l = |u_l| + |v_l|, \qquad r(a, j) = \tfrac{1}{2}\,\mathrm{Tr}_{K_0/\mathbb{Q}}\,\frac{z([j]P_0)}{z([a]P_0)}.$$

With the fixed constant $A_{\mathrm{ht}}^{(0)} = H_6(S_{\mathrm{ht}})$, its final tests are

$$|r(a, j) - n| < \epsilon_l, \qquad 4\epsilon_l < 4^{-A_{\mathrm{ht}}^{(0)}|\kappa F(n)|^{24}}.$$

These are strict tests on exactly computable rationals and integers.

The reference program S21 uses exact quadratic-field arithmetic and capped integer operations. The ideal register intermediate representation S22 computes the same predicate on every five-natural tuple. Its independent all-input argument covers normalized signed arithmetic, origin and zero-ordinate cases, equal-abscissa addition, and every early rejection. Nine exact polynomial

identities verify the nonexceptional projective formulas and preservation of $Y^2 = X^3 + AXZ^4$. Scalar loops decrease by halving; the precision loop divides by four; factorial iteration is bounded. This is a proof about the ideal intermediate representation, rather than an assertion that a budgeted simulator always finishes.

For example, if $\epsilon = U/V > 0$ with natural U, V and integer $B \geq 0$,

$$4\epsilon < 4^{-B} \quad \Longleftrightarrow \quad 4^B \leq \left\lfloor \frac{V-1}{4U} \right\rfloor .$$

This permits a finite cap. For positive L, $\sigma(x) = \min(x, L+1)$ commutes with saturated addition and multiplication. Powers and factorials can be computed with this saturation. Capping an exponent is safe for base at least two because an exponent above L already makes the value exceed L. Base zero, base one and exponent zero are separate cases. This proves equality with the fixed-height expression without requiring its full evaluation when the available precision is too small.

10.1 The natural polynomial and the power interface

The saved precision polynomial has 1 305 811 natural variables, the first five being inputs. Its SHA256 is

`f3e163f5c60b972c98b48647274f956dd7d78428e27ddc0543574a5f18120b8c.`

Source S23 specializes a fixed universal natural halting polynomial at the code of a wrapper. The wrapper inverts four Cantor pairs and then runs the unchanged precision machine. Its pairing equations are

$$2t_i = (c_i + t_{i+1})(c_i + t_{i+1} + 1) + 2t_{i+1} \quad (0 \leq i < 4), \qquad t_4 = c_4.$$

Their triangular intervals prove unique natural unpairing. The new polynomial sums the square of the old halting polynomial and the four squared differences above.

The universal compiler packs a finite run of a fixed m-state increment/decrement/branch program into base-B integers. Acceptance means reaching the unique final halt with all registers zero; the fixed interpreter includes a finite cleanup stage. Choose

$$c > 0, \quad K = 2^c, \quad B = 4K > m, \quad T = B^q, \quad F_q = 1 + B + \cdots + B^q, \quad D = (K-1)F_q,$$

and require every input to be below K. For natural integers, say X is masked by Y when every set binary bit of X occurs in Y. Require each register history R_j to be masked by D and satisfy $R_j < T$. Each state history S_i is masked by F_q, and

$$\sum_{i=0}^{m-1} S_i = F_q, \qquad S_{\text{halt}} = T.$$

Thus register digits are below K, terminal register digits are zero, and state digits are zero or one. Since $B > m$, the state sum has no carries and selects exactly one state at each time.

The nonzero mask N_j is determined by

$$KN_j = (R_j + D) \text{ AND } (KF_q).$$

Indeed each digit $r + K - 1 < 2K < B$ has its K-bit set exactly when $r > 0$. For a decrement/branch instruction at state i testing register j, impose

$$V_i = S_i \text{ AND } N_j, \qquad S_i = V_i + W_i.$$

These are the nonzero and zero branch histories. Form the packed increment indicator A_j from states incrementing j, the decrement indicator C_j from the appropriate V_i, and each incoming-state indicator I_i from the states or branches directed to i. They have zero-one digits, since there is just one active instruction and branch at each time. The transition equations are

$$R_j + BC_j = \text{initial}_j + BR_j + BA_j, \qquad S_i = \mathbf{1}_{i=0} + BI_i.$$

Every register coefficient on either side is at most $K < B$; uniqueness of base-B digits gives the initial conditions and each exact transition. The possible extra final slot vanishes because halt has no outgoing instruction and the terminal registers are zero. For $q = 0$ the same equations require the initial state already to be halt and the inputs to vanish. Conversely, any finite accepting run fits after choosing c sufficiently large, and supplies all these masks and equations.

10.2 The literal alpha and power formulas

The alpha system below is `Exp_Matrices.alpha_equations`, with equivalence `alpha_equivalence`; the power system is `Exponential_Relation.exp_repr` and its polynomial expansion `exp_dioph` in the pinned AFP source [57]. Their variables range over natural numbers. The alpha recurrence itself is integer-valued; the bounds below justify its natural-valued uses. Define $U_n = \alpha_b(n)$ by

$$U_0 = 0, \qquad U_1 = 1, \qquad U_{n+2} = bU_{n+1} - U_n \quad (n \geq 0).$$

For $b \geq 3$ this recurrence gives $0 \leq U_n < U_{n+1}$ and $U_n \geq n$. For $n \geq 1$, the matrix

$$M = \begin{pmatrix} b & -1 \\ 1 & 0 \end{pmatrix}, \qquad M^n = \begin{pmatrix} U_{n+1} & -U_n \\ U_n & -U_{n-1} \end{pmatrix}$$

has determinant one. Thus adjacent terms are coprime and $U_n^2 - bU_nU_{n-1} + U_{n-1}^2 = 1$. Parameter congruence follows directly from the recurrence: $b \equiv b' \pmod{d}$ implies $\alpha_b(n) \equiv \alpha_{b'}(n) \pmod{d}$. In particular $\alpha_2(n) = n$. This is parameter congruence, not merely congruence between indices.

Every natural solution of $x^2 + y^2 = 1 + bxy$, $b > 2$, is an unordered adjacent pair. To prove this, order it as $X > Y$. If $Y = 0$, $X = 1$. Otherwise $Z = bY - X$ satisfies $XZ = Y^2 - 1$, hence $0 \leq Z < Y$, and (Y, Z) satisfies the same equation. Descent reaches $(1, 0)$; reversing the steps generates exactly the adjacent pairs. No ordering assumption on the original pair is needed.

For $k > 0$, reduction of M^k modulo U_k is the invertible scalar $U_{k+1}I$. Dividing an index by k and using strict growth proves $U_k \mid U_m$ if and only if $k \mid m$. More precisely, for $m > 0$,

$$U_k^2 \mid U_m \quad \Longleftrightarrow \quad kU_k \mid m. \tag{C.1}$$

For $k = 0$ both sides are false. For $k > 0$ write $m = kq$ and expand $M^{kq} = (U_kM - U_{k-1}I)^q$ modulo U_k^2. Its lower-left entry is $qU_k(-U_{k-1})^{q-1}$, so coprimality proves (C.1), including $U_k = 1$.

For $m > 0$ put $v = U_{m+1} - U_{m-1} = \operatorname{tr}(M^m)$. Cayley–Hamilton gives $M^{2m} \equiv -I \pmod{v}$. Extend the recurrence by $U_{-j} = -U_j$. Every $n \geq 0$ has $n = 2lm + \varepsilon j$ with $l \geq 0$, $0 \leq j \leq m$ and $\varepsilon \in \{1, -1\}$; then $U_n \equiv (-1)^l \varepsilon U_j \pmod{v}$. The value sign need not equal the index sign. If $b \geq 4$, $v = bU_m - 2U_{m-1} > 2U_m$.

The actual alpha formula has natural witnesses r, s, t, u, v, w, x, y and the following fifteen conditions:

$$\begin{gathered} b > 3, \quad u^2 + t^2 = 1 + but, \quad s^2 + r^2 = 1 + bsr, \quad r < s, \quad u^2 \mid s, \quad v + 2r = bs, \\ w \equiv b \pmod{v}, \quad w \equiv 2 \pmod{u}, \quad w > 2, \quad x^2 + y^2 = 1 + wxy, \\ 2X < u, \quad 2X < v, \quad x \equiv X \pmod{v}, \quad 2N < u, \quad x \equiv N \pmod{u}. \end{gathered} \tag{C.2}$$

A strict inequality $a < b$ is $b = a+1+d$; divisibility is $b = ad$; congruence is $A+Mu = B+Mv$, with fresh natural witnesses. At zero modulus the last equation says equality, so there is no exception hidden in this replacement.

For soundness, the unordered Pell result gives $u = U_k$, $k > 0$, and $x = \alpha_w(n)$. The ordered pair gives $s = U_m, r = U_{m-1}$, $m > 0$. By (C.1), $u \mid m$. Parameter congruence gives $x \equiv U_n \pmod{v}$ and $x \equiv n \pmod{u}$. Reflect n to $j \in [0, m]$. Since $2X < v$ and $2U_j < v$, the congruence $X \equiv \pm U_j \pmod{v}$ forces $X = U_j$, with the negative case forcing both zero. Then $j \le X$, $2j < u$ and $2N < u$ force $N = j$ from $N \equiv \pm j \pmod{u}$. Thus $X = U_N$.

Conversely choose an odd $u = U_k > 2U_N$. Such terms are unbounded, since adjacent terms are coprime and the sequence grows. Set $t = U_{k+1}$, $m = ku$, $s = U_m$, $r = U_{m-1}$ and $v = bs - 2r$. Equation (C.1) gives $u^2 \mid s$, and the required strict bounds hold. The moduli u, v are coprime: a common divisor is odd, divides s and $2r$, hence divides both adjacent terms. CRT gives $w \equiv 2 \pmod{u}$, $w \equiv b \pmod{v}$ with $w > 2$. Take $x = \alpha_w(N)$, $y = \alpha_w(N+1)$. Parameter congruence verifies every remaining condition. This includes $N = X = 0$.

The literal power formula separates $Q = 0, R = 0, P = 1$ and $Q = 0, R > 0, P = 0$. For $Q \ge 1$ it imposes

$$a_0 = \alpha_{Q+4}(R+1), \quad b = a_0 + Q^2 + 2, \quad m + Q^2 + 1 = bQ, \quad P < m,$$
$$a_1 = \alpha_b(R), \quad a_2 = \alpha_b(R+1), \quad P + ba_1 \equiv Qa_1 + a_2 \pmod{m}. \tag{C.3}$$

Induction gives $a_0 \ge (Q+3)^R$, so $m > Q^R$ and $m > 0$. Let $E_R = QU_R - bU_R + U_{R+1}$. The recurrence and $bQ \equiv Q^2 + 1 \pmod{m}$ give $E_0 = 1$ and $E_{R+1} \equiv QE_R \pmod{m}$. Thus (C.3) says $P \equiv Q^R \pmod{m}$. Both values lie in $[0, m)$, forcing equality. Conversely equality supplies every witness. This includes $R = 0$, $Q = 1$ and $0^0 = 1$.

These formulas make the binomial and bit operations polynomial. For $B = 2^{n+1}$, the equation

$$(1+B)^n = B^k(Bq + C) + r, \quad C < B, \quad r < B^k$$

selects exactly $\binom{n}{k}$, since all its base-B coefficients are below B, including zero for $k > n$. The identity $(1+t)^n = \prod_{i:\,\mathrm{bit}_i(n)=1}(1+t^{2^i})$ over $\mathbb{F}_2$ makes odd binomial coefficients equivalent to bit inclusion. Bit disjointness of a, b is equivalent to inclusion of a in $a+b$: at the least shared bit there is no incoming carry and that bit is cleared. Finally $x = z + a, y = z + b$ with z, a, b pairwise bit-disjoint says exactly $z = x$ AND y.

10.3 Interpreter, lowering and five-input specialization

The universal program numbering is $e + 1 = 2^w(2D + 1)$. At $w = 0$ it halts. Otherwise the four base-2^w digits at positions $4p, \ldots, 4p + 3$ give the opcode, register and two targets at state p. Opcode one increments, opcode two decrements or takes its zero branch; every other opcode halts. Missing digits are zero, so decoding is total.

The interpreter stores a finite tape as 0 or $\mathrm{cons}(h, t) = 2^h(2t + 1)$. Repeated halving decodes a nonempty head and tail; the tail is smaller than the code. To update a register, save the finite reversed prefix, replace the entry and restore the prefix. Absent entries are zero. The reserved jump register remains zero. Copies restore their source; aliased addition snapshots its operand before overwriting it. Division maintains $x = qy + r$ and subtracts only when $r \ge y$; the interpreter's divisors are at least two. These invariants prove that one interpreter loop executes exactly one decoded instruction, with finite work. A simulated halt enters cleanup and clears all registers. An infinite simulation never enters cleanup. Thus its all-zero halt is equivalent to virtual halting, without requiring the virtual final tape itself to be zero.

The precision IR uses normalized signed pairs for integer arithmetic and pairs over the basis $1, \sqrt{2}$ for field arithmetic. Norm zero implies the field element is zero: after clearing denominators, a primitive solution of $a^2 = 2b^2$ would have both entries even. Every division is guarded by that exact test. The Jacobian point formulas satisfy the curve equation and the affine chord or tangent formulas after multiplying denominators. Equal abscissas split into doubling and opposite-point branches; zero ordinate doubles to the identity. The fixed projective point is the same affine P_0. These are polynomial identities, independent of the generator theorem.

The scalar loop halves its remaining count. The precision loop divides its remainder by four. The capped arithmetic computes exactly $\min(x, L + 1)$; zero bases, unit bases and zero exponents are separate. For bases at least two an exponent exceeding L already exceeds the cap. Thus the ideal IR terminates and agrees with the stated rational predicate on all five natural inputs, including every early rejection.

Each primitive lowering uses fresh auxiliary registers, snapshots overlapping sources, and preserves the live-register map at the end of each finite block. Saturating decrement gives natural subtraction, and repeated addition gives multiplication. All variable divisions are guarded. Induction on completed loop iterations proves the same claim for loops. Acceptance enters cleanup; rejection enters a permanent self-loop. Hence the primitive machine halts exactly on acceptance, although the ideal decision program is total.

For $\mathrm{pair}(a, b) = (a + b)(a + b + 1)/2 + b$, unpairing maintains remainder $=$ input $- s(s + 1)/2$ and subtracts $s + 1$ while $s <$ remainder. At exit $0 \le$ remainder $\le s$, so $b =$ remainder, $a = s - b$. Four repetitions put the five codes into registers zero through four. Fresh scratch lies above the old register range and is cleared; old work registers remain zero. Relocating every target by the prefix length therefore preserves the precision machine.

The polynomial specialization fixes the universal program input to this wrapper code, maps the old packed input to variable five, and maps every old variable $i \ge 2$ and private-block start by $i \mapsto i + 7$. The four pairing variables occupy five through eight, disjoint from the five inputs. The appended four squared equations force their unique Cantor values. Thus, for the exact saved polynomial C,

$$\forall \mathbf{c} \in \mathbb{N}^5, \qquad \exists \mathbf{z} \in \mathbb{N}^{1305806}\ C(\mathbf{c}, \mathbf{z}) = 0 \quad \Longleftrightarrow \quad \mathrm{accepts_naturals}(\mathbf{c}) = \mathrm{True}. \tag{C.4}$$

The natural compiler proof has no height, restricted-converse or rational-model premise. Its use of the height expression concerns only computation of that fixed integer expression.

10.4 Translation to point codes

The typing order is independent of the natural predicate. First, `member` forces a canonical code in D, and raw addition on two canonical elliptic inputs forces their canonical sum, including the origin and two-torsion cases. The fixed scalar graph gives $[m]Q$ on D. In `multiply`, the three root membership calls are forced conjuncts inside `height_certificate`. Thus the four scalar shifts are in D before they are passed, unchanged, to the 27 slope calls and the six test-group calls within each slope. This supplies the hypotheses of every raw addition in those groups. Only after globally typed multiplication is established do the four square products and three successive additions justify `natural`. Their outputs are canonical by the raw-addition postcondition, so no separate output membership assumption is needed. The signed-encoding and exceptional branches start their addition chains at this natural code, the literal origin, or the fixed signed generator. An inactive product branch is not assumed true. The fixed generator equations are conjuncts outside the internal products in all three templates that use them. These observations give the base cases for the source-DAG induction below. Apply (C.4) using the exact signed index model. Induction

on the actual source DAG gives each node the code of its signed integer value. Natural guards constrain source variables, not intermediate sums or products. Raw addition receives canonical codes; globally typed multiplication justifies the natural guards. Fixed generator equations and scalar templates supply constants.

The power-value recipe has three signed code inputs, seventy-nine private natural witnesses, and an output code for the value of the power-template polynomial. That value is not independently required to vanish. Its saved root equality is $[17751579, 15]$. The precision parent has 54056 explicit variables and 15845 disjoint private blocks of seventy-nine variables. No parent variable node accesses a private coordinate. Replacing each block by its private power-value call therefore preserves the full existential assignment. Only the final parent root is equated to the identity.

At each primitive formula, equality is a difference, conjunction is a sum of squares, and disjunction is a product. A topological induction, including the globally conjoined fixed equations, proves equality of polynomial values under substitution, not merely equisatisfiability of an unrelated formula. This is the required link to the saved AST.

11 A variable-shift relation for ordinary values

We prove the ordinary-value deduction from the exact index model, the compiled precision relation, the height theorem and PC. These are supplied by Sections 3, 10, 4 and Theorem 5.2, respectively. Capital letters below denote five-coordinate point codes; their lower-case indices are ordinary integers. The external variable t is any rational number.

11.1 Approximation for every fixed shift

Define $\mathrm{Close}(N, \mathcal{K}, t)$ by choosing A, L, J, linking all five signed codes to precision inputs in the order (n, κ, a, l, j), and imposing $|t - r(a, j)| < \epsilon_l$ on those same coordinates. Put

$$B(n, \kappa) = H_6(S_{\mathrm{ht}})|\kappa F(n)|^{24}.$$

The precision contract gives

$$\mathrm{Close}(\iota(n), \iota(\kappa), t) \Longrightarrow |t - n| < e^{-B(n,\kappa)}, \tag{10}$$

$$\mathrm{Close}(\iota(n), \iota(\kappa), n) \quad \text{for every } n \notin \mathcal{C},\ \kappa \neq 0. \tag{11}$$

The first follows from $|t - n| < 2\epsilon_l < 4^{-B}/2 < e^{-B}$. For the second, positive multiples of P_0 approach the identity simultaneously in the compact group $E(\mathbf{R}) \times E(\mathbf{R})$. This follows, for instance, by partitioning the compact closure of the cyclic subgroup into arbitrarily small neighborhoods and subtracting two sufficiently close multiples. If the resulting positive differences were bounded, a fixed nonzero multiple would be the identity; infinite order excludes this. Choose l first to make the positive ϵ_l small enough. Since the derivative of $[n]$ at the identity is n [60, IV.2.3(a), p. 122], $r(a, na) \to n$ along positive multiples a approaching it. Choose such an a and put $j = na$. This works for every already fixed nonzero shift κ, regardless of its size.

11.2 The six local tests

For $N = \iota(n)$, $\mathcal{K} = \iota(\kappa)$, form the code for κn, choose $H = \iota(h)$, $U = \iota(s_0)$, and put $V_0 = H - \mathcal{K}$, $V_1 = U - \iota(\kappa n)$. Require $\kappa \neq 0$, $H, V_0 \neq O$. Define $w(Q) = 1/x(Q)$ for $Q \neq O$, $w(O) = 0$. For $t \notin \mathcal{C}$

use

$$D_1 = w(H), \qquad D_2 = w(U), \qquad D_3 = \sigma w(V_0), \qquad D_4 = \sigma w(V_1),$$
$$D_5 = z(U)/z(H) - t, \qquad D_6 = \sigma(z(V_1)/z(V_0)) - t.$$

Here σ is conjugation in K_0. Choose $\eta, q \in \mathbb{Q}$ and impose

$$\eta^2 = 1, \qquad R_0 = \eta 2730^6 F(t), \qquad a_0 = 1 - R_0 q^2 \neq 0,$$
$$R = R_0 a_0^2, \qquad C_j = D_j/F(t), \qquad W = \prod_{j=1}^{6}(1 + RC_j^2) \neq 0,$$
$$\Theta = \mathrm{Tr}_{K_0/\mathbb{Q}}(W^{-1}) \neq 0, \qquad B_{\mathrm{pole}} = R/\Theta^2, \qquad \Phi(B_{\mathrm{pole}}).$$

Call this relation $\mathrm{Test}(N, \mathcal{K}, t)$. All quotients are guarded polynomially. There is one common denominator $F(t)$, one product W, one trace and one pole instance.

Lemma 11.1 (Divisibility)**.** *If* $\mathrm{Test}(\iota(n), \iota(\kappa), t)$ *holds, then* $M_{S_{\mathrm{ht}}}(t) \mid \kappa F(n)$.

Proof. Let $p \mid M_{S_{\mathrm{ht}}}(t)$. Exactly one $c \in \mathcal{C}$ has $\delta = v_p(t - c) > 0$ odd. The contact differences are units, so $v_p(F(t)) = \delta$; the field is unramified at p. Put $b = v_p(a_0)$. If $q = 0$, then $b = 0$. Otherwise $\delta + 2v_p(q)$ is odd and nonzero, so $b = \min(0, \delta + 2v_p(q)) \leq 0$. In either case $\alpha = v_p(R) = \delta + 2b$ is odd. At each place above p,

$$v(1 + RC_j^2) = \min(0, \alpha + 2(v(D_j) - \delta)).$$

Let $\lambda = \max_{v|p} v(W) \leq 0$. Then $v_p(\Theta) \geq -\lambda \geq 0$. The pole input has odd valuation $\alpha - 2v_p(\Theta)$, which is positive by Lemma 5.1. Therefore $\alpha > 0$ and $\lambda > -\alpha/2$.

At a place attaining λ, each nonpositive factor valuation is at least λ. Consequently

$$v(D_j) \geq \delta - 3\alpha/4 = \delta/4 - 3b/2 > 0 \quad (1 \leq j \leq 6).$$

Integral valuations are at least one, at the same place for all tests. The first four put H, U in one formal group and V_0, V_1 in the conjugate formal group. At the identity we set both $z(O) = 0$ and $\log(O) = 0$; thus the cases $U = O$ and $V_1 = O$ obey the same ratio identities below. Their denominators are nonzero by the guards on H and V_0. For a nonzero point in either formal group,

$$\log(Q) = z(Q)(1 + e_Q), \qquad v(e_Q) \geq 1.$$

Here $p > 324$ and $v(p) = 1$. By [60, IV.5.5 and IV.6.4, pp. 129–132], the logarithm is an injective isometry on $v(z) \geq 1$. The integral invariant differential gives in relative degree $r - 1$ the bound $(r - 1)v(z) - v_p(r) \geq 1$ for every $r \geq 2$. Thus the guarded nonzero points H, V_0 have nonzero logarithms. The fifth test first makes $z(U)/z(H)$ integral and congruent to t; multiplication by the units $1+e_Q$ preserves that congruence. The relation $[s_0]H = [h]U$ then gives $\log(U)/\log(H) = s_0/h$. Likewise the sixth gives $(s_0 - \kappa n)/(h - \kappa) \equiv t$. Multiplying by the integer denominators and subtracting yields $v_p(\kappa(n - t)) \geq 1$. Since $t \equiv c$, we get $p \mid \kappa(n - c)$, hence $p \mid \kappa F(n)$. The squarefree product divides as asserted. No unit assumption on κ was used. □

Lemma 11.2 (One suitable shift)**.** *For every* $n \in \mathbb{Z} \setminus \mathcal{C}$ *there is* $\kappa \neq 0$ *with* $\mathrm{Test}(\iota(n), \iota(\kappa), n)$.

Proof. Choose the sign so $R_0 = \eta 2730^6 F(n) > 0$ and let S' be its finite prime support. Each sufficiently small local formal neighborhood has finite index in the compact local group [60, VII.6.3, p. 200]. A positive common multiple of the quotient orders gives κ with $[\kappa]P_0$ in every chosen neighborhood. Take

$$H = [2\kappa]P_0, \quad U = [2\kappa n]P_0, \quad V_0 = [\kappa]P_0, \quad V_1 = [\kappa n]P_0.$$

The formal parameter and the derivative of $[n]$ make each C_j integral above S' after sufficiently shrinking the neighborhoods. Infinite order preserves the nonzero guards.

Choose rational q integral at S' and arbitrarily close to $1/\sqrt{R_0}$ in $\mathbf{R}$, without equality. Denominators $R_0 m + 1$ with nearest integer numerators give such choices, adjusting by one if equality occurs. At S', a_0 is a unit and $W \equiv 1$, $\Theta \equiv 2$ modulo $p^{v_p(R_0)}$. Integral trace justifies this also at the ramified place two. The pole input has valuation at least four at two and at least six at the other primes of S_0. Outside S', its valuation is $2v_p(a_0) - 2v_p(\Theta)$ and is even, so its reduced denominator is a square. In both real embeddings, R tends to zero positively, $W \to 1$ and $\Theta \to 2$. All guards hold and $0 < B_{\mathrm{pole}} < 1/2$ for a sufficiently close choice. Theorem 5.2 supplies the pole witnesses. □

11.3 Exact ordinary value

Define $\mathrm{Val}(N, t)$ by membership of N and the disjunction

$$\bigvee_{c \in \mathcal{C}} (N = \iota(c) \wedge t = c) \quad \vee \quad \exists \mathcal{K}\, (\mathrm{Close}(N, \mathcal{K}, t) \wedge \mathrm{Test}(N, \mathcal{K}, t)).$$

The same five coordinates $\mathcal{K}$ are used in both calls. In an accepting main branch $n \notin \mathcal{C}$, $\kappa \neq 0$, and Lemma 11.1 and the height estimate imply

$$h(t) \leq H_6(S_{\mathrm{ht}}) |\kappa F(n)|^{24} = B(n, \kappa).$$

For $t = a/b \neq n$ in reduced form, $|a - nb| \geq 1$, hence

$$|t - n| \geq 1/b \geq e^{-h(t)} \geq e^{-B(n,\kappa)},$$

contradicting (10). Thus $t = n$. Conversely choose κ by Lemma 11.2, then use (11) for that same shift. Private witness blocks are disjoint, so the two calls can be satisfied simultaneously. The six remaining integers have exactly their displayed branches.

Therefore $\mathrm{Val}(\iota(n), t)$ is exactly the graph $t = n$. Existentially quantifying N gives ordinary $\mathbb{Z}$ in $\mathbb{Q}$. S01 and S16 multiply the exception and main-branch polynomials and conjoin membership by a sum of squares. The literal substitutions in Section 2 and Appendix A3 identify this polynomial with the saved root P_*. This proves the zero-set part of Lemma 1.4.

12 The cutoff and polynomial size

The literal constants in S03 define the catalogue product H_{cat}, its maximum $H_{\max}$, $N_s = P_s H_{\mathrm{cat}}$ with $P_s = \prod_{p \leq 324} p$, the least formal multiple M_0, and $c_* = 4M_0$. They take

$$R = \max(D, H_{\max}, c_*, 10^{12}), \quad A_{\mathrm{ht}} = 2 \cdot 10^{13} \max(1, C_{\mathrm{cm}}, Q_R^{17}), \quad T = (48 A_{\mathrm{ht}})^{13}.$$

The pole clearing constant is $\ell_0 = 12 P_s^{2916} H_{\mathrm{cat}}^{732}$. The symbol $F_0 = 324!$ is distinct from the six-contact polynomial $F(u)$.

Proposition 12.1 (Literal cutoff)**.** *The fixed prescription satisfies*

$$R = D, \qquad T = (96 \cdot 10^{13})^{13} D^{2210 G_0^2 D^3} < 2^{2^{2^{9013}}}.$$

This is a statement about the prescribed integer, independently of its semantic role in excluding false multiplication triples.

Proof. We give the bounds of S25. Exact small calculations give $10^5 < W < 2^{512}$ and $2^{2000} < F_0 < 2^{2^{2250}}$. For $q \geq 10^5$, the unconditional progression estimate [53, Theorem 1.2, (1.10)–(1.11), Corollary 6.17 with $A = 1$] applies for $x \geq \exp(0.03\sqrt{q}(\log q)^3)$, including the possible exceptional-zero case. At $X = 2^{W^3}$ its error is at most $X/(160 \log X)$ and the hypothesis follows from $0.03\sqrt{W}(\log W)^3 < (\log 2)W^3$. Hence $\vartheta(X; W, a) > X/(2W)$, giving more than $X/(2W^4) > 22$ primes in each unit class. No prime at most 324 belongs to such a class. Thus

$$H_{\max} < X, \quad H_{\mathrm{cat}} \leq X^{22W}, \quad \log_2 N_s < 2^{2054}.$$

For every odd prime dividing N_s, take the actual reduction order of P_0 at each place above it, and let their product be m_0. All preceding positive multiples stay in the dyadic formal subgroup, since P_0 starts there and the subgroup is closed under addition. A nonidentity point there has negative abscissa valuation, so its reciprocal is defined; the constructor accepts the identity directly. Thus every earlier search step is defined without a nontorsion premise. At the bad place $s \equiv 3 \pmod 7$, the point reduces to $(2, 6)$ on $y^2 = x^3$ and is nonsingular because $2y \not\equiv 0$. The other place $s \equiv 4 \pmod 7$ has good reduction. Take the actual order of the reduced point in the nonsingular reduction group, using [60, VII.2.1, p. 188]. At a good place the residue degree is at most two, so Hasse's bound [60, V.1.1, p. 138] gives order at most $(p+1)^2 < p^3$ for odd p. The singular smooth group over $\mathbb{F}_7$ has order seven. There are at most two places over each rational prime, so $m_0 \leq N_s^6$. If $[N_s^4 m_0]P_0$ is the identity, the search has already succeeded. Otherwise its reciprocal coordinate is defined and the following estimates apply. At each odd place, $[m_0]P_0$ is formal, and multiplication by N_s^4 gives

$$v(1/x([N_s^4 m_0]P_0)) \geq 2 + 8v_p(N_s).$$

At two normalize $v_s(s) = 1$, so $v_s(2) = 2$. Initially $v_s(x) = -6$, $v_s(y) = -9$, and $v_s(z) = 3 > 2$. The log isometry therefore gives $v_s(z([2]Q)) = v_s(z(Q)) + 2$ on this subgroup, and hence

$$v_s(1/x([N_s^4 m_0]P_0)) \geq 6 + 16v_2(N_s).$$

Writing this reciprocal as $a + bs$, extraction of a and b uses division by 2 and $2s$ and loses at most three units of v_s. Thus both rational coefficients lie in $N_s^4 \mathbb{Z}_{(N_s)}$, where denominators in $\mathbb{Z}_{(N_s)}$ are prime to N_s. This proves the required divisibility and guard. Therefore the literal search succeeds by $N_s^4 m_0$ and $M_0 \leq N_s^{10}$. This uses actual finite orders, not numerical upper bounds incorrectly treated as multiples of those orders.

Consequently $c_* < 2^{2^{2059}}$, whereas $2^{2^{8008}} < D < 2^{2^{9010}}$, so $R = D$. Next $C_{\mathrm{cm}} < 2^{50000D^2}$ and

$$\log_2 Q_D^{17} = 170 G_0^2 D^3 \log_2 D > 50000 D^2.$$

The maximum is therefore Q_D^{17}, giving the identity for T. Finally $G_0 < 2^{37}$, $E_0 < D^3$, $\log_2 Q_D < D^5$, $\log_2 A_{\mathrm{ht}} < D^7$, and $\log_2 T < D^8 < 2^{2^{9013}}$. □

12.1 Quantitative syntax bounds

Use coefficient ℓ^1 norm for integer polynomials and the weighted pair norm $\|a+bs\| = \|a\|_1 + 2\|b\|_1$. They are submultiplicative. The division-polynomial recurrences give

$$\deg f_n \le (n+1)^3, \qquad \|f_n\| \le 2^{100(n+1)^3}.$$

For the even recurrence the exponent slack is $4m^3 - 24m - 27 \ge 9$ for $m \ge 3$; for the odd recurrence it is at least ten for $m \ge 2$. This absorbs the fixed factors and addition.

The following facts have been checked for the actual saved expression: the absolute values of all literal fixed constants and all occurring division-polynomial indices are at most T; the specified syntax-height traversal has root height at most 256; each model recipe has at most 2^{20} constraints and each composition recipe has at most 2^{20} terms. The check accounts for all 23 templates, seven recipes, private witness blocks, 173 scalar constant nodes and 50 finite constructors. Its exact artifact identity and independent replay are recorded in the source map. Together with the constructor bounds below, this discharges the quantitative part of B. The all-input arithmetic interpretation of the same constituents is the separate substitution induction of Section 10.

Put $B_* = T + 1$. Coordinates of $\iota(n)$ for $|n| \le T$ have absolute numerator and positive denominator at most 2^{K_*}, with $K_* = 2^{30}(T+1)^3$. Evaluate the division polynomials at P_0 and use the fixed coordinate formulas; at most eight quadratic-field operations are needed, each taking an available height bound H to at most $3H^4$. There are at most $8(T+1)^3$ guard factors. The resulting bounds for each finite constructor are

$$\deg \le B_*^4, \qquad \log_2 \max(1, \|\cdot\|_1) \le B_*^8.$$

Give variables syntax height zero, constants height one, and add one for each arithmetic operation or finite constructor. A template call adds one plus the callee's intrinsic height and the maximum argument height. The substitution inequalities

$$\deg P(Q) \le \deg P \max(1, \max_i \deg Q_i), \qquad \|P(Q)\|_1 \le \|P\|_1 \max(1, \max_i \|Q_i\|_1)^{\deg P}$$

give degree and logarithmic norm at most B_*^{8h} at positive height h, with the bounded sums in the actual recipes treated as in S26. This general call rule applies to polynomial arguments. In each outer composition recipe the arguments are coordinate variables, so the call is only a renaming, even when coordinates repeat. For any such map σ,

$$\deg P(Y_{\sigma(i)}) \le \deg P, \qquad \|P(Y_{\sigma(i)})\|_1 = \sum_\beta \left| \sum_{\alpha \mapsto \beta} c_\alpha \right| \le \sum_\alpha |c_\alpha| = \|P\|_1.$$

Thus these calls retain the callee's majorant height before charging the sum-of-squares or product connective. The seven recipe heights are $228, 233, 235, 251, 252, 254, 256$. A rule charging an extra unit for each of these mere renamings would describe a looser syntax-depth bound, not this proved degree-and-norm majorant. For the model wrappers use the safe height bound four for the generator equations and root differences; the callee contribution dominates this bound. The literal height-two claim for those equations in S26 is not used. This height is a majorant for the compact constructor syntax, not for an expanded monomial tree. The height-256 bound yields

$$\deg P_* \le (T+1)^{2048}, \qquad |[X^\alpha]P_*| \le 2^{(T+1)^{2048}}.$$

Since T is an integer, $T + 1 \le U = 2^{2^{2^{9013}}}$. Hence

$$(T+1)^{2048} \le U^{2048} = 2^{2^{2^{9013}+11}} < 2^{2^{2^{9014}}}.$$

As $9014 < 65536 = 2 \uparrow\uparrow 4$, the degree is less than $2 \uparrow\uparrow 7$ and the coefficient magnitude less than $2 \uparrow\uparrow 8$. The root recipe declares 292 890 231 904 witnesses. Together with the preceding estimates, these complete the bounds for the same P_* in Theorem 1.3.

13 Consequences of the integer definition

The proof of Theorem 1.3 combines the following results.

Result	Mathematical inputs	Proof location
Index group	Two-isogeny descent, reductions, finite height search	Section 3
Six-contact bound	Quaternionic model, tame pullback, compatible-isogeny height theorem, theta comparison	Section 4
Pole completeness	Simultaneous descent, restricted converse, forward rank-one theorem	Section 5
Multiplication	Index group, height bound, selected primes, pole completeness, finite guard	Section 6
Precision relation	Decision program, lowering, power and packed histories	Section 10
Ordinary value	Shared-shift Close and Test, height and rational separation	Section 11
Bounded P_*	Literal substitutions and quantitative syntax	Sections 2, 12; Appendix A3

13.1 Three corollaries

We apply the exact equality between the rational zero projection of P_* and ordinary $\mathbb{Z}$ proved in Theorem 1.3.

Corollary 13.1 (Negation of Mazur's finite-component conjecture)**.** *Mazur's finite-connected-components conjecture for the real-topological closure of rational points, Conjecture 3 of [86, 92], is false. Its stronger Conjecture 2 is false as well.*

Proof. We recall Mazur's projection argument, originating in [86] and repeated in [92, p. 166]. Conjecture 3 in [92, p. 165] asserts that the closure of the rational points of any variety over $\mathbb{Q}$ in its real locus has finitely many connected components. In that source a variety means a reduced scheme of finite type; irreducibility is not required. Put $r = 292\,890\,231\,904$ and

$$V = \left(\{P_*(t, w_1, \ldots, w_r) = 0\} \subset \mathbf{A}^{r+1}_{\mathbb{Q}}\right)_{\mathrm{red}}, \qquad C = \overline{V(\mathbb{Q})} \subset V(\mathbf{R}).$$

Reduction changes neither the rational nor the real points. Let $\pi : V(\mathbf{R}) \to \mathbf{R}$ be projection to t. Theorem 1.3 gives $\pi(V(\mathbb{Q})) = \mathbb{Z}$. Since $\mathbb{Z}$ is closed in $\mathbf{R}$ and π is continuous, $C \subset \pi^{-1}(\mathbb{Z})$, so $\pi(C) \subset \mathbb{Z}$. Every integer already occurs on $V(\mathbb{Q})$, hence $\pi(C) = \mathbb{Z}$. The image of any connected component of C is a connected subset of $\mathbb{Z}$, therefore a singleton. Points projecting to distinct integers belong to distinct components. Thus C has infinitely many connected components, contradicting the stated conjecture. Conjecture 2 in the same source would give the homotopy type of a finite simplicial complex and hence only finitely many connected components, so it implies Conjecture 3 and is contradicted too. □

Corollary 13.2 (A universal rational Diophantine polynomial)**.** *There exist $m \geq 1$ and a fixed polynomial $U \in \mathbb{Z}[e, n, x_1, \ldots, x_m]$ such that, for every $e, n \in \mathbb{N}$,*

$$\exists \mathbf{x} \in \mathbb{Q}^m \; U(e, n, \mathbf{x}) = 0 \quad \Longleftrightarrow \quad \textit{machine } e \textit{ halts on input } n.$$

Here e and n are ordinary integer parameters, with $\mathbb{N}$ embedded in $\mathbb{Q}$, and are not elliptic-curve coordinates. In particular, every computably enumerable subset of $\mathbb{N}$ is represented by a suitable fixed value of e in this one polynomial.

Proof. Fix an effective enumeration of machines. The Davis–Putnam–Robinson–Matiyasevich theorem, [93, Basic Theorem, p. 3], applied to the universal halting relation gives a fixed $D \in \mathbb{Z}[e, n, z_1, \ldots, z_s]$, with $s \geq 1$, such that, for $e, n \in \mathbb{N}$,

$$\text{machine } e \text{ halts on } n \quad \Longleftrightarrow \quad \exists \mathbf{z} \in \mathbb{Z}^s \; D(e, n, \mathbf{z}) = 0.$$

To use integer witnesses in this statement, replace each nonnegative witness by a sum of four integer squares. In the positive-integer convention of the cited source, apply the theorem to the predicate that machine $a - 1$ halts on $b - 1$, substitute $a = e + 1$, $b = n + 1$, and use one plus that sum for each witness; see also [93, p. 2]. A dummy witness ensures $s \geq 1$ if needed. For each z_i take a separate block $\mathbf{w}_i$ of r rational variables, where r is the witness count of P_*. Define

$$U(e, n, \mathbf{z}, \mathbf{w}_1, \ldots, \mathbf{w}_s) = D(e, n, \mathbf{z})^2 + \sum_{i=1}^{s} P_*(z_i, \mathbf{w}_i)^2.$$

A rational zero makes every summand zero. Theorem 1.3 then forces every z_i to be an integer and yields the displayed halting condition. Conversely, integer witnesses for D have separate rational witness blocks for P_*, giving a zero of U. All blocks are private and the polynomial is fixed independently of e, n. No variable, degree or coefficient bound proved for P_* is asserted for this new polynomial U. □

OpenAI already obtained Turing degree $0'$ for this set in [94, Corollary 7.1, pp. 56–57]; that paper explicitly does not assert many-one completeness. The stronger direction below is $0' \leq_m \mathrm{HTP}(\mathbb{Q})$, obtained from the integer definition through the universal polynomial. The reverse many-one reduction is the standard enumeration of rational solutions.

Corollary 13.3 (The many-one degree of $\mathrm{HTP}(\mathbb{Q})$)**.**

$$\mathrm{HTP}(\mathbb{Q}) \equiv_m 0', \qquad 0' = \{e \in \mathbb{N} : \textit{machine } e \textit{ halts on input } e\}.$$

Here $\mathrm{HTP}(\mathbb{Q})$ *is the set of codes of integer-coefficient polynomials, of arbitrary finite arity, having a rational zero, under a fixed effective coding.*

Proof. Given e, specialize the fixed polynomial of Corollary 13.2 at its ordinary integer parameters to obtain $U(e, e, \mathbf{x})$. Computing its coefficient list and code is effective, and

$$e \in 0' \quad \Longleftrightarrow \quad \ulcorner U(e, e, \mathbf{x}) \urcorner \in \mathrm{HTP}(\mathbb{Q}).$$

This proves $0' \leq_m \mathrm{HTP}(\mathbb{Q})$.

Conversely, from the code of a polynomial f construct a machine that ignores its input and enumerates all rational tuples of the appropriate arity. It evaluates f using exact rational arithmetic and halts as soon as it finds a zero. Its index $j(f)$ is computable from the finite code of f, and

$$f \in \mathrm{HTP}(\mathbb{Q}) \quad \Longleftrightarrow \quad j(f) \in 0',$$

where the left side uses the code of f. Thus $\mathrm{HTP}(\mathbb{Q}) \leq_m 0'$. Invalid polynomial codes, if present in the coding convention, are sent to a fixed nonhalting index. Using rational-coefficient polynomial codes gives the same result, since clearing their finitely many denominators is effective and preserves rational zeros. The two reductions prove the assertion. □

More generally, an integer Diophantine problem translates to a rational one by adjoining P_* for every integer witness, with private witness blocks. Natural witnesses can first be expressed as sums of four integer squares. Bounds for the resulting polynomial require their own composition calculation; the bounds for P_* do not automatically transfer.

For example, a machine that enumerates ZFC proofs and halts upon finding a contradiction has a rational Diophantine halting condition by Corollary 13.2. This equivalence does not assert that a contradiction exists, and no additional explicit polynomial is specified here.

14 The base residual compiler

This section records the base residual compiler used as a construction stage for the quartic in Section 15. Its degree-eight sum-of-squares encoding fixes the arithmetic interfaces and conventions; it is not a third final polynomial specification. The construction uses a joint comparison of four elliptic indices, a Lucas equation, and the five-contact height estimate proved below. The archived counts describe this base compiler before its reciprocal revision and quadratic lift.

Proposition 14.1 (Finite specification of the base compiler)**.** *The prescription below specifies a polynomial*

$$F_8 \in \mathbb{Z}[X, V_1, \ldots, V_{15671}]$$

of degree exactly eight, with exactly 129497 *nonzero monomials and maximum absolute coefficient*

$$685783949153203534233601 < 2^{80}.$$

These statements are finite identities about the displayed construction and its integer coefficient table.

Theorem 14.2 (Arithmetic interpretation)**.** *For every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (v_1, \ldots, v_{15671}) \in \mathbb{Q}^{15671}, \qquad F_8(x, v_1, \ldots, v_{15671}) = 0.$$

The required height estimate is proved below; Theorem 4.2 uses a different contact set. The seven-variable pole lemma below is proved from Theorem 5.3; it is not inferred from Theorem 5.2, which has different variables and a restricted completeness domain. The finite coefficient calculation does not establish the height bound. The supplied unpublished Freitag archive is the provenance of this construction and of the original coefficient table. It is not cited here as an arithmetic theorem. The published inputs already used in Sections 3 and 6 remain available, together with the quaternion trace theorem specified below.

14.1 The height input and the exact pole predicate

Put

$$S_* = \{p : p \leq 76176\}, \qquad \mathcal{D} = \{0, 1, 2401, -343/32\},$$

$$H_* = 2(4608^{165888} + 6^{100}) \left(\prod_{p \leq 76176} p \right)^{24} + 128.$$

For $a \in \mathbb{Q} \setminus \mathcal{D}$, let $M_5(a)$ be the product of the primes outside S_* at which the point $(a : 1)$ has positive odd contact order with one of $\infty, 0, 1, 2401, -343/32$. Explicitly, choose local homogeneous coordinates $(u : v)$ and subtract $\min(v_p(u), v_p(v))$ from the valuations of v and of $u - \delta v$ for $\delta \in \mathcal{D}$.

Theorem 14.3 (The five-contact height estimate)**.** *For every $a \in \mathbb{Q} \setminus \mathcal{D}$,*

$$h(a) \le H_* M_5(a)^{24}.$$

The displayed constant satisfies $\log_2 H_* < 4783227$.

Proof. Write $s = a$, $M(s) = M_5(a)$, $H = H_*$, $B_* = \prod_{p \le 76176} p$, and $\mathcal{C} = \{\infty, 0, 1, 2401, -343/32\}$. We prove the geometric comparison, the non-CM estimate through the modular level, and the CM estimate over the same fiber field. All heights use the original stable Faltings normalization.

Recovering the contact line from published quotients Let $B/\mathbb{Q}$ be the indefinite quaternion algebra of discriminant 210, $\mathcal{O}$ a maximal order, X its maximal-level Shimura curve, and W its Atkin–Lehner group. The standard genus formula gives

$$g(X) = 1 + \frac{\varphi(210)}{12} = 5,$$

because the elliptic-point terms vanish: 5 splits in $\mathbb{Q}(i)$ and 7 splits in $\mathbb{Q}(\sqrt{-3})$, so neither quadratic field embeds in B. The group W is $(\mathbb{Z}/2\mathbb{Z})^4$ with involutions w_m indexed by square classes of divisors of 210.

The published models of [100, Theorem 4.2 and Table 2, pp. 938–939] have the following coefficient rows (a, b, c, d, e) for $y^2 = ax^4 + bx^3 + cx^2 + dx + e$:

m	a	b	c	d	e
30	-43	-686	-2915	-1372	-172
42	-43	600	-986	-600	-43
70	-43	256	-130	-768	-387
105	-43	2	505	12	-1548

The cited theorem identifies each with $X/\langle w_m \rangle$. The genus formula and the Atkin–Lehner group law used here also appear in the same published paper, Proposition 5.2 and the paragraph following Proposition 5.3, pp. 940–941.

Here is a deduction of all the required branch information, rather than an additional assumption about it. Each of these four genus-one quotients has eight fixed points upstairs, by Riemann–Hurwitz. A stabilizer for the W action on a characteristic-zero smooth curve is cyclic, and therefore has order at most two. Each nonempty fixed set of an involution is a union of W-orbits of size eight. Riemann–Hurwitz for $X \to X/W$ now gives

$$8 = 16(2g(X/W) - 2) + \sum_{1 \ne w \in W} \#\,\mathrm{Fix}(w).$$

The four known fixed sets already contribute 32. Thus $g(X/W) = 0$, the total is 40, and there is exactly one further orbit of eight fixed points. Inertia generators around the branch points have product one. The four classes $30, 42, 70, 105$ are independent in W, and their product is the class 210. Hence there are exactly five branch points, each of index two, with distinct labels $30, 42, 70, 105, 210$. Since the involutions are defined over $\mathbb{Q}$ and each label occurs only once, each branch point is rational. Three of them trivialize the genus-zero quotient.

Normalize $b_{210} = \infty$, $b_{105} = 0$, $b_{70} = 1$, and write $b_{42} = \lambda$, $b_{30} = \mu$. On a genus-one quotient the residual group $(\mathbb{Z}/2)^3$ has translation subgroup of order four, necessarily the full two-torsion group after choosing an origin. Its quotient has the same Jacobian j-invariant: it is multiplication by two over the algebraic closure. The remaining double cover of X/W branches at the four points other than b_m. Therefore the four quartic invariants determine the four Legendre invariants of these four-point subsets.

For a quartic put

$$I = 12ae - 3bd + c^2, \quad J_q = 72ace + 9bcd - 27ad^2 - 27b^2e - 2c^3, \quad j = \frac{6912I^3}{4I^3 - J_q^2}.$$

The exact computation gives the Legendre parameters $1/2401, 32/375, 1/225, 5/1029$, in row order. Enumerating the six cross-ratio values of each of the first two parameters and testing the last two invariants leaves exactly

$$\lambda = 2401, \qquad \mu = -343/32.$$

The script `check_height.py` reproduces this rational computation. Every nonzero determinant of the five primitive contact forms has prime support in $\{2, 3, 5, 7\}$.

Integral geometry and the degree 576 field Choose $\mu_0 \in \mathcal{O}$ with $\mu_0^2 = -210$. The fixed-maximal-order statement of [29, Section 1, pp.591–592] supplies such an element in this chosen $\mathcal{O}$, and states that the associated involution preserves $\mathcal{O}$. Use the maximal moduli stack $\mathcal{X}$ for this order and involution. Yuan's maximal-level theorem [101, Section 2.1, pp. 230–231] makes it regular, proper and semistable over $\mathbb{Z}$, with a universal abelian surface, and smooth over $R_0 = \mathbb{Z}[1/210]$. Its generic stabilizer is $\{\pm 1\}$. We now construct the entire fine principal-level-three model over R_0. This construction uses the maximal-level statement only: an $\mathcal{O}$-linear level map, without an isomorphism requirement, would not define the required full marking.

The full marking at the ramified prime Over $\mathcal{X}_{R_0}$ consider the functor of triples

$$(A, i, \eta), \qquad \eta : (\mathcal{O}/3\mathcal{O})_S \xrightarrow{\sim} A[3],$$

where η is an isomorphism of finite etale group schemes respecting the left $\mathcal{O}$-action. Morphisms preserve i and η. Since 3 is invertible, $A[3]$ is finite etale of rank 3^4 [116, Theorem 8.2 and Remark 8.4, pp. 115–116]. At a geometric point, $V_3(A)$ is a faithful module of dimension four over $\mathbb{Q}_3$ for the division algebra B_3 of dimension four; it therefore has rank one over B_3. Every full left $\mathcal{O}_3$-stable lattice L in B_3 is principal. Indeed, choose $x \in L \setminus \{0\}$ with least valuation

$$w(x) = \tfrac{1}{2} \operatorname{ord}_3 \operatorname{nrd}(x).$$

Such a least value exists since the lattice is bounded and the value group is $\frac{1}{2}\mathbb{Z}$. For $y \in L$, one has $w(yx^{-1}) \geq 0$, hence $y \in \mathcal{O}_3 x$; stability proves the reverse inclusion. The valuation and maximal-order assertions are [98, 13.3.4 and 13.3.11, pp. 206–207]; the factor $1/2$ is retained. Consequently $T_3(A)/3$, and therefore $A[3]$, is the regular left $\mathcal{O}/3\mathcal{O}$-module. This argument uses the division algebra at 3, not a split matrix order.

Etale-locally on the base both group schemes are constant. Their isomorphisms form a finite etale scheme, and the equalities expressing $\mathcal{O}$-linearity on a finite integral basis select an open and closed subscheme. Descent gives the representable finite etale $\mathrm{Isom}_{\mathcal{O}}$ cover $\mathcal{Y} \to \mathcal{X}_{R_0}$. The preceding lattice calculation proves that every geometric fiber is nonempty. The radical of $\mathcal{O}_3$ has

residue field $\mathbb{F}_9$ and square $3\mathcal{O}_3$ up to a unit, so $\mathcal{O}/3\mathcal{O}$ has 81 elements, radical of size 9, and 72 units. The automorphism group of its regular left module is the unit group of the opposite ring. Thus this cover has degree 72 over the stack. It is proper and smooth over R_0 and carries the pullback of the universal abelian scheme.

Rigidity and projectivity of this same cover A geometric stabilizer is finite, since $\mathcal{X}$ is proper Deligne–Mumford. Its action on $T_3(A)$ is faithful [116, Lemma 12.2, p. 122]. A stabilizer preserving η acts in $1 + 3M_4(\mathbb{Z}_3)$, which has no nonidentity torsion. To see this, an element of prime order l would have the form $1 + 3^a M$, with $a \geq 1$ and $M \not\equiv 0 \pmod 3$. If $l \neq 3$, its lth power has nonzero first term $l3^a M$ modulo 3^{a+1}. If $l = 3$, its cube minus 1, divided by 3^{a+1}, is

$$M + 3^a M^2 + 3^{2a-1} M^3 \not\equiv 0 \pmod 3.$$

Both contradict the assumed order. All geometric stabilizers of $\mathcal{Y}$ are trivial. This also gives rigidity on arbitrary bases: its inertia is finite and unramified, the identity section is both open and closed, and its complement has empty geometric fibers. Hence the entire inertia is the identity section. The stack is therefore an algebraic space Y, by [28, Tag 03YR], and the same universal abelian scheme descends to it.

Here projectivity needs an argument; proper algebraic spaces need not be schemes. Each geometric fiber of Y/R_0 is a proper smooth algebraic-space curve, and is a scheme by [28, Tag 0ADD]. Every generic component maps finitely and nonconstantly to the genus-five coarse curve already identified above from [100]. Riemann–Hurwitz gives genus at least five on each such component. The algebraic-space Stein factorization, with the finite-etale component assertion [28, Theorem 76.36.5 and Lemma 76.36.9, Tag 0A18], gives a finite etale component base R_0'/R_0, over which the fibers are geometrically connected. Smooth proper genus is locally constant [28, Tag 0E83]. Every component of this finite etale base meets the generic fiber, so every geometric fiber component of Y has genus at least two. The line Ω^1_{Y/R_0} is consequently ample on every fiber. The fiberwise ampleness theorem for proper algebraic spaces [28, Tag 0D3A] makes it relatively ample near every base point. Tag 0D32 then makes Y a scheme with a relatively ample line. Finally Tag 0B45, implication (6) to (1), gives projectivity, since the affine base itself has an ample line bundle. This proves smooth proper projective representability of the actual full-marking functor over R_0.

The complete generic moduli problem It remains to identify the whole generic fiber, rather than an unspecified form of its complex curve. Let $V = B$ with its regular left B-action and the alternating form ψ of the selected positive polarization. It satisfies

$$\psi(bx, y) = \psi(x, b^* y), \qquad b^* = \mu_0^{-1} \bar{b} \mu_0.$$

The space of compatible alternating forms is one-dimensional over $\mathbb{Q}$. After splitting B, write $V = W_1 \otimes W_0$ with both factors of dimension two. The involution on $\mathrm{End}(W_1)$ is adjoint to a symmetric form, and a compatible alternating form is that symmetric form tensored with an alternating form on W_0. The latter space is one-dimensional. Its similitude group is the right-multiplication copy of $(B^{\mathrm{op}})^\times$; after splitting it is $\mathrm{GSp}(W_0)$, hence the connected type-C_1 case of [21, Example 8.6 and Proposition 8.7, p. 325].

We use the full Hodge-orbit condition of [21, Theorem 7.4 and its preceding definition, pp. 321–322]. The general implication printed in Proposition 8.14 and Theorem 8.17 requires the author's correction; the polarization condition alone does not suffice in general. For this exact datum the additional condition follows directly. Over $\mathbb{R}$, the centralizer of the left algebra acts on W_0 as

$\mathrm{GL}_2(\mathbb{R})$. An admissible complex structure is J on W_0 with $J^2 = -1$. For $v \neq 0$, (v, Jv) is a basis, and its matrix g satisfies $J = gJ_0g^{-1}$. Thus $1 \otimes g$ lies in the actual centralizer $G(\mathbb{R})$, commutes with the fixed left B-action, and conjugates the entire Hodge map $a + bi \mapsto 1 \otimes (a + bJ)$ to the fixed map. Both half-planes occur. Positive rational multiples refer here only to representatives that are actual polarizations on the same abelian variety; the rational comparison retains the full $\mathbb{Q}^\times$-class of the alternating form and allows either sign of similitude multiplier. No positive-multiplier subgroup replaces $G(\mathbb{R})$.

The rational homology of a surface in this moduli problem has rank one over B, so a B-linear identification with V exists. The one-dimensional calculation of forms makes it a similitude, and the preceding calculation verifies the missing Hodge-orbit condition for this identification. A basis b_i of B can be encoded by the tensors $\psi(x, b_i y)$, as in loc. cit., 8.16. The determinant condition is the corresponding trace condition. These checks give all the conditions of 7.4 without using the erroneous general implication from 8.17. The reflex field is $\mathbb{Q}$: the cocharacter class is the usual $(1, 0)$ class for this quaternionic inner form of GL_2. The standard involution identifies the opposite group with the conventional $B^\times$ description and preserves the order, center, normalizer and principal congruence subgroup, compatibly with the maximal-level datum.

An integral $\mathcal{O}$-linear Tate isomorphism is automatically a similitude, by the same one-dimensional calculation. At 3 its multiplier is a unit, since both integral forms are perfect; it is not required to be 1 modulo 3. Every isomorphism modulo 3 lifts, since $\mathcal{O}_3^\times \to (\mathcal{O}_3/3\mathcal{O}_3)^\times$ is surjective. Thus the full marking has stabilizer exactly $U(3)$, with all its multipliers and geometric components. Fixing one Weil-pairing value would give a different functor.

Integral lattices and the canonical rational model For a complex marked surface choose integral $\mathcal{O}$-linear Tate isomorphisms at every finite prime, matching η at 3. At split primes these exist by the matrix-unit decomposition of a stable lattice; at division primes the valuation argument above applies. Their $U(3)$-orbit is independent of the choices. Together with the rational polarization class and B-action this gives the quadruple in Milne's moduli description. Conversely the finite-adelic orbit specifies a full lattice in the rational Tate space. An isogenous abelian variety realizes this lattice: after multiplying by an integer, take the quotient by the corresponding finite torsion subgroup. Stability gives the integral $\mathcal{O}$-action, and the same orbit gives the marking at 3.

The compatible rational polarization class has a unique positive representative perfect on this lattice. Locally its dual-lattice discrepancy is scalar; a positive rational scalar cancels the finitely many valuations, and two such scalars have ratio a positive rational unit at every prime, hence ratio 1. Equivalently, the canonical principal-polarization assertion [29, p. 592] gives this polarization from the integral order action and fixed involution. That assertion precedes the prime-to-discriminant level restrictions in Buzzard's Section 2; we do not apply that section's level-representability theorem at 3. Restoring this positive representative does not rescale the marking or fix its unit multiplier. A rational quasi-isogeny carrying the entire integral Tate lattice to itself is an actual isomorphism, since any nontrivial finite kernel or cokernel is detected on some prime-power torsion lattice. These constructions are inverse.

The comparison is algebraic. On each arithmetic component use the natural integral lattice local system with its Hodge structures. An arithmetic stabilizer preserving the lattice has positive polarization multiplier and is a unit at every finite prime, so its multiplier is 1. The polarization is therefore integral and flat on that component. The family equivalence [21, Theorem 14.8, p. 352] gives the abelian scheme and its $\mathcal{O}$-action algebraically. The flat full marking is algebraic as well: its graph is a union of connected components in the finite etale isomorphism cover. The fine functor yields a morphism to $Y_{\mathbb{C}}$. The preceding object classification makes it bijective on geometric points,

hence an isomorphism of smooth complex curves. Indeed it has degree one in characteristic zero, and the local discrete valuation rings agree. This uses every component and commutes with forgetting level.

Finally the comparison commutes with every automorphism of $\mathbb{C}/\mathbb{Q}$. The order action, polarization, torsion subgroup quotients and markings are algebraic, and the adelic comparison is transported through torsion, not through a supposed Galois action on rational Betti homology. The type-A/C canonical model criterion in [21, pp. 354–355] therefore identifies $Y_{\mathbb{Q}}$ with the entire canonical $U(3)$ curve. Uniqueness and the Hecke-map statement, loc. cit., 13.6–13.7, identify its maximal-level forgetful map and canonical right normalizer action. The CM and reciprocity results underlying that published criterion remain its named inputs. The author's corrections to 8.14/8.17 and to proof intermediates in 5.22 and 11.2 are recorded in the full-cover source packet; no corrected general implication or Hausdorff argument is silently substituted here.

Thus Y is the required entire principal-level-three scheme with its universal surface. Its degree over the stack is 72, while over the generic coarse curve the signs $\{\pm 1\}$ identify opposite markings, giving degree 36. No field extension or extra level has entered this construction. At primes dividing 210, the height argument below continues to use the maximal stack and its Hodge line, not an asserted smooth fine model at those primes.

Work first over $R_0 = \mathbb{Z}[1/210]$. The effective level-change group has order 36, prime to every residue characteristic of R_0. Its generic action is free. For a finite tame group acting on a smooth relative curve, the fixed locus of an element is smooth locally along its fixed points; an isolated special fixed point consequently lifts. Thus no new fixed point occurs here, and $Y_{R_0} \to X_{R_0}$ is finite etale. The W action extends to the smooth stable model of X. Its stabilizers remain cyclic, of order at most two. Distinct involution fixed loci cannot intersect; each fixed locus is etale over the base. Consequently X_{R_0}/W is a smooth genus-zero curve with five disjoint branch sections of index two. The three rational sections already chosen identify it with $\mathbb{P}^1_{R_0}$. Their values are the five elements of $\mathcal{C}$.

Set $K = \widehat{\mathcal{O}}^\times$, $U = U(3)$, $Z = \mathbb{A}_f^\times$, and $N = N_{B(\mathbb{A}_f)^\times}(\widehat{\mathcal{O}})$. Normalization preserves $3\widehat{\mathcal{O}}$, so U is normal in N. The canonical adelic action is defined over the reflex field $\mathbb{Q}$ [102, 2.2.5 and 2.3.1, pp. 269–270]; this refers to the entire canonical model, not an individually selected complex component. Every central finite idele is a rational scalar times a unit idele. At three its scalar residue is 1 or -1, which a rational sign removes. Thus $Z \subset \mathbb{Q}^\times U$ acts trivially on Y.

At three the quaternion algebra is ramified. Its maximal order modulo three has 81 elements, radical of size 9, and hence 72 units. It follows that

$$1 \longrightarrow (\mathcal{O}/3\mathcal{O})^\times/\{\pm 1\} \longrightarrow \Gamma = N/(ZU) \longrightarrow W \longrightarrow 1, \qquad |\Gamma| = 36 \cdot 16 = 576.$$

Above an unbranched point the actions of W and of the effective level-change kernel are free. The fiber therefore is a torsor of size 576 for this constant group. The Galois permutations commute with its regular action, so belong to its regular centralizer, also of order 576. The normal splitting field L consequently satisfies $[L : \mathbb{Q}] \mid 576$: the Galois group is a subgroup of that regular centralizer. In particular $[L : \mathbb{Q}] \leq 576$.

All fiber surfaces and their three-torsion markings are defined over L. For two such surfaces, a homomorphism annihilating the three-torsion factors through [3]. Hence reduction embeds

$$\operatorname{Hom}(A, B)/3 \operatorname{Hom}(A, B) \hookrightarrow \operatorname{Hom}(A[3], B[3]).$$

The finite Galois image on the integral Hom lattice is trivial modulo three. The principal congruence subgroup modulo three in an integral general linear group is torsion-free, so this image is trivial. Thus every geometric homomorphism between fiber surfaces is defined over L.

At a prime outside 210, tame local coordinates for the quotient are $w = z^2$ up to a unit at a branch section, and etale elsewhere. A base point with even contact gives no ramification after an unramified extension. The universal abelian scheme extends at the resulting integral point of Y_{R_0}. Therefore

$$\operatorname{rad}(D_L N_{\mathrm{bad}}) \mid B_* M(s). \tag{R}$$

This assertion includes contact with infinity. It is not a statement obtained merely from the numerical prime support of pairwise determinants.

For the geometric assertion, use the rational PEL description of the same full-three curve. A finite-adelic normalizer move fixes the rational Hodge object and replaces its integral lattice by a commensurable one. One integer clears the finitely many local denominators; the finite torsion quotient in the full-cover construction then gives an isogeny $f : A \longrightarrow A_\sigma$ between the actual conjugate fiber surfaces. This complex isogeny is geometric over $\overline{\mathbb{Q}}$. Indeed, torsion points of both algebraic surfaces are algebraic, so f and each of its $\mathrm{Aut}(\mathbb{C}/\overline{\mathbb{Q}})$-conjugates agree on all source torsion. Torsion is Zariski dense and the target is separated, so these morphisms agree. The invariant graph descends by the affine-ideal argument: on affine charts its unique reduced Gröbner basis has invariant coefficients, hence coefficients in $\overline{\mathbb{Q}}$.

In the non-CM case the full geometric endomorphism algebra is B. More explicitly, the rank-one left-B Hodge calculation gives $B \otimes K_0^{\mathrm{op}}$, where $K_0 = B \cap \mathbb{R}[J]$ is either $\mathbb{Q}$ or an imaginary quadratic field. The latter is the CM alternative, so $K_0 = \mathbb{Q}$ here. The Hodge equivalence and the geometric descent just used identify this with $\mathrm{End}^0_{\mathbb{Q}}(A)$; transport through f gives the same conclusion for A_σ.

Let F be the chosen abstract quadratic field, with specified embeddings $i : F \hookrightarrow \mathrm{End}^0(A)$ and $i_\sigma : F \hookrightarrow \mathrm{End}^0(A_\sigma)$. Put $j(x) = f^{-1} i_\sigma(x) f$. Skolem–Noether, in the form of [105, Corollary 7.1.5, p. 97], gives $b \in B^\times$ such that $j(x) = b^{-1} i(x) b$ for every $x \in F$. Consequently $g = fb^{-1}$ satisfies $g i(x) = i_\sigma(x) g$. Choose a positive integer n clearing its denominators in $\mathrm{Hom}(A, A_\sigma) \otimes \mathbb{Q}$. Then $h = ng$ is an actual homomorphism with the same intertwining identity. Clearing the denominators of its rational inverse shows that h is an isogeny. The simultaneous Hom argument above places h, both rational F-actions, and their intertwining identity over the same L.

Thus each non-CM conjugate surface is of GL_2-type over L, and these surfaces admit pairwise F-linear isogenies over L. This assertion does not choose coherent isogenies or construct a compatible Galois system. The separate scalar normalizer maps below supply the B-compatible maps used in the paragraph "The scalar lift with controlled support"; the subsequent paragraph "The compatible modular representations" constructs the representations and checks their cross-prime traces. Those arguments, including their support conditions, are separate inputs. The bounded descent to a GL_2-type variety over $\mathbb{Q}$ is also a separate conclusion.

The metric comparison, with its exact additive budget Put $t = 32s$, $(a_1, a_2, a_3, a_4) = (0, 32, 76832, -343)$ and $P(t) = \prod_i (t - a_i)$. The sections

$$q_0 = (dt)^2 / P(t), \qquad q_1 = t q_0$$

are holomorphic bicanonical sections on X, with no common zero. At a finite branch point use $t - a_i = z^2$; at infinity use $t = z^{-2}$. At infinity q_0 vanishes to order two and q_1 does not vanish.

For each prime let $\delta_p = \max_{i<j} v_p(a_i - a_j)$, so

$$J = \prod_p p^{\delta_p} = 2^{10} 3^2 5^3 7^4 = 2765952000.$$

Both J^2q_i extend across the maximal regular stack. Indeed, at the generic point of a vertical component semistability makes p a uniformizer, and the relative differential satisfies $v(du) \geq v(u)$. Writing $r_i = v(t - a_i)$ gives $v(dt) \geq \max r_i$. If $v(t) < 0$, all r_i equal $v(t)$ and both sections have nonnegative valuation. Otherwise all $r_i \geq 0$ and at most one exceeds δ_p. In either case $2\max r_i - \sum r_i \geq -2\delta_p$. This proves the required vertical integrality; regularity handles codimension two. This extension takes place on the maximal stack itself. It uses no integral coarse-model pullback from Yuan's Lemma 2.1 or the coarse arithmetic-intersection formula of his Theorem 2.2.

For the archimedean estimate write $r_i = |t - a_i|$, $U_0 = 1 + \sum_i r_i$ and

$$ds_\rho^2 = \frac{U_0}{\prod_i r_i}|dt|^2, \qquad C_{\mathrm{met}} = 16(12 \cdot 76832 + 9)^3 = 12540153541944538512.$$

Its pullback is smooth and positive at every branch point and infinity. The curvature formula, $\Delta U_0 = \sum_i r_i^{-1}$ and $|\nabla U_0|^2 \leq 16$ give

$$-K_\rho \geq \frac{\sum_i \prod_{j \neq i} r_j}{2U_0^3}.$$

When $|t| \leq 2 \cdot 76832 + 2$, at most one $r_i < 1/2$, and $U_0 \leq 12 \cdot 76832 + 9$. The displayed bound is at least C_{met}^{-1}. Outside this disk, $|t|/2 \leq r_i \leq 3|t|/2$ and $U_0 \leq 7|t|$, giving the stronger bound $1/1372$. The maximum principle against the curvature -1 metric implies $ds_\rho^2/C_{\mathrm{met}} \leq ds_{\mathrm{hyp}}^2$. Yuan's Petersson norm is $\|d\tau\| = 2\,\mathrm{Im}\,\tau$. Consequently

$$\|q_0\|_{\mathrm{Pet}} \leq 4C_{\mathrm{met}}/U_0, \qquad \|q_1\|_{\mathrm{Pet}} \leq 4C_{\mathrm{met}}|t|/U_0 \leq 4C_{\mathrm{met}}.$$

Yuan's isometry [101, Theorem 1.1, p. 229, and metric definitions, pp. 228–229] identifies the square of the Hodge bundle with $210\omega_{\mathcal{X}}^{\otimes 2}$ with the induced metric. The Hodge height in his normalization equals the original stable Faltings height plus $\log \pi$ in dimension two. Thus

$$h_{\bar{\omega}^{\otimes 2}}(y) = 2h_F(A_y) + 2\log \pi + \log 210.$$

The sign of $\log 210$ is positive: the sublattice $210\omega^{\otimes 2}$ has height $h_{\bar{\omega}^{\otimes 2}} - \log 210$. Evaluate on an extension where the point extends to the proper maximal stack; its universal abelian scheme computes stable height. The integral sections and the supremum estimate yield

$$h(s) \leq 2h_F(A_y) + 2\log \pi + \log 210 + 2\log J + \log(4C_{\mathrm{met}}) + \log 32.$$

The integer comparison $210J^2(4C_{\mathrm{met}})32 \cdot 16 < 2^{160}$, together with $\pi < 4$ and $\log 2 < 3/4$, bounds the additive term by $120 < 128$. Therefore

$$h(s) \leq 2\max(0, h_F(A_y)) + 128. \tag{G}$$

A level bound for the non-CM fibers This argument allows the scalar character order and coefficient field to vary. It does not assert bounded same-field descent to a rational GL_2-type variety. Let s be a rational point outside the five contacts, let $L/\mathbb{Q}$ be the normal fiber field constructed above, and let A/L be a non-CM fiber surface. Thus $[L : \mathbb{Q}] \mid 576$, all geometric homomorphisms between conjugate fibers are defined over L, and $\mathrm{End}^0(A_{\overline{\mathbb{Q}}}) = B$, the quaternion division algebra of discriminant 210. Put

$$R = B_* M(s), \qquad T = \{p : p \mid R\}.$$

The geometric argument gives that L is unramified outside T and A has good reduction outside the primes above 210. All conjugate QM structures lie in the same normalizer orbit.

Good reduction after full torsion. First a QM surface with the indicated division action has potentially good reduction at every finite place. Indeed, pass to an extension with semistable reduction, using [42, Theorem 3.4, p. 406]. Every endomorphism extends uniquely to the Néron model by its mapping property. On the identity component of the geometric special fiber it preserves the maximal torus $\mathcal{T}$: this torus is the maximal connected affine subgroup, and every homomorphism from a torus to an abelian variety is zero. If its rank t is positive, restriction gives a unital homomorphism

$$B \longrightarrow \mathrm{End}^0(\mathcal{T}) = M_t(\mathbb{Q}), \qquad 1 \le t \le 2.$$

The kernel is a two-sided ideal of the simple algebra B, and the map is unital, so it is injective. For $t = 1$ dimensions prohibit this; for $t = 2$ it would be an isomorphism from a division algebra to $M_2(\mathbb{Q})$, also impossible. Hence $t = 0$. The semistable identity component is abelian, which is good reduction. These Néron-model and reduction conventions are those of [42, §3, pp. 405–406].

Set $E = L(A[11])$. We next justify that $E/\mathbb{Q}$ is Galois. For each σ, the normalizer relation supplies a compatible isogeny ${}^{\sigma}A \to A$ of degree prime to 11. Here is the prime-to-11 detail. In the analytic description $A_z = \mathbb{C}^2/(\mathcal{O}(z,1)^t)$ of the published 2021 edition of Voight [105, equations (43.1.2)–(43.1.3)], a positive normalizer element $b = \left(\begin{smallmatrix} a & b' \\ c & d \end{smallmatrix}\right)$ gives the scalar compatible map $(cz+d)I : A_{bz} \to A_z$. When $\mathcal{O}b \subset \mathcal{O}$, its degree is $[\mathcal{O} : \mathcal{O}b] = \mathrm{Nrd}(b)^2$. At 11 the algebra is split and the maximal-order normalizer is $\mathbb{Q}_{11}^* \,\mathrm{GL}_2(\mathbb{Z}_{11})$. Rescale b rationally to make it an 11-adic unit, then clear the remaining denominators by an integer prime to 11. This produces the claimed isogeny. Changes of level marking use isomorphisms. All these geometric homomorphisms are L-defined by construction. Consequently the 11-torsion representations of A and its conjugates over L have the same kernel. Thus every conjugate of E equals E.

The faithful action on $A[11]$ now gives

$$[E : \mathbb{Q}] \le 576 |\,\mathrm{GL}_4(\mathbb{F}_{11})| < 2^{10}11^{16} < 2^{66}. \tag{LR1}$$

For residue characteristic different from 11, full rational 11-torsion gives semistable reduction over E by [42, Theorem 3.5, p. 406]. The preceding torus argument makes this good reduction. In particular it applies at each of $2, 3, 5, 7$.

The scalar lift with controlled support. Choose the isogenies $\mu_\sigma : {}^{\sigma}A \to A$ over L from the normalizer construction above, retaining their full B-linearity. This can be checked on every component in the rational PEL description of Section 14.1. For n normalizing $\widehat{\mathcal{O}}$, the points (h, a) and (h, an) have the same rational left B-module V and Hodge structure, with commensurable lattices Λ and Λ_n. The identity on V commutes with B. At 11 the normalizer has the form $n_{11} = qu$, with $q \in \mathbb{Q}_{11}^{\times}$ and $u \in \mathcal{O}_{11}^{\times}$, so $\Lambda_{n,11} = q\Lambda_{11}$. Multiplication by $11^{-v_{11}(q)}$, followed by an integer prime to 11 clearing the remaining denominators, gives $r \in \mathbb{Q}^*$ with $r\Lambda_n \subset \Lambda$ and equality at 11. The Hodge-lattice equivalence turns $r\,\mathrm{id}_V$ into a B-linear isogeny from the translated surface to the original one, of degree prime to 11. The accepted integral-lattice identification retains the actual $\mathcal{O}$-labels; changes of the full-three marking are $\mathcal{O}$-linear isomorphisms. The two-surface Hom argument places these maps over the same L. In the analytic chart of [105, equations (43.1.2)–(43.1.3)], the action on every A_z is $i_z = i_\infty$. The displayed map $\lambda I : A_{bz} \to A_z$, with $\lambda = cz + d$, therefore satisfies $\lambda I\, i_{bz}(x) = i_z(x)\lambda I$ for $x \in B$; $\lambda(bz, 1)^t = b(z, 1)^t$ verifies its lattice direction. Take $\mu_1 = 1$ and inflate this finite choice to $G_{\mathbb{Q}}$. Their discrepancies $c(\sigma, \tau)$ commute with B, so lie in $\mathbb{Q}^*$. Put $a_\sigma = (\deg \mu_\sigma)^{-1/4} > 0$ and $\nu_\sigma = a_\sigma \mu_\sigma$, with trivial Galois action on the added algebraic coefficients. The rational degree used above satisfies $D(cu) = c^4 D(u)$ for $c \in \mathbb{Q}^*$. Its multiplicativity and the positive fourth-root convention give normalized discrepancy $c(\sigma, \tau)/|c(\sigma, \tau)|$. Thus the discrepancies of the ν_σ form a sign cocycle $\varepsilon \in Z^2(G_{\mathbb{Q}}, \{\pm 1\})$, inflated from $\mathrm{Gal}(L/\mathbb{Q})$.

The sign class is unramified, hence zero, locally outside T. At each odd prime where its Brauer invariant is $1/2$, choose an odd Dirichlet character modulo p; at 2 use the odd character modulo 4 if required. Their product χ is tame at every odd prime, has local conductor at most 4 at 2, and has order

$$m \le 2 \prod_{\substack{p \in T \\ p \neq 2}} (p-1) \le 2R.$$

For constant coefficient groups, the boundary in $1 \to \{\pm 1\} \to \mu_\infty \xrightarrow{2} \mu_\infty \to 1$ is the obstruction to a character square root. Local reciprocity identifies this obstruction with the value at -1: the cyclic torsion factor has precisely this obstruction, while the valuation and principal-unit factors admit character square roots. Thus the boundary of χ equals $[\varepsilon]$ at every finite place. Brauer reciprocity and the global-to-local injection give equality globally. A pointwise square root, corrected by a sign cochain, gives

$$b : G_{\mathbb{Q}} \longrightarrow \mu_{2m}, \qquad b^2 = \chi, \qquad \delta b = \varepsilon. \tag{LR2}$$

Only this *sign* cochain is asserted to take root-of-unity values; the full scalar splitting also involves the a_σ.

On G_L, b is a conjugation-invariant character. Outside T its inertia characters are quadratic. Only finitely many are nontrivial; twist by the quadratic character of the product of those odd primes to kill them. At each odd prime there is a unique nontrivial quadratic inertia character. This changes no square or coboundary in (LR2), and introduces possible ramification only at $2 \in T$. The kernel field L' of $b|G_L$ is normal over $\mathbb{Q}$ and

$$[L' : \mathbb{Q}] \le 2m[L : \mathbb{Q}] \le 2304R. \tag{LR3}$$

The compatible modular representations. Choose a number field containing all normalization scalars, the values of b, and a splitting field of B. For each coefficient prime ℓ and an embedding into $\overline{\mathbb{Q}}_\ell$, set

$$\rho_\ell(\sigma) = b(\sigma)^{-1} a_\sigma \mu_\sigma \circ \sigma.$$

The cocycle identity makes this a representation commuting with B; after splitting B, matrix units give $\rho_\ell = r_\ell \oplus r_\ell$ with $\dim r_\ell = 2$. On $G_{L'}$, ρ_ℓ is the Tate action of A. Every geometric endomorphism is defined over $L \subset L'$, and $\mathrm{End}^0(A_{\overline{\mathbb{Q}}}) = B$. Faltings' semisimplicity and endomorphism theorems therefore apply to the actual Tate action on $G_{L'}$. After splitting B, its commutant is M_2; the original matrix-unit factor has scalar commutant and is semisimple. The same calculation after every finite coefficient extension proves that this original r_ℓ is absolutely irreducible for every ℓ. The representation is unramified outside $T \cup \{\ell\}$, by (LR2) and the reduction and field properties. Apply Lemma 4.1 with $t_\sigma = a_\sigma / b(\sigma)$. It gives a primitive weight-two newform f whose Tate realization is exactly the original r_5. Write N for the level of this final form, after the finite character in that lemma has been absorbed. The finite character δ absorbed in the lemma is not identified with the controlled character $\chi = b^2$; the original L', b and r_ℓ remain those in (LR2)–(LR3) and the estimates below. The comparison at every coefficient prime uses the original family, not a replacement of the controlled field or its ramification set. Choose an auxiliary finite Galois field K_{aux} defining the finitely many isogenies μ_σ, their twists, and the scalar choices. Choose a positive integer m clearing all quasi-isogeny denominators. After excluding a finite set S of rational primes, including those ramified in K_{aux}, these data spread to abelian schemes and homomorphisms $m\mu_\sigma$. The auxiliary field K_{aux} and comparison set S do not replace L' or T.

Let $p \notin S$ and choose a place of K_{aux} above p. An arithmetic Frobenius lift σ_p acts on the residue field by $x \mapsto x^p$. Reduction identifies ${}^{\sigma_p}A$ with the p-Frobenius twist of the reduced surface

A_v, so

$$e_p = \overline{m\mu_{\sigma_p}} \circ \mathrm{Fr}_{p,A_v} \in \mathrm{End}(A_v).$$

Here Fr_{p,A_v} is relative p-Frobenius. Replacing it by Frobenius for the full residue-field cardinality would give a different power of σ_p. The rational degree polynomial of an endomorphism is its characteristic polynomial on every prime-to-p rational Tate module, by [116, Proposition 12.9 and the preceding definition, p. 125]. Thus $q_p = \mathrm{Tr}(e_p)/m \in \mathbb{Q}$ is independent of the Tate prime. This division by m retains the original quasi-isogeny.

For $\ell \neq p$, reduction on ℓ-power torsion is compatible with these spread homomorphisms. Indeed, the torsion is finite étale and becomes a disjoint union of sections over the strict henselization. It therefore identifies $\mu_{\sigma_p} \circ \sigma_p$ on rational Tate modules with e_p/m. Since $\rho_\ell = r_\ell \oplus r_\ell$, this gives

$$\mathrm{Tr}\, r_\ell(\sigma_p) = \frac{1}{2}\, \iota_\ell(a_{\sigma_p}/b(\sigma_p))\, q_p.$$

The factor $1/2$ comes from the two equal summands of the full Tate representation.

Fix a coefficient prime ℓ. Choose a number field F_{coeff} containing the original scalar and splitting field and the coefficients of the primitive form f. Choose an embedding into $\overline{\mathbb{Q}}_5$ extending the fixed one used to identify the original r_5 with the realization of f, and an embedding into $\overline{\mathbb{Q}}_\ell$ extending the fixed embedding for the original r_ℓ. At ℓ, work over a common finite coefficient extension containing both coefficient images. Before either embedding,

$$\tau_p = \tfrac{1}{2}(a_{\sigma_p}/b(\sigma_p))q_p \in F_{\mathrm{coeff}}.$$

Outside the finite set also containing $\ell, 5$ and the level of f, the identification at five gives $\iota_5(\tau_p) = \iota_5(a_p(f))$. Injectivity gives equality in F_{coeff}, and the embedding at ℓ gives equality of the traces of the original r_ℓ and the corresponding classical Tate realization $U_{f,\ell}$.

Use the established absolute irreducibility of the original r_ℓ. Chebotarev and Brauer–Nesbitt identify r_ℓ with $U_{f,\ell}^{\mathrm{ss}}$ over the common coefficient extension. No independent semisimplicity assertion about $U_{f,\ell}$ is needed: its semisimplification is one simple two-dimensional factor, so $U_{f,\ell}$ has composition length one and is itself irreducible. Hence $r_\ell \simeq U_{f,\ell}$ as representations of the full $G_{\mathbb{Q}}$. The finite comparison exclusions neither restrict this isomorphism to a subgroup nor enlarge T.

For each rational prime p one may choose $\ell \neq p$ and restrict this isomorphism to its decomposition group. In particular, the same form f may still be compared with r_3 at the prime 5 and with r_5 at the other places. The comparison asserts no new numerical conductor bound. The modular level is the conductor of these realizations by the published result recalled below. Li's Fricke relation and Shimura's Jacobian construction, as in the baseline modularity argument, also place r_5 in the Tate module of $J_1(N)$.

The primitive level equals the local conductor. Let $f \in S_2(\Gamma_0(N), \eta)$ be the primitive newform obtained after absorbing the finite character twist, and let $U_{f,\ell}$ be its two-dimensional Tate realization for the chosen coefficient embedding. Here η is the nebentypus of f; it is not identified with the controlled character $\chi = b^2$ in (LR2). Carayol's published conductor result for normalized primitive newforms gives

$$a_p(U_{f,\ell}) = v_p(N), \qquad p \neq \ell. \tag{CL1}$$

Indeed, [120, §2.1, p. 36] permits arbitrary nebentypus and coefficient number fields, and weight two. Section 2.4 on p. 37 states the conductor assertion also at primes dividing N; the conclusion following the full Theorem 4 in §4.1 on p. 42 proves that assertion. We use the full theorem, not its

weaker version with an extraordinary-cuspidal exclusion. Thus the assertion includes dyadic primes and nonsquarefree level. The cohomological realization in that source is dual to the Tate realization; duality preserves the Artin conductor. The unramified norm twist in its Hecke-correspondence convention likewise changes no conductor exponent.

For each rational prime p, choose an auxiliary prime $\ell \neq p$. The preceding compatible-family comparison identifies the actual r_ℓ constructed from A, a_σ, and b with $U_{f,\ell}$ for this same final newform and coefficient embedding. Restriction to $G_{\mathbb{Q}_p}$ preserves this isomorphism, hence preserves inertia-invariant dimensions and the Swan conductor. Consequently

$$a_p(r_\ell) = a_p(U_{f,\ell}) = v_p(N), \qquad p \neq \ell. \tag{CL2}$$

The conductor includes the tame term and Swan term, or equivalently the full Weil–Deligne conductor with its monodromy contribution. No semisimplification on inertia is taken. Applying (CL2) with an auxiliary prime different from p controls every prime divisor of N; no local compatibility theorem at $p = \ell$ is required.

The conductor bound. Let $F = E\mathbb{Q}(\chi)$, where $\mathbb{Q}(\chi)$ means the kernel field of the character, not its value field. On G_F, b is a conjugation-invariant quadratic character. Its kernel gives a Galois extension $F'/\mathbb{Q}$ of degree at most two over F. For $p = 2, 3, 5, 7$, let $e_E, e_{F'}$ denote local ramification indices over $\mathbb{Q}_p$. From (LR1), $v_p(e_E) \leq 65$. At odd p both added extensions are tame, whereas at 2 each adds ramification index at most two. Consequently

$$v_p(e_{F'}) \leq 65 \ (p = 3, 5, 7), \qquad v_2(e_{F'}) \leq 67. \tag{LR4}$$

The possibly unbounded tame ramification index causes no problem. For a finite Galois extension of $\mathbb{Q}_p$ with ramification index e and different exponent δ, an Eisenstein polynomial over the maximal unramified subfield gives

$$\delta \leq e - 1 + ev_p(e).$$

Indeed the derivative terms have distinct valuations modulo e, and its leading derivative term has the stated valuation. In upper numbering the different formula is

$$\delta/e = 1 - 1/e + \int_0^\infty (1 - |G^u|^{-1})\, du.$$

Before the last positive break the integrand is at least $1/2$. Thus (LR4) bounds this break for $F'/\mathbb{Q}_p$ by 136. These ramification and conductor conventions are those of [43, §§4.1–4.3, pp. 155–159]. Over F' the scalar correction is trivial and A has good reduction. Above that break, therefore, the two-dimensional representation is trivial on upper wild groups. Its Swan conductor is at most 272, and its tame conductor is at most two. Hence

$$a_p(r) \leq 274 < 300 \qquad (p = 2, 3, 5, 7).$$

For $p > 7$, $L/\mathbb{Q}$ is tame because its degree divides 576, A/L has good reduction, and b is trivial on wild inertia: on that pro-p group its square is tame and its remaining image has order at most two. Thus $a_p(r) \leq 2$ there; it is zero outside T. Using the coefficient-prime comparison above proves

$$N \leq 210^{300} R^2. \tag{LR5}$$

The original constant is preserved. Apply the accompanying modular-Jacobian transfer calculation with $C = 210^{300}$ and (LR3)–(LR5). It gives

$$h_F(A) < 2^{24}210^{3600}R^{24} < 2^{28824}R^{24} < 4608^{165888}R^{24}. \tag{LR6}$$

The last comparison uses $210 < 2^8$, $4608 > 2^{12}$, and $28824 < 1990656 = 12 \cdot 165888$. Together with the separately proved CM bound and geometric height comparison, (LR6) yields exactly

$$h(s) \le (2(4608^{165888} + 6^{100})B_*^{24} + 128)M(s)^{24}.$$

No contact, coefficient constant, or radical exponent has been changed.

Transfer from the level, allowing unbounded coefficient degree. Let A/L be the non-CM quaternionic surface. Write $R \ge 1$ for the squarefree product of the prescribed rational bad primes, including 2, and assume $[L : \mathbb{Q}] \le 576$. The modularity construction provides a weight-two newform of level N and a field L' over which the Tate modules of A and $J = J_1(N)$ share an irreducible constituent. In the notation of the sign-splitting construction, the auxiliary Dirichlet character has order

$$m \le 2 \prod_{\substack{p \mid R \\ p \ne 2}} (p-1) \le 2R.$$

The sign-splitting cochain has values in μ_{2m}. If one instead uses the full scalar splitting map, it also includes the square root of the degree map; that factor equals one on G_L. Thus in either notation the restriction to G_L is a character valued in μ_{2m}, and its kernel field satisfies

$$d' = [L' : \mathbb{Q}] \le 2m[L : \mathbb{Q}] \le 2304R. \tag{LT1}$$

The quadratic twist used to remove extra ramification does not increase this bound, since 2 divides $2m$. There is no bound here on m independent of R.

Faltings' Hom theorem, followed by scalar extension in the Hom space, gives a nonzero L'-homomorphism $A \to J$. Absolute simplicity of A and Poincaré reducibility therefore give an L'-isogeny $A \times Z \to J$ for an abelian variety Z/L'. Put $g = \dim J$; then $g \ge 2$. The modular genus estimate [104, (5.9), p. 13] gives $g \le N^2/24$. The standard Belyi map has degree at most N^2 and [38, Theorem 1.1.1 and §2.3] gives

$$h_F(J) \le 13000000N^{12}. \tag{LT2}$$

All heights in this argument use the original stable Faltings metric.

Suppose the companion local argument has proved $N \le CR^2$, where $C \ge 1$. Then

$$g \le C^2R^4/24, \qquad h_F(J) \le 13000000C^{12}R^{24}.$$

Set $l = \log C$ and $r = \log R$. The published Gaudron–Rémond Theorem 1.9(1) [103, pp. 7–8], combined with half-logarithmic isogeny-height variation, gives

$$h_F(A) + h_F(Z) \le h_F(J) + V, \quad V = 2g^2\left(8g^2\log(7g) + \log d' + \log\max(1, \log d', h_F(J))\right). \tag{LT3}$$

Here the source variety for the isogeny estimate is $J_{L'}$. The three logarithms in (LT3) satisfy respectively

$$\log(7g) \le 2l + 4r, \quad \log d' < 8 + r, \quad \log\max(1, \log d', h_F(J)) < 17 + 12l + 24r.$$

For the last assertion, the bound in (LT2) also dominates 1 and $\log d'$; use $8 + r \le 9R^{24}$. It follows that

$$V \le \frac{C^8 R^{16}}{24^4}(32l + 64r) + \frac{C^4 R^8}{288}(25 + 12l + 25r). \tag{LT4}$$

Using $l \le C^4/4$, $r \le R^8/8$, and also $r \le R^{16}/16$ in the second summand gives

$$\frac{V}{C^{12}R^{24}} \le \frac{16}{24^4} + \frac{25 + 3 + 25/16}{288} < 1. \tag{LT5}$$

The Bost lower bound in [104, (2.2), p. 5] gives $h_F(Z) \ge -\frac{1}{2}\dim Z \log(2\pi^2) > -\frac{3}{2}g$. Consequently

$$h_F(A) < (13000000 + 1 + 1)C^{12}R^{24} < 2^{24}C^{12}R^{24}. \tag{LT6}$$

This calculation needs no uniform bound on the degree of the newform's coefficient field and no assertion that all its geometric factors are isogenous to A.

For the companion conductor estimate one may take $C = 210^{300}$. Since $210 < 2^8$ and $4608 > 2^{12}$,

$$2^{24}C^{12} = 2^{24}210^{3600} < 2^{28824} < 2^{1990656} < 4608^{165888}. \tag{LT7}$$

Thus the non-CM height inequality is strictly within the original coefficient budget, once the stated modularity and level estimate have been established. In the five-contact application $R = B_* M(s)$, so the radical exponent is exactly 24, as required.

The CM height estimate over the same field Let A/L be an abelian surface of CM type, let B be a quaternion division algebra over $\mathbb{Q}$ embedded in $\mathrm{End}^0_{\overline{L}}(A)$, and suppose that all geometric endomorphisms of A are defined over L. Assume

$$d = [L : \mathbb{Q}] \le 576, \qquad \mathrm{rad}(|D_L|) \le R, \qquad R \ge 1.$$

All heights below are stable Faltings heights in Faltings's original normalization. We prove

$$h_{\mathrm{F}}(A) < 854R < 6^{100}R^{24}. \tag{12}$$

No assertion about the existence or degree of L is included in this lemma.

First, $A_{\overline{L}}$ is isogenous to E^2 for a CM elliptic curve E. Here is a dimension argument which makes the alternative cases explicit. If A were geometrically simple, its division endomorphism algebra would act faithfully on the four-dimensional rational homology. Its dimension therefore divides four. As it contains B, it would equal B; but B has no commutative semisimple subalgebra of dimension four, contrary to CM type. If the two elliptic factors were not isogenous, their endomorphism algebra would be a product of commutative fields and could not contain B. Finally, if E had no CM, the endomorphism algebra would be $M_2(\mathbb{Q})$, and the embedding of the four-dimensional division algebra B would be an impossible isomorphism.

The elliptic factors and the isogeny may be taken over L. Indeed, the rational idempotents in $\mathrm{End}^0(A)$ are defined over L, and the image of an integral multiple of such an idempotent is an elliptic subvariety over L. Endomorphisms and homomorphisms of the resulting factors extend to rational endomorphisms of A by the inclusions and rational projections; these are fixed by $\mathrm{Gal}(\overline{L}/L)$. Thus the factor isogenies, and all endomorphisms of E, are defined over L.

Put $F = \mathrm{End}^0(E)$, an imaginary quadratic field, and $\mathcal{O} = \mathrm{End}(E)$. Differentiation gives an injective unital map $F \longrightarrow \mathrm{End}_L(\mathrm{Lie}\, E) = L$: it is nonzero, and every nonzero elliptic endomorphism in characteristic zero is separable. Consequently $F \subset L$.

We next pass to the maximal order without extending L. Choose $f \geq 1$ with $f\mathcal{O}_F \subset \mathcal{O}$, choose an integral basis α_1, α_2 of $\mathcal{O}_F$, and form the finite subgroup scheme

$$K = \sum_{i=1}^{2} (f\alpha_i)(E[f]) \subset E.$$

Each $f\alpha_i$ is defined over L, so K and the quotient $E' = E/K$ are defined over L. Under a complex uniformization $E(\mathbb{C}) = \mathbb{C}/\Lambda$, this subgroup is $\mathcal{O}_F\Lambda/\Lambda$; thus $E'(\mathbb{C})$ has lattice $\mathcal{O}_F\Lambda$ and endomorphism ring $\mathcal{O}_F$. The ring cannot be larger, since the rational endomorphism algebra remains F. This is also a direct special case of the same-field maximal-order theorem of Rémond, but the construction just given is the proof used here. It follows that $A' = (E')^2$ is L-isogenous to A.

Write Δ_F for the fundamental discriminant of F. The discriminant tower formula implies that every rational prime dividing Δ_F divides D_L. A quadratic fundamental discriminant has odd prime exponents one and exponent at two at most three. Therefore

$$|\Delta_F| \leq 4 \operatorname{rad}(|D_L|) \leq 4R.$$

The multiplier four includes the possible factor $2^3/2$.

For completeness we give an elementary analytic bound on $h_{\mathrm{F}}(E')$. CM elliptic curves have potentially good reduction. After a finite extension with everywhere good reduction, the elliptic discriminant formula of Silverman, reproduced as equation (6.2) in [41], gives

$$h_{\mathrm{F}}(E') = \frac{1}{[L' : \mathbb{Q}]} \sum_{\sigma : L' \hookrightarrow \mathbb{C}} \left(-\log(2\pi) - \log|\eta(\tau_\sigma)|^2 - \tfrac{1}{2} \log \operatorname{Im} \tau_\sigma \right).$$

Here L' is used only to evaluate the stable height; it is not the field used in the isogeny estimate below. The discriminant term is zero because reduction is good. We use the individual-embedding formula (6.2), not the class-average assertion of Proposition 6.1, so no hypothesis on Weierstrass coefficients lying in $\mathbb{Q}(j)$ is needed.

For each embedding choose a reduced quadratic parameter of discriminant Δ_F. If $y = \operatorname{Im} \tau$, then

$$\frac{\sqrt{3}}{2} \leq y = \frac{\sqrt{|\Delta_F|}}{2a} \leq \frac{\sqrt{|\Delta_F|}}{2}, \qquad a \in \mathbb{Z}_{\geq 1}.$$

Put $r = |e^{2\pi i \tau}| < 1/16$. The product defining η gives

$$-\log|\eta(\tau)|^2 \leq \frac{\pi y}{6} + 2 \sum_{n \geq 1} \frac{r^n}{1 - r^n} \leq \frac{\pi y}{6} + \frac{2r}{(1-r)^2} < \frac{\pi \sqrt{|\Delta_F|}}{12} + \frac{32}{225}.$$

Also $-\frac{1}{2} \log y \leq \frac{1}{4} \log(4/3) < 1/12$. Discarding the negative term $-\log(2\pi)$ and using $\pi < 4$ shows

$$h_{\mathrm{F}}(E') < \sqrt{|\Delta_F|} + 1 \leq 2\sqrt{R} + 1 \leq 3R.$$

The height is additive on products, hence $h_{\mathrm{F}}(A') < 6R$.

Apply [103, Theorem 1.9(1)] over L, with source A' and target A and with no extension of the base field. For a surface, its quantity $\Xi(A')$ is

$$\Xi(A') = \left(14^{32} d \max(1, \log d, h_{\mathrm{F}}(A')) \right)^8.$$

The general assertion supplies an isogeny $A' \to A$ of degree at most $\Xi(A')^2$; no identification of CM type with that paper's separate "MM" condition is made or needed. The isogeny-height inequality then gives

$$h_{\mathrm{F}}(A) \le 6R + 8\log\left(14^{32}\,576\max(1,\log 576,6R)\right).$$

One can see the last inequality directly from invariant differentials: after semistable base change, pullback on determinant differentials has integral cokernel, while its archimedean squared norm is multiplied by the isogeny degree. Thus the height difference is half the logarithm of that degree minus a nonnegative normalized cokernel length.

Finally, $\log 14 < 3$, $\log 576 < 7$, and $\max(1,\log 576,6R) \le 7R$. Since $\log 7 < 2$ and $\log R \le R$,

$$h_{\mathrm{F}}(A) < 6R + 8(96 + 7 + 2 + \log R) \le 854R.$$

As $6^4 = 1296 > 854$ and $R \ge 1$, this proves (12). In particular the original $6^{100}R^{24}$ allowance absorbs this CM estimate with the field degree still exactly bounded by 576.

The unconditional numerical certificate and scalar detector The specified integer H satisfies

$$\log_2 H < 4783227.$$

Indeed the exact sieve enumerates 7499 primes in S_*, with largest prime 76163, and their product has bit length 109445. Hence $B_* < 2^{109445}$. With $A_0 = 4608^{165888}$, we have $6^{100} < A_0$ and $128 < A_0$, whence $H < 8A_0B_*^{24}$. Since $4608 < 2^{13}$,

$$\log_2 H < 3 + 165888 \cdot 13 + 24 \cdot 109445 = 4783227.$$

This is a check of the number H, not a proof that H is a height constant.

For completeness, the scalar product used by the original polynomial is

$$r(\beta) = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1).$$

At a prime $p > 76176$, if $v_p(\beta) < 0$, its valuation is $5v_p(\beta)$ and so detects odd contact with infinity. If β is p-integral and has odd contact with a finite member of $\mathcal{C}$, only that factor has positive valuation; every other factor is a unit. The extra root $1/2$ introduces only exceptional collision primes (notably factors of 4801 and 718). Thus every prime of $M(\beta)$ has odd valuation in $r(\beta)$. If a separate comparison proves $M(\beta)^{48} \mid s \ne 0$, the five-contact estimate just proved gives $h(\beta) \le H\sqrt{|s|}$.

Combining the non-CM and CM estimates with (R) and (G) gives

$$h(s) \le 2(4608^{165888} + 6^{100})(B_*M(s))^{24} + 128 \le H_*M(s)^{24},$$

because $M(s) \ge 1$. The numerical certificate above proves the remaining assertion about H_*. □

Define $\Psi(b)$ by seven rational witnesses t, U, e, p, q, s, h and four equations, with $B = 17b$:

$$\begin{aligned} U(1+bt^2) &= b(1-bt^2), & p^2 &= B(1-e)(1+3e), \\ h &= e(B-U^2), & Bq^2 &= (s^2-Bh)(s^2+3Bh). \end{aligned} \tag{13}$$

Lemma 14.4 (Exact seven-variable pole statement)**.** *If $b \ne 0$ and $\Psi(b)$, then $v_p(b)$ is not a negative odd integer for every $p \notin \{2,3,17\}$. Conversely, $\Psi(b)$ holds whenever $b > 0$, the denominator of b in lowest terms is a square, and b is a square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$.*

Proof. These are Lemmas 14.5 and 14.7. Their proof gives the simultaneous two-twist construction explicitly, and applies the existing Theorem 5.3 and Lemma 5.4. It introduces no new simultaneous-realization assumption. □

The lemma includes the degenerate values of the seven witnesses permitted by (13). No inverse condition on e, h, s, or U may be inserted into this particular polynomial. For $b \neq 0$, the first equation itself implies $1 + bt^2 \neq 0$. The predicate is soluble at $b = 0$; the norm argument below accounts for that case where necessary.

14.1.1 The seven-witness parity predicate and its finite correction

Put $B = 17b$. The predicate used here is exactly the existence of $t, U, e, p, q, s, h \in \mathbb{Q}$ satisfying

$$U(1 + bt^2) = b(1 - bt^2), \tag{14}$$

$$p^2 = B(1 - e)(1 + 3e), \tag{15}$$

$$h = e(B - U^2), \tag{16}$$

$$Bq^2 = (s^2 - Bh)(s^2 + 3Bh). \tag{17}$$

Denote it by $\Psi(b)$. No nonzero conditions are added to this formula.

Lemma 14.5 (The exact one-sided assertion)**.** *If $b \neq 0$ and $\Psi(b)$, then $v_\ell(b)$ is not a negative odd integer for any prime $\ell \notin \{2, 3, 17\}$.*

Proof. Suppose $k = v_\ell(b) < 0$ is odd. Equation (14) forces $1 + bt^2 \neq 0$: if it vanished, its right side would be $2b \neq 0$. If $t = 0$, then $U = b$. Otherwise $v_\ell(bt^2)$ is odd and nonzero, so $(1 - bt^2)/(1 + bt^2)$ is a unit. Thus $v_\ell(U) = k$ in all cases, and $v_\ell(B - U^2) = 2k$.

If $e = 0$, equation (15) gives $p^2 = B$, impossible. Write $j = v_\ell(e)$. If $j < 0$, the right side of (15) has valuation $k + 2j$, which is odd. If $j > 0$, it has valuation k, again odd. Consequently $j = 0$. This includes the possibilities $p = 0$, $e = 1$, and $e = -1/3$. Now $h \neq 0$ and $v_\ell(Bh) = 3k$ is odd. If $s \neq 0$, its square has even valuation, different from $3k$, so the two factors on the right of (17) have the same finite valuation. Their product has even valuation and is nonzero. The same conclusion holds when $s = 0$, since the product is $-3(Bh)^2$. The left side is either zero or has odd valuation, a contradiction. □

The omitted guards therefore cause no loss of the asserted soundness. At $b = 0$, the equations require $U = p = h = s = 0$ and permit arbitrary t, e, q. For $b \neq 0$, the six-witness source formula is recovered exactly by $e_{\text{old}} = Be$, $p_{\text{old}} = Bp$, $q_{\text{old}} = Bq$, with s, t, U unchanged. Its first curve residual is B^2 times (15)'s residual and its second is B times (17)'s residual after (16). Division by B reverses this transformation.

Lemma 14.6 (Simultaneous realization from the existing restricted converse)**.** *Use Theorem 5.3, the restricted converse* RC *for $E_l : y^2 = x(x - l)(x + 3l)$ with $l > 0$ squarefree and $l \equiv 11 \pmod{12}$. Let $\beta, \theta > 0$ be squarefree, with disjoint supports avoiding $2, 3$, such that $\beta \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, and each is a local square at every prime dividing the other. Then there is a positive squarefree n, prime to 6θ, for which both E_n and $E_{n\theta}$ have rational points with nonzero abscissa in $\beta\mathbb{Q}^{\times 2}$. The case $\theta = 1$ is allowed.*

Proof. Here are the descent steps, so that no separate realization theorem is being assumed. Write $B_0 = \operatorname{supp}(\beta)$, $J = \operatorname{supp}(\theta)$, and $d = |B_0| \geq 1$. For a support prime q_i, put $s_i = [-1/q_i]$, $\lambda_i = [3/q_i]$, and define over $\mathbb{F}_2$

$$(Kv)_i = \sum_{j \neq i} [q_j/q_i](v_j + v_i).$$

Add $d-1$ prime slots I of type $(s,\lambda)=(0,0)$ and one slot q_0 of type $(1,1)$. Let the B_0-by-I symbol matrix have columns e_i+e_d for $1\le i<d$. Set all other symbols incident with I to zero, prescribe $[q_0/q_b]$ as the corresponding row sum, and obtain reverse symbols by reciprocity. The q_0–J symbols remain free. Mutual squareness gives zero row sums for both directed B_0–J blocks; hence $K\mathbf{1}_{B_0}=0$ on both supports.

Choose a with ones at one fixed member of B_0 and at q_0. The matrix $K^\sharp=K+\lambda a^t$ kills $\mathbf{1}$ on both sides. After deleting q_0, its smaller-support matrix has the form

$$M_0=\begin{pmatrix} D & C \\ C^t & 0 \end{pmatrix}, \qquad \mathbf{1}^tD\mathbf{1}=1.$$

The columns of C span the even-sum hyperplane. If $M_0(x,y)^t=0$, then $x=c\mathbf{1}$; multiplication of the first equation by $\mathbf{1}^t$ gives $c=0$, and injectivity of C gives $y=0$. This also proves the case $d=1$, where $M_0=(1)$. Adding J preserves M_0. Each free q_0–J bit toggles just one new diagonal entry. The enlarged determinant is multilinear in these bits, with full-product coefficient $\det M_0=1$, so some Boolean choice makes it nonzero. CRT and Dirichlet's theorem realize the prescribed symbols by distinct fresh primes, with types imposed modulo 12. Thus $n=\prod_{B_0\cup I\cup\{q_0\}} q$ and $n\theta$ are squarefree and congruent to 11 modulo 12.

The projection $T=1-\mathbf{1}_{B_0}a^t$ satisfies $K=K^\sharp T$. Nonsingularity of the deleted matrix gives $\ker K^\sharp=\langle\mathbf{1}\rangle$, and therefore

$$\ker K=\langle\mathbf{1},\mathbf{1}_{B_0}\rangle, \qquad \lambda\notin\operatorname{im}K$$

on both supports. For the latter assertion, $Kx=\lambda$ would make $Tx-\mathbf{1}_{B_0}$ constant, contradicted by applying a^t.

For clarity, the local descent needed here is the manuscript's full two-descent in coordinates $(x+3l,x-l)$. The two torsion classes are $(l,1)$ and $(3,-l)$. At an odd good prime, the local image consists of the four pairs with even valuations; at a prime dividing l, the two torsion classes span it by their independent valuation vectors. At 3 it is the set of all first-coordinate classes with square second coordinate. At 2 both coordinate valuations are even: the only problematic case $x-l=2u$, u odd, would give $(y/2)^2\equiv l \pmod 8$, impossible for $l\equiv 3 \pmod 4$. When both differences are divisible by 4, their difference $4l$ and the cubic equation give even valuations for both. At infinity the first coordinate is positive, and the positive-second-coordinate subgroup has index two. Its possible representatives consequently are $(3^\delta\prod q_i^{u_i},\prod q_i^{v_i})$ with necessary equations

$$Ku+\operatorname{diag}(\lambda)v=\delta\lambda, \quad (K+\operatorname{diag}(s))v=0, \quad (\lambda+s)^tv=0.$$

Reciprocity gives $K+K^t=ss^t+\operatorname{diag}(s)$. Hence $\ker(K+\operatorname{diag}(s))=\ker K^t\cap s^\perp$ is a line. The functional λ^t is nonzero on this line, because $\lambda\notin\operatorname{im}K$ and $\lambda^t\mathbf{1}=0$, whereas $s^t\mathbf{1}=1$. The displayed equations force $v=0$, $\delta=0$, and $u\in\ker K$. There are at most eight Selmer classes. The class $(\beta,1)$ is actually local everywhere: it is the identity at 2 and infinity, the first torsion class at 3, satisfies the torsion comparison at support primes by $K\mathbf{1}_{B_0}=0$, and is unramified at the other odd primes. It is independent of the two torsion classes since B_0 is nonempty and $q_0\notin B_0$. Thus each two-Selmer group has dimension exactly three.

Lemma 5.4 gives full two-primary Selmer corank one. Explicitly, the Kummer sequence gives $r+\dim\mathrm{Sha}[2]=1$; writing $\mathrm{Sha}[2^\infty]=(\mathbb{Q}_2/\mathbb{Z}_2)^t\oplus F$, the alternating Cassels–Tate pairing on the finite quotient makes $\dim F[2]$ even, so $F=0$ and $r+t=1$. Apply RC to n and $n\theta$, and the manuscript's published forward analytic-rank-one theorem. Both curves have rank one and finite Sha, hence $\mathrm{Sha}[2]=0$. Kummer exactness now represents $(\beta,1)$ by actual rational points. These are not two-torsion, so their abscissas are nonzero of class β. When $\theta=1$ the same point can be used twice. □

Lemma 14.7 (The completeness scope actually used)**.** *Assume only the existing* RC *and the published descent and forward-rank facts used above. If $b > 0$ has square reduced denominator and is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, then $\Psi(b)$ holds.*

Proof. Choose $t \in \mathbb{Q}$ close to a solution of $bt^2 = 1$ at infinity and at $2, 3, 17$. In (14) this makes U arbitrarily small at those places. We can and do require $V = B(B - U^2) > 0$ and V locally square at all three primes. Explicit sufficient conditions are $U^2 < B$ in $\mathbb{R}$, $v_2(U^2/B) \geq 3$, and $v_3(U^2/B), v_{17}(U^2/B) \geq 1$. The denominator tends locally to $2 \neq 0$ and is positive globally.

Let β, θ be the positive squarefree representatives of B, V. The number $B = 17b$ is square at 2, a nonsquare unit class at 3, and has odd valuation at 17. Thus $\beta \equiv 17 \pmod{24}$ and $\beta > 1$; $\theta \equiv 1 \pmod{24}$ and $17 \nmid \theta$. At any prime $\ell \mid \beta$ outside $2, 3, 17$, the valuation of b is odd and therefore positive, since its reduced denominator is square. The same quotient calculation as in soundness gives $v_\ell(U) = v_\ell(b)$; hence $1 - U^2/B$ is a square unit. This proves that V, and hence θ, is square at every prime of β, including 17 by construction. The supports are disjoint. Conversely, if $\ell \mid \theta$, then $\ell \notin \{2, 3, 17\}$ and $v_\ell(B)$ is even. Odd valuation of $B(B - U^2)$ forces $v_\ell(B) = 2v_\ell(U)$ and strict cancellation, so $B/U^2 \in 1 + \ell\mathbb{Z}_\ell$ is square. This proves the other mutual-square condition. If $U = 0$, then $\theta = 1$ and this last argument is unnecessary.

Apply the preceding lemma. Write $V = \theta c^2$ and rescale the second point by $(x, y) \mapsto (c^2 x, c^3 y)$ to put it on E_{nV}. If the first point has abscissa Ba^2, divide both curve parameters by a^2 and the point coordinates by a^2, a^3. The resulting points have abscissas B, Bs^2 with $s \neq 0$, on E_{e_0} and $E_{e_0 V}$ respectively. Write their ordinates as y_1, y_2. Set

$$e = e_0/B, \qquad p = y_1/B, \qquad q = y_2/(B^2 s), \qquad h = e(B - U^2).$$

Direct substitution verifies all four equations. Every division made in choosing these witnesses is by a nonzero number; none is an additional equation or guard in the original polynomial. □

Finite correction and the unguarded norm root. At 2 use valuation parity followed by unit coordinates $[-1], [5]$; at 3 and 17 use valuation parity followed by the nonsquare-unit bit. The columns for $2, 3, 5, 7, 11, 13, 17$ are exactly

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

The accompanying exact computation verifies rank seven and all 128 distinct subset signatures. Rational solutions of $e^2 - e = 0$ are exactly $0, 1$. Therefore the original eight bits and two products

$$A = (1 + e_2)(1 + 2e_3)(1 + 12e_{13})(1 + 16e_{17}), \quad D = (1 + 4e_5)(1 + 6e_7)(1 + 10e_{11})(1 - 2\eta)$$

give exactly $AD = (-1)^\eta \prod p^{e_p}$, always nonzero. For nonzero integral r, choose $\delta = \operatorname{sign}(r)$ and the unique subset product μ having the same local signature as δr. Taking $AD = \delta\mu$ makes ADr positive and locally square at $2, 3, 17$.

Choose J_0 sufficiently deep at all places over primes dividing μr, then S in a sufficiently divisible subgroup satisfying the anchor. Since $f = z^2$ times a formal unit, the coefficients of $f(J_0), f(S), z(S)/z(J_0)$ can attain any prescribed finite lower valuation bounds. This remains true

at ramified and bad primes by working in sufficiently deep formal subgroups. Thus at $\ell \mid r$ all normalized leaves are integral, and at $\ell \mid \mu$ with $\ell \nmid r$ all nonconstant leaves can be made divisible by ℓ. The norm tree $N_r(a,b) = a^2 + rb^2$ has its constant leaf 1 on the all-left path; its root W is a unit at these primes. In the actual application $r = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1)$ with $\beta = X^2+2$ and $X \in \mathbb{Z}$. It is nonzero and $|r| > 1$ since $X^2 \neq 2399$ and $\beta \geq 2$. Consequently $W \neq 0$, and $b = ADrW^2$ is positive, locally square at $2, 3, 17$, and has square reduced denominator. Completeness applies.

For soundness no global guard $W \neq 0$ is needed. At any prime where $v_\ell(r)$ is odd, the two nonzero summands of $a^2 + rb^2$ have different valuation parity. Such a norm vanishes only if both inputs vanish. Recursing to the constant leaf 1 proves $W \neq 0$ at every prime relevant to an odd-contact argument. Outside the fixed exceptional set, AD is a unit and $v_\ell(ADrW^2) = v_\ell(r) + 2v_\ell(W)$. The exact original norm argument therefore applies unchanged. If r has no relevant odd valuation, there is no local conclusion to prove. No guard or witness has been inserted, and the original polynomial and coefficient bound are unchanged.

14.2 Elliptic charts and the anchor

Use the normalized curve and point

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$

$$P_* = (57/8 - 7\sqrt{2}/2, -165/8 + 417\sqrt{2}/32), \qquad G = 2E(K) = \mathbb{Z}P_*.$$

The identification of G follows from Proposition 3.1 by the unit change of variables $\lambda = 3 + 2\sqrt{2}$. Write $f = 1/x$ and $z = -x/y$. A nonzero point of G is represented by eight field variables u, v, w, z, f, h, x, y satisfying

$$f = u^2, \quad v^2 + df = 1, \quad w^2 + v^2 = 2, \quad h = vw, \quad zh + u = 0, \quad xf = 1, \quad yz + x = 0. \tag{18}$$

Each field equality means its two rational coefficient equalities.

Lemma 14.8 (The sparse point chart)**.** *The equations* (18) *parametrize precisely the nonzero points of G, and their reciprocal functions are defined.*

Proof. It gives $x = u^{-2}$, $y = vw/u^3$ and the three square root differences $x = (1/u)^2$, $x - d = (v/u)^2$, $x + d = (w/u)^2$. Here is a direct verification that these squares give precisely the required doubles. Write $e_1 = 0, e_2 = d, e_3 = -d$ and choose $r_i^2 = x - e_i$ with $r_1 r_2 r_3 = -y$. Put

$$B = r_1r_2 + r_1r_3 + r_2r_3, \qquad X' = x + B, \qquad Y' = -(r_1 + r_2)(r_1 + r_3)(r_2 + r_3).$$

Since $X' - e_i = (r_i + r_j)(r_i + r_k)$, the point (X', Y') lies on E. None of the pair sums is zero, because the e_i are distinct. Its tangent slope is $-(r_1 + r_2 + r_3)$, and the duplication equations give $[2](X', Y') = (x, y)$. Conversely, if $(x, y) = [2](a, b)$ is nonzero, then $b \neq 0$ and direct duplication gives

$$x - e_i = \left(\frac{(a - e_i)^2 - (e_i - e_j)(e_i - e_k)}{2b} \right)^2.$$

Choose the signs of these square roots to have product y, and recover u, v, w, h, z, f from (18). The group G is torsion-free, so its nonzero points have $x, y \neq 0$. Thus the sparse chart and the membership equations in Section 3 have the same nonzero image. The inverse relation $xf = 1$ excludes zero denominators. In particular u, h, z, y are nonzero. □

Subtraction uses the equations

$$\ell x - y = j, \quad \ell a + b = j, \quad x' = \ell^2 - x - a, \quad \ell x' + y' = j, \quad f'x' = 1, \quad z'y' + x' = 0$$

for $(x, y) - (a, b)$, with an inverse for the rational norm of $x - a$. Doubling uses the usual tangent formula with its nonzero denominators.

The baseline duplication estimates give

$$4h(x(Q)) - \log 392 \le h(x(2Q)) \le 4h(x(Q)) + \log 676.$$

Since $h(x(P_*)) = \log 41$, passage to the canonical height gives $2\widehat{h}(P_*) \ge \log 41 - \frac{1}{3}\log 392 > 1$, and hence

$$h(x(nP_*)) \ge n^2 - 6 \qquad (n \ne 0). \tag{19}$$

The deliberately weaker constant six suffices throughout this section.

Let $\mathcal{T}(a)$ be the seven-variable formula

$$u_0^2 - 2u_1^2 - 3u_2^2 + 6u_3^2 = 1, \qquad (a - 2u_0)^2 - 8v_1^2 - 12v_2^2 + 24v_3^2 = 4. \tag{20}$$

Then $\mathcal{T}(\mathbb{Q}) = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. Indeed the quaternion algebra $(2, 3)$ is ramified exactly at $2, 3$ and split at infinity. Koenigsmann's trace-sum theorem, Proposition 6 of [99], identifies the sum of two norm-one trace sets with the intersection of these two local rings. Eliminating the scalar coordinate of the second norm-one element gives exactly (20), with no translation of a.

Introduce twenty-five rational variables

$$c_0 = 3, \qquad c_{i+1} = c_i^2 \quad (0 \le i < 24), \qquad C = c_{24}. \tag{21}$$

Thus $C = 3^{2^{24}}$, but this large integer is not a coefficient in F_8. For $S = sP_* \ne O$ and $f(S) = f_0 + f_1\sqrt{2}$, impose $Ca_i = f_i$ and $\mathcal{T}(a_i)$, for $i = 0, 1$. At the prime above 3, P_* reduces to the nonidentity point $(\sqrt{2}, 0)$ of order two, and $v_3(z(2P_*)) = 1$. The formal group therefore gives $v_3(f(sP_*)) = 2 + 2v_3(s)$ whenever this valuation is positive. The anchor forces

$$|s| \ge 3^{2^{23}-1} > (20H_*)^2. \tag{22}$$

For the last inequality use Theorem 14.3, $\log_2 3 > 3/2$, and $3(2^{23}-1) = 25165821 > 4(5+4783227) = 19132928$. The allowed indices contain a nonzero subgroup: sufficiently divisible indices make $f(sP_*)/C$ integral at both 2 and 3. The anchor can therefore be imposed together with any finite list of formal divisibilities.

14.2.1 The literal seven-witness semilocal predicate

For $t \in \mathbb{Q}$, let $\mathcal{T}(t)$ mean that there are $u_0, u_1, u_2, u_3, v_1, v_2, v_3 \in \mathbb{Q}$ with

$$u_0^2 - 2u_1^2 - 3u_2^2 + 6u_3^2 = 1, \qquad (t - 2u_0)^2 - 8v_1^2 - 12v_2^2 + 24v_3^2 = 4.$$

Then

$$\mathcal{T}(t) \quad \Longleftrightarrow \quad t \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}.$$

Indeed, use the quaternion convention $i^2 = 2$, $j^2 = 3$ and $ij = -ji$. Its reduced norm is $x_0^2 - 2x_1^2 - 3x_2^2 + 6x_3^2$ and its reduced trace is $2x_0$. Put $v_0 = (t - 2u_0)/2$. The two displayed equations say exactly that t is a sum of two reduced traces of norm-one elements. No eighth variable is needed, since v_0 is recovered by this rational expression with fixed nonzero denominator 2.

In the notation of [99, Definition 4 and Proposition 6], this is $T_{2,3} = S_{2,3}+S_{2,3}$. That proposition identifies $T_{a,b}$ with the intersection of the local integer rings at the ramified places, with a separate real interval only when infinity is ramified. Here the Hilbert symbols are -1 at 2 and 3, and 1 elsewhere. At 3 this follows from $(2/3) = -1$; at every other odd prime both parameters are units. At 2, writing $2 = 2^1 \cdot 1$ and $3 = 2^0 \cdot 3$, the usual exponent is $0 + 1 \cdot (3^2 - 1)/8 + 0 = 1$. The algebra splits over $\mathbb{R}$ because both parameters are positive. Thus the ramification set is precisely $\{2, 3\}$, which proves the claimed equivalence in both directions. This uses Koenigsmann's sum of two trace sets, not a predicate with an additional finite offset.

14.2.2 The original anchor and its exact index subgroup

Write $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, and use the literal normalized curve and generator

$$E : y^2 = x^3 - (11 - 6\rho)x, \qquad P = \left(\frac{57}{8} - \frac{7}{2}\rho, -\frac{165}{8} + \frac{417}{32}\rho\right).$$

As in the original construction, use Proposition 3.1, transported by the unit change of variables, to obtain $2E(K) = \mathbb{Z}P$. For a nonzero $S = nP$, put $f(S) = 1/x(S) = f_0 + f_1\rho$. Retain the exact constant chain

$$c_0 = 3, \qquad c_{j+1} = c_j^2 \quad (0 \le j < 24), \qquad C = c_{24}.$$

It forces $C = 3^{2^{24}}$, with 25 shared rational variables and 25 quadratic-or-linear equations. The anchor introduces two variables a_i, imposes $Ca_i = f_i$, and imposes $\mathcal{T}(a_i)$ for each $i \in \{0, 1\}$. It has 16 private witnesses and six quadratic equations, exactly as in the saved module.

Set $N = 2^{24}$ and $m = N/2 = 2^{23}$. We claim that, for nonzero n,

$$\text{the anchor holds on } nP \quad \Longleftrightarrow \quad n \in 2 \cdot 3^{m-1}\mathbb{Z}.$$

First consider the place above 3. It is unramified quadratic, with residue field $\mathbb{F}_9$ and normalized valuation $v(3) = 1$. For rational a, b one has $v(a + b\rho) = \min(v_3(a), v_3(b))$: after dividing by the smaller power of 3, linear independence of $1, \bar{\rho}$ over $\mathbb{F}_3$ precludes cancellation. The curve has good reduction there. Direct calculation gives

$$\bar{P} = (\bar{\rho}, 0), \qquad z(2P) = \frac{3396356159088 - 4135204317756\rho}{3287811625151}, \qquad z = -x/y.$$

The reduction of P has order two. The denominator in this expression is a 3-unit and both numerator coefficients have valuation one. Consequently nP is in the formal kernel exactly when n is even, and $v(z(2P)) = 1$.

For completeness, the valuation of every such multiple can be computed without an unproved uniformity assertion. The integral formal group in the parameter z has inverse $-z$, so its multiplication series are odd. Their linear coefficient is the corresponding integer. Thus, for a 3-unit integer u, $[u]z = uz + O(z^3)$ preserves positive valuation, whereas $[3]z = 3z + O(z^3)$ increases valuation by exactly one at depth at least one. The higher terms have valuation at least $3v(z) > v(z) + 1$. Apply this repeatedly to $[n/2](2P)$, allowing negative integers by negation. It gives

$$v(z(nP)) = 1 + v_3(n), \qquad v(f(nP)) = 2 + 2v_3(n) \quad (2 \mid n).$$

The second equality follows from $f = z^2(1-(11-6\rho)f^2)$ on the formal branch. The integral formal-group construction and its linear multiplication coefficient are given in [60, IV§1, Example IV.2.2.3 and Proposition IV.2.3(a)]. No minimal-model assertion at the bad prime 2 is used below.

At the place above 2, normalize $v(2) = 1$, so $v(\rho) = 1/2$. The two summands of a rational pair have valuations of opposite integer parity after multiplication by 2, and hence cannot cancel. The displayed coordinates give

$$v(x(P)) = -3, \qquad v(y(P)) = -9/2, \qquad v(z(P)) = 3/2.$$

Thus P lies on the formal branch at infinity, even though reduction at 2 is bad. Every integral multiple lies in this formal subgroup. One can see the integral branch directly by setting $w = -1/y$: the equation becomes $w = z^3 - (11 - 6\rho)zw^2$, which has a unique solution $w \in z^3\mathcal{O}[[z]]$; its group law is the integral formal group just cited. In particular $f(nP)$ is integral at 2. Since

$$v(a + b\rho) = \min(v_2(a), v_2(b) + 1/2),$$

its rational coefficients are both 2-integral. As C is a 2-unit, the two anchor conditions at 2 impose no further restriction on n.

At 3, the two anchor conditions are equivalent to $v(f(nP)) \geq N$. Positive valuation of f implies negative valuation of x, hence membership in the formal kernel, so n is even. The preceding exact formula makes the remaining condition $v_3(n) \geq m - 1$. Conversely these two conditions give both required coordinate divisibilities at 3, and the condition at 2 already holds. The proved characterization of $\mathcal{T}$ supplies its rational witnesses, proving the subgroup claim.

In particular, anchor indices contain a nonzero finite-index subgroup and have positive representatives. They can be combined with any finite collection of requirements that indices be divisible by fixed integers: take a common multiple with $2 \cdot 3^{m-1}$. More generally, any finitely many formal-neighborhood conditions whose preimages are finite-index subgroups can be imposed in the same way. This statement does not claim compatibility with arbitrary nonzero residue conditions.

The old and new reciprocal coordinates are equivalent for this anchor. With $\lambda = 3 + 2\rho$, one has $f_{\mathrm{old}} = \lambda^2 f$ and

$$\begin{pmatrix} f_{\mathrm{old},0} \\ f_{\mathrm{old},1} \end{pmatrix} = \begin{pmatrix} 17 & 24 \\ 12 & 17 \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \end{pmatrix}.$$

This matrix has determinant one and integral inverse, so it preserves coordinatewise membership in $C(\mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)})^2$ in both directions.

14.2.3 The unchanged numerical cutoff

In this calculation $H = H_*$ is the constant in Theorem 14.3.

The anchor gives $|n| \geq 3^{2^{23}-1}$. Conditional only on the separately audited original height bound $\log_2 H < 4783227$, it follows that $|n| > (20H)^2$. Indeed $\log_2 3 > 3/2$ and $\log_2 20 < 5$, and

$$3(2^{23} - 1) = 25165821 > 19132928 = 4(5 + 4783227).$$

Thus the original 24 squarings suffice whenever that exact height input is established. This calculation does not replace the required five-contact height theorem by a six-contact theorem or by a larger constant. No part of this proof changes the saved polynomial.

14.3 The exact norm tests

For a nonzero rational contact r and rational tests X_i of positive orders L_i, apply the binary operation $N_r(a, b) = a^2 + rb^2$ to the ordered leaves $1, X_1/r^{L_1}, \ldots, X_k/r^{L_k}$ in the following fixed order. First pair 1 with X_1/r, where $L_1 = 1$. Group the remaining tests by increasing order,

retaining their original order within each group. Pair successive tests within each group and carry any last unpaired test. Then pair successive resulting nodes at each level and carry any last node until only a root W remains. The polynomial equations for the bottom pairs are

$$rw - r - X_1^2 = 0, \qquad r^{2L}w - X_i^2 - rX_j^2 = 0;$$

an unpaired normalized leaf satisfies $r^L w - X_i = 0$. Higher nodes satisfy $w - a^2 - rb^2 = 0$. Powers of r are computed by binary squaring. Add an inverse for r, a variable for W^2, and the predicate $\Psi(rW^2)$.

At a prime with $R = v_p(r)$ odd, the two summands in every norm node have distinct valuation parity. There is no cancellation. The constant leaf is on the all-left path, so $v_p(W) \le 0$ if $R < 0$. Consequently $v_p(rW^2) > 0$ first implies $R > 0$. It also gives $v_p(W) > -R/2$. Descending any norm node, a bound $v_p(N_r(a,b)) > -cR$ with $c < 1$ gives $v_p(a) > -cR/2$ and $v_p(b) > -(c+1)R/2$. Every normalized leaf therefore has valuation greater than $-R$. Thus $v_p(X_i) > (L_i - 1)R \ge L_i - 1$, or $v_p(X_i) \ge L_i$. The root cannot vanish at such a prime. Indeed $N_r(a,b) = 0$ forces $a = b = 0$ over $\mathbb{Q}_p$, and descent would make its constant leaf zero. Thus Lemma 14.4 applies even though the polynomial has no separately imposed inverse for W.

For completeness at a positive prime $r = p$, tests of orders L_i can be made so deep that every nonconstant normalized leaf is p-integral and the root W is a p-adic unit. Then rW^2 has square reduced denominator. If r is a local square at $2, 3, 17$, the completeness half of Lemma 14.4 applies.

14.4 Forty-one contacts and one common comparison point

Set

$$R = 47027 = 31 \cdot 37 \cdot 41, \quad W_0 = 25878772920, \quad c = 18484837801, \quad V_0 = 32W_0.$$

For nine rational targets a_i, put $A_i = Ra_i + 3$ and $w_i = A_i v - u$. Use the forty-one nonzero contacts

$$v/V_0, \qquad u - \delta v \ (\delta \in \mathcal{D}), \qquad -(w_i - \delta v) \ (\delta \in \mathcal{D},\ 1 \le i \le 9). \tag{23}$$

For integer a_i, the prime-pattern argument of Section 6 applies on $v = 32W_0(c + W_0 y)$, $u = c + W_0 x$ and gives positive ordinary prime values for every distinct contact, all congruent to c modulo W_0. Repeated targets use the same prime values. These are local squares at $2, 3, 17$. The precise application uses at most forty-one distinct affine forms and their successor exclusions. At $31, 37, 41$ there are at most eight slopes since every A_i reduces to 3; at $p \ge 43$ there are at most forty. Choosing two slopes outside that set proves local admissibility. The small-prime conditions are built into c. The full uniform fiber sieve and same-prime order argument are given in Subsection 14.4.1 below. No conditions relating orders at different selected primes are needed.

An odd normalized contact of u/v or w_i/v outside S_* forces some raw contact in (23) to have odd valuation. To see this, let m be the minimum of the two coordinate valuations. For a finite contact of odd depth, if m is even its distinguished linear form has odd valuation; if m is odd another finite contact form has valuation m. At an odd contact with infinity the same argument uses the denominator if m is even and a finite form if m is odd. All differences of distinct finite contact points are units outside S_*. The signs and V_0 do not affect parity there.

For nonzero points $Q_i = n_i P_*$, $1 \le i \le 4$, take $Q_0 = P_*$, one common $J = jP_* \ne O$, and four rational proxies b_i. At each raw contact introduce $X_0 = hP_*$ and $X_i = m_i P_*$ and the subtraction charts $Y_i = X_i - 4Q_i$, $0 \le i \le 4$. The points $4Q_i$ are computed by two doublings. Put

$$L(Z) = \sum_{j=0}^{15} \frac{(-1)^j \binom{2j}{j} D^j}{4j+1} Z^{4j+1}.$$

This polynomial has degree sixty-one; the first omitted logarithm term has degree sixty-five. Test order one for

$$f(X_i)\ (0 \le i \le 4), \quad f(J^\sigma), \quad f(Y_i^\sigma)\ (0 \le i \le 4),$$

and order sixty-four for

$$L(z(X_i))/L(z(X_0)) - b_i\ (1 \le i \le 4), \qquad z(Y_i^\sigma)/z(J^\sigma)\ (0 \le i \le 4).$$

The inverses in these formulas are imposed explicitly. For each field test $a + b\sqrt{2}$ of order L_i, use the rational tests $(\theta^2 - 2)b$ and $a + \theta b$ of that order, together with $\theta^2 - 2$ of order one. There is one common rational θ for all forty-one comparisons. This gives twenty-three rational tests of order one and eighteen of order sixty-four.

At an odd raw contact the norm tests select a place above p, and the conjugate parameter ratios imply $h \equiv 4$ and $m_i \equiv 4n_i \pmod{p^{64}}$. In particular h is a unit. The logarithm expansion and its degree sixty-five remainder give

$$b_i \in \mathbb{Z}_p, \qquad b_i \equiv n_i \pmod{p^{64}}. \tag{24}$$

This uses the same formal-group estimates as Section 6.

For completeness choose a consistent orientation at each distinct selected prime. Let $d_{1,p}, d_{2,p}$ be the two reduction orders. The ordinary-prime conditions give $\gcd(d_{1,p}, d_{2,p}) = 4$ and $p \nmid d_{1,p} d_{2,p}$. Choose $j = \operatorname{lcm}_p d_{2,p}$. For each occurrence at p choose

$$h \equiv 0 \pmod{d_{1,p}}, \qquad h \equiv 4 \pmod{d_{2,p}}, \qquad h \equiv 4 \pmod{p^{64+v_p(j)}}.$$

These congruences are compatible. Avoid $h = 0, 4, -4$, and set $X_i = hn_iP_*$. They give all the required formal conditions and parameter ratios relative to the single point J. Increase the last precision if necessary and choose θ last by the Chinese remainder theorem. Repeated occurrences of a prime use the same orientation. Every subtraction denominator is nonzero, and the formal-logarithm unit calculation at p proves $L(z(X_0)) \neq 0$.

14.4.1 The original forty-one-prime pattern

Let

$$\mathcal{D} = \{0, 1, 2401, -343/32\}, \quad R = 47027 = 31 \cdot 37 \cdot 41, \quad W = 25878772920 = 4 \prod_{q \le 29} q,$$

$$c = 18484837801, \qquad V_0 = 32W = 828120733440.$$

For integers $a_1, \dots, a_k$, $k \le 9$, put $A_i = Ra_i + 3$, $w_i = A_i v - u$. There are $5 + 4k$ displayed positions

$$v/V_0, \qquad u - \delta v, \qquad u - (A_i - \delta)v \quad (\delta \in \mathcal{D},\ 1 \le i \le k). \tag{25}$$

For every lower bound these positions can simultaneously be positive primes beyond that bound, congruent to c modulo W, with the following property. For the curve $E : y^2 = x^3 - (3 - \sqrt{2})^2 x$, both reductions above each prime are good, and their cardinalities d_1, d_2 satisfy

$$\gcd(d_1, d_2) = 4, \qquad p \nmid d_1 d_2.$$

This is a same-prime statement and asserts no cross-prime gcd condition.

Proof. Use $u = c + Wx$, $v = V_0(c + Wy)$. Put

$$\mathcal{S} = \mathcal{D} \cup \bigcup_i (A_i - \mathcal{D}), \quad K_s = V_0 s \in W\mathbb{Z}, \quad \Psi_0 = c + Wy, \quad \Psi_s = c + Wx - K_s(c + Wy).$$

Remove duplicate targets. The remaining slopes are all distinct. Indeed, the nonzero integral differences in $\mathcal{D} - \mathcal{D}$ are $\pm 1, \pm 2400, \pm 2401$, none divisible by 31. The integral sums in $\mathcal{D} + \mathcal{D}$ are $0, 1, 2, 2401, 2402, 4802$, with residues $0, 1, 2, 14, 15, 28$ modulo 31, none equal to $A_i \equiv 3$. These facts exclude intersections between secondary families and between primary and secondary families. The $m = 1 + |\mathcal{S}| \leq 41$ forms have pairwise independent linear parts. Repeated targets may be restored afterwards with repeated values.

At $q \mid W$, the forms equal c; direct reduction gives $c \equiv 1 \pmod 8$, $c \equiv 3 \pmod 7$, and $c \equiv 1 \pmod q$ at all other odd primes $q \leq 29$. Thus the forms and their successors are units at every odd $q \mid W$. For $q > 29$, the affine map from (x, y) to (u, v) is invertible. At $q = 31, 37, 41$, all $A_i = 3$ modulo q, so $|\mathcal{S} \bmod q| \leq 8$. For $q \geq 43$ this cardinality is at most 40. In either case choose distinct residues b, e outside the slopes. Put $\gamma = V_0^{-1}$ modulo q and, reversing b, e if necessary, require $e - b \neq -\gamma$. Then

$$v = (e - b)^{-1}, \quad u = bv$$

satisfies

$$u - sv = v(b - s) \neq 0, \quad u - sv + 1 = v(e - s) \neq 0, \quad \gamma v \neq 0, \quad \gamma v + 1 \neq 0.$$

This proves all the value and successor local conditions. No condition on a successor is imposed at 2.

Add an unused integer coordinate z, and choose a fixed rectangular box Ω inside the cone $v > 0$, $u/v > \max(0, |A_1|, \ldots, |A_k|) + 2413$ in the homogeneous leading coordinates. All forms on $X\Omega$ lie between αX and βX for positive fixed constants. Fix $Q > 82$ containing every prime divisor of W and every nonzero $K_s - K_t$. Impose one good residue at every prime up to Q. For a cutoff $L > Q$, exclude $\Psi_\nu = 0, -1$ at every additional prime up to L. The surviving proportion is

$$\gg_{\mathbf{a}} \prod_{Q < q \leq L} (1 - 2m/q) \gg_{\mathbf{a}} (\log L)^{-2m},$$

by [97, Theorem 1, modulus 2].

For fixed L, apply [50, Main Theorem, p. 1763] to every surviving lattice residue in the convex box. The singleton partition in [50, Definition 1.5 and Lemma 1.6] bounds complexity by $m - 2 \leq 39$. The inputs MN(39) and GI(39) follow from [51, Theorem 1.1] and Section I. At imposed primes the local factor is $(1 - 1/q)^{-m}$; at the other primes it is $1 + O(m^2/q^2)$. The singular product therefore has a positive lower bound independent of L. All coefficient and modulus bounds are fixed before X tends to infinity. Summing the finitely many asymptotics and removing proper prime powers gives

$$\#\{\text{prime patterns}\} \geq c_{\mathbf{a}} \frac{X^3}{(\log L)^{2m} (\log X)^m}. \tag{26}$$

The constant $c_{\mathbf{a}} > 0$ is independent of L; the threshold for X may depend on it. Proper powers contribute at most $O_{\mathbf{a}}(X^{5/2} (\log X)^{m+1})$ before removing von Mangoldt weights, since each fixed value has $O_{\mathbf{a}}(X^2)$ fibers.

We give the uniform upper estimates needed to impose the order condition. For forbidden sets of cardinality $\rho(q)$ in a two-dimensional box of side N, put $g(q) = \rho(q)/q^2$, $h(q) = g(q)/(1 - g(q))$, and extend multiplicatively on squarefree support. CRT gives

$$A_d = N^2 g(d) + R_d, \qquad |R_d| \ll Nd + d^2.$$

With $G(Z) = \sum_{d \le Z} h(d)$, take

$$\lambda_d = \frac{\mu(d)}{G(Z) \prod_{q \mid d}(1 - g(q))} \sum_{\substack{e \le Z/d \\ (e,d)=1}} h(e) \quad (d \le Z).$$

Then $\lambda_1 = 1$, $|\lambda_d| \le d$, and the Selberg quadratic form equals $G(Z)^{-1}$. To verify the last assertion, use $y_r = \sum_{r \mid d} \lambda_d g(d) = \mu(r)h(r)/G(Z)$ and $1/g(n) = \sum_{r \mid n} 1/h(r)$. The number of sifted points is thus

$$\ll N^2/G(Z) + NZ^6 + Z^8. \tag{27}$$

For $\rho(q) = 2q - 1$, supported on $q \equiv 1 \pmod 4$ with at most one prime omitted, $G(Z) \gg_Q \log Z$, uniformly in that omission. For $\rho(q) = (m-1)q$ at all $q > Q$, $G(Z) \gg_{Q,m} (\log Z)^{m-1}$. These follow from [97, Theorem 1, moduli 4 and 2]. Truncation is justified by using $z = Z^{1/512}$ and the probability distribution $h(d)/\prod_{q \le z}(1 + h(q))$: its expected $\log d$ is at most $40 \sum_{q \le z} (\log q)/q < \frac{1}{2} \log Z$ for large Z. Thus at least half the mass lies below Z. Removing one norm-sieve prime changes its product by a factor at least $1/2$.

For a selected prime, $(7/p) = -1$ and $p \equiv 1 \pmod 8$. The two reductions of $3 - \sqrt{2}$ have product 7 and opposite square classes, so their traces are opposite. Their discriminants $64d^6$ are units. The automorphism $(x, y) \mapsto (-x, iy)$ is rational over $\mathbb{F}_p$. By the Tate-module injection [60, III.7.4], its rational centralizer is $\mathbb{Q}(i)$: at the auxiliary prime 3 the centralizer has dimension two because $T^2 + 1$ is irreducible. Frobenius is integral by [60, V.2.3.1(b)]; hence it is $a + bi \in \mathbb{Z}[i]$. The trace and determinant formulas [60, III.8.6] give

$$a^2 + b^2 = p, \qquad d_1 = p + 1 - 2a, \qquad d_2 = p + 1 + 2a.$$

Full rational two-torsion gives $4 \mid d_1, d_2$, and $v_2(d_1 + d_2) = 2$, so their gcd has 2-part exactly 4. Also $0 < d_i < 2p$ and $d_i \ne p$, so $p \nmid d_1 d_2$. An odd common divisor ℓ divides a and $p + 1$, and forces $b^2 \equiv -1 \pmod \ell$. The successor conditions exclude $\ell \le L$.

The number of bad prime values of any fixed form is

$$\ll_{\mathbf{a}} X/(L \log X) + X^{7/8} + \sqrt{X} \log X. \tag{28}$$

Indeed, for $L < \ell \le X^{1/8}$, write $a = \ell r$, $b = b_0 + \ell s$, with at most two b_0 modulo ℓ. The box has side $N \asymp \sqrt{X}/\ell \gg X^{3/8}$. Use the norm case of (27) with $Z = X^{1/100}$; its error is uniformly $o(N^2/\log X)$. Sum $O(X/(\ell^2 \log X))$. For $\ell > X^{1/8}$, $a \ne 0$ gives $O(\sqrt{X}/\ell)$ choices with no additive constant, and the b count is $O(\sqrt{X}/\ell + 1)$. Summation proves the remaining terms.

On a fiber $\Psi_0 = p$, use (x, z); the other forms are $c + Wx - K_s p$. On a fiber $\Psi_{s_0} = p$, use (y, z) and

$$x = K_{s_0} y + \frac{p - c(1 - K_{s_0})}{W}, \qquad \Psi_s = p + (K_{s_0} - K_s)\Psi_0.$$

The quotient is integral for an actual fiber. For $Q < q \le X^{1/100} < p$, the remaining zero lines are distinct and parallel, with total cardinality $(m-1)q$. The second case of (27) bounds each fiber by $O_{\mathbf{a}}(X^2/(\log X)^{m-1})$, independently of p, L. Multiply by (28) and sum over m positions. The bad patterns are at most

$$C_{\mathbf{a}} \frac{X^3}{L(\log X)^m} + O_{\mathbf{a}}\left(X^{23/8}/(\log X)^{m-1} + X^{5/2}/(\log X)^{m-2}\right).$$

Choose L so large that $C_{\mathbf{a}}/L$ is less than half $c_{\mathbf{a}}/(\log L)^{2m}$, then choose X large. A good pattern survives, proving the assertion. □

14.4.2 A shared auxiliary point with private shifts

Let $\mathcal{P}$ be the finite set of distinct selected primes, all larger than 76176. At each p fix an orientation of its two places, and let $d_{p,1}, d_{p,2}$ be the corresponding group cardinalities. Set

$$j = \operatorname{lcm}_{p\in\mathcal{P}} d_{p,2} > 0, \qquad J = [j]P_*.$$

For each p, independently choose an integer h_p satisfying

$$h_p \equiv 0 \pmod{d_{p,1}}, \qquad h_p \equiv 4 \pmod{d_{p,2}}, \qquad h_p \equiv 4 \pmod{p^{64+v_p(j)}}. \tag{29}$$

The first two congruences are compatible because their modulus gcd is 4. The last is compatible because $p \nmid d_{p,1}d_{p,2}$. The resulting arithmetic progression contains an h_p outside $\{0, 4, -4\}$. Put

$$X_0 = [h_p]P_*, \qquad X_i = [h_p n_i]P_*, \qquad Y_i = [(h_p - 4)n_i]P_*, \quad n_0 = 1.$$

These are private for each contact occurrence, whereas J is common. Use the same orientation at repeated occurrences of p.

At the primary place all X_i are formal; at the opposite place J and all Y_i are formal. The formal logarithm is a homomorphism and an isometry on $p\mathbb{Z}_p$ by [60, IV.6.4 and its proof; VII.2.1–2.2]. The hypotheses hold because p is odd, split, and of good reduction. For any integers k, j whose multiples of P_* lie in the formal group,

$$\frac{\log([k]P_*)}{\log([j]P_*)} = \frac{k}{j}.$$

To see this without defining a logarithm of P_* itself, apply the homomorphism law to $[j]([k]P_*) = [k]([j]P_*)$. Nonzero formal multiples have nonzero logarithm. Therefore

$$v_p\left(\frac{z(Y_i)}{z(J)}\right) = v_p\left(\frac{(h_p - 4)n_i}{j}\right) \geq 64, \qquad \frac{\log(X_i)}{\log(X_0)} = n_i.$$

This proves the required local comparison conditions. Although other $d_{q,2}$ may be divisible by p, their effect is exactly the term $v_p(j)$ in (29); no cross-prime gcd assertion is needed.

For the original truncation

$$L(z) = z\sum_{r=0}^{15} \frac{(-1)^r\binom{2r}{r}}{4r+1}(Dz^4)^r,$$

the full logarithm has the same series extended over all $r \geq 0$. This follows by integrating $(1 + 4Dz^4)^{-1/2}$, the invariant differential in the parameter $z = -x/y$. If $v_p(z) \geq 1$ and $p > 76176$, every omitted term has valuation at least $65v_p(z)$: its integer numerator can only increase valuation, and for $n = 4r + 1 \geq 65$, $nv_p(z) - v_p(n) \geq 65v_p(z)$. For $n < p$ this is immediate; for $n \geq p$ use $n - 65 \geq v_p(n)$. Thus $L(z)/\log(z) \in 1 + p^{64}\mathbb{Z}_p$. The required primary truncated-log ratios are congruent to the integers n_i modulo p^{64}.

Conversely, in an accepted comparison at any $p > 76176$, its conjugate parameter-ratio tests imply

$$m_i - 4n_i \equiv 0 \pmod{p^{64}}, \qquad h - 4 \equiv 0 \pmod{p^{64}},$$

since j is a nonzero integer. Hence h is a p-unit. The same logarithm identity and truncation estimate give $L(X_i)/L(X_0) \equiv m_i/h \equiv n_i \pmod{p^{64}}$ and show that this ratio is integral. The rational proxy test then gives the claimed integrality and congruence for every b_i.

14.4.3 The common root approximation and parity boundary

For each distinct selected p, retain the chosen square root ρ_p of 2. After constructing the private points, each field test $A + B\sqrt{2}$ of order k has $v_p(A + B\rho_p) \geq k$. Choose a common integer θ sufficiently close to ρ_p at every selected prime that $v_p(\theta - \rho_p) \geq k - v_p(B)$ for every nonzero coefficient B, and at least one. There are finitely many requirements; CRT supplies such an integer. Since $\theta + \rho_p$ is a unit,

$$v_p((\theta^2 - 2)B) \geq k, \qquad v_p(A + \theta B) \geq k.$$

Repeated primes require the same root and impose compatible precision bounds. Conversely, the root test and these two rational tests imply the field test at the selected root. This verifies the circuit's one common θ, not a separate root variable per contact.

Every raw contact parameter in the complete assignment is a positive prime congruent to c modulo W. It is therefore a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The binary norm batch has unit root at its contact prime, because normalized leaves are integral and its leftmost constant leaf is 1. Its root is positive and nonzero in the real embedding. The argument $r\mathcal{W}^2$ consequently is positive, has square reduced denominator, and is locally square at $2, 3, 17$. These are the exact hypotheses of the original specialized parity-completeness input. This verifies its consumption but does not constitute a new proof of that input, which is supplied here by Lemma 14.7. In particular no integrality of the comparison points at these three small primes is required here.

14.5 Scalar height and the defining relation

Put $\beta = X^2 + 2$ and

$$r(\beta) = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

This is nonzero for every rational X, since $X^2 = 2399$ has no rational solution. Introduce $J_0 \neq O$ and apply a scalar norm batch to the two coefficients of each of $f(J_0), f(S), z(S)/z(J_0)$, of orders $1, 1, 1, 1, 48, 48$. In this batch replace the pole argument by $\delta\mu r W^2$, where $\delta \in \{1, -1\}$ and μ is a squarefree product of $2, 3, 5, 7, 11, 13, 17$. Precisely, eight Boolean variables produce

$$A = (1 + e_2)(1 + 2e_3)(1 + 12e_{13})(1 + 16e_{17}), \quad B = (1 + 4e_5)(1 + 6e_7)(1 + 10e_{11})(1 - 2\eta),$$

and the pole argument is $ABrW^2$. These factors have no valuation outside S_*. Every odd five-contact of β gives an odd valuation of $r(\beta)$ there. At infinity its valuation is five times that of β; at a finite contact all other factors are units. Thus the scalar tests force $M_5(\beta)^{48} \mid s$, and Theorem 14.3 gives

$$h(\beta) \leq H_* \sqrt{|s|}. \tag{30}$$

For integral X, the scalar r is a nonzero integer. The square classes of $2, 3, 5, 7, 11, 13, 17$ form a basis of $\mathbb{Q}_2^*/\mathbb{Q}_2^{*2} \times \mathbb{Q}_3^*/\mathbb{Q}_3^{*2} \times \mathbb{Q}_{17}^*/\mathbb{Q}_{17}^{*2}$. Choose δ and μ so that $\delta\mu r$ is positive and a square at these three places. Choose J_0 in the required formal subgroups and then choose S sufficiently divisible relative to J_0 at all primes dividing $r\mu$, together with the anchor conditions. All normalized nonconstant leaves are then integral at primes dividing r, and vanish at primes dividing μ but not r. The root is a unit at these primes. Consequently $\delta\mu r W^2$ has square reduced denominator, and Lemma 14.4 supplies the scalar pole witnesses.

Take $A = nP_*, S = sP_*, T = tP_*, Z = zP_* \neq O$, and use $(\beta, \xi, \tau, \zeta)$ as the four proxies. Impose

$$\tau^2 - \xi(\xi - 2)\tau\zeta + (\xi - 2)^2\zeta^2 = 1. \tag{31}$$

The nine rational targets in (23) are a nonzero common denominator e and the eight coefficients of $ex(A), ex(S), ex(T), ex(Z)$. Require its forty-one comparison batches, the scalar batch, and the anchor just described.

If every associated five-contact radical has its Kth power dividing a nonzero integer N, Theorem 14.3 and $Ra_i + 3 = u/v + w_i/v$ give

$$h(a_i) < 2H_*|N|^{24/K} + 14, \qquad h(x(Q)) < 7H_*|N|^{24/K}$$

for each of the four points. The second bound follows by applying the first to e and both coefficients of $ex(Q)$ and absorbing the constant smaller than fifty into H_*.

Lemma 14.9 (Lucas growth)**.** *If $s, t, z \in \mathbb{Z}$, $|s| = a \geq 7$, $z \neq 0$, and*

$$t^2 - s(s-2)tz + (s-2)^2z^2 = 1,$$

then $|t| > \exp(a-3)$. Conversely each such s admits positive integers t, z satisfying this equation.

Proof. Put $w = \operatorname{sgn}(s)(s-2)z$ and $\rho = (a + \sqrt{a^2-4})/2$. The equation becomes $t^2 - atw + w^2 = 1$. Every positive norm-one unit of $\mathbb{Z}[\rho]$ is a power of ρ: multiply it by a power into $[1, \rho)$, write the result as $v = A + B\rho$, and use $B = (v - v^{-1})/(\rho - \rho^{-1}) \in [0, 1) \cap \mathbb{Z}$. Thus $v = 1$. If $U_0 = 0$, $U_1 = 1$ and $U_{k+1} = aU_k - U_{k-1}$, the identities for $\rho^{\pm k}$ give $|w| = U_k$ and $|t| \in \{U_{k-1}, U_{k+1}\}$ for some $k \geq 1$. Modulo $a-2$ one has $U_k \equiv k$, and modulo $a+2$ one has $U_k \equiv (-1)^{k-1}k$. The divisibility of w by $|s-2|$ therefore gives $k \geq a-2$ when $s > 0$ and $k \geq a+2$ when $s < 0$. Induction gives $U_k \geq (a-1)^{k-1}$, whence $|t| \geq (a-1)^{a-4} > e^{a-3}$ for $a \geq 7$. For existence take $k = a-2$, $z = U_k/(a-2)$ if $s > 0$, and $k = a+2$, $z = U_k/(a+2)$ if $s < 0$, in both cases with $t = U_{k+1}$. □

Proof of Theorem 14.2. Suppose the relation holds, and let $B = \max(|n|, |s|, |t|, |z|)$ and $\Delta = t^2 - s(s-2)tz + (s-2)^2z^2 - 1$. If $\Delta \neq 0$, the congruences (24) imply that every coordinate contact radical to the sixty-fourth power divides Δ. Since $|\Delta| < 5B^4$, the index-height and shared-height bounds give

$$B^2 - 6 < 7H_*|\Delta|^{3/8} < 14H_*B^{3/2}.$$

Hence $B < (14H_* + 6)^2 \leq (20H_*)^2$, contradicting the anchor. Thus $\Delta = 0$. Lemma 14.9 gives $|t| > \exp(|s| - 3)$ because $z \neq 0$.

If $\beta \neq n$, put $N = |\operatorname{num}(\beta - n)|$ in lowest terms. The same congruences give

$$B^2 < 7H_*N^{3/8} + 6, \qquad N \leq 2e^{h(\beta)}B,$$

whence $\log B < \frac{8}{13}\log(16H_*) + \frac{3}{13}h(\beta)$. Writing $a = |s| > (20H_*)^2$, (30) implies $h(\beta) < a/20$ and $\log(16H_*) < a/20$. Therefore $\log B < 11a/260 < a - 3$, another contradiction. We obtain $\beta = n \in \mathbb{Z}$, and $X \in \mathbb{Q}$, $X^2 = \beta - 2 \in \mathbb{Z}$ imply $X \in \mathbb{Z}$ by reducing its denominator.

Conversely let $X \in \mathbb{Z}$ and set $A = \beta P_*$. Choose the scalar height data and then an anchored S with all required divisibilities. Lemma 14.9 gives positive integers t, z satisfying the index equation; set $T = tP_*$ and $Z = zP_*$. Choose an integer e clearing their abscissa coefficients. The ordinary prime pattern provides (23). Set the proxies equal to their integer indices, choose J and the private points as above, and finally choose the common θ. The norm and pole completeness statements supply all remaining witnesses. No subsequent choice changes the target points. □

14.6 The base compiler prescription and finite verification

Every equation above is implemented with named arithmetic intermediates so that each rational residual has degree at most four. A field product uses $(a+b\sqrt{2})(c+d\sqrt{2}) = (ac+2bd)+(ad+bc)\sqrt{2}$. Scalar inverses are equations $uv = 1$. Field inverses use the two coefficient equations of the field product being one; subtraction has the separate scalar inverse of its abscissa-difference norm. The logarithm uses Horner evaluation in $(Dz^4)^2$ with block length two. The two doublings of the fixed point P_* also use arithmetic intermediates; inserting its large precomputed fourth multiple as a coefficient would give a different polynomial. The chart is exactly (18), including the intermediate $h = vw$.

For each nonzero rational residual, multiply by the least common multiple of its coefficient denominators, then divide all coefficients by their positive greatest common divisor. Denote the resulting integral rows by $R_1, \dots, R_{15107}$. In the variable order specified by `degree8-original/build.py` and its residual table, define

$$F_8 = \sum_{i=1}^{15107} R_i^2. \tag{32}$$

This equation is an exact finite prescription for every coefficient. The residual table, its full collected expansion, and the construction certificate accompany the manuscript. No huge constant is treated as an additional free parameter. In particular, all twenty-five variables in (21) belong to the stated witness count.

The recursive module counts are

module	new variables	residuals
pole	7	4
anchor	16	6
scalar batch, including pole	34	31
forty-one-test batch, including pole	58	55
one joint comparison, including batch	373	360
root, including all its calls	15646	15082
shared anchor chain	25	25

The root has 303 direct variables and 285 direct residuals; its calls are one anchor, one scalar batch, and forty-one comparisons. Thus $303+16+34+41\cdot373 = 15646$ and $285+6+31+41\cdot360 = 15082$. Expansion of (32) gives the finite assertions of Proposition 14.1. Over $\mathbb{Q}$, a sum of squares vanishes exactly when every summand vanishes, so the residual interpretation proves the theorem. This computation checks the polynomial and its compilation. The proof of the height estimate is the arithmetic argument given above.

15 An explicit quartic defining the integers

This is the second final polynomial of the manuscript. The original huge polynomial remains unchanged. The base residual compiler in Section 14 supplies construction machinery and inherited arithmetic inputs; it is not a third final polynomial.

Theorem 15.1 (An explicit quartic definition)**.** *With precisely the inherited arithmetic hypotheses and qualifications of Section 14 and Appendix A2, the coefficient file below defines one integer polynomial Φ of total degree 4, with 72 925 nonzero monomials in 4 973 variables. For every $q \in \mathbb{Q}$,*

$$q \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists (w_1, \dots, w_{4972}) \in \mathbb{Q}^{4972}, \qquad \Phi(q, w_1, \dots, w_{4972}) = 0.$$

There are 4 972 *existential witnesses besides the original input. The coefficient bound is*

$$\|\Phi\|_\infty = 685783949153203534233890 < 2^{80}.$$

15.1 Literal specification and coefficient file

The complete coefficient table is `quartics/quartic-4973/polynomial.expanded.txt`. After one comment line, each line gives a nonzero signed integer coefficient followed by a nondecreasing list of variable indices. Repeated indices give their multiplicities as exponents; a coefficient alone is the constant. Thus `-6 0 2 2` denotes $-6X_0X_2^2$. The original input is X_0. Like terms are collected, and the rows are in lexicographic index-list order. The occurrence-frequency relabeling is recorded in `quartics/quartic-4973/variable-map.json` and fixes index zero. The gzip companion has exactly the same uncompressed bytes. The coefficient table has 1 622 053 bytes, including its comment and line endings, and SHA-256

`094f6a7ed17e7cc8d5d17a2f809127bbaffd0fe343cb2a32d18ac97e61279876`

This specifies every coefficient without an auxiliary choice of witnesses.

15.2 Arithmetic source and retained foundations

The selected source uses the following retained height, contact and reciprocal-logarithm proofs before the new source equivalence is established. An element of K is represented by two rational coefficients, with

$$(a_0, a_1)(b_0, b_1) = (a_0b_0 + 2a_1b_1, a_0b_1 + a_1b_0), \qquad a^{-1} = \frac{(a_0, -a_1)}{a_0^2 - 2a_1^2} \quad (a \neq 0).$$

The inverse denominator is nonzero by the irrationality of $\sqrt{2}$.

Theorem 15.2 (A sharper five-contact height estimate)**.** *Keep S_*, $\mathcal{D}$ and M_5 from Theorem 14.3, and put*

$$B_* = \prod_{p \le 76176} p, \qquad A_{19} = 2^{109}210^{1800}, \qquad H_{19} = 2A_{19}B_*^{19} + 128.$$

For every $a \in \mathbb{Q} \setminus \mathcal{D}$,

$$h(a) \le H_{19}M_5(a)^{19}, \qquad \log_2 H_{19} < 2093966.$$

Here h is the absolute logarithmic Weil height.

Proof. We use the geometric construction and compatible representations in the proof of Theorem 14.3, not merely its final exponent-24 inequality. Write $R = B_*M_5(a)$ and $C = 210^{300}$. The geometric comparison there is

$$h(a) \le 2\max(0, h_F(A)) + 128.$$

All h_F in this proof use the original stable Faltings metric. The CM estimate in that proof is $h_F(A) < 854R$.

In the non-CM case the same construction gives an absolutely simple quaternionic surface A/L, a normal field $L'/\mathbb{Q}$, and the final primitive weight-two newform f, after the finite twist has been absorbed, with

$$[L : \mathbb{Q}] \le 576, \quad d' = [L' : \mathbb{Q}] \le 2304R, \quad N \le CR^2, \quad \mathrm{rad}(N) \mid R.$$

Its two-dimensional compatible representation r_ℓ satisfies

$$V_\ell(A)|_{G_{L'}} \simeq (r_\ell \oplus r_\ell)|_{G_{L'}}, \qquad (\det r_\ell)^2|_{G_{L'}} = \chi_{\mathrm{cyc}}^2|_{G_{L'}}.$$

These are (LR3)–(LR5) and the determinant calculation in the existing modularity argument. There is no bounded-degree hypothesis on the coefficient field of f.

Let ϵ be the nebentypus of this final form. In the Tate-module convention, with arithmetic Frobenius, $\det r_\ell = \epsilon\chi_{\mathrm{cyc}}$; see [107, §12.5, (12.5.1)–(12.5.2) and Remark 12.5.1, pp. 119–120]. Equivalently, at a good prime its characteristic polynomial is $X^2 - a_p X + p\epsilon(p)$. A consistent contragredient convention replaces the finite character by its inverse, with the same kernel and order. The character ϵ^2 is trivial on $G_{L'}$, so its image order divides d'. Consequently

$$t = \mathrm{ord}(\epsilon) \le 2d' \le 4608R.$$

This bounds the character order, not the coefficient-field degree.

Put $H = \ker\epsilon \subset (\mathbb{Z}/N\mathbb{Z})^*$ and

$$\Gamma_H = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N) : d \bmod N \in H \right\}.$$

Since $\epsilon(-1) = 1$, this group contains $-I$. Let $X_H = X_1(N)/H$ for the diamond action and $J_H = \mathrm{Jac}(X_H)$. The curve, diamond action and quotient are defined over $\mathbb{Q}$. The rational model here parametrizes a point of exact order N, up to H, and is the model specified by [109, §2.1, (2.1.8)]; it introduces no cyclotomic field-degree factor.

Write $q : J_1(N) \to A_f$ for the newform quotient. Its pulled-back differentials are spanned by all Galois conjugates of f; see [37, Theorem 1(i),(iv), p. 526] and [108, §3.1]. Every $h \in H$ fixes those differentials, since its eigenvalue on each conjugate is a conjugate of $\epsilon(h) = 1$; the diamond convention is [108, §3.2.1]. Homomorphisms of abelian varieties in characteristic zero are determined by their differentials. Hence $q \circ \langle h \rangle = q$. For

$$e_H = \frac{1}{|H|} \sum_{h \in H} \langle h \rangle \in \mathrm{End}^0_{\mathbb{Q}}(J_1(N))$$

we therefore have $qe_H = q$.

Let H_{eff} be the image of H in $\mathrm{Aut}(X_1(N))$, and let $\pi : X_1(N) \to X_H$ be the quotient. On Jacobians,

$$\pi^*\pi_* = \sum_{h \in H_{\mathrm{eff}}} h, \qquad \pi_*\pi^* = [|H_{\mathrm{eff}}|].$$

The averages over H and H_{eff} agree. In particular the trivial action of -1 is not counted twice in the first identity. The image of e_H is thus $\mathbb{Q}$-isogenous to J_H, and its restriction of q is surjective. Poincare reducibility makes A_f a $\mathbb{Q}$-isogeny factor of J_H.

The accepted realization of r_ℓ in A_f puts it in $V_\ell(J_H)$. Over L', Faltings' Hom theorem and extension of scalars give a nonzero homomorphism $A \to J_H$. Absolute simplicity and Poincare reducibility give an L'-isogeny $A \times Z \to J_H$. In particular $g = \dim J_H \ge 2$. The Jacobian has its canonical principal polarization; no bound on a polarization degree of A or Z is being assumed.

The map $X_H \to X(1)$ has degree

$$d_H = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_H] = Nt \prod_{p \mid N} (1 + 1/p).$$

There is no factor of two here, because $-I \in \Gamma_H$. It is branched only over the cusp and the two elliptic points, so $\deg_B(X_H) \le d_H$. The congruence-curve genus bound gives $g \le 1 + d_H/12 \le d_H$, using $g \ge 2$. These degree and genus conventions, and the resulting height inequality, are recorded in [104, §4, (4.1)–(4.2), p. 8]. For every prime $p \ge 5$, $(1 + 1/p)^6 \le p$, while the possible factors at two and three have product at most two. It follows that

$$d_H \le 9216 C R^{19/6} < T, \qquad T = 2^{14} C R^{19/6}, \qquad g < T.$$

By [38, Theorem 1.1.1 and §2.3],

$$h_F(J_H) \le 13000000\, d_H^5 g < 2^{24} T^6.$$

This is the generic Belyi-curve estimate; no squarefree-level restriction or trivial-nebentypus assumption enters it.

Apply [103, Theorem 1.9(1), pp. 7–8] with source $(J_H)_{L'}$ and target $A \times Z$. Combining its degree bound with half-logarithmic isogeny-height variation gives

$$h_F(A) + h_F(Z) \le h_F(J_H) + V, \quad V = 2g^2 \left(8g^2 \log(7g) + \log d' + \log \max(1, \log d', h_F(J_H))\right).$$

The Bost lower bound in [104, (2.2), p. 5] gives $h_F(Z) > -3g/2$. Put $u = \log T$. Then

$$V \le 16 T^4 (2 + u) + 2T^2 (25 + 7u) \le T^6.$$

Indeed $\log d' < 8 + \log R \le 8 + u$, and the last logarithm in V is less than $17 + 6u$. For $T \ge 16$ the inequalities $2 + \log T \le T^2/32$ and $25 + 7 \log T \le T^4/4$ follow at 16 and remain true by differentiation. Thus

$$h_F(A) < (2^{24} + 1) T^6 + \tfrac{3}{2} T < 2^{25} T^6 = A_{19} R^{19}.$$

The earlier CM estimate is also less than this bound. The geometric comparison and $M_5(a) \ge 1$ prove the assertion. Finally $B_* < 2^{109445}$, $210 < 2^8$ and $A_{19} B_*^{19} > 64$ give

$$H_{19} < 4 A_{19} B_*^{19} < 2^{\,2 + 109 + 1800 \cdot 8 + 19 \cdot 109445} = 2^{2093966}.$$

□

Lemma 15.3 (Thirty-seven contacts for rational coefficients)**.** *Keep $K = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, $D = d^2$, and $E : y^2 = x^3 - Dx$ from Section 14. Let A, S, T, Z be nontorsion points of $E(K)$. Order the eight rational coefficients of $x(A), x(S), x(T), x(Z)$ as $a_1, \dots, a_8$. Put*

$$R_{\mathrm{c}} = 47027, \quad W_0 = 25878772920, \quad c = 18484837801, \quad V_0 = 32 W_0,$$

$$A_i = R_{\mathrm{c}} a_i + 3, \quad w_i = A_i v - u.$$

Use the following thirty-seven nonzero raw contacts:

$$r_0 = v/V_0, \qquad u - \delta v \quad (\delta \in \mathcal{D}), \qquad -(w_i - \delta v) \quad (1 \le i \le 8,\ \delta \in \mathcal{D}). \tag{33}$$

Here and below the order within $\mathcal{D}$ is $(0, 1, 2401, -343/32)$. The common five positions are numbered $0, \dots, 4$; the other positions are numbered $5, \dots, 36$ in the displayed order.

Every normalized odd contact of u/v or w_i/v outside S_* is detected by an odd valuation at a raw position belonging to that slope. If the radicals for both slopes $u/v, w_i/v$ have their eth powers dividing an integer $N \ne 0$, then, with $H = H_{19}$,

$$h(a_i) < 2H|N|^{19/e} + 14.$$

Consequently the two coefficients of any one of the four abscissas give

$$h(x(Q)) < 4H|N|^{19/e} + 30 < 7H|N|^{19/e} \quad (Q \in \{A, S, T, Z\}) \tag{34}$$

whenever all its required positions have those divisibilities.

Conversely, for any fixed eight rational coefficients, the inherited ordinary-prime construction supplies these raw contacts with every finite contact a positive selected prime and r_0 a positive rational square times a selected prime. Repeated positions may have equal values. These contacts have the positivity and local squareclasses used by Lemmas 15.10 and 15.11.

Proof. Nonzero raw contacts imply $v \neq 0$ and exclude the four finite branch values for every slope. At a prime outside S_* all differences of distinct members of $\mathcal{D}$ are units. Write m for the minimum of the valuations of the two coordinates of one slope. If an odd finite normalized contact has m even, its distinguished raw linear form has odd valuation. If m is odd, another finite form has valuation m. For an odd contact at infinity use the denominator in the first case and a finite form in the second. The unit V_0 and the signs do not change parity. This proof uses rational coordinates and also covers negative valuations; it does not presuppose that a contact is prime.

Apply Theorem 15.2 to the two slopes. The identity $R_c a_i + 3 = u/v + w_i/v$ and the elementary logarithmic height inequalities give the first bound, since $\log(6R_c) < 14$. For $x(Q) = a + b\sqrt{2}$ use $h(x(Q)) \leq h(a) + h(b) + h(\sqrt{2}) + \log 2$. This proves (34), since $H > 128$ and $|N| \geq 1$.

For the converse choose a positive integer D_0 with $D_0 a_i \in \mathbb{Z}$ for every i, after fixing the eight coefficients. This integer is a choice in the existence proof, not a variable of the source. Set

$$T_0 = V_0 D_0^2, \qquad p_0 = c + W_0 y, \qquad u = c + W_0 x, \qquad v = T_0 p_0.$$

For each distinct slope $b \in \mathcal{D} \cup \bigcup_i (A_i - \mathcal{D})$ put $K_b = T_0 b \in W_0\mathbb{Z}$ and $\Psi_b = u - K_b p_0$. Together with p_0, these are at most thirty-seven integer affine forms, each congruent to c modulo W_0. Distinct slopes give independent linear parts. Exact duplicate slopes are removed only while applying the prime theorem and restored at all their original positions afterward.

The value and successor admissibility in Section 14.4.1 remains valid. At primes dividing W_0 use the same residue c. If a prime $\ell > 29$ divides D_0, all K_b vanish modulo ℓ, and $u = p_0 = 1$ avoids zero and minus one. If $\ell > 29$ does not divide D_0, the change from (u, p_0) to (u, v) is invertible. For $\ell = 31, 37, 41$ there are at most eight slope residues, since $R_c a_i \equiv 0$; for $\ell \geq 43$ there are at most thirty-six. Choose distinct missing residues b_0, e_0 and order them so that $e_0 - b_0 \neq -T_0^{-1}$. Then

$$v = (e_0 - b_0)^{-1}, \qquad u = b_0 v$$

make every Ψ_b different from zero and minus one, and make $p_0 = v/T_0$ different from zero and minus one. Interchanging the two missing residues permits the required ordering, because ℓ is odd.

Choose a fixed positive cone $u/v > \max_i |A_i| + 2413$, $v > 0$. Enlarge the fixed initial cutoff past 82, every prime dividing D_0, and the nonzero determinants of the distinct forms. The inherited prime-pattern proof then applies with its same unused third coordinate and with complexity at most 35. For its fiber upper estimates, fixing $\Psi_{b_0} = p$ gives the same one-variable forms $\Psi_b = p + (K_{b_0} - K_b)p_0$. All coefficients, the cone, the denominator D_0 and the cutoff are fixed before the box grows. Thus the proof uses precisely the inherited fixed-system prime and successor estimates, not a uniform estimate in a growing denominator. It gives the positive ordinary primes and the same-prime elliptic-order conditions used there.

Now $r_0 = D_0^2 p_0$. At a prime $\ell \mid D_0$ outside S_*, both u and p_0 are units, $v_\ell(v) = 2v_\ell(D_0)$, and $v_\ell(A_i) \geq -v_\ell(D_0)$. Hence $w_i = A_i v - u$ is a unit. Both slopes have negative valuation $-2v_\ell(D_0)$,

and their normalized infinite depths are even. No new normalized odd contact is created at such a denominator prime. At p_0, the odd valuation of r_0 is one.

For each occurrence choose its private comparison witnesses sufficiently deep that its norm root is a unit at its assigned prime. The remaining denominator valuations in the pole argument occur in the square of that root. The raw contacts are positive, their reduced denominators are squares, and they are squares in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$ by the retained residue c and the square factor D_0^2. Lemmas 15.10 and 15.11 supply the selected ordinary-pole witnesses. Equal prime occurrences use the same orientation and may use separate private points. No condition on reduction orders at different primes has been added. □

Lemma 15.4 (Reciprocal logarithm and its error)**.** *Put $f = 1/x$, $z = -x/y$, and define*

$$a_0 = 1, \qquad a_1 = -\frac{2}{5}, \qquad a_j = \frac{4j-5}{4j+1}a_{j-1} \quad (j \geq 2), \qquad G(w) = \sum_{j\geq 0} a_j w^j.$$

On the formal branch at the identity,

$$\log_E(z) = zG(Df^2).$$

For every $J \geq 0$ put $G_J(w) = \sum_{j=0}^{J} a_j w^j$. For every integer $10 \leq e \leq 64$, set $m = \lceil e/4 \rceil$ and

$$G_{m-1}(w) = \sum_{j=0}^{m-1} a_j w^j, \qquad L_e(z,f) = zG_{m-1}(Df^2).$$

Let $p > 76176$ be split and choose either place above it. If the point is nonidentity and lies in the formal group at that place, so $r = v_p(z) \geq 1$ and $v_p(f) > 0$, then

$$v_p(\log_E(z) - L_e(z,f)) \geq (4m+1)r, \qquad \frac{L_e(z,f)}{\log_E(z)} \in 1 + p^e\mathbb{Z}_p.$$

Proof. The curve equation gives

$$f = z^2(1 - Df^2), \qquad w = Df^2, \qquad Dz^4 = \frac{w}{(1-w)^2}.$$

Coefficient comparison in the defining recurrence gives

$$4w(1-w)G'(w) + (1+w)G(w) = 1 - w.$$

Indeed the coefficient of w is $5a_1 + a_0 = -1$, and each later coefficient gives $(4j+1)a_j = (4j-5)a_{j-1}$. Differentiating the curve identity yields $z\,dw/dz = 4w(1-w)/(1+w)$. Consequently

$$\frac{d}{dz}\{zG(w)\} = \frac{1-w}{1+w} = (1 + 4Dz^4)^{-1/2}.$$

The square root has constant term one. This is the normalized invariant differential for z, and $zG(w)$ has zero constant term, which proves the formal identity.

At the stated place, D is a unit and the formal branch gives $v_p(f) = 2r$, hence $v_p(w) = 4r$. The denominator of a_j, for $j \geq 1$, divides $5\prod_{k=2}^{j}(4k+1)$ and therefore divides $(4j+1)!$. Thus

$$v_p(a_j) \geq -v_p((4j+1)!) \geq -\frac{4j+1}{p-1}.$$

For $j \geq m$ with $4j+1 < p$, the coefficient is p-integral and $v_p(a_j w^j) \geq 4mr$. Otherwise $j \geq 19044$, so $3j \geq 4m$ and $(4j+1)/(p-1) \leq j$. Therefore

$$v_p(a_j w^j) \geq 4jr - j \geq 4mr.$$

These lower bounds tend to infinity. They control the entire infinite tail, including terms whose denominators are divisible by p. Multiplication by z proves the first estimate. Formal logarithm isometry gives $v_p(\log_E(z)) = r$, and proves the relative estimate. □

Lemma 15.5 (Global nonvanishing)**.** *For every $f \in K$ and every $J \geq 2$,*

$$G_J(Df^2) \neq 0.$$

In particular $L_e(z, f) \neq 0$ whenever $z \neq 0$ and $10 \leq e \leq 64$.

Proof. Let ν_2 be the rational exponent-of-two valuation and put

$$v(a + b\sqrt{2}) = \tfrac{1}{2}\nu_2(a^2 - 2b^2) = \min\{\nu_2(a), \nu_2(b) + \tfrac{1}{2}\}.$$

The two terms in the norm have valuations of opposite parity when finite. Norm multiplicativity and the displayed minimum prove the valuation identities. Its finite values lie in $\frac{1}{2}\mathbb{Z}$, and $v(D) = 0$. Every a_j with $j \geq 1$ has valuation one, because $a_1 = -2/5$ and all subsequent recurrence factors have odd numerator and denominator. If $f = 0$, the assertion is immediate. If $t = v(f) \geq 0$, the constant term of $G_J(Df^2)$ is uniquely least in valuation. If $t < 0$, then $t \leq -1/2$ and the nonconstant valuations $1+2jt$ strictly decrease; the last is at most $1-J < 0$, below the constant term as well. In either case a uniquely least term proves nonvanishing. No good-reduction assumption at two is used. □

15.3 Conventions and inherited hypotheses

Retain the field and curve of Section 14:

$$K = \mathbb{Q}(\sqrt{2}), \qquad d = 3 - \sqrt{2}, \qquad D = d^2, \qquad E:\ y^2 = x^3 - Dx.$$

For an affine point with nonzero abscissa write $f(P) = 1/x(P)$ and $p(P) = y(P)/x(P)$. The rational function $z = -x/y$ extends with value zero at O and $T_0 = (0, 0)$; its other poles are $(d, 0)$ and $(-d, 0)$. The letter J denotes a common comparison point. We write $\jmath$ for $1/z(J)$, to distinguish that field element from a point or an integer index. The implementation conjugates the complete alternate-comparison identity: its boundary value is $1/\overline{z(J)}$, and its primary and target coefficients in that identity are conjugated as well.

The following arguments retain all inherited qualifications of Section 14 and Appendix A2. In particular, the restricted converse, rank and original arithmetic assertions retain their stated proof status. The historical prime-pattern source used arXiv:1009.3998v5 with its erratum. Here its inverse-theorem premise is supplied by Section I, with the published transference inputs [50, 51]; their stated hypotheses remain in force. The modularity step uses Lemma 4.1 and the published prime-five statement in [106, p. 624]. Its hypotheses, weight-two conclusion and finite twist are checked for the actual representation in that lemma. The source-status map in [1] separates this input from its proof ancestry and the preserved GL2-type/Serre reconstruction. The Yuan inputs used here are checked against Section 2.1 and Theorem 1.1 of the published Chinese article, including its metric definitions. The stack argument above does not use the integral coarse-model Lemma 2.1 or Theorem 2.2; no equivalence of the complete Chinese and English proofs is asserted.

The stable Faltings metric, arithmetic-Frobenius, diamond-quotient and final-twist conventions are unchanged. None of the finite source or polynomial checks strengthens these assumptions.

In particular we use, in exactly their previous scopes, the generator and torsion statements $2E(K) = \mathbb{Z}P_*$ and $E(K)_{\mathrm{tors}} = E[2](K)$; the formal-group logarithm and valuation statements; the fixed-system prime-pattern theorem with arbitrary prescribed finite exclusions; the height, scalar, anchor and Lucas lemmas; and the restricted-converse and simultaneous-realization inputs used for the pole predicate. The one additional explicit trace application below is identified in Lemma 15.12.

Put

$$B_* = \prod_{p \le 76176} p, \qquad H = 2(2^{109}210^{1800})B_*^{19} + 128, \qquad \log_2 H < 2093966.$$

This is H_{19} of Theorem 15.2. The source parameters remain high order 40, beta order 12, scalar orders $(1, 1, 1, 1, 32, 32)$ and anchor constant $C = 3^{2^{25}}$.

We also retain Lemma 15.4. Explicitly, write

$$\sqrt{1-w} \sum_{j \ge 0} \frac{\binom{2j}{j}}{4^j(4j+1)} w^j = \sum_{j \ge 0} a_j w^j, \qquad G_m(w) = \sum_{j=0}^{m} a_j w^j.$$

The polynomials used here are G_9 and G_2. They are nonzero at every argument Df^2 with $f \in K$, by the previously proved dyadic argument. For a nonidentity formal-group point at either place above a split prime $p > 76176$, the approximation $-G_9(Df^2)/p(P)$ has relative order at least 40, and $-G_2(Df^2)/p(P)$ has relative order at least 12. Actual formal membership is required: small z alone is not the premise of this lemma.

15.3.1 The retained scalar, anchor and beta interfaces

The scalar input is still $\beta = X^2 + 2$. The scalar batch tests the two rational coefficients each of $f(J_0), f(S), z(S)/z(J_0)$, with orders $(1, 1, 1, 1, 32, 32)$ in that order. Its original pole predicate is retained. The anchor constant is computed by

$$c_0 = 3, \qquad c_{i+1} = c_i^2 \quad (0 \le i < 25), \qquad C = c_{25}.$$

Thus the integer $C = 3^{2^{25}}$ is a witness value, not a polynomial coefficient. The four-witness anchor below consumes this same terminal value. The height targets are the eight rational abscissa coefficients of Lemma 15.3.

For every beta-bearing comparison, the common and beta quotient equations are explicitly

$$\Lambda p_H = G_H, \qquad B_A = C_A p_A, \qquad B_A \Lambda = G_A.$$

The global nonvanishing in Lemma 15.5 forces $\Lambda, p_H, B_A, p_A \neq 0$. Hence C_A is the ratio of the beta and common approximate logarithms, with their two minus signs cancelled. The beta residual is $C_A - 2\beta$. In full comparisons G_H has order 40 and G_A order 12; in beta-only comparisons both have order 12. The Lucas numerators use the normalized equations below, whose permitted zero-ordinate branches must not be excluded by this beta argument.

15.4 Selective comparisons and the local elliptic chart

Lemma 15.6 (Three-mode coverage)**.** *Number the unchanged raw contacts from* 0 *to* 36*. Retain full comparisons at* $0, \dots, 4$ *and* $21, \dots, 28$*, beta-only comparisons at* $5, \dots, 12$*, and Lucas-only comparisons at* $13, \dots, 20$ *and* $29, \dots, 36$*. The contacts carrying Lucas information cover the three points* S, T, Z*, and those carrying beta information cover* A, T*.*

Proof. The first five positions are shared. Each of A, S, T, Z has its original block of eight finite positions, in that order. Thus the Lucas set contains the five shared positions and all three finite blocks of S, T, Z; the beta set contains the five shared positions and the two finite blocks of A, T. The unchanged two-slope rational-height argument applies separately to these sets. No position is deleted when two slopes, raw values, or assigned primes coincide. The two maxima used below are consequently $B_L = \max(|s|, |t|, |z|)$ and $B_{AT} = \max(|n|, |t|)$. □

Lemma 15.7 (Doubled formal comparisons)**.** *At a split good prime above the cutoff, if $v(z(R)) > 0$, then either R or $R + T_0$ is in the formal group. Whenever the relevant rational function is defined,*

$$v(\log_E([2]R)) = v(z(R)),$$

with infinite valuation for $R = O$ or T_0.

Proof. For an affine nontorsion point, $z(R)^2 = x(R)/(x(R)^2 - D)$. If $v(x) = 0$, the right side has nonpositive valuation. If $v(x) < 0$, the point is formal. If $v(x) > 0$, translation by T_0 gives $x' = -D/x$ and $z' = -z$, so the translated point is formal with the same parameter valuation. Doubling kills T_0 and is an isometry on the formal group at an odd good prime. The asserted equality follows from the retained logarithm isometry. The two other nonzero two-torsion points are poles of z and are not in the domain. □

Private primary points may therefore lie in $E(K)$ rather than $2E(K)$. If $[2]P = [2\alpha(P)]P_*$, their coefficients $\alpha(P)$ may be half-integral. They are integral at each consumed odd prime. One compares logarithms only after writing an actual integer-multiple identity between formal points; torsion is not discarded by equating these half-indices. The primary f tests still put the primary points themselves in the formal group. An alternate test of $z(P - Q)/z(J)$, together with the opposite-place $f(J)$ test, implies $v(z(P - Q)) > 0$, and the lemma supplies the needed doubled logarithm.

Lemma 15.8 (Joint primary and difference graph)**.** *Let $Q = (a, aq)$ be nontorsion and $\jmath \neq 0$. Set*

$$F_Q = 1/a, \quad c = a + DF_Q, \quad r = \jmath/q, \quad s_Q = c/q, \quad t_Q = 1/q^2, \quad u = rs_Q.$$

The three field equations

$$k = (A - r)p, \tag{35}$$

$$kp + up - 2Drf - 4Dt_QA = 0, \tag{36}$$

$$2D(A + r)f + s_Qk + 4Dt_QA + \jmath q = 0 \tag{37}$$

give the exact graph $f = 1/x(P)$, $p = y(P)/x(P)$, $A = \jmath z(P - Q)$ for affine primary points P with $x(P) \neq 0$ and finite $z(P - Q)$. In particular this primary chart excludes O and T_0. The equations permit $A = 0$ and do not require inverses of A, k or of a difference parameter.

Proof. Set $x = p^2 + Df$. The target identity is $c^2 - q^4 = 4D$. Reversible linear combinations of the displayed equations give

$$AU = \jmath H_2, \qquad AV = \jmath H_1,$$

where

$$U = q(x + Df) + cp, \quad V = q^2(x - Df) - 4D, \quad H_1 = q(x + Df) - cp, \quad H_2 = x - Df - q^2.$$

Direct expansion yields

$$H_2V - H_1U = -4Dq^2(fp^2 + Df^2 - 1).$$

Thus $xf = 1$ and $P = (x, xp)$ is on E. If $U = V = 0$, then for $v \in \{d, -d\}$ one has $p = -2v/q$ and $x = v(a + v)/(a - v)$, precisely $P = Q + (v, 0)$. At either point $H_2 = (4D - q^4)/q^2 \neq 0$: equality $q^4 = 4D$ would imply $N_{K/\mathbb{Q}}(q)^2 = 28$, impossible for a rational norm. These genuine poles cannot satisfy the graph equations. Else U or V is nonzero and $A/\jmath$ is uniquely determined. The chord identities

$$HU = NH_2, \qquad HV = NH_1, \quad H = (p - q)(1 + DfF_Q), \quad N = pq(1 - DfF_Q) + 2D(f + F_Q)$$

identify it with $z(P - Q)$ on the chord domain and hence at every regular point. At $P = Q$ and $Q + T_0$ the value is zero. Opposition is covered by $V \neq 0$. At $p = 0$, the curve has $x = \pm d$, $U = 2qx \neq 0$, and the graph still gives its unique regular value. Conversely substituting any point and its regular difference value reverses these identities and recovers all three equations with $k = (A - r)p$. □

The last $p = 0$ cases are needed below. They are excluded at consumed places by the positive primary-f valuation, but need not be excluded globally for every normalized numerator.

15.5 Private conics, packed field tests and the ordinary pole

Each contact r has its own rational variables θ, s_c, S_c with

$$S_c = s_c^2, \qquad \theta^2 - rS_c = 2.$$

In particular $r, s_c \neq 0$. For a field value $F_i = a_i + b_i\sqrt{2}$ of required order e_i, define

$$Y_i = a_i + \theta b_i, \qquad T_i = 2a_iY_i - a_i^2 + 2b_i^2 = Y_i^2 - (\theta^2 - 2)b_i^2, \qquad r^{2e_i}W_i = T_i.$$

The upper tree uses successive pairs $N_r(u, v) = u^2 + rv^2$, starting with the literal leaf 1 followed by the W_i in stable order, carrying an unpaired final child unchanged. Denote its root by W.

Lemma 15.9 (Local conic and tree deductions)**.** *Assume the ordinary pole gives no negative odd valuation to nonzero rW^2 outside its fixed exceptional primes. At any $p > 76176$ with odd $R = v_p(r)$, the equations above imply $R > 0$ and choose one root $\rho \in \mathbb{Q}_p$ of 2 such that $v_p(a_i + \rho b_i) \geq e_i$ for every retained field.*

Proof. The conic makes $\delta = v_p(\theta^2 - 2) = R + 2v_p(s_c)$ odd. It cannot be negative, since then $v_p(\theta^2)$ would be odd; it cannot be zero. Thus $\delta > 0$, θ is a unit, and one root ρ satisfies $v_p(\theta - \rho) = \delta$ while the other factor is a unit. The summands in $N_r(u, v)$ have opposite valuation parity and cannot cancel. The literal leaf proves $W \neq 0$. If $R < 0$, ascent along that leaf gives $v_p(W) \leq 0$, contradicting the pole conclusion. Hence $R > 0$. The pole conclusion gives $v_p(W) > -R/2$. Descent through the tree uses

$$v_p(N_r(u, v)) > -cR \implies v_p(u) > -cR/2, \quad v_p(v) > -(c + 1)R/2,$$

with $0 \leq c < 1$, and gives $v_p(W_i) > -R$. Consequently $v_p(T_i) > (2e_i - 1)R \geq 2e_i - 1$, so $v_p(T_i) \geq 2e_i$. The two summands of T_i also have opposite parity, yielding $v_p(Y_i) \geq e_i$ and $\delta + 2v_p(b_i) \geq 2e_i$. It follows that $v_p((\theta - \rho)b_i) \geq e_i$ and hence the claimed field order. Zero summands are treated with infinite valuation. No integrality of a_i, b_i and no global squareclass of F_i was assumed. □

The proof uses $v_p(T_i) \geq 2e_i$, not the generally unjustified stronger bound $2e_iR$. It is independent of the elliptic meaning of the fields.

Lemma 15.10 (Conic completeness)**.** *In the retained integral-input construction, each contact $r = d_0^2 p_0 > 0$ admits the needed private conic and arbitrarily deep approximation to its chosen split branch, independently of the other contact occurrences.*

Proof. Here $p_0 \equiv 1 \pmod 8$ and $d_0 \neq 0$ is a p_0-unit. The ring $\mathbb{Z}[\sqrt{2}]$ is norm Euclidean: subtracting nearest integers from two coefficients leaves $|a|, |b| \leq 1/2$ and $|a^2 - 2b^2| \leq 1/2 < 1$. The split prime ideal therefore gives integers a, b with $a^2 - 2b^2 = p_0$, after multiplying by the norm-minus-one unit if needed. Since $b \neq 0$, $(\theta_0, s_0) = (a/b, 1/(d_0 b))$ is a rational conic point. The parametrization

$$\Delta = \frac{2(rs_0 - \theta_0 t)}{t^2 - r}, \qquad s_c = s_0 + \Delta, \quad \theta = \theta_0 + t\Delta$$

and rational approximation give simultaneous approximation at the finite set of required places. There is no rational pole $t^2 = r$. Near the selected root, $\delta = 1 + 2v_{p_0}(s_c)$ can be made arbitrarily large. After all field coefficients are fixed, choose it so that $\delta + 2v_{p_0}(b_i) \geq 2e_i$ for every i. The original field orders then make all W_i integral; the constant path makes W a p_0-unit. At the real place the tree is positive, even if some W_i are negative. The extra approximations used for the following pole may be imposed at the same time. No conic variable is shared between occurrences. □

Proposition 15.11 (Linear-unit ordinary pole)**.** *Set $g = 3 - 2\theta$. Replace only the ordinary pole module by*

$$\ell = gW, \quad z_\ell = \ell^2, \quad c_r = re, \qquad p_r^2 - 17r - 34c_r + 51ec_r = 0,$$
$$w = 17e - c_r z_\ell, \quad j_s = s_r^2, \quad k_r = rq_r,$$
$$j_s^2 + 34j_s w - 867w^2 - 17q_r k_r = 0.$$

It has the one-sided pole conclusion used in Lemma 15.9. It has integral-input completeness under the same inherited realization and rank hypotheses. The scalar-height pole module is unchanged.

Proof. At an odd prime with odd $v_p(r)$, the conic argument makes θ integral, and $(3-2\theta)(3+2\theta) = 1 - 4rs_c^2$ is a unit. Thus g is a unit. Suppose $b = rW^2 \neq 0$ has negative odd valuation outside $2, 3, 17$. The first conic $p_r^2 = 17r(1-e)(1+3e)$ forces e to be a unit: positive, negative or infinite valuation of e would make the right side have odd valuation. Since $v_p(\ell) = v_p(W)$, one has $v_p(w) = v_p(b) < 0$, an odd integer. In

$$(s_r^2 - 17w)(s_r^2 + 51w) = 17rq_r^2$$

the left side is nonzero with even valuation, since the valuations of s_r^2 and w differ in parity. The right side is zero or has odd valuation. This contradiction proves the asserted exclusion, including zero ordinate branches of the first conic. If $W = 0$, the conclusion has no nonzero pole input; explicit witnesses are $\ell = z_\ell = p_r = q_r = k_r = 0$, $e = 1$, $c_r = r$, $w = 17$, $s_r = 17$, $j_s = 289$.

For completeness retain the generalized fixed-U realization argument under its original hypotheses. For $b > 0$ with square reduced denominator and square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, it applies to a prescribed U_0 when

$$U_0^2/(17b) < 1 \text{ at infinity}, \quad v_2(U_0^2/(17b)) \geq 3, \quad v_3(U_0^2/(17b)), v_{17}(U_0^2/(17b)) \geq 1,$$

and $v_p(U_0) = v_p(b)$ at every other odd-support prime of b. Indeed, put $B = 17b$ and $V = B(B - U_0^2)$, and let $\beta_{\mathrm{sf}}, \eta_{\mathrm{sf}}$ be their positive squarefree representatives. The displayed conditions make V positive and a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. Consequently

$$\beta_{\mathrm{sf}} > 1, \quad \beta_{\mathrm{sf}} \equiv 17 \pmod{24}, \quad \eta_{\mathrm{sf}} \equiv 1 \pmod{24}, \quad 17 \nmid \eta_{\mathrm{sf}}.$$

At an odd-support prime of B outside $2, 3, 17$, the square reduced denominator makes $v_p(b)$ positive, and the hypothesis on U_0 gives $v_p(U_0^2/B) > 0$. Thus $1-U_0^2/B$ is a square unit, so η_{sf} is a square in $\mathbb{Q}_p$ and $p \nmid \eta_{\mathrm{sf}}$. The same conclusion at 17 follows from the displayed smallness condition. Conversely, at a prime dividing η_{sf} the necessary strict cancellation in $B - U_0^2$ gives $v_p(B) = 2v_p(U_0)$ and makes B/U_0^2 a square unit. Thus β_{sf} is a square in $\mathbb{Q}_p$. In particular $\gcd(\beta_{\mathrm{sf}}, \eta_{\mathrm{sf}}) = 1$, and both mutual-square conditions hold. If $U_0 = 0$, then $V = B^2$ and $\eta_{\mathrm{sf}} = 1$, so the converse condition is empty. These are precisely the hypotheses consumed by the inherited simultaneous-realization proof; no new rank or converse statement is substituted.

Now set $b = rW^2$ and $U_0 = rW^2 g$. In the complete construction $r = d_0^2 p_0$ and W is a p_0-unit, so $v_{p_0}(b) = 1$ and all other valuations of b are even. Hence its reduced denominator is a square. The retained local squares of r at $2, 3, 17$ also make b a square at those places. After the fields are fixed, approximate the private conic also near $(\theta, s_c) = (3/2, 1/(2\sqrt{r}))$ at infinity, $2, 3, 17$. These local points exist by the original contact construction. The continuous function $(r/17)W(\theta)^2 g(\theta)^2$ vanishes there, so it satisfies the displayed smallness conditions while the selected prime still makes W a unit. The local-unit argument gives the required odd-support valuations of U_0. The resulting old three-conic witnesses satisfy

$$p_o^2 = 17b(1-e)(1+3e), \quad h_o = e(17b - U_0^2), \quad (s_o^2 - 17bh_o)(s_o^2 + 51bh_o) = 17bq_o^2.$$

Here r, W, b are nonzero. Set

$$p_r = p_o/W, \quad w = h_o/b, \quad s_r = s_o/b, \quad q_r = q_o/(r^2W^3).$$

These substitutions give $w = e(17 - r\ell^2)$ and the final conic in the displayed normalized system; the other coordinates are its definitions. Thus the witness conversion uses only stated nonzero quantities. Equivalently the unnormalized form has $L = r\ell$, $z_L = r^2 z_\ell$, $w_L = r^2 w$, $s_L = r s_r$, $q_L = r^2 q_r$, $j_L = r^2 j_s$, $k_L = r^2 k_r$, with e, p_r, c_r unchanged. All divisions in the reverse scaling are powers of the conic-proved nonzero r. This also proves that normalization loses no zero branch. □

Lemma 15.12 (Four-witness anchor)**.** *For rational a, b, both belong to $\mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$ if and only if $9 + 3(a^2 + b^2)$ is a norm-one trace for the quaternion algebra $(2, 3)$. Consequently the original two-coefficient anchor is equivalent to*

$$CH - 9C^2 - 3f_0^2 - 3f_1^2 = 0, \qquad H^2 - 8U^2 - 12V^2 + 24W_a^2 - 4C^2 = 0.$$

Proof. By Definition 4, Observation 5 and Facts (a), (b), (c), (e) in the proof of Proposition 6 of [99], the algebra is ramified exactly at $2, 3$, the trace set is unrestricted at the other places, and it is integral at its two ramified places $2, 3$, contains integral traces with residues 1 at 2 and 0 at 3, and is determined by these local trace conditions. These are the explicit trace facts used here, in addition to the older sum-of-traces criterion. If $m = \min(v_p(a), v_p(b)) < 0$, then at 3 the sum of squares has valuation $2m$, while at 2 it has valuation $2m$ for unequal valuations and $2m + 1$ for equal finite valuations. Thus $9 + 3(a^2 + b^2)$ would be nonintegral, proving one implication, with zeros covered by infinite valuation. Conversely integral a, b give a trace congruent to zero at 3. At 2 it is odd or belongs to $4\mathbb{Z}_2$. For the latter case put $x_0 = h/2 \in 2\mathbb{Z}_2$, where h is this trace. If $x_0 \equiv 2 \pmod 4$, then $(x_0^2 - 1)/3 \equiv 1 \pmod 8$ is a square, giving $x_0^2 - 3v^2 = 1$. If $x_0 \equiv 0 \pmod 4$, then $1 - x_0^2/6$ is a square, and $u = v = 1$, $w^2 = 1 - x_0^2/6$ gives $x_0^2 - 2u^2 - 3v^2 + 6w^2 = 1$. The local trace criterion supplies rational witnesses. Finally divide the two homogeneous equations by C^2, which is allowed because the retained constant C is nonzero. This is an equality of existential projections, not a rational formula choosing quaternion witnesses. □

15.6 Local and normalized Lucas tests

Write

$$\mathcal{F}(s,t,z) = t^2 - s(s-2)tz + (s-2)^2z^2 - 1.$$

Lemma 15.13 (Local Lucas residual)**.** *The primary and alternate logarithm comparisons, before any proxy equality is used, give local integral quotients C_s, C_t, C_z congruent to the integer root indices modulo p^{40}. An order-40 test of $\mathcal{F}(C_s, C_t, C_z)$ implies $p^{40} \mid \mathcal{F}(s,t,z)$.*

Proof. The implication follows by substitution into the polynomial with integer coefficients. One may evaluate the field residual using

$$V = (C_s - 2)C_z, \quad D_L = C_sC_t - V, \quad E_L = C_t^2 - VD_L - 1.$$

These are quadratic equations, and $E_L = \mathcal{F}(C_s, C_t, C_z)$. They replace the three proxy comparisons and the global proxy equation. They do not impose $E_L = 0$ as a field equality. Removal of the proxy coordinates requires their exclusive incidence in that equation and those comparison arguments. Integer completeness and the reverse input implication use the two height contradictions below; no conversion of arbitrary old proxy witnesses is asserted. □

Lemma 15.14 (Normalized local Lucas evaluation)**.** *Let $\Lambda p_H = G_H$ be the common negative-logarithm equation. For the three Lucas numerators replace their quotient blocks and the preceding evaluation by*

$$\begin{aligned} B_i\Lambda = G_i \quad (i = s,t,z), \qquad & PB_s = p_s, \\ (U+V)p_t = PB_t, \qquad & K_Lp_t = (1-2P)B_t, \\ Vp_z = (1-2P)B_z, \qquad & E_L = U^2 - K_LV - P^2. \end{aligned}$$

An order-40 test of E_L has the same required integer consequence. Every integer input still admits the changed witnesses.

Proof. The global nonvanishing of the retained G_i implies $\Lambda, p_H, B_s, B_t, B_z \neq 0$. Also $p_t \neq 0$: otherwise the two equations with left factor p_t would force both $P = 0$ and $1 - 2P = 0$. At a consumed place all primary points are formal, so their p_i are nonzero. Put $C_i = B_i/p_i$. Then $P = 1/C_s$ and direct substitution gives

$$E_L = P^2\mathcal{F}(C_s, C_t, C_z).$$

Since C_s is locally integral, the implication $v(E_L) \geq 40 \Rightarrow v(\mathcal{F}(C_s, C_t, C_z)) \geq 40$ is valid even when C_s is not a unit. No local denominator has been discarded.

For completeness the integer $s \neq 0$ and all root points are fixed before the prime-pattern theorem is used. Its prescribed finite-exclusion version allows every chosen prime to avoid the divisors of s. Then C_s is a unit at the consumed place. The old order-40 residual implies the new one, and the displayed equations assign the new witnesses. This imposes no compatibility between group orders at different primes.

Globally p_s or p_z may vanish at an occurrence not being consumed in the local argument. Such points have $x = \pm d$ and are covered by the regular extension in Lemma 15.8; they must not be excluded by reusing an earlier blanket nonzero assertion. At a consumed place $C_s = 2$ gives $P = 1/2$ and $V = K_L = 0$, since both p_t and p_z are nonzero there. No division by $1 - 2P$ is needed. Globally the branch $P = 1/2$, $p_z = 0$ forces $K_L = 0$ but need not force $V = 0$; the latter value may be chosen in the completeness construction. □

The retained field lists can now be stated without proxy coordinates. In a full occurrence there are the five primary f values, the common $\overline{f(J)}$, the beta residual, E_L, and five alternates. In a beta-only occurrence there are two primary f values, the common field, the beta residual and two alternates. In a Lucas-only occurrence there are four primary f values, the common field, E_L and four alternates. These lists have lengths $13, 6, 10$. The formal fields all have order one. The beta residual and beta alternate have order 12; the Lucas residual and Lucas alternates have order 40. The denominator alternate has order $e_0 = 40$ in full and Lucas-only occurrences, and $e_0 = 12$ in beta-only occurrences. Its G precision is the same e_0.

Theorem 15.15 (The two height contradictions)**.** *Retain the inherited scalar and contact-height conclusions, and the integer root indices n, s, t, z of the chosen carrier, with $s, z \neq 0$. Let $\beta = X^2+2$ and $\beta' = 2\beta$. Suppose the local tests above give $C_i \equiv n_i \pmod{p^{e_i}}$ at all consumed contacts, with beta residual $C_A - \beta'$. Put $H_* = 5H$. Suppose*

$$n_Q^2 - 6 < 7H_* N^{19/e}, \qquad h(\beta') < a/40, \qquad \log(40H_*) < a/40, \qquad a = |s| > (24H_*)^{10},$$

where the first inequality is the contact-height bound whenever the appropriate radical powers divide N. Then the changed arithmetic source has the same integer-input conclusion as Version 3.

Proof. If $\Delta = \mathcal{F}(s,t,z) \neq 0$, every relevant order-40 radical power divides $|\Delta|$. By three-mode coverage, all three heights in $B_L = \max(|s|,|t|,|z|)$ satisfy the assumed inequality. Since $|\Delta| < 5B_L^4$, the same elementary estimate as in Version 3 gives

$$B_L^{2-76/40} < 24H_*.$$

This contradicts $B_L \geq a > (24H_*)^{10}$. Therefore $\Delta = 0$. The displayed lower bound on a gives $|s| \geq 7$, and $z \neq 0$ was retained. The Lucas lemma therefore gives the stronger individual conclusion $|t| > \exp(a-3)$, not merely a bound on $\max(|t|,|z|)$.

If $\beta' \neq n$, put $N = |\operatorname{num}(\beta' - n)|$. The beta contacts cover A, T, so with $B = \max(|n|,|t|)$ one has

$$B^2 < 7H_* N^{19/12} + 6, \qquad N \leq 2e^{h(\beta')}B.$$

Thus $B^{2-19/12} < 40H_* e^{(19/12)h(\beta')}$ and

$$\log B < \frac{12+19}{40(24-19)}a = \frac{31}{200}a < a - 3,$$

contrary to $B \geq |t|$. Hence $\beta' = n$ is an integer. $\beta' = 2\beta = 2(X^2+2)$; writing $X = u/v$ in lowest terms gives $v^2 \mid 2$, hence again $v = 1$. □

15.7 General root carrier

Lemma 15.16 (Integer doubled root indices)**.** *Let $R = (1+\sqrt{2}, 2\sqrt{2})$, so direct duplication gives $2R = P_*$. Every $Q \in E(K)$ has a unique integer n_Q with $2Q = n_Q P_*$. It is nonzero exactly when Q is nontorsion. For such a point,*

$$h(x(Q)) > n_Q^2/4 - 4.$$

Proof. The inherited generator assertion gives $2Q = n_Q P_*$. The torsion assertion makes its kernel exactly $E[2](K)$, so equivalently $Q = n_Q R + \tau$, $\tau \in E[2](K)$. Uniqueness follows from the infinite order of P_*. The unchanged height bounds give

$$n_Q^2 - 6 \leq h(x(2Q)) \leq 4h(x(Q)) + \log 676.$$

Since $(6 + \log 676)/4 < 4$, the stated inequality follows. □

The root points A, S, T, Z, J_0, J may now be any nontorsion points of $E(K)$. Their indices always mean the integer doubled indices of the lemma. They are distinct from the half-integer coefficients used for private points in the earlier carrier convention.

Lemma 15.17 (Small general-root charts)**.** *For A, S, T, Z retain x, y and the actual first-doubling square $q = x^2$, and impose $y^2 = xq - Dx$. For S also retain $xf = 1$, $yz + x = 0$. Keep the nonzero tangent denominator, or the direct tangent equation of Lemma 15.19. For each of J_0, J use f, z, k with*

$$k = fz, \qquad f = z^2 - Dk^2,$$

with $z \neq 0$ enforced by its retained inverse. These charts reconstruct exactly the required nontorsion root points.

Proof. The first equations are the curve equation with a reused square. The nonzero tangent denominator excludes all points with $y = 0$, hence all torsion. The two extra S equations preserve its reciprocal and parameter. In the other chart, $z \neq 0$ implies $f \neq 0$: $f = 0$ would force $k = 0$ and then $z = 0$. Set $x = 1/f$, $p = -1/z$, $y = xp$. Dividing the second equation by z^2 gives $fp^2 = 1 - Df^2$, so $P = (x, y)$ lies on the curve and $xy \neq 0$. Conversely every nontorsion point gives these witnesses. These are rational maps on the stated nonzero domains. The new equations cannot justify deleting an unrelated guard without its incidence proof. □

For clarity, the inherited scalar argument still applies to the new indices. At each odd contact prime of β outside S_*, the constant-left scalar norm tree excludes negative odd raw valuation and gives all six assigned coefficient orders. The first four put S, J_0 in the formal group at every place above that prime. Writing $2S = sP_*$ and $2J_0 = j_0P_*$, with nonzero integer j_0, the last two give $v_p(s/j_0) \geq 32$, hence $v_p(s) \geq 32$. The logarithm comparison uses the exact identity $[2j_0]S = [2s]J_0$. It is valid also at inert primes: both rational coefficients are tested and the extension is unramified outside two. The unchanged scalar raw polynomial and its small-prime correction detect every prime of $M_5(\beta)$, including an odd infinite contact. Consequently

$$M_5(\beta)^{32} \mid s, \qquad h(\beta) \leq HM_5(\beta)^{19} \leq H|s|^{19/32}.$$

The height theorem's domain condition holds. The value $\beta = X^2 + 2$ is none of $0, 1, -343/32$, and $\beta = 2401$ would require the nonsquare integer 2399 to be a rational square. Doubling both indices therefore changes neither the scalar ratio nor its divisibility argument.

Take $H_* = 5H$ and $\beta' = 2\beta$. The unchanged contact-height bound $h(x(Q)) < 7HN^{19/e}$ and the preceding lemma give

$$n_Q^2 - 6 < 28HN^{19/e} + 7 < 35HN^{19/e} = 7H_*N^{19/e}.$$

The scalar ratio is still s/j_0, since doubling both indices cancels the factor two. Its order-32 tests give $h(\beta) \leq H|s|^{19/32}$. The full anchor at both 2 and 3 is retained. Its 3-adic consequence, using $2S = sP_*$, is

$$s \in 2 \cdot 3^{2^{24}-1}\mathbb{Z} \setminus \{0\}, \qquad \log_2 |s| > 25165822.5.$$

Here formal doubling at 3 is an isometry and the retained calculation is $v_3(f(sP_*)) = 2 + 2v_3(s)$. The 2-adic part of the anchor is not replaced by this consequence. Since $\log_2 H_* < 2093969$, one has

$$|s| > (24H_*)^{10}, \qquad |s| > (40H_*)^{32/13}, \qquad h(2\beta) < |s|/40, \quad \log(40H_*) < |s|/40.$$

For example the first two required binary cutoffs are respectively $10 \cdot 2093974$ and $(32/13) \cdot 2093975$, both below the displayed anchor lower bound. Thus Theorem 15.15 applies with unchanged orders. Both beta-bearing comparators must now test $C_A - 2\beta$; the scalar definition $\beta = X^2 + 2$ and its raw scalar polynomial remain unchanged.

15.7.1 Doubled targets

Lemma 15.18 (Targets $2Q_i$ and P_*)**.** *The general-root construction may use targets $2Q_i = n_iP_*$ and fixed denominator target $P_* = 2R$, requiring only one doubling of each variable root. It preserves the local integer congruences and integer completeness.*

Proof. For soundness write $2X_0 = hP_*$, $2X_i = m_iP_*$ and $2J = j_dP_*$. The root J is nontorsion, so $j_d \in \mathbb{Z} \setminus \{0\}$. The doubled differences have indices $h-2$ and $m_i - 2n_i$. Lemma 15.7 and the alternate tests give

$$v_p((h-2)/j_d) \geq e_0, \qquad v_p((m_i - 2n_i)/j_d) \geq e_i,$$

where $e_0 = 40$ in full/Lucas mode and $e_0 = 12$ in beta-only mode. Since j_d is integral, $h \equiv 2$ and $m_i \equiv 2n_i$ at those orders. In particular h is a unit at the odd prime. The exact identity $[2h]X_i = [2m_i]X_0$ permits logarithm comparison on the primary formal group, giving ratio m_i/h. The retained relative errors, with $e_i \leq e_0$, give $C_i \equiv n_i \pmod{p^{e_i}}$.

For completeness let $N_{p,1}, N_{p,2}$ be the two full finite group cardinalities supplied by the inherited ordinary-prime pattern. It gives $\gcd(N_{p,1}, N_{p,2}) = 4$ and $p \nmid N_{p,1}N_{p,2}$. Each reduction has exactly four rational two-torsion points, so its doubling image has order $M_{p,i} = N_{p,i}/4$. Consequently $M_{p,1}$ and $M_{p,2}$ are coprime and prime to p. Every reduction of a doubled global point is killed by the corresponding $M_{p,i}$; no global reduction-surjectivity or cyclicity claim is needed. Put $j_c = \operatorname{lcm}_p M_{p,2}$ and $J = j_cP_*$, whose doubled index is $j_d = 2j_c$. Separately for each contact choose

$$k \equiv 0 \pmod{M_{p,1}}, \quad k \equiv 1 \pmod{M_{p,2}}, \quad k \equiv 1 \pmod{p^{40+v_p(j_c)}}, \qquad k \notin \{0, 1, -1\}.$$

The three moduli are compatible. Set $X_0 = kP_*$ and $X_i = kn_iP_*$. The primary points are formal at the first place. Their differences from the chosen targets, and J, are formal at the second. The alternate logarithm ratios are $(k-1)/j_c$ and $(k-1)n_i/j_c$, while the primary ratios are n_i. This supplies every required order. The finite exclusions avoid zero points and the old open-chart exceptional cases. There is no simultaneous CRT condition on different contact occurrences. □

The fixed target can be specified directly by

	1	$\sqrt{2}$
$x(P_*)$	$57/8$	$-7/2$
$q = y(P_*)/x(P_*)$	$-87/41$	$129/164$
$F_Q = 1/x(P_*)$	$456/1681$	$224/1681$
$s_Q = (x(P_*) + DF_Q)/q$	$-55123/14637$	$19301/58548$
$t_Q = 1/q^2$	$7624/14161$	$39904/127449$

Direct substitution verifies the curve and target identities. The source stores x, q, F_Q and the conjugates of s_Q, t_Q. Its remaining coefficients are affine in $\jmath$:

$$r = \jmath\left(-\frac{232}{357} + \frac{86}{357}\sqrt{2}\right), \qquad u = \jmath\left(\frac{291673}{127449} - \frac{29440}{42483}\sqrt{2}\right),$$

when $\jmath = 1/\overline{z(J)}$. The ten constant bindings and four affine bindings replace the fixed derivation, subject to full incidence checks. Their isolated coefficient calculation does not bound the complete sum of squares.

15.8 Exact implementation refinements

Lemma 15.19 (Direct tangent equation)**.** *For a curve point with $q = x^2$, replace the inverse and slope equations $(2y)h = 1$, $\lambda = (3q - D)h$ by $2y\lambda = 3q - D$. This preserves all retained rational coordinates.*

Proof. The new equation and the curve force $y \neq 0$. If $y = 0$, then $x = 0$ or $x^2 = D$, and $3q - D$ is respectively $-D$ or $2D$, a contradiction. The unique inverse is restored by $h = (2y)^{-1}$. There is also the polynomial restoration

$$h = \frac{2\lambda(3q - 2D) - 9xy}{4D^2},$$

because $(3q - D)(3q - 2D) - 9x(x^3 - Dx) = 2D^2$. For a retained second doubling its input is on the curve: with $t = X - x$, $Y_0 = -y - \lambda t$, the identity

$$\begin{aligned} Y^2 - X^3 + DX = y^2 - x^3 + Dx + t(2y\lambda - 3x^2 + D) \\ + t^2(\lambda^2 - X - 2x) + (Y - Y_0)(Y + Y_0) \end{aligned}$$

proves curve preservation from the first doubling equations. Target-two sources have only the first four variable-root doublings; no second set of savings or equations is present in that branch. □

Lemma 15.20 (Shared field-square evaluation)**.** *For the common field $F = a + b\sqrt{2} = \overline{f(J)}$, bind once $c = a^2 + 2b^2$, $d_s = 2ab$. In each designated order-one packed field replace*

$$Y = a + \theta b, \qquad PW = 2aY - a^2 + 2b^2$$

by $PW = c + \theta d_s$, with the same $P = r^2$. This is an exact projection preserving W, every higher norm node and the pole witnesses.

Proof. Substitution gives $2a(a + \theta b) - a^2 + 2b^2 = c + \theta d_s$. The inverse map restores $Y = a + \theta b$, without division. All zero cases are included. The two coefficients describe F^2, but their packed use still enforces the original order-one condition on F; they must not be relabelled as a new order-one test of F^2. The same identity permits reuse of an already available primary-f square in its own packed numerator, after checking that square's defining rows. □

Lemma 15.21 (First packed denominator and norm fusion)**.** *For the first order-one primary field only, replace its virtual denominator $W_0 = T_0/r^2$ by $W_0 = T_0/r$, and its first norm $N = 1 + rW_0^2$ by*

$$T_0 = 2a_0Y_0 - a_0^2 + 2b_0^2, \qquad rN - r - T_0^2 = 0.$$

Keep every other denominator, the conic, all higher tree nodes and the ordinary pole. The arithmetic input projection is unchanged.

Proof. The earlier tree proof still gives $v_p(W_0) > -R$, so $v_p(T_0) > 0$. More generally $v_p(T) \geq 2e - 1$ suffices for an order-e field test: opposite parity gives $v_p(Y) \geq e$ and $\delta + 2v_p(b) \geq 2e - 1$, hence $\delta + v_p(b) \geq e + (\delta - 1)/2 \geq e$. Thus the first field still has order one, and every other deduction is unchanged. For completeness the old depth-two choice has $v_{p_0}(T_0) \geq 2$ and $R = 1$, so the new virtual leaf is integral and the root remains a unit. For fixed nonzero r, all new tree values remain rational polynomials in θ, allowing the same exceptional-place pole approximation.

After that arithmetic change, the displayed fusion is exact: set $T_0 = rW_0$ forward and $W_0 = T_0/r$ backward. The retained conic proves $r \neq 0$. These maps preserve N and all later nodes, including $N = 0$ or $T_0 = 0$ and negative rational r. The unguarded case $r = 0$ would be invalid and is not admitted. The old r^2 coordinate remains for other leaves. The changed field is primary $f(H)$, not the separately shared common field $\overline{f(J)}$. □

15.9 Composition, completeness and finite realization

For integral input first set $\beta = X^2 + 2$ and take $A = \beta P_*$, whose doubled index is 2β. Choose J_0, S together in the old subgroup with the required scalar and full anchor divisibilities. Their doubled-index ratio is unchanged. The Lucas existence lemma then gives t, z for the now fixed s, and take $T = tR$, $Z = zR$. Fix the eight abscissa coefficients and denominator data before using the prime-pattern theorem. Exclude the divisors of the fixed nonzero s as well as every earlier finite exceptional set.

Next choose one common comparison point J using the stated least common multiple, then choose the private points separately for each occurrence, using precisely the target-two CRT rule. Construct all joint-graph, reciprocal-G, normalized-ratio and shared-square fields. Only then choose each private conic point and its norm/pole witnesses by simultaneous approximation. Repeated primes retain their common orientation but separate conic and private-point witnesses. The constructions above give every required equation. In the reverse direction the conic and pole decode field orders at every odd-contact prime, without an ordinariness assumption. The graph and doubled logarithm arguments give integer-index congruences, and the two height contradictions give the integer input. The exact fusions preserve the specified retained coordinates; arithmetic source replacements use the full soundness and completeness argument rather than a claim about arbitrary old private witness tuples.

Finally each selected source must be compiled afresh into rational quadratic residual equations with the original input coordinate retained. For real, hence rational, values the all-weights-one sum of their squares vanishes exactly when every residual vanishes. Clearing nonzero rational row denominators preserves those zero sets and gives an integral polynomial of degree at most four. This elementary last step does not prove the inherited arithmetic inputs. The exact source, complete flattening map, allowed algebraic transformations, and full coefficient collection must be checked for the single selected endpoint. None of the local identities, allocation counts or isolated coefficient bounds in this appendix is an assertion of that final check.

Theorem 15.22 (The selected arithmetic source)**.** *With precisely the inherited arithmetic hypotheses and source-version qualifications stated above, the source consisting of the selected equations has rational input projection $\mathbb{Z}$. Its input coordinate is the original rational number X.*

Proof. First suppose that rational witnesses satisfy the source. At every prime above the cutoff with an odd raw-contact valuation, the private conic, linear-unit pole and constant-left norm tree give the required field orders, by Lemmas 15.9 and 15.21. This argument starts at arbitrary odd-contact primes and does not assume that they are ordinary. The joint graph supplies actual primary points and their regular difference parameters. Positive primary-f valuations put the primary points in the formal group; the common conjugate-f test and each alternate test allow Lemma 15.7 to be applied to the doubled difference. Lemma 15.18, with denominator order 12 in beta-only mode and 40 otherwise, gives the integer-index congruences for each retained numerator. The beta quotient equations and Lemma 15.14 therefore give the precise local tests used in Theorem 15.15.

The retained scalar system gives its order-32 divisibility, while the four-witness anchor is equivalent to the original two-place condition. The general-root height and anchor estimates establish every cutoff of that theorem with $H_* = 5H$. The three-mode position sets cover exactly the two maxima needed in its proof. It follows first that $\mathcal{F}(s, t, z) = 0$, then that $2\beta = n \in \mathbb{Z}$. Write $X = u/v$ in lowest terms with $v > 0$. Since $2X^2 = 2\beta - 4$ is integral, v^2 divides 2; hence $v = 1$ and $X \in \mathbb{Z}$.

Conversely fix any $X \in \mathbb{Z}$. Set $\beta = X^2 + 2$ and $A = \beta P_*$. Choose the scalar pair S, J_0 in the retained subgroup using the inherited scalar and full anchor construction, and choose t, z by the

inherited Lucas existence lemma for the resulting nonzero integer doubled index s. Set $T = tR$, $Z = zR$. All four roots and the two common roots are nontorsion and admit the general-root charts. Fix their rational abscissa coefficients and denominator data before applying Lemma 15.3. Its fixed-system prime theorem permits the finite exclusion of divisors of s, as required by the normalized Lucas construction.

Choose one common J by the least common multiple in Lemma 15.18; at each occurrence choose its private points using that lemma's three compatible congruences. No congruences between different occurrences are imposed. The joint graph, exact reciprocal evaluators, beta quotients and normalized Lucas equations then have the stated witnesses and local orders. The explicit order of choices in Section 15.9 now chooses each private conic after its field coefficients are fixed. Conic approximation supplies the integral packed leaves and a unit norm root; the fixed-parameter pole realization supplies its remaining witnesses. This remains true with the first denominator changed to r. The shared square and direct tangent equations are exact identities with the reconstruction maps proved above. Thus every equation of the selected source has a rational solution at this input. This proves both implications without identifying arbitrary old and new private witness tuples. □

The finite evaluations used here can themselves be written quadratically. For precision 40 bind four field pairs

$$w = Df^2, \qquad w_2 = w^2, \qquad w_3 = w_2 w, \qquad U_6,$$

with the fourth pair defined by the quadratic equation

$$U_6 = \frac{a_3 + a_4 w + a_5 w_2 + w_3(a_6 + a_7 w + a_8 w_2 + a_9 w_3)}{a_6}.$$

The quadratic output expression is

$$G_9(w) = a_0 + a_1 w + a_2 w_2 + a_6 w_3 U_6, \qquad a_6 = -\frac{418}{16575} \neq 0.$$

For precision 12 bind only $w = Df^2$ and use $G_2(w) = 1 + a_1 w + a_2 w^2$. Substitute these expressions directly into $\Lambda p_H = G_H$ and the beta or normalized-Lucas equations $B_i \Lambda = G_i$. Every product is quadratic in the bound rational coordinates. Expansion gives each required a_j coefficient exactly. The scale is one; no finite-error estimate is used as a substitute for these polynomial identities.

15.10 Exact source specification and audit boundary

The source just proved has 4 935 rational coordinates, including input zero, and 4 513 residual equations. Its distinct name in `quartics/quartic-4973/source-entry.json` is

`target2-tangent-first-norm-normalized-linear-pole-k40-q12-a25-h19-s32-4a82d9a8dc1b`

The compressed source `quartics/quartic-4973/source-residuals.jsonl.gz` has SHA-256

`94f8a1972cbfc0cf6493d418e628a18482307dec5c427db72e86d3eacb1a6f5d`

Its uncompressed bytes have SHA-256

`4a82d9a8dc1b5c92eb40a49751c0a6aae073c8f765ca5e276a74802fd56b84be`

The source contains 11 modules. Its complete construction map is `quartics/quartic-4973/construction-map.json`. The independent source audit checks the full module equations, exact ordered expression lists and contact positions, root and scalar interfaces, all retained guards, the anchor terminal coordinate and complete flattening. It is recorded in `quartics/quartic-4973/source-audit.json`. The source-auditor SHA-256 is

Those checks establish finite source identities. The source's rational input projection is proved in Theorem 15.22, with the inherited qualifications unchanged. A source digest is not an arithmetic proof, and the historical base compiler remains construction machinery rather than another final polynomial.

15.11 Quadratic reductions preserving the rational input projection

We describe the algebraic reductions applied to the new, fixed source. All scalar coordinates range over $\mathbb{Q}$, and coordinate zero is the original input X. Each field equation denotes its two rational coefficient equations. These reductions prove equality of rational input projections to that source; its arithmetic interpretation is Theorem 15.22.

Definitions, row operations, and interfaces. An equation $cv - F = 0$, with $c \in \mathbb{Q}^*$ and F independent of v, permits substitution $v = F/c$ in every other equation and deletion of v. Restriction is the forward solution map, and adjoining $v = F/c$ is its inverse. Each use below retains degree at most two after substitution. The initial quadratic lift uses the reverse operation: introduce targets for products of two available coordinates until the source residuals are quadratic. Evaluating these product definitions and forgetting their targets give the two solution maps. A previously defined product can be reused only with its retained defining equation.

An operation replacing R_i by $aR_i + bR_j$, while retaining R_j, has inverse $R_i = (R_i' - bR_j)/a$ when $a \neq 0$ and $i \neq j$. Thus every recorded linear row operation preserves the simultaneous zero set. A shear replaces an old witness by $v = v' + cw$, with fixed rational c and another witness w retained. Its forward and reverse coordinate maps are $v' = v - cw$ and $v = v' + cw$. This is an invertible rational coordinate change and preserves quadratic degree.

Every removed internal coordinate must have no occurrence outside the replaced rows, unless the stated substitution is made in *every* such row. Retained coordinates keep their identities and values. All new coordinates are fresh, and input coordinate zero is never a target of deletion or redefinition. These conditions apply to the current system after all preceding transformations.

Elimination modulo retained equations. For a recorded relation elimination, a current defining row has the form $x - N(y) = 0$ up to nonzero rational scale, with N quadratic and independent of the deleted witness x. For each other affected row $F_j(x, y)$ the certificate gives an exact polynomial identity

$$F_j(N(y), y) = G_j(y) + \sum_k M_{jk}(y) H_k(y),$$

up to the recorded nonzero row scales. Every H_k is an unchanged, x-free row, and every new row G_j is at most quadratic. Forgetting x takes an old solution to a new one, since the H_k vanish. Conversely adjoin $x = N(y)$: the same identities recover every affected $F_j = 0$. Complete occurrence checks ensure that no other row uses x. This is an identity over $\mathbb{Q}$, not a division by a possibly zero field value. The retained support rows are never replaced in this operation.

The recorded contact-power circuits. The circuit replacements preserve the boundary values r^2, r^{24}, r^{80}. The replacement definitions are

$$r_2 = r^2, \quad r_4 = r_2^2, \quad r_8 = r_4^2, \quad r_{16} = r_8^2, \quad r_{24} = r_{16} r_8, \quad r_{40} = r_{24} r_{16}, \quad r_{80} = r_{40}^2.$$

Their exact outputs agree with the old acyclic product definitions. For either direction retain r and the boundary outputs, and assign each internal target its displayed power of r. These maps work also for $r = 0$ and use no inverse. Each deleted internal target has no use outside the replaced definitions; every new internal target is fresh. The certificate checks those conditions and the exact output identities in the current system. Only circuit events with these retained powers are included in this description.

A quadratic presentation of the parity block. This replacement concerns only its rational equations, with unchanged input b. Its old private variables t, U, e, p, q, s, h satisfy

$$U(1 + bt^2) = b(1 - bt^2), \qquad p^2 = 17b(1 + 2e - 3e^2),$$
$$h = e(17b - U^2), \qquad 17bq^2 = (s^2 - 17bh)(s^2 + 51bh).$$

Use instead twelve private rational variables $u, v, a, z, e, p, c, w, s, q, j, k$ and nine quadratic equations

$$z = u^2, \quad a = bv, \quad z + va = b^2, \quad c = be,$$
$$p^2 - 17b - 34c + 51ec = 0, \qquad w - 17bc + cz = 0,$$
$$j = s^2, \qquad k = bq, \qquad j^2 + 34jw - 867w^2 - 17qk = 0.$$

For the forward map put $\delta = 1 + bt^2$. It is nonzero: otherwise $bt^2 = -1$ and the first old equation gives $0 = 2b$, a contradiction. Keep e, p, s, q and set

$$u = U, \quad v = 2bt/\delta, \quad a = bv, \quad z = U^2, \quad c = be, \quad w = bh, \quad j = s^2, \quad k = bq.$$

The identity $[b(1-bt^2)]^2 + b(2bt)^2 = b^2(1+bt^2)^2$ proves the first conic; substitution proves the other equations.

Conversely $u^2 + bv^2 = b^2$. Set $h = e(17b - z)$ and keep e, p, s, q. If $b = 0$, take $U = t = 0$. If $b \neq 0$ and $u + b \neq 0$, take $U = u$ and $t = v/(u + b)$. In this case the first old residual, multiplied by $(u + b)^2$, is

$$(u - b)(u + b)^2 + b(u + b)v^2 = (u + b)(u^2 + bv^2 - b^2) = 0.$$

If $b \neq 0$ and $u = -b$, then $v = 0$; take $U = b$ and $t = 0$. In every case $U^2 = z$, and the remaining old equations follow from $c = be$, $w = bh$, $j = s^2$, and $k = bq$. The possible sign change of U is harmless because all old witnesses and deleted lifting coordinates are private. These maps prove equality of existential projections; they are not asserted to be inverse on the private witness tuples. This rule is applied only at the certificate's height-parity blocks; the ordinary comparison pole is the linear-unit source predicate already proved above. The actual multiplicity is given in the following census.

15.12 Finite realization and proof of the quartic theorem

The fresh quadratic lift introduces only defining product equations. Their evaluation and forgetting maps preserve rational solutions while fixing the original input. The certificate then applies exactly the operations below, under the current-row and complete-incidence hypotheses just proved. The resulting file `quartics/quartic-4973/residuals.json.gz` contains 4 552 nonzero integer residuals of degree at most two. Every weight in `quartics/quartic-4973/weights.json` is one, and

$$\Phi(X_0, \ldots, X_{4972}) = \sum_{i=1}^{4552} R_i^2.$$

At rational values each square is nonnegative, so this sum vanishes if and only if all residuals vanish. Nonzero rational row normalization preserves each row's zero set before this integral sum is formed.

The certificate `quartics/quartic-4973/certificate.jsonl.gz` records 551 transformations:

height-parity replacements	1
contact-power circuit replacements	13
invertible linear row operations	266
rational definition eliminations	121
invertible rational shears	2
eliminations modulo retained equations	148

Its SHA-256 is

`81ca768459a13b76dbdf4bfe81e94fbab9ab99f14a5731f6b1a1b03e28c1df2b`

The independent verifier `quartics/quartic-4973/verify_certificate.py` has SHA-256

`b7180b01029b3d9364d94914401a4f1f66b190762f10b99a38e9fd4f49437ffe`

It checks the named source binding, fresh lift, every recorded polynomial identity and reconstruction map, defining order, current-row supports, every occurrence of each removed coordinate, final quadratic degrees, and the bijective variable map fixing input zero. It then compares the complete collected integer expansion, including every cross term. The endpoint verification receipt is `quartics/quartic-4973/evidence/final-verification.json`.

Proof of Theorem 15.1. Theorem 15.22 proves both rational-input implications for the selected arithmetic source. For any fixed rational input, every defining lift and every recorded transformation has the forward and reverse solution maps proved above. Their composition preserves the existence of rational witnesses and keeps that input fixed. Hence the final residual system has rational input projection exactly $\mathbb{Z}$. The all-one sum of its squares vanishes precisely on its simultaneous zero set, so the same equivalence holds for Φ.

The literal coefficient table is the verified complete integral expansion of that sum. Its nonzero degree-four term makes its degree exactly four; the collected coefficients give the stated monomial count and maximum. The checked variable map gives the stated ambient coordinates, one of which is the input. Thus all numerical claims and the input-projection equivalence concern this one polynomial. □

The finite checks do not replace the inherited rank, height, prime-pattern, formal-group, norm or pole arguments. Their qualifications remain those stated in the source proof and Appendix A2.

16 Arithmetic interfaces for the smaller quartic

Lemma 16.1 (The original scalar certificate at order thirty-two)**.** *Let $K = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, $D = d^2$, and $E : y^2 = x^3 - Dx$, with the point P_* of Section 14. Put*

$$\beta = X^2 + 2, \qquad r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1), \qquad X \in \mathbb{Q}.$$

For $h_0, a \in \mathbb{Z} \setminus \{0\}$ and points $J_0 = h_0 P_*$ and $S = aP_*$, take the two rational coefficients, in the basis $1, \sqrt{2}$, of $f(J_0), f(S), z(S)/z(J_0)$ in that order. Denote them by $X_1, \dots, X_6$, and put

$$(m_1, \dots, m_6) = (1, 1, 1, 1, 32, 32), \qquad l_i = X_i / r^{m_i}.$$

Use the same ordered norm tree as the original scalar construction:

$$\begin{aligned} N_r(u,v) &= u^2 + rv^2, \\ n_0 &= N_r(1, l_1), \quad n_1 = N_r(l_2, l_3), \quad n_3 = N_r(l_5, l_6), \\ W &= N_r(N_r(n_0, n_1), N_r(l_4, n_3)). \end{aligned}$$

Let

$$c \in \left\{ \pm \prod_{p \in I} p : I \subseteq \{2, 3, 5, 7, 11, 13, 17\} \right\}.$$

If $b = crW^2 \neq 0$ satisfies the original pole predicate $\Psi(b)$, then

$$M_5(\beta)^{32} \mid a, \qquad h(\beta) \leq H_* |a|^{3/4}.$$

Here M_5 and the exact constant H_* are those of Theorem 14.3; in particular $1 \leq H_* < 2^{4783227}$.

For every integral X, the scalar certificate is complete with successively chosen positive indices h_0, a. One may require h_0 to be divisible by any prescribed positive integer and a to be arbitrarily divisible and arbitrarily large. After S has been chosen, the normalized-leaf conditions and the nonvanishing of W persist when h_0 is replaced by every integer in one suitable congruence class $h \equiv h_0 \pmod{L}$. The modulus L may also be required to be divisible by any prescribed positive integer.

Proof. The value r is nonzero: $\beta \geq 2$, and $\beta = 2401$ would give $X^2 = 2399$, which has no rational solution. At a prime $p > 76176$, the multiplier c is a unit. Every prime in $M_5(\beta)$ has odd valuation in r, by the contact computation in the proof of Theorem 14.3. One can also observe that a negative valuation of $\beta = X^2 + 2$ is even, so only its finite contacts contribute here.

Write $R = v_p(r)$. The pole obstruction in Lemma 14.4 and $b \neq 0$ make the odd valuation $R + 2v_p(W)$ positive. In a norm node $u^2 + rv^2$ the two nonzero summands have opposite valuation parity, so there is no cancellation. The constant leaf on the all-left path first excludes $R < 0$. Then $v_p(W) > -R/2$. Descending the tree, the implication

$$v_p(N_r(u,v)) > -tR, \quad t < 1 \quad \Longrightarrow \quad v_p(u) > -tR/2, \quad v_p(v) > -(t+1)R/2$$

shows that every leaf l_i has valuation greater than $-R$. Thus $v_p(X_i) > (m_i - 1)R \geq m_i - 1$, and integrality of the valuation gives $v_p(X_i) \geq m_i$. This is the original norm argument with the two assigned orders changed to thirty-two.

The order-one tests put J_0 and S in the formal group at each place above p. Such a completion is unramified, and the formal logarithm is an isometry on that group. Since $[h_0]S = [a]J_0$, it follows that

$$v_{\mathfrak{p}}\left(\frac{z(S)}{z(J_0)} \right) = v_p(a/h_0).$$

Both rational coefficients of this ratio have valuation at least thirty-two, so $v_p(a/h_0) \geq 32$. As h_0 is a nonzero integer, $v_p(a) \geq 32$. This proves $M_5(\beta)^{32} \mid a$. Theorem 14.3 now gives $h(\beta) \leq H_* M_5(\beta)^{24} \leq H_* |a|^{3/4}$. No exponent-nineteen height estimate is used.

For completeness, let $X \in \mathbb{Z}$. The integer r is nonzero and $|r| > 1$. The original seven-prime squareclass basis, proved in the finite-correction argument following Lemma 14.7, permits a choice of c such that $cr > 0$ and cr is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. Let $\mathcal{S}$ be the finite prime support of the integer cr.

At the places above $\mathcal{S}$, choose h_0 sufficiently divisible that J_0 is in arbitrarily deep formal subgroups. This remains possible after requiring any fixed divisibility of h_0, because these subgroups

have finite index in the local elliptic groups. Having fixed this nonzero J_0, choose a sufficiently divisible that $S = aP_*$ is still deeper. The coefficients of $f(J_0), f(S)$ and $z(S)/z(J_0)$ then attain every finite lower valuation bound needed here. This includes ramified and bad primes, by working in sufficiently deep formal subgroups. We arrange that all normalized leaves are integral when $p \mid r$ and divisible by p when $p \mid c$ but $p \nmid r$. The index a may simultaneously exceed any fixed bound.

At $p \mid r$, the root of the displayed tree is congruent to one modulo p, because $r \equiv 0$ and the all-left leaf is one. At $p \mid c$, $p \nmid r$, all other leaves vanish modulo p, so the root is again congruent to one. Thus W is a unit on $\mathcal{S}$, and in particular is nonzero. At these primes $v_p(crW^2) = v_p(cr) \geq 0$; outside $\mathcal{S}$ its valuation is even. Hence b has square reduced denominator, is positive, and has the same local squareclasses as cr at two, three, and seventeen. Lemma 14.7 supplies the scalar pole witnesses.

Finally, with S fixed, these scalar tests are rational functions of J_0 defined at the chosen point. Their coefficient maps are continuous in the product of the completions above each prime in $\mathcal{S}$. The finite lower valuation bounds persist in sufficiently small neighborhoods. A common finite-index subgroup of the integer orbit gives a congruence $h \equiv h_0 \pmod{L}$ contained in all those neighborhoods. Enlarging L preserves that property. The same unit calculation proves the asserted stability for the final shared point. □

Lemma 16.2 (The fixed multiplier seventeen with preparation zero)**.** *Let $b \in \mathbb{Q}^\times$, put $B = 17b$, and define $\Psi_0(b)$ by the existence of $e, p, q, s \in \mathbb{Q}$ satisfying*

$$p^2 = B(1-e)(1+3e), \qquad h = e(B-b^2), \qquad Bq^2 = (s^2 - Bh)(s^2 + 3Bh),$$

where h is the displayed expression, not a further free choice. Then $\Psi_0(b)$ excludes negative odd valuations of b at every prime outside $\{2, 3, 17\}$. Conversely, it holds whenever b has square reduced denominator, is positive and square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and

$$b < 17, \qquad v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2.$$

No additional nonzero condition on e, p, q, s, h is imposed.

Proof. Taking the preparation witnesses $t = 0$ and $U = b$ in (13) gives exactly the displayed equations. The obstruction therefore follows from Lemma 14.5, including its degenerate witness cases.

For completeness put

$$V = B(B - b^2) = B^2(1 - b/17).$$

The real inequality makes V positive. The valuation bounds make $1 - b/17$ a square at two, three, and seventeen: its difference from one has valuation at least three at two and at least one at the two odd primes. Thus V is square at these three places.

Let μ and θ be the positive squarefree representatives of B and V. The prescribed squareclasses of b give $\mu \equiv 17 \pmod{24}$; in particular $\mu > 1$ and its support contains seventeen but neither two nor three. Also $\theta \equiv 1 \pmod{24}$, and seventeen is absent from its support. At a prime $\ell \mid \mu$ outside the three exceptions, $v_\ell(b)$ is positive and odd, since the reduced denominator of b is square. Hence $1 - b/17$ is a square unit there, so V and θ are local squares. The same assertion at seventeen was already arranged. These facts make the supports disjoint.

Conversely, if $\ell \mid \theta$, then $\ell \notin \{2, 3, 17\}$. If $v_\ell(b) > 0$, the valuation of V is $2v_\ell(b)$; if $v_\ell(b) < 0$, it is $3v_\ell(b)$, again even because the denominator is square. Thus an odd valuation of V forces $v_\ell(b) = 0$ and $1 - b/17 \equiv 0 \pmod{\ell}$. It follows that $B = 17b \equiv 17^2 \pmod{\ell}$ is a nonzero square unit. This proves the other mutual-squareness condition.

Lemma 14.6 now gives the two rational points in the squareclass of B. To make the conversion explicit, write $V = \theta c^2$ and scale the second point to the curve with parameter nV. If the first abscissa is Ba^2, scale both points and parameters by a so that their abscissas are B and Bs^2, with $s \neq 0$, and their curve parameters are e_0 and $e_0 V$. For their ordinates y_1, y_2, set

$$e = e_0/B, \qquad p = y_1/B, \qquad q = y_2/(B^2 s), \qquad h = e(B - b^2).$$

The first curve equation gives $p^2 = B(1-e)(1+3e)$. Since $e_0 V = B^2 h$, the second gives $Bq^2 = (s^2 - Bh)(s^2 + 3Bh)$. These are the required witnesses. The only arithmetic existence theorem used in this specialization is the existing simultaneous-realization lemma, with its stated restricted-converse and descent dependencies. □

Lemma 16.3 (Real bounds independent of later progression moduli)**.** *Let G be the closure of the integer orbit*

$$\{([n]P_*, [n]P_*^{\sigma}) : n \in \mathbb{Z}\} \subset E(\mathbb{R}) \times E^{\sigma}(\mathbb{R}),$$

and let G^0 be its identity component. Fix a finite collection of rational functions formed from nonzero integral multiples and fixed translates of either elliptic coordinate, with all denominators nonzero as rational functions on the relevant curve. Suppose their pole set is contained in a finite union of inverse images of finite sets under these nonzero multiplication-and-translation maps on either factor. There is a bound C such that every nonconstant integer progression contains arbitrarily large positive indices at which all these functions are defined and their real absolute values are at most C. The bound is chosen from the functions before the progression modulus is specified.

If $m = |G/G^0|$, $m \mid L$, and $m \mid h_0$, the progression $h \equiv h_0 \pmod{L}$ is dense in G^0, and its positive tail meets every nonempty open subset of G^0 infinitely often.

Proof. The real elliptic groups are compact Lie groups. Their closed subgroup G is a compact abelian Lie group with finitely many components, and G^0 is a torus. The projection of G^0 onto each elliptic identity circle is surjective. Indeed the projected integer orbit is infinite, since P_* has infinite order and each real embedding is injective on the group of algebraic points. If the connected projection were trivial, the projected G would be finite; a nontrivial connected closed subgroup of a circle is the whole circle.

Each nonzero multiplication map on a circle has finite fibers. Consequently each of the prescribed inverse images of a finite set has Haar measure zero in every component of G. The finite union of pole sets cannot cover a component. Choose in each component a nonempty open neighborhood whose compact closure avoids the poles. Continuity bounds all the fixed rational functions on these finitely many compact sets by one constant C.

For a progression $h = r + kn$, its closure is $[r]P_* + [k]G$. Multiplication by the positive integer k maps G^0 onto itself, so this closure contains an entire component of G. The positive tail of the progression is dense in its closure: compactness gives unbounded positive multiples tending to the identity, so each negative multiple is also a limit of positive multiples. It therefore meets the chosen neighborhood in that component at arbitrarily large positive indices.

If $m \mid k$, then $[k]G = G^0$. If also $m \mid r$, the coset is G^0 itself. Apply this with $k = L$ and $r = h_0$ for the final assertion. □

17 The shared-point quartic in 962 variables

This section gives a new complete quadratic compilation of the shared-point construction in the public agentic notes [9]. The shared comparison points, reciprocal-trace detectors, fixed height

scales, direct Lucas test, and affine anchor change are arithmetic and algebraic improvements recorded in those notes. The simultaneous prime theorem used to choose the shared points is Sun's Theorem 4.1, in the precise form of Theorem A8.1 below. Our contribution here is an independently specified gate program, its source verification, and further changes of coordinates and circuit identities leading to a 962-variable polynomial after 32 certified eliminations. The definition does not depend on an unavailable coefficient table. The relevant passages in the public notes are shared-point construction, fixed height scales, and quadratic compilation and anchor translation.

We retain the arithmetic conclusions and conventions of Section 14, including the restricted-converse input to Lemma 14.4. In this section the height estimate is Theorem 14.3, with exponent 24 and its original constant H_*; it is not the exponent-19 estimate used for the preceding quartic. Thus

$$1 \le H_* < 2^{4783227}.$$

The exact finite compilation and coefficient assertions below require no arithmetic conjecture. Identifying the rational input projection with $\mathbb{Z}$ uses the stated arithmetic results and the simultaneous prime theorem.

The point, pole, and correction variables below specify the arithmetic source predicate. The final compilation subsection gives explicit maps to its implementation coordinates.

17.1 Points, local parameters, and quotient domains

Use the normalized curve and generator already established in Section 14:

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2, \quad E : y^2 = x^3 - Dx,$$

$$P_* = (57/8 - 7\sqrt{2}/2, -165/8 + 417\sqrt{2}/32), \qquad 2E(K) = \mathbb{Z}P_*.$$

The last assertion follows from Proposition 3.1 by the unit change of variables recorded there. Conjugation of $K/\mathbb{Q}$ is denoted by σ. All field equations below mean both rational component equations in the basis $1, \sqrt{2}$.

The source uses nine nonidentity points

$$A, S, T, Z, H, U_A, U_S, U_T, U_Z \in 2E(K).$$

Each is represented as the double of an affine halfpoint $(\alpha, \eta) \in E(K)$, using four rational halfpoint coordinates. The equations for a double are

$$\eta^2 = \alpha^3 - D\alpha, \qquad 2\eta m = 3\alpha^2 - D, \qquad x = m^2 - 2\alpha, \qquad y = m(\alpha - x) - \eta. \tag{38}$$

Products of degree greater than two in this display are evaluated by the quadratic gates specified below. Set

$$Q_0 = P_*, \quad (Q_1, Q_2, Q_3, Q_4) = (A, S, T, Z), \quad (U_0, U_1, U_2, U_3, U_4) = (H, U_A, U_S, U_T, U_Z).$$

Compute $[4]Q_i$ by two doublings, or by fixed rational arithmetic for Q_0, and compute

$$V_i = U_i - [4]Q_i \quad (0 \le i \le 4).$$

For the subtraction of (x', y') from (x, y) we retain an inverse of $x - x'$ in K and use

$$m = (y + y')/(x - x'), \quad x'' = m^2 - x - x', \quad y'' = m(x - x'') - y.$$

Lemma 17.1 (Domains of the point quotients)**.** *The equations* (38) *force* $\eta \neq 0$ *and describe a finite nonidentity point of* $2E(K)$*. Every such point has nonzero abscissa and ordinate. The guarded differences* V_i *have the same properties. Consequently* $f(Q) = 1/x(Q)$ *and* $z(Q) = -x(Q)/y(Q)$ *are defined and nonzero at every point where they are used here.*

Proof. If $\eta = 0$, the curve equation gives $\alpha = 0$ or $\alpha^2 = D$. The right side of the slope equation is then $-D$ or $2D$, respectively, a contradiction. The ordinary duplication formulas therefore apply. They give

$$x([2](\alpha, \eta)) = \left(\frac{\alpha^2 + D}{2\eta}\right)^2 \neq 0,$$

because $D = d^2$ and the numerator is positive at either real embedding. An ordinate-zero point has abscissa $0, d$, or $-d$. The latter two are not squares in K: their rational norm is 7, whereas the norm of a square is a rational square. Thus the doubled ordinate is nonzero. A guarded affine difference is finite and is again in $2E(K)$, so the same argument applies. Conversely any nonidentity point of $2E(K)$ has an affine halfpoint of nonzero ordinate and is represented by these formulas. □

Write

$$a_0 = 1, \quad a_1 = -2/5, \quad a_j = \frac{4j-5}{4j+1} a_{j-1} \ (j \geq 2), \qquad G_J(w) = \sum_{j=0}^{J} a_j w^j,$$

and, for $e = 16, 52$, put

$$\ell_e(Q) = z(Q) G_{e/4-1}(Df(Q)^2). \tag{39}$$

Lemma 15.5 applies to both truncation degrees 3 and 12. Thus all these values are nonzero globally.

Lemma 17.2 (The logarithm estimate at unramified places)**.** *Let* $p > 76176$ *be a rational prime and* $\mathfrak{p} \mid p$ *a place of* K*. If a nonidentity point* Q *is in the formal group at* $\mathfrak{p}$*, normalize* $v(p) = 1$ *and put* $r = v(z(Q))$*. Then* $r \geq 1$*, and for* $e \in \{16, 52, 64\}$*,*

$$v(\log_E Q) = r, \qquad v(\log_E Q - \ell_e(Q)) \geq (e+1)r, \qquad \ell_e(Q)/\log_E Q \in 1 + p^e \mathcal{O}_{K_\mathfrak{p}}.$$

Here ℓ_{64} *is defined by the same formula with* G_{15}*. Ratios with numerator precision* e *and denominator precision* e' *in this set have the valuation of their full logarithm ratio and relative error of valuation at least* $\min(e, e')$*.*

Proof. The local extension is unramified, including when p is inert. The curve has good reduction and D is a unit. On the formal branch,

$$f = z^2(1 - Df^2), \qquad w = Df^2, \qquad v(w) = 4r.$$

The recurrence gives $4w(1-w)G'(w) + (1+w)G(w) = 1 - w$ and hence the same identity $\log_E Q = zG(w)$ as in Lemma 15.4. For $j \geq 1$ the denominator of a_j divides $(4j+1)!$, so

$$v(a_j) \geq -\frac{4j+1}{p-1}.$$

Set $m = e/4 \leq 16$. If $j \geq m$ and $4j + 1 < p$, then $v(a_j w^j) \geq 4mr$. Otherwise $j \geq 19044$ and

$$v(a_j w^j) \geq 4jr - j \geq 3jr \geq 4mr.$$

The lower bounds tend to infinity, proving the estimate for the entire tail. Every nonconstant term of $G(w)$ has positive valuation, so the leading term gives $v(\log_E Q) = r$. Division proves the relative estimate, and division of two relative estimates proves the ratio assertion. No step uses residue degree one. □

17.2 The complete rational source

The ordinary input is X, and $\beta = X^2 + 2$. Define

$$(k_1, k_2, k_3, k_4) = (16, 52, 52, 52), \qquad L_i = \frac{\ell_{k_i}(U_i)}{\ell_{52}(H)}, \qquad L_i' = \sigma\Big(\frac{\ell_{k_i}(V_i)}{\ell_{52}(V_0)}\Big).$$

The sixteen field tests D_j, with their orders m_j, are exactly

$$\begin{array}{c|c} f(U_0), \dots, f(U_4),\ \sigma f(V_0), \dots, \sigma f(V_4) & 1 \quad (10) \\ L_i - L_i' \ (1 \le i \le 4) & 16, 52, 52, 52 \\ L_1 - \beta & 16 \\ F(L_2, L_3, L_4) & 52, \end{array} \tag{40}$$

where

$$F(U, V, W) = V^2 - U(U-2)VW + (U-2)^2 W^2 - 1. \tag{41}$$

There is no rational proxy for an integer point label.

For $j = 1, \dots, 4$ write $x(Q_j) = x_{j,0} + x_{j,1}\sqrt{2}$ and set

$$\mathcal{A}_0 = 1, \quad \mathcal{A}_{2j-1} = x_{j,0}^2, \quad \mathcal{A}_{2j} = (1 + x_{j,1}^2)^2.$$

These are positive nonzero rational squares. In fact $x(Q_j)$ is positive at both real embeddings by Lemma 17.1, so $x_{j,0} > 0$. Retain one rational coordinate λ and put

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad F_5(T) = \prod_{\theta \in \Theta} (T - \theta), \qquad t_r = \mathcal{A}_r \lambda, \quad \delta_r = t_r - 1.$$

For every $0 \le r \le 8$, require $t_r \notin \Theta \cup \{1\}$ and define

$$R_r = \mathcal{A}_r^{-5} F_5(t_r)/\delta_r^6, \qquad \widetilde{D}_{r,j} = D_j/\delta_r^{m_j+1}, \tag{42}$$

$$W_r = \prod_{j=1}^{16} \Big(1 + R_r(\widetilde{D}_{r,j}/R_r^{m_j})^2\Big), \qquad T_r = \mathrm{Tr}_{K/\mathbb{Q}}(W_r^{-1}), \qquad B_r = R_r/T_r^2. \tag{43}$$

Require $W_r, T_r \neq 0$. For $B = B_r$ retain four rational witnesses e_r, s_r, p_r, q_r and impose

$$\begin{aligned} C_r &= 17 B_r, \qquad h_r = e_r(C_r - B_r^2), \\ p_r^2 &= C_r(1 - e_r)(1 + 3e_r), \\ C_r q_r^2 &= (s_r^2 - C_r h_r)(s_r^2 + 3 C_r h_r). \end{aligned} \tag{44}$$

The first two expressions are arithmetic definitions. The last two are the preparation-zero specialization of the original pole predicate. There is no nonzero guard on e_r, s_r, p_r, q_r, or h_r.

The scalar block shares the point H. Define

$$r_\beta = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

It is nonzero for rational X: the only positive possible root not excluded by $\beta \ge 2$ would require $X^2 = 2399$. Let $Y_1, \dots, Y_6$ be, in order, the two rational components of $f(H), f(S), z(S)/z(H)$, and set

$$(e_1, \dots, e_6) = (1, 1, 1, 1, 32, 32), \qquad y_i = Y_i / r_\beta^{e_i}, \qquad N_r(u, v) = u^2 + r v^2.$$

The scalar norm tree has no padding leaf. It is

$$\begin{aligned} n_0 &= N_{r_\beta}(1, y_1), \quad n_1 = N_{r_\beta}(y_2, y_3), \quad n_3 = N_{r_\beta}(y_5, y_6), \\ W_\beta &= N_{r_\beta}\big(N_{r_\beta}(n_0, n_1), N_{r_\beta}(y_4, n_3)\big). \end{aligned} \tag{45}$$

Use eight Boolean coordinates ϵ_-, ϵ_p, for $p \in \{2, 3, 5, 7, 11, 13, 17\}$, impose $\epsilon^2 - \epsilon = 0$, and set

$$c = (1 - 2\epsilon_-) \prod_p (1 + (p-1)\epsilon_p), \qquad b_\beta = c r_\beta W_\beta^2, \qquad C_\beta = 17 b_\beta.$$

Require $b_\beta \neq 0$. Retain five rational witnesses $t_\beta, e_\beta, s_\beta, p_\beta, q_\beta$ and impose

$$(1 + b_\beta t_\beta^2) U_\beta = b_\beta (1 - b_\beta t_\beta^2), \qquad h_\beta = e_\beta (C_\beta - U_\beta^2), \tag{46}$$

$$p_\beta^2 = C_\beta (1 - e_\beta)(1 + 3e_\beta), \qquad C_\beta q_\beta^2 = (s_\beta^2 - C_\beta h_\beta)(s_\beta^2 + 3 C_\beta h_\beta). \tag{47}$$

Here U_β, h_β are arithmetic values. The preparation equation forces $1 + b_\beta t_\beta^2 \neq 0$: otherwise $b_\beta t_\beta^2 = -1$ but its right side is $2b_\beta \neq 0$.

Finally fix the integers

$$C = 3^{2^{26}}, \qquad \mathsf{S} = C^2 = 3^{2^{27}}, \qquad \kappa = \mathsf{S}^2 = 3^{2^{28}}. \tag{48}$$

The symbol S in the coefficient files means S; it is not the point S and is not a quantified coordinate. Write $f(S) = f_0 + f_1\sqrt{2}$, retain three private rational witnesses u, v, w, and impose

$$u^2 + 6v^2 - 2w^2 - 3(f_0^2 + f_0 f_1 - f_1^2)^2 + 2\mathsf{S}^2 = 0. \tag{49}$$

This completes the rational source. Every rational expression has the displayed guard or a domain implication already proved. Equations (44) and (47) retain their degenerate witness values.

Lemma 17.3 (The three-witness anchor)**.** *Equation* (49) *is equivalent to* $f_0/C, f_1/C \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. *For the nonzero integer label* a *of* $S = [a]P_*$ *it implies*

$$0 \neq a \in 2\,3^{2^{25}-1}\mathbb{Z}, \qquad |a| > (21 H_*)^{13/2} > (20 H_*)^4.$$

Its permitted labels contain arbitrarily divisible positive integers.

Proof. Put $R = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. The quaternion algebra $(2, 3)_{\mathbb{Q}}$ ramifies exactly at 2 and 3. The quadratic embedding criterion [105, Proposition 14.6.7] shows, for rational $t \neq 0$, that $1 - 2/(3t^2)$ is represented by $2x^2 + 3y^2 - 6z^2$ precisely when it is nonsquare in both $\mathbb{Q}_2$ and $\mathbb{Q}_3$. This last condition is exactly $t \in R$. Indeed at 2 a positive valuation of t gives odd valuation of the expression, valuation zero gives residue 3 modulo 8, and negative valuation gives a square in $1 + 8\mathbb{Z}_2$. At 3, nonnegative valuation of t gives odd negative valuation, while negative valuation gives a square in $1 + 3\mathbb{Z}_3$. The expression is nonzero because $t^2 = 2/3$ has no rational solution. Multiplying by $3t^2$ gives

$$t \in R \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \; u^2 + 6v^2 - 2w^2 - 3t^2 + 2 = 0.$$

The case $t = 0$ has witnesses $(0, 0, 1)$. A rational-square value of $1 - 2/(3t^2)$ is excluded by the division-algebra criterion as well: a nonzero trace-zero quaternion cannot square to a nonzero rational square in a division algebra.

The form $A^2 + AB - B^2$ has no nontrivial zero over $\mathbb{F}_2$ or $\mathbb{F}_3$. Scaling a pair therefore proves $v_p(a^2 + ab - b^2) = 2\min(v_p(a), v_p(b))$ for $p = 2, 3$. Apply the preceding three-witness formula to $(f_0^2 + f_0 f_1 - f_1^2)/C^2$ and multiply its witnesses by C^2. This gives exactly (49).

The local calculation in Section 14.2.2 applies with exponent 2^{26}: the reduction of P_* at 3 has order two and $v_3(z(2P_*)) = 1$. For even nonzero a it gives $v_3(f([a]P_*)) = 2 + 2v_3(a)$. At 2 the rational components of $f([a]P_*)$ are integral. The anchor is consequently equivalent to $0 \neq a \in 2\,3^{2^{25}-1}\mathbb{Z}$. Its minimum possible absolute value has binary logarithm greater than

$$1 + \tfrac{3}{2}(2^{25} - 1) = 50331647.5,$$

whereas $\log_2((21H_*)^{13/2}) < \frac{13}{2}(5+4783227) = 31091008$. The last comparison with $(20H_*)^4$ follows from $21^{13} > 20^8$ and $H_* \geq 1$. The subgroup description proves the last assertion. □

17.3 Local tests and the height implication

Lemma 17.4 (One place for every test)**.** *Let $R \in \mathbb{Q}^*$, $D_j \in K$, and positive integers m_j define*

$$W = \prod_j (1 + R(D_j/R^{m_j})^2), \quad T = \mathrm{Tr}(W^{-1}), \quad B = R/T^2,$$

with $W, T \neq 0$ and a preparation-zero pole certificate for B. If $p > 76176$ and $v_p(R)$ is odd, then $v_p(R) > 0$ and some $\mathfrak{p} \mid p$ has $v_\mathfrak{p}(D_j) \geq m_j$ for every j.

Proof. Set $\alpha = v_p(R)$. The nonconstant term in each nontrivial factor has odd valuation $\alpha + 2(v(D_j) - m_j\alpha)$; unramifiedness makes these valuations integral. The factor valuation is its minimum with zero. Thus $\lambda_\mathfrak{p} = v_\mathfrak{p}(W) \leq 0$. Put $\lambda = \max_{\mathfrak{p}|p} \lambda_\mathfrak{p}$. Integrality of trace gives $v_p(T) \geq -\lambda$. Hence

$$v_p(B) \leq \alpha + 2\lambda.$$

The left side is odd and, by Lemma 14.4, not negative. It is positive, so $\alpha > 0$ and $\lambda > -\alpha/2$. At a place attaining λ, each factor has valuation at least λ, because all factor valuations are nonpositive. For nonzero D_j this implies

$$v(D_j) > (m_j - 3/4)\alpha \geq m_j - 3/4.$$

Integrality gives $v(D_j) \geq m_j$; a zero D_j satisfies it as well. □

Lemma 17.5 (The two-anchor comparison)**.** *Write the integer labels of (A, S, T, Z) as (n, a, t, b). If the first fourteen tests in (40) have their specified orders at a place $\mathfrak{p} \mid p$, $p > 76176$, then*

$$L_i \in \mathcal{O}_{K_\mathfrak{p}}, \qquad L_i \equiv n_i \pmod{p^{k_i}}, \qquad (n_1, n_2, n_3, n_4) = (n, a, t, b).$$

This holds at split and inert primes.

Proof. Write $H = [h]P_*$ and $U_i = [u_i]P_*$. The order-one tests put H, U_i in the formal group at $\mathfrak{p}$ and V_0, V_i in the formal group at the conjugate place. Positive valuation of $f = 1/x$ gives formal membership on this good model. The guarded points give $h \neq 0, 4$. Their full logarithm ratios, after conjugating the second, are

$$r_i = u_i/h, \qquad r_i' = (u_i - 4n_i)/(h - 4).$$

These identities follow from the homomorphism property on equal formal multiples, and do not require P_* itself to be formal. At least one of $h, h-4$ is a p-unit. Its ratio is integral, and Lemma 17.2 gives the same valuation to its truncated ratio. The cross test makes the other truncated ratio integral too. Both relative errors are therefore absolute errors, and $L_i \equiv r_i \equiv r_i'$ modulo p^{k_i}. Multiplication by $h, h-4$ and subtraction give $4L_i \equiv 4n_i$. Divide by four. At an inert prime conjugation preserves the single local valuation, so the same argument applies. □

Lemma 17.6 (Height transfer)**.** *Let M_r be the odd-contact radical of t_r for the five contacts Θ, omitting primes at most 76176. If M_r^k divides a nonzero integer N for every r, then*

$$h(x(Q_i)) < 7H_*|N|^{24/k} \qquad (1 \le i \le 4).$$

At every prime counted in M_r, the source imposes all sixteen field tests with their specified orders at one place above that prime.

Proof. The fractional linear map $\phi(t) = 2 + 1/t$ carries Θ to $\infty, 0, 1, 2401, -343/32$. Its integral matrix has determinant -1. All primes dividing the relevant fixed coefficients and contact differences, including differences from 1, lie in $\{2, 3, 5, 7, 11, 37, 109, 439, 2399\}$. Thus outside the cutoff $M_r = M_5(\phi(t_r))$, and

$$h(t_r) \le h(\phi(t_r)) + \log 3 \le (H_* + \log 3)M_r^{24}.$$

At a prime of M_r, δ_r is a unit. Exactly one contact factor has odd positive valuation, while $\mathcal{A}_r^{-5}$ is a rational square. Therefore $v_p(R_r)$ is odd. Lemma 17.4 applied to $\widetilde{D}_{r,j}$ gives the asserted orders for D_j.

Set $T_0 = (H_* + \log 3)|N|^{24/k}$. The case $r = 0$ gives $h(\lambda) \le T_0$, and the other cases give $h(\mathcal{A}_r\lambda) \le T_0$. Hence $h(\mathcal{A}_r) \le 2T_0$. For the first component, $h(x_{i,0}) \le T_0$. For $x_{i,1} = a/b$ in lowest terms,

$$h(1 + x_{i,1}^2) = \log(a^2 + b^2) \ge 2h(x_{i,1}),$$

so $h(x_{i,1}) \le T_0/2$. Finally

$$h(x(Q_i)) \le h(x_{i,0}) + h(x_{i,1}) + \tfrac{3}{2}\log 2 < 7H_*|N|^{24/k}.$$

□

Lemma 17.7 (The two reduced group cardinalities)**.** *Let $p > 76176$ be prime and let $\pi \in \mathbb{Z}[i]$ have norm p, with $\pi \equiv 9 \pmod{16}$ and $\pi \equiv 1 + 3i \pmod 7$. The two reductions of E/K at the places above p have group cardinalities $N(\pi - 1)$ and $N(\pi + 1)$, in some order.*

Proof. The congruences give $p \equiv 1 \pmod 8$ and $p \equiv 3 \pmod 7$. Thus p splits in K and $(7/p) = -1$. The reductions of $d = 3 - \sqrt{2}$ have product 7 and opposite square classes. The two curves $y^2 = x^3 - d^2x$ are consequently opposite quadratic twists of $y^2 = x^3 - x$ and have opposite traces.

The Frobenius calculation already given in Section 14.4.1 applies: the order-four automorphism is defined over $\mathbb{F}_p$, its rational centralizer is $\mathbb{Q}(i)$, and integrality puts Frobenius in $\mathbb{Z}[i]$. Write it as $a+bi$, so $a^2+b^2 = p$ and the two cardinalities are $p+1-2a$ and $p+1+2a$. Full rational two-torsion makes both cardinalities divisible by four. Since $p \equiv 1 \pmod 4$, this forces a odd and b even. The real part of π is odd and its imaginary part is even. Unique factorization in $\mathbb{Z}[i]$ therefore makes $a + bi$ one of $\pm\pi, \pm\overline{\pi}$; the associates with factor i have the wrong parity. Hence the unordered cardinalities are exactly $N(\pi - 1), N(\pi + 1)$. □

Theorem 17.8 (Arithmetic interpretation of the source)**.** *With the inherited arithmetic results and Theorem A8.1, the rational source of Section 17.2 has input projection exactly $\mathbb{Z}$.*

Soundness. Let $\mathcal{B} = \max(|n|, |a|, |t|, |b|)$ and $\Delta = F(a, t, b) \in \mathbb{Z}$. Polynomial substitution preserves congruences between integral local elements. Lemmas 17.5 and 17.6 therefore give $M_r^{52} \mid \Delta$ whenever $\Delta \ne 0$. The index-height estimate (19) yields

$$\mathcal{B}^2 - 6 < 7H_*|\Delta|^{24/52}, \qquad |\Delta| < 5\mathcal{B}^4,$$

the latter because $\mathcal{B} \ge |a| > 7$. Consequently

$$\mathcal{B}^{2/13} < (7 \cdot 5^{6/13} + 6) H_* < 21 H_*.$$

For the last strict inequality, $5^6 7^{13} < 15^{13}$. This contradicts Lemma 17.3. Thus $\Delta = 0$, and Lemma 14.9, with $b \neq 0$, gives $\log \mathcal{B} > |a| - 3$.

Lemma 16.1 applies to the exact tree (45) and gives

$$h(\beta) \le H_* |a|^{3/4} < |a|/20.$$

If $\beta \neq n$, put $N = |\operatorname{num}(\beta - n)|$ in lowest terms. The order-16 test gives $M_r^{16} \mid N$. Therefore

$$\mathcal{B}^2 < 7 H_* N^{3/2} + 6, \qquad N \le 2 e^{h(\beta)} \mathcal{B},$$

and $\log \mathcal{B} < 2 \log(32 H_*) + 3 h(\beta)$. The anchor also gives $\log(32 H_*) < |a|/20$, so the right side is less than $|a|/4 < |a| - 3$, a contradiction. Hence $\beta = n \in \mathbb{Z}$. Since $X \in \mathbb{Q}$ and $X^2 = n - 2 \in \mathbb{Z}$, the reduced denominator of X is one. □

Completeness. Fix $X \in \mathbb{Z}$, set $n = X^2 + 2$, and choose the Boolean correction c so that $c r_\beta > 0$ is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, as in the finite squareclass computation accompanying Lemma 14.4. Let $\mathcal{S}$ be the prime support of $c r_\beta$. The scalar completeness construction in Lemma 16.1 first chooses $h_0 > 0$ divisible by 64 with $[h_0]P_*$ sufficiently deep at these primes. It then chooses a positive label a sufficiently divisible relative to h_0 and satisfying the anchor. The normalized scalar leaves are integral at primes dividing r_β and divisible at primes dividing c but not r_β. Choose positive t, b with $F(a, t, b) = 0$ by Lemma 14.9. This fixes the four root points and all scales $\mathcal{A}_r$.

Temporarily put $H = [h]P_*$ and $U_i = [h n_i]P_*$. All comparison tests are rational functions of $[h]P_*$ with nonzero denominators at $h = h_0$. Their poles lie above finite sets under fixed nonzero multiplication and translation maps on the two real factors, as required by the real progression lemma. Continuity preserves the scalar conditions at every place over $\mathcal{S}$ and bounds below the valuations of all sixteen comparison tests at the places over $2, 3, 17$, throughout a progression

$$h \equiv h_0 \pmod{L}, \qquad 64 \mid L.$$

One obtains L by putting sufficiently small open formal subgroups inside the finitely many translated neighborhoods and taking a common multiple of their indices. Lemma 16.3 gives, independently of future progression moduli, real bounds for the tests attained by arbitrarily large positive h in each nonconstant progression. The direct polynomial F introduces no denominator and has the same boundedness property.

Choose a positive integer D_0 such that every $D_0^2 \theta / \mathcal{A}_r$ is integral, and put

$$v = D_0^2 w^2, \qquad \lambda = u/v.$$

The contact positions are the integer forms $u - (D_0^2 \theta / \mathcal{A}_r) w^2$. Delete repeated forms, but retain their incidences in the nine detectors. The five forms within one detector are distinct, and the zero contact is the common form u. Apply Theorem A8.1 to these forms. Prescribe arbitrarily high divisibility of w, a sufficiently large lower bound for u/w^2, and primes larger than 76176 and all prime factors of the fixed scales and contact differences. In its odd squarefree avoidance modulus include every odd prime factor of $3 \cdot 17 L$ except 7. Writing this modulus as M, the prescribed Gaussian classes are

$$\pi_p \equiv 9 \pmod{16}, \qquad \pi_p \equiv 1 + 3i \pmod{7}, \qquad \pi_p \equiv i \pmod{M}.$$

Let the resulting Gaussian generators be π_p, and write $d_-(p) = N(\pi_p - 1)$ and $d_+(p) = N(\pi_p + 1)$ for the cardinalities of the two reduced elliptic groups, orienting the places by Lemma 17.7. These cardinalities annihilate the reductions of P_*; their exact point orders are not needed. They satisfy

$$v_2(d_-) = 6, \quad v_2(d_+) = 2, \quad \gcd(d_-(p), d_+(q)) = 4$$

for every pair of selected primes. No odd prime factor of L divides either order. For 7 this follows separately from $d_+ \equiv 6$ and $d_- \equiv 2 \pmod 7$. Set $M_- = \operatorname{lcm}_p d_-(p)$ and $M_+ = \operatorname{lcm}_p d_+(p)$. The congruences

$$h \equiv 0 \pmod{M_-}, \quad h \equiv 4 \pmod{M_+}, \quad h \equiv h_0 \pmod{L}$$

are compatible: their pairwise gcds are $4, 64, 4$, respectively. Choose a sufficiently large positive h in this progression attaining the real bounds. It avoids $0, 4, -4$, so all point and subtraction guards hold. At a selected prime, H, U_i are formal at the place of order d_-, while V_0, V_i are formal at its conjugate place. Their exact ratios are n_i. Lemma 17.2 gives all sixteen required orders, including the order-52 polynomial test because $F(a, t, b) = 0$.

We check that the nine preparation-zero poles can be filled. For each detector,

$$R_r = \frac{\prod_{\theta \in \Theta}(u - (\theta/\mathcal{A}_r)v)}{v^5 \delta_r^6}.$$

The denominator is a rational square. At the five distinct contact primes of this detector R_r has valuation one, and elsewhere its valuation is even. The primes do not divide v: otherwise two distinct forms in the same detector would give the same prime value, impossible since $w > 0$ and their coefficients are distinct. The lower bound for the primes excludes the fixed scale and difference denominators. At either place over a contact prime a factor of W_r has valuation zero or negative, and its inverse has residue one or zero, respectively. At the successful place every inverse factor has residue one. Consequently T_r has residue one or two and is a unit. It follows that B_r has positive odd valuation at its contact primes and even valuation elsewhere. Its reduced denominator is a square.

The prescribed Gaussian classes make every contact prime square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The last product formula therefore makes B_r square in each of those fields. The uniform real bounds and $\lambda \longrightarrow +\infty$ give

$$R_r \sim \mathcal{A}_r^{-6} \lambda^{-1}, \quad W_r = 1 + O(\lambda^{-3}), \quad T_r = 2 + O(\lambda^{-3}), \quad B_r \sim (4\mathcal{A}_r^6 \lambda)^{-1}.$$

In particular $0 < B_r < 17$. At a small prime $p = 2, 3, 17$, put $\nu_p = v_p(v)$ and $\eta_{p,r} = v_p(\mathcal{A}_r)$. The common form u is a selected prime greater than these small primes, so $v_p(u) = 0$ and $v_p(\lambda) = -\nu_p$. The fixed neighborhoods give $v(D_j) \geq -c_p$. For sufficiently large prescribed ν_p one has

$$v_p(R_r) = \nu_p - 6\eta_{p,r}, \qquad v(\widetilde{D}_{r,j}) \geq -c_p + (m_j + 1)(\nu_p - \eta_{p,r}).$$

The nonconstant factor term has valuation at least $3\nu_p + (10m_j - 8)\eta_{p,r} - 2c_p$, tending to infinity. Thus T_r is as close to 2 as needed, including at the ramified prime 2, and

$$v_p(B_r) = \nu_p - 6\eta_{p,r} - 2v_p(2).$$

Prescribing the divisibility of w before applying the prime theorem therefore ensures $v_2(B_r) \geq 3$, $v_3(B_r) \geq 1$, $v_{17}(B_r) \geq 2$. Lemma 16.2 supplies the four witnesses in every coordinate pole.

Finally the progression preserved all scalar leaf conditions. The constant leaf on the all-left path makes W_β a unit at every prime dividing r_β; the divisibility of the other leaves makes it a unit at primes dividing c but not r_β. Since $|r_\beta| > 1$, it is nonzero. Therefore $b_\beta = c r_\beta W_\beta^2$ is positive, has square reduced denominator, and is locally square at $2, 3, 17$. Lemma 14.4 supplies the scalar preparation and pole witnesses for this final H. The anchor already holds for S. All witnesses have now been supplied without changing the root points. □

17.4 The quadratic compilation in 962 variables

We compile the shared-point source above using different point charts, pole coordinates, and arithmetic circuits. Every change has explicit rational witness maps. The arithmetic source, including its simultaneous prime input, is unchanged. Throughout this subsection

$$K = \mathbb{Q}(\sqrt{2}), \qquad D = (3-\sqrt{2})^2 = 11 - 6\sqrt{2}, \qquad \mathsf{S} = 3^{2^{27}}.$$

An equation in K means its two rational component equations. The fixed integer S is a coefficient, not a coordinate. An assignment of a quadratic expression to one rational coordinate adds one quadratic equation. Affine expressions require no coordinate.

Theorem 17.9 (The 962-variable quartic)**.** *The complete table specified below defines*

$$P_{962} \in \mathbb{Z}[X_0, \dots, X_{961}]$$

with degree four, exactly 55409 *nonzero coordinate monomials, integer coefficient content one, and coefficient height*

$$H(P_{962}) = 64\mathsf{S}^2 = 64 \cdot 3^{2^{28}} < 2^{429496736}.$$

It is the sum of squares of 919 *quadratic residuals and has the same rational input projection as the shared-point source of Theorem 17.8. Consequently, under that theorem's arithmetic hypotheses, for every $x \in \mathbb{Q}$,*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists X_1, \dots, X_{961} \in \mathbb{Q}, \quad P_{962}(x, X_1, \dots, X_{961}) = 0.$$

The 962 *variables include the input and* 961 *existential witnesses.*

A quadratic chart for the nine points. For each point introduce $t, h \in K$ and set

$$u = t^2, \qquad X = h^2, \qquad u(X - u/4) = D, \qquad Y = h(u/2 - X). \tag{50}$$

The three assignments use six rational coordinates. The displayed constraint contributes two quadratic residuals. The four rational components of t, h replace the four halfpoint components.

Lemma 17.10 (Exact point-chart maps)**.** *The chart* (50) *describes precisely the finite doubles used in the shared-point source, with the same rational witnesses up to the following invertible changes. It preserves all required nonzero point-coordinate domains. The rational component X_0 of X is positive.*

Proof. For an old halfpoint (a, b) with $b^2 = a^3 - Da$ and its tangent equation, both a, b are nonzero. Indeed $b = 0$ would give $a = 0$ or $a^2 = D$, whereas the tangent numerator $3a^2 - D$ would be $-D$ or $2D$. Then $a = 0$ is excluded by the curve equation. Define

$$t = b/a, \qquad h = (a^2 + D)/(2b) = a/t - t/2.$$

The curve equation gives $t^2 = a - D/a$. Substitution gives $t^2(h^2 - t^2/4) = D$. The doubling formulas give exactly $X = h^2$ and $Y = h(t^2/2 - h^2)$.

Conversely, a chart solution has $t \neq 0$ because $D \neq 0$. Set

$$a = th + t^2/2, \qquad b = at, \qquad m = h + t.$$

If $a = 0$, then $h = -t/2$, contradicting the chart equation. Thus $b \neq 0$. Direct expansion gives

$$a^2 - t^2 a = D, \quad b^2 = a^3 - Da, \quad 2bm = 3a^2 - D, \quad m^2 - 2a = h^2,$$

and $m(a-h^2)-b = h(t^2/2-h^2)$. The substitutions are inverse, so they recover exactly the old finite double. Lemma 17.1 therefore supplies its nonzero abscissa and ordinate, and the same remains true for the unchanged guarded differences and subsequent doublings.

Both real embeddings of D are positive. Since $t \neq 0$, the chart equation implies $h^2 = t^2/4 + D/t^2 > 0$ under both embeddings. Hence X_0, the average of these two positive values, is positive. □

The five affine subtraction formulas keep their original inverses of the differences of abscissas. Neither a subtraction guard nor a point occurrence is deleted. Relative to the direct halfpoint doubling circuit, the chart saves two rational arithmetic coordinates for each of the nine points.

The logarithm polynomial. Retain the source's coefficients

$$a_0 = 1, \quad a_1 = -2/5, \quad a_j = a_{j-1}(4j-5)/(4j+1).$$

For $P(w) = \sum_{j=0}^{12} a_j w^j$ write

$$P(w) = a_{12} Q(w)^2 + R(w), \qquad \deg Q = 6, \quad Q \text{ monic}, \quad \deg R \leq 5. \tag{51}$$

This identity is constructed without any witness division. Starting with the leading coefficient of Q equal to one, match degrees eleven down to six successively. At each step the new coefficient is determined by division by $2a_{12} \neq 0$. The remainder then has degree at most five. After computing w^2, w^3, every term through w^6 is quadratic in $1, w, w^2, w^3$. Assigning Q and then $a_{12}Q^2+R$ therefore uses four field assignments, rather than five. The eight order-52 logarithms save sixteen rational coordinates. The order-16 logarithms keep their exact degree-three polynomials. No truncation order changes.

Usually the final logarithm is $\ell = zP(Df^2)$, with $f = 1/x$ and $z = -x/y$. For U_0 and V_0 only, omit the separate z coordinates and impose the direct quotient equations

$$y\ell = -xP(Df^2). \tag{52}$$

The polynomial output is an affine expression in assigned coordinates, so these are quadratic component equations. The ordinate is nonzero by Lemma 17.1; hence (52) is exactly the old logarithm. The source's logarithm-nonvanishing result and every logarithm-ratio domain remain unchanged. In the scalar block replace $z(H)\rho = z(S)$ by

$$x(H)\rho = -y(H)z(S). \tag{53}$$

Both $x(H), y(H)$ are nonzero, so this has the same unique value $\rho = z(S)/z(H)$.

Products, tests, and detector evaluation. For each of the three orders, compute the elementary symmetric coefficients of $\prod_j (1 + qD_j^2)$ by a balanced product tree. At an internal node with coefficient vectors a_i, b_j, assign each component of $\sum_{i+j=k} a_i b_j$ as one quadratic expression. This is the ordinary coefficient-convolution identity. It preserves every coefficient and every test, including its assigned order.

The degree-ten detector polynomial is evaluated in two blocks:

$$\sum_{j=0}^{10} c_j q^j = \sum_{j=0}^{4} c_j q^j + q^5 \left(\sum_{j=0}^{5} c_{j+5} q^j \right). \tag{54}$$

Compute q^2, q^3, q^4, q^5 using four scalar assignments, assign the two components of the parenthesized block, and assign the two final components directly. The low block is not assigned separately. Every final summand is quadratic in the available coordinates. This uses eight rational coordinates. The degree-two and degree-four polynomials use their direct power sums, with their two component sums assigned once. These formulas are identities over K.

The Lucas expression is regrouped as

$$t^2 - a(a-2)tb + ((a-2)b)^2 - 1 = t^2 - (at)C + C^2 - 1, \qquad C = (a-2)b.$$

Thus one field multiplication is removed without changing the tested polynomial. The fixed quintic is evaluated by its exact coefficients using t^2; it remains $F_5(t) = t(t + 1/2)(t + 1)(t - 1/2399)(t + 32/407)$.

Weight quotients and their domains. Fix a detector and write A for its scale, $\delta = A\lambda - 1$, and e for its weight. The scales are

$$1, \quad x(Q)_0^2, \quad (1 + x(Q)_1^2)^2 \qquad (Q = A, S, T, Z).$$

They are all positive by Lemma 17.10. Retain the inverse $\delta i = 1$, and replace the separate fifth-power product and inverse of F_5 by

$$F_5(A\lambda)e = (A\delta)^4(A\delta). \tag{55}$$

Here $(A\delta)^4$ denotes the output of its retained power-assignment gates; the displayed product is quadratic in the available coordinates. The right side is nonzero, so this equation forces both $F_5(A\lambda)$ and e nonzero. In particular the removed inverse of F_5 is recovered uniquely. Define the low-order weight by

$$\delta q_1 = ei^2,$$

where i^2 is likewise a retained square-assignment output. This gives the unchanged $q_1 = e/\delta^3$. Retain the repeated-squaring values e^6, e^{24}, e^{96} and set

$$c = q_1 e^6, \qquad q_{16} = ce^{24}, \qquad q_{52} = ce^{96}.$$

These are the old $q_1 e^{30}$ and $q_1 e^{102}$, with one shared product.

If the detector product is $W = w_0 + w_1\sqrt{2}$, retain $w_0 j = 1$ and use

$$w_0 n = (w_0^2 - 2w_1^2)/2, \qquad ev = in, \qquad B = nv. \tag{56}$$

The trace guard is essential and is retained. It gives $W \neq 0$ in K, so $\mathrm{Norm}(W) \neq 0$ and $n = \mathrm{Norm}(W)/(2w_0) \neq 0$. Equations (55)–(56) therefore give

$$B = \frac{i}{e} \left(\frac{\mathrm{Norm}(W)}{2w_0} \right)^2 \neq 0,$$

the original detector input. The reverse map restores $R = i/e$, its old inverse intermediates, and n^2 by their unique rational values. No inverse is inferred from an unguarded zero-over-zero quotient.

The scalar norm and finite corrections. The scalar contact polynomial r and its inverse $ri = 1$ remain. Writing $f = f(H)_0$, $u = f(H)_1$, and $g = f(S)_0$, retain the leaf $\ell = ui$ and replace two norm-tree nodes by

$$rn_0 = r + f^2, \qquad rn_1 = u\ell + g^2.$$

with the existing leaf ui retained. Since $r \neq 0$, these equal $n_0 = 1 + r(fi)^2$ and $n_1 = (ui)^2 + r(gi)^2$. All other nodes and all orders in the scalar norm tree are unchanged.

Replace the eight Boolean correction choices and their seven product coordinates by $c_0, \ldots, c_7 \in \mathbb{Q}$ satisfying

$$c_0^2 = 1, \qquad (c_j - c_{j-1})(c_j - p_j c_{j-1}) = 0 \quad (1 \leq j \leq 7), \tag{57}$$

where $(p_1, \ldots, p_7) = (2, 3, 5, 7, 11, 13, 17)$. Induction shows $c_j \neq 0$ and $c_j/c_{j-1} \in \{1, p_j\}$. Thus c_7 is exactly an allowed signed squarefree correction. Conversely every original choice determines such a chain. Recover the original Booleans by

$$\epsilon_- = (1 - c_0)/2, \qquad \epsilon_j = \frac{c_j/c_{j-1} - 1}{p_j - 1}.$$

All denominators are nonzero. The scalar preparation coordinate, its defining quotient, and the explicit inverse of its pole input are retained.

The ten pole blocks. Write b for a pole input, $C = 17b$, and U for its prepared value. For the nine coordinate poles $U = b$. In the scalar pole U retains the source's preparation formula and its denominator domain. Every C is nonzero: this was proved above for coordinate detectors, and the scalar input has its retained inverse.

Replace the old witness e_0 by $z = Ce_0$. Introduce the five intermediates

$$P = p^2, \quad F = U^2, \quad H = z(C - F), \quad V = s^2, \quad Q = q^2,$$

and impose

$$CP = (C - z)(C + 3z), \qquad CQ = (V - H)(V + 3H). \tag{58}$$

Every equation is quadratic. Under $z = Ce_0$, the first residual is C times the old residual $p^2 - C(1 - e_0)(1 + 3e_0)$, and the second is unchanged because $H = Ce_0(C - U^2)$. The inverse map $e_0 = z/C$ is defined. This saves one arithmetic coordinate at each pole, without changing the pole predicate or any preparation condition.

Counts and the final polynomial. The initial allocation is

	coordinates	structural residuals
input	1	0
nine point charts	36	18
common slope	1	0
nine coordinate poles	36	18
scalar pole with preparation	5	2
private anchor	3	1
signed correction chain	8	8
total	90	47

The complete program has 904 arithmetic coordinates and their 904 defining equations. Hence the raw counts are 994 coordinates and 951 quadratic residuals.

The private anchor uses exactly the old affine change

$$u = \mathsf{S} + (U - 3V + 3W)/4, \quad v = \mathsf{S}/2 + V/4, \quad w = 3\mathsf{S}/2 + W/4.$$

Its determinant is $1/64$. If $T = f_0^2 + f_0 f_1 - f_1^2$, sixteen times the old anchor after substitution is

$$-48T^2 + U^2 - 6UV + 6UW + 15V^2 - 18VW + 7W^2 + 8\mathsf{S}U.$$

No anchor condition or witness is removed.

Each certified compiler elimination solves a row $aY + q = 0$ with $a \in \mathbb{Q}^*$ and q independent of Y, substitutes $Y = -q/a$ everywhere, and removes that row. It is accepted only if every remaining residual is still quadratic. The reverse map reinserts the unique displayed value of Y. There are exactly 32 such eliminations, no row additions, and no omission of an unused coordinate. The input is retained. Consequently

$$90 + 904 - 32 = 962, \qquad 47 + 904 - 32 = 919.$$

Normalize the remaining rows to primitive integer polynomials $q_1, \ldots, q_{919}$ and define

$$P_{962}(X_0, \ldots, X_{961}) = \sum_{\nu=1}^{919} q_\nu^2. \tag{59}$$

The complete coefficient table is `quartics/quartic-962/polynomial.json`, with SHA-256

`6b798de7d00ac3c7d725ca0a69d39f1eac1ddb02fee2756cb6323770b952af32`.

An entry with coefficient c, exponent `sPower` k, and coordinate list $[i_1, \ldots, i_d]$ means $c\mathsf{S}^k X_{i_1} \cdots X_{i_d}$. Every entry is included and repeated indices give powers. This defines an ordinary integer polynomial.

Proof of Theorem 17.9. For a source solution apply the point-chart, correction-chain and pole substitutions, and evaluate all new assignment and quotient coordinates in dependency order. The identities and domain proofs above show that every structural residual vanishes. Conversely apply the inverse substitutions, restore every deleted quotient by its proved nonzero denominator, and evaluate the source assignments. Every source point, logarithm, test, detector input, scalar value and anchor value is recovered. These maps fix X_0. The affine anchor change, certified eliminations and nonzero row normalizations preserve the same projection. Over $\mathbb{Q}$, (59) vanishes exactly when all 919 residuals vanish. This proves the projection statement, and Theorem 17.8 gives its conditional arithmetic interpretation.

Independent exact convolution gives 55409 distinct monomials. Of their coefficients, 55401 are S-independent integers, seven are $c\mathsf{S}$ with $0 < |c| \le 768$, and one is $64\mathsf{S}^2$. The largest absolute S-independent coefficient has bit length 1526. Thus no coefficient vanishes on specialization and $64\mathsf{S}^2$ is the exact height. It occurs at X_{79}^2, where $X_{79} = U$. The coefficient of X_0^4 is one, proving degree four and integer content one. Finally $\log_2(64\mathsf{S}^2) < 6 + (8/5)2^{28} < 429496736$ because $3^5 < 2^8$. □

The frozen package includes the builder, raw systems, all 904 gate definitions, the exact elimination and normalization certificate, compact coordinate map, final residuals, complete coefficient table, source-equivalence proofs and independent reviews. Its manifest has SHA-256

`36fca72cb0eb126193ffa43d890ff09d5e29cbf1274561d2632ed1e227163787`.

The independent checker verifies every gate, all 894 certificate operations, the complete sum-of-squares expansion, support, integrality, content and height. A fresh isolated reproduction gives the same seven deterministic artifacts byte for byte. These finite checks establish the stated algebraic facts; they do not prove any inherited arithmetic input.

18 Why a single defining polynomial must have degree at least four

The argument in this section comes from the agentic notes [9, section on unconditional lower bounds]. We give its specialization to $\mathbb{Q}$, including the treatment of singular cubics, cones and geometric reducibility. It is independent of the arithmetic inputs to our constructions.

For $F \in \mathbb{Q}[T, Y_1, \dots, Y_r]$, write

$$S_F = \{t \in \mathbb{Q} : \exists\, y \in \mathbb{Q}^r,\ F(t, y) = 0\}.$$

All degrees below are total degrees, including the input variable T.

Theorem 18.1 (Degree obstruction)**.** *If $S_F = \mathbb{Z}$, then $\deg F \geq 4$.*

We use two published results. The integral form of Hilbert's irreducibility theorem says that an irreducible polynomial $g(T, Y) \in \mathbb{Q}(T)[Y]$ admits infinitely many specializations $T = t \in \mathbb{Z}$ which preserve its degree in Y and are irreducible over $\mathbb{Q}$. Deleting finitely many values also avoids its coefficient poles and any prescribed nonzero polynomial in t. This integral formulation is recalled in the introduction of Müller [8]; merely obtaining specializations in $\mathbb{Q}$ would not suffice here. The second result is the characteristic-zero cubic theorem of Colliot-Thélène and Salberger [5, §1, Proposition 1.3, pp. 520–521]: a geometrically integral cubic hypersurface of dimension at least two over a characteristic-zero field is unirational over that field if it has a rational point and is not a cone. This includes singular cubics; the proof treats the codimension-one singular case and uses [6, Remark 2.3.1, p. 56] for the regular-in-codimension-one case. We apply it only over $\mathbb{Q}$, after removing every vertex direction. The exact source and quotient-hypothesis connection is recorded in [7]. In particular, no unirationality assertion is made for a cone over a plane cubic.

Lemma 18.2. *A nonconstant rational function $R \in \mathbb{Q}(U_1, \dots, U_m)$ cannot take only integer values at the rational points of a nonempty Zariski-open subset of $\mathbb{A}^m_{\mathbb{Q}}$ on which it is defined.*

Proof. Shrink the open subset to a nonempty principal open subset and choose a connected real open ball in it, avoiding the denominator of R. Rational points are dense in this ball. If their images belong to the closed set $\mathbb{Z}$, continuity puts the image of the entire ball in $\mathbb{Z}$. Connectedness then makes R constant on the ball. Its numerator minus that constant times its denominator vanishes on a real open set, hence is the zero polynomial. This contradicts nonconstancy. □

Consequently, if an integral variety $X/\mathbb{Q}$ is unirational and $f \in \mathbb{Q}(X)$ is nonconstant, the rational values of f cannot all be integers. Indeed, compose f with a dominant rational map from projective space to X, and restrict to a nonempty affine open subset of its domain. Dominance keeps this composite nonconstant, so Lemma 18.2 applies.

Lemma 18.3. *If $\deg F \leq 2$ and $S_F \subseteq \mathbb{Z}$, then $|S_F| \leq 2$.*

Proof. Otherwise choose rational zeros a, b, c with distinct input coordinates. If they lie on an affine line, the restriction of F to that line has three zeros and degree at most two, so vanishes identically. The input is nonconstant on the line and takes every rational value, a contradiction.

If the points are not collinear, let L be their rational affine plane. If $F|_L = 0$, a line in L with nonconstant input gives the same contradiction. If $F|_L$ factors over $\mathbb{Q}$ into affine linear polynomials, its zero set consists of at most two rational lines. One contains two of the three selected points and again has nonconstant input. A nonzero polynomial of degree at most one cannot vanish at three noncollinear points.

It remains to consider an irreducible quadratic on L. Its projective conic has a smooth rational point: a reduced irreducible quadratic in characteristic zero has at most one geometric singular point, while the three chosen affine points are distinct. A conic with a smooth rational point is rational. Explicitly, intersecting it with the pencil of lines through that point gives a rational parametrization by taking the second intersection point. The input function is nonconstant because it takes three different values. Composing it with this parametrization contradicts Lemma 18.2. □

Proof of Theorem 18.1. Suppose $S_F = \mathbb{Z}$ and $\deg F \leq 3$. The polynomial is neither zero nor a nonzero constant. Lemma 18.3 excludes degree at most two. It also excludes a reducible cubic: every nonconstant irreducible factor then has degree at most two, and the finite union of their projections cannot equal $\mathbb{Z}$. Thus F is irreducible of degree three.

Homogenize F to H on a rational vector space E, using X_0 as the homogenizing coordinate and X_1 for the input coordinate. Thus on $X_0 \neq 0$ the input is X_1/X_0. The homogeneous polynomial H is irreducible over $\mathbb{Q}$. If it is geometrically reducible, its geometric irreducible factors form a single Galois orbit with equal degrees. In characteristic zero they occur without repetition. Since the total degree is three, there are exactly three conjugate linear factors. Every rational zero lies on each of their hyperplanes. Take rational zeros normalized by $X_0 = 1$ with inputs zero and one. Their whole affine joining line lies on all three hyperplanes and its input takes every value in $\mathbb{Q}$, a contradiction. We may therefore assume H is geometrically integral.

Define the rational vertex subspace

$$W = \{v \in E : D_v H = 0 \text{ identically}\}, \qquad D_v H(x) = \frac{d}{ds} H(x + sv)\Big|_{s=0}.$$

These are linear conditions on v. In characteristic zero, $D_v H = 0$ is equivalent to $H(x+sv) = H(x)$ for all x, s. Thus H descends to a cubic $\overline{H}$ on E/W. After a basis of W is extended to one of E, this simply deletes the coordinates on which H does not depend. The quotient has no remaining vertex direction. This assertion holds over the algebraic closure as well, since the kernel of a matrix over $\mathbb{Q}$ commutes with extension of scalars. Hence its projective hypersurface is not a cone.

For $v \in W$ and a rational zero x with $x_0 = 1$, translation by sv gives rational zeros whose inputs, whenever defined, are

$$\frac{x_1 + sv_1}{1 + sv_0}.$$

Unless $v_1 = x_1 v_0$, this is a nonconstant fractional linear function of s, with cofinite image in $\mathbb{Q}$. Its image cannot be contained in $\mathbb{Z}$. Applying the resulting equality first at input zero and then at input one gives $v_0 = v_1 = 0$ for every $v \in W$. Therefore X_0 and X_1 descend to independent linear forms on E/W.

Choose quotient coordinates beginning with these forms. The affine chart $X_0 = 1$ of $\overline{H} = 0$ has exactly the same input projection as $F = 0$. To see both inclusions, project rational points from E to E/W, and in the reverse direction use a rational linear section of that projection. This section preserves X_0 and X_1, and H is constant along each fiber.

If the quotient projective hypersurface has dimension at least two, [5, Proposition 1.3] applies. It is geometrically integral, has a rational point in $X_0 = 1$, is not a cone, and its ground field $\mathbb{Q}$ has characteristic zero. It is therefore unirational. Its input function is nonconstant, since its rational values include all integers. The consequence of Lemma 18.2 then contradicts that its input projection is contained in $\mathbb{Z}$.

If the quotient has dimension one, its affine chart is defined by an irreducible polynomial $G(T, Y) \in \mathbb{Q}[T, Y]$. If its degree in Y is zero, its input projection is finite. If that degree is one, solving the linear equation for Y gives a rational point above every rational T outside a finite

set, also impossible. If its degree in Y is at least two, Gauss's lemma makes it irreducible in $\mathbb{Q}(T)[Y]$. Integral Hilbert irreducibility supplies an integer t for which $G(t, Y)$ remains irreducible of that same degree. It has no rational root, contradicting $S_G = \mathbb{Z}$. Finally, a zero-dimensional projective quotient has only finitely many rational points and cannot have input projection $\mathbb{Z}$. These exhaust the possibilities and prove the theorem. □

Remark 18.4. This lower bound concerns a single polynomial equation and counts degree in every variable. It allows systems of quadratic equations; over $\mathbb{Q}$, their sum of squares is quartic. It puts no lower bound here on the number of variables in a degree-four definition.

A1 Dependency order in the arithmetic argument

The even coefficient bound uses stationary-base selection as well as integrality. The odd minimum uses a fixed nonzero witness and the prior genus-index lower bound. Support removal is applied only to the actual unramified specialization. The functional bound and determinant clearing are proved separately, and the same corrected coordinate is used at every address. These constructions are given in Sections 7–9 and Appendices A5–A7.

The conversion to ordinary integers uses the canonical model and total multiplication, followed by the exact computation relation. Only the shared-shift separation argument in Section 11 identifies the external rational variable with the ordinary value of its integer index. The literal substitution and size calculation then apply to the same expression P_*.

A2 Source identities and scope

The following catalogue records the base collection of 83 source identifiers; parity, construction and uniformity supplements appear separately. The released construction files and their exact identities are recorded in `construction/SOURCE_MANIFEST.json`. Current proof-source selection and versioned supporting arguments are indexed in `research/version14/SUPPORTING-NOTES.md`.

ID	Source basename	Use
S01	`ordinary-integer-polynomial.js.txt`	Outer construction
S02	`rational-circuit-translator.js.txt`	Substitution semantics
S03	`finite-polynomial-expressions.js.txt`	Fixed constructors
S04	`variable-shift-polynomial-library.js.txt`	Close and Test
S05	`elliptic-generator-proof.txt`	Generator certificate
S06	`explicit-elliptic-data.txt`	Rank, torsion, codes
S07	`verify-elliptic-generator.js.txt`	Exact finite algorithm
S08	`six-contact-height-proof.txt`	Contacts and constants
S09	`z-in-q-height-independent-audit.txt`	Height proof
S10	`explicit-pole-predicate.txt`	Pole polynomial
S11	`z-in-q-pole-completeness-audit.txt`	Simultaneous descent
S12	`parity-frozen-4eaee875.tex`	Raw local matrices
S13	`multiplication-certificate-correctness-proof.txt`	Multiplication proof
S14	`twenty-two-prime-pattern-proof.txt`	Selected primes
S15	`z-in-q-variable-shift-bridge.txt`	Ordinary-value graph
S16	`z-in-q-ordinary-integer-assembly-proof.txt`	Outer composition

ID	Source basename	Use
S17	`z-in-q-restricted-converse-assembly.txt`	RC assembly inputs
S18	`ring-limits-frozen-4eaee875.tex`	Finite models and limits
S19	`paired-functional-and-uniformity-proof.txt`	Paired functional
S20	`determinant-transfer-audit.txt`	Clearing and transfer
S21	`variable-shift-precision.py.txt`	Exact decision program
S22	`precision-register.js.txt`	Ideal IR and lowering
S23	`specialize-precision.js.txt`	Wrapper specialization
S24	`integer-halting-correctness-proof.txt`	Natural compiler
S25	`z-in-q-explicit-cutoff-bound.txt`	Literal cutoff
S26	`z-in-q-polynomial-size-bound-proof.txt`	Syntax/norm argument
S27	`z-in-q-numerical-polynomial-bounds.txt`	Tower comparison
S28	`determinant-switch-audit.txt`	Switches and outer rank
S29	`ring-determinants-frozen-4eaee875.tex`	Ring data and derivatives
S30	`preliminaries-frozen-4eaee875.tex`	Selmer coranks and lengths
S31	`fixed-certificate-constants.js.txt`	Exact least-multiple search
S32	`pointwise-frozen-4eaee875.tex`	Distinct point indices

In particular S30, S18, S29 and S32 are, respectively, Sections 2, 4, 5 and 9 of OpenAI's *A pointwise 2-converse for elliptic curves with rational two-torsion* [90], frozen at `openai/math` commit `4eaee87548250b2682affda7d05d947c05554bae`. The source filenames are provenance labels for those sections, not independent publications or independent arithmetic assumptions.

The supplement manifest also records the revised fixed-degree CM proof, the complete natural compiler proof, actual arithmetic and analytic producer arguments, finite-Artin and determinant-count clarifications, the fixed BFH witness application, exact substitution checks, and the new fixed-form, weighted-theta, operator and normalization derivations. For the height estimate, the derivation in Section 4 supersedes the historical arithmetic-height arguments in S08–S09 and the earlier CM supplements. Those historical alternatives are not additional premises. Primary theorem applications remain explicit in the mathematical text. The version records identify the source files used in the construction.

A3 Literal finite-constructor specification

The supplementary file `ordinary-integer-definition.json` is part of this specification. It is a finite directed acyclic description, not an expanded monomial list. The following rules determine one polynomial from its bytes. Diagnostic labels and intended-theorem strings add no equations.

The root is `integers`. Variable zero is the external input and variables one through 292890231904 are its rational witnesses. A nonnegative point reference r means $X_r, \dots, X_{r+4}$. References $-1, -2, -3$ respectively mean

$$(1,0,0,0,0), \quad (0, g_0, -145, g_1, g_2), \quad (0, g_0, -145, -g_1, -g_2).$$

Every model recipe conjoins $8g_0 - 1641 = 0$, $4g_1 + 15465 = 0$, $32g_2 - 87483 = 0$.

A3.1 Scalar nodes and fixed expressions

Read nodes in their stored topological order. A `const` is its decimal integer; `var` is its local variable; `add` and `mul` are ordinary polynomial operations. For a `call` to a template with p parameters, substitute the specified node polynomials for variables $0, \dots, p-1$, and X_{w+j} for variable $p+j$, where w is the witness start. The value is the callee's entire root polynomial.

A `fixed_integer` evaluates a finite expression using literals, references to preceding definitions, addition, multiplication, nonnegative powers, maximum, factorial, the catalogue product or maximum, and the least formal multiple. The catalogue uses the first 22 primes in increasing order in each unit class. The searches are deterministic and terminate by Section 12. They produce constants, not quantified witnesses.

A `psi_component` uses the division-polynomial recurrence in Section 2, pair multiplication $(a,b)(c,d) = (ac + 2bd, ad + bc)$ and $A = (-11243, 7950)$. It returns component zero or one; every index is fixed. The positive fixed point codes use

$$x_n = x(P_0) - \frac{\psi_{n+1}(P_0)\psi_{n-1}(P_0)}{\psi_n(P_0)^2}, \qquad y_n = \frac{\psi_{2n}(P_0)}{2\psi_n(P_0)^4}.$$

Nontorsion makes these fixed denominators nonzero. Negative indices negate both ordinate components.

A `finite_guard_product` ranges over $(a,b,c) \in [-T,T]^3 \cap \mathbb{Z}^3$ with $ab \neq c$, with fixed T. Write the fifteen coordinates of the three codes as reduced A_j/B_j, $B_j > 0$, and put $D = \prod_j B_j$. Its factor is

$$\sum_{j=1}^{15} (DX_j - (D/B_j)A_j)^2.$$

Multiply these factors, with empty product one. This is an integer polynomial with precisely the finite false-triple zero set used above.

A3.2 Recipe constraints and witness scope

A model-evaluation recipe sums squares of the generator equations, listed natural calls, scalar-constant calls, add/multiply/power-value calls, and the five coordinate differences of `rootEquality`. A source-variable row with a non-null natural witness start contributes exactly that natural call. Operation kinds zero, one and two select addition, multiplication and power value. Source-node and provenance fields add no equations.

A sum-of-squares composition sums squares of its callee polynomials; a product composition multiplies them. Parameter `[var,i]` means X_i. Each private block uses its recorded start and length. The checked ranges are disjoint from parameters and other private blocks. These rules include globally conjoined fixed equations in active callees.

A3.3 The saved templates and the mathematical predicates

The following dictionary identifies the relations encoded by the saved templates, in addition to the decoding rules above. Point arguments are canonical five-coordinate codes. Field values have their two rational components in the basis $(1, \sqrt{2})$; conjugation negates the second component. An inverse equation guards each displayed quotient. Conjunction is encoded by a sum of squares, and disjunction by a product, with the private witness blocks specified in the file. The identities are identities of zero sets over $\mathbb{Q}$; squaring a callee polynomial can change the literal polynomial without changing the relation it defines.

Saved template	Relation and constraints
`member`	Membership in the canonical code domain $D = 2E(K)$ of Section 3.
`raw_add`, `raw_z`, `raw_inverse_x`	The group-law relation on typed inputs, $z = -x/y$, and $w = 1/x$, respectively, with $z(O) = w(O) = 0$.
`log25`	The two-component equality to the displayed scaled degree-25 polynomial L in Section 6.
`pole`	$Q_{\mathrm{pole}}(B)^2$, where Q_{pole} is the eleven-witness polynomial of Section 5. The outer square comes from the singleton conjunction added by the rational builder.
`test_group`	The twelve field tests of Section 6, including $A_i - \sigma(A'_i)$ for $i = 1, 2, 3$. Its four input points are $[c_*]P_0, [c_*]P_1, [c_*]P_2, [c_*]P_3$. It types the first four auxiliary points and obtains the second four by subtraction; its two denominator points and logarithms are nonzero.
`slope`	The normalized twelve-test products in six private groups, their traced inverses, and $\Phi(R/\Theta^2)$ from Section 6, with all contact and denominator guards.
`height_certificate`	The 27 slopes summing in triples to the nine homogeneous coordinate quantities in that section, together with membership of the three input points.
`multiply.scalar`, `multiply.guard`, `multiply`	The fixed $[c_*]$ graph; the inverse of the finite false-triple product; and their conjunction with the height certificate at the fixed ℓ_0.
`natural`	Four model squares, formed by multiplication, whose sum is the input code, as in Section 10.
`value.signed_encode`	$N = \iota(n)$ and a natural code $C = \iota(c)$ with $n = c/2$ for even c, and $n = -(c+1)/2$ for odd c. The two branches use $c = 2q$ and $c = 2q + 1$ with $q \geq 0$.
`value.abs`	$a = \lvert x \rvert$: a is a sum of four rational squares and $a^2 = x^2$.
`value.close_coordinates`	The strict coordinate window of Section 11, using the typed points A, L, J, $r = \mathrm{Tr}(z(J)/z(A))/2$ and $\epsilon = \lvert(z(L))_0\rvert + \lvert(z(L))_1\rvert > 0$, with $\lvert t - r\rvert < \epsilon$. This template alone does not assert the precision relation.
`value.test`	Exactly $\mathrm{Test}(N, \kappa, t)$ of Section 11: $C_1 = [\kappa n]P_0$, $V_0 = H - \kappa P_0$, $V_1 = U - C_1$, the six displayed raw-z tests, their normalization by $F(t)$, and the single guarded pole input. These ratios use z, not L.
`value.exceptions`	The six branches $N = \iota(c)$ and $t = c$, $c \in \mathcal{C}$, with the constant points computed by the fixed group-law chains.
`scalar.integer_constant_`i	The fixed integer scalar graph requested by the translated precision computation.

The composition recipes make the remaining identifications explicit. `power_value` translates the source power template with its three input codes and output code; `precision` translates the five-natural-input precision polynomial, in the order (n, κ, a, ℓ, j) after signed encoding. The recipe `close` introduces A, L, J, forms those five signed encodings, conjoins `precision`, and calls `value.close_coordinates`. Thus it is the full Close. The recipe `main` introduces one shared κ and con-

joins `close` and `value.test` at exactly that same code. The product recipe `value_union` joins this branch with `value.exceptions`; `value` also imposes `member` on N. Finally `integers` existentially introduces N and calls `value` with external coordinate $t = X_0$. These are precisely Val and its input projection. The ordered calls and private ranges are checked by `auditOrdinaryIntegerPolynomial` in `construction/ordinary-integer-polynomial.js`. `scripts/check_construction.py` regenerates the saved expression byte for byte. The dictionary supplies the mathematical identification; those finite checks verify the stored syntax and wiring, not the arithmetic theorems used to prove the predicates.

The following parameter and witness counts are extracted from the exact specification, without expanding its polynomial.

Template	Parameters	Witnesses	Root node
`member`	5	7	149
`raw_add`	15	6	308
`raw_z`	7	1	68
`raw_inverse_x`	7	0	54
`log25`	4	0	229
`pole`	1	11	106
`test_group`	44	162	515
`slope`	22	1,298	2681
`height_certificate`	36	35,106	260
`multiply.scalar`	10	1	136
`multiply.guard`	15	1	22
`multiply`	15	35,134	83
`natural`	5	140,604	79
`value.signed_encode`	10	140,654	138
`value.abs`	2	4	23
`value.close_coordinates`	16	53	271
`value.test`	11	35,282	775
`value.exceptions`	6	406	487
`scalar.integer_constant_0`	10	1	136
`scalar.integer_constant_1`	10	1	136
`scalar.integer_constant_2`	10	1	136
`scalar.integer_constant_3`	10	1	136
`scalar.integer_constant_4`	10	1	136

Recipe	Parameters	Witnesses	Total variables
`power_value`	20	17,786,698	17,786,718
`precision`	25	292,889,492,836	292,889,492,861
`close`	11	292,890,196,199	292,890,196,210
`main`	6	292,890,231,486	292,890,231,492
`value_union`	6	292,890,231,892	292,890,231,898
`value`	6	292,890,231,899	292,890,231,905
`integers`	1	292,890,231,904	292,890,231,905

The five scalar-constant templates use the values in `library.fixedConstants`. Large program

literals are retained byte-for-byte in the supplementary specification. Replacing them by unspecified sufficiently large constants would change this definition.

A4 Primary theorem applications

Input	Exact application boundary
Selmer parity [61, Theorem 1.4]	$E/\mathbb{Q}$, $p = 2$, full 2^∞-Selmer corank; no finite-Sha hypothesis.
Kato [69]	Fixed weight-two form and symbol, integral transfer and fixed-valuation smoothing; rational divisibility is Theorem 13.4(2), with oddness, rational irreducibility and the unipotent condition.
BFH and Hoffstein–Luo [70, 71]	Actual signed-prime twist with trivial character and arbitrary finite splitting set. Only the nonvanishing clause has the four-prime bound before the extra prime.
Waldspurger and Ramsey [79, 80]	Full good-Hecke eigenspace, same dyadic squareclass, and algebraic vectors by flat field extension. No arbitrary-ring base change in weight 3/2.
CM norms and heights [74, 73]	Split level, conductors coprime to level and the new prime, no extra units, unaveraged sums and one fixed parametrization.
Conductor-one reduction Section A35.3	Maximal CM order, one inert prime, fixed orientations and same-place ideal action. The geometric, CM and good-reduction inputs remain separate.
Duality [54, 56]	Actual derived Kummer conditions and their homotopies. Support change only after the coefficient representation is unramified at the removed place.
Forward finiteness [66, 67]	Only at known analytic nonvanishing/simple zeros or a known nontorsion Heegner trace.
Isogeny ratio [54, proof of I.7.3]	Finite Sha, dual sum/difference maps, canonical heights and pull-back measures; no BSD leading-term assumption.
Height [106, 38, 39, 40, 41, 48]	The same rationally descended r_5: absolute irreducibility, finite ramification, de Rham weights and oddness; then the published prime-five input, finite twist, explicit conductor and modular-Jacobian transfer. CM order class number is bounded using the original Hom field.
Primes [50, 51, 53] and Section I	Fixed finite-complexity system after fixing sieve cutoff, and the unconditional explicit progression estimate.

The released specification, natural precision polynomial and construction files are identified in `construction/SOURCE_MANIFEST.json`. For the selected supporting proof arguments, see `research/version14/SUPPORTING-NOTES.md`. A source identity is not a proof verdict.

A5 Fixed-form arithmetic and the even coefficient source

This appendix supplies the constructions used in (3.1)–(3.3). Throughout, $E = E_\ell$ is fixed, with the lattice, parametrization and period convention of Section 7. Constants may depend on these

data and on the prescribed finite list of local types. They do not depend on the number or sizes of the later good conductor primes.

A5.1 One retained class, including discriminant $4h$

All numbered primary references in this subsection are to Kato [69].

Fix a non-CM elliptic curve $E/\mathbf{Q}$ of conductor N, its weight-two newform f, $T = T_2E$, and $V = T \otimes \mathbf{Q}_2$. In the application $E = E_\ell$ is fixed first. Put $\Lambda = \mathbf{Z}_2[[\Gamma]] = \mathbf{Z}_2[[t]]$, where Γ is the real cyclotomic Galois group. Let $\kappa : G_{\mathbf{Q}} \to \Gamma \subset \Lambda^\times$ be the quotient map, and set $A = \Lambda[1/2]$. We use arithmetic Frobenius throughout. The action on the cyclotomic group-ring coefficient is the inverse deck action.

The precise output.

Proposition A5.1 (One retained Kato system and its two comparisons)**.** *Use the fixed curve, lattice and cyclotomic convention just specified. Fix $e \geq \max\{3, v_2(N)\}$ large enough for the fixed initial local data; the smoothing choice is (K5). Let G be a finite elementary abelian two-group, let $\psi : G_{\mathbf{Q}} \to G$ be a continuous homomorphism, and let $\chi_x = \chi_0(x \circ \psi)$, for $x \in \widehat{G}$, be a finite family of quadratic characters with one fixed real sign and fixed local squareclasses at the primes dividing $2N$. Since the characters of G separate its elements, $\psi(c) = 1$ for complex conjugation c, and $\psi|_{G_{\mathbf{Q}_v}} = 1$ at every $v \mid 2N$. All varying conductor primes are good odd primes for E. Let F be the squarefree least common multiple of the odd parts of the primitive conductors. There is a single retained-smoothed class*

$$Z_F \in 2^{-c_E} H^1(G_{\mathbf{Q},S}, T \otimes \Lambda[G](\chi_0\psi\kappa^{-1})), \qquad S = \mathrm{prime}(2NF), \tag{K1}$$

where c_E is fixed before F and G. There is also one $F_E \in \Lambda$ with $F_E(0) \neq 0$, fixed before F, G, such that at every height-one prime $\mathfrak{p}$ of A,

$$\ell_{A_\mathfrak{p}} H^2(C_{S,x})_\mathfrak{p} \leq \ell_{A_\mathfrak{p}} \left(H^1(C_{S,x})/AF_E(Z_F)_x \right)_\mathfrak{p}. \tag{K2}$$

Here $C_{S,x}$ is the full rational global Iwasawa complex of the constituent. For an odd squarefree signed parameter h giving $\chi_x = \chi_h$, central reciprocity for this same class gives

$$v_2 \exp^*(F_E(Z_F)_x(0)) = v_2 L_*(h) + \sum_{q|F,\ q\nmid h} v_2 \#E^h(\mathbf{F}_q) + 2e + v_2 F_E(0) + \epsilon_h, \tag{K3}$$

when the value is nonzero. The numbers ϵ_h belong to a bounded set depending only on the fixed curve, period conventions and local types at two. A zero central L-value gives zero dual exponential. Formula (K3) does not assert that its remaining unit is one. No part of (K1)–(K3) compares integral lattices of varying newforms.

Proof.

The retained route and its comparison. All classes in this subsection retain the smoothing integers of (K5). The comparison used below is the exact identity (E.1), obtained by integral weighted corestriction and the norm relation of [69, Proposition 8.12]. We do not divide by the smoothing operators, or compare a chosen generator with a separately normalized zeta class for the varying newform. In particular the central factor is s_F of (K6), whose valuation is $2e$ for every F. The use of power-series zero finiteness in paragraph 7 establishes nonvanishing of the supplied Euler system in both generic sign components; it is not a comparison of two Euler systems.

1. One original lattice and one symbol combination. Kato (8.1.3), p. 180, constructs retained-smoothed classes

$$ {}_{c,d_s}z_m(f,1,1,\xi,S_m) \in H^1(\mathbf{Z}[\zeta_m,1/2],L_f(1)), \qquad S_m=\mathrm{prime}(2mN), \tag{K4}$$

where $L_f = V_{\mathcal{O}_\lambda}(f)$ is the image lattice of the fixed form. For $\xi \in \mathrm{SL}_2(\mathbf{Z})$, section 8.9, p. 185, explicitly constructs the trace homomorphism into the integral induced lattice

$$\mathbf{Z}_2[\mathrm{Gal}(\mathbf{Q}(\zeta_m)/\mathbf{Q})] \otimes V_{2,\mathbf{Z}_2}(Y_1(N)).$$

Section 8.11 projects this lattice onto L_f. Auxiliary modular curve levels used in the trace do not change its target lattice. Choose a rational identification $L_f(1)\otimes\mathbf{Q}_2 = V$. One fixed 2^{c_E} sends its lattice into T.

By Kato 13.6–13.7, p. 227, the symbol span has a nonzero projection to each Betti sign line. The union of the two rational kernels does not contain the whole span. Choose an integral finite linear combination ξ with both projections nonzero, clearing its fixed denominators once. Norm relations and reciprocity are linear, so the same combination works at every m. Although (K4) is written with one symbol, it henceforth denotes this fixed combination.

2. Retain one smoothing choice for the whole family. Fix $e \geq \max\{3, v_2(N)\}$, enlarged only for fixed initial data, and take

$$c = d_s = 1 + 2^e\, 3N_{\mathrm{odd}}F. \tag{K5}$$

These integers exceed one, are prime to $6NF$, are one modulo N and F, and are one modulo every dyadic quadratic conductor under consideration. Distinctness or primality of c, d_s is not a hypothesis of Kato's construction. The exponents (u,v) in (4.2.4), p. 143, are $(1,1)$ when $k = 2, r = r' = 1$. The retained-smoothed version of Theorem 6.6, p. 163, therefore has central scalar

$$s_F = (c^2-c)(d_s^2-d_s) = c(c-1)d_s(d_s-1), \qquad v_2(s_F) = 2e. \tag{K6}$$

No smoothing factor is divided out. The prime factors of cd_s exclude some Euler-system indices in Kato 13.1; they are not added to the inverted arithmetic support S, as is explicit in (8.1.3) and in the definition of H^q in section 8.2. In particular their number creates no local H^2 correction.

3. The integral j_* condition survives ramified transfer. For a number field K, let $X_K = \mathrm{Spec}\,\mathcal{O}_K[1/2]$ and $j_K : \mathrm{Spec}\,K \to X_K$. At finite coefficients $M = T/2^aT$, the low-degree Leray sequence gives an injection

$$H^1(X_K, j_{K,*}M) \longrightarrow H^1(K,M)$$

whose image is the kernel of all inertia restrictions outside two. Indeed the higher-direct-image stalk formula and Leray sequence [55, Tags 03Q9 and 03QA] identify the stalks of $R^1j_{K,*}M$ at a finite place with $H^1(I_v,M)$; the edge map is restriction to these stalks. The generic-point map j_K is affine, hence quasi-compact and quasi-separated, as required by the stalk formula. Kato's section 8.2, pp. 180–181, defines the integral groups by the inverse limit of these finite-coefficient groups, so this assertion applies to the classes (K4) before rationalization.

Let L/K be finite, with absolute Galois groups $H \subset G_K$. For a class $z \in H^1(H,M)$ and an inertia subgroup $I \subset G_K$, the Mackey formula is

$$\mathrm{res}^{G_K}_I \,\mathrm{cor}^{G_K}_H\, z = \sum_{g\in I\backslash G_K/H} \mathrm{cor}^{I}_{I\cap gHg^{-1}}\, \mathrm{res}^{gHg^{-1}}_{I\cap gHg^{-1}}(gz). \tag{K7}$$

The intersection in each summand is the upper inertia subgroup for the corresponding place of L. If all upper inertia restrictions vanish, every summand is zero. This does not require L/K to be unramified. It is a finite double-coset identity at each finite coefficient precision. Uniqueness under the Leray injection makes the extensions across X_K compatible in a. Thus the integral j_* condition survives corestriction, including at primes ramified in a quadratic twisting field.

4. Primitive twisting is an actual Euler system. Fix a constituent $\chi = \chi_h$ of primitive conductor $D = |\Delta_h| = 2^a d$, d odd, and a basis e_χ of $\mathbf{Z}_2(\chi)$. Put $\Sigma = \mathrm{prime}(2Ndcd_s)$. For every positive integer m with $(m, cd_s) = 1$, set

$$K_m = \mathbf{Q}(\zeta_m), \quad L_m = \mathbf{Q}(\zeta_{Dm}), \quad y_m = \mathrm{cor}_{L_m/K_m}(z_{Dm} \otimes e_\chi). \tag{K8}$$

The character is trivial over L_m, so tensoring is an integral map there. Formula (K7) places (K8) in the precise j_* group required by Kato 13.1 on its index set $\mathrm{prime}(m) \cap \Sigma = \{2\}$. The larger definition is permitted by (8.1.3), Sections 8.9–8.11 and Proposition 8.12: $S_{Dm} = \mathrm{prime}(2NDm)$ is disjoint from $\mathrm{prime}(cd_s)$, since $d \mid F$ and (K5) holds. Each class uses the original f, ξ, c, d_s. We will also use this same construction on the different index set required by [89, Section 0.1], as checked below.

For $m \mid m'$, transitivity of corestriction reduces the norm of (K8) to Proposition 8.12, p. 186. For a deck automorphism g, equivariance gives

$$\mathrm{cor}((gz) \otimes e_\chi) = \chi(g)^{-1} g\, \mathrm{cor}(z \otimes e_\chi).$$

Consequently the factor at a newly introduced good prime q is

$$E_q(\chi) = 1 - \frac{a_q \chi(q)}{q} \sigma_q^{-1} + \frac{1}{q} \sigma_q^{-2}. \tag{K9}$$

In Kato 13.1, pp. 224–225, the polynomial is $P_q(X) = \det(1 - \mathrm{Fr}_q X \mid T\chi)$ and the norm factor is $P_q(q^{-1}\sigma_q^{-1})$, exactly (K9). Repeated prime powers introduce no factor because Proposition 8.12 has a product over newly added support primes. Thus the subfamily on Kato 13.1's index set is an Euler system of the stated lattice, not a class inferred from proportionality of rank-one Iwasawa modules. The required norm relations on the 1999 index set are verified separately below.

For $m = 2^n$ with $n \geq a$, the field L_m can be replaced by $\mathbf{Q}(\zeta_{d2^n})$: apply repeated-two norm compatibility from exponent $n + a$ to n. This introduces no factor 2^a. The construction therefore covers discriminant $4h$ on the same cofinal tail. At the coefficient level it retains the full character $\chi_2\chi_{\mathrm{odd}}$. A dyadic character nontrivial on Γ changes its action by the integral automorphism $1+t \mapsto -(1+t)$, or $t \mapsto -2 - t$. A dyadic sign character instead changes the finite real-sign component. Neither operation is a division by a sign projector.

5. One coefficient map produces every constituent. For n large enough to contain the dyadic characters, put $A_{F,n} = \mathrm{Gal}(\mathbf{Q}(\zeta_{F2^n})/\mathbf{Q})$. Apply Shapiro to z_{F2^n}, then the map on group bases

$$\Phi_n : \mathbf{Z}_2[A_{F,n}] \longrightarrow \mathbf{Z}_2[\Gamma_n \times G], \qquad [s] \longmapsto \chi_0(s)[\bar{s}][\psi(s)]. \tag{K10}$$

With inverse deck actions, both sides transform under $g \in G_{\mathbf{Q}}$ by the inverse of its group image. On the target the extra scalar is $\chi_0(g)$; since $\chi_0^{-1} = \chi_0$, (K10) is equivariant. This is an integral coefficient homomorphism. Shapiro restriction to an inertia subgroup is the direct sum of the upper restrictions; the same zero restrictions used in (K7) therefore remain zero after (K10). In particular the map acts on actual integral classes. The map in (K10) is defined on this finite group for the

following reason. The constituent indexed by the trivial character of G is χ_0, and $x \circ \psi = \chi_x \chi_0$ for every $x \in \widehat{G}$. All these quadratic Dirichlet characters have conductor dividing $F2^n$ once n is large enough. If an element of $G_{\mathbf{Q}}$ acts trivially on $\mathbf{Q}(\zeta_{F2^n})$, every character of G is therefore one on its image under ψ. Character separation makes that image the identity. Thus both χ_0 and ψ factor through $A_{F,n}$. Explicitly, for $g \in G_{\mathbf{Q}}$,

$$\Phi_n([g^{-1}s]) = \chi_0(g)\,[\bar{g}]^{-1}[\psi(g)]^{-1}\Phi_n([s]).$$

This verifies equivariance on each integral group basis before evaluation at any character. The quotient to the real tower sends each basis element to its image; its matrix has entries zero and one. The transition maps and character evaluations are also maps of integral coefficient modules. None of these operations is the idempotent $|G|^{-1}\sum_g x(g)[g]$.

The maps Φ_n commute with the norm transition maps. Proposition 8.12 has no new prime when n increases, so their images define the single class Z_F in (K1). The finite-coefficient degree-zero groups are finite and satisfy Mittag–Leffler. The inverse-limit exact sequence [56, Corollary 4.1.3, p. 96] therefore has zero $\varprojlim^1 H^0$ term, so it gives this degree-one class in completed cochains. The quotient from the full cyclotomic tower to Γ is the integral quotient of group bases, equivalently corestriction to the real subfield. No factor $(1+\tau)/2$ occurs.

After evaluation at x, (K10) is the coefficient form of the unnormalized corestriction weighted by χ_x. Factor that corestriction through the primitive conductor field in (K8). Transitivity and Proposition 8.12 give the exact identity

$$(Z_F)_x = \prod_{q \mid F/d} E_q(\chi_x)\, y_{2^\infty}. \tag{E.1}$$

It is an identity of the given classes. The same smoothing choice appears on both sides. There is no group-order denominator and no comparison with a canonical zeta class of $f \otimes \chi_x$.

For clarity, at a constituent write $\chi_h = \chi_2\chi_{\mathrm{odd}}$ and $|\Delta_h| = 2^a d$. Before taking the cofinal limit, the corresponding scalar coefficient map is exactly

$$[s] \longmapsto \chi_2(s)\chi_{\mathrm{odd}}(s)[s|_{\Gamma_n}], \qquad n \geq \max\{a, 2\}. \tag{E.2}$$

Thus the dyadic character is part of the module action, including when the field discriminant is $4h$.

6. Central trace and period normalization. First use (E.1) to remove the unused tame primes. The central specialization of (K8) is the corestriction of $z_D \otimes e_\chi$ from $\mathbf{Q}(\zeta_D)$ to $\mathbf{Q}$, with omitted support prime$(2Nd)$. For $D = 1$ use the trivial character and the same omitted support. This follows from the same repeated-two norm identity even when $D = 4|h|$. One must do this before using a Gauss basis; the Gauss sum of an imprimitive character modulo a larger cyclotomic level can be zero.

Theorem 9.7, p. 189, identifies the dual exponential of z_D with the retained-smoothed de Rham zeta element. Dual exponential commutes with corestriction, whose de Rham map is field trace. Thus the coefficient of e_χ is the exact unnormalized sum

$$\sum_{g \in \mathrm{Gal}(\mathbf{Q}(\zeta_D)/\mathbf{Q})} \chi(g)\, g(\exp^* z_D). \tag{K12}$$

Theorem 6.6, p. 163, evaluates precisely this sum. In weight two, $r = r' = 1$, its Betti sign is $\epsilon = \chi(-1)$ and its power of $2\pi i$ is zero. If e_B^ϵ is a fixed rational generator of that Betti line and $\delta(f, 1, \xi)^\epsilon = b_\epsilon e_B^\epsilon$, then $b_\epsilon \in \mathbf{Q}^\times$ is fixed and

$$\mathrm{per}_f\big((\mathrm{K12})\big)^\epsilon = s_F L_{\mathrm{prime}(2Nd)}(f, \chi, 1)\, b_\epsilon e_B^\epsilon. \tag{K13}$$

Choose the original de Rham generator so that its Betti period on this line is $\Omega_E^\epsilon e_B^\epsilon$, absorbing a fixed rational comparison in b_ϵ. Formula (K13) says that the scalar multiplying e_χ in (K12) is $s_F b_\epsilon L_{\mathrm{prime}(2Nd)}(f,\chi,1)/\Omega_E^\epsilon$. Its Galois character is χ.

To descend it, take the invariant basis $\sqrt{h}\,e_\chi$ of the quadratic character line. With the usual complex convention $\Omega_E^- = i|\Omega_E^-|$ and $\sqrt{h} = i\sqrt{|h|}$ for $h<0$, division by $\sqrt{h}$ gives, up to the already fixed rational period choice, exactly

$$s_F b_\epsilon \frac{L_*(h)}{h} \prod_{p\mid 2N} (\text{the omitted local factor at } p). \tag{K14}$$

The signed integer h is odd, so h^{-1} has valuation zero. This computation also explains the usual Gauss-basis description: for the primitive character, $\tau(\chi)^2 = \Delta_h$ and $\tau(\chi)e_\chi$ is invariant; replacing it by $\sqrt{h}e_\chi$ costs the fixed dyadic factor $\sqrt{\Delta_h/h}$. It does not cost $\varphi(D)$ or any power of the number of prime divisors of h. The discriminant-$4h$ case has ratio two in this last basis change, and the direct basis calculation (K14) already includes it. One can verify the normalization without choosing an idempotent. Write $\beta = \exp^* z_D$ in the original de Rham line and put $\beta_\chi = \sum_g \chi(g) g\beta$. Then $u\beta_\chi = \chi(u)\beta_\chi$ for every u in the finite cyclotomic group. Thus $\beta_\chi \otimes e_\chi$ is invariant, and its coordinate in the invariant basis $\sqrt{h}\,e_\chi$ is $\beta_\chi/\sqrt{h}$. Theorem 6.6 uses precisely β_χ, rather than $\beta_\chi/\varphi(D)$. Theorem 9.7 applies here since $1 \le r = r' = 1 \le k-1 = 1$; it has no restriction that p be odd. The dual form f^* in Theorem 6.6 equals f, because the elliptic newform has rational coefficients.

In the real-period convention, $L_*(h) = \sqrt{|h|}L(E^h,1)/|\Omega_E^\epsilon|$, up to the fixed rational period comparison already included in b_ϵ. For $h>0$, $\Omega_E^+\sqrt{h} = |\Omega_E^+|\sqrt{|h|}$; for $h<0$, $\Omega_E^-\sqrt{h} = -|\Omega_E^-|\sqrt{|h|}$. Consequently

$$\frac{L(E^h,1)}{\Omega_E^\epsilon\sqrt{h}} = \frac{L_*(h)}{h}$$

with the same fixed period comparison. This proves the signed factor in (K14), not only its valuation. For a primitive quadratic character, $\tau(\chi)^2 = \Delta_h$ and $\tau(\chi)/\sqrt{h}$ equals a fixed choice of $\sqrt{\Delta_h/h}$. Changing from its Gauss basis to the displayed invariant basis multiplies the coordinate by this ratio, which is one or two up to sign. In particular the discriminant-$4h$ case adds no factor depending on the number of odd conductor primes.

Identifying this generator with a minimal invariant differential (or its dual, according to the fixed convention) for E^h at two has bounded valuation: odd squarefree h belongs to one of finitely many classes in $\mathbf{Q}_2^\times/\mathbf{Q}_2^{\times 2}$, and changing a representative by a unit square scales the quadratic twist differential by a two-adic unit. The local factors at $2N$ also form a fixed finite nonzero list. At a good active prime the local L-factor is one; at an unused good prime (K9) at $t=0$ is $\#E^h(\mathbf{F}_q)/q$. This proves (K3), including all its uniformity claims and the assertion about zero values.

7. Every hypothesis of Kato 13.4(2). Apply the theorem to the integral system (K8), after multiplying by the fixed lattice comparison if needed. Its hypotheses are:

- Each generic sign component is nonzero. For a finite-order cyclotomic character ρ, corestriction and the ρ-sum combine into the single $(\chi\rho)$-weighted sum on the original class. Theorems 6.6 and 9.7 apply with the same ξ, c, d_s. Their smoothing factors are nonzero, since $c, d_s > 1$ cannot be roots of unity. Theorem 13.5(2), p. 227, gives only finitely many zero L-values among characters supported on the fixed finite set prime$(2Nd)$. There are infinitely many ρ of each parity. A torsion Iwasawa class can have a nonzero specialization at only finitely many such characters, since a nonzero annihilating power series has only finitely many cyclotomic zeros. Hence both generic sign components are nonzero.

- The two real eigenspaces of $T\chi$ have rank one.
- The theorem uses inverse Frobenius in its purity condition. On $T\chi$ its good eigenvalues have complex absolute value $q^{-1/2}$, so its integer w is -1.
- $V\chi$ is irreducible; twisting preserves irreducibility of the non-CM rational Tate representation.
- The image over $\mathbf{Q}(\mu_{2^\infty})$ contains an open subgroup of $\mathrm{SL}_2(\mathbf{Z}_2)$. Intersecting with $\ker\chi$ keeps an open subgroup, which contains a nonidentity unipotent. It fixes a one-dimensional subspace and gives condition (v).

These are all rational hypotheses. The residual irreducibility, integral cokernel and $p \neq 2$ conditions of part (3) are not used. Theorem 13.4(1) gives torsion $\mathbf{H}^2(T\chi)$, and part (2) gives, at every height-one prime away from two,

$$\ell\mathbf{H}^2(T\chi)_{0,\mathfrak{p}} \leq \ell(A_\mathfrak{p}/J_\mathfrak{p}), \qquad \mathbf{H}^2_0 = \ker(\mathbf{H}^2 \to \mathbf{H}^2_{\mathrm{loc}}). \tag{K15}$$

The finiteness and Euler characteristic (12.2.1)–(12.2.2), p. 220, give generic rank one in $\mathbf{H}^1$. Its localization away from two is torsion-free. Indeed a torsion element for a scalar prime would come from invariants of the corresponding finite-dimensional rational coefficient algebra at that prime; its coefficient action is abelian, whereas the open commutator subgroup on $V\chi$ has no fixed vectors. The coefficient exact sequence excludes these invariants. The scalar specialization need not be a finite-order character. Thus $\mathbf{H}^1_\mathfrak{p}$ is free of rank one. Localization commutes with Hom for this finitely presented module. Writing $y_{2^\infty} = ae$ in a local basis gives $J_\mathfrak{p} = (a)$. Formula (K15) is exactly the index bound for the supplied class, not for an arbitrarily rescaled generator.

The published prime-two Euler-system input. For $p = 2$ we use the rational, away-from-two case of [89, Theorem 0.8, pp. 317–318], with the stronger hypothesis in its condition 0.6(ii), to justify the application of [69, Theorem 13.4(2)]. For our non-CM elliptic representation the image of $G_{\mathbb{Q}^{\mathrm{ab}}}$ contains an open subgroup of $\mathrm{SL}_2(\mathbb{Z}_2)$, as stated in [89, Example 0.6.2, pp. 317–318], using Serre's open-image theorem. A quadratic character χ is trivial on $G_{\mathbb{Q}^{\mathrm{ab}}}$. Thus the same subgroup acts on $V\chi$ and contains a matrix $\left(\begin{smallmatrix}1 & b\\ 0 & 1\end{smallmatrix}\right)$ with $b \neq 0$. Its fixed space over $\mathbb{Q}_2$ has dimension one. This supplies the unipotent in $G_{\mathbb{Q}^{\mathrm{ab}}}$, not merely in the cyclotomic subgroup. The latter subgroup also contains this open SL_2 subgroup, so $V\chi$ remains absolutely irreducible over $\mathbb{Q}(\mu_{2^\infty})$. Indeed, a line invariant under both upper and lower nonidentity unipotents cannot exist, even over an algebraic closure. No inference from irreducibility over $G_\mathbb{Q}$ alone is needed.

We check the index set and the supplied class in that theorem explicitly. Put $N_K = 2Nd$ and $N'_K = cd_s$. These integers are coprime by (K5), since $d \mid F$. Section 0.1 of [89] requires every index m with $N_K \mid m$ and $(m, N'_K) = 1$. The enlarged definition (K8) supplies all these classes. They originate in the integral j_* group over $\mathbf{Z}[\zeta_m, 1/2]$ and map to full cohomology over $\mathbf{Z}[\zeta_m, 1/N_K]$. The varying construction support S_{Dm} is not an instruction to invert all its primes in this last scheme. Apply the fixed lattice map 2^{c_E} from (K4) if needed, identically at every level.

For a prime $\ell \nmid N'_K$, if $\ell \mid m$ then $S_{Dm\ell} = S_{Dm}$, so Proposition 8.12 gives norm factor one. If $\ell \nmid m$, the condition $N_K \mid m$ implies $\ell \nmid 2Nd$, and hence $S_{Dm\ell} \setminus S_{Dm} = \{\ell\}$. The equivariance identity before (K9) gives the factor $P_\ell(\ell^{-1}\sigma_\ell^{-1}) = E_\ell(\chi)$, with $P_\ell(X) = 1 - a_\ell\chi(\ell)X + \ell X^2$. These are exactly the two cases of (0.1.1) in [89]. Smoothing primes are excluded by N'_K and need not be inverted.

For the class in Section 0.7 of that source take its parameter $d = 1$; this parameter is distinct from our odd conductor part d. For every $n \geq 1$,

$$S_{DN_K2^n} = S_{D2^n} = \mathrm{prime}(2Nd).$$

Proposition 8.12 therefore gives

$$\operatorname{cor}_{\mathbf{Q}(\zeta_{DN_K2^n})/\mathbf{Q}(\zeta_{D2^n})} z_{DN_K2^n} = z_{D2^n}.$$

Transitivity of corestriction, through the two intermediate fields in the common upper field, now yields

$$\operatorname{cor}_{\mathbf{Q}(\zeta_{N_K2^n})/\mathbf{Q}(\zeta_{2^n})} y_{N_K2^n} = \operatorname{cor}_{\mathbf{Q}(\zeta_{D2^n})/\mathbf{Q}(\zeta_{2^n})}(z_{D2^n} \otimes e_\chi) = y_{2^n}.$$

No Euler factor occurs because the supports are equal. No field degree occurs because this is corestriction of the actual upper class, not corestriction of a restriction. These equalities commute with transition maps, and $n \geq 1$ is a cofinal tail. Write ξ_K for the Section 0.7 class, to distinguish it from the modular-symbol combination ξ. It is exactly the full-cohomology image of y_{2^∞}, multiplied by 2^{c_E} if the fixed lattice map was needed. That factor is a unit at every height-one prime under consideration, so its coordinate ideal is precisely $J_\mathfrak{p}$. The preceding nonvanishing argument supplies a nontorsion class on each sign component, purity gives $w = -1$, and oddness gives condition $0.6(\mathrm{iv}_\mathfrak{p})$. At the height-one primes used here, 2 is invertible and the localized Iwasawa algebra is a DVR. Theorem 0.8 then bounds the coordinate ideal of this actual retained class. Its strong residual and integral-cokernel assumptions concern primes containing 2, which are not used here.

We also account for the difference between the two published cohomology conventions. Write C_j for the j_* Iwasawa complex of [69, Sections 8.2 and 12.2], and C_S for *étale* cochains on the open arithmetic scheme obtained by inverting 2 and every odd prime in the actual ramification support S, including the fixed twist, as in [89, Sections 0.1 and 0.5]. The support modulus called N in that source is taken to contain these primes; it is not merely the conductor of the untwisted curve. Work in one rational sign component and localize at $\mathfrak{p}$. Their quotient is the direct sum of the odd local complexes (K16). Each differential has nonzero determinant, so its degree-one cohomology vanishes. Hence $H^1(C_j) \simeq H^1(C_S)$ and $H^2(C_j) \hookrightarrow H^2(C_S)$. At an odd prime q, the unramified local term has cohomological dimension one after inverting 2. The localization triangle identifies the degree-two cohomology of (K16) with full local $H^2(\mathbb{Q}_q, T\chi)$ in Iwasawa cohomology. The map from $H^2(C_S)$ is the usual local restriction. Exactness therefore identifies $H^2(C_j)$ with the kernel of all those odd local restrictions. Intersecting with the kernel of restriction at 2 gives

$$\ker\bigl(H^2(C_j) \to H^2_{\mathrm{loc},2}\bigr) \simeq \ker\bigl(H^2(C_S) \to \bigoplus_{q\in S} H^2_{\mathrm{loc,q}}\bigr).$$

This is precisely the comparison of the subscript-zero groups in the two sources. The H^1 identification preserves the actual Euler-system class and its coordinate ideal J. Thus Theorem 0.8 gives (K15) in the convention used here. The argument for (K16) below is local and does not use (K15). We do not invoke the odd-prime-only Perrin–Riou argument or an unchecked version of Rubin's theorem at 2.

8. Paying for the full global complex. At an odd prime put $W_q = H^1(I_q, V\chi)$. The rational cone from j_* to full cochains is

$$[\, W_q \otimes A \xrightarrow{1-\sigma_q^{-1}\operatorname{Fr}_{W_q}} W_q \otimes A\,], \qquad \deg = (1,2). \tag{K16}$$

Its determinant D_q is nonzero, since the cyclotomic Frobenius exponent is nonzero. The map is injective even after localization at a prime dividing D_q; it is not being reduced modulo that prime. Thus the cone contributes only degree-two length $v_\mathfrak{p}(D_q)$. The long exact sequence shows that full H^2 increases by at most their sum and that H^1 is unchanged. This inequality does not require vanishing of the next j_* group.

At a good inactive prime, $W_q = V\chi(-1)$, so $D_q = E_q(\chi)$ with exactly the convention of (K9). At a good active prime, ramified quadratic inertia acts by -1; its rational cohomology vanishes and $D_q = 1$. This includes every active base prime, regardless of how many there are. At an old bad prime retain $H^1(I_q, V\chi)$ itself; replacing it by $V^{I_q}\chi(-1)$ would in general be false.

Choose a fixed integral F_2 divisible over A by the characteristic elements of $\mathbf{H}^2_{\mathrm{loc}}(T\chi)$ for the finitely many allowed local types at two and finite sign components. Its central value can be chosen nonzero: derived local control, and the absence of local Iwasawa H^3, identify its rational central H^2 with $H^2(\mathbf{Q}_2, V\chi) = 0$. The last equality follows from local duality and finiteness of local elliptic torsion. Hence the characteristic elements have no factor t. Kato (12.2.3), p. 220, gives the required finiteness and torsion of this local Iwasawa module.

At the finitely many old bad primes, choose stable lattices in the W_q and multiply all the determinants D_q for the finitely many local twisting types. Clear their fixed scalar denominators. Their central values are nonzero, since the cokernel of $1 - \mathrm{Fr}_{W_q}$ is $H^2(\mathbf{Q}_q, V\chi) = 0$ by the same local duality. Call this product F_{bad} and set $F_E = F_2 F_{\mathrm{bad}}$.

The quotient $\mathbf{H}^2/\mathbf{H}^2_0$ injects into the local $\mathbf{H}^2_{\mathrm{loc}}$, so F_2 pays for it in (K15). Multiplying a class in a free rank-one module by a nonzero scalar adds that scalar's valuation to its index. Formula (E.1) already pays for every unused good determinant in (K16); F_{bad} pays for the remaining old ones. Adding these length inequalities proves (K2). Here is the length calculation with all contributions separated. Fix a height-one prime $\mathfrak{p}$ of A and write $R = A_{\mathfrak{p}}$. Let M be the localized degree-one module for the primitive j_* system, let $y = y_{2\infty}$, and write $\mathrm{ind}_R(y) = \ell_R(M/Ry)$. The preceding freeness argument gives $M \simeq R$. Put $Q_\chi = \prod_{q|F/d} E_q(\chi)$. All primes in F/d are good: the family has identical local characters at primes dividing N, so a bad prime cannot ramify in one constituent and be unramified in another. The cone (K16) has zero degree-one cohomology, hence identifies M with the localized full degree-one module. Its long exact sequence and (K15) give

$$\begin{aligned}
\ell_R H^2(C_{S,x})_{\mathfrak{p}} &\le \ell_R \mathbf{H}^2(T\chi)_{0,\mathfrak{p}} + \ell_R \mathbf{H}^2_{\mathrm{loc}}(T\chi)_{\mathfrak{p}} + \sum_{q\in S,\ q\neq 2} v_{\mathfrak{p}}(D_q) \\
&\le \mathrm{ind}_R(y) + v_{\mathfrak{p}}(F_2) + v_{\mathfrak{p}}(F_{\mathrm{bad}}) + v_{\mathfrak{p}}(Q_\chi) \\
&= \mathrm{ind}_R(F_E Q_\chi y) = \mathrm{ind}_R(F_E(Z_F)_x).
\end{aligned}$$

The second line uses $D_q = 1$ at active good primes, and $D_q = E_q(\chi)$ at inactive good primes. It does not charge an extra fixed constant for each active prime.

For completeness, nonvanishing at the center of the fixed correction factors follows before multiplying over local types. Local derived control and $H^3_{\mathrm{Iw}} = 0$ identify $\mathbf{H}^2_{\mathrm{loc}}/t\mathbf{H}^2_{\mathrm{loc}}$ rationally with $H^2(\mathbf{Q}_2, V\chi)$. Local duality identifies the latter with the dual of $H^0(\mathbf{Q}_2, V\chi)$, which is zero because local elliptic torsion is finite. Nakayama at (t) therefore excludes a factor t from its characteristic element. At an old odd prime the cokernel of $1 - \mathrm{Fr}_{W_q}$ is $H^2(\mathbf{Q}_q, V\chi) = 0$, by the same duality. Its determinant at $t = 0$ is nonzero. Taking products over the finite list of local types, followed by one fixed power of two to clear scalar denominators, gives $F_E \in \Lambda$ with $F_E(0) \neq 0$. This operation is independent of the binary dimension and all later good conductor primes.

9. Uniformity and the boundary of this proposition. The only lattice denominator is fixed before the family. The smoothing integers may grow, but their central valuation is $2e$. The multiplier F_E depends only on the original local types. Transfer, coefficient push, real quotient and norm compatibility are integral. The proof never divides by a conductor degree, $|G|$, a class number, or a Fourier projector. It keeps one class through (K1), (K2), and (K3). □

We now use this same class in the positive complex. The preceding argument supplies its integral construction, its rational height-one divisibility and its exact central reciprocity normalization.

Proposition A5.2 (One integral even scalar for the family)**.** *Retain Proposition A5.1 and assume, as for the present curve, that $E[2](\mathbb{Q}) \simeq (\mathbb{Z}/2\mathbb{Z})^2$. Let C^+ be the fiber of the ordinary global cochains to the ordinary real cochains, with real degree zero retained. It has a compatible finite free model under the coefficient changes used here. For $B = \mathcal{O}[G]$, it satisfies $C^+ \otimes B \simeq B^2[-1]$. The fixed real invariant generator, the positive lift of the retained class, and the original determinant basis define the single scalar u^{ev} by (E.4), with*

$$u^{\mathrm{ev}} \in \Lambda[1/2][G], \qquad 2^{c_E} u^{\mathrm{ev}} \in \Lambda[G].$$

Here c_E may be enlarged for the fixed data but is independent of G and the later good conductor primes. The assertion requires the common real sign and local squareclasses at $2N$ in Proposition A5.1.

Proof. Let C^+ be the fiber of ordinary global cochains to ordinary real cochains. Real H^0 is included. Over a finite Artin quotient, continuous cochains are filtered unions of finite free function modules, hence flat and therefore free. A residual splitting into finite cohomology and disks lifts by the nilpotent perturbation series to a finite free model with actual homotopies. Minimal models at successive quotients are related by invertible matrices: a homotopy equivalence reduces to an isomorphism of minimal residual complexes, and Nakayama applies. Lifting these matrices and comparison homotopies makes the models compatible. Their acyclic, termwise surjective comparison cones have acyclic inverse limit. Here is the compatibility argument. At successive finite coefficient quotients, reduction of a minimal finite free model is a minimal model of the reduced cochains. Two such models are homotopy equivalent; reducing a homotopy inverse modulo the maximal ideal shows that the comparison is degreewise invertible. Lift these invertible matrices to the next quotient and change bases. The model differentials then reduce to the preceding differentials exactly. The comparison maps to the original cochains agree after reduction up to chain homotopy. Since reduction of continuous cochains is termwise surjective, lift that homotopy and adjust the upper comparison by its boundary. The comparison maps are now exactly compatible and remain quasi-isomorphisms.

Let $\mathcal{K}_n$ be their acyclic comparison cones. Their transition maps are termwise surjective. If s denotes the transition shift on the product, successive lifting gives the exact sequence of complexes

$$0 \longrightarrow \varprojlim_n \mathcal{K}_n \longrightarrow \prod_n \mathcal{K}_n \xrightarrow{1-s} \prod_n \mathcal{K}_n \longrightarrow 0.$$

Products of complexes of modules preserve acyclicity, so the long exact cohomology sequence makes $\varprojlim_n \mathcal{K}_n$ acyclic. The inverse-limit comparison is therefore a quasi-isomorphism from the bounded finite free limit model to the continuous coefficient complex. This also identifies its derived coefficient changes with the finite-quotient constructions. This proves the needed perfection and coefficient change.

The residual positive complex. Write $K_n = \mathbf{Q}(\zeta_{2^{n+2}})^+$, and let S_n consist of the places above two and a fixed finite set of odd rational primes. Put $r_n = 2^n$ and $s_n = \#S_n$. We first compute with the scalar coefficient $\mathbf{F}_2 = \mu_2$.

The ordinary two-class group of K_n is trivial. Indeed this holds for $K_0 = \mathbf{Q}$. Assuming it for K_{n-1}, let H be the Hilbert two-class field of K_n, with the real places split. The canonical extension H/K_{n-1} is Galois with a finite two-group G_H. Only the unique dyadic prime can ramify, and its inertia I has order two. The quotient of G_H^{ab} by the image of I gives an everywhere unramified abelian two-extension of K_{n-1} with real places split, hence is trivial by induction. Thus G_H^{ab} is

cyclic, generated by an element of order at most two. The Frattini quotient of G_H has one generator, so G_H is cyclic and has order at most two. Since it contains K_n, we obtain $H = K_n$. The S_n-class group is a quotient, so its two-part also vanishes.

The signatures of units span all r_n real signs. For $n \geq 1$ put $m = n + 2$, $\zeta = \zeta_{2^m}$, and

$$u = 1 + \zeta + \zeta^{-1} = \zeta^{-1}\frac{\zeta^3 - 1}{\zeta - 1}.$$

This is a unit: the numerator and denominator generate the same dyadic prime ideal in the cyclotomic field. Its real conjugates are $1 + 2\cos(2\pi a/2^m)$ for odd $0 < a < 2^{m-1}$. The negative ones have $2^m/3 < a < 2^{m-1}$, and their number is

$$2^{m-2} - \left\lfloor \frac{2^{m-1}}{3} + \frac{1}{2} \right\rfloor,$$

an odd integer. Hence $N_{K_n/\mathbf{Q}}(u) = -1$. The signature vector of u has augmentation one in the local ring $\mathbf{F}_2[\mathrm{Gal}(K_n/\mathbf{Q})]$ and is therefore a unit. Its Galois translates span all signatures. At $n = 0$ use -1.

Kummer theory now gives $\dim H^1(G_{K_n,S_n}, \mathbf{F}_2) = r_n + s_n$, and its signature map to $\mathbf{F}_2^{r_n}$ is onto. The map in degree zero is the diagonal injection. The positive complex is the fiber of ordinary global cochains to the sum of ordinary real cochains, so its long exact sequence gives $\dim H^1(C_n^+) = r_n + s_n - 1$. The vanishing two-part of the S_n-class group identifies global H^2 with $\mathrm{Br}(\mathcal{O}_{K_n,S_n})[2]$. The local invariant description has $r_n + s_n$ coordinates with one sum relation. Projection to the r_n real coordinates is onto, because $s_n \geq 1$, and has kernel of dimension $s_n - 1$. Together with signature surjectivity this gives

$$\dim H^1(C_n^+) = 2^n + s_n - 1, \qquad \dim H^2(C_n^+) = s_n - 1, \qquad H^i(C_n^+) = 0 \quad (i \neq 1, 2).$$

The last assertion uses the usual global-to-real comparison in degrees greater than two; degree zero vanishes by the diagonal injection.

For each fixed support, s_n is bounded. The dyadic prime is unique; each fixed odd prime has open decomposition group in the real cyclotomic $\mathbf{Z}_2$-extension, since its image in $\mathbf{Z}_2^\times/\{\pm 1\}$ has nonzero two-adic logarithm. Use now the finite free model and coefficient change already constructed. Its scalar reduction is a bounded finite free $\mathbf{F}_2[[t]]$-complex, and its finite cyclotomic specializations are derived quotients by t^{2^n}. Smith decomposition shows that torsion disks and their Tor terms contribute bounded dimensions, while a free cohomology generator contributes 2^n. The displayed dimensions therefore give generic rank one in degree one and zero in all other degrees.

For the present curve, $E[2]$ is the direct sum of two trivial $\mathbf{F}_2$-representations. Thus the residual positive complex has rank two in degree one only. The residue field of the local ring $B = \mathcal{O}[G]$ is $\mathbf{F}_2((t))$; the augmentation ideal of the finite two-group lies in its maximal ideal. A minimal finite free B-model has zero residual differential, so the just-computed dimensions force exactly two generators in degree one and no others. Consequently

$$C^+ \otimes B \simeq B^2[-1]. \tag{E.3}$$

This uses the local ring B, not semisimplicity of $\mathbf{F}_2[G]$.

The common real sign is needed here. The real cyclotomic character and ε are trivial on complex conjugation, so the real action is the fixed action on $T_2E \otimes \chi_0$, extended to the group ring. Its invariant module is the fixed free rank-one $\mathbb{Z}_2$-lattice extended to $\Lambda[G]$, and then to $\mathcal{O}[G]$. Thus a generator can be chosen before G varies. Real H^1 is killed by two over these coefficient rings,

so the positive lift below costs the same single factor two for every binary dimension. Choose a generator of the fixed real invariant lattice and its connecting class a. Since real H^1 is killed by two, $2^{c_E+1}F_EZ_F$ has an integral positive lift b. For a basis θ of the original inverse determinant line, define

$$u^{\mathrm{ev}}\theta = 2^{-c_E}\, a \wedge b. \tag{E.4}$$

The wedge does not depend on the choice of the positive lift. Indeed, the difference of two lifts is in the image of real degree-zero cohomology by the cone exact sequence. That module is generated by the chosen real invariant generator, so the difference is a multiple of a and wedges to zero with a. After rationalization set $z = F_EZ_F$ and choose any lift $\widetilde{z}$ of z. Formula (E.4) then reads

$$u^{\mathrm{ev}}\theta = 2a \wedge \widetilde{z}.$$

Thus the factor 2^{c_E} used to make the lift integral cancels exactly. Increasing the fixed clearing exponent does not change the rational scalar. Character evaluation and derived central specialization preserve this equality and its cone maps. The class whose dual exponential occurs in (E.7) is consequently the same $F_E(Z_F)_h$ as in (K3), and the single remaining factor two is exactly the $+1$ in (E.7).

From the full cohomology bound to the same determinant scalar. Here is the rational divisibility step explicitly. Fix a character x of G, a height-one prime $\mathfrak{p}$ of A, and $R = A_{\mathfrak{p}}$. Put $z = F_E(Z_F)_x$, without changing its normalization. The full global degree-one module is free of rank one over R by the argument following (K15) and the injectivity of the maps in (K16). Write it as Re, and write $z = \alpha e$. The higher real cohomology vanishes over R, since two is invertible. Global degree-zero cohomology is zero: an invariant would restrict to an invariant of the open nonabelian subgroup on the elliptic representation, on which the scalar coefficient character is trivial. Thus the positive-cone exact sequence gives

$$0 \longrightarrow Ra \longrightarrow H^1(C_x^+ \otimes R) \longrightarrow Re \longrightarrow 0, \qquad H^2(C_x^+ \otimes R) = H^2(C_{S,x} \otimes R).$$

The element a here is the image of the fixed real invariant generator in (E.4), and it is primitive in this positive degree-one module, because the quotient Re is free. Choose a lift $\widetilde{e}$ of e; then $(a, \widetilde{e})$ is a basis. The equality already proved from (E.4) becomes

$$u_x^{\mathrm{ev}}\theta_x = 2\alpha\, a \wedge \widetilde{e}.$$

Let $d_1, \dots, d_r$ be the nonzero Smith entries of a finite free model of $C_x^+ \otimes R$. All its cohomology outside degrees one and two is zero, its degree-one cohomology is free, and its degree-two cohomology is torsion. Its inverse determinant lattice, under the rational identification with $\bigwedge^2 H^1$, is therefore generated, up to an R-unit, by $(\prod_i d_i)a \wedge \widetilde{e}$. It follows that

$$\begin{aligned} v_{\mathfrak{p}}(u_x^{\mathrm{ev}}) &= v_{\mathfrak{p}}(\alpha) - \sum_i v_{\mathfrak{p}}(d_i) \\ &= \ell_R(H^1(C_{S,x} \otimes R)/Rz) - \ell_R H^2(C_{S,x} \otimes R) \;\geq\; 0, \end{aligned} \tag{E.4a}$$

where two is a unit and the inequality is exactly (K2). This uses the full H^2 corrected in (K16), rather than only Kato's $\mathbf{H}^2_0$.

The intersection of the height-one valuation rings of the normal domain A gives $u_x^{\mathrm{ev}} \in A$ for every x. Since $|G|$ is a power of two, the character idempotents belong to $A[G]$ and these memberships give $u^{\mathrm{ev}} \in A[G]$. This establishes membership only; it is not a bound for the denominator from

character inversion. The uniform bound comes separately from (E.3): in its free rank-two $\mathcal{O}[G]$-module the integral vectors a, b of (E.4) have integral wedge, so $2^{c_E}u^{\mathrm{ev}} \in \mathcal{O}[G]$ with the fixed exponent already used for the original lattice. Coefficientwise intersection in the original group basis gives

$$A \cap \mathcal{O} = \Lambda, \qquad 2^{c_E}u^{\mathrm{ev}} \in \Lambda[G]. \tag{E.5}$$

Indeed an element of A has bounded powers of two as denominators in its power-series coefficients; membership in $\mathcal{O}$ makes their minimum valuation nonnegative. The coefficientwise intersection retains c_E, without a term $v_2|G|$. □

A5.2 Central saturation and the numerical lower bound

Proposition A5.3 (Central calibration of the even scalar)**.** *Use the scalar and determinant basis of Proposition A5.2. At a corank-zero constituent E^h with finite* Ш$[2^\infty]$*, its nonzero central value satisfies*

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + \epsilon_h, \qquad |\epsilon_h| \le C_{\mathrm{loc}},$$

with one constant for the fixed local types. At such a constituent $u_h^{\mathrm{ev}}(0) = 0$ if and only if $L_*(h) = 0$. Consequently every admissible h with $L_*(h) \neq 0$ satisfies $v_2 L_*(h) \ge 2\omega(h) + s(h) - c_E - C_{\mathrm{loc}}$: the forward rank-zero theorem in Lemma 5.5 supplies the preceding corank and finiteness hypotheses in this case. The central saturation corrections are those in (E.6)–(E.7); no primitivity after inverting t is assumed to survive central specialization.

Proof. At a corank-zero constituent put

$$g = \ell_{\mathbb{Z}_2} E^h(\mathbb{Q})[2^\infty], \quad g_v = \ell_{\mathbb{Z}_2} E^h(\mathbb{Q}_v)[2^\infty], \quad r_i = \ell_{\mathbb{Z}_2} H^i(\mathbf{R}, T_h) \quad (i = 1, 2).$$

Let i be the index length of the image of global H^1 in the free rank-one quotient $L_2 = H^1(\mathbb{Q}_2, T_h)/H^1_f(\mathbb{Q}_2, T_h)$. Orthogonal Kummer duality, with its real terms, gives

$$\ell H^2_{\mathrm{glob}} = s(h) - i + \sum_v g_v + r_2 - g. \tag{E.6}$$

Write $K = \ker(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}})$, let k be its torsion length and m its free index. If $P = H^1(C_h^+)$, p_0 is its torsion length and a_s the saturation index of the real image in P/P_{tors}, exactness gives

$$\ell \operatorname{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) = r_1 - g + k - m, \qquad k = p_0 + a_s.$$

For the numerical calculation suppose also $L_*(h) \neq 0$. The wedge in (E.4) adds $a_s + 1 + v_{\mathrm{free}}(z) - m$. Conversion to the inverse determinant adds $p_0 - \ell H^2(C_h^+)$. Substitution of these equalities cancels m and a_s exactly:

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 \exp^* z - s(h) - \sum_v g_v + 2g + 1 - r_1 - e_2, \tag{E.7}$$

where e_2 is the valuation of $\exp^*$ on a generator of L_2. Primitivity of the real image after inverting t has not been specialized to $t = 0$.

Here are the real local convention and the exact length sequence used in (E.6). Put $T = T_2(E^h)$ and $D = E^h[2^\infty]$. Since $E^h[2]$ is rational, complex conjugation J satisfies $J \equiv 1 \pmod 2$. The integral projectors $(1 \pm J)/2$ give $T = T^+ \oplus T^-$. At the real place use the bounded Kummer complex

$$U_{\mathbf{R}} = \tau_{\le 1} R\Gamma(\mathbf{R}, T) = [\, T^+ \oplus T^- \xrightarrow{J-1} T^- \,]$$

in degrees zero and one, with its map to ordinary real cochains. Modulo 2^n its degree-zero cohomology is $T^+/2^n \oplus (T^-/2^n)[2] = E^h(\mathbf{R})[2^n]$, and its degree-one image is the real Kummer line $T^-/2$. The latter can also be checked using a rectangular period lattice: dividing the nonidentity real component, represented by half the imaginary period, produces the minus-line cocycle. Thus this line is its exact annihilator under finite real Tate duality. The ordinary completion $E^h(\mathbf{R})^\wedge_2[-1]$ alone would lose the first summand of degree-zero cohomology.

At a finite place take the derived Kummer map from $E^h(\mathbb{Q}_v)^\wedge_2[-1]$. Its reduction has full local degree-zero cohomology and injective Kummer degree-one map. If C_{Sel} denotes the resulting ordinary global Selmer cone, these assertions identify

$$H^1(C_{\mathrm{Sel}} \otimes^{\mathbf{L}} \mathbb{Z}/2^n) = \mathrm{Sel}_{2^n}(E^h/\mathbb{Q}).$$

At corank zero the discrete Selmer group is finite, the Mordell–Weil rank is zero, and its order is that of $\mathrm{Sha}[2^\infty]$. The orthogonal-Kummer specialization of finite Poitou–Tate duality [56, 5.7.1.4–5.7.1.7], followed by the compact/discrete limits, gives

$$0 \longrightarrow L_2/J_2 \longrightarrow \mathrm{Sel}_{2^\infty}(E^h/\mathbb{Q})^\vee \longrightarrow H^2_{\mathrm{glob}} \longrightarrow \bigoplus_{v\in S_f} H^2(\mathbb{Q}_v, T) \oplus H^2(\mathbf{R}, T) \longrightarrow H^0(\mathbb{Q}, D)^\vee \longrightarrow 0,$$

where J_2 is the image of global H^1. There is no real degree-one quotient: the compact real Kummer image is all of $H^1(\mathbf{R}, T)$, and its discrete orthogonal Kummer group is zero. At finite $v \neq 2$ the compact Kummer image is likewise all of local H^1, while at two its quotient is L_2. The real degree-two term is retained. The finite-level images satisfy Mittag–Leffler, and high-degree localization at the real place terminates the sequence.

This also makes H^2_{glob} finite. The cone sequence then makes $H^2(C_{\mathrm{Sel}})$ finite, and universal coefficients give

$$0 \to E^h(\mathbb{Q})/2^n \to \mathrm{Sel}_{2^n}(E^h/\mathbb{Q}) \to H^2(C_{\mathrm{Sel}})[2^n] \to 0.$$

Comparison with the Kummer sequence gives $\ell H^2(C_{\mathrm{Sel}}) = s(h)$, with no extra real torsion. Equivalently the cone sequence is

$$0 \to L_2/J_2 \to H^2(C_{\mathrm{Sel}}) \to H^2_{\mathrm{glob}} \to \bigoplus_{v\in S_f} H^2(\mathbb{Q}_v, T) \oplus H^2(\mathbf{R}, T) \to H^0(\mathbb{Q}, D)^\vee \to 0.$$

Its alternating lengths are exactly (E.6).

For the positive cone, $H^2_{\mathrm{glob}} \to H^2(\mathbf{R}, T)$ is onto: any real component's image in $H^0(\mathbb{Q}, D)^\vee$ can be canceled by a component at two, since restriction $H^0(\mathbb{Q}, D) \to H^0(\mathbb{Q}_2, D)$ is injective. Thus its exact sequences are

$$0 \to T^+ \xrightarrow{a} P \to K \to 0, \qquad 0 \to \mathrm{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) \to H^2(C_h^+) \to H^2_{\mathrm{glob}} \to H^2_{\mathbf{R}} \to 0.$$

If k and p_0 are the torsion lengths of K and P, m is the free index of K in global H^1, and a_s is the saturation index of the real line in P/P_{tors}, these sequences give

$$\ell\,\mathrm{coker}(H^1_{\mathrm{glob}} \to H^1_{\mathbf{R}}) = r_1 - g + k - m, \qquad k = p_0 + a_s.$$

The wedge $2^{-c_E} a \wedge b$ contributes $a_s + 1 + v_{\mathrm{free}}(z) - m$, and the inverse determinant comparison contributes $p_0 - \ell H^2(C_h^+)$. Substitution leaves $1 + v_{\mathrm{free}}(z) + i - s(h) - \sum_v g_v + 2g - r_1$. Finally $v_2 \exp^* z = v_{\mathrm{free}}(z) + i + e_2$, proving (E.7) without specializing a primitivity assertion from the punctured family. Apply (K3) to the retained class $z = F_E(Z_F)_h$ used in the wedge above. The

primitive-conductor trace and the invariant $\sqrt{h}\,e_{\chi_h}$ basis have already been computed in paragraph 6, including discriminant $4h$. Absorb the fixed smoothing and old-support factors into ϵ_{rec}. Then

$$v_2 \exp^* z = v_2 L_*(h) + \sum_{q \text{ unused}} v_2 \# E^h(\mathbf{F}_q) + \epsilon_{\mathrm{rec}}, \qquad |\epsilon_{\mathrm{rec}}| \le C_E.$$

At unused good primes $g_q = v_2 \# E^h(\mathbf{F}_q)$, so these terms cancel. At an active good prime the inertia invariants in $E^h[2^\infty]$ are $E[2]$, of length two for the present curve. The remaining local types have fixed bounds, including global torsion by its injection at two. Thus

$$v_2 u_h^{\mathrm{ev}}(0) = v_2 L_*(h) - 2\omega(h) - s(h) + \epsilon_h, \qquad |\epsilon_h| \le C_{\mathrm{loc}}. \tag{E.8}$$

The central rational positive complex has only rank-two H^1, so this is specialization of (E.4). Corank-zero duality also shows $u_h^{\mathrm{ev}}(0) = 0$ if and only if $L_*(h) = 0$. At $L_*(h) \neq 0$, Lemma 5.5 supplies the corank-zero and finite-Sha hypotheses just used. Equations (E.5) and (E.8), already with $G = 1$, yield

$$v_2 L_*(h) \ge 2\omega(h) + s(h) - c_E - C_{\mathrm{loc}}. \tag{E.9}$$

No binary-family dimension is needed for this lower bound. □

A5.3 The actual even form and its phase

Proposition A5.4 (An actual even form with rational normalized phase)**.** *Fix a sign σ, unit squareclasses at a finite support S_0 containing $2N$, and a positive squarefree reference D_0 in that filter with $L_*(\sigma D_0) \neq 0$. For every fixed finite extra exclusion set B prime to D_0, there is an algebraic holomorphic cuspidal form of weight $3/2$, normalized by $a(D_0) = 1$, with the scalar characters ψ, η constructed below and $\psi/\eta = \chi_{-\sigma}$. It has the prescribed coefficient support. For every retained positive squarefree D in the original S_0 filter,*

$$a(D) = z_D r_D, \qquad z_D \in \mu_{2^\infty}, \quad r_D \in \mathbb{Q}, \qquad r_D^2 = L_*(\sigma D)/L_*(\sigma D_0).$$

It satisfies the all-index good-prime recurrence (E.12) outside $S_0 \cup B$. Nonzero retained squarefree coefficients have integral $v_2 a(D)$, and $v_2 a(D) - \omega(D)$ has a finite attained minimum. The form and the initial sieve are fixed before this minimum is chosen. Additional exclusions do not authorize a new minimum.

Proof. Fix a sign σ and unit squareclasses at a finite support S_0 containing two and every prime dividing N. Let $D_0 > 0$ be squarefree in these prescribed classes, with $L_*(\sigma D_0) \neq 0$. In (E.10)–(E.11), D ranges over positive squarefree parameters in these same unit squareclasses at every prime of S_0. Fix also a finite extra exclusion set B with $\gcd(D_0, \prod_{p \in B} p) = 1$.

A nonzero reference functional and its scalar local type. Use the global Shimura–Waldspurger packet with underlying PGL_2 representation $E^{(\sigma)}$ in the normalization of Waldspurger's Corollary 2. The nonzero central value at σD_0 supplies the member with a nonzero D_0-Whittaker functional. Equivalently, take the nonzero theta lift associated with $E^{(\sigma D_0)}$ and the additive character of frequency D_0; the change of additive character identifies its underlying packet with the preceding one. The nonvanishing and the local identification are the content of Waldspurger's global theta theorem and its Fourier-functional calculation [83, Theorem 1 and Proposition 1] and [79, Proposition 18 and its proof, pp. 466–467]. The positive real frequency selects the holomorphic member of lowest weight $3/2$. The lift is in the unary-orthogonal cuspidal spectrum.

Here is a local construction of a scalar type that retains the reference functional. At $p \in S_0$, write $\ell = \ell_{D_0}$ and choose v with $\ell(v) \neq 0$. Normalize Haar measure in

$$P_R v = \frac{1}{\mathrm{vol}(p^{-R}\mathbb{Z}_p)} \int_{p^{-R}\mathbb{Z}_p} e_p(-D_0 b)\pi(n(b))v \, db.$$

Then $\ell(P_R v) = \ell(v)$. Let $M(u)$ be the unit action normalized by the Weil multiplier. Explicitly, if the chosen torus lifts have cocycle $(u, v)_p$, their theta-cubed multiplier γ has two-power root-of-unity values and satisfies $\gamma(u)\gamma(v) = (u, v)_p \gamma(uv)$; put $M(u) = \gamma(u)^{-1}\pi(\widetilde{m}(u))$. It is therefore an action of $\mathbb{Z}_p^\times$ and $M(u)\pi(n(b))M(u)^{-1} = \pi(n(u^2 b))$. Changing u to its inverse changes none of the following support conditions. Direct substitution gives

$$\ell(M(u)P_R v) = \mathbf{1}_{u^2 \equiv 1 \pmod{p^R}} \ell(M(u)v).$$

The integral is a finite sum on this smooth vector.

The normalized $M(-1)$ acts by a scalar $\delta \in \{1, -1\}$. Choose a continuous unit character α of two-power order with $\alpha(-1) = \delta$. Such a character exists: for odd p, -1 lies in the two-primary factor of the finite cyclic group of residue-field units; for $p = 2$ use the factor $\{1, -1\}$ of $\mathbb{Z}_2^\times$. Take R large enough that both v and α are constant on the small neighborhoods appearing above. The projection

$$w = \int_{\mathbb{Z}_p^\times} \alpha(u)^{-1} M(u) P_R v \, d^\times u$$

satisfies $M(u)w = \alpha(u)w$. Its ℓ-value is nonzero: the neighborhoods of 1 and -1 contribute the same nonzero value. For odd p the solution set is $1 + p^R\mathbb{Z}_p$ together with $-1 + p^R\mathbb{Z}_p$. For $p = 2$ and $R \geq 3$ it is

$$(1 + 2^{R-1}\mathbb{Z}_2) \ \cup \ (-1 + 2^{R-1}\mathbb{Z}_2).$$

Thus the four residue classes modulo 2^R form two neighborhoods; there are not four independently assignable phases. For completeness, the noncancellation in this projection is an exact calculation. Normalize multiplicative Haar measure to give $\mathbb{Z}_p^\times$ volume one, and put $K_R = 1 + p^R\mathbb{Z}_p$ for odd p, and $K_R = 1 + 2^{R-1}\mathbb{Z}_2$ for $p = 2$. Increase R until $M(k)v = v$ and $\alpha(k) = 1$ for $k \in K_R$. On K_R the integrand after applying ℓ is $\ell(v)$. On $-K_R$ it is $\alpha(-1)^{-1}\ell(M(-1)v) = \delta^{-1}\delta\ell(v)$. It vanishes elsewhere by the displayed additive projection formula. Consequently

$$\ell(w) = 2\,\mathrm{vol}(K_R)\ell(v) \neq 0.$$

This calculation takes place in characteristic zero; reduction at two is not involved. For the later support projection choose m so that the required unit squareclass is a union A of residue classes modulo p^m. Its operator is

$$\Pi_A = \frac{1}{p^m} \sum_{b \bmod p^m} \left(\sum_{a \in A} e_p(-ab/p^m) \right) \pi(n(b/p^m)).$$

On the Fourier functional of integral frequency D it acts as $\mathbf{1}_A(D)$. It therefore preserves the reference value and annihilates every excluded frequency. Conjugation by $M(u)$ permutes the summands and replaces A by $u^2 A = A$, proving the asserted commutation. Its denominators are fixed before the form is normalized.

The vector $P_R v$, and then w, is fixed by upper unipotents $n(\mathbb{Z}_p)$, because D_0 is a unit. Smoothness makes w fixed by lower unipotents at sufficiently deep level. Together with the unit eigencharacter, the elementary decomposition of $K_0(p^m)$ gives a scalar K_0 type for sufficiently large m. The

Weil normalization contributes only two-power roots of unity, so the resulting classical character has two-power order. The forced values at -1 give the global parity required for a weight-3/2 scalar character: the product of the finite central actions is the inverse of the holomorphic archimedean action, and the theta-cubed multiplier has the same product relation on rational $-I$. Thus the quotient character satisfies $\psi(-1) = 1$. At 2 one may, if desired, choose α of arbitrarily large two-power conductor before increasing R; this does not enlarge the prime support.

Apply the finite additive Fourier idempotent for the prescribed unit-squareclass subset modulo a sufficiently high power of p. This idempotent keeps the D_0 coefficient. Its support condition is stable under multiplication by unit squares, so it commutes with the unit action and retains a scalar type, at a possibly deeper level. At every prime outside S_0 take the spherical vector. The packet there is unramified; its standard squarefree coefficient c_p^0 is 1 at both valuations zero and one [79, Section I.3, pp. 377–378]. This includes the primes dividing D_0. Local Whittaker uniqueness and the tensor-product factorization show that the resulting global reference coefficient is still nonzero.

The exact full-space comparison. Let the resulting scalar character be ψ and put $\eta = \psi\chi_{-\sigma}$. Both have two-power order. Let V be the full good-Hecke eigenspace of this scalar character, at the level just constructed. Keep all old multiplicities. In Waldspurger's notation its integral-weight eigenform and character are

$$\phi_0 = f_E \otimes \eta, \qquad \chi_0 = \psi\chi_{-1}, \qquad \phi_0 \otimes \chi_0^{-1} = f_E \otimes \chi_\sigma.$$

This also fixes the signs in the packet convention: Waldspurger's positive index D twists the last representation by χ_D. The Hecke eigenvalues in V are $\eta(p)a_p(E)$.

The nonzero space implies hypothesis (H1) by Waldspurger's Proposition 2 [79, Section I.2, p. 377]. In explicit terms, at a principal-series place its two inducing characters take value 1 at -1. Corollary 2 [79, pp. 483–484] applies to every $f \in V$ and any positive squarefree pair whose ratio is a square in every $\mathbb{Q}_p$ at a prime dividing the level. Our D, D_0 satisfy this condition because the level has prime support S_0. The final paragraph of that proof removes (H2) for this same-$\mathbb{Q}_2$-squareclass comparison. No assumption of non-supercuspidality at two is required.

The printed coefficient identity and its dyadic hypothesis. For clarity, write $\mathcal{L}_D = L(E^{(\sigma D)}, 1)$ and let N_{W} denote the level in Waldspurger's notation, not the conductor of E. In [79, Corollary 2, pp. 483–484] take

$$k = 3, \quad \chi = \psi, \quad \chi_0 = \psi\chi_{-1}, \quad \rho_0 = \rho_E \otimes \eta, \quad n_1 = D_0, \quad n_2 = D.$$

Then $\rho_0 \otimes \chi_0^{-1} = \rho_E \otimes \chi_\sigma$. The corollary, in its printed order, is

$$a_f(D_0)^2 \mathcal{L}_D \, \psi(D/D_0) D^{1/2} = a_f(D)^2 \mathcal{L}_{D_0} D_0^{1/2}.$$

The passage from the automorphic central argument 1/2 to the classical argument 1 is the same for both terms. Dividing only by the nonzero reference value gives (E.10). In particular this is a literal square identity, with neither complex conjugation nor an unspecified absolute-value normalization.

Every prime of N_{W} belongs to S_0 at this point of the construction. The prescribed local classes make D/D_0 a square in $\mathbb{Q}_p^\times$ at each such prime, including $p = 2$. In the proof of the corollary, the local coefficient functions at these primes have equal values at D, D_0. Outside the level, the squarefree arguments have valuation zero or one and the unramified coefficient function has value one in both cases. The real coefficient contributes exactly the power $(D/D_0)^{(k-2)/4}$ before squaring. The quotient of epsilon factors is computed on pp. 483–484: its Hilbert-symbol product disappears

by reciprocity and its remaining character is $\chi(D/D_0) = \psi(D/D_0)$. Thus these local terms do not leave an additional prime-dependent factor in the displayed identity.

The final paragraph on p. 484 explicitly explains the use of (H2): it is needed in that argument only to compare Fourier indices in different squareclasses of $\mathbb{Q}_2^\times$. Our pair is in one squareclass. Hence the corollary as stated requires (H1) here, but does not impose (H2), a nonsupercuspidal condition at two, or a new dyadic conductor restriction. The nonzero scalar eigenspace supplies (H1) by [79, Proposition 2, p. 377]. Moreover the proof on p. 483 extends its linear relation to every vector in the full eigenspace before squaring. The resulting identity therefore applies simultaneously to old multiplicities, and later to both Galois-conjugated and character-twisted vectors at a common level. At that later level the same squareclass test is repeated; its prime support is still S_0. Putting $n_1 = D_0, n_2 = D$ in the corollary, with weight $3/2$, gives

$$a_f(D)^2 = \psi(D/D_0)\,\frac{\sqrt{D}\,L(E^{(\sigma D)},1)}{\sqrt{D_0}\,L(E^{(\sigma D_0)},1)}\,a_f(D_0)^2. \tag{E.10}$$

The period cancels, so the quotient in this formula is the quotient of the specified L_*-values. No product over the primes dividing D or D_0 is missing: the unramified factors at squarefree valuations are exactly those checked above.

For clarity, (E.10) gives proportional coefficient functionals on the whole space. If $\ell_0(f) = a_f(D_0)$ and $\ell_D(f) = a_f(D)$, then $\ell_D^2 = C\ell_0^2$ as a polynomial identity on V. If $C = 0$, then $\ell_D = 0$. Otherwise, choose a square root c of C. The polynomial ring on the finite-dimensional vector space V is an integral domain, so $(\ell_D - c\ell_0)(\ell_D + c\ell_0) = 0$ implies $\ell_D = c\ell_0$ or $\ell_D = -c\ell_0$. The scalar is common to every vector. This argument does not assume that V is one-dimensional.

An algebraic vector and the Galois comparison. Ramsey's section model [80, Definition 3.1 and Theorem 4.2] realizes the half-integral-weight space as sections of the indicated divisor sheaf on $X_1(4N_{\mathrm{aux}})$. Here N_{aux} is supported on S_0 and includes the chosen local depths; it need not equal the conductor N of E. Over characteristic-zero fields, flat field extension commutes with these sections. Cuspidality, scalar character, support conditions and prescribed good-Hecke eigenvalues are algebraic linear conditions. Although infinitely many coefficients or primes can occur in their description, a finite subset cuts out the same subspace, by finite dimensionality.

One need not impose Petersson orthogonality as an additional transcendental linear condition. The elliptic good-Hecke eigenspace is automatically orthogonal to unary theta series: at a good prime, the unary eigenvalues have absolute value $p+1$, whereas $|\eta(p)a_p(E)| \le 2\sqrt{p} < p+1$. The unary space and its orthogonal complement are stable under the Hecke operators and their adjoints. Thus their elliptic eigenspaces are disjoint. The nonzero reference functional on the complex space is consequently nonzero on its algebraic model. Choose such an algebraic g and scale it to $a(D_0) = 1$. This uses flat field extension only, not Ramsey's arbitrary-ring base-change assertion in weight $3/2$; see [80, Remark 4.4]. Here the field-extension assertion can be checked without a cohomology-vanishing theorem. Work over a number field K containing the character values and the roots of unity specifying the cusps and the support idempotents. A finite affine cover of the modular curve computes global sections as the kernel of the difference of restriction maps to pairwise intersections. Tensoring with a field extension is flat and preserves that kernel. Thus the sections in Ramsey's Definition 3.1 commute with $K \subset \overline{\mathbb{Q}} \subset \mathbf{C}$, also in weight $3/2$. Impose vanishing at the cusps, the diamond character, the support idempotents, and the kernels of $T(p^2) - \eta(p)a_p(E)$ inside this finite-dimensional space. The span in its dual of all the defining linear equations has a finite basis. These conditions therefore commute with the same field extensions. Since the reference functional is nonzero after extension to $\mathbf{C}$, it is nonzero before extension; a vector in a K-basis has nonzero

reference value and may be divided by that value. This also places every coefficient of the chosen form in one number field. No assertion about base change to a residue field, nor about vanishing of H^1 in weight $3/2$, is used.

Let $\tau \in \mathrm{Aut}(\overline{\mathbb{Q}}/\mathbb{Q})$. Choose an odd integer e describing its action on all values of η and ψ, and put $\nu = \eta^{(e-1)/2}$. Coefficientwise conjugation gives a form g^τ with character ψ^e and good eigenvalues $\eta(p)^e a_p(E)$. One can reduce this conjugation assertion to integral weight by multiplying by the ordinary theta series: it has rational coefficients, and its fourth power has trivial scalar character. The algebraic divisibility condition at the cusps is precisely the one in Ramsey's section model, so division after conjugation is again a half-integral-weight form. The coefficientwise twist $g \otimes \nu$ has character $\psi\nu^2 = \psi^e$ and exactly these good eigenvalues. The identities hold because ψ/η is quadratic. The twist is modular at a level still supported on S_0; this follows as well by writing it as the finite additive translation sum using the Gauss sum of the primitive character inducing ν. The full multiplier, cusp and conjugation calculation is in Section A14.

Pass to a common such level. Both forms lie in its full good-Hecke eigenspace, so the proportionality of its D and D_0 functionals gives

$$\tau(a(D)) = \frac{\nu(D)a(D)}{\nu(D_0)a(D_0)} = \nu(D/D_0)a(D).$$

There is no vanishing reference denominator: D_0 is prime to S_0 and $a(D_0) = 1$. No rationality of the central-value quotient has been assumed in obtaining this equality.

Write the Dirichlet character as a product of its finite prime-power factors η_p. For every p dividing its conductor choose $u_p \in \mathbb{Z}_p^\times$ with $u_p^2 = D/D_0$, and set

$$z_D = \prod_{p \mid \mathrm{cond}(\eta)} \eta_p(u_p).$$

Then z_D is a two-power root of unity, $z_D^2 = \eta(D/D_0)$, and

$$\tau(z_D) = z_D^e, \qquad \nu(D/D_0) = z_D^{e-1}.$$

Changing the selected odd representative of e has no effect on the last identity, because D/D_0 is a local square at the conductor. The preceding Galois identity now shows that $a(D)/z_D$ is fixed by every τ, hence lies in $\mathbb{Q}$. Different choices of the local square roots change z_D by a sign, which changes only the sign of this rational number.

Finally, $\psi/\eta = \chi_{-\sigma}$ is trivial on D/D_0: its conductor is supported at two and the ratio is a square there. Thus $\psi(D/D_0) = z_D^2$. Dividing (E.10) by z_D^2 gives

$$a(D) = z_D r_D, \qquad z_D \in \mu_{2^\infty}, \qquad r_D \in \mathbb{Q}, \qquad r_D^2 = \frac{L_*(\sigma D)}{L_*(\sigma D_0)}. \tag{E.11}$$

The argument also covers the zero coefficient case. In particular, every nonzero coefficient in (E.11) has integral normalized two-adic valuation.

Recurrence, later exclusions, and valuations. Only now impose the extra finite coprimality sieve B. The zero-or-retain Fourier sieve is a finite additive translation operator. It preserves cuspidality, keeps every coefficient whose index is prime to B, and commutes with good $T(p^2)$ for $p \notin S_1 = S_0 \cup B$. In particular it keeps $a(D_0) = 1$. The squarefree comparison (E.11) was proved before this enlargement, so it requires no extra squareclass hypotheses at the primes of B. After

the sieve it is asserted at the retained indices. The standard good $T(p^2)$ formula, with $\psi^2 = \eta^2$, gives

$$a(p^2n) = \{\eta(p)a_p(E) - \psi(p)(-n/p)\}a(n) - p\eta(p)^2a(n/p^2), \qquad p \notin S_1. \tag{E.12}$$

For squarefree D, write $a(Dp^{2r}) = \eta(p)^r b_r$. The first multiplier is $a_p(E) - (\psi/\eta)(p)(-D/p)$; subsequent multipliers satisfy $b_{r+1} = a_p(E)b_r - pb_{r-1}$. All are integers, including when $p \mid D$. Iteration over the good primes gives the same assertion for every retained square shell after removal of the η phase. More explicitly, for $p \notin S_1$ set $A_{p,0}(D) = 1$, $A_{p,1}(D) = a_p(E) - (\psi/\eta)(p)(-D/p)$, and $A_{p,r+1}(D) = a_p(E)A_{p,r}(D) - pA_{p,r-1}(D)$ for $r \geq 1$. Then, for every retained index Dm^2 with D squarefree,

$$a(Dm^2) = z_D\eta(m)r_D \prod_{p^r \| m} A_{p,r}(D).$$

The formula includes $p \mid D$, when the initial Legendre symbol is zero. When applying the recurrence at one prime, the other prime-square factors do not alter that Legendre symbol. This proves the formula by induction on the prime factors of m, with $a(n/p^2) = 0$ when $p^2 \nmid n$. All factors $A_{p,r}(D)$ are integers. Thus every coefficient used by the extraction and shell operators of Proposition A6.5 is a rational number times a two-power root of unity, including coefficients with non-squarefree index. In the chosen residue field such a root reduces to one. A rational coefficient of valuation zero reduces to the element 1 of its prime field $\mathbb{F}_2$. Consequently the fresh coefficient-one reading of that proposition detects precisely equality in the fixed minimum-depth bound. It uses the same g, D_0 and normalization throughout.

Combining (E.9) and (E.11) makes the values $v_2a(D) - \omega(D)$ at nonzero retained squarefree coefficients a bounded-below subset of $\mathbb{Z}$. It is nonempty because D_0 is retained, and therefore has an attained finite minimum. The form and the initial finite sieve are fixed before this minimum is chosen. Restricting later fresh primes by further exclusions does not authorize choosing another minimum. □

A6 Actual odd theta forms and their operators

We now construct the forms in (3.8), with the same h_*, D_w, L and actual Kummer stages as in Section 7. Write $m = |h_*|$. An admissible D is positive, squarefree, prime to the fixed support and in the prescribed unit classes. Then $K_D = \mathbb{Q}(\sqrt{Dh_*})$ is imaginary, split at $2N$, has no extra units, and Dh_* is an odd fundamental discriminant. The conductor-one point is the unaveraged sum

$$P_D = \sum_{[\mathfrak{a}] \in \mathrm{Pic}(\mathcal{O}_{K_D})} \chi_{h_*}(\mathrm{N}\,\mathfrak{a})\,\pi(x_D^{[\mathfrak{a}]}). \tag{O.1}$$

It is distinct from the conductor-$|h|$ ring-class point used in Section 9.

A6.1 A fixed witness and one compatible sequence

Proposition A6.1 (One witness and a compatible detecting sequence)**.** *For the fixed $E = E_\ell$ under the full-corank-one hypothesis of Theorem 5.3, use the prepared positive unit filter containing 1. There exist a negative odd fundamental discriminant h_* with $L(E^{h_*}, 1) \neq 0$, a positive fundamental discriminant D_w in that filter with $L'(E^{D_w}, 1) \neq 0$, and a fixed integer L such that LP_D belongs to the D-twist rational line for every admissible D and $Q = LP_{D_w}$ is nontorsion. There is one sequence of actual stages with $M \to \infty$, good primes ℓ_0, and additive maps*

$$\lambda_M : E(\mathbb{F}_{\ell_0^2}) \longrightarrow \mathbb{Z}/2^M\mathbb{Z}, \qquad v_2\lambda_M(\overline{Q}) = c_w < M,$$

where c_w is fixed. At these stages Frobenius squared is trivial on $E[2^M]$, $(D_w/\ell_0) = 1$, $(h_*/\ell_0) = -1$, and λ_M has Frobenius sign $+1$. Any specified finite prime exclusion is permitted. Additional field prescriptions require a compatible common lift; no preservation of c_w under arbitrary later field prescriptions is asserted.

Proof. We use the positive unit-squareclass filter containing one, with its fixed support enlarged before choosing the witness. The rational Selmer parity theorem [61, Theorem 1.4, p. 568], applied over $\mathbb{Q}$ with $p = 2$, gives root number -1 for E under the full-corank-one assumption. Apply clause (ii) of [70, pp. 543–544] to its weight-two newform, with every old prime in the splitting set, to obtain a negative fundamental discriminant h_* with $L(E^{(h_*)}, 1) \neq 0$. Including two makes h_* odd and excludes the extra-unit fields. Now add the primes of h_* to that set and apply clause (i) to obtain a positive fundamental discriminant D_w with $L'(E^{(D_w)}, 1) \neq 0$. Splitting at every old prime puts D_w in the filter containing one and makes it prime to h_*. These choices precede every detecting prime and every theta precision.

The conductor-one genus character for $K_D = \mathbb{Q}(\sqrt{Dh_*})$ cuts out $F_D = \mathbb{Q}(\sqrt{D}, \sqrt{h_*})$. Thus the unaveraged sum P_D in (O.1) belongs to $E(F_D)$. Its rational character components are the D- and h_*-components. In the degenerate case $D = 1$ these are the trivial and h_* characters of the actual quadratic group. The h_*-component has rank zero by the forward rank-zero theorem. For D_w, the unaveraged height formula [73, Theorem 1.1] has Rankin derivative $L'(E^{(D_w)}, 1)L(E^{(h_*)}, 1) \neq 0$, so P_{D_w} is nontorsion. Here the conductor is one, all primes of N split, and the extra-unit fields have been excluded.

We make the fixed multiplier explicit. Choose distinct good rational primes p_1, p_2 for E and put

$$B_j = \prod_{f=1}^{4} \#E(\mathbb{F}_{p_j^f}), \qquad B = B_1 B_2.$$

For every number field of degree at most four, prime-to-p_j torsion injects into a reduction group with residue degree at most four. The two primes therefore show that B kills its entire torsion subgroup. Put $B_{\mathrm{cusp}} = C_\pi$, where C_π annihilates the image of the cuspidal subgroup defined after (4.1). This includes every Atkin–Lehner translation $\pi(W\infty)$, which need not vanish merely because $\pi(\infty) = 0$. For $\Delta_D = \mathrm{Gal}(F_D/\mathbb{Q})$, the integral character operator

$$4e_D = \frac{4}{|\Delta_D|} \sum_{\gamma \in \Delta_D} \chi_D(\gamma)\gamma$$

is defined also when $D = 1$. The difference $4P_D - 4e_D P_D$ is torsion, since its only possible rational component is the rank-zero h_*-component. Hence $4BP_D = B(4e_D P_D)$ transforms exactly by χ_D. Choose L to be a common multiple of the previously required fixed multiplier and $4BB_{\mathrm{cusp}}$, for example their product. This retains every earlier descent and local-condition factor. Then LP_D is a rational point of $E^{(D)}$ under the standard twist identification, for every admissible D. In particular

$$Q = LP_{D_w} \in A(\mathbb{Q}), \qquad A = E^{(D_w)},$$

is nontorsion. No assertion that the original P_D is $E(K_D)$-rational is required.

Choose compatible division points Q_n with $2^n Q_n = Q$ and $2Q_{n+1} = Q_n$, and let $c_Q(g) = (gQ_n - Q_n)_n \in T_2 A$ be the Kummer cocycle. Write ρ_A for the Tate representation and $H = \ker \rho_A$. On H, this cocycle is a homomorphism. Its image is compact, and hence a closed $\mathbb{Z}_2$-submodule of $T_2 A$. Moreover

$$c_Q(ghg^{-1}) = \rho_A(g)c_Q(h) \qquad (h \in H),$$

so its rational span is invariant under the Tate image.

The curve A is non-CM. Open image [85] supplies both irreducibility of its rational Tate representation and a central scalar $z \neq 1$ in its Tate image. If $c_Q(H) = 0$, the rational cocycle descends to that image. Commuting z with g gives

$$(z-1)c_Q(g) = (g-1)c_Q(z).$$

Since $z-1$ is invertible over $\mathbb{Q}_2$, this is a coboundary. It contradicts the Kummer injection $A(\mathbb{Q})\otimes\mathbb{Q}_2 \hookrightarrow H^1(\mathbb{Q}, V_2A)$ and nontorsion of Q. Thus the translation image is nonzero. Irreducibility makes its rational span all of V_2A. It consequently contains $2^{a_0}T_2A$ for some a_0.

Let $F_0/\mathbb{Q}$ be any fixed finite Galois auxiliary extension, including $\mathbb{Q}(\sqrt{D_w}, \sqrt{h_*})$ and any fixed ring-class data to be retained with the action of complex conjugation. Put $H_0 = H \cap G_{F_0}$. It has finite index d in H. For $h \in H$, $h^d \in H_0$ and $c_Q(h^d) = d\, c_Q(h)$. Hence $c_Q(H_0)$ still contains 2^aT_2A for some fixed a. This argument uses no independence between F_0 and the Tate or division-point fields.

Fix actual complex conjugation τ. Since $D_w > 0$, the real twist identification identifies its matrix on T_2A with its matrix C on $T = T_2E$. We have $C^2 = 1$ and $\det C = -1$. Its dual $+1$-eigenspace therefore contains a primitive integral row $b : T \to \mathbb{Z}_2$, with $bC = b$ and $b(T) = \mathbb{Z}_2$. Choose $t_0 \in T$ with $b(t_0) = 1$, and choose $h \in H_0$ with $c_Q(h) = v = 2^at_0$. Set $\sigma = h\tau$. If $d_\tau = c_Q(\tau)$, then $(1+C)d_\tau = 0$, because $\tau^2 = 1$. Consequently

$$\rho_A(\sigma) = C, \qquad c_Q(\sigma^2) = (1+C)(v + d_\tau) = (1+C)v, \qquad b\, c_Q(\sigma^2) = 2^{a+1}.$$

Put $c_w = a + 1$. The element σ and this finite valuation are fixed before precision varies. Its restriction to F_0 is that of τ: it fixes $\sqrt{D_w}$ and negates $\sqrt{h_*}$. Since h fixes $\sqrt{D_w}$, the matrix of σ on T_2E is also C.

For each $M > c_w$, take the finite Galois extension generated by F_0, $E[2^M]$, $A[2^M]$, and all 2^M-division points of Q. Restrict this same actual σ to the extension. Finite Chebotarev supplies infinitely many good rational primes ℓ_0 in its conjugacy class. Choose a place realizing this representative, and exclude two, all bad primes, the support of D_wh_* and any other finite set. Thus, with the torsion basis transported by this place,

$$\mathrm{Fr}_{\ell_0} \equiv C \pmod{2^M}, \qquad (D_w/\ell_0) = 1, \qquad (h_*/\ell_0) = -1.$$

The determinant also gives $\ell_0 \equiv -1 \pmod{2^M}$. In particular $\mathrm{Fr}_{\ell_0}^2$ acts trivially on $E[2^M]$.

Here is the finite-field map without a normalization ambiguity. Put $n = 2^M$ and write F for Frobenius on the reduction of E. For $P \in E(\mathbb{F}_{\ell_0^2})$, choose R with $nR = P$ and set

$$\kappa_M(P) = F^2R - R \in E[n] = T/nT.$$

Changing R by n-torsion does not change this value. The map is additive, with kernel $nE(\mathbb{F}_{\ell_0^2})$. It is onto: $F^2 - 1$ is a nonzero separable isogeny, so for $S \in E[n]$ one can solve $(F^2-1)R = S$ and take $P = nR$. Also $\kappa_M(FP) = F\kappa_M(P)$, by choosing FR as a division point. Therefore $\lambda_M = b\kappa_M$ has Frobenius sign $+1$.

Good reduction identifies the reduction of the global division cocycle with this exact map. The chosen place makes the twist identification defined over $\mathbb{F}_{\ell_0}$, since $(D_w/\ell_0) = 1$. Thus the fixed witness satisfies

$$\lambda_M : E(\mathbb{F}_{\ell_0^2}) \longrightarrow \mathbb{Z}/2^M\mathbb{Z}, \qquad v_2\lambda_M(\overline{Q}) = c_w < M. \tag{O.2}$$

Choosing increasing M and excluding previous primes gives one sequence of actual stages. Subsequent finite prime exclusions do not alter c_w. New field prescriptions require a compatible common lift and, if added after this choice, are not asserted to preserve this particular constant. □

A6.2 Local functions and the theta sum

Proposition A6.2 (The actual odd theta form and its recurrence)**.** *Fix an actual stage of Proposition A6.1 with $M \geq 2$. Use $m = |h_*|$, the same L, the Eichler order of level N in the quaternion algebra ramified at ℓ_0, ∞, and the local Schwartz functions specified below, including (O.3). With $c_0 = |\operatorname{GL}_4(\mathbb{F}_3)| = 24\,261\,120$, the sum H_M in (O.4) is well defined modulo 2^M in the character ring. It gives an actual holomorphic form*

$$g_M = \sum_n a_M(n) q^n, \qquad a_M(n) = H_M(mn), \qquad g_M \in M_{3/2}(\Gamma_0(K), \psi),$$

with level supported only at the specified coefficient primes. The characters are exactly $\psi = \eta_B(m/\cdot)$ and $\eta = \eta_B \chi_{h_*}$, so $\psi/\eta = \chi_{-1}$. For every prime outside that support, the recurrence (O.6) holds at every integer index, with nonintegral indices interpreted as zero. Changing the lifts of P_M loses no precision. A specified further local coefficient sieve retains this conclusion outside the enlarged support.

Proof. In this appendix $M = M_\theta$; take $M \geq 2$. Retain the same $m = |h_*|$, L, and detecting prime ℓ_0 as in (O.2). Since $\operatorname{Fr}_{\ell_0}^2 = 1$ on $E[2^M]$, evaluation at arithmetic Frobenius squared identifies the finite-field Kummer quotient with $T_2E/2^M T_2E$. Equivariance and $bC = b$ imply $\lambda_M(\operatorname{Fr}_{\ell_0} P) = \lambda_M(P)$. The determinant gives $\ell_0 \equiv -1 \pmod 4$; hence $(m/\ell_0) = (-1/\ell_0)(h_*/\ell_0) = 1$.

The finite-field evaluation map. Write $k_0 = \mathbb{F}_{\ell_0^2}$ and let F denote its arithmetic Frobenius. For $P \in E(k_0)$, choose $Q_P \in E(\overline{k_0})$ with $2^M Q_P = P$. Then

$$\kappa_M(P) = FQ_P - Q_P \in E[2^M]$$

is independent of the choice of Q_P, since F is the identity on $E[2^M]$. This is an additive map. It kills $2^M E(k_0)$, and its kernel is exactly that subgroup: a zero value means that Q_P itself lies in $E(k_0)$. The finite group $E(k_0)/2^M E(k_0)$ has order $|E(k_0)[2^M]| = 2^{2M}$, as does $E[2^M]$, so the induced injection is an isomorphism. Composing κ_M with the fixed row b modulo 2^M gives λ_M. The arithmetic Frobenius over $\mathbb{F}_{\ell_0}$ commutes with F and acts on the torsion by C. Hence $\kappa_M(\operatorname{Fr}_{\ell_0} P) = C\kappa_M(P)$ and $bC = b$ proves the asserted Frobenius sign. For the fixed witness, specialization of a compatible division point identifies this evaluation with the cocycle value at σ^2, whose b-value is 2^{a+1}. Thus the visibility valuation is fixed as the precision grows.

By [98, Main Theorem 14.6.1], there is a quaternion algebra $B/\mathbb{Q}$ ramified exactly at ℓ_0 and infinity. Let R be an Eichler order of level N. Choose an enhanced supersingular reference curve $\mathcal{E}_0$ with endomorphism order R, as in [75, Proposition 3.6, pp. 441–443]. For an enhanced supersingular curve $\mathcal{E}$, use the right R-module $\operatorname{Hom}(\mathcal{E}_0, \mathcal{E})$ convention of [75, Proposition 3.3 and Remark 3.5(a), pp. 440–441]. Writing its completed lattice as $z\widehat{R}$ gives $X = B^\times \backslash \widehat{B}^\times / \widehat{R}^\times$, with endomorphism order $R_z = B \cap z\widehat{R}z^{-1}$ and automorphism group $R_z^\times$. Here the enhancement is the cyclic subgroup of order N; arbitrary prime-power exponents, including at two, are allowed. In the source's Section 3 setup one may choose its auxiliary prime outside $N\ell_0$. By [75, Remark 3.5(b), p. 441], Frobenius acts on this set by right multiplication by a division uniformizer at ℓ_0 and by 1 elsewhere, and this action is an involution. Consequently every corresponding geometric coarse point is fixed by Frobenius squared. Its closed-point residue degree therefore divides two, so it is a point of $X_0(N)(\mathbb{F}_{\ell_0^2})$. This asserts coarse-point rationality; no chosen scalar-Frobenius elliptic model is needed.

Define $P_M(z) = \lambda_M(L\pi_{\ell_0}(z))$. Here the specialization is legitimate because $\ell_0 \nmid N$: the modular curve has good reduction and the map to the good-reduction elliptic curve extends over the local

base. The good Hecke correspondences also extend. Writing π as the Abel–Jacobi map based at infinity followed by the elliptic quotient gives, for $p \nmid N\ell_0$,

$$\sum_{z' \text{ a } p\text{-neighbor of } z} \pi_{\ell_0}(z') = a_p(E)\pi_{\ell_0}(z).$$

Indeed the quotient intertwines T_p with $[a_p(E)]$ and $T_p(\infty) = (p+1)\infty$. For an Atkin–Lehner involution W_{p^e} the analogous identity has eigenvalue w_{p^e} and the constant $\pi(W_{p^e}\infty)$, which L kills. Frobenius acts on the supersingular set as the right division-uniformizer operation. Consequently P_M has the asserted good Hecke and Atkin–Lehner eigenvalues modulo 2^M, and uniformizer eigenvalue $+1$. Choose arbitrary integral lifts of this finite function.

The finite Schwartz function. All local functions below are zero off their specified supports. Set $V = B^0$ and $Q_B(x) = \mathrm{Nrd}(x) = -x^2$. For $p \nmid 2Nm\ell_0$ use $1_{R_p^0}$. For odd $p^e \| N$, use the split integral unit support. The cyclic level quotient selects one of the two eigensummands and its root $r \in \mathbb{Z}_p^\times$. The projectors $(x+r)/(2r)$ and $(r-x)/(2r)$ are integral, so reduction and Hensel lifting show that the selected root modulo any fixed power of p is locally constant. Weight it by a unit character $\alpha_p(r)$ of finite 2-power order with

$$\alpha_p(-1) = w_{p^e}\chi_{h_*}(p)^e. \tag{O.3}$$

Such a character exists: at an odd prime the cyclic 2-primary part of $\mathbb{F}_p^\times$ admits a character taking -1 to -1, and the trivial character supplies the other parity. Conjugation by $R_p^\times$ preserves the cyclic quotient and its selected summand. Multiplication of x by a unit u preserves the summands and replaces r by ur, giving multiplier $\alpha_p(u)$.

At two require $e_x = (1+x)/2 \in R_2$ and $x^2 = r^2 \in \mathbb{Z}_2^{\times 2}$. The eigenvalues $a = (1+r)/2$ and $a' = (1-r)/2$ of e_x differ by the unit r. Its projectors $(e_x - a')/r$ and $(a - e_x)/r$ are integral and remain distinct modulo two. If $2^e \| N$, the cyclic quotient is a free rank-one $\mathbb{Z}/2^e$-module. Of these complementary projectors exactly one acts as the identity on it, since the only idempotents of $\mathbb{Z}/2^e$ are zero and one. This defines the selected root without a choice of its parity. For every odd u,

$$e_{ux} = ue_x + (1-u)/2.$$

This affine transformation preserves the two projectors individually, so the selected root changes to ur, even when the two eigenvalue parities switch. The same root character rule therefore applies at two. A character with either prescribed parity exists, using the trivial character or the character modulo four. If $2 \nmid N$ there is no root character at two.

For $p \mid m$, require $x \in M_2(\mathbb{Z}_p)^0$ and $v_p Q_B(x) = 1$. The reduction of x is a nonzero nilpotent: zero reduction would force $p^2 \mid Q_B(x)$. All nonzero nilpotents are conjugate to $X_0 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. The centralizer of X_0 in $\mathrm{GL}_2(\mathbb{F}_p)$ is $\{\left(\begin{smallmatrix} a & b \\ 0 & a \end{smallmatrix}\right) : a \neq 0\}$ and has square determinant. Thus

$$\epsilon_p(gX_0g^{-1}) = (\det g/p)$$

is well-defined. Conjugation by u multiplies it by $(\det u/p)$, and scalar multiplication by $u \in \mathbb{Z}_p^\times$ multiplies it by (u/p), as follows from conjugation of X_0 by $\mathrm{diag}(u,1)$.

At ℓ_0 require $x \in R_{\ell_0}^0$ with nonzero reduction $\bar{x} = t\beta$ in the trace-zero line of $\mathbb{F}_{\ell_0^2}$, and use (t/ℓ_0). Order units act trivially on the residue field. A division uniformizer acts by Frobenius, which sends β to $-\beta$, giving multiplier -1. Unit scaling gives (u/ℓ_0). Moreover β^2 is a nonsquare in $\mathbb{F}_{\ell_0}$: otherwise $\beta = \pm a$ for $a \in \mathbb{F}_{\ell_0}$, contrary to its nonzero trace-zero condition. Both -1 and β^2

are nonsquares, so $Q_B(x) = -t^2\beta^2$ is a nonzero square. Since $(m/\ell_0) = 1$, extracted indices have $(D/\ell_0) = 1$.

These supports are compact and open, and all weights are locally constant. Additional fixed unit-squareclass conditions multiply them by locally constant characteristic functions. They are preserved by $x \mapsto ux$, which multiplies the norm by u^2. Let $\mathcal{O}$ be the valuation ring at a place above two of the field generated by the fixed root-character values at primes dividing N. All other weights are quadratic, so this field is independent of M and ℓ_0. With $f = \prod f_p$, the preceding calculations give

$$f(u^{-1}xu) = \chi_{h_*}(\operatorname{Nrd} u)^{-1} f(x) \quad (u \in \widehat{R}^\times), \qquad f(vx) = \alpha(v) f(x) \quad (v \in \widehat{\mathbb{Z}}^\times).$$

The ramified local quadratic character on units at $p \mid m$ is $(\cdot/p)$; all other order-unit character factors are trivial. The scaling character has

$$\alpha(-1) = \Big(\prod_{p^e \| N} w_{p^e} \chi_{h_*}(p)^e \Big) \Big(\prod_{p \mid m} (-1/p) \Big)(-1) = 1.$$

For the last equality, $m \equiv 3 \pmod 4$ and the first product is $w(E^{h_*}) = +1$. More explicitly, $w(E) = -\prod w_{p^e}$ and, since $(h_*, N) = 1$, $w(E^{h_*}) = w(E)\chi_{h_*}(-N)$. The latter twist formula also follows by multiplying the local unramified-twist factors $\chi_{h_*}(p)^{v_p N}$ at $p \mid N$ and the factors $(-1/p)$ at the newly ramified good primes.

Let η_B denote the classical character associated with α in the inverse-idele convention. The formula $\eta_B(d) = \prod_p \alpha_p(d)$ for positive integers d prime to the support is verified in the transformation calculation below.

Set $c_0 = |\operatorname{GL}_4(\mathbb{F}_3)| = 24\,261\,120$, so $v_2(c_0) = 9$, and define

$$H_M(n) = c_0 \sum_{z \in X} \frac{P_M(z)\chi_{h_*}(\operatorname{Nrd} z)}{|R_z^\times|} \sum_{x \in V(\mathbb{Q}),\ Q_B(x) = n} f(z^{-1}xz). \tag{O.4}$$

Left multiplication embeds each $R_z^\times$ into $\operatorname{GL}_4(\mathbb{Z})$; reduction modulo three is injective on finite subgroups. Indeed, a prime-order element $1 + 3^a A$ in its kernel, with $a \geq 1$ and A nonzero modulo three, has nonzero first term in its prime-th power modulo 3^{a+1}, or modulo 3^{a+2} when its order is three, a contradiction. Thus $|R_z^\times| \mid c_0$. The right-conjugation identity proves independence of the right representative. On the left, rational reduced norms are positive and the global quadratic idele character is trivial on them. The transformation law below shows that each summand is an actual holomorphic theta series with the same level and character; $c_0/|R_z^\times|$ is an integer. Consequently the coefficients lie in the fixed character ring $\mathcal{O}$. Changing any lift of P_M by 2^M changes every coefficient by an element of $2^M\mathcal{O}$. In particular there is no loss of precision from an automorphism denominator.

A6.3 Transformation law and recurrence

Use the standard global additive character, self-dual measures, and the Gaussian $e^{-2\pi Q_B}$ at infinity. Weil's operator formulas [84, I.13] give multiplication by $\mathbf{e}_p(bQ_B)$ for $n(b)$, scaled pullback $f_p(x) \mapsto \gamma_{Q_B}(u) f_p(ux)$ for $m(u)$ with u a unit, and a constant times Fourier transform for the Weyl element. The upper unipotents $n(\mathbb{Z}_p)$ fix f_p, because Q_B is integral on its support. If L_p' is any lattice of translations fixing f_p, its Fourier transform has support in $(L_p')^\vee$. Choose A_p with $p^{A_p} Q_B((L_p')^\vee) \subseteq \mathbb{Z}_p$. Fourier conjugation then proves that $n^-(p^{A_p}\mathbb{Z}_p)$ fixes f_p. This includes $p = 2$. At two take $A_2 \geq 2$ and use the comparison with the ordinary theta multiplier given below; no splitting of

the metaplectic cover over all diagonal units is asserted. For clarity, the local normalizations and the lower-unipotent argument are as follows. Take $\mathbf{e}_\infty(t) = e^{2\pi i t}$ and $\mathbf{e}_p(t) = e^{-2\pi i\{t\}_p}$, so that $\prod_v \mathbf{e}_v(r) = 1$ for $r \in \mathbb{Q}$. For $Q(x) = x^{\mathsf{T}}Ax$ put $B_Q(x,y) = Q(x+y) - Q(x) - Q(y) = 2x^{\mathsf{T}}Ay$ and set

$$\mathcal{F}_Q\phi(y) = \int \phi(x)\mathbf{e}_p(B_Q(x,y))\,dx_Q, \qquad dx_Q = |\det(2A)|_p^{1/2}\,dx.$$

Here dx is the coordinate product measure with $\mathrm{vol}(\mathbb{Z}_p^3) = 1$. At the real place use the same formula $dx_Q = |\det(2A)|_\infty^{1/2}dx$ with coordinate Lebesgue measure. Thus $\mathcal{F}_Q^2\phi(x) = \phi(-x)$. If $\phi(x+\lambda) = \phi(x)$ for $\lambda \in L'$, translation in this integral shows that its Fourier transform vanishes unless $\mathbf{e}_p(B_Q(\lambda, y)) = 1$ for every $\lambda \in L'$. This is precisely the asserted support in $(L')^\vee$. Since this dual lattice is compact, an integer A_p with $p^{A_p}Q((L')^\vee) \subset \mathbb{Z}_p$ exists. The operator for $n^-(c)$ is a Fourier conjugate of multiplication by $\mathbf{e}_p(-cQ)$; consequently it fixes ϕ whenever $c \in p^{A_p}\mathbb{Z}_p$. A scalar used to normalize the Weyl operator cancels with its inverse in this conjugation.

All the specified local functions have such a translation lattice. At an odd level prime the unit-root condition and its character depend on a finite residue quotient. At two the square-unit condition is a congruence modulo eight; the integral projectors selecting the cyclic quotient, and the finite root character, again depend on a finite residue quotient. At a prime dividing m, valuation one and the nonzero-nilpotent sign depend on finite residue data. The division-prime function is already a residue function. Each support is contained in a lattice and is a finite union of cosets of a sufficiently small lattice. This proves the Schwartz–Bruhat and translation assertions for the actual weights.

For the reference $Q_0 = \sum X_i^2$, the dyadic dual of $\mathbb{Z}_2^3$ is $\frac{1}{2}\mathbb{Z}_2^3$ and $\mathrm{vol}_{dx_{Q_0}}(\mathbb{Z}_2^3) = 2^{-3/2}$. In particular its Fourier transform is $2^{-3/2}1_{\frac{1}{2}\mathbb{Z}_2^3}$, not $1_{\mathbb{Z}_2^3}$. The lower-unipotent conclusion nevertheless holds at depth two, because $4Q_0(\frac{1}{2}\mathbb{Z}_2^3) \subset \mathbb{Z}_2$. These positive measure factors affect neither the diagonal eigenvalue quotient nor the counting coefficients of the theta sum.

The diagonal factor can be computed directly. For $Q_B = \sum_{i=1}^3 a_i X_i^2$, the one-dimensional Weil-index identity

$$\frac{\gamma(ua_i)}{\gamma(a_i)} = \frac{\gamma(u)}{\gamma(1)}(u, a_i)_p$$

and multiplication under orthogonal sums give, relative to the ordinary theta multiplier cubed, the factor $(u, a_1a_2a_3)_p$. These identities are [84, II.25 and II.28]; the latter obtains the Hilbert symbol from the four-dimensional quaternion norm form. For $B = (-a, -b)$ the norm on B^0 is $aX^2 + bY^2 + abZ^2$, whose determinant is $(ab)^2$. There is therefore no additional determinant character for Q_B.

The normalized diagonal factor. The Weil-index identity used here can be derived without choosing Gauss-sum signs. The norm form of the quaternion algebra (u, a) is $\langle 1, -u, -a, ua\rangle$. By multiplicativity under orthogonal sums and $\gamma(-q) = \gamma(q)^{-1}$, its index is

$$\frac{\gamma(1)\gamma(ua)}{\gamma(u)\gamma(a)}.$$

For a split quaternion algebra the norm form is hyperbolic and its index is one; for the division algebra its index is minus one by [84, II.28, Proposition 4]. These are exactly the two values of $(u, a)_p$. Solving this equality gives

$$\frac{\gamma(ua)}{\gamma(a)} = \frac{\gamma(u)}{\gamma(1)}(u, a)_p.$$

Applying it to three diagonal entries gives the ordinary one-dimensional theta factor cubed times $(u, \det Q_B)_p$. This derivation includes $p = 2$. A change of diagonalization changes the determinant only by a square, so it changes no symbol. For Q_B/m the determinant becomes $m^{-3} \det Q_B$, whose squareclass is m. Thus the extra factor is precisely $(u, m)_p$. This establishes the determinant-character contribution in (O.5) with the same additive character as the original theta multiplier.

An operator construction of the common double cover. Here is a direct justification that the same lifts may be used in the comparison. It also fixes the signs of the diagonal operators. Work over $k = \mathbb{Q}_p$, including $p = 2$, or $k = \mathbb{R}$, with the additive character just specified. For a nondegenerate ternary Q write $\gamma_Q = \gamma(Q)$ and define

$$\begin{aligned} U_Q(b)\phi(x) &= \mathbf{e}_k(bQ(x))\phi(x), \\ D(a)\phi(x) &= |a|_k^{3/2}\phi(ax), \\ W_Q = \gamma_Q \mathcal{F}_Q, \qquad M_Q(a) &= r_Q(a)D(a), \qquad r_Q(a) = \frac{\gamma(aQ)}{\gamma(Q)}. \end{aligned}$$

The Weyl matrix in this convention is $w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. The opposite Fourier-sign convention changes both compared representations together.

The Fourier formula of [84, I.14, Theorem 2], or its completion-of-the-square calculation, says for $b \neq 0$ that

$$\int_{k^3} \mathbf{e}_k\big(bQ(z) + B_Q(z, v)\big)\, dz_Q = |b|_k^{-3/2}\gamma(bQ)\mathbf{e}_k(-b^{-1}Q(v)).$$

The integral has the oscillatory meaning of that theorem. Applying it to a Schwartz function gives the operator identity

$$\mathcal{F}_Q U_Q(b)\mathcal{F}_Q = \gamma(bQ)U_Q(-b^{-1})D(b^{-1})\mathcal{F}_Q^{-1}U_Q(-b^{-1}).$$

Thus no choice of an individual square root of a Gauss sum is needed. The index identities proved above also give

$$\gamma(Q)^2 = (\det Q, -1)_k\gamma(1)^6, \qquad \gamma(1)^4 = (-1, -1)_k, \qquad r_Q(a)r_Q(b) = (a, b)_k r_Q(ab).$$

For the middle equality apply the quaternion-norm formula to $\langle 1, 1, 1, 1\rangle$; the first follows by applying the same formula to $\langle 1, -a_i, 1, -a_i\rangle$ for each diagonal entry a_i of Q.

Consequently the displayed operators satisfy

$$\begin{gathered} U_Q(b)U_Q(c) = U_Q(b + c), \qquad M_Q(a)U_Q(b)M_Q(a)^{-1} = U_Q(a^2 b), \\ M_Q(a)M_Q(b) = (a, b)_k M_Q(ab), \qquad W_Q M_Q(a) W_Q^{-1} = M_Q(a^{-1}), \\ W_Q^2 = (-1, -1)_k M_Q(-1), \\ W_Q U_Q(b) W_Q = (-b, -1)_k U_Q(-b^{-1})M_Q(-b^{-1})W_Q U_Q(-b^{-1}) \quad (b \neq 0). \end{gathered}$$

For the last equality, the preceding Fourier identity leaves the scalar $\gamma(Q)^2\gamma(bQ)^2$; the index-square formula reduces it to $(-b, -1)_k$. Every scalar in these relations is independent of the ternary form. Interpreting a scalar -1 as the central element ε gives one common double cover of $\mathrm{SL}_2(k)$. Indeed these relations reduce every word to one of $\varepsilon^e U(x)M(a)$ or $\varepsilon^e U(x)M(a)WU(y)$, with $e \in \{0, 1\}$. The ordinary Bruhat decomposition makes their projections unique, apart from ε^e; the operators send ε to $-\mathrm{id}$, so the central kernel has exactly two elements. This realizes the rank-three pullback of Weil's metaplectic operators, rather than assuming a section over the diagonal units.

For $Q_0 = \langle 1,1,1\rangle$ these operators are the tensor products of the three ordinary one-dimensional theta operators. For either Q or Q_0, changing a lift multiplies its operator by the same -1. Hence the quotient of their eigenvalues is independent of the lift. At two the relation $M_Q(-1)^2 = (-1,-1)_2 \,\mathrm{id} = -\,\mathrm{id}$ explicitly retains the order-four lift of $-I$.

On a unit a the quotient of the two diagonal index factors is

$$\frac{r_Q(a)}{r_{Q_0}(a)} = (a, \det Q)_k.$$

Together with $f_p(ax) = \alpha_p(a) f_p(x)$, this is exactly the eigenvalue quotient $\alpha_p(a)(a, \det Q)_p$ used below. There is no remaining eighth-root factor at two.

The metaplectic comparison at two and the extracted character. We specify how the preceding index calculation fixes the multiplier, including its dyadic part. Let

$$Q_0 = X_1^2 + X_2^2 + X_3^2, \qquad \phi_{0,p} = 1_{\mathbb{Z}_p^3}, \qquad \theta(\tau) = \sum_{n\in\mathbb{Z}} e^{2\pi i n^2 \tau}.$$

Use the same additive character and metaplectic lifts for the rank-three Weil representations of Q_B and Q_0. The latter theta series is literally $\theta(\tau)^3$; this fixes the meaning of the ordinary theta multiplier cubed, including its eighth roots of unity. For either representation the nontrivial central element of the metaplectic cover acts by -1. Consequently the ratio of two eigenvalues computed using the same lift is independent of that lift. In particular a central factor in multiplying the three lifts in the displayed triangular factorization cancels in this ratio. We compare eigenvalues, and do not choose a group section of the cover over $\Gamma_0(2^{A_2})$.

Here is the comparison on the three factors. Both upper unipotents act trivially on the indicated vectors. Choose A_p large enough that both lower unipotents do also. This is the Fourier-support argument above for f_p; for $\phi_{0,p}$ it requires no positive depth at an odd prime and depth two at two, since the bilinear dual of $\mathbb{Z}_2^3$ for Q_0 is $\frac{1}{2}\mathbb{Z}_2^3$ and $4Q_0(\frac{1}{2}\mathbb{Z}_2^3) \subset \mathbb{Z}_2$. The Weyl scalar cancels in conjugating an upper unipotent to a lower one, so it introduces no additional eigenvalue. On a diagonal unit u, the scaling part acts by $\alpha_p(u)$ on f_p and by one on $\phi_{0,p}$. The quotient of the Weil-index parts, by the preceding calculation, is

$$\frac{\gamma(uQ_B)/\gamma(Q_B)}{(\gamma(u)/\gamma(1))^3} = (u, \det Q_B)_p. \tag{O.5a}$$

If the inverse Weil-index convention is used for the diagonal operator, both numerator and denominator are inverted; the quotient is unchanged because the Hilbert symbol has order two. Thus this computation does not leave an undetermined inverse or a dyadic root of unity. These operator comparisons are precisely the scaled pullback, Fourier conjugation and scalar relations of [84, I.13–I.14 and II.25–II.28].

Here $\det Q$ means the determinant of the matrix in $Q(x) = x^{\mathsf{T}} A x$. The associated bilinear Gram matrix is $2A$. If that convention is used instead, both Q and Q_0 acquire the same factor 2^3 in their determinants, which cancels in the comparison. In particular no extra squareclass of two is introduced at the dyadic place.

For a general ternary form Q in this argument, at each prescribed prime take $A_p \geq 1$ and increase A_p also to the conductor of $u \mapsto (u, \det Q)_p$. The eigenvalue relative to the ordinary theta multiplier, on a matrix whose upper-left entry is a, is then

$$\alpha_p(a)(a, \det Q)_p. \tag{O.5b}$$

Indeed the triangular factorization has just the two trivial unipotent factors and this diagonal factor, and its lift ambiguity cancels as explained above. This also verifies multiplication: the upper-left entry of a product is congruent to the product of the upper-left entries modulo p^{A_p}, so both local characters in (O.5b) multiply. The argument therefore applies to every matrix in the indicated congruence subgroup at a prescribed prime, not only to a choice of generators. At an unramified prime outside the support, the full spherical invariance established below treats matrices with nonunit upper-left entry as well.

For Q_B the determinant squareclass is one at every place, including two. Thus (O.5b) has no determinant factor. For the extracted form $Q'_B = Q_B/m$ it is $m^{-3} \equiv m$. At an odd prime $p \mid m$ and a unit u, and at two, respectively, the additional factors are explicitly

$$(u, m)_p = (u/p)^{v_p(m)}, \qquad (u, m)_2 = (-1)^{((u-1)/2)((m-1)/2)}.$$

Here m is odd and $m \equiv 3 \pmod 4$, so the second factor is $(-1)^{(u-1)/2}$. It is already trivial on $1 + 4\mathbb{Z}_2$. At every odd prime outside m the factor on units is one. There is consequently no new level prime, and there is no omitted dyadic character.

Finally take a positive integer d prime to the complete level. The finite operator used in the classical transformation is the one at γ^{-1}, with upper-left entry d. The root-character factor is therefore $\prod_p \alpha_p(d) = \eta_B(d)$. Hilbert reciprocity gives

$$\prod_{p \mid 2m} (d, m)_p = \prod_{q \mid d} (m/q)^{v_q(d)} = (m/d). \tag{O.5c}$$

Indeed the infinite symbol is one because $m, d > 0$, all other finite symbols are one, and at an odd $q \mid d$ the symbol is $(m/q)^{v_q(d)}$. This proves the displayed product without discarding its dyadic factor. Both sides extend to the same even Dirichlet character. Since $h_* = -m$, division by the genus character now gives

$$\frac{\eta_B(d)(m/d)}{\eta_B(d)\chi_{h_*}(d)} = (-1/d).$$

Thus the characters in (O.5), with the stated inverse-idele convention for η_B, are exactly the characters supplied by these local Weil operators.

Take A_p also at least the conductor exponent of α_p. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $c \in p^{A_p}\mathbb{Z}_p$ and a a unit, use

$$\gamma = n^-(c/a)m(a)n(b/a).$$

The scalar is the diagonal theta factor times $\alpha_p(a)$. In passing to the classical transformation law one applies the finite operator at γ^{-1}, whose upper-left entry is d. Thus the classical Dirichlet character is $\eta_B(d) = \prod_p \alpha_p(d)$, not its inverse. Its value at -1 is one, as required by the weight-3/2 theta multiplier.

There are no additional level primes hidden in this argument. Outside the prescribed support, p is odd, B_p is split, and $R_p^0 = \mathfrak{sl}_2(\mathbb{Z}_p)$ is self-dual for the bilinear form of $Q_B = -a^2 - bc$. Its determinant is a unit. Hence the Fourier transform of $1_{R_p^0}$ equals itself; the unramified Weil constant is one, as also follows from [84, II.27] by integrating over this self-dual lattice. Upper unipotents, diagonal units, and the Weyl element fix it. These generate $\mathrm{SL}_2(\mathbb{Z}_p)$, so the exponent there is zero. At the finitely many prescribed primes the preceding A_p suffice. Conjugation by any $z \in \widehat{B}^\times$ is an isometry of Q_B, commutes with the Weil representation, and preserves the self-dual measure. The same level and character therefore work for every summand indexed by z.

[84, III, Theorem 6] applies to this Schwartz–Bruhat function on $V(\mathbb{A})$ and proves invariance of its rational theta sum. The normalized archimedean expression is $\sum_x f(x)e^{2\pi i\tau Q_B(x)}$: the factor

$y^{3/4}$ in the Weil action is removed in passing to the classical form. It has holomorphic weight $3/2$. At a rational cusp, the slash operation replaces the finite Schwartz function by its finite Weil transform. Its support meets $V(\mathbb{Q})$ in a finite union of rational lattice cosets, with bounded denominators. Positive definiteness gives nonnegative rational exponents of bounded denominator and normal convergence for $\Im\tau > 0$. This proves holomorphy at every cusp. Theorem 6 alone asserts automorphic invariance; this last argument supplies the holomorphy conclusion used here.

Extraction. The support at every $p \mid m$ has $v_p Q_B = 1$. All nonzero indices are integral and divisible by m. Repeat the preceding construction for $Q'_B = Q_B/m$ with the same finite weights and the Gaussian for Q'_B. This quadratic form is integral on the support. Away from m it differs by a unit, so the good-prime self-duality argument remains valid. Its determinant squareclass is $m^{-3} \equiv m$, giving the extra Dirichlet character $(m/\cdot)$. Hence

$$g_M(\tau) = \sum_n H_M(mn) q^n \in M_{3/2}(\Gamma_0(K), \psi).$$

Here K has only the stated prime support, and

$$\psi = \eta_B(m/\cdot), \qquad \eta = \eta_B \chi_{h_*}, \qquad \psi/\eta = \chi_{-1}. \tag{O.5}$$

This directly constructs the extracted form without a half-integral-weight extraction operator. The inverse in this classical passage can also be checked before any character notation is introduced. Put $g_\tau = n(x)m(\sqrt{y})$ for $\tau = x + iy$, so $g_\tau i = \tau$. For $\gamma \in \mathrm{SL}_2(\mathbb{Q})$ write $\gamma g_\tau = g_{\gamma\tau} k_\infty$ with $k_\infty \in \mathrm{SO}_2(\mathbb{R})$. In the adelic group,

$$(g_{\gamma\tau}, 1_f) = \gamma\,(g_\tau, \gamma_f^{-1})(k_\infty^{-1}, 1_f).$$

This equality also holds with consistent metaplectic lifts. The global rational lifts are compatible for the two forms: [84, II.30, Proposition 5] gives $\prod_v \gamma_v(Q) = \prod_v \gamma_v(aQ) = 1$ for rational Q, a, and hence the local normalizing factors in W_Q and $M_Q(a)$ have product one. The product formula also gives $\prod_v |\det(2A)|_v^{1/2} = 1$, so the product of the stated self-dual measures is the measure for which the rational lattice has covolume one. Apply the rational invariance of [84, III.41, Theorem 6] to the displayed adelic identity. The archimedean Gaussian for either positive ternary form has the same weight $3/2$; therefore its factor from k_∞^{-1} is the same as that for θ^3. The relative finite factor is evaluated at γ_f^{-1}, whose upper-left entry is d. Thus the relative classical character is $\prod_p \alpha_p(d)(d, \det Q)_p$, over the prescribed primes. This proves the inverse-idele convention directly. It compares the transformation identities and does not divide by a value of a theta function.

For the extracted form there is likewise no measure or degree normalization in its coefficients. With exactly the same finite weights, replace only the quadratic form and archimedean Gaussian by Q_B/m and $e^{-2\pi Q_B/m}$. The summand indexed by x now has exponent $Q_B(x)/m$. The original support makes $Q_B(x)$ an integer with valuation one at every prime dividing m, so these exponents are nonnegative integers. Coefficient n is exactly the original sum over $Q_B(x) = mn$, namely $H_M(mn)$. The self-dual measure changes by $|m|_v^{-3/2}$ locally, but a theta coefficient is a discrete sum and contains no Haar measure; this change is used only in the Fourier operators. Outside the original support it is a unit rescaling of the same unimodular local form, and at primes dividing m the lower-unipotent depth can be increased within the existing support. The determinant changes by m^{-3}, whose squareclass is m, so (O.5c) supplies exactly $(m/\cdot)$. In particular no new level prime, factor of m, or dyadic root of unity is introduced by extraction.

All-index recurrence and its phases. Let p avoid the full support and put $s = \eta_B(p)$. Locally use $R_p = M_2(\mathbb{Z}_p)$ and $V_p = \mathfrak{sl}_2$. The $p+1$ neighbors are indexed by lines of $\mathbb{F}_p^2$. For integral x with nonzero reduction and $Q_B(x) = n$, the number of preserved lines is $1 + (-n/p)$, including the nilpotent case. If $x \in pR_p^0$, all $p+1$ lines are preserved; the excess over the preceding formula is p.

A nonintegral x in a neighbor lies in $p^{-1}R_p^0$. Write $x = y/p$ with y primitive and $Q_B(y) = p^2 n$. Then $\bar{y}$ is nonzero nilpotent. In a basis with $\bar{y} = X_0$, write $y = \left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right)$. Here $p \mid a, c$, b is a unit, and $p^2 \mid -a^2 - bc$ forces $p^2 \mid c$. Exactly the neighbor belonging to the kernel line makes y/p integral. The integral and nonintegral parts therefore exhaust the sum with no genericity assumption on n.

Scaling by p at all other places multiplies the weight by s. Thus the primitive nonintegral contribution is $s^{-1}H_M(p^2 n) - H_M(n)$, and the integral contribution is $(1+(-n/p))H_M(n) + psH_M(n/p^2)$. Their sum is

$$s^{-1}H_M(p^2 n) + (-n/p)H_M(n) + psH_M(n/p^2).$$

Here and below a coefficient at a nonintegral index is zero. For completeness, if N_{ij} counts neighbor kernels from i to j and $w_i = |\operatorname{Aut}(i)|$, counting actual isogenies and dualizing gives $N_{ij}w_j = N_{ji}w_i$, or $N_{ij}/w_i = N_{ji}/w_j$. This proves the self-adjointness with the mass $1/w_i$ appearing in H_M. A neighbor changes the norm character by $\chi_{h_*}(p)$. Moving the neighbor operator onto P_M and using its actual modular-parametrization identity consequently gives

$$H_M(p^2 n) + s(-n/p)H_M(n) + ps^2 H_M(n/p^2) = s\chi_{h_*}(p)a_p(E)H_M(n) \pmod{2^M\mathcal{O}}.$$

This computation explains both powers of s even when s is not quadratic.

Put $a_M(n) = H_M(mn)$ and $\eta = \eta_B\chi_{h_*}$. Since $(-mn/p) = (m/p)(-n/p)$ and $\eta(p)^2 = s^2$, one obtains

$$a_M(p^2 n) = \{\eta(p)a_p(E) - \psi(p)(-n/p)\}a_M(n) - p\eta(p)^2 a_M(n/p^2) \pmod{2^M\mathcal{O}}. \tag{O.6}$$

The proof covers $p \mid n$ and $p^2 \mid n$ explicitly. At zero the coefficients vanish by the local support, and negative indices vanish by positive definiteness. An added finite coprimality or unit-squareclass sieve is realized by the corresponding local characteristic function for Q'_B; the same modularity proof applies. At primes outside the enlarged support the three indices $p^2 n, n, n/p^2$ have the same retain-or-zero status. Hence these sieves preserve the displayed recurrence. □

A6.4 The unaveraged point comparison

CM and specialization inputs. For the conductor-one comparison, the fixed-type ideal action, its CM reciprocity convention and forward inert reduction are recorded in Section A35. Common good models and the same-place specialization of the labelled prime-to-$\ell_0 N$ ideal quotients, including the selected level and residue orientations, are proved in Section A35. The enhanced supersingular class-set and Frobenius inputs remain separate.

The actual weighted coefficient comparison.

Proposition A6.3 (The unaveraged weighted theta comparison)**.** *Put $m = |h_*|$ and $F = \mathbf{Q}(\sqrt{Dh_*})$, where D is a positive squarefree parameter in the fixed odd filter. Assume $(D/\ell_0) = 1$ at the chosen actual stage, so that ℓ_0 is inert in F. Keep the local Schwartz functions, the actual function $P_M(z) = \lambda_M(L\pi_{\ell_0}(z))$, and the integer c_0 of (O.4). Then a 2-power root of unity $\zeta_M(D)$ satisfies*

$$H_M(mD) = c_0\, 2^{\omega(N)}\zeta_M(D)\lambda_M(\overline{LP_D}) \pmod{2^M}. \tag{O.7}$$

The root of unity belongs to the field generated by the fixed characters at the primes of N. No class number is divided out.

Proof. The filter makes Dh_* an odd fundamental discriminant, with F split at every prime dividing $2N$. Its unit group is $\{\pm 1\}$. The condition $Q_B(x) = mD$ is equivalent to $x^2 = Dh_*$. Together with $(1+x)/2 \in R_{z,2}$ and the other integrality conditions, this says exactly that

$$\mathcal{O}_F \longrightarrow R_z, \qquad \sqrt{Dh_*} \longmapsto x$$

is an optimal embedding. Optimality follows because the source order is already maximal. All nonzero local weights on such an embedding are roots of unity. The valuation-one condition at each prime of h_* holds because $(D, h_*) = 1$; at ℓ_0 the residue of x is nonzero. The fixed unit filters hold by admissibility.

First classify the local embeddings. Away from $N\ell_0$ there is one local orbit: an $\mathcal{O}_{F,p}$-stable lattice is locally a fractional ideal, hence principal, and this also covers ramified primes dividing Dh_*. At $p^e \| N$, the algebra is split. The two integral idempotents decompose both lattices of the Eichler lattice pair. Its quotient is cyclic of order p^e, so its elementary exponents lie wholly in one summand. The two possibilities give exactly two orientations. This applies at two using the integral spectral projectors specified in the construction. At ℓ_0, the maximal order of the division algebra has residue field $\mathbf{F}_{\ell_0^2}$. The two residue-field embeddings of the unramified quadratic order give the two orientations. Consequently there are $2^{\omega(N)+1}$ local orientation tuples.

Fix one tuple. Choose a rational embedding $\iota : F \hookrightarrow B$ and write $x_0 = \iota(\sqrt{Dh_*})$. Such an embedding exists because F is a field at the two ramified places of B, namely ℓ_0 and infinity [98, Proposition 14.6.7, $(v) \Rightarrow (iii)$]. Choose z_0 realizing the selected integral local embeddings. The adelic representatives with this tuple are exactly

$$\widehat{F}^\times z_0 \widehat{R}^\times.$$

Their quotient by the rational centralizer $F^\times$ is

$$F^\times \backslash \widehat{F}^\times / (\widehat{F}^\times \cap z_0 \widehat{R}^\times z_0^{-1}) = F^\times \backslash \widehat{F}^\times / \widehat{\mathcal{O}}_F^\times = \mathrm{Pic}(\mathcal{O}_F).$$

Thus this is an actual class-group torsor, not an average or a quotient of a class-group orbit.

For an ideal-class representative $a \in \widehat{F}^\times$, the element a centralizes x_0, and therefore

$$f((az_0)^{-1} x_0 (az_0)) = f(z_0^{-1} x_0 z_0).$$

On the other hand,

$$\chi_{h_*}(\mathrm{Nrd}(az_0)) = \chi_{h_*}(\mathrm{N}_{F/\mathbf{Q}}\, a)\chi_{h_*}(\mathrm{Nrd}\, z_0).$$

The first factor is exactly the genus character in (O.1). To see that it is well-defined on the class group, for $D > 1$ the biquadratic field $\mathbf{Q}(\sqrt{D}, \sqrt{h_*})$ has discriminant $D\, h_*\, (Dh_*) = (Dh_*)^2$. Its relative discriminant over F is therefore one. The character obtained by restricting χ_{h_*} to G_F is unramified and, under class field theory, is $a \mapsto \chi_{h_*}(\mathrm{N}_{F/\mathbf{Q}}\, a)$. For $D = 1$ it is the trivial character. Ideal representatives can be chosen prime to h_*, giving exactly the elementary norm formula in (O.1). Reversing the Artin convention has no effect because this character is quadratic.

We use the direct conductor-one reduction argument in Section A35.3. Its retained inputs are CM ideal-class reciprocity, potentially good CM reduction and inert-prime supersingularity, extension and specialization of endomorphisms and prime-to-ℓ_0 subgroup quotients, and the geometric and Tate-lattice description of supersingular pairs used above. The latter input, including its Frobenius compatibility, remains separate from the reduction argument. The proof computes the labelled target torsor using the pointwise centralizer $F^\times$, independently of reduction surjectivity. Ideal representatives prime to $N\ell_0$ then give actual subgroup quotients whose reduction, at the

one fixed place, agrees with the same ideal action. Thus reduction is an equivariant bijection onto the full orientation tuple whose ℓ_0 sign is induced by the fixed CM type. Its multiplicity is one. Restricting to any underlying supersingular pair gives the same bijection on its fiber. This uses maximal $\mathcal{O}_F$, every split prime-power level (including powers of two), and $\ell_0 \nmid N\,\mathrm{disc}(F)$. The other level orientations have geometric Atkin–Lehner anchors; the opposite ℓ_0 sign is obtained by Frobenius, not by claiming it is in the image of the same fixed-type reduction map.

Choose the reduction place compatibly with (O.2), and choose z_0 as the reduction of the actual reference Heegner point x_D in (O.1). Writing

$$\chi_F([a]) = \chi_{h_*}(\mathrm{N}_{F/\mathbb{Q}}\, a), \qquad z_a = [az_0],$$

equivariance gives the exact finite-sum identity

$$\sum_{[a]\in \mathrm{Pic}(\mathcal{O}_F)} \chi_F([a]) P_M(z_a) = \lambda_M\left(\overline{L \sum_{[a]\in \mathrm{Pic}(\mathcal{O}_F)} \chi_F([a])\pi(x_D^{[a]})}\right) = \lambda_M(\overline{LP_D}) \quad \text{in } \mathbb{Z}/2^M\mathbb{Z}.$$

Here the supersingular coarse points and their parametrization values are defined over $\mathbb{F}_{\ell_0^2}$, so every term is in the domain of the same λ_M. Additivity, specialization, and multiplication by the fixed integer L commute with this finite sum. If the Artin action is written with inverses, reindexing $[a] \mapsto [a]^{-1}$ leaves it unchanged since χ_F is quadratic. Changing the reference point within its torsor multiplies the sum only by χ_F of that reference class.

The weights are constant on each embedding orbit used in the following count. Indeed a unit of the definite order R_z has positive integral reduced norm with integral inverse, so its norm is one. The conjugation law for f therefore gives weight one for its action. Along the class-group torsor the Schwartz factor is constant because a centralizes x_0, and the remaining norm factor is exactly $\chi_F([a])$. Hence the finite-sum identity just proved applies with the single phase $\chi_{h_*}(\mathrm{Nrd}\, z_0) f(z_0^{-1} x_0 z_0)$.

For each fixed z, the stabilizer of an embedding in $R_z^\times$ is $\mathcal{O}_F^\times = \{\pm 1\}$. Each embedding orbit thus has $|R_z^\times|/2$ elements. The denominator $|R_z^\times|$ in (O.4) leaves the factor $1/2$ per embedding class. The vectors x and $-x$ are not identified by this stabilizer: at ℓ_0 their residues define the two different embeddings of the residue field, whereas conjugation by an order unit acts trivially on that field. They already lie in the two different ℓ_0 orientation tuples counted above. The contribution of the reference orientation tuple is

$$\frac{c_0}{2}\, \zeta_M(D) \lambda_M(\overline{LP_D}), \qquad \zeta_M(D) = \chi_{h_*}(\mathrm{Nrd}\, z_0) f(z_0^{-1} x_0 z_0),$$

up to the harmless quadratic reference-class factor just described. All nonzero factors in this displayed phase are fixed-character values or signs. Thus it is a 2-power root of unity; its field does not grow with D, M, or ℓ_0.

It remains to account for the other orientations; they are not all in the image of the one sign component of the reduction map. At $p^e \| N$, use the local Atkin–Lehner normalizer. It reverses the selected root, so multiplies the Schwartz weight by

$$\alpha_p(-1) = w_{p^e} \chi_{h_*}(p)^e.$$

It multiplies the norm-weighted parametrization value by the same sign: its reduced norm has valuation e, and the actual parametrization transforms by w_{p^e} after L kills the fixed cuspidal translation. The two signs multiply to one. This normalizer has trivial components at the other primes, so it changes no other orientation or local weight.

At ℓ_0, a division-algebra uniformizer exchanges the two residue embeddings. It sends the trace-zero residue β to $-\beta$, multiplying its weight by $(-1/\ell_0) = -1$. Its reduced norm has valuation one and hence contributes $\chi_{h_*}(\ell_0) = -1$. Its action on supersingular pairs is Frobenius. Since λ_M has Frobenius sign $+1$ and the parametrization is defined over $\mathbf{Q}$, this introduces no further sign. Again the two factors multiply to one. The uniformizer operation used here may be described with arithmetic or geometric Frobenius: they have the same action on the supersingular coarse points and their parametrization values over $\mathbb{F}_{\ell_0^2}$. The identity $\lambda_M(FP) = \lambda_M(P)$ from (O.2) consequently applies to the very operation that exchanges the residue embeddings. Its norm has valuation one, and χ_{h_*} is unramified at ℓ_0, so its norm-character value is exactly $\chi_{h_*}(\ell_0) = -1$, independently of the uniformizer's unit part. Likewise, at $p^e \| N$, the norm of the Atkin–Lehner normalizer has valuation e and χ_{h_*} is unramified at p. Its norm-character value is exactly $\chi_{h_*}(p)^e$. These observations fix the norm signs in both orientation changes.

It follows that every one of the $2^{\omega(N)+1}$ tuples has the same contribution as the reference tuple. Multiplying by their number gives precisely $c_0 2^{\omega(N)}$, as asserted. This counts individual embeddings with their stabilizer; a conjugate-pair formulation of the geometric correspondence must not be used to insert or remove another factor of two.

Finally, $c_0/|R_z^\times|$ is an integer for every z. Thus a change of the chosen integer lift of $P_M(z)$ by 2^M changes the actual coefficient by an element of $2^M\mathcal{O}$. Every preceding equality is consequently valid at the original retained precision. No denominator depending on the class number, the discriminant, or the stage has entered.

Precision in the orientation count. The factor $1/2$ in the orbit count does not require dividing a congruence modulo 2^M by two. Indeed, $-1 \in R_z^\times$ implies $2 \mid |R_z^\times|$, and the chosen integer c_0 is divisible by $|R_z^\times|$. Hence $c_0/2$ is an integer. Choose lifts of all $P_M(z)$ before summing. Each orbit of embeddings contributes $(c_0/|R_z^\times|)(|R_z^\times|/2)P_M(z) = (c_0/2)P_M(z)$, with its prescribed unit weight. Replacing any lift by a multiple of 2^M changes this expression by a multiple of 2^M. The comparison with the reduced point may therefore be made at precision M, separately for each orientation. Summing the $2^{\omega(N)+1}$ orientations gives the stated multiplier $c_0 2^{\omega(N)}$ without any loss of precision. In particular, write $c_\theta = 9 + \omega(N)$, the valuation of the fixed integer $c_0 2^{\omega(N)}$. At the fixed witness D_w, (O.2) and (O.7) imply

$$v_2 a_M(D_w) = c_\theta + c_w \qquad \text{whenever } M > c_\theta + c_w.$$

The phase is a unit and the congruence error has strictly larger valuation, so this is an equality, not merely a lower bound. Discarding the finitely many earlier stages is sufficient. Thus the visibility used in choosing the odd minimum retains a fixed witness and a fixed valuation; neither the class number nor the changing prime ℓ_0 introduces a new loss of precision.

□

Proposition A6.4 (Visibility and the native genus index)**.** *Use the actual odd form and unaveraged point of Propositions A6.2 and A6.3. There is one constant C_{det}, independent of the actual stage and admissible D, for which (O.8) holds. Explicitly, for a known simple-zero D, $v_2 a_M(D) \geq \min\{M, j_{\mathrm{gen}}(D) - C_{\mathrm{det}}\}$. For every admissible D, the strict inequality $v_2 a_M(D) < M$ implies $\mathrm{ord}_{s=1} L(E^D, s) = 1$ and $j_{\mathrm{gen}}(D) \leq v_2 a_M(D) + C_{\mathrm{det}}$. If an admissible odd-sign D is not a simple zero, its coefficient vanishes modulo 2^M. No genus index of a torsion point is defined or used in this assertion.*

Proof. For a known simple-zero D, write $P_D = m_D G + T$ in its native rank-one free lattice. The same L kills T. An integral finite-field Kummer map cannot decrease $v_2(m_D)$, so (O.7) gives the

lower bound by the native index. The rational D-twist subgroup has index at most two in the native genus lattice by the calculation preceding Proposition 7.15. Absorb this and the fixed initial convention in C_{det}. The first inequality below holds for a known simple-zero D; the visibility implication holds for every admissible D:

$$\begin{aligned} v_2 a_M(D) &\geq \min\{M, j_{\mathrm{gen}}(D) - C_{\mathrm{det}}\}, \\ v_2 a_M(D) < M \implies \mathrm{ord}_{s=1} L(E^D, s) = 1, &\quad j_{\mathrm{gen}}(D) \leq v_2 a_M(D) + C_{\mathrm{det}}. \end{aligned} \tag{O.8}$$

If an odd-sign D is not a simple zero, the height formula gives a torsion genus point, killed by L, and its coefficient is zero modulo 2^M. Combined with the first inequality on simple-zero parameters, this gives the all-D truncated bound used below, without assigning a rank-one lattice index to a torsion point. □

A6.5 Integer divisions and actual trace operators

Proposition A6.5 (Arithmetic realization of the coefficient rules)**.** *Use the rational full-two-torsion graph conventions of Definitions 7.4 and 7.5. For an even source of Proposition A5.4, use its fixed minimum depth. For an odd source, first fix a finite operator budget $|U| \leq W$, take $R \geq 2$, and choose an actual stage satisfying the depth inequalities of Proposition A7.3. Then the integer divisions (O.9)–(O.11) are valid at every Fourier index, with the stated truncation for the odd source. The fresh coefficient-one reading is $\overline{(t(\mathrm{Fr}_q)\theta H_U)[1]} = \mathcal{A}(Uq)$. On all tested nonempty configurations, these actual symbols depend only on their typed graph and satisfy the rules of Lemma 7.6: a pair difference contracts the endpoints and adds their types and every external incidence; whole-prime omission uses the same source, phase, depth and stage; and an isolated primary vertex of neutral type and neutral external incidences makes the symbol zero. Every contraction has nonempty output. Stationary character tests are carried along in the same product rule. This statement asserts neither values at untested odd-stage weights nor an identity at an empty contraction output.*

Proof. All new primes below avoid the full coefficient support S, including every late sieve and the current ℓ_0. Here S is the actual coefficient support. The coefficients of g vanish at all indices not prime to S. For the odd source this is forced at $2, N, h_*, \ell_0$ by the local Schwartz support and the extraction at $|h_*|$. At every additional coefficient-sieve prime it is imposed by an exact zero-or-retain Fourier sieve, which commutes with the good recurrence outside the enlarged support. This is the vanishing used in the condition $(m, S) = 1$ in the unary series below. Later stationary prime-selection tests may enrich the common label alphabet without resieving the intrinsic odd source; such tests do not enlarge this coefficient-support condition. Use the minimum-depth inequalities proved in Appendix A7. The general bookkeeping assigns weight one to a split prime of $\mathbb{Q}(E[2])$ and one-half to a prime with transposition Frobenius, called simple. In our actual full-two-torsion family every prime is split, $w(U) = |U|$, and the depth δ is an integer. Thus only matching parity, meaning $e_U = 0$, is needed here; the displayed general formula also records where the unused simple-prime term would enter. For a finite set U, put

$$n_U = \lceil \delta + w(U) \rceil, \quad e_U = n_U - \delta - w(U), \qquad B_U = 2^{-n_U} \prod_{p \in U} (U_p - \eta(p) V_p) g,$$

where $U_p(\sum a(n)q^n) = \sum a(pn)q^n$ and $V_p g(\tau) = g(p\tau)$. These are operations on the actual form g. We verify the level at a new odd prime using the classical theta multiplier. Write the old level as $4A$, with $p \nmid 4A$, and put $\beta = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$. For $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(4Ap)$ one has

$$\beta\gamma\beta^{-1} = \begin{pmatrix} a & pb \\ c/p & d \end{pmatrix} \in \Gamma_0(4A).$$

The square-root factor in the theta multiplier is unchanged on evaluating the latter matrix at $p\tau$. The quotient of its arithmetic factor by that of γ is $(c/p\,/d)/(c/d) = (p/d)$. Cubing the multiplier leaves this quadratic factor unchanged. Hence $V_p g$ has level $4Ap$ and character $\psi(p/\cdot)$.

At this level the normalized sum

$$U_{p^2} h(\tau) = \frac{1}{p^2} \sum_{r \bmod p^2} h((\tau + r)/p^2)$$

preserves the same weight and character. Indeed, put $\alpha_r = \left(\begin{smallmatrix} 1 & r \\ 0 & p^2 \end{smallmatrix}\right)$. For $\gamma \in \Gamma_0(4Ap)$ choose $r' \equiv (b + rd)/(a + rc) \pmod{p^2}$. Then

$$\alpha_r \gamma = \gamma'_r \alpha_{r'}, \qquad \gamma'_r = \begin{pmatrix} a + rc & (b + rd - (a + rc)r')/p^2 \\ p^2 c & d - cr' \end{pmatrix} \in \Gamma_0(4Ap).$$

The map $r \mapsto r'$ is a permutation, since $a + rc$ is a unit modulo p and the fractional linear map has determinant one. Writing $d' = d - cr'$, its nebentypus factor is unchanged since $d' \equiv d \pmod{4Ap}$. Its theta arithmetic factor is unchanged as well: $d' \equiv d \pmod{c}$, $d' \equiv d \pmod 4$, and $(p^2 c/d') = (c/d') = (c/d)$. These are the usual Kronecker-symbol identities for the theta multiplier on $\Gamma_0(4)$. Its square-root factor, evaluated at $(\tau + r')/p^2$, is $(c\tau + d)^{1/2}$. Thus the sum transforms with the same theta multiplier cubed and the same character. The formulas also hold for the triangular matrices, and holomorphy at the cusps is preserved by the finite rational translates. On Fourier expansions $U_{p^2} V_p = U_p$. Consequently both terms of $U_p - \eta(p) V_p$ have level exponent at most one at p and character $\psi(p/\cdot)$. Iteration introduces no higher exponent at previously added primes.

The same computation applies to the unary corrections. Start with the odd-character unary theta series at the old support, including its fixed coprimality sieve. Its J-series is its transform by V_J, for squarefree $J \subset U$. A prime $p \mid J$ therefore has exponent at most one and the extra character $(p/\cdot)$; a prime $p \nmid J$ is absent from its original level. Multiplication by the ordinary theta series introduces no odd level prime. This proves the precise level and oldform alternatives used in (O.12).

Write an index as Jrm^2 with $J \subset U$ and r squarefree, prime to U. Products of prime sets are understood here. The originating squarefree parameter in the coefficient of g is $D = (U \setminus J)r$, not the squarefree part Jr of the tested index. Remove the phase $\eta(p)^s$ in (O.6) and put

$$b_{-1} = 0, \quad b_0 = 1, \quad b_1 = a_p - \kappa, \quad b_{s+1} = a_p b_s - p b_{s-1}, \quad \kappa = (\psi/\eta)(p)(-D/p).$$

For $p \in U \setminus J$ the factor $b_s - b_{s-1}$ is odd; for $p \in J$ the factor $b_{s+1} - b_s$ is even. Consequently, for the even source,

$$v_2 B_U[Jrm^2] \geq w(r) + \tfrac{1}{2}\#\{p \in J : p \text{ is simple}\} - e_U. \tag{O.9}$$

The right side is an integer on the allowed filter and is nonnegative. In our full-two-torsion application all primes are split and $w = \omega$. For the odd source, fix an integer $R \geq 1$ and a weight bound W first. At one sufficiently advanced actual stage, simultaneously for all U with $w(U) \leq W$ and every Fourier index, (O.9) holds with its right side replaced by its minimum with R. Thus the untruncated bound on a prescribed bounded range of squarefree weights follows by choosing R and then the stage. For the unary-removal and two-trace tests below, choose $R \geq 2$ after fixing the operator budget. For a split prime $a_p \equiv p + 1 \pmod 4$. The recurrence gives the same residue on each divided shell,

$$(b_{s+1} - b_s)/2 \equiv c_p(D) := \{a_p - (\psi/\eta)(p)(-D/p) - 1\}/2 \pmod 2. \tag{O.10}$$

The two integral shell calculations. Here every prime is split in the rational two-torsion field, so $a_p \equiv p+1 \pmod 4$. For $p \mid D$ one has $\kappa = 0$. Reducing the recurrence modulo two gives $b_{s+1} = b_{s-1}$, with $b_0 = 1$ and $b_1 = 0$. Thus $b_s - b_{s-1}$ is odd for all $s \geq 0$, including $s = 0$ with $b_{-1} = 0$. For $p \nmid D$, one has $\kappa \in \{1, -1\}$, so every b_s is odd. Write $d_s = b_{s+1} - b_s$. The recurrence modulo four gives $d_s \equiv p d_{s-1} \pmod 4$ for $s \geq 1$. Each d_s is even, and multiplication of an even residue modulo four by an odd number fixes that residue. Consequently

$$d_s \equiv d_0 = a_p - \kappa - 1 \pmod 4 \qquad (s \geq 0).$$

This proves (O.10) without dividing a residue-field identity by two. For an index Jrm^2, each $p \in J$ therefore supplies an actual factor two, whereas the factors for $p \in U \setminus J$ are integral units modulo two. The squarefree input has $\omega(D) = |U| - |J| + \omega(r)$. Its depth bound plus these $|J|$ factors gives $\delta + \omega(D) + |J| = n_U + \omega(r)$ before division by 2^{n_U}. All removed character factors are units. For the odd source the same argument is a congruence modulo $2^M \mathcal{O}_{\mathrm{coef}}$ before that division. It therefore retains precision $M - n_U$ afterward; the subsequent division in (O.11) retains precision $M - n_U - 1$. Choose $i \in \{0, 1\}$ with $\eta\chi_{-4}^i$ odd and let

$$\Theta_J = \sum_{\substack{m>0 \\ (m,S)=1}} \eta(m)\chi_{-4}(m)^i m q^{Jm^2}.$$

The Poisson calculation in Section A13 gives this genuine unary weight-3/2 form. Its scalar character is $\eta\chi_{-4}^i(-J/\cdot)$ and every $p \mid J$ enters the level at exponent one. In matching parity the shell recurrence gives $B_U[Jm^2] - B_U[J]\eta(m) \in 2\mathcal{O}_{\mathrm{coef}}$. Thus

$$H_U = \frac{B_U - \sum_J B_U[J]\Theta_J}{2} \tag{O.11}$$

is integral; in opposite parity take $H_U = B_U$. This is division by the integer two, not reduction modulo the maximal ideal of a ramified coefficient ring. For odd-stage forms, the coarse bound and bounded-weight stabilization proved below pay every fixed denominator before choosing the stage.

Write $\mathfrak{m}_{\mathrm{coef}}$ for the maximal ideal of the chosen coefficient valuation ring $\mathcal{O}_{\mathrm{coef}}$. In the following trace argument a reduced expansion means coefficientwise reduction modulo $\mathfrak{m}_{\mathrm{coef}}$. The preceding divisions by integer powers of two have already been justified in $2^a\mathcal{O}_{\mathrm{coef}}$; reduction modulo the maximal ideal alone would not justify them. No saturation of the integral Fourier module is assumed, and no identification of its abstract quotient with all reduced expansions is needed. Multiply (O.11) and the undivided corrections by $\theta = 1 + 2\sum_{m>0} q^{m^2}$. They are actual weight-two forms. For a fixed stage and U, choose a finite complete coefficient-field extension containing all constituents and retain their old multiplicities and diamond translates in the finite-dimensional characteristic-zero space. Use modular Galois representations on cuspidal constituents and the two character summands on Eisenstein constituents. Holomorphy alone does not exclude the latter. Diamond translates remain integral: their correction coefficients have the form

$$\tfrac{1}{2}B_U[J]\{\psi_U(u) - \eta(u)\chi_{-4}(u)^i(-J/u)\},$$

whose character ratio is quadratic. Good Hecke operations preserve integrality by $(T_q f)[n] = f[qn] + q(\langle q\rangle f)[n/q]$. Finitely many Fourier coordinates inject this space into a finite coordinate space. The module generated by these translates is therefore finite over the coefficient DVR, and closed. Chebotarev and continuity make scalar trace and determinant operators limits of good Hecke and diamond operations, so they preserve this module. No integral eigenspace projector is used.

Reading an added vertex. For the actual full-two-torsion family, all primes are split, $\delta \in \mathbb{Z}$, and $n_U = \delta + |U|$. Put $F_U = \theta H_U$ and let $q \notin S \cup U$ be a fresh prime, where S includes every late sieve and the current moving theta prime. Write $u = \prod_{p \in U} p$ and let $\mathcal{A}(D)$ denote the normalized valuation symbol on the tested range: it is one exactly when $v_2 a(D) = \delta + \omega(D)$ on the filter. For an odd source, that range has a fixed weight bound chosen before the valid actual stage, as specified before (F.7). No finite-stage value at larger weights is asserted; the compatible limit tables used later are a separate object. Expanding the operators, every term containing a V_p has zero coefficient at q, and no unary index Jm^2 with $J \subset U$ equals q. Therefore

$$B_U[q] = 2^{-n_U} a(uq), \qquad H_U[q] = 2^{-n_U - 1} a(uq).$$

At coefficient one the diamond term in the good Hecke formula vanishes. Since $F_U \equiv H_U \pmod{2\mathcal{O}_{\mathrm{coef}}}$,

$$\overline{(t(\mathrm{Fr}_q) F_U)[1]} = \overline{H_U[q]} = \overline{2^{-\delta - |U| - 1} a(uq)} = \mathcal{A}(uq) \in \mathbb{F}_2.$$

Here residues lie in the coefficient residue field, with $\mathbb{F}_2$ its prime subfield. Every permitted phase is a two-power root of unity and reduces to one. For an odd source, fix the finite operator budget $|U| \le W$ and $R \ge 2$ first; the stage choice in (O.9) may be taken with $M_\theta \ge \max(0, \delta + W) + R$. Thus $M_\theta - n_U - 1 \ge 1$, so the normalized finite-stage error is still divisible by the integer two. The same reading holds with value zero off the fixed or moving coefficient support.

The diamond correction above is a sum of integral multiples of $\theta \Theta_J$; its fresh coefficient q is zero after reduction. It introduces neither an extra phase nor a denominator in this reading. All readings and subsequent omissions use this same g, η, ψ, δ and actual stage; only the set U changes.

For any fixed Fourier index n, Chebotarev approximation lets one test a finite product of trace operators using distinct fresh primes $q_1, \dots, q_a$ avoiding n and the level. The good Hecke formula then reads the coefficient at $nq_1 \cdots q_a$ of the original vector. Different indices may use different fresh primes; the conclusion is an identity of coefficientwise reduced expansions.

The determinant identity uses the actual diamond formula, not merely the reduction of its eigenvalues. For a good prime r, $d(\mathrm{Fr}_r) = r\langle r \rangle$. The formula above shows that $(d(\mathrm{Fr}_r) - 1) F_U$ reduces to a sum of integral multiples of the undivided corrections $\theta B_U[J] \Theta_J$: the scalar $r\psi_U(r) - 1$ lies in $\mathfrak{m}_{\mathrm{coef}}$. A fresh trace kills every such correction. Indeed, to test at n choose $q \nmid nJ$; then nq cannot have squarefree part J, and $\theta \equiv 1 \pmod{\mathfrak{m}_{\mathrm{coef}}}$. This remains true after further fresh traces. Continuity and the closed integral module therefore give, on the reduced trace orbit of F_U,

$$d(u)t(v) = t(v), \qquad t(u)t(v) = t(uv) + t(uv^{-1}), \qquad t(1) = 0.$$

The middle equality follows from the characteristic-zero matrix identity $\mathrm{tr}(A)\,\mathrm{tr}(B) = \mathrm{tr}(AB) + \det(B)\,\mathrm{tr}(AB^{-1})$, using the determinant identity just proved; the last follows from $\mathrm{tr}(1) = 2$.

Two fresh primes outside $S \cup U$ contribute two to the squarefree weight in (O.9). Thus $B_U[nq_1q_2]$ is divisible by four, no unary correction occurs there, and $H_U[nq_1q_2]$ remains divisible by two. Here $R \ge 2$ and the actual-stage budget were fixed before the test. It follows that $t(g)t(h)F_U = 0$ after reduction. Applying the middle identity to $t(gh)t(h)$ gives $t(gh^2)F_U = t(g)F_U$. Invariance under all squares, and continuity, give factorization through the elementary quadratic quotient unramified outside $S \cup U$. Over $\mathbb{Q}$ this quotient is finite: its squareclasses are generated by -1, 2, and the primes in that support. This does not assert abelianity of the characteristic-zero Galois representations.

At $p \in U$, first keep the summands of F_U separately. The block θB_U and each unary block with $p \in J$ have level exponent at most one and primitive quadratic character at p. The unary blocks with $p \notin J$ are degeneracies of forms at a level omitting p. This distinction is preserved

when retaining all old multiplicities. In particular, the latter blocks cannot acquire a Steinberg constituent merely by being viewed at a larger level.

Enlarge the characteristic-zero coefficient field to realize all the cuspidal newform constituents and the Eisenstein character pairs of these blocks. On a constituent and all its degeneracies, define $t(g)$ and $d(g)$ to be the scalar trace and determinant of its two-dimensional Galois representation, acting as the identity on its multiplicity space. Distinct characteristic-zero constituents with the same good eigenvalues have the same trace and determinant by Chebotarev, so these definitions agree on coinciding summands. In particular they give well-defined operators on their sum. No decomposition modulo two, and no residual irreducibility assertion, enters this construction.

For a cuspidal eigenform f, let Π_f denote its unitary adelic normalization: at a good prime q its Satake parameters α_q, β_q satisfy $a_q = \sqrt{q}(\alpha_q + \beta_q)$ and $\alpha_q\beta_q = \chi(q)$. Apply [82, §§0.3–0.7, Theorem A, pp. 409–411] to $\Pi_f^\vee$, and set $\rho_f = \sigma(\Pi_f^\vee)^\vee$. Here $F = \mathbb{Q}$, the real weight is 2, and the coefficient place lies above 2. The degree of F is odd, so the additional finite discrete-series-place hypothesis for even degree does not apply. The local prime p is odd. Carayol's reciprocity sends geometric Frobenius to a uniformizer, and his ω_p is $|\cdot|_p$. His normalized-induction formula is

$$\sigma(\mathrm{Ind}(\xi_1, \xi_2)) = \xi_1^{-1}\omega_p^{-1/2} \oplus \xi_2^{-1}\omega_p^{-1/2}.$$

Thus if $\Pi_{f,p} = \mathrm{Ind}(\mu_1, \mu_2)$, ρ_f on arithmetic Frobenius has eigenvalues $\sqrt{p}\,\mu_1(p), \sqrt{p}\,\mu_2(p)$ whenever these characters are unramified. In particular its good polynomial is $X^2 - a_qX + q\chi(q)$. Both the automorphic contragredient and the dual on the Galois side are part of this convention.

For a block with ramified character, its local conductor is exactly one: it is at most the level exponent and at least the conductor of its determinant. Casselman's newvector theorem and conductor calculation [81, Theorem 1, pp. 302–306] show that a special representation has conductor one only for an unramified twist; a ramified twist has conductor at least two. A supercuspidal representation has conductor at least two. Consequently $\Pi_{f,p} = \mathrm{Ind}(\mu_1, \mu_2)$ with μ_1 unramified and μ_2 of conductor one, after interchanging the characters if necessary. Its newvector space is a line. On inertia the ramified character is the prescribed quadratic character.

We check the Fourier normalization of the operator on this line. Use normalized induction, so that

$$v\left(\begin{pmatrix} a & x \\ 0 & d \end{pmatrix} g\right) = \mu_1(a)\mu_2(d)|a/d|_p^{1/2} v(g).$$

Put $K_0(p) = \{g \in \mathrm{GL}_2(\mathbb{Z}_p) : g_{21} \in p\mathbb{Z}_p\}$. Use its character $k \mapsto \mu_2(k_{22})$. This is the lower-right-character convention. Casselman's upper-left-character convention is carried to it by $h_p = \left(\begin{smallmatrix} 0 & 1 \\ -p & 0 \end{smallmatrix}\right)$, which normalizes $K_0(p)$ and interchanges its diagonal entries under conjugation. We label the inducing characters so that μ_1 is unramified, using the interchange isomorphism for irreducible principal series. Normalize Haar measure by $\mathrm{vol}(K_0(p)) = 1$. The character-weighted double-coset operator for $K_0(p)\,\mathrm{diag}(p, 1)K_0(p)$, with value one at $\mathrm{diag}(p, 1)$, is

$$H_p = \sum_{b \in \mathbb{Z}/p\mathbb{Z}} \Pi_{f,p}\left(\begin{pmatrix} p & b \\ 0 & 1 \end{pmatrix}\right).$$

Take the newvector supported on $BK_0(p)$ with $v(1) = 1$. Evaluating at 1 gives

$$(H_p v)(1) = p\mu_1(p)p^{-1/2} = \sqrt{p}\,\mu_1(p).$$

This is the Fourier operator $U_p^{(2)}$. Indeed, in the Whittaker model use an additive character of conductor $\mathbb{Z}_p$. Writing $a(n) = \mathrm{diag}(n, 1)$ and $n(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$,

$$a(n)\begin{pmatrix} p & b \\ 0 & 1 \end{pmatrix} = n(nb)a(np) \quad (n \in \mathbb{Z}_p), \qquad (H_pW)(a(n)) = pW(a(np)).$$

Normalize the archimedean Whittaker function by $W_\infty(a(y)) = y\exp(-2\pi y)$ for $y > 0$. Comparison of its positive-n term with $yf(iy)$ gives the finite Whittaker coefficient a_n/n. The displayed equality therefore sends a_n to a_{pn}. Hence $U_p^{(2)} = \sqrt{p}\,\mu_1(p)$ on the new line, exactly the arithmetic-Frobenius eigenvalue on the unramified Galois line. This scalar is algebraic; the square roots only express the unitary normalization. Old multiplicities at primes other than p carry the same scalar. Casselman's oldspace dimension formula [81, Corollary, p. 306, and Theorem 4, p. 313] does not introduce an additional multiplicity at this conductor-one prime. For Eisenstein constituents, the two Galois characters and the divisor-sum Fourier coefficients give the same equality directly; Carayol's cuspidal theorem is not applied to them.

Choose an inertia element τ on which the quadratic character is -1, and a Frobenius lift ϕ. On a ramified block, a basis of the two character lines gives

$$\rho(\tau) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \rho(\phi) = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}, \qquad U_p^{(2)} = \lambda, \quad d(\phi) = \lambda\mu.$$

For $\rho(g) = \left(\begin{smallmatrix} a & b \\ c & e \end{smallmatrix}\right)$, the two inertia differences are

$$t(g\tau) - t(g) = -2e, \qquad t(g\tau\phi) - t(g\phi) = -2e\mu.$$

It follows in characteristic zero that

$$U_p^{(2)}\{t(g\tau\phi) - t(g\phi)\} = d(\phi)\{t(g\tau) - t(g)\}. \tag{O.12}$$

On a block coming from level prime to p, the representation is unramified and both differences vanish. This proves the identity on its entire old multiplicity space even if $U_p^{(2)}$ is not scalar or semisimple on that space. It therefore proves (O.12) on the actual sum containing F_U.

These are identities on characteristic-zero forms before any reduction. To relate them to the integral trace module above, choose finitely many Fourier coordinates that inject its ambient space. Every generator of that module has integral coordinates; hence the module is a submodule of a finite free module over the complete coefficient DVR. It is finite and closed. Simultaneous Chebotarev approximation in the finitely many Galois representations makes $t(g)$ and $d(g)$ limits of good Hecke and diamond operators on this space. They preserve this integral module, so the identity can be evaluated on the integral F_U and then reduced. This does not require integral projectors onto constituents. In particular no division by the two in the displayed inertia differences is used.

Apply (O.12) at coefficient one. Put $V = U \setminus \{p\}$, $v = \prod_{r\in V} r$, so $n_U = n_V + 1$. For a fresh prime r set

$$C_p(vr) = \frac{a_p - (\psi/\eta)(p)(-vr/p) - 1}{2} \in \mathbb{Z}, \qquad c_p(vr) = C_p(vr) \bmod 2.$$

The two surviving operator terms at pr and the good recurrence give

$$H_U[pr] \equiv \eta(p)C_p(vr)\,2^{-n_V-1}a(vr) \pmod{2\mathcal{O}_{\mathrm{coef}}}.$$

For the even source this is an equality before reduction. For the odd source the error is in $2^{M_\theta - n_U - 1}\mathcal{O}_{\mathrm{coef}}$, which is contained in $2\mathcal{O}_{\mathrm{coef}}$ at our chosen stage. Choose fresh q, q_τ, r_0, r_1 representing respectively $g, g\tau, g\phi, g\tau\phi$ in one sufficiently precise joint quotient for the U and V forms and all retained characters. The omitted form uses the same g, η, ψ, δ and stage, and its level omits the entire rational prime p. Thus the two V symbols at vr_0, vr_1 agree. Their Legendre signs at

p are opposite, so $c_p(vr_0) + c_p(vr_1) = 1$ in $\mathbb{F}_2$. The coefficient-one reading and the reduced trace identity now give

$$\mathcal{A}_U(Uq_\tau) + \mathcal{A}_U(Uq) = \mathcal{A}_V(Vr_0) \qquad \text{in } \mathbb{F}_2.$$

Here the subscript records the operator set, not a new coefficient source or minimum. The factor $\eta(p)$ reduces to one; no factor two is cancelled in (O.12). For every retained quadratic character χ unramified at p, the representative of the omitted vertex satisfies

$$\chi(g\phi) = \chi(g)\chi(\phi) = \chi(\mathrm{Fr}_q)\chi(\mathrm{Fr}_p).$$

For a fixed-support character this is the product of the two vertex labels. For the character of a third vertex $r \in V$, it adds the two bits (r/q) and (r/p) in $\mathbb{F}_2$. Reciprocity then gives the reverse bits from the same endpoint types. All additional stationary tests were included in the joint quotient, so they obey this formula as well. This proves the full product-label omission rule, including every external incidence.

A fresh vertex with trivial label in the elementary extension for $S \cup U$ has the same reduced reading as $t(1)$, hence gives zero. An isolated primary vertex has precisely this fixed label and these zero incidence bits. A contracted cluster of primary vertices has the same property when its external incidences are neutral. The result is still a nonempty configuration, including the one-vertex case; no contraction deletes the last vertex.

Finally the fixed-U factorization gives dependence on the entire finite typed graph. For two disjoint prime realizations of the same graph, choose a fresh bridge copy with the prescribed types and matching incidences to both realizations. CRT and Dirichlet provide the primes successively, and reciprocity determines their reverse bits. Replace the vertices of the first copy by bridge vertices one at a time, then the bridge vertices by those of the second copy. At each step the other vertices form the same fixed U, and the changed vertex has the same elementary Frobenius data for $S \cup U$; its symbol therefore agrees. If the original realizations overlap, first replace one by a fresh copy avoiding both original prime sets, with incidences matching that first realization. A vertex and its own replacement are never tested together, so their mutual bit can be chosen freely. The resulting two disjoint realizations admit the preceding bridge argument without conflicting prescriptions. Only graphs of the original bounded size are evaluated; extra bridge incidences are prime-selection conditions. Thus these are rules for the actual arithmetic graph symbols, at the already chosen valid stage. The simultaneous finite graph budget and stabilization are treated below. □

A7 Nonvanishing, normalization and the final family

The uses of the coefficient and address propositions have the following order. First apply only their even-source case, whose depth is already fixed by Proposition A5.4. This gives the rough companion and the genus lower bound. These in turn establish the odd minimum and its finite-stage inequalities. Only then apply the odd case of Proposition A6.5 and the two-symbol address construction. In particular the proof of the rough companion does not use the later odd minimum. We give the choice order underlying (3.3)–(3.9). This also separates the two valuation arguments: the even upper transfer uses the common Kato scalar, while the final contradiction uses the common Heegner coordinate of Section 9.

A7.1 An absolute bound of five new primes

The theorem of Hoffstein–Luo [71, p. 436] supplies fundamental discriminants with nonzero central value and at most four prime factors. Its finite splitting set is arbitrary. The bound four is absolute; no uniform size threshold for those discriminants is needed.

Proposition A7.1 (Nonvanishing with five primes and finite local prescriptions)**.** *Let $A/\mathbb{Q}$ be a fixed modular elliptic curve for this application, let S contain $2, 3$ and all its bad primes, and let B be a finite prime exclusion set. In every nonempty signed unit-squareclass filter at S of even root number, there is a signed odd squarefree parameter u, prime to $S \cup B$, with $L(A^u, 1) \neq 0$ and $\omega(u) \leq 5$. Only the intermediate parameter supplied by Hoffstein–Luo is required to be fundamental; the final field discriminant may be $4u$. The number five is independent of the prescribed finite set. No uniform bound on the sizes of the selected primes is asserted.*

Proof. Let $A/\mathbb{Q}$ be any fixed modular elliptic curve for this application, S contain two, three and all its bad primes, and B be any finite exclusion set. Fix a nonempty signed unit filter of even root number. By CRT and Dirichlet choose one signed prime q_0 in its classes, outside $S \cup B$. Apply the theorem to the actual curve A^{q_0}, putting

$$S' = S \cup B \cup \{|q_0|\} \cup \{p : p \mid N_{A^{q_0}}\}$$

in its splitting set. The theorem gives a fundamental d with $\chi_d(p) = 1$ for all $p \in S'$ and at most four prime factors. The functional equation forces $d > 0$, since A^{q_0} has even root number and every conductor prime splits. Splitting at two gives $d \equiv 1 \pmod 8$. Thus $q_0 d$ has all original unit classes, is prime to B, is squarefree, and has at most five prime factors. If $q_0 d \equiv 3$ or $7 \pmod 8$, its field discriminant is $4q_0 d$; only d was required to be fundamental in the theorem. This is compatible with the coefficient construction (E.2). □

Here h is restricted to the prepared positive odd-sign unit filter containing one, as in (3.3), so $h \equiv 1 \pmod 8$. The multiset of dummy labels and its size are fixed before h; its realizing primes are chosen after h and avoid its support.

For the rough companion the curve itself varies: write a for the bounded dummy product and take $A = E^{ha}$. Choose q_0 so $k_0 = aq_0$ is negative, 1 modulo eight and a square at every prime of $2Nh$, with the required exclusions. Then k_0 is odd fundamental, prime to N_{E^h}, and

$$w(E^{hk_0}) = w(E^h)\chi_{k_0}(-N_{E^h}) = +1.$$

This proves the sign directly for the actual application. Apply Hoffstein–Luo to E^{hk_0} with every conductor, old, h, dummy and forbidden prime in its splitting set. The resulting $k' = k_0 d$ adds at most five new primes. The constants and sizes in the analytic theorem may depend on this varying curve and the whole finite set; the bound five does not.

A7.2 The even upper transfer

Proposition A7.2 (A uniformly bounded partitionable companion)**.** *Fix E and the finitely many prepared even sources, local types and unit seeds used for (3.3). There are constants A, C_E, fixed before h and before any additional finite prime exclusion set, with the following property. Let h be positive squarefree in the prepared odd-sign unit filter containing 1, prime to its fixed support; in particular $h \equiv 1 \pmod 8$. There is a negative odd fundamental discriminant k', prime to $2Nh$ and the additional exclusions, square at every place dividing $2Nh$, for which hk' is partitionable in the sense of Definition 7.8, $L_*(hk') \neq 0$, and*

$$\omega(k') \leq A, \qquad v_2 L_*(hk') \leq 2\omega(h) + 2\omega(k') + s(hk') + C_E.$$

Here A is the fixed dummy-label count plus five. Dummy types and their number are fixed first, but their distinct prime realizations are selected after h and the exclusions. The proof uses the one-symbol stationary-address conclusion of Lemma 7.11, whose proof is given in Proposition A7.6.

Proof. For one fixed even source write (E.11) as

$$v_2 L_*(\sigma D) = 2v_2 a(D) + \kappa_e.$$

The lower bound (E.9) and a nonzero reference coefficient imply that the nonempty set of integers $v_2 a(D) - \omega(D)$ has a finite attained minimum δ_e. A unit of the normalized symbol means precisely equality to this minimum. It therefore gives

$$L_*(\sigma D) \neq 0, \qquad v_2 L_*(\sigma D) = 2\omega(D) + C_{e,\mathrm{unit}}, \quad C_{e,\mathrm{unit}} = 2\delta_e + \kappa_e. \tag{F.1}$$

This constant belongs to the fixed source, not to a curve later introduced solely to obtain nonvanishing.

For the negative types used in the rough companion, the unsigned index is 7 modulo eight. Its total $(-1/D)$ label is nonzero, so every unit seed is nonempty. A seed of least positive size has no dependence on internal edge bits: a nonzero edge derivative would be a smaller nonempty unit. Its labels therefore suffice. If a list with the same total label contains the first $r-1$ seed anchors and a nonempty remainder, contract that remainder into the last anchor. Its total label is forced to be the remaining seed label. This proves partitionability regardless of how many old h-primes are in the remainder.

For a partitionable d already known to have Selmer corank zero, the one-symbol address construction keeps every prime of d stationary and gives arbitrarily large cubes d_x, $d_0 = d$, with unit symbol at each $x \neq 0$. The scalar in (E.5) is common to that cube. By (E.8) and (F.1),

$$v_2 u_x^{\mathrm{ev}}(0) \leq B_e := C_{e,\mathrm{unit}} + C_{\mathrm{loc}} \quad (x \neq 0).$$

Choose $M - c_E > B_e$. The constant coefficient of $2^{c_E} u^{\mathrm{ev}} \in \Lambda[G_b]$ has sign-value bits of total Boolean degree at most $2^{M-1} - 1$. For a larger cube dimension, the Boolean comparison in Section 9 gives $x \neq 0$ with $v_2(u_x^{\mathrm{ev}}(0) - u_0^{\mathrm{ev}}(0)) \geq M - c_E$. If the base were zero or had valuation greater than B_e, this would contradict the upper bound at x. Therefore

$$v_2 L_*(d) - 2\omega(d) - s(d) \leq C_{e,\mathrm{unit}} + 2C_{\mathrm{loc}}. \tag{F.2}$$

The required dimension may depend on d; this bound does not. It precedes the odd minimum and uses no Heegner transfer.

Prepare these sources and nonempty seeds for every total type at the fixed support. There are finitely many. A bounded supply of dummy anchors and reserves works for all of them. For each h realize that supply by distinct primes avoiding the entire fixed support and h. Put every dummy and every other already fixed forbidden prime in the exclusion set of the preceding nonvanishing construction. It gives k' with $\omega(k') \leq A$, where A is the fixed dummy count plus five. For the total type that actually occurs, hk' is partitionable. Nonvanishing gives corank zero by the forward theorem, so (F.2) applies. Taking the maximum over the finitely many prepared constants proves

$$v_2 L_*(hk') \leq 2\omega(h) + 2\omega(k') + s(hk') + C_E. \tag{F.3}$$

No new even minimum is chosen after h or its exclusion set. Those excluded primes are not added as relaxed places in a Selmer complex. □

A7.3 The odd minimum is chosen after its lower bound

Proposition A7.3 (The odd minimum and every finite operator budget)**.** *Fix the original odd filter, the detecting sequence of Proposition A6.1, and one nonprincipal ultrafilter on its stages.*

The arithmetic lower bound (3.5), the native comparisons (3.4), and Proposition A7.2 give (F.4)–(F.5) before any odd minimum is chosen. The bounded-weight visible depths therefore have an attained finite integer minimum δ_o. For every fixed weight cutoff, the simultaneous inequality (F.6) holds on a set in this same ultrafilter. For every finite operator budget and $R \geq 2$, one may choose a stage where all the integer divisions and residual errors required in Proposition A6.5 are valid at every Fourier index. On a fixed squarefree-output range the unit symbol defined before (F.7) is exhaustive, and its value 1 *implies (F.7), with the one constant $C_o = \delta_o + C_{\mathrm{det}}$. No minimum is changed on omitting a vertex or adding stationary tests, and no single finite stage is claimed to serve every unbounded weight range.*

Proof. The arithmetic result (3.5) and the two native-lattice height comparisons (3.4) give, using (F.3),

$$2j_{\mathrm{gen}}(h) \geq 2\omega(h) + s(h) - C_{\mathrm{gen}}, \qquad C_{\mathrm{gen}} = 2A + C_R + C_E - v_2 L_*(h_*) + C_H. \tag{F.4}$$

The variable $s(hk')$ cancels. This estimate precedes the theta prime, precision, odd minimum, final field and binary dimension.

Combining (F.4) with (O.8) gives on all admissible D

$$v_2 a_M(D) \geq \min\{M, \omega(D) - C_{\mathrm{coarse}}\}, \qquad C_{\mathrm{coarse}} = C_{\mathrm{gen}}/2 + C_{\mathrm{det}}. \tag{F.5}$$

At visible coefficients (O.8) first gives a simple zero, allowing (F.4); at the other odd-sign parameters the coefficient is zero modulo 2^M. This avoids a circular use of the future minimum.

Fix one nonprincipal ultrafilter on the actual stages with $M = M_\theta \to \infty$. Consider the integers r for which a bounded-weight sequence of visible D_M has $v_2 a_M(D_M) - \omega(D_M) = r$ on a set in the ultrafilter. The fixed witness D_w and (O.2),(O.7) make this set nonempty. Equation (F.5) bounds it below. Let its attained minimum be δ_o. For every fixed W, on a set in the same ultrafilter,

$$v_2 a_M(D) \geq \omega(D) + \delta_o \quad \text{for every visible } D,\ \omega(D) \leq W. \tag{F.6}$$

Otherwise choose a violating D_M at each bad stage. One of the finitely many integers between $-C_{\mathrm{coarse}}$ and $\delta_o - 1$ would occur on an ultrafilter-large set, a contradiction. Invisible coefficients satisfy the required truncated inequality once $M > W + \delta_o$.

For a fixed finite operator budget $|U| \leq W_U$, put $N_0 = \max(0, \delta_o + W_U)$ and fix $R \geq 2$. Choose an integer $T \geq W_U + 1$ with $T - C_{\mathrm{coarse}} \geq N_0 + R$, then choose a stage in the ultrafilter-large set where (F.6) holds through weight T, with $M \geq N_0 + R$ and $M > T + \delta_o$. For the index Jrm^2 put $D = (U \setminus J)r$. The integral shell recurrence has an error in $2^M\mathcal{O}$. If $\omega(D) > T$, (F.5) gives

$$v_2\big(2^{n_U} B_U[Jrm^2]\big) \geq N_0 + R.$$

If $\omega(D) \leq T$, (F.6) for visible coefficients and $v_2 a_M(D) \geq M$ for invisible ones instead give

$$v_2\big(2^{n_U} B_U[Jrm^2]\big) \geq \min\{M, n_U + \omega(r)\}.$$

Consequently $v_2 B_U[Jrm^2] \geq \min\{R, \omega(r)\}$ for every Fourier index and every U in the budget at this one stage. At $r = 1$ the normalized originating coefficient $2^{|J|-n_U} a_M(U \setminus J)$ is integral, so (O.10) proves the unary congruence modulo the integer two. The additional division in (O.11) leaves error valuation at least $R - 1 \geq 1$. In particular $M - n_U - 1 \geq 1$ for every reading and first-shell omission in the budget. It does not select one stage for infinitely many budgets.

More precisely, fix a squarefree-output cutoff L. On a set of stages in the same ultrafilter, (F.6) holds through L and $M_\theta > \delta_o + L$. Define $\mathcal{A}_{M_\theta, L}(D)$ only for positive squarefree D with $\omega(D) \leq L$: it is zero off the full coefficient filter, one when $v_2 a_{M_\theta}(D) = \delta_o + \omega(D)$ on that filter, and zero

otherwise. This is exhaustive on the tested range: visible coefficients satisfy (F.6), while invisible ones have valuation at least $M_\theta > \delta_o + \omega(D)$. In particular a unit in this range is visible and satisfies

$$\operatorname{ord}_{s=1} L(E^D, s) = 1, \qquad j_{\mathrm{gen}}(D) \le \omega(D) + C_o, \quad C_o = \delta_o + C_{\mathrm{det}}. \tag{F.7}$$

For the preceding operator budget take $L \ge W_U + 1$ and include its valid-stage requirements; all fresh-vertex and omission outputs then lie in this range. The constant C_o is independent of L and of the stage. No value is assigned by this definition at unbounded weights of one finite stage. No minimum is changed when a vertex is omitted or a stationary condition is later added. □

A7.4 Nonempty rational seed and a common finite stage

Proposition A7.4 (Nonempty seeds after fixing the complementary twist)**.** *Choose the negative even unit k with the fresh ramification and exclusions specified in Section 7, and fix $K = \mathbb{Q}(\sqrt{k})$ before the address dimension. Then E/K has full two-primary Selmer corank one. Transport the even coefficient source to E^k, normalize at index 1, and use its positive square-unit filter. On that filter its coefficients have $v_2 a_{\mathrm{new}}(D) \ge \omega(D)$, and the source has a nonempty K-split unit seed. If the intrinsic odd limit symbol is a unit at the empty configuration, E already has a simple zero. Otherwise it too has a nonempty unit seed. In this remaining case both seeds can be read through the actual common alphabet, whose projections onto the two intrinsic alphabets are separately surjective. No surjectivity onto their product is required.*

Proof. Choose the final negative even unit k as in Section 7, with fresh ramification outside the old odd label and torsion fields. Fix $K = \mathbb{Q}(\sqrt{k})$ before the binary dimension. The forward theorem gives corank zero for E^k; the sum-and-difference isogeny, on compact Selmer modules tensored with $\mathbb{Q}_2$, gives full corank one for E/K. The intrinsic odd symbol omits the primes of k. Later ℓ_0 avoids them. Fresh ramification implies that every old odd label still lifts to a K-split label. Additional stationary tests can have relations; use their actual common alphabet and its separate surjective projections.

Transport to the fixed complementary twist. Put $d_k = |k|$, so that $k = -d_k < 0$. The old source has sign -1; its Fourier coefficient at the positive index d tests E^{-d}. Choose the new source for E^k with sign $+1$, normalize its coefficient at 1 to be 1, and impose a support containing the old support and all primes dividing k. For the following comparison D is positive and squarefree, is prime to that support, and lies in the square unit classes of 1 required by the new filter. In particular D is a square in every old support completion, so $d_k D$ and d_k lie in the same old filter. Waldspurger comparison on the two sources gives

$$\begin{aligned} v_2 a_{\mathrm{new}}(D) &= v_2 a_{\mathrm{old}}(d_k D) - v_2 a_{\mathrm{old}}(d_k) \\ &\ge \omega(D). \end{aligned}$$

The equality is a valuation equality; the two sources may have different root-of-unity phases. The normalized central-value ratios agree, since both are

$$\sqrt{D}\,\frac{L(E^{kD}, 1)}{L(E^k, 1)}.$$

Fixed period constants cancel in each ratio. The old unit condition at d_k, and $\omega(d_k D) = \omega(d_k) + \omega(D)$, prove the inequality. The signed twist is kD throughout; no Fourier coefficient at a negative index is used. The final family already satisfies these unit classes. In the nonempty-seed argument

the fresh prime p' is chosen to match all fixed labels of p, so pp' satisfies the same filter, including its dyadic condition.

For a unit of this normalized source equality holds, so $v_2 L_*(h_x k) \le 2\omega(h_x) + C_{e,\mathrm{final}}$ with one fixed period-normalization constant. Its K-split seed must be nonempty. Suppose otherwise. The integral coefficient valuations then improve the preceding lower bound to $\omega(D)+1$ on every nonempty product of allowed K-split primes. For a fresh good K-split prime p outside the support, apply the trace identity to $f = \theta H_{\{p\}}$:

$$U_p^2 t(\phi) = U_p^3 + d(\phi)U_p, \qquad (t(\phi)f)[p^2] = f[p^3] + (d(\phi)f)[p].$$

Here and in this calculation all congruences are modulo the ideal $2\mathcal{O}_{\mathrm{coef}}$ before passage to the residue field. Write $a_p = a_p(E^{(k)})$. We check every theta convolution term in the three coefficients. Put

$$B = B_{\{p\}} = \frac{(U_p - \eta(p)V_p)g}{2}, \qquad b_1 = B[1] = \frac{a(p)}{2}, \quad b_p = B[p],$$
$$H = H_{\{p\}} = \frac{B - b_1\Theta_1 - b_p\Theta_{\{p\}}}{2}.$$

The inherited depth is zero, and H is integral by (O.11). Also $H[0] = 0$, since the cuspidal source and the unary series have zero constant term. The ordinary lower bound gives $b_1 \in \mathcal{O}_{\mathrm{coef}}$. Under the assumed absence of a nonempty K-split unit seed one has the stronger assertion

$$b_1 \in 2\mathcal{O}_{\mathrm{coef}}. \tag{G.4a}$$

Indeed, if p lies in the coefficient filter, the improved bound is $v_2 a(p) \ge 2$; otherwise the actual filter gives $a(p) = 0$. No condition that p itself belong to that filter is imposed.

Let $\delta_B, \delta_1, \delta_p$ be the scalars by which $d(\phi)$ acts on the full weight-two blocks $\theta B, \theta\Theta_1, \theta\Theta_{\{p\}}$, respectively. They are integral units. The determinant is the cyclotomic character times the weight-two nebentypus on every cuspidal constituent and on the Eisenstein character pair, independently of old multiplicity. For this positive-twist source $\psi/\eta = \chi_{-1}$ and $i = 0$. The half-integral-weight characters of B and $\Theta_{\{p\}}$ are therefore both

$$\psi(p/\cdot) = \eta(-p/\cdot).$$

Multiplication by the same θ preserves this equality. Consequently $\delta_B = \delta_p$ for the actual Frobenius lift ϕ, including its action on the ramified character. The ratio of the characters of $\theta\Theta_{\{p\}}$ and $\theta\Theta_1$ is quadratic. Their cyclotomic determinant factors are identical, so $\delta_p/\delta_1 \in \{1,-1\}$ and $\delta_p - \delta_1 \in 2\mathcal{O}_{\mathrm{coef}}$. We must not simply equate these two scalars. On the actual characteristic-zero module the exact identity is

$$d(\phi)f = \delta_p f + \frac{(\delta_p - \delta_1)b_1}{2}\,\theta\Theta_1.$$

Now $H[p] = 0$, since $B[p] = b_p$, $\Theta_{\{p\}}[p] = 1$ and $\Theta_1[p] = 0$. Integrality of H implies $f[p] \in 2\mathcal{O}_{\mathrm{coef}}$. Also $\theta\Theta_1[p] = 2\sum_{0<j^2<p}\Theta_1[p-j^2]$ is even. The displayed determinant identity therefore already proves $(d(\phi)f)[p] \in 2\mathcal{O}_{\mathrm{coef}}$ using only the ordinary integrality of b_1. The following direct expansion checks the individual cross terms under the no-seed assumption in force here.

For $n_j = p - j^2 > 0$, one has $p \nmid n_j$ and $\Theta_{\{p\}}[n_j] = 0$. We claim that

$$B[n_j] \in 2\mathcal{O}_{\mathrm{coef}}. \tag{G.4b}$$

If n_j meets the old support, the coprimality sieve makes this coefficient zero. In particular $2 \in S$, so every odd j gives an even n_j and vanishes in this way. Otherwise write $n_j = rm^2$, with r

squarefree. Since $p \nmid n_j$, the V_p term is zero and $B[n_j] = a(prm^2)/2$. The integral good-prime shell recurrences express $a(prm^2)$ as an integral multiple, up to unit character factors, of $a(pr)$. If $r > 1$, the ordinary coefficient lower bound gives $v_2 a(pr) \geq \omega(pr) = 1 + \omega(r) \geq 2$, or the coefficient is zero off the filter. This uses no assertion that the primes dividing r split in K. If $r = 1$, the same shell recurrences express $B[m^2]$ as an integral multiple of b_1, and (G.4a) applies. This proves (G.4b). Also every coefficient of $b_1\Theta_1$ is even by (G.4a).

Now expand $\theta = 1 + 2\sum_{j>0} q^{j^2}$ before discarding any term. Linearity of the determinant operator on the three actual blocks gives the exact expression

$$\begin{aligned}(d(\phi)f)[p] = {} & \frac{\delta_B B[p] - \delta_p b_p}{2} \\ & + \sum_{0<j^2<p} (\delta_B B[p-j^2] - \delta_1 b_1 \Theta_1[p-j^2] - \delta_p b_p \Theta_{\{p\}}[p-j^2]).\end{aligned} \tag{G.4c}$$

There is no constant-theta $\Theta_1[p]$ term since p is not a square, and $\Theta_{\{p\}}[p] = 1$. The first line is zero because $B[p] = b_p$ and $\delta_B = \delta_p$. Every summand in the second line is even by (G.4a)–(G.4b) and the vanishing of its last term. In particular representations $p = j^2 + m^2$, which can occur when $p \equiv 1 \pmod 4$, are covered by the case $r = 1$ and are not omitted. Thus $(d(\phi)f)[p] \in 2\mathcal{O}_{\mathrm{coef}}$.

The other two readings have no determinant operator. Since H is integral, for every positive integer N,

$$f[N] = H[N] + 2\sum_{0<j^2\leq N} H[N-j^2] \equiv H[N] \pmod{2\mathcal{O}_{\mathrm{coef}}}. \tag{G.4d}$$

This includes $N = p^3$ and every term in its theta convolution. Choose a fresh K-split prime p' matching ϕ in a sufficiently precise finite Galois quotient for the retained integral module, and matching all fixed rational characters. The good Hecke formula at $p' \neq p$ identifies $(t(\phi)f)[p^2]$ modulo two with $f[p^2p']$, hence with $H[p^2p']$ by (G.4d). Neither unary series has support at p^2p', and the recurrence at $p \mid pp'$ has $\kappa = 0$. Consequently

$$H[p^2p'] = \frac{a(p^3p') - \eta(p)a(pp')}{4} = \frac{\eta(p)(a_p - 1)a(pp')}{4}.$$

The product pp' lies in the square filter even if p alone does not. Its two primes split in K, so the no-seed assumption gives $v_2 a(pp') \geq 3$. This last coefficient is therefore even. The trace identity and (G.4c)–(G.4d) now imply $H[p^3] \in 2\mathcal{O}_{\mathrm{coef}}$. The following computation uses precisely this unary-subtracted coefficient.

The coefficient behind the mod-eight contradiction. Write $a_p = a_p(E^k)$. The transported source tests positive twists of E^k. Thus $\psi/\eta = \chi_{-1}$ and, at the squarefree index 1, $\kappa = (\psi/\eta)(p)(-1/p) = 1$. Also η is odd, so the unary correction uses $i = 0$. With $a(1) = 1$ the recurrence gives

$$a(p^2) = \eta(p)(a_p - 1), \qquad a(p^4) = \eta(p)^2\{a_p(a_p - 1) - p\}.$$

The inherited depth is zero. Hence

$$B_{\{p\}}[p] = \frac{\eta(p)(a_p - 2)}{2}, \qquad B_{\{p\}}[p^3] = \frac{\eta(p)^2((a_p - 1)^2 - p)}{2}.$$

Only $\Theta_{\{p\}}$ contributes at p^3, with coefficient $\eta(p)p$. Therefore the actual integer division gives

$$H_{\{p\}}[p^3] = \frac{\eta(p)^2(a_p - 1)(a_p - p - 1)}{4}.$$

This expression is integral because full rational two-torsion gives $a_p \equiv p+1 \pmod 4$. The positive-twist character convention is essential in this computation. Under the assumed absence of a nonempty unit seed, the trace identity makes this coefficient zero in the residue field. Since its phase is a unit and $a_p - 1$ is odd, it follows that $a_p \equiv p+1 \pmod 8$. The contradiction is obtained only after this conditional deduction. Choose p also with $(k/p) = 1$, $p \equiv 1 \pmod 4$ and $(\ell/p) = -1$, outside the finite support. These prescriptions are compatible: K has a ramified prime outside the ramification of $\mathbf{Q}(i, \sqrt{\ell})$, so $K \cap \mathbf{Q}(i, \sqrt{\ell}) = \mathbf{Q}$. The corresponding quadratic residue classes are therefore nonempty and Dirichlet's theorem supplies infinitely many such primes. Since k is a square modulo p, the reductions of $E^{(k)}$ and E_ℓ are isomorphic. A half of a nonzero two-torsion point with root e_i on a split cubic requires both differences $e_i - e_j$ to be squares. For the roots $0, \ell, -3\ell$, one of these required differences is, respectively, $-\ell, \ell, -4\ell$. All three are nonsquares under the chosen conditions. Thus none of the three nonzero points of order two has a half over $\mathbf{F}_p$. If the two-primary subgroup had order greater than four, its structure as a finite abelian group with exactly four elements killed by two would give an element of order four, and hence such a half. Its order is consequently exactly four. This gives $v_2(p + 1 - a_p) = 2$, contradicting the congruence modulo eight above. This proves the required nonempty even seed. If the intrinsic odd symbol already has a unit at the empty configuration, its detection formula gives a simple zero for E and the converse is finished. Otherwise take a least nonempty odd unit seed. Thus the remaining two-symbol address argument has two nonempty seeds. □

Proposition A7.5 (Compatible finite graph tables on one ultrafilter)**.** *Use δ_o and the ultrafilter of Proposition A7.3. First fix the abstract type of the bounded elementary label groups and their projection data on a set in this same ultrafilter. For every finite cutoff W, the valid odd stages give finitely many possible symbol tables on the compatible typed graphs with at most $W+1$ vertices. One table occurs on a set in that ultrafilter. These selected tables agree on their common restrictions, and their union defines the intrinsic odd symbol on all compatible finite typed graphs. It retains its bounded-weight witness and the nonempty-output coefficient rules. After fixing k, its stationary tests may be included in a common alphabet with separately surjective projections on the same ultrafilter, without changing δ_o. Every specified finite collection of tests has a common actual stage. No stage common to all cutoffs is selected.*

Proof. Here the finite graph argument has a definite arithmetic order. The moving theta prime contributes only a bounded number of quadratic label coordinates. The root characters at the old level were fixed before M_θ; growing Kummer precision introduces no growing-order character on the variable primes. Thus the elementary label groups, their projections and their finite addition tables have bounded size. Choose their abstract type on the same ultrafilter as δ_o. For each fixed weight cutoff W, restrict first to the ultrafilter-large set of stages where the preceding budget and precision conditions hold for every $|U| \le W$, with $R = 2$. On those stages the coefficient reading, factorization, contraction and isolation rules hold on all configurations of at most $W + 1$ vertices, with the stated nonempty-output restriction. They therefore define actual symbol truth tables on the compatible typed graph data. There are finitely many such tables; select the one occurring on a set in the same ultrafilter. Two selected tables agree on their common restriction by taking the finite intersection of their valid-stage sets. Their union defines the odd limit symbol on compatible finite typed graphs of every size; it is not a total valuation symbol at any one finite stage. Every seed or contraction identity uses only finitely many such graphs and is inherited at a cutoff containing all of them. The bounded-weight witness remains a witness in these tables. No stage common to all cutoffs is selected. After fixing k, include its tests and choose the common projection data on the same filter, without redefining the intrinsic odd minimum. Stabilize the finite alphabet and its two projections in this way. □

Proposition A7.6 (Stationary and simultaneous address constructions)**.** *Use the definitions and realization freedoms in Definitions 7.4, 7.5, 7.10 and Lemma 7.7, and the coefficient rules of Proposition A6.5. Fix* $b \geq 1$*. Either take one symbol on a nonempty set of always active terminals with a nonzero polynomial in their pair bits, or take one or two symbols on terminals with nonzero linear activations, including a terminal of every nonzero activation, such that each symbol separately has a nonzero terminal polynomial at every* $x \neq 0$*. In the latter case the symbols may read different compatible projections of the common alphabet and fixed stationary external data. Require the active data to satisfy each prescribed filter at every nonzero address. Starting from fixed compatible typed terminal and stationary data, one can adjoin finitely many primary auxiliaries with nonzero linear activations and choose only incidences involving them so that all the symbols equal* 1 *at every* $x \neq 0$*. All original data stay fixed, no auxiliary is present at zero, and the required neutral external incidences and splitting conditions are retained. Only nonempty configurations and nonempty contraction outputs are used. A nonzero iterated difference, hence a partitionability certificate, suffices for the required nonzero terminal polynomial. For the two final symbols the zero-total packets below supply these terminal hypotheses using only separate projection surjectivity. For the even stationary application, the original prime representatives may be fixed at the outset. For a stabilized odd symbol, first fix the abstract data and the finite budget, then choose an actual stage carrying them, and only then choose prime representatives. Previously fixed representatives may be retained when compatible with the full type at that stage; arbitrary earlier primes are not asserted to satisfy a later moving condition.*

Proof.

Total labels of the final seeds. Both final symbols have projected total label zero. The intrinsic odd filter contains 1: its fixed unit squareclasses, positive sign, and moving condition $(D/\ell_0) = 1$ are those of 1. The even source for E^k is likewise normalized in the unit filter containing 1. Every nonzero coefficient, and hence every unit seed, lies in its respective filter. Additional stationary tests at primes dividing k refine the common alphabet but are forgotten by the intrinsic odd projection. They do not add conditions to that odd symbol.

Take repeated packets with total zero in the common elementary label group, including sufficiently many copies of every label. For either projection separately, keep the first $r - 1$ anchors of its unit seed and contract the nonempty remainder into the last anchor. Since the total projected label is zero, the remainder has the last seed label. This argument uses surjectivity of each projection separately. It does not require that a fixed old odd seed admit lifts whose total is zero in the common alphabet. The whole packet already has that total, and contractions preserve it. Thus the final twists satisfy the stationary tests as well as both coefficient filters. Concretely, choose a packet with an even number of copies of every common label, and enough copies to provide the anchors of either seed under its respective projection. Its total label is zero. For every nonzero activation λ, take a disjoint copy of this whole packet and give all its vertices activation λ. At every $x \neq 0$ there are exactly 2^{b-1} active packets. The active list is therefore nonempty, has total common label zero, and contains enough anchors for each projected seed separately. Its terminal polynomial for each symbol is nonzero by the contraction certificate, regardless of the initially fixed bits. It also contains a terminal of each required activation. This constructs the terminal input of the address lemma explicitly. No simultaneous lifting of a prescribed pair of seed labels is needed. Every added network vertex is primary, so the total common label remains zero at each address.

Network operators and their degree. We spell out the simultaneous argument, including the product over all addresses. Write ∂_e for the finite difference in a terminal pair bit e. Products

of these operators commute and $\partial_e^2 = 0$. The contraction rule identifies a forest product with its contracted graph. A cycle product is zero: after a spanning path is contracted its last edge is internal and changes neither a type nor any external incidence. No contraction removes the last vertex.

A network of primary auxiliary vertices acts on any of the symbols by an operator in the ∂_e. To see this directly, expand in the bits incident to the auxiliary vertices. Contract the chosen forest edges. A component with no terminal becomes an isolated primary vertex, and its contribution vanishes by primary isolation. A component meeting several terminals gives their forest difference. If Q is the grounded auxiliary Laplacian and c_a is its attachment column at terminal a, the degree-zero and degree-one terms are

$$\det(Q)\,1 + \sum_{a<b} c_a^{\mathsf{t}} \operatorname{adj}(Q) c_b\, \partial_{ab}.$$

All other terms have difference degree at least two. The displayed coefficients are the rooted-forest minors, by Cauchy–Binet on the incidence matrix. They are polynomial identities, so apply to singular Q as well. Distinct networks have no edges between their auxiliary vertices. Their operators multiply: expanding both forests gives their union, and unions containing a terminal cycle give zero by the preceding observation. This also shows that the operator is the same for both projected symbols.

Fix a target $x_* \neq 0$. Choose a basis $\lambda_1, \dots, \lambda_b$ of the dual address space with $\lambda_i(x_*) = 1$ for all i. Such a basis is obtained from one functional taking value one by adding it to a basis of the space of functionals vanishing at x_*. Group i has three primary vertices, each activated by λ_i. In its physical three-dimensional coordinate space let $e_i = (1,1,1)$ and choose u_i, v_i spanning its orthogonal complement. The vector e_i has norm one. Prescribe the symmetric matrix Q by its bilinear matrix I_b on the e_i and

$$H(T,B) = \begin{pmatrix} 0 & T \\ T^{\mathsf{t}} & B \end{pmatrix}$$

on the remaining coordinates. In physical coordinates each $h_i = Qe_i = e_i$ is supported on its own group. Attach this column to a terminal activated by λ_i, which exists by hypothesis. For an active set J the sum of these active attachment columns is $Q_J \mathbf{1}_J$. Thus the prescribed principal matrix Q_J really is the grounded graph Laplacian on that address, including its diagonal entries.

For a selected terminal pair a, b active at x_*, let their activations be α, β. Attach an extra column c, dual to u_1, to both. If $\alpha \neq \beta$, attach the same column to a terminal of activation $\alpha + \beta$; it is absent at x_*. The extra columns sum to zero at every address since $\alpha + \beta + (\alpha + \beta) = 0$. If $\alpha = \beta$, the two columns already cancel. Hence they do not alter Q_J.

For the singular network set $T = I + P$, where P is the cyclic shift, and give B one diagonal entry one. On every proper group subset, the restricted cyclic shift is a union of paths, so T_J is invertible and $\det Q_J = \det(T_J)^2 = 1$. On all groups, the kernels of T and T^{t} are generated by the all-one vector. The one nonzero diagonal entry of B forces the lower component of a vector in $\ker H$ to vanish. Thus $\ker Q$ is precisely the line generated by $r = \sum_i u_i$, and $\operatorname{adj}(Q) = rr^{\mathsf{t}}$. The anchor columns pair to zero with r, while $c^{\mathsf{t}} r = 1$. Only a, b receive active extra columns at x_*. Consequently its network operator $S_{x_*,e}$, for $e = ab$, has the expansions

$$S_{x_*,e}|_{x_*} = \partial_e + \text{terms of degree at least two}, \qquad S_{x_*,e}|_x = 1 + \text{terms of positive degree} \quad (x \neq x_*).$$

For $b = 1$, $T = 0$ and $B = (1)$ give exactly the same calculation.

For a varying network set $T = I + L$, with L the consecutive lower shift, and $B_\varepsilon = \varepsilon E_{bb}$ for $\varepsilon \in \{0, 1\}$. Every T_J is invertible and both operators $V^{\varepsilon}_{x_*,e}|_x$ have constant coefficient one. The

upper-left block of H^{-1} is $T^{-\mathsf{t}}B_\varepsilon T^{-1}$. Its change on the selected columns is $((T_J^{-1})_{b1})^2$, interpreted as zero when an endpoint is absent. This is one exactly when the entire path is present, which means $x = x_*$. Anchor columns have zero change. Therefore

$$(V^0_{x_*,e} + V^1_{x_*,e})|_{x_*} = \partial_e + \text{terms of degree at least two},$$
$$(V^0_{x_*,e} + V^1_{x_*,e})|_{x} = \text{terms of degree at least two} \quad (x \neq x_*).$$

One choice for both symbols at every nonzero address. For each $x \neq 0$ let $P_{1,x}, P_{2,x}$ be the two nonzero terminal polynomials, of degrees $d_{1,x}, d_{2,x}$, and let $H_{s,x}$ be their highest homogeneous parts. Work in the ordinary polynomial ring in the terminal pair variables for this address. Since it is a domain, $H_{1,x}H_{2,x} \neq 0$. Choose a coefficient-one monomial

$$M_x = \prod_e X_e^{n_{x,e}}, \qquad n_{x,e} \in \{0,1,2\}, \qquad \sum_e n_{x,e} = d_{1,x} + d_{2,x}.$$

For $n_{x,e} = 2$ install one singular network $S_{x,e}$; it acts on both symbol factors. For $n_{x,e} = 1$ install one varying network $V_{x,e}^{\varepsilon_{x,e}}$. Use disjoint auxiliary sets for all these networks. Let $\Gamma_y(\varepsilon)$ be the resulting graph at address y, and consider the finite sum in $\mathbf{F}_2$

$$\sum_\varepsilon \prod_{y\neq 0} \prod_{s=1}^{2} A_s(\Gamma_y(\varepsilon)). \tag{G.1a}$$

The total degree available to all terminal polynomials is $D = \sum_{y\neq 0}(d_{1,y} + d_{2,y})$.

Each singular network at x has zero constant at x, so its action costs at least one difference in each of the two factors there, hence at least two in (G.1a). For a varying network, summing over its ε cancels the all-constant term. Its first-degree terms are independent of ε away from its target, and their difference at the target is ∂_e. Thus every surviving term costs at least one difference, and a cost-one term acts at its target on exactly one of the two factors. The sum of all these minimum costs is exactly D. Any higher-degree action, or any additional action at a wrong address, exceeds D and annihilates at least one terminal polynomial. Hence only the minimum-cost terms survive.

At address x, a variable with exponent two is therefore differentiated once in each factor; a variable with exponent one is differentiated in exactly one factor, summed over the two possibilities. The resulting scalar is precisely $[M_x](H_{1,x}H_{2,x}) = 1$. Lower-degree parts cannot contribute. Taking the product over all $x \neq 0$ proves that (G.1a) equals one. At least one choice of the finitely many ε consequently makes every factor in that product one. This is one graph for both symbols at every nonzero address. Boolean reduction of $H_{1,x}H_{2,x}$ before selecting M_x would destroy this exponent-two accounting and is not used. Here is the stationary and one-symbol specialization, so that the application in Proposition A7.2 uses the same proved construction. If all terminals are always active, attach each column $h_i = Qe_i$ to any one fixed terminal. Since h_i is supported on its own group, their restricted sum is $Q_J\mathbf{1}_J$ at every address. The two additional columns c at a selected terminal pair cancel directly; no balancing terminal of activation $\alpha + \beta$ is needed. The same singular program therefore isolates that pair difference at its target address. For one symbol, choose a coefficient-one top monomial of each nonzero P_x and put a separate singular program at x for each variable in that monomial. Its own programs already spend $\deg P_x$ contraction degrees at x; any extra degree kills the term. Thus exactly the selected top coefficient survives, and it is one. A constant nonzero P_x needs no program. This proves the stationary one-symbol case as well as the moving one-symbol case. In all cases the primary realization of Lemma 7.7 changes only incidences with the newly adjoined vertices. These are finite constructions for each b, with no assertion of one infinite prime array.

□

Proposition A7.7 (One actual stage for the final family)**.** *In the nonempty-seed case of Proposition A7.4, fix the final k, K and the stabilized common alphabet. For every sufficiently large b, the address construction is realized at one actual theta stage by a finite set Q of distinct good odd primes split in K, avoiding the fixed support and the specified finite exclusions. With nonzero activations λ_q it gives exactly the family in (3.9),*

$$h_x = \prod_{q \in Q} (q^*)^{\lambda_q(x)}, \qquad h_0 = 1, \qquad q^* = (-1)^{(q-1)/2} q.$$

All fixed local types are retained, and for every $x \neq 0$,

$$\operatorname{ord}_{s=1} L(E^{h_x}, s) = 1, \quad L_*(h_x k) \neq 0, \qquad j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + C_o, \quad v_2 L_*(h_x k) \leq 2\omega(h_x) + C_e.$$

The constants precede b. The finite graph budget and then the actual theta precision are chosen after b. The theta prime remains a coefficient prime and exclusion, not a relaxed Selmer prime.

Proof. For each chosen b, this construction has a finite budget W_b. Enlarge it to bound every graph product and squarefree output in its finite auxiliary tests, as well as every operator set. Contractions only decrease the number of vertices. Thus every final h_x lies in the tested range of (F.7); this enlargement changes no fixed constant. Only now take one actual theta stage in the finite intersection of the selected table sets through $W_b + 1$, the valid-stage set for the full budget $|U| \leq W_b$ with $R = 2$, and the required precision and finite exclusion sets. In particular (F.6) holds through the threshold T used for that budget. Realize its prescribed forward residue bits by distinct primes using CRT and Dirichlet; quadratic reciprocity supplies the reverse bits. There is no uniform bound on W_b, no intersection over all budgets, and no product-surjectivity assumption on the alphabets. The resulting one family satisfies (3.9), with constants fixed before b. The theta prime ℓ_0 is a coefficient prime and an exclusion, not an active relaxed Selmer prime or a moving arithmetic prime r_i.

Finite intersection and the upper bound. The required stage conditions form a finite list once the address program is fixed. Let $\mathcal{U}$ be the chosen ultrafilter and let $A_1, \ldots, A_t \in \mathcal{U}$ be the selected truth-table sets used by that program, including all intermediate graphs. Let $B \in \mathcal{U}$ impose (F.6) through the threshold T for its operator budget and impose $M \geq \max(0, \delta_o + W_b) + 2$ and $M > T + \delta_o$. Finite prime exclusions are included in B. Then $B \cap \bigcap_{i=1}^{t} A_i \in \mathcal{U}$, hence is nonempty. Choose a stage there before realizing the prime labels. The coefficient rules hold for every realization at this stage, not merely for a previously selected finite list of Fourier indices. The address construction makes each nonzero address a unit of both sources. For the odd source this says exactly

$$v_2 a_M(h_x) = \delta_o + \omega(h_x) < M.$$

The strict inequality follows from the tested-range precision condition. Applying (O.8) gives the simple zero and $j_{\mathrm{gen}}(h_x) \leq \omega(h_x) + \delta_o + C_{\mathrm{det}}$. The normalized even source gives $v_2 L_*(h_x k) \leq 2\omega(h_x) + C_{e,\mathrm{final}}$. Thus one may take $C_o = \delta_o + C_{\mathrm{det}}$ and $C_e = C_{e,\mathrm{final}}$, both fixed before b. Only the number of tests, the threshold T, and the chosen stage depend on the address dimension. □

A7.5 One coordinate in the final contradiction

The family in Proposition A7.7 is now used with the same coordinate of Lemma 8.11 and its central calibration in Proposition 8.10. The denominator and coefficient comparisons are those of

Propositions 9.8 and 9.9; the zero-base assertion is Lemma 8.12. For this fixed K, (3.4), (F.7), and the even bound just proved give

$$2j_{\mathrm{rc}}(h_x,k)\le 4\omega(h_x)+C_J,\qquad C_J=2C_o+C_{e,\mathrm{final}}-v_2L_*(h_*)+C_{H,\mathrm{final}}$$

at every $x\neq 0$. The finite support comparison uses the same class, duality map, bases and coefficient stages. Its integral correction

$$\eta=\prod_q(1+v_q(1+g_q)),\qquad l_q/e_q^2=1+2v_q,\qquad \widetilde{u}=\eta u_{\mathrm{rel}}$$

is formed before characters. The actual integral determinant triangle gives $\widetilde{u}_x=\epsilon_x u_{\mathrm{prim},x}$ with $\epsilon_x\in\Lambda^\times$, rather than independently chosen units in $\mathcal{O}^\times$. Therefore the central identity is

$$\begin{aligned}v_2\widetilde{u}_x(0)=2j_{\mathrm{rc}}(h_x,k)-4\omega(h_x)-s(h_x)-s(h_xk)\\ -\sum_{p\mid k}t_p+\epsilon'_x,\qquad 0\le t_p\le 2,\quad |\epsilon'_x|\le C_{\mathrm{cal}}.\end{aligned}\tag{F.8}$$

The nonnegative finite Sha terms give one upper bound $v_2\widetilde{u}_x(0)\le B$ for all $x\neq 0$, independent of b. At the unknown base no finiteness of Sha is assumed. The retained central class and corank-one argument instead give $\widetilde{u}_0(0)=0$ if the base derivative vanishes.

The fixed-field construction of Section 9 has

$$c_{\mathrm{det}}=a+4(h_K+2c_E+2t_K),\quad L_T(M)=16(d_K+5)M,$$
$$f_x\widetilde{u}_x=d_x,\qquad c=v_2f_0(0)\le c_{\mathrm{det}}.$$

Choose $M>\max(c_{\mathrm{det}},B+2c_{\mathrm{det}},0)$, then $b>(L_T(M)+2)(2^M-1)$ large enough for the address construction, then its finite graph budget and actual theta stage M_θ. The residual comparison tests coefficients of f from $-L_T(M)$ through zero and only the constant coefficient of d. It gives $x\neq 0$ with

$$v_2(\widetilde{u}_x(0)-\widetilde{u}_0(0))\ge M-2c>B.$$

The estimate uses both $v_{\mathcal{O}}(f_x)\le c$ and $\widetilde{u}_x\in\Lambda[1/2]$; neither a bound on the whole numerator nor a finite-Artin inverse of t is asserted. This contradicts (F.8) under the supposed base zero. The Rankin factorization and $L(E^k,1)\neq 0$ give the asserted simple zero of E.

This completes the proof of the restricted converse, Theorem 5.3, from the displayed classical inputs and the arithmetic, coefficient and determinant constructions. The conversion from that converse to pole witnesses, the integer model, the natural compiler and finally the ordinary rational value is the separate argument in the main text.

A8 Simultaneous primes with coprime adjacent norms

This appendix gives the prime-pattern argument needed for the construction. Its strategy is due to Sun [10]. We supply the local calculations, the two ranges of the sieve, the normalization of the ray-class densities, and the order in which the parameters are chosen. For the analytic input we prove the special case for fixed ray class fields of $\mathbb{Q}(i)$ directly. The ideal-norm method follows Frei and Tinková [13]; no theorem from that work is assumed. The proof uses the published fixed-field prime estimate of Mitsui, the one-parameter equidistribution and factorization results of Green–Tao, and published ordinary-prime transference results. The field and its ray modulus are fixed

before every asymptotic limit. The exact source versions checked are identified in the bibliography and the accompanying theorem-application inventory.

Write $\mathrm{N}(z) = z\bar{z}$ for $z \in \mathbb{Z}[i]$, and put $d_-(z) = \mathrm{N}(z-1)$ and $d_+(z) = \mathrm{N}(z+1)$. An Artin or Frobenius element in this appendix is arithmetic Frobenius: on residue fields it sends x to $x^{\mathrm{N}\mathfrak{p}}$.

Theorem A8.1 (The simultaneous prime input)**.** *Let $c_1, \dots, c_k$ be distinct integers, where $k \geq 2$. Let $q \geq 1$, let M be an odd squarefree positive integer with $7 \nmid M$, and let $H, Y > 0$. There are positive integers u, w such that*

$$q \mid w, \qquad u/w^2 > H, \qquad p_j = u + c_j w^2 > Y \quad (1 \leq j \leq k),$$

where every p_j is prime. Each p_j has a Gaussian prime generator π_j of norm p_j such that

$$\pi_j \equiv 9 \pmod{16}, \qquad \pi_j \equiv 1 + 3i \pmod{7}, \qquad \pi_j \equiv i \pmod{M}.$$

No odd prime divides both $d_-(\pi_j)$ and $d_+(\pi_j)$. For $j \neq h$, no odd prime divides both $d_\epsilon(\pi_j)$ and $d_\eta(\pi_h)$, for any $\epsilon, \eta \in \{-, +\}$. Moreover,

$$v_2(d_-(\pi_j)) = 6, \quad v_2(d_+(\pi_j)) = 2, \quad \gcd(d_-(\pi_j), d_+(\pi_h)) = 4 \quad \textit{for all } j, h,$$

and all these norms are coprime to $7M$.

The quantifiers concern fixed data. No estimate uniform in q, M, k or the c_j is asserted. We first prove the result for $q = 1$. Replacing the coefficients by $q^2 c_j$ and then replacing the parameter v by $w = qv$ proves the general case, since the ratio before this substitution can be made greater than $q^2 H$.

A8.1 Local counts and the medium primes

Fix a nonzero integer κ and write

$$Q_\kappa = a^2 + b^2 - c^2 - d^2 - \kappa w^2, \quad G_\epsilon = (a+\epsilon)^2 + b^2, \quad H_\eta = (c+\eta)^2 + d^2, \qquad \epsilon, \eta \in \{1, -1\}.$$

We use $\chi(\ell) = (-1/\ell)$ for odd primes ℓ.

Lemma A8.2 (Finite-field counts)**.** *If $\ell \nmid 2\kappa$, then $Q_\kappa = 0$ has $\ell^4 - 1$ nonzero solutions over $\mathbb{F}_\ell$. At most $4\ell^2$ of its solutions satisfy $G_\epsilon = H_\eta = 0$. If $\chi(\ell) = -1$, the latter set has exactly one element.*

Let $\gamma_1, \dots, \gamma_k$ be distinct integers, including 0 and $-\kappa$, and put $L_j = a^2 + b^2 + \gamma_j w^2$. Outside a fixed set of primes containing the divisors of $2\kappa \prod_{j<h}(\gamma_j - \gamma_h)$, the proportion $g(\ell)$ of nonzero zeros of Q_κ on which some L_j vanishes is

$$g(\ell) = \frac{(\ell-1)(\ell-\chi(\ell))[2\nu_0 + (k-2)(\ell-\chi(\ell))] + \nu_0^2 - 1}{\ell^4 - 1}, \qquad \nu_0 = \ell + (\ell-1)\chi(\ell). \tag{60}$$

Consequently $g(\ell) = (k + 2\chi(\ell))/\ell + O_k(\ell^{-2})$.

Proof. The norm $x^2 + y^2$ has ν_0 representations of zero and $\ell - \chi(\ell)$ representations of every nonzero element. This follows either by factoring over $\mathbb{F}_\ell$ when $\chi = 1$, or by the norm map from $\mathbb{F}_{\ell^2}$ when $\chi = -1$. Summing these fibres for Q_κ gives ℓ^4 solutions including the origin. In the split case, each of the equations $G_\epsilon = 0$ and $H_\eta = 0$ is a union of two lines. On each of the four pairs of lines the remaining equation is $-2\epsilon a + 2\eta c - \kappa w^2 = 0$, leaving at most ℓ^2 choices. In the inert case these equations force $(a, b, c, d) = (-\epsilon, 0, -\eta, 0)$, and then $w = 0$.

For $w \neq 0$, the conditions $L_j = 0$ are disjoint. For the two distinguished coefficients one norm is zero and the other is nonzero, giving $2\nu_0(\ell - \chi)$ choices. Each other coefficient gives $(\ell - \chi)^2$ choices. There are $\ell - 1$ possible nonzero w. For $w = 0$, both norms are zero, giving $\nu_0^2 - 1$ nonzero vectors. This proves (60). In particular, when $k = 2$ and $\chi = -1$, its value is $2/(\ell^2 + 1)$, not zero. □

Lemma A8.3 (Primitive congruence count)**.** *Let ψ be a smooth nonnegative compactly supported function away from the origin, and let m be squarefree and prime to 2κ. For each $\ell \mid m$ prescribe an arbitrary subset Ω_ℓ of the nonzero zeros of Q_κ over $\mathbb{F}_\ell$. The weighted number of primitive integral vectors on $Q_\kappa = 0$ with these reductions equals*

$$C_\kappa \sigma_\infty(\psi) B^3 \prod_{\ell \mid m} \frac{|\Omega_\ell|}{\ell^4 - 1} + O_{\kappa,\psi,\varepsilon}(B^{5/2+\varepsilon} m^{5/2+\varepsilon}). \tag{61}$$

The constant C_κ is positive and independent of m and the subsets.

Proof. Apply Browning–Loughran [11], Theorem 4.1, to Q_κ in five variables. It is isotropic, since $(1,0,1,0,0)$ is a zero, and nonsingular, since $\kappa \neq 0$. Its discriminant has no prime divisor outside 2κ. The theorem permits $n = 5$ and arbitrary nonzero residue subsets; its error is uniform in the modulus and in these subsets. Impose primitivity by $\sum_{h \mid (a,b,c,d,w)} \mu(h)$. If $(h, m) > 1$ there are no permitted vectors. Otherwise the subsets become $h^{-1}\Omega_\ell$, of the same cardinalities, and the scale becomes B/h. The sum of the error terms is convergent. Extending the main-term sum to infinity costs $O(B)$. At a good prime the unrestricted local density is

$$\sigma_\ell = \frac{1 - \ell^{-4}}{1 - \ell^{-3}}.$$

Indeed, nonsingular primitive zeros lift with four free coordinates, and the vectors divisible by ℓ contribute the geometric factor ℓ^{-3}. The primitive density is therefore $1 - \ell^{-4}$. This turns the factor $|\Omega_\ell|/\ell^4$ of the unprimitive count into $|\Omega_\ell|/(\ell^4 - 1)$ in (61). The remaining finitely many local factors are positive because the form has a nonsingular rational zero. Empty subsets give the assertion directly. □

In using this count, the two norms lie between fixed positive multiples of B^2, and $|w| \leq \tau B$, where $0 < \tau \leq \tau_0$. A smooth majorant can be chosen with $\sigma_\infty(\psi) \ll \tau$ uniformly in τ: integrate first in the two radial norm variables, on which the derivative of their difference is bounded away from zero. The constants in error terms may depend on derivatives of this majorant, and hence on τ.

Lemma A8.4 (Medium-prime bound)**.** *For fixed coefficients and sufficiently large fixed L, the contribution of prime patterns with a common adjacent-norm divisor $L < \ell \leq B^{1/100}$ is at most*

$$\frac{C\tau B^3}{(\log B)^k L \log L} + O_\tau(B^{27/10}). \tag{62}$$

The count is unweighted by logarithms, and C is independent of L, τ. The bound covers both distinct positions and the two signs at one position.

Proof. For two positions, use Q_κ with their fixed nonzero coefficient difference and impose $G_\epsilon = H_\eta = 0 \pmod{\ell}$. For squarefree d supported on good primes other than ℓ, impose also that some L_j vanishes at every prime dividing d. The preceding lemmas give a sieve sequence

$$A_{\ell,d} = X_\ell g(d) + r_{\ell,d}, \qquad X_\ell \ll \tau B^3/\ell^2, \qquad |r_{\ell,d}| \ll_{\tau,\varepsilon} B^{5/2+\varepsilon}(\ell d)^{5/2+\varepsilon}.$$

Enlarge the fixed exceptional set so that $0 < g(p) < 1$ outside it. Set $h(p) = g(p)/(1 - g(p))$ and extend h multiplicatively to squarefree integers. The elementary Selberg upper sieve has denominator

$$\mathcal{G}_\ell(z) = \sum_{\substack{d < z \text{ squarefree} \\ (d, \ell \prod_{p \in S} p) = 1}} h(d) \gg (\log z)^k.$$

Here is the dimension calculation. Williams [16], Theorem 1 with fixed modulus 4 and residues $1, 3$, gives Mertens' estimates in the two reduced classes. Taking logarithms gives $\sum_{p\le y} g(p) = k\log\log y + O(1)$ and $\sum_{p\le y} g(p)\log p \ll \log y$. For the latter bound only the elementary Chebyshev estimate is needed: $\prod_{n<p\le 2n} p \mid \binom{2n}{n} < 4^n$ gives $\vartheta(x) \ll x$ by dyadic summation, and partial summation gives $\sum_{p\le y}(\log p)/p \ll \log y$. In the Euler product $\prod_{p\le y}(1+h(p))$, normalize the weights $h(d)$ to a probability distribution. Its expected value of $\log d$ is $\sum_{p\le y} g(p)\log p$. Taking y to be a sufficiently small fixed power of z leaves at least half of the mass on $d < z$, by Markov's inequality. Removing one prime ℓ changes the estimate by a bounded factor.

For completeness the sieve weights may be taken as

$$\lambda_d = \frac{\mu(d)}{\mathcal{G}_\ell(z)} \prod_{p\mid d} (1-g(p))^{-1} \sum_{\substack{r<z/d \ (r,d)=1 \ r \text{ squarefree} \\ (r,\ell\prod_{p\in S} p)=1}} h(r).$$

They have $\lambda_1 = 1$, $|\lambda_d| \ll_\varepsilon d^\varepsilon$, and direct expansion gives $\sum_{d,e} \lambda_d\lambda_e g([d,e]) = 1/\mathcal{G}_\ell(z)$. Thus the number with no prime divisor below z in any L_j is at most

$$\frac{C\tau B^3}{\ell^2(\log z)^k} + O_{\tau,\varepsilon}(B^{5/2+\varepsilon}\ell^{5/2+\varepsilon}z^{7+4\varepsilon}).$$

All the prime values under consideration exceed z. Put $z = B^{1/100}$ and sum over $L < \ell \le B^{1/100}$. The main term uses $\sum_{\ell>L}\ell^{-2} \ll 1/(L\log L)$, another consequence of the same Chebyshev bound; for example $\varepsilon = 1/1000$ makes the error smaller than $B^{27/10}$.

At one position, an odd common divisor of $(a-1)^2+b^2$ and $(a+1)^2+b^2$ forces $a = 0$ and $b^2 = -1 \pmod\ell$. There are at most 2ℓ such triples (a, b, w) over $\mathbb{F}_\ell$. For $w \ne 0$ the zero sets of the k prime forms are disjoint, and the ordinary three-dimensional local sieve density is $(k+\chi(\ell))/\ell + O_k(\ell^{-2})$. Counting residue classes in a smooth box gives the corresponding three-dimensional congruence estimate. The same Selberg weights and the same, weaker error in (62) apply. □

A8.2 The large-prime tail

We use Lemmas 2.1 and 4.1 of Browning–Heath-Brown [12]. The adjacent-norm polynomials are inhomogeneous, so we give the fixed-coordinate argument rather than invoke their homogeneous quadric-sieve theorem directly. Their Lemma 2.1 says that coprime polynomials in $\mathbb{Z}[X_1, \dots, X_m]$ of bounded degree and coefficient height at most H have at most

$$O\left(\frac{V\log(VH)}{L\log L} + \frac{V\log(VH)}{B_{\min}}\right) \tag{63}$$

integer points in a box of side parameters $B_i \ge 2$ at which a prime greater than $L \ge 2$ divides every polynomial; $V = \prod B_i$. "Coprime" here excludes a common constant prime factor as well as a common nonconstant factor. Their Lemma 4.1 covers the zeros modulo a squarefree q of a rank-m quadratic form, with $m \ge 3$, by at most $(3m)^{\omega(q)}q^{m-2}$ lattices of determinant q^{m-1}. The number with largest Euclidean successive minimum exceeding T is

$$O_m((3m)^{\omega(q)}q^{2m-3}T^{-m}). \tag{64}$$

Every prime dividing q must be odd and preserve the rank of the form. The lattices in that lemma contain $q\mathbb{Z}^m$; in particular their largest successive minima are at most q.

Lemma A8.5 (Large common divisors on the quadric)**.** *Fix the implied constant in the box of size $O(B)$. For fixed $\kappa \neq 0$ and $\epsilon, \eta \in \{1, -1\}$, uniformly for $B, L \geq 2$, the number of integral vectors in this box satisfying $Q_\kappa = 0$ and $\ell \mid G_\epsilon, H_\eta$ for some prime $\ell > L$ is*

$$O_{\kappa,\alpha}\left(\frac{B^{3+\alpha}}{L \log L} + B^{11/4+\alpha}\right) \qquad (\alpha > 0). \tag{65}$$

Proof. Sign changes of a, c reduce to $\epsilon = \eta = 1$. Put $x_0 = a + c$, $x_1 = a - c$, $x = b - d$, $z = b + d$, and $R = \kappa w^2 - xz$. Then $x_0 x_1 = R$. Parity restrictions can be discarded in an upper bound. For a fixed nonzero value of either x_0 or x_1, the triple (x, z, w) determines the other coordinate uniquely whenever it gives an integral solution. We shall use this injection when discarding divisibility conditions.

First suppose that $\ell \mid x_0, x_1$. Then $\ell^2 \mid R$. For each fixed integer $v = O(B^2)$, the equation $R = v$ has $O_{\kappa,\alpha}(B^{1+\alpha})$ triples: for each w, count the factorizations $xz = \kappa w^2 - v$ by the divisor bound; the zero right-hand side occurs for at most two w and contributes $O(B)$ each. If $R \neq 0$, it has $O_\alpha(B^\alpha)$ factorizations $x_0 x_1 = R$. Since $0 < |h| \ll B^2/\ell^2$ and $\ell \ll B$, summation gives

$$O_{\kappa,\alpha}\left(B^{3+2\alpha} \sum_{\ell > L} \ell^{-2}\right) = O_{\kappa,\alpha}(B^{3+2\alpha}/(L \log L)).$$

The same bound for triples with $R = 0$, followed by the $O(B)$ choices with $x_0 x_1 = 0$, gives $O_{\kappa,\alpha}(B^{2+\alpha})$. We may discard all these $R = 0$ solutions before proceeding.

In the remaining cases choose $t = x_1$ or $t = x_0$ such that $\ell \nmid 2t$. Take $t = x_1$ first. Here $G_1 - H_1 = \kappa w^2 + 2t$ and $4t^2 G_1 = K$. Thus the two norm congruences are equivalent to divisibility of

$$P = \kappa w^2 + 2t, \qquad K = (\kappa w^2 - xz + t^2 + 2t)^2 + t^2(x + z)^2.$$

The identity

$$K - (t^2 + x^2)(t^2 + z^2) = P(P + 2t^2 - 2xz)$$

shows that it suffices to sieve the coprime pair $P, (t^2 + x^2)(t^2 + z^2)$. They have no common non-constant factor over $\mathbb{Q}$, because the first depends nontrivially only on w, whereas the second is independent of w. Their contents have no common prime, since the second has coefficient 1 on $x^2 z^2$. For $t = x_0$ we still have $4t^2 G_1 = K$, whereas $t(G_1 - H_1) = \kappa(t + 2)w^2 - 2xz$. The pair is therefore

$$P' = \kappa(t + 2)w^2 - 2xz, \qquad K.$$

If $t \neq -2$, P' is an irreducible ternary quadratic form of rank three and K has nonzero constant term $t^2(t + 2)^2$; homogeneity rules out divisibility. If $t = -2$, neither x nor z divides K. The coefficient 1 of $x^2 z^2$ in K excludes every common content prime. In both charts the fixed value t is nonzero.

In either chart, $t \mid R$. Put

$$S_R = \{p : p \mid 2\kappa\}, \qquad q = \prod_{p \mid t \ p \notin S_R} p.$$

The matrix of R has determinant $-\kappa/4$, so R has rank three at every prime in this product, and those primes are odd. Also $q \ll B$. For a fixed q, every eligible nonzero $t = O(B)$ is supported on $S_R \cup \{p : p \mid q\}$. For every $\delta > 0$, Rankin's bound gives

$$\#\{t : q(t) = q,\ 0 < |t| \ll B\} \ll B^\delta \prod_{p \in S_R \text{ or } p \mid q} (1 - p^{-\delta})^{-1} \ll_{\kappa,\delta} B^\delta q^\delta \ll_{\kappa,\delta} B^{2\delta}.$$

For the middle inequality, remove the finitely many primes with $p^{-\delta} > 1/2$ and bound the remaining factors by $2^{\omega(q)} \ll_\delta q^\delta$. Renaming the small exponent gives $O_{\kappa,\alpha}(B^\alpha)$ choices of t.

Apply (64) with $m = 3$ to $q \mid R$. If $\lambda_1 \le \lambda_2 \le \lambda_3$ are the successive minima, then $\lambda_3 \le q \ll B$ and $\lambda_1\lambda_2\lambda_3 \asymp q^2$. A basis comparable to these minima therefore gives $O(\prod_i(1+B/\lambda_i)) = O(B^3/q^2)$ points in the box. The lattices with $\lambda_3 > T$ contribute at most

$$B^\alpha \sum_{q \ll B} O\left(9^{\omega(q)} q^3 T^{-3} \frac{B^3}{q^2}\right) \ll_{\kappa,\alpha} B^{5+2\alpha} T^{-3}.$$

For the other lattices choose a basis with lengths comparable to the successive minima. The inverse basis matrix bounds the new coordinates by $O(B/\lambda_i)$, so the resulting rectangular box has volume $O(B^3/q^2)$ and minimum side parameter $\gg B/T$. Enlarge its side parameters to at least 2, as required in (63); since $\lambda_i \ll B$, this changes volume only by a fixed factor. The substitution into the polynomial pair is invertible over $\mathbb{Q}$. For $\ell \nmid 2t$ its determinant, of absolute value q^2, is invertible modulo ℓ too. Thus no new common content prime outside $2t$ appears. Remove common content supported on $2t$, which cannot affect the primes being counted. Coefficient heights are powers of B with fixed exponents. Formula (63) bounds this lattice's contribution by

$$O\left(\frac{B^3 \log B}{q^2 L \log L} + \frac{B^2 T \log B}{q^2}\right).$$

There are at most $9^{\omega(q)} q$ lattices. Since $9^{\omega(q)} \ll_\alpha q^\alpha$, summing over $q \ll B$ and the $O_{\kappa,\alpha}(B^\alpha)$ values of t gives $O_{\kappa,\alpha}(B^{3+3\alpha}/(L \log L) + B^{2+3\alpha} T)$. Choose $T = B^{3/4}$ and rename the small exponent to obtain (65). □

Proposition A8.6 (Weighted obstruction tail)**.** *For fixed distinct c_j there are $L_* \ge 2$ and $\tau_0, C > 0$ such that, for every fixed $0 < \tau \le \tau_0$, uniformly for $L \ge L_*$, the logarithmically weighted number of pairs*

$$N/2 \le u \le N, \qquad 1 \le w \le \tau\sqrt{N}, \qquad p_j = u + c_j w^2 \text{ all prime},$$

having any of the odd common divisors forbidden in Theorem A8.1 greater than L, is at most

$$\frac{C\tau N^{3/2}}{L \log L} + O_\tau(N^{3/2 - 1/800}). \tag{66}$$

The weight is $\prod_j \log p_j$, and the bound allows every Gaussian generator, so it also holds for any prescribed choice of generators.

Proof. Take $B = \sqrt{N}$ and $L_0 = B^{1/100}$, and choose τ_0 so that every prime value in the displayed region is between fixed positive multiples of B^2. Thus $\prod_j \log p_j \ll (\log B)^k$. Choose L_* above the fixed exceptional primes in the medium-prime bound. When $L \le L_0$, that bound gives

$$\frac{C\tau B^3}{L \log L} + O_\tau(B^{27/10} (\log B)^k)$$

for $L < \ell \le L_0$. Its proof is uniform in this interval of L: the remainder sum is bounded by the sum over all $\ell \le L_0$. When $L > L_0$ this part is empty.

For distinct positions, apply Lemma A8.5 at L_0 with $\alpha = 1/1000$. After multiplication by the weight, the bound is

$$O\left(\left(\frac{B^{3+1/1000}}{B^{1/100} \log L_0} + B^{11/4+1/1000}\right)(\log B)^k\right) = O(B^{3-1/200}).$$

This bounds all the remaining primes, also when $L > L_0$. At one position, an odd divisor of both adjacent norms divides a and b^2+1. The coprime pair a, b^2+1 in variables (a,b,w), inserted in (63) at L_0, gives

$$O\left(\left(\frac{B^3\log B}{L_0\log L_0}+B^2\log B\right)(\log B)^k\right)=O(B^{3-1/200}).$$

No factor τ is needed in these power-saving errors.

For a fixed pair of positions, the vector (a,b,c,d,w) determines $u=a^2+b^2-c_jw^2$; at one position (a,b,w) does so. Thus the vector counts bound the pattern counts. There are finitely many pairs of positions and signs. Each split rational prime has eight Gaussian elements of that norm, so counting all generator choices changes the bound by at most a fixed factor. Every vector used in the medium-prime count is primitive, since a common integer divisor greater than one would have its square divide a prime norm. All logarithmic factors in the remainders are absorbed for sufficiently large B, with thresholds depending only on the fixed data and τ. Weakening the error to $B^{3-1/400}=N^{3/2-1/800}$ proves the claim. □

A8.3 Gaussian ray weights and their progression main terms

Let $E=\mathbb{Q}(i)$ and let H be a fixed positive integer divisible by 16. Let K/E be the ray class field for $H\mathbb{Z}[i]$. This field is Galois over $\mathbb{Q}$, since complex conjugation preserves both the modulus and its principal ray subgroup. Put

$$A=\mathrm{Gal}(K/E)\simeq(\mathbb{Z}[i]/H\mathbb{Z}[i])^\times/\{1,-1,i,-i\},\qquad G=\mathrm{Gal}(K/\mathbb{Q}).$$

We use arithmetic Frobenius throughout. Let $C\subseteq A$ be invariant under complex conjugation, so that C is a union of conjugacy classes in G. Define $\Lambda'_C(p)=\log p$ for unramified rational primes whose Frobenius lies in C, and put $\Lambda'_C(n)=0$ otherwise. In particular this function is supported on rational primes which split in E.

For ideals coprime to H let $[\mathfrak{a}]\in A$ denote their ray class, and set

$$T_C(n)=\sum_{\substack{\mathrm{N}\mathfrak{a}=n\\(\mathfrak{a},H)=1}}\Lambda_E(\mathfrak{a})1_C([\mathfrak{a}]).$$

Here $\Lambda_E(\mathfrak{p}^j)=\log\mathrm{N}\mathfrak{p}$ and it is zero on all other ideals. Every unramified split rational prime has two conjugate prime ideals of norm p, so

$$T_C(p)=2\Lambda'_C(p)\qquad(p\nmid 2H).\tag{67}$$

The remaining difference between T_C and $2\Lambda'_C$ is supported on rational prime powers of exponent at least two and the finitely many excluded primes. Its total absolute value up to X is $O_H(\sqrt{X}\log^2(2X))$. This bound follows by summing the at most two prime ideals above each rational prime and their powers; no density theorem is used in this comparison.

For $\chi\in\widehat{A}$ put

$$c_C(\chi)=|A|^{-1}\sum_{a\in C}\overline{\chi(a)},\qquad T_\chi(n)=\sum_{\mathrm{N}\mathfrak{a}=n}\Lambda_E(\mathfrak{a})\chi(\mathfrak{a}),$$

where each character is extended by zero on ideals not coprime to H. Finite Fourier inversion gives the exact identity

$$T_C(n)=\sum_{\chi\in\widehat{A}}c_C(\chi)T_\chi(n).\tag{68}$$

The field and all these characters are fixed before X tends to infinity.

Lemma A8.7 (Fixed-field Hecke Siegel–Walfisz estimate)**.** *Let $E=\mathbb{Q}(i)$, let χ be a fixed finite-order ray character of E with finite modulus $\mathfrak{m}$, and let $B>0$. There is $c=c(\chi,B)>0$ such that, uniformly for $X\ge 3$, $1\le Y\le X$, positive integers $q\le(\log X)^B$, and Dirichlet characters ρ modulo q,*

$$\sum_{\mathrm{N}\mathfrak{a}\le Y}\Lambda_E(\mathfrak{a})\chi(\mathfrak{a})\rho(\mathrm{N}\mathfrak{a})=r_{\chi,\rho}Y+O_{\chi,B}\big(X\exp(-c\sqrt{\log X})\big). \tag{69}$$

Here $\Lambda_E(\mathfrak{p}^j)=\log\mathrm{N}\mathfrak{p}$ for a prime ideal $\mathfrak{p}$ and $j\ge1$, and it is zero otherwise; characters are extended by zero on ideals not coprime to their moduli. The number $r_{\chi,\rho}$ is one if the primitive Hecke character inducing $\chi(\rho\circ\mathrm{N})$ is trivial, and zero otherwise. There is also $c_B>0$ such that, with the same ranges of X,Y,q,ρ,

$$\sum_{n\le Y}\Lambda(n)\rho(n)=1_{\rho\ \text{principal}}Y+O_B\big(X\exp(-c_B\sqrt{\log X})\big). \tag{70}$$

In particular, either error is $O(X(\log X)^{-D})$ for every fixed $D>0$, with the corresponding dependence on D allowed.

Proof. We use the published fixed-field estimate of Mitsui [2, §1, Lemma 15, p. 12]. In his notation, for a fixed number field K, a fixed $A>0$, and any ray character ϑ whose finite modulus has norm at most $(\log x)^A$, it states

$$\sum_{\mathrm{N}\mathfrak{p}\le x}\vartheta(\mathfrak{p})\log\mathrm{N}\mathfrak{p}=E(\vartheta)x+O_{K,A}\big(x\exp(-c_{K,A}\sqrt{\log x})\big), \tag{71}$$

where $E(\vartheta)=1$ for the principal character and is zero otherwise. The sum is over prime ideals of every residue degree. Mitsui defines these characters and $E(\vartheta)$ on p. 2, includes imprimitive characters, and states the uniformity in the character and its modulus immediately before Lemma 14 on p. 11. Infinite primes may occur in the ray modulus; only its finite part enters the norm bound.

For completeness, the elementary passage from Mitsui's proved smoothed estimate (Lemma 14, pp. 11–12) to (71) also applies to complex characters. Put

$$P(x,\vartheta)=\sum_{\mathrm{N}\mathfrak{p}\le x}\vartheta(\mathfrak{p})\log\mathrm{N}\mathfrak{p}\log\frac{x}{\mathrm{N}\mathfrak{p}}.$$

Lemma 14 gives $P(x,\vartheta)=E(\vartheta)x+O_{K,A}(xe^{-c_1\sqrt{\log x}})$. Subtract its values at xe^h and x, where $h=e^{-c_1\sqrt{\log x}/2}$, and divide by h. The resulting left side is the prime sum in (71) plus boundary terms of absolute value at most

$$\sum_{x<\mathrm{N}\mathfrak{p}\le xe^h}\log\mathrm{N}\mathfrak{p}\ll_K(xh+1)\log x.$$

Indeed, for each integer norm there are at most $[K:\mathbb{Q}]$ prime ideals. The main term is $E(\vartheta)x(e^h-1)/h=E(\vartheta)x+O(xh)$, and the error divided by h is $O_{K,A}(xe^{-c_1\sqrt{\log x}/2})$. Reducing the positive exponent constant proves (71).

Prime powers of exponent at least two contribute $O_K(\sqrt{x}\log^2(2x))$: sum over exponents $j\ge2$ and use the elementary bound $O_K(x^{1/j})$ for the number of prime ideals of norm at most $x^{1/j}$. Consequently (71) holds with the prime sum replaced by $\sum_{\mathrm{N}\mathfrak{a}\le x}\Lambda_K(\mathfrak{a})\vartheta(\mathfrak{a})$.

The character $\rho\circ\mathrm{N}$ on E has ray modulus (q): if $\alpha\equiv1\pmod{q\mathbb{Z}[i]}$, then $\mathrm{N}\alpha\equiv1\pmod q$. Thus $\chi(\rho\circ\mathrm{N})$ has a modulus dividing $\mathfrak{m}(q)$, whose norm is $\mathrm{N}\mathfrak{m}\,q^2$. For $\sqrt{X}\le Y\le X$, the bounds $q\le(\log X)^B$ and $\log Y\ge\frac12\log X$ imply

$$\mathrm{N}\mathfrak{m}\,q^2\le(\log Y)^{2B+1}$$

for all sufficiently large X, with the threshold depending only on $\mathfrak{m}$ and B. Apply the preceding estimate with $K = E$, $x = Y$, and $A = 2B+1$. Its error is at most the error in (69), after reducing c. For $1 \le Y < \sqrt{X}$, the elementary bound $\sum_{\mathrm{N}\mathfrak{a}\le Y} \Lambda_E(\mathfrak{a}) \ll_E Y \log^2(2Y)$ suffices, and the remaining bounded values of X are absorbed in the implied constant.

The principal character at the chosen modulus is exactly the character induced by the trivial primitive character, giving the stated $r_{\chi,\rho}$. Equivalently one can work at the primitive conductor and remove the extra Euler factors: the corresponding prime-power towers contribute only $O_\chi(\log(q+2)\log(Y+2))$, which is absorbed in the error.

For (70), take $K = \mathbb{Q}$ and use the ray character attached to ρ, allowing the real infinite place for odd characters. Its finite modulus norm is q. The same proof uses $A = B+1$ for $Y \ge \sqrt{X}$ and the elementary bound for smaller Y. Finally, $e^{-c\sqrt{\log X}} \ll_{c,D} (\log X)^{-D}$ for every fixed $D > 0$. □

The preceding lemma gives the following consequence, including $Y = 0$ by the empty-sum convention. For every $B, D > 0$, every fixed finite-order ray character χ of E, and every Dirichlet character ρ of modulus $q \le (\log X)^B$,

$$\sum_{\mathrm{N}\mathfrak{a}\le Y} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a})\rho(\mathrm{N}\mathfrak{a}) = r_{\chi,\rho}Y + O_{H,B,D}(X(\log X)^{-D}) \quad (0 \le Y \le X), \tag{72}$$

where $r_{\chi,\rho} = 1$ if the induced Hecke character $\chi(\rho\circ\mathrm{N})$ is trivial and is zero otherwise. The omitted Euler factors only change the sum by a polylogarithmic quantity. The constant may be ineffective. We require uniformity in ρ, q, Y, but not in H.

Lemma A8.8 (Progression density from the fixed-field input)**.** *For $(a,q) = 1$ define $J_q = K\cap\mathbb{Q}(\zeta_q)$ and let $\sigma_a(\zeta_q) = \zeta_q^a$. For fixed $B, D > 0$, uniformly for $q \le (\log X)^B$ and $0 \le Y \le X$,*

$$\sum_{\substack{n\le Y\\ n\equiv a \pmod q}} \Lambda'_C(n) = \frac{\#\{g \in C : g|_{J_q} = \sigma_a|_{J_q}\}}{[K\mathbb{Q}(\zeta_q):\mathbb{Q}]}Y + O_{H,B,D}(X(\log X)^{-D}).$$

Proof. Orthogonality of Dirichlet characters and (68) show that the coefficient of Y for T_C is

$$\frac{1}{\varphi(q)}\sum_{\chi\in\widehat{A}} c_C(\chi)\sum_{\rho \bmod q}\overline{\rho(a)}r_{\chi,\rho}.$$

The finite sum over χ has size depending only on H, while the average over ρ cancels its own factor $\varphi(q)$ in the error.

To evaluate this coefficient without a convention about inverse Artin symbols, put $L = K\mathbb{Q}(\zeta_q)$ and $B_q = \mathrm{Gal}(L/E)$. Restriction identifies B_q with

$$\{(g,s) \in A \times \mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) : g|_{J_q} = s|_{J_q}\}.$$

The characters $\chi(g)\rho(s)$ restrict to characters of B_q. Their average on B_q is one precisely when their pullback by Artin reciprocity is the trivial Hecke character; otherwise it is zero. This is exactly $r_{\chi,\rho}$. Expanding $1_C(g)$ and $1_{s=\sigma_a}$ by Fourier inversion therefore identifies the displayed coefficient with the proportion of elements of B_q satisfying $g \in C$ and $s = \sigma_a$. Its numerator is the numerator in the lemma and its denominator is $|B_q| = [L:E]$. Dividing by two using (67) changes the denominator to $[L:\mathbb{Q}]$. The prime-power error is smaller than the stated error. This proves the formula. □

Choose a fixed integer W divisible by the conductor of K^{ab}. It is in particular divisible by 4. For a unit $b \pmod W$ put

$$\mu_C(b) = \frac{\#\{g \in C : g|_{K^{\mathrm{ab}}} = \sigma_b|_{K^{\mathrm{ab}}}\}}{[K : K^{\mathrm{ab}}]},$$
$$f_{C,b,W}(n) = \frac{\varphi(W)}{W}\Lambda'_C(Wn+b),$$
$$\lambda_{b,W}(n) = \frac{\varphi(W)}{W}\Lambda'(Wn+b).$$

The prime-supported ordinary von Mangoldt function is denoted by Λ'. Replacing it by the usual Λ costs only the same negligible prime-power error. The centered weight is

$$h_{C,b,W} = f_{C,b,W} - \mu_C(b)\lambda_{b,W}. \tag{73}$$

Here $0 \le \mu_C(b) \le 1$, so $|h_{C,b,W}(n)| \le \lambda_{b,W}(n) = O_W(\log(2N))$ on $[N]$.

Corollary A8.9 (Centered sums on progressions)**.** *For every $D > 0$,*

$$\sum_{n\in P} h_{C,b,W}(n) = O_{H,W,D}(N(\log N)^{-D})$$

uniformly over arithmetic progressions P contained in $[N]$.

Proof. Write P as a residue class modulo q intersected with an interval. If $q > (\log N)^{D+3}$, its cardinality is at most $N/(\log N)^{D+3} + 1$, and the pointwise logarithmic bound proves the claim directly. Otherwise the condition $n \in P$ becomes one residue class $a \pmod{Wq}$ and an interval for $Wn + b$.

If $(a, Wq) > 1$, every prime in this residue class divides Wq. There are at most $O(\log(Wq))$ such primes, with total logarithmic weight polylogarithmic in N, which is negligible here. Suppose instead that $(a, Wq) = 1$. Since $\mathbb{Q}(\zeta_{Wq})$ is abelian and already contains K^{ab}, one has $K \cap \mathbb{Q}(\zeta_{Wq}) = K^{\mathrm{ab}}$. Also $a \equiv b \pmod W$, so the lemma's main coefficient is

$$\frac{\mu_C(b)}{\varphi(Wq)}.$$

For the ordinary prime weight, the Siegel–Walfisz main coefficient is $1/\varphi(Wq)$. The two terms of (73) therefore have equal main terms. Apply the progression lemma at both interval endpoints, with an error exponent larger than D and with $X = O_W(N)$, and subtract. This proves the claimed bound. All moduli used are at most a fixed power of $\log N$, and W is fixed throughout this argument. □

A8.4 Transfer from Gaussian ideal norms to polynomial nilsequences

Throughout this subsection $E = \mathbb{Q}(i)$, $\mathcal{O}_E = \mathbb{Z}[i]$, and H_0 is a fixed positive integer. A ray character χ modulo $H_0\mathcal{O}_E$ is extended by zero to ideals not coprime to $H_0\mathcal{O}_E$. The trivial character on all ideals is included by taking $H_0 = 1$. We use the conventions of [3]: G is connected, simply connected and nilpotent, Γ is a discrete cocompact subgroup, and the filtration consists of closed connected rational subgroups. Lipschitz norms include the supremum norm.

Lemma A8.10 (Gaussian norm transfer)**.** *Fix positive integers m, d. There are $\kappa, \sigma > 0$, depending only on m, d, and constants $C, X_0, \delta_0 > 0$, depending on H_0, m, d, with the following property. Suppose*

$$X \geq X_0, \qquad X^{-\sigma} \leq \delta \leq \delta_0, \qquad 1 \leq Q \leq \delta^{-\kappa}, \qquad 1 \leq q \leq \delta^{-\kappa}, \quad q \in \mathbb{Z}.$$

Let G/Γ have dimension m, a filtration $G_\bullet$ of degree at most d, and a Q-rational adapted Mal'cev basis. Suppose $g \in \mathrm{poly}(\mathbb{Z}, G_\bullet)$ is totally δ-equidistributed on $[X]$. If $F : G/\Gamma \to \mathbb{C}$ is Lipschitz and $\int F = 0$, then for every $0 \leq A < B \leq X$ and every $b \in \mathbb{Z}$,

$$\left| \sum_{\substack{A < \mathrm{N}\mathfrak{a} \leq B \\ \mathrm{N}\mathfrak{a} \equiv b \pmod q}} \chi(\mathfrak{a}) F(g(\mathrm{N}\mathfrak{a})\Gamma) \right| \leq C X \delta^{\kappa} \|F\|_{\mathrm{Lip}}. \tag{74}$$

There is no lower bound on $B - A$ or on the number of progression terms. Inserting $\log \mathrm{N}\mathfrak{a}$ on the left costs at most an additional factor $O(\log(2X))$.

We prove the lemma using only the one-parameter quantitative Leibman theorem [3, Theorem 2.9], together with elementary polynomial and lattice arguments. No multiparameter equidistribution theorem is used.

Lemma A8.11 (Quadratic coefficient extrapolation)**.** *Let $X \geq 1$, $P(t) = a_0 + a_1 t + a_2 t^2 \in \mathbb{Z}[t]$, $a_2 \neq 0$, and suppose*

$$|a_i| \leq B_0 X^{1-i/2}, \qquad L \geq X^{1/2}/B_0, \qquad B_0 \geq 2.$$

For every scalar polynomial $p : \mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ of degree at most d, there is an integer $1 \leq k \leq C_d B_0^{C_d}$ such that

$$\|kp\|_{C^\infty[X]} \leq C_d B_0^{C_d} \|p \circ P\|_{C^\infty[L]}.$$

Here, for $p(n) = \sum_j \alpha_j \binom{n}{j}$, we use $\|p\|_{C^\infty[X]} = \max_{1 \leq j \leq d} X^j \|\alpha_j\|_{\mathbb{R}/\mathbb{Z}}$.

Proof. Write $\binom{P(t)}{j} = \sum_\ell s_{j,\ell} \binom{t}{\ell}$. These coefficients are integers since the left side is integer-valued. Its degree is $2j$, and comparison of leading coefficients gives

$$s_{j,2j} = \frac{(2j)!}{j!} a_2^j \neq 0, \qquad |s_{j,\ell}| \ll_d B_0^j X^{j-\ell/2}.$$

For the bound, expand powers of P and change between the monomial and binomial bases. The change-of-basis matrices have fixed rational coefficients. Descending induction in ℓ shows that every term contributing to $s_{j,\ell}$ has weighted input degree at least ℓ, which gives the displayed estimate.

If $p = \sum_j \alpha_j \binom{n}{j}$ and $p \circ P = \sum_\ell \beta_\ell \binom{t}{\ell}$, then

$$\beta_{2j} = \sum_{h=j}^{d} s_{h,2j} \alpha_h.$$

Put $k_j = \prod_{h=j}^{d} s_{h,2h}$. Descending induction in j gives

$$X^j \|k_j \alpha_j\|_{\mathbb{R}/\mathbb{Z}} \ll_d B_0^{O_d(1)} \|p \circ P\|_{C^\infty[L]}.$$

Indeed the β_{2j} term is bounded using $X^j L^{-2j} \leq B_0^{2j}$, and a higher coefficient is multiplied by $s_{h,2j} = O_d(B_0^h X^{h-j})$, canceling its X^{-h} error. All multipliers used to clear higher rows are integers of size $B_0^{O_d(1)}$. Multiplying to the common nonzero integer $|k_1|$ proves the assertion. The constant coefficient α_0 is not used. □

Lemma A8.12 (Small horizontal character obstruction)**.** *There are $c_0, \sigma_0 > 0$, depending only on m, d, such that the following holds for sufficiently large X and sufficiently small $\delta \geq X^{-\sigma_0}$. Suppose the basis is T-rational and a nontrivial horizontal character η satisfies*

$$|\eta| \leq T, \qquad \|\eta \circ g\|_{C^\infty[X]} \leq T, \qquad T \leq \delta^{-c_0}.$$

Then g is not totally δ-equidistributed on $[X]$.

Proof. Choose representatives of the nonconstant binomial coefficients of $\eta \circ g$ satisfying $|\alpha_j| \leq TX^{-j}$. On the initial interval of length $\lfloor X/(100dT) \rfloor$ its values lie in an arc of length less than $1/10$, translated by its constant coefficient. Choose a Lipschitz function on $\mathbb{R}/\mathbb{Z}$ which equals one on that arc, has mean zero and bounded Lipschitz norm. Its composition with η has Lipschitz norm at most $T^{C(m,d)}$, by the Mal'cev metric estimates in [3, Appendix A]. Its Haar mean is zero: a nontrivial horizontal character maps Haar measure onto Haar measure on the circle. Choose c_0 with $C(m,d)c_0 < 1/4$ and $c_0 < 1/4$. The initial interval is longer than δX, while the discrepancy one exceeds $\delta T^{C(m,d)}$. Taking σ_0 sufficiently small absorbs the floors. This contradicts total equidistribution. □

Proof of Lemma A8.10. Fix a constant C_* which dominates the exponents in Lemma A8.11, the degree-$2d$ case of [3, Theorem 2.9], and the preceding obstruction. Choose

$$0 < \kappa < \min\{(1000C_*^2)^{-1}, c_0/(12C_*)\}, \qquad 0 < \sigma < \min\{(1000C_*^2)^{-1}, \sigma_0\}, \qquad 2\kappa\sigma < 1/4.$$

These choices depend only on m, d. Decrease δ_0, depending on H_0, m, d, whenever a fixed multiplicative constant must be absorbed.

We first record a pullback consequence. Suppose P and L satisfy Lemma A8.11 with $B_0 \leq \delta^{-5\kappa}$. Then $(g(P(t))\Gamma)_{t\in[L]}$ is δ^κ-equidistributed. Otherwise the quantitative Leibman theorem gives a nontrivial η with

$$|\eta|, \ \|\eta \circ g \circ P\|_{C^\infty[L]} \ll \delta^{-C_*\kappa}.$$

The basis is $\delta^{-\kappa}$-rational as required. The composed sequence has degree at most $2d$ for the stretched filtration $\widetilde{G}_j = G_{\lceil j/2 \rceil}$, with the same adapted basis. This follows from the elementary one-parameter coordinate characterization in [3, Lemma 6.7]. Extrapolation gives a nonzero integer multiple $k\eta$ with

$$|k\eta|, \ \|k\eta \circ g\|_{C^\infty[X]} \ll \delta^{-6C_*\kappa},$$

with C_* chosen sufficiently large initially. It remains nontrivial. The choice $6C_*\kappa < c_0/2$ and a reduction of δ_0 make this contradict Lemma A8.12. Only ordinary equidistribution of this pullback is needed.

Every nonzero Gaussian ideal has exactly four generators. The function $z \mapsto \chi((z))$, extended by zero when z is not coprime to H_0, is periodic modulo $H_0\mathbb{Z}[i]$ and invariant under units. Thus the sum in (74) is exactly

$$\frac{1}{4} \sum_{\substack{z=x+iy\neq 0 \\ A<x^2+y^2\leq B \\ x^2+y^2\equiv b \pmod q}} \chi((z))F(g(x^2+y^2)\Gamma).$$

The factor $1/4$ also applies to ideals with generators on an axis; we have introduced no angular fundamental domain.

Set $\rho = \delta^\kappa$ and $M = \lfloor \rho\sqrt{X}/H_0 \rfloor$. On each horizontal row $y \in \mathbb{Z}$, and in each residue class of x modulo H_0, partition the points into consecutive blocks of M points with step H_0. The character

is constant on a block. The intersection of the annulus with a horizontal row consists of at most two intervals, so at most four blocks per residue class meet an endpoint. There are $O(\sqrt{X})$ relevant rows. Consequently the boundary blocks contain

$$O(H_0\sqrt{X}\,M) = O(\rho X)$$

points. Estimate their contribution trivially. Excluding $z = 0$, if necessary, costs one more bounded summand, absorbed by ρX.

Every remaining contributing block lies wholly in the annulus. Split its index modulo q. The norm congruence is constant on each resulting class. Such a class has the form

$$x = x_0 + H_0 q n, \quad 1 \le n \le L, \qquad P(n) = (x_0 + H_0 q n)^2 + y^2, \qquad L \ge \frac{\rho\sqrt{X}}{4H_0 q}.$$

The hypotheses on δ and σ ensure $M \ge 2q$ and absorb the floors for large X. A harmless endpoint adjustment gives $|x_0| \le 2\sqrt{X}$. The coefficients of P satisfy

$$|a_0| \ll X, \qquad |a_1| \ll H_0 q\sqrt{X}, \qquad a_2 = H_0^2 q^2.$$

Since H_0 is fixed and $q \le \delta^{-\kappa}$, these bounds and the lower bound for L fit the pullback consequence with $B_0 \le \delta^{-5\kappa}$ after decreasing δ_0. Hence the sum on the class is $O(L\delta^{\kappa}\|F\|_{\mathrm{Lip}})$. The classes partition the interior points, of which there are $O(X)$; this elementary bound follows from the enclosing square. Summing and adding the boundary contribution proves (74).

The number of row endpoints is independent of $B - A$, explaining why no lower interval length is required. Finally apply Abel summation to the uniform bound with variable upper endpoint to insert $\log \mathrm{N}\mathfrak{a}$ at cost $O(\log(2X))$. □

Remark A8.13. For Type II, apply the lemma in dimension $2m$ to $(g_1(n), g_2(n))$ and $F_1 \otimes \overline{F_2}$, when this entire product orbit is totally equidistributed. Mean-zero factors ensure that the product test function has mean zero; they do not imply equidistribution of the product orbit. Use the smaller exponents for dimensions m and $2m$. In dimension zero mean-zero test functions vanish, so the assertion is immediate.

The constraints $rn \equiv b \pmod q$ and $r'n \equiv b \pmod q$ either are inconsistent or specify one residue class modulo a divisor of q. Intersecting their interval restrictions only changes A, B. Thus all Type I and Type II truncations are covered by the lemma. If $\delta = (\log X)^{-B_1}$ and $Q, q \le (\log X)^{C_1}$, any desired saving $X(\log X)^{-A_1}$ follows once $B_1\kappa > \max\{C_1, A_1\}$, with H_0 fixed.

A8.5 Gaussian ideal Vaughan decomposition and Type I/II estimates

We give the ideal decomposition and the subsequent integer-parameter recombination in full. The architecture is that of [51, Section 3], with the Gaussian ideal-norm adaptation attributed to Frei–Tinkova. The analytic inputs used here are the separately proved Lemma A8.10 and the published one-parameter results [3, Theorem 2.9 and Lemma 4.5]. In particular, no theorem of Frei–Tinkova is an assumed input.

Fix the Gaussian ray modulus H_0. All ideals in this subsection are nonzero integral ideals of $\mathbb{Z}[i]$. Write $\mathrm{N}\mathfrak{a}$ for the absolute ideal norm, μ_E for ideal Mobius, and $\Lambda_E(\mathfrak{p}^j) = \log \mathrm{N}\mathfrak{p}$ for $j \ge 1$, with Λ_E zero on other ideals. A ray character χ is extended by zero on ideals not coprime to H_0. It is completely multiplicative and satisfies $|\chi| \le 1$. We use a Lipschitz norm that includes the supremum norm. All nilmanifolds have connected, simply connected nilpotent covering group, a cocompact lattice, and an adapted rational Mal'cev basis, with the filtration and polynomial-sequence conventions of [3].

Lemma A8.14 (Ideal multiplicities and moments)**.** *Let $a_E(n) = \#\{\mathfrak{a} : \mathrm{N}\mathfrak{a} = n\}$. For $X \geq 1$,*

$$a_E(n) \leq \tau(n), \tag{75}$$

$$\sum_{n \leq X} a_E(n)^2 \leq X(1+\log X)^3, \qquad \sum_{n \leq X} \frac{a_E(n)^2}{n} \leq (1+\log X)^4. \tag{76}$$

Moreover

$$A_E(X) := \#\{\mathfrak{a} : \mathrm{N}\mathfrak{a} \leq X\} = \frac{\pi}{4} X + O(\sqrt{X}+1). \tag{77}$$

In particular $A_E(X) \ll X$, and the number of ideals of norm in $(B/2, B]$ is comparable to B for all sufficiently large B.

Proof. At a split rational prime p one has $a_E(p^k) = k+1$; at an inert prime it is one for even k and zero for odd k; and $a_E(2^k) = 1$. Multiplicativity proves (75). Prime by prime, $(k+1)^2 \leq \binom{k+3}{3}$, so $\tau(n)^2 \leq d_4(n)$. Summing over ordered quadruples with product at most X, and summing the last coordinate first, bounds $\sum_{n \leq X} d_4(n)$ by $X(1+\log X)^3$. Bounding each of the four harmonic sums separately gives $\sum_{n \leq X} d_4(n)/n \leq (1+\log X)^4$. Every ideal has exactly four Gaussian generators. Counting lattice points in the disk of radius $\sqrt{X}$, and dividing by four, proves (77): the unit squares meeting its boundary lie between circles of radii $\sqrt{X} - \sqrt{2}$ and $\sqrt{X} + \sqrt{2}$ and number $O(\sqrt{X}+1)$. □

Lemma A8.15 (Exact ideal Vaughan identity)**.** *Let $N \geq 8$, $U = V = N^{1/3}$, $L = \log(2N)$, and let $f : \mathbb{N} \longrightarrow \mathbb{C}$ satisfy $|f| \leq 1$ and vanish outside $(N/2, N]$. Then*

$$\sum_{\mathfrak{a}} \Lambda_E(\mathfrak{a}) \chi(\mathfrak{a}) f(\mathrm{N}\mathfrak{a}) = T_0 - T_1 + T_2, \tag{78}$$

where

$$\begin{aligned} T_0 &= \sum_{\mathrm{N}\mathfrak{d} \leq U} \mu_E(\mathfrak{d}) \chi(\mathfrak{d}) \sum_{\mathfrak{b}} \log(\mathrm{N}\mathfrak{b}) \chi(\mathfrak{b}) f(\mathrm{N}\mathfrak{d}\,\mathrm{N}\mathfrak{b}), \\ T_1 &= \sum_{\mathrm{N}\mathfrak{d} \leq UV} \alpha_{\mathfrak{d}} \chi(\mathfrak{d}) \sum_{\mathfrak{b}} \chi(\mathfrak{b}) f(\mathrm{N}\mathfrak{d}\,\mathrm{N}\mathfrak{b}), \\ T_2 &= \sum_{V < \mathrm{N}\mathfrak{d} \leq N/U} \beta_{\mathfrak{d}} \chi(\mathfrak{d}) \sum_{\mathrm{N}\mathfrak{b} > U} \mu_E(\mathfrak{b}) \chi(\mathfrak{b}) f(\mathrm{N}\mathfrak{d}\,\mathrm{N}\mathfrak{b}), \end{aligned}$$

with

$$\alpha_{\mathfrak{d}} = \sum_{\substack{\mathfrak{b}\mathfrak{c} = \mathfrak{d} \\ \mathrm{N}\mathfrak{b} \leq U,\ \mathrm{N}\mathfrak{c} \leq V}} \mu_E(\mathfrak{b}) \Lambda_E(\mathfrak{c}), \qquad \beta_{\mathfrak{d}} = \sum_{\substack{\mathfrak{c} \mid \mathfrak{d} \\ \mathrm{N}\mathfrak{c} > V}} \Lambda_E(\mathfrak{c}).$$

Both coefficients have absolute value at most L.

Proof. Convolution on ideals satisfies $\mu_E * 1 = \mathbf{1}_{\mathfrak{a}=(1)}$, $1 * \Lambda_E = \log \mathrm{N}$, and $\mu_E * \log \mathrm{N} = \Lambda_E$, as one checks on prime powers. Splitting μ_E at U and Λ_E at V gives the exact identity

$$\Lambda_E = \mu_{\leq U} * \log \mathrm{N} + \Lambda_{\leq V} - \mu_{\leq U} * 1 * \Lambda_{\leq V} + \mu_{>U} * 1 * \Lambda_{>V}. \tag{79}$$

The second term vanishes on the support of f. Complete multiplicativity of χ, including its zero extension, permits separation of each convolution factor and yields (78) with exactly the stated cutoffs. Finally $\sum_{\mathfrak{c} \mid \mathfrak{d}} \Lambda_E(\mathfrak{c}) = \log \mathrm{N}\mathfrak{d}$ and every summand is nonnegative. This proves the coefficient bounds, with no discarded endpoint or error term. □

Proposition A8.16 (Quantitative extraction)**.** *There are absolute constants $c_0 > 0$ and N_0 such that the following holds. Under Lemma A8.15, suppose $N \geq N_0$, $0 < \epsilon \leq 1$, and*

$$\left| \sum_{\mathfrak{a}} \Lambda_E(\mathfrak{a}) \chi(\mathfrak{a}) f(\mathrm{N}\mathfrak{a}) \right| \geq \epsilon N L. \tag{80}$$

Put $\theta = c_0 \epsilon^4 / L^{24}$. At least one of the following holds.

1. *Some dyadic $R \leq 2N^{1/3}$ has at least θR integers $r \in (R/2, R]$ satisfying*

$$\left| \sum_{\mathfrak{b}} \log(\mathrm{N}\mathfrak{b}) \chi(\mathfrak{b}) f(r\mathrm{N}\mathfrak{b}) \right| \geq \theta N / r.$$

2. *Some dyadic $R \leq 2N^{2/3}$ has at least θR integers $r \in (R/2, R]$ satisfying the same inequality without the logarithm.*

3. *There are dyadic R, B with*

$$N^{1/3} < R \leq 2N^{2/3}, \qquad N/2 < RB < 4N, \tag{81}$$

and at least θR^2 pairs $r, r' \in (R/2, R]$ satisfying

$$\left| \sum_{B/2 < \mathrm{N}\mathfrak{b} \leq B} f(r\mathrm{N}\mathfrak{b}) \overline{f(r'\mathrm{N}\mathfrak{b})} \right| \geq \theta B. \tag{82}$$

Proof. One of $|T_i|$ is at least $\epsilon NL/3$. For T_0 put

$$G(r) = \frac{r}{NL} \left| \sum_{\mathfrak{b}} \log(\mathrm{N}\mathfrak{b}) \chi(\mathfrak{b}) f(r\mathrm{N}\mathfrak{b}) \right|;$$

for T_1 use r/N times the absolute unweighted inner sum. In either case Lemma A8.14 gives $G \ll 1$ and

$$\epsilon \ll \sum_{r \leq N^e} \frac{a_E(r) G(r)}{r}, \qquad e = 1/3 \text{ or } 2/3.$$

Cauchy–Schwarz and (76) imply $\sum_{r \leq N^e} G(r)^2 / r \gg \epsilon^2 / L^4$. There are $O(L)$ dyadic cells, so one satisfies

$$\sum_{R/2 < r \leq R} G(r)^2 \gg R\epsilon^2 / L^5.$$

Since $G \ll 1$, at least $cR\epsilon^2/L^5$ terms have $G(r) \geq c\epsilon / L^{5/2}$. Both the count and the claimed amplitudes follow after decreasing c_0.

For T_2, split both ideal norms into dyadic cells. There are $O(L^2)$ rectangles, and a nonzero cell satisfies (81) and $B \geq N^{1/3}/4$. A rectangle contributes at least $c\epsilon N/L$; we use the weaker bound $c\epsilon N/L^2$. Grouping $\mathfrak{d}$ by $r = \mathrm{N}\mathfrak{d}$ writes this contribution as

$$T = \sum_{R/2 < r \leq R} A_r \sum_{\mathfrak{b} \in \mathcal{B}} c_{\mathfrak{b}} f(r\mathrm{N}\mathfrak{b}), \qquad \mathcal{B} = \{\mathfrak{b} : B/2 < \mathrm{N}\mathfrak{b} \leq B\}.$$

Here $|A_r| \le La_E(r)$ and $|c_{\mathfrak{b}}| \le 1$; the restriction $\mathrm{N}\mathfrak{b} > U$ belongs to $c_{\mathfrak{b}}$. All product cutoffs remain in f, so the domain is rectangular. By (76), $\sum_r |A_r|^2 \ll RL^5$. Thus, with $F_{r,\mathfrak{b}} = f(r\mathrm{N}\mathfrak{b})$,

$$T_* := \sum_r \left| \sum_{\mathfrak{b} \in \mathcal{B}} c_{\mathfrak{b}} F_{r,\mathfrak{b}} \right|^2 \gg \epsilon^2 RB^2/L^9. \tag{83}$$

Put $M = |\mathcal{B}| \asymp B$. Expansion and Cauchy–Schwarz in the ordered pair $(\mathfrak{b}, \mathfrak{b}')$ give

$$\begin{aligned} T_*^2 &= \left| \sum_{\mathfrak{b},\mathfrak{b}'} c_{\mathfrak{b}} \overline{c_{\mathfrak{b}'}} \sum_r F_{r,\mathfrak{b}} \overline{F_{r,\mathfrak{b}'}} \right|^2 \\ &\le M^2 \sum_{\mathfrak{b},\mathfrak{b}'} \left| \sum_r F_{r,\mathfrak{b}} \overline{F_{r,\mathfrak{b}'}} \right|^2 = M^2 \sum_{r,r'} \left| \sum_{\mathfrak{b}} F_{r,\mathfrak{b}} \overline{F_{r',\mathfrak{b}}} \right|^2 . \end{aligned} \tag{84}$$

Consequently

$$\mathbb{E}_{r,r'} \left| \mathbb{E}_{\mathfrak{b} \in \mathcal{B}} F_{r,\mathfrak{b}} \overline{F_{r',\mathfrak{b}}} \right|^2 \gg \epsilon^4/L^{18}.$$

Every summand is at most one. Thresholding gives (82) and its stated count, with the weaker common parameter $\theta = c_0\epsilon^4/L^{24}$. The character and Mobius coefficients have disappeared. All conjugates in (84) are necessary. □

Progressions and interval restrictions. In the application,

$$f(n) = 1_{N/2 < n \le N}\, 1_{u < n \le v}\, 1_{n \equiv b \pmod q} F(g(n)\Gamma).$$

For Type I the congruence $rn \equiv b \pmod q$ is either insoluble or one class modulo $q/\gcd(r,q)$. In Type II its intersection with $r'n \equiv b \pmod q$ is either empty or one class modulo

$$\operatorname{lcm}(q/\gcd(r,q), q/\gcd(r',q)),$$

which divides q. The remaining restrictions are interval intersections. Lemma A8.10 covers them without a lower bound on their lengths. On the product nilmanifold the function is $F(x)\overline{F(y)}$, whose Haar integral is $|\int F|^2 = 0$. After normalizing $\|F\|_{\mathrm{Lip}} = 1$, its Lipschitz norm for the sum metric is at most two. Thus all mean-zero, modulus and interval hypotheses for product norm transfer are satisfied.

Lemma A8.17 (Integer-parameter recombination)**.** *For each m, d there are $C_{\mathrm{rec}}, K_{\mathrm{rec}} \ge 2$ with the following property. Let $0 < \eta < \eta_0$, $N \ge \eta^{-K_{\mathrm{rec}}}$, and let g have degree at most d on an m-dimensional nilmanifold with an η^{-1}-rational adapted basis. Each of the following implies that $(g(n)\Gamma)_{n \le N}$ is not totally $\eta^{C_{\mathrm{rec}}}$-equidistributed.*

1. *For at least ηR values $r \in (R/2, R]$, where $R \le 2N^{2/3}$, the orbit $(g(rn)\Gamma)_{n \le \lceil N/r \rceil}$ is not totally η-equidistributed.*
2. *For at least ηR^2 pairs $r, r' \in (R/2, R]$, where $N^{1/3}/4 \le R \le 2N^{2/3}$ and $N/2 < RB < 4N$, the product orbit $(g(rn), g(r'n))_{n \le B}$ is not totally η-equidistributed.*

Proof. All constants in this proof depend only on m, d. Failure of total equidistribution gives a progression $a + q[\ell] \subseteq [M]$ with $\ell \ge \eta M$, $q \le 2/\eta$, and $|a| \le 2M$ on which ordinary η-equidistribution fails. Apply [3, Theorem 2.9], in dimension m or $2m$. A nonzero horizontal character has magnitude and restricted smoothness norm at most η^{-C_1}, after shrinking η_0 to absorb fixed factors.

Here is the required removal of a possibly large shift. Write the scalar phase $p(a+qt)$ in monomials. Multiplication by $d!$ clears conversion from its binomial coefficients. Inverting $t = (n-a)/q$ and multiplying by q^d clears every remaining denominator. The error in the coefficient of n^i is bounded by

$$C_d \eta^{-C_1} \sum_{j=i}^{d} q^{d-j} |a|^{j-i} \ell^{-j} \le \eta^{-C_2} M^{-i}, \qquad i \ge 1. \tag{85}$$

The integer parts disappear modulo one. Thus a nonzero integer multiple of the character has magnitude and $C^\infty[M]$ norm at most η^{-C_2}. This argument does not treat $a = O(M)$ as a bounded-height rational parameter. There are at most η^{-C_3} possible resulting characters, since their lattice has rank at most $2m$. Pigeonholing therefore fixes one character on a set of density at least η^{C_4}.

In Type I write $\psi(g(n)) = \sum_{j=0}^{d} \beta_j n^j$ modulo one. For the selected set $S \subseteq (R/2, R]$, coefficient conversion gives

$$\|d!\beta_j r^j\|_{\mathbb{R}/\mathbb{Z}} \le \eta^{-C_5} (R/N)^j, \qquad 1 \le j \le d. \tag{86}$$

If $R \le \eta^{-C_6}$ for a sufficiently large fixed C_6, choose one $r \in S$ and multiply the character by $d!r^d$. Multiplication of the jth bound by r^{d-j} gives magnitude and $C^\infty[N]$ norm bounded by a fixed power of η^{-1}. This treats every small dyadic scale, including $R = 1$.

Otherwise apply [3, Lemma 4.5] to $r \mapsto d!\beta_j r^j$ on $[\lceil R \rceil]$. The selected set has density at least a fixed power of η, and its values lie in an arc of length at most $2\eta^{-C_5}(R/N)^j$. Taking K_{rec} large, the bound $R \le 2N^{2/3}$ makes this length at most half the density. The choice of C_6 also places the recurrence length beyond the fixed inverse density powers. The published recurrence lemma supplies nonzero integers t_j, $|t_j| \le \eta^{-C_7}$, such that

$$\|t_j (d!)^2 \beta_j\|_{\mathbb{R}/\mathbb{Z}} \le \eta^{-C_8} N^{-j}. \tag{87}$$

The extra factorial clears the leading-coefficient conversion. Taking the product of the t_j and the fixed factorials gives a single nonzero horizontal character of magnitude and $C^\infty[N]$ norm at most η^{-C_9}.

For Type II the pigeonholed character is $\psi_1 \oplus \psi_2$, not both zero. Write the two scalar phases with coefficients β_j, β'_j. The information replacing (86) is

$$\|d!(\beta_j r^j + \beta'_j r'^j)\|_{\mathbb{R}/\mathbb{Z}} \le \eta^{-C_5} (R/N)^j. \tag{88}$$

Interchange the two coordinates if necessary so that $\psi_1 \neq 0$. Fix an r' with at least $\eta^{C_4} R$ partners. Equation (88) puts $d!\beta_j r^j$ in an arc centered at $-d!\beta'_j r'^j$. The same published recurrence lemma permits this arbitrary center and gives (87) for ψ_1. Its large-length requirement follows from $R \ge N^{1/3}/4$ and a sufficiently large K_{rec}. There is no substitution of $(r-r_0)^j$ for $r^j - r_0^j$.

Finally let Ψ denote the nonzero character just obtained. Its binomial expansion shows that for $1 \le n \le c\eta^{2C_9+4}N$ the phase $\Psi(g(n))$ stays within $1/10$ of $\Psi(g(0))$. Choose a mean-zero Lipschitz circle function equal to one on that arc and compose it with Ψ. Its mean is zero because a nonzero horizontal character pushes Haar measure to circle Haar measure. Its Lipschitz norm is bounded by a fixed power of η^{-1} by [3, Appendix A]. Choose C_{rec} larger than the length and Lipschitz exponents, and then enlarge K_{rec} so the initial interval is nonempty. Its average one contradicts total $\eta^{C_{\mathrm{rec}}}$-equidistribution. This proves both assertions. All exponent constants are finite combinations of the published one-parameter constants and degree-d polynomial algebra. □

Proposition A8.18 (Power-form Gaussian ideal minor-arc bound)**.** *For each m, d there is $c_{\mathrm{G}} > 0$ with the following property. Fix H_0 and a finite-order Gaussian ray character χ of modulus H_0, extended by zero. Let G/Γ have dimension m, filtration degree d, and a $\mathbb{Q}$-rational adapted basis,*

with $Q \geq 2$. Suppose $(g(n)\Gamma)_{n\leq N}$ is totally δ-equidistributed, where $0 < \delta < 1$. If F is Lipschitz and $\int F = 0$, then for every interval progression $P \subseteq [N]$ of common difference at most Q,

$$\left| \sum_{\mathrm{N}\mathfrak{a}\leq N} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a})1_P(\mathrm{N}\mathfrak{a})F(g(\mathrm{N}\mathfrak{a})\Gamma) \right| \ll_{H_0,m,d} QN\log(2N)^7\delta^{c_{\mathrm{G}}}\|F\|_{\mathrm{Lip}}. \tag{89}$$

No lower bound on the length of P or on δ is required.

Proof. Normalize $\|F\|_{\mathrm{Lip}} = 1$. Take the smaller positive exponents κ, σ in Lemma A8.10 for dimensions m and $2m$, and decrease κ to at most one. Partial summation gives the same transfer bound with an extra factor $\log(2M)$ for a $\log \mathrm{N}\mathfrak{b}$ weight: the unweighted lemma holds for every partial interval, and $\int_1^M dt/t \leq \log(2M)$. Let $C_{\mathrm{rec}}, K_{\mathrm{rec}}$ be as in Lemma A8.17, and choose

$$a_* = \frac{1}{4C_{\mathrm{rec}}}, \qquad c_* = \frac{a_*\kappa}{32}, \qquad s_* = \min\left\{\frac{1}{100}, \frac{1}{4a_*K_{\mathrm{rec}}}, \frac{\sigma}{12a_*}\right\}. \tag{90}$$

First assume $N^{-s_*} \leq \delta \leq \delta_0$, with the fixed cutoff δ_0 sufficiently small, and restrict the sum to $(N/2, N]$. We claim that its absolute value is

$$\ll QNL^7\delta^{c_*}, \qquad L = \log(2N). \tag{91}$$

The elementary bound is $O(NL)$. Thus the claim is immediate when $QL^6\delta^{c_*} \geq 1$, and in the other case $Q \leq \delta^{-c_*}$ and $L \leq \delta^{-c_*/6}$. If (91) fails with a sufficiently large implied constant, Proposition A8.16 applies with ϵ a fixed large multiple of $QL^6\delta^{c_*}$; if this exceeds the permitted range, the elementary bound already proves the claim. The extracted density and normalized amplitude are at least a fixed positive constant times δ^{4c_*}, hence at least δ^{5c_*} after decreasing δ_0.

Set $\eta = \delta^{a_*}$. If an extracted orbit were totally η-equidistributed, Lemma A8.10, with the progression analysis above, would bound the extracted sum by

$$O(M\delta^{a_*\kappa}L^e), \qquad e = 0 \text{ or } 1.$$

Here $M = \lceil N/r \rceil \geq N^{1/3}/2$ in Type I and $M = B \geq N^{1/3}/4$ in Type II. The rationality and modulus hypotheses hold because $c_* < a_*\kappa$. The lower-parameter hypothesis holds because $a_*s_* \leq \sigma/12$; increasing the fixed threshold for N absorbs the constants two and four. Since $a_*\kappa = 32c_*$, the displayed bound is smaller than $M\delta^{5c_*}$, even for $e = 1$. This is a contradiction. Thus every extracted orbit fails total η-equidistribution. Its density is at least η, since $a_* > 5c_*$. Also $N \geq \eta^{-K_{\mathrm{rec}}}$ by $a_*s_*K_{\mathrm{rec}} \leq 1/4$, and $Q \leq \eta^{-1}$. Lemma A8.17 now contradicts total δ-equidistribution, because $\eta^{C_{\mathrm{rec}}} = \delta^{1/4} > \delta$. This proves (91). Fixed values $\delta \geq \delta_0$ are covered by the elementary bound with a larger constant.

We spell out the passage through all scales. Use exact real endpoints $M_j = 2^{-j}N$ and retain shells with $M_j \geq 2\sqrt{\delta}N$. The discarded norms are at most $2\sqrt{\delta}N$ and their total absolute weight is $O(\sqrt{\delta}NL)$ by Lemma A8.14. There is no rounding of ideal-norm cutoffs. On $[\lfloor M_j \rfloor]$ the orbit is totally $2\sqrt{\delta}$-equidistributed: every progression of the requisite length has at least δN terms for sufficiently large N, and the original discrepancy bound is stronger. For $\delta \geq N^{-s_*/4}$ the lower-parameter condition for the shell bound holds on every retained scale, since $M_j \geq N^{1-s_*/8}$ up to fixed constants. The progression is merely intersected with the shell; its common difference remains at most Q. Summing the shell estimates and using $\sum_j M_j \leq 2N$ gives

$$O(QNL^7\delta^{c_*/2}). \tag{92}$$

For $1/N < \delta < N^{-s_*/4}$ replace δ by $\delta^{s_*/8} \geq N^{-s_*/8}$. Total equidistribution at the original parameter implies it at this larger one. The exponent in (92) becomes $c_* s_*/16$. For $\delta \leq 1/N$, singleton progressions imply $|F(g(n)\Gamma)| \leq \delta$ at each integer n, so the elementary $O(NL)$ weight bound suffices. All remaining bounded N are absorbed by the fixed constant. Thus one may take

$$c_{\mathrm{G}} = c_* s_*/16 > 0. \tag{93}$$

Its dependence is only on m, d and the corresponding norm-transfer exponents; H_0 enters constants and thresholds, not this exponent. □

Corollary A8.19 (Ordinary prime term)**.** *Under the nilmanifold hypotheses of Proposition A8.18, there is $c_{\mathbb{Q}} > 0$, depending only on m, d, such that*

$$\left| \sum_{n \leq N} \Lambda'(n) 1_P(n) F(g(n)\Gamma) \right| \ll_{m,d} QN \log(2N)^7 \delta^{c_{\mathbb{Q}}} \|F\|_{\mathrm{Lip}}, \tag{94}$$

where $\Lambda'(p) = \log p$ on rational primes and is zero otherwise. The same bound holds for ordinary von Mangoldt Λ.

Proof. For positive integers the convolution identities $\mu * 1 = 1_{n=1}$, $1 * \Lambda = \log$, and $\mu * \log = \Lambda$ give (79) with exactly the displayed norm cutoffs replaced by integer cutoffs. Its coefficients are bounded by $\sum_{c|d} \Lambda(c) = \log d$. Integer multiplicity is one, so both moment bounds in Lemma A8.14 remain valid. The two Cauchy–Schwarz calculations (83) and (84) apply to the resulting finite integer sums and give Proposition A8.16 with the same conservative parameter $c_0 \epsilon^4 / L^{24}$. Lemma A8.17 concerns only integer dilation parameters and is unchanged.

The unweighted replacement for norm transfer follows directly from total equidistribution. If an interval progression $J \subseteq [X]$ has $|J| \geq \zeta X$, then

$$\left| \sum_{n \in J} F(g(n)\Gamma) \right| \leq |J| \zeta \|F\|_{\mathrm{Lip}} \leq X \zeta \|F\|_{\mathrm{Lip}}.$$

If $|J| < \zeta X$, its cardinality and the supremum norm give the same bound. This treats every modulus and every truncated interval. Partial summation supplies the logarithmic weight; the identical long-progression/short-progression dichotomy treats the mean-zero product function $F \otimes \overline{F}$ on the product nilmanifold. Thus the exponent choices and all shell steps in the proof of Proposition A8.18 apply, with any sufficiently small fixed positive transfer exponents, to give (94) for Λ. No extension of a Mobius theorem to Λ is assumed.

Finally proper prime powers contribute at most $O(\sqrt{N} \log(2N)^2 \|F\|_{\mathrm{Lip}})$, by counting at most $N^{1/k}$ possible bases for each $2 \leq k \leq \log_2 N$. Shrink $c_{\mathbb{Q}}$ below $1/4$. If $\delta > 1/N$, this error is absorbed because $N\delta^{c_{\mathbb{Q}}} \geq N^{1-c_{\mathbb{Q}}}$. If $\delta \leq 1/N$, singleton equidistribution and $\sum_{n \leq N} \Lambda'(n) \leq N \log(2N)$ prove the bound directly. This gives the assertion for Λ' as well. □

A8.6 Fixed-modulus affine interpolation

Fix integers $W \geq 1$ and $0 \leq b < W$. The modulus W remains fixed as the size parameter tends to infinity. We first justify the affine interpolation needed to apply ideal-norm estimates to a W-tricked sequence.

Lemma A8.20 (Linear coefficient extrapolation)**.** *Let $X \geq 1$ and let $p : \mathbb{Z} \to \mathbb{R}/\mathbb{Z}$ have degree at most d. Suppose $P(t) = u + vt$ with $u, v \in \mathbb{Z}$, $v \neq 0$, $|u| \leq CX$, $|v| \leq E$, and $L \geq X/E$, where $E \geq 2$. There is an integer $1 \leq k \leq C_d E^{C_d}$ such that*

$$\|kp\|_{C^\infty[X]} \leq C_d(C+E)^{C_d}\|p \circ P\|_{C^\infty[L]}.$$

Proof. Expand $\binom{u+vt}{j}$ in the binomial basis in t. All coefficients are integers, the diagonal coefficient is v^j, and the coefficient in row ℓ is $O_d((C+E)^j X^{j-\ell})$. Starting in degree d, clear the nonzero diagonal entries successively. Their product is a nonzero integer bounded by $E^{O_d(1)}$. At degree j use $X^j/L^j \leq E^j$; a higher-row error is multiplied by $O_d((C+E)^h X^{h-j})$, canceling its X^{-h} factor. Multiplication to the common product of diagonal entries proves the claim, exactly as in Lemma A8.11. □

Lemma A8.21 (Affine interpolation preserves quantitative distribution)**.** *Fix dimension and degree bounds m, d. There are $a, \sigma > 0$, depending only on m, d, and thresholds depending on W, m, d, with the following property. Suppose a nilmanifold and its rational filtration satisfy the Green–Tao conventions, its adapted basis is Q-rational, and $g \in \mathrm{poly}(\mathbb{Z}, G_\bullet)$. Suppose also that*

$$N^{-\sigma} \leq \delta \leq \delta_0(W, m, d), \qquad Q \leq \delta^{-a},$$

and that g is totally δ-equidistributed on $[N]$. There is $\widetilde{g} \in \mathrm{poly}(\mathbb{Z}, G_\bullet)$ satisfying

$$\widetilde{g}(Wn + b) = g(n) \quad (n \in \mathbb{Z})$$

which is totally δ^a-equidistributed on $[WN + b]$.

Proof. Use the one-parameter coordinate characterization in [3, Lemma 6.7]. Each second-kind Mal'cev coordinate of $g(n)$ is a polynomial, with the prescribed degree bound from the filtration. Evaluate those same polynomials at real arguments and apply the inverse coordinate map, obtaining $g_{\mathbb{R}}(t)$. Set

$$\widetilde{g}(t) = g_{\mathbb{R}}((t-b)/W).$$

Affine composition preserves the coordinate degree bounds, so the same characterization shows that this is a polynomial sequence for the original filtration. It gives the stated identity on integer n. Equivalently, one may use the Taylor product with real powers; those powers exist in a connected simply connected nilpotent group.

Put $X = WN + b$ and $\epsilon = \delta^a$. Suppose the asserted total ϵ-equidistribution fails. Then on a progression

$$u + v[L] \subseteq [X], \qquad L \geq \epsilon X, \quad u, v \in \mathbb{Z}, \quad 1 \leq v \leq 2/\epsilon, \quad |u| \leq 2X,$$

the sequence is not ϵ-equidistributed. By the one-parameter quantitative Leibman theorem [3, Theorem 2.9], there is a nontrivial horizontal character η satisfying

$$|\eta| \ll \epsilon^{-C}, \qquad \|\eta \circ \widetilde{g} \circ (u + vt)\|_{C^\infty[L]} \ll \epsilon^{-C}.$$

The rationality hypothesis holds since $Q \leq \epsilon^{-1}$. Lemma A8.20 gives a positive integer $k \ll \epsilon^{-C}$ such that

$$\|k\eta \circ \widetilde{g}\|_{C^\infty[X]} \ll \epsilon^{-C}.$$

Forward composition with the fixed integer polynomial $Wn + b$ gives

$$\|k\eta \circ g\|_{C^\infty[N]} \ll_{W,d} \epsilon^{-C}.$$

For this last bound expand $\binom{Wn+b}{j}$ in the binomial basis; its matrix is integral, and X is comparable to WN. The character $k\eta$ remains nontrivial.

Let C_{aff} bound all exponents in these estimates, including the character size. Fix explicitly

$$0 < a < \min\{1, c_0/(2C_{\mathrm{aff}})\}, \qquad 0 < \sigma < \min\{\sigma_0, 1/(2a)\},$$

where c_0, σ_0 are from Lemma A8.12. Decrease δ_0 to absorb fixed W-dependent constants. The resulting character bounds contradict that lemma and total δ-equidistribution of g. The inequality $a\sigma < 1/2$ ensures $\epsilon X \gg N^{1/2}$, so all progression parametrizations and floor estimates used above hold for large N. This proves the assertion on the entire interval $[WN + b]$, including progressions outside the residue class b modulo W. □

The Gaussian input in the following lemma is furnished by Proposition A8.18; its rational counterpart is Corollary A8.19.

Lemma A8.22 (W-tricked minor-arc interface)**.** *For $E = \mathbb{Q}(i)$ and a fixed ray character χ, put*

$$T_\chi(t) = \sum_{\mathrm{N}\mathfrak{a}=t} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a}).$$

Assume the mean-zero ideal estimate

$$\left| \sum_{\substack{\mathrm{N}\mathfrak{a}\le X \\ \mathrm{N}\mathfrak{a}\in R}} \Lambda_E(\mathfrak{a})\chi(\mathfrak{a})F_0(h(\mathrm{N}\mathfrak{a})\Gamma) \right| \ll Q_1 X \log(2X)^7 \eta^{c_E} \|F_0\|_{\mathrm{Lip}} \tag{95}$$

for a positive c_E, whenever h is totally η-equidistributed, the basis is Q_1-rational, and R is any interval progression of modulus at most Q_1. Then, under the hypotheses of Lemma A8.21, for every Lipschitz F_0 with Haar mean zero and every interval progression $P \subseteq [N]$ of modulus $q \le Q$,

$$\left| \sum_{n\in P} T_\chi(Wn + b)F_0(g(n)\Gamma) \right| \ll_{H_0,W,m,d} QN \log(2N)^7 \delta^{ac_E} \|F_0\|_{\mathrm{Lip}}. \tag{96}$$

The same implication holds for $E = \mathbb{Q}$ and the corresponding ordinary von Mangoldt estimate.

Proof. Take $X = WN + b$ and the sequence $\widetilde{g}$ constructed above. The conditions $t = Wn + b$ and $n \in P$ specify an interval progression R of modulus Wq. Apply (95) with $Q_1 = \max\{Q, Wq, 2\} \ll_W Q$ and $\eta = \delta^a$. The ideal sum then equals the left side of (96) exactly, while $X \ll_W N$ and $\log(2X) \ll_W \log(2N)$. No assertion about fractional-argument distribution has been assumed: it was proved in Lemma A8.21. □

Corollary A8.23 (Centered ray-prime weights)**.** *Suppose a conjugation-invariant fixed Gaussian ray condition has the checked finite-character expansion*

$$\Lambda'_C(t) = \frac{1}{2}\sum_\chi c_\chi T_\chi(t) + e(t), \qquad \sum_{t\le X} |e(t)| \ll_{H_0} \sqrt{X}\log(2X)^2.$$

Assume both the Gaussian estimate (95) *and its ordinary rational counterpart, with positive exponents $c_E, c_{\mathbb{Q}}$. Let G/Γ and g satisfy the hypotheses of Lemma A8.21, and let F_0 be Lipschitz with Haar mean zero. Then any fixed linear combination of the W-tricked ray-prime weight and the*

W-tricked ordinary-prime weight obeys, for every interval progression $P \subseteq [N]$ of modulus at most Q,

$$\left|\sum_{n\in P} w_N(n)F_0(g(n)\Gamma)\right| \ll QN\log(2N)^7\delta^{c_W}\|F_0\|_{\mathrm{Lip}}, \qquad c_W = a\min\{c_E, c_{\mathbb{Q}}\} > 0,$$

for mean-zero F_0, in a regime $\delta \geq N^{-\sigma_1}$ and $Q \leq \delta^{-a}$ with fixed $\sigma_1 > 0$. Constants may depend on H_0, W and the fixed linear combination. In particular this applies to the proposed centering

$$w_N(n) = \frac{\varphi(W)}{W}\Lambda'_C(Wn+b) - \mu_C(b)\lambda_{b,W}(n),$$

once the normalization of $\lambda_{b,W}$, the scalar $\mu_C(b)$ and the displayed character expansion have been fixed and checked.

Proof. Apply Lemma A8.22 to the finitely many characters and to the rational von Mangoldt function. The Gaussian factor $1/2$ accounts for the two conjugate prime ideals above an eligible split rational prime. Replacing ordinary von Mangoldt by its prime-only version also has a proper-prime-power error $O_W(\sqrt{N}\log(2N)^2)$. Thus the stated estimate holds with the additional error $O_{H_0,W}(\sqrt{N}\log(2N)^2\|F_0\|_\infty)$. Choose $\sigma_1 \leq \sigma$ so small that $\sigma_1 c_W < 1/4$. Then $N\delta^{c_W} \geq N^{3/4}$ throughout the claimed regime, and the square-root errors are absorbed. All normalization factors are fixed constants because W is fixed. This proves the minor-arc estimate but does not assert the centered progression estimate (97), which must be proved separately from the arithmetic main terms. □

Remark A8.24. Fix the arithmetic moduli first, then the positive exponents in this subsection. They do not depend on the later factorization exponent B. Proposition A8.25 chooses B large enough to satisfy the resulting upper complexity condition. Its output M is polylogarithmic in N, so its equidistribution parameter also obeys $\delta \geq N^{-\sigma_1}$ for sufficiently large N. The affine interpolation argument is motivated by the corresponding step of Frei–Tinkova, but uses no theorem from that work as an input.

A8.7 Quantitative factorization for centered weights

We isolate the analytic implication needed after the arithmetic progression main terms have been established. Throughout, $L = \log(2N)$, all nilpotent groups are connected and simply connected, and all filtrations are by closed connected rational subgroups. Constants may depend on fixed arithmetic moduli and on the fixed dimension and degree bounds.

Proposition A8.25 (Factorization implication)**.** *Fix dimension and degree bounds m, d. Let $w_N : [N] \to \mathbb{C}$ satisfy the following three hypotheses.*

1. *For a fixed $K_w \geq 0$, $|w_N(n)| \ll L^{K_w}$.*
2. *For every $A_0, C_0 > 0$, uniformly over all interval progressions $P \subseteq [N]$ of common difference at most L^{C_0},*

$$\left|\sum_{n\in P} w_N(n)\right| \ll_{A_0,C_0} NL^{-A_0}. \tag{97}$$

No lower bound on $|P|$ is imposed here.

3. *There are fixed $c > 0$, $C_1, K_1 \geq 0$ such that if a nilmanifold of dimension at most m and filtration degree at most d has a Q-rational adapted Mal'cev basis, h is a polynomial sequence totally δ-equidistributed on $[N]$, and F_0 is Lipschitz with Haar mean zero, then*

$$\left|\sum_{n\in P} w_N(n)F_0(h(n)\Gamma)\right| \ll N\delta^c Q^{C_1}L^{K_1}\|F_0\|_{\mathrm{Lip}} \tag{98}$$

for every interval progression $P \subseteq [N]$ of modulus at most Q and length at least N/Q. It is sufficient for this hypothesis to hold only when

$$\delta \geq N^{-\sigma}, \qquad Q \leq \delta^{-\kappa} \tag{99}$$

for fixed positive σ, κ. An estimate allowing arbitrary progression lengths is stronger than required.

Then, for every $A, C_{\mathrm{init}} > 0$, uniformly over initial nilmanifolds of dimension at most m and degree at most d with an $L^{C_{\mathrm{init}}}$-rational adapted basis, all polynomial sequences g, and all Lipschitz F,

$$\left|\sum_{n\leq N} w_N(n)F(g(n)\Gamma)\right| \ll_{A,C_{\mathrm{init}}} NL^{-A}\|F\|_{\mathrm{Lip}}. \tag{100}$$

Proof. Normalize $\|F\|_{\mathrm{Lip}} = 1$ and put

$$D = A + K_w + 3, \qquad M_0 = L^{\max\{1,C_{\mathrm{init}}\}}.$$

Leave a factorization exponent B unspecified for the moment. By [3, Theorem 1.19], there is an integer M with

$$M_0 \leq M \ll_{B,m,d} M_0^{E(B,m,d)}$$

and a factorization $g = \varepsilon g'\gamma$ into polynomial sequences. Here ε is (M,N)-smooth, $\gamma(n)\Gamma$ has period $q \leq M$, and g' is totally M^{-B}-equidistributed in a rational subgroup quotient. The new adapted basis consists of M-rational combinations of the initial one.

For each residue j modulo q, choose a reduced rational representative γ_j of $\gamma(j)\Gamma$. Split that residue class into $\lceil ML^D\rceil$ consecutive blocks of nearly equal cardinality. For sufficiently large N after B is fixed, the resulting progressions $P_{j,k}$ satisfy

$$\#\{P_{j,k}\} \ll M^2L^D, \qquad |P_{j,k}| \gg \frac{N}{M^2L^D}, \qquad \operatorname{diam}(P_{j,k}) \ll \frac{N}{ML^D}. \tag{101}$$

Their common difference is q. Indeed each original residue class has cardinality comparable to N/q, and rounding the balanced block sizes changes these estimates only by an absolute factor, since all powers of M are polylogarithmic in N once B is fixed.

Choose $n_{j,k} \in P_{j,k}$ and put $a_{j,k} = \varepsilon(n_{j,k})$. The right-invariant ambient metric and smoothness imply

$$\begin{aligned}|F(\varepsilon(n)g'(n)\gamma_j\Gamma) - F(a_{j,k}g'(n)\gamma_j\Gamma)| \\ \leq d(\varepsilon(n),\varepsilon(n_{j,k})) \ll L^{-D} \qquad (n \in P_{j,k}).\end{aligned}$$

There is no factor depending on the coordinates of $g'(n)$ in this estimate: right invariance removes it. The total weighted freezing error is therefore $O(NL^{K_w-D}) = O(NL^{-A-3})$.

We use the quantitative conjugation claim proved in [51, p. 547], whose inputs are the rational-subgroup, rational-element and metric comparisons of [3, Appendix A]. There are fixed $a > 0$, $C_2 \geq 1$, depending only on m, d, such that, with

$$H_j = \gamma_j^{-1} G' \gamma_j, \quad \Lambda_j = H_j \cap \Gamma, \quad h_j(n) = \gamma_j^{-1} g'(n) \gamma_j,$$

the quotient H_j/Λ_j has an adapted basis of rationality at most M^{C_2}, h_j is totally M^{-aB+C_2}-equidistributed, and

$$F_{j,k}(x\Lambda_j) = F(a_{j,k}\gamma_j x \Gamma) \quad \text{satisfies} \quad \|F_{j,k}\|_{\mathrm{Lip}} \leq M^{C_2}, \qquad \|F_{j,k}\|_\infty \leq 1.$$

We have enlarged C_2 to absorb fixed multiplicative constants. These exponents do not depend on B. In applying this published claim, one must account for the commensurable lattice resulting from rational conjugation; conjugation is not asserted to be an isometry of the original quotients. Its finite-index and metric losses are included in a, C_2.

Write $F_{j,k} = m_{j,k} + F^0_{j,k}$ by subtracting its Haar mean on H_j/Λ_j. Then $|m_{j,k}| \leq 1$, $\int F^0_{j,k} = 0$, and $\|F^0_{j,k}\|_{\mathrm{Lip}} \leq 2M^{C_2}$. This subtraction is needed even if the initial function F happened to have Haar mean zero. For a dimension-zero factor the centered function simply vanishes.

Put

$$U = C_2 + D + 5, \qquad Q_* = M^U, \qquad b_* = aB - C_2.$$

Since $M \geq L$, Q_* dominates the rationality, cell modulus and $N/|P_{j,k}|$. Applying (98) to all centered cell functions and using (101) gives a total of at most

$$CNL^{K_1} M^{-b_* c + C_1 U + C_2 + 2 + D}. \tag{102}$$

Every cell is counted here. In particular, the bound N in the minor-arc hypothesis has not been replaced by an unsupported cell cardinality.

We now fix B sufficiently large that

$$b_* > 1, \qquad b_* c \geq C_1 U + C_2 + 2 + D + A + K_1 + 4, \qquad b_* \kappa \geq U + 1 \tag{103}$$

when the restricted regime (99) is used. The last condition is omitted for an unrestricted minor-arc estimate. These inequalities involve only fixed exponents and not $E(B, m, d)$. They make (102) $O(NL^{-A-4})$. They also ensure $Q_* \leq (M^{-b_*})^{-\kappa}$. The other regime condition, $M^{-b_*} \geq N^{-\sigma}$, holds for all sufficiently large N: after fixing B the upper bound on M is a fixed power of $\log N$.

Only at this point use that upper bound, writing $M \leq L^{E_*}$ after enlarging the fixed exponent E_*. There are at most $O(L^{2E_*+D})$ cells of modulus at most L^{E_*}. Apply (97) with

$$C_0 = E_* + 1, \qquad A_0 = A + 2E_* + D + 4.$$

Summing the constant terms $m_{j,k} \sum_{n \in P_{j,k}} w_N(n)$ gives $O(NL^{-A-4})$. Combined with the centered estimates and freezing error, this proves (100). □

Remark A8.26 (Order of parameter choices). The order in the proof is

$$(A, K_w, C_{\mathrm{init}}, c, C_1, K_1, a, C_2, \kappa, \sigma) \longrightarrow (D, U) \longrightarrow B \longrightarrow E_* \longrightarrow (A_0, C_0) \longrightarrow N_0.$$

Thus there is no requirement that B dominate the factorization upper exponent $E(B, m, d)$. The centered progression estimate allows every logarithmic saving, so its exponent can be chosen after E_* is known. The minor-arc input must have a fixed power form such as (98); unspecified polylogarithmic specializations alone would not justify this argument.

Corollary A8.27 (Fixed-modulus centered nilsequence estimate)**.** *Fix the Gaussian ray field and a conjugation-invariant union C of ray classes as above. Let W be a fixed multiple of the conductor of $K^{\mathrm{ab}}/\mathbb{Q}$, and let b be a unit residue modulo W. For every $A, C_{\mathrm{init}} > 0$ and fixed dimension and degree bounds, uniformly over the nilmanifolds and polynomial sequences in Proposition A8.25 with an $(\log(2N))^{C_{\mathrm{init}}}$-rational adapted basis,*

$$\left|\sum_{n\le N} h_{C,b,W}(n)F(g(n)\Gamma)\right| \ll_{H,W,A,C_{\mathrm{init}},m,d} N(\log(2N))^{-A}\|F\|_{\mathrm{Lip}}.$$

The constants are uniform in b and in the polynomial orbit. No uniformity as W varies is asserted.

Proof. We check the three hypotheses of Proposition A8.25. The pointwise inequality $|h_{C,b,W}| \le \lambda_{b,W}$ gives its first hypothesis with $K_w = 1$. Corollary A8.9 gives its second hypothesis for every A_0, C_0, in fact for every interval progression.

For the third hypothesis, take $H_0 = H$ in Corollary A8.23. Its character expansion is (68), with the factor $1/2$ and the prime-power error supplied by (67) and the bound following that equation. Its two assumed minor-arc estimates are Proposition A8.18 and Corollary A8.19. Apply that corollary to the fixed linear combination defining $h_{C,b,W}$ in (73). It gives (98) with $c = c_W$, $C_1 = 1$ and $K_1 = 7$ in the regime $\delta \ge N^{-\sigma_1}$, $Q \le \delta^{-a}$. The affine-interpolation result initially restricts δ to a fixed sufficiently small interval. For larger δ, the bound $|h_{C,b,W}| \ll_W \log(2N)$ gives (98) directly after enlarging its fixed constant. For a dimension-zero quotient the mean-zero test function vanishes. For all other dimensions and degrees within the fixed bounds, take the minimum of the finitely many positive exponents c_W, a, σ_1 and the maximum of the fixed constants. Thus the restricted regime (99) holds with fixed positive κ, σ, uniformly throughout the required class.

Proposition A8.25 now proves the displayed estimate. There are only finitely many unit residues modulo the fixed W, so their constants can be bounded by a single constant. In particular, division by N and the relabeling $W = V$, $N = L$ give the fixed-complexity estimate (104). □

A8.8 From Gaussian nilsequence estimates to the polynomial count

Fix the Gaussian ray field $K/\mathbb{Q}(i)$ and the conjugation-invariant ray classes used in the preceding argument. Let D be divisible by the conductor of $K^{\mathrm{ab}}/\mathbb{Q}$. For each relevant union C of classes, let Λ'_C be its rational prime weight and let $\mu_C(b)$ be the conditional density in (109). In particular $0 \le \mu_C(b) \le 1$. We use the following conclusion of the preceding Gaussian norm, progression and factorization arguments, as assembled in Corollary A8.27: for every fixed multiple V of D, every unit $b \pmod V$, every fixed bounded-complexity nilsequence $F(g(n)\Gamma)$, and every $B > 0$,

$$\begin{aligned} |\mathbb{E}_{n\le L} h_{C,V,b}(n)F(g(n)\Gamma)| &\ll_{V,B,\mathrm{complexity}} (\log L)^{-B}, \\ h_{C,V,b}(n) &= \frac{\varphi(V)}{V}\big(\Lambda'_C(Vn+b) - \mu_C(b)\Lambda'(Vn+b)\big). \end{aligned} \tag{104}$$

The bound is uniform in the polynomial orbit and functions with the stated complexity bound. The field, its ray modulus and V are fixed in this statement. We do not assume any uniform rate as V varies.

Published transference inputs. We use Green–Tao, *Linear equations in primes* [50], Proposition 6.4 and Appendix D for the ordinary prime majorant, Proposition 7.1 and Appendix C for its generalized von Neumann bound, and Lemma 4.4 with its subsequent normal-form reduction. These are statements about the ordinary prime majorant and arbitrary functions dominated by

it; they do not assert an Artin-prime theorem. For the inverse step we use the relative inverse theorem [50, Proposition 10.1], together with the local proof of the bounded inverse theorem in Section I. The latter supplies GI(k) for every fixed $k \geq 3$, with the normalized interval norm and a fixed finite family of same-step linear nilsequences. Its supporting modules give the required constructions; the retained published inputs are specified there. The relative theorem applies to a real $f : [L] \to \mathbb{R}$, dominated by a nonnegative measure on $\mathbb{Z}/Q\mathbb{Z}$, where $Q \in [C_*L, 2C_*L]$ is prime and $C_* \geq 20$. For target norm $U^{k+1}[L]$, it requires $(k+2)2^{k+1}$-pseudorandomness in the sense of [50, Definitions 6.1–6.3], including both the full linear-forms condition and the correlation condition. A fixed positive norm lower bound then gives a fixed positive correlation with a 1-bounded Lipschitz linear nilsequence from a fixed finite family of k-step nilmanifolds. The family, metrics, Lipschitz bound and correlation threshold depend only on the fixed parameters, not on L or the function. Complex test functions in the bounded inverse theorem can be replaced by a real or imaginary part, losing only a fixed factor in the correlation. We use the normalized zero-extension interval norm, as in [50, Appendix B, (B.13)–(B.15)] and Section I.

Lemma A8.28 (Full ordinary-prime majorant with a fixed modulus factor)**.** *Fix positive integers E, T, an integer $S \geq 2$, and $C_* \geq 20$. Suppose $z = z(L) \to \infty$, $z \leq \log\log\log L$, and $V = E\prod_{p\leq z} p$. For each L let $b_1, \dots, b_T$ be any unit residues in $\{0, \dots, V-1\}$ and let $Q \in [C_*L, 2C_*L]$ be prime. There is a nonnegative S-pseudorandom measure ν on $\mathbb{Z}/Q\mathbb{Z}$ and a fixed $\alpha > 0$ such that, on $[L^{3/5}, L]$,*

$$\alpha \sum_{i=1}^{T} \frac{\varphi(V)}{V} \Lambda'(Vn + b_i) \leq \nu(n).$$

All error terms and domination constants are uniform in these unit residues. More explicitly, $\mathbb{E}\nu = 1 + o(1)$, and for $d, m \leq S$ and every finite-complexity system of m affine-linear forms in d variables with integer linear coefficients bounded by S,

$$\mathbb{E}_{x\in(\mathbb{Z}/Q\mathbb{Z})^d} \prod_{j=1}^{m} \nu(\psi_j(x)) = 1 + o(1),$$

uniformly also in their affine constant terms. For each $2 \leq m \leq S$ there is $\tau_m : \mathbb{Z}/Q\mathbb{Z} \to \mathbb{R}_{\geq 0}$ such that, for all shifts, including repetitions,

$$\mathbb{E}_x \prod_{j=1}^{m} \nu(x + h_j) \leq \sum_{i<j} \tau_m(h_i - h_j), \qquad \mathbb{E}\tau_m^q = O_{S,T,C_*,E,q}(1) \quad (1 \leq q < \infty).$$

Proof. We spell out the fixed-factor and normalization points in the construction of [50, Proposition 6.4 and Appendix D]. Fix the smooth even function

$$\chi(t) = \begin{cases} \exp(1 - (1-t^2)^{-1}), & |t| < 1, \\ 0, & |t| \geq 1, \end{cases} \qquad c_\chi = \int_0^1 |\chi'(t)|^2\, dt > 0.$$

We retain c_χ rather than impose a second normalization on χ. Put $\gamma = 1/(32S)$, $R = L^\gamma$, and

$$\Lambda_{\chi,R,2}(a) = \log R \left(\sum_{d|a} \mu(d)\chi(\log d/\log R) \right)^2, \qquad \widetilde{\nu}(n) = \frac{1}{Tc_\chi} \frac{\varphi(V)}{V} \sum_{i=1}^{T} \Lambda_{\chi,R,2}(Vn + b_i).$$

Set $\nu(n) = (1 + \widetilde{\nu}(n))/2$ on $[L] \subset \mathbb{Z}/Q\mathbb{Z}$ and $\nu(n) = 1$ outside it. These functions are nonnegative. For a prime $Vn + b_i > R$, the corresponding divisor square is $\log R$. Since $V = L^{o(1)}$, eventually $V \leq L$ and $\log(Vn + b_i) \leq 4\log L$ for $n \leq L$. Thus $\alpha = \gamma/(8Tc_\chi)$ proves the displayed domination.

Here is why the divisor-sum estimates are uniform with this fixed γ. In the proof of [50, Theorem D.3], a product of $m \le S$ squares introduces $2m$ divisors, each at most R. The residue-class lattice count has error $O_d(qL^{d-1})$, where $q \le R^{2m}$ is their common modulus. Summing its absolute values gives $O_{S,\chi}(R^{4m}L^{d-1}(\log R)^m) = o(L^d)$. This count is independent of the coefficients and affine constants of the forms. Consequently γ need not shrink with V. The remaining Euler-product argument is [50, Appendix D, (D.4)–(D.23), especially Proposition D.4]. For the present forms its constants are uniform: at $p \mid V$ every nonempty divisor event is empty, by the unit-residue condition; at $p \nmid V$, each nonzero linear form has one-equation density $1/p$. For the linear-forms condition, choose z larger than all nonzero two-by-two linear-coefficient minors that can occur. Every such minor has absolute value at most $2S^2$. Then every pair of distinct linear parts has rank two at $p > z$, so every nonvertical local event has density $O_S(p^{-2})$. No bound on affine constants is used here. The finitely many smaller primes are either in V or absorbed in the fixed initial threshold. The Taylor and Euler-product bounds therefore have constants depending only on S, χ; their exceptional prime factor is $\exp(O_S(\sum_{p\le z} p^{-1/2}))$. Here we include every $p \le z$ in the exceptional set, also for a one-form product; this covers its altered vertical local events. This times $(\log R)^{-1/20}$ tends to zero under the stated bound on z.

For all sufficiently large z, the prime divisors of E are at most z. Thus $\varphi(V)/V = \prod_{p\le z}(1-1/p)$, regardless of the exponents in E. The local factors at these primes are exactly $(1-1/p)^{-m}$ and cancel this normalizing factor to the mth power. The factors at $p > z$ have product $1 + o(1)$. The sieve factor for each square is c_χ by [50, Lemma D.2], and is cancelled by our factor c_χ^{-m}. Expanding the average over the T residues therefore gives the required $1 + o(1)$ for every subproduct of $\widetilde{\nu}$ on a convex region. To pass to the cyclic group, represent its coordinates in $[1, Q]$ and partition into the finitely many convex regions recording each wraparound and whether each form lies in $[L]$. The number of regions is bounded in terms of S, C_*, uniformly in affine constants. Expand $\nu = 1 + \frac{1}{2}1_{[L]}(\widetilde{\nu} - 1)$ on these regions. Every nonempty centered product has mean $o(1)$ by inclusion–exclusion. This proves the full cyclic linear-forms condition and its mass case.

For clarity we also verify the correlation part of Appendix D. On each of the finitely many wraparound intervals, expand the factors of ν and then their averages over the T residues. Distinct shifted forms $V(n + h_j) + b_{i_j}$ have exceptional primes outside z only among the divisors of $V(h_j - h_{j'}) + b_{i_j} - b_{i_{j'}}$. The local upper bound and the error bound in [50, Appendix D, (D.25) and the following estimates], followed by the arithmetic–geometric mean inequality over pairs, bound the result by a sum of fixed multiples of

$$H_A(u; a, b) = \exp\left(A \sum_{\substack{p>z \\ p \mid Vu+a-b}} p^{-1/2}\right), \qquad Vu + a - b \neq 0,$$

where A is fixed in terms of S. Terms with fewer than two active factors are covered by adding a fixed positive constant. All small-prime error factors are uniformly bounded by the preceding choice of z. Sum these bounds over $a, b \in \{b_1, \dots, b_T\}$ and over the possible integer lifts $u, u - Q, u + Q$ of a cyclic difference. This defines $\tau_m(u)$ for nonzero cyclic differences, after increasing a fixed multiplicative constant.

These weights have every fixed moment bounded. For fixed q, the exponential with Aq is bounded by a constant depending on A, q times $\prod_{p>z,\, p\mid Vu+a-b}(1 + p^{-1/4})$. Expanding this product reduces its average over any of the above integer intervals to

$$O_{A,q,C_*}\left(\sum_{d\ge 1} d^{-5/4} + \frac{1}{L}\sum_{d \ll_{C_*} VL} d^{-1/4}\right) = O_{A,q,C_*}(1).$$

Indeed all such divisors are coprime to V, so the Chinese remainder count is at most $O_{C_*}(L/d+1)$, uniformly in $a-b$. The second term is $O(V^{3/4}L^{-1/4}) = o(1)$. The finitely many summands and lifts preserve the fixed-moment bound. An integer equality $Vu + a - b = 0$, with $0 \le a, b < V$, requires $u = 0$ and $a = b$. To include repeated cyclic shifts, define instead $\tau_m(0) = (1 + \|\nu\|_\infty)^S$. The divisor bound $d(n) = n^{o(1)}$ gives $\|\nu\|_\infty = L^{o(1)}$, uniformly here, so this changes every fixed moment by $L^{-1+o(1)}$. For completeness that divisor bound follows by fixing $\epsilon > 0$: for all sufficiently large primes, $a + 1 \le p^{\epsilon a}$; the finitely many smaller primes contribute a fixed constant after slightly enlarging ϵ. Apply this to the prime factorization of $n \ll VL$ and then let ϵ decrease. This proves the correlation condition, including repetitions, and completes the construction. □

An auxiliary cutoff, distinct from the polynomial cutoff. Put

$$V_z = D \prod_{\substack{p \le z \\ p \nmid D}} p.$$

Here z is an auxiliary cutoff used only to prove the next linear replacement statement. It is not the cutoff v defining W, A in (112). For every fixed integer $s \ge 2$, one can choose $z = z(L) \to \infty$ sufficiently slowly that

$$\max_{C,\ (b,V_z)=1} \|h_{C,V_z,b}\|_{U^s[L]} = o(1). \tag{105}$$

The maximum is over the fixed finite list of ray-class unions and the finitely many unit residues for the current V_z.

Here are the normalization and limit details for the inverse step. For a required norm order $s \ge 2$, put $r = \max(s, 4)$ and $k = r - 1$. Choose a fixed integer $S \ge (r+1)2^r$ at least as large as the requirements of the intended generalized von Neumann bound. Choose the fixed ambient constant $C_* > 2^r$, $C_* \ge 20$, large enough for that bound. The number T of ordinary prime weights to be dominated at any one application is fixed by the finite system. All these choices precede a norm threshold δ.

For sufficiently large z our V_z has the form $E \prod_{p \le z} p$ with E fixed. Apply Lemma A8.28. It gives one fixed $\alpha > 0$ such that $f = \alpha 1_{[L^{3/5}, L]} h_{C,V_z,b}$ is dominated by the constructed ν on $[L]$, uniformly in the relevant unit residues. This uses $|h_{C,V,b}| \le (\varphi(V)/V)\Lambda'(Vn + b)$. For the inverse step one may take $T = 1$ and use a separate majorant for each residue; its constants are uniform. The fixed larger T is used when several weights enter the generalized von Neumann bound.

The removed initial interval has $U^s[L]$ norm $o(1)$. The weights are $O(\log L)$ since $V_z = L^{o(1)}$, and cubes whose initial vertex lies in that interval form an $O(L^{-2/5})$ fraction. Thus the 2^sth power of the removed norm is $O((\log L)^{2^s} L^{-2/5})$. The same interval contributes $o(1)$ to correlations with bounded tests. If the unscaled norm is at least $\delta > 0$, eventually $\|f\|_{U^s[L]} \ge \alpha\delta/2$.

Take $Q \in [C_*L, 2C_*L]$ prime and write $I = [L] \subset \mathbb{Z}/Q\mathbb{Z}$. The common ambient choice is valid for all interval norms up to order r. The finite-group Gowers–Cauchy–Schwarz inequality gives monotonicity of the group norms and $\|1_I\|_{U^s(\mathbb{Z}/Q\mathbb{Z})} \ge |I|/Q$, while $\|1_I\|_{U^r(\mathbb{Z}/Q\mathbb{Z})} \le 1$. Hence, for the zero extension $\widetilde{f}$,

$$\|f\|_{U^r[L]} = \frac{\|\widetilde{f}\|_{U^r(\mathbb{Z}/Q\mathbb{Z})}}{\|1_I\|_{U^r(\mathbb{Z}/Q\mathbb{Z})}} \ge \|f\|_{U^s[L]} \frac{L}{Q} \ge \frac{\alpha\delta}{4C_*}.$$

This does not require monotonicity with constant one for normalized interval norms. Apply [50, Proposition 10.1] with $k = r - 1$ and threshold $\min(\alpha\delta/(4C_*), 1/2)$; Section I supplies GI($k$). The measure has the required $(k+2)2^{k+1}$-pseudorandomness because it is S-pseudorandom. We obtain a

fixed positive correlation with a bounded linear nilsequence from a fixed finite family, independently of L, z, b and the ray-class union. This contradicts (104) along a sufficiently slow cutoff.

Here is the order of that cutoff choice. The finite set of required norm orders, the majorant complexities, C_*, T, χ, α and E are fixed first. For each order enumerate the tests $\delta = 1, 1/2, 1/3, \ldots$. Their inverse-theorem families and positive thresholds depend only on these fixed parameters and on δ. At stage j, choose a cutoff z_j larger than all fixed excluded primes and large enough that the limiting linear-forms errors for the finitely many bounded linear-coefficient types and subproducts are below $1/j$, uniformly in affine constants. Then fix V_{z_j} and choose L_j so that all remaining majorant errors are below $1/j$, the truncation bounds hold, and the nilsequence bounds for the first j tests and all unit residues modulo V_{z_j} are smaller than their fixed correlation thresholds. Enlarge L_j to ensure $z_j \le \log\log\log L$ for $L \ge L_j$. The uniform correlation-moment bounds of the lemma hold throughout; they are not inferred from the linear-forms errors. On these stages the full pseudorandom family tends to its required limit, so the qualitative relative inverse theorem applies to any hypothetical sequence violating (105). Take the maximum of the finitely many nilmanifold thresholds at each test. Their fixed smooth metrics are comparable to fixed Mal'cev metrics, and a linear orbit with any base point is a polynomial orbit. Thus the existing nilsequence estimate covers the supplied tests.

When this construction is used at the refined scale $L \asymp N/V_{z_j}$ below, enlarge a physical threshold N_j so that every constant multiple of N/V_{z_j} used by the normal-form reduction is at least L_j for $N \ge N_j$. The cutoff is constant on each physical-scale stage. No equation $z = z(N/V_z)$ is assumed. This proves (105) for the required orders.

Lemma A8.29 (Fixed-system linear replacement)**.** *Let $\Psi = (\psi_1, \ldots, \psi_t) : \mathbb{Z}^d \to \mathbb{Z}^t$ have fixed nonzero integer linear parts of finite complexity. Allow its constant terms to vary with N, subject to $\|\Psi\|_N \le L_0$ for a fixed L_0. Uniformly over convex regions $\mathcal{K} \subseteq [-N, N]^d$ with $\Psi(\mathcal{K}) \subseteq [1, L_0 N]^t$,*

$$\sum_{\mathbf{n} \in \mathcal{K} \cap \mathbb{Z}^d} \left(\prod_{i=1}^{t} \Lambda'_{C_i}(\psi_i(\mathbf{n})) - \prod_{i=1}^{t} \mu_{C_i}(\psi_i(\mathbf{n}) \bmod D) \Lambda'(\psi_i(\mathbf{n})) \right) = o(N^d). \tag{106}$$

The same conclusion holds for a mixed product in which any selected factors have already been replaced by their ordinary-prime models.

Proof. We can discard points where some $\psi_i \le N^{9/10}$. For a nonconstant fixed linear form their number is $O_{\Psi, L_0}(N^{d-1}(N^{9/10}+1))$; the product of weights is $O((\log N)^t)$. This is $o(N^d)$, uniformly in the constant terms. Apply the auxiliary V_z-trick to the variables: $\mathbf{n} = V_z \mathbf{m} + \mathbf{r}$. If one output residue is not a unit modulo V_z, its prime weight vanishes on the remaining range, since the output is larger than every prime dividing V_z. For an admissible residue vector write

$$\psi_i(V_z \mathbf{m} + \mathbf{r}) = V_z \widetilde{\psi}_{i,\mathbf{r}}(\mathbf{m}) + b_{i,\mathbf{r}}, \qquad 1 \le b_{i,\mathbf{r}} \le V_z.$$

The new linear coefficients are the original ones. Its normalized constant terms remain bounded at scale N/V_z. Because $D \mid V_z$, the factor μ_{C_i} is constant on this residue vector.

Telescope the difference of the two products. After normalization by $(\varphi(V_z)/V_z)^t$, each term has one factor h_{C_i, V_z, b_i}; all other factors are nonnegative and at most λ_{V_z, b_j}. Introduce a common ordinary-prime majorant for these at most t weights and truncate as above. Multiply every factor by a common fixed $\alpha_0 > 0$ so that each has absolute value at most that majorant, and apply the normal-form extension of Green–Tao Lemma 4.4 and Proposition 7.1. Here α_0 depends only on the fixed system, the chosen complexities and the fixed conductor, not on z or the residue vector. The small factor has the required norm by (105). Dividing the resulting estimate by α_0^t

changes only a fixed constant. The normal-form extension adds only a fixed number of variables with fixed coefficients; its convex-region reduction is the one following Lemma 4.4, and divides out their box volume. The error on each residue vector is therefore $o((N/V_z)^d)$, uniformly in the vector and convex region. The discarded initial segment stays outside the retained output range since $V_z = N^{o(1)}$.

One can make this reduction explicit. For each pair $i \neq j$ choose an integer vector v_{ij} with $\dot{\psi}_j(v_{ij}) = 0$ and $\dot{\psi}_i(v_{ij}) \neq 0$; fixed nonproportional linear parts ensure such vectors exist with fixed size. Replace $\mathbf{m}$ by $\mathbf{x} + \sum_{i \neq j} y_{ij} v_{ij}$ and allow each new y_{ij} to range over an integer interval of length comparable to N/V_z. The block y_{ij} for fixed i certifies normal form. The region defined by the original convex constraint on this sum and the new box is convex. The integer change $(\mathbf{x}, \mathbf{y}) \mapsto (\mathbf{x} + \sum y_{ij} v_{ij}, \mathbf{y})$ is bijective, so its sum is exactly the original sum times the number of points in the new box. Division by that number gives the asserted error. If there is only one form, choose an integer vector v with $\dot{\psi}_1(v) \neq 0$ and replace $\mathbf{m}$ by $\mathbf{x} + yv$. The same box and bijection argument gives the reduction in this case.

No growing factor is lost when summing residue vectors. If $\mathcal{R}_z$ is the set of admissible vectors, then

$$\frac{|\mathcal{R}_z|}{V_z^d} \left(\frac{V_z}{\varphi(V_z)} \right)^t = \prod_{p | V_z} \beta_p(\Psi) = O_{\Psi, L_0}(1). \tag{107}$$

Exclude the finitely many primes dividing a chosen nonzero coefficient of each linear part, and a chosen nonzero two-by-two minor for each pair of linear parts. At every remaining prime each output hyperplane has probability p^{-1}, and any two meet with probability p^{-2}. Hence $\beta_p = 1 + O_t(p^{-2})$, uniformly in the constant terms. This also covers $t = 1$, when there are no pair minors and the good-prime factor is exactly one. The finitely many remaining factors are bounded by $(p/(p-1))^t$. The Chinese remainder theorem proves the equality, also for the fixed prime powers dividing D. Multiplying the per-residue error by this bounded mass gives $o(N^d)$. The proof works unchanged for the mixed products. All auxiliary thresholds can be chosen uniformly over bounded constant terms and convex regions, because the published majorant and generalized von Neumann conclusions have this uniformity. □

The exact local cube needed in the appendix. Now fix the appendix cutoff v, and put W, A, n, R as in its proof. This is a new limiting argument; the auxiliary cutoff in the proof of Lemma A8.29 has already been eliminated from that lemma's statement. Let

$$f_{C,b}(x) = \frac{\varphi(W)}{W} \Lambda'_C(Wx + b) 1_{R \leq x \leq n}, \qquad g_{C,b} = f_{C,b} - \mu_C(b).$$

For fixed $a \geq 2$ and $C_1 < C_2 < C_3$, uniformly over $q \leq A^{C_1}$ and $nA^{-C_3} \leq J \leq nA^{-C_2}$,

$$\|g_{C,b}\|_{U^a_{q[J]}}^{2^a} \ll_{a, C_1, C_2, C_3, \rho, D} \frac{1}{\log v} + o_v(1). \tag{108}$$

The norm uses the difference multiset $q(h - h')$, $h, h' \in [J]$, on $\mathbb{Z}/n\mathbb{Z}$.

To prove this, remove the cutoff $x < R$ at cost $O((\log n)^{2^a} R/n) = o_v(1)$. Expand the centered cube and partition its domain according to every wrap-around integer m_ω. The relevant affine forms are precisely

$$W \left(x + q \sum_{r=1}^{a} \omega_r (h_r - h'_r) - m_\omega n \right) + b, \qquad \omega \in \{0, 1\}^a.$$

They have pairwise nonproportional linear parts. Each region is convex; there are $O_a(1)$ regions because $qJ/n < 1$. For fixed v, the number of possible q is finite and J/n stays in a fixed positive compact interval. Apply Lemma A8.29 to every subproduct and region. Its error is $o_v(n^{2a+1})$ uniformly in J, and division by nJ^{2a} gives $o_v(1)$. Since every output is b modulo D, all its conditional densities are the same $\mu_C(b)$. Thus the centered Artin cube equals $\mu_C(b)^{2^a}$ times the centered ordinary-prime cube, plus $o_v(1)$.

The ordinary cube is evaluated exactly as in Tao–Ziegler [14], Proposition 6, published pages 39–40, using the ordinary-prime linear correlation theorem, for example the published formulation in Bienvenu [4], Theorem 1.1, pages 79–80. The factors at $p \mid W$ cancel the normalization. For $p \nmid Wq$, pairwise independent gradients give local factor $1 + O_a(p^{-2})$, uniformly in the wrap constants. For $p \mid q$, $p > v$, the factor is $1 + O_a(p^{-1})$. Since $\log q = O_{C_1,\rho,D}(v)$, there are $O(v/\log v)$ such prime divisors. Their product is $1 + O(1/\log v)$, and the other large-prime factors give $1 + O(1/v)$. Centering cancels the common main term. Prime powers, if the ordinary theorem is stated for Λ, contribute $o_v(1)$ by their $O(\sqrt{n}\log^2 n)$ total weight and the fixed linear-fiber bound. This proves (108), and hence supplies (112) without an Artin linear-correlations theorem.

Completion by the published polynomial transfer. The remainder of the proof of Proposition A8.30, starting with its arbitrary-function dense-model step, now applies unchanged to these fixed Gaussian ray fields. Specifically use the published Tao–Ziegler [14] Theorem 11 and Corollary 3, then Theorem 9 and Theorem 13. The model for an arbitrary $f \le \lambda_{W,b}$ is used; one does not assert that an Artin weight has model one. Equation (108) shows that its bounded model, minus $\mu_C(b)$, has small global norms. The model is bounded by a fixed constant depending on the chosen sieve exponent. Divide its difference from $\mu_C(b)$ by a fixed bound for that difference before applying Theorem 9, whose hypothesis is a bound by one. After restoring this fixed factor, concatenation and model approximation give the required local norm bounds. None of these scaling constants depends on v.

Keep the original parameter order: local box order and coefficient bound first, then $B_0 > C + 2$, then the global order a, and finally small ρ. Only then let v increase. The quantities $M_0 = \lfloor T/W \rfloor$, $M_1 = \lfloor M_0/A \rfloor$ and $J = \lfloor A^{-2B_0 a} M_1^2 \rfloor$ satisfy the size hypotheses recorded in the appendix when $T/\sqrt{N} \ge 4\sqrt{W}/A$. For each fixed v the estimates are uniform for T in that allowed range. The final assertion $T = o(\sqrt{N})$ is imposed only along the subsequent diagonal $v = v(N) \to \infty$, not while v is frozen.

After $u = Wx + b$, $w = Wm + c$, the polynomials are $c_j W m^2 + 2c_j cm + (b + c_j c^2 - b_j)/W$, with distinct leading coefficients. The bounded-model transfer kills every term containing a centered factor. Output-boundary strips have width $O(T^2) = o(N)$. For completeness, dominate them first by strips of width ϵN. Here is the needed localization of the ordinary majorant estimate. This argument also covers negative coefficients c_j. For the true integer count use an enlarged period $n_* = \lceil BN/W \rceil$, where $B > 2 + \max_j |c_j|$ is fixed. We may assume $T \le \sqrt{N}$. Nonpositive outputs have prime weight zero, and every positive output is at most $(1 + \max_j |c_j|)N$. Discard also tuples for which some positive output is at most $W\lceil n_*^\rho \rceil + W$. For each w there are $O_k(Wn_*^\rho + W)$ such u, so their total weight is $O_k((Wn_*^\rho + W)T(\log N)^k) = o_v(NT)$, uniformly in the allowed T range. All remaining true outputs have their ordinary prime majorants on $[n_*]$ without wrap-around. Use the same localization calculation on the original period n to bound the artificial wrapped terms of the cyclic count. The parameters n_* and n differ by a fixed factor, so both applications satisfy the same polynomial-forms hypotheses. In the following calculation n denotes either chosen ambient period. Write $\nu_0(x)$ for the majorant factor with constant shift if there is one, and set $\nu_0 = 1$ otherwise. There is at most one such factor. Put $F(x) = \mathbb{E}_m \prod_i \nu_i(x + P_i(m))$, omitting that factor.

Tao–Ziegler, Theorem 15 in Appendix A (published page 57), applied to $\nu_0 F$, $\nu_0 F^2$ and ν_0, gives

$$\mathbb{E}_x \nu_0(x)(F(x)-1)^2 = o_{v\to\infty}(1) + o_v(1).$$

In the square use independent variables m, m'; the differences between all its nonconstant shift polynomials are nonconstant, including those from distinct copies. Weighted Cauchy–Schwarz now shows, for every interval I in the x variable,

$$\mathbb{E}_x 1_I(x)\nu_0(x)F(x) = \mathbb{E}_x 1_I(x)\nu_0(x) + o_{v\to\infty}(1) + o_v(1).$$

For fixed relative interval length the one-dimensional divisor-sum mean estimate gives $\mathbb{E}_x 1_I \nu_0 = |I|/n + o_{v\to\infty}(1) + o_v(1)$; it follows either directly from the ordinary majorant construction or from its linear-forms estimate on that fixed interval. The normalized prime weights are bounded by fixed multiples of their sieve majorants, with constants depending on the fixed sieve exponent. Thus a strip of width ϵN has total weight $[O(\epsilon) + o_{v\to\infty}(1) + o_{v,\epsilon}(1)]NT$, with that fixed domination factor included in the implied constants. The errors here have the same iterated order as the rest of the polynomial transfer. Then take $\epsilon \downarrow 0$ as described below. This avoids a spurious factor $(\log N)^k$ from a pointwise bound. The same positive majorant treats interval endpoints.

Choose increasing cutoffs v_j for which the global/local errors and $4\sqrt{W(v_j)}/A(v_j)$ tend to zero, and then choose size thresholds valid for all the finitely many residues, step values and transfer parameters at that cutoff. At stage j impose the boundary estimates for every $\epsilon \in \{1, 1/2, \dots, 1/j\}$, preserving all earlier tests. Let $v(N) = v_j$ between successive thresholds and take $\omega(N) \geq 4\sqrt{W(v(N))}/A(v(N))$, tending to zero sufficiently slowly. This gives the uniform assertion for $\omega(N)\sqrt{N} \leq T = o(\sqrt{N})$. Indeed, for an arbitrary such T, first fix k; eventually $T^2/N < 1/k$ and the estimate for the strip of width N/k applies. Take the upper limit as $N \to \infty$ and then $k \to \infty$. There is no requirement that $T^2/N < 1/j(N)$ along the chosen stages. The main term is precisely (111); the admissible residue mass is bounded, so summation does not amplify errors. The $u \in [N/2, N]$ version follows by using that convex interval directly, or by subtraction with thresholds enlarged for $N/2$.

This bridge uses the separately established Gaussian nilsequence input and published ordinary-prime transference. It uses no statement or estimate from Frei–Tinková as a theorem.

A8.9 Fixed Artin conditions in quadratic patterns

Here the field is fixed throughout the limiting argument. Let K be the ray class field of $\mathbb{Q}(i)$ for $H\mathbb{Z}[i]$, where $16 \mid H$. Put $G = \mathrm{Gal}(K/\mathbb{Q})$, and let $C_1, \dots, C_k$ be conjugation-invariant subsets of $\mathrm{Gal}(K/\mathbb{Q}(i))$, regarded as conjugacy-invariant subsets of G. Let D be divisible by the conductor of K^{ab}. For a unit $b \pmod D$, define

$$\mu_j(b) = \frac{\#\{g \in C_j : g|_{K^{\mathrm{ab}}} = \sigma_b|_{K^{\mathrm{ab}}}\}}{[K : K^{\mathrm{ab}}]}, \qquad \sigma_b(\zeta_D) = \zeta_D^b, \tag{109}$$

and set $\mu_j(b) = 0$ for nonunits. Define $\Lambda'_j(p) = \log p$ for unramified primes with arithmetic Frobenius in C_j, and zero otherwise.

Proposition A8.30 (Fixed-field polynomial count)**.** *There is a function $\omega(N) \to 0$, depending on the fixed data, such that whenever $\omega(N)\sqrt{N} \leq T = o(\sqrt{N})$,*

$$\sum_{u\leq N} \sum_{1\leq w\leq T} \prod_{j=1}^{k} \Lambda'_j(u + c_j w^2) = \mathfrak{S}\, NT + o(NT), \tag{110}$$

where

$$\mathfrak{S} = \Big(\frac{D}{\varphi(D)}\Big)^k \mathbb{E}_{u,w \bmod D} \prod_j \mu_j(u + c_j w^2) \prod_{p \nmid D} \beta_p, \quad \beta_p = \mathbb{E}_{u,w \bmod p} \prod_j \frac{p}{p-1} 1_{p \nmid u + c_j w^2}. \tag{111}$$

The same statement on $N/2 \le u \le N$ has main term $\mathfrak{S}NT/2$.

Proof. The Gaussian linear replacement of Lemma A8.29 and its local-cube consequence (108) supply the analytic input. We record the normalization and polynomial-transfer parameters here.

Choose a small positive ρ and first keep v fixed. Put

$$W = D \prod_{\substack{p \le v \\ p \nmid D}} p, \quad A = W^{1/\rho}, \quad n = \lfloor N/W \rfloor, \quad R = n^\rho,$$

and, for unit $b \pmod W$, put

$$f_{j,b}(x) = \frac{\varphi(W)}{W} \Lambda'_j(Wx + b) 1_{R \le x \le n}, \qquad g_{j,b} = f_{j,b} - \mu_j(b).$$

View these functions on $\mathbb{Z}/n\mathbb{Z}$, with representatives $1, \dots, n$. If $a \ge 2$ and $C_1 < C_2 < C_3$ are fixed, (108) gives

$$\|g_{j,b}\|^{2^a}_{U^a_{q[J]}} \ll_a (\log v)^{-1} + o_v(1), \quad q \le A^{C_1}, \quad nA^{-C_3} \le J \le nA^{-C_2}. \tag{112}$$

The implicit constant may also depend on the other fixed complexities and the fixed ray field. The norm, wrap-around regions, conditional densities, and order of the auxiliary cutoff are all treated explicitly in Section A8.8.

We now use the arbitrary-function dense-model theorem of Tao–Ziegler [14]. Their published Theorem 11 and Corollary 3 give a single bounded model accurate in both global Gowers norms and the local polynomial box norms. Their Theorem 9 is the bounded-function concatenation statement, and their Theorem 13 bounds the polynomial average by one of those local norms. All majorants here are the ordinary prime majorants, since $0 \le \Lambda'_j \le \Lambda'$. The fixed factor D in W is allowed in their divisor-sum construction: for sufficiently large v, $W = E \prod_{p \le v} p$ with a fixed E whose prime factors are at most v. A unit residue makes every small-prime divisibility condition identically false, including at the extra prime powers. At the remaining primes the divisor-sum local factors are unchanged. Polynomial coefficient bounds absorb the fixed factor E. Thus the polynomial-forms estimates for the majorant have the same $1 + o(1)$ conclusion.

The following choice records the size hypotheses of these results. First choose their local box order d' and a coefficient exponent C; choose an integer $B_0 > C + 2$, then a sufficiently large global order a, and finally a sufficiently small ρ. Set $\rho_* = \rho/B_0$ and $\mathcal{A} = A^{B_0}$. Put $M_0 = \lfloor T/W \rfloor$, $M_1 = \lfloor M_0/A \rfloor$. For $T/\sqrt{N} \ge 4\sqrt{W}/A$ one has

$$A^{-1}\sqrt{n} \le M_0 \le \sqrt{n}, \qquad A^{-3}\sqrt{n} \le M_1 \le \sqrt{n}.$$

The model at cutoff n^{ρ_*} dominates the function already truncated at n^ρ. Corollary 3 supplies approximation for the degree-one local step polynomials, whose coefficients are $O(A^C)$, and simultaneously for $q \le A^{B_0 a}$ with $J = \lfloor A^{-2B_0 a} M_1^2 \rfloor$. In particular

$$nA^{-2B_0 a - 7} \le J \le nA^{-2B_0 a}.$$

This is within (112). Let $f'_{j,b}$ be the model supplied by Corollary 3 at cutoff n^{ρ_*}. Its bound is $0 \le f'_{j,b} \le B_* - 1$, where $B_* \ge 2$ depends on the fixed ray field, complexities and on ρ_*, but

not on v, n, b or the polynomials within the stated coefficient bounds. This is the bounded model of Tao–Ziegler's Proposition 4: its construction first rescales the prime weight by a fixed positive constant and then divides the model by that constant. In particular, it need not be bounded by one. Since $0 \le \mu_j(b) \le 1$, the function

$$F_{j,b} = \frac{f'_{j,b} - \mu_j(b)}{B_*}$$

is bounded in magnitude by one. The triangle inequality, the global part of Corollary 3, and (112) show that its required global norms tend to zero under the diagonal limit below. Apply Theorem 9 to $F_{j,b}$ with parameters $d = d_0 = 2$, $M = M_1$, $A = \mathcal{A}$, and global order a. Its smallness threshold is independent of $\mathcal{A}$; B_* has already been fixed before v is chosen. Multiplication by B_* and the local part of Corollary 3 therefore show that the local norm of $g_{j,b}$ tends to zero. Theorem 13, with the original parameter A and coarse scale M_0, makes every polynomial average containing a centered factor $o(1)$. These are the reasons for choosing the complexities before v.

For the final W-trick write $u = Wx + b$ and $w = Wm + c$. If b_j is the residue of $b + c_j c^2$ modulo W, the resulting integer polynomial is

$$c_j W m^2 + 2c_j c m + (b + c_j c^2 - b_j)/W.$$

Its leading coefficient differences are nonzero, and its coefficients are $O(W)$. Expand the product into its means and centered factors and apply the preceding conclusion. End intervals of length $O(T^2) = o(N)$ are controlled by the weighted second-moment localization and the fixed-strip argument in Section A8.8. Summing residue classes does not amplify the error: their normalized admissible mass is $\prod_{p|W} \beta_p = O(1)$. Indeed, outside the fixed coefficient-difference primes,

$$\beta_p = \left(\frac{p}{p-1}\right)^k \frac{(p-1)(p-k+1)}{p^2} = 1 + O_k(p^{-2}).$$

This also proves convergence of the product in (111). Let N tend to infinity for each fixed v and then let v tend to infinity sufficiently slowly. A diagonal choice supplies $\omega(N)$ and yields (110). Subtract the corresponding formula with upper limit $N/2$, enlarging ω if necessary, to obtain the last assertion. □

Remark A8.31 (Published numbering). The labels in the published 2018 paper differ from those in arXiv 1603.07817v1. The relevant dictionary is

arXiv	Theorem 1.4	Theorem 2.5	Theorem 3.1	Corollary 3.9	Theorem 4.1
published	Theorem 4	Theorem 9	Theorem 11	Corollary 3	Theorem 13

The published locations are respectively pages $4, 21, 28, 38, 41$. The preprint's equations (3.5) and (4.1) are published equations (36) and (57). The dense model used above is the theorem's assertion for an arbitrary function bounded by the prime weight. The special model 1 for the ordinary prime weight is not an assertion about Artin weights.

A8.10 The ray-class lower bound

For ray class fields we use the published existence and reciprocity statements in Kopp–Lagarias [15], Theorems 1.1, 1.3 and 7.2, and the modulus exact sequence of Theorem 6.5, specialized to the

maximal order $\mathbb{Z}[i]$. There are no real places. This replaces an appeal to unpublished course notes. In this principal ideal domain the ray group for the finite modulus $H\mathbb{Z}[i]$ is

$$(\mathbb{Z}[i]/H\mathbb{Z}[i])^\times/\{1,-1,i,-i\}. \tag{113}$$

It also follows directly by sending a residue r to the ideal class of (r): surjectivity is principality, and the kernel consists exactly of unit residues. The Artin map sends this ideal class to arithmetic Frobenius. Conjugation acts by $r \mapsto \bar{r}$.

Choose a fixed L_1 larger than all coefficient-difference primes and larger than 11. Let H_0 be divisible by $16 \cdot 7M$ and by all odd primes at most L_1. Prescribe a unit residue $r \pmod{H_0}$ which is 9 modulo 16, $1+3i$ modulo 7, and i modulo every other odd prime dividing H_0. Compatible higher powers can be chosen arbitrarily. The associated rational-prime Frobenius condition is the two-element conjugacy class $C_0 = \{h_r, h_{\bar{r}}\}$ in the Galois group G_0 of the ray field K_0 over $\mathbb{Q}$. The two elements are distinct: an equality of their ray classes would require $r \equiv \epsilon\bar{r} \pmod{H_0}$ for a Gaussian unit ϵ; modulo 16 this forces $\epsilon = 1$, contradicted modulo 7. The field is Galois over $\mathbb{Q}$, because conjugation preserves the modulus and the defining ray subgroup.

Let D_0 be the conductor of K_0^{ab} and choose a unit $b_0 \pmod{D_0}$ inducing the restriction of h_r. The value in (109) is then $\mu_0 = |C_0|/|G_0'| > 0$. For $L \geq L_1$ let $T_L = \prod_{L_1 < \ell \leq L,\, \ell \nmid H_0} \ell$ and let K_L be the ray field for $H_0 T_L \mathbb{Z}[i]$. For each new prime write

$$B_\ell = (\mathbb{Z}[i]/\ell\mathbb{Z}[i])^\times, \qquad N_\ell = \ker(B_\ell \xrightarrow{\mathrm{N}} \mathbb{F}_\ell^\times).$$

The norm map is onto and $|N_\ell| = \ell - \chi(\ell)$.

Lemma A8.32 (Abelian fibres and their surviving proportion)**.** *Put $N_L = \prod_{\ell \mid T_L} N_\ell$ and $G_L = \mathrm{Gal}(K_L/\mathbb{Q})$. Then*

$$N_L \subseteq G_L', \qquad G_L'/N_L \simeq G_0', \qquad K_L^{\mathrm{ab}} = K_0^{\mathrm{ab}}\mathbb{Q}(\zeta_{T_L}).$$

Let C_L consist of those elements above C_0 whose normalized Gaussian residue z satisfies $\mathrm{N}(z-1)\mathrm{N}(z+1) \not\equiv 0 \pmod{\ell}$ at every $\ell \mid T_L$. For the density μ_L of this set,

$$\mu_L(b) \geq \mu_0 \prod_{\ell \mid T_L} \left(1 - \frac{4}{\ell - 1}\right) \qquad (b \equiv b_0 \pmod{D_0},\ (b, D_0 T_L) = 1). \tag{114}$$

Proof. In the CRT presentation of (113), the unit group embeds in the H_0 factor, because it already embeds modulo 16. Once the representative above r or $\bar{r}$ is normalized, the kernel of reduction to the base ray group is exactly $\prod B_\ell$. There is no further quotient by the four units.

The map $z \mapsto z/\bar{z}$ from B_ℓ onto N_ℓ is surjective. For split primes this is $(x,y) \mapsto (x/y, y/x)$. For inert primes it is $z \mapsto z^{1-\ell}$ in the cyclic group $\mathbb{F}_{\ell^2}^\times$. These maps are commutators with complex conjugation, so $N_L \subseteq G_L'$. The norm on each new residue factor gives the cyclotomic action $\zeta_\ell \mapsto \zeta_\ell^{\mathrm{N}(z)}$. This is also seen by applying arithmetic Frobenius to ζ_ℓ; it is compatible with the ray map by reciprocity. The reduction map gives $K_0 \subseteq K_L^{N_L}$. The cyclotomic action just computed is trivial on N_L, so also $\mathbb{Q}(\zeta_{T_L}) \subseteq K_L^{N_L}$. The fields K_0 and $\mathbb{Q}(\zeta_{T_L})$ are disjoint over $\mathbb{Q}$. Indeed, K_0 is unramified at every prime dividing T_L, whereas the inertia groups at these primes generate the Galois group of $\mathbb{Q}(\zeta_{T_L})/\mathbb{Q}$. Thus their intersection is trivial. Moreover, the kernel computation above gives

$$[K_L^{N_L} : \mathbb{Q}] = [K_0 : \mathbb{Q}] \prod_{\ell \mid T_L} \frac{|B_\ell|}{|N_\ell|} = [K_0 : \mathbb{Q}]\varphi(T_L).$$

This is the degree of $K_0\mathbb{Q}(\zeta_{T_L})$, proving

$$K_L^{N_L} = K_0\mathbb{Q}(\zeta_{T_L}), \qquad G_L/N_L \simeq G_0 \times \prod_{\ell \mid T_L} \mathbb{F}_\ell^\times.$$

Since $N_L \subseteq G_L'$, taking commutators in this quotient gives $G_L'/N_L \simeq G_0'$ and $K_L^{\mathrm{ab}} = K_0^{\mathrm{ab}}\mathbb{Q}(\zeta_{T_L})$.

Fix a nonzero norm $s \in \mathbb{F}_\ell^\times$. For $z = a + bi$ in that norm fibre, $\mathrm{N}(z \pm 1) = s + 1 \pm 2a$. For each sign, vanishing fixes a and leaves at most two values of b. Thus at most four of the $\ell - \chi(\ell)$ elements are excluded. The conditions are invariant under conjugation, so C_L is a union of rational conjugacy classes. In every compatible abelian fibre its numerator in (109) is at least $|C_0| \prod(\ell - \chi(\ell) - 4)$, and its denominator is $|G_0'| \prod(\ell - \chi(\ell))$. Their quotient proves (114). □

Proposition A8.33 (Finite-cutoff lower bound)**.** *There is a constant $c > 0$ independent of L such that the singular series in Proposition A8.30 for the fixed field K_L and the set C_L at every position satisfies*

$$\mathfrak{S}_L \geq c(\log L)^{-4k}.$$

Proof. At the base modulus retain just $u = b_0$, $w = 0 \pmod{D_0}$. At each new prime retain all pairs on which the k norm values are units. Apply (114) separately to each position. The norm-fibre denominators cancel as shown in the previous lemma; a factor inverse to the degree of K_L is not left in the bound. The normalized proportion of the retained pairs at each new prime is $\beta_\ell = 1 + O_k(\ell^{-2})$. Each is positive, since $w = 0$, $u \neq 0$ is allowed. Their product, together with the tail Euler product, is bounded below by a fixed positive number. Williams' theorem with modulus 1, or its product for the two reduced classes modulo 4, gives

$$\prod_{\ell \mid T_L} (1 - 4/(\ell - 1))^k \gg (\log L)^{-4k}.$$

The base modulus contributes another fixed positive factor. □

A8.11 The final order of limits

Proof of Theorem A8.1. Fix the coefficients, M, and the base modulus first. Choose one fixed $L \geq \max(L_1, L_*)$ so large that

$$\frac{C}{L \log L} < \frac{c}{32(\log L)^{4k}},$$

where C, c are from the obstruction and lower-bound propositions. Now fix K_L, C_L. Only after this choice do we use the asymptotic polynomial theorem or let its small-prime cutoff tend to infinity.

Choose numbers $\tau_j \downarrow 0$, all at most τ_0. Choose successively N_j so large that for $N \geq N_j$ $\omega(N) \leq \tau_j/2$, that $\tau_j\sqrt{N} \geq 2$, and that the error term in (66) is less than $c\tau_j N^{3/2}/(32(\log L)^{4k})$. This is possible for each fixed τ_j. For $N_j \leq N < N_{j+1}$ use $\tau(N) = \tau_j$ and define $T = \lfloor \tau(N)\sqrt{N} \rfloor$. Then $T = o(\sqrt{N})$ and $T \geq \tau\sqrt{N}/2$. Apply Proposition A8.30 to this piecewise choice of T. For all sufficiently large N its lower count is at least $cNT/(4(\log L)^{4k})$ and exceeds the full obstruction tail. Hence a pattern remains with no forbidden odd common divisor greater than L. The ray conditions exclude every odd divisor at most L from each individual adjacent norm. At 7 the adjacent norms of $1+3i$ are 2 and 6; at every other odd base prime the adjacent norms of i are 2. Thus no forbidden odd common divisor remains at all.

The residue modulo 16 gives, on writing $a = 9 + 16A$ and $b = 16B$,

$$(a+1)^2 + b^2 \equiv 4 \pmod 8, \qquad (a-1)^2 + b^2 \equiv 64 \pmod{128}.$$

These yield the exact two-adic valuations and hence the cross-sign gcd 4. The residue modulo M gives coprimality with M. Finally $u \geq N/2$ and $w \leq \tau\sqrt{N}$ imply $u/w^2 \geq 1/(2\tau^2) \to \infty$. For sufficiently small τ_0, all $u + c_j w^2$ are between fixed positive multiples of N, so they exceed Y when N is large. The dilation at the start imposes $q \mid w$. □

A9 The odd-product quartics in 523 and 540 variables

The following construction incorporates the November 7 contribution of Gabriel Conant, developed with ChatGPT. We give its arithmetic argument, including the connection to the results proved above. Variable counts include the input. The older quartics remain as source and proof baselines.

Theorem A9.1 (The Version 9 quartics)**.** *There are specified integer polynomials P_{523} and P_{540}, of total degree four in respectively 523 and 540 variables, such that*

$$q \in \mathbb{Z} \iff \exists v \in \mathbb{Q}^{522} : P_{523}(q, v) = 0 \iff \exists w \in \mathbb{Q}^{539} : P_{540}(q, w) = 0.$$

The first has 35,163 nonzero monomials and largest coefficient magnitude $64 \cdot 3^{2^{27}}$. Its coefficient table uses the fixed integer $S_0 = 3^{2^{26}}$. The second has 35,213 nonzero monomials and largest coefficient magnitude $3^{2048} + 1005072314631333 < 2^{3247}$. Its complete decimal table occupies 708,848 bytes. Both have content one.

The proof is the composition of the source equivalences, local and global arguments, and exact polynomial certificates below. The intermediate 554-variable construction is included to separate the odd-product argument from the additional formal-depth refinement. It is not a further claimed improvement over the two polynomials in the theorem.

A9.1 Inherited inputs and the exact changes of convention

Use the normalized curve and generator of Section 17.1, with $P = P_*$, not a new generator. The coefficient basis is $(1, \sqrt{2})$ and conjugation sends $a + b\sqrt{2}$ to $a - b\sqrt{2}$. At unramified finite places $v(p) = 1$; at ramified places the integral valuation has value group $\mathbb{Z}$. Heights are absolute logarithmic Weil heights, with the local degree weights divided by $[K : \mathbb{Q}] = 2$. Throughout this section set $\mathcal{H} = H_{19}$ of Theorem 15.2.

The source is the system in Section 17.2, with the following explicit changes. Replace the comparison targets $[4]P_*, [4]Q_i$ by P_*, Q_i; replace the orders 16 and 52 by 12 and 40 before the further refinement below. Retain the sixteen-test product, the same five contacts, the scalar norm tree with orders $(1, 1, 1, 1, 32, 32)$, the Boolean correction and the preparation-zero pole equations. Change β to $2X^2 + 3$; use the point $S - P$ only in the scalar and physical-anchor blocks; and change the fixed coefficient S to $S_0 = 3^{2^{26}}$. The point-chart replacement, the three new scales, and the final order-one/order-forty test list are given below. All unchanged quotient guards and permitted zero values of pole witnesses are retained. This description, the displayed equations below, and the pinned rational straight-line program specify the source without a choice of arithmetic witnesses.

Input	Exact result used and translation
Elliptic labels and domains	Section 17.1, Lemma 17.1, and (19): $2E(K) = \mathbb{Z}P_*$, nonzero coordinate domains, and the same lower bound $h(x([n]P_*)) \geq n^2 - 6$.
Height of rational contacts	Theorem 15.2, with $\mathcal{H} < 2^{2093966}$, applied through the fractional linear map and contact calculation of Lemma 17.6. Only that calculation is reused; its exponent 24 is replaced by 19.
Pole obstruction and realization	Lemmas 14.4, 14.6, and 17.4. These include the restricted-converse dependency proved in the earlier arithmetic sections; it has not been discarded.
Scalar and anchor	Lemma 16.1, with its proof applied to $2X^2 + 3$ and $S - P$, and Lemma 17.3 with the smaller exponent calculated just below.
Local comparisons	Lemma 17.2 and the recurrence of Lemma 15.4; orders and target changes are justified below, not inferred by renaming indices.
Global choices	Theorem A8.1, Lemma 17.7, Proposition A8.30, Section A8.11, and Lemma 16.3. All fixed data precede the asymptotic prime choice.
Final contradictions	Lemma 14.9 and the explicit new height inequalities below.

The published results behind these inputs retain the versions and convention qualifications in the theorem-application inventory. The new contribution does not supply an assumed replacement for the restricted converse, the height theorem, or the Gaussian prime argument.

Lemma A9.2 (Direct targets and smaller anchor)**.** *For the changed source above, the local comparison and completeness arguments remain valid with direct targets. The scalar anchor on $R = [a-1]P_*$ implies*

$$0 \neq a - 1 \in 2\,3^{2^{24}-1}\mathbb{Z}, \qquad \log_2 |a-1| > 25165823.5, \qquad h(\beta) \leq \mathcal{H}|a-1|^{19/32}.$$

It permits arbitrarily divisible positive scalar indices $a - 1$.

Proof. For direct targets set $V_0 = H - P_*$ and $V_i = U_i - Q_i$, with $H = [h]P_*$, $U_i = [u_i]P_*$ and $Q_i = [n_i]P_*$. The quotient guards give finite nonidentity points. At a consumed prime the full logarithm ratios are u_i/h and $(u_i - n_i)/(h-1)$. One of $h, h-1$ is a unit. As in Lemma 17.5, the cross congruence first makes both approximate ratios integral, and then makes their relative errors absolute. Subtraction of the two integral congruences gives $L_i \equiv n_i$, without division by four. The logarithm recurrence and denominator bound in Lemma 15.4 give the same estimates at orders 12 and 40. The test list below strengthens the input order to 40 in the final refinement.

For completeness, the reduced elliptic groups have all four rational two-torsion points, since $p > 76176$ and the three roots $0, \pm d$ are distinct. Thus their doubling images have order $d_\pm(p)/4$. The point P_* is itself a double, so these integers annihilate its reductions. The prime theorem and Lemma 17.7 therefore give, on setting $M_\pm = \operatorname{lcm}_p(d_\pm(p)/4)$,

$$\gcd(M_-, M_+) = 1, \quad \gcd(M_-, L) = 16, \quad \gcd(M_+, L) = 1.$$

Here $64 \mid L$, the exact two-parts are 2^6 and 2^2 before division by four, and all odd prime divisors of L were avoided. Consequently $h \equiv 0 \pmod{M_-}$, $h \equiv 1 \pmod{M_+}$ and $h \equiv h_0 \pmod{L}$ are compatible when $64 \mid h_0$. The real-progression argument supplies large positive choices and avoids the finite forbidden point and quotient loci. The later prime-separation lemma justifies adding the factor Π^{19}.

For the anchor, repeat the proof of Lemma 17.3 with $C = 3^{2^{25}}$ and $\mathsf{S} = C^2 = S_0$. Its quaternion criterion and anisotropic binary form are unchanged. Thus the two rational coefficients of $f(R)/C$

belong to $\mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. The exact local calculation of Section 14.2.2 now gives $a-1 \in 2\,3^{2^{24}-1}\mathbb{Z} \setminus \{0\}$, and

$$\log_2 |a-1| \geq 1 + (2^{24}-1)\log_2 3 > 1 + \tfrac{3}{2}(2^{24}-1) = 25165823.5.$$

For the scalar proof the new label is at least three, its five-contact product never vanishes, and all its negative valuations at the consumed odd primes are even. These are exactly the properties of X^2+2 used in Lemma 16.1; they hold for $2X^2+3$ by the calculation below. That proof therefore gives $M_5(\beta)^{32} \mid a-1$. Applying Theorem 15.2 gives the asserted exponent 19/32. Its finite correction and continuity arguments apply to the same norm tree with the scalar point R; choosing $a-1$ sufficiently divisible meets the anchor and any further prescribed divisibility. □

A9.2 An exact replacement of the point chart

Put $K = \mathbb{Q}(\sqrt{2})$, $\delta = 3 - \sqrt{2}$ and $D = \delta^2 = 11 - 6\sqrt{2}$. The old chart, with letters in K, is

$$u = t^2, \qquad x = h^2, \qquad u(x - u/4) = D, \qquad y = h(u/2 - x). \tag{115}$$

The new chart is

$$x = h^2, \qquad j^2 = x^2 - D, \qquad y = hj. \tag{116}$$

An equation over K means its two rational component equations. The last expression in (116) need not be assigned its own coordinates.

Lemma A9.3. *If $U, V \in K$, $U \neq 0$, and $V^2 = U^3 + 4DU$, then U is a square in K.*

Proof. The ring $\mathcal{O} = \mathbb{Z}[\sqrt{2}]$ is a PID: rounding both coefficients of an element of K to nearest integers leaves a remainder of absolute norm at most $1/2$, giving a Euclidean algorithm. Its units are $\pm(1+\sqrt{2})^n$, and every totally positive unit is a square. For completeness, multiply a positive unit by a power of $1+\sqrt{2}$ to place its first real embedding in $[1, 1+\sqrt{2})$. Its other embedding is the positive or negative reciprocal. Writing the unit as $a + b\sqrt{2}$ then forces $a, b \in \{0, 1\}$, and the norm condition and excluded upper endpoint leave only the unit 1. The two signs of the fundamental unit under the real embeddings give the assertion about totally positive units.

Let $m = v_{\mathfrak{p}}(U)$, with integral normalization. Away from $(\sqrt{2})$ and (δ) the two terms U^3 and $4DU$ have valuations $3m$ and m. If m is odd, their unequal minimum is odd, a contradiction. At $(\sqrt{2})$ these valuations are $3m$ and $m+4$, with equality only at the even value $m = 2$. Again m cannot be odd. At (δ) they are $3m$ and $m+2$, with equality only at $m = 1$. In that exceptional case write $U = \delta u$ with u a local unit. Then

$$V^2 = \delta^3 u(u^2 + 4).$$

The residue field is $\mathbf{F}_7$, in which -4 is a nonsquare. Thus the right side has valuation three, also impossible. Every finite valuation of U is even.

Under each real embedding, $D > 0$ and $U(U^2 + 4D) = V^2$, so $U > 0$. By the PID property, $U = \gamma^2 \eta$ for $\gamma \in K^*$ and a unit η. This unit is totally positive, hence a square. □

Proposition A9.4. *The charts* (115) *and* (116) *have exactly the same projected pairs $(x, y) \in K^2$. Every solution of the new chart has $h, j, x, y \neq 0$, and x is positive under both real embeddings.*

Proof. Given (115), put $j = t^2/2 - h^2$. Then

$$j^2 = h^4 - t^2h^2 + t^4/4 = h^4 - D,$$

and $y = hj$. Conversely, given (116), set $U = 2(h^2 + j)$ and $V = 2hU$. The identity $j^2 = h^4 - D$ gives

$$U^2 - 4h^2U + 4D = 0,$$

hence $V^2 = U^3 + 4DU$. Also $U \neq 0$, since $U = 0$ would imply $D = 0$. By Lemma A9.3, choose $t \in K$ with $t^2 = U$. It follows that

$$t^2(h^2 - t^2/4) = D, \qquad h(t^2/2 - h^2) = hj,$$

as required.

If $h = 0$, then $j^2 = -D$, contradicting either real embedding. If $j = 0$, then $h^4 = D$, so $h^2 = \pm\delta$. Taking norms yields $\mathrm{N}(h)^2 = 7$, impossible for $\mathrm{N}(h) \in \mathbb{Q}$. The remaining assertions follow immediately. □

The proof is specific to this field and this value of D. It does not use the generally false assertion that square abscissa characterizes doubles on every elliptic curve. It also avoids adding a Mordell–Weil rank or generator hypothesis to the chart replacement.

A9.3 Using the chart without ordinate variables

All nine point modules use the new chart. It has six rational coordinates: the two components of each of h, j, x, subject to four quadratic component equations. The old module had ten coordinates, including the components of t, h, u, x, y. Thus the change saves four coordinates per point, provided each downstream ordinate is consumed as the expression hj.

For the difference of chart points Q, R, retain the existing inverse equations for $x_Q - x_R$ and define the slope by

$$(x_Q - x_R)m = h_Q j_Q + h_R j_R.$$

Its numerator and defining equation are quadratic in the available coordinates. Define the difference abscissa by the existing square of m, and its ordinate by

$$y_{Q-R} = m(x_Q - x_{Q-R}) - h_Q j_Q.$$

This again is a quadratic assignment. The retained inverse ensures that the slope has exactly its old value.

For a chart point,

$$z = -x/y = -h/j.$$

Thus the local parameter is defined by $jz = -h$. When a logarithm value is combined with its parameter, replace $y\ell = -xG$ by $j\ell = -hG$, where G is the unchanged truncated logarithm polynomial. At the scalar ratio replace

$$x_H\rho = -y_H z_S \qquad \text{by} \qquad h_H\rho = -j_H z_S.$$

All cancellations are reversible by Proposition A9.4. Difference-point denominators and all other guards remain unchanged. The fixed target has

$$h = -1 + \tfrac{7}{4}\sqrt{2}, \qquad j = \tfrac{39}{8} - \tfrac{9}{2}\sqrt{2};$$

these give exactly its previous abscissa and ordinate.

Together with Proposition A9.4, these identities give rational witness maps in both directions. To reconstruct the old chart, choose the square root supplied by Lemma A9.3; then evaluate all old assignments. The choice of sign does not alter (x, y) or any downstream value.

A9.4 Shorter weight circuits

For each of nine detectors, write e for its nonzero weight and q for its order-one weight. The required outputs are $q_{12} = qe^{22}$ and $q_{40} = qe^{78}$. The old construction uses seven pure powers of e and two final quotients. Instead construct

$$a_1 = e^{-1}, \quad a_3 = a_1^2/e, \quad a_7 = a_3^2/e, \quad a_{14} = a_7^2, \quad a_{29} = a_{14}^2/e, \quad a_{55} = a_{29}^2/a_3,$$

and impose

$$a_{29}q_{12} = qa_7, \qquad a_{55}q_{40} = q_{12}e.$$

These are six power coordinates and two output coordinates. They give the same outputs and save one coordinate per detector.

The nonzero condition is inherited, rather than presumed from a quotient notation. In the source, w_0 has a retained inverse and

$$w_0 n = (w_0^2 - 2w_1^2)/2, \qquad (e\Delta)b = n^2.$$

Since $w_0 \neq 0$ and 2 is a nonsquare in $\mathbb{Q}$, the first equation gives $n \neq 0$. The second gives $e\Delta \neq 0$. Therefore every Laurent denominator above is nonzero. Conversely all Laurent coordinates are uniquely defined for each source solution. The changes introduce no new zero branches and exclude no source solution.

A9.5 Scalar preparation with one fewer coordinate

The scalar pole uses a nonzero input b and a preparation witness t, with

$$U = \frac{b(1 - bt^2)}{1 + bt^2}.$$

The old circuit stores $b^{-1}, t_2 = t^2, bt_2, U$. Replace these four coordinates by t_2, T, V, with

$$t_2 = t^2, \qquad b(T - t_2) = 1, \qquad TV = 2,$$

and use the affine expression $U = V - b$. The equation $b(T - t_2) = 1$ forces $b \neq 0$, and $TV = 2$ forces $T \neq 0$. Consequently $T = t^2 + b^{-1}$, $1 + bt^2 = bT \neq 0$, and

$$U = 2/T - b = \frac{b(1 - bt^2)}{1 + bt^2}.$$

Conversely every old solution extends uniquely by these values of T, V. Thus the modification saves one coordinate and preserves all downstream quantities. It also supplies the nonzero condition formerly imposed by the separate coordinate b^{-1}.

A9.6 An exact product-height identity

Put

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E : y^2 = x^3 - Dx,$$

and put

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right).$$

The model calculation in Section 14 proves that $P = P_*$ has infinite order and $2E(K) = \mathbb{Z}P_*$; this is the same normalized curve and generator, not the earlier model of Proposition 3.1. Write h for the absolute logarithmic Weil height. The following lemma uses the displayed coordinates and doubling identity, together with the infinite order of P.

Lemma A9.5. *For odd integers $n_1, \dots, n_r$,*

$$\mathrm{h}\left(\prod_{i=1}^{r} x([n_i]P)\right) = \sum_{i=1}^{r} \mathrm{h}(x([n_i]P)).$$

Proof. Let $T_0 = (0,0)$. At a finite place v with $v(D) = 0$, use the finite quotient $E(K_v)/E_1(K_v)$, where E_1 denotes the formal neighborhood of infinity for this integral equation. Here we use the integral formal group construction of [60, Chapter IV, Sections 1–3]. Its parameter is $z = -x/y$, and $x = z^{-2}(1 + O(z))$, $y = -z^{-3}(1 + O(z))$ identify this neighborhood with the maximal ideal. It is an open subgroup; the proper curve $E(K_v)$ is compact, so its quotient is finite. This argument does not assume that the equation is minimal at the place above two. For a finite nonidentity point Q, membership in E_1 is equivalent to $v(x(Q)) < 0$. Since

$$x(Q + T_0) = -D/x(Q),$$

positive valuation of $x(Q)$ is equivalent to $Q + T_0 \in E_1$. The image of T_0 in this quotient has order two.

If an odd multiple of P belongs to E_1, the image of P has odd order. None of its multiples can then equal the image of T_0, so all $x([n_i]P)$ have nonpositive valuation. If no odd multiple belongs to E_1, all these abscissae have nonnegative valuation. Thus the valuations have a common weak sign. This argument includes the place above two.

The only place with $v(D) > 0$ is $\mathfrak{d} = (3 - \sqrt{2})$ above seven. The displayed integral model is minimal there, since $v_{\mathfrak{d}}(-D) = 2 < 4$. Its reduction is $y^2 = x^3$ over $\mathbb{F}_7$. Substituting $\sqrt{2} = 3$ shows that P reduces to $(1, 6)$, a nonsingular point. By [60, Proposition VII.2.1, pp. 188–190], all multiples therefore belong to the subgroup $E_0(K_{\mathfrak{d}})$ of points with nonsingular reduction. A positive valuation of x would force a positive valuation of y, and hence reduction to the singular point $(0, 0)$. It cannot occur. The valuations again have a common weak sign.

At either real embedding, each $[n_i]P$ is a double of a non-torsion point. The duplication formula makes its abscissa a nonzero square. Its ordinate is nonzero, so the equation of E gives $x([n_i]P) > d$ at that embedding. Both real values $3 \pm \sqrt{2}$ of d exceed one.

At a finite place, numbers whose valuations have the same weak sign satisfy

$$\log \max\left(1, \left|\prod_i x([n_i]P)\right|_v\right) = \sum_i \log \max(1, |x([n_i]P)|_v).$$

At both real places the same equality holds because all the factors exceed one. Summing with normalized local weights proves the assertion. □

The product here is an element of K. It is not its norm down to $\mathbb{Q}$. Norms can cancel a pole at one split place against a zero at the other, whereas the proof above works separately at every place.

Corollary A9.6. *Suppose the four root points $Q_i = [n_i]P$ have odd indices, and write*

$$C = \prod_{i=1}^{4} x(Q_i) = C_0 + C_1\sqrt{2}.$$

The original nine height-detector scales may be replaced by

$$1, \qquad C_0^2, \qquad (1 + C_1^2)^2.$$

The root-height estimate retains the original constant $7\mathcal{H}$, where $\mathcal{H}$ is the exponent-19 height constant of the source.

Proof. Both real values of C are positive, so $C_0 > 0$. All three scales are positive nonzero rational squares. If a nonzero integer N is divisible by the kth powers of the three contact radicals, the unchanged detector estimate gives

$$\mathrm{h}(\lambda),\ \mathrm{h}(C_0^2\lambda),\ \mathrm{h}((1+C_1^2)^2\lambda) \le T := (\mathcal{H}+\log 3)|N|^{19/k}.$$

The elementary rational height identities yield

$$\mathrm{h}(C_0) \le T, \qquad \mathrm{h}(C_1) \le T/2.$$

Indeed, height subadditivity first bounds the heights of the two scales by $2T$; also $\mathrm{h}(C_0^2) = 2\,\mathrm{h}(C_0)$ and $\mathrm{h}((1+C_1^2)^2) \ge 4\,\mathrm{h}(C_1)$. Hence

$$\mathrm{h}(C) \le \mathrm{h}(C_0) + \mathrm{h}(C_1) + \tfrac{3}{2}\log 2 \le \tfrac{3}{2}T + \tfrac{3}{2}\log 2 < 7\mathcal{H}|N|^{19/k}.$$

Here $\mathcal{H} \ge 1$. Lemma A9.5 bounds each root height by $\mathrm{h}(C)$. In particular, the inherited lower bound, with $B = \max_i |n_i|$, gives

$$B^2 - 6 < 7\mathcal{H}|N|^{19/k}. \tag{117}$$

The old completeness proof uses its scales only after the roots have been fixed, and requires fixed positive rational squares. Thus its prime selection and pole construction apply to this shorter list. □

A9.7 Enforcing odd indices by quadratic equations

Put $\epsilon = 1 + \sqrt{2}$. The existing chart for a point Q has K-coordinates

$$x = h^2, \qquad j^2 = x^2 - D, \qquad Q = (x, hj).$$

To this chart add $t, r \in K$ and the equations

$$t^2 = 2(x + j), \qquad th + x + j = \epsilon r^2. \tag{118}$$

These are four rational quadratic equations in four additional rational coordinates. No extra inverse witnesses are needed.

Proposition A9.7. *The projected points of the augmented chart are exactly $P + 4E(K)$, that is, the odd multiples of P.*

Proof. Use the squareclass homomorphism

$$\alpha : E(K) \longrightarrow K^*/K^{*2}, \qquad \alpha(O) = 1, \quad \alpha(T_0) = [-D] = [-1], \quad \alpha((a,b)) = [a] \quad (a \neq 0).$$

For clarity, its homomorphism property follows from the line-intersection identity

$$x(R+S)x(R)x(S) = \left(\frac{x(R)y(S) - x(S)y(R)}{x(R) - x(S)}\right)^2$$

when the displayed abscissae are nonzero and distinct. Doubling follows from $x(2R) = ((x(R)^2 + D)/(2y(R)))^2$; addition by T_0 follows from $x(R + T_0) = -D/x(R)$. Inverse pairs and identity summands are immediate. If $R + S = T_0$, then $x(R)x(S) = -D$. These cases cover the exceptional pairs.

Since $2E(K) = \mathbb{Z}P$ and $2R_0 = P$, every point of $E(K)$ has a unique expression $mR_0 + T$, where $m \in \mathbb{Z}$ and $T \in E[2](K)$. The two-torsion is generated by T_0 and $T_d = (d, 0)$, so

$$\alpha(mR_0 + aT_0 + bT_d) = [\epsilon]^m[-1]^a[d]^b.$$

The classes $[\epsilon], [-1], [d]$ are independent. A square $\epsilon^c(-1)^a d^b$, with $a, b, c \in \{0, 1\}$, would have rational square norm $(-1)^c 7^b$. This forces $c = b = 0$, and the real embeddings then force $a = 0$. Consequently

$$\alpha(R) = [\epsilon] \quad \Longleftrightarrow \quad R \in R_0 + 2E(K). \tag{119}$$

Given the augmented chart, set $u = t^2 = 2(x + j)$ and

$$a = th + u/2, \qquad b = at.$$

Expansion gives

$$u(x - u/4) = D, \qquad a^2 - ua = D, \qquad b^2 = a^3 - Da.$$

In particular t, a, b are nonzero. The tangent slope at $R = (a, b)$ is $h + t$; duplication gives $2R = (x, hj) = Q$. Equation (118) says $a = \epsilon r^2$, so $\alpha(R) = [\epsilon]$. By (119), $Q \in P + 4E(K)$.

Conversely, for $Q = (2m + 1)P$, take $R = (2m + 1)R_0 = (a, b)$. Then $a, b \neq 0$, and $a = \epsilon r^2$ for some $r \in K$. Set

$$t = b/a, \qquad h = (a^2 + D)/(2b), \qquad x = h^2, \qquad j = t^2/2 - x.$$

The same duplication identities give $Q = (x, hj)$, $j^2 = x^2 - D$, and both equations (118). □

Only the four root points receive this augmentation. The shared point and the four comparison points retain their old charts: their indices need not be odd. The total cost is sixteen coordinates and sixteen quadratic residuals. The projected-point formulation matters, since the added equations can select a sign representation of the old h, j chart.

A9.8 Odd labels, scalar bounds, and completeness

Write the root points as

$$A = [n]P, \qquad S = [a]P, \qquad T = [t]P, \qquad Z = [b]P.$$

Here S denotes a point; the fixed coefficient remains S_0. Replace the old input expression by

$$\beta = 2X^2 + 3.$$

For rational X, integrality of β is equivalent to integrality of X: if $X = u/v$ in lowest terms, then $\beta \in \mathbb{Z}$ forces $v^2 \mid 2$, and hence $v = 1$. For integral X, β is odd.

In the scalar norm block and the physical anchor, replace S by

$$R = S - P = [a - 1]P.$$

The comparison and product-height blocks still use S. The retained subtraction guard excludes $a = \pm 1$, which is harmless for completeness. The original scalar contact product is

$$\beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

It never vanishes for rational X: $\beta \geq 3$, and $\beta = 2401$ would give $X^2 = 1199$, which is not a rational square. At an odd prime, a negative valuation of β is twice that of X, as required by the scalar argument.

A9.8.1 Soundness

Put $c = |a - 1|$. Applying the scalar and anchor lemmas to R gives

$$\mathrm{h}(\beta) \le \mathcal{H} c^{19/32}, \qquad \log_2 c > 25{,}165{,}823.5, \qquad 1 \le \mathcal{H} < 2^{2{,}093{,}966}.$$

The anchor is still $S_0 = 3^{2^{26}}$. These explicit bounds imply

$$c > (80\mathcal{H})^{10}, \qquad |a| \ge c - 1 \ge c/2 > (40\mathcal{H})^{10},$$

and

$$\mathrm{h}(\beta) < c/80 \le |a|/40, \qquad \log(40\mathcal{H}) < c/80 \le |a|/40. \tag{120}$$

For example $10(7 + 2{,}093{,}966) < 25{,}165{,}823.5$, and $80 < 2^7$. The first estimate in (120) follows from $c^{13/32} > 80\mathcal{H}$; the logarithmic estimate is weaker still.

The local comparison lemmas retain their original conclusions. Let

$$F = t^2 - a(a-2)tb + (a-2)^2 b^2 - 1, \qquad B = \max(|n|, |a|, |t|, |b|).$$

For $B > 4$, the triangle inequality gives $|F| \le 2B^4 + 6B^3 + 5B^2 + 1 < 5B^4$. If $F \ne 0$, the three radicals divide it to order 40, so (117) gives

$$B^{1/10} < 7\mathcal{H}\, 5^{19/40} + 6 < 24\mathcal{H}.$$

This contradicts the anchor. Thus $F = 0$. Since the point Z is finite and nonidentity, $b \ne 0$. Lemma 14.9 now gives $\log B > |a| - 3$.

If $n \ne \beta$, the nonzero numerator N of $n - \beta$, in reduced form, satisfies $|N| \le 2e^{\mathrm{h}(\beta)} B$, and the radicals divide it to order 12. Equation (117) and $B > 4$ give

$$B^{5/12} < 40\mathcal{H} e^{(19/12)\,\mathrm{h}(\beta)},$$

so

$$\log B < \frac{12}{5}\log(40\mathcal{H}) + \frac{19}{5}\,\mathrm{h}(\beta) < \frac{31}{200}|a| < |a| - 3.$$

This is the other contradiction. Therefore $n = \beta \in \mathbb{Z}$, and $X \in \mathbb{Z}$.

A9.8.2 Completeness

Fix $X \in \mathbb{Z}$, and set $n = \beta$. Perform the old scalar preparation: choose its finite correction and then a positive shared index h_0 divisible by 64 and sufficiently deep at the fixed small primes. Choose a positive scalar index s sufficiently divisible to satisfy the scalar leaves and physical anchor for $[s]P$. We may require $6 \mid s$ and take s arbitrarily large. Set $a = s + 1$. Then a is odd, $a \equiv 1 \pmod 3$, and $S - P = [s]P$ is exactly the prepared scalar point.

Define

$$U_0 = 0, \qquad U_1 = 1, \qquad U_{j+1} = aU_j - U_{j-1}, \qquad k = 2(a-2),$$

and take $b = U_k/(a-2)$ and $t = U_{k+1}$. Modulo $a - 2$, induction gives $U_j \equiv j$, so $b \in \mathbb{Z}$. Modulo two the sequence has period $0, 1, 1$; since $k \equiv 1 \pmod 3$, both U_k and U_{k+1} are odd. The positive odd denominator $a - 2$ shows that b is odd. The Lucas identity gives

$$t^2 - a(a-2)tb + (a-2)^2 b^2 = 1.$$

Thus all four root indices n, a, t, b are positive and odd. Proposition A9.7 supplies their chart witnesses.

Fix these roots and the three scales of Corollary A9.6. Local continuity of the scalar and comparison functions at the prepared shared point gives the same fixed modulus L, with $64 \mid L$. Only then choose the contact primes by the inherited simultaneous-prime theorem. The direct-target construction chooses a common index h satisfying the primary and opposite reduction-order congruences and $h \equiv h_0 \pmod{L}$, with the required real bounds. Its comparison points are $H = [h]P$ and $U_i = [hn_i]P$. Their shifted points have indices $h-1$ and $(h-1)n_i$.

All quotient-domain arguments remain valid. The product scales are fixed positive squares, so the original pole witnesses apply unchanged. The scalar block is preserved by the congruence modulo L. This gives a rational solution for every integral X and completes the arithmetic proof of the primary construction.

A9.9 Replacing five logarithms by asymmetric formal depth

This section is used only for the 523-variable construction. Let

$$U_0 = H = [h]P, \quad U_i = [u_i]P, \quad V_0 = H - P, \quad V_i = U_i - Q_i,$$

where $Q_i = [n_i]P$ are the four roots. Put $f = 1/x$ and $z = -x/y$. Keep the alternate truncated logarithms $\ell_{40}(V_i) = z(V_i)G_9(Df(V_i)^2)$, but replace every primary truncated logarithm by $z(U_i)$. Thus

$$L_i = z(U_i)/z(H), \qquad L_i' = \overline{\ell_{40}(V_i)/\ell_{40}(V_0)}.$$

The bar denotes conjugation in K. The sixteen tests now have the following orders:

1	$f(U_1), \dots, f(U_4),\ \overline{f(V_0)}, \dots, \overline{f(V_4)}$
40	$f(H),\ L_1 - L_1', \dots, L_4 - L_4',\ L_1 - \beta,$ the direct Lucas residual.

The scalar tests retain their original order 32. The stronger beta tests imply all former order-12 conclusions.

A9.9.1 Local soundness

At a consumed unramified place above $p > 76176$, a formal point Q with $r = v(z(Q)) \geq 1$ satisfies

$$\log_E(Q)/z(Q) \in 1 + p^{4r}\mathcal{O}. \tag{121}$$

This is the same formal-logarithm expansion used in the source, starting its tail at the first nonconstant term. Namely $v(f(Q)) = 2r$, and for its coefficients a_j the factorial estimate is $v(a_j) \geq -(4j+1)/(p-1)$. If $4j + 1 < p$, the coefficient is integral. For the remaining $j \geq 19044$, the displayed bound still makes the jth term have valuation at least $4r$. Both $z(Q)$ and its full logarithm have valuation r.

The new test on $f(H)$ gives $v(z(H)) \geq 20$. The order-one tests put the other points on the formal branch. The full logarithm ratios at the primary and conjugate places are

$$r_i = u_i/h, \qquad r_i' = (u_i - n_i)/(h-1).$$

At least one of $h, h-1$ is a p-unit, so at least one of these ratios is integral. Even before knowing the depth of U_i, the weak unit estimate (121) and the alternate truncation estimate show that L_i, L_i' have the same valuations as r_i, r_i'. The order-40 cross test transfers integrality to both sides. Now L_i integral implies $v(z(U_i)) \geq v(z(H)) \geq 20$. The primary logarithmic corrections are therefore 1

modulo p^{80}. The alternate corrections are 1 modulo p^{40}. Since the ratios are integral, these relative estimates become absolute:

$$L_i \equiv r_i \equiv r_i' \pmod{p^{40}}.$$

Multiplying by h and $h-1$ and subtracting gives $L_i \equiv n_i \pmod{p^{40}}$. At an inert prime conjugation preserves the local valuation, so the same proof applies. The height contradictions of Section A9.8 remain unchanged.

A9.9.2 A stronger choice within the existing prime-pattern argument

For completeness, the common index must acquire extra primary divisibility. The following observation ensures compatibility with the opposite reduction orders.

Lemma A9.8. *The contact primes in the source completeness argument may be chosen so that no selected prime divides any opposite reduced group order.*

Proof. After merging repeated contact forms and fixing all scale and congruence data, the prime-pattern proof uses distinct forms $p_j = u + c_j w^2$, with fixed coefficients c_j. Integer dilations and ray congruences change only fixed constants. It counts $u \in [N/2, N]$ and $1 \le w \le T$, where $T = \lfloor \tau(N)\sqrt{N} \rfloor$ and $\tau(N) \to 0$ can be chosen arbitrarily slowly in the final diagonal argument. Its weighted count, after the prescribed obstruction removal, retains a fixed positive proportion of NT. These are the fixed-field weighted count in Proposition A8.30 and the final diagonal argument of Section A8.11, including its obstruction removal.

If there are k forms, impose also

$$\frac{\tau(N) N^{1/6}}{(\log N)^k} \longrightarrow \infty.$$

This is compatible with the same diagonal choice: for each fixed positive value of τ, its threshold can be added to the existing finite list of thresholds. The strip $w < N^{1/3}$ has at most $O(N^{4/3})$ integer pairs and weight at most $O((\log N)^k)$ per pair. Its total weight is $o(NT)$, so an admissible pattern remains outside it.

For these patterns $w^2 \ge N^{2/3}$ and $w^2 = o(N)$. All p_j are asymptotic to the same u. Hasse's bound [60, Theorem V.1.1, p. 138] gives $d_+(p_j) = p_j + O(\sqrt{N})$ for the opposite reduced group order. Every ratio $d_+(p_j)/p_i$ is eventually strictly between $1/2$ and 2. Hence $p_i \mid d_+(p_j)$ would force equality. If $i \ne j$, this contradicts

$$|p_i - p_j| \ge c_* N^{2/3}, \qquad c_* := \min_{i \ne j} |c_i - c_j| > 0.$$

If $i = j$, equality is impossible because $4 \mid d_+(p_i)$ whereas p_i is odd. □

A9.9.3 Completeness with deeper primary points

Keep the choice order in Section A9.8: the roots, scales and continuity modulus L precede the prime selection. Choose primes using Lemma A9.8, beyond all prime factors of L and the fixed scale data. Put

$$M_\pm = \operatorname*{lcm}_p (d_\pm(p)/4), \qquad \Pi = \prod_p p.$$

The direct-target prime theorem gives

$$\gcd(M_-, M_+) = 1, \quad \gcd(M_-, L) = 16, \quad \gcd(M_+, L) = 1.$$

Lemma A9.8 gives $\gcd(\Pi, M_+) = 1$, and the cutoff gives $\gcd(\Pi, L) = 1$. Thus

$$h \equiv 0 \pmod{\Pi^{19} M_-}, \qquad h \equiv 1 \pmod{M_+}, \qquad h \equiv h_0 \pmod{L}$$

are compatible, since $64 \mid h_0$. The inherited real-progression argument supplies a sufficiently large h in this progression with the same real bounds.

The point P is globally a double, so multiplication by the primary doubling-image order first makes it formal, with z-valuation at least one. Multiplication by p^{19} raises this valuation by nineteen, using the formal logarithm isometry of [60, Theorem IV.6.4, p. 132]; these completions are unramified, $v(p) = 1$, and $1 > 1/(p-1)$. Consequently $H = [h]P$ and $U_i = [hn_i]P$ have primary depth at least twenty, and their parameter ratios equal n_i to order eighty. At the opposite places, $V_0 = [h-1]P$ and $V_i = [(h-1)n_i]P$ are formal and their retained logarithm ratios equal n_i to order forty.

The pole-completeness estimates still apply to these fixed test orders. In the source notation their large-real-parameter decay is λ^{-3} for each order m, and the small-prime lower bound is $3\nu + (10m-8)\eta - 2c$, which still tends to infinity with the preparatory divisibility. Scalar conditions and the anchor are preserved by $h \equiv h_0 \pmod{L}$. This proves completeness for the optional refinement.

A9.10 Polynomial construction and exact verification

The generators apply three changes to the earlier point-chart source: the four odd-coset charts, the product and three scales, and the shifted scalar point with $\beta = 2X^2 + 3$. The depth variant first changes the logarithms and test orders as in Section A9.9. Each field equation is represented by two rational component equations. Intermediate products are assigned coordinates as needed to keep every residual quadratic.

The full-log source has 577 raw coordinates; the pinned upstream compiler reduces this to 561. Seven further certified eliminations give 554. The depth source has 540 raw coordinates; compilation gives 532 and nine certified eliminations give 523. Every elimination is justified by exact polynomial identities and reversible substitutions preserving the input. Clearing residual denominators and taking the sum of their squares gives the displayed integer quartics. Over $\mathbb{Q}$, the sum vanishes exactly when every residual does.

The constant-chain versions add $c_0, \ldots, c_{16}$ with

$$c_0 = 3^{1024}, \qquad c_{i+1} = c_i^2 \quad (0 \le i < 16).$$

These witnesses have unique rational values and $c_{16} = S_0$. Substitution for S_0 leaves every residual quadratic. Direct expansion gives the 3,247-bit maximum coefficient bound.

The accompanying archive contains the complete polynomial tables, residual systems, coordinate maps, elimination certificates, pinned source generators, arithmetic notes and independent checkers. Each final package has a standard-library-only `verify_all.py`. It replays all eliminations, verifies the unique constant lift, independently expands the sums of squares, and compares every coefficient. Both full packages have passed fresh independent runs. A separate source replay rebuilds the two compiled precursors and checks the match to their frozen residual systems.

These finite checks certify the counts, degrees and stated algebraic transformations. The arithmetic assertions in Sections A9.6–A9.9 received separate adversarial AI review; they have not received independent human peer review or a complete formal proof check. The inherited arithmetic inputs are the results of this manuscript identified in Section A9.1; no theorem from the contribution archive is assumed as an additional arithmetic input.

Proof of Theorem A9.1. The chart identities and Lemma A9.2 identify the exact source used by the generators with the arithmetic system described here. Proposition A9.7 enforces the odd root indices, so Corollary A9.6 supplies the required height bound. Section A9.9.1 verifies the local implications for the final test orders. Section A9.8.1 then forces the input to be an integer. Conversely, for an integer input, Sections A9.8.2 and A9.9.3, using Lemma A9.8, construct witnesses for every source equation and retained guard. The certified substitutions preserve this projection in both directions. The sum of squares vanishes over $\mathbb{Q}$ precisely when all its residuals vanish. Finally the 17-coordinate constant chain has exactly one rational solution and ends at S_0, so it preserves the same projection. This proves the statements for both polynomials. □

The complete tables are in the files

`quartics/quartic-523-symbolic-coefficients.json`
`quartics/quartic-540-fully-expanded.txt`.

Their SHA-256 digests, respectively, are

`6a3685980de4512bdc7f3a781d0c0d4b32cce5829ec83f03f318264931bdbbe3`
`83e7a31e11f9ad2edc2f67935e50bff437d87ef4b77d37b6a4391e8117770680`.

The reproduction command is `pythonscripts/check_odd_product.py`; the complete source, certificates and repaired, non-mutating verifiers are in `quartics/quartic-523-540/`.

A10 A quartic in 498 variables

We now reduce the 523-variable construction of Theorem A9.1 to 498 variables. The input remains one rational variable X; every other variable remains existentially quantified over $\mathbb{Q}$. The changes below use the same normalized curve, the same integer labels, the same three height detectors, and the same published and manuscript inputs as Section A9. In particular, they do not replace the restricted converse, the exponent-nineteen height theorem, or the simultaneous-prime argument by new assumptions.

Theorem A10.1. *There is a specified integer polynomial P_{498} of total degree four in 498 variables such that*

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{v} \in \mathbb{Q}^{497} : P_{498}(X, \mathbf{v}) = 0.$$

The polynomial has 24,669 nonzero monomials, content one, and largest absolute coefficient $16 \cdot 3^{2^{26}}$. Its symbolic coefficient table uses the fixed integer $C = 3^{2^{25}}$, not an additional variable. There is also a specified integer quartic P_{514} with

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{w} \in \mathbb{Q}^{513} : P_{514}(X, \mathbf{w}) = 0.$$

It has 24,716 nonzero monomials and content one. Its largest absolute coefficient is

$$3^{2048} + 1005072314676622 < 2^{3247}.$$

The proof distinguishes two sorts of changes. Some are rational changes of private witnesses with a rational inverse. Others alter the arithmetic certificate while preserving both soundness and the existence of witnesses for every integer input. For the latter we prove those two assertions separately; we do not claim equality of all intermediate witness assignments.

A10.1 Conventions and retained inputs

Throughout put

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad d_* = 3 - \rho, \quad D = d_*^2 = 11 - 6\rho, \quad E : y^2 = x^3 - Dx.$$

The point $P = P_*$ is the same normalized generator as in Section 17.1 and Section A9:

$$P = (c_*, a_* b_*), \qquad a_* = -1 + \tfrac{7}{4}\rho, \quad b_* = \tfrac{39}{8} - \tfrac{9}{2}\rho, \quad c_* = a_*^2 = \tfrac{57}{8} - \tfrac{7}{2}\rho.$$

Thus $b_*^2 = c_*^2 - D$. In particular a_*, b_*, c_*, D are nonzero. We use the inherited equality $2E(K) = \mathbb{Z}P$, the nonzero point domains of Lemma 17.1, and the inherited estimate $\mathrm{h}(x([n]P)) \geq n^2 - 6$. Here h is the absolute logarithmic Weil height. At every unramified place above a rational prime p we normalize $v(p) = 1$. The coefficient basis is always $(1, \rho)$ and conjugation sends ρ to $-\rho$.

Retain the four odd-coset root charts of Proposition A9.7, with labels

$$Q_1 = [n]P, \quad Q_2 = S = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad n, a, t, b \text{ odd}.$$

The letter b in this display is an integer point label. We use $\mathfrak{b}$ below for a rational input to a pole predicate. Put $Q_0 = P$, $U_0 = H = [h]P$, $U_i = [u_i]P$, and $V_i = U_i - Q_i$. The scalar point is

$$R = S - P = [s]P, \qquad s = a - 1,$$

and the input label is $\beta = 2X^2 + 3$.

We retain the pole obstruction and pole-completeness statements identified in Section A9.1, specifically Lemmas 14.4, 14.6, and 17.4, including their existing source dependencies. We also retain the three scales of Corollary A9.6 and its product-height estimate. Write $\mathcal{H} = H_{19}$, so that

$$1 \leq \mathcal{H} < 2^{2093966}, \qquad B^2 - 6 < 7\mathcal{H}|N|^{19/k} \tag{122}$$

whenever the three contact radicals divide a nonzero integer N to order k, where

$$B = \max(|n|, |a|, |t|, |b|).$$

This is exactly (117); none of its scales is replaced by a field norm in the present construction.

The local formal-group inputs are the integral formal group in $z = -x/y$, its multiplication series, and its logarithm. The integral construction is the one in [60, IV§1, Example IV.2.2.3 and Proposition IV.2.3(a)]. At the unramified places used below, $p > 76176$, and the logarithm is an isometry on depth at least one by [60, Theorem IV.6.4, p. 132]; its hypothesis is $1 > 1/(p-1)$. The coefficient and truncation estimates are those of Lemma 15.4 and its unramified extension in Lemma 17.2. We spell out their new applications below. No minimal-model assertion at the prime above two is needed.

A10.2 Mixed logarithmic precision

Let $f = 1/x$ and $z = -x/y$, as before. Define

$$a_0 = 1, \quad a_1 = -\tfrac{2}{5}, \quad a_j = \frac{4j-5}{4j+1} a_{j-1} \quad (j \geq 2), \qquad G_J(T) = \sum_{j=0}^{J} a_j T^j.$$

Thus the source truncations are $\ell_{12} = zG_2(Df^2)$ and $\ell_{40} = zG_9(Df^2)$. Keep every primary logarithm equal to the uncorrected parameter z. Only the alternate numerator for V_1 is shortened:

$$L_i = \frac{z(U_i)}{z(H)}, \qquad L_1' = \frac{\overline{\ell_{12}(V_1)}}{\ell_{40}(V_0)}, \qquad L_i' = \frac{\overline{\ell_{40}(V_i)}}{\ell_{40}(V_0)} \quad (2 \le i \le 4). \tag{123}$$

The only denominator logarithm remains $\ell_{40}(V_0)$, with its original nonzero domain. In particular this change introduces no new logarithm denominator.

The sixteen field tests and their orders are now

$$\begin{array}{r|l} 1 & f(U_1), \dots, f(U_4),\ \overline{f(V_0)}, \dots, \overline{f(V_4)} \\ 12 & L_1 - L_1',\ L_1 - \beta \\ 40 & f(H),\ L_2 - L_2',\ L_3 - L_3',\ L_4 - L_4',\ F(L_2, L_3, L_4), \end{array} \tag{124}$$

where

$$F(A, T, B) = T^2 - A(A-2)TB + (A-2)^2B^2 - 1.$$

This changes the two order-forty input tests in the 523-variable source to order twelve. The three other label comparisons and the Lucas test remain of order forty.

Lemma A10.2. *At every consumed unramified place above $p > 76176$, the tests* (124) *imply*

$$L_1 \equiv n \pmod{p^{12}}, \qquad L_i \equiv n_i \pmod{p^{40}} \quad (2 \le i \le 4), \qquad (n_2, n_3, n_4) = (a, t, b).$$

The primary points U_i have depth at least twenty after integrality of the ratios has been established.

Proof. The order-one tests put the indicated points in the formal group. The test on $f(H)$ gives $v(z(H)) \ge 20$. For a formal point Q of depth $r \ge 1$, the unchanged logarithm expansion gives

$$\log_E(Q)/z(Q) \in 1 + p^{4r}\mathcal{O}.$$

Its truncation after G_2 has relative error of order at least twelve, and its truncation after G_9 has relative error of order at least forty. To check these particular applications, the first omitted indices are respectively $j = 3$ and $j = 10$ in the source recurrence. For $4j+1 < p$ the coefficients are integral. For the remaining indices the source factorial bound $v_p(a_j) \ge -(4j+1)/(p-1)$ is dominated by $4jr$ and gives the same stated lower bounds. Both truncating factors are units.

The full ratios are the rational numbers

$$r_i = \frac{u_i}{h}, \qquad r_i' = \frac{u_i - n_i}{h - 1}.$$

At least one of $h, h-1$ is a p-unit, so at least one of these two ratios is integral. Before any improved depth is inferred for U_i, the unit factors just described show $v(L_i) = v(r_i)$ and $v(L_i') = v(r_i')$. Each cross-test has positive order, so integrality transfers to both approximate ratios and both full ratios. This first step is essential.

Now L_i integral gives $v(z(U_i)) \ge v(z(H)) \ge 20$. The primary relative errors have order at least eighty. The alternate relative errors have order twelve for $i = 1$ and forty for $i \ge 2$: in the first case the denominator still has order forty, so it does not reduce the numerator's order twelve. Since the ratios are integral, these relative errors give absolute congruences. Together with the cross-tests they yield $L_i \equiv r_i \equiv r_i'$ to the asserted order. Multiply by h and $h-1$ and subtract. The result is $L_i \equiv n_i$, without division by either integer. At an inert prime conjugation preserves the single local valuation, so this argument has the same conclusion there. □

For clarity, the detector's product is unchanged as a mathematical operation. For a test D_j of order m_j, a scale $\mathcal{A}$, and $t = \mathcal{A}\lambda$, $\delta = t - 1$, its factors are

$$1 + \mathcal{R}\left(\frac{D_j}{\delta^{m_j+1}\mathcal{R}^{m_j}}\right)^2, \qquad \mathcal{R} = \mathcal{A}^{-5}F_5(t)/\delta^6, \qquad F_5(t) = \prod_{\theta\in\Theta}(t-\theta),$$

with the same Θ as (42). On writing $e = (\mathcal{A}\delta)^5/F_5(t)$, the multiplier of D_j^2 is $q_{m_j} = q_1 e^{2m_j-2}$, where $q_1 = e/\delta^3$. For each $m \in \{1, 12, 40\}$, form the coefficients of

$$\prod_{j:m_j=m}(1 + TD_j^2)$$

and evaluate at $T = q_m$. Their degrees are respectively $9, 2, 5$. Their product is exactly the same sixteen-factor detector expression. The smaller G_2 numerator and these three grouped products explain the changed quadratic circuit. Multiplying in K means the two rational bilinear identities in the fixed basis. Every reassociation or grouping here follows from distributivity, with no division or new witness domain. The retained coefficient tables and exact quadratic elimination identities specify the resulting circuit.

A10.3 Five rational half-trace guards

For each $0 \le i \le 4$ write the subtraction denominator as

$$d_i = x(U_i) - x(Q_i) = A_i + B_i\rho.$$

The old field inverse used two private rational coordinates. Replace those two equations by the single equation

$$A_i\gamma_i = 1, \qquad \gamma_i \in \mathbb{Q}. \tag{125}$$

No other equation used the removed inverse coordinates. This saves one coordinate in each of five blocks. It strengthens the domain; there is no claim that every old trace-zero assignment is retained.

Soundness has an explicit rational map. From (125) we have $A_i \neq 0$, hence $d_i \neq 0$ and $A_i^2 - 2B_i^2 \neq 0$. Restore the old field inverse as

$$d_i^{-1} = \frac{A_i - B_i\rho}{A_i^2 - 2B_i^2}.$$

All old subtraction outputs and all their consumers are unchanged.

For completeness fix the four root points and the scalar preparation before choosing contact primes, as in Section A9.8.2. Increase the allowed finite prime cutoff to exclude every prime at which any of $P, Q_1, \ldots, Q_4$ has nonintegral abscissa. This is a finite restriction on already fixed points. Choose any one of the split contact primes supplied by Lemma 17.7, and denote its primary and opposite places by $v, \bar{v}$. The unchanged CRT construction makes every U_i formal at v and makes it reduce to Q_i at $\bar{v}$. All five Q_i have finite integral reduction. Consequently

$$v(d_i) < 0, \qquad \bar{v}(d_i) \ge 0.$$

Conjugation exchanges the two places, so $v(\overline{d_i}) = \bar{v}(d_i) \ge 0$. The two terms of $d_i + \overline{d_i} = 2A_i$ have unequal valuations; the sum is nonzero. Thus all five inverses $\gamma_i = 1/A_i$ exist for the same completeness witnesses. This includes $i = 0$. It imposes no unit condition on the root labels and no condition on the permitted zero pole witnesses.

A10.4 Normalizing the three detector poles

Here $\mathfrak{b}$ is one detector's rational pole input. In the compiler's rational coordinates the seven relevant equations are

$$\begin{aligned} p^2 &= p_2, & 17\mathfrak{b}p_2 + 3e^2 - 34e\mathfrak{b} - 289\mathfrak{b}^2 &= 0,\\ u_2 &= \mathfrak{b}^2, & 17e\mathfrak{b} - eu_2 - 17g &= 0,\\ s_0^2 &= s_2, & q_0^2 &= q_2,\\ & & 17\mathfrak{b}q_2 + 867g^2 - 34gs_2 - s_2^2 &= 0. \end{aligned}$$

In particular the stored g is the source's quantity $Bh/17$, not Bh. The following substitution respects that factor:

$$e = \mathfrak{b}E_0, \quad p = \mathfrak{b}P_0, \quad s_0 = \mathfrak{b}T_0, \quad q_0 = \mathfrak{b}^2Q_0,$$
$$p_2 = \mathfrak{b}^2P_2, \quad s_2 = \mathfrak{b}^2T_2, \quad q_2 = \mathfrak{b}^4Q_2, \quad u_2 = \mathfrak{b}^2, \quad g = \mathfrak{b}^2G_0.$$

The replacement consists of six quadratic equations:

$$\begin{aligned} P_0^2 &= P_2, & 17\mathfrak{b}P_2 + 3E_0^2 - 34E_0 - 289 &= 0,\\ 17E_0 - E_0\mathfrak{b} - 17G_0 &= 0, & T_0^2 &= T_2,\\ Q_0^2 &= Q_2, & 17\mathfrak{b}Q_2 + 867G_0^2 - 34G_0T_2 - T_2^2 &= 0. \end{aligned} \tag{126}$$

The division by $\mathfrak{b}$ in the reverse coordinate change is legitimate in both systems. Indeed the unchanged detector equations, with its actual product $W = w_0 + w_1\rho$, include

$$w_0\iota = 1, \qquad 2w_0\nu = w_0^2 - 2w_1^2,$$

and an equation whose specialization at $\mathfrak{b} = 0$ is $\nu^2 = 0$. Thus $\mathfrak{b} = 0$ would give $\nu = 0$ and $w_0^2 = 2w_1^2$. Rationality then forces $w_0 = w_1 = 0$, contradicting $w_0\iota = 1$.

Under the displayed substitution the old u_2 equation is an identity. The old p_2, first-conic, g, and s_2 equations are $\mathfrak{b}^2$ times their replacements; the old q_2 and second-conic equations are $\mathfrak{b}^4$ times their replacements. These identities prove mutually inverse rational maps fixing every outside coordinate. They remain valid when any of e, p, s_0, q_0, g is zero. Applying the change to three detectors saves three coordinates. The scalar pole with its preparation variable is not altered by this normalization.

A10.5 One coefficient in the physical anchor

Put $C = 3^{2^{25}}$ and write the actual scalar value as $f(R) = f_0 + f_1\rho$. The previous fixed parameter was $S_0 = C^2$. The old three-witness anchor encoded $f_0/C, f_1/C \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}$. On the present integer orbit it is enough to impose that condition on f_1/C.

Lemma A10.3. *For every nonzero even integer s, write $f([s]P) = f_0 + f_1\rho$. Both coefficients are 2-integral, and*

$$v_3(f_1) = 2 + 2v_3(s), \qquad v_3(f_0) \geq 3 + 2v_3(s).$$

In particular $f_1 \neq 0$, and

$$f_1/C \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)} \quad \Longleftrightarrow \quad f_0/C, f_1/C \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)}.$$

Proof. At 3 the field is unramified quadratic with residue field $\mathbb{F}_9 = \mathbb{F}_3[\rho]/(\rho^2 - 2)$. The exact starting calculation in Section 14.2.2 is

$$z(2P) = \frac{3396356159088 - 4135204317756\rho}{3287811625151}, \qquad z(2P)/3 \equiv 2 + \rho \pmod 3.$$

The second congruence follows by reducing the displayed rational coefficients after division by three. Its square is ρ in $\mathbb{F}_9$. Write $s = 2 \cdot 3^k u$ with $3 \nmid u$, allowing negative u. The integral formal group has inverse $-z$, and hence odd multiplication series. Thus

$$[u]z = uz + O(z^3), \qquad [3]z = 3z + O(z^3)$$

with integral coefficients. At depth $r \geq 1$, division of the second identity by 3^{r+1} leaves an error of valuation at least $2r - 1 > 0$. Multiplication by three therefore preserves the leading residue. Multiplication by u multiplies it by the nonzero residue of u. Consequently

$$z([s]P)/3^{k+1} \equiv u(2 + \rho) \pmod 3.$$

The exact curve relation is $f = z^2(1 - Df^2)$. On the formal branch its second factor is one modulo three, and $u^2 = 1$ in $\mathbb{F}_3$. Therefore

$$f([s]P)/3^{2k+2} \equiv \rho \pmod 3.$$

Independence of $1, \rho$ over $\mathbb{F}_3$ proves the two asserted coefficient valuations, including the possible zero value of f_0.

At the place above two normalize $v(2) = 1$. The literal point has $v(x(P)) = -3$, $v(y(P)) = -9/2$, and $v(z(P)) = 3/2$. It lies in the integral formal subgroup, which contains all its integral multiples. One may construct that subgroup directly from $w = z^3 - Dzw^2$, where $w = -1/y$, as in Section 14.2.2; this uses no assertion that the displayed equation is minimal at two. Thus $f([s]P)$ is integral. The equality

$$v(a + b\rho) = \min\{v_2(a), v_2(b) + \tfrac{1}{2}\} \quad (a, b \in \mathbb{Q})$$

shows that both rational coefficients are 2-integral. Since C is a 2-unit, the equivalence now follows from the coefficient valuations at three. □

The three-witness criterion in the proof of Lemma 17.3 applies to every rational t:

$$t \in \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)} \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \; u^2 + 6v^2 - 2w^2 - 3t^2 + 2 = 0. \tag{127}$$

To recall its exact input, $(2, 3)_{\mathbb{Q}}$ ramifies precisely at two and three. The embedding criterion [98, Proposition 14.6.7, p. 232] tests the nonsquareness of $1 - 2/(3t^2)$ at those places when $t \neq 0$. At two, positive valuation of t gives odd negative valuation of the expression, valuation zero gives residue three modulo eight, and negative valuation gives a square in $1 + 8\mathbb{Z}_2$. At three, nonnegative valuation gives odd negative valuation of the expression, whereas negative valuation gives a square in $1 + 3\mathbb{Z}_3$. The expression is nonzero because $t^2 = 2/3$ has no rational solution. Multiplication by $3t^2$ gives (127); for $t = 0$ the witnesses are $(0, 0, 1)$. These are exactly the field, local conditions, and exceptional case already present in that source proof.

Apply this criterion to $t = f_1/C$, multiply its witnesses by C, and then replace the old w by $w + C$. The resulting anchor is

$$u^2 + 6v^2 - 2w^2 - 4Cw - 3f_1^2 = 0. \tag{128}$$

It uses the same three private rational witnesses and removes the old coordinate for $f_0^2 + f_0 f_1 - f_1^2$. Since a is odd and R is nonidentity, $s = a - 1$ is a nonzero even integer. By Lemma A10.3, old and new anchors have the same projection onto all their nonprivate coordinates. In each direction use (127) to choose the private witnesses; they need not be the same witnesses in both anchors. The affine translation of w is invertible.

The exact allowed subgroup is unchanged:

$$s \in 2\,3^{2^{24}-1}\mathbb{Z} \setminus \{0\}, \qquad \log_2 |s| \geq 1 + (2^{24} - 1)\log_2 3 > 25165823.5. \tag{129}$$

Indeed $v_3(f_1) \geq v_3(C) = 2^{25}$ is precisely $2+2v_3(s) \geq 2^{25}$. The subgroup permits arbitrarily divisible positive s, as required by completeness.

A10.6 A shorter scalar norm tree

Keep the exact contact polynomial and finite correction

$$r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1), \qquad c \in \left\{ \pm \prod_{p \in I} p : I \subseteq \{2, 3, 5, 7, 11, 13, 17\} \right\}.$$

Here $r \neq 0$ for rational X. The only positive contact not excluded by $\beta \geq 3$ is $\beta = 2401$, which would imply $X^2 = 1199$, impossible over $\mathbb{Q}$. The old inverse equation for r and its power chain remain in place.

Write

$$f(H) = h_0 + h_1\rho, \qquad z(R)/z(H) = z_0 + z_1\rho, \qquad q = h_0^2 - 2h_1^2 = \mathrm{Norm}_{K/\mathbb{Q}}(f(H)).$$

Use $N_r(u, v) = u^2 + rv^2$ and replace the old six-leaf scalar tree by

$$W = N_r\left(N_r(1, q/r),\ N_r(z_0/r^{32}, z_1/r^{32})\right). \tag{130}$$

The scalar pole input is still exactly $\mathfrak{b} = crW^2$, with the original preparation and nonzero condition. This tree changes that input's value, so completeness must supply new pole witnesses.

For an explicit quadratic encoding, let $i_{32} = r^{-32}$ be the retained power-chain output and introduce eight rational coordinates $q, A, U, V, V_2, B, B_2, W$ with equations

$$\begin{aligned} q &= h_0^2 - 2h_1^2, & r(A - 1) &= q^2, \\ U &= z_0 i_{32}, & V &= z_1 i_{32}, \\ V_2 &= V^2, & B &= U^2 + rV_2, \\ B_2 &= B^2, & W &= A^2 + rB_2. \end{aligned} \tag{131}$$

The existing W^2 coordinate is reattached to this W with its actual fixed rational normalization. Thirteen old leaf and norm coordinates disappear, giving a net saving of five at this stage. The final elimination of q is given below.

Lemma A10.4. *The new scalar certificate implies $M_5(\beta)^{32} \mid s$ and $\mathrm{h}(\beta) \leq \mathcal{H}|s|^{19/32}$.*

Proof. Let $p > 76176$ be a prime of the same five-contact radical as in Lemma 16.1, and put $\eta = v_p(r)$. The unchanged contact calculation makes η odd; the correction c is a p-unit. The original pole obstruction, applied to the retained nonzero input crW^2, gives

$$\eta + 2v_p(W) > 0.$$

At a node $u^2 + rv^2$, nonzero summands have opposite valuation parity, so they cannot cancel. Zero summands are harmless with valuation $+\infty$. If $\eta < 0$, the constant-left path in (130) gives $v_p(A) \le 0$ and $v_p(W) \le 0$, a contradiction. Thus $\eta > 0$ and $v_p(W) > -\eta/2$. Descending the two levels gives

$$v_p(A) > -\eta/4, \quad v_p(B) > -3\eta/4,$$
$$v_p(q/r) > -5\eta/8, \quad v_p(U) > -3\eta/8, \quad v_p(V) > -7\eta/8.$$

Therefore

$$v_p(q) > 3\eta/8 > 0, \qquad v_p(z_0) > (32 - 3/8)\eta, \qquad v_p(z_1) > (32 - 7/8)\eta. \tag{132}$$

Since η is a positive integer, rational integral valuations give $v_p(z_0), v_p(z_1) \ge 32$.

At a split prime $v_p(q)$ is the sum of the two valuations of $f(H)$; at an inert prime it is twice the single valuation. Positivity of $v_p(q)$ therefore supplies at least one place v above p with $v(f(H)) > 0$. We claim only existence of one such place. At that place H is formal and $v(z(H)) \ge 1$. By the two coefficient bounds,

$$v(z(R)) \ge v(z(H)) + 32 > 0.$$

This need not make R itself formal: it may lie near $(0, 0)$. We handle both branches explicitly.

Put $m = v(x(R))$. The retained point domains give $x(R)y(R) \ne 0$. Since D is a unit and

$$z(R)^2 = \frac{x(R)}{x(R)^2 - D},$$

positive valuation of $z(R)$ excludes $m = 0$. For $m > 0$ its valuation is $m/2$, and for $m < 0$ it is $-m/2$. The exact duplication identity

$$x([2]R) = \frac{(x(R)^2 + D)^2}{4x(R)(x(R)^2 - D)} \tag{133}$$

then gives, for either sign of m,

$$v(x([2]R)) = -2v(z(R)) < 0, \qquad v(z([2]R)) = v(z(R)).$$

Its numerator and denominator are nonzero: with $m \ne 0$ their relevant terms have unequal valuations, and p is odd. Also $[2]R$ is nonidentity because $s \ne 0$ and P has infinite order.

Both H and $[2]R$ are now in the formal subgroup where the logarithm is a homomorphism and isometry. Apply it to the exact global identity

$$[h]([2]R) = [2s]H.$$

The points are nontorsion, so their nonzero formal logarithms permit comparison of valuations. Since 2 is a p-unit,

$$v_p(h) + v(z([2]R)) = v_p(s) + v(z(H)), \qquad v_p(s/h) \ge 32.$$

The nonzero integer h has nonnegative p-valuation, hence $v_p(s) \ge 32$. This proves the claimed divisibility at every prime of $M_5(\beta)$. The same exponent-nineteen height theorem as in Lemma A9.2 gives the asserted height bound. □

For completeness use the old scalar preparation with its stronger six-leaf conditions. For integral X, r is a nonzero integer. Choose c, a sufficiently divisible positive h_0, and then a still deeper positive s satisfying the anchor, exactly as in the proof of Lemma 16.1. At primes dividing r, both coefficients of $f(H)$ are divisible by r locally, so q/r is integral; the two retained ratio leaves are integral. At primes dividing c but not r, the same leaves are divisible by the prime. The constant-left tree is consequently one modulo every prime on the support of cr. It is nonzero and a unit at those primes. Thus crW^2 is positive, has square reduced denominator, and has the same local squareclasses as cr at two, three, and seventeen. The original pole-completeness lemma supplies fresh scalar pole witnesses. The new norm and ratio functions are continuous at the chosen points, so the same finite-neighborhood argument provides a congruence class of h_0 preserving all required conditions.

A10.7 A reciprocal chart for the scalar difference

This replacement is used only for $R = S - P$. The five comparison subtractions retain their strengthened guards (125). Write the incoming scalar point as $S = (x, hj)$ with $x = h^2$ and $j^2 = x^2 - D$. Put

$$d = c_*x + D, \quad N = a_*j - b_*h, \quad M = -2a_*Dh(x + c_*) - b_*j(c_*x - D).$$

Both real embeddings give $c_* > 0$, $D > 0$, and $x \geq 0$; hence $d \neq 0$. Introduce $g, k \in K$, retain the existing outputs $f, z \in K$, and introduce a rational guard σ. The final equations are

$$dg = N, \qquad a_*dk + 2Dh + a_*b_*j - 2Db_*g = 0, \qquad f = g^2, \qquad f_0\sigma = 1, \qquad kz = -g. \tag{134}$$

Each field equation means its two rational component equations. Every component equation is quadratic. Since $f_0 = g_0^2 + 2g_1^2$, the guard is precisely $g \neq 0$ over rational coordinates, and imposes no unmentioned sign choice.

We give the identities and all exceptional cases needed to identify these outputs with the actual group difference. If

$$G = dg - N, \quad K_0 = d^2k - M, \quad L = a_*dk + 2Dh + a_*b_*j - 2Db_*g,$$

then expansion using $a_*^2 = c_*$ and $b_*^2 = c_*^2 - D$ gives

$$dL = a_*K_0 - 2Db_*G. \tag{135}$$

Thus the first two equations of (134) are exactly equivalent to $g = N/d$, $k = M/d^2$, because $a_*d \neq 0$. The latter expressions satisfy

$$k^2 + Dg^4 = 1, \qquad (a_*j - b_*h)(a_*j + b_*h) = (x - c_*)(c_*x + D). \tag{136}$$

For example the first is the polynomial identity $M^2 + DN^4 = d^4$ modulo $h^2 = x$, $j^2 = x^2 - D$ and the two fixed-constant identities. This also specifies an exact denominator-cleared check.

The guard gives $g \neq 0$. If $k = 0$, the first identity in (136) gives $d_*g^2 = 1$ or -1. Taking rational norms gives $7\,\mathrm{Norm}(g)^2 = 1$, impossible for a rational norm. Thus $k \neq 0$, and all displayed outputs are well defined. If $x \neq c_*$, use the ordinary subtraction slope $m = (hj + a_*b_*)/(x - c_*)$. Substitution and clearing denominators give

$$m^2 - x - c_* = 1/g^2, \qquad m(x - 1/g^2) - hj = k/g^3.$$

These are the abscissa and ordinate of $S-P$. Therefore the retained outputs are exactly $f(R)=g^2$ and $z(R)=-g/k$.

If $x=c_*$, the incoming chart forces $h=\pm a_*$ and $j=\pm b_*$. Equal choices of signs give $S=P$ and $g=0$, excluded by the guard. The other choices give $S=-P$. For $h=a_*, j=-b_*$ the explicit values are

$$g=\frac{-2a_*b_*}{c_*^2+D}, \qquad k=\frac{c_*^4-6c_*^2D+D^2}{(c_*^2+D)^2}.$$

They give the abscissa and ordinate of $-2P$ by the ordinary duplication formula. Simultaneously negating h, j negates g, k and leaves $1/g^2$, k/g^3, and $-g/k$ unchanged. Thus this branch also represents the true difference, not an arbitrary value of an old zero-over-zero slope.

In the complete present system that extra branch is impossible. It would give $R=-2P$ and $v_3(f_1(R))=2$ by Lemma A10.3, whereas (128) requires $v_3(f_1(R))\geq 2^{25}$. It follows that every final solution has $x\neq c_*$. Restore the old inverse of $x-c_*$, slope, slope square, and ordinate products by their ordinary rational formulas. All retained outputs coincide. Conversely every old solution extends to (134) by the displayed formulas, with $\sigma=1/f_0$; its actual nonzero $f(R)=g^2$ ensures $f_0>0$. This is an equivalence on all nonprivate coordinates of the complete system, with the anchor explicitly used to exclude the additional branch.

For the coordinate count one may first use $d_2=d^2$ and $d_2k=M$. That intermediate reciprocal chart replaces eight old scalar difference coordinates by the six components of g, d_2, k and one rational guard, saving one. Identity (135) then removes the two components of d_2. Forward, divide $dL=0$ by the already nonzero d; backward, reinsert the unique $d_2=d^2$ and use $a_*K_0=0$. This second step is a direct rational equivalence requiring no new exceptional-point argument. The combined saving is three.

A10.8 A shorter weight circuit with both outputs retained

In each detector let e be its weight e, $q=q_1$, and keep both outputs q_{12}, q_{40}. The grouped evaluations already contain $Q=q^3$ and $T=q_{12}^2$. Replace the six old intermediate powers of e by five rational coordinates $\alpha, \gamma, u, v, \tau$ and impose

$$e\alpha=1, \quad e\gamma=\alpha^2, \quad \gamma u=eq, \quad \gamma v=u^2, \quad Qq_{12}=v^2, \quad q\tau=eT, \quad vq_{40}=T\tau. \tag{137}$$

These are seven quadratic equations for seven coordinates when the two retained outputs are counted. The old circuit had eight.

The first equation gives $e\neq 0$. The unchanged definition of q is $q\delta^3=e$, with the powers of δ represented by their retained coordinates, so $q\neq 0$ as well. The equations now determine, in order,

$$\alpha=e^{-1}, \quad \gamma=e^{-3}, \quad u=qe^4, \quad v=q^2e^{11}, \quad q_{12}=qe^{22}.$$

The division by $Q=q^3$ is legitimate. The retained equation for T then gives $T=q^2e^{44}$, and the last two equations give

$$\tau=qe^{45}, \qquad q_{40}=qe^{78}.$$

In particular $v\neq 0$ whenever it is used as a divisor. This is a successive computation, not a circular specification involving T.

The forward map from the old circuit assigns these five Laurent monomials and retains q_{12}, q_{40}, Q, T, and every other coordinate. For the reverse map restore the old six powers as $e^{-1}, e^{-3}, e^{-7}, e^{-14}, e^{-29}, e^{-55}$. The old output equations then give the same qe^{22} and qe^{78}. The replaced internal powers have no outside consumers. Thus these maps preserve every outside coordinate, not merely the input. Applying this equivalence to all three detectors saves three coordinates.

A10.9 Removing the scalar norm coordinate

One further elimination uses coordinates already present in the order-forty test $f(H)$. With $f(H) = h_0 + h_1\rho$, its square is stored through

$$t = h_0^2 + 2h_1^2, \qquad u = h_0h_1.$$

Here u is half the ρ-coefficient, a normalization that matters. The scalar coordinate $q = h_0^2 - 2h_1^2$ from (131) has only one other consumer, the equation $r(A-1) = q^2$. Use the identity

$$(h_0^2 - 2h_1^2)^2 = (h_0^2 + 2h_1^2)^2 - 8(h_0h_1)^2 = t^2 - 8u^2. \tag{138}$$

Delete q and its defining row and replace the consumer by

$$r(A-1) = t^2 - 8u^2.$$

This is still quadratic. Forward, forget q. Backward, reinsert $q = h_0^2 - 2h_1^2$. Identity (138) proves both rows in the old system and makes these maps mutually inverse. No division, nonzero hypothesis, or extra arithmetic input is used. The proof of Lemma A10.4 applies by this unique reinsertion; it does not require retaining q as a quantified variable.

A10.10 Soundness of the combined construction

Let a rational solution of the final system be given. The inherited odd charts identify the four root points with odd integer labels. The reciprocal chart supplies the actual $R = S - P$, is nonidentity, and, together with the anchor, excludes $S = -P$. The explicit maps above therefore restore all unchanged point domains and quotient guards needed by the earlier arguments. The removed scalar norm coordinate also has its unique rational value.

Put $\kappa = |s| = |a-1|$. By (129) and Lemma A10.4,

$$\log_2 \kappa > 25165823.5, \qquad \mathrm{h}(\beta) \le \mathcal{H}\kappa^{19/32}.$$

These are the exact bounds consumed in the Version 9 height argument. For completeness we repeat the numerical deductions. Since

$$10(7 + 2093966) < 25165823.5, \qquad 80 < 2^7,$$

we have

$$\kappa > (80\mathcal{H})^{10}, \qquad |a| \ge \kappa - 1 \ge \kappa/2 > (40\mathcal{H})^{10},$$

and

$$\mathrm{h}(\beta) < \kappa/80 \le |a|/40, \qquad \log(40\mathcal{H}) < \kappa/80 \le |a|/40. \tag{139}$$

The first inequality uses $\kappa^{13/32} > 80\mathcal{H}$; the logarithmic inequality is weaker. In particular all labels are in the large range needed below.

Set

$$F = t^2 - a(a-2)tb + (a-2)^2b^2 - 1, \qquad B = \max(|n|, |a|, |t|, |b|) > 4.$$

The triangle inequality gives $|F| < 5B^4$. If $F \neq 0$, Lemma A10.2 and the unchanged detector implication make the three contact radicals divide F to order forty. Equation (122) then gives

$$B^{1/10} < 7\mathcal{H}\,5^{19/40} + 6 < 24\mathcal{H},$$

contradicting $B \geq |a| > (40\mathcal{H})^{10}$. Hence $F = 0$. The point Q_4 is nonidentity, so its integer label $b \neq 0$. Lemma 14.9 therefore gives $\log B > |a| - 3$.

If $n \neq \beta$, let N be the nonzero reduced numerator of $n - \beta$. Then $|N| \leq 2e^{\mathrm{h}(\beta)}B$, and the same three radicals divide N to order twelve. Using $B^2 - 6 > (5/8)B^2$ for $B > 4$, equation (122) gives

$$B^{5/12} < 40\mathcal{H}e^{(19/12)\,\mathrm{h}(\beta)}.$$

Consequently

$$\log B < \frac{12}{5}\log(40\mathcal{H}) + \frac{19}{5}\,\mathrm{h}(\beta) < \frac{31}{200}|a| < |a| - 3,$$

a contradiction. Thus $n = \beta \in \mathbb{Z}$. If $X = u/v$ in lowest terms with $v > 0$, integrality of $2X^2 + 3$ forces $v^2 \mid 2$, so $v = 1$. Every rational solution therefore projects to an integer input.

A10.11 Completeness with all replacements at once

Fix an integer X and put $n = \beta = 2X^2 + 3$, a positive odd integer. First perform the scalar preparation with the original six-leaf bounds, which, as shown above, suffice for the new scalar tree. Choose its finite correction and a sufficiently divisible positive shared index h_0, with $64 \mid h_0$. Choose a positive scalar index s still more divisible, large enough for the physical anchor, and with $6 \mid s$. Put $a = s + 1$. The actual prepared scalar point is $R = [s]P = S - P$. The new anchor has exactly the old allowed subgroup, and the new scalar tree has the required unit root. Fresh scalar pole witnesses therefore exist.

To choose the two other root labels, define

$$U_0 = 0, \quad U_1 = 1, \quad U_{j+1} = aU_j - U_{j-1}, \qquad k = 2(a-2), \quad b = U_k/(a-2), \quad t = U_{k+1}.$$

Induction gives $U_j \equiv j \pmod{a-2}$, so b is an integer. Modulo two the sequence has period $0, 1, 1$. Since $a \equiv 1 \pmod 3$, $k \equiv 1 \pmod 3$, and both U_k, U_{k+1} are odd. The positive odd denominator $a - 2$ makes b odd. The Lucas identity gives $F(a, t, b) = 0$. Thus all four root indices are positive and odd, and Proposition A9.7 supplies their chart witnesses.

Fix these root points and the three positive rational square scales of Corollary A9.6. The new scalar functions and the mixed alternate numerator are rational functions defined at the prepared points. Their required finite local conditions persist in a sufficiently small neighborhood. As in the original argument, choose a common continuity modulus L, with $64 \mid L$, so that $h \equiv h_0 \pmod L$ preserves them. All these choices precede the contact-prime selection. Increase the finite prime cutoff to include the fixed root denominators needed for the five half-trace guards.

Choose the contact primes using the same simultaneous-prime theorem and Lemma A9.8. Put

$$M_{\pm} = \mathrm{lcm}_p(d_{\pm}(p)/4), \qquad \Pi = \prod_p p.$$

The retained direct-target construction supplies

$$\gcd(M_-, M_+) = 1, \quad \gcd(M_-, L) = 16, \quad \gcd(M_+, L) = 1, \quad \gcd(\Pi, M_+) = \gcd(\Pi, L) = 1.$$

Thus the same three congruences are compatible:

$$h \equiv 0 \pmod{\Pi^{19}M_-}, \qquad h \equiv 1 \pmod{M_+}, \qquad h \equiv h_0 \pmod L.$$

Use the inherited real-progression argument to choose a sufficiently large positive h in this progression, avoiding the finitely many forbidden point and quotient loci. Set $U_i = [hn_i]P$ and $V_i = [(h-1)n_i]P$.

At every primary place the points H, U_i have depth at least twenty: the doubling-image order first makes them formal, and multiplication by p^{19} raises depth by nineteen under the same formal logarithm isometry. At opposite places the points V_i are formal. Hence L_i agrees with its integer label to order at least eighty; the alternate ratio agrees to order twelve for $i = 1$ and forty for the other three labels. All tests (124) hold. The same primary/opposite reductions prove that all five half-traces are nonzero, by the argument following (125). The chosen scalar label has $S \neq \pm P$, so its reciprocal chart has the ordinary rational lift described above.

The pole-realization estimates are valid for these fixed orders $1, 12, 40$. In the source notation their large-real-parameter decay is still λ^{-3}; the small-prime lower bound is $3\nu + (10m - 8)\eta - 2c$, which tends to infinity under the same preparatory divisibility. No new prime-pattern assertion is used. Obtain the detector pole witnesses and apply the rational normalization (126). Apply the weight-circuit maps (137), the reciprocal denominator cancellation, and the norm-square elimination. These exact maps preserve all outside coordinates. They therefore give one simultaneous rational solution of the final system for the chosen integer X.

This proves completeness in the required order of choices. In particular, it does not combine separately chosen primes or incompatible continuity moduli from different optimizations.

A10.12 Exact polynomial, counts, and coefficient lift

Each equation above is represented by its rational component rows; intermediate products in the retained straight-line program are represented by their specified coordinates. Normalize each nonzero rational residual by clearing denominators and dividing its integer content, and let P_{498} be the sum of the squares of those 467 normalized residuals. This is an integer polynomial. A sum of rational squares vanishes exactly when every residual vanishes. The soundness and completeness just proved therefore establish Theorem A10.1.

The following count records the actual transported, fully reduced systems; it does not count unreduced formulas in isolation.

System or replacement	Total variables	Quadratic residuals
The retained Version 9 baseline	523	492
Mixed input precision and its checked compilation	519	488
Three normalized detector poles	516	485
Five half-trace guards	511	480
Single-coefficient translated anchor	510	479
Shorter scalar norm tree	505	474
Scalar reciprocal chart with d^2 retained	504	473
Three weight circuits retaining q_{12}, q_{40}	501	470
Cancellation of the scalar d^2 coordinates	499	468
Norm-square coordinate elimination	498	467

The mixed-depth compilation uses only exact quadratic identities, constant-pivot substitutions, and invertible shears in shared coefficient coordinates. A constant-pivot substitution inserts its unique polynomial value in the reverse direction; a shear inserts its inverse linear transformation. Both maps fix the input. The final norm-square elimination is the explicit identity (138). Thus no compiler reduction introduces a variable denominator or alters an arithmetic guard. The exact certificates also check that private coordinates deleted in the other replacements have no undeclared consumers. In particular, the half-trace witnesses occur only in their guard equations; the removed weight powers occur only in their old weight circuit; the removed scalar d^2 coordinates occur only in their definitions and k equations; and the removed norm occurs only in its definition and constant-node equation.

The frozen residual file is `candidate498-ir.json`, with SHA-256

`7682a1229149c75e27322e86dec424449b55321b06c48e468f90fa78bcebb61a`.

The complete sparse coefficient table `polynomial498.json` has SHA-256

`817fec936dee3b358fc8b5b0b9a5570694adf009b134432467ef0be914aeedc4`.

Its exact expansion has 24,669 monomials and degree four; all 498 variables occur. A coefficient of absolute value one proves content one. The only quadratic-in-C coefficient is $16C^2$, at the square of the translated anchor witness w, with no lower-C term at that monomial. All other coefficients are independent of C or linear in C and are smaller in absolute value. The coefficient comparison is carried out symbolically; no expansion of $3^{2^{25}}$ is needed. It gives the exact maximum $16C^2 = 16 \cdot 3^{2^{26}}$.

There is also an ordinary-coefficient lift with sixteen additional coordinates:

$$c_0 = 3^{1024}, \qquad c_{j+1} = c_j^2 \quad (0 \le j < 15), \qquad C = c_{15}.$$

The recurrence has exactly one rational solution and ends at $3^{1024 \cdot 2^{15}} = 3^{2^{25}}$. Replace $-4Cw$ in (128) by $-4c_{15}w$. All residuals remain quadratic. This gives a specified quartic in 514 variables with the same input projection and 513 existential witnesses. It has 483 quadratic residuals and 24,716 monomials, with content one and maximum coefficient $3^{2048} + 1005072314676622 < 2^{3247}$. Its coefficient table, `polynomial514.json`, has SHA-256

`d247a27709afda5f2be626390cffa1a175c3dfcbee1a5c832a1c457b32629bd5`.

The lift is unique, so it requires no additional arithmetic argument.

The exact checks establish the residual identities, the witness counts, and every coefficient of the displayed sums of squares. They are separate from the arithmetic arguments above. The inherited source qualifications of Section A9.1 remain in force; a finite polynomial check is not a new proof of the restricted converse, the height theorem, or the simultaneous-prime input.

A11 A quartic presentation with 394 variables

The transformations in this section give the following specified polynomials. The arithmetic inputs and their scope are those retained in Section A10.1.

Theorem A11.1. *There is a specified integer polynomial P_{394} of total degree four in 394 variables such that*

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{v} \in \mathbb{Q}^{393} : P_{394}(X, \mathbf{v}) = 0.$$

It has 6,338 nonzero monomials, content one, and largest absolute coefficient $16 \cdot 3^{2^{26}}$. Its symbolic coefficient table uses $C = 3^{2^{25}}$ as a fixed integer, not an additional variable. There is also a specified integer quartic P_{411} with

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{w} \in \mathbb{Q}^{410} : P_{411}(X, \mathbf{w}) = 0.$$

It has 6,388 nonzero monomials, content one, and all coefficients have absolute value less than 2^{3247}. Its largest coefficient has exactly 3,247 bits; its exact value is included in the coefficient artifact.

Some transformations below are inverse rational changes of private witnesses, fixing every outside coordinate. Others change the arithmetic certificate or enlarge a point domain; for those we prove soundness and existence of witnesses for every integer input separately.

A11.1 Conventions and retained arithmetic inputs

Put $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $d_0 = 3 - \rho$, $D = d_0^2$, and $E : y^2 = x^3 - Dx$. Retain $2E(K) = \mathbb{Z}P$, with

$$P = (c_*, a_* b_*), \quad a_* = -1 + 7\rho/4, \quad b_* = 39/8 - 9\rho/2, \quad c_* = a_*^2.$$

The roots $Q_i = [n_i]P$ have nonzero odd indices, with $(n_1, n_2, n_3, n_4) = (n, a, t, b)$. In the intermediate source the auxiliary point is $H = [h]P \neq O$, and $U_i = [u_i]P$, $V_0 = H - P$, $V_i = U_i - Q_i$. We will allow U_i and V_i to be O for $i > 0$. The proof below still excludes $V_0 = O$. The scalar point is $R = Q_2 - P = [s]P$, where $s = a - 1$ is nonzero and even.

The retained root-height implication is

$$B^2 - 6 < 7\mathcal{H}|N|^{19/k}, \qquad B = \max_i |n_i|, \quad 1 \le \mathcal{H} < 2^{2093966}, \tag{140}$$

whenever the kth powers of the three contact radicals divide the nonzero integer N. Their scales are $1, C_0^2, (1 + C_1^2)^2$, where $\prod_i x(Q_i) = C_0 + C_1\rho$. The retained scalar anchor gives

$$\log_2 |s| > 25165823.5. \tag{141}$$

These are precisely the root-height and anchor statements proved earlier; neither is inferred from a finite coefficient calculation.

A11.2 Rational logarithms and the two order-ten tests

Write $f = 1/x$, $z = -x/y$, and extend these regular functions by $f(O) = z(O) = 0$. Define

$$\begin{aligned}
Q(w) &= 1 - \frac{1526}{1073}w + \frac{513}{1073}w^2, \\
N(w) &= 1 - \frac{9776}{5365}w + \frac{2941}{3219}w^2 - \frac{1024}{13949}w^3 - \frac{1920}{237133}w^4 - \frac{1024}{711399}w^5 - \frac{1408}{5928325}w^6 + \frac{2432}{103152855}w^8, \\
\mathcal{R}(w) &= N(w)/Q(w).
\end{aligned}$$

The exact identity

$$QG_9 - N = \frac{2970726}{254443709}w^{10} - \frac{41697666}{6361092725}w^{11}$$

proves that this replacement has the same order-forty local accuracy. Indeed, on $[0, 1]$ the denominator is at least $60/1073$ and the Bernstein coefficients of the numerator are

$$1, \ \frac{4143}{5365}, \ \frac{260069}{450660}, \ \frac{27829}{67340}, \ \frac{4636683}{16599310}, \ \frac{82391}{474266}, \ \frac{2245591}{23713300}, \ \frac{955437}{23713300}, \ \frac{83904}{10113025}.$$

They are all positive and give $N \ge 83904/10113025$. Every nonidentity point of $2E(K)$ has $0 < Df^2 < 1$ at both real embeddings; at O its value is zero. Thus it is defined and nonzero on these doubled-point domains. For the later enlarged H domain and its difference H-P, we instead use the global numerator and denominator nonvanishing certificates below.

At an unramified place above $p > 76176$, a nonidentity formal point of depth $r = v(z) \ge 1$ has $v(Df^2) = 4r$. All coefficient denominators above are supported at primes at most37. Consequently $v(\mathcal{R} - G_9) \ge 40r$. The earlier full-tail estimate gives $v(\log_E(Q) - z\mathcal{R}(Df^2)) \ge 41r$. The point O has exact zero error. We retain the G_2 numerator at V_1, of relative precision12, and use $\mathcal{R}$ at V_0, V_2, V_3, V_4.

Set $\ell_0 = z(V_0)\mathcal{R}(Df(V_0)^2)$ and define

$$L_i = z(U_i)/z(H), \quad L_1' = \overline{z(V_1)G_2(Df(V_1)^2)/\ell_0}, \quad L_i' = \overline{z(V_i)\mathcal{R}(Df(V_i)^2)/\ell_0} \quad (i > 1).$$

The nonzero denominator is proved below, before these quotients are used. The tests have orders1 on $f(U_i)$ and $\overline{f(V_i)}$, order40 on $f(H)$, order10 on $L_1 - L_1', L_1 - \beta$, and order40 on the other three cross-tests and

$$F(L_2, L_3, L_4), \qquad F(A, T, B) = T^2 - A(A-2)TB + (A-2)^2B^2 - 1.$$

Conjugation and the place conventions are unchanged.

For completeness, spell out the zero-safe local argument. Let $A(Q)$ be the analytic unit defined by $\log_E(Q) = z(Q)A(Q)$, with $A(O) = 1$. Define $T_2 = G_2/A$ and $T_{40} = \mathcal{R}/A$ by their power series. These units equal1 at O. Their errors at nonidentity formal points have the stated orders. The primary ratio is exactly $(u_i/h)A(H)/A(U_i)$, also when $u_i = 0$. The alternate ratio is $(u_i - n_i)/(h-1)$ times the corresponding conjugated unit ratio. At least one of $h, h-1$ is a p-unit. Therefore one full ratio is integral, and the positive-order cross-test transfers integrality to both approximate and full ratios. Only now infer depth at least20 for a nonzero U_i; if $U_i = O$, its error is zero. The primary error has order80. Multiplication by $h, h-1$ and subtraction give

$$L_1 \equiv n \pmod{p^{10}}, \qquad L_i \equiv n_i \pmod{p^{40}} (i > 1).$$

No quotient $\log_E(O)/z(O)$ is used.

A11.3 Odd roots, shared reciprocals and identity-inclusive charts

The retained finite chart for a root is $x = h_Q^2$, $j_Q^2 = x^2 - D$, $Q = (x, h_Q j_Q)$. It has $h_Q j_Q \neq 0$ by the real embeddings and $\mathrm{Norm}(d_0) = 7$. Replace its old odd-index witnesses by $A \in K, u \in \mathbb{Q}$ with

$$A^2 - 2(x + j_Q)A - D = 0, \qquad \mathrm{Norm}(A) = -u^2.$$

Here is the reverse implication, including the squareclass choice. Put $q = x + j_Q$ and $t = (A - q)/h_Q$. The equation gives $t^2 = 2q$. The point (A, At) lies on E and its tangent slope is $h_Q + t$, so its double is exactly $(x, h_Q j_Q)$. The inherited squareclass decomposition writes its abscissa class as $[\epsilon]^m[-1]^e[d_0]^f$, where $\epsilon = 1 + \rho$ and $e, f \in \{0, 1\}$. The norm condition forces m odd and $f = 0$. Thus $[A] = [\epsilon]$ or $[-\epsilon]$. In the latter case replace A, t by $2q - A, -t$; the product of the two abscissae is $-D$, so the new class is $[\epsilon]$. Choosing the corresponding square root recovers the old odd witnesses without changing the root. Conversely the old witnesses t, r give $A = th_Q + q = \epsilon r^2$, $u = \mathrm{Norm}(r)$. This is equality of exterior projections, not a unique private inverse.

For the primary ratios keep one shared field value $S = j_H/h_H = -1/z(H)$. The finite-point equations become

$$h_H S = j_H, \qquad j_i L_i = h_i S.$$

These remove the four private primary parameters $z(U_i)$ without changing their ratios. The scalar equation can be fused in the same way. If its old equations are $Tz_R + a_* N = 0$ and $h_H \varrho + j_H z_R = 0$, replace them by

$$T\varrho - a_* NS = 0.$$

The inverse is $z_R = -\varrho/S$; $S \neq 0$ follows from the retained finite H chart.

For $i > 0$ an identity-inclusive primary chart instead uses

$$f_i = r_i^2, \qquad J_i^2 + Df_i^2 = 1, \qquad J_i L_i = r_i S.$$

When $r_i \neq 0$, set $h_i = 1/r_i, j_i = J_i/f_i$ to recover the old finite chart. When $r_i = 0$, interpret $U_i = O$. Both choices $J_i = \pm 1$ give that same point. The factor J_i cannot vanish: it would give

$Dr_i^4 = 1$, and taking real signs and then norms gives $7\,\mathrm{Norm}(r_i)^2 = 1$, impossible over $\mathbb{Q}$. Thus $z(U_i) = -r_i/J_i$ and the primary ratio is correct also at O.

Write the corresponding fixed root as $Q_i = (c, ab)$ with $c = a^2$, $b^2 = c^2 - D$. Define

$$d = c + Df_i, \quad dg = aJ_i - br_i, \quad u = cr_i + bg, \quad dk = bJ_i - 2au, \quad f(V_i) = g^2, \quad kz(V_i) = -g.$$

The factor d is positive at both real embeddings. Reduction by the primary conic gives $k^2 + Dg^4 = 1$, so $k \neq 0$ by the same norm argument. On the ordinary subtraction domain these expressions give $V_i = (1/g^2, k/g^3) = U_i - Q_i$. They have the following exact boundary values. If $r_i = 0$, then $g = J_i/a$, $k = -bJ_i/c$, and $V_i = -Q_i$. If $g = 0$, substitution gives $(cr_i^2 - 1)(c + Dr_i^2) = 0$, hence $U_i = Q_i$ and $V_i = O$. The remaining equal-abscissa case is the regular doubling limit for $U_i = -Q_i$. Equality with the group-law morphism follows from its equality on the dense ordinary domain and regularity of these formulas.

For H and the denominator difference retain the finite chart, or its following exact rescaling. With $x = h_H^2$, put

$$d = c_*x + D, \quad N = a_*j_H - b_*h_H, \quad dg = N, \quad T = -2Dh_H - a_*b_*j_H + 2Db_*g.$$

The scalar reciprocal elimination is the identity $L = a_*dZ - zK - a_*G$, where $G = dg - N$, $K = a_*dk - T$, $Z = kz + g$, $L = Tz + a_*N$. Since $a_*d \neq 0$, reinserting $k = T/(a_*d)$ recovers the old equations.

For V0 one may also eliminate j_H by writing $j_H = h_HS$ and $g = h_Hv$. The resulting exact equations are

$$S^2 = x - Df(H), \quad (c_*x + D)v = a_*S - b_*, \quad f(V_0)f(H) = v^2,$$

$$(-2D - a_*b_*S + 2Db_*v)\ell_0 + a_*(a_*S - b_*)\mathcal{R}(Df(V_0)^2) = 0.$$

Both maps divide only by the already nonzero h_H and x. These exact maps precede the separately justified enlargement of H below.

The denominator cannot be the identity. The old argument using $V_1 \neq O$ is no longer available. Instead suppose $V_0 = O$. The reciprocal chart gives $H = P$, and

$$f(H) = \frac{8(57 + 28\rho)}{1681}$$

is a unit at every consumed place over $p > 76176$. The one-place detector lemma would force its order-forty test at any contact prime, a contradiction. Thus all three contact radicals equal1. Equation (140) with $N = 1$ gives $B^2 - 6 < 7\mathcal{H}$, contrary to $B \geq |a| > (40\mathcal{H})^{10}$ from the unchanged anchor. This argument precedes every interpretation of an LP quotient. Now $V_0 \neq O$, its parameter is nonzero, and the established positivity of $\mathcal{R}$ gives $\ell_0 \neq 0$. The local ratio proof above therefore applies to all new solutions. It proves soundness directly, without asserting that identity-branch witnesses lift to the old affine source.

A11.4 Scaled biquadratic comparison charts

For one root write $Q = (c, ab)$, $c = a^2$ and $b^2 = c^2 - D$. The reciprocal comparison chart above uses r, J, f, g, k, u, F, with $f = r^2$, $F = g^2$. Its retained squared outputs are $Y_U = f^2$ and $Y_V = F^2$, as elements of K, not separate component squares. Replace the seven field values by five:

$$V = ar, \quad Q' = ag, \quad I = aJ, \quad K' = ak, \quad W = rg.$$

The new equations are

$$\begin{aligned} VQ' &= cW, & V^2 + (Q')^2 + DW^2 + 2bW &= 1,\\ aI &= bV + cQ' + DVW, & aK' &= -bQ' - cV - DQ'W,\\ IL &= VS, & K'z &= -Q',\\ DcY_U + I^2 &= c, & DcY_V + (K')^2 &= c. \end{aligned}$$

Every displayed equation is quadratic in retained coordinates. In particular c is an existing coordinate, so cY is a quadratic product.

Here is an algebraic proof including all zero branches. Put $d = c + Dr^2$, $e = c + Dg^2$, and define

$$T = c(r^2 + g^2) + Dr^2g^2 + 2brg - 1,$$

$$G = dg - aJ + br, \quad E' = ak + bg + er, \quad H' = dk - bJ + 2a(cr + bg).$$

Modulo only $c = a^2$ and $b^2 = c^2 - D$, direct expansion gives

$$\begin{aligned} a^2(J^2 + Dr^4 - 1) &= dT - G(dg + br + aJ),\\ a^2(k^2 + Dg^4 - 1) &= eT + E'(ak - bg - er),\\ dE' &= aH' + DrT - bG. \end{aligned}$$

The factors a,c,D are nonzero, and d,e are positive at both real embeddings. An old solution has G=H'=0, hence T=E'=0, proving all new equations and the second reciprocal conic. Conversely set $r = V/a$, $g = Q'/a$, $J = I/a$, $k = K'/a$; the product equation forces W=rg. The new equations give T=G=E'=0, so the three identities recover both conics and $H' = 0$. Set $f = r^2$, $F = g^2$ and $u = cr + bg$. Comparing each new Y equation with its recovered conic gives $Y_U = r^4$ and $Y_V = g^4$. Dividing the ratio equations by a restores the old ratios. Thus these are inverse rational maps, fixing every outside coordinate.

There is no division by r,g,J,k,W,L,z or either test value. When r=0 the first point is O and its ratio is zero; when g=0 the difference is O and its parameter is zero. The biquadratic equation prevents both from vanishing. These are exactly the already accepted identity branches. This transformation replaces the intermediate inverse-addition change; it is not composed with a second copy of that elimination.

A11.5 Enlarging the auxiliary H point domain

After the exact H.j rescaling, the two coordinates H.h occur only in the two component equations x=H.h^2. Delete them and keep

$$xf = 1, \qquad S^2 = x - Df.$$

They define the finite point H=(x,xS) of E(K), with x nonzero, but no longer require H to be a double. The rank and torsion calculation in the generator proof gives

$$H = [m]R_0 + T, \quad P = 2R_0, \quad m \in \mathbb{Z}, \quad T \in E[2](K).$$

Write h=m/2 only for its rational free index. It is not notation for multiplication by a half-integer on the elliptic curve.

We first exclude all finite points in $E[2]$, $P + E[2]$ and $-P + E[2]$ before interpreting ratios. The points O and (0,0) are excluded by the finite chart. Put $d_0 = 3 - \rho$, $c_* = x(P)$. The exact norm and translation identities are

$$\mathrm{Norm}(c_*) = 41^2/64, \quad \mathrm{Norm}(c_* - d_0) = 17^2/64, \quad \mathrm{Norm}(c_* + d_0) = 63^2/64,$$

$$x(P+(0,0)) = -D/c_*, \quad x(P+(d_0,0)) = d_0(c_*+d_0)/(c_*-d_0),$$
$$x(P+(-d_0,0)) = -d_0(c_*-d_0)/(c_*+d_0).$$

Each of these abscissae and its inverse is integral outside $\{2,3,7,17,41\}$. The same is true for the nonzero torsion abscissae, and negating P changes none of these abscissae. Thus f(H) is a unit at every consumed place. The retained order40 f(H) test would then rule out every contact prime. All three radicals would equal1, contradicting the root-height bound and scalar anchor exactly as in the preceding V0 exclusion. This proof needs no ratio involving H. Consequently m is neither0 nor2, H and H-P are nontorsion, S is nonzero, and H is neither P nor -P.

In the rescaled difference equations put

$$d = c_*x + D, \quad N = a_*S - b_*, \quad dv = N, \quad T' = -2D - a_*b_*S + 2Db_*v.$$

The former positivity proof for d no longer applies. If d=0, the curve equation gives $x = -D/c_*$, $S = \pm b_*/a_*$, so H is one of the excluded points $\pm P + (0,0)$. If N=0, it gives $(x-c_*)(x+D/c_*) = 0$, also excluded. Thus d,N,v are nonzero. Direct subtraction now gives

$$x(H-P) = 1/(xv^2), \qquad y(H-P) = T'/(a_*dxv^3), \qquad z(H-P) = -a_*dv/T'.$$

There is no unresolved equal-abscissa case, since $x = c_*$ means H=P or -P. Since H-P is nontorsion, its ordinate and T' are nonzero. In particular the retained equation $f(V_0)f(H) = v^2$ has its intended interpretation, even if either real abscissa is negative.

The old positivity interval for the rational logarithm is no longer enough. A stronger certificate is available: N(w) above has no root modulo71, and Q(w) has no root modulo7. At the respective prime all coefficients are integral and the leading coefficient is a unit. These assertions are finite substitutions through all residues. Both primes split in K, with square roots of2 equal to12,59 modulo71 and3,4 modulo7. A K-root would give a local root, necessarily integral, and then a forbidden residue root. Hence N and Q never vanish in K. The actual equation $T'\ell_0 + a_*N\mathcal{R}(Df(V_0)^2) = 0$ therefore implies $\ell_0 \neq 0$ and gives the correct denominator logarithm. No sign assertion for this logarithm is needed.

The formal ratio argument changes by using integral group identities:

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i-n_i)]V_0.$$

They kill T. Applying the formal logarithm only to formal points gives ratios u_i/h and $(u_i-n_i)/(h-1)$. At every consumed odd prime h and h-1 are integral and differ by1. The earlier transfer and subtraction proof therefore applies, including zero numerators. At opposite split places use the same conjugation as before; at an inert place it preserves the valuation.

The scalar transfer uses the same integral-index convention. For $R = [s]P$ use

$$[m](2R) = [4s]H.$$

If the scalar coefficient test has depth d_s, formal logarithm isometry gives $v_p(s/(m/2)) \geq d_s$. Since m/2 is p-integral, $v_p(s) \geq d_s$. This includes the selected depth24. The proof that 2R is formal when R is near O or (0,0) is unchanged. Thus both the comparison and scalar height arguments survive the larger domain. Completeness uses the original H in $2E(K)$ and forgets its two square witnesses. It need not realize any newly admitted half-index or torsion translate.

A11.6 A positive scalar input and the depth-twenty-four certificate

Use the selected label and contact polynomial

$$\tau = 2X + 1, \qquad \beta = 51\tau^2 + 2504, \qquad r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1).$$

For every rational X, $\beta \geq 2504$, hence $r > 0$. For integral X, τ is odd and β is a positive odd integer. Conversely, the final conclusion must be that β is an *odd* integer. If $\tau = u/v$ in lowest terms, integrality gives $v^2 \mid 51$; as 51 is squarefree, $v = 1$. Oddness of β then makes τ odd and $X = (\tau - 1)/2$ integral. Mere integrality would not suffice: $X = -1/2$ gives the even integer $\beta = 2504$. The unchanged odd chart for $Q_1 = [n]P$ supplies the required oddness once $n = \beta$ has been proved.

At a consumed prime $p > 76176$, 51 is a unit. A negative valuation of β is $2v_p(\tau)$, hence even. The five contact factors then have the same valuation, so an odd valuation of r must be positive. This argument makes no assertion at the exceptional primes3 and17.

For integral X the following simultaneous local calculation holds. Since τ is odd, $\beta \equiv 3 \pmod 8$; the two even contact factors each have valuation one and equal odd parts modulo8. The other factors have residues3,7,5. Therefore $v_2(r) = 2$ and $r/4 \equiv 1 \pmod 8$. At3, $\beta \equiv 2$ and the first four factors have product1 modulo3. Furthermore

$$(2\beta - 1)/3 = 34\tau^2 + 1669 \equiv 1 + \tau^2 \pmod 3,$$

which is nonzero. Thus $v_3(r) = 1$. At17, the contact residues are 5,4,1,10,9 with product15, a nonzero square. Thus $v_{17}(r) = 0$.

An intermediate one-choice construction used $c = 33033$ when $3 \mid \tau$ and $c = 33$ otherwise. Both are square units at2 and17, and their unit parts at3 are respectively1 and2, correcting the two possible classes of r. The selected construction removes even this choice and sets

$$c = r + 4896, \qquad 4896 = 2^5 \cdot 3^2 \cdot 17.$$

At2 the quotient $c/r = 1 + 4896/r$ belongs to $1 + 8\mathbb{Z}_2$; at3 and17 it belongs to $1 + p\mathbb{Z}_p$. Hence cr is a square in all three local fields. Also c, r are positive integers. At a consumed scalar contact prime, positivity of the odd valuation $v_p(r)$ was proved before using c; since4896 is a unit there, c is a unit there too. We do not assert that c is a unit at *every* large prime. Its additional finite prime support is included in the preparation before the continuity modulus is chosen.

The physical input square $Y = X^2$ is retained. In each other consumer, the old occurrence of Y is replaced by $102Y + 102X + 76$, since $2(102Y + 102X + 76) + 2403 = 51(2X + 1)^2 + 2504$. This substitution must include every scalar test and height-input consumer, not the defining row $Y = X^2$ itself.

For the deterministic correction the two old physical rows, in their fixed normalization, are

$$cs_0 - 20619812F = 0, \qquad (7895415c + 20619812F)V - 20619812b_0 = 0.$$

The retained contact expression has $r = -4(s_0 + 7895415)$. Substitution and multiplication by -4 give $rcV + 82479248b_0 = 0$. With the already retained coordinate $r_2 = r^2$, the selected row is

$$(r_2 + 4896r)V + 82479248b_0 = 0.$$

This is quadratic and preserves the physical sign of the scalar input. The free correction coordinate and its product coordinate have no remaining consumers. In the reverse arithmetic interpretation the correction is defined to be $r + 4896$; the removed finite-choice equation is not reimposed.

Write $q = \mathrm{Norm}_{K/\mathbb{Q}}(f(H))$ and $z(R)/z(H) = z_0 + z_1\rho$. The selected scalar root is

$$B_s = \frac{z_0^2 + rz_1^2}{r^{32}}, \qquad W = \frac{B_s^2 + rq^2}{r^{32}}, \qquad \mathfrak{b} = crW^2. \tag{142}$$

Here $q \neq 0$ and $r > 0$, so $W > 0$. Forward powers compute r^{32}; there is no need to retain an inverse-power tower or r^{64}. The square q^2 is computed without a separate norm coordinate: if $f(H) = f_0 + f_1\rho$, use

$$q^2 = (f_0^2 + 2f_1^2)^2 - 8(f_0 f_1)^2.$$

The factor 8 is essential: the second stored quantity is a product, not the full ρ-coefficient.

Here is the changed local application. At a consumed prime let $\eta = v_p(r) > 0$ be odd. The correction is a unit there. The pole obstruction gives $\eta + 2v_p(W) > 0$. The two terms in each numerator of (142) have opposite valuation parity, so they cannot cancel; assign valuation $+\infty$ to a zero term. Successively this gives

$$v_p(B_s) > 63\eta/4, \quad v_p(q) > 61\eta/4, \quad v_p(z_0) > 191\eta/8, \quad v_p(z_1) > 187\eta/8.$$

Thus both rational coefficient valuations are at least24. Positive valuation of the norm supplies a place with $v(f(H)) > 0$; there H is formal and $v(z(R)/z(H)) \geq 24$. Small $z(R)$ alone does not imply that R is formal. If $e = v(x(R))$, then $e \neq 0$, and the duplication identity gives, for either sign of e,

$$v(x(2R)) = -2v(z(R)), \qquad v(z(2R)) = v(z(R)).$$

For the enlarged H domain apply the formal logarithm to $[m](2R) = [4s]H$, where $h = m/2$, to obtain $v_p(s/(m/2)) \geq 24$. Since $m/2$ is p-integral at this odd prime, $v_p(s) \geq 24$. This uses integer group multiples only. The same height theorem therefore gives

$$\mathrm{h}(\beta) \leq \mathcal{H}|s|^{19/24}.$$

The exact inequality $(5/24)\,25165823.5 > 2093966 + 7$ and (141) imply $\mathrm{h}(\beta) < |s|/80 \leq |a|/40$. The retained anchor also gives $\log(48\mathcal{H}) < |a|/40$.

For completeness choose c first. The existing preparation permits arbitrary finite coefficient-depth conditions over the support of cr. At a prime dividing r, impose depth $8v_p(r)$ on the coefficients of $f(H)$ and depth $24v_p(r)$ on z_0, z_1. Then W is integral there. At primes dividing c but not r, sufficiently positive depths give integrality as well. The actual pole input crW^2 is positive, has square reduced denominator and the prescribed local squareclasses. These are the retained pole-completeness hypotheses; W need not be a unit. Fresh pole witnesses are supplied for this new input.

The scalar coefficient guard may be imposed on $f(R)_1$ instead of $f(R)_0$. In the reciprocal chart $f(R) = g^2$ one has $f(R)_0 = g_0^2 + 2g_1^2$ and $f(R)_1 = 2g_0g_1$. A nonzero second coefficient therefore implies a positive first coefficient and recovers the old guard. Completeness uses the established exact formula $v_3(f([s]P)_1) = 2 + 2v_3(s)$ for nonzero even s, so this second coefficient is nonzero on every prepared scalar orbit point. This is an arithmetic replacement of the guard, not an equivalence for arbitrary pairs g_0, g_1.

A11.7 The two final height contradictions

The local order-forty tests force the contact radicals to divide $F = t^2 - a(a-2)tb + (a-2)^2b^2 - 1$ to order40. If $F \neq 0$, then $|F| < 5B^4$ and (140) contradicts the retained lower bound $B \geq |a| > (40\mathcal{H})^{10}$. Thus $F = 0$. The nonzero odd root label b and the retained Lucas lemma give

$\log B > |a| - 3$. If $n \neq \beta$, its nonzero reduced numerator N satisfies $|N| \leq 2e^{\mathrm{h}(\beta)}B$ and is divisible by the three contact radicals to order10. Since $B^2 - 6 > (5/8)B^2$ for $B > 4$,

$$B^{1/10} < 48\mathcal{H}e^{(19/10)\,\mathrm{h}(\beta)}, \qquad \log B < 10\log(48\mathcal{H}) + 19\,\mathrm{h}(\beta) < 29|a|/40 < |a| - 3.$$

This contradiction proves $n = \beta$. The unchanged root chart makes n odd, so the parity-sensitive input argument above gives $X \in \mathbb{Z}$. The first contradiction uses order40; lowering only the scalar depth does not weaken that test.

A11.8 Polynomial circuit reductions

For retained field values Y_i, each order block evaluates $\prod_i (1 + qY_i)$. In order1 group the nine factors as

$$[V_0, U_1, U_2], \quad [V_1, U_3, U_4], \quad [V_2, V_3, V_4],$$

where labels denote the squared test values with their existing conjugations. Order10 uses its pair; order40 retains its pair and triple. For a triple a, b, c the polynomial is

$$1 + q(a + b + c) + q^2(ab + ac + bc) + q^3abc.$$

Its coefficients can be shared among the three detectors. If an old private coordinate is $d = bc$ and $e_2 = a(b + c) + d$, eliminate it by

$$d = e_2 - a(b + c), \qquad e_3 = ae_2 - a^2(b + c).$$

For the chosen first members the square a^2 already has a retained quadratic expression from the logarithm block. Substitution therefore keeps the new residual quadratic. The reverse map reinserts the displayed d. Every other use is substituted as well, including affine-normalized W coordinates. Such a coordinate has the explicit inverse $W_{\mathrm{old}} = (W_{\mathrm{new}} - r)/c$ with a fixed nonzero rational c. These are polynomial reconstructions apart from division by constants. They do not alter the field test values or pole conditions.

For the order weights let E and q be the retained nonzero scale and order-one weight. Use

$$Ea = q, \quad Eb = a^2, \quad qc = b^2, \quad cd = q^2q^3, \quad cq_{10} = d^2, \quad cq_{40} = q_{10}^4.$$

The last fourth power is evaluated through its retained square. These equations give $c = q^3/E^6$, $d = q^2E^6$, $q_{10} = qE^{18}$ and $q_{40} = qE^{78}$. All divisions are licensed by the unchanged norm/trace guards. The expression $q^5 = q^2q^3$ removes the former fifth-power coordinate without increasing degree.

An exact elimination is certified by a residual identity

$$F_{\mathrm{new}} = F_{\mathrm{old}}|_{x=q} + \sum_j A_j G_j,$$

where the retained G_j vanish and the old defining equation has a nonzero constant pivot in x. Forgetting x proves the forward map; reinserting x=q proves the reverse map. Checking all consumers is part of this argument, not an inference from a lower variable count.

A11.9 Factored poles and the preparation conic

For a normalized detector, the old second-conic rows include

$$17E - Eb - 17F = 0, \quad t_2 = t^2, \quad q_2 = q^2, \quad 17bq_2 + 867F^2 - 34Ft_2 - t_2^2 = 0.$$

Replace $F = tv$, $q = tQ$, $q_2 = t^2Q_2$ and use

$$Q_2 = Q^2, \qquad 17bQ_2 + 867v^2 - 34tv - t^2 = 0, \qquad 17E - Eb - 17tv = 0.$$

The old conic is t^2 times the new conic, so every new solution reconstructs an old one, including t=0. For completeness the retained construction supplies positive semantic b. If t!=0 divide by t; if t=0 the old equation forces F=q=0, so the zero lift works. The same argument applies to the scalar pole after its exact fixed normalization. Its stored input has the opposite sign from the semantic positive input; positivity must not be inferred from the stored symbol.

For scalar preparation, only the square N of the prepared value is consumed outside its private block. The old equations are

$$t_2 = t^2, \quad \mathfrak{b}(T - t_2) = 1, \quad TW = 2, \quad N = (W - \mathfrak{b})^2.$$

Replace them by

$$U^2 = N, \qquad \mathfrak{b}J = 1, \qquad JN + Y^2 = \mathfrak{b}.$$

Forward take $U = W - \mathfrak{b}$, $J = T - t_2$, $Y = tW$. Conversely, if $d = \mathfrak{b} + U \neq 0$, take $W = d$, $t = Y/d$, $t_2 = t^2$, $T = t_2 + J$. The conic identity gives TW=2. If d=0, then Y=0 and $N = \mathfrak{b}^2$; choose $W = 2\mathfrak{b}$, $t = t_2 = 0$, $T = J$. This changes only the sign of a private unsquared value and preserves N. The inverse guard is retained, and no rational point on an unrelated conic is being assumed.

A11.10 Specializing the scalar preparation to zero

Write $q = \mathrm{Norm}(f(H))$, and write the actual scalar ratio as $z(R)/z(H) = z_0 + z_1\rho$. The retained expressions are

$$B = (z_0^2 + rz_1^2)/r^{32}, \qquad W = (B^2 + rq^2)/r^{32}, \qquad \mathfrak{b} = crW^2.$$

Here r, c are positive and $q \neq 0$, so $W, \mathfrak{b} > 0$ at every rational solution, independently of a private preparation inverse. The earlier preparation-zero lemma applies when $\mathfrak{b} < 17$, its reduced denominator is a square, it is locally square at $2, 3, 17$, and its valuations there are at least $3, 1, 2$, respectively. We now arrange these conditions at the same points used in the final construction.

Let G be the closure of the simultaneous real orbit of P and its conjugate, let G^0 be its identity component, and put $m_0 = |G/G^0|$. The real-progression lemma says that a progression with both residue and modulus divisible by m_0 has dense positive tail in G^0. The functions f, z extend continuously to the identity with value zero. Choose a multiple M of $64m_0$ whose multiples give all desired formal depths at primes in $S = \mathrm{supp}(cr) \cup \{2, 3, 17\}$. In particular require coefficient depth 8η for $f(H)$ when $\eta = v_p(r) > 0$, and positive coefficient depth at17. Density permits $H_0 = [h_0]P$, $h_0 > 0$, $M \mid h_0$, as close to the two real identities as needed to make $q(H_0)^2/r^{31}$ arbitrarily small. Its real z values are nonzero.

With H_0 fixed, choose a multiple N_0 of $6m_0$ imposing the anchor and all required scalar finite depths. At $p \mid r$ require coefficient depth 24η for $z(R)/z(H_0)$; impose positive depths also at17 and at primes dividing c but not r. The formal subgroups have finite index, so arbitrary further depths can be imposed this way. Density of the positive progression $s \equiv 0 \pmod{N_0}$ permits s arbitrarily

large and $R = [s]P$ arbitrarily close to both real identities. The two real ratios, hence their rational coefficients z_0, z_1, become small. Choose them so that $\mathfrak{b}(H_0, R) < 17/4$. This also permits the size part of the anchor; set $a = s + 1$ as before.

At $p \mid r$ these depths give $v_p(q) \geq 16\eta$, $v_p(B) \geq 16\eta$ and $v_p(W) \geq 0$. At $p \mid c$, $p \nmid r$, the imposed positive depths again give $v_p(W) \geq 0$. Outside this support $v_p(\mathfrak{b}) = 2v_p(W)$, so the reduced denominator of $\mathfrak{b}$ is a square. Its squareclasses at $2, 3, 17$ are those of cr. Since $v_2(cr) = 4$, $v_3(cr) = 2$, the first two required bounds hold. At17 both r, c are units; the separately imposed depths give $v_{17}(q), v_{17}(B) \geq 2$, hence $v_{17}(W) \geq 4$ and $v_{17}(\mathfrak{b}) \geq 8$. Inclusion of17 does not depend on it belonging to the support of cr.

Keep R fixed. A nonempty real neighborhood $U \subset G^0$ of H_0 has $0 < \mathfrak{b}(H, R) < 17$. A sufficiently divisible continuity modulus L, divisible by $64m_0$, preserves all the finite conditions under $h \equiv h_0 \pmod{L}$. Choose the remaining labels, root points and height scales before the contact primes, and enlarge L for their finite continuity conditions. The finitely many comparison pole sets have measure zero in G^0 by the real-progression lemma. Choose a nonempty open U', with compact closure in U, avoiding them. All comparison functions are now uniformly bounded on that closure, by bounds fixed before any contact prime is chosen.

The original simultaneous-prime and separation argument is valid for this enlarged fixed L. Its compatible congruences are

$$h \equiv 0 \pmod{\Pi^{19} M_-}, \qquad h \equiv 1 \pmod{M_+}, \qquad h \equiv h_0 \pmod{L}.$$

Their solution progression has residue and modulus divisible by m_0. Its positive tail is therefore dense in G^0 and meets U' at arbitrarily large indices. Choose the final H there. It has every required primary and opposite-place depth, all scalar finite depths, the real comparison bounds and $0 < \mathfrak{b} < 17$ simultaneously. The preparation-zero lemma supplies fresh pole witnesses with prepared value exactly $\mathfrak{b}$. A zero factor in the factored pole is handled by the zero lift already proved above.

For clarity, the physical elimination is also exact. With its fixed positive normalization K, the semantic input is $\mathfrak{b} = -Kx$ and the consumed square is $K^2 n$. The old rows are

$$U^2 = K^2 n, \qquad 1 + KxJ = 0, \qquad Kx + K^2 Jn + V^2 = 0.$$

Replace all three by $n = x^2$ and forget U, J, V. At any new solution $\mathfrak{b} > 0$ as proved independently above; restore $U = \mathfrak{b}$, $J = 1/\mathfrak{b}$, $V = 0$. Each old row is then zero, and every outside value is unchanged. The opposite direction for the integer input projection is the simultaneous construction, not a claim that an arbitrary old preparation witness can be specialized.

A11.11 Normalizing the remaining scalar pole

After the preparation-zero replacement, let x again denote the physical scalar input coordinate, so $\mathfrak{b} = -Kx > 0$ and $x \neq 0$. This nonvanishing was proved from the unchanged point and scalar root rows; it does not depend on the preparation square or on the private pole equations. Assign weight one to the five private slots (e, h, p, q, v) and weight two to their square slots (p_2, q_2). Replace each old slot of weight j by x^j times its new slot, and discard the preparation-square slot n, setting it equal to x^2. Every one of the five remaining pole residuals is then x^2 times its new residual. For example the first conic and the h equation have respectively their ex and x^2 terms replaced by e and1, and

$$17ex + Ken - 17hv = 0 \quad \longmapsto \quad 17e + Kxe - 17hv = 0.$$

The square rows remain $p_2 = p^2$, $q_2 = q^2$, and the factored second-conic coefficient list is unchanged. In the physical indexing of the reviewed component, the weight-one slots are $51, 52, 53, 54, 532$, the weight-two slots are $527, 534$, and 531 is n. These are implementation indices, not mathematical constants.

From an old solution divide each private slot by its indicated power of x. From a new solution multiply by that power and restore $n = x^2$. These maps are inverse and fix every outside coordinate. The sign of x is immaterial, and no division uses a private pole value. In particular all zero branches of p,q,v or h are preserved. These seven slots occur only in the five displayed pole rows, and n has only its former definition and these consumers. Thus there is no untransformed outside use of a rescaled slot.

A11.12 Rescaling the scalar anchor

The odd scalar root S has the unchanged chart $S = (x, hj)$, $x = h^2$, $j^2 = x^2 - D$, hence $S = [a]P$ for an odd integer a. Write the retained reciprocal difference equation as

$$(c_* x + D)g = a_* j - b_* h, \qquad g = g_0 + g_1\rho.$$

Its denominator is positive in both real embeddings. We replace the old inverse of the irrational coefficient of f(S-P) by $g_0\tau = 1$. This implies $g \neq 0$ before using the anchor. The subtraction chart identifies $g^2 = f(S - P)$: at $S = P$ the quotient would be zero, which is excluded, and at $S = -P$ the explicit duplication formula gives the same value $f(-2P)$. Thus $S - P = [s]P$ has a nonzero even index $s = a - 1$ without any prior anchor divisibility assertion.

The earlier scalar coefficient lemma applies to every such s and gives

$$v_3(f_1([s]P)) = 2 + 2v_3(s).$$

Its hypotheses contain no anchor equation. Indeed $z(2P)/3 \equiv 2+\rho \pmod 3$; multiplication by units and then by powers of3 in the integral formal group preserve the stated leading term. The curve equation makes the leading residue of $f/3^{2+2v_3(s)}$ a nonzero rational multiple of ρ. Independence of1 and ρ in the residue field proves the displayed coefficient valuation, also for negative s. Consequently $F = f_1([s]P) = 2g_0g_1 \neq 0$, in particular $g_1 \neq 0$. This is the noncircular justification for restoring the old inverse.

The old three-row subsystem is

$$F = 2g_0g_1, \quad F\sigma = 1, \quad u^2 + 6v^2 - 2w^2 - 3F^2 - 4Cw = 0.$$

The new subsystem is

$$g_0\tau = 1, \qquad U^2 + 6V^2 - 2W^2 - 12g_1^2 - 4C\tau W = 0.$$

Forward put $(U, V, W) = (u, v, w)/g_0$, $\tau = 1/g_0$ and forget F. Backward put $(u, v, w) = g_0(U, V, W)$, restore $F = 2g_0g_1$ and $\sigma = 1/F$. If $q_{\mathrm{old}}, q_{\mathrm{new}}$ denote the anchor residuals, the exact identity is

$$q_{\mathrm{old}}|_{(u,v,w)=g_0(U,V,W),\ F=2g_0g_1} - g_0^2 q_{\mathrm{new}} = 4Cg_0W(g_0\tau - 1).$$

These inverse maps fix the input and all other coordinates. They permit zero values of U,V,W. They are equivalences for the full point system, not for an isolated anchor with arbitrary g: that isolated equation would admit $g_1 = 0$, which the coefficient lemma just excluded.

When C remains a fixed integer coefficient, both new residuals are quadratic and their squared sum contributes $16C^2\tau^2W^2$. For the ordinary companion compute C by the quadratic chain $c_0 =$

3^{1024}, $c_j = c_{j-1}^2$ for $1 \le j \le 15$, so that $c_{15} = 3^{2^{25}} = C$. Introduce a separate multiplication slot T_C and equation

$$T_C = C\tau.$$

Use $-4T_C W$ in the anchor. Treating a computed C as a coefficient would instead make $C\tau W$ cubic, so this extra slot is necessary for that companion. Both directions restore T_C uniquely; no nonvanishing or point-domain argument is changed. This slot and its gate, together with the sixteen constant-chain slots, are included in the ordinary companion.

A11.13 Completeness and finite certification

Fix an integer X and compute the positive deterministic scalar correction. Choose the preliminary shared point and then the deep nonzero scalar point by the simultaneous real and finite-place construction just proved. Choose the odd Lucas labels, then their actual root points and positive root scales. Fix a continuity modulus for all the new rational functions. Only after these choices select the contact primes and solve the retained primary/opposite congruences. Take a sufficiently large common index h in the prescribed real open set, avoiding the finitely many old exceptional point loci. This gives $U_i = [hn_i]P$ and $V_i = [(h-1)n_i]P$, all nonidentity in this completeness construction. The new rational logarithms have the required local precision, and the original real pole estimates and arbitrary finite preparation depths give positive semantic pole inputs. Obtain fresh pole witnesses with scalar preparation equal to its input, apply the factored-conic maps, and reconstruct each finite private circuit. These are simultaneous choices; no claim is made that an arbitrary old pole witness survives a changed logarithm or changed scalar root.

A11.14 The specified polynomials and coefficient tables

Clear denominators and divide each nonzero quadratic residual by its integer content. Define P_{394} to be the sum of the squares of the 364 resulting integer residuals. Over the rational numbers this sum vanishes exactly when every residual vanishes. The soundness and simultaneous completeness arguments above therefore prove its stated integer projection. The sixteen constant-chain equations and the $T_C = C\tau$ equation give the ordinary companion: its 381 normalized residual squares define P_{411}, with the same integer projection.

Expansion of the specified residual lists gives respectively 6,338 and 6,388 distinct nonzero monomials. Both polynomials have total degree four and content one. For the symbolic polynomial the largest absolute coefficient is $16C^2 = 16 \cdot 3^{2^{26}}$; for the ordinary polynomial it has 3,247 bits and is strictly less than 2^{3247}. These counts and bounds are computed from the complete coefficient lists, not by adding component savings. Every listed coordinate occurs.

The coefficient artifacts use consecutive coordinate indices, with index one for X. The symbolic JSON coefficient table has 973,221 bytes, and the ordinary JSON coefficient table has 982,235 bytes. Their plain text tables have 212,067 and 214,957 bytes, respectively. The tables, residual lists, coordinate maps and exact maximum coefficients specify the two polynomials independently of their factored construction.

A12 A quartic presentation with 299 variables

We retain the arithmetic inputs, qualifications and source editions of Sections A10.1 and A11. The polynomial of the main construction is unchanged. The following are smaller presentations of the same rational input projection.

Theorem A12.1. *There is a specified integer polynomial P_{299} of total degree four in 299 variables such that*

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{v} \in \mathbb{Q}^{298} : P_{299}(X, \mathbf{v}) = 0.$$

It has 6,630 nonzero monomials, content one, and largest absolute coefficient $186624 \cdot 3^{2^{26}}$. *Its coefficient artifact writes $C = 3^{2^{25}}$ as a fixed integer, not a quantified variable. There is also a specified integer quartic P_{314} with*

$$X \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{w} \in \mathbb{Q}^{313} : P_{314}(X, \mathbf{w}) = 0.$$

It has 6,672 nonzero monomials and content one. Every coefficient has absolute value strictly less than 2^{6496}; its largest coefficient has exactly 6,496 bits. The compact ordinary decimal coefficient table occupies 264,430 bytes in its specified ASCII format. The separate table retaining source coordinate identifiers occupies 222,890 bytes.

The ordinary coefficient bound here is different from the bound for P_{411}. We prove the new bound for the new artifact. Some changes below are inverse changes of private witnesses; others change an arithmetic certificate or point domain. For the latter we prove soundness and completeness separately.

A12.1 Conventions and retained interfaces

Put $K = \mathbb{Q}(\rho)$, $\rho^2 = 2$, $d = 3 - \rho$, $D = d^2 = 11 - 6\rho$, and $E : y^2 = x^3 - Dx$. Retain $E(K) = \mathbb{Z}R_0 + E[2](K)$ and $2E(K) = \mathbb{Z}P$, where

$$R_0 = (1 + \rho, 2\rho), \quad P = (c_*, a_* b_*), \quad a_* = -1 + 7\rho/4, \quad b_* = 39/8 - 9\rho/2, \quad c_* = a_*^2.$$

Thus $b_*^2 = c_*^2 - D$. Write $f(Q) = 1/x(Q)$ and $z(Q) = -x(Q)/y(Q)$, with both functions extended regularly by zero at O. The roots are $Q_i = [n_i]P$ with labels $(n_1, n_2, n_3, n_4) = (n, a, t, b)$. Their nonzero odd integral labels will be restored before height or logarithm arguments. The scalar point is $Q_2 - P = [s]P$, where $s = a - 1$.

The retained inputs include the squareclass calculation, formal expansions and logarithm estimates, simultaneous contact-prime construction with its real-density condition, preparation-zero pole criterion, root-height implication, Lucas argument and scalar-depth-24 argument. In particular the three positive rational square scales are

$$A_0 = 1, \quad A_1 = C_0^2, \quad A_2 = (1 + C_1^2)^2, \qquad \prod_i x(Q_i) = C_0 + C_1\rho.$$

The same root-height conclusion is

$$B^2 - 6 < 7\mathcal{H}|N|^{19/k}, \qquad B = \max_i |n_i|, \quad 1 \le \mathcal{H} < 2^{2093966},$$

when the kth powers of the three required contact radicals divide the nonzero integer N. The independently restored scalar anchor gives $B > (40\mathcal{H})^{10}$, excluding all three radicals equal to one without using H or auxiliary logarithms. The retained preparation-zero equations exclude negative odd valuations of a pole input at required large primes. Their completeness interface uses the same small-prime squareclasses and depths, square reduced denominator, and real interval $(0, 17)$. We verify the changed consumers of these inputs below.

A12.2 The fixed-difference chart for H

First replace the old $x(H)$ pair by the equations

$$c_*(f+F)+Dv^2+2b_*v=1, \qquad v=ft, \qquad F=vt. \tag{143}$$

The common primary normalizer is the fixed affine expression

$$\mathcal{S}=(b_*+c_*t+Dv)/a_*. \tag{144}$$

Its substitution into every primary, logarithm and scalar comparison row preserves quadratic degree. If $f=0$, the last two equations force $v=F=0$, contradicting the first. Set $x=1/f$. They give $v^2=fF$ and $a_*f\mathcal{S}-b_*f-(c_*+Df)v=0$. Here is an unconditional certificate for the recovered curve. Put $d_1=c_*+Df$, $B_1=b_*f+d_1v$, $A_1=a_*f\mathcal{S}-B_1$, and $K_1=c_*(f+F)+Dv^2+2b_*v-1$. Then

$$c_*f(f\mathcal{S}^2+Df^2-1)=fd_1K_1+c_*d_1(v^2-fF)+(b_*^2-c_*^2+D)f^2+A_1(a_*f\mathcal{S}+B_1).$$

Thus $\mathcal{S}^2=x-Df$ and the old fixed-difference equations hold. This direction divides only by c_*f. Conversely, the old full chart gives $t=v/f$ and the same affine expression for $\mathcal{S}$; its previously established exceptional-point exclusion supplies the forward implication. All outside values are preserved.

Next use the already needed outputs $U=f(H)^2$, $V=f(H-P)^2$. Store W,v,t,U,V, put $Q=(1-DW-2b_*v)/c_*$, and impose

$$W=v^2, \qquad UV=W^2, \qquad Q^2=U+V+2W, \qquad t(U+W)=Qv. \tag{145}$$

On $UQ\neq 0$, set $f=(U+W)/Q$, $F=(V+W)/Q$. If $E_1=UV-W^2$ and $E_2=Q^2-U-V-2W$, then

$$\begin{aligned}(U+W)^2-UQ^2&=-UE_2-E_1,\\ (V+W)^2-VQ^2&=-VE_2-E_1,\\ (U+W)(V+W)-WQ^2&=-WE_2+E_1.\end{aligned}$$

These give $f^2=U$, $F^2=V$, $fF=W$, $f+F=Q$, $ft=v$, and $F=vt$, restoring (143).

The extra branches are classified before this reconstruction. For $U=0$ one has $W=v=0$, $Q=1/c_*$, $V=1/c_*^2$. For $Q=0$ one has $U=V=-v^2$ and

$$Dv^2+2b_*v-1=0, \qquad v=\frac{3+2\rho}{4} \quad \text{or} \quad v=\frac{-36+16\rho}{49}.$$

These are units at every place above $p>76176$: $c_*+b_*=12-8\rho$ has norm16 and $c_*-b_*=(9+4\rho)/4$ has norm49/16. Section A12.5 excludes these branches by their unchanged V_0 unit leaf and the pre-H height/anchor contradiction. No division by UQ is used to prove its own legitimacy.

A12.3 Root parity and the shifted anchor

The root chart $c_i=a_i^2$, $b_i^2=c_i^2-D$ represents (c_i,a_ib_i). Its abscissa is nonzero; for the final presentation this follows from the retained product-height inverse as proved in Section A12.6. The squareclass map has independent values

$$\alpha(R_0)=[1+\rho], \qquad \alpha(T_0)=[-1], \qquad \alpha(T_d)=[d].$$

Indeed norms in a putative square relation force the exponents of $1+\rho$ and d even, and the real embeddings exclude the remaining class $[-1]$. Thus the square abscissa already puts each root in $2E(K) = \mathbb{Z}P$, with nonzero integral label, before an odd-chart condition.

Keep the scalar reciprocal equations

$$(c_*c_2 + D)g = a_*b_2 - b_*a_2, \qquad g = g_0 + g_1\rho, \qquad g_0\tau = 1.$$

The denominator is nonzero at both real embeddings. The subtraction identity gives $f(Q_2-P) = g^2$. At equal abscissae, $Q_2 = P$ contradicts the guard and $Q_2 = -P$ gives the exact duplication value for $-2P$. Hence $s = n_2 - 1 \neq 0$ is an integer before height or logarithms. Write $F_s = 2g_0g_1$.

Let X_n be the ρ coefficient of $x([n]P)$. For $n \neq 0$,

$$v_3(X_n) = \begin{cases} 0 & n \text{ odd}, \\ -2 - 2v_3(n) & n \text{ even}, \end{cases} \qquad v_2(X_n) = -1 \quad (n \text{ odd}). \tag{146}$$

At3 the reduction of P is $(\rho, 0)$ over $\mathbb{F}_9$, which proves the odd case. For $n = 2 \cdot 3^k u$, u a3-unit, the formal expansion gives $z([n]P)/3^{k+1} \equiv u(2+\rho) \pmod 3$. Since $x(z) = z^{-2} + O(z^2)$ and $(2+\rho)^{-2} = 3/2 - \rho \equiv 2\rho \pmod 3$, the scaled abscissa has nonzero imaginary coefficient modulo3. This proves the even case and $X_n \neq 0$. At2, $z(P) = (232 + 86\rho)/357$ has component valuations3 and1. The integral odd multiplication series preserves these values: its error has field valuation at least9/2 when $v(2) = 1$. For $z = A + B\rho$ with those valuations the imaginary coefficient of z^{-2} is $-2AB/(A^2 - 2B^2)^2$, of valuation-1. The Laurent correction has larger valuation. This uses integral formal coefficients, not good reduction at2.

The retained scalar-coefficient lemma gives $v_3(F_n) = 0$ for odd n, $2 + 2v_3(n)$ for even nonzero n, and dyadic integrality. It is applied with both parities. Put

$$q = 12(X_A^2 + X_T^2) + 108X_Z^2, \qquad T = F_s + Cq, \tag{147}$$

and use the private anchor

$$u^2 + 6v_a^2 - 2w_a^2 - 4Cw_a - 3T^2 = 0. \tag{148}$$

The inherited quaternion criterion, including parameter zero, supplies rational witnesses exactly when T/C is integral at2 and3. Its3-integrality forces all three summands in $F_s/C + q$ to be3-integral. Their possible finite valuations lie in the disjoint classes $2\mathbb{Z}$, $1 + 4\mathbb{Z}$, $3 + 4\mathbb{Z}$; the binary sum uses anisotropy modulo3 and (146). A negative minimum cannot cancel. Thus X_A, X_T, X_Z are3-units, so the three root labels are odd. Also F_s/C is3-integral and is dyadically integral by the scalar lemma. The old unshifted anchor can therefore be restored with new private witnesses before H or height. An odd s would make F_s a3-unit, so s is even and the S-root label $s+1$ is odd.

For an odd root, its half point $[n_i]R_0 = (A, B)$ has $A = (1+\rho)r^2$. Duplication gives

$$a_{\text{half}} = (A^2 + D)/(2B), \qquad b_{\text{half}} = (B/A)^2/2 - c_i,$$

with $a_{\text{half}}^2 = c_i$ and the correct product ordinate. After changing both root signs if necessary, the old half-chart equations are $A^2 - 2(c_i + b_{\text{half}})A - D = 0$ and $\mathrm{N}(A) + \mathrm{N}(r)^2 = 0$. In the final A,T,Z modules the sign freedom is $a_i \mapsto -a_i$, with reconstructed $b_i \mapsto -b_i$. For S negate a_2, b_2, g_0, g_1, τ together. These actions fix the actual points, F_s, all detector values and the private unscaled anchor witnesses. The scalar rows change at most by an overall sign.

Conversely the old odd roots make q integral at2 and3 by (146), so the same quaternion criterion supplies the shifted anchor. No prime or H choice changes. The exact private unscaling used here

multiplies the old anchor witnesses and its parameter B by g_0, with inverse multiplication by τ. It gives $T = g_0 B = F_s + Cq$ and (148). This exact step assumes no private anchor witness nonzero.

More explicitly, let the old private residuals be $R_Q = Q - q$, $R_B = B - 2g_1 - CQ\tau$, and $R_A = U^2 + 6V^2 - 2W^2 - 4C\tau W - 3B^2$. Let $N_T = T - 2g_0 g_1 - Cq$ and let N_A be the new anchor residual. With $G = g_0\tau - 1$, pullback by the forward map gives the unconditional identities

$$N_T = g_0 R_B + C R_Q + CQG, \qquad N_A = g_0^2 R_A + 4C g_0 W G.$$

The reverse map is $Q = q$, $B = \tau T$, $U = \tau u$, $V = \tau v_a$, $W = \tau w_a$. Its pullbacks are

$$R_Q = 0, \qquad R_B = \tau N_T + 2g_1 G, \qquad R_A = \tau^2 N_A.$$

Thus every private anchor consumer is restored with its literal normalization, and all nonprivate coordinates are fixed.

A12.4 Contact poles and their conditional domains

For each scale put

$$R_j = F_5(A_j\lambda), \quad E_j R_j = 1, \qquad F_5(t) = t(t+1/2)(t+1)(t-1/2399)(t+32/407).$$

The retained contact gates make R_j affine in existing coordinates. With $W_j = G_{j,0}G_{j,1}G_{j,2}$ retain only

$$\mathrm{N}(W_j) = 2\,\mathrm{Re}(W_j) z_j, \qquad B_j(R_j + E_j + 2) = z_j^2. \tag{149}$$

There is no quantified inverse of $\mathrm{Re}(W_j)$. At a required contact prime, $\alpha = v_p(R_j) > 0$ is odd and $1 + R_j$ is a unit. The groups below give $W_j \neq 0$. The norm row then forces its real component and z_j nonzero: $w_0^2 - 2w_1^2 = 0$ over $\mathbb{Q}$ forces both components zero. Thus

$$T_j^{\mathrm{tr}} = \mathrm{Tr}(W_j^{-1}) = 1/z_j, \qquad B_j = \frac{R_j}{(T_j^{\mathrm{tr}})^2 (1 + R_j)^2}.$$

These domains are needed only at required contact primes. A detector with radical one has empty local obligations; no global trace inverse is asserted for it. The case $R_j = -1$ likewise cannot occur at a required contact.

In completeness the ratio between R_j and the earlier contact rational function is $A_j^5 (t-1)^6$, a rational square and a unit at selected contacts. Hence R_j has valuation one there and even valuation outside the prescribed odd support. Both denominators in B_j change valuations by even integers. This includes additional primes introduced by later witnesses.

A12.5 Common-scale groups and the successful place

Use the physical tests

$$H_0 = f(H), \quad H_i = L_{i+1} - \overline{LP_{i+1}}\ (1 \le i \le 3), \quad H_4 = \mathcal{L}(L_2, L_3, L_4),$$

$$M_0 = L_1 - \overline{LP_1}, \quad M_1 = L_1 - \beta(X), \quad P_i = f(U_i)^2, \quad Z_1 = \overline{f(V_1)^4}, \quad T_i = \overline{f(V_i)^8}.$$

Here $\beta(X) = 204(X^2 + X) + 2555$ is the retained comparison polynomial. Conjugation is part of the physical tests. The Lucas polynomial is

$$\mathcal{L}(A, T, Z) = T^2 - A(A-2)TZ + (A-2)^2 Z^2 - 1.$$

The affine recentering $C_L = (A-2)Z$, $V_L = AT$ gives $\mathcal{L} + 1 = T^2 + C_L^2 - C_L V_L$ without changing its value.

For a nonzero common rational γ, store

$$ZH_i = H_i^4/\gamma^{160}, \quad ZM_i = M_i^4/\gamma^{40}, \quad ZP_2 = P_2^2/\gamma^{160}, \quad ZP_3 = P_3^2/\gamma^{80}, \quad ZP_4 = P_4^2/\gamma^{80},$$

and $ZP_1 = P_1^2$. These definitions are quadratic using the retained square inputs and common powers. For each detector,

$$G_0 = 1 + RZH_1 + R^2ZH_2 + EU + E^4Z_1 + \gamma^{16}E^8, \tag{150}$$

$$G_1 = 1 + RZH_3 + R^2ZH_4 + R^3ZP_2 + E^4ZP_4 + EZP_1 + E^8T_0, \tag{151}$$

$$G_2 = 1 + RZM_0 + R^2ZM_1 + R^3ZP_3 + E^8T_2 + E^4T_3 + ET_4. \tag{152}$$

Here $U = H_0^2$ is the actual squared-H output even before H is reconstructed. The product, trace and pole use these same final groups.

At a consumed place set $\alpha = v(R) > 0$ odd and $c_\gamma = v(\gamma) \in \mathbb{Z}$. For a reciprocal point $h^2 = f(1 - Df^2)$, a nonzero f has even valuation: the leading valuation is $v(f)$ when positive and $3v(f)$ when negative. Zero leaves are omitted. Thus the relevant powers have periods4,8,16 before formality or logarithms. In the generic squared-H branch its algebraic reconstruction gives the same curve fact; its exceptional branches have the fixed signed values already classified. Other high and middle tests are literal fourth powers, independently of their later ratio meanings.

In term order, the valuation periods and coefficients of α are

$$\begin{array}{l|l} G_0 & (4,1),(4,2),(4,-1),(8,-4),(16,-8) \\ G_1 & (4,1),(4,2),(8,3),(8,-4),(8,-1),(16,-8) \\ G_2 & (4,1),(4,2),(8,3),(16,-8),(16,-4),(16,-1). \end{array}$$

All gamma exponents are divisible by the relevant periods. Within each row the classes modulo16 are disjoint and exclude zero for every odd α. Every group therefore has a unique minimum valuation, is nonzero, and has valuation at most zero. This covers arbitrary c_γ and every zero-leaf pattern.

Let ℓ_p be the maximum valuation of W over places above p. Trace integrality gives $v_p(T^{\mathrm{tr}}) \geq -\ell_p$. The pole obstruction and (149) give

$$0 \leq v_p(B) = \alpha - 2v_p(T^{\mathrm{tr}}) \leq \alpha + 2\ell_p.$$

Its odd parity makes $\ell_p > -\alpha/2$. At a place attaining the maximum, every group, and then every individual term, has valuation greater than $-\alpha/2$. This is one place for all tests, also at an inert unramified prime.

The guard gives $c_\gamma > 15\alpha/32$. The high and middle terms give

$$4v(H_i) > 160c_\gamma - (r + 1/2)\alpha, \qquad 4v(M_i) > 40c_\gamma - (r + 1/2)\alpha \quad (r = 1 \text{ or } 2).$$

For $\alpha = 1$ the worst bounds are157.5 and37.5; for $\alpha = 3$ they are312.5 and72.5; for $\alpha \geq 5$ they exceed72.5α and16.25α. Thus $v(H_i) \geq 40$ and $v(M_i) \geq 10$. The primary bounds are

$$\begin{aligned} 4v(f(U_2)) &> 160c_\gamma - 3.5\alpha, \\ 4v(f(U_3)) &> 80c_\gamma - 3.5\alpha, \\ 4v(f(U_4)) &> 80c_\gamma + 3.5\alpha. \end{aligned}$$

Using even valuations, these give respectively40,20,22 for the reciprocal depths. The other low terms force positive valuation of every required nonzero primary or alternate reciprocal value.

On either exceptional squared-H branch T_0 is a unit at every consumed place. Its E^8T_0 term has valuation-8α, contradicting the common strict bound. Either branch would therefore make all radicals one, contrary to the pre-H root-height/anchor bounds. The unchanged fixed-point exclusions from Section A11 then restore the full H/V0 domain before logarithms. Here their required hypothesis is only positive valuation of $f(H)$, rather than the former stronger order40 test. A nonidentity torsion value of H, or a value in either of the fixed translates $\pm P + E[2](K)$, has the same fixed unit obstruction used there at every consumed large prime. The new term EU, where $U = f(H)^2$ after reconstruction, gives $2v(f(H)) > \alpha/2 > 0$ and excludes those fixed values. The identity is already outside the reconstructed affine H chart. Thus weakening the old H test does not weaken these domain exclusions; no high-depth logarithm estimate is used in them.

A12.6 Reciprocal point charts and their orientation

For each of the four root indices use reciprocal coordinates

$$f = 1/x(U), \quad h = -y(U)/x(U)^2, \qquad F = 1/x(V), \quad k = -y(V)/x(V)^2.$$

The pair $(0,0)$ denotes O. Retain the square caches $Y = f^2$, $B = F^2$. In particular the sign of h and k is negative. The reduced chart uses only f, h, F, s, Y, B and imposes

$$h^2 = f - DfY, \tag{153}$$

$$e = s^2 - c + D(f + F) = 0, \tag{154}$$

$$q = 2sh - c(f - F) - 1 + D(2Y + fF) = 0. \tag{155}$$

Here s is a private slope, unrelated to the scalar integer $a - 1$ when that integer is denoted by s elsewhere. All displayed equations have degree at most two in the quantified coordinates. Products such as fY use the retained cache Y. Define, without adding a coordinate,

$$k = s(f - F) - h.$$

Put $r_1 = s(f - F) - h - k$, $r_2 = s(h - k) - 1 + D(Y + fF + B)$ and $r_4 = h^2 - f + Df^3$. Exact expansion, using the two caches, gives

$$r_1 = 0, \qquad r_2 = q - (f - F)e, \tag{156}$$

$$k^2 - F + DF^3 = r_4 - (f - F)q + (f - F)^2 e. \tag{157}$$

Thus both reciprocal cubics and the three slope equations hold before a ratio, valuation or logarithm is interpreted.

These cubics describe actual points of $E(K)$. For $f \neq 0$ the inverse is $x = 1/f$, $y = -h/f^2$; for $f = 0$ the cubic gives $h = 0$ and the point is O. This chart omits $(0,0) \in E(K)$ but includes the other two finite points of order two. Those latter points will be excluded by the low tests before division by a ratio denominator. Translation by $(0,0) \in E(K)$ sends the represented point to $(-Df, -Dh)$, including the case in which the represented point is O.

The oriented difference is

$$U - V = (c, y_R), \qquad y_R = Dh - s(Df + c). \tag{158}$$

For completeness, this assertion includes every branch of the equations. If $f \neq F$, the line with slope s joins $(-Df, -Dh)$ and $(-DF, Dk)$, so the usual secant formula gives the stated difference.

If $f = F$, the first slope equation gives $k = -h$. The case $h = 0$ is impossible: the cubic gives either $f = 0$ or $Df^2 = 1$, whereas the second slope equation would give $3Df^2 = 1$. Hence $h \neq 0$, and that equation gives $2sh = 1 - 3Df^2$, exactly the tangent slope at $(-Df, -Dh)$. The tangent formula again gives (158). If $U = O$ the equations give $Fc = 1$ and $R = -V$; if $V = O$ they give $fc = 1$ and $R = U$. Both points cannot be O.

There is also a division-free check on the root curve:

$$y_R^2 - c^3 + Dc = D^2 r_4 - D(Df + c)(r_2 - sr_1) + e(c + Df)(c + DF). \tag{159}$$

Consequently the retained square equation $a_i^2 = c_i$ recovers the old root chart once $c_i \neq 0$: put $b_i = y_R/a_i$, obtaining $b_i^2 = c_i^2 - D$.

The nonzero assertion uses no auxiliary ratio. In the literal system the successive root products are

$$c_1c_2 = v_{337} + 2\rho v_{338}, \quad c_1c_2c_3 = 2(v_{339} + 2\rho v_{340}), \quad c_1c_2c_3c_4 = 2(v_{341} + 2\rho v_{342}).$$

If any c_i vanished, both last components would vanish. The unchanged height-product chain would then give $v_{343} = v_{391} = v_{392} = v_{393} = v_{394} = 0$, contradicting its retained inverse equation. Thus every c_i and a_i is nonzero. The squareclass calculation now places every root in $2E(K)$. This supplies the root chart used in the parity and pre-H arguments above; it does not require the auxiliary points to lie in $2E(K)$.

For the scalar root retain both old coordinates a_2, b_2 and the old equation $b_2^2 = c_2^2 - D$. Add the quadratic orientation equation

$$a_2b_2 = Dh - s(Df + c_2). \tag{160}$$

The scalar subsystem consumes this particular ordinate. A root curve equation alone would allow its negative and would not identify its scalar point with $U_2 - V_2$. The physical caches for this chart are $Y = v_{257} + 2\rho v_{258}$ and $B = v_{196} + 2\rho v_{197}$. Every old scalar consumer of a_2, b_2, c_2 is retained.

Set $N_0 = \mathcal{S}$, the fixed affine expression in (144). After the H domain is restored its meaning is $-1/z(H)$. It is not a new quotient or a new coordinate in the pre-H equations. The primary equations are

$$hL + fN_0 = 0, \qquad (1 - DY)L + hN_0 = 0. \tag{161}$$

The retained alternate equation is

$$(1 - DB)z = s(f - F) - h = k. \tag{162}$$

It remains quadratic after substituting for k. At O these equations force $L = 0$ or $z = 0$, respectively, rather than allowing a free ratio. The raw alternate equation $kz = F$ is not imposed. Its recovery requires the global order of the proof. With $A = 1 - DB$ and $g = Az - k$,

$$A(kz - F) = kg + (k^2 - F + DFB). \tag{163}$$

The right side vanishes by the curve and regular ratio equations. One may cancel A only after the low tests have been applied. At a consumed place they give $v(F) > 0$ for a nonidentity point, so A is a local unit. At the identity $F = 0$ and $A = 1$. Thus $A \neq 0$ in K, and the raw ratio follows.

The order is therefore the same for all four root indices. First restore the cubics, oriented group differences and nonzero root charts. Next restore root parity and the scalar-anchor contradiction to empty contact support. Apply the same-place low tests using the actual cubics and literal caches. Only then cancel $1 - DB$ and interpret the ratios. In particular the scalar auxiliary pair is not passed to an older argument which assumed an unavailable square root of f or F.

A12.7 Auxiliary point domains and the H-depth bootstrap

We give the domain argument for all four pairs simultaneously. At a good consumed place D is a unit. On $h^2 = f - Df^3$, a nonzero f has even valuation. If $v(f) > 0$, $2v(h) = v(f)$; if $v(f) < 0$, $2v(h) = 3v(f)$; and if $v(f) = 0$ there is nothing to prove. The identity has $f = 0$ and every positive-depth condition is automatic. Consequently the valuations of f^2, f^4, f^8 have periods $4, 8, 16$ on all of $E(K)$ represented by this chart. This proves the periods used in the group calculation before H has been restored. The low tests exclude the finite points of order two, whose reciprocal abscissae are units at these places.

After the fixed H/V0 domain and the raw ratios have been restored, write

$$H = mR_0 + T_H, \quad U_i = k_iR_0 + T_i, \quad V_i = (k_i - 2n_i)R_0 + T_i, \quad V_0 = (m-2)R_0 + T_H,$$

where all the T's are in $E[2](K)$. The correct identities, which kill torsion without imposing a doubled-point domain, are

$$[2m]U_i = [2k_i]H, \qquad [2(m-2)]V_i = [2(k_i - 2n_i)]V_0. \tag{164}$$

The restored H exclusions make H, V_0 nontorsion. Taking formal logarithms gives full ratios

$$\frac{u_i}{h_0}, \qquad \frac{u_i - n_i}{h_0 - 1}, \qquad u_i = k_i/2, \quad h_0 = m/2.$$

The half-indices are integral at each consumed odd prime. At the identity the formal unit $\log(Q)/z(Q)$ means its regular extension, whose value is one. No quotient $0/0$ is taken.

One of $h_0, h_0 - 1$ is a unit. The corresponding full ratio is integral. The depth-one formal unit estimates and the positive cross-test transfer integrality to both approximate ratios and both full ratios. In particular L_2, L_3, L_4 are integral at the common place. Their retained Lucas test has positive valuation and its value at $(0,0,0)$ is -1. At least one of these three L_i is therefore a unit. The primary bounds proved above give formal depth at least10 for its point U_i. The exact identity $z(U_i) = L_i z(H)$ then gives depth at least10 for H. All integral primary L_i give the same depth for every nonidentity primary point.

This bootstrap precedes the order40 estimate. At depth10 the relative primary logarithm error has valuation at least40. The alternate estimates have order40 for the degree-nine approximants and order12 for the middle approximants. Their denominators are units at the consumed place. Since the full ratios are integral, these relative errors are also absolute errors. Multiplying the two ratios by h_0 and $h_0 - 1$ and subtracting gives the retained congruences to the integer root labels n_i, with their required orders10 or40. No assertion that the global auxiliary labels u_i are integers is used.

The larger auxiliary domain also requires global nonvanishing of the rational approximant numerator and denominator. In its variable $w = Df^2$ their coefficient lists are

$$\begin{aligned} N(w) : {} & 1, -9776/5365, 2941/3219, -1024/13949, \\ & -1920/237133, -1024/711399, -1408/5928325, 0, 2432/103152855, \\ Q(w) : {} & 1, -1526/1073, 513/1073. \end{aligned}$$

The unchanged certificates modulo71 for N and modulo7 for Q apply to every element of K. Both primes split in K; the respective coefficients are integral, with unit leading coefficient, and their reductions have no root. A root with negative valuation is excluded by its unique leading term, and an integral root is excluded by reduction. Thus the old nonzero products and inverses remain justified. A positivity interval restricted to doubled real points is not used here.

The retained root-height, Lucas and scalar-depth arguments can now be applied with their original inputs. The roots are the same odd integer multiples, the congruences have the required precision, the scalar point has the orientation in (160), and the scalar depth24 transfer is unchanged. They conclude that the distinguished input is an integer. This proves soundness of the changed arithmetic certificates.

A12.8 Simultaneous completeness of the common scale

We check that a single choice of witnesses meets all the new tests. Start with the original simultaneous construction for an integer input, with the same positive progression and the same contact-prime restrictions. Use its depth19 preparation. The resulting primary formal depths are at least20, the high tests have valuation at least40, and the middle tests have valuation at least12. The fixed H/V0 exclusions remain part of this construction.

At each contact prime $\alpha = v(R) = 1$. Prescribe $v(\gamma) = 1$. In the three groups the high-term valuations are at least1 or2 and the middle-term valuations at least9 or10. The primary terms have lower bounds159,3,83,76 in root order; the guard has lower bound8. The alternate low terms are unchanged and have positive valuation. Every group is therefore one plus an element of the maximal ideal at the successful place. At the other place above a split contact prime, the inverse of each group has residue zero or one. Thus the trace of the product of inverses has residue one or two and is a unit. The pole value has positive odd valuation one.

At each preparation prime $p = 2, 3, 17$, put $\eta_j = v_p(A_j)$, prescribe $v_p(\gamma) = v_p(\lambda) = -\nu_p$, and enlarge ν_p before choosing the contact primes. Since $v_p(R) = 5(\eta_j - \nu_p)$, the coefficients multiplying the bounded test values have valuations

$\gamma^{-160}R$	$155\nu_p + 5\eta_j$
$\gamma^{-160}R^2$	$150\nu_p + 10\eta_j$
$\gamma^{-160}R^3$	$145\nu_p + 15\eta_j$
$\gamma^{-40}R$	$35\nu_p + 5\eta_j$
$\gamma^{-40}R^2$	$30\nu_p + 10\eta_j$
$\gamma^{-80}R^3$	$65\nu_p + 15\eta_j$
$\gamma^{-80}E^4$	$100\nu_p - 20\eta_j$
$\gamma^{16}E^8$	$24\nu_p - 40\eta_j$.

Each tends to $+\infty$ with ν_p. These are all the new coefficient forms. The previous finite preparation therefore works after increasing its bounds. At the real places impose $\lambda < \gamma < 2\lambda$. The corresponding powers of λ are $-155, -150, -145, -35, -30, -65, -100, -24$. They all tend to zero uniformly in this interval. The groups tend to one, their trace to two, and the pole value tends to $1/(4R)$ through positive values. This gives the real conditions required by the unchanged local-global pole certificates.

Away from the contact support, R has even valuation. Since the other factors in its pole value are squares, new primes in the denominator of γ create no odd pole support. This also explains why γ may be chosen on the anchor quadric below rather than independently. The argument requires its specified local valuations and real interval, not an absence of other denominator primes.

The choices have a fixed order. First prepare the scalar and root witnesses and all finite and real bounds. Choose the contact primes and then λ. Choose the anchor point and hence γ with the required local conditions. Finally choose H in the original progression, deep enough at the finitely many selected places and in the retained real neighborhoods. The bounds above are uniform in the anchor choice. Thus this last H choice meets all three groups simultaneously.

A12.9 Coupling the scale to the quaternion anchor

Let T be the shifted scalar anchor already constructed. Writing the old quaternion coordinates as u, v, w, put $z = w + C$. Its equation is

$$u^2 + 6v^2 - 2z^2 = N, \qquad N = 3T^2 - 2C^2. \tag{165}$$

Here $N \neq 0$, since $N = 0$ would make the rational square $(T/C)^2$ equal $2/3$. Use u to define $\gamma = 1/u$. The independent common-scale chain is replaced by the eight coordinates specified by

$$a_2 = u^2, \quad a_4 = a_2^2, \quad a_8 = a_4^2, \quad a_{16} = a_8^2, \quad a_{16}J = 1, \quad g_{40}a_8 = J^2, \quad g_{80} = g_{40}^2, \quad g_{160} = g_{80}^2.$$

These equations force $u \neq 0$, $J = \gamma^{16}$ and $g_j = \gamma^j$ for $j = 40, 80, 160$. They recover the former common-scale coordinates uniquely and prove soundness of this replacement.

We prove that the extra choice of u does not obstruct completeness. In the original scalar preparation require also the allowed condition $F_s \in 8\mathbb{Z}_2$. The same positive progression and further formal depths remain available. For the three odd roots $v_2(X_i) = -1$, and

$$q = 3(2X_A)^2 + 3(2X_T)^2 + 27(2X_Z)^2 \equiv 1 \pmod 8.$$

The shifted anchor therefore has $t = T/C$ odd at2 and integral at3. The retained quaternion criterion supplies one rational point P on (165). Fix it before choosing the contact primes.

The required local sets on this quadric are nonempty. Choose $\nu_2, \nu_{17} \geq 1$ and $\nu_3 > v_3(C) = 2^{25}$, enlarged to meet all earlier preparation bounds. At2 take

$$u = 2^{\nu_2}, \qquad v = T/2, \qquad z = (C/2)\sqrt{4 - 3t^2 + 2(u/C)^2}.$$

The radicand is one modulo8. At3 take

$$u = 3^{\nu_3}, \quad v = 0, \quad z = C\sqrt{1 - 3t^2/2 + (u/C)^2/2};$$

the radicand is one modulo3. At17 take

$$u = 17^{\nu_{17}}, \quad v = T/\sqrt{2}, \quad z = C\sqrt{1 + u^2/(2C^2)}.$$

Here2 is a square in $\mathbb{Q}_{17}$ and the last radicand is one modulo17. Each construction solves the quadric exactly.

At a contact prime $p > 76176$ exclude the finitely many denominator primes of T, and prescribe $u = p^{-1}$. Setting $V = pv$, $Z = pz$ leaves $6V^2 - 2Z^2 = p^2N - 1$. Modulo p the right side is -1. The binary form $Z^2 - 3V^2$ represents every nonzero residue: it is a split product if3 is a square and a surjective finite-field norm otherwise. The resulting solution is nonsingular, so Hensel lifting gives the desired local point. At the real place take $1/(2\lambda) < u < 1/\lambda$, choose v sufficiently large, and solve for a positive real z. Hence $\lambda < \gamma < 2\lambda$.

For clarity, weak approximation on this particular quadric can be proved without adding quantified witnesses. Let $Q(u, v, z) = u^2 + 6v^2 - 2z^2$ and let B be its associated symmetric bilinear form, so $B(X, X) = Q(X)$. For a direction d with $Q(d) \neq 0$ put

$$\Phi(d) = P - \frac{2B(P, d)}{Q(d)}d.$$

Then $Q(\Phi(d)) = N$. Every point X on the quadric with $Q(X - P) \neq 0$ is obtained by taking $d = X - P$. This image meets every nonempty local open set. Indeed the excluded tangent section

has equation $B(P,X)=N$. If P and X are not proportional, choose $h \perp X$ with $B(P,h)\neq 0$ and use the local curve

$$X(s)=\sqrt{1-s^2Q(h)/N}\,X+sh.$$

It stays on the quadric and leaves that section to first order. If P is proportional to X on the section, then $P=X$. The tangent plane is nondegenerate because $N\neq 0$; choose h there with $Q(h)\neq 0$. The same curve leaves the section to second order. The square root exists for sufficiently small s, also over $\mathbb{Q}_2$.

Finally approximate the three direction coordinates simultaneously by rationals. After clearing the local denominators with an integer D_0, impose the required finite congruences modulo an integer M. For integers $L\equiv 1 \pmod{M}$ tending to infinity, coordinates of the form $(A+Mk)/(D_0L)$ retain those congruences and have real mesh $M/(D_0L)$ tending to zero. The Chinese remainder theorem chooses A for each coordinate. Continuity of Φ on the chosen local neighborhoods now supplies a rational point in all the required open sets. Its u gives precisely the valuations and real interval required for γ. This proves simultaneous completeness of the coupled anchor.

A12.10 Exact private changes and the finite polynomial

We relate the preceding arithmetic argument to the literal quadratic presentation. Coordinates represent elements of K by their rational components; each displayed field equation contributes its two component equations. All rational coefficients are cleared and each equation is divided by its integer content before taking the sum of squares. These operations do not change its zero set.

For the reciprocal replacement, an old root chart with scaled private coordinates V_{old}, I_{old}, Q_{old}, K_{old} has $c=a_i^2\neq 0$. Its actual reciprocal coordinates are

$$f=V_{\mathrm{old}}^2/c,\quad h=-I_{\mathrm{old}}V_{\mathrm{old}}/c,\quad F=Q_{\mathrm{old}}^2/c,\quad k=-K_{\mathrm{old}}Q_{\mathrm{old}}/c.$$

The old identities $I_{\mathrm{old}}^2=c(1-DY)$ and $K_{\mathrm{old}}^2=c(1-DB)$, together with $I_{\mathrm{old}}L=V_{\mathrm{old}}N_0$ and $K_{\mathrm{old}}z=-Q_{\mathrm{old}}$, give the new curve and both regular ratio equations, including the identity cases. The actual equality $U-V=Q_i$ supplies a rational secant slope or its rational tangent limit. For the scalar pair it also supplies (160). No new local neighborhood, root sign, anchor choice or H choice is needed for this change. Conversely (159) restores the deleted root ordinates and the preceding soundness proof applies to the enlarged auxiliary domain. We assert equality of input projections, not a bijection to all the old private square-root fibers.

The subsequent deletion of each alternate ordinate is an exact change on the full zero set. Its inverse is $k=s(f-F)-h$, and (157) and the globally justified (163) recover every deleted equation. This accounts for six rational coordinates in the A, T and Z pairs. The scalar pair saves two further coordinates while retaining its root ordinate and adding the orientation equation.

The remaining private replacements are exact power chains or polynomial changes. For a local inverse $ER=1$, a shorter chain may specify

$$EJ=R^2,\quad E_4J=E,\quad E_8=E_4^2,\quad E_{19}J=E_8^2,$$
$$RE_{39}=E_{19}^2,\quad RE_{79}=E_{39}^2,\quad E_{158}=E_{79}^2,\quad E_{157}=RE_{158}.$$

It forces $J=R^3$ and exactly the indicated powers of E. There is no zero or sign branch because $ER=1$. The final common-scale presentation needs only the local coordinates for E, R^2, R^3, E^4, E^8, with shared powers specified above. Its equivalence as an arithmetic certificate was proved in the common-place and completeness arguments, rather than inferred from coordinate counting.

The sparse Lucas change replaces its private coordinate $C_{\mathrm{old}} = AZ$ by $C_{\mathrm{new}} = C_{\mathrm{old}} - 2Z$. The inverse is $C_{\mathrm{old}} = C_{\mathrm{new}} + 2Z$, and the defining polynomial becomes

$$F + 1 = T^2 + C_{\mathrm{new}}^2 - C_{\mathrm{new}}(AT).$$

Both substitutions are affine in the private coordinates. The cached product AT makes the displayed equation quadratic. Every external Lucas value is unchanged. This change reduces terms, not the number of witnesses.

For reference the successive total-variable counts of the selected construction are

$$394,\ 392,\ 390,\ 372,\ 369,\ 366,\ 326,\ 319,\ 318,\ 315,\ 314,\ 313,\ 307,\ 301,\ 299.$$

The early exact eliminations, affine H chart and squared-H replacement are justified above. The count326 replaces separate pole blocks by the three groups; the final group argument proves their required arithmetic properties directly. The count315 precedes common scaling. Common scaling removes twelve local coordinates, adds nine shared coordinates and two scaled test coordinates, giving314. The coupled quaternion anchor replaces those nine by eight, giving313. Reciprocal A, T and Z root charts give307, alternate-ordinate elimination gives301, and the oriented scalar auxiliary chart gives299. Intermediate alternatives with different depth requirements are not needed for this selected proof.

The literal final system has279 quadratic component equations in299 active rational variables, including the input. Its polynomial is their unweighted sum of squares. Over $\mathbb{Q}$ it vanishes exactly when every component equation vanishes. Exact expansion, collection by exponent vector and removal of zero coefficients give6,630 monomials, total degree four, content one, and maximum absolute coefficient $186624C^2 = 186624 \cdot 3^{2^{26}}$. Every coordinate is active. The coefficient artifacts specify the exponent-vector order and the integer coefficient representation, so these finite assertions refer to a single polynomial, not just to its zero set.

A12.11 The ordinary coefficient presentation

The large fixed coefficient can be represented with fourteen constant coordinates. Put

$$(a_0, \ldots, a_{13}) = (2049, \underbrace{2048, \ldots, 2048}_{11 \text{ entries}}, 0, 0)$$

and impose $c_0 = 3^{a_0}$ and $c_j = 3^{a_j} c_{j-1}^2$ for $1 \le j \le 13$. If $c_j = 3^{e_j}$ then $e_0 = 2049$ and $e_j = 2e_{j-1} + a_j$. After the eleven2048 steps $e_{11} = 2^{23}$, and the two last squares give $e_{13} = 2^{25}$. Thus $c_{13} = C$. Each coordinate is uniquely determined; no square-root sign is introduced.

Add one more coordinate

$$Q = 48v_{83}^2 + 48v_{87}^2 + 432v_{89}^2.$$

Only two rows of the symbolic system contain C. Replace them by

$$\begin{aligned} v_{55}^2 + 6v_{56}^2 - 2v_{57}^2 - 4c_{13}v_{57} - 3v_{903}^2 &= 0, \\ v_{903} - 2v_{543}v_{544} - c_{13}Q &= 0. \end{aligned}$$

Both are quadratic. Substitution of the unique constants and Q gives the exact symbolic rows. Projection and unique extension therefore give inverse maps on full rational solution sets and fix the input. There are $299 + 14 + 1 = 314$ variables and $279 + 14 + 1 = 294$ quadratic equations.

Apply the same primitive integer normalization and unweighted sum of squares. The ordinary polynomial has 6,672 nonzero monomials and content one. Let K_0 denote the constant coefficient

of the symbolic sum of squares, an explicitly tabulated positive integer of1,514 bits. The exact largest absolute coefficient of the ordinary expansion is

$$3^{4098} + K_0, \qquad 2^{6495} \le 3^{4098} + K_0 < 2^{6496}.$$

This proves the asserted ordinary coefficient bound. The source-coordinate table retains the original coordinate identifiers and has222,890 bytes. A compact table renames the active variables consecutively by an explicit bijection fixing the input; its byte count is a separate artifact property. That renaming changes neither the equations nor the coefficient statistics. The proof and the coefficient tables above establish Theorem A12.1.

A13 Unary theta series with imprimitive sieves

We give the unary-series proof used in the theta construction. Poisson summation is the Schwartz–Gaussian specialization of Weil, I.19, Theorem 4 and I.20 [153]; the multiplier convention is Ramsey equation (1.1), p. 600 [80]. Let S contain 2, the conductors of η and the old characters, and all late sieve primes, including ℓ_0. Choose i so that $\lambda_0 = \eta\chi_{-4}^i$ is odd. Choose C divisible by its conductor and all primes in S, with no prime factors outside S, and induce λ_0 to a character λ modulo C, extended by zero off the units. It need not be primitive. Put $e(t) = \exp(2\pi i t)$ and

$$F_\lambda(z) = \sum_{m>0} \lambda(m) m e(m^2 z).$$

Then, in the theta-multiplier convention used above,

$$F_\lambda \in S_{3/2}(\Gamma_0(4C^2), \lambda\chi_{-4}).$$

For squarefree positive J prime to S, dilation gives

$$F_\lambda(Jz) \in S_{3/2}(\Gamma_0(4C^2J), \lambda\chi_{-4}(J/\cdot)).$$

This is an available level, not a minimal-level assertion. In particular each prime of J occurs to exponent one, and no fresh prime outside J occurs.

Write $\Theta(z,w) = \sum_{n\in\mathbb{Z}} e(n^2 z + nw)$ and, for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(4)$, put $j(\gamma, z) = \epsilon_d^{-1}(c/d)(cz+d)^{1/2}$, with the fixed theta branch. Then $j^2 = \chi_{-4}(d)(cz+d)$. Poisson summation for a Gaussian gives

$$\vartheta(-1/\tau, u/\tau) = (-i\tau)^{1/2} \exp(\pi i u^2/\tau)\vartheta(\tau, u),$$

where $\vartheta(\tau,u) = \sum_n \exp(\pi i n^2\tau + 2\pi i n u)$, and direct substitution gives invariance under $\tau \mapsto \tau + 2$. With $\tau = 2z$, the associated matrix is $\left(\begin{smallmatrix} a & 2b \\ c/2 & d \end{smallmatrix}\right) \in \Gamma(2)$. The Euclidean algorithm with even row additions gives the generators $T^2, ST^2S^{-1}, -I$. Composition of the two identities, with the scalar fixed at $w = 0$, yields

$$\Theta(\gamma z, w/(cz+d)) = j(\gamma, z) \exp\left(\frac{\pi i c w^2}{2(cz+d)}\right) \Theta(z, w). \tag{166}$$

For $\gamma \in \Gamma_0(4C^2)$ set $D = cz + d$. Differentiate in w and then set $w = Dv$:

$$\Theta_w(\gamma z, v) = D j(\gamma, z) e^{\pi i c D v^2/2} \big(\Theta_w(z, Dv) + \pi i c v \Theta(z, Dv)\big).$$

Directly shifting the summation variable proves

$$\Theta(z, w + 2mz) = e(-m^2 z - mw)\Theta(z, w),$$
$$\Theta_w(z, w + 2mz) = e(-m^2 z - mw)(\Theta_w(z, w) - 2\pi i m\Theta(z, w)).$$

Take $v = r/C$ and $m = cr/(2C) \in \mathbb{Z}$. Since $Dv = 2mz + dr/C$, the derivative corrections cancel. The remaining exponential is

$$e^{\pi i c D r^2/(2C^2)} e(-m^2 z - mdr/C) = e(-cdr^2/(4C^2)) = 1,$$

including the full dyadic divisor $4C^2 \mid c$. Hence

$$\Theta_w(\gamma z, r/C) = Dj(\gamma, z)\Theta_w(z, dr/C).$$

For $\widehat{\lambda}(r) = C^{-1} \sum_{a \bmod C} \lambda(a)e(-ar/C)$, Fourier inversion and oddness give

$$F_\lambda(z) = (4\pi i)^{-1} \sum_{r \bmod C} \widehat{\lambda}(r)\Theta_w(z, r/C).$$

The transformed coefficient is $\lambda(dn) = \lambda(d)\lambda(n)$, because d is a unit modulo C. Thus

$$F_\lambda(\gamma z) = \lambda(d)Dj(\gamma, z)F_\lambda(z) = \lambda(d)\chi_{-4}(d)j(\gamma, z)^3 F_\lambda(z).$$

This gives the character, not its inverse, also for imprimitive λ. The finite Fourier transform is only a complex-analytic proof device: the original coefficients remain the integral algebraic numbers $\lambda(m)m$.

For cusp vanishing take any $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $c > 0$. Put $L = cC$, $A = 2i/(cD)$, so $\Re A > 0$. With $F_{\mathrm{full}} = 2F_\lambda$,

$$F_{\mathrm{full}}(\gamma z) = \sum_{r \bmod L} \lambda(r)e(ar^2/c) \sum_{n \equiv r \ (L)} ne^{-\pi A n^2}.$$

The outer coefficient is L-periodic. The Fourier transform of $xe^{-\pi A x^2}$ is $-i\xi A^{-3/2} e^{-\pi \xi^2/A}$. Poisson summation on each coset therefore gives, with a fixed nonzero scalar β,

$$D^{-3/2} F_{\mathrm{full}}(\gamma z) = \beta \sum_{k \in \mathbb{Z}} kS(k)e(cdk^2/(4L^2))e(c^2 k^2 z/(4L^2)),$$

where $S(k) = \sum_{r \bmod L} \lambda(r)e(ar^2/c + kr/L)$ is bounded. The exponents are $k^2/(4C^2)$; the zero coefficient vanishes. Normal convergence and decay prove cusp vanishing. Infinity follows from the original series; $c < 0$ follows on replacing γ by $-\gamma$.

For dilation, use $\gamma_J = \left(\begin{smallmatrix} a & bJ \\ c/J & d \end{smallmatrix}\right)$ and $J\gamma z = \gamma_J(Jz)$. The ratio $j(\gamma_J, Jz)/j(\gamma, z) = (J/d)$ proves the stated character and level. Dilation sends rational cusps to rational cusps. Multiplication by ordinary theta gives weight two since $j^4 = (cz + d)^2$. The support of C also proves the fresh-prime omissions required by the application.

A14 Coefficient conjugation and finite theta twists

Let $4 \mid L$, let g be an algebraic-coefficient cusp form of weight $3/2$ and character ψ at level L, and let $\tau \in \mathrm{Aut}(\overline{\mathbb{Q}})$. In Ramsey's geometric theta convention (Definition 3.1 [80]), g/θ^3 is a section on $X_1(L)$ with its prescribed divisor and cusp conditions. These conditions commute with field extension and conjugation; its diamond character becomes $\tau\psi$. The chosen Tate cusp may not be

rational. If $\tau(\zeta_L) = \zeta_L^a$, its conjugate marking differs by diamond a, so the two expansions differ by a nonzero constant. Rescaling by this constant gives exactly the coefficientwise conjugate at the fixed marking, where theta has rational coefficients. This preserves all cusp conditions. Only flat field extension is used, not a weight-at-least-five vanishing theorem.

Ramsey's Theorem 6.2 [80], at weight 3/2, gives the good-prime Hecke formula

$$a(p^2n) + \psi(p)(-n/p)a(n) + p\psi(p^2)a(n/p^2),$$

with a nonintegral-index coefficient interpreted as zero. Its non-character factors are rational. Conjugation therefore sends eigenvalues $\eta(p)a_p(E)$ to $\tau(\eta(p))a_p(E)$.

For a primitive character ν of conductor m, set

$$h(z) = G(\overline{\nu})^{-1} \sum_{t \bmod m} \overline{\nu}(t) g(z + t/m).$$

The primitive Gauss identity, including its zero at indices not prime to m, gives $h = \sum_n \nu(n)a(n)q^n$; for $m = 1$ this is g. For $\gamma \in \Gamma_0(Lm^2)$ choose $t' \equiv d^2 t \pmod m$. Matrix multiplication gives

$$n(t/m)\gamma = \gamma_t n(t'/m), \qquad \gamma_t = \begin{pmatrix} a + tc/m & b + (td - at')/m - ctt'/m^2 \\ c & d - ct'/m \end{pmatrix} \in \Gamma_0(L).$$

Put $d' = d - ct'/m$. The theta multiplier is unchanged: $d' \equiv d \pmod L$ and modulo four, and $(c/d') = (c/d)$. Indeed for $c \neq 0$ write $c = sr^2$ with signed squarefree s. The relevant quadratic character has conductor dividing $4|s|$, which divides c/m: at odd primes in s, $v_p(c) - v_p(m) \geq 1$; at two, $v_2(c) \geq 2 + 2v_2(m)$ gives the required exponent two, or three when $v_2(c)$ is odd. The signed Kronecker convention handles negative c, d. For $c = 0$, periodicity and the original $-I$ law give the same assertion. The square-root factors agree because $c(z + t'/m) + d' = cz + d$. Reindexing multiplies the weights by $\nu(d)^2$, so h has character $\psi\nu^2$.

Rational translations preserve cusp vanishing: after cusp scaling their residual upper-triangular matrix rescales the imaginary part by a positive rational number. Choose a common cusp width for the finitely many translations. Thus h is cuspidal at level supported on Lm. For an imprimitive presentation in the application, use the inducing primitive character; the prescribed unit support of g makes the coefficient product identical, and the conductor is supported on the old set S_0. No new fresh-prime factor is introduced.

Direct substitution in the Hecke formula gives $T(p^2)h = \nu(p)^2\lambda_p h$ for $p \nmid Lm$. For $\nu = \eta^{(e-1)/2}$, e odd, and ψ/η quadratic, one has $\psi\nu^2 = \psi^e$ and $\nu(p)^2\eta(p) = \eta(p)^e$. Apply the full good-Hecke-eigenspace proportionality already proved in the theta section, at a common level supported on S_0. With its squareclass conditions, D_0 prime to S_0 and $a(D_0) = 1$, the reference denominator is the nonzero $\nu(D_0)$; consequently

$$\tau(a(D)) = \nu(D/D_0)a(D).$$

The full-eigenspace proportionality and unit support remain explicit inputs from the earlier theta construction. No rationality of an L-value quotient is assumed.

Part I
The polynomial inverse theorem and its local joins

The following internal modules prove the polynomial inverse theorem used above, including the degree–rank induction and its current selection joins. All finite data and standardness conventions

are retained. Module names below identify internal sections, not external theorem assumptions.

A15 Assembly of the precise inverse-theorem input

The following sections assemble the inverse theorem from the internal local results listed below and their stated published inputs.

A15.1 Exact target and conventions

The target, denoted GI(k), is the following unchanged linear formulation. For each fixed integer $k \geq 3$ and each $0 < \delta \leq 1$, there are a finite family $\mathcal{M}_{k,\delta}$ of nilmanifolds G/Γ, each with connected simply connected nilpotent group of step at most k and a discrete cocompact lattice, fixed smooth metrics, and constants $C, c > 0$, such that every 1-bounded $f : [N] \to \mathbb{C}$ with $\|f\|_{U^{k+1}[N]} \geq \delta$ correlates with a 1-bounded test F of Lipschitz constant at most C along a linear orbit:

$$\left|\mathbb{E}_{n\in[N]} f(n)F(g^n x)\right| \geq c.$$

The family and constants depend on k, δ, not on N, f or the orbit parameters. Strict source thresholds can be used with $\delta/2$ to give this weak-inequality version. No effective numerical bounds are asserted.

For $d \geq 2$, the interval norm is the norm of the zero extension to a cyclic group of sufficiently large order divided by the norm of the interval indicator, as in the published GTZ definition on p. 1232. We use the same normalization at every inverse step. On a finite abelian group A, write

$$\|v\|_{U^d(A)}^{2^d} = \mathbb{E}_{x,t_1,\dots,t_d\in A} \prod_{\omega\in\{0,1\}^d} \mathcal{C}^{|\omega|} v(x + \omega \cdot t),$$

where $\mathcal{C}$ is complex conjugation. Our correlation convention is $|\mathbb{E} f\psi|$; conjugating the test function converts the other common convention without changing its degree or bounds.

All limit nilcharacters in the assembly are internal in the explicitly specified sense of the local Lemma 13.2 and tautification units: the vertical identity holds for all limit group elements and all limit points, with the natural extension of a standard vertical character. Every constructor used below must supply that domain, rather than only standard-point equivariance. This is an explicit application check, not a consequence of bounded Lipschitz continuity alone.

The manuscript currently requires GI(20) and GI(39). The proposed relative inverse replacement also requires GI($r-1$) for each fixed $r = \max(s, 4)$ chosen before the norm threshold and asymptotic variable. A polynomial orbit without a same-step linearization or an N-dependent family does not satisfy this target.

A15.2 Intermediate interfaces that must be supplied

The intermediate conclusions and their hypotheses are listed below. Each is proved in the indicated internal section or supplied by the explicitly cited published input.

1. The compatible bounded polynomial inverse theorem of order 2, proved in Section A16 relative to its retained published forward bracket theorem and Bohr-set inputs. It gives the fixed finite family and normalized interval U^3 norm. The needed trivial-subgroup specialization of the false quadratic extension is proved locally, and the periodic connected polynomial construction replaces the affected skew-iterate route.

2. The nilcharacter approximation/selection interface: from positive correlation with a bounded degree-d nilsequence, obtain a degree-d unit-vector nilcharacter and a scalar component giving positive correlation. For a dense family of parameters, make the choices in a limit fashion and, after passage to a dense subfamily if necessary, keep the finite dimensions, complexities, bounds and positive correlation standard and uniform. Section 4 below gives the finite-net and unit-vector construction used at this interface using the previous finite inverse family. Later selection applications use Section A31.

3. Linearization at inverse order s: for a dense family of degree $s-1$ nilcharacters χ_h correlated with $\Delta_h f$, obtain one multidegree $(1, s-1)$ nilcharacter $\chi(h, n)$ with which $\Delta_h f$ has degree-at-most-$s-2$ correlation for many h. This is the precise Theorem 7.1 interface. The modified Proposition 7.3 is supplied by `local-proof-sol-lp02-03-r3`; the quotient and rank-lower comparison by `lp04-001`; the final orbit by the Section A24; and the degree conversions, unit-vector recovery, finite rank iteration and rank-zero endpoint by `rank-iteration-001`. The frozen Sections 9–11 reconstruction proves the one-step frequency premise at every newly constructed rank family, conditional on its precise one-variable D and F contracts. Its theorem takes the same uniform unit-vector degree–rank family and accumulated multidegree factor as the rank iteration. It reconstructs all core/linear/independent data anew; none is inherited. Its fixed unit padding is removed by projecting the first factor of the entire LP04 matrix, so the matrix is exactly $\chi_h \otimes \overline{\zeta_h}$ required by the next rank step. Section A24 supplies the fixed coordinate isometry, with the zero-index groups explicit. All these steps retain the full-limit vertical identities and uniform bounds.

   For clarity, the invariant concerns represented families: ordinary unit-vector tests with exact global vertical identities, fixed standard bounds and finitely many standard models and characters. The initial positive-degree selection is made among these representations, by the parameter-level selection argument for the preceding inverse order; it does not upgrade an arbitrary weak nilcharacter or preserve a prescribed symbol. Degree-to-rank conversion retains the represented function and restricts its character, so this invariant starts at (m, m), where $m = s-1$, with accumulated factor $A = 1$.

   Distinguish the fixed inclusion J that pads the incoming χ_h in Sections 9–11 from the final Section 12 inclusion I, for which $Z(h, n) = I\zeta_h(n)$ on the retained shift set. The full ordinary-top character in Section 12 is chosen before the LP04 comparison and restricts to its cancelling rank-top character. The comparison thus supplies the entire vector

$$W_h = (J\chi_h) \otimes \overline{\zeta_h}$$

   on one common model of degree-rank at most $(m, R-1)$. If $QJ = \mathrm{id}$ is the coordinate projection on the first factor, then

$$(Q \otimes \mathrm{id})W_h = \chi_h \otimes \overline{\zeta_h} =: V_h.$$

   This fixed bounded linear map preserves the common model, internal parameter dependence and standard Lipschitz bound. The projection recovering ζ_h from Z is instead the one associated with I. In particular, this conclusion concerns every matrix entry, not a separately selected scalar entry.

   The identity $\chi_{h,a} = \sum_b \zeta_{h,b}(\chi_{h,a}\overline{\zeta_{h,b}})$ uses the unit norm of the whole vector ζ_h and therefore uses exactly V_h in the correlation recovery. The finite ordinary restart dictionary preserves the global unit and vertical identities after internal selection; the ordinary tests chosen in

the proof of E.5 are extended only after their finite selection. Tensoring the accumulated factor with Z preserves its unit norm and multidegree $(1, m)$. Sections 9–11 then reconstruct all frequency data for this new family. No ordinary degree drop is claimed before the rank endpoint: at rank zero the degree-rank axiom makes the remaining factor ordinary degree at most $m-1$, which is then absorbed into the lower-degree test. The geometric, analytic and source premises of all these component constructions remain those of their stated contracts.

The centered fractional-part convention in this chain is $(-1/2, 1/2]$, a fixed fundamental cube permitted by Section 12. The reconstruction uses the affine-offset coefficient transfer supplied by the local multifactor lemma, as connected below.

4. Symmetrization at inverse order s: from that input obtain one degree-s nilcharacter Θ and a bounded nilsequence $\Psi(h, n)$ supported on

$$J = \{(i, j) : i + j \le s - 1\} \ \cup \ \{(i, s - i) : 2 \le i \le s\}, \tag{167}$$

such that the relevant $\chi(h, n)$ is a bounded finite linear combination of components of $\Theta(n + h)\otimes\overline{\Theta(n)}\otimes\Psi(h, n)$. This is the required Theorem 7.4 output, supplied by the locally authored Section A28. It takes the degree-at-most-$s-2$ correlation actually supplied above; the printed strict hypothesis is not used. Its proof binds the corrected multilinearization, literal block symmetry, local Lemma 13.2 and later shift/sign calculations. It also retains the precise stronger Proposition 8.4, E.11 and E.13 applications. The precise plain Proposition 8.4 input is supplied by the accepted local `plain-quadruples-001`: select the same scalar coordinate of $\chi(h, \cdot)\otimes\phi_h$ and retain that entire lower factor. For $H \subset [-N, N]$ of density α, correlation δ and global scalar bound B, that proof gives standard positive density and correlation depending only on α, δ, B. Its output has signs $++--$ and shifts in the second and fourth factors by $h_1 - h_4$, exactly as used in Section 13. The functions have global polynomial representations; the interval cutoff belongs to f. The proof removes its remaining bounded cutoff by an inequality, not by an asserted identity. Finite component selection and the internal version preserve standard uniform constants. This is an explicit substitution of the supplying proof for the published Proposition 8.4 invocation in the frozen Section 13 proof, lines 88–125. That earlier file remains unchanged and does not itself refer to the later local proof. Its intervals $I_1 = [-N, N]$ and $J = [N]$, scalar component, global test domain, and quadruple count on the N^3 scale are exactly the local theorem's data. The reparametrization $(h_1, h_2, h_3, h_4) = (h_0 + a, h_0 + b, h_0 + a + b, h_0)$ is therefore unchanged; the subsequent pigeonhole has $h_0 \in I_1$ and $a, b \in [-2N, 2N]$. The E.11 and E.13 inputs require the actual multivariable Appendix D.5 factorization and all-box reconstruction; a one-variable D.4 proof does not supply that premise. This obligation is supplied by the local multifactor theorem, as connected below.

5. Splitting for a union of two finite downsets: a fixed bounded nilsequence on their union is uniformly approximable, to any fixed positive error, by a finite sum of products of nilsequences supported on the two downsets, with fixed finite complexity and bounds. This is the E.4 application used explicitly in the next section.

6. The converse correlation bound: a bounded scalar function with positive correlation against a bounded degree-d nilsequence has positive $U^{d+1}[N]$ norm, with a bound for each fixed complexity and correlation. The local `converse-001` proves the required implication using the published compact cube constraint, an explicitly normalized smooth average, a uniform Fourier bound and Gowers–Cauchy–Schwarz. It uses accepted exact same-step linearization for polynomial tests. It does not import the disputed averaging-domain transition in Green–

Tao 11.2 or invoke the inverse theorem being proved. Its exact source application is the one stated in Section A30.

7. Countability, up to filtered isomorphism, of the fixed rational filtered nilmanifolds of bounded degree, and equivalence of fixed smooth metrics on a compact nilmanifold. The published countability assertion is on p. 1250 with its Mal'cev-coordinate justification. Its exact scope must include the taut rational objects constructed by this project.

8. The local increment linearization theorem: an exact polynomial nilsequence of degree s has a same-step linear representation on fixed geometry with a fixed Lipschitz comparison. This is supplied by the internal $P_1 \rtimes \mathbb{R}$ construction, rather than by the defective full-P_0 suspension in Appendix C.

The increment unit supplies the exact sequence and finite-family join used here. It does not assert the stronger clause in the original Proposition C.2 about one universal projection into the original, unnormalized nilmanifold for every polynomial: bounded normalization is absorbed in the test function F_q. That distinction changes no part of the target in section 1.

The affine-transfer and multivariable-factorization connections are as follows.

First, the one-variable weak factorization uses a progression $A+b[M]$ with $|A| = O(N)$, standard bounded nonzero integer b, and $M \asymp N$. Apply the local inverse-affine lemma with one variable to a real lift of the horizontal polynomial. Its input is the positive binomial coefficient bound on $m \mapsto f(A + bm)$. In the inverse substitution, the coefficient from input degree j to positive output degree i has size $O(N^{j-i})$ and a standard denominator independent of A. One standard integer multiplier clears the integer portions; the residual $O(N^{-j})$ becomes $O(N^{-i})$. Thus the same original horizontal character, multiplied by a standard positive integer, has the required ambient coefficient bound. It remains nonzero. No bound on the constant coefficient is asserted or needed. This is the exact proof-level transfer at DF002 lines 132–145, distinct from its right-coset periodicity argument.

Second, at Section 13 lines 327–343 take the local multifactor bias theorem with dimension $s + 1$, degree s, and its scale equal to $8N$. The input is the actual reflected unit nilcharacter σ_ε and reflected convex polytope obtained by the orthant selection. Every positive progression is the full $[8N]$; the auxiliary lower-degree test is constant. Its conclusion supplies a globally represented lower-degree nilsequence agreeing with σ_ε on all of $[-8N, 8N]^{s+1}$. The inverse reflection is an integer linear substitution, so the same full-box conclusion holds for σ. This does not extend an identity from only the selected positive orthant by assumption.

Third, after the displayed diagonal pullback in Section 13, the actual unit tensor-product representative $\xi(h, n)$ of the antisymmetric symbol satisfies $s![\xi] = 0$ on $[-8N, 8N]^2$. By the definition of the zero symbol, the coordinates of $\xi^{\otimes s!}$ are lower-degree nilsequences on that box; their finite product of models is one common lower-degree model. Apply the local torsion theorem in dimension two, degree s, scale $8N$, and positive standard tensor power $s!$. It gives a globally represented degree-at-most-$(s-1)$ nilsequence equal to ξ on that same symmetric box. Only afterward restrict to $[-N, N] \times [N]$, as in Section 13 lines 345–363.

There is also a distinct converse use at Section 13 lines 298–305. For $s \geq 4$, put $d = s - 3 \geq 1$ in the accepted local converse and apply it to the uniformly selected lower-degree test correlated with the chosen coordinate of α'. This gives the positive $U^{s-2}[N]$ bound there, uniformly after the stated selections. For $s = 3$, a degree-zero nilsequence is constant: a fixed positive correlation with a bounded constant directly implies a fixed positive $|\mathbb{E}_{n\in[N]}\alpha'(n)|$: if the correlation is at least $c > 0$ and the constant has modulus at most M, this mean has modulus at least c/M. Zero extension has group mean $(N/|Q|)\mathbb{E}_{[N]}\alpha'$, and the interval indicator has U^1 quantity $N/|Q|$, so normalization

gives exactly that interval mean. No degree-one test is inserted into this case.

Closure of fixed-degree nilsequences under conjugation, scalar multiplication and finite products uses the actual product nilmanifold and product filtration. The top degree is the maximum of the input degree bounds. This elementary object identification is also required when it is used in the induction below.

A15.3 Correct splitting of the symmetrization remainder

We first give the lattice detail needed by the splitting application. It uses the integral Hall-coordinate lemma in the internal LP04 section, not a general assertion that disjoint rational Lie subgroups automatically separate points after quotienting by a lattice.

Lemma A15.1 (Splitting on two finite multidegree downsets)**.** *Let $J_1, J_2 \subset \mathbb{N}^a$ be finite downsets. Every bounded Lipschitz nilsequence of multidegree supported on $J_1 \cup J_2$ is uniformly approximable, to each fixed positive error, by a finite sum of products of nilsequences supported on J_1 and J_2. The construction uses fixed geometries and bounds before the polynomial coefficients or the interval length vary, for each fixed input geometry and test bound.*

Proof. Use bounded tautification to set the polynomial's value at zero to the identity and restrict the ambient group to the group generated by its positive coordinate groups. For each positive $j \in J_1 \cup J_2$ choose a finite generating set of $\Gamma \cap G_j$. Its real generated group is G_j, by the rational subgroup assertion. Introduce one formal letter of weight j for each such generator. Take the free real Lie algebra on these letters and discard every individual-letter multidegree whose total vector weight is outside $J_1 \cup J_2$. This is an ideal: the complement of a downset is upward closed. Every letter has positive total weight, so the surviving length is bounded by $\max_{j \in J_1 \cup J_2} |j|$.

Let U be the exponential group and Λ the group generated by the exponential letters. The Hall-coordinate lemma proves that Λ is a lattice and that all weight filtration groups are rational. Sending each letter to its chosen original lattice generator gives a filtered Lie homomorphism $\phi : U \to G$ with $\phi(\Lambda) \subseteq \Gamma$. It maps U_j onto G_j: the letters of weight exactly j already generate G_j as a real group. Lift each Taylor coefficient of the normalized original polynomial through this surjection. The multivariate Taylor theorem, B.9, makes their ordered binomial product an adapted polynomial $\widetilde{g}$ with $\phi\widetilde{g} = g$. Pulling the test back along $U/\Lambda \to G/\Gamma$ therefore preserves the sequence exactly. Its Lipschitz bound increases by a fixed geometric factor only.

For $\nu = 1, 2$ let I_ν be the connected ideal subgroup spanned by the surviving Hall words whose weight is outside J_ν. The quotient U/I_ν has degree support in J_ν. The integral Hall lemma says that its quotient lattice is obtained by deleting precisely those coordinates. No surviving Hall word is deleted by both quotients, since its weight belongs to $J_1 \cup J_2$. Consequently the map

$$U/\Lambda \longrightarrow U/(I_1\Lambda) \times U/(I_2\Lambda)$$

is injective. In detail, if two cosets have equal images, their difference has integral coordinates in each projected Hall system. The union of the retained coordinate sets is the full Hall system, so every coordinate of the difference is integral. The difference is in Λ. This checks the lattice condition as well as the Lie algebra condition.

The pullbacks of the two algebras of Lipschitz functions on these compact quotients contain the constants, are closed under conjugation, and together separate points by that injection. The complex Stone–Weierstrass theorem approximates the pulled-back test uniformly by the algebra they generate. Each element of that algebra is a finite sum of products with one factor from each quotient, since each of the two individual algebras is closed under multiplication. Composing with

$\widetilde{g}$ gives the asserted splitting. For an internal bounded Lipschitz test, first approximate by one standard Lipschitz test on the fixed compact manifold, using Definition 5.1, and then perform this standard finite approximation. For a fixed uniform Lipschitz and norm bound, compactness of that class in the uniform norm supplies a finite net and thus fixed finite bounds. Only this fixed-bound form is used below. If the positive groups are trivial, the sequence is constant and the assertion follows with one constant product. □

The coordinate order is (h, n). Set

$$A = \{(0, j) : 0 \le j \le s-1\}, \qquad B = \{(i, j) : j \le s-2,\ i + j \le s\}.$$

Both are downsets and $A \cup B = J$. Indeed the only point of J with $j = s-1$ is $(0, s-1)$, covered by A. Every other point of J has $j \le s-2$ and lies in B. Conversely a point of B of total degree s has $i \ge 2$, so belongs to the second part of (167); all its other points lie in the first part. This proves equality in both directions.

Apply interface 5 to the actual Ψ with these two downsets. A sequence supported on A is independent of h: its polynomial map has every h-direction derivative in the trivial first-coordinate group. It is a degree-at-most-$s-1$ function $\psi(n)$ of n alone. For a factor supported on B, fixing h gives a polynomial nilsequence in n of degree at most $s-2$, since all higher n-direction groups are trivial. If normalization of this section at fixed h is needed, the accepted bounded tautification theorem supplies it with uniform geometry and parameter dependence.

Here is the needed approximation-to-correlation connection. After uniform selection, suppose the preceding correlation is at least a standard $\epsilon_0 > 0$ on a dense set of h, and the lower-degree test factors have magnitude at most a standard M. Choose the uniform splitting error less than $\epsilon_0/(2M)$, also absorbing the fixed finite component and coefficient bounds into M. The error in each normalized correlation is then at most $\epsilon_0/2$. A finite sum remains, so one summand has a positive standard correlation on a dense subfamily by finite pigeonhole. The same finite choice of components of Θ can be made on a dense subfamily. The result is two fixed bounded scalar nilsequences θ, θ' of degree at most s such that

$$n \longmapsto f(n+h)\overline{f(n)}\theta(n+h)\theta'(n) \tag{168}$$

is degree-at-most-$s-2$ biased for a dense set of h. The B-factor is absorbed into that lower-degree test. The A-factor $\psi(n)$ is multiplied into the unshifted component forming θ'. No scalar component is divided by, or assumed to have modulus one.

The printed source on p. 1272 uses the opposite coordinate restrictions while claiming independence of h. The corrected downsets and the approximation estimate above are the explicit replacement for that step; its historical printed form is preserved in the findings record.

A15.4 Uniform nilcharacter selection and fixed geometries

Here is the selection needed at the start of each induction step. Use the finite-family inverse statement already obtained at the previous order; the finite-family extraction below is performed after each order, before proceeding to the next. This avoids assuming that an arbitrary external assignment of nilmanifolds to parameters is an internal family.

On one fixed taut degree-d nilmanifold, the top group is central and its quotient by its lattice is a compact torus T. For each integral character η of T, a unit-vector smooth function with that vertical character exists as follows. Cover the compact quotient by T by finitely many trivializing charts. Choose smooth real functions ρ_j supported inside the charts with $\sum_j \rho_j^2 = 1$, using a

smooth partition of unity and normalizing its vector by its positive Euclidean norm. In the jth trivialization define the jth component by $\rho_j(b)e(\eta(t))$, and extend it by zero off that chart. These components are smooth, have the exact character, and their squared magnitudes sum to one. The central rational quotient and its lattice are justified by the accepted quotient geometry in the local Lemma 13.2 unit. This is the explicit construction used in published Lemma 6.4.

Components of such unit-vector functions, including frequency zero, form a conjugation-stable multiplicative family: use tensor products for multiplication and conjugation for conjugation. They contain constants and separate points. Distinct points on the same T-fibre are separated by an integral character, with a nonzero component of the unit vector. Points on different fibres are separated by a smooth frequency-zero scalar unit function pulled back from the base. Stone–Weierstrass therefore gives uniform approximation of every continuous test by a finite linear combination of these components.

For the fixed norm and Lipschitz bound in the previous inverse theorem, Arzela–Ascoli gives a finite uniform net of tests. Apply the preceding approximation to each element of that net with error smaller than a fixed fraction of the positive inverse correlation. There are finitely many models and net elements, so the total list of unit-vector test maps, dimensions, frequencies, coefficient bounds and Lipschitz bounds is fixed. Pigeonholing a scalar component of an approximating sum preserves a fixed positive correlation. No component is assumed to have modulus one; the entire vector has norm one.

For each finite interval and each of its good shifts, ordinary choice selects one polynomial orbit and one item from this fixed finite list. Taking the ultralimit of these finite assignments gives an internal family. A finite pigeonhole step restricts to a dense set of shifts with one common fixed model and fixed test map if required. The unit and vertical identities hold at every representative and hence for all limit arguments, with the natural extension of the chosen standard character. This establishes the selection interface needed here from the preceding finite inverse theorem. It does not replace separate selection applications later in Sections 8–13, which remain with their actual consumers.

For the countability interface, use the rational Mal'cev-coordinate description of a lattice in a connected simply connected nilpotent Lie group, as invoked in published Section 3 and p. 1250. In each finite dimension the rational structure constants have countably many possibilities. The rational subspaces defining the finitely many filtration groups are specified by finite matrices with rational entries, again a countable choice. In the rational structure determined by the lattice, a finite list of rational logarithmic generators specifies the lattice. Thus there are at most countably many such filtered lattice data up to isomorphism. All local constructions here have finite rational filtrations and discrete cocompact lattices, so belong to this class. On each compact quotient any two fixed smooth Riemannian metrics are bi-Lipschitz equivalent: compare their positive quadratic forms on the compact unit tangent bundle. These are the exact countability and metric comparisons used in the extraction below.

A15.5 The finite-group inequality needed for the last induction step

Lemma A15.2. *For bounded complex functions a, b on a finite abelian group Q and an integer $d \geq 1$,*

$$\mathbb{E}_{h\in Q}\|a(\cdot+h)b(\cdot)\|_{U^d(Q)}^{2^d} \leq \|a\|_{U^{d+1}(Q)}^{2^d}\|b\|_{U^{d+1}(Q)}^{2^d}.$$

Proof. For $t = (t_1, \dots, t_d)$ write $C_a(x;t) = \prod_\omega \mathcal{C}^{|\omega|}a(x+\omega\cdot t)$, and define C_b likewise. Expanding the left side and using the bijection $(x,h) \mapsto (x, y = x+h)$ gives

$$\mathbb{E}_t(\mathbb{E}_y C_a(y;t))(\mathbb{E}_x C_b(x;t)).$$

Its absolute value is at most $\left(\mathbb{E}_t|\mathbb{E}_y C_a(y;t)|^2\right)^{1/2}\left(\mathbb{E}_t|\mathbb{E}_x C_b(x;t)|^2\right)^{1/2}$. Each squared expression is the $(d+1)$-cube average: expand the square and substitute its second base point as the first plus a new increment. It therefore equals the corresponding U^{d+1} norm to the power 2^{d+1}. This proves the inequality. The original left side is nonnegative because it is an average of powers of Gowers seminorms. □

Choose a cyclic group Q with size between C_sN and $2C_sN$ for a fixed sufficiently large C_s, and zero-extend all functions supported on $[N]$. The normalized interval and group norms differ by the fixed indicator factor. For each fixed order that factor is bounded above by one and below by $N/|Q|$, using the finite-group Gowers–Cauchy–Schwarz inequality. Thus a fixed positive interval norm gives a fixed positive group norm, and conversely; the constants depend only on the fixed ambient ratio and order. Taking C_s larger than 2^{s+1} ensures the same ambient group is valid for all interval orders used here. The zero-extended product in (168) is exactly the group product for $|h| \le N$; there is no wrapped overlap since the ambient size is larger than $4N$.

By interface 6, (168) has a uniformly positive $U^{s-1}[N]$ norm for a dense set of h. Let $a = f\theta$ and $b = \overline{f}\theta'$, both zero-extended. The average over all group shifts of $\|a(\cdot+h)b\|_{U^{s-1}(Q)}^{2^{s-1}}$ is bounded below by a fixed positive constant: the contributing set of interval shifts has positive density also in Q, and the other summands are nonnegative. Lemma A15.2 and the fixed bound on b imply $\|a\|_{U^s(Q)}$ is positive by a standard amount, hence so is $\|f\theta\|_{U^s[N]}$. The inverse theorem at order $s-1$ now makes $f\theta$ correlate with a degree-at-most-$s-1$ nilsequence ψ. The actual product $\theta\psi$ is a degree-at-most-s nilsequence on the product filtered nilmanifold. Thus f is s-biased.

A15.6 Starting and completing the induction

The bounded induction base is Section A16. It starts from the exact forward bracket theorem, proves the affected trivial-subgroup specialization, constructs a periodic connected polynomial test and transfers its correlation directly to the normalized interval.

For completeness the initial derivative step has the same normalization control. The cube identity on the fixed ambient cyclic group gives

$$\mathbb{E}_h\|\Delta_h f\|_{U^s(Q)}^{2^s} = \|f\|_{U^{s+1}(Q)}^{2^{s+1}}.$$

If the interval norm is positive by a standard amount, so is the right side. Since f is bounded, all summands have a fixed upper bound. The nonzero differences occur only on the shifts represented by $[-N, N]$, so a dense subset of those shifts has a uniformly positive $U^s[N]$ norm. Apply the inverse theorem of order $s-1$, then interface 2, to obtain the family required by interface 3. Interfaces 3–5 and the preceding section complete the induction step. The bounded limit-N case is included: a bounded interval has standard finite size, and any function on it is represented by a Lipschitz function along a rational circle rotation, with a standard finite Lipschitz bound. Thus the induction proves the limit polynomial inverse conclusion for every limit N, conditional on the listed intermediate interfaces and base.

We spell out the finite-family extraction rather than replace it by pointwise compactness. Enumerate the rational filtered nilmanifolds of degree at most s up to isomorphism as $M_1, M_2, \ldots$, with one fixed smooth metric on each, using interface 7. If the finite polynomial inverse statement failed at some fixed $\delta > 0$, for each j choose a 1-bounded $f_j : [N_j] \to \mathbb{C}$ with norm at least δ, but correlation at most $1/j$ against every polynomial nilsequence from the first j manifolds with 1-bounded test and Lipschitz norm at most j. Its ultralimit has positive norm, so the limit inverse conclusion gives a fixed standard filtered nilmanifold, bounded Lipschitz limit test and positive standard correlation. Normalize the test by a fixed bound to make it 1-bounded. It is isomorphic

to some fixed M_{j_0}; fixed metric equivalence transfers its Lipschitz bound to a standard constant. Representatives of the polynomial and test satisfy the same bounds with one slightly larger standard constant on an ultrafilter-large set. For j larger than that constant and j_0, the selected correlation upper bound tends to zero, contradicting the positive lower bound. This yields the finite-family polynomial statement with all parameters fixed independently of N.

Finally interface 8 converts every member of the finite polynomial family to a fixed nilmanifold of step at most s, with a fixed Lipschitz bound and exactly equal sequence. A finite maximum of those bounds and the unchanged positive correlation give the exact linear target in section 1. This final join depends on the reviewed increment construction; it does not import the failed full-polynomial suspension claim.

A15.7 Conclusion and published inputs

The preceding argument assembles the local results and their stated hypotheses. The affine coefficient transfer, multivariable factorization, bias and torsion conclusions are used through the explicit substitutions above.

The published forward bracket and Bohr theorems, quantitative equal-side equidistribution, rational-coordinate results and compact cube constraint retain their stated scope. The conclusion has the normalized norm, quantifier order, fixed finite family and same-step linear output specified at the start.

A16 The polynomial GI(2) base from the published bracket theorem

This section derives the exact polynomial interval base from the forward bracket-quadratic theorem and Bohr regularity in Green–Tao, *An inverse theorem for the Gowers $U^3(G)$ norm*, Proceedings of the Edinburgh Mathematical Society 51 (2008), 73–153. It replaces the needed use of the false quadratic-extension assertion by an explicit finite-box proof. It also constructs the bracket tests directly on connected polynomial models, so neither Theorem 12.9 nor its erroneous skew-shift iterate parameters are a theorem premise. The forward Bohr/bracket inverse theorem and its unchanged proof inputs remain published inputs, not newly proved by this unit. The complete inverse theorem is assembled in Section A15.

A16.1 Exact target and published input

The required base is this statement. For every $\delta > 0$ there are finitely many fixed rational filtered nilmanifolds of degree at most two, with connected simply connected groups, and constants $L < \infty, c > 0$ such that every complex 1-bounded $f : [N] \to \mathbb{C}$ with $\|f\|_{U^3[N]} \geq \delta$ correlates by at least c with $F(g(n)\Gamma)$ on one of them. The polynomial g is adapted, $\|F\|_\infty \leq 1$, and $\mathrm{Lip}(F) \leq L$. All geometry and constants are fixed before N and f. The interval norm is zero extension to a sufficiently large cyclic group divided by the norm of the interval indicator.

The published forward input is Theorem 10.11, pp. 124–125, especially (10.7) and (10.9). For each fixed sufficiently small $\eta > 0$, and each prime $P > 2$, a function $u : \mathbb{Z}/P\mathbb{Z} \to \mathbb{C}$ with $|u| \leq 1$ and $\|u\|_{U^3} \geq \eta$ yields a frequency set S of size at most $D(\eta)$, a proper progression Q containing a Bohr set $B(S, \rho_0)$ with $\rho_0 \geq \rho_*(\eta) > 0$, and one bracket quadratic

$$\phi(n) = c + \sum_j a_j\{\alpha_j n\} + \sum_{j,k} a_{jk}\{\alpha_j n\}\{\alpha_k n\} \pmod 1.$$

Here $\alpha_j = \xi_j/P$ represents a frequency in S, $\{t\} \in [-1/2, 1/2)$, and coefficients are unrestricted real numbers. For every nonempty $A \subseteq Q$ there is a shift h_A such that

$$|\mathbb{E}_{n\in A} u(n + h_A)\mathrm{e}(-\phi(n))| \geq b_*(\eta) > 0.$$

The actual powers and exponential constants printed in 10.11 are stronger than needed; only the stated positive/finiteness dependence on η is consumed. Its proof uses the forward Theorem 2.7, Corollary 10.6 and Proposition 10.10. The affected special case in the latter is supplied next.

A16.2 The exact specialization replacing quadratic extension

The final PDF's Lemma 10.7, pp. 122–123, invokes quadratic extension from a subgroup. The author notice of December 2023 identifies Proposition 3.2, p. 88, as false. The later use needed here is Proposition 10.10, where the progression is proper and $H = \{0\}$. The notice numbers these items 10.6 and 10.8 in its preprint version; we bind to the final PDF numbering and argument.

Lemma A16.1. *On a finite box $B = \prod_{j=1}^{d}\{0, \dots, L_j - 1\}$, any $v : B \to \mathbb{R}/\mathbb{Z}$ whose third additive differences vanish whenever the whole cube lies in B has the form*

$$v(l) = c + \sum_j b_j l_j + \sum_j d_j \binom{l_j}{2} + \sum_{j<k} e_{jk} l_j l_k.$$

Consequently a locally quadratic phase on a proper progression with trivial subgroup has the polynomial coordinate representation needed in Proposition 10.10, without extension to the ambient group.

Proof. Coordinates of length one can be omitted. Each mixed second difference is constant on its rectangular domain: every coordinate difference of it is a vanishing third difference, and the domain is connected by coordinate steps. The same holds for each available diagonal second difference when $L_j \geq 3$. Choose their constants as e_{jk} and d_j, setting $d_j = 0$ when $L_j = 2$. Subtract these quadratic terms. Every remaining first difference is constant because its differences in every available coordinate vanish. When $L_j = 2$ its domain has only one slice in direction j, so mixed differences suffice. Subtract the resulting linear function; the remainder is constant by connectedness. The coordinate map of a proper progression is injective and pulls back the local quadratic identities to the box. This proves the claim. Division by two in $\mathbb{R}/\mathbb{Z}$, if one wants the symmetric coefficient convention of (10.5), is possible; no uniqueness is used. □

In Proposition 10.10 the independent vectors supplied by Corollary 10.6 express each progression coordinate as a real linear combination of $\{\xi n/P\}$ on that progression. Substituting the lemma's polynomial gives precisely its bracket quadratic. The later localization and shift in the proof of 10.11 use this specialization, not the extension of a phase from a nontrivial subgroup. The remaining published forward inverse and Bohr-set applications are retained as explicit inputs. This does not claim to prove a corrected general version of 3.2.

A16.3 A smooth periodic bracket test with positive correlation

Use Lemmas 8.1–8.2 to choose a regular Bohr set $B = B(S, \rho) \subseteq Q$, where ρ has a positive lower bound depending only on η and may be taken at most $1/8$. Its density in $\mathbb{Z}/P\mathbb{Z}$ is at least $b_0(\eta) > 0$. Apply the displayed (10.9) input with $A = B$ and write its shift as h.

For $d = |S| > 0$, choose a fixed $0 < \epsilon < 1/(100d)$ sufficiently small in terms of $b_*(\eta)$ and d. Let $\chi : \mathbb{R}/\mathbb{Z} \to [0, 1]$ be one on $|x| \leq \rho(1 - \epsilon)$, zero on $|x| \geq \rho(1 + \epsilon)$, and linear on the intervening

strips. Its Lipschitz bound is at most $(2\epsilon\rho)^{-1}$, with an inessential larger fixed bound permitted. In particular it vanishes on a fixed neighbourhood of the fractional-part seam at $1/2$. Put

$$W(n) = \prod_{j=1}^{d} \chi(\alpha_j n)^{1+4d}, \qquad \Psi(n) = W(n)\mathrm{e}(-\phi(n)).$$

The difference $W - 1_B$ vanishes outside the regularity boundary shell $B(S, \rho(1+\epsilon)) \setminus B(S, \rho(1-\epsilon))$ and is bounded by one. Bohr regularity bounds the shell by $Cd\epsilon|B|$ with an absolute fixed C. Choose ϵ so that $Cd\epsilon \le b_*(\eta)/2$. It follows that

$$\left|\mathbb{E}_{n\in\mathbb{Z}/P\mathbb{Z}} u(n+h)\Psi(n)\right| \ge \tfrac{1}{2} b_*(\eta) b_0(\eta) =: c_0(\eta) > 0. \tag{169}$$

When S is empty the phase is constant and this conclusion already uses a constant test; that case needs no cutoff construction.

Every factor in Ψ is periodic with period P, because $P\alpha_j$ is integral. We now represent this same function for all integers, not merely signed representatives of $\mathbb{Z}/P\mathbb{Z}$.

A16.4 Connected polynomial models for every bracket factor

Write the coefficients below as $a = q + s$, with $q \in \mathbb{Z}$ and $|s| \le 1/2$. A linear bracket factor is

$$\chi(\alpha n)\mathrm{e}(a\{\alpha n\}) = \chi(\alpha n)\mathrm{e}(s\{\alpha n\})\mathrm{e}(q\alpha n).$$

It is the test $F(x, y) = \chi(x)\mathrm{e}(sx + y)$ along the linear polynomial $(\alpha n, q\alpha n)$ on $\mathbb{R}^2/\mathbb{Z}^2$. The seam is harmless because χ vanishes there. This test is 1-bounded and has Lipschitz bound $O(1 + \operatorname{Lip}\chi)$, uniformly in a, α.

A quadratic bracket factor satisfies the exact identity

$$\begin{aligned}&\chi(\alpha n)^2\chi(\gamma n)^2\mathrm{e}(a\{\alpha n\}\{\gamma n\})\\ &= [\chi(\alpha n)\chi(\gamma n)\mathrm{e}(s\{\alpha n\}\{\gamma n\})]\mathrm{e}(q\alpha\gamma n^2)[\chi(\gamma n)\mathrm{e}(-q\alpha n[\gamma n])][\chi(\alpha n)\mathrm{e}(-q\gamma n[\alpha n])].\end{aligned}$$

Indeed the omitted term in expanding the integer part is $q[\alpha n][\gamma n] \in \mathbb{Z}$, where $[t] = t - \{t\}$. The first bracket is a test on a two-torus with uniformly bounded Lipschitz constant. The pure quadratic phase is a test $\mathrm{e}(z)$ along $z = q\alpha\gamma n^2$ on $\mathbb{R}/\mathbb{Z}$, using the degree-two filtration $G_0 = G_1 = G_2 = \mathbb{R}$, $G_3 = 0$.

For the remaining bracket use the connected simply connected Heisenberg group with coordinates (u, v, w) and law

$$(u, v, w)(u', v', w') = (u + u', v + v', w + w' + uv').$$

Its lattice is $\mathbb{Z}^3$ in these coordinates; its degree-two filtration has $G_0 = G_1 = G$ and $G_2 = \{(0, 0, w)\}$. On a right lattice quotient, reduce v first, then u. The resulting coordinates are

$$(\{u\}, \{v\}, w - u[v] \pmod 1).$$

Define a test on this fundamental cube by $F_H(\{u\}, \{v\}, z) = \chi(\{v\})\mathrm{e}(z)$. It is well-defined and globally Lipschitz. Across the u faces the central coordinate does not change; the central faces are periodic; across the v faces the cutoff is identically zero in a neighbourhood. Within the remaining finitely many fixed charts its Lipschitz constant is bounded by a fixed geometric multiple of $1+\operatorname{Lip}\chi$.

Evaluate this test on the polynomial $p(n) = (\beta n, \alpha n, 0)$. The exact value is

$$F_H(p(n)\Gamma) = \chi(\alpha n)\mathrm{e}(-\beta n[\alpha n]).$$

For completeness its left first difference is $(\beta h, \alpha h, -\alpha\beta hn)$, its next difference is $(0, 0, -\alpha\beta hk)$, and its third difference is the identity. These satisfy the degree-two filtration for every integer shift, so this is an adapted polynomial without any appeal to an incorrect formula for linear skew iterates. Take $\beta = q\gamma$ and interchange α, γ for the two remaining factors.

Apply the linear construction to each coefficient of $-\phi$ and the quadratic construction to all ordered pairs, inserting zero coefficients if needed. Each frequency receives one cutoff from the linear factor and $4d$ from the quadratic factors, exactly the power in W. The constant phase is harmless. Products give a fixed connected rational polynomial model of degree at most two, of dimension at most $2d + 9d^2$. The product test is 1-bounded and has a finite Lipschitz bound depending only on d, ρ, ϵ. Arbitrarily large coefficients occur only in polynomial orbit parameters. There are finitely many geometric product types for $d \le D(\eta)$.

Thus the model represents the periodic function $\Psi(n)$ exactly for all $n \in \mathbb{Z}$. Individual polynomial factors need not be P-periodic; their product is periodic by the exact bracket identity. That distinction is used in the next change of variables.

A16.5 The normalized interval transfer

Fix an integer $C > 16$ and choose a prime P with $CN < P < 2CN$. Zero-extend the given interval function to u on $\mathbb{Z}/P\mathbb{Z}$. The interval-indicator Gowers norm is at least its mean N/P, so $\|u\|_{U^3} \ge \delta/(2C)$. Choose once and for all a positive η at most this bound and within the smallness range of the published forward theorem. All preceding data now depend only on δ.

Using the exact periodicity of Ψ in (169), change variables to obtain

$$\left|\mathbb{E}_{n\in\mathbb{Z}/P\mathbb{Z}} u(n)\Psi(n-h)\right| \ge c_0(\eta).$$

The zero extension gives

$$\left|\mathbb{E}_{n\in[N]} f(n)\Psi(n-h)\right| \ge (P/N)c_0(\eta) > 0.$$

The translated sequence is represented by the same test on the same fixed model, with polynomial orbit $g(n-h)$. Translation preserves adapted polynomiality, either by the derivative criterion or B.5. No signed-representative split, discontinuous interval indicator or additional N-dependent geometry is introduced. Norm, Lipschitz and correlation bounds have precisely the required fixed dependence. Bertrand's postulate applies to the integer $CN > 1$, so the prime choice also covers every positive N.

A16.6 Source comparison and bounded related-use findings

The unpublished author notice identifies the false general quadratic extension, but is used here only to locate the affected proof. The needed trivial-subgroup specialization is proved in Section 2. Its published source locator is Proposition 3.2 on p. 88, correcting the p. 89 metadata typo in assembly001's RELATED-USES record.

The actual publisher image of Example 12.3 on p. 136 gives $T(x, y) = (x + \alpha, y + \beta + mx)$ but prints an iterate omitting mnx and using $mn(n + 1)\alpha/2$. Direct iteration gives

$$T^n(x, y) = (x + n\alpha, y + n\beta + mnx + m\alpha\tbinom{n}{2}).$$

Already $n = 1$ distinguishes these formulas. The later pure quadratic choice in Lemma 12.11 inherits the printed iterate convention. Rather than silently change its parameters, Section 4 constructs all needed factors directly on connected polynomial models. This preserves the qualitative base theorem's strength and avoids those printed iterates entirely.

The cutoff's derivative bound also depends on its transition width. This proof retains the finite factors $(\epsilon\rho)^{-1}$ instead of importing the universal numerical bounds stated for those tests in Lemma 12.11. The required application needs only finite bounds after η is fixed.

The base uses the forward chain $2.7 \to 10.6 \to 10.10 \to 10.11$, the affected trivial-subgroup use of 10.7, Bohr regularity 8.1–8.2, and the elementary construction 12.2–12.11 as consumed by the interval base. The chain notation records its published dependencies. The converse half of 2.7, the finite-field-only extension use and unrelated applications are not premises of this unit. The base uses the precise bracket theorem and the local construction, with the converse proved separately.

A17 Tautification and the shifted nilcharacter construction

The geometric input is the rational-subgroup and quotient argument in Section A18.

A17.1 The exact change of filtration

All Lie groups and rationality conventions are those of [130, Sections 3–6]. A rational subgroup is connected, and its intersection with the ambient lattice is cocompact. The three domain filtrations used here are the degree filtration on $\mathbb{Z}^a$, the multidegree filtration on $\mathbb{Z}^a$, and the degree-rank filtration on $\mathbb{Z}$ from [130, Definition 6.17]. For limit objects, defining relations are interpreted by their natural ultrapower extension as in [130, Appendix A.2–A.3, pp. 1324–1325]. Thus vertical equivariance quantifies over every limit point of the nilmanifold and every limit element of the top group, using the extension of the standard vertical character. We do not infer this exact internal identity from an identity asserted only at standard points. For these three cases respectively put

$$H = G_1, \qquad H = \langle G_{e_1}, \dots, G_{e_a} \rangle_{\mathbb{R}}, \qquad H = G_{(1,0)}.$$

In the middle case the connected real generated subgroup equals the group generated by the connected subgroups: the latter is the connected analytic subgroup with the generated Lie algebra, and such subgroups in a simply connected nilpotent group are closed. The rational-subgroup lemma in Section A18 shows that H is rational.

Theorem A17.1. *Suppose G/Γ is a rational filtered nilmanifold with finite downset of degrees J for one of these three domain filtrations. Every polynomial nilsequence $F(g(n)\Gamma)$ admits an exactly equal representation on $H/(H \cap \Gamma)$ with filtration*

$$G'_0 = H, \qquad G'_i = G_i \quad (i > 0).$$

For degree-rank, 0 denotes $(0, 0)$. This filtration is taut and has the same positive-degree groups and no larger degree support, hence the same allowed degree bound. Its test-function Lipschitz bound depends only on the original fixed filtered nilmanifold and original Lipschitz bound, not on the polynomial coefficients. For a nilcharacter of positive degree d, the same standard vertical character on G_d is preserved. These assertions hold uniformly for limit maps and families of polynomial coefficients with fixed geometric data.

Proof. As in Section A18, first restrict to G_0 and its lattice if necessary. Choose $g(0) = q\gamma$ with q in a fixed compact representative set, and set

$$h(n) = q^{-1}g(n)\gamma^{-1}, \qquad F_q(x\Gamma) = F(qx\Gamma).$$

Then $h(0) = 1$, $F_q(h(n)\Gamma) = F(g(n)\Gamma)$, and h is polynomial for the old filtration. Indeed every G_i is normal in G_0; use the constant map $g(0)^{-1}$, product closure in [130, Corollary B.4], and filtered conjugation by q. The Lipschitz bound is uniform because q lies in a fixed compact set.

We show that $h(n)$ belongs to H. In the degree case the difference $h(n)h(0)^{-1}$ is in G_1 for every $n \in \mathbb{Z}^a$, by the derivative condition. In the degree-rank case the same statement holds with $G_{(1,0)}$. In the multidegree case connect 0 to n by a finite path of positive or negative coordinate steps. At a step in the jth coordinate, $h(v \pm e_j)h(v)^{-1} \in G_{e_j}$. Multiplying these increments along the path and using $h(0) = 1$ proves the assertion. This argument applies to every integer input and puts no bound on its size.

Every original positive group is contained in H. For degree and degree-rank this follows from nesting and the least positive index. For multidegree, any positive index dominates some e_j. Consequently the displayed new groups are well-defined subgroups of H, are descending, and satisfy the commutator axioms inherited from G, including those with index zero because $H \subseteq G_0$. Their intersections with $H \cap \Gamma$ equal their intersections with Γ and are lattices. In degree-rank all the required equalities $G_{(d,0)} = G_{(d,1)}$ for $d > 0$ remain unchanged. The new group at zero is exactly the required positive group or group generated by the minimal positive groups, so the new filtration is taut.

All positive iterated derivatives of h have the same required group membership as before, and the zeroth-order membership was just checked. Thus h is polynomial for the new filtration. Restrict F_q through the fixed smooth inclusion $H/(H \cap \Gamma) \longrightarrow G/\Gamma$. The inclusion is an embedding of compact manifolds, so this costs only a fixed Lipschitz comparison factor and gives the asserted exact sequence.

For a nilcharacter let $\eta : G_d \longrightarrow \mathbb{R}$ be its standard vertical character. The proof in Section A18 applies verbatim with G_d in place of its top cube group: for each $t \in G_d \cap \Gamma$, the continuous map $q \mapsto \eta(qtq^{-1})$ on connected G_0 is integer-valued by the unit-norm vertical identity at $q\Gamma$. It is therefore constant. The lattice spans G_d as a connected real group, so $\eta(qtq^{-1}) = \eta(t)$ for all $t \in G_d$ and $q \in G_0$. Hence F_q and its restriction have the same standard vertical character.

For families, choose one bounded standard representative section of the fixed compact quotient. Such a section can be formed from finitely many local smooth sections, taking the first available chart. Apply this one section to each constant coefficient. The resulting choices are limit maps when the coefficients are limit maps. Discontinuities in the parameter do not enter a Lipschitz bound in the orbit variable. All metric constants and groups remain fixed. The exact identities can be imposed on representatives on an ultrafilter-large set by the witness argument of Section A18; the ordinary proof then passes to the limit. □

This also proves equality of the classes of nilsequences under the old and taut conventions, at each finite degree support: one inclusion is the theorem and the reverse inclusion is immediate. It therefore preserves the equivalence relation defined by a lower-degree nilsequence. It is not merely an assertion that the top-degree function has a similar replacement.

Corollary A17.2. *In a taut degree or degree-rank filtration of degree at most (d, r) or d, the ambient group has nilpotency class at most d. In a taut multidegree filtration supported on a finite downset J, its class is at most $D = \max_{a \in J} |a|$. If the support is contained in the downset below one positive index d, the group G_d is central.*

Proof. For degree and degree-rank, brackets of j elements of the ambient first positive group belong respectively to G_j or $G_{(j,0)}$, which vanish when $j > d$. For multidegree, expand an iterated Lie bracket using the generating Lie algebras $\mathfrak{g}_{e_i}$. A bracket of length j lies in some $\mathfrak{g}_a$ with $|a| = j$, hence vanishes for $j > D$. The generated Lie algebra has the asserted nilpotency class, and so does its simply connected group. Finally $[G_d, G_{e_i}] \subseteq G_{d+e_i} = 1$ in the multidegree case. The same argument uses the first positive group in the other two cases. Since these groups generate the ambient group, G_d is central. □

A17.2 A complete shifted nilcharacter construction

Theorem A17.3. *For any of the three domain filtrations above, let χ be a nilcharacter of positive degree d. Let $i > 0$ and let u belong to the domain group at index i. Then*

$$n \longmapsto \chi(n+u) \otimes \overline{\chi(n)}$$

is a nilsequence with support strictly below d. The representing geometric data and Lipschitz bound can be fixed independently of u within this domain group. For limit u the representation and test function can be chosen in a limit fashion. This supplies the shift assertion required in [130, Lemma E.8(iv)] for the three domains used in the proof.

Proof. By Theorem A17.1 represent the same χ on a taut nilmanifold G/Γ, with the same degree and vertical character. Write its polynomial map as g. In the degree-rank case use $i = (1, 0)$: any nonzero element of a positive domain group requires this index, and if the given domain group is trivial then $u = 0$ also belongs to this first group. Put $U(v) = (v, 1)$ and $D(v) = (v, v)$ in G^2. For each index j let L_j be the connected subgroup generated by

$$U(G_{j+i}) \quad \text{and} \quad D(G_j).$$

These groups are rational by Section A18, and are descending. The generator commutators have the following memberships:

$$\begin{aligned}
[D(G_j), D(G_k)] &\subseteq D(G_{j+k}),\\
[D(G_j), U(G_{k+i})] &\subseteq U(G_{j+k+i}),\\
[U(G_{j+i}), U(G_{k+i})] &\subseteq U(G_{j+k+2i}) \subseteq U(G_{j+k+i}).
\end{aligned}$$

Passing to generated Lie algebras and using BCH, as in Section A18, proves $[L_j, L_k] \subseteq L_{j+k}$, including index zero. Thus L_0/Δ, with $\Delta = L_0 \cap \Gamma^2$, is a rational filtered nilmanifold of degree at most d. In the degree-rank case its additional equalities hold: $L_{(a,0)} = L_{(a,1)}$ follows from $G_{(a,0)} = G_{(a,1)}$ and $G_{(a+1,0)} = G_{(a+1,1)}$ for $a > 0$.

The difference $a_u(n) = g(n+u)g(n)^{-1}$ is polynomial for the shifted filtration $(G_{j+i})_j$. Its zeroth-order value belongs to G_i. Every further iterated derivative is an iterated derivative of g with the derivative in direction u innermost, so the defining index sum has the additional term i. Therefore $U(a_u(n))$ and $D(g(n))$ are polynomial for (L_j). By [130, Corollary B.4], so is their product

$$P_u(n) = (g(n+u), g(n)).$$

This verifies the actual orbit, not only its coordinate projections.

As $i > 0$, $G_{d+i} = 1$ and $L_d = D(G_d)$. Corollary A17.2 shows that G_d is central in G_0, hence L_d is central in L_0. The function

$$T(x\Gamma, y\Gamma) = F(x\Gamma) \otimes \overline{F(y\Gamma)}$$

restricted to L_0/Δ is invariant under $D(G_d)$, since its two vertical characters cancel. The rational central-quotient argument in Section A18 therefore gives a nilsequence on L_0/L_d with the exact value $T(P_u(n)\Delta)$. Its filtration group at d is trivial and all groups outside the downset below d were already trivial. Its support is consequently strictly below d. The same argument gives a fixed Lipschitz bound on the quotient.

The quotient may not yet be taut. Apply Theorem A17.1 to this lower-degree nilsequence, preserving its exact values without enlarging its degree support. The groups L_j and their quotient depend on the index i but not on the element u. The map $u \mapsto P_u(0)$ is a limit map when g and u are limit data. The one fixed bounded section used in Theorem A17.1 then gives a limit choice of the final normalized representation, with uniform Lipschitz bounds independent of u. This proves all the assertions. □

For the erratum's needed case the domain is $\mathbb{Z}^2$ with multidegree, $d = (1, s-1)$, $i = (1, 0)$ and $u = (k, 0)$. The resulting support is contained in

$$\{a : a \le (0, s-1)\} \ \cup \ \{a : a \le (1, s-2)\},$$

uniformly in k. This is exactly the support needed before applying the published splitting and vertical Fourier lemmas in the proof of the new difference decomposition. Those applications, and the subsequent two Cauchy–Schwarz steps, are separate proof obligations.

A17.3 Scope of the correction

The tautification theorem supplies the general representation repair suggested, but not fully proved, in Section 2 of the author erratum. It also proves the top centrality and nilpotency-class assertions for the taut conventions. The shifted construction above supplies the needed instance of E.8(iv) without leaving its auxiliary rationality or central quotient implicit. This unit does not claim to have audited every filtration introduced in the published proof. In particular, E.8(v), E.9, Lemma 12.1, the Section 12 quotient, and the final modified Proposition 7.3 argument still need their own exact applications or replacements.

A18 A local replacement for the filtration argument in GTZ Lemma 13.2

This section proves the multilinearity lemma and its auxiliary filtration construction.

A18.1 Statement and conventions

We use the nilmanifold, polynomial-map and nilcharacter conventions of [132, Sections 3, 4 and 6]. In particular all Lie groups below are connected, simply connected and nilpotent; filtration groups are connected rational Lie subgroups. A rational subgroup J means that $J \cap \Gamma$ is a lattice in J. The multidegree ordering is coordinatewise. Write $\mathbf{1} = (1, \dots, 1) \in \mathbb{N}^s$ and $|a| = \sum_j a_j$. The group G_a is trivial when $a \not\le \mathbf{1}$. We use $\mathrm{e}(t) = \exp(2\pi i t)$ and complex conjugation componentwise in tensor products. A vector-valued nilcharacter has a unit-vector-valued Lipschitz test function and a real vertical character, as in [132, Definition 6.1]. For limit objects, every defining relation is read in its natural ultrapower extension, using [132, Appendix A, Definition A.2 and the following discussion, and Definition A.3, pp. 1324–1325]. In particular the vertical identity is required for every limit point $x \in {}^*(G/\Gamma)$ and every $t \in {}^*G_{\mathbf{1}}$, with the standard homomorphism η extended to ${}^*\eta$. This is the internal nilcharacter convention used in this proof. We do not infer this exact

identity from an identity only at standard points; bounded Lipschitz continuity alone would not justify that inference.

Theorem A18.1 (The unchanged multilinearity statement)**.** *Let $s, m \geq 1$ be fixed standard integers. Let χ be a nilcharacter of multidegree at most $\mathbf{1}$ on $\mathbb{Z}^s$, or a limit nilcharacter with that description in the sense of [132, Sections 5–6]. For fixed homomorphisms $L_1, L_1', L_2, \ldots, L_s : \mathbb{Z}^m \longrightarrow \mathbb{Z}$, the three pullbacks below are nilcharacters of degree at most s and satisfy*

$$[\chi(L_1 + L_1', L_2, \ldots, L_s)]_{\mathrm{Symb}^s} = [\chi(L_1, L_2, \ldots, L_s)]_{\mathrm{Symb}^s} + [\chi(L_1', L_2, \ldots, L_s)]_{\mathrm{Symb}^s}.$$

The same assertion holds in each coordinate. More explicitly, each component of

$$\chi(x + x', y_2, \ldots, y_s) \otimes \overline{\chi(x, y_2, \ldots, y_s)} \otimes \overline{\chi(x', y_2, \ldots, y_s)} \tag{170}$$

is a polynomial nilsequence of degree at most $s-1$ on $\mathbb{Z}^{s+1}$. The representing nilmanifold is fixed by the original filtered nilmanifold; its test-function Lipschitz bound is bounded in terms of that fixed data and the original Lipschitz bound, independently of the polynomial coefficients.

The symbol notation abbreviates the degree-s equivalence relation of [132, Definitions 6.6 and 6.22]. Thus the explicit assertion about (170) supplies the asserted equality, with no approximation or loss in degree. It is enough to prove the ordinary statement with uniform fixed-data bounds and then take limit sequences. We supply that uniformity below. We do not assume G_0 equals the group generated by its positive multidegree groups.

A18.2 Subgroups and lattices used in the construction

We use the following basic nilpotent Lie-group facts recorded in [132, Section 3, pp. 1242–1243]: the exponential is a homeomorphism; the connected subgroup generated by a set is the exponential of the generated Lie algebra; and, for a finite set A, its discrete generated group is cocompact in its connected real generated group. The last assertion is applied only when A lies in an already specified discrete lattice. Here are the compactness details needed to apply these facts.

Lemma A18.2. *Let M be a connected simply connected nilpotent Lie group with lattice Λ. Let $J_1, \ldots, J_t$ be connected rational subgroups. Then the connected subgroup generated by the J_i is closed, simply connected, and rational in M. This also applies to diagonal or coordinate embeddings of such subgroups into a finite product of M.*

Proof. First a lattice Λ_J in a connected Lie group J is finitely generated. Choose a proper left-invariant path metric, and R such that every point is within distance R of some lattice point; compactness of J/Λ_J, followed by inversion, gives such an R. Follow a path from the identity to a lattice point in steps of length at most one and choose nearby lattice points, using the prescribed endpoints. Consecutive chosen lattice points differ by an element in the compact ball of radius $2R + 1$. Its intersection with the discrete lattice is finite and generates the lattice.

The connected real subgroup generated by this finite generating set is all of J. To justify this last assertion, let U be that subgroup. It contains Λ_J, so J/U is compact. A connected subgroup of a simply connected nilpotent group is closed, and its homogeneous space is a Euclidean space. One can see the latter directly by a chain of Lie subalgebras from $\mathfrak{u}$ to $\mathfrak{j}$, each of codimension one and an ideal in the next: the normalizer of a proper subalgebra of a nilpotent Lie algebra strictly contains it, so such a chain is obtained by choosing one normalizing vector at each step. Successive

exponential coordinates give the Euclidean coordinates on J/U. For completeness, the normalizer assertion follows by taking the first member of the ascending central series not contained in the subalgebra; the preceding member is contained in it, so any vector in that member but outside the subalgebra normalizes it. Hence compactness forces $U = J$.

Choose a finite generating set for each $J_i \cap \Lambda$. Their union A lies in Λ and generates all the J_i as a connected real group. Its abstract generated group is discrete because it lies in Λ. The finite-set fact from [132, p. 1243] therefore makes it a lattice in $\langle A\rangle_{\mathbb{R}}$. Consequently the larger group $\Lambda \cap \langle A\rangle_{\mathbb{R}}$ is also a lattice there. A Lie subalgebra of a nilpotent Lie algebra exponentiates to a closed simply connected subgroup, which proves the other assertions. For a diagonal or coordinate embedding, apply the same argument to the images of the finite generating sets, which lie in the product lattice. □

We will also use the elementary central-quotient consequence explicitly. If J is such a group, Δ its lattice, and C a connected central rational subgroup, then J/C is connected simply connected nilpotent and the image of Δ is a lattice. Cocompactness follows from the continuous image of J/Δ. For discreteness, if lattice cosets tend to the identity coset, choose local lifts tending to the identity and write the lattice points as those lifts times elements of C. Reduce those elements modulo the lattice $C \cap \Delta$ into a fixed compact set. The adjusted lattice points now lie in a compact subset of J, so only finitely many occur. Thus their cosets cannot accumulate at the identity. Simple connectedness follows either from exponential coordinates or the homotopy sequence with connected kernel C.

A18.3 Bounded normalization at zero

Write $\chi(n) = F(g(n)\Gamma)$, with the data in Theorem A18.1. Initially restrict to the fixed subgroup G_0 and its lattice $\Gamma_0 = G_0 \cap \Gamma$, restricting F through the smooth inclusion $G_0/\Gamma_0 \longrightarrow G/\Gamma$. This only changes a fixed metric comparison constant. In the following notation $G = G_0$ and $\Gamma = \Gamma_0$.

Choose a fixed compact set of representatives for G/Γ. Write $g(0) = q\gamma$, with q in that compact set and $\gamma \in \Gamma$. Set

$$h(n) = q^{-1}g(n)\gamma^{-1} = q^{-1}\big(g(n)g(0)^{-1}\big)q, \qquad F_q(z\Gamma) = F(qz\Gamma). \tag{171}$$

Then $h(0) = 1$ and $F_q(h(n)\Gamma) = F(g(n)\Gamma)$ exactly. Each G_a is normal in G_0, because $[G_0, G_a] \subseteq G_a$. The constant map with value $g(0)^{-1}$ is polynomial into the original filtration. Product closure of polynomial maps [132, Corollary B.4], followed by the filtered conjugation by q, shows that h is polynomial into the same multidegree filtration. No conjugation of the lattice is made.

The maps $z\Gamma \mapsto qz\Gamma$ are smooth with uniformly bounded derivatives for q in the fixed compact set. Thus F_q has a uniform Lipschitz bound. If $F(tz) = \mathrm{e}(\eta(t))F(z)$ for $t \in G_{\mathbf{1}}$, then

$$F_q(tz) = \mathrm{e}(\eta_q(t))F_q(z), \qquad \eta_q(t) = \eta(qtq^{-1}). \tag{172}$$

This is a homomorphism on $G_{\mathbf{1}}$. Its restriction to $G_{\mathbf{1}} \cap \Gamma$ is integer-valued: put $z = \Gamma$ in (172) and use both $F_q(t\Gamma) = F_q(\Gamma)$ and the fact that $F_q(\Gamma)$ is a unit vector. In fact $\eta_q = \eta$. For each fixed $t \in G_{\mathbf{1}} \cap \Gamma$, the continuous map $q \mapsto \eta(qtq^{-1})$ on connected G_0 has integer values, so it equals its value $\eta(t)$ at the identity. The lattice generates $G_{\mathbf{1}}$ as a connected real group, by the argument in Lemma A18.2. Two continuous real homomorphisms agreeing on that lattice therefore agree on $G_{\mathbf{1}}$. This proves the assertion for every $q \in G_0$. In particular the normalization preserves the same standard vertical character, including for limit choices of q.

A18.4 The descending auxiliary filtration

For $a \in \{0,1\}^s$ with $a \neq 0$, define the following homomorphisms on G_a whenever the indicated first coordinate occurs:

$$A_a(u) = (u,u,u) \quad (a_1 = 0), \qquad B_a(u) = (u,u,1), \quad C_a(u) = (u,1,u) \quad (a_1 = 1).$$

For $1 \le i \le s$ let H_i be the connected subgroup of G^3 generated by these images for all a with $|a| \ge i$. Set $H_{s+1} = \{1\}$ and

$$K = H_1, \qquad K_0 = K_1 = H_1, \qquad K_i = H_i \ (i \ge 1). \tag{173}$$

Every group just defined is fixed by the original filtration, not by g. Lemma A18.2 makes it a connected closed simply connected rational subgroup of G^3. Put $\Delta = K \cap \Gamma^3$. The sequence (K_i) is descending by its definition.

We check the commutator law. For generators of types A, A, A, B, and A, C, the coordinatewise commutator has respectively type A, B, C and belongs to the group with index $a + b$. The weight adds. If two generators both have first index one, every possibly nontrivial coordinate commutator belongs to $G_{a+b} = \{1\}$ because $(a+b)_1 = 2$. Thus the B, B, B, C, and C, C commutators are trivial. These assertions hold for their Lie algebras as well. Bracketing generated Lie algebras and using the Jacobi identity gives

$$[\mathfrak{h}_i, \mathfrak{h}_j] \subseteq \mathfrak{h}_{i+j} \quad (i,j \ge 1).$$

Here iterated brackets remain in the indicated space: each bracket is a sum of iterated brackets of the original generators, and its total original weight is at least $i + j$. BCH, which terminates for a nilpotent group, then gives $[H_i, H_j] \subseteq H_{i+j}$. The index-zero assertions also hold since $K_0 = H_1$: for example $[K_0, K_j] \subseteq H_{j+1} \subseteq H_j = K_j$. Hence (K_i) is a rational degree filtration, is taut at zero, and has degree at most s.

The top group has a particularly simple exact description:

$$K_s = H_s = \{(uv, u, v) : u, v \in G_{\mathbf{1}}\}. \tag{174}$$

Indeed $a = \mathbf{1}$ is the only index of weight s and has first coordinate one. The group $G_{\mathbf{1}}$ is abelian since $[G_{\mathbf{1}}, G_{\mathbf{1}}] \subseteq G_{2\mathbf{1}} = \{1\}$. Thus its B and C images commute and their product is exactly the displayed connected group. It is central in K, also following from the commutator law.

A18.5 The exact polynomial orbit

Choose once and for all a total order of $\{0,1\}^s$ extending its partial order. By [132, Lemma B.9], the normalized map has the Taylor expansion

$$h(n) = \prod_{a \in \{0,1\}^s \setminus \{0\}} u_a^{n^a}, \qquad u_a \in G_a, \qquad n^a = \prod_{j=1}^{s} n_j^{a_j}. \tag{175}$$

The absent constant coefficient equals $h(0) = 1$. Define

$$P(x, x', y) = (h(x + x', y), h(x, y), h(x', y)), \qquad y = (y_2, \dots, y_s).$$

This is exactly the ordered product of the following factors, in the order of (175). If $a_1 = 0$, its factor is

$$A_a(u_a)^{\prod_{j=2}^{s} y_j^{a_j}}.$$

If $a_1 = 1$, its factor is

$$B_a(u_a)^x \prod_{j=2}^{s} y_j^{a_j} C_a(u_a)^{x'} \prod_{j=2}^{s} y_j^{a_j}.$$

The latter identity is exact because the powers of the same element u_a commute. In either case the integer exponent has degree $|a|$ and the fixed group element lies in $K_{|a|}$. A map $v^{Q(n)}$ of this form is polynomial into (K_i): its j-fold derivative is $v^{\Delta^j Q(n)}$, which lies in K_j for $j \le \deg Q$, and is the identity for $j > \deg Q$. Here derivatives commute within the cyclic one-parameter subgroup generated by v, and $K_{\deg Q} \subseteq K_j$ is used in the first case. The values also lie in K_0. Product closure [132, Corollary B.4] now proves that

$$P \in \mathrm{poly}(\mathbb{Z}_{\mathbb{N}}^{s+1} \longrightarrow K_{\mathbb{N}}). \tag{176}$$

This argument proves the assertion for all integer inputs, not just a box. In particular it verifies that the orbit lies in the normalized ambient group K, which the exact-weight construction at index zero did not ensure.

A18.6 The lower-degree quotient and its test function

Let $\overline{K} = K/K_s$, let $\pi : K \longrightarrow \overline{K}$ be the quotient map, and let $\overline{\Delta} = \pi(\Delta)$. The central-quotient argument after Lemma A18.2 shows that $\overline{K}/\overline{\Delta}$ is a compact nilmanifold with connected simply connected nilpotent group. Give it the filtration $\overline{K}_i = \pi(K_i)$. It is taut and has degree at most $s-1$. For each $i \le s$, $K_s \subseteq K_i$, so

$$\overline{\Delta} \cap \pi(K_i) = \pi(\Delta \cap K_i).$$

The right side is cocompact in $\pi(K_i)$, by the compact quotient of $K_i/(\Delta \cap K_i)$; it is discrete as a subgroup of $\overline{\Delta}$. Thus the quotient filtration is rational at every index. Equation (176) remains polynomial after applying π, directly from the derivative definition.

Define on K/Δ

$$T_q(z_1, z_2, z_3) = F_q(z_1\Gamma) \otimes \overline{F_q(z_2\Gamma)} \otimes \overline{F_q(z_3\Gamma)}.$$

It is well-defined and Lipschitz, by restriction from $(G/\Gamma)^3$. For $(uv, u, v) \in K_s$, equation (172) multiplies this tensor by

$$\mathrm{e}\big(\eta_q(uv) - \eta_q(u) - \eta_q(v)\big) = 1.$$

Therefore T_q is invariant under K_s and descends to a function $\overline{T}_q$ on $\overline{K}/\overline{\Delta}$.

For explicit control of its regularity, fix a smooth metric on K/Δ and average it over the compact central torus $K_s/(K_s \cap \Delta)$. The averaged metric and the original fixed metric are uniformly equivalent. The quotient metric makes the quotient a Riemannian submersion. A function constant on its fibers and Lipschitz upstairs is Lipschitz downstairs with the same bound for these metrics, by horizontal path lifting. All the metric comparisons are fixed by the original filtered nilmanifold. The bound for T_q is at most a fixed multiple of the bound for F_q, since its three factors have norm one. The earlier compact normalization therefore makes the Lipschitz bound for $\overline{T}_q$ uniform in g.

Finally the expression (170) is exactly

$$\overline{T}_q\big(\pi(P(x, x', y))\overline{\Delta}\big).$$

This proves the claimed degree at most $s-1$ representation.

For clarity, the individual factors also have valid degree-s representations. In one copy of G, take the connected group J_i generated by G_a for $|a| \ge i \ge 1$, and set $J_0 = J_1$. The same rationality and commutator arguments give a taut rational degree-s filtration on J_1, with $J_s = G_{\mathbf{1}}$. The

Taylor factors of h and of each of its linear pullbacks are polynomial for this filtration by the same integer-polynomial argument. Restricting F_q gives a unit-vector-valued Lipschitz test function with vertical character η_q. Hence those factors are nilcharacters of degree at most s.

Substitute the homomorphism $n \mapsto (L_1(n), L_1'(n), L_2(n), \dots, L_s(n))$ into the lower-degree representation. Derivatives, or the explicit polynomial factors above, show that degree does not increase. The definition of symbol equivalence proves Theorem A18.1. Relabeling coordinates gives the other coordinate identities.

A18.7 Limit data and the actual published consumers

For a limit nilcharacter the filtered nilmanifold and its metric data are standard fixed data, and the polynomial and Lipschitz test functions are limit sequences with bounded Lipschitz norm. Choose the bounded representatives q separately in each member of those sequences; this gives a limit map with values in the same compact set. All groups K_i, the lattice, the quotient and the metric constants constructed above are unchanged along the sequence. The same standard vertical character is retained by the preceding argument. Representatives for the test functions may be taken to satisfy their defining unit-norm and vertical-character identities on an ultrafilter-large set of indices: otherwise, choosing a violating point, and for the second identity a violating group element, at every bad index would contradict the exact identity of the limit function at the resulting limit points. This uses the all-limit quantifiers just specified and the evaluation rule in [132, Definition A.3]; it would be invalid with only standard-point quantifiers. Discard those bad indices. The formulas now hold at every remaining member, and the test-function bounds are uniform. Their limit is therefore a limit polynomial nilsequence of degree at most $s-1$ in precisely the required sense. No compactness argument selects a nilmanifold depending on an unbounded polynomial coefficient.

The first published use, immediately after Lemma 13.2 on p. 1318, expands $[\widetilde{\chi}(n+h, \dots, n+h)]$ by applying the identity in each coordinate. Before any use of symmetry the exact expansion is

$$[\widetilde{\chi}(n+h, \dots, n+h)] = \sum_{\omega \in \{0,1\}^s} [\widetilde{\chi}(z_{\omega_1}, \dots, z_{\omega_s})], \qquad z_0 = n,\ z_1 = h.$$

Every map used here is a fixed integer homomorphism from $\mathbb{Z}^2$. Thus the coefficient counting and subsequent symmetry step in that argument receive exactly the same multilinearity statement; no stronger inverse theorem is asserted by this local repair.

The further explicit appeals in Lemmas 13.3, 13.4 and 13.5 on pp. 1319–1322 likewise use addition, subtraction and repeated copies of their variables. These are integer homomorphisms from the displayed finite Cartesian powers of $\mathbb{Z}$, so the theorem applies without a range or normalization change. Constant shifts in those arguments still require the separate symbol-shift assertion in E.8; this proof does not discharge it. The repaired lemma supplies the multilinearity input to Theorem 7.4, then to the published inverse-theorem argument. Other corrections, including the false original Proposition 8.3 and its replacement, remain separate obligations. Source and current-manuscript consumer spans are recorded in the accompanying contracts.

A19 Difference decomposition and the additive-quadruple estimate

A19.1 Inputs and conventions

We use the natural-ultrapower convention of GTZ Appendix A and the shift theorem in Section A17. A bound $\gg 1$ is positive and standard. Write N for an unbounded limit integer, $J = [N]$ and $I = [-N, N] \cap {}^*\mathbb{Z}$. All changes of variables are integer identities at the representative indices. The author erratum guides the proof. Its Proposition 3.1 and modified Proposition 7.3 are the conclusions here, rather than assumed premises.

The shift theorem gives a lower-support sequence with fixed geometry and uniform Lipschitz bound. It gives neither a separated finite decomposition nor a correlation conclusion. Products, conjugates, and translates use product manifolds, conjugated tests, and translated polynomial orbits, respectively. A bounded internal scalar depending only on parameters is degree zero in the averaging variable.

A19.2 Difference decomposition

Proposition A19.1 (LP02)**.** *For standard $s \geq 3$, let χ be a multidegree $(1, s-1)$ nilcharacter. For every standard $\epsilon > 0$ each component of $D_k(h, n) = \chi(h+k, n) \otimes \overline{\chi(h, n)}$ is approximated uniformly in k, h, n to error ϵ by*

$$\sum_{j=1}^{L} \Psi_{j,k,a_j}(n) B_{j,k}(h, n).$$

Here L is standard, Ψ_{j,k,a_j} is a component of a unit-vector degree-$(s-1)$ nilcharacter, and $B_{j,k}$ is supported on the downset below $(1, s-2)$. The families are internal with standard finite geometry lists and uniform bounds.

Proof. Apply the project shift theorem, Theorem 3, with positive index $(1, 0)$ and shift $(k, 0)$. The support is in the union of the downsets $J_0 = \downarrow (0, s-1)$ and $J_1 = \downarrow (1, s-2)$. We now give the fixed geometric split explicitly. Write G/Γ for the taut lower-support representation from the shift theorem and $J = J_0 \cup J_1$. For each positive $a \in J$ take a rational basis of $\mathfrak{g}_a$ for the rational structure of Γ. This exists because G_a is rational. Take a free rational Lie algebra with one generator of weight a for each basis vector. Kill all homogeneous words whose total weight is outside J. The excluded weights form an upward-closed set, so these words generate a homogeneous ideal. All generator weights are positive. Thus word length is bounded by $\max_{a \in J} |a| = s - 1$ in the quotient, which is finite-dimensional and nilpotent. Let $\mathfrak{u} = \bigoplus_{a \in J \setminus \{0\}} \mathfrak{u}[a]$ denote it and let U be its simply connected real group. Set $\mathfrak{u}_b = \bigoplus_{a \geq b} \mathfrak{u}[a]$. This is a descending rational multidegree filtration with the required bracket rule.

Sending generators to their basis vectors defines a rational homomorphism $\rho : U \to G$. The killed words map to zero by the filtration rule in G. It maps each positive layer onto G_b: the generators tagged b span $\mathfrak{g}_b$, and all weights at least b map into that layer. At zero it is onto because G is taut and generated by its coordinate groups $G_{(1,0)}$ and $G_{(0,1)}$. Thus the polynomial lifts exist by published GTZ Corollary B.10, applied to this layerwise-surjective filtered map, and can be selected internally at the representative indices.

Choose a homogeneous rational basis of $\mathfrak{u}$ and let L_0 be its integer span. There is a positive integer D such that $\Lambda = \exp(DL_0)$ is a lattice and $\rho(\Lambda) \subset \Gamma$. Here are the scaling details. In this basis BCH is a finite rational polynomial with linear part addition. For two inputs in DL_0, terms of degree at least two lie in DL_0 once D clears all the coefficient denominators. Inversion negates the

logarithm. Thus this set is a subgroup, discrete in exponential coordinates. It is cocompact: order the basis by increasing total weight and successively reduce each weight's logarithm coordinates modulo D. Right multiplication by an element of the corresponding homogeneous lattice leaves lower weights unchanged, and at the current weight is ordinary addition. The resulting bounded box of logarithms is a compact set of representatives. The same argument applies to every filtration layer and every homogeneous quotient. To impose $\rho(\Lambda) \subset \Gamma$, use rational Mal'cev coordinates of the second kind adapted to Γ, as in GTZ Section 3. The coordinates of $\rho(\exp x)$ are finite rational polynomials in the chosen coordinates of x, with zero constant term. Taking D divisible by their denominators makes the coordinates of $\rho(\exp(Dx))$ integral for $x \in L_0$. This gives the required lattice inclusion. Increasing D preserves the BCH and compactness arguments.

For $i = 0, 1$, quotient U by the connected subgroup with Lie algebra the sum of homogeneous weights outside J_i. These sums are ideals by the downset property. The image lattices are the exponential lattices in the surviving homogeneous bases. The pair of quotient maps separates points of U/Λ: if two points have the same image in both, the logarithm of their group difference lies in the projected lattice in each quotient. Each homogeneous weight survives in at least one quotient. Hence every coordinate of that logarithm belongs to $D\mathbb{Z}$, so the difference lies in Λ. This proves separation on the quotient spaces, rather than just intersection-triviality of the two ideals.

The pullbacks of Lipschitz tests from the two compact quotients generate a conjugation-closed algebra containing constants and separating points. Stone–Weierstrass gives a uniform approximation by finite sums of products of these tests. The lifted tests of the shift family have a common Lipschitz and sup bound on this fixed compact manifold, since the map to G/Γ is fixed and smooth. A standard finite $\epsilon/4$ net for this bounded equicontinuous class gives finitely many standard product lists, each chosen to tolerance $\epsilon/4$. Select the first net entry within $\epsilon/4$ using an internal sup-norm comparison at representative indices. Use a smaller net radius if needed. Thus the splitting error is at most $\epsilon/2$, uniformly in k, h, n. Pad the lists with zero factors to a standard common length.

In the J_0 quotient all surviving generators have zero first weight and strictly positive second weight. The restricted one-variable degree filtration is therefore taut: its degree-zero and degree-one groups coincide. The highest layer is central by the bracket law and the degree bound. The orbit is independent of h, since the first-variable derivative lies in the trivial first-coordinate group. Restrict to $h = 0$ and apply published Lemma E.5 with this taut one-variable geometry of degree $s-1$. Use the ordinary functions constructed in its proof on printed p. 1353. For each of the finitely many standard J_0 tests, fix ordinary Lipschitz unit-vector tests with global vertical frequency and ordinary top-invariant coefficient functions. Extend these functions to the ultrapower after making these finite choices. The resulting tests have exact unit norm and equivariance at all limit points and all limit top-group elements, and internal selection from the finite lists preserves both identities. Its test decomposition gives linear functionals of unit-vector nilcharacters times degree-at-most-$(s-2)$ tests. Expand the functionals in coordinates and absorb their coefficients and lower-degree factors into the J_1 factors. The product still has support J_1. Choose individual Fourier tolerances after the finite lists and their sup bounds so that the added error is at most $\epsilon/2$. On fixing h, k, the lower factor has degree at most $s-2$ in n by the $(0,1)$ derivative index. All geometry lists are fixed and finite, and all orbit data and choices are internal. □

A19.3 The two averages

Proposition A19.2 (LP03)**.** *Under the hypotheses of GTZ Theorem 7.2, for standard $s \geq 3$, there are $\gg N^3$ additive quadruples in H for which*

$$\chi_{h_1}(n) \otimes \chi_{h_2}(n + h_1 - h_4) \otimes \overline{\chi_{h_3}(n)} \otimes \overline{\chi_{h_4}(n + h_1 - h_4)}$$

is $(s-2)$-biased on J. The witnessing tests have standard bounded complexity and positive standard correlation.

Proof. Extend f by zero outside J. GTZ A.11 and A.12 select the lower-degree correlation tests internally on the dense internal set $H \subset I$. The relation includes a selected positive lower bound $1/j$ with standard integer j. Select global representatives of the lower-degree tests and of the restricted nilcharacters by the same internal selection, storing the representations as part of the witness. Only their original values on J enter the unsmoothed identities. The bounded-complexity descriptions are a countable union of internal sets. The boundedness principle A.5 gives uniform standard complexity and a common positive lower bound. Expand the bounded-dimensional tensor contraction in coordinates. Retain a positive-density subset of H with a fixed witnessing coordinate pair by the standard finite pigeonhole. Write U_h, V_h for the conjugates of those coordinates of $\chi(h,\cdot)$ and χ_h. Then there are scalar lower-degree tests c_h such that

$$\left|\mathbb{E}_{n\in J} f(n+h)\overline{f(n)}U_h(n)V_h(n)c_h(n)\right| \geq \eta > 0$$

on the retained set, with standard uniform bounds. Align the phase by multiplying c_h by a unit scalar. Set $c_h = 0$ outside that set and extend V_h boundedly there. This is an internal choice. The real average over $h \in I$ is $\gg 1$.

Apply Cauchy–Schwarz in n to eliminate $\overline{f(n)}$. It gives

$$\mathbb{E}_{n\in J}\,|\mathbb{E}_{h\in I} f(n+h)U_h(n)V_h(n)c_h(n)|^2 \gg 1.$$

Expand the square and put $h' = h+k$, with $K = [-2N, 2N] \cap {}^*\mathbb{Z}$. The exact new parameter cutoff is $1_{h+k\in I}$, degree zero in n. On $I \times K$ the new normalized average is $|I|/|K|$ times the old one. With $c_{h,k} = 1_{h+k\in I} c_h \overline{c_{h+k}}$ this gives

$$\left|\mathbb{E}_{h\in I, k\in K, n\in J} f(n+h)\overline{f(n+h+k)}U_h(n)\overline{U_{h+k}(n)}V_h(n)\overline{V_{h+k}(n)}c_{h,k}(n)\right| \gg 1.$$

The U product is a component of D_k. Take LP02 with error less than half the correlation divided by a common bound for the other factors. A term of the standard finite sum retains a positive standard correlation. Absorb its lower factor into $c_{h,k}$.

For its surviving coordinate $\Psi_{k,a}(n)$ use the identity

$$\Psi_{k,a}(n) = \sum_b [\Psi_{k,a}(n)\overline{\Psi_{k,b}(n+h)}]\Psi_{k,b}(n+h).$$

The identity uses the unit vector norm of $\Psi_k(n+h)$. Individual coordinates need not have modulus one. The bracket is a component of the conjugate transpose of the shift tensor. The one-variable degree case of the project shift theorem gives degree at most $s-2$, uniformly and internally in h, k, using the finite geometries of LP02. Select a surviving term by the same finite averaging argument. Absorb the bracket into $d_{h,k}$ and write the remaining coordinate as $P_k(n+h)$. The lower test is still exactly zero unless both $h, h+k$ are retained shifts.

Set $m = n+h$ and $M = [-2N, 2N] \cap {}^*\mathbb{Z}$. The change to the full M average, with the exact cutoff, multiplies the normalized correlation by $N/|M|$. We have

$$\left|\mathbb{E}_{m\in M, k\in K} f(m)\overline{f(m+k)}P_k(m)\mathbb{E}_{h\in I} 1_J(m-h)V_h(m-h)\overline{V_{h+k}(m-h)}d_{h,k}(m-h)\right| \gg 1.$$

The outer factor depends only on (m,k) and is bounded. The second Cauchy–Schwarz in (m,k) gives $\gg 1$ for the mean squared modulus of the inner average. Expand with independent $h, h' \in I$. Put

$$(h_1, h_2, h_3, h_4) = (h, h'+k, h+k, h'), \qquad n = m-h.$$

The shift substitution is injective, with inverse $(h, h', k) = (h_1, h_4, h_3 - h_1)$. It has $h_1 + h_2 = h_3 + h_4$. The first cutoff restricts n to J, where $m = n + h \in M$ automatically. The second is $1_J(n + h_1 - h_4)$. The four factors in the new average are

$$V_{h_1}(n)V_{h_2}(n + h_1 - h_4)\overline{V_{h_3}(n)V_{h_4}(n + h_1 - h_4)}.$$

The lower factor is $d_{h,k}(n)\overline{d_{h',k}(n + h_1 - h_4)}$ times the remaining cutoff. The ratio $N/|M|$ is comparable to one. The parameter vanishing is exact unless all four shifts are retained. This proof keeps zero extension throughout.

To absorb the cutoff, choose fixed $0 < \lambda < 1/4$. On the circle $\mathbb{R}/16\mathbb{Z}$ take a piecewise linear test equal to one on $[\lambda, 1 - \lambda]$, zero off $[-\lambda, 1 + \lambda]$ in $[-8, 8)$, between zero and one, with Lipschitz bound $O(1/\lambda)$. Evaluate it at $(n + h_1 - h_4)/N$. This is a one-step nilsequence in n, with fixed geometry and internal orbit. Its argument lies in $[-2, 3]$ and does not wrap. The normalized mean absolute error against the cutoff is $O(\lambda + 1/N)$ uniformly in all shifts: only two endpoint bands of total $O(\lambda N + 1)$ integers differ. Multiplying by bounded factors changes only the constant. Choose λ after the positive correlation bound so the error loses at most half. Since $s \geq 3$, the one-step test is absorbed into degree at most $s - 2$. Parameter vanishing persists.

If the outer correlation has modulus at least $\beta > 0$ and each inner average has modulus at most B, at least a fraction $\beta/(2B)$ of the inner averages have modulus at least $\beta/2$. The shift triple box has size comparable to N^3. By injectivity, these yield $\gg N^3$ additive quadruples in H. Conjugating the scalar correlations undoes the initial choice of V_h and preserves lower degree. The resulting scalar correlations are coordinates of the claimed tensor, giving bias on the original interval J. □

A19.4 Scope

The published proof inputs used here are GTZ Appendix A, Corollary B.10, E.5, the correlation conventions, and the stated hypotheses of Theorem 7.2. The project shift input and these exact applications await separate nonauthor review. This unit supplies the additive-quadruple result, not the subsequent frequency analysis, the other affected symbol rules, or the final GI theorem.

A20 One-variable factorization and approximate additive linearization

This section proves the one-variable factorization connections, using the final published sources [GTZ], [GT12], [GT14], and [GT08] identified in its references. Only one-variable factorization is asserted here.

A20.1 Objects, source differences, and the required outputs

All nilpotent Lie groups below are standard, connected and simply connected; all lattices and filtrations are standard and rational. Subgroups described as rational are connected, and their intersection with the ambient lattice is cocompact in that subgroup. We use the natural ultrapowers of these objects, with the internal-function convention of [GTZ, Appendix A]. A bounded quantity is bounded by some standard constant. In the finite combinatorial proof all constants depend only on the stated positive density and error constants. Passing that finite proof to an ultrapower therefore preserves uniformity and internal choices. Braces denote the representative in $(-1/2, 1/2]$.

An $\mathbb{N}$-filtration $G_\bullet$ may have $G_1 \neq G_0 = G$. Its positive groups satisfy $[G_i, G_j] \subseteq G_{i+j}$ and are rational. The polynomial convention is the ordinary degree filtration on the source. In fixed

adapted Mal'cev coordinates, write the coordinates of a polynomial map as a finite polynomial in n or $\binom{n}{j}$. Coordinate degree is bounded in terms of the fixed filtered group. For a smooth factor, "small Taylor coefficients" means that its constant coefficient is bounded and its coefficient of index $j \geq 1$ is $O(N^{-j})$, in the group Taylor convention of [GTZ, Lemma B.9]. Equivalent coordinate bounds are justified below.

The statement of [GTZ, D.2] cannot be used as a literal equivalence: on $\mathbb{R}/\mathbb{Z}$, $g(n) = n/N$ is equidistributed on $[N]$ but is not equidistributed on $[\lfloor N/2 \rfloor]$. For example its first Fourier average on that half interval tends to $2i/\pi$, whereas its full-interval average tends to zero. Its nontrivial horizontal character is $1/N$-Lipschitz. We instead use the one-variable weak factorization [GT12, Proposition 9.2], whose proof uses the quantitative theorem on an actual dense progression. The published correction [GT14, pp.1175–1176] explicitly retains the one-variable results. Its minor corrections on pp.1182–1183 are part of this source choice.

The properness requirement printed in the second bullet of [GTZ, D.4] is removed. Only a weak-factorization step that starts with failed total equidistribution forces a proper subgroup. Also, a rational periodic factor is not asserted to take values in Γ.

Theorem A20.1 (One-variable factorization, DF-D4)**.** *Let N be unbounded and let $g \in {}^*\mathrm{poly}(\mathbb{Z}, G_\bullet)$. There are a standard connected rational subgroup $P \leq G$ and internal polynomial maps $\epsilon, p, \gamma \in {}^*\mathrm{poly}(\mathbb{Z}, G_\bullet)$ such that*

$$g(n) = \epsilon(n)p(n)\gamma(n) \qquad (n \in {}^*\mathbb{Z}).$$

The map p takes values in ${}^*P$, with its induced filtration. On $[N]$, ϵ is bounded, is $O(1/N)$-Lipschitz, and has the small Taylor coefficient bounds above. There is a standard discrete subgroup $\Lambda \geq \Gamma$ with $\gamma(n) \in {}^*\Lambda$ for every n, and a standard integer $q > 0$ with $\gamma(n+q)\Gamma = \gamma(n)\Gamma$ for every n. For every standard finite-index subgroup $\Delta \leq P \cap \Gamma$, $p(n)\Delta$ is totally equidistributed in P/Δ on $[N]$. The subgroup P is allowed to equal G or to be trivial.

Theorem A20.2 (Approximate four-function linearization, DF-F1)**.** *Let N be unbounded, let $j \geq 1$ be standard, and put $e = N^{-j}$. Let $H \subseteq [-N, N] \cap {}^*\mathbb{Z}$ be internal, and let $\xi_i : H \to {}^*\mathbb{T}$, $1 \leq i \leq 4$, be internal functions. Suppose that for at least cN^3 ordered quadruples in H^4 satisfying $h_1 + h_2 = h_3 + h_4$ one has*

$$\|\xi_1(h_1) + \xi_2(h_2) + \xi_3(h_3) + \xi_4(h_4) - a\|_{\mathbb{T}} \leq Ce,$$

where $c > 0$ and $C < \infty$ are standard and $a \in {}^*\mathbb{T}$. For any specified index i there are standard $K < \infty$, $c' > 0$, an internal $H' \subseteq H$ with $|H'| \geq c'N$, internal $\alpha_k \in {}^*\mathbb{T}$, $\beta_k \in {}^*\mathbb{R}$, and $\delta \in {}^*\mathbb{T}$ such that

$$\xi_i(h) = \delta + \sum_{k=1}^{K} \{\alpha_k h\}\beta_k + O(e) \quad (h \in H'). \tag{F}$$

The implicit constant is standard and uniform over H'. The bracket sum restricted to H' is an exact Freiman homomorphism: it preserves every relation $h_1 + h_2 = h_3 + h_4$ with all four points in H'. No initial Freiman property, boundedness of the real β_k, or regularity of the input functions is assumed. All choices may be made internally.

A20.2 Weak factorization and coefficient bounds

Lemma A20.3 (Weak step, DF-D3)**.** *Suppose first that $G_1 = G$. If $g(n)\Gamma$ is not totally equidistributed on $[N]$, it admits the factorization of Theorem A20.1 with p taking values in a standard proper connected rational subgroup, without the final equidistribution requirement. All other bounds and the discrete-overgroup and period assertions hold.*

Proof. Failed total equidistribution supplies an internal dense progression and a standard Lipschitz test with noninfinitesimal discrepancy. Thus for some standard $0 < \delta < 1/2$ it fails total δ-equidistribution in the quantitative convention of [GT12]. Reduce δ further to include the rationality constant of the fixed Mal'cev basis. Apply the one-variable case of [GT12, Proposition 9.2, pp.522–523] with this δ, the actual degree of $G_\bullet$, and the actual dimension of G. Its bounds are standard. The same argument works for an internal Lipschitz test with a standard norm bound: a finite uniform net in the standard Lipschitz ball first supplies a standard test with positive discrepancy. This uses compactness of G/Γ.

Here are the stronger bounds extracted from that source's construction, including the distinctions needed for later use. Write $g(0) = c\lambda$ with c in a fixed bounded fundamental set and $\lambda \in {}^*\Gamma$. Then $\widetilde{g} = c^{-1}g\lambda^{-1}$ has value 1 at zero. Left translation by bounded c changes a Lipschitz norm by a standard factor, and right multiplication by λ^{-1} changes no point of G/Γ. Thus the same failed-total-equidistribution argument applies to $\widetilde{g}$, with a possibly smaller standard δ.

The source's horizontal-character step gives a nonzero standard integer vector k with

$$\psi(\widetilde{g}(n)) = \sum_{r\geq 1} t_r \binom{n}{r}, \qquad \|k \cdot t_r\|_{\mathbb{T}} = O(N^{-r}). \tag{D1}$$

Here $k \cdot \psi$ is the real lift of the horizontal homomorphism; in particular it kills all commutators. Standardness of k follows because the quantitative bound leaves finitely many integer vectors. The source obtains (D1) by applying its Theorem 2.9 to the witnessing progression and undoing that progression by its Lemma 8.4. This use does not assert the false single-scale equivalence described above.

Choose integers a_r nearest to $k \cdot t_r$. The coordinate support condition in [GT12, Lemma 6.7] says that t_r lies in the coordinate subspace for G_r. If k vanishes on that subspace, take $a_r = 0$ and $v_r = w_r = 0$. Otherwise fix a rational vector u_r in that subspace with $k \cdot u_r = 1$, and put

$$v_r = a_r u_r, \qquad w_r = (k \cdot t_r - a_r)u_r.$$

Define ϵ_0 and γ_0 by Mal'cev coordinates

$$\psi(\epsilon_0(n)) = \sum_r w_r \binom{n}{r}, \qquad \psi(\gamma_0(n)) = \sum_r v_r \binom{n}{r}, \qquad p = \epsilon_0^{-1}\widetilde{g}\gamma_0^{-1}. \tag{D2}$$

The same coordinate-support lemma and closure of polynomial maps under products show that these are polynomial for the original filtration. The real horizontal lift vanishes identically on p, since its coefficient at every index is $k\cdot t_r - (k\cdot t_r - a_r) - a_r = 0$. Consequently p lies in the connected kernel of that real homomorphism. This is a standard proper rational subgroup. A kernel of the circle-valued homomorphism alone would not establish connectedness; (D2) supplies the real zero required here.

Each w_r is $O(N^{-r})$. The vectors v_r have a common standard denominator, but their numerators need not be bounded. This denominator statement is all that is required. By [GT12, A.11–A.12], γ_0 is uniformly rational and periodic modulo Γ with a standard period. The more explicit common discrete overgroup argument below also applies. Set $\epsilon = c\epsilon_0$ and $\gamma = \gamma_0\lambda$. These are the correct ordered factors of the original g. Multiplication by bounded c preserves the stated small-coordinate bounds, and the added λ lies in the original lattice. The next lemma verifies all closure and Taylor assertions used here and in the descent. □

Lemma A20.4 (Closure of the controlled factors)**.** *In a fixed rational nilpotent group, finite products of bounded polynomial maps with coefficient bounds $O(N^{-r})$ retain those bounds and are*

$O(1/N)$-Lipschitz on $[N]$. These bounds imply the group Taylor bounds in Theorem A20.1. A finite collection of rational lattices commensurable with Γ, together with sets of points whose rational coordinates have a fixed common denominator, is contained in a standard discrete subgroup $\Lambda \geq \Gamma$. Any fixed-degree polynomial map into this group is periodic modulo Γ with a standard period.

Proof. Coordinate products, inverses, and changes between first-kind and Mal'cev coordinates are fixed polynomials by [GT12, A.2–A.3]. In a coordinate polynomial $\sum a_r n^r$, the bounds $a_r = O(N^{-r})$ are equivalent to bounded coefficients after substituting $n = Nt$. Products of such polynomials retain that property. Converting back to the binomial basis gives at index r a fixed linear combination of coefficients of monomial indices $\ell \geq r$, hence $O(N^{-r})$. Constant coefficients stay bounded. On the fixed compact coordinate range for $0 \leq t \leq 1$, the group-coordinate maps have bounded derivatives; differentiation in n contributes N^{-1}. This proves the Lipschitz assertion, with any fixed standard metric.

For clarity about the group Taylor convention, one may assign formal weight r to each coefficient of n^r in the logarithm of a coordinate polynomial. The recursive Taylor extraction by $g(0)$, then $g_1^{\binom{n}{1}}$, then $g_2^{\binom{n}{2}}$, and so on uses only fixed BCH polynomials. Its coefficient $\log g_r$ contains no monomial of formal weight less than r. Indeed after removing indices below r, each surviving polynomial term of formal weight w has ordinary degree at most w. The residual group map is the identity at $0, \ldots, r-1$, as a formal identity in its input coefficients. Its logarithmic terms of weight $w < r$ are therefore polynomials of degree less than r with r zeros, and vanish. BCH products add weights, so this assertion is preserved at each extraction step. The remaining finitely many monomials have weight at least r, and are $O(N^{-r})$. This proves the group Taylor assertion, including under products and fixed rational embeddings of subgroups.

For the discrete-overgroup assertion, use a rational basis adapted to the lower central series and let its weights be $w_i \geq 1$. All points of each given lattice have logarithmic coordinates with bounded denominators, as do the specified coordinate sets, by the fixed polynomial coordinate changes. Choose a standard integer D divisible by these denominators and by the denominators of all coefficients in the finite BCH formula in this basis. Increase D for the finitely many basis conversions if necessary. The set

$$L = \sum_i D^{-(2w_i-1)} \mathbb{Z} X_i$$

is closed under BCH. For a nonlinear term with $t \geq 2$ inputs of weights $w_{i_1}, \ldots, w_{i_t}$ contributing to output weight w_k, one has $w_k \geq \sum w_{i_a}$. Therefore the difference of denominator exponents is

$$(2w_k - 1) - \sum_a (2w_{i_a} - 1) \geq t - 1 \geq 1.$$

The extra power of D clears that BCH coefficient. Linear terms and negation also preserve L. Thus $\exp L$ is a standard discrete subgroup containing all the required points and Γ. It is cocompact, and $[\exp L : \Gamma]$ is finite. This argument also proves commensurability claims needed when a lattice of a rational subgroup is enlarged inside G.

For the period in the required right-coset convention, write the Mal'cev coordinate vector of $\gamma(n)$ as a polynomial of fixed degree D_0. Its values have a common standard coordinate denominator, since $\gamma(n) \in {}^*\Lambda$. Interpolating at $0, \ldots, D_0$ shows that its coefficients have a common standard denominator too. Their numerators may be arbitrary internal integers. By the rational polynomial formulas for multiplication and inversion, the vector

$$R(n, r) = \psi(\gamma(n)^{-1} \gamma(n + r))$$

is a polynomial in the two integer variables n, r with a common standard coefficient denominator A. The identity $R(n,0) = 0$ is a polynomial identity, so every monomial has a positive power of r. For the standard integer $q = A$, every coordinate of $R(n,q)$ is an integer for every internal integer n. Adapted Mal'cev coordinates identify Γ with the integer-coordinate points. Hence $\gamma(n)^{-1}\gamma(n+q) \in {}^*\Gamma$, exactly the assertion $\gamma(n+q)\Gamma = \gamma(n)\Gamma$. This proves the conclusion of [GT12, A.12] directly; its printed left quotient $\gamma(n+q)\gamma(n)^{-1}$ is not the quotient used here. □

A20.3 Minimal subgroup descent, with all finite covers retained

Proof of Theorem A20.1. First address the filtration convention. If G_1 is proper, write $g(0) = c\lambda$ as above and set $p = c^{-1}g\lambda^{-1}$. Then $p(0) = 1$ and

$$p(n) = \lambda(g(0)^{-1}g(n))\lambda^{-1} \in G_1.$$

The groups G_r are normal in G_0, so conjugation preserves them. The induced filtration on G_1 has zeroth and first group both G_1, and the positive groups remain G_r. Hence the one-variable result with the convention $G_0 = G_1$ applies there. This also shows directly that a controlled representation exists: $\epsilon = c$, $\gamma = \lambda$, and the represented subgroup is G_1. If $G_1 = G$, the identity-factor representation is already enough. Trivial groups cause no exception.

Call a subgroup P a representing subgroup if a factorization with all conclusions of Theorem A20.1 except total equidistribution exists. Choose one with least dimension among the nonempty collection of standard rational representing subgroups. This uses a least element of the finite set $\{0, \dots, \dim G\}$, not an infinite iteration or a uniform choice over all finite covers.

Suppose that for a standard finite-index $\Delta \le P \cap \Gamma$ the orbit $p\Delta$ is not totally equidistributed. The induced filtration of P is rational for Δ, since passing to a finite-index lattice does not change its rational Lie algebra. If $P \cap G_1$ is proper in P, the normalization just given already represents p in that proper subgroup, with a bounded constant on the left and a Δ-element on the right. Otherwise apply Lemma A20.3 in P/Δ to obtain

$$p = \epsilon_2 p_2 \gamma_2$$

with p_2 in a standard proper rational $P_2 < P$. Rationality relative to Δ is rationality relative to $P \cap \Gamma$, so P_2 is rational in G. The actual new factorization is

$$g = (\epsilon\epsilon_2)p_2(\gamma_2\gamma). \tag{D3}$$

The order of both products is necessary in a noncommutative group. Lemma A20.4 proves the small-coefficient assertions for the left product, in the ambient coordinates. For the right product, the rational lattice coordinates of γ_2 relative to Δ have common standard denominators in the original rational structure. The same lemma supplies a single discrete overgroup containing its values, the old overgroup, and Γ. It also supplies the standard period modulo Γ. Thus (D3) is a representation by the lower-dimensional subgroup P_2, contrary to minimality.

No choice of Δ was imposed in that minimality argument. Consequently the chosen p is totally equidistributed for every standard finite-index $\Delta \le P \cap \Gamma$, with the same factors. This is the required quantifier order. The argument only produces standard constants for this particular internal orbit and fixed standard geometry. It does not assert a prescribed effective bound uniform over all internal orbits. □

A20.4 A finite combinatorial proof of the linearization contract

We prove a finite assertion with constants depending only on c, C and then transfer. Take $N \geq 1$, $0 < e \leq 1$, and suppose the hypotheses of Theorem A20.2 hold in finite sets. Put $M = \lceil 1/e \rceil$ and $Q = 100N+1$. Round each $\xi_i(h)$ in the circle to $y_i(h) \in M^{-1}\mathbb{Z}/\mathbb{Z}$, with error at most $1/(2M) \leq e/2$. Since $Me \leq 2$, the approximate sum can take only a bounded number, depending on C, of values in this grid. For at least $c_1 N^3$ of the original quadruples its rounded value is a fixed a_0 in that grid. Here $c_1 > 0$ depends only on c, C.

Embed $[-N, N]$ in $B = \mathbb{Z}/Q\mathbb{Z}$ and identify the target grid with $A = \mathbb{Z}/M\mathbb{Z}$. In $B \times A$ form the four sets

$$A_1 = \{(h, y_1(h))\}, \quad A_2 = \{(h, y_2(h))\}, \quad A_3 = \{(-h, y_3(h))\}, \quad A_4 = \{(-h, y_4(h))\}. \tag{F1}$$

The signed first coordinates are essential. The selected quadruples now give $c_1 N^3$ representations of $(0, a_0)$ in $A_1 + A_2 + A_3 + A_4$. The initial additive relation has no wraparound, since $4N < Q$. Fourier inversion and Holder's inequality in this finite abelian group give

$$r_{A_1+A_2+A_3+A_4}(0, a_0) \leq \prod_{i=1}^{4} E(A_i)^{1/4}, \qquad E(A_i) = |\{x_1 + x_2 = x_3 + x_4 : x_a \in A_i\}|.$$

For completeness, with unnormalized Fourier transforms the inverse formula has factor $|B \times A|^{-1}$; Holder bounds the sum of four transforms by the fourth moments, each of which equals $|B \times A| E(A_i)$. Those factors cancel. Each $E(A_i) \leq |H|^3 \leq 27N^3$. It follows that $E(A_1) \geq c_2 N^3$ for a positive $c_2 = c_2(c, C)$. The same reasoning can isolate any A_i; reflecting its first coordinate back if necessary gives the graph of the specified y_i. We henceforth call that function y and its graph A_1.

Apply [GT08, Theorems 5.2 and 5.3, p.95] to A_1 in the actual finite group $B \times A$. Their hypotheses follow from the energy bound and $|A_1| \leq 3N$. They give a subset Γ' of its graph and a constant $K_0 = K_0(c, C)$ such that

$$|\Gamma'| \geq N/K_0, \qquad |9\Gamma' - 8\Gamma'| \leq K_0 N. \tag{F2}$$

We use these published theorems themselves, not the special correlation premise in [GT08, Proposition 5.4].

Graph refinement with the actual cyclic target

Let $V = \{a \in A : (0, a) \in 8\Gamma' - 8\Gamma'\}$. The map $\Gamma' \times V \to 9\Gamma' - 8\Gamma'$ given by addition is injective: equality determines the first graph coordinate, then its unique second coordinate, and finally a. Hence $|V| \leq K_0^2$. There is a set S_A of at most $O(1 + \log K_0)$ characters of A for which

$$V \cap \{a : \|\chi(a)\|_{\mathbb{T}} < 1/4 \ \forall \chi \in S_A\} = \{0\}.$$

Indeed, for any nonzero a a uniformly random character takes values uniformly among the roots of unity of its order; at least one third of those values have distance at least $1/4$ from zero. Choose enough independent characters that the union bound over $V \setminus \{0\}$ is less than one. This proves the required separation without identifying A with the dual of B, an identification that is unavailable here.

Partition each coordinate of the character vector on A into intervals of length at most $1/100$. A cell contains a subset $\Gamma'' \subseteq \Gamma'$ of size $c_3 N$. In any element of $8\Gamma'' - 8\Gamma''$, every selected character of the second coordinate has a real representative of absolute value less than $16/100 < 1/4$ when its

first coordinate is zero. It follows that the only element of this difference set with first coordinate zero is $(0,0)$. Therefore $4\Gamma'' - 4\Gamma''$ is a graph. Write H'' for the projection of Γ'', and let

$$\zeta : 2H'' - 2H'' \longrightarrow A$$

be the function whose graph is $2\Gamma'' - 2\Gamma''$. It is an exact Freiman homomorphism on this domain: two sums of two of its graph points lie in $4\Gamma'' - 4\Gamma''$, where the first coordinate determines the second. Also $\zeta(0) = 0$. Thus $\zeta(x+y) = \zeta(x) + \zeta(y)$ whenever $x, y, x+y$ lie in its domain. This proves precisely the generalization of [GT08, Lemma 9.2] needed here.

A proper progression and an explicit linear bracket formula

The set H'' has density at least $c_3/101$ in B. By [GT08, Lemma 6.4, pp.97–98] there is a bounded set of characters S with $B(S, 1/4) \subseteq 2H'' - 2H''$. Add the faithful character $x \mapsto x/Q$ to S, and shrink the radius to $\rho = 1/16$. Write $d = |S|$, a bounded integer. [GT08, Lemma 10.4, p.121] supplies a proper centered progression

$$P = \Big\{ \sum_{a=1}^{r} l_a v_a : |l_a| < L_a \Big\} \subseteq B(S, \rho), \qquad |P| \geq c_4 Q,$$

with $r \leq d$, and linearly independent vectors $\Phi(v_a) = (\{\chi(v_a)\})_{\chi \in S}$. The coset subgroup in that lemma is zero because S contains a faithful character. Its proof also gives

$$\Phi\left(\sum_a l_a v_a \right) = \sum_a l_a \Phi(v_a) \quad \text{on } P. \tag{F3}$$

This is the lift of a lattice progression inside the real cube, not an unjustified inference from reduction modulo one. Delete inactive generators with no nonzero integer coefficient allowed; all remaining v_a lie in P. Every partial sum along a coordinate path to a point of this centered box stays in P. The local additivity of ζ consequently gives

$$\zeta\left(\sum_a l_a v_a \right) = \sum_a l_a \zeta(v_a).$$

Choose real lifts b_a of $\zeta(v_a)$ and real dual vectors u_a to the independent vectors $\Phi(v_a)$ on their span. Using (F3), define $b_\chi = \sum_a b_a (u_a)_\chi$. Then, for every $p \in P$,

$$\zeta(p) = \sum_{\chi \in S} b_\chi \{\chi(p)\} \pmod 1. \tag{F4}$$

This is a linear calculation. It uses neither the quadratic-extension theorem nor the quadratic classification lemma in [GT08]. The author notice concerning that extension theorem is therefore not a new assumption here.

Translation, fixed carries, and the Freiman restriction

Averaging translates of P in B gives an $x \in B$ with $|H'' \cap (x+P)| \geq |H''||P|/Q \geq c_5 N$. For $h = x + p$ and $h_0 = x + p_0$ in this intersection, the graph relation gives $y(h) - y(h_0) = \zeta(p - p_0)$, since $\Gamma'' - \Gamma''$ is contained in $2\Gamma'' - 2\Gamma''$. Both p, p_0 lie in the domain of ζ. Its Freiman identity applied to $(p - p_0) + p_0 = p + 0$ gives

$$y(h) = \delta_x + \zeta(p).$$

For each $\chi \in S$, write $\chi(h) = \alpha_\chi h$ on the cyclic group. The difference

$$\{\chi(h-x)\} - \{\chi(h)\} + \{\chi(x)\}$$

is an integer in a fixed set of at most three values. Restrict to a cell on which all these carries are fixed. This costs a factor at most 3^d and, by (F4), gives

$$y(h) = \delta + \sum_{\chi \in S} b_\chi \{\alpha_\chi h\}.$$

Finally partition each $\{\alpha_\chi h\}$ into a fixed number of intervals of length at most $1/10$, and retain a largest cell H'. For any integer relation $h_1 + h_2 = h_3 + h_4$ in H', the corresponding sum of four fractional parts, with signs $+, +, -, -$, is an integer of absolute value at most $1/5 < 1$. It is therefore zero in $\mathbb{R}$ for each character. The real weighted sum is an exact Freiman homomorphism on H'. Undoing the rounding changes the displayed value by at most $e/2$. All density losses depend only on c, C and all sets and choices are finite. This proves the finite assertion, and transfer with $e = N^{-j}$ proves Theorem A20.2. A bounded rank can be made a fixed standard rank by passing to its ultrapower value or by adding zero terms.

A20.5 Consumer boundary and related-use scope

Theorem A20.1 supplies exactly the one-variable factorization consumed by the Sections 9–11 repair. That repair still must derive its nonzero coset integral and its degree-rank horizontal inclusion for the actual Taylor quotients. In particular, it may multiply a character by a standard denominator to kill the rational periodic contribution; it may not silently erase that contribution. No such application is accepted by this packet.

Theorem A20.2 applies separately at each of the finitely many required indices j, including $j = 1$, and to any chosen one of the four functions. The number of applications is standard. Dense refinements and all losses remain standard only when that finite iteration is recorded by the consumer. The theorem supplies no global Freiman extension beyond H'.

The related-use audit is bounded by uses of the single-scale equivalence, properness in D.4, composition of factorization factors, deletion of rational Taylor contributions, and the graph/Bohr/bracket joins in F.1. The multi-variable D.5 and its consumers in Appendix E require their own exact application review. They are not inferred from the one-variable theorem here. Likewise, this packet does not accept Gaussian's pending Sections 9–11 argument or any downstream assembled inverse theorem.

A21 Equal-scale factorization and the bias and torsion joins

This section proves the multivariable factorization needed for E.11 and E.13. The only quantitative equidistribution input is the final published equal-side theorem in [GT14, pp.1176–1177], with the published minor corrections to [GT12]. No unpublished arbitrary-side extension is imported.

A21.1 Conventions and exact statements

Fix standard positive integers k, d and an unbounded integer N. All groups, rational structures, lattices, coordinate bases and finite dimensions are standard, with natural ultrapower extensions. A nilpotent Lie group is connected and simply connected. The rational degree filtration $G_\bullet$ has finite length; the source $\mathbb{Z}^k$ has its total-degree filtration. Multiindices have nonnegative integer entries,

and $|\alpha| = \sum_i \alpha_i$. All vertical identities for a limit nilcharacter hold at every limit group element and point, as in the accepted tautification-003 convention. We do not substitute equivariance only at standard points.

In fixed Mal'cev coordinates, small polynomial coefficients mean bounded constant coefficient and coefficient $O(N^{-|\alpha|})$ at positive index α in either the monomial or product-binomial basis. The constants are standard and may depend on the actual fixed geometric data and orbit. Total equidistribution on $[N]^k$ means equidistribution on every product of dense arithmetic progressions in $[N]$, with standard Lipschitz tests. It also applies to internal tests with a common standard Lipschitz bound: finite uniform nets in a compact Lipschitz ball reduce those tests to standard ones, with the desired accuracy chosen before taking the limit.

Theorem A21.1 (Equal-scale factorization, MF-D4)**.** *For $g \in {}^*\mathrm{poly}(\mathbb{Z}^k, G_\bullet)$ there are a standard connected rational subgroup $P \le G$ and internal polynomial maps ϵ, p, γ such that*

$$g(n) = \epsilon(n)p(n)\gamma(n) \qquad (n \in {}^*\mathbb{Z}^k). \tag{1}$$

The map p lies in P with its induced filtration, and is totally equidistributed on $[N]^k$ in P/Δ for every standard finite-index $\Delta \le P \cap \Gamma$, with the same p and P. The coefficients of ϵ are small as specified above; it is bounded and $O(1/N)$-Lipschitz on every fixed standard dilate $[-AN, AN]^k$. Its group Taylor coefficient at positive index α is $O(N^{-|\alpha|})$ as well. There is a standard discrete subgroup $\Lambda \ge \Gamma$ containing all values of γ, and a standard $q > 0$ such that

$$\gamma(n + qe_i)\Gamma = \gamma(n)\Gamma \qquad (n \in {}^*\mathbb{Z}^k, 1 \le i \le k). \tag{2}$$

There is no properness requirement on P, and no claim that γ is Γ-valued. All factors in (1) are defined on the full integer lattice.

Write $I = [-N, N] \cap {}^*\mathbb{Z}$.

Theorem A21.2 (Actual bias and torsion outputs, MF-E11/E13)**.** *Let $d \ge 2$ be standard, and let χ be a globally represented unit vector nilcharacter of degree d on ${}^*\mathbb{Z}^k$. If a scalar coordinate of χ has nonzero correlation with a degree at most $d-1$ nilsequence after multiplication by the indicator of a convex polytope in I^k and a product of dense progressions in $[N]$, then χ agrees on all of I^k with a globally represented degree-at-most-$(d-1)$ vector nilsequence. The same conclusion holds if $\chi^{\otimes a}$ agrees on I^k with a degree-at-most-$(d-1)$ vector nilsequence for a standard integer $a \ge 1$.*

These assertions give exactly the lower-degree conclusions of [GTZ, E.11] and [GTZ, E.13] on their required symmetric box. They include finite scalar-coordinate reductions for vector correlations. They do not require an extrapolation from an unrelated subprogression or the deletion of a rational Taylor coefficient.

A21.2 A weighted inverse-affine estimate

The following estimate also supplies the large-offset transition behind [GT12, Proposition 9.2]. Merely inserting an offset of size N into the height bound of its Lemma 8.4 would not give the required standard bound.

Lemma A21.3 (Large offsets, bounded steps, MF-AFFINE)**.** *Let $f(n)$ be a real polynomial of fixed total degree at most D in k variables. Let A_i be internal integers with $|A_i| = O(N)$ and let b_i be nonzero integers of bounded absolute value. Put $\widetilde{f}(m) = f(A_1 + b_1m_1, \dots, A_k + b_km_k)$. Suppose every positive product-binomial coefficient $\widetilde{f}_\beta$ satisfies $\|\widetilde{f}_\beta\|_{\mathbb{T}} = O(N^{-|\beta|})$. There is a standard positive integer r, depending only on D, k and the step bounds, such that every positive coefficient of rf has the same bound $O(N^{-|\alpha|})$ modulo one.*

Proof. Invert the substitution over $\mathbb{R}$ and expand

$$\prod_i \binom{(n_i - A_i)/b_i}{\beta_i} = \sum_{\alpha \le \beta} c_{\alpha\beta} \binom{n}{\alpha}. \tag{3}$$

Expanding each falling factorial first in monomials shows that the denominators divide a fixed integer formed from $\beta_i!$ and $b_i^{\beta_i}$. Conversion of ordinary powers to binomial coefficients uses integer Stirling coefficients. Thus one standard r clears all denominators, independently of the integer numerators A_i. The coefficient of n^ν before the last conversion is a polynomial in the A_i of total degree at most $|\beta| - |\nu|$. Consequently

$$|c_{\alpha\beta}| = O(N^{|\beta|-|\alpha|}), \qquad r c_{\alpha\beta} \in {}^*\mathbb{Z}. \tag{4}$$

The case $\beta = 0$ contributes only to $\alpha = 0$. For each $\beta > 0$ write $\widetilde{f}_\beta = z_\beta + e_\beta$, where z_β is an integer and $e_\beta = O(N^{-|\beta|})$. In the coefficient of index $\alpha > 0$ in rf, the contributions $r c_{\alpha\beta} z_\beta$ are integers, while

$$\sum_{\beta \ge \alpha} r c_{\alpha\beta} e_\beta = O(N^{-|\alpha|}). \tag{5}$$

There are only a fixed number of indices. This proves the claim. Negative steps cause no change after taking a positive common denominator. The uncontrolled constant coefficient causes no positive-index error. □

A21.3 From a dense rectangle to the published equal-side theorem

Lemma A21.4 (Weak factorization in finitely many parameters)**.** *Suppose $G_0 = G_1 = G$. If $g\Gamma$ is not totally equidistributed on $[N]^k$, it has a factorization (1) with p in a standard proper connected rational subgroup, retaining all conclusions of Theorem A21.1 except the final equidistribution condition.*

Proof. Failure supplies a product of progressions $P_i = \{a_i + b_i t : 1 \le t \le L_i\} \subseteq [N]$ with $L_i \ge \delta N$, standard $\delta > 0$, and a bounded Lipschitz test with noninfinitesimal discrepancy. Their integer steps obey $1 \le b_i \le 2/\delta$ and their offsets are $O(N)$. Subtract the Haar mean, normalize the test and rotate a constant complex phase so that the real mean discrepancy is positive.

In the index rectangle $\prod[L_i]$, partition each axis into blocks of a common length $L = \lfloor \eta N \rfloor$, leaving a final remainder of length less than L. Choose the standard $\eta > 0$ small compared with the discrepancy, δ, k and the test norm. The discarded proportion is at most $\sum_i L/L_i \le k\eta/\delta + o(1)$, so its contribution is less than one quarter of that discrepancy. At least one full cube of side L has a positive standard discrepancy. On this cube the polynomial is $\widetilde{g}(m) = g(A_1 + b_1 m_1, \dots, A_k + b_k m_k)$ with $|A_i| = O(N)$.

Apply the exact equal-side theorem of [GT14, pp.1176–1177] at length L, using a standard error parameter below this discrepancy and below the reciprocal rationality bound of the fixed adapted basis. The result gives a nonzero bounded horizontal integer character ξ for which all positive coefficients of the real horizontal lift of $\xi\widetilde{g}$ have circle norm $O(L^{-|\alpha|}) = O(N^{-|\alpha|})$. Only finitely many such characters are possible, so the selected character is standard. Lemma A21.3 replaces it by a nonzero standard multiple and gives the same coefficient bounds for ξg on the original variables. This is the precise rectangle-to-equal-cube join; the original unequal-side theorem is not used.

Normalize $g(0) = c\lambda$ with bounded c and $\lambda \in {}^*\Gamma$ and apply the preceding argument to $c^{-1} g \lambda^{-1}$, whose failure is preserved by bounded left translation. Its coordinates are $\sum_{\alpha>0} t_\alpha \binom{n}{\alpha}$, with t_α

supported in $G_{|\alpha|}$ by [GT12, Lemma 6.7]. Write the real horizontal lift as $u \cdot \psi$, with u a fixed integer vector. Choose integers a_α nearest to $u \cdot t_\alpha$. On a support where u is nonzero choose a fixed rational vector v_α of that support with $u \cdot v_\alpha = 1$, and define

$$w_\alpha = (u \cdot t_\alpha - a_\alpha)v_\alpha, \quad z_\alpha = a_\alpha v_\alpha, \quad \psi(\epsilon_0(n)) = \sum w_\alpha \binom{n}{\alpha}, \quad \psi(\gamma_0(n)) = \sum z_\alpha \binom{n}{\alpha}. \tag{6}$$

If u vanishes on a support, set all these vectors and its integer to zero. Coordinate support and polynomial group closure make these maps polynomial for the original filtration. The real lift is identically zero on $p = \epsilon_0^{-1} c^{-1} g \lambda^{-1} \gamma_0^{-1}$. Thus p lies in its standard proper connected rational kernel. The original factors are $\epsilon = c\epsilon_0$ and $\gamma = \gamma_0 \lambda$. The w_α have precisely the required small bounds; the z_α have a common standard denominator, without a numerator bound. Controlled products and periods are checked next. □

A21.4 Controlled products, periods, and descent

Scale all variables by $n_i = Nt_i$. Fixed coordinate polynomials with small coefficients become polynomials in t with bounded coefficients. The fixed BCH and coordinate-change polynomials preserve this property under finite products, inverses and fixed rational subgroup embeddings. Conversion between monomial and product-binomial bases is triangular in componentwise indices, so the coefficient of $\binom{n}{\alpha}$ is $O(N^{-|\alpha|})$. Differentiating the scaled coordinate maps on the compact box $[-A, A]^k$ gives the claimed $O(1/N)$ Lipschitz bound on every fixed dilate. A real-coordinate polynomial extension of ϵ has the same bounds, which will be used below.

These estimates also imply the group Taylor bounds. Assign formal weight $|\alpha|$ to each logarithmic coefficient of n^α and weight zero to the bounded constant term. Group Taylor extraction in any fixed admissible order uses fixed BCH polynomials. Inductively every term of formal weight w has coordinate polynomial total degree at most w; removing a Taylor factor at index β preserves this property because its binomial power has degree $|\beta|$. The coefficient at index α cannot contain a term of formal weight less than $|\alpha|$: such a term has total degree less than $|\alpha|$ and has zero coefficient at that index. This is the same triangular Taylor extraction as [GTZ, B.9], now with product-binomial indices. Each surviving BCH monomial contributes at least $N^{-|\alpha|}$ decay. In particular no estimate is inferred solely from separate degree in individual variables.

The common discrete overgroup argument is independent of the number of parameters. In a rational lower-central basis of weights w_i, choose a standard D clearing every coordinate denominator and every fixed BCH coefficient. As in DF002, the additive lattice $\sum_i D^{-(2w_i-1)}\mathbb{Z}X_i$ is BCH-closed: a nonlinear term with t inputs has output denominator exponent exceeding the input sum by at least $t-1$. Its exponential is a standard discrete overgroup $\Lambda \geq \Gamma$ containing the rational factors and any finite product of them.

Here is the full multivariable period argument, in the right-coset direction. Each coordinate of $\gamma(n)$ is a polynomial of fixed degree whose values have a common standard denominator. Tensor-product interpolation on a fixed integer grid gives a standard common denominator for all its coefficients. For every i the vector

$$R_i(n, r) = \psi(\gamma(n)^{-1}\gamma(n + re_i))$$

is a rational coefficient polynomial with a common standard denominator. It is identically zero when $r = 0$. Choose a standard positive q divisible by all these denominators. Each monomial then has a positive power of r, so $R_i(n, q)$ is integral for every internal integer vector n. This is exactly (2). This proof does not use the wrongly oriented quotient in the printed proof of [GT12, A.12]. The corresponding one-variable sublemma of DF002 has a separate independent root review.

To finish Theorem A21.1, first normalize the constant term and restrict to G_1 if $G_1 \neq G$. The identity

$$c^{-1}g(n)\lambda^{-1} = \lambda(g(0)^{-1}g(n))\lambda^{-1} \in G_1$$

holds in any number of variables, since first derivatives lie in G_1 and the positive groups are normal in G_0. Thus a controlled representation exists with a taut positive filtration. Choose a standard rational representing subgroup of least dimension. For any standard finite-index lattice $\Delta \leq P \cap \Gamma$, failure of total equidistribution in P/Δ permits Lemma A21.4, after the same normalization if its induced first group is proper. It supplies $p = \epsilon_2 p_2 \gamma_2$ in a proper rational $P_2 < P$. The correct new identity is

$$g = (\epsilon\epsilon_2)p_2(\gamma_2\gamma).$$

The preceding controlled-product and discrete-overgroup arguments verify every required bound and period. Rationality relative to Δ is the same rational structure as relative to $P \cap \Gamma$. This contradicts minimal dimension. It proves total equidistribution for every standard finite-index cover with the same factors. Only a minimum among $0, \ldots, \dim G$ was selected; no infinite sequence of refinements or dependence of the chosen factors on the later cover is hidden here.

A21.5 One quotient for all periodic residue classes

We now work in a taut ambient degree-d group G, so G_d is central. Tautification-003, Theorem 1, supplies this exact representation for each nilsequence used below with the same positive-degree groups and the same standard vertical character. Products of these representations remain taut.

Apply Theorem A21.1, and choose a standard representative $\lambda_\rho \in \Lambda$ for each value of the right coset of $\gamma(n)$ on a residue class $n \equiv \rho \pmod q$, $\rho \in (\mathbb{Z}/q\mathbb{Z})^k$. Such representatives can be standard: Λ/Γ is a fixed finite set, and periodicity makes the coset constant on each class. Let

$$\Gamma_0 = \bigcap_{\lambda \in \Lambda} \lambda\Gamma\lambda^{-1}, \qquad \Delta = P \cap \Gamma_0. \tag{7}$$

The first intersection is the kernel of the action on the finite coset set, so it is a standard normal finite-index subgroup of Λ contained in Γ. Thus Δ is finite index in $P \cap \Gamma$, and p is totally equidistributed on P/Δ. For every ρ the function

$$x \longmapsto F(zx\lambda_\rho\Gamma) \tag{8}$$

is well-defined on this *same* quotient: for $\delta \in \Delta$, $\lambda_\rho^{-1}\delta\lambda_\rho \in \Gamma$. No deletion of a rational horizontal coefficient is involved.

Lemma A21.5 (A nonzero integral from the actual cutoff)**.** *Let F be an internal bounded Lipschitz scalar test of standard norm on G/Γ. If*

$$|\mathbb{E}_{n \in I^k} 1_\Omega(n) 1_{P_1 \times \cdots \times P_k}(n) F(g(n)\Gamma)| > c > 0 \tag{9}$$

for a convex polytope $\Omega \subseteq I^k$ and dense progressions $P_i \subseteq [N]$, then for some residue ρ and bounded $z = \epsilon(n_0)$ one has

$$\left| \int_{P/\Delta} F(zx\lambda_\rho\Gamma)\, d\mu(x) \right| > c' > 0. \tag{10}$$

Here c' is standard and positive. The factors and quotient are those just constructed for the same actual g.

Proof. Partition the progression product into its finitely many nonempty residue classes modulo q. Each intersection is again a product of dense progressions, by the elementary congruence calculation for each axis. Their steps are bounded by the old step times q. Rotate a common complex phase in (9); one class retains a real sum at least $c_0 N^k$ for a standard $c_0 > 0$. Fix this lower bound before selecting a mesh.

Use index coordinates on that progression product. The inverse image of Ω intersected with the index rectangle is convex and lies in a box of side $O(N)$. Partition it by cubes of side $\lfloor \eta N \rfloor$. The cubes meeting its boundary have at most $O_k(\eta N^k) + O_k(N^{k-1})$ integer points, with constants also depending on the fixed step bounds. Indeed place disjoint unit cubes at those integer points and enclose them in the $O_k(\eta N + 1)$ neighborhood of the convex boundary; apply the published boundary-volume estimate [GT10, Corollary A.2, p.1810]. For a convex set with empty interior, first thicken by an arbitrarily small ball and then take the decreasing limit of its boundary neighborhoods. Choose standard η small enough that the discarded boundary contribution is below $c_0 N^k/4$. The total number of points in the full cubes is at most N^k. Some cube contained in the convex set therefore gives a dense progression product R whose phase-adjusted average is at least $3c_0/4$.

The choice of η can also ensure that $|F(\epsilon(n)x\lambda_\rho\Gamma) - F(\epsilon(n_0)x\lambda_\rho\Gamma)|$ is below $c_0/4$ for all $n, n_0 \in R$ and all $x \in P/\Delta$. This follows from the $O(1/N)$ variation of ϵ, its bounded image, the fixed standard representative λ_ρ, and compactness of P/Δ. Choose the mesh only after these standard constants and the original positive discrepancy are known. For $n \in R$, (1) and the fixed right coset give $F(g(n)\Gamma) = F(\epsilon(n)p(n)\lambda_\rho\Gamma)$. Freeze ϵ at n_0. Total equidistribution of the same p in P/Δ on R identifies the resulting average with the integral in (10), up to an infinitesimal error. Internal-test uniformity follows from the finite Lipschitz net specified at the start. This proves (10). □

A21.6 Exact reconstruction on the symmetric box

Let $G = G^\chi \times G^0$ be the taut product of a degree-d nilcharacter representation and any degree-at-most-$(d-1)$ auxiliary representation. The latter has trivial degree-d group. Write F_χ for the original unit vector test, and let η be its real standard vertical character. Set

$$K = P_d = P \cap G_d.$$

We justify the rational intersection with the original lattice explicitly. Fix a rational Mal'cev basis for G/Γ. A connected subgroup H with compact $H/(H \cap \Gamma)$ has rational Lie algebra in this basis. Indeed, [GT12, Lemma A.2] makes the coordinates of $\log \gamma$ rational for $\gamma \in \Gamma$. The Lie algebra $\mathfrak{l}$ generated by $\log(H \cap \Gamma)$ therefore has a rational basis, by selecting finitely many independent iterated brackets. If $\mathfrak{l}$ were proper in $\mathfrak{h}$, put it in a maximal proper subalgebra $\mathfrak{m}$. The last lower central term not contained in $\mathfrak{m}$ supplies a vector outside $\mathfrak{m}$ normalizing it. Thus maximality makes $\mathfrak{m}$ an ideal, and $\mathfrak{h}/\mathfrak{m}$ has dimension one, since otherwise a line would be a proper nonzero subalgebra. The quotient functional exponentiates by BCH to a surjection $H \to \mathbb{R}$ killing $H \cap \Gamma$, contradicting compactness. Hence $\mathfrak{l} = \mathfrak{h}$. Apply this argument to P and G_d. Their Lie-algebra intersection has a rational basis by rational linear algebra. The global exponential homeomorphism and BCH identify K with its exponential, so K is closed and connected. Equip K with its rational lower central filtration. Choose a standard Q bounding the finitely many rational coordinates. [GT12, Definition 2.5 and Proposition A.10] then give a Mal'cev basis for $K/(K \cap \Gamma)$, so this actual intersection is a lattice. The trivial subgroup case is immediate. Since G_d is central in the taut ambient group, K is central too. The coset map from $(K \cap \Gamma)/(K \cap \Delta)$ to $(P \cap \Gamma)/\Delta$ is injective. Thus $K \cap \Delta$ has finite index in $K \cap \Gamma$ and is a lattice in K, without requiring normality of Δ. The central-quotient argument in LP13-002, Section 2, now applies with its rationality premise

established: the image $\overline{\Delta}$ of Δ is a lattice in $\overline{P} = P/K$. The quotient filtration kills $P_d = K$ and hence has degree at most $d-1$. All these objects depend only on the fixed standard geometry.

Suppose that

$$\eta(\mathrm{pr}_\chi K) = 0. \tag{11}$$

For each ρ and bounded real-coordinate parameter $t \in [-1/5, 1/5]^k$, define, initially on P/Δ,

$$H_\rho(x,t) = F_\chi(\widetilde{\epsilon}_\chi(10Nt)\, x_\chi\, (\lambda_\rho)_\chi \Gamma_\chi), \tag{12}$$

where $\widetilde{\epsilon}$ is the real-coordinate polynomial extension. This is well-defined by (7). It is invariant under K by the all-limit vertical identity and (11); centrality permits its element to pass across $\widetilde{\epsilon}$ and λ_ρ. Hence (12) descends to $\overline{P}/\overline{\Delta}$. The scaled-coefficient estimates imply a standard Lipschitz bound in t, uniformly in N. Smooth multiplication on fixed compact ranges gives the same conclusion in x. Invariance descends this bound to the quotient metric; any fixed standard Riemannian metric is equivalent on this compact nilmanifold.

Choose a fixed Lipschitz function $B(t)$ on $\mathbb{T}^k$ equal to one on $[-1/10, 1/10]^k$ and zero outside $(-1/5, 1/5)^k$. Using centered representatives inside that chart, define $\widetilde{H}_\rho(x,t) = B(t)H_\rho(x,t)$ there and zero outside. It is a globally bounded Lipschitz vector function. The zero collar makes the definition independent of the chart boundary. The functions are limit functions with a standard bound, since their coordinate-level construction has uniform bounds before taking the ultrapower.

For each $\rho \in (\mathbb{Z}/q\mathbb{Z})^k$ choose a standard Lipschitz function b_ρ on $\mathbb{T}^k$ with $b_\rho(\sigma/q) = 1_{\rho=\sigma}$. For example use products of triangular bumps of radius $1/q$; for $q = 1$ use the constant function one. The vector function

$$\mathcal{H}(x,t,z) = \sum_\rho b_\rho(z)\widetilde{H}_\rho(x,t) \tag{13}$$

is globally bounded Lipschitz with a standard bound. Its orbit is

$$n \longmapsto (\overline{p}(n)\overline{\Delta}, n/(10N),\, n/q). \tag{14}$$

This orbit has degree at most $d-1$: its first component has that degree and both tori have degree one, with $1 \le d-1$. For every $n \in I^k$, exactly the correct residue value is sampled in (13), and $B(n/(10N)) = 1$. The identities (1), (2), and (12) give

$$\chi(n) = \mathcal{H}((14)). \tag{15}$$

This is an equality of the actual vector functions on the entire symmetric box. It is not an extrapolation of an equality on just one progression. No unit condition is imposed on $\mathcal{H}$ away from the orbit; a lower-degree vector nilsequence does not require one.

A21.7 Bias and torsion applications on the same objects

Proof of Theorem A21.2. For the bias assertion, choose the scalar coordinate and the lower-degree scalar test occurring in the actual nonzero correlation. If a vector correlation is given, a fixed finite coordinate choice supplies one such scalar nonzero average; its density loss is standard. Form the taut product representation $G^\chi \times G^0$ and the product test $F = (F_\chi)_i\overline{F_0}$, using the sign in that correlation. It has vertical character η on the degree-d group because G^0_d is trivial. Apply Theorem A21.1 and Lemma A21.5 to this actual product orbit and test, obtaining a nonzero integral (10).

Translation of Haar measure on P/Δ by any $u \in K$ preserves that integral. The integrand is multiplied by $e(\eta(\mathrm{pr}_\chi u))$, since K is central in the ambient taut group. The integral is nonzero,

so this phase is one for every standard $u \in K$. The real homomorphism η is continuous and K is connected; its image is therefore a connected subset of $\mathbb{Z}$, hence zero. Its natural ultrapower extension vanishes on every limit element too. This is (11). Equations (12)–(15), with the *original* vector test F_χ, now prove the asserted all-box lower-degree representation.

For the torsion assertion, let Ψ be the given degree-at-most-$(d-1)$ vector nilsequence equal to $\chi^{\otimes a}$ on I^k. Put its finitely many coordinates into one common product nilmanifold of that degree. Since χ is a unit vector, pointwise on $[N]^k$ one has

$$\sum_j (\chi^{\otimes a})_j(n)\overline{\Psi_j(n)} = 1.$$

At least one scalar term has a nonzero standard average, with lower bound at least the reciprocal of the fixed tensor dimension in absolute value. Use that term as the test on the product of the original χ geometry and the auxiliary geometry of Ψ. Its top vertical character is $a\eta$. Lemma A21.5 applies with the full positive cube and no nontrivial cutoff. The same Haar argument gives $a\eta(\mathrm{pr}_\chi K) = 0$ as a real homomorphism. As $a \geq 1$, (11) follows. The same reconstruction (12)–(15) uses the original F_χ, not its tensor power, and proves the required conclusion.

In the symbol formulation of [GTZ, E.13], $a[\chi] = 0$ means exactly that the tensor power is equivalent to the constant nilcharacter. By the definition of equivalence, its coordinates are lower-degree nilsequences on the specified box. This is precisely the Ψ premise just used, with a finite common representation. No torsion-free conclusion in degree one is asserted; the time and residue torus factors explain the requirement $d \geq 2$. □

A21.8 Scope of the factorization

This proof supplies the finite-parameter factorization actually needed by E.11/E.13 and their all-symmetric-box reconstruction. It does not assert the false horizontal Taylor closeness printed in D.5. The one-variable D.6 argument must retain its denominator correction in the degree-rank horizontal quotient. The inverse-affine Lemma A21.3 also validates the large-offset coefficient transition in the one-variable weak-factorization source. The affine-transfer application and the E.11/E.13 substitutions are made explicitly in Section A15.

A22 Sections 9–11 at every rank step

This section separates two analytic hypotheses, F and D, from the algebraic, selection and representation arguments proved here. Their supplying results are proved in Sections A20 and A21. The universal group uses the explicit fixed-rank weighted presentation below. No final inverse theorem is used as a premise.

A22.1 The family at an arbitrary rank

Fix a standard $s \geq 3$, $m = s - 1$ and $1 \leq R \leq m$, and an unbounded limit integer N. All groups, lattices, metrics and finite dimensions below are standard unless a natural ultrapower is displayed or understood. Braces use $(-1/2, 1/2]$ throughout this packet. The Section 12 constructor allows either fixed half-open fundamental cube; use this same choice there. Let $H \subseteq [-N, N] \cap {}^*\mathbb{Z}$ be dense and internal. The input consists of a bounded function f on $[N]$, a global unit-vector multidegree $(1, m)$ nilcharacter $A(h, n)$, and an internal family of unit-vector degree–rank (m, R) nilcharacters χ_h. The latter have a fixed finite list of rational models and standard uniform Lipschitz bounds. For every $h \in H$, $\Delta_h f$ has uniformly positive degree-at-most-$m-1$ correlation with $A(h, \cdot) \otimes \chi_h$,

with bounded internal witnessing tests. Every vertical identity is on the full natural extension of its group and compact quotient, not merely their standard points.

This is exactly the new input in rank-iteration001, Section 3. In particular no core/petal representation or independence statement is inherited from the preceding rank. They will be reconstructed. The corrected LP03 gives, after any dense internal restriction of H, many additive quadruples $h_1 + h_2 = h_3 + h_4$ for which

$$\chi_{h_1}(n) \otimes \chi_{h_2}(n + h_1 - h_4) \otimes \overline{\chi_{h_3}(n)} \otimes \overline{\chi_{h_4}(n + h_1 - h_4)} \tag{177}$$

is $(m-1)$-biased, with standard uniform witnesses. This hereditary statement means reapplying LP03 to the restricted correlation hypothesis. It does not assert that a previously chosen dense set of quadruples survives restriction.

Analytic contract F. For internal functions $X_1, \dots, X_4 : H \to {}^*\mathbb{T}$ and $\epsilon = N^{-j}$, if $X_1(h_1) + \cdots + X_4(h_4) = O(\epsilon)$ on many additive quadruples, there are a dense internal H', a standard finite K, and internal $\alpha_k \in {}^*\mathbb{T}$, $\beta_k \in {}^*\mathbb{R}$, $\delta \in {}^*\mathbb{T}$ such that

$$X_4(h) = \sum_{k=1}^{K} \{\alpha_k h\}\beta_k + \delta + O(\epsilon) \quad (h \in H').$$

The errors have a common standard bound. A permutation of the four positions changes which function is recovered. This is the precise Appendix F.1 input; the signed-graph and no-wrap argument is a separate analytic supplier.

Analytic contract D. For a polynomial sequence in one variable in a fixed rational filtered nilmanifold there is a factorization $g = \varepsilon g' \gamma$ with ε bounded, Lipschitz $O(1/N)$ and positive-index Taylor coefficients $O(N^{-i})$; g' takes values in a connected rational subgroup Q and is totally equidistributed on every standard finite cover of $Q/(Q \cap \Gamma)$; and γ takes values in a fixed discrete overgroup $\widehat{\Gamma} \supseteq \Gamma$ and is periodic modulo Γ with standard period. The subgroup Q is allowed to equal G. All factors remain adapted to the ordinary degree filtration. Only this one-variable contract is needed. Its quantitative published supplier and corrected proof are separate.

Theorem A22.1. *Subject to these two analytic contracts and the stated LP03 input, a dense internal subfamily has the following representation. There are fixed standard dimension blocks C, L, B, fixed filtered lattice-compatible maps from a weighted universal group U, and one integral top character η on $U_{(m,R)}$. Up to a fixed coordinate isometry ι,*

$$\iota\chi_h(n) = F_h(g_h(n)\Gamma, g_{0,h}(n)\Gamma_0) \qquad (n \in [N]).$$

The lower model has degree–rank at most $(m, R-1)$. The functions F_h are unit vectors with uniform standard bounds and exact top-character law; they need not be equal as functions. The horizontal coefficients are fixed core frequencies, linear petals $\{\alpha_a h\}\beta_a$ for $a \in L$, and independent petals for $a \in B$. The linear petals are Freiman homomorphisms on the retained set. For almost every additive quadruple, the core, all four independent-petal blocks and any three linear-petal blocks are independent modulo $O(N^{-i})$ separately at each weight i.

For every weight-m, length-R iterated commutator w, the real number $\eta(w)$ is zero if w contains two petals, or contains one letter from B. These are precisely the annihilations required by LP04. They hold for every new rank family satisfying the initial hypotheses, without inherited frequency information. The construction of the final two-variable test is the separate Section 12 join.

A22.2 Selection and the corrected universal representation

We use the following elementary internal-choice rule, which also makes the countability uses precise. For an increasing internal sequence of witness sets S_k, suppose each member of an internal parameter set has a witness in some standard S_k. The least successful index is an internal integer. It is uniformly bounded: otherwise the internal nonempty sets on which it exceeds $1, 2, \dots$ have a common element by countable saturation, contrary to pointwise standard boundedness. At representative indices choose the first witness in a fixed choice order. This gives internal choices with one standard bound. Countable saturation here follows by diagonalizing nested finite intersections: at representative index v, choose a witness for the largest $k \le v$ whose first k conditions hold there. This k tends to infinity along the ultrafilter. The unsupported diagonal $x_{v,v}$ alone would not establish this conclusion.

We also need the "almost all" form. For nested internal bad sets B_k in a hyperfinite set X, if every B_k has infinitesimal relative size, the same diagonal construction chooses an unbounded K for which B_K has infinitesimal relative size and contains every standard B_k. Consequently, if their countable union is not contained in a sparse internal set, some B_k is dense. Applying finite pigeonhole inside that B_k fixes an integer coefficient vector and a standard error bound. This is the exact quantifier step used for failed frequency independence below.

First restrict to one of the fixed models for χ_h. Bounded tautification and normalization use their separately proved project interfaces. In a fixed model, only finitely many integral vertical characters can occur under a common Lipschitz bound. Indeed, along a fixed unit-speed top-torus direction X, the identity $F(\exp(tX)x) = \mathrm{e}(t\eta(X))F(x)$ and $\|F(x)\| = 1$ give $2\pi|\eta(X)| \le C_X \operatorname{Lip}(F)$ by taking small t. On an integral basis this bounds all integer character coordinates. Finite pigeonhole therefore fixes the character.

For each weight i, choose finitely many elements of $\Gamma \cap G_{(i,0)}$ whose logarithms generate its rational Lie algebra. Such elements are obtained by clearing denominators in an adapted rational Mal'cev basis; their real generated subgroup is $G_{(i,0)}$. Use one formal letter of weight i for each. Define U by killing all Lie words of weight greater than m and all weight-m words of length greater than R. Set

$$U_{(d,r)} = \exp \operatorname{span}\{\text{surviving words of weight} > d, \text{ or weight} = d \text{ and length} \ge \max(r, 1)\}$$

for positive d, with $U_{(0,0)} = U$. This is the exact LP04 weighted presentation before the additional petal relations. Its Hall-coordinate lemma supplies the lattice generated by the letters and its filtered intersections. The upward-closed word conditions prove every bracket inclusion. In particular $U_{(d,0)} = U_{(d,1)}$.

Mapping each letter to its chosen lattice element gives a filtered homomorphism $\phi : U \to G$. The killed words vanish by the degree–rank bracket law in G; the map sends the generated lattice into Γ. It is onto each $G_{(i,0)}$. In $\mathrm{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$ the images of the weight-i letters span the real vector space and generate a full sublattice. Thus the induced map of compact horizontal tori is onto. Select internal preimages of $\mathrm{Taylor}_i(g_h\Gamma)$ to get frequencies $\xi_{h,i,a} \in {}^*\mathbb{T}$. Pull the fixed top character back by ϕ and initially take G_0 trivial. This proves the exact Definition 9.11 representation, including lattice compatibility and all-limit equivariance, for every new family.

Here and below horizontal Taylor coefficients are homomorphisms on the polynomial-sequence group, are unchanged by translating n, and for $c \in G_{(i,1)}$ the sequence $c^{\binom{n}{i}}$ has only the ith horizontal coefficient c and zero others. These facts use the iterated-difference commutator rule B.7: all noncommutative correction terms have rank at least two and disappear modulo $G_{(i,2)}$; an extra difference has weight above i. Binomial differences give the last assertion. Polynomial products use B.4. No claim that arbitrary Taylor factors commute is used.

If $\xi_{i,a} = \sum_b c_{i,a,b}\xi'_{i,b}$ with standard integers $c_{i,a,b}$, map the new weight-i letter $e'_{i,b}$ to $\prod_a e_{i,a}^{c_{i,a,b}}$ in a fixed order. The universal relations and lattice inclusion hold, giving $\psi : U' \to U$. In the abelian horizontal quotient this is exactly the indicated integer matrix. Replace ϕ by $\phi\psi$ and η by $\eta\psi$; the original orbit and test are unchanged. When these matrices depend on h, the preceding bounded-choice rule and finite pigeonhole fix them on a dense internal subfamily before this step.

A22.3 Frequency regularization with the actual quantifiers

Fix one weight and put $\epsilon = N^{-i}$. A partial description consists of fixed core frequencies, independent-candidate petals b_h, bracket-linear petals l_h, fixed rational frequencies, and $O(\epsilon)$ errors, with all original frequencies standard integer combinations of them. Every list is standard finite, and every family is internal. Dense restriction fixes the integer matrices and error bounds. Put all original frequencies in b_h to obtain an initial partial description.

Minimize lexicographically the numbers of b petals, then l petals, then core frequencies, among all such partial descriptions on dense subsets. Bracket-linear petals are required to be individually Freiman homomorphisms. This costs only a finite dense restriction: place each $\{\alpha h\}$ in one of a fixed finite collection of intervals of width less than $1/4$. For an additive quadruple the difference of the four fractional parts is an integer of absolute value less than one, hence zero. Multiplication by any real β preserves that identity. The intervals may be anywhere in the fundamental cube; none is required to surround zero.

If independence of the core, four b blocks and the first three l blocks fails on more than a sparse set of quadruples, the bad-set rule above fixes one nonzero standard integer relation with error $O(\epsilon)$ on many quadruples. There are three exhaustive cases.

If it contains only core coefficients, solve for a core with nonzero coefficient a. Choose internal ath roots of the other core frequencies. Division introduces a residue in the finite group $a^{-1}\mathbb{Z}/\mathbb{Z}$ and an $O(\epsilon)$ error. Move them to the rational and error lists. This reduces the core count without changing either petal count.

If the relation involves l but no b, it is a relation on three free parameters. Fix two by averaging over the many triples. On a dense set of the remaining parameter, solve for one l coefficient. Divide the remaining bracket-linear coefficients as real numbers β/a, and root the core coefficients. Freeze the finite rational carry and add the values at the two fixed parameters to the core. A further fixed-box restriction restores the Freiman laws for the divided bracket-linear terms. This reduces the l count without changing the b count.

If a b coefficient occurs, use a symmetry of $h_1 + h_2 = h_3 + h_4$ to place a nonzero one in position four. The Freiman laws allow elimination of the fourth l block after this permutation. Group the relation as $X_1(h_1) + \cdots + X_4(h_4) = O(\epsilon)$, with X_4 containing that b coefficient and fixed core terms. Contract F expresses X_4 as finitely many bracket-linear terms plus a constant and error. Solve for the chosen b coefficient, using roots of the other b and core frequencies and dividing the real bracket-linear coefficients. Freeze the finite rational carry, enlarge the other lists, and refine into fixed boxes as above. The original frequencies are still integer combinations of the new lists, because the old frequencies are integer multiples of their chosen roots. The b count has decreased. Each case contradicts the lexicographic minimum.

Thus the required independence holds for almost all quadruples. Apply this argument successively for the finitely many weights $1, \ldots, m$. Dense restrictions preserve previously obtained independence: their bad sets remain $o(N^3)$, while the retained parameter set has standard positive density. Standard integer relations, rational carries, error constants, finite dimensions and fixed model choices are selected internally with the explicit bounded-choice rule. This proves the regularization step of 10.5–10.7 conditional only on contract F, with its needed family uniformity.

A22.4 Removing rational and small coordinates without losing unit norm

Apply the integer change-of-basis construction to the regularized frequency lists. We now remove each rational or small coordinate in turn. The remaining top character and lattice map must be updated; forgetting them would not give a new representation.

For a small coordinate $\xi = O(N^{-i})$, choose its small real lift and put $c = \phi(e_{i,a})^{\xi} \in G_{(i,1)}$. Set $g'_h(n) = c^{-\binom{n}{i}} g_h(n)$. B.4 proves polynomiality and horizontal homomorphism shows that exactly the selected coordinate has been removed. On $[N]$, $c^{\binom{n}{i}}$ is a bounded smooth function of $u = n/(2N)$. Indeed its logarithm is $\binom{2Nu}{i}\xi \log \phi(e_{i,a})$; its coefficients and all derivatives on a fixed compact interval are bounded because $N^i\xi = O(1)$. Extend this logarithm to the unit circle by using its real formula on the lift $(-1/4, 3/4)$ and multiplying by a fixed smooth cutoff equal to one on $[0, 1/2]$ and zero near the lift's endpoints. Let $c_h(u)$ be its exponential. Replace the test by

$$F'_h(x, x_0, u) = F_h(c_h(u)x, x_0).$$

It is globally a unit vector and uniformly Lipschitz. The top image $\phi(U_{(m,R)})$ is central, so the same exact vertical law holds everywhere. Add the circle orbit $n/(2N)$ to the lower model. It has degree one, which is at most $(m, R-1)$ because $m \geq 2$. Thus this is an exact representation, not an approximation to the original sequence.

For a rational coordinate $q \in \mathbb{Q}/\mathbb{Z}$, put $\gamma = \phi(e_{i,a})^q$ and $g'_h(n) = g_h(n)\gamma^{-\binom{n}{i}}$. Again only that horizontal coordinate is removed. The element γ is rational relative to Γ. The subgroup $\widehat{\Gamma} = \langle \Gamma, \gamma \rangle$ is discrete: in rational Mal'cev coordinates the finite nilpotent collection formulas express every word as a polynomial in integer exponents of a finite set of generators and their finitely many commutators; clearing their fixed rational denominators gives a discrete coordinate grid. Since Γ is cocompact, its index in $\widehat{\Gamma}$ is finite. Its normal core $\Gamma' = \bigcap_{\delta \in \widehat{\Gamma}} \delta\Gamma\delta^{-1}$ is therefore a finite-index subgroup, normalized by γ.

Choose a standard b with $\gamma^b \in \Gamma$ and a standard period q_0 such that $\binom{n+q_0}{i} \equiv \binom{n}{i} \pmod{b}$. For example a sufficiently divisible multiple of $b\,i!$ works, by the integer binomial difference formula. Put $\gamma_r = \gamma^{\binom{r}{i}}$ for $r \pmod{q_0}$. Then

$$F_h(g_h(n)\Gamma, x_0) = F_h(g'_h(n)\gamma_r\Gamma, x_0) \quad (n \equiv r \pmod{q_0}).$$

The test $W_{r,h}(x\Gamma', x_0) = F_h(x\gamma_r\Gamma, x_0)$ is well-defined, unit, and has the same top-frequency law. Also $W_{0,h}(x\Gamma', x_0) = F_h(x\Gamma, x_0)$ is such a global unit test.

To interpolate the finitely many residue tests, choose disjoint small arcs around r/q_0 in $\mathbb{T}$. In the rth arc choose a smooth angle θ_r which is zero at r/q_0 and $\pi/2$ near its boundary. Set $a_r = \cos\theta_r$ there and zero outside; set $b = \sin\theta_r$ there and $b = 1$ outside all arcs. The supports of the a_r are disjoint and $\sum_r a_r^2 + b^2 = 1$. The test

$$\left(\sum_r a_r(u) W_{r,h}(x, x_0),\ b(u) W_{0,h}(x, x_0) \right)$$

is globally Lipschitz, unit and has the same vertical character. At the orbit $u = n/q_0$ it is exactly $(\chi_h(n), 0)$. This fixed zero padding is explicit; we do not assert that an arbitrary scalar interpolation stays unit or that the old dimension has been retained. Correlation and subsequent matrix comparisons recover the original family by the fixed coordinate projection.

The map from the remaining universal group must now land in the covering lattice Γ'. Choose a standard M for which every remaining lattice generator's Mth power lies in Γ'. Send each new letter to that Mth power, and divide its horizontal frequency by M. Pull back the character by the

same map. For linear petals choose the division as $\{\alpha h\}(\beta/M)$; for core coordinates choose fixed roots and for independent coordinates choose internal roots. Repeating this finite process produces a common standard denominator. It is not silently dropped, as in the displayed conclusion of the printed proof of 10.10.

Independence is preserved: multiply an alleged standard integer relation between divided frequencies by the common denominator to contradict the old independence. After dividing linear petals, refine again into the fixed short boxes to make their real fractional-part sums exactly additive, hence their divided values Freiman. This is valid at arbitrary box locations. All rational covers, maps, dimensions, and periods are fixed on a dense internal subfamily before use. Small-coordinate curve bounds are uniform by the internal bounded-choice rule. A finite composition of paddings is one fixed coordinate isometry. We have proved the precise representation part of Theorem A22.1, without replacing the varying F_h by a single equal function through an approximation argument.

A22.5 The horizontal consequence of factorization, with denominators

Here we derive the needed D.6 conclusion from contract D. View a degree–rank group as ordinarily filtered by $G_i = G_{(i,0)}$. We retain the finer horizontal spaces $\mathrm{Horiz}_i(G) = G_{(i,1)}/G_{(i,2)}$. For a polynomial orbit $g\Gamma$ put

$$\Xi_i = \{\lambda \in \widehat{\mathrm{Horiz}_i(G/\Gamma)} : \lambda(\mathrm{Taylor}_i(g\Gamma)) = O(N^{-i}) \pmod 1\}.$$

Characters here have standard integer coordinates. Their real linear lifts define $\Xi_i^{\perp}$ inside the real horizontal vector space.

Suppose F is a bounded Lipschitz test with nonzero standard average. Take one scalar component if needed, and factor $g = \varepsilon g'\gamma$. On a progression of the standard periodic modulus, choose a bounded standard representative $\gamma_P \in \widehat{\Gamma}$ of the constant coset of γ. Set $Q_P = \gamma_P^{-1} Q \gamma_P$ and $g_P = \gamma_P^{-1} g' \gamma_P$. The finite-cover equidistribution clause in D, applied to the common finite-index lattice, makes g_P totally equidistributed in $Q_P/(Q_P \cap \Gamma)$ on every retained dense progression. Rationality of γ_P gives rationality of Q_P.

Partition into those residue progressions and then into intervals of length δN for a sufficiently small standard δ. Averaging selects a piece with an average bounded below by the original fixed threshold. On that piece freeze $\varepsilon(n)\gamma_P$ at one point. The uniform Lipschitz error is $O(\delta)$, so equidistribution gives

$$\left| \int_{P/(P\cap\Gamma)} F(\varepsilon_0 x)\, d\mu(x) \right| > c > 0 \tag{178}$$

for the connected rational $P = Q_P$ and bounded ε_0. This also holds for an internal test with a standard Lipschitz bound: compactness gives a finite standard uniform net in that Lipschitz ball. Apply total equidistribution to each net entry and then let the fixed net error tend to zero. Thus no unjustified standardness of the selected F_h or of the frozen bounded translate is being used.

We claim $\pi_i(P \cap G_i) \supseteq \Xi_i^{\perp}$. Let λ be an integer character annihilating the rational space $\pi_i(P \cap G_i)$. Then $\lambda(\mathrm{Taylor}_i(g_P)) = 0$. Conjugation changes no horizontal coefficient since $[G, G_i] \subseteq G_{(i,2)}$. Hence the horizontal homomorphism identity gives

$$\lambda(\mathrm{Taylor}_i(g)) = \lambda(\mathrm{Taylor}_i(\varepsilon)) + \lambda(\mathrm{Taylor}_i(\gamma)) \pmod 1.$$

The first term is $O(N^{-i})$. The second is rational of bounded standard denominator: derivatives of γ lie in $\widehat{\Gamma} \cap G_i$, whose horizontal image is a finite-index over-lattice of that of $\Gamma \cap G_i$. Thus for some standard positive q, $q\lambda \in \Xi_i$. Every annihilator of the rational image has this property. Taking real annihilators proves the claim. In particular no false assertion that γ takes values in Γ or has zero horizontal coefficient is needed.

A22.6 Applying the coset conclusion to the actual quadruple

After all the preceding dense restrictions, reapply hereditary LP03. There are $\gg N^3$ biased quadruples. The exceptional sets for frequency independence have size $o(N^3)$. Permutations preserving the additive quadruple equation make the omitted linear-petal position any of the four; the union of the four exceptional sets is still sparse. Fix a biased quadruple with all four required independence statements.

Use the representations of its four nilcharacters and the lower bias test to form a product group

$$\mathcal{G} = G^{(1)} \times G^{(2)} \times G^{(3)} \times G^{(4)} \times G^{(0)}.$$

The first four groups have degree–rank at most (m, R); the last has degree at most $(m, R-1)$, including all four lower representation models and the ordinary degree-at-most-$m-1$ bias witness. The stronger strict inequality printed for this last group is unnecessary and is not assumed. Translations in n preserve horizontal Taylor coefficients by their homomorphism/difference identity. Zero extensions in (177) introduce only an interval-intersection mask. Approximate this mask in normalized L^1 to error δ by a bounded Lipschitz circle test in $n/(cN)$, with fixed large c and Lipschitz constant $O(1/\delta)$. Its two transition strips have total length $O(\delta N)$; multiplying by the uniformly bounded remaining tests controls the average error. Choosing δ below the fixed bias threshold preserves positive bias. This circle has degree one and belongs in $G^{(0)}$ because $m \geq 2$.

The resulting product test has a nonzero scalar average and exact vertical law

$$F(\phi_1(t_1)x_1, \dots, \phi_4(t_4)x_4, x_0) = \mathrm{e}(\eta(t_1) + \eta(t_2) - \eta(t_3) - \eta(t_4))F(x_1, \dots, x_4, x_0).$$

All factors, including the cutoff, are uniformly Lipschitz on a fixed standard product geometry. Apply the preceding factorization consequence to obtain a connected rational $P \subseteq \mathcal{G}$ and the nonzero coset integral (178).

A22.7 Rational duality and the two commutator conclusions

For each weight i, let $V_{123,i}$ be the real space spanned by the three-coordinate diagonal images of core letters and by each individual coordinate image of every petal letter in positions $1, 2, 3$. Define $V_{124,i}$ likewise. The projection of $\pi_i(P \cap \mathcal{G}_i)$ to these three factors contains the indicated V.

Here is the exact duality argument. If containment failed, rationality gives a rational linear functional that vanishes on the projected space but not on V. Extend it by zero in the omitted factors and clear denominators to make it an integer character λ. Because the full space contains $\Xi_i^{\perp}$, this character is in the rational span of Ξ_i. After multiplication by another standard integer it belongs to Ξ_i itself. Evaluating it on the Taylor coefficient gives a relation

$$\sum_{a \in C} (c_{1a} + c_{2a} + c_{3a})\xi_{C,a} + \sum_{k=1}^{3} \sum_{a \in L \cup B} c_{ka}\xi_{h_k,a} = O(N^{-i}).$$

The selected independence forces every petal coefficient and each displayed core sum to vanish. That says the character vanishes on V, a contradiction. The 124 proof is identical. This uses rational real subspaces, not an unjustified identification of an arbitrary real annihilator with one integral member of Ξ_i.

Similarly the projection onto all four main factors contains the space $V_{B,i}$ generated by diagonal core images and by the individual-coordinate images of every letter in B. In the analogous integer relation eliminate the fourth L coefficient using $\xi_{h_4,a} = \xi_{h_1,a} + \xi_{h_2,a} - \xi_{h_3,a}$ for $a \in L$. Independence makes the four B coefficients vanish separately and the sum of core coefficients vanish. This is

exactly annihilation of $V_{B,i}$. No numerical range for B is borrowed from the conflicting printed order of the L and B blocks.

Let w be any bracketing of R letters of weights summing to m. First suppose it has two petal occurrences. For one occurrence lift its first-coordinate image, with zero second and third coordinates, from $V_{123,i}$ to $P \cap \mathcal{G}_i$. For the second lift its first-coordinate image with zero second and fourth coordinates from $V_{124,i}$. Lift the remaining letters so that their first horizontal coordinate is the required one. Take the same iterated commutator of these lifts.

In the first factor it equals $\phi_1(w)$ exactly. Replacing one lift by its horizontal error inserts an element of $G_{(i,2)}$ in place of $G_{(i,1)}$, raising the total rank to at least $R+1$ at weight m; every such term vanishes. Higher BCH terms have larger weight and vanish as well. In factor two, either selected petal lift has such an error; factor three has the first, and factor four the second. Therefore those coordinates are the identity. In factor zero, an R-fold commutator of the prescribed weights lies in its trivial (m, R) group. We obtain

$$(\phi_1(w), 1, 1, 1, 1) \in P. \tag{*}$$

This reasoning applies to each position of a repeated letter and to every bracketing, not only a single left-nested commutator.

If w has a B letter, use $V_{B,i}$ to lift that occurrence with all other main coordinates zero horizontally. Lift the remaining occurrences by their first coordinates and repeat the same rank-error calculation. This proves (*) as well. It works for $R = 1$; the other top coordinates then already vanish, and the lower model has trivial weight-m group.

Finally (*) gives a real-zero conclusion. The element is central in $\mathcal{G}$. Since P is a connected subgroup of a simply connected nilpotent group, its entire one-parameter subgroup belongs to P. Left translation by its tth power leaves Haar measure on $P/(P \cap \Gamma)$ invariant and commutes with the bounded coset translate. The vertical law multiplies the nonzero integral by $\mathrm{e}(t\eta(w))$. Hence $\mathrm{e}(t\eta(w)) = 1$ for every real t, so $\eta(w) = 0$ as a real number. Using only $t = 1$ would give an integer condition and would be insufficient. This proves both annihilations in Theorem A22.1.

A22.8 The LP04 and rank-family joins

The output has the exact LP04 hypotheses: the corrected universal weighted group, an integral top character with all required real-zero annihilations, fixed lattice-compatible ϕ, matched horizontal Taylor tori, and a unit representation with a lower degree–rank factor. Its petal quotient kills the whole ideal generated by the bad words at every weight; the vanishing just proved concerns the top restriction needed to descend the character. These distinct roles are not identified.

The Section 12 join chooses its stronger integral extension to the ordinary top before invoking LP04 comparison, and gives one global multidegree $(1, m)$ nilcharacter equal to a fixed coordinate embedding of ζ_h. LP04 then supplies the whole matrix $(\iota\chi_h) \otimes \overline{\zeta_h}$ with strictly lower degree–rank. Projecting the first tensor factor through the fixed coordinate projection gives $\chi_h \otimes \overline{\zeta_h}$ with the same bound and exact values. Thus the unit padding used in rational removal does not weaken the actual matrix input required by rank-iteration001.

All selections, roots, covers, maps and tests were made internally on representatives; each retained dimension, character and bound is standard. There are finitely many weights and coordinates at any one rank, so all density and correlation losses remain positive standard numbers. The next rank family from rank-iteration001 again has exactly the hypotheses of Section 1. Repeating the present reconstruction at that new rank uses its own new frequency data. No independence is transported through a Stone–Weierstrass approximation or assumed for the newly selected test. There are m rank steps, so the same finite-loss conclusion holds across the full iteration. This is

the required source-to-consumer compatibility, conditional on the two separately supplied analytic contracts and the stated hypotheses of the internal proof units used here. Representation equality is required on $[N]$. Whenever a global sequence is used, take the displayed polynomial model as its extension. Small-frequency removal need not preserve a previous extension outside $[N]$; the interval mask in the quadruple argument is precisely what makes that change harmless.

A22.9 Bounded correction and consistency record

The source scope is final GTZ Sections 9–11, the immediate pp. 1307–1309 consumer, the D.4–D.6 interface, B.4/B.7, and the selection statements in Appendix A. The attached record pins exact passages and the current project consumers. Concrete changes are:

1. State the fixed rank R explicitly in Definition 9.1's second truncation relation, where the printed formula writes r without the fixed rank subscript. The printed weight cutoff $\sum i_j \geq s$ is already equivalent to $\sum i_j > m$ and is not an error. Rendering the source rather than relying on text extraction establishes this distinction.

2. Retain the common division from 10.9 in 10.10 and restore Freiman laws after real coefficient division. Fixed arbitrary-location boxes suffice.

3. Rational removal uses a finite cover and explicit same-character unit completion with a fixed isometry, rather than unproved unit interpolation.

4. Keep varying uniformly Lipschitz F_h; total boundedness does not make them literally equal as claimed in the immediate p. 1308 narrative.

5. Use named core/linear/independent blocks, because p. 1302 reverses the order used in the statement of 10.10 and Theorem 11.1.

6. Treat the repeated clauses (iii),(iv) of Theorem 11.1 as duplicates; there are two distinct annihilation conclusions.

7. Use degree–rank at most $(m, R-1)$ for the lower product model, not an unneeded strict inequality below that index.

8. Retain the rational Taylor term in D.5 and clear its denominator in the annihilator argument. The factorization subgroup may equal the ambient group. The separate analytic supplier addresses the D.2 total-equidistribution issue and F.1 signed-graph/no-wrap details.

9. The real-zero character argument uses all real powers in the connected subgroup. One integral phase identity alone is insufficient.

10. Reapply LP03 after each dense restriction; intersect many biased quadruples with almost-all independence, not two unrelated dense sets.

These are authored repairs and scoped dependencies, not a declaration that all historical versions or the entire inverse theorem have been checked. The original published source and all earlier frozen packets are preserved.

A23 GTZ-LP-04: the petal quotient and degree–rank construction

This section proves the algebraic quotient, its exact lattice and character statements, and the comparison needed by [141, Lemma 12.1], conditional on the explicitly stated frequency input. That frequency input is supplied in Section A22. The erratum [145] motivated the construction but is not a theorem premise. The rational-subgroup and tautification arguments are in Sections A18 and A17.

A23.1 Indices, free groups, and the actual objects

Fix integers $m \geq 1$, $1 \leq R \leq m$, and finite disjoint sets of letters C_i, L_i, B_i for $1 \leq i \leq m$. Their union is $\mathcal{A}$. The letters in C_i are core, those in L_i retained linear petals, and those in B_i discarded petals. Every letter in these three sets has weight i. A bracket word has length ℓ equal to its number of entries, weight w equal to the sum of their weights, and a multidegree recording the number of occurrences of each individual letter. Repeated entries count repeatedly. All bracketings are allowed, including length-one words. Write p for the number of petal entries, and b for the number of discarded entries. We use $[x, y] = x^{-1}y^{-1}xy$, real powers $x^t = \exp(t \log x)$, and $\mathrm{e}(t) = \exp(2\pi i t)$.

Let $\mathcal{F}$ be the free real Lie algebra on symbols X_a, $a \in \mathcal{A}$. Its usual length grading and its individual-letter multigrading are retained. Take the homogeneous ideal spanned by all words for which

$$w > m \quad \text{or} \quad (w = m \text{ and } \ell > R), \tag{179}$$

and let $\mathfrak{u}$ be the quotient. These forbidden multidegrees are closed under addition of any nonzero multidegree, so their span is indeed an ideal. All surviving words have length at most m. Thus $\mathfrak{u}$ is finite-dimensional and nilpotent. Let $U = \exp \mathfrak{u}$, with multiplication given by the finite BCH polynomial, and set $e_a = \exp X_a$. The group is connected and simply connected. These are the weighted universal objects intended in [141, Definition 9.1, p. 1279], with the top rank in the relation fixed to be R, not a running rank index.

Use $\mathrm{DR} = \{(d, r) : d, r \in \mathbb{N},\ 0 \leq r \leq d\}$ with lexicographic order. Put $U_{(0,0)} = U$. For $d > 0$ let

$$\mathfrak{u}_{(d,r)} = \operatorname{span}_{\mathbb{R}}\{W : w(W) > d \text{ or} [w(W) = d,\ \ell(W) \geq \max(r, 1)]\}, \qquad U_{(d,r)} = \exp \mathfrak{u}_{(d,r)}. \tag{180}$$

In particular the first alternative has *no* length restriction. For ranks larger than d use the notation $U_{(d,r)} = U_{(d+1,0)}$ only as the next-index convention; do not put such pairs into the index monoid. Write $U_d = U_{(d,0)}$. The top group is

$$T = U_{(m,R)} = \exp \operatorname{span}_{\mathbb{R}}\{W : w(W) = m,\ \ell(W) = R\}.$$

Lemma A23.1. *The groups in* (180) *form a taut degree–rank filtration of degree at most* (m, R). *Each is connected and normal in* U, *and* T *is central.*

Proof. Bracketing words adds weights, lengths, and individual-letter multidegrees. Each displayed span is an ideal, since bracketing with a nonzero word increases weight. It is a Lie subalgebra and BCH therefore identifies its exponential with a connected subgroup. Nesting follows directly from lexicographic order: passing to a larger degree removes all lower-weight words, and at fixed degree passing to a larger rank increases the length cutoff. For positive indices $(d, r), (e, t)$, a bracket of two allowed words either has weight greater than $d + e$, or has weight $d + e$ and length at least $\max(r, 1) + \max(t, 1) \geq \max(r + t, 1)$. Hence the Lie bracket lies in $\mathfrak{u}_{(d+e,r+t)}$. The BCH commutator formula has this bracket and further brackets, all in the same ideal, which proves the

group commutator condition. At index zero use normality. Every generator has weight at least one, so $U_{(1,0)} = U$; also $U_{(d,0)} = U_{(d,1)}$ for all $d > 0$. The truncation gives the claimed degree support. Bracketing a top word with any generator raises its weight above m, proving centrality of T. □

A23.2 An integral coordinate lemma

Here is the integral premise used, with its scope separated from the subsequent quotient argument. Hall collection says that in a free nilpotent group on a finite ordered alphabet, every element has a unique collected expression in the basic commutators with integer exponents; the length-j basic commutators give a basis of its jth lower-central quotient. This published statement is recorded in [142, Definition 5.4.3, Theorem 5.4.5 and the final paragraph of §5.4, pp. 96–98]. The weight in that reference is bracket *length*, not the additional weight w used here.

The corresponding basic Lie brackets form a homogeneous basis of the free nilpotent Lie algebra. To spell out the connection, antisymmetry and Jacobi reorder a bracket $[c_i, c_j]$ until $i > j$, and when $c_i = [c_s, c_t]$ until $j \geq t$. Jacobi replaces the offending bracket by brackets with the smaller inner entry moved outward. Induction on total length and then the finite Hall ordering gives spanning by basic brackets. Independence follows by mapping the free Lie algebra to the real tensor of the lower-central graded Lie ring of the free group: the bracket is induced by the group commutator, and Hall's basis assertion makes the images of those basic brackets independent in each length. This also respects the individual letter multigrading because the reordering identities preserve every count.

Lemma A23.2 (Upward-closed monomial quotients)**.** *In a free nilpotent Lie algebra of class at most m, discard an upward-closed set of nonzero individual-letter multidegrees. Let V be the exponential of the quotient and Λ the discrete group generated by the surviving* $\exp X_a$*. The surviving Hall commutators give unique real coordinates*

$$(t_1, \dots, t_v) \longmapsto c_1^{t_1} \cdots c_v^{t_v}$$

on V, and Λ is exactly the set with all $t_i \in \mathbb{Z}$. It is a lattice. A further upward-closed set of multidegrees defines a rational connected normal subgroup V_S. The quotient map q between two such monomial quotients obeys, for each such S,

$$q(\Lambda \cap V_S) = q(\Lambda) \cap q(V_S). \tag{181}$$

Proof. Order basic commutators by increasing length. For a basic group commutator c, BCH gives $\log c = W_c +$ terms of strictly greater length, where W_c is the corresponding basic Lie bracket. Each correction uses all the letters of c and possibly more occurrences of them. A discarded basic word therefore vanishes in the quotient, including every BCH correction. The surviving leading brackets are a basis, since the discarded Lie ideal is the direct sum of precisely the discarded multigraded components. Consequently the logarithm of the displayed coordinate map is triangular by length with diagonal entries one. Solving successively in each length gives a polynomial inverse and proves unique real coordinates.

Hall collection expresses every word in the original $\exp X_a$ and their inverses as a product of basic group commutators with integer exponents. After quotienting, discarded factors are identities, so it gives the claimed integer coordinates. Conversely every surviving basic commutator is itself a word in the generators, which proves the other inclusion. In particular the subgroup is discrete. Suffix spans in the length-ordered basis are ideals, since every nonzero bracket increases length. Right multiplication by c_i^{-n} changes coordinate i by $-n$, leaves all earlier coordinates unchanged,

and can change only later coordinates. Reduce the coordinates successively modulo integers. Every right coset has a representative with coordinates in $[0,1]^v$, a compact set, proving cocompactness.

For V_S, quotient once more by its multigraded ideal. The same triangular coordinate argument says that an element belongs to V_S exactly when all coordinates outside S vanish. Thus its lattice consists of the integer coordinates on the basic commutators in S, and the same compact-coordinate argument proves rationality. A quotient deletes discarded basic factors and keeps the remaining ones in their original order. Membership in $q(V_S)$ forces all retained coordinates outside S to vanish. An element of $q(\Lambda) \cap q(V_S)$ therefore lifts as the identical integer product of retained basic commutators in S. This proves (181). □

The truncation (179) is upward-closed, as are all the thresholds (180). The lemma proves that $\Gamma = \langle e_a : a \in \mathcal{A} \rangle_{\mathbb{Z}}$ is a lattice in U, and every $U_{(d,r)}$ is rational. It also proves the lattice statement below without treating it as a general property of arbitrary rational quotients.

A23.3 The corrected kernel and the descended character

Call a word bad when $p \geq 2$ or $b \geq 1$. Let $I \subseteq \mathfrak{u}$ be the span of all bad Lie words of every weight. Let G^* be the connected real subgroup generated by all bad iterated *group* commutators in the e_a, again of every weight and including length one. Equivalently, the real generated subgroup contains every real power of each such commutator. This convention is essential: a discrete generated subgroup would not be the kernel of a map of simply connected Lie groups.

Let $\widetilde{\mathfrak{u}}$ be the free Lie algebra on $C_i \cup L_i$ with relations (179) and the vanishing of all words with at least two petal entries. Give $\widetilde{G} = \exp \widetilde{\mathfrak{u}}$ the filtration (180) on its surviving words. Put $\widetilde{\Gamma} = \langle \widetilde{e}_a : a \in C_i \cup L_i \rangle_{\mathbb{Z}}$. Define $q : U \to \widetilde{G}$ by $e_a \mapsto \widetilde{e}_a$ on retained letters and $e_a \mapsto 1$ on discarded letters.

Theorem A23.3. *The homomorphism q is well-defined and surjective, and*

$$\ker q = G^* = \exp I.$$

Thus G^* is connected, rational and normal; it need not be central. For every degree–rank index a, including zero,

$$q(U_a) = \widetilde{G}_a \simeq U_a/(G^* \cap U_a), \tag{182}$$

$$q(\Gamma \cap U_a) = \widetilde{\Gamma} \cap \widetilde{G}_a. \tag{183}$$

Suppose $\eta : T \to \mathbb{R}$ is a continuous homomorphism integral on $T \cap \Gamma$, and for every bad group commutator c of weight m and length R one has $\eta(c) = 0$. Then there is a unique continuous $\widetilde{\eta} : \widetilde{G}_{(m,R)} \to \mathbb{R}$ satisfying

$$\widetilde{\eta}(q(t)) = \eta(t) \quad (t \in T), \qquad \widetilde{\eta}(\widetilde{\Gamma} \cap \widetilde{G}_{(m,R)}) \subseteq \mathbb{Z}.$$

Proof. Bad multidegrees are upward-closed, so I is an ideal. The universal property of the free Lie algebra identifies $\mathfrak{u}/I$ with $\widetilde{\mathfrak{u}}$: killing discarded letters kills every word involving one, and killing all words with two petals gives exactly the other defining relations. There are no additional relations because both presentations impose the same homogeneous ideal on the same free Lie algebra. Exponentiation of this surjective Lie map gives q, and its kernel is $\exp I$ by the global exponential diffeomorphisms.

BCH puts the logarithm of every bad group commutator in I. Conversely take the bad Hall commutators. Their logarithms have the bad basic Lie brackets as leading terms, with only higher-length bad terms. The triangular matrix on the finite-dimensional bad subspace is invertible. Their

logarithms therefore span I, and the connected real subgroup they generate is $\exp I$. This proves the asserted description of G^*, not just its normal closure. All these spaces are spanned by rational basic words. Lemma A23.2 proves rationality and $q(\Gamma) = \widetilde{\Gamma}$.

The Lie map deletes bad multigraded components and preserves weight and length of every surviving word. Applying (180) before or after the map therefore gives identical spaces. Exponentiating proves (182), and Lemma A23.2 proves (183).

The direct multigrading gives

$$I \cap \log T = \operatorname{span}_{\mathbb{R}}\{W : W \text{ bad},\ w(W) = m,\ \ell(W) = R\}.$$

There is no cancellation between a bad multidegree and a good one. At weight m the BCH correction terms in a group commutator have larger weight and vanish. Thus the bad group commutators of weight m and length R are precisely the exponentials of their Lie brackets. The assumption on η, and real linearity of a continuous homomorphism on the vector group T, prove that η vanishes on $G^* \cap T$. It consequently descends through (182) at the top index. Finally (183) lifts every top lattice point downstairs to one upstairs, where η is integral. This proves the last assertion. □

Why the omitted lower weights matter. With one core letter x and one discarded letter b, both of weight one, and $m = 3, R = 3$, the quotient kills b itself. A subgroup generated only by bad weight-three words cannot contain b. Moreover $[b, x]$ survives in U, so the actual kernel is not central. This example tests both false shortcuts.

The precise analytic input. In the notation of [141], take $m = s - 1$, $R = r_*$, $|C_i| = D_{*,i}$, $|L_i| = D'_{\mathrm{lin},i}$ and $|B_i| = D'_{\mathrm{ind},i}$. Use the *final output* of Lemma 10.10, after rational-core and small frequencies have been removed by Lemma 10.9. Then the two hypotheses on η in Theorem A23.3 are exactly Theorem 11.1(i),(ii): all weight-m, length-R words with two petals or one beyond-linear petal are annihilated. The repeated (iii),(iv) add nothing. For a bracketing convention restricted to left-normed words, Jacobi expresses any other bracket of these same entries as a linear combination of such words; their petal counts and total weight are unchanged. At the central top layer this is also an exact identity of logarithms, so the vanishing extends to all bracketings used here.

Theorem 11.1 requires the hypotheses of Theorem 7.2, the limit family of degree–rank (m, R) nilcharacters, and the sunflower representation with its common standard dimension vector and vertical frequency. Its proof uses the additive-quadruple bias conclusion of Proposition 7.3 and the stated independence from Lemma 10.10. In the repaired project, that bias conclusion must come from LP-03, not the false original Proposition 8.3. No use of Theorem 1.3 is made to supply it. This unit assumes these inputs; merely partitioning generators does not prove the required vanishing.

A23.4 Vertical test functions and petal actions

Lemma A23.4. *For the descended integral character there is a finite-dimensional unit-vector-valued smooth function on $\widetilde{G}/\widetilde{\Gamma}$ of vertical frequency $\widetilde{\eta}$. Its dimension and Lipschitz bound depend only on the fixed filtered group, lattice, metric and character.*

Proof. The central torus $\widetilde{T}/(\widetilde{T} \cap \widetilde{\Gamma})$ acts freely. Choose finitely many local smooth sections of the resulting principal torus bundle and smooth real functions a_j subordinate to their domains, with $\sum_j a_j^2 > 0$ everywhere on the base. Normalize them so $\sum_j a_j^2 = 1$. In a local trivialization (z, t) put the jth component equal to $a_j(z)\mathrm{e}(\widetilde{\eta}(t))$, extending by zero outside its support. The support stays inside the chart, so the extension is smooth. Transition functions change only the component's phase. The vector has norm one, and all components have the same vertical frequency.

Compactness gives a finite Lipschitz bound. If the torus or character is trivial the same construction, or the constant function, applies. □

The quotient Lie algebra is a direct sum $\mathfrak{c} \oplus P$, where $\mathfrak{c}$ consists of words with no petals and P of words with one. Here P is an abelian ideal: a bracket of two such words has two petals and is zero. Thus $\widetilde{G}_{\mathrm{petal}} = \exp P$ is abelian, normal and rational by Lemma A23.2. Decompose $P = \bigoplus_{a \in L} P_a$ by the unique petal letter. Core conjugation preserves each P_a. For a tuple $t = (t_a) \in \mathbb{R}^L$, the map fixing core generators and sending X_a to $t_a X_a$ on linear petals is a Lie endomorphism M_t: it preserves all defining ideals. It scales P_a by t_a, so on the petal group, written multiplicatively,

$$M_t M_u = M_{tu}, \qquad M_t(p) M_u(p) = M_{t+u}(p), \qquad M_t(gpg^{-1}) = g M_t(p) g^{-1}.$$

The last identity follows because M_t commutes with every core adjoint action and conjugation by P acts trivially on P. This proves the precise identities used in (12.8)–(12.10), without treating M_t as an automorphism when some $t_a = 0$. For integral tuples t, M_t preserves $\widetilde{\Gamma}$, because it sends each generator to an integer power. It also preserves the petal lattice by its subgroup invariance. These are the lattice facts needed by the subsequent shear semidirect products.

A23.5 The comparison filtration and corrected Lemma 12.1

We record sufficient input explicitly. Let G/Λ, G^0/Λ^0 be fixed taut rational degree–rank nilmanifolds of degrees at most (m, R) and strictly below (m, R), respectively. Let $\phi : U \to G$ be filtered and send Γ into Λ. Suppose a_h, c_h are polynomial sequences and F_h are unit-vector-valued uniformly Lipschitz functions on $G/\Lambda \times G^0/\Lambda^0$ such that

$$F_h(\phi(t)x, z) = \mathrm{e}(\eta(t)) F_h(x, z) \quad (t \in T). \tag{184}$$

Let $E_{h,i} = \prod_{a \in C_i \cup L_i \cup B_i} e_a^{\xi_{h,a}}$ in a fixed order, and suppose the horizontal Taylor coefficient of $a_h \Lambda$ is that of $\phi(E_{h,i})$ in $G_{(i,1)}/G_{(i,2)}$ modulo the horizontal lattice. Let

$$b_h(n) = \prod_{i, a \in C_i} \widetilde{e}_a^{\xi_{h,a}\binom{n}{i}} \prod_{i, a \in L_i} \widetilde{e}_a^{\xi_{h,a}\binom{n}{i}}.$$

These are exactly the core and retained-petal factors in (12.3)–(12.5). Using the function from Lemma A23.4, set

$$\chi_h(n) = F_h(a_h(n)\Lambda, c_h(n)\Lambda^0), \qquad \widetilde{\chi}_h(n) = \widetilde{F}(b_h(n)\widetilde{\Gamma}).$$

Proposition A23.5. *Under these hypotheses, every component of $\chi_h \otimes \overline{\widetilde{\chi}_h}$ is a nilsequence of degree–rank strictly below (m, R). The representing filtered nilmanifold is fixed independently of h and the coefficients; its Lipschitz bounds depend only on the fixed data and original bounds.*

Proof. For $d > 0$ put $\bar{r} = \max(r, 1)$ and let $\sigma(d, \bar{r})$ denote the successor of $(d, \bar{r})$ in the positive degree–rank list. In the product $G \times G^0 \times \widetilde{G}$, let $H_{(d,r)}$ be the connected subgroup generated by

$$J_{(d,r)} = G_{\sigma(d,\bar{r})} \times G^0_{(d,\bar{r})} \times \widetilde{G}_{\sigma(d,\bar{r})},$$
$$D(U_{(d,\bar{r})}) = \{(\phi(u), 1, q(u)) : u \in U_{(d,\bar{r})}\}.$$

Set $H_0 = H_{(1,0)}$. This specifies the zero group; it is not an arbitrary full product. All generator subgroups are rational. For the diagonal image, the image of the appropriate lattice generates its

real subgroup by the Hall coordinates and $\phi(\Gamma) \subseteq \Lambda$. The rational-generated subgroup lemma of Section A18 applies.

Nesting is direct. For positive indices i, j, commutators of two diagonal generators lie in $D(U_{i+j})$. A commutator involving a J_i generator has its first and third coordinates in the successor group at $i+j$, since $\sigma(i)+j \succeq \sigma(i+j)$, with the convention replacing rank zero by one. Its middle coordinate is in G^0_{i+j}. The same argument works for two J generators. Taking generated Lie algebras and BCH gives $[H_i, H_j] \subseteq H_{i+j}$. At zero use $H_0 = H_{(1,0)} = H_{(1,1)}$ and nesting. Thus this is a taut rational degree–rank filtration. Its top group is exactly

$$H_{(m,R)} = D(T), \tag{185}$$

because every group in $J_{(m,R)}$ is trivial.

We next check actual polynomial membership, including constant coefficients and the distinction between a horizontal vector and its torus class. Write $a_h(0) = v_h \lambda_h$ with v_h in a fixed compact representative set and $\lambda_h \in \Lambda$. Replace $a_h(n)$ by $v_h^{-1} a_h(n) \lambda_h^{-1}$, and replace $F_h(x, z)$ by $F_h(v_h x, z)$. Do the same in the second coordinate. Values of χ_h are unchanged, the new sequences start at the identity, and Lipschitz bounds stay uniform. Since G is taut, conjugation by v_h acts trivially on every $G_{(i,1)}/G_{(i,2)}$: its commutator is in $G_{(i+1,1)} \subseteq G_{(i,2)}$. The horizontal torus conditions remain unchanged. Equation (184) remains unchanged since the top group of the taut group is central. Tautify the original targets before constructing the total-frequency representation, using Section A17 and then [141, Lemma 9.12]; no conjugated, potentially irrational replacement of ϕ is being silently used here.

For each i, choose $\lambda_{h,i} \in \Lambda \cap G_i$ lifting the horizontal lattice difference between the new a_h and $\phi(E_{h,i})$. Such a lift exists by the definition of the horizontal lattice. Multiply $a_h(n)$ on the right by the inverse of $\prod_i \lambda_{h,i}^{\binom{n}{i}}$. This does not change its orbit. The factor is polynomial by the Taylor criterion [141, Lemma B.9]. The horizontal Taylor map is a homomorphism [141, Definition 9.6 and the argument following it], so now the horizontal equalities hold in the vector groups themselves.

Taylor-expand $a_h(n) = \prod_{i \geq 1} a_{h,i}^{\binom{n}{i}}$. Then $a_{h,i} \phi(E_{h,i})^{-1} \in G_{(i,2)}$. Hence $(a_{h,i}, 1, q(E_{h,i})) \in H_{(i,1)}$. The Taylor criterion and product closure [141, Lemma B.9 and Corollary B.4] show that

$$n \mapsto \left(a_h(n), 1, \prod_i q(E_{h,i})^{\binom{n}{i}} \right)$$

is polynomial for H. Both its last coordinate and b_h have the same horizontal Taylor coefficients, since multiplication of factors is addition in each horizontal quotient and discarded generators have image one. Their ratio is polynomial and all its horizontal Taylor coefficients vanish. Its Taylor coefficient at index i therefore lies in $\widetilde{G}_{(i,2)}$. These groups, with zeroth group $\widetilde{G}_{(1,2)}$, form a degree filtration by the same commutator check. The ratio starts at one; Lemma B.9 now shows that it is polynomial for this smaller filtration. Embedded in the third coordinate it is polynomial in H. The second-coordinate map $(1, c_h(n), 1)$ is also polynomial in H. Product closure proves that the exact triple $(a_h(n), c_h(n), b_h(n))$ is polynomial in H.

Restrict $F_h \otimes \widetilde{F}$ to $H/(H \cap (\Lambda \times \Lambda^0 \times \widetilde{\Gamma}))$. On $D(t)$ its vertical multiplier is $\mathrm{e}(\eta(t) - \widetilde{\eta}(q(t))) = 1$ for every $t \in T$. It therefore descends through the central rational subgroup (185). The central quotient and lattice argument in Section A18 applies here: this kernel is contained in every nontrivial indexed subgroup, unlike the earlier kernel G^*. The top group becomes trivial, so the quotient has degree–rank strictly below (m, R). Restriction, quotient charts, and the compact translations cost only constants from the fixed geometry. This proves the proposition. □

When $R = 1$, strictly below $(m, 1)$ means ordinary degree at most $m - 1$. For $R > 1$ it means degree–rank at most $(m, R - 1)$, not necessarily degree at most $m - 1$. The first sentence of the published proof of Lemma 12.1 claims the latter unnecessarily strong conclusion. The proved former conclusion is exactly what the following rank-induction step consumes.

A23.6 Uniformity, limits, and the remaining Section 12 interfaces

Every group, ideal, Hall list, lattice and quotient above depends only on m, R and the fixed finite labelled dimension data. All formulas are finite polynomials, Lie-group maps, or choices from fixed lattice and compact representative sections. A fixed ordering chooses the first lift of a lattice class; a fixed finite atlas chooses a bounded representative. Applying these standard choices to representative sequences gives limit maps. The derivative and vertical identities are required on their full ultrapower domains, as in Section A17; identities only at standard points are insufficient. Thus no bound depends on N, on h, or on the real polynomial coefficients. The character is fixed before choosing a vertical test function. No assertion of a bound uniform over unbounded characters is made. Taking the dense subfamily with fixed geometric data and character is the separate selection step in Lemma 10.10 and Section 12, using Appendix A.

The ordinary degree filtration used to build the final semidirect product is $\widetilde{G}_d = \widetilde{G}_{(d,0)}$. The petal ideal and scalar-action identities proved above give precisely the normality, commutativity, and integral stability used on pp. 1312–1314. In particular a bracket with two petal entries is zero at *every* weight. This assertion would fail if one quotiented only by the old top-weight kernel.

There is a separate unit-norm issue at the cutoff in (12.11) and the paragraph following it. A compactly supported scalar cutoff times a unit-vector function is not itself unit-vector-valued. Here is an exact completion for that use. On the final compact nilmanifold choose a smooth vertical-equivariant local unit vector V on the torus-coordinate chart, agreeing with $\widetilde{F}(g\widetilde{\Gamma})$ there. Choose a smooth function θ of the torus coordinates equal to zero on the closed inner box and equal to $\pi/2$ outside a slightly larger box still inside the chart. These coordinates are unchanged by the top vertical action. Construct a global unit-vector function V_0 of the same vertical character by Lemma A23.4, applied to the final top torus. Then

$$(\cos\theta\, V,\ \sin\theta\, V_0)$$

is smooth, unit-vector-valued, has the required vertical character, and equals $(V, 0)$ on the inner box. The first component extends by zero, since its support stays inside the chart. Choose θ constant near both boundaries, so both extensions are smooth. This preserves the required values for the dense parameter set after shrinking to the inner box. Its geometry and bounds are fixed before N. The final semidirect-product filtration and polynomial orbit must still be checked at their displayed definitions; they are not consequences of the character descent alone.

A23.7 Related applications of the quotient construction

The failure mechanism is restriction of a generated ideal or filtration to too few weights, followed by an unjustified quotient or lattice identification. The audit scope is the universal degree–rank definitions and the consumers of their commutator/Taylor quotients in Sections 9–12. It includes the zero-index and repeated normalization assertions in those passages.

1. Definition 9.1, p. 1279: use the fixed $R = r_*$ in the top relation, and count the ℓ actual entries. Equations (179) and (180), including all higher weights, give the intended object. Lemmas A23.1 and A23.2 prove its filtration and lattice, rather than inheriting typographical indices.

2. Definition 9.6, pp. 1281–1282: use $\mathrm{Horiz}_i = G_{(i,1)}/G_{(i,2)}$, and $G_{(1,2)} = G_{(2,0)}$. The vector/torus distinction is discharged in Proposition A23.5, not by replacing a congruence by equality.

3. Definition 9.11 and Lemma 9.12, pp. 1284–1287: their G_i means $G_{(i,0)}$. First tautify, then choose lattice generators and construct the filtered map. This order preserves a genuine lattice map; the proof does not assume arbitrary real conjugation preserves the lattice.

4. Lemma 9.14 and Corollary 9.15, pp. 1287–1288, are explicitly stated there not to be used later. Their comparison-filtration argument has the same zero-index and horizontal-lattice issues. The construction in Proposition A23.5 supplies that mechanism with q trivial when the character is zero. Their broader reuse requires this version, not the unqualified original paragraph.

5. Lemma 10.8, p. 1297: the vertical domain is $U_{(m,R)}$, not the larger ordinary degree group U_m. This unit consistently defines vertical frequencies only on that top group. The source argument's other linear-algebra steps are not newly reproved here.

6. Lemma 10.10 and Theorem 11.1, pp. 1299–1307: use the final core/linear/independent decomposition and the exact two vanishing assertions. The bias and independence premises remain named upstream inputs. The repeated (iii),(iv) of Theorem 11.1 are redundant. Reordering linear and independent petals requires relabelling every associated index, not changing which generators are annihilated.

7. Quotient construction, p. 1309: Theorem A23.3 proves the full kernel, every filtered quotient, and the top-lattice saturation needed for integral vertical descent. The kernel is not asserted central.

8. Lemma 12.1, pp. 1309–1311: Proposition A23.5 proves the rank drop with actual constant normalization, horizontal lattice lifts, exact polynomial orbit and conjugate vertical cancellation. The degree-only strengthening is not used or claimed.

9. Petal actions, pp. 1311–1314: the proved abelian ideal and its individual-petal splitting verify (12.8)–(12.10), including zero and negative scaling parameters and integral lattice stability. The displayed final multidegree filtration, chart gluing, and polynomial orbit remain explicit downstream construction checks for LP-00/LP-07. The unit-norm cutoff completion above removes the identified local norm obstruction.

The list concerns this quotient mechanism and its stated applications; it does not assert a classification of all possible errors in the cited source.

A24 The Section 12 semidirect-product join

This section supplies the final geometric construction on published GTZ pp. 1313–1315 from the explicit petal quotient and comparison interface of LP04. The analytic frequency input, rank iteration and other inverse theorem interfaces remain separate. We neither invoke the final inverse theorem to prove an intermediate construction nor import the unpublished erratum as a theorem. All quotients below are right lattice quotients.

A24.1 Exact input and output

Fix the data of LP04: integers $m \geq 1$, $1 \leq R \leq m$, fixed finite sets of core and retained petal letters with positive integer weights, and the weighted free nilpotent quotient

$$G = \exp \mathfrak{g}, \qquad \mathfrak{g} = \mathfrak{c} \oplus \mathfrak{p}.$$

The homogeneous relations discard all words of weight greater than m, all weight-m words of length greater than R, and all words containing at least two petals. The last relation holds at every weight. Thus $\mathfrak{p}$ is an abelian ideal. Write $P = \exp \mathfrak{p}$, $G_j = G_{(j,0)}$, $P_j = P \cap G_j$, $G_0 = G_1 = G$, and $P_0 = P_1 = P$. Let Γ be the lattice generated by the exponentials of the letters, and $\Lambda = P \cap \Gamma$. LP04's Hall-coordinate lemma supplies the exact integer lattice coordinates and rationality of G_j, P_j. Put $T = G_{(m,R)}$, and let $\eta : T \to \mathbb{R}$ be the descended continuous character integral on $T \cap \Gamma$.

There are ℓ retained petal letters. For $t \in V = \mathbb{R}^\ell$, let $M_t : P \to P$ scale the summand with unique petal letter a by t_a. We use the following proved LP04 identities, for all real parameters:

$$M_t(p)M_u(p) = M_{t+u}(p), \qquad M_t(gpg^{-1}) = gM_t(p)g^{-1}, \qquad M_t(P_j) \subseteq P_j. \tag{186}$$

For integral z, $M_z(\Lambda) \subseteq \Lambda$. These are endomorphisms; M_0 is not asserted to be invertible.

The family to be represented has the form

$$b_h(n) = g_0(n)M_{\{\alpha h\}}(g_1(n)), \qquad \widetilde{\chi}_h(n) = \widetilde{F}(b_h(n)\Gamma). \tag{187}$$

Here $g_0 : \mathbb{Z} \to G$ and $g_1 : \mathbb{Z} \to P$ are polynomial for the ordinary weight filtrations; $\alpha \in \mathbb{R}^\ell$ is unrestricted. Braces denote one fixed half-open fundamental cube of $V/\mathbb{Z}^\ell$, for example $[-1/2, 1/2)^\ell$. The factors used in GTZ are ordered products $e_a^{\xi_a \binom{n}{i}}$ for core letters of weight i, and the commuting product $\prod_{a \text{ petal}} e_a^{\beta_a \binom{n}{w(a)}}$ for g_1. Core coefficients are independent of h. Identifying an arbitrary input family with this specific factorization is an upstream obligation; it is not inferred from the names of the factors. For the actual coefficients in (12.3)–(12.4), the algebraic identification is exact: $M_t(\exp(uX_a)) = \exp(t_a u X_a)$, and M_t is a homomorphism on the abelian petal group. Hence applying $M_{\{\alpha h\}}$ to the displayed g_1 gives precisely the product in (12.4). No reordering past a core factor is involved. The hypotheses that produce these coefficient data are still the separate analytic input.

The function $\widetilde{F}$ is a choice made below, before applying the LP04 comparison. It is not an arbitrary previously fixed function having only T-equivariance. This order of choices is necessary.

Theorem A24.1. *One can choose $\widetilde{F}$ on the fixed G/Γ, with unit vector norm and the required T-frequency η, such that the following holds. There is a fixed connected simply connected nilpotent group H, a lattice Δ, and a rational taut $\mathbb{N}^2$-filtration supported on the downset below $(1, m)$. The nilpotency class is at most $m + 1$. There is a finite list of fixed smooth unit-vector functions on H/Δ, all with the same top vertical character, with fixed finite Lipschitz bounds.*

For every α, g_0, g_1 as above and every finite nonempty parameter set $S \subseteq \mathbb{Z}$, some $S' \subseteq S$ has $|S'| \geq |S|/J$, where the fixed integer J depends only on ℓ, and one function F' in this list satisfies

$$F'(O(h, n)) = (\widetilde{\chi}_h(n), 0) \quad (h \in S',\ n \in \mathbb{Z}). \tag{188}$$

Here O is a polynomial orbit of multidegree at most $(1, m)$. The zero padding is a fixed isometric coordinate inclusion, so projection onto the original coordinates gives literal equality with $\widetilde{\chi}_h$. No scalar component is divided by.

All constants, manifolds and functions are independent of α, the polynomial coefficients, S and N. The conclusion transfers to the full internal ultrapower domains and preserves a dense set of limit parameters. It supplies this construction and its LP04 comparison join; it does not by itself establish the other premises of Theorem 7.1 or GI.

A24.2 Extend the character before choosing the test

Lemma A24.2. *The character η extends to a continuous character $\bar{\eta} : G_m \to \mathbb{R}$ integral on $G_m \cap \Gamma$.*

Proof. The weight-m group is central. Its surviving Hall commutators have weight m and lengths at most R. At this weight the logarithm of each group commutator is exactly its Lie bracket: every correction has greater weight and vanishes. By LP04's exact Hall coordinates, these logarithms form a real basis and their integer span is precisely the logarithm of $G_m \cap \Gamma$. The subset of length R is a basis of T, and its integer span is $T \cap \Gamma$. Assign to those basis elements the values of η, and assign zero to all remaining weight-m basis elements. These assigned values are integers. Linear extension on this central vector group proves the assertion. The construction depends only on the fixed Hall ordering and η. □

The central torus $G_m/(G_m \cap \Gamma)$ acts freely on G/Γ. For any integral character of this torus there is a smooth equivariant unit-vector function of finite dimension. Here is the construction we use twice. Choose finitely many local smooth sections of the torus quotient and real smooth functions a_j supported in their domains with $\sum a_j^2 > 0$ everywhere. Such functions can be obtained by smooth Euclidean bumps in a finite collection of relatively compact local charts whose smaller charts cover the compact base. Replace a_j by $a_j/(\sum a_i^2)^{1/2}$. In the jth local trivialization (x, z) define the jth component to be $a_j(x)\mathrm{e}(\bar{\eta}(z))$ and extend it by zero. The support lies inside its chart. Thus each component is globally smooth, has the stated vertical transformation law, and the squared vector norm is $\sum a_j^2 = 1$. Local sections can be seen directly: choose a linear complement to the central Lie subspace. Centrality gives $\exp(X + Z) = \exp X \exp Z$, so the complement gives smooth local lifts from the group quotient. The image lattice is discrete by the Hall coordinates with the central weight-m coordinates deleted. Small coordinate neighborhoods with disjoint lattice translates give the same local lifts on the compact quotient.

Choose this function as $\widetilde{F}$. It has the original frequency η on T and now also the required frequency on all of G_m. Every cancellation in LP04's comparison uses only the restriction to T, so this choice preserves that proof. The comparison does not require a specific previously selected $\widetilde{F}$.

Adverse test. T can be strictly smaller than P_m. For example, with $m = R = 2$ and just one petal generator of weight two, T is trivial whereas $G_m = P \simeq \mathbb{R}$. The unit-vector function $(\cos 2\pi x, \sin 2\pi x)$ on $\mathbb{R}/\mathbb{Z}$ satisfies the vacuous T-law but has no scalar vertical character for all translations. Consequently T-equivariance alone cannot be cited as the final multidegree frequency. The integral extension and new choice above are the repair, not an assertion that every old test already had the stronger property.

A24.3 Both semidirect products and their signs

Set $K = G \ltimes P$ with multiplication

$$(g, p)(g', p') = (gg',\ (g')^{-1} p g'\, p'). \tag{189}$$

Associativity follows either directly from conjugation or from the injective homomorphism $(g, p) \mapsto (g, gp)$ into $G \times G$. Its image is the closed subgroup of pairs agreeing modulo P. The inverse is $(g^{-1}, gp^{-1}g^{-1})$. Its underlying manifold is $G \times P$, so K is connected and simply connected.

Define $\rho_t(g, p) = (gM_t(p), p)$. To check multiplication, applying ρ_t to (189) gives first component $gM_t(p)g'M_t(p')$, by (186). Multiplying the two images has the same first component. Its second is

$$M_t(p')^{-1}(g')^{-1} p g' M_t(p') p' = (g')^{-1} p g' p',$$

since all terms surrounding p' are in the abelian group P. Thus ρ_t is a homomorphism. Further, $\rho_t\rho_u = \rho_{t+u}$ and $\rho_0 = \mathrm{id}$, so ρ_t is an automorphism with inverse ρ_{-t}. This conclusion uses the additive shear action, not invertibility of M_t.

Set $H = V \ltimes_\rho K$ with the right-action convention

$$(t,k)(t',k') = (t+t', \rho_{t'}(k)k'). \tag{190}$$

In three coordinates this reads

$$(t,g,p)(t',g',p') = (t+t',\ gM_{t'}(p)g',\ (g')^{-1}pg'p').$$

The action identity proves associativity, and the inverse of (t,k) is $(-t, \rho_{-t}(k^{-1}))$. Smoothness follows from the polynomial BCH formulas and the linear maps on $\mathfrak{p}$. The manifold $V \times G \times P$ is connected and simply connected.

For a complete nilpotence and filtration check, use Lie coordinates $(t,X,Y) \in V \oplus \mathfrak{g} \oplus \mathfrak{p}$. The bracket is

$$[(t,X,Y),(t',X',Y')] = \big(0,\ [X,X'] + M_{t'}Y - M_tY',\ [X,Y'] + [Y,X']\big). \tag{191}$$

Indeed the embedding of K has derivative $(X,Y) \mapsto (X, X+Y)$, giving the two bracket terms of K. The derivative of ρ_t is $(X,Y) \mapsto (X + M_tY, Y)$. Conjugation by $(t,1)$ on K is ρ_{-t} in convention (190), giving the signs displayed.

Assign bidegree $(0,i)$ to core words of weight i in the X copy, $(1,i)$ to petal words of weight i in the X copy, $(0,i)$ to petal words of weight i in the Y copy, and $(1,0)$ to V. Every nonzero term of (191) has the sum of the input bidegrees. For example, $[V,Y_i]$ lies in the first petal copy of degree $(1,i)$, and brackets involving two petal words vanish. All weights are at most m and the first degree is at most one. Each generator has positive total degree, so every bracket of length greater than $m+1$ vanishes. This proves nilpotence of H of class at most $m+1$.

A24.4 The complete filtration and the lattice

Define $H_{a,j} = 1$ for every $a > 1$ and every $j \geq 0$. The remaining positive-j formulas, together with the zero-index groups, are

$$\begin{aligned} &H_{0,0} = H, && H_{a,j} = 1 \quad (a>1),\\ &H_{1,0} = \{(t,p,1) : t \in V, p \in P\}, && H_{1,j} = \{(0,p,1) : p \in P_j\},\\ &H_{0,j} = \{(0,g,p) : g \in G_j, p \in P_j\}. \end{aligned}$$

For $j > m$ these groups are trivial. These definitions are also the sums of the homogeneous Lie spaces whose bidegrees dominate the index. Thus the groups are connected, descending and normal; brackets lie in the sum-index subgroup by (191), and the BCH commutator terms do also. To spell out the boundary cases, two $H_{1,0}$ elements commute, $[H_{1,0}, H_{0,j}] \subseteq H_{1,j}$, and $[H_{1,i}, H_{0,j}] \subseteq H_{1,i+j}$. Brackets with index zero are covered by normality. The groups $H_{1,0}$ and $H_{0,1} = K$ generate H, which proves tautness. In particular

$$Z_* := H_{1,m} = \{(0,p,1) : p \in P_m\} \tag{192}$$

is central. Importantly $H_{0,1}$ contains the full $G_1 = G$ in its first group coordinate, not merely P.

Let $\Lambda_K = \Gamma \ltimes \Lambda$ with law (189). Since P is normal, Γ normalizes Λ. Therefore this is a subgroup, including inverses. For $z \in \mathbb{Z}^\ell$, both ρ_z and ρ_{-z} preserve Λ_K, by the integral LP04 action. Consequently

$$\Delta = \mathbb{Z}^\ell \ltimes_\rho \Lambda_K = \{(z,\gamma,\lambda) : z \in \mathbb{Z}^\ell, \gamma \in \Gamma, \lambda \in \Lambda\}$$

is a subgroup of H. It is discrete in the product manifold topology.

Here is explicit cocompactness. Right-multiply (t,k) by $(-z,1)$, where z reduces t into $[0,1]^\ell$. This gives $(t-z,\rho_{-z}k)$. For an arbitrary $(g,p)\in K$, choose $\gamma\in\Gamma$ with $g\gamma^{-1}$ in a fixed compact representative set for G/Γ. Right multiplication by $(\gamma^{-1},1)$ gives $(g\gamma^{-1},\gamma p\gamma^{-1})$. Now reduce the second coordinate into a fixed compact representative set for P/Λ by right multiplication by $(1,\lambda^{-1})$. The resulting compact product set represents every right coset of Λ_K. Applying this after the t reduction gives a compact product set representing every Δ coset. No bound on the intermediate conjugate or on z is needed.

The same argument inside $G_j \ltimes P_j$, using the already proved lattices $\Gamma\cap G_j$ and $\Lambda\cap P_j$, proves rationality of $H_{0,j}$. The intersections for $H_{1,j}$ are exactly $\Lambda\cap P_j$, and for $H_{1,0}$ exactly $\mathbb{Z}^\ell\times\Lambda$. These are cocompact by LP04 and the product construction. This checks all filtered lattice intersections, including zero and trivial groups. The topology, filtration, lattice and smooth metric on the compact quotient are fixed before any orbit coefficients are chosen.

A24.5 Polynomial orbit and exact reduction

Define

$$O(h,n)=(0,g_0(n),g_1(n))(\alpha h,1,1)\Delta. \tag{193}$$

We verify polynomiality in the full group before taking the quotient. The source filtration on $D=\mathbb{Z}^2$ is $D_{0,0}=\mathbb{Z}^2$, $D_{1,0}=\mathbb{Z}\times\{0\}$, $D_{0,1}=\{0\}\times\mathbb{Z}$, and $D_{a,j}=\{0\}$ at every other nonzero index. It is an $\mathbb{N}^2$-filtered abelian group. The embeddings $g\mapsto(0,g,1)$ and $p\mapsto(0,1,p)$ are homomorphisms taking the weight-j groups to $H_{0,j}$. Each independent-of-h factor is therefore polynomial on $\mathbb{Z}^2$ with its coordinate multidegree filtration: h differences vanish, and n differences have the asserted $H_{0,j}$ membership. The map $(h,n)\mapsto(\alpha h,1,1)$ is a homomorphism in h with values in $H_{1,0}$, has zero second h difference, and is constant in n. Thus it too is polynomial for this same filtration. Published Corollary B.4, GTZ p. 1334, says that polynomial maps between the same index-filtered groups are closed under pointwise multiplication. Here both source and target are the precise $\mathbb{N}^2$-filtered groups just checked. Applying it to these three factors proves polynomiality of (193), including all mixed derivatives and all integer inputs. This is the only external polynomial-product theorem used in this join.

For the explicit inputs preceding Theorem A24.1, their univariate polynomiality can also be checked before this use. An r-fold difference of $e_a^{c\binom{n}{i}}$ is $e_a^{c\Delta^r\binom{n}{i}}$ because all its values lie in one abelian one-parameter subgroup. It vanishes for $r>i$; for $r\le i$ it lies in $G_i\subseteq G_r$. This holds for arbitrary integer increments, including negative ones. Products are polynomial by the same Corollary B.4; for g_1 all factors already commute in P. No Taylor expansion with a potentially unjustified centrality hypothesis is imported for this check.

Write $z_h=\alpha h-\{\alpha h\}\in\mathbb{Z}^\ell$. Formula (190) gives a representative of (193) as

$$(\alpha h, g_0(n)M_{\alpha h}(g_1(n)), g_1(n)).$$

Right multiplication by $(-z_h,1,1)\in\Delta$ changes this to

$$(\{\alpha h\}, g_0(n)M_{\{\alpha h\}}(g_1(n)), g_1(n)). \tag{194}$$

The equality follows from the additive identity in (186). It is exact for every h,n; there is no illegal commutation of a fractional-part operation with a nonabelian product.

A24.6 Fixed charts, unit completion and uniform bounds

Projection onto t gives a smooth map $\pi : H/\Delta \to V/\mathbb{Z}^\ell$. If a box $U \subset V$ has width less than one in each coordinate, projection is injective on U. On the corresponding open torus box there is an exact smooth trivialization

$$U \times K/\Lambda_K \longrightarrow \pi^{-1}(U \bmod \mathbb{Z}^\ell), \qquad (t, k\Lambda_K) \longmapsto (t,k)\Delta. \tag{195}$$

Indeed two representatives in U differ by an element of Δ only if their integer t displacement is zero; their k coordinates then differ precisely by right multiplication in Λ_K. Surjectivity follows by reducing the t coordinate to its unique lift in U. The formulas are smooth on this chart.

Choose finitely many fixed closed boxes I_j covering the closed fundamental cube $[-1/2, 1/2]^\ell$, with fixed slightly larger boxes U_j of widths less than one. They may extend beyond the cube's boundary; each U_j still projects injectively. Choose a smooth function θ_j on the torus which is zero on I_j modulo integers and $\pi/2$ outside a smaller open box with closure contained in U_j. It can be constructed by a fixed Euclidean bump in U_j, equal to one on I_j, followed by multiplication of its complement by $\pi/2$. The chart margins are fixed positive real numbers, not functions of N. At least one box contains $\{\alpha h\}$ for at least $|S|/J$ of the parameters, where J is the fixed number of boxes. Fix that box and call this subfamily S'.

On its chart define

$$W(t, (g,p)\Lambda_K) = \widetilde{F}(g\Gamma).$$

The projection $K/\Lambda_K \to G/\Gamma$ is well-defined and smooth, since the first coordinate of (189) is a homomorphism and maps Λ_K into Γ. Thus W is a smooth local unit vector. Define a character on the actual group (192) by

$$\eta'(0,p,1) = \bar{\eta}(p) \qquad (p \in P_m).$$

It is integral on $Z_* \cap \Delta$ by Lemma A24.2. Left multiplication by $(0, p_*, 1) \in Z_*$ changes the chart's first group coordinate to $p_* g$, keeps t fixed, and hence multiplies W by $\mathrm{e}(\bar{\eta}(p_*))$. This uses equivariance on P_m, not merely on $T \cap P$.

The torus $Z_*/(Z_* \cap \Delta)$ acts freely on H/Δ: centrality makes the stabilizer at every point exactly $Z_* \cap \Delta$. It is compact by the rationality already proved. The quotient by Z_* has coordinates $V \times (G/P_m) \times P$, and the image of Δ has coordinates $\mathbb{Z}^\ell \times q(\Gamma) \times \Lambda$. Here $q(\Gamma)$ is discrete by LP04's Hall-coordinate quotient, so the image lattice is discrete and the quotient base is compact. Central complements and small coordinate neighborhoods give local lifts exactly as in Section 2. The finite local-section construction from that section therefore gives a global smooth unit vector W_0 with frequency η'. Set

$$F_j' = (\cos(\theta_j \circ \pi) W, \sin(\theta_j \circ \pi) W_0). \tag{196}$$

The first component is extended by zero outside the chart. It vanishes on a neighborhood of the chart boundary, so this extension is smooth. Outside the chart the second coefficient is one. Inside, the squared norm is $\cos^2\theta_j + \sin^2\theta_j = 1$. Both components have the same vertical frequency because Z_* fixes π. Therefore F_j' is a genuine unit-vector nilcharacter test, including outside the box.

For $h \in S'$, the representative (194) uses the unique lift $\{\alpha h\} \in I_j$. Its cutoff angle is zero, so (196) gives (188) for every integer n. The second component is exactly zero there, not an approximation.

Every chart used has compact closure in a fixed larger chart, and its fiber K/Λ_K is compact. The displayed projections and their derivatives are bounded on these sets. The finite bump functions, $\widetilde{F}$, W_0, and the fixed metrics have bounded derivatives. Taking the maximum over the finite list proves

a finite Lipschitz bound depending only on the fixed groups, character and charts. Unbounded polynomial coefficients change only the orbit; they do not enter these derivative bounds. This also proves fixed finite dimension. If $\ell = 0$ the torus is a point, use one chart and $\theta = 0$; all arguments still apply. Trivial top groups or zero characters likewise cause no exception. This completes the ordinary proof of Theorem A24.1.

A24.7 Internal domains and the LP04 consumer

All groups, charts, test functions and characters above are standard objects after fixing the selected dimension data and character. All algebraic identities hold for every real parameter and every group point. Taking representative sequences gives internal polynomial maps, including for unbounded limit coefficients. The standard tests extend to the whole ultrapower of the compact quotient. Their vertical laws then hold for every limit element of Z_* and every limit point, by transfer of the actual universal identity. We are not inferring all-limit equivariance from an identity checked only on standard points.

Finite pigeonhole over the fixed list of boxes preserves positive standard density of a limit parameter family. Equivalently apply the ordinary cardinality inequality to representatives and pass to the ultralimit. After choosing the resulting fixed box, the nilcharacter $\chi(h, n) = F'_j(O(h, n))$ is defined on all of ${}^*\mathbb{Z}^2$, not only on the dense subfamily. Its exact restriction is the fixed isometric inclusion of $\widetilde{\chi}_h$ for all limit integers n there.

Choose $\widetilde{F}$ by Section 2 before invoking LP04's comparison proposition. That proposition's top cancellation is unchanged because $\bar{\eta}|_T = \eta$. It produces the same strictly lower degree–rank comparison for $\chi_h \otimes \overline{\widetilde{\chi}_h}$. For clarity, unit norm gives the exact component recovery identity

$$(\chi_h)_a = \sum_b (\widetilde{\chi}_h)_b \big((\chi_h)_a \overline{(\widetilde{\chi}_h)_b}\big).$$

On the selected subfamily, replace $(\widetilde{\chi}_h)_b$ by the corresponding component of the single $\chi(h, n)$. The terms in parentheses are exactly the lower degree–rank sequences supplied by LP04. The sum is finite and its coefficients are fixed; no division or nonvanishing of an individual component is used. The zero padding in (188) changes none of these terms.

At $R = 1$, lower degree–rank means degree at most $m - 1$. At $R > 1$ the subsequent rank iteration remains an upstream/downstream assembly obligation; we do not incorrectly replace it by an immediate ordinary degree drop. Likewise the frequency decomposition leading to (187), the initial correlation, and the remaining Theorem 7.1 induction premises are not established by this geometric join.

A24.8 Bounded related-use and consistency audit

The search scope is the Section 12 construction and its immediate LP04 comparison and Theorem 7.1 consumers, published pp. 1308–1315. We also check the polynomial-product supplier B.4 at p. 1334. This is an authored bounded audit, not a claim of independent source-first coverage of GI.

1. The smaller degree–rank character domain versus the ordinary top group is repaired by the integral extension and the new test choice. The repeated final frequency on p. 1315 is explicitly restricted to P_m and uses $\bar{\eta}$. A favorable quotient-character review alone does not supply this extension.

2. The illustrative $H_{0,1}$ on p. 1312 restricts its first group coordinate to the petal subgroup, although the orbit contains the core $e_2^{\beta n}$. For nonzero generic β its n difference is outside that

alleged group. The general p. 1314 definition uses full G_j; we use and prove that definition, not the defective illustration.

3. Zero or negative scaling parameters do not make every M_t invertible. The actual invertible automorphisms are the shears ρ_t, whose inverse and integer lattice preservation are proved here.

4. The lattice in p. 1314 is a subgroup of H, not of the original G. Every filtered lattice intersection and the exact right-coset reduction are checked above; neither discreteness alone nor a real endomorphism establishes the claim.

5. The cutoff on pp. 1312 and 1315 cannot alone be a unit-vector test or guarantee equality on the selected box. Fixed inner boxes, positive chart margins and (196) supply unit norm, literal equality and coefficient-independent Lipschitz bounds.

6. In particular a variable box of arbitrarily small width would not supply a fixed Lipschitz bound. The predetermined finite cover removes this possible parameter dependence at a controlled density loss.

7. The p. 1308 claim that pigeonhole puts the fractional parts in a short interval around zero is not valid for an arbitrary dense family. For $\alpha = 1/2$ and odd h, our fractional parts are all $-1/2$. Our fixed cover allows a box at any location, including across the chosen fundamental-cube boundary, and does not need a near-zero assertion.

8. Formula (12.11) is checked as right multiplication by an actual lattice element; the displayed identities do not silently change the object or introduce a different representative in the local test.

9. A lower rank comparison is not automatically a lower ordinary degree comparison. The consumer paragraph above preserves the exact LP04 conclusion and leaves the rank iteration to its actual proof.

Earlier printed statements and earlier frozen project packets are retained; this proposal does not relabel them as having contained the new argument. Any other current copies of the smaller-domain frequency claim or the illustrative filtration must be corrected or marked historical when this proof is incorporated. The attached occurrence record identifies these specific copies and the new canonical assertions for review.

A25 Degree conversion and the finite rank iteration

This section proves the degree conversions and the assembly of the rank steps from the precisely stated one-step geometric/frequency interface. The analytic sunflower and commutator hypotheses are supplied in Section A22. Every internal nilcharacter below has exact unit norm and vertical equivariance for all limit elements under the natural ultrapower interpretation.

A25.1 The actual degree-to-rank conversion

Let G/Γ be a connected simply connected nilpotent Lie group with lattice and rational degree filtration $(G_i)_{i\geq 0}$ supported on $0 \leq i \leq m$. Apply accepted bounded tautification first, so $G_0 = G_1 = G$ and polynomial orbits may be normalized at zero without changing the represented sequence. Write $\mathfrak{g}_i = \log G_i$. All the vector spaces here are rational for the rational structure determined by Γ.

For $(d,r) \in \mathrm{DR} = \{(d,r) \in \mathbb{N}^2 : 0 \le r \le d\}$ with $d > 0$ define $\mathfrak{h}_{(d,r)}$ to be the real span of all iterated Lie brackets of length $\ell \ge 1$ with entries $X_j \in \mathfrak{g}_{i_j}$, $i_j \ge 1$, for which

$$w := \sum_{j=1}^{\ell} i_j > d, \qquad \text{or} \qquad w = d \text{ and } \ell \ge \max(r,1). \tag{197}$$

Take every bracketing and set $\mathfrak{h}_{(0,0)} = \mathfrak{g}$, $H_{(d,r)} = \exp \mathfrak{h}_{(d,r)}$. Only finitely many nonzero brackets occur, since their weight exceeds m when they can no longer contribute and every entry has positive weight.

Lemma A25.1. *These are connected rational groups forming a taut degree–rank filtration supported on (m,m). For each $d > 0$, $H_{(d,0)} = H_{(d,1)} = G_d$. Hence the original degree-polynomial orbits are exactly the degree–rank polynomial orbits for this refinement. A degree-m nilcharacter is a degree–rank (m,m) nilcharacter with the same function and with the top character restricted.*

Proof. All bracket spans in (197) are rational, because brackets of rational vectors are rational. Bracketing with $\mathfrak{g}_1$ raises weight by at least one, so each span is an ideal of $\mathfrak{g}$. It is therefore a Lie subalgebra, and its exponential is a connected normal subgroup, which is closed by the global exponential homeomorphism. Choose a rational Mal'cev basis for G/Γ compatible with its rational structure. Each bracket span has a rational basis in these coordinates. Its lower central series is rational, since brackets of rational vectors are rational. Choose one standard Q bounding the finitely many rational structure constants and basis coefficients. Green–Tao, *Annals of Mathematics* 175 (2012), Definition 2.5 and Proposition A.10, applied to this subgroup with its lower central filtration, supplies a Mal'cev basis relative to its actual intersection with Γ. That intersection is therefore discrete and cocompact. This application uses Lie-algebra rationality in Definition 2.5; it does not assume the lattice conclusion.

The lexicographic nesting is immediate at a fixed d as r increases. If d increases, every new allowed weight was strictly above the old d, so its span is contained in the old one as well. For two positive indices $(d,r),(e,t)$, brackets either have weight greater than $d+e$, or weight exactly $d+e$ and length at least $\max(r,1)+\max(t,1) \ge \max(r+t,1)$. Thus the Lie bracket is in $\mathfrak{h}_{(d+e,r+t)}$. The BCH commutator and its higher terms lie there too, using the ideal property. At index zero use normality. This verifies every pair of indices, not only equal indices.

For $r = 0$ or 1, every allowed bracket has weight at least d and thus lies in $\mathfrak{g}_d$. Conversely the length-one brackets with $i_1 = d$ include all of $\mathfrak{g}_d$. This proves $H_{(d,0)} = H_{(d,1)} = G_d$. In particular $H_{(1,0)} = G$. If $d > m$ all brackets vanish. At degree m every nonzero bracket has length at most its weight m, which gives the asserted support.

The domain filtration on $\mathbb{Z}$ has $\mathbb{Z}$ at indices $(0,0),(1,0)$ and zero at all other indices, including $(1,1)$, as in Definition 6.17. The derivative criterion for an adapted map therefore asks for its ath differences to lie in $H_{(a,0)}$; these groups equal G_a. This proves the equality of the polynomial classes directly, matching published (6.8). The new top $H_{(m,m)}$ is a rational subgroup of G_m. Restricting the original vertical character is integral on its lattice and preserves all exact vertical identities and unit norm. No enlargement of its character domain is asserted. □

A25.2 The total-degree conversion used in related applications

For a rational multidegree filtration $(G_a)_{a \in \mathbb{N}^q}$ supported on a finite downset J, first use bounded tautification to restrict to $H = \langle G_{e_1}, \dots, G_{e_q} \rangle$. Let

$$K_0 = H, \qquad \mathfrak{k}_i = \sum_{a \in J:\ |a| \ge i} \mathfrak{g}_a, \qquad K_i = \exp \mathfrak{k}_i \quad (i \ge 1).$$

Each $\mathfrak{g}_a$ is an ideal in $\mathfrak{g}_0$, so the sum is a rational ideal. It equals the Lie algebra of the connected generated group. The brackets satisfy $[\mathfrak{k}_i, \mathfrak{k}_j] \subseteq \mathfrak{k}_{i+j}$, because $[\mathfrak{g}_a, \mathfrak{g}_b] \subseteq \mathfrak{g}_{a+b}$ and $|a+b| = |a| + |b|$. The index-zero condition follows from normality. Also $K_1 = H$ and K_i is trivial above $D = \max_{a \in J} |a|$. The rational subgroup lemma verifies every lattice intersection.

For the diagonal restriction $n \mapsto g(n, \dots, n)$ of a multivariable polynomial orbit, use its Taylor expansion. A factor with coefficient $u_a \in G_a$ has scalar exponent $\prod_j \binom{n}{a_j}$ of ordinary degree $|a|$. Such a factor is adapted to $K_\bullet$ since $u_a \in K_{|a|}$; its successive differences have the same one-parameter coefficient and the required scalar polynomial degree. The product is polynomial by B.4. This checks the actual diagonal map rather than treating the domain filtrations as identical. For a rectangle $J = \{a : a \le d\}$, the only index of total weight $|d|$ is d, so $K_{|d|} = G_d$. A multidegree-d nilcharacter therefore restricts diagonally to a degree-$|d|$ nilcharacter with exactly the same top character. For a nonrectangular downset only the nilsequence degree assertion has been proved, and no unwarranted common top character is claimed.

A25.3 The precise one-step interface

Fix $s \ge 3$, write $m = s - 1$, and let $1 \le R \le m$. Suppose f is bounded on $[N]$, H is a dense internal set of shifts, $A(h, n)$ is a single unit multidegree $(1, m)$ nilcharacter, and (χ_h) is an internal uniformly bounded family of unit degree–rank (m, R) nilcharacters. Suppose that for every retained h, $\Delta_h f$ has positive degree-at-most-$m-1$ correlation with $A(h, \cdot) \otimes \chi_h$.

The one-step premise used here is the following exact output, after restricting to a dense internal subset and fixing uniform standard bounds. It must be supplied by the corrected additive-quadruple result, the actual Section 10 frequency representation, the two conclusions of Theorem 11.1, accepted LP04 and the final Section 12 join:

1. There is a unit multidegree $(1, m)$ nilcharacter $Z(h, n)$, defined globally, and a family of unit-vector sequences $\zeta_h(n)$ such that $Z(h, n) = \iota \zeta_h(n)$ for retained h, with one fixed coordinate isometry ι. Zero padding is allowed.

2. The entire matrix-valued function $V_h = \chi_h \otimes \overline{\zeta_h}$ is a nilsequence of degree–rank at most $(m, R-1)$, with standard uniform geometry and Lipschitz bounds and internal parameter dependence.

The second clause states a rank drop; for $R > 1$ it does not state an ordinary-degree drop. The final Section 12 constructor must choose its test equivariant on the full ordinary top before invoking LP04's comparison. Equivariance only on $G_{(m,R)}$ does not by itself justify its larger multidegree top. Those obligations remain in the separately authored geometric join and are not hidden in this premise.

A25.4 Recovering the next family without dividing by components

The unit-vector identity gives, for every component a and all relevant integers,

$$(\chi_h)_a(n) = \sum_b (\zeta_h)_b(n) \big((\chi_h)_a(n) \overline{(\zeta_h)_b(n)} \big). \tag{198}$$

The terms in parentheses are components of V_h. Replace the first factor by its corresponding coordinate in $Z(h, n)$ and form the unit nilcharacter $A'(h, n) = A(h, n) \otimes Z(h, n)$ on the product model. The product filtration has the same multidegree bound $(1, m)$.

Choose a scalar component and a degree-at-most-$m-1$ test witnessing the incoming correlation, with fixed bounds and a fixed positive threshold after the usual finite selection. Inserting (198)

expresses it as a fixed finite sum of correlations of $\Delta_h f$ against components of $A'(h,\cdot)$, the same ordinary lower-degree test, and a component of V_h. At least one summand has a positive fixed correlation. Finite pigeonhole restricts to a dense subset with fixed component choices if necessary. No scalar component is inverted, or assumed nonzero or of modulus one.

On each fixed rational degree–rank model for V_h, apply the unit-character approximation construction to its top group at $(m, R-1)$. That group is central after tautification, since bracketing with $G_{(1,0)}$ raises the first coordinate above m. Its lattice is rational. Smooth local sections and a finite unit partition on the central torus produce exact-character unit-vector tests. The complex linear span of their components is a unital conjugation-stable algebra separating points; Stone–Weierstrass and a finite uniform net of the bounded Lipschitz test class give a finite approximation by components of such tests. This is the same explicit compact construction in the assembly's selection section, now on the stated degree–rank model. It uses centrality and rationality, not an identification of this top with the larger ordinary-degree top.

Choose the approximation error so that its correlation with the other fixed bounded factors is less than half the surviving threshold. Another finite selection supplies a unit degree–rank $(m, R-1)$ nilcharacter χ'_h for which $\Delta_h f$ has positive degree-at-most-$m-1$ correlation with $A'(h,\cdot)\otimes\chi'_h$. Finite choices and orbit choices are made on representatives and then passed to the ultralimit, retaining internal dependence and uniform bounds. Every test has its exact standard vertical identity before extension, so the all-limit convention is preserved. This is the output required by the next rank step.

A25.5 Initialization, termination and the actual endpoint

Start with the degree-m family obtained from the preceding inverse order. Lemma A25.1 makes it a degree–rank (m, m) family with the same functions. Take $A = 1$, viewed as a constant unit multidegree $(1, m)$ nilcharacter. The original positive correlation is degree-at-most-$m-1$ correlation with the lower test equal to one.

Apply the one-step premise and the preceding recovery argument at $R = m, m-1, \ldots, 1$. There are exactly m steps, a fixed standard number. A finite sequence of positive standard density losses still leaves a positive standard density; the same holds for positive correlation and finite complexity bounds. Products of the successive multidegree $(1, m)$ nilcharacters retain that bound on their fixed product model.

At the last step the residual family has support at most $(m, 0)$. Its extra degree–rank axiom says $G_{(m,0)} = G_{(m,1)}$. But $(m, 1) > (m, 0)$, so the latter group is trivial. All groups of larger degree are also trivial. Thus the ordinary filtration $(G_{(i,0)})_i$ has degree at most $m-1$. The residual nilcharacter, viewed merely as a bounded nilsequence, can now be multiplied into the already present ordinary degree-at-most-$m-1$ correlation test. The product degree is still at most $m-1$.

We obtain one internal unit multidegree $(1, m) = (1, s-1)$ nilcharacter $A_{\text{final}}(h, n)$ such that $\Delta_h f$ has degree-at-most-$s-2$ correlation with it on a dense set of shifts. This is the exact Theorem 7.1 endpoint needed by the separately authored symmetry proof. It is conditional on the one-step premise being supplied at every input rank and every constructed family. Acceptance of the finite iteration alone does not establish those analytic premises or the full inverse theorem.

A25.6 Source findings and bounded related-use scope

The actual final printed p. 1262 was inspected visually. In Example 6.11 the multidegree-to-degree group is printed with $|a| \le i$, which includes G_0 at every i and does not give bounded degree. For

example $G_0 = G_1 = \mathbb{R}$, $G_2 = 0$ would give the group $\mathbb{R}$ at every index. The required rule is $|a| \geq i$, proved in Section 2 above.

The same example's degree-to-rank rule prints the equality of summed weights to the fixed bound d, rather than to the running index d'. The desired identity $H_{(d',0)} = G_{d'}$ then need not hold. For instance the linear filtration $G_0 = G_1 = \mathbb{R}$, $G_2 = 0$ with allowed degree bound $d = 2$ yields zero at $(1, 0)$ under that printed rule, whereas $G_1 = \mathbb{R}$. A nonconstant linear polynomial is then lost. Section 1 supplies the current-index rule and proves every filtration, lattice and polynomial assertion used here. The printed length condition is $\ell \geq \max(r', 1)$; that sign is not itself an error.

The bounded related-use scope is Example 6.11, the polynomial identity (6.8), Lemma 6.21's degree conversions, the initialization and rank-zero termination of Theorems 7.1–7.2, and the diagonal/total-degree conversions used in E.9 and the Section 13 assembly. Separate authors of the latter consumers have been notified. No claim is made here to have read every unrelated use of a filtration in the source or every historical correction. These formula errors are not evidence that the intended inverse theorem is false. Earlier source bytes and decisions are preserved.

A26 Scaling symbols and symmetric multilinearisation with the factorial retained

This section proves the multilinearization and scaling assertion E.8(v), including the required divisibility. Its tautification input is proved in Section A17. The unpublished erratum motivates the factorial correction but supplies no theorem premise.

A26.1 Conventions and preliminary constructions

We use the rational, connected, simply connected nilpotent groups, polynomial maps, finite-dimensional unit-sphere-valued nilcharacters and symbol equivalence of [148, Sections 3–6 and Appendix E]. For a vector-valued function u, $u^{\otimes r}$ means its ordinary tensor power if $r \geq 0$, with $u^{\otimes 0} = 1$, and $\overline{u}^{\otimes(-r)}$ if $r < 0$. Conjugation is always written explicitly. Equivalence at multidegree d means that every component of $u \otimes \overline{v}$ is a nilsequence with support in $\{a : a \leq d,\ a \neq d\}$; all inequalities on multiindices are coordinatewise. In the degree case this is degree at most $d - 1$. We use the group law on symbols proved in published E.7 and E.8(i)–(iii). Their proofs use tensor products, conjugation, contraction of a unit vector with its conjugate, and a rational top-group quotient. With tautification, the top group is central, so those proofs apply to the present objects.

Initially all data are ordinary, and all integers d, k, q are fixed standard integers. The last section states exactly how the constructions pass to limits and parameter families. A multidegree nilcharacter on a limit subset comes with a representing global polynomial orbit and test function; we use that representation to extend it, and only then restrict back.

Apply Section A17 to the given representation. We can work on a taut group and with an exactly equal represented sequence whose polynomial map g satisfies $g(0) = 1$. Positive groups and the vertical character are unchanged. For a degree filtration $G_0 = G_1$; for multidegree G_0 is generated by the G_{e_i}. In particular G_d is central when the support is contained below one positive d. Write $\mathfrak{g}_a = \log G_a$ with its rational structure induced by Γ.

Lemma A26.1 (Monomial coordinates)**.** *For multidegree support $a \leq d$, the normalised map has an expression*

$$g(n) = \prod_{0<a\leq d} \exp(X_a n^a), \qquad X_a \in \mathfrak{g}_a, \quad n^a = \prod_i n_i^{a_i}. \tag{1}$$

Fix any order refining total weight. For a degree-D map on $\mathbb{Z}^k$, replace $a \le d$ by $0 < |a| \le D$ and require $X_a \in \mathfrak{g}_{|a|}$. Every such ordered product is polynomial for the indicated filtration.

Proof. For a degree-D input, use the multidegree target $\mathfrak{g}_a = \mathfrak{g}_{|a|}$: coordinate derivatives are particular degree derivatives, so the input is polynomial for this target. Published Lemma B.9 gives ordered binomial monomials with coefficient at index a in G_a. Expand the finite BCH formula. A product of scalar binomial polynomials with labels $a_1, \dots, a_r$ has monomials n^b only for $b \le a_1 + \cdots + a_r$. Its Lie-bracket coefficient belongs to $\mathfrak{g}_{a_1+\cdots+a_r} \subseteq \mathfrak{g}_b$. Thus $\log g(n) = \sum_{0<b\le d} Y_b n^b$, $Y_b \in \mathfrak{g}_b$; terms outside the support vanish and the constant is zero. To obtain (1), process increasing total monomial weight: choose X_b to match the coefficient at n^b in the BCH logarithm of the partial ordered product. All commutators of previous factors have larger total monomial weight, and their coefficients at b belong to $\mathfrak{g}_b$. Consequently the adjustment is in $\mathfrak{g}_b$ and does not change any previously matched weight. The finite support makes this algorithm terminate. The argument with $\mathfrak{g}_{|b|}$ is identical. For the converse each factor is a polynomial map: in its one-parameter abelian subgroup an iterated difference of n^a vanishes if its coordinate order exceeds a, and otherwise the coefficient stays in $\mathfrak{g}_a \subseteq \mathfrak{g}_b$ at derivative index b. Apply the product theorem [148, Corollary B.4]. This use does not invoke E.8, E.9 or GI. □

We will use rational connected subgroups and rational central quotients. Their generated subgroups and central quotients are the geometric construction proved in Section A18. We spell out the additional rational-image check used below. For each rational G_a, choose finitely many elements $\gamma_{a,j}$ of $\Gamma \cap G_a$ whose logarithms span $\mathfrak{g}_a$. Such a choice exists: the real span of the logarithms of a nilpotent subgroup is a Lie algebra. Indeed conjugation by integer powers of any of its elements gives a polynomial in that integer, and finite differences extract its iterated brackets; BCH then gives the assertion. Its connected subgroup contains the lattice, so cocompactness forces it to be G_a, by the homogeneous-space argument of Section A18. The same polynomial/BCH argument gives the rational Lie structure $\operatorname{span}_{\mathbb{Q}} \log \Gamma$; all filtration spaces are rational for this structure.

For the scaling image below, the vectors $(q^{|a|} \log \gamma_{a,j}, \log \gamma_{a,j})$ exponentiate to the actual lattice elements $(\gamma_{a,j}^{q^{|a|}}, \gamma_{a,j})$. For the multilinear image, multiply each $\log \gamma_{a,j}$ by an integer N so that its rational second component in (8) lies in the lattice L constructed next. Its first component then exponentiates to $\gamma_{a,j}^N$. These finitely many lattice elements have logarithms spanning each image Lie algebra $\mathfrak{k}_b$. The finite-generator fact in [148, p. 1243] makes their generated discrete subgroup cocompact in K_b. Thus K_b is rational for the actual product lattice, not merely for an abstract rational form. This also checks $b = 0$.

The following explicit lattice choice records the extra finite-symmetry requirement needed below. In a rational nilpotent Lie algebra with a finite group P of rational automorphisms, choose a full rational lattice L_0 and replace it by $\sum_{\sigma\in P} \sigma L_0$. Clear rational bracket denominators and the finitely many BCH denominators by replacing L_0 by $L = ML_0$. Then $\mathrm{BCH}(L, L) \subseteq L$ and $-L = L$. The group $\exp L$ is discrete. It is cocompact: an adapted rational central-series basis, after integer scaling, lies in L; successive rounding of its coordinates gives a compact set of coset representatives. The same construction makes all specified rational filtration subspaces rational. If a rational linear functional on a specified rational central subspace is prescribed, increase M to make it integral on L intersected with that subspace. This gives a P-invariant lattice, not just a rational structure invariant under P. Each choice uses only fixed geometric data.

A26.2 Scaling and divisibility on the actual orbit

Theorem A26.2 (Corrected E.8(v))**.** *Let $d \neq 0$, and let $\chi(n) = F(g(n)\Gamma)$ have multidegree at most d. For every standard integer q,*

$$[\chi(qn)] = q^{|d|}[\chi(n)]. \tag{2}$$

For degree $D > 0$ on $\mathbb{Z}^k$, the exponent is D. The equivalence is on the whole domain, hence on every indicated restriction. Its geometric and Lipschitz bounds are independent of the polynomial coefficients.

Proof. Use (1). Form the finite positive Rees Lie algebra

$$\mathcal{R} = \bigoplus_{0<a\leq d} \mathfrak{g}_a t^a, \qquad [Xt^a, Yt^b] = [X,Y]t^{a+b},$$

where a component outside the downset is zero. This is a Lie algebra: it is a subalgebra of $\mathfrak{g} \otimes \mathbb{R}[t_1, \dots, t_k]$ modulo the ideal spanned by monomials outside the downset. Define the rational Lie map

$$A_q(Xt^a) = (q^{|a|}X, X) \in \mathfrak{g} \oplus \mathfrak{g}.$$

Its bracket identity is exact because $q^{|a|}q^{|b|} = q^{|a+b|}$, including negative and zero q. Let $\mathfrak{k}_b$ be the image of $\bigoplus_{a\geq b,\ a\neq 0} \mathfrak{g}_a t^a$, and put $K_b = \exp \mathfrak{k}_b$. These are rational, descending, and $[\mathfrak{k}_b, \mathfrak{k}_c] \subseteq \mathfrak{k}_{b+c}$. Also $\mathfrak{k}_0 = \sum_i \mathfrak{k}_{e_i}$, so the filtration is taut. The ambient group is K_0, with lattice $K_0 \cap \Gamma^2$. Every factor of

$$(g(qn), g(n)) = \prod_a \exp(n^a A_q(X_a t^a))$$

is polynomial at index a, and hence so is their product. The top group has Lie algebra $\mathfrak{k}_d = \{(q^{|d|}X, X) : X \in \mathfrak{g}_d\}$. For $c = q^{|d|}$, the restricted function

$$(x, y) \longmapsto F(x) \otimes \overline{F(y)^{\otimes c}} \tag{3}$$

is invariant under K_d: the two vertical phases are $c\eta(X)$ and $-c\eta(X)$. This calculation uses $F^{\otimes c}$, with the signed convention above, on the same second coordinate. The rational central quotient K_0/K_d has the quotient filtration with its top group zero. Thus (3) along the displayed orbit is a lower-degree nilsequence, which is precisely (2). Restriction and descent preserve a fixed Lipschitz bound since the geometric maps and compact quotient are fixed. For degree D use $\bigoplus_{0<|a|\leq D} \mathfrak{g}_{|a|} t^a$ and the filtration by $|a| \geq j$. The top image is the same graph with exponent D; all other steps agree. At $q = 0$ the first orbit coordinate is 1, and (3) is the constant $F(1)$; no 0^0 or division is used. □

Corollary A26.3 (Divisibility, with the representative specified)**.** *For positive degree and every standard $Q \geq 1$, there is a nilcharacter ψ on the same domain with $[\chi] = Q[\psi]$.*

Proof. In (1) replace each X_a by $Q^{-|a|}X_a$ and call the resulting map g_Q. It is polynomial on the same group and lattice. With the same F put $\xi(n) = F(g_Q(n)\Gamma)$. Then $g_Q(Qn) = g(n)$ exactly, factor by factor in the same order. Theorem A26.2 gives $[\chi] = Q^D[\xi]$, where $D = |d|$ or the degree bound, and $\psi = \xi^{\otimes Q^{D-1}}$ gives the asserted equation. The test function still has unit norm; its frequency and finite target dimension change only by the displayed standard tensor power. In particular this is divisibility of the actual symbol, not division of an arbitrarily selected multilinear representative after construction. □

A26.3 Raw multilinearisation and the factorial graph

Let $d = (d_1, \dots, d_k) \neq 0$, $D = |d|$, and divide $\{1, \dots, D\}$ into ordered blocks B_i of size d_i. For a subset J put $w(J) = (|J \cap B_i|)_i$ and $a! = \prod_i a_i!$; set $C = d!$. Construct

$$\widetilde{\mathfrak{g}} = \bigoplus_{\varnothing \neq J \subseteq [D]} \mathfrak{g}_{w(J)} \epsilon_J, \qquad [X\epsilon_J, Y\epsilon_L] = \begin{cases} [X,Y]\epsilon_{J \cup L}, & J \cap L = \varnothing, \\ 0, & J \cap L \neq \varnothing. \end{cases} \tag{4}$$

This is a subalgebra of $\mathfrak{g} \otimes \mathbb{R}[\epsilon_1, \dots, \epsilon_D]/(\epsilon_1^2, \dots, \epsilon_D^2)$; Jacobi therefore holds. There is no empty component: normalisation $g(0) = 1$ is what permits its removal. Put

$$\widetilde{\mathfrak{g}}_b = \bigoplus_{1_J \geq b} \mathfrak{g}_{w(J)} \epsilon_J. \tag{5}$$

These are descending rational Lie subalgebras, their bracket has the required index sum, and they vanish when any $b_i > 1$. The zero group is generated by the first-coordinate groups since each nonempty J contains a coordinate. Thus (5) is taut. Its top group is central. The block permutation group $P = \prod_i S_{d_i}$ preserves (4), permutes the groups in (5) by the same permutation of their indices, and acts trivially on the top copy $\mathfrak{g}_d \epsilon_{[D]}$. Choose a P-invariant lattice $\widetilde{\Gamma}$ as in Section A26.1, making

$$\widetilde{\eta}(\exp(X\epsilon_{[D]})) = \eta(\exp X) \tag{6}$$

integral on its top lattice. This is possible since η is integral on $\Gamma \cap G_d$ and hence rational on the rational top Lie algebra.

Choose a Lipschitz unit-sphere-valued function $\widetilde{F}$ with frequency $\widetilde{\eta}$. Here is the needed existence construction, with no invariance under P assumed. The central torus acts freely on $\widetilde{G}/\widetilde{\Gamma}$: its stabilizer is precisely the top lattice. On finitely many local trivialisations of this torus bundle, take the character (6) times a smooth base bump. Choose the bumps so that their supports cover the compact base and they vanish near the chart boundaries. The vector of these functions has no common zero; divide by its norm. This gives a globally Lipschitz unit vector of the required frequency, with a fixed finite dimension. It is the partition-of-unity construction underlying published (6.3).

For the coefficients of (1), define

$$\widetilde{g}(z) = \prod_{0 < a \leq d} \exp\left(a! \sum_{w(J)=a} z_J X_a \epsilon_J \right), \qquad z_J = \prod_{j \in J} z_j, \qquad \widetilde{\chi}(z) = \widetilde{F}(\widetilde{g}(z)\widetilde{\Gamma}). \tag{7}$$

Each factor is polynomial for (5). To see this without commuting distinct coefficients, fix a; the summands in its exponent commute because their Lie coefficients are the same X_a, so (7)'s factor is a product of the individual monomial factors. Then use product closure. Thus $\widetilde{\chi}$ has multidegree at most $(1, \dots, 1)$. Let Δn repeat n_i throughout B_i.

Lemma A26.4 (Raw diagonal identity)**.** *The diagonal $\widetilde{\chi}(\Delta n)$ is equivalent at multidegree d to $\chi(n)^{\otimes C}$. In particular for $k = 1$ the multiplier is $d!$.*

Proof. On the positive Rees algebra $\mathcal{R}$ of Section A26.2, set

$$T(Xt^a) = \left(X, a! \sum_{w(J)=a} X\epsilon_J \right) \in \mathfrak{g} \oplus \widetilde{\mathfrak{g}}. \tag{8}$$

This is a rational Lie map. Indeed for each U with $w(U) = a + b$ the number of ordered disjoint decompositions $U = J \sqcup L$ with $w(J) = a, w(L) = b$ is $\prod_i \binom{a_i+b_i}{a_i}$. Multiplying by $a!b!$ gives $(a+b)!$. Thus $[T(Xt^a), T(Yt^b)] = T([X,Y]t^{a+b})$ exactly. If $a + b$ is outside the support, both sides vanish by $[\mathfrak{g}_a, \mathfrak{g}_b] \subseteq \mathfrak{g}_{a+b} = 0$.

Define $\mathfrak{k}_b = T(\bigoplus_{a \geq b, a \neq 0} \mathfrak{g}_a t^a)$, $K_b = \exp \mathfrak{k}_b$. As before these give a descending, rational, taut filtration on $K_0 \subseteq G \times \widetilde{G}$, with lattice $\Lambda = K_0 \cap (\Gamma \times \widetilde{\Gamma})$. They give the actual comparison orbit

$$(g(n), \widetilde{g}(\Delta n)) = \prod_a \exp\left(n^a T(X_a t^a)\right). \tag{9}$$

No asserted equality of top weights is being used to generate a non-descending filtration: every defining sum includes all $a \geq b$. The top is exactly

$$K_d = \{(\exp X, \exp(CX\epsilon_{[D]})) : X \in \mathfrak{g}_d\}. \tag{10}$$

It is rational and central. The function on the comparison nilmanifold is

$$\mathcal{F}(x, y) = F(x)^{\otimes C} \otimes \overline{\widetilde{F}(y)}. \tag{11}$$

Its two vertical phases on (10) are $C\eta(\exp X)$ and $-\widetilde{\eta}(\exp(CX\epsilon_{[D]})) = -C\eta(\exp X)$. Therefore (11) descends to a Lipschitz function F' on K_0/K_d with the projected lattice Λ' and quotient filtration. Writing k'_a for the projected $\exp T(X_a t^a)$ gives the corrected identity

$$\chi(n)^{\otimes C} \otimes \overline{\widetilde{\chi}(\Delta n)} = F'\left(\prod_a (k'_a)^{n^a} \Lambda'\right). \tag{12}$$

The right side has top group zero, so it has support strictly below d. For $k = 1$, (11) contains $F^{\otimes d!}$ and (12) contains $\chi^{\otimes d!}$. Dropping either factor would leave the uncancelled vertical phase $(1 - d!)\eta$ and would not prove the claimed equivalence.

The diagonal itself is a degree-d nilcharacter: coarsen (5) by taking the direct sum of components with $w(J) \geq a$ at index a. This is descending, rational and taut, its top is still the $J = [D]$ component, and (7) on the diagonal is polynomial for it. This verifies the exact symbol domain used in (12), without assuming that arbitrary linear maps preserve arbitrary multidegree filtrations. □

A26.4 Literal symmetry and division before construction

The map $\widetilde{g}$ is equivariant under block permutations: $\widetilde{g}(\sigma z) = \sigma \widetilde{g}(z)$ after fixing the compatible action convention. An arbitrary $\widetilde{F}$ need not be invariant, so equivariance alone does not prove that $\widetilde{\chi}$ is symmetric. We supply the missing transition explicitly.

Let V be the finite-dimensional Hermitian target of $\widetilde{F}$ and $m = |P| = C$. Regard $\operatorname{Sym}^m V$ as the symmetric subspace of $V^{\otimes m}$ with its induced norm, and write $\odot$ for symmetrised tensor product. On the compact nilmanifold put

$$B(x) = \bigodot_{\sigma \in P} \widetilde{F}(\sigma x), \qquad F_{\mathrm{sym}}(x) = B(x)/\|B(x)\|. \tag{13}$$

The product of nonzero vectors in the symmetric algebra is nonzero: after a choice of basis this algebra is a polynomial ring over $\mathbb{C}$, an integral domain. Hence $\|B(x)\| > 0$ everywhere. Continuity and compactness give a positive minimum, so (13) is Lipschitz. Reindexing the finite product proves exact P-invariance. Since P fixes the top group, (13) has vertical frequency $m\widetilde{\eta}$. Consequently

$$\chi_{\mathrm{sym}}(z) = F_{\mathrm{sym}}(\widetilde{g}(z)\widetilde{\Gamma}) \tag{14}$$

is a literally block-symmetric multidegree-$(1,\dots,1)$ nilcharacter. On a block diagonal, $\sigma\widetilde{g}(\Delta n) = \widetilde{g}(\Delta n)$. All factors in (13) then equal the same unit vector v, and $v^{\odot m} = v^{\otimes m}$ has norm one. Thus the diagonal of (14) is exactly $\widetilde{\chi}(\Delta n)^{\otimes m}$ in the symmetric subspace, not merely a proportional vector with an uncontrolled scalar. Lemma A26.4 therefore gives

$$[\chi_{\mathrm{sym}}(\Delta n)] = Cm[\chi(n)]. \tag{15}$$

Theorem A26.5 (E.9 and E.10 with a symmetric output)**.** *For a multidegree-d nilcharacter χ there is a literally block-symmetric multidegree-$(1,\dots,1)$ nilcharacter $M(\chi)$ whose block diagonal is equivalent to χ. The one-variable case is the asserted symmetric E.9 output. The assertion holds on any given limit subset after restriction.*

Proof. If $d \neq 0$, first use Corollary A26.3 with $Q = Cm = C^2$ to choose the specified representative ρ with $[\chi] = Cm[\rho]$. Apply (4)–(14) to this ρ, including its actual polynomial coefficients, lattice, function and vertical frequency. Equation (15) now identifies its output diagonal with $[\chi]$. We do not divide the already symmetrised function and assume that its symmetry survives. If $C = 1$, all constructions and division remain valid. Coordinates with $d_i = 0$ have empty blocks; the normalised polynomial is independent of them. If all $d_i = 0$, the original sequence is a constant unit vector and that vector on the zero-variable domain gives the assertion exactly. □

A26.5 The exact input to Section 13

Let $s \geq 2$ be fixed and let $\chi(h,n)$ be the multidegree-$(1,s-1)$ nilcharacter occurring immediately before published (13.4). First choose ρ with $[\chi] = s[\rho]$ by Corollary A26.3, in that multidegree symbol group. Then set $\widetilde{\chi} = M(\rho)$ from Theorem A26.5. It has s variables, multidegree $(1,\dots,1)$ and literal symmetry in its last $s-1$ variables. Its actual diagonal gives

$$[\chi(h,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)} = s[\widetilde{\chi}(h,n,\dots,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)}. \tag{16}$$

Here is the comparison of symbol groups in (16). For support $a \leq (1,s-1)$, give the same group the degree filtration generated by the G_a with $|a| \geq j$. Commutators satisfy the index law, rationality follows from the rational generated-subgroup lemma, and the orbit is polynomial by its monomial expansion. At total weight s the only possible index is $(1,s-1)$, so deleting that top index leaves degree at most $s-1$. Thus both multidegree equivalences used above become degree-s equivalences, with the same functions and the same input (h,n). This proves precisely (13.4), including its multiplier s.

The sequence $\Theta(n) = \widetilde{\chi}(n,\dots,n)$ is a degree-at-most-s nilcharacter by the same coarsening with the binary cube. Fixed linear pullbacks $n \mapsto n+h$ preserve that degree filtration and preserve lower-degree equivalence; one may either use the monomial expansion or published Corollary B.5 and E.8(vi) in its degree-only form. No full symmetry in all s variables is asserted here. Section 13's later symmetrisation argument is exactly what must address the first variable. Its uses of multilinearity require the separately proved replacement for Lemma 13.2; that result is not inferred from (16). Likewise the subsequent bias and density arguments remain separate.

A26.6 Uniformity and limit interpretation

All nilpotent groups, rational subspaces, finite tensor dimensions, integer powers and permutation groups above are fixed by the original geometric data, d, k, and the specified standard divisibility/scaling integers. They do not depend on orbit coefficients or interval length. The bounded-basepoint normalisation supplied by Section A17 has a uniform Lipschitz constant. Subgroup

restrictions, rational quotients and metrics are fixed compact geometric maps. The lattice and the test function in (6)–(7) are selected once from those fixed data. The positive minimum in (13) is for that fixed test function on that fixed compact space, not an infimum over a varying family of independently chosen functions. For a varying original test function with bounded Lipschitz norm, the raw comparison (11) varies, but the auxiliary $\widetilde{F}$ and (13) can still be fixed because their frequency and group are fixed.

Every coefficient change is a finite BCH expression or a rational scalar multiple in a fixed Lie algebra. Apply it coordinatewise to a representing sequence to obtain a limit map. The finite formulas, exact identities, positive fixed lower bound in (13), and uniform Lipschitz estimates pass to the natural ultrapowers used in published Appendix A. Where geometric data range over a standard finite list, perform this construction for each member of that list. No selection depending on an unbounded interval length or uncontrolled pointwise choice is introduced. All vertical identities here have that natural-ultrapower meaning: they hold for every limit point and every limit top-group element, by transfer of the exact ordinary identity. An identity tested only on standard points would not suffice; no such restriction is used here.

The preceding constructions supply the applications E.8(v),(vii), E.9, E.10 and (13.4) through the definition of Θ, including literal symmetry in the last $s-1$ variables. The scope is these applications, not a classification of all corrections to GTZ.

A27 The plain additive-quadruple implication on a finite interval

Scope. This local proof supplies the exact plain correlation statement of GTZ Proposition 8.4, p. 1277, needed by the Section 13 join. It does not use Proposition 8.3 or infer plain bias from lower-degree bias. The argument is the elementary two-step Cauchy–Schwarz pattern; the finite domains and constants are written explicitly. No source error in Proposition 8.4 is asserted.

A27.1 Precise finite assertion

Put $J=\{1,\dots,N\}$, $I=\{-N,\dots,N\}$, and suppose $H\subseteq I$ has $|H|\ge\alpha|I|$ for fixed $\alpha>0$. Let $f:J\to\mathbb{C}$ be 1-bounded and extend it by zero to $\mathbb{Z}$. For each $h\in H$ let $\chi_h:\mathbb{Z}\to\mathbb{C}$ be globally bounded by a fixed $B>0$, and assume

$$\left|\mathbb{E}_{n\in J} f(n+h)\overline{f(n)}\chi_h(n)\right|\ge\delta>0.$$

The values outside J are part of the data and are not truncated.

Theorem A27.1. *Write $a=\alpha\delta/3$ and $\eta=a^4/4$. There are at least*

$$\frac{\eta}{2B^4}(4N+1)(2N+1)^2$$

distinct additive quadruples $(h_1,h_2,h_3,h_4)\in H^4$, $h_1+h_2=h_3+h_4$, for which

$$\left|\mathbb{E}_{n\in J}\chi_{h_1}(n)\chi_{h_2}(n+h_1-h_4)\overline{\chi_{h_3}(n)\chi_{h_4}(n+h_1-h_4)}\right|\ge\eta/2.$$

In particular their number is a positive fixed multiple of N^3.

A27.2 Two exact Cauchy–Schwarz steps

For each $h \in H$ choose a scalar c_h of modulus one so the given average multiplied by c_h is real and at least δ. Set $u_h = c_h \chi_h$ on H, and $u_h = 0$ for every $h \notin H$. The functions remain defined on all integers. Then

$$\mathbb{E}_{h\in I}\mathbb{E}_{n\in J} f(n+h)\overline{f(n)}u_h(n) \geq \alpha\delta.$$

Let $M = J + I = \{1-N, \dots, 2N\}$, so $|M| = 3N$. Set $m = n + h$. The preceding average is exactly

$$3\mathbb{E}_{m\in M} f(m)\mathbb{E}_{h\in I}\overline{f(m-h)}u_h(m-h).$$

Terms with $m - h \notin J$ vanish because f was zero-extended. Cauchy–Schwarz in m, using $|f| \leq 1$, therefore yields

$$\mathbb{E}_{m\in M}\left|\mathbb{E}_{h\in I}\overline{f(m-h)}u_h(m-h)\right|^2 \geq a^2. \tag{199}$$

Expand the square and put $h' = h + k$, $n = m - h$. Let $K = \{-2N, \dots, 2N\}$. The expanded left side equals

$$\frac{N}{|M|}\frac{|K|}{|I|}\mathbb{E}_{k\in K}\mathbb{E}_{n\in J}\overline{f(n)}f(n-k)T(k,n), \qquad T(k,n) = \mathbb{E}_{h\in I}u_h(n)\overline{u_{h+k}(n-k)}.$$

This is an equality with the following domain checks. The factor $\overline{f(n)}$ forces $n \in J$, and then $m = n + h \in M$ automatically for $h \in I$. The condition $h + k \in I$ is supplied by $u_{h+k} = 0$ outside $H \subseteq I$. Thus no additional cutoff in $n-k$ is inserted into T. Its values there are legitimate because the original functions are globally defined. The remaining factor $f(n-k)$ stays in the bounded weight.

The prefactor is positive and at most two. By (199), the absolute value of the displayed average without prefactor is at least $a^2/2$. Cauchy–Schwarz in (k, n), dropping only the bounded weight $\overline{f(n)}f(n-k)$ by an inequality, gives

$$\mathbb{E}_{k\in K}\mathbb{E}_{n\in J}|T(k,n)|^2 \geq a^4/4 = \eta.$$

Expand this square with independent $h, h' \in I$. It is the average over $(k, h, h') \in K \times I^2$ of

$$C(k,h,h') = \mathbb{E}_{n\in J}u_h(n)u_{h'+k}(n-k)\overline{u_{h'}(n)u_{h+k}(n-k)}.$$

The average is real and at least η, while $|C| \leq B^4$. Consequently at least a fraction $\eta/(2B^4)$ of triples have $|C| \geq \eta/2$. Indeed the complementary contribution to the average of $|C|$ is at most $\eta/2$, and the contribution of the selected fraction is at most that fraction times B^4.

Such a triple has $h, h', h+k, h'+k \in H$, since otherwise $C = 0$. Set

$$(h_1, h_2, h_3, h_4) = (h, h'+k, h', h+k).$$

It is an additive quadruple and $h_1 - h_4 = -k$. The map is injective, with inverse $h = h_1$, $h' = h_3$, $k = h_4 - h_1$. The factors c_h contribute only a constant phase of modulus one to C, so they do not change its absolute value. This proves the theorem, including the exact unweighted average on J.

A27.3 Vector families, internal limits and actual consumer

For vector-valued χ_h of fixed finite dimension D, a positive Euclidean correlation implies that some component has magnitude at least $\delta/\sqrt{D}$. Finite pigeonhole retains a component on a subfamily of density at least α/D. Apply the scalar theorem there. The scalar output is a component of the

full tensor product, so it implies plain vector bias. A finite list of dimensions can first be padded into one fixed dimension. For a bounded f with a fixed bound other than one, divide it by that bound, changing δ by a fixed factor only.

Apply the finite assertion to representatives of an internal family, with uniform standard bounds, density and correlation. Choices of components and phases can be made for each finite family and then passed to the ultralimit. All constants above are positive standard numbers, so the selected quadruples have positive standard density on the N^3 scale and the correlation is positive by a standard amount. This is exactly the plain Proposition 8.4 output.

In Section 13 the input to this theorem is the entire bounded product that includes the selected lower nilcharacter. Keeping it inside χ_h throughout the calculation preserves that lower information. The theorem requires no degree bound and introduces no lower-degree test factor. It therefore supplies the stronger plain statement that cannot be replaced merely by the degree-biased LP03 output.

The bounded application audit checks the original J, the enlarged $M = J + I$, the parameter range K, the exact inverse reindexing and the global domain of χ_h. A possible artificial cutoff $1_J(n-k)$ is unnecessary: $f(n-k)$ is removed by the second Cauchy–Schwarz inequality, not by claiming an equality after erasing its support. This resolves the domain question without classifying the published Proposition 8.4 as false or defective. The modified Proposition 7.3, its nilsequence-specific factorization and the remaining Section 13 hypotheses are outside this theorem's scope.

A28 The Section 13 correlation and symmetry join

This section proves the symmetrization conclusion from the internal multilinearization theorem and its stated published inputs. The signed identities retain the actual unconjugated α in published (13.10).

A28.1 Target and exact inputs

Fix a standard integer $s \geq 3$ and an unbounded limit integer N. Write $I_t = [-tN, tN] \cap {}^*\mathbb{Z}$, $J = [N]$, $r = s-1$ and $m = s-2$. All geometric data and finite dimensions are standard. Vertical identities hold for every limit point and every limit top-group element under the natural ultrapower extension, as in the project tautification and LP13 units. Tensor products have their Hilbert tensor norms. Every nilcharacter is unit-vector-valued; a scalar coordinate need not have modulus one.

Theorem A28.1 (Theorem 7.4 on the consumed domain)**.** *Let f be bounded on J, extended by zero outside J, and let $H \subseteq I_1$ be a dense limit subset. Suppose that $\chi(h,n)$ is a globally represented multidegree-$(1, s-1)$ nilcharacter and $\Delta_h f$ has degree-at-most-$(s-2)$ correlation with $\chi(h,\cdot)$ on J for every $h \in H$, with GTZ's correlation convention. Then there are a globally represented degree-at-most-s unit nilcharacter Θ and a bounded globally represented nilsequence $\Psi(h,n)$ supported on*

$$\mathcal{J} = \{(i,j) : i + j \leq s-1\} \cup \{(i, s-i) : 2 \leq i \leq s\} \tag{1}$$

such that, on $I_1 \times J$, each component of $\chi(h,n)$ is a bounded finite linear combination of components of

$$\Theta(n+h) \otimes \overline{\Theta(n)} \otimes \Psi(h,n). \tag{2}$$

All families and bounds are standard in the limit sense, independently of the interval length. No effective numerical constants are asserted.

We use the following exact premises. The project LP05-001 supplies a unit nilcharacter $\widetilde{\chi}$ of cube multidegree $(1,\dots,1)$, literally symmetric in its last r variables, with

$$[\chi(h,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)} = s[\widetilde{\chi}(h,n,\dots,n)]_{\mathrm{Symb}_s(*\mathbb{Z}^2)}. \tag{3}$$

The project LP13 theorem supplies multilinearity of the symbol of $\widetilde{\chi}$ under fixed integer linear substitutions, in each slot. The project tautification-003 Theorem 3 supplies the uniform shift tensor of a positive-degree nilcharacter, lowering degree by one in the ordinary degree case. It applies to limit shifts and limit families.

The unchanged published inputs are the definitions and elementary symbol arithmetic E.3, E.7 and E.8(i)–(iii),(vi); nilcharacter approximation and selection E.5/E.6 and Appendix A; Proposition 8.4 with its plain-correlation hypothesis; the converse Proposition 5.6; and E.11/E.13. Their exact source applications are recorded separately. In particular Proposition 8.4 is a distinct input. The weaker degree-$(s-2)$ biased-quadruple conclusion of Proposition 7.3 alone would not retain the lower factors needed here. No corrected assertion is supplied by citing the unpublished erratum.

For an ordered list of linear forms define

$$T(L_1;L_2,\dots,L_s) = [\widetilde{\chi}(L_1,L_2,\dots,L_s)]_{\mathrm{Symb}_s}.$$

The semicolon separates the distinguished first slot. We use the notation n^j in a slot list for j repeated entries, not a power of n. All formulas between such symbols are on the full indicated limit lattice until a box is explicitly named.

A28.2 Plain correlation and retention of the lower factors

By E.5/E.6 in the ordinary one-dimensional degree ordering and Appendix A selection, after restricting to a still dense internal subset of H, choose degree-at-most-$(s-2)$ unit nilcharacters φ_h so that $\Delta_h f$ plainly correlates with $\chi(h,\cdot)\otimes\varphi_h$. The family is internal, with a standard finite geometry list, bounded dimensions and common Lipschitz bounds. Here the selection is into the countable increasing complexity presentation of the nilcharacter class. An internal image contained in that union is contained in a standard stage by Appendix A compactness; otherwise nested nonempty internal sets of parameters outside successive stages give an image outside the union. Positive correlations can likewise be made uniform by A.6. Pad targets by zero coordinates to one fixed dimension. Outside the retained set use a fixed constant unit nilcharacter, and keep all parameter indicators separately. This preserves the unit norm needed for the identities below. Conjugate the selected lower nilcharacter if needed to match the published correlation convention. A finite pigeonhole also fixes a scalar coordinate of $\chi(h,\cdot)\otimes\varphi_h$ retaining plain correlation on a dense set. Thus the scalar version of Proposition 8.4 suffices; the full unit vectors remain available behind these coordinates.

Apply published Proposition 8.4 to this actual family. Its hypotheses are plain correlation on J, internal dependence, bounded functions and a dense subset of I_1; all hold. Its conclusion gives bias for many additive quadruples (h_1,h_2,h_3,h_4) in that subset, with tensor signs $+,+,-,-$ and the shift $n+h_1-h_4$ on the second and fourth factors. Parameterise

$$(h_1,h_2,h_3,h_4) = (h_0+a, h_0+b, h_0+a+b, h_0).$$

This is an injective correspondence with the retained quadruples. $h_0\in I_1$ and $a,b\in I_2$. Pigeonhole in h_0 to obtain one fixed h_0 and a dense set of pairs $(a,b)\in I_2^2$. Define the global unit nilcharacter

$$\tau(a,b,n) = \chi(h_0+a,n)\otimes\chi(h_0+b,n+a)\otimes\overline{\chi(h_0+a+b,n)}\otimes\overline{\chi(h_0,n+a)}. \tag{4}$$

It has total degree at most s, not degree three for general s.

For any coordinate of a unit nilcharacter v, the exact identity

$$v_j(n+a) = \sum_k (v_j(n+a)\overline{v_k(n)})v_k(n) \tag{5}$$

follows from $\sum_k |v_k(n)|^2 = 1$. The bracket is a coordinate of the shift tensor. Apply the project shift theorem to φ_{h_0+b} and φ_{h_0}, with shift a. These brackets have degree at most $s-3$, uniformly in all parameters. Conjugate (5) when the shifted factor is conjugated. Thus the original bias implies degree-at-most-$(s-3)$ bias of

$$\tau(a,b,n) \otimes \varphi_{h_0+a}(n) \otimes \varphi_{h_0+b}(n) \otimes \overline{\varphi_{h_0+a+b}(n)} \otimes \overline{\varphi_{h_0}(n)}. \tag{6}$$

Selection gives internal scalar degree-at-most-$(s-3)$ tests with common bounds and finite geometries. A standard finite choice of coordinates retains a positive density of the pairs. Phase-align their scalar averages by a bounded parameter scalar, and set the test equal to zero on other pairs. These factors are constant in n, so they have degree zero. We obtain fixed scalar coordinates t of τ and u, v, w, z of the displayed φ factors with

$$\left|\mathbb{E}_{a,b\in I_2,n\in J} t(a,b,n)u_a(n)v_b(n)\overline{w_{a+b}(n)z(n)}p_{a,b}(n)\right| \gg 1. \tag{7}$$

The scalar $p_{a,b}$ is a uniformly bounded internal lower-degree family. Finite coordinate selection, rather than division by a coordinate, justifies this passage even if individual coordinates vanish.

A28.3 The three Cauchy–Schwarz eliminations

Every average in this section has $n \in J$. All bounds are standard. Applying Cauchy–Schwarz in (a,n) to remove the bounded factor $u_a\overline{z}$ in (7), and expanding the squared inner average, gives

$$\left|\mathbb{E}_{a,b,b'\in I_2,n\in J} t(a,b,n)\overline{t(a,b',n)}v_b(n)\overline{v_{b'}(n)w_{a+b}(n)}w_{a+b'}(n)p_{a,b}(n)\overline{p_{a,b'}(n)}\right| \gg 1. \tag{8}$$

Put $c = a+b+b'$. Keep $b, b' \in I_2$, take $c \in I_6$, and retain $\mathbf{1}_{c-b-b'\in I_2}$. This exact indicator is independent of n. The normalized average changes by the factor $|I_2|/|I_6|$, comparable to one. No three-term sum of elements of I_2 is asserted to lie in I_3. Pigeonhole in c to fix $c_0 \in I_6$. Set

$$\alpha(b,b',n) = \tau(c_0-b-b',b,n) \otimes \overline{\tau(c_0-b-b',b',n)}, \qquad V_b(n) = v_b(n)w_{c_0-b}(n). \tag{9}$$

In particular the second factor of V_b is not conjugated: its sign follows from the last two w factors in (8). Absorb the parameter cutoff into the scalar lower test. Equation (8) becomes, for a fixed coordinate a_0 of the actual α,

$$\left|\mathbb{E}_{b,b'\in I_2,n\in J} a_0(b,b',n)V_b(n)\overline{V_{b'}(n)}q_{b,b'}(n)\right| \gg 1. \tag{10}$$

The functions V_b need only be bounded; no inverse or unit norm for them is used. Cauchy–Schwarz in (b',n) removes $\overline{V_{b'}}$ and produces the product $a_0(b_1,b',n)\overline{a_0(b_2,b',n)}V_{b_1}\overline{V_{b_2}}$ with the product of the corresponding q tests. Cauchy–Schwarz in (b_1,b_2,n) removes the remaining V factors. Its expanded four-term product is

$$a_0(b_1,b_1',n)\overline{a_0(b_2,b_1',n)a_0(b_1,b_2',n)}a_0(b_2,b_2',n). \tag{11}$$

Its scalar test is the product of four q tests with the same signs, still degree at most $s-3$, internal and uniformly bounded. Pigeonhole in (b_2, b_2'), calling the fixed values $(b_0, b_0') \in I_2^2$. The resulting expression is a coordinate of the global unit nilcharacter

$$\alpha'(b, b', n) = \alpha(b, b', n) \otimes \overline{\alpha(b_0, b', n)} \otimes \overline{\alpha(b, b_0', n)} \otimes \alpha(b_0, b_0', n). \tag{12}$$

Hence $\alpha'(b, b', \cdot)$ is degree-at-most-$(s-3)$ biased on J for many $(b, b') \in I_2^2$. To see the last assertion quantitatively, if the final average has modulus at least $\delta > 0$ and each inner average is at most B, a proportion at least $\delta/(2B)$ has modulus at least $\delta/2$. All selections and parameter cutoffs above depend internally on the parameters. Products and conjugates do not increase the ordinary nilsequence degree or destroy the fixed finite complexity bounds.

A28.4 Exact symbols and the lower-slot remainder

Constants such as h_0, c_0, b_0, b_0' are limit integers. Give the original multidegree group its total-degree filtration generated by all components whose total weight is *at least* the index, as verified in LP05. Apply the uniform project ordinary-degree shift theorem on the full slot domain to a constant translation in the relevant coordinate. Its difference has degree at most $s-1$, and stays lower-degree under any fixed integer linear pullback. Thus a constant added to a slot can be removed in the degree-s symbol. A slot consisting only of a constant becomes zero. LP13 additivity gives $T(\dots, 0, \dots) = 2T(\dots, 0, \dots)$, so that symbol is zero. This argument does not choose an implicit subgroup basepoint for a fixed-slot orbit. The coarsening uses $|a| \geq j$; the opposite inequality printed in Example 6.11 is not used.

From (3),(4), linearity in the first slot, and last-block symmetry,

$$\begin{aligned} [\tau(a, b, n)] &= s\big(T(b; (n+a)^r) - T(b; n^r)\big) \\ &= s \sum_{j=1}^{r} \binom{r}{j} T(b; a^j, n^{r-j}). \end{aligned} \tag{13}$$

In the first line $(n+a)^r$ denotes repeated slots. The leading term is $srT(b; a, n^m)$. Let

$$R_\tau(a, b, n) = s \sum_{j=2}^{r} \binom{r}{j} T(b; a^j, n^{r-j}). \tag{14}$$

Every summand has at most $s-3$ slots equal to n. Substitute $a = c_0 - b - b'$ in (9), remove c_0 as just justified, and expand the j copies of $-b-b'$ by LP13. The leading symbol of α is

$$-sr\, T(b - b'; b + b', n^m). \tag{15}$$

This has the order of the two distinguished entries dictated by (13). Interchanging them would require an unproved first-slot symmetry and is not permitted. The remainder is the difference of the two substitutions in (14), each fully expanded into entries in $\{b, b', n\}$ and still with at most $s-3$ copies of n.

Apply the mixed difference (12). At the level of degree-s symbols, constant slots vanish; a monomial with no b or no b' cancels. In (15) the pure terms $T(b; b, n^m)$ and $T(b'; b', n^m)$ therefore disappear. There is a globally represented unit nilcharacter β such that

$$[\alpha'(b, b', n)] = -sr\big(T(b; b', n^m) - T(b'; b, n^m)\big) + [\beta(b, b', n)]. \tag{16}$$

Here $[\beta]$ is an explicit finite integer linear combination of symbols $T(v_1; \dots, v_s)$, with every $v_i \in \{b, b', n\}$ and at most $s-3$ slots equal to n. More precisely, expand (14) after the two substitutions

$(a,b) = (-b-b',b)$ and $(-b-b',b')$, subtract the second, then delete the monomials missing b or missing b'. Tensor the corresponding actual pullbacks of $\widetilde{\chi}$ with their signed integer multiplicities. This constructs β and proves (16). Thus the needed remainder assertion is stronger than merely a separate bound on its degree in n.

Define the actual unit nilcharacter

$$\sigma(b,b',h_1,\dots,h_m,n) = \bigotimes_{\omega\in\{0,1\}^m} \mathcal{C}^{|\omega|}\alpha'\left(b,b',n+\sum_{j=1}^{m}\omega_j h_j\right), \tag{17}$$

where $\mathcal{C}$ is conjugation. Every shift has its own h_j. It has total degree at most s by linear pullback and tensor arithmetic. Apply the alternating sum to (16). In a term with fewer than m slots equal to n, expand each such slot into n and the available h_j. If a subset S of the shift labels is used, its coefficient contains

$$\sum_{\omega:\,\omega_j=1\ (j\in S)} (-1)^{|\omega|} = 0 \quad \text{unless } S=[m].$$

There are too few slots to use all m labels, so every β term vanishes. For the leading term there are exactly $m!$ surviving bijections of labels to slots, each with coefficient $(-1)^m$. Last-block symmetry identifies their values. With the minus sign in (16), the exact result is

$$[\sigma] = (-1)^{s-1}s!\big(T(b;b',h_1,\dots,h_m) - T(b';b,h_1,\dots,h_m)\big). \tag{18}$$

The unused coordinate n on the right is still a coordinate of the same domain. Formula (18) keeps the actual definitions (9),(12),(17). If one instead conjugates α and α' explicitly, its coefficient changes sign; silently using the other object is not an identification.

A28.5 From correlation to a zero symbol on the correct box

By Proposition 5.6, the positive degree-at-most-$(s-3)$ bias obtained after (12) implies a positive $U^m[J]$ norm for a scalar coordinate of α', for many (b,b'). Appendix A uniformity gives a standard positive lower bound on a dense internal set. Equivalently the average of the vector $U^m[J]^{2^m}$ quantities is $\gg 1$. Since the dimension is standard finite, at least one fixed scalar coordinate contributes $\gg 1$. For $s=3$, this is the converse at degree zero and U^1, so no extra one-step test has been inserted.

Expand the interval Gowers norm using its zero-extension definition and the same normalization as GTZ. One can choose the auxiliary cyclic group of order a fixed multiple of N greater than $2(m+1)N$. For a nonzero cube all vertices lie in J, so its differences have unique lifts $h_j \in I_1$, and no lifted sum can wrap around that cyclic group. The normalizing denominator is a positive standard constant multiple of the count of such cubes. Thus the expansion gives a coordinate of (17) with a positive standard average, up to a fixed positive normalizing factor, on the convex polytope

$$\Omega = \{(b,b',h,n) : b,b' \in I_2, \quad 1 \le n+\omega\cdot h \le N \text{ for all } \omega\in\{0,1\}^m\}. \tag{19}$$

In particular $\Omega \subseteq I_8^{s+1}$, and $\left|\mathbb{E}_{x\in I_8^{s+1}}\mathbf{1}_\Omega(x)\sigma(x)\right| \gg 1$. All defining inequalities in (19) are affine linear; their number depends only on s. The selection of biased pairs is used only for a lower bound on the nonnegative norm average, not as an extra arbitrary cutoff in (19).

Published E.11 asks for dense subprogressions of a positive interval. Partition I_8^{s+1} into its finitely many sign orthants and coordinate-zero hyperplanes. Since σ is bounded, the latter have normalized mass $O_s(1/N)$ and contribute $o(1)$. By the triangle inequality one open orthant has positive standard weighted average. For its signs $\varepsilon_i \in \{-1,1\}$, reflect $x_i = \varepsilon_i y_i$. The reflected

polytope is convex and the reflected nilcharacter $\sigma_\varepsilon(y) = \sigma(\varepsilon y)$ has degree at most s. Take every progression in E.11 to be the full positive interval $[8N]$. The hypothesis is plain bias, hence bias of degree strictly below s. E.11, with dimension $s+1$, degree $s \geq 3$ and scale $8N$, gives $\sigma_\varepsilon \in \mathrm{Nil}_{<s}(I_8^{s+1})$. Reflecting back gives

$$[\sigma]_{\mathrm{Symb}_s(I_8^{s+1})} = 0. \tag{20}$$

Only one orthant was used to obtain bias; E.11 itself supplies the conclusion on the full symmetric box. No extrapolation from positive points is implicit.

Pull (18),(20) back along the fixed integer linear map

$$(h, n) \longmapsto (b, b', h_1, \ldots, h_m, n_{\mathrm{aux}}) = (h, n, n, \ldots, n, 0).$$

It maps I_8^2 into I_8^{s+1}. Lower-degree nilsequences pull back with the same degree, so

$$s!\big(T(h; n^r) - T(n; h, n^{r-1})\big) = 0 \quad \text{in } \mathrm{Symb}_s(I_8^2). \tag{21}$$

The sign in (18) can be removed because both sides equal zero. Apply E.13 on this symmetric box, with dimension two, degree $s \geq 3$ and the fixed positive integer $s!$. Its torsion-free conclusion gives

$$T(h; n^r) = T(n; h, n^{r-1}) \quad \text{in } \mathrm{Symb}_s(I_8^2). \tag{22}$$

We restrict to $I_1 \times J$ only after this use. E.13 is not being applied directly to an asymmetric rectangle.

A28.6 Exact finite-component factorization

Set $\Theta(n) = \widetilde{\chi}(n, \ldots, n)$, the same object as in LP05. It is a unit degree-at-most-s nilcharacter. Expand $[\Theta(n+h)] - [\Theta(n)]$ by LP13. Terms with exactly one h equal

$$T(h; n^r) + rT(n; h, n^{r-1}) = sT(h; n^r) = [\chi(h, n)] \tag{23}$$

on $I_1 \times J$, by (22),(3). Let $R(h, n)$ be the tensor product of the actual pullbacks of $\widetilde{\chi}$ for all subsets of the s slots of cardinality at least two, putting h in those slots and n elsewhere. There is no permutation of the distinguished first slot in this definition. Every factor has multidegree at most $(i, s-i)$ for its subset size i: coarsen the cube filtration by counting the coordinates assigned to h and n, using all weights at least the index. The polynomial orbit, rationality and top frequency are as in the LP05 block coarsening. Their product is therefore supported on the union of these downsets, contained in $\mathcal{J}$.

By the symbol equality just proved, the unit-vector expression

$$W(h, n) = \chi(h, n) \otimes \overline{\Theta(n+h)} \otimes \Theta(n) \otimes R(h, n) \tag{24}$$

is a degree-at-most-$(s-1)$ nilsequence on $I_1 \times J$. By the definition of the restricted nilsequence class, it has a global representation whose restriction is (24); no global equality beyond this rectangle is claimed. Such a degree filtration gives multidegree support $i + j \leq s - 1$ by assigning the group at (i, j) its total-degree group. Choose that representation and set $\Psi = \overline{R} \otimes W$. It has support in (1) by the product filtration and is uniformly bounded.

Finally tensor (24) with $\Theta(n+h) \otimes \overline{\Theta(n)} \otimes \overline{R}$. Contract the matching pairs using their unit norms. Each contraction is a fixed linear map between fixed finite tensor spaces. The result is exactly χ, so (2) holds component by component with fixed bounded coefficients. This does not divide by any scalar coordinate or replace symbol equality by an unproved scalar equality. It proves Theorem A28.1.

A28.7 Bounds and source comparison

All finite coordinate selections, tensor dimensions, permutation counts and parameter boxes depend only on fixed degree and geometric data. Appendix A selection/uniformity supplies the finite family of lower tests before the averaging manipulations. The three Cauchy–Schwarz steps use common standard bounds, and each pigeonhole retains a standard positive amount. Fixed parameter values may be unbounded limit integers; the project shift theorem supplies uniform constructions for exactly that case. All algebraic identities are finite and transfer coordinatewise, with the all-limit vertical convention retained. E.11 and E.13 have the exact degree, dimension, progression and symmetric-box domains stated above.

The proof retains, rather than erases, the published factorial history. Theorem 7.4 on both pp. 1271 and 1315 prints the stricter hypothesis "$< s-2$-correlates", whereas the preceding Theorem 7.1 supplies degree-at-most-$(s-2)$ correlation. The target here deliberately assumes the latter, and Section 2 proves the required selection and shift step under that weaker hypothesis. The printed Theorem 7.4 is not a premise. In addition to the repeated shift index and alternating-derivative sign in (13.14)–(13.17), the first two slots in printed Lemma 13.4 are interchanged relative to (13.3),(13.10). This changes the sign in (13.13). Formula (18) is the resulting coefficient for the actual unconjugated objects. The published simplification in Lemma 13.3 also retains an a where cancellation requires b; its tail has degree at most $s-3$ in n. The scalar lower-factor bookkeeping and the I_6 substitution above give their own exact formulas. These source discrepancies are not counterexamples to the final zero relation. Their required related uses and current representations must be checked independently before a complete correction claim.

This packet proves a local composition relative to its stated project and published premises. It does not reprove E.11/E.13's equidistribution inputs, Proposition 5.6's published converse sources, Proposition 8.4, the frequency analysis, or the final finite-family inverse-theorem assembly. Those are direct dependencies or separate consumers, not silently discharged here.

A29 Same-step linearization using the increment polynomial group

This section proves the same-step linearization construction. It uses the bounded normalization and tautification proved in Section A17, and the published algebraic polynomial-map results specified below. It does not assume GI(s).

A29.1 The required conclusion

Let G/Γ be a connected simply connected nilpotent Lie group modulo a lattice, with a rational degree filtration $G_\bullet$ of degree at most $s \geq 1$. Every G_i is connected and rational. Let a scalar polynomial nilsequence be $F(g(n)\Gamma)$, with F complex-valued, 1-bounded and Lipschitz. The polynomial g is adapted to $G_\bullet$ in the right-difference convention $\partial_u g(n) = g(n+u)g(n)^{-1}$.

Theorem A29.1. *For this fixed filtered nilmanifold there is a fixed s-step nilmanifold $\widetilde{G}/\widetilde{\Gamma}$ with a fixed smooth metric, and a constant C depending only on the original fixed geometric data, with the following property. For every polynomial g and every such F, there are $a \in \widetilde{G}$ and a 1-bounded Lipschitz function $\widetilde{F} : \widetilde{G}/\widetilde{\Gamma} \longrightarrow \mathbb{C}$ such that*

$$F(g(n)\Gamma) = \widetilde{F}(a^n\widetilde{\Gamma}) \quad (n \in \mathbb{Z}), \qquad \mathrm{Lip}(\widetilde{F}) \leq C\,\mathrm{Lip}(F).$$

Fixed smooth metric comparisons are included in C. The same construction applies to limit polynomial maps and bounded Lipschitz limit functions, with fixed nilmanifold and uniform constants.

Consequently a finite-family polynomial inverse theorem of degree s implies the finite-family linear inverse theorem of step s, without changing the input Gowers norm or losing correlation.

If $\mathrm{Lip}(F) = 0$, the constant test function has a constant extension and the displayed bound is understood literally. A point nilmanifold can be used for a constant sequence. Step at most s is the meaning of s-step throughout.

By Section A17, restrict to a fixed taut representation with $G_0 = G_1 = G$ and replace the polynomial by h, where $h(0) = 1$. The exact identity is

$$g(0) = q\gamma, \qquad h(n) = q^{-1}g(n)\gamma^{-1}, \qquad F_q(x\Gamma) = F(qx\Gamma), \qquad F_q(h(n)\Gamma) = F(g(n)\Gamma).$$

Here q lies in a fixed compact set and $\mathrm{Lip}(F_q)$ is bounded by a fixed multiple of $\mathrm{Lip}(F)$. The underlying taut filtered nilmanifold is fixed independently of g. It therefore suffices to prove the theorem for this normalized h and F_q. In what follows all geometric data refer to this fixed taut representation.

A29.2 Polynomial groups and their Lie algebras

For $i \geq 0$ put

$$P_i = \mathrm{poly}(\mathbb{Z}_{\mathbb{N}} \longrightarrow G_{\bullet+i}).$$

The zeroth group of the shifted filtration is G_i; it need not be taut. The group operation is pointwise multiplication. By [151, Corollary B.4 and Proposition B.6], these are groups with

$$[P_i, P_j] \subseteq P_{i+j}, \qquad P_i \triangleleft P_0. \tag{200}$$

Those results concern abstract filtered groups and require no equality of the first two groups. Their conditions hold because the shifted groups inherit nesting and the commutator law from $G_\bullet$.

Here is an explicit finite-dimensional description, including the smooth real shift needed below. Let $\mathfrak{g}_i = \log G_i$ and define

$$V_i = \left\{ X(t) = \sum_{j=0}^{s-i} X_j t^j : X_j \in \mathfrak{g}_{j+i} \right\} \quad (0 \leq i \leq s), \qquad V_i = 0 \quad (i > s).$$

Use pointwise Lie bracket. Bracketing coefficients gives

$$[V_i, V_j] \subseteq V_{i+j}. \tag{201}$$

For instance, a bracket of a degree-a coefficient in $\mathfrak{g}_{a+i}$ and a degree-b coefficient in $\mathfrak{g}_{b+j}$ belongs to $\mathfrak{g}_{a+b+i+j}$ and has polynomial degree $a + b$. Thus V_i is a finite-dimensional nilpotent Lie algebra for every i. Its simply connected group is obtained by the terminating BCH formula, or equivalently by pointwise exponentiation.

Lemma A29.2. *The pointwise exponential identifies that group with P_i. In particular P_i is connected and simply connected, and its smooth topology can be described either by the coefficients of $\log p(t)$ or by the Taylor coefficients in [151, Lemma B.9].*

Proof. If $i > s$, the zeroth group of the shifted filtration is trivial, so both P_i and V_i are trivial. Fix $0 \leq i \leq s$ and put $m = s - i$. We use the connectedness of every G_j stated in the setup.

First recall explicitly why the global exponential coordinates used here are smooth. For a connected nilpotent Lie group H, let $Q_a(x) = x^a$, where a is a positive integer. In left tangent trivializations, differentiating the a factors gives

$$DQ_a(x) = \sum_{k=0}^{a-1} \mathrm{Ad}(x)^{-k}.$$

The infinitesimal adjoint action lowers the lower central series. An exponential neighborhood generates H, so $\mathrm{Ad}(x)$ induces the identity on each successive quotient of that series. Consequently $DQ_a(x)$ induces multiplication by a on each quotient and is invertible. The usual local Lie exponential is smooth with derivative the identity at zero. Given $X \in \mathfrak{h}$, choose a so large that X/a lies in this local chart. The identity

$$\exp_H(Y) = Q_a(\exp_H(Y/a))$$

then proves that the exponential has invertible derivative near X. For simply connected nilpotent H, its global bijectivity is the exponential homeomorphism recalled in [151, Section 3]. Its local smooth inverses therefore agree to give a smooth global inverse. This argument uses the local Lie exponential, differentiation of group operations and the inverse function theorem. It does not infer smoothness from the word *homeomorphism*.

By [151, Proposition B.6], P_i is a group under pointwise multiplication. By [151, Lemma B.9], its elements have the unique increasing-order Taylor expansion

$$p(n) = \prod_{j=0}^{m} u_j^{\binom{n}{j}}, \qquad u_j \in G_{i+j}, \qquad n \in \mathbb{Z}.$$

The coefficients are recovered by the triangular formulas

$$u_0 = p(0), \qquad u_j = \left(\prod_{k<j} u_k^{\binom{j}{k}} \right)^{-1} p(j).$$

The order of the displayed inverse product is part of the formula. Transfer the product manifold structure of $\prod_{j=0}^{m} G_{i+j}$ to P_i using these coordinates. Each evaluation at an integer is smooth. Evaluation of a product or inverse followed by the recovery formulas proves smoothness of the group operations. The recovered coefficients lie in the prescribed embedded subgroups; smoothness into those subgroups follows from their local embedded charts. The same formulas identify this topology with the pointwise topology.

Evaluation at $0, \dots, m$ is an injective group homomorphism $P_i \longrightarrow G^{m+1}$. Hence P_i is nilpotent, with class at most the class of G. Each closed connected subgroup of a simply connected nilpotent group is simply connected. In exponential coordinates it is the exponential of its Lie subalgebra: the finite nilpotent BCH formula makes that exponential image a connected subgroup, and uniqueness of the connected subgroup with that Lie algebra gives the assertion. Thus the product manifold above, and therefore P_i, is connected and simply connected. This step does not assert that P_0 has class at most s.

At the identity, differentiating the Taylor parametrization gives

$$(X_0, \dots, X_m) \longmapsto \left(n \longmapsto \sum_{j=0}^{m} \binom{n}{j} X_j \right), \qquad X_j \in \mathfrak{g}_{i+j}.$$

The binomial-to-monomial matrix is triangular with nonzero diagonal. A contribution to degree b comes only from $j \geq b$, and $\mathfrak{g}_{i+j} \subseteq \mathfrak{g}_{i+b}$. Descending triangular inversion preserves these same coefficient spaces. This identifies the tangent space with exactly V_i. Evaluation maps are smooth homomorphisms, so their derivatives preserve Lie brackets. The finitely many evaluations just used determine every tangent vector. They therefore identify the bracket with the pointwise bracket on V_i.

Uniqueness of one-parameter subgroups now gives

$$\mathrm{ev}_n(\exp_{P_i} X) = \exp_G(X(n)) \qquad (n \in \mathbb{Z}).$$

Apply the smooth-global-exponential argument above to P_i. Its inverse assigns to p exactly the Lie polynomial whose values are $\log_G(p(n))$. Conversely every $X \in V_i$ gives an element of P_i. This proves the pointwise exponential identification and the smooth logarithmic coefficient coordinates. The group law is determined by multiplication in G at the integer arguments. Under the ordinary nilpotent BCH identification it is the pointwise BCH law. The coordinate spaces and finite evaluation maps depend only on the fixed filtration and i. □

For real u, define T^u on V_i by $T^u X(t) = X(t+u)$. Coefficient expansion and nesting show that this preserves V_i. It is a smooth one-parameter group of Lie algebra automorphisms, since the bracket is pointwise. Exponentiating defines a smooth automorphism group on P_i, which at integer u is the ordinary shift of polynomial sequences. Its infinitesimal action is ordinary differentiation, and

$$\frac{d}{dt} V_i \subseteq V_{i+1}. \tag{202}$$

Indeed the coefficient of t^j in $X'(t)$ belongs to $\mathfrak{g}_{j+1+i}$.

A29.3 The lattice and the same-step semidirect product

Define

$$\Lambda_i = \{p \in P_i : p(n) \in \Gamma \text{ for every } n \in \mathbb{Z}\}.$$

It is a subgroup by pointwise multiplication. We give the needed lattice argument to avoid relying on the disputed step estimate in Appendix C.

Lemma A29.3. *Λ_i is a discrete cocompact subgroup of P_i. Integer shifts preserve Λ_i.*

Proof. It is enough to consider $i \leq s$ and put $d = s - i$; for larger i both groups are trivial. Discreteness follows from evaluation at $0, \dots, d$. A lattice-valued polynomial close enough to the identity at these finitely many points equals the identity at all those points, since Γ is discrete. Taylor uniqueness then makes it the identity polynomial.

For cocompactness, use induction to write

$$p = \gamma_j h_j c_j,$$

where $\gamma_j \in \Lambda_i$, c_j ranges over a fixed compact subset of P_i, and h_j is the identity at $0, \dots, j-1$. For $j = 0$ take $\gamma_0 = c_0 = 1$ and $h_0 = p$. Taylor expansion shows $h_j(j) \in G_{j+i}$. Choose a fixed compact left fundamental set for the lattice $G_{j+i} \cap \Gamma$ in G_{j+i} and write $h_j(j) = \lambda_j v_j$, where λ_j is in that lattice and v_j is in the compact set. Define

$$\ell_j(t) = \lambda_j^{\binom{t}{j}}, \qquad v_j^*(t) = v_j^{\binom{t}{j}}.$$

Both are in P_i, by the same one-parameter difference check as above; $\ell_j \in \Lambda_i$ because $\binom{n}{j}$ is an integer for every $n \in \mathbb{Z}$. Now put

$$\gamma_{j+1} = \gamma_j \ell_j, \qquad h_{j+1} = \ell_j^{-1} h_j (v_j^*)^{-1}, \qquad c_{j+1} = v_j^* c_j.$$

These formulas preserve the factorization and make h_{j+1} the identity at $0, \dots, j$. The possible v_j^* form a compact subset of P_i, so the possible c_{j+1} remain in a fixed compact set. At $j = d+1$, Taylor uniqueness gives $h_{d+1} = 1$. Thus $p = \gamma c$ with c in a fixed compact set, which proves left cocompactness; inversion gives right cocompactness. Finally an integer shift preserves both polynomiality and all integer lattice values, and its inverse does also. □

Now use the *increment* group P_1, not P_0, and set

$$\widetilde{G} = P_1 \rtimes_T \mathbb{R}, \qquad (p,u)(q,v) = (p\, T^u q, u+v), \qquad \widetilde{\Gamma} = \Lambda_1 \rtimes_T \mathbb{Z}. \tag{203}$$

The real action just constructed makes $\widetilde{G}$ a connected simply connected Lie group, diffeomorphic to $P_1 \times \mathbb{R}$. Its Lie algebra is $V_1 \rtimes \mathbb{R}\tau$, where $[\tau, X] = X'$. Give this Lie algebra the filtration

$$\mathfrak{l}_0 = \mathfrak{l}_1 = V_1 \oplus \mathbb{R}\tau, \qquad \mathfrak{l}_j = V_j \quad (j \geq 2).$$

Equations (201) and (202) prove $[\mathfrak{l}_i, \mathfrak{l}_j] \subseteq \mathfrak{l}_{i+j}$ for all indices. In particular $\mathfrak{l}_{s+1} = 0$. Thus $\widetilde{G}$ is nilpotent of step at most s. This explicit shifted-filtration calculation is the same-step bound used in the theorem.

The subgroup $\widetilde{\Gamma}$ is discrete, because its clock coordinate is integral and Λ_1 is discrete. It is cocompact as well. Given (p,u), left-multiply by $(1, -\lfloor u \rfloor)$, then left-reduce the resulting polynomial coordinate modulo Λ_1 using Lemma A29.3. The resulting representative lies in the product of a fixed compact set in P_1 and $[0,1]$. This proves left cocompactness; inversion proves right cocompactness. The operations used for left reduction belong to $\widetilde{\Gamma}$ by the integer-shift assertion of the lemma.

A29.4 The orbit identity and fixed compact fiber

Let $\theta : \widetilde{G}/\widetilde{\Gamma} \longrightarrow \mathbb{R}/\mathbb{Z}$ be the map induced by the clock coordinate, and set $K = \theta^{-1}(0)$. It is a fixed compact submanifold canonically identified with P_1/Λ_1: every element of the fiber can be written $(p,0)\widetilde{\Gamma}$, and two such representatives agree precisely when their polynomial coordinates differ on the right by an element of Λ_1. Evaluation at zero gives a smooth map

$$\pi : K \longrightarrow G/\Gamma, \qquad \pi((p,0)\widetilde{\Gamma}) = p(0)\Gamma. \tag{204}$$

It is well-defined because $\lambda(0) \in \Gamma$ for $\lambda \in \Lambda_1$. Since K is compact, π is Lipschitz for fixed smooth metrics.

The normalized polynomial $h \in P_0$ has $h(0) = 1$. Define its right increment by

$$b(t) = h(t)^{-1} h(t+1).$$

The left increment $h(t+1)h(t)^{-1}$ lies in P_1 directly from the iterated-difference definition. Proposition B.6 gives $P_1 \triangleleft P_0$, so conjugating that left increment by $h(t)^{-1}$ shows $b \in P_1$. This verifies the orientation of the increment actually used, rather than substituting a left increment into a right-telescoping product.

Put $a = (b,1) \in \widetilde{G}$. For every integer n define $b_n(t) = h(t)^{-1} h(t+n)$. The exact cocycle identity

$$b_{n+m}(t) = b_n(t)\, b_m(t+n)$$

and $b_0 = 1$ prove, for positive and negative integers alike,

$$a^n = (b_n, n).$$

Modulo $\widetilde{\Gamma}$, right multiplication by $(1, -n)$ gives $a^n\widetilde{\Gamma} = (b_n, 0)\widetilde{\Gamma} \in K$. Consequently

$$\pi(a^n\widetilde{\Gamma}) = b_n(0)\Gamma = h(n)\Gamma, \qquad n \in \mathbb{Z}. \tag{205}$$

The last equality uses the normalization $h(0) = 1$.

A29.5 Test functions, uniformity and the inverse-theorem join

The function $F_q \circ \pi$ is 1-bounded on K and has a Lipschitz constant bounded by a fixed multiple L of $\mathrm{Lip}(F)$. Extend its real and imaginary parts separately by the explicit McShane formula

$$E_u(x) = \inf_{y \in K} \bigl(u(y) + L\, d(x, y)\bigr).$$

This equals u on K and is L-Lipschitz: the triangle inequality proves the Lipschitz assertion, and the original Lipschitz inequality proves equality at points of K. Clip each extension to $[-1, 1]$, combine them as a complex function, and apply metric projection onto the closed unit disk. This projection Q is 1-Lipschitz: the nearest-point inequalities for the disk give

$$\langle x - Qx, Qy - Qx\rangle \le 0, \qquad \langle y - Qy, Qx - Qy\rangle \le 0,$$

and therefore $\|Qx - Qy\|^2 \le \langle Qx - Qy, x - y\rangle \le \|Qx - Qy\|\|x - y\|$. It also fixes the original values on K. Thus the result $\widetilde{F}$ is 1-bounded, agrees with $F_q \circ \pi$ on K, and is at most $\sqrt{2}L$-Lipschitz. Equation (205) proves the exact equality in Theorem A29.1.

The nilmanifold, lattice, compact fiber, evaluation map and all metric constants depend only on the fixed original filtered nilmanifold and s. The possibly unbounded polynomial coefficients occur only in the allowed translation element a and in the bounded representative q affecting the test function. They do not alter the geometry or its Lipschitz bound. For limit data the same algebraic formulas and one fixed bounded representative section give limit choices of b and a. Apply the explicit extension formula to each representative test function. Its uniform bound and Lipschitz constant give a bounded Lipschitz limit function on the same standard nilmanifold. The ordinary orbit identities hold at every member of the defining sequences and therefore at every limit integer.

Suppose now that a polynomial inverse theorem supplies, for fixed s and δ, a finite family of rational degree-s filtered nilmanifolds, a fixed Lipschitz bound, and a positive correlation lower bound against 1-bounded test functions. Apply the theorem to each member of that finite family, taking the maximum of its finitely many fixed metric constants. There is still a finite family, now of step at most s with linear orbits, and the correlation is unchanged because the represented sequences are equal pointwise. No comparison of Gowers norms is involved in this join. This proves the final assertion of Theorem A29.1.

The construction also covers a point input group: P_1 is trivial and the clock group yields a circle with a constant test function, or it may be replaced by a point. For $s = 1$ the increment group consists of constant polynomials in an abelian group and the shift action is trivial, so $\widetilde{G}$ is abelian, as the required bound predicts.

A29.6 Comparison with the failed full-group construction

The original Appendix C used $P_0 \rtimes \mathbb{R}$. For the taut abelian degree-s filtration on $\mathbb{R}$, the commutator of its clock with $p(t) = \frac{1}{2}\binom{t}{s}$, repeated s times, is the nonzero constant $1/2$. That is a commutator of length $s+1$, so that group is not s-step. This adverse example is recorded in the separate finding packet. The present construction uses P_1 and its increment b, reducing the polynomial degree available in the clock action by one. Its Lie filtration above verifies the required step bound for all inputs, not only this example. It changes the construction, while retaining the same-step linear conclusion needed by GI(s).

The finite-family polynomial inverse theorem is still a separate premise. This unit does not prove its rank reduction, symmetrization, or ultralimit-to-finite-family extraction. Those dependencies remain explicit in the theorem assembly.

A30 A normalized local converse for the GI assembly

This section derives the precise correlation-to-Gowers-norm implication used by the assembly. It imports the published compact cube constraint, Green–Tao, *Linear equations in primes*, Proposition 11.5, p. 1799, with its definitions in 11.4 and proof in Appendix E. It uses the internal increment linearization for polynomial tests. It does not import Proposition 11.2's averaging-domain transition, assume an inverse theorem, or claim to reprove the cube constraint.

A30.1 Available geometric assertion and needed conclusion

Fix an integer $d \geq 1$ and a fixed d-step nilmanifold $M = G/\Gamma$ with a smooth metric. The available cube constraint says there is a compact subset $\Sigma \subseteq M^{\{0,1\}^{d+1}\setminus\{0\}}$ and a continuous map $P : \Sigma \to M$ such that, for every $g \in G$, $x \in M$, $n \in \mathbb{Z}$ and $h \in \mathbb{Z}^{d+1}$, the nonzero vertices $(g^{n+\omega\cdot h}x)_{\omega\neq 0}$ lie in Σ and

$$g^n x = P((g^{n+\omega\cdot h}x)_{\omega\neq 0}).$$

No bound on g or the integers is imposed. In our application G is connected simply connected and at most d-step; these satisfy the published nilmanifold hypothesis.

Theorem A30.1. *For fixed d, M, L and every $\kappa > 0$ there is $c > 0$ such that every 1-bounded $v : [N] \to \mathbb{C}$ and 1-bounded L-Lipschitz test $F : M \to \mathbb{C}$ satisfying*

$$|\mathbb{E}_{n\in[N]} v(n)F(g^n x)| \geq \kappa$$

obey $\|v\|_{U^{d+1}[N]} \geq c$. The bound is uniform in N, g, x, F. The same conclusion holds for a fixed finite family of rational degree-at-most-d polynomial models, using the exact same-step increment linearization and taking a finite maximum of its bounds.

A30.2 A finite tensor approximation uniform in the test

The algebra generated by coordinate functions that are Lipschitz on M separates points on the compact set Σ, contains constants and is closed under conjugation. Complex Stone–Weierstrass gives, for every fixed continuous F and $\epsilon > 0$, an approximation

$$F(P(z)) = \sum_{a=1}^{A} c_a \prod_{\omega\neq 0} H_{\omega,a}(z_\omega) + R(z), \qquad |R(z)| \leq \epsilon,$$

where the factors are 1-bounded Lipschitz functions. Scaling each factor into the unit disk is absorbed into the finite coefficients c_a; no coefficient is discarded. By Arzela–Ascoli the class $\|F\|_\infty \le 1$, $\mathrm{Lip}(F) \le L$ is compact in the uniform norm. A finite uniform net, approximated as above, makes A, $\sum_a |c_a|$ and all factor bounds depend only on M, L, ϵ. Choose a net radius $\epsilon/2$ and approximate its members to $\epsilon/2$ to retain the displayed total error.

For each orbit, put $b_{\omega,a}(t) = H_{\omega,a}(g^t x)$. The cube constraint gives

$$F(g^n x) = \sum_a c_a \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h) + O(\epsilon) \tag{206}$$

for every integer n, h, uniformly. The factors are 1-bounded; their dependence on the orbit will no longer matter in the analytic estimate.

A30.3 Normalized smooth averaging and a uniform Fourier bound

Choose once and for all a smooth $\sigma : \mathbb{R} \to [0,1]$ supported on $[-1,2]$ and equal to one on $[0,1]$. Define

$$S_N = \sum_{h \in \mathbb{Z}} \sigma(h/N), \qquad \mu_N(h) = \sigma(h/N)/S_N.$$

Then $N \le S_N \le 3N+1 \le 4N$, and μ_N is a probability measure with finite support. Average (206) against the product probability measure $\prod_j \mu_N(h_j)$ on all integer h. The error remains at most ϵ, with no missing normalizing scalar. Thus

$$F(g^n x) = \sum_a c_a B_a(n) + O(\epsilon), \quad B_a(n) = S_N^{-(d+1)} \sum_{h \in \mathbb{Z}^{d+1}} \prod_j \sigma(h_j/N) \prod_{\omega \neq 0} b_{\omega,a}(n + \omega \cdot h).$$

Take $Q = C_d N$ with a fixed integer $C_d > 10(d+2)$, and identify representatives of $\mathbb{Z}/Q\mathbb{Z}$ with a centered interval. The sampled cutoff $w_N(h) = \sigma(h/N)$ has support strictly inside that interval. Write its discrete Fourier expansion as $w_N(h) = \sum_r \widehat{w}_N(r) e(rh/Q)$ with normalized coefficients. There is a bound

$$\sum_{r \in \mathbb{Z}/Q\mathbb{Z}} |\widehat{w}_N(r)| \le C'_d \tag{207}$$

independent of N. To prove it, the zero coefficient and every coefficient are at most $S_N/Q = O_d(1)$. For $r \neq 0$, summation by parts twice gives

$$|1 - e(r/Q)|^2 |\widehat{w}_N(r)| \le Q^{-1} \sum_h |\Delta^2 w_N(h)| \le C/(NQ).$$

The final bound follows from the second derivative of the fixed smooth cutoff, with only $O(N)$ possibly nonzero terms. For the distance $r_* \in [1, Q/2]$ to zero modulo Q, $|1 - e(r/Q)| \ge 4r_*/Q$. Thus $|\widehat{w}_N(r)| \le C_d/r_*^2$ and summation proves (207). Smooth zero extension of σ ensures that the same finite-difference bound holds at its support boundaries.

A30.4 The exact Gowers estimate

Zero-extend v to $\mathbb{Z}/Q\mathbb{Z}$. Extend each bounded integer function $b_{\omega,a}$ to the cyclic group so that it agrees on all integer arguments that occur when $n \in [N]$ and $h_j \in [-N, 2N]$. Those arguments lie between $-(d+1)N$ and $(2d+3)N$; the choice of C_d places the whole interval strictly inside

the centered representative range. There is therefore no conflicting cyclic value. The normalized correlation with B_a is exactly

$$\frac{Q}{N}\left(\frac{Q}{S_N}\right)^{d+1}\mathbb{E}_{n,h\in\mathbb{Z}/Q\mathbb{Z}}v(n)\prod_j w_N(h_j)\prod_{\omega\neq 0}b_{\omega,a}(n+\omega\cdot h).$$

The prefactor is bounded above by a constant depending only on d. This identity comes from an all-integer smooth average; no hard box average is replaced by a larger summation domain.

Expand the product of cutoffs by Fourier series. The sum of absolute values of its coefficients is at most $(C'_d)^{d+1}$. In each term, the factor $e(\sum_j r_j h_j/Q)$ can be absorbed into the nonzero vertex functions. For each j, choose an index $k\neq j$, which exists since $d+1\geq 2$, and use $h_j=(n+h_j+h_k)-(n+h_k)$. Multiply the functions at vertices e_j+e_k and e_k by the corresponding opposite linear phases. Their absolute bounds remain one. The Gowers–Cauchy–Schwarz inequality for the full $(d+1)$-cube gives the absolute bound $\|v\|_{U^{d+1}(\mathbb{Z}/Q\mathbb{Z})}$ for every such term. Complex conjugations are absorbed into the vertex functions, leaving the norm and absolute bounds unchanged. Hence

$$|\mathbb{E}_{[N]}vB_a|\leq K_d\|v\|_{U^{d+1}(\mathbb{Z}/Q\mathbb{Z})}\leq K_d\|v\|_{U^{d+1}[N]}.$$

The last inequality uses the prescribed normalized interval norm: the cyclic norm is the interval norm multiplied by the cyclic norm of the interval indicator, which is at most one. If necessary enlarge C_d also to exceed 2^{d+1} so this single ambient group realizes that interval normalization.

Combining the finite tensor approximation with this estimate yields

$$|\mathbb{E}_{[N]}v(n)F(g^nx)|\leq K_d\Big(\sum_a|c_a|\Big)\|v\|_{U^{d+1}[N]}+\epsilon.$$

Choose $\epsilon<\kappa/2$. This proves Theorem A30.1 with a positive bound depending only on d,M,L,κ. A fixed bound other than one for v or F is handled by dividing by that bound first. This proof is complex-valued throughout; it does not need a real-valued dual-norm convention.

A30.5 Polynomial input and source comparison

For the actual assembly input, a scalar polynomial nilsequence has degree $d=s-2\geq 1$ on fixed rational geometry. The accepted increment linearization001 represents it exactly on a fixed group of step at most d, with a fixed Lipschitz multiplier and unchanged norm bound. Applying the theorem there and taking finite maxima over a finite family gives precisely the positive normalized $U^{s-1}[N]$ norm required by the last induction step. This is a converse implication and has no inverse-theorem premise.

The literal published proof of 11.2 on p. 1801 defines an average over $h\in[N]^{d+1}$ with $\sigma(h_j/N)$, while $\sigma=1$ on $[0,1]$. Its later passage from (11.6) to (11.7) widens the shift domain to a cyclic group with the same cutoff supported on $[-1,2]$. That changes the set of summands. The local proof above instead starts with the normalized all-integer smooth average and tracks its exact cyclic prefactor. Whether an earlier source correction already supplied this interpretation has not been established. The observation is a bounded proof-transition issue, not a counterexample to Proposition 11.2 or the converse theorem.

The applications are the normalized average, its dual-norm estimate, Corollary 11.6 and the present converse. Published Proposition 11.5 remains the compact cube-constraint input. The full inverse theorem is assembled in Section A15.

A31 Uniform selection of represented lower tests

Fix standard $s \geq 3$, $d = s - 2 \geq 1$, a nonempty internal interval $J = [N]$, and an internal parameter set H. The actual represented unit families in the inverse-theorem construction are retained throughout. Suppose the lower-test correlation premise holds for every h. In its tensor convention, a bounded scalar component u_h has positive standard-size correlation with a scalar nilsequence of degree at most d. Apply the ordinary-test approximation in GTZ Lemma 6.4, with error smaller than half this correlation divided by $\max(1, \|u_h\|_\infty)$. Its finite component sum, with finite coefficient sum, contains a component of an ordinary globally equivariant unit-vector test retaining positive standard-size correlation. Conjugation preserves that class. Compatible polynomial Taylor coefficients may be limit coefficients. This proves pointwise existence before uniformity.

Enumerate the countable rational filtered model types, equip each with one fixed smooth metric, and enumerate their integral vertical characters and finite target dimensions. Let the ordinary set T_M contain the first M tags, all ordinary global unit tests with Lipschitz and supremum bounds at most M, all compatible Taylor coefficients without any bound on them, the finite coordinate and conjugation choices, and reciprocal correlation thresholds with positive integer denominator at most M. Choose these sets increasing in an ordinary disjoint union T. The natural extension of the single ordinary sequence $M \mapsto T_M$ supplies an internal exhaustion. This is stronger than an arbitrary countable family of internal sets. Each pointwise witness belongs to ${}^*T_M$ for some standard M.

The relation $R(h, t)$ says that the modulus of the normalized finite correlation sum is at least the encoded threshold. Evaluation and finite sums are internal; norm one and global equivariance are already conditions on the ordinary parameter set and transfer. No external predicate of positive standard size enters R. The least internal stage supporting a witness is pointwise bounded by a standard integer, hence bounded by one standard M_0 on H. Indeed, otherwise the internal sets E_M of points requiring a stage greater than M would be decreasing and nonempty. Choose decreasing representatives $E_{M,i}$ and put $S_M = \{i : E_{M,i} \neq \varnothing\}$. The S_M are decreasing and ultrafilter-large. Put

$$k_i = \max(\{k \leq i : i \in S_k\} \cup \{0\})$$

and choose a point of $E_{k_i,i}$ when $k_i > 0$. For each fixed M, on $S_M \cap \{i \geq M\}$ it belongs to $E_{M,i}$. Its limit belongs to every E_M, contradicting pointwise standard-stage existence. Internal choice inside ${}^*T_{M_0}$ now supplies the witness map. This is the bounded-stage argument of GTZ Appendix A, Definitions A.5 and A.10 and Lemma A.11; the countability of model tags is as on p. 1250. No compactness of the unbounded orbit coefficients is used.

Pad the finitely many target dimensions with zeros. If H has positive density, finite pigeonhole fixes the coordinate and convention choices on a positive-density internal subset. Outside it use a constant unit representative and keep the indicator separate. This is the later rank-step selection. For the earlier two-family application, keep both original strong represented families $\chi(h, n)$ and $\chi_h(n)$ unchanged and apply the same argument to

$$u_{h,a,b}(n) = \Delta_h f(n)\, \overline{\chi(h, n)_a}\, \overline{\chi_h(n)_b},$$

with conjugations as required by the tensor convention. No polynomial dependence on h is needed. Finite pigeonhole fixes the coordinate pair. Multiplication of the selected test by the internal unit phase of its nonzero correlation aligns its average; extend this scalar test by zero outside the retained subset. The phase has degree zero. The resulting lower tests and correlation threshold have common standard bounds. This selects a lower correlation witness, not a replacement of

either incoming family, and does not upgrade arbitrary weak nilcharacters to strong ones. The following shift and whole-matrix reconstruction arguments are the ones proved in the preceding inverse-theorem sections.

A32 Integral endomorphisms and polarized automorphisms

We supply the integral-lattice step in the marked-surface construction. The inputs are Milne's published chapter [116], especially §8 (torsion and Tate modules), §12 (decomposition, degree polynomial and divisibility), and §17 (Rosati positivity). Work over an algebraically closed field of characteristic different from 3.

For an abelian surface A, T_3A is free of rank four and $T_3A/3T_3A = A[3]$: successively lift torsion points for surjectivity, and shift a compatible sequence for the kernel. A unital action of the division quaternion algebra over $\mathbf{Q}_3$ on V_3A is faithful and has rank one. An endomorphism acting trivially on the Tate module is zero. Indeed, after isogeny decomposition, a nonzero map between simple isogenous factors is an isogeny and cannot kill torsion of unbounded order.

Finite rational rank alone does not prove finite generation of integral endomorphisms. Let S be simple, let M be generated freely by $e_1, \dots, e_r$ in $\mathrm{End}(S)$, put $V = \mathbf{Q}M$ and $D = V \cap \mathrm{End}(S)$. The degree polynomial is a positive integer on every nonzero point of D. A neighborhood of zero where its absolute value is less than one therefore contains no such point. Applying this to differences makes D uniformly discrete. A compact fundamental parallelepiped for M meets D finitely, so D/M is finite and $N_M D \subset M$ for some positive integer N_M.

The matrices $T_3(e_i)$ are independent over $\mathbf{Q}_3$. Scale a purported relation to coefficients $a_i \in \mathbf{Z}_3$, not all zero, and choose integers $n_i \equiv a_i \pmod{3^m}$. The endomorphism $\sum_i n_i e_i$ kills $S[3^m]$. The divisibility lemma gives

$$3^{-m} \sum_i n_i e_i \in D, \qquad N_M n_i / 3^m \in \mathbf{Z}.$$

For nonzero a_j, taking $m > v_3(N_M) + v_3(a_j)$ contradicts the latter inclusion. Consequently $r \le (2 \dim S)^2$. All finite independent sets satisfy this bound, so the rational endomorphism space is finite dimensional. Choose M spanning it and repeat the lattice argument to prove that the entire integral ring is finitely generated. For a general abelian variety, decomposition embeds its endomorphisms into a finite matrix of Hom groups between simple factors. Nonisogenous factors contribute zero; composing with a fixed isogeny embeds each remaining Hom group into an endomorphism group of a simple factor. Thus the integral endomorphisms again form a lattice.

A pointed polarization-preserving automorphism u satisfies $u^\dagger u = 1$. Rosati positivity makes $q(u) = \mathrm{Tr}(u^\dagger u)$ positive definite. Such automorphisms lie on the compact sphere $q(u) = \mathrm{Tr}(1) = 4$, whose intersection with the lattice is finite. This proves the required finiteness; translations are excluded by the pointed-morphism convention.

A33 Descent of the marked abelian scheme

Set $R = \mathbf{Z}[1/30]$. The earlier moduli construction supplies the smooth projective full-five scheme X_5, its universal polarized surface and order action, with all required determinant components. The geometric module calculation supplies the three-markings. The order-linear Isom sheaf is finite etale: after an etale trivialization of torsion, it is the finite subset of bijections preserving addition and finitely many order generators. Stacks Tag 03RV represents it, and the geometric calculation proves surjectivity. Hence $X_{35} \to X_5$ is finite etale. A finite map is projective (0B3I);

composition is projective here over the affine base (01W7, Lemma 29.44.14). Thus X_{35} is smooth projective and regular.

Let $G_5 = (\mathcal{O}/5\mathcal{O})^\times$ change the five-marking. For every nonidentity g, its equalizer with the identity is closed and has no geometric point by the earlier automorphism calculation, hence is empty. A section of the constant group is locally a fixed g. The action is therefore free on arbitrary test schemes, including nonreduced ones, without inverting $|G_5|$. Tag 07S7 represents $Y = X_{35}/G_5$ and makes $q : X_{35} \to Y$ a torsor. Descent of finite and etale morphisms (02LA, 02VN) makes it finite etale surjective.

Smoothness descends (036U). The pullback of the diagonal of Y/R is the disjoint union of the closed graphs of G_5, so 02L6 makes Y/R separated. The proper cover makes it universally closed (03GN) and quasi-compact, hence proper. The norm of an ample line bundle descends existence of an ample line bundle (0BD4); properness gives projectivity (0B45). This norm proves projectivity of the quotient, not a replacement of its polarization by a multiple.

Before duality, use the regular-base projectivity assertion in BLR, §8.4, proof of Corollary 6, p. 234, for these regular noetherian bases of finite type over R. The universal abelian scheme over X_{35} is projective. Its relabeling descent datum is effective by Stacks 0CCJ. Full faithfulness (023Q) descends zero, addition, inverse and their identities. Properness and smoothness descend (02L1, 02VL), and geometric connectedness follows after choosing a point of the nonempty covering fibre. The result is an abelian scheme A/Y, again projective by the regular-base assertion.

BLR §8.2, Theorem 1 and Proposition 4, pp. 210–211, and §8.4, Theorem 5, p. 234, now apply to these projective flat finitely presented families with reduced irreducible geometric fibres and zero sections. They give the Picard functor, normalized Poincare bundle and dual. The fibrewise definitions of Pic^0 and Pic^τ commute with base change, so canonically

$$(A^\vee) \times_Y X_{35} \simeq (A_{35})^\vee.$$

The actual principal polarization is invariant under relabeling and descends by 023Q. It remains an isomorphism (02L4) and is a polarization on each geometric fibre. Every order endomorphism and the three-marking descend similarly, with identities checked after q.

An automorphism of a full-three object trivial on geometric fibres acts trivially on finite etale five-torsion, since maps of finite constant sets are detected at geometric points. After adding a five-marking it is the identity by full-five rigidity, and therefore is the identity before the cover. For an arbitrary full-three object, its five-marking sheaf is a G_5-torsor; its map to X_{35} descends uniquely to Y. Conversely pull back the descended universal object. These constructions are inverse after a faithfully flat cover and hence inverse. This identifies the full-three moduli problem, without an extra five-marking factor in its degree.

A34 The finite forgetful map and its Picard norm

Diamond–Im, §9.3, pp. 77–79 [107], supplies smooth proper models of $X_1(N)$ and $X_0(N)$ over $\mathbf{Z}[1/N]$ and the diamond-group quotient; for $N \le 4$ the coarse curves coincide. Thus the actual forgetful map $u : X_1(N) \to X_0(N)$ is finite surjective over $\mathbf{Q}$. On quotient affine charts, the monic orbit polynomial makes elements integral over the invariants; finite type gives finiteness and lying-over gives surjectivity. The ineffective element -1 is harmless. Compose with the finite surjective modular map to the elliptic curve, obtaining $f : X_1(N) \to A$. At a closed point of the smooth target, its local ring is a DVR and the finite source module is torsion-free, hence free. Therefore f is finite locally free of positive constant degree d.

Its norm is

$$N_f(\mathcal{L}) = \det(f_*\mathcal{L}) \otimes \det(f_*\mathcal{O})^{-1}.$$

Locally trivialize $\mathcal{L}$ over the finite source. Changing a generator by a unit u changes its determinant trivialization by $\det(m_u)$. These are exactly the function norms of Stacks 0BD2 and the transition functions of 0BCY. They prove multiplicativity; no multiplicativity of determinant for arbitrary tensor products of vector bundles is asserted. Restriction of scalars and exterior powers here commute with arbitrary base change. Moreover

$$N_{f_T}(\mathcal{L} \otimes \pi_X^*\mathcal{M}) = N_{f_T}(\mathcal{L}) \otimes \pi_A^*(\mathcal{M}^{\otimes d}).$$

The norms therefore pass to the quotient by line bundles from T and after fppf sheafification define a natural transformation of relative Picard sheaves. Representability and Yoneda give a rational group morphism. After algebraic closure it sends the connected identity component into the connected identity component, so restricts to Pic^0. No rational point of $X_1(N)$ or universal rational line bundle is required. On divisor line bundles it is pushforward, by the valuation formula for the determinant norm (Silverman II.3.6(c),(d)). It is thus exactly the morphism used in the earlier point-difference proof of surjectivity.

A35 Labelled CM points and their common-place reduction

We give the CM and reduction arguments used in the conductor-one sum. Let F be imaginary quadratic with maximal order $\mathcal{O}$, fix a complex CM type, let q be an odd inert unramified prime, and let N be prime to $q\,\mathrm{disc}(F)$ with all its prime divisors split in F. In the conductor-one application $\mathcal{O}^\times = \{\pm 1\}$. For each $p^e \| N$ choose one prime $\mathfrak{p}$ over p, and set $\mathfrak{n} = \prod \mathfrak{p}^e$. Then $\mathcal{O}/\mathfrak{n} \simeq \mathbf{Z}/N\mathbf{Z}$. A labelled CM point consists of an elliptic curve, its actual $\mathcal{O}$-action of the fixed type, and the cyclic subgroup $A[\mathfrak{n}]$. Isomorphisms preserve these labels.

A35.1 Ideal action and reciprocity

Complex uniformization and its CM refinement (Silverman VI.4.1, VI.4.1.1, VI.5.1.1 and VI.5.3, with the corrected formulation) identify such curves with $\mathbf{C}/\Lambda$, where Λ is a fractional $\mathcal{O}$-ideal. Labelled isomorphisms correspond to multiplication by $F^\times$. The ideal action of an integral ideal $\mathfrak{a}$ is

$$\mathbf{C}/\Lambda \longrightarrow \mathbf{C}/\mathfrak{a}^{-1}\Lambda;$$

its kernel is $A[\mathfrak{a}]$ and its degree is $\mathrm{N}\mathfrak{a}$. For $\mathfrak{a}$ prime to N, it transports the specified level subgroup. Every ideal class has a representative prime to any prescribed finite set: choose an integral ideal I in the inverse class and, by the ideal Chinese remainder theorem, an $x \in I$ with $v_\mathfrak{p}(x) = v_\mathfrak{p}(I)$ at that set. Then $(x)I^{-1}$ is integral and prime to the set.

The order class polynomial is monic integral (Fowler, §2.1, pp. 612–613; Silverman C.11.2(a)), so all these j-invariants are algebraic integers. Silverman C.11.2(c),(d) gives the Hilbert class field $H = F(j(\Lambda))$. The Frobenius formula in C.11.4, at primes where the chosen models have good reduction, is

$$j(\Lambda)^{\mathrm{Frob}_\mathfrak{p}} = j(\Lambda\mathfrak{p}^{-1}).$$

To use it for all ideal classes, choose H-models for each of the finitely many j-values (Silverman III.1.4(c)) and exclude the ideal primes below every bad prime of these models, together with the primes dividing Nq. Choose class representatives prime to this enlarged set and factor them into prime ideals. Iterating the formula gives the class action. This does not assert good reduction at q

yet. An equality of j-invariants lifts to an isomorphism of the curves (Silverman III.1.4(b)); because the Galois action fixes F and hence the CM type, faithfulness of the tangent character in complex uniformization makes it an $\mathcal{O}$-linear isomorphism. The selected level subgroup is then preserved. This asserts invariance of labelled isomorphism classes, not that arbitrary choices of isomorphisms are defined over H.

In the convention of Cai–Shu–Tian, the point indexed by $\mathfrak{b}$ is the isogeny $\mathbf{C}/\mathfrak{b} \to \mathbf{C}/(\mathfrak{b}\mathfrak{n}^{-1})$. Thus it is $[\mathfrak{b}^{-1}]$ times the reference point in the preceding convention. Their Artin compatibility (pp. 2524, 2527 and 2531) agrees after this one global inversion of class indices. Quadratic characters are unchanged by that inversion. All reductions below take place at one chosen place. The forward CM reduction criterion of Gross–Zagier, pp. 198–199 [66], gives supersingularity when $(\mathrm{disc}(F)/q) = -1$, including $q = 3$; no converse lifting theorem is needed.

A35.2 A simultaneous good model

Consider the finite diagram of curves, order actions, selected level groups and ideal isogenies needed in the class sum, with isogeny degrees prime to qN. The integral j-invariants and Silverman VII.5.5 give potentially good reduction for each curve. A common finite extension gives all good models over one complete DVR R of residue characteristic q, and defines the finitely many maps and order generators. These elliptic schemes are Neron models (BLR §1.4, Proposition 2, p. 19). The Neron mapping property, BLR §1.2, Definition 1 and Proposition 8, pp. 12,15, extends every generic morphism. Uniqueness extends all composition and order-action identities.

The special-fibre $\mathcal{O}$-action remains injective. For nonzero α, its dual composite is multiplication by a positive integer (Silverman III.6.1 and III.6.2(a)); the dual is defined over the same field by uniqueness. After extension this identity still holds, and multiplication by that integer is nonzero on the special elliptic curve, of degree its square (III.6.2(d)), even if q divides the integer.

For n prime to q, the differential of $[n]$ is a unit. The etale criterion for maps between smooth curves (EGA IV_4, Corollary 17.11.2(c'), p. 84) makes it etale. It is proper by the graph and projection argument, and quasi-finite; hence finite (EGA III_1, Corollary 4.4.11). Therefore $A[n]$ is finite etale. Over the strict henselization it is constant, and specialization of geometric points is a bijection compatible with all endomorphisms, by EGA IV_4, Propositions 18.5.14–15. Simultaneous kernels are thus preserved as subsets of a constant group scheme.

Let $f : A \to B$ be one of the isogenies of degree d prime to q. Its dual extends, and their composites are $[d]$. Their differentials are therefore units. Thus f is etale, proper and quasi-finite, hence finite etale. Its degree is constantly d and it is surjective: finite etale rank is locally constant (EGA IV_4, Proposition 18.2.3 and Remark 18.2.7(i)), and the elliptic target over the local base is connected. If f is the ideal quotient for $\mathfrak{a}$, then $d \in \mathfrak{a}$, so its kernel lies in $A[d]$. Equality there with the common kernel of finitely many generators of $\mathfrak{a}$ holds generically and hence on the constant components; the special kernel is exactly $\overline{A}[\mathfrak{a}]$. The map

$$\overline{A} \times \ker \overline{f} \longrightarrow \overline{A} \times_{\overline{B}} \overline{A}, \qquad (x, t) \longmapsto (x, x + t)$$

is an isomorphism, with inverse $(x, y) \mapsto (x, y - x)$. Consequently $\overline{B}$ represents the quotient by this subgroup.

Choose u, v with $ud + vN = 1$. On N-torsion, $[u]\widehat{f}$ is the inverse of f, preserving all order eigensummands and therefore every chosen prime-power level subgroup, including powers of 2. On q^r-torsion use compatible inverses of d modulo q^r; this gives an isomorphism of q-divisible groups without an etaleness claim for them. Finally the tangent identity for the $\mathcal{O}$-linear map f, whose differential is nonzero, shows that the residue character $\mathcal{O}/q \to \overline{\mathbf{F}}_q$ is the same on both ends. It is not replaced by its conjugate.

A35.3 The oriented conductor-one correspondence

Use the enhanced Eichler class-set description and rational Skolem–Noether calculation proved in the quaternionic sections above. The source set of labelled points of the fixed CM type and level choices is a torsor under $\mathrm{Pic}(\mathcal{O})$. We now verify the same description of the oriented supersingular target without assuming surjectivity of reduction.

Away from Nq, an $\mathcal{O}_p$-stable lattice is a principal fractional ideal locally, also at primes ramifying in F. At $p^e \| N$, the two integral idempotents of $\mathcal{O}_p = \mathbf{Z}_p \times \mathbf{Z}_p$ split the lattice chain. A cyclic quotient of order p^e forces the two exponents to be $(e, 0)$ or $(0, e)$; a mixed pair would not give a cyclic quotient. This holds at $p = 2$ as well. At q, write the division algebra as

$$B_q = F_q + F_q\Pi, \qquad \Pi a = \bar{a}\Pi,$$

with Π^2 a central uniformizer times a unit. Conjugation by a maximal-order unit acts trivially on the residue field $\mathbf{F}_{q^2}$, whereas conjugation by Π is its nontrivial automorphism. These are the two local embedding classes, distinguished by the residue orientation and parity of the norm valuation; they must not be identified by a normalizer quotient.

Rational Skolem–Noether fixes a labelled embedding of F. Its stabilizer is the centralizer $F^\times$, not the normalizer. With a fixed reference representative z_0, the locally compatible representatives are $\widehat{F}^\times z_0 \widehat{R}^\times$. The intersection with $\widehat{F}^\times$ is $\widehat{\mathcal{O}}^\times$, since the order is maximal. Thus the fixed-orientation target is

$$F^\times \backslash \widehat{F}^\times / \widehat{\mathcal{O}}^\times = \mathrm{Pic}(\mathcal{O}).$$

It is a torsor independently of reduction. Reduce an actual reference CM point to obtain y_0. The injective action is optimal because $\mathcal{O}$ is maximal, and the residue character gives the chosen orientation. The simultaneous good-model argument proves equivariance of reduction: prime-to-qN ideal quotients multiply the Tate lattices by $\mathfrak{a}^{-1}$, preserve the q-divisible group and level choices, and preserve the tangent orientation. A nonempty equivariant map between these two torsors is a bijection, with multiplicity one.

All choices at primes dividing N are obtained by the prime-power Atkin–Lehner quotients and duals. Their degrees are prime to q, so the same characteristic-zero constructions reduce compatibly, including powers of 2. The opposite residue orientation at q is the Frobenius image of the geometric reference point, or equivalently is obtained from the conjugate CM type. It is not an extra image of the same fixed-type torsor. Frobenius and inverse Frobenius have the same action on the underlying $\mathbf{F}_{q^2}$-defined unlabelled pair, but this does not identify their labelled embeddings.

Consequently the class-character sum on this fixed component is the reduction, followed by the same additive functional, of the characteristic-zero sum:

$$\sum_{[\mathfrak{a}]} \chi(\mathfrak{a}) P_M(\mathfrak{a} z_0) = \lambda_M \left(\overline{L \sum_{[\mathfrak{a}]} \chi(\mathfrak{a}) \pi(x_D^{[\mathfrak{a}]})} \right).$$

The same place, the same map π and the same additive functional are used throughout. Global inversion of Artin indices does not change a quadratic character. No stabilizer division is hidden here: for a fixed underlying pair the embedding stabilizer is $R_z^\times \cap F^\times = \mathcal{O}^\times = \{\pm 1\}$. Its orbit has size $|R_z^\times|/2$, and weighting by $1/|R_z^\times|$ gives $1/2$, also at the extra-automorphism curves. The embeddings x and $-x$ have opposite residue orientations, so are not conjugate by a unit. For the other components, the already proved Schwartz and parametrization factors $w_{p^e}\chi_{h^*}(p)^e$ occur twice and multiply to one. In the actual application $q = \ell_0 \equiv 3 \pmod 4$ and $\chi_{h^*}(q) = -1$; the residue weight is $(-1/q) = -1$, the norm character is -1, and the functional's Frobenius factor is $+1$. This is the factor $c_0 2^{\omega(N)}$ in the conductor-one identity above.

The actual application has $F = \mathbf{Q}(\sqrt{Dh^*})$ with fundamental discriminant and maximal order, $q = \ell_0$, every prime dividing $2N$ split, and no extra units. These are exactly the hypotheses used here. No corresponding assertion for nonmaximal orders or arbitrary conductors has been inserted.

Part II
The 233-variable family and its predecessors

A36 Integral evaluation and the effective contact height

A36.1 Conventions and the evaluation estimate

If K is a number field, $\mathcal{O}_K$ denotes its ring of integers. For a nonzero ideal $\mathfrak{a} \subset \mathcal{O}_K$, write $\mathrm{N}\mathfrak{a} = \#(\mathcal{O}_K/\mathfrak{a})$. If E is a projective $\mathcal{O}_K$-module with Hermitian metrics at its complex embeddings, $\widehat{\deg}(E)$ means the arithmetic degree of its determinant. In rank one, for a nonzero integral section s,

$$\widehat{\deg}(E) = \log \#(E/\mathcal{O}_K s) - \sum_{\sigma : K \hookrightarrow \mathbb{C}} \log \|s\|_\sigma .$$

The definition is independent of the choice of s by the product formula. Both embeddings in a complex conjugate pair are included.

Throughout, h_{F} is the original stable Faltings height. For a curve X/K with semistable reduction, its Jacobian height is

$$h_{\mathrm{F}}(\mathrm{Jac}\, X) = \frac{\widehat{\deg} H^0(\mathcal{X}, \omega_{\mathcal{X}/\mathcal{O}_K})}{[K : \mathbb{Q}]}, \tag{208}$$

where $\mathcal{X}$ is a minimal regular semistable model. The metric on differentials at a complex embedding is

$$\langle \omega, \eta \rangle = \frac{i}{2} \int_{X(\mathbb{C})} \omega \wedge \overline{\eta}.$$

This is the determinant metric corresponding to the original Faltings normalization; see [122, Sections 2.1 and 2.3] and [124, Section 2.1]. Passing to a finite extension on which the curve is semistable does not change the stable height.

A *Belyi map* is a finite morphism from a smooth projective geometrically connected curve to $\mathbb{P}^1$, unramified outside $0, 1, \infty$. Let

$$\pi : X \longrightarrow \mathbb{P}^1$$

be such a map, of degree d, where X has genus $g \geq 1$. Write t for the coordinate on the target. Choose a number field K over which X and π are defined, X is semistable, and every point of $\pi^{-1}(2)$ is rational. Let $\mathcal{Y}$ be the normalization of $\mathbb{P}^1_{\mathcal{O}_K}$ in $K(X)$. It is finite and flat over $\mathbb{P}^1_{\mathcal{O}_K}$. On the affine t-chart write its algebra as B, a finite projective $\mathcal{O}_K[t]$-algebra of rank d, and put

$$C = B/(t-2)B \subset K^d, \qquad O = \mathcal{O}_K^d. \tag{209}$$

The inclusion in K^d uses the d distinct points of the split fibre. Every element of C is integral over $\mathcal{O}_K$, so $C \subset O$. Let $\mathfrak{D}(C)$ be the discriminant ideal of its trace pairing.

Theorem A36.1. *With this notation,*

$$h_{\mathrm{F}}(\mathrm{Jac}\, X) \leq \frac{\log \mathrm{N}\mathfrak{D}(C)}{2[K : \mathbb{Q}]} + g \log \frac{2}{\sqrt{\pi}}. \tag{210}$$

The auxiliary field K is used to split a finite fibre and to compute a stable height. Its degree will cancel from the estimates. No upper bound for $[K:\mathbb{Q}]$ is assumed in this theorem.

A36.2 Trace-dual lattices and regular differentials

Lemma A36.2. *Let $C \subset O = \mathcal{O}_K^d$ be an order with $C \otimes K = K^d$, and let*

$$C^\vee = \{x \in K^d : \mathrm{Tr}_{K^d/K}(xC) \subset \mathcal{O}_K\}.$$

For any set I of g coordinate positions, set $F = \mathrm{pr}_I(C^\vee) \subset K^g$. Then F is a projective $\mathcal{O}_K$-module containing $\mathcal{O}_K^g$. With the standard Euclidean metric on K^g at every embedding,

$$\widehat{\deg}(F) \le \tfrac{1}{2}\log \mathrm{N}\mathfrak{D}(C).$$

Proof. Work locally at a finite prime $\mathfrak{p}$. The trace pairing on K^d is the ordinary dot product, and O is self-dual. Thus $O \subset C^\vee$, and dualizing the inclusion $C \subset O$ shows

$$\mathrm{length}(C^\vee/O) = \mathrm{length}(O/C) = \tfrac{1}{2} v_{\mathfrak{p}}(\mathfrak{D}(C)).$$

For example, if a basis matrix for C relative to O is U, then the trace Gram matrix is $U^{\mathsf{T}}U$, which proves the last equality. These equalities are valid locally even when C is not globally free.

Projection gives a surjection $C^\vee/O \twoheadrightarrow F/\mathcal{O}_K^g$. The latter quotient therefore has no larger length at any prime. A finite torsion-free module over the Dedekind ring $\mathcal{O}_K$ is projective. Finally, the standard basis of $\mathcal{O}_K^g$ has determinant norm one, so

$$\widehat{\deg}(F) = \log \#(F/\mathcal{O}_K^g) \le \tfrac{1}{2}\log \mathrm{N}\mathfrak{D}(C).$$

□

Lemma A36.3. *Let $\mathcal{X}$ be a minimal regular semistable model of X. The evaluations of a differential in $E = H^0(\mathcal{X}, \omega_{\mathcal{X}/\mathcal{O}_K})$ at the points of $\pi^{-1}(2)$, expressed in the coordinate t, lie in $C^\vee$. Equivalently, there is an integral map*

$$E \longrightarrow C^\vee, \qquad \omega \longmapsto \left(\frac{\omega}{dt}(P)\right)_{P \in \pi^{-1}(2)}.$$

Proof. First, normalization is finite because the base is excellent. A normal surface is Cohen–Macaulay, and a finite morphism from such a surface to a regular surface is flat by miracle flatness; see [121, Tag 00R4]. Consequently B is finite projective over $\mathcal{O}_K[t]$.

Take a common regular model $\mathcal{Z}$ dominating $\mathcal{X}$ and $\mathcal{Y}$. Pullback under a blowup of regular arithmetic surfaces takes regular relative differentials to regular relative differentials. One can also read this directly from the canonical-sheaf formula for a blowup; see [122, Lemma 2.3.1]. Hence a differential in E is regular on $\mathcal{Z}$. The birational morphism $\mathcal{Z} \to \mathcal{Y}$ is an isomorphism at every codimension-one point of the normal surface $\mathcal{Y}$.

The dualizing module of a Cohen–Macaulay normal surface is reflexive: it is torsion-free and satisfies Serre's condition (S_2). Thus sections defined at all codimension-one points extend across the codimension-two complement; see [121, Tags 0AWQ and 0AVT, Lemmas 31.13.12–31.13.14]. The differential therefore defines a section of the relative dualizing sheaf of $\mathcal{Y}$.

Finite duality identifies this sheaf on the affine chart with

$$\mathrm{Hom}_{\mathcal{O}_K[t]}(B, \mathcal{O}_K[t])\, dt;$$

see [121, Tags 0BUL and 0C0I]. This statement uses the dualizing *module*; it does not require that the normal surface have an invertible dualizing sheaf. Because B is finite projective, specializing the dual at $t = 2$ gives $\mathrm{Hom}_{\mathcal{O}_K}(C, \mathcal{O}_K)\,dt$. On the generic fibre the identification sends a differential $f\,dt$ to the functional $b \mapsto \mathrm{Tr}(fb)$. After splitting the fibre, its coordinates are exactly the values $f(P) = \omega/dt(P)$. This proves the assertion. □

Proof of Theorem A36.1. Riemann–Hurwitz gives

$$2g - 2 = -2d + \deg \mathrm{Ram}(\pi) \le d - 3,$$

since each of the three branch fibres contributes at most $d - 1$. In particular $d > 2g - 2$. A nonzero holomorphic differential cannot vanish at all d distinct points of $\pi^{-1}(2)$. Therefore the evaluation map has rank g. Choose g coordinate positions I for which projection remains injective. By Lemmas A36.2 and A36.3, we obtain an integral, generically invertible map

$$\phi : E \longrightarrow F = \mathrm{pr}_I(C^\vee), \qquad \widehat{\deg}(F) \le \tfrac{1}{2} \log \mathrm{N}\mathfrak{D}(C).$$

Fix a complex embedding of K. The disc $D = \{t : |t - 2| < 1/2\}$ is simply connected and avoids the branch locus. Its inverse image is a disjoint union of d analytic sheets, each mapped isomorphically to D. On one such sheet write $\omega = f(t)\,dt$. The mean-value inequality implies

$$|f(2)|^2 \le \frac{4}{\pi} \int_D |f(t)|^2 \, dx\, dy \le \frac{4}{\pi} \|\omega\|_{L^2}^2.$$

Hence the norm of each evaluation functional is at most $2/\sqrt{\pi}$. In an orthonormal basis of $E \otimes_\sigma \mathbb{C}$, each row of the evaluation matrix has this same bound on its Euclidean norm. Hadamard's determinant inequality gives

$$\| \det \phi \|_\sigma \le (2/\sqrt{\pi})^g.$$

The determinant is nonzero and integral. Taking its arithmetic degree yields

$$\widehat{\deg}(E) \le \widehat{\deg}(F) + \sum_\sigma \log \| \det \phi \|_\sigma \le \tfrac{1}{2} \log \mathrm{N}\mathfrak{D}(C) + [K : \mathbb{Q}] g \log(2/\sqrt{\pi}).$$

Now apply (208). □

A36.3 The discriminant of the fibre

Let S be a finite set of rational primes, and put

$$R_S = \prod_{p \in S} p, \qquad \omega(S) = \#S.$$

Assume that the normalization defining the Belyi cover is finite étale over

$$\mathrm{Spec}\, \mathcal{O}_{K_0}[S^{-1}, t, t^{-1}, (t-1)^{-1}], \tag{211}$$

where K_0 is a field of definition. This property persists after any finite field extension: finite étale base change is finite étale, and a finite étale algebra over a normal base is normal. Thus it agrees with the normalization of the new generic cover on this open set. In particular the auxiliary splitting field in Section 1 introduces no new vertical branch primes outside S.

Lemma A36.4. *Under* (211)*, the order C in* (209) *satisfies*

$$\frac{\log \mathrm{N}\mathfrak{D}(C)}{[K:\mathbb{Q}]} \le d\,(\log 2 + \log R_S + \omega(S)\log d)\,.$$

Proof. The determinant of the trace pairing of $B/\mathcal{O}_K[t]$ defines an effective discriminant divisor on the base. Formation of the trace pairing commutes with specialization. Its intersection with the section $t = 2$ is therefore $\log \mathrm{N}\mathfrak{D}(C)$. The coefficient of this divisor at a codimension-one point is $\sum_j f_j r_j$, where e_j, f_j, r_j are the ramification index, residue degree, and different exponent of the corresponding extensions of discrete valuation rings. They satisfy $\sum_j e_j f_j = d$. These are the usual local trace-discriminant and different identities; see [123, Chapter III, Sections 3 and 6] and [122, Section 4.2].

At a horizontal branch divisor the residue characteristic is zero. The different exponent is $r_j = e_j - 1$, so the discriminant coefficient is at most d. The section $t = 2$ meets $t = 0$ with normalized intersection $\log 2$, and meets neither $t = 1$ nor $t = \infty$. Thus the horizontal contribution is at most $d \log 2$.

Let $\mathfrak{p} \mid p$ be a prime of K, and write $e_{\mathfrak{p}} = v_{\mathfrak{p}}(p)$. Lenstra's different bound, in the form [122, Proposition 4.1.3], gives

$$r_j \le e_j - 1 + e_j e_{\mathfrak{p}} \left\lfloor \frac{\log d}{\log p} \right\rfloor .$$

This bound also applies to the possibly imperfect residue field at the generic point of a vertical fibre. Its discriminant coefficient is consequently at most

$$d\left(1 + e_{\mathfrak{p}} \left\lfloor \frac{\log d}{\log p} \right\rfloor\right).$$

Only $p \in S$ can contribute. The intersection of the vertical fibre over $\mathfrak{p}$ with the section has value $f_{\mathfrak{p}} \log p$. Using

$$\sum_{\mathfrak{p}\mid p} e_{\mathfrak{p}} f_{\mathfrak{p}} = [K:\mathbb{Q}], \qquad \sum_{\mathfrak{p}\mid p} f_{\mathfrak{p}} \le [K:\mathbb{Q}],$$

and $\lfloor \log d / \log p \rfloor \log p \le \log d$ proves the estimate. □

Theorem A36.5. *For a Belyi cover satisfying* (211)*,*

$$h_{\mathrm{F}}(\mathrm{Jac}\, X) \le \frac{d}{2}\,(\log 2 + \log R_S + \omega(S)\log d) + g \log \frac{2}{\sqrt{\pi}}. \tag{212}$$

Proof. Combine Theorem A36.1 and Lemma A36.4. □

A36.4 The integral modular cover

Let N be a positive integer and let $H \subset (\mathbb{Z}/N\mathbb{Z})^\times$ contain -1. Write

$$X_H = X_1(N)/H, \qquad t_H = [(\mathbb{Z}/N\mathbb{Z})^\times : H], \qquad J_H = \mathrm{Jac}\, X_H.$$

The action is the diamond action. Assume that X_H has positive genus, and use the rational map $t = j/1728$ to the projective line.

Lemma A36.6. *The map $X_H \to \mathbb{P}^1_t$ has degree*

$$d_H = N t_H \prod_{p\mid N} (1 + p^{-1}).$$

It is unramified outside $0, 1, \infty$ and admits a finite étale model over

$$U = \mathrm{Spec}\, \mathbb{Z}[1/(6N), t, t^{-1}, (t-1)^{-1}].$$

Proof. Over U consider

$$E_t: \quad y^2 = x^3 - \frac{3t}{t-1}x - \frac{2t}{t-1}. \tag{213}$$

Its discriminant and j-invariant are

$$\Delta(E_t) = \frac{1728t^2}{(t-1)^3}, \qquad j(E_t) = 1728t.$$

Thus it is an elliptic curve over U. Since N is invertible, its scheme of points of exact order N is finite étale. The constant group H acts freely on this scheme: if $aP = P$ and P has exact order N, then $a = 1$ in $(\mathbb{Z}/N\mathbb{Z})^\times$. The quotient is finite étale. This follows also from the classification of finite étale covers by locally constant finite sets; freeness is enough, even when a residue characteristic divides $|H|$.

To identify its generic fibre, fix an elliptic curve with j-invariant different from 0 and 1728 over an algebraic closure. Its automorphism group is $\{1, -1\}$. The geometric fibre of $X_1(N)$ is the set of primitive torsion points modulo this automorphism group. Since $-1 \in H$, the fibre of X_H is precisely the set of primitive torsion points modulo H. Different isomorphisms to (213) differ by ± 1 and hence give the same H-orbit. In particular quadratic twisting introduces no extra double cover. This identifies the finite étale model with the normalization of X_H over U.

There are $N^2 \prod_{p|N}(1 - p^{-2})$ primitive points. Division by $|H| = \varphi(N)/t_H$ gives the asserted degree. Equivalently, the congruence subgroup contains $-I$, so its $\mathrm{SL}_2(\mathbb{Z})$ index is already the degree of the coarse map to $X(1)$; see [124, Section 4]. The only possible branch points of that map are the two elliptic points and the cusp, which become $0, 1, \infty$ in the t coordinate. □

Lemma A36.7. *Let V be an excellent discrete valuation ring of characteristic zero, let E/V be an elliptic scheme, and let n be a positive integer. Write $E[n] = \operatorname{Spec} A$, and let $\widetilde{A}$ be its normalization in its generic finite étale algebra. Under the generic trace pairing, the complementary modules satisfy*

$$A^\vee = n^{-1}A, \qquad \widetilde{A}^\vee \subseteq n^{-1}\widetilde{A}.$$

The different of every field component of $\widetilde{A}/V$ therefore contains n. The same assertion holds for every intermediate field of such a component.

Proof. Multiplication $[n] : E \to E$ is finite flat and a local complete intersection morphism. Its derivative in invariant differential bases is multiplication by n. Consequently its relative differential module has zeroth Fitting ideal (n). For a finite flat local complete intersection, this Kähler different is the different; see [121, Tag 0BWB, Lemmas 49.12.3 and 49.12.6].

Base change along the zero section gives $E[n]/V$. The relative differential module and its zeroth Fitting ideal commute with base change, while the morphism remains finite flat and a local complete intersection. Its different is therefore (n). Its relative dualizing module is invertible. Under the trace pairing this is the complementary fractional ideal $A^\vee$, whose inverse is the different; see [121, Tag 0BW0, Lemmas 49.8.2–49.8.3, and Tag 0BWB, Lemma 49.12.4]. Thus $A^\vee = n^{-1}A$ as fractional ideals, not merely as abstract modules.

Normalization is finite, and inclusion reverses trace duals:

$$\widetilde{A}^\vee \subseteq A^\vee = n^{-1}A \subseteq n^{-1}\widetilde{A}.$$

Hence $n\widetilde{A}^\vee \subseteq \widetilde{A}$, which says exactly that the different contains n. Finally, for normal discrete valuation rings $V \subseteq W \subseteq U$, the different formula

$$\mathfrak{D}_{U/V} = \mathfrak{D}_{U/W}\mathfrak{D}_{W/V}U$$

shows that $v_U(\mathfrak{D}_{U/V}) \le v_U(n)$ implies $v_W(\mathfrak{D}_{W/V}) \le v_W(n)$, because the different exponent of U/W is nonnegative. Apply this in each field component and at each place. These finite-duality and different arguments do not require a perfect residue field. □

Corollary A36.8. *Let g_H be the genus of X_H. Then*

$$h_{\mathrm{F}}(J_H) \le \frac{d_H}{2}\left(\log 12 + \log N + 2\log d_H\right) + g_H \log\frac{2}{\sqrt{\pi}}. \tag{214}$$

Proof. Use the auxiliary field K from Theorem A36.1. At a prime $\mathfrak{p}$ of K over $p > 3$, localize the t-line at the generic point of the special fibre. The family (213) is an elliptic scheme over this discrete valuation ring: its discriminant $1728t^2/(t-1)^3$ is a unit there. By Lemma A36.6, the generic cover for X_H is the primitive N-torsion cover modulo H. Each of its field components is an intermediate field of a field generated by a single N-torsion point. Lemma A36.7 therefore bounds every different exponent above $\mathfrak{p}$ by the valuation of N. The coefficient of the vertical discriminant divisor is at most

$$d_H v_{\mathfrak{p}}(N).$$

After summing over these primes and dividing by $[K:\mathbb{Q}]$, the contribution is at most $d_H \log N$.

At $p = 2, 3$, the estimate in Lemma A36.4 gives at most $d_H(\log 6 + 2\log d_H)$. The horizontal intersection at $t = 2$ contributes at most $d_H \log 2$. Thus the normalized logarithmic discriminant of the fibre order is at most

$$d_H(\log 12 + \log N + 2\log d_H).$$

Apply Theorem A36.1. In particular no degree of a full torsion splitting field enters this estimate. □

A36.5 An abelian surface occurring as an isogeny factor

Lemma A36.9. *Let J be an abelian variety of dimension $g \ge 2$ over a number field L', and let A be an abelian surface which is an isogeny factor of J over L'. Put $d' = [L':\mathbb{Q}]$ and*

$$H_J = \max(1, h_{\mathrm{F}}(J) + 3g/2).$$

Then

$$h_{\mathrm{F}}(A) \le H_J + 8\left(32\log 14 + \log d' + \log\max(1, \log d', H_J)\right). \tag{215}$$

Proof. Apply the construction in the published proof of [126, Lemma 16.2(1), pp. 97–98] over the fixed field L' to J. It gives an L'-isogenous abelian variety J_0, a product over L' of simple abelian varieties, with $h_{\mathrm{F}}(J_0) \le h_{\mathrm{F}}(J)$. Since A is an L'-isogeny factor of J_0, uniqueness of simple factors up to L'-isogeny, with multiplicities, selects a subproduct B that is L'-isogenous to A. Write $J_0 = B \times E$; then $\dim B = 2$ and $\dim E = g - 2$. Additivity of stable Faltings height on products and Bost's lower bound in the original normalization give

$$h_{\mathrm{F}}(B) = h_{\mathrm{F}}(J_0) - h_{\mathrm{F}}(E) \le h_{\mathrm{F}}(J) + 3(g-2)/2 \le H_J.$$

Here the lower bound is $h_{\mathrm{F}}(E) \ge -\log(\pi\sqrt{2})\dim E > -3\dim E/2$ when $\dim E > 0$, and $h_{\mathrm{F}}(E) = 0$ when $\dim E = 0$; see [39, §2.3 and Corollary 8.4]. The direct supplier for the product construction is the published 2023 proof just cited; that proof in turn uses Rémond's 2022 height results [125] and his 2017 results. This use does not remove those underlying dependencies.

Apply [126, Theorem 1.9(1)] with source B and target A, both over L'. In dimension two it supplies an isogeny of degree at most

$$\left(14^{32} d' \max(1, \log d', h_{\mathrm{F}}(B))\right)^{16}.$$

Stable Faltings height increases under an isogeny of degree D by at most $\frac{1}{2}\log D$; see [39, §2.3]. Replacing $h_{\mathrm{F}}(B)$ by H_J proves (215). □

The field L' in this lemma is the field on which the surface factor is realized. It need not split a fibre of a modular curve, and it need not be the auxiliary field K of Theorem A36.1. The evaluation estimate is a stable height bound valid after arbitrary extension; the degree entering (215) is only $[L' : \mathbb{Q}]$.

A36.6 The five-contact height estimate

For $a = u/v \in \mathbb{Q}$ in lowest terms, with $v > 0$, put $h(a) = \log \max(|u|, v)$. Define

$$S_* = \{p \text{ prime} : p \le 76176\}, \quad B_* = \prod_{p \in S_*} p, \quad \mathcal{D} = \{0, 1, 2401, -343/32\}.$$

For $a \notin \mathcal{D}$, let $M_5(a)$ be the product of the primes $p \notin S_*$ for which at least one of

$$v_p(v), \quad v_p(u), \quad v_p(u - v), \quad v_p(u - 2401v), \quad v_p(32u + 343v)$$

is positive and odd. Every prime occurs only once. The first valuation is the contact at infinity.

We use the following consequence of the internal arithmetic construction in Sections 1.

Lemma A36.10 (Modular realization)**.** *For each $a \in \mathbb{Q} \setminus \mathcal{D}$, put $R = B_* M_5(a)$. There is an abelian surface A such that*

$$h(a) \le 2\max(0, h_{\mathrm{F}}(A)) + 128. \tag{216}$$

Either $h_{\mathrm{F}}(A) < 854R$, or there are a normal number field $L'/\mathbb{Q}$ and an even nebentypus ε of a primitive weight-two newform, of level N, such that

$$\begin{aligned} &[L' : \mathbb{Q}] \le 2304\varphi(R), \qquad N \le 210^{300} R^2, \qquad \operatorname{rad}(N) \mid R, \\ &\varepsilon^2|_{G_{L'}} = 1, \qquad A_{L'} \text{ is an isogeny factor of } \operatorname{Jac}(X_1(N)/\ker\varepsilon)_{L'}. \end{aligned} \tag{217}$$

Proof. The geometric comparison and CM alternative are proved in Theorem 15.2 and its proof. Retain the sharper scalar-character order from (LR2): $m \le 2\prod_{p|R,\ p\ \mathrm{odd}}(p-1) = 2\varphi(R)$. The construction (LR3) gives $[L' : \mathbb{Q}] \le 2m[L : \mathbb{Q}]$ and $[L : \mathbb{Q}] \le 576$, hence $[L' : \mathbb{Q}] \le 2304\varphi(R)$. Equations (LR4)–(LR5) give the all-prime level bound and its support. The determinant and diamond-quotient argument in the proof of Theorem 15.2 gives the same even character, its square triviality over L', and the indicated isogeny factor over that field. No coefficient-field degree or extra cyclotomic extension is used. □

Here R is squarefree and divisible by 6. The degree bound in (217) retains the totient from the scalar character: its order satisfies $m \le 2\prod_{p|R,\ p\ \mathrm{odd}}(p-1) = 2\varphi(R)$, and the construction gives $[L' : \mathbb{Q}] \le 1152m$. The square of the nebentypus is trivial over L' by the determinant identity for the compatible representations. The newform quotient factors through the diamond quotient because every element of $\ker\varepsilon$ fixes all conjugates of its differentials. No bound on the degree of the newform's coefficient field is assumed.

Theorem A36.11. *Use the modular realization just proved. Put*

$$D_0 = 4608 \cdot 210^{300}, \qquad C_3 = 19 D_0 B_*^3 \log B_*.$$

Then $C_3 < 2^{330770}$ *and*

$$h(a) \le C_3 M_5(a)^3 (1 + \log M_5(a)) \quad (a \in \mathbb{Q} \setminus \mathcal{D}). \tag{218}$$

In particular,

$$h(a) < 2^{330789} M_5(a)^{7/2}. \tag{219}$$

For every fixed $\alpha > 3$ *there is an effective constant* C_α *such that* $h(a) \le C_\alpha M_5(a)^\alpha$.

Proof. Set $M = M_5(a)$ and $R = B_* M$. In the non-CM alternative, let $H = \ker \varepsilon$ and $t_H = \operatorname{ord} \varepsilon$. Since ε^2 is trivial on $G_{L'}$, the image of ε^2 has order dividing $[L' : \mathbb{Q}]$. Therefore $\operatorname{ord}(\varepsilon) \le 2[L' : \mathbb{Q}]$, and hence

$$t_H \le 2[L' : \mathbb{Q}] \le 4608 \varphi(R).$$

The exact index identity in Lemma A36.6 gives

$$\begin{aligned} d_H &\le 4608 \cdot 210^{300} R^2 \varphi(R) \prod_{p \mid R} (1 + p^{-1}) \\ &= D_0 R^3 \prod_{p \mid R} (1 - p^{-2}) < D_0 R^3. \end{aligned} \tag{220}$$

Also $2 \le g_H \le d_H$, because the Jacobian contains a surface factor and the cover is a Belyi map.

Write $\ell = \log R$. The exact numerical verification in Section A36.8 gives

$$D_0^2 < B_*, \qquad 144 < B_*, \qquad \ell \ge 1.$$

Consequently

$$\log N \le \tfrac{5}{2}\ell, \qquad \log d_H \le \tfrac{7}{2}\ell, \qquad \log 12 \le \tfrac{1}{2}\ell.$$

Using $\log(2/\sqrt{\pi}) < 1$ in Corollary A36.8, we obtain

$$h_{\mathrm{F}}(J_H) \le 5 d_H \ell + d_H \le 6 d_H \ell. \tag{221}$$

Thus the number H_J in Lemma A36.9 is at most

$$H_* = 8 D_0 R^3 \ell.$$

Let $d' = [L' : \mathbb{Q}]$. Since $d' \le 2304 R$, $2304 < e^8$, $\ell \ge 1$, and $\log D_0 \le \ell$,

$$\log d' < 9\ell, \qquad \log H_* \le 9\ell.$$

Moreover $H_* > \max(1, \log d')$. Lemma A36.9, with $\log 14 < 3$, now gives

$$\begin{aligned} h_{\mathrm{F}}(A) &\le H_* + 8(96 + 18\ell) \\ &\le H_* + 912\ell \le 9 D_0 R^3 \ell. \end{aligned}$$

The CM bound $854R$ is smaller than the same expression. Combining with (216), and using $D_0 R^3 \ell > 128$, gives

$$h(a) \le 19 D_0 R^3 \ell.$$

Finally $\log R = \log B_* + \log M \le (\log B_*)(1+\log M)$, since $\log B_* > 1$ and $M \ge 1$. This proves (218).

The explicit numerical bound uses

$$19 < 2^5, \qquad D_0 < 2^{2413}, \qquad B_* < 2^{109445}, \qquad \log B_* < 2^{17}.$$

Consequently

$$C_3 < 2^{5+2413+3\cdot 109445+17} = 2^{330770}.$$

For $u \ge 0$, the maximum of $(1+u)e^{-u/2}$ occurs at $u = 1$ and is less than two. Taking $u = \log M$ gives a constant smaller than 2^{330771}, and in particular proves the convenient bound (219). More generally, for each $\eta > 0$, $(1+u)e^{-\eta u}$ is bounded on $[0,\infty)$ by an explicitly computable number. Apply this with $\eta = \alpha - 3$. □

A36.7 Disjoint sets of contacts

Corollary A36.12. *Fix $r \ge 1$ and put*

$$\Theta_r = \bigcup_{j=0}^{r-1} \left(3j + \left\{0, -\tfrac{1}{2}, -1, \tfrac{1}{2399}, -\tfrac{32}{407}\right\}\right).$$

There is an explicitly determined finite exceptional set Σ_r such that, if $M_r(t)$ is the product of primes outside Σ_r at which some $t-\theta$, $\theta \in \Theta_r$, has positive odd valuation, then

$$h(t) \ll_r M_r(t)^{3/r}(1+\log M_r(t)) \ll_r M_r(t)^{7/(2r)} \qquad (t \in \mathbb{Q} \setminus \Theta_r).$$

All implied constants are effective under the stated modular realization input.

Proof. Apply Theorem A36.11 to $\phi_j(t) = 2 + 1/(t-3j)$ for $0 \le j < r$. Their matrices have determinant -1, so $h(t) \le h(\phi_j(t)) + O_r(1)$. In addition to S_*, exclude the primes dividing contact coefficients and the nonzero determinants of pairs of primitive contact forms. The $5r$ rational contacts are distinct. Outside the resulting fixed set, different contact forms cannot vanish at the same residue point. Thus the radicals $M_j(t)$ for the different blocks have disjoint prime support, and $\prod_j M_j(t) = M_r(t)$. Removing the exceptional primes from a block changes its height inequality by at most an effective fixed factor.

Multiplying the r inequalities gives

$$h(t)^r \ll_r M_r(t)^3 \prod_{j=0}^{r-1}(1+\log M_j(t)).$$

Each logarithmic factor is at most $1+\log M_r(t)$. Taking rth roots proves the first bound. Applying (219) in each block proves the second. The primitive contact forms, their pairwise determinants, and the fixed exceptions can be constructed directly as in Sections A37. □

Four translated sets of five contacts therefore give height exponent $7/8$ with twenty distinct finite contacts. This statement concerns logarithmic height; replacing it by an estimate for multiplicative height would be a different assertion.

A36.8 Numerical constants in the contact-height bound

Here B_* is the squarefree prime product defined above and $D_0 = 4608 \cdot 210^{300}$. The integer comparisons used in the preceding estimates are

$$D_0^2 < B_*, \qquad 144 < B_*, \qquad D_0 < 2^{2413}, \qquad B_* < 2^{109445}.$$

These are finite comparisons of the displayed integers. They can be checked by multiplying the primes in the definition of B_*; primality up to its stated cutoff is decided by trial division up to the square root. The constant estimates follow by substituting these four comparisons into the preceding displayed bounds. No numerical comparison determines the ineffective aggregate-height constant or a defining value of N.

A37 Disjoint contacts and height amplification

A37.1 The height hypothesis and the conclusion

For $t = m/n \in \mathbb{Q}$ in lowest terms with $n > 0$, put $h(t) = \log \max(|m|, n)$. Valuations are normalized by $v_p(p) = 1$. Let S_0 be the set of primes at most 76176, and put

$$H = 2^{110} 210^{1800} \left(\prod_{p \in S_0} p \right)^{19} + 128.$$

The arithmetic premise used in this paper is the following five-contact height bound from Sections 1: for $s = u/v \in \mathbb{Q} \setminus \{0, 1, 2401, -343/32\}$ in lowest terms,

$$h(s) \le H M_5(s)^{19}, \tag{222}$$

where $M_5(s)$ is the product of primes outside S_0 for which at least one of

$$v_p(v), \quad v_p(u), \quad v_p(u - v), \quad v_p(u - 2401v), \quad v_p(32u + 343v)$$

is positive and odd. The contact at infinity is represented by v. All height conclusions below are implications of (222). No stronger height premise is assumed.

For $r \ge 1$, define

$$\Theta = \left\{ 0, -\frac{1}{2}, -1, \frac{1}{2399}, -\frac{32}{407} \right\}, \qquad \Theta_r = \bigcup_{j=0}^{r-1} (3j + \Theta).$$

Introduce the following $5r$ primitive integral linear forms:

$$\begin{aligned}
L_{j,0}(m, n) &= m - 3jn, \\
L_{j,1}(m, n) &= 2m + (1 - 6j)n, \\
L_{j,2}(m, n) &= m + (1 - 3j)n, \\
L_{j,3}(m, n) &= 2399m - (7197j + 1)n, \\
L_{j,4}(m, n) &= 407m + (32 - 1221j)n.
\end{aligned}$$

Their zeros m/n are exactly Θ_r. Write these forms as $L_i = a_i m + b_i n$, using a single index when the pair (j, k) is immaterial.

Definition A37.1. The finite set Σ_r consists of S_0 and all prime divisors of the nonzero integers

$$a_i, \qquad a_i b_{i'} - a_{i'} b_i \ (i \neq i'), \qquad a_i + b_i.$$

Put

$$E_r = \prod_{p \in \Sigma_r \setminus S_0} p, \qquad C_r = \log \max(3, 9r - 10), \qquad H_r = C_r + H E_r^{19}.$$

For $t \in \mathbb{Q} \setminus \Theta_r$, let $M_r(t)$ be the product of primes $p \notin \Sigma_r$ for which $v_p(t - \theta)$ is positive and odd for some $\theta \in \Theta_r$.

Every integer used to define Σ_r is nonzero. Indeed, the five elements of Θ are distinct and lie in $[-1, 1/2399]$, so their translates by $3j$ are disjoint. Also $1 \notin \Theta_r$: for $j = 0$ this is immediate, and for $j \geq 1$ the least element is $3j - 1 \geq 2$. Every a_i is positive. In particular, all quantities in Definition A37.1 are explicitly computable from r.

Theorem A37.2. *Assume* (222). *For every positive integer r and every $t \in \mathbb{Q} \setminus \Theta_r$,*

$$h(t) \leq C_r + H E_r^{19} M_r(t)^{19/r} \leq H_r M_r(t)^{19/r}. \tag{223}$$

Moreover, $M_r(1) = 1$.

A37.2 Disjoint radicals and the proof

Lemma A37.3. *Let $t = m/n$ be in lowest terms. At a prime $p \notin \Sigma_r$, at most one of the integers $L_i(m, n)$ is divisible by p. If $p \mid n$, none of them is divisible by p. Consequently $M_r(t)$ is the product of the pairwise coprime radicals $M_{r,j}(t)$ obtained by using only the five forms with first index j.*

Proof. If $p \mid n$, then m and each a_i are units modulo p. Thus $L_i(m, n)$ is a unit. If $p \nmid n$ and two forms vanish modulo p, their determinant vanishes modulo p, contrary to the definition of Σ_r. Since a_i and n are units in the latter case,

$$v_p\left(t + \frac{b_i}{a_i}\right) = v_p(L_i(m, n)).$$

The assertions about the radicals follow. □

Proof of Theorem A37.2. For $0 \leq j < r$, put

$$\phi_j(t) = 2 + \frac{1}{t - 3j} = \frac{2m + (1 - 6j)n}{m - 3jn}.$$

The numerator and denominator are relatively prime: the matrix

$$A_j = \begin{pmatrix} 2 & 1 - 6j \\ 1 & -3j \end{pmatrix} \quad \text{has determinant } -1.$$

Its inverse is

$$A_j^{-1} = \begin{pmatrix} 3j & 1 - 6j \\ 1 & -2 \end{pmatrix}.$$

The operator norm for the maximum norm is $\max(3, 9j - 1)$ for $j \geq 0$, interpreting the formula at $j = 0$ as 3. Hence

$$h(t) \leq h(\phi_j(t)) + C_r. \tag{224}$$

The five projective contact forms for $\phi_j(t)$ are exactly $L_{j,0},\dots,L_{j,4}$, up to signs. In particular, the hypothesis $t \notin \Theta_r$ excludes every forbidden value and pole of ϕ_j. Primes removed by enlarging S_0 to Σ_r occur at most once in the radical, regardless of their valuations. Therefore

$$M_5(\phi_j(t)) \le E_r M_{r,j}(t).$$

If $h(t) \le C_r$, the result is immediate. Otherwise (222) and (224) give, for every j,

$$0 < h(t) - C_r \le H E_r^{19} M_{r,j}(t)^{19}.$$

Multiplying these r inequalities and applying Lemma A37.3 proves the first inequality in (223). The second follows from $M_r(t) \ge 1$. Finally, every prime dividing a contact value $L_i(1,1)$ belongs to Σ_r, so $M_r(1) = 1$. □

Remark A37.4. This argument uses the product of disjoint odd-valuation radicals. It does not replace an odd-valuation radical by an ordinary radical. At the removed primes, an arbitrary positive valuation contributes at most one prime factor to the comparison with M_5.

A37.3 A compatible rational normalization

The following identities describe how the larger contact set can enter a local valuation construction. They are algebraic identities; they do not assert that an arbitrary circuit using them has the projection property of a defining polynomial.

Let $d = 5r$ and set $\epsilon = 1$ for odd r, $\epsilon = 2$ for even r. Thus $d + \epsilon$ is even and $1 \le \epsilon \le 2$. Put

$$F_r(t) = \prod_{\theta \in \Theta_r} (t - \theta).$$

Fix a positive rational square a, take $t = a\lambda$, and write $\delta = t - 1$. When $\delta F_r(t) \ne 0$, define

$$R = \frac{F_r(t)}{a^d \delta^{d+\epsilon}}, \qquad e = \frac{(a\delta)^d}{F_r(t)}, \qquad q = \frac{1}{R\delta^4}. \tag{225}$$

The corresponding cleared identities are

$$F_r(t)e = (a\delta)^d, \qquad q\delta^{4-\epsilon} = e, \qquad e\delta^{\epsilon} = R^{-1}. \tag{226}$$

Proposition A37.5. *The following assertions hold.*

1. *Every solution of the first identity in* (226) *has* $F_r(t) \ne 0$. *If* $q = 0$ *in the first two identities, then* $\delta = 0$ *and* $M_r(t) = 1$.
2. *If* $p \mid M_r(t)$, *then* δ *is a* p*-adic unit and* $v_p(R)$ *is odd. In particular,* $v_p(q) = -v_p(R)$.
3. *For* $\lambda \to +\infty$,
$$R \sim a^{-(d+\epsilon)}\lambda^{-\epsilon} > 0, \qquad q \sim a^{d+\epsilon-4}\lambda^{\epsilon-4} \longrightarrow 0.$$
4. *At a fixed prime where* $v_p(t) < 0$ *and is smaller than every valuation of a nonzero contact, one has*
$$v_p(q) = d\, v_p(a) - (4-\epsilon)v_p(t).$$
For fixed a, *this tends to* $+\infty$ *as* $v_p(\lambda)$ *tends to* $-\infty$.

Proof. If $F_r(t) = 0$, then $t \in \Theta_r$, whence $\delta \neq 0$ and $(a\delta)^d \neq 0$, a contradiction. If $q = 0$, the second identity gives $e = 0$, and the first gives $\delta = 0$; Theorem A37.2 gives $M_r(t) = 1$.

At a consumed prime, exactly one contact difference has positive odd valuation, all other contact differences are units, and $t - 1$ is a unit. These facts follow from the determinant and $L_i(1, 1)$ conditions in Σ_r. Since $v_p(a)$ is even, the valuation $v_p(F_r(t)) - d\, v_p(a)$ is odd. This proves the second assertion. The third follows by comparing the leading terms in (225). For the fourth, every contact difference and δ has valuation $v_p(t)$. Thus $v_p(e) = d\, v_p(a)$, and the second identity in (226) proves the formula. □

Proposition A37.6. *Suppose $\lambda = u/v$ with v a positive integral square and $u \in \mathbb{Z}$, and suppose the d numbers*

$$p_\theta = u - \frac{\theta}{a} v \qquad (\theta \in \Theta_r)$$

are positive primes. Assume they avoid Σ_r and every prime dividing the numerator or denominator of a. If they are distinct, then R is positive for λ sufficiently large and its reduced denominator is a square. Its primes of odd valuation are exactly the p_θ, with valuation one. If in addition each p_θ is a square in $\mathbb{Q}_2$, $\mathbb{Q}_3$, and $\mathbb{Q}_{17}$, then so is R.

Proof. Direct substitution gives

$$R = \frac{\prod_{\theta \in \Theta_r} p_\theta}{v^d \delta^{d+\epsilon}}.$$

The denominator is a rational square. We check that it is a unit at each p_θ. Such a prime cannot divide v: otherwise it also divides u and hence all the distinct primes $p_{\theta'}$, since all contact and scale denominators are units there. Thus $a\lambda$ reduces to θ at this prime, and δ reduces to $\theta - 1 \neq 0$. Its denominator is therefore a unit, and the valuation of R is one. At every other prime the valuation is even. This proves the assertion about the reduced denominator and the odd valuations. Positivity follows from $d + \epsilon$ even and the positive numerator. The final assertion follows because a product of local squares divided by a nonzero square is a square. □

The prime-pattern hypothesis in Proposition A37.6 is the same finite-pattern requirement used in the arithmetic construction: the contacts and scales are fixed first, rational coefficients are cleared by a fixed square denominator, and a finite set of primes is avoided by Section A41 and Theorem A8.1. Increasing r changes this finite list and the associated constants. An explicit numerical cutoff must consequently be recalculated.

A37.4 The divisibility exponents required by height arguments

Lemma A37.7. *Let $k \in \mathbb{Z}_{>0}$, $\alpha > 0$, $H_* > 0$, and suppose a radical M satisfies $h(s) \leq H_* M(s)^\alpha$ at two nonzero rational values $s = \lambda$ and $s = T^2\lambda$, where $T \neq 0$. If $N \neq 0$ is an integer and*

$$M(\lambda)^k \mid N, \qquad M(T^2\lambda)^k \mid N,$$

then

$$h(T) \leq H_* |N|^{\alpha/k}.$$

The two radicals need not be relatively prime.

Proof. The elementary height inequalities give

$$2h(T) = h(T^2) \leq h(T^2\lambda) + h(\lambda).$$

Each radical is at most $|N|^{1/k}$, proving the result. □

Corollary A37.8. *Assume* (222). *Suppose* $B \geq 1$, $A > 0$, $J \geq 0$, $T, \lambda \in \mathbb{Q}^{\times}$, *and*

$$B^2 \leq Ah(T) + J.$$

Assume both λ *and* $T^2\lambda$ *avoid* Θ_r. *Let an integer residual* Δ *satisfy*

$$|\Delta| \leq 5B^4, \qquad M_r(\lambda)^k \mid \Delta, \qquad M_r(T^2\lambda)^k \mid \Delta.$$

If $rk > 38$, *then all solutions with* B *sufficiently large in terms of* r, k, A, J, H_r *satisfy* $\Delta = 0$.

Proof. If $\Delta \neq 0$, Theorem A37.2 and Lemma A37.7 imply

$$B^2 \leq AH_r 5^{19/(rk)} B^{76/(rk)} + J.$$

Since $76/(rk) < 2$, this bounds B above by a constant depending only on the displayed data. □

Corollary A37.9. *Under the same hypotheses, suppose an integer* $N \neq 0$ *instead satisfies*

$$|N| \leq 2\exp(h(\beta))B, \qquad M_r(\lambda)^{\mu} \mid N, \qquad M_r(T^2\lambda)^{\mu} \mid N.$$

If $2r\mu > 19$, *then*

$$\log B \leq \frac{19}{2r\mu - 19}\, h(\beta) + C,$$

where C *depends only on* r, μ, A, J, H_r.

Proof. Put $\gamma = 19/(r\mu) < 2$. The preceding argument gives

$$B^2 \leq AH_r 2^{\gamma} \exp(\gamma h(\beta)) B^{\gamma} + J.$$

If $B^2 \leq 2J$, the assertion follows by increasing C. Otherwise absorb J into $B^2/2$, take logarithms, and divide by $2 - \gamma$. The coefficient of $h(\beta)$ becomes $\gamma/(2-\gamma) = 19/(2r\mu - 19)$. □

The least positive integers meeting these two strict inequalities are

$$k_r = \left\lfloor \frac{38}{r} \right\rfloor + 1, \qquad \mu_r = \left\lfloor \frac{19}{2r} \right\rfloor + 1.$$

Some useful values are:

r	1	2	3	4	5	6	7	10	13	20	39
$5r$	5	10	15	20	25	30	35	50	65	100	195
k_r	39	20	13	10	8	7	6	4	3	2	1
μ_r	10	5	4	3	2	2	2	1	1	1	1

These are requirements on divisibility of integer residuals. A formal approximation formula must separately be proved accurate enough to produce those divisibilities. Lower exponents can shorten such a formula, but the enlarged product F_r requires additional arithmetic operations. Neither this table nor Theorem A37.2 is a coordinate count for a polynomial definition of the integers.

A37.5 Exact finite verification

The accompanying standard-library program `check_contact_height_amplification.py` reconstructs the linear forms for the values of r in the table, verifies their primitivity and pairwise distinct zeros, and factors every integer defining Σ_r. It records the exceptional primes larger than 76176, the exact product E_r, and the height-transformation norm. It also checks the integer threshold inequalities and the normalized valuation identities. This calculation verifies the finite algebra; the arithmetic premise (222) remains explicit.

A38 Uniform aggregate trace height

A38.1 Functions with one torsion zero and one pole

Throughout the geometric argument the base field is algebraically closed of characteristic zero. Let E be an elliptic curve, let O be its identity, and write $\mathcal{R} = \mathrm{End}(E)$. This ring is commutative and has no zero divisors. Let $T \in E$ have finite order $N > 1$. Suppose $f \in k(E)$ satisfies

$$\mathrm{div}(f) = m[T] - m[O] \qquad (m > 0).$$

Thus f has its only zero at T and its only pole at O. Such a function exists when $m = N$, because $N(T - O)$ is a principal divisor on an elliptic curve.

Lemma A38.1. *Let $P \in E$ have infinite order. Let $n_1, \dots, n_r$ be nonzero integers prime to N, and let $\alpha_1, \dots, \alpha_r \in \mathcal{R}$. At every point U of E, a pole of one function $f([n_i]P + \alpha_i U)$ cannot coincide with a zero of another. Consequently, if some $\alpha_i \neq 0$, the product of these functions has a pole and is nonconstant.*

Proof. Suppose

$$[n_i]P + \alpha_i U = O, \qquad [n_j]P + \alpha_j U = T.$$

The first equation requires $\alpha_i \neq 0$; a constant factor cannot be a zero or a pole because its argument is nontorsion. Applying the two endomorphisms and subtracting gives

$$(n_i\alpha_j - n_j\alpha_i)P = -\alpha_i T.$$

The right side is torsion. A nonzero endomorphism is an isogeny and cannot take a nontorsion point to a torsion point. Hence $n_i\alpha_j = n_j\alpha_i$.

Put $g = \gcd(|n_i|, |n_j|)$, $u = n_i/g$, and $v = n_j/g$. There are integers a, b with $au + bv = 1$. Set $\gamma = a\alpha_i + b\alpha_j$; the preceding equality gives $\alpha_i = u\gamma$ and $\alpha_j = v\gamma$. Therefore $R = [g]P + \gamma U$ satisfies $[u]R = O$ and $[v]R = T$. The order of R divides $|u|$, which is prime to N, whereas T has order $N > 1$. This is impossible. Finally, every nonzero α_i is surjective, so the corresponding factor has a pole, and the absence of a zero at that point keeps it a pole of the product. □

Two nontorsion points P_1, P_2 will be called independent over $\mathcal{R}$ modulo torsion if $\alpha P_1 + \beta P_2$ torsion implies $\alpha = \beta = 0$.

Lemma A38.2. *Suppose P_1, P_2 are independent over $\mathcal{R}$ modulo torsion. If n, m are nonzero integers and α, β are nonzero endomorphisms, the functions $f([n]P_1 + \alpha U)$ and $f([m]P_2 + \beta U)$ have no common pole.*

Proof. A common pole would give $\alpha U = -[n]P_1$ and $\beta U = -[m]P_2$. Commutativity then gives $n\beta P_1 - m\alpha P_2 = O$, contradicting independence. □

A38.2 Excluding abelian translates in an aggregate fiber

Fix nonzero scalars c_1, c_2 and an integer $r \geq 1$. On E^{2r} consider the rational function

$$F(Q_1, \dots, Q_r, Q'_1, \dots, Q'_r) = c_1 \prod_{i=1}^{r} f(Q_i) + c_2 \prod_{i=1}^{r} f(Q'_i).$$

Theorem A38.3. *Let P_1, P_2 be independent over $\mathcal{R}$ modulo torsion. Suppose every n_i, m_i is nonzero and prime to N. Put*

$$q = ([n_1]P_1, \dots, [n_r]P_1, [m_1]P_2, \dots, [m_r]P_2).$$

No positive-dimensional abelian translate through q is contained in a level set of F.

Proof. Suppose $q + B$ is such a translate, with B a nonzero abelian subvariety of E^{2r}. There is a nonzero homomorphism $\phi : E \to B$. For example, a nonzero coordinate projection $B \to E$ is surjective; dualizing and composing with an inverse to a polarization up to multiplication produces such a map. Write its coordinates as $(\alpha_1, \dots, \alpha_r, \beta_1, \dots, \beta_r)$, not all zero. The restriction of F to $q + \phi(E)$ is

$$c_1 \prod_i f([n_i]P_1 + \alpha_i U) + c_2 \prod_i f([m_i]P_2 + \beta_i U).$$

Within either product a pole survives by Lemma A38.1. A pole of one product is not a pole of the other by Lemma A38.2. Since at least one endomorphism is nonzero, at least one product has a pole, and it remains a pole of the sum. This contradicts constancy. □

The same proof permits any finite number of blocks based on points that are pairwise independent over $\mathcal{R}$ modulo torsion. The integers in each block need only be prime to the order of the unique torsion zero of its chosen function.

A38.3 Sums of abscissas without a parity restriction

Give E a Weierstrass equation and write x for its abscissa. This function has its only pole at O, of order two. Let $w_1, \dots, w_r, w_1', \dots, w_r'$ be positive rational numbers and let c_1, c_2 be nonzero algebraic numbers. Define

$$F_+(Q_1, \dots, Q_r, Q_1', \dots, Q_r') = c_1 \sum_i w_i x(Q_i) + c_2 \sum_i w_i' x(Q_i').$$

Lemma A38.4. *Let $P \in E$ have infinite order, let all integers n_i be nonzero, and let $\alpha_i \in \mathcal{R}$. If some $\alpha_i \neq 0$, the function*

$$\sum_i w_i x([n_i]P + \alpha_i U)$$

has a pole and is nonconstant.

Proof. Choose a pole U_0 of a nonconstant summand, say the ith. If another summand, the jth, has a pole there, then $[n_i]P + \alpha_i U_0 = [n_j]P + \alpha_j U_0 = O$. Applying the endomorphisms and subtracting gives $(n_i\alpha_j - n_j\alpha_i)P = O$ and hence $n_i\alpha_j = n_j\alpha_i$.

Let z be a uniformizer at O with $x = z^{-2} + O(z^{-1})$, and use $z(U - U_0)$ locally at U_0. Let λ_j be the nonzero tangent multiplier of α_j at O. The leading coefficient of the jth summand at this pole is $w_j\lambda_j^{-2}$, and the endomorphism identity gives $\lambda_j = (n_j/n_i)\lambda_i$. Consequently the coefficient of z^{-2} in the full sum is

$$\lambda_i^{-2} \sum_{j:\, [n_j]P + \alpha_j U_0 = O} w_j (n_i/n_j)^2.$$

The sum is a nonzero positive rational number. The pole therefore cannot cancel. □

Theorem A38.5. *Suppose P_1, P_2 are independent over $\mathcal{R}$ modulo torsion, and all n_i, m_i are nonzero. No positive-dimensional abelian translate through*

$$q = ([n_i]P_1)_{i=1}^r \times ([m_i]P_2)_{i=1}^r$$

is contained in a level set of F_+.

Proof. Restrict to an elliptic subgroup direction exactly as in Theorem A38.3. Every nonconstant block sum has a pole by Lemma A38.4. A pole in one block cannot occur in the other block, because the same argument as Lemma A38.2 would give a nontrivial endomorphism relation between P_1 and P_2. At least one block is nonconstant, so their weighted sum is nonconstant. □

This result uses no torsion zero of x and imposes no parity condition. Its positivity assumption concerns the fixed rational weights within each block, not the values of the abscissas.

Theorem A38.6 (Several blocks)**.** *Let $P_1, \dots, P_s$ be nontorsion points on E such that each pair is independent over $\mathcal{R}$ modulo torsion. Fix positive integers r_j, positive rational weights w_{ji}, and nonzero algebraic scalars c_j. On $E^{\sum_j r_j}$ put*

$$F_{\mathrm{sum}}((Q_{ji})) = \sum_{j=1}^{s} c_j \sum_{i=1}^{r_j} w_{ji} x(Q_{ji}).$$

For any tuple $q = ([n_{ji}]P_j)$ with all $n_{ji} \neq 0$, no positive-dimensional abelian translate through q is contained in a fiber of F_{sum}.

There is a product version. For each j suppose $\mathrm{div}(f_j) = m_j[T_j] - m_j[O]$*, where T_j has order $N_j > 1$. Replace F_{sum} by*

$$F_{\mathrm{product}}((Q_{ji})) = \sum_{j=1}^{s} c_j \prod_{i=1}^{r_j} f_j(Q_{ji}).$$

The same conclusion holds if every n_{ji} is prime to N_j.

Proof. Restrict to a nonzero homomorphism $E \to B$ in any proposed positive-dimensional subgroup B. Some block has a nonconstant coordinate. Within that block a pole survives by Lemma A38.4, or by Lemma A38.1 in the product case. A pole cannot occur in two distinct blocks: their two pole equations would contradict the pairwise independence of their base points. The pole therefore survives in the full sum, contradicting constancy. □

The scalars c_j need not be real or positive. Pairwise independence suffices; no independence assumption on the entire collection is required.

A38.4 A linear height comparison

We use the following consequence of Von Buhren's family height theorem [127, Theorem 1.1]. Let A be an abelian variety, S a projective variety, and $V \subseteq A \times S$ an integral closed subvariety. For a fixed subgroup $\Gamma \subseteq A(\overline{\mathbb{Q}})$ of finite rank, there are constants $\mu, \nu \geq 0$ such that

$$\widehat{h}_A(a) \leq \mu h_S(s) + \nu \tag{227}$$

whenever $a \in \Gamma$ lies in the fiber V_s and does not lie on a positive-dimensional abelian translate contained in that fiber. The slope μ can be computed from projective geometric data. The theorem does not provide an effective value for ν.

Corollary A38.7. *Assume the data of Theorem A38.3 are algebraic numbers. There are constants $M > 0$ and $M_0 \geq 0$, independent of all the integers n_i, m_i, such that*

$$\sum_{i=1}^{r} \bigl(\widehat{h}_E([n_i]P_1) + \widehat{h}_E([m_i]P_2)\bigr) \leq Mh(F(q)) + M_0$$

for every tuple in that theorem. Here h is the absolute logarithmic height on $\mathbb{P}^1$. One may choose an effective geometric slope M after fixing compatible height normalizations. No effective M_0 is asserted. The same conclusion holds with F_+ and the hypotheses of Theorem A38.5, allowing all nonzero labels.

Proof. Let $A = E^{2r}$ and take for $V \subseteq A \times \mathbb{P}^1$ the closure of the graph of F. This is integral and projective. Let Γ be the subgroup generated by P_1 in each of the first r coordinates and by P_2 in each of the last r coordinates. It is finitely generated.

Every indicated q is in the domain of F, since no $[n_i]P_1$ or $[m_i]P_2$ is the identity. If an abelian translate through q were contained in the projective fiber $V_{F(q)}$, it would meet the domain of F at q. Its dense open intersection with that domain would make F constant on the translate. Theorem A38.3 rules this out. This argument also handles possible boundary components introduced by closing the graph; none through q changes the conclusion.

Apply (227), using a symmetric ample product line bundle on A. Its canonical height is a fixed positive multiple of the sum of the coordinate canonical heights. Rescaling gives the stated inequality. For F_+ the same proof uses Theorem A38.5 to exclude the exceptional translates. □

For either function of Theorem A38.6 the same argument gives a uniform inequality

$$\sum_{j,i} \widehat{h}_E([n_{ji}]P_j) \leq Mh(F(q)) + M_0.$$

The subgroup is generated by the finitely many base points placed in their respective coordinates. The constants are fixed before any of the integer labels are chosen.

A38.5 The quadratic-field elliptic curve

Let

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad d' = 3 + \sqrt{2}, \quad E:\ y^2 = x^3 - d^2x,$$

and let σ be the nonidentity automorphism of $K/\mathbb{Q}$. The point used in the rational source is

$$R_0 = (1 + \sqrt{2}, 2\sqrt{2}), \qquad P = 2R_0 = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

For this section its infinite order is the only group property needed; no assertion about the full Mordell–Weil group is required.

Lemma A38.8. *The displayed point P has infinite order.*

Proof. Substitution verifies $R_0 \in E(K)$, and the usual tangent formula gives the displayed double. At the prime $\mathfrak{p} = (\sqrt{2})$ normalize $v_{\mathfrak{p}}(\sqrt{2}) = 1$, so $v_{\mathfrak{p}}(2) = 2$. The displayed coordinates give

$$v_{\mathfrak{p}}(x(P)) = -6, \qquad v_{\mathfrak{p}}(y(P)) = -9, \qquad v_{\mathfrak{p}}(-x(P)/y(P)) = 3.$$

The parameter $z = -x/y$ places P in the formal group of this integral Weierstrass equation. Its multiplication series have integral coefficients and linear term $[m](z) = mz + O(z^2)$. For odd m

and $v(z) \geq 3$, the linear term uniquely has least valuation, so $v([m]z) = v(z)$. For multiplication by two, the linear term has valuation $v(z)+2 < 2v(z)$, and therefore $v([2]z) = v(z)+2$. No successive multiplication by an odd integer and by a power of two can give zero. Every nonzero integer has this form, proving the claim. □

Lemma A38.9. *Over a field containing $\sqrt{d}$ and $\sqrt{d'}$, identify E and E^σ with $E_0 : Y^2 = X^3 - X$ by*

$$(X, Y) = \left(\frac{x}{d}, \frac{y}{d\sqrt{d}}\right), \qquad (X', Y') = \left(\frac{x'}{d'}, \frac{y'}{d'\sqrt{d'}}\right).$$

The images P_1, P_2 of P, P^σ are independent over $\mathrm{End}(E_0) = \mathbb{Z}[i]$ modulo torsion.

Proof. The squareclasses of d, d' are independent in $K(i)$. Indeed, a positive element of the real field K that is a square in $K(i)$ is already a square in K: writing a square root as $a+ib$ gives $ab = 0$, and the purely imaginary case would be negative at a real embedding. Neither d nor d' is a square in K, since its norm to $\mathbb{Q}$ is 7. Their product 7 is not a square in K: expanding $(a+b\sqrt{2})^2 = 7$ for rational a, b gives $ab = 0$ and then requires either 7 or $7/2$ to be a rational square.

It follows that $K(i, \sqrt{d}, \sqrt{d'})/K(i)$ has an automorphism fixing $\sqrt{d}$ and negating $\sqrt{d'}$. It fixes P_1, negates P_2, and fixes every endomorphism in $\mathbb{Z}[i]$. If $\alpha P_1 + \beta P_2$ were torsion, applying the automorphism and subtracting would make $2\beta P_2$ torsion. Infinite order forces $\beta = 0$, and then $\alpha = 0$. □

Theorem A38.10. *For any fixed $r \geq 1$, there are constants $M_+ > 0$ and $M_{+,0} \geq 0$ such that, for all nonzero integers $n_1, \ldots, n_r$,*

$$\sum_{i=1}^{r} \widehat{h}_E([n_i]P) \leq M_+ h\left(\frac{1}{2}\,\mathrm{Tr}_{K/\mathbb{Q}} \sum_{i=1}^{r} x([n_i]P)\right) + M_{+,0}. \tag{228}$$

There are also constants $M_\times > 0$, $M_{\times,0} \geq 0$ such that, when all the labels are odd,

$$\sum_{i=1}^{r} \widehat{h}_E([n_i]P) \leq M_\times h\left(\frac{1}{2}\,\mathrm{Tr}_{K/\mathbb{Q}} \prod_{i=1}^{r} x([n_i]P)\right) + M_{\times,0}. \tag{229}$$

Both slopes can be chosen effectively from the fixed geometry; the additive constants are not determined effectively here.

Proof. On E_0, the coordinate X has divisor $2[(0,0)] - 2[O]$. Its zero is a nonzero point of order two, so the index condition in Theorem A38.3 is exactly oddness. Under the isomorphisms of Lemma A38.9, the rational function in the height on the right of (229) is

$$F = \frac{d^r}{2} \prod_{i=1}^{r} X_i + \frac{(d')^r}{2} \prod_{i=1}^{r} X_i'.$$

For the sum, the corresponding function is $F_+ = (d/2)\sum_i X_i + (d'/2)\sum_i X_i'$ and Theorem A38.5 applies without parity restrictions. Apply Corollary A38.7 to both functions with $m_i = n_i$. Canonical heights are unchanged by these pointed isomorphisms, and conjugation preserves absolute canonical height. The coordinate height sum there is twice the sum in (229); divide by two. □

Notice that the inequality is linear. Finiteness of each individual fiber alone would not give this conclusion. The additive constant is the remaining effectiveness issue; it is not an additional mathematical conjecture.

A38.6 A trace-height consequence over number fields

Corollary A38.11. *Let K/F be a finite extension of number fields, let E/K be an elliptic curve, and let $P \in E(K)$ have infinite order. For every embedding $\sigma : K \hookrightarrow \overline{F}$ over F, suppose there is a pointed isomorphism $\phi_\sigma : E^\sigma \to E_0$ to a fixed elliptic curve over $\overline{F}$. Assume that the points $P_\sigma = \phi_\sigma(P^\sigma)$ are pairwise independent over* $\mathrm{End}(E_0)$ *modulo torsion.*

For every fixed positive integer r there are constants $M > 0$, $M_0 \geq 0$ such that, for all nonzero integer labels,

$$\sum_{i=1}^{r} \widehat{h}_E([n_i]P) \leq Mh\left(\mathrm{Tr}_{K/F} \sum_{i=1}^{r} x([n_i]P)\right) + M_0.$$

The slope can be chosen effectively from the fixed geometric data; the additive constant has the same ineffective status as in Corollary A38.7.

Proof. Under a pointed isomorphism of Weierstrass models the abscissa transforms as $x^\sigma = a_\sigma x_0 + b_\sigma$, with $a_\sigma \neq 0$. The function in the height is therefore

$$\sum_\sigma a_\sigma \sum_i x_0([n_i]P_\sigma) + r \sum_\sigma b_\sigma.$$

Theorem A38.6 excludes constant translates through the indicated points for the nonconstant part. Adding the fixed scalar does not change that conclusion. The graph-family argument gives a linear bound for the sum of the canonical heights in all the embedding blocks. Absolute canonical height is unchanged by conjugation and pointed isomorphism, so this sum is $[K : F]$ times the left side of the displayed inequality. Divide by $[K : F]$. □

This corollary is a height comparison under explicit independence hypotheses. It does not assert a Diophantine definition of the ring of integers of K.

A39 Plain formal parameters and rational approximations

A39.1 Uniform logarithm truncations of arbitrary order

Let $E : y^2 = x^3 - Dx$ over a field of characteristic zero, with $D \neq 0$. Write $f = 1/x$ and $z = -x/y$ near the identity. The equation gives

$$f = z^2(1 - Df^2).$$

Define rational numbers

$$c_0 = 1, \qquad c_1 = -\frac{2}{5}, \qquad c_j = c_{j-1}\frac{4j-5}{4j+1} \quad (j \geq 2),$$

and put

$$Q_J(T) = \sum_{j=0}^{J-1} c_j T^j, \qquad L_J(Q) = z(Q)Q_J(Df(Q)^2) \quad (J \geq 1).$$

Lemma A39.1. *The formal series*

$$L(Q) = z(Q) \sum_{j \geq 0} c_j (Df(Q)^2)^j$$

is the normalized formal logarithm of the displayed equation. For every odd prime p and $j \geq 1$,

$$v_p(c_j) \geq -\lfloor \log_p(4j+1) \rfloor.$$

Proof. The invariant differential is $\omega = dx/(2y) = z\,df/(2f)$. Put $T = Df^2$ and $S(T) = \sum c_j T^j$. From $z^2 = f/(1-T)$, differentiation shows that $d(zS(T)) = \omega$ is equivalent to

$$4T(1-T)S'(T) + (1+T)S(T) = 1 - T.$$

The constant coefficient is $c_0 = 1$, the linear coefficient is $5c_1 = -2$, and every subsequent coefficient is precisely the stated recurrence. The resulting series is $z + O(z^5)$ and hence is the normalized formal logarithm.

For $j \geq 1$ its coefficient is exactly

$$c_j = 2\prod_{i=1}^{j} \frac{4i-5}{4i+1}.$$

For an odd prime power p^r, each of the two arithmetic progressions in this quotient has length j and invertible step four. The counts of terms divisible by p^r therefore differ by at most one. No term has absolute value exceeding $4j+1$. Summing the possible loss of one valuation over the prime powers at most $4j+1$ proves the claimed lower bound. □

Theorem A39.2. *Let $F/\mathbb{Q}_p$ be unramified, where p is odd, normalize $v(p) = 1$, and suppose D is integral. For any nonzero point Q in the formal group, so that $v(z(Q)) \geq 1$, and every integer $J \geq 1$,*

$$v(L_J(Q)) = v(z(Q)),$$
$$v(L(Q) - L_J(Q)) \geq v(z(Q)) + 3J.$$

In particular, the conclusion is uniform in J and does not require $p > J$ or $p > 4J$.

Proof. The relation between f and z gives $v(f(Q)) = 2v(z(Q))$. For every $j \geq 1$ and odd p one has

$$\lfloor \log_p(4j+1) \rfloor \leq j.$$

Indeed $3^{j+1} > 4j+1$ for $j \geq 1$, by induction. Consequently Lemma A39.1 gives

$$\begin{aligned} v(z(Q)c_j(Df(Q)^2)^j) &\geq v(z(Q)) + 4jv(z(Q)) - \lfloor \log_p(4j+1) \rfloor \\ &\geq v(z(Q)) + 3j. \end{aligned}$$

All nonconstant terms are strictly smaller in absolute value than the leading term $z(Q)$, proving the first equality. The same estimates show that the infinite series converges, and that its tail beginning at $j = J$ has the stated valuation. □

Corollary A39.3. *In the setting of Theorem A39.2, suppose $Q = [n]P$ and $R = [m]P$ are nonzero formal points, where m, n are nonzero integers. For truncation lengths $J_1, J_2 \geq 1$, put $k = 3\min(J_1, J_2)$. Then*

$$v\left(\frac{L_{J_1}(Q)}{L_{J_2}(R)} - \frac{n}{m}\right) \geq v(n/m) + k.$$

If n/m is integral, the right side is at least k.

Proof. Formal additivity applied to $[m]Q = [n]R$ gives $L(Q)/L(R) = n/m$. The leading-term estimate in the theorem makes both logarithms nonzero. Write $L_{J_1}(Q) = L(Q)(1+\varepsilon_Q)$ and $L_{J_2}(R) = L(R)(1+\varepsilon_R)$; their two errors have valuations at least k. Subtracting the exact ratio proves the assertion. □

For the field $\mathbb{Q}(\sqrt{2})$, every odd completion is unramified and $D = 11 - 6\sqrt{2}$ is integral. Thus these statements apply at every contact prime $p > 76176$, independently of the chosen truncation length. If M is a power of two with $M \geq 2$, the choice $J = M/2$ has relative precision at least $3M/2$, which is stronger than precision M. Arbitrarily long truncations therefore leave the fixed contact-height constant and its exceptional-prime set unchanged.

A39.2 Nonvanishing at the real embeddings

Lemma A39.4. *For every integer $J \geq 1$, one has $Q_J(T) > 0$ for $0 \leq T \leq 1$.*

Proof. All c_j for $j \geq 1$ are negative. For $r \geq 1$, the recurrence telescopes to

$$\sum_{j=0}^{r} c_j = -\frac{4r-1}{2} c_r > 0.$$

The identity holds at $r = 1$ and its successive differences are c_r. Because $0 \leq T^j \leq 1$ and $c_j < 0$, it follows that $Q_J(T) \geq Q_J(1) > 0$; $J = 1$ is immediate. □

Corollary A39.5. *For $K = \mathbb{Q}(\sqrt{2})$ and $D = (3 - \sqrt{2})^2$, let $Q \in 2E(K)$ be nonzero. Then $L_J(Q) \neq 0$ for every $J \geq 1$.*

Proof. The duplication identity

$$x(2R) = \left(\frac{x(R)^2 + D}{2y(R)} \right)^2$$

shows that the abscissa of a nonzero double is positive at both real embeddings. Its ordinate cannot be zero: the two nonzero ordinate-zero abscissas are $\pm(3 - \sqrt{2})$, neither a square in K since their norm is seven, while abscissa zero is excluded by the duplication formula. At either embedding the curve equation now implies $x(Q) > \sqrt{D}$. Thus $0 < Df(Q)^2 < 1$ and $z(Q) \neq 0$. Lemma A39.4 proves the assertion. □

Consequently any rational system whose logarithm denominators are evaluated at nonzero doubles may use these truncations at arbitrary finite length. Increasing the length introduces no new nonvanishing condition at those points.

A39.3 A finite integer coefficient specification

For a polynomial G, write $\|G\|_1$ for the sum of the absolute values of its coefficients. In a finite free algebra over $\mathbb{Q}$ or an integer polynomial ring, choose a basis and let M_A be the matrix of multiplication by A. Use the maximum column sum of the coefficient ℓ^1 norms as its matrix norm. This norm is submultiplicative, has $\|I\| = 1$, and satisfies $\|cI\| = |c|$ for rational scalars.

Compatible weights mean nonnegative weights on base variables and positive weights on the algebra generators for which multiplication and reduction by the monic defining relations do not increase the sum of the input weights. The weighted degree of the zero element is $-\infty$; all other weighted degrees are nonnegative.

Define

$$D_J = \prod_{i=1}^{J-1} (4i + 1), \qquad G_J(T) = D_J Q_J(T) \in \mathbb{Z}[T].$$

Empty products are one. Its coefficients are explicitly

$$[T^0]G_J = D_J, \qquad [T^j]G_J = 2\left(\prod_{i=1}^{j}(4i-5)\right)\left(\prod_{i=j+1}^{J-1}(4i+1)\right) \quad (1 \le j < J).$$

This finite product-and-sum definition specifies the ordinary integer polynomial without requiring a list of all its coefficients.

Proposition A39.6. *For $J \ge 2$, the degree of G_J is $J-1$ and*

$$D_J \le (4J)^{J-1}, \qquad \|G_J\|_1 \le 2J(4J)^{J-1}.$$

If $J = 2^{b-1}$, with $b \ge 2$, these imply

$$D_J < 2^{(b+1)J}, \qquad \|G_J\|_1 < 2^{(b+1)J}.$$

If A is an element of a finite free algebra with $\|M_A\| \le 2^B$, $B \ge 0$, then

$$\|M_{G_J(A)}\| \le \|G_J\|_1\, 2^{(J-1)B}.$$

If the algebra has the multiplicative weighted filtration associated with compatible positive generator weights, then $\operatorname{wdeg}(G_J(A)) \le (J-1)\max(0, \operatorname{wdeg}(A))$.

Proof. Every nonconstant coefficient is a product of $J-1$ integers of absolute value at most $4J$, times two. The same bound without two holds for the constant coefficient. Summing the coefficient bounds gives the displayed estimate. Its binary exponent when $J = 2^{b-1}$ is

$$1 + (b-1) + (J-1)(b+1) = (b+1)J - 1.$$

The operator estimate follows from the triangle inequality and submultiplicativity, and the weight estimate follows term by term. □

For a rational argument A/B, its homogeneous numerator is the explicit polynomial

$$G_J^{\mathrm{hom}}(A,B) = \sum_{j=0}^{J-1} [T^j]G_J\, A^j B^{J-1-j}.$$

Thus $Q_J(A/B) = G_J^{\mathrm{hom}}(A,B)/(D_J B^{J-1})$ on $B \ne 0$. This form is convenient for a finite algebra over an integer polynomial ring. Its degree and operator-norm estimates follow from the same proof with the larger of the two argument bounds.

A40 Prime selection and simultaneous denominator preparation

A40.1 The precise analytic input

Fix distinct integers $c_1, \dots, c_k$, with $k \ge 2$, and put

$$p_j = u + c_j w^2.$$

The fixed congruences, Gaussian ray conditions and reduction-order avoidance conditions in Theorem A8.1 are imposed throughout. A fixed dilation of w can be absorbed into the integers c_j. All such data are chosen before the parameter X tends to infinity.

We derive the following quantitative statement internally. Proposition A8.30 supplies the fixed Gaussian ray-class count. Proposition A8.33 bounds the finite-cutoff singular series below by a positive multiple of $(\log L)^{-4k}$; Proposition A8.6 bounds the full forbidden common-divisor tail by $C\tau X^{3/2}/(L\log L)+O_\tau(X^{3/2-1/800})$. Choose L first so that this tail is smaller than the positive cutoff mass, then fix the field and ray data, then choose the diagonal thresholds as in Section A8.11. Each surviving weight is a product of k prime logarithms. This proves the following statement for the fixed compatible Gaussian data used here; no uniformity in growing fields or moduli and no general-Galois preprint estimate is invoked. There are constants $c, C>0$ and a sufficiently slowly decreasing function $\tau(X)>0$, with $\tau(X)\to 0$, such that, for

$$T=\lfloor \tau(X)\sqrt{X}\rfloor,$$

the admissible pairs in $X/2\le u\le X$, $1\le w\le T$ carry nonnegative weights $\omega_X(u,w)$ satisfying

$$\sum_{u,w}\omega_X(u,w)\ge cXT, \qquad 0\le \omega_X(u,w)\le C(\log X)^k. \tag{230}$$

The weights vanish outside admissible pairs. The lower bound is after the removal of all self-order and cross-order obstructions, not merely after a finite list of congruences.

For clarity, the order of limits in the cited argument is essential. Its finite cutoff, denoted by L there, is fixed first. The surviving weight is at least

$$\frac{c_1XT}{8(\log L)^{4k}}$$

for a positive constant c_1 and all sufficiently large X. This gives the first inequality in (230) with a fixed positive constant. Each surviving weight is a product of k prime logarithms, up to fixed normalizations, giving the second inequality. The diagonal construction of τ allows additional lower thresholds which hold eventually for every fixed positive value of τ.

All assertions below that select prime patterns depend on these cited estimates. The elementary statements about weights, integers and valuations are proved here.

A40.2 Removing a thin strip

Lemma A40.1. *Assume* (230). *Let* $0<\alpha<1/2$ *be fixed. The function* τ *may be chosen sufficiently slowly that admissible pairs with* $w\ge X^\alpha$ *still have total weight at least* $cXT/2$ *for every sufficiently large* X.

Proof. The strip $w<X^\alpha$ contains at most $X^{1+\alpha}+X$ integer pairs. Its weight is at most

$$C(X^{1+\alpha}+X)(\log X)^k.$$

Choose the diagonal thresholds so that

$$\frac{\tau(X)X^{1/2-\alpha}}{(\log X)^k}\longrightarrow\infty.$$

This is compatible with every prior threshold: for a fixed positive value τ_j, the displayed quotient tends to infinity, so the start of the jth diagonal interval can be increased to meet this additional condition. Also require $\tau(X)\sqrt{X}\ge 2$, so $T\ge\tau(X)\sqrt{X}/2$. The strip weight divided by XT tends to zero. Subtracting it from (230) proves the assertion. □

Theorem A40.2. *For each selected prime p_j, let d_j be a positive integer such that*

$$d_j = p_j + O(\sqrt{p_j}), \qquad 4 \mid d_j,$$

where the implied constant is uniform in j and the pattern. Under the cited analytic input, the pattern can be chosen so that

$$p_i \nmid d_j \qquad (1 \le i, j \le k).$$

The fixed congruence and avoidance conditions remain satisfied, all selected primes tend to infinity, and u/w^2 tends to infinity.

Proof. Apply Lemma A40.1 with $1/4 < \alpha < 1/2$. Since $w^2 \le T^2 = o(X)$, every p_j/u tends to one, uniformly over the selected patterns. The same is true of d_j/u. Consequently $1/2 < d_j/p_i < 2$ for sufficiently large X. Divisibility would therefore imply $d_j = p_i$.

For $i \ne j$, let $c_* := \min_{i \ne j} |c_i - c_j| > 0$. Then

$$|p_i - p_j| \ge c_* w^2 \ge c_* X^{2\alpha},$$

whereas $|d_j - p_j| = O(\sqrt{X})$. Because $2\alpha > 1/2$, equality is impossible for sufficiently large X. For $i = j$, the divisibility $4 \mid d_i$ excludes $d_i = p_i$, since p_i is an odd prime. The other conditions follow from the support of the weights and from $u \ge X/2$, $w^2 = o(X)$. □

For the elliptic curves used in the defining-polynomial constructions, Hasse's bound supplies the displayed estimate for d_j, and the fixed Gaussian congruences make the opposite order divisible by four. Thus Theorem A40.2 gives precisely the separation used to impose additional formal depth on the primary points. The application of this observation to those constructions is already present in Sections 1; the argument above states its quantitative source and permitted limits.

A40.3 Simultaneous denominator preparation

Let S be a finite set of rational primes, and let $a_1, \dots, a_s \in \mathbb{Q}^\times$ be fixed.

Lemma A40.3. *There is a positive integer D such that, whenever u is a prime outside S and w is a positive integer, the numbers*

$$\lambda = \frac{u}{D^2 w^2}, \qquad t_i = a_i \lambda, \qquad \delta_i = t_i - 1$$

satisfy $v_p(\delta_i) = v_p(t_i) < 0$ for every $p \in S$ and every i. The integer D can also satisfy any prescribed finite set of divisibility conditions.

Proof. Choose D with $2v_p(D) > \max_i v_p(a_i)$ for every $p \in S$, and enlarge it to satisfy the other fixed divisibilities. Since $v_p(u) = 0$ and $v_p(w) \ge 0$,

$$v_p(t_i) = v_p(a_i) - 2v_p(D) - 2v_p(w) < 0.$$

The two terms of $t_i - 1$ have distinct valuations, so its valuation equals $v_p(t_i)$. All requirements concern a finite set of primes. □

This choice can be made after fixing the scales and the finite preparation set, but before selecting primes. Multiplying the denominator dilation only changes fixed quadratic coefficients of the contact forms. The coefficient of the zero contact is zero, so u itself is one of the selected primes; the lower cutoff keeps it outside S. This is within the fixed-dilation scope of the cited theorem.

A40.4 Powers of one local weight

For nonzero rational R, δ, define

$$q_m = R^{1-2m}\delta^{-2m-2}, \qquad q = q_1 = R^{-1}\delta^{-4}, \qquad m \geq 1.$$

Proposition A40.4. *The alternative weights $q'_m = q^{2m-1}$ satisfy*

$$q'_m = q_m \delta^{-6(m-1)}.$$

At every valuation where δ is a unit, q'_m and q_m have the same valuation. At every valuation where $v(\delta) \leq 0$, $v(q'_m) \geq v(q_m)$. In particular, a lower bound for $v(q_m D^2)$ remains true with q'_m in place of q_m.

Proof. The exponent of R in q^{2m-1} is $1-2m$ and its exponent of δ is $4-8m$. Subtracting the exponent $-2m-2$ in q_m gives $-6(m-1)$. Taking valuations proves the assertions. □

Corollary A40.5. *Consider factors*

$$1 + \sum_{j=0}^{r-1} q_{m_j}^{2^j} D_j^{2^{j+1}}$$

in a local test. At contact places where $v(R)$ is odd and δ is a unit, replacing q_{m_j} by q^{2m_j-1} preserves every valuation and the distinct-valuation argument for its summands. At the finite places prepared by Lemma A40.3, any prescribed positive lower bounds on the summand valuations are preserved. If, as $\lambda \to +\infty$, one has $R \asymp \lambda^{-1}$, $\delta \asymp \lambda$, and all D_j bounded at the real embeddings, the modified factors tend to one.

Proof. The finite-place conclusions follow term by term from Proposition A40.4. At a contact place, the valuation of the jth nonconstant summand is 2^j times an odd integer; these valuations and zero are pairwise distinct. At infinity, $q = O(\lambda^{-3})$, hence $q^{2m_j-1} = O(\lambda^{-3(2m_j-1)})$. Each nonconstant summand therefore tends to zero. □

The corollary is a statement about the specified local tests and their simultaneous preparation. A complete defining-polynomial construction must still prove the pole realization and index-height steps, specify its equations, and check their possible zero strata. Those separate arithmetic requirements are unaffected by this normalization.

A41 Finite avoidance and simultaneous scalar preparation

A41.1 Finite avoidance and the common congruence progression

For a Gaussian integer π, set

$$d_-(\pi) = N_{\mathbb{Q}(i)/\mathbb{Q}}(\pi - 1), \qquad d_+(\pi) = N_{\mathbb{Q}(i)/\mathbb{Q}}(\pi + 1).$$

Theorem A8.1 permits any fixed finite list of distinct integral coefficients, any fixed divisibility of the square-denominator parameter, and any fixed lower bounds for the prime values and for the slope. Its avoidance modulus is odd, squarefree, and prime to seven. The fixed Gaussian residues are 9 modulo 16 and $1+3i$ modulo seven; at every prime dividing the avoidance modulus the residue is i. The theorem excludes every odd common divisor between the two order norms at the same position and between any two order norms at distinct positions.

Lemma A41.1. *Let $L \geq 1$ be divisible by 64, and fix distinct integers $c_1, \dots, c_k$, with $k \geq 2$. Prescribe $q \geq 1$ and real numbers $Y, H_0 > 0$. Theorem A8.1 supplies positive integers u, w such that*

$$q \mid w, \qquad u/w^2 > H_0, \qquad p_j = u + c_j w^2 > Y$$

are primes, with Gaussian generators whose order norms satisfy

$$v_2(d_-(\pi_j)) = 6, \qquad v_2(d_+(\pi_j)) = 2,$$
$$\gcd(d_-(\pi_j), d_+(\pi_h)) = 4 \qquad (1 \leq j, h \leq k),$$

and are all prime to the odd part of L. The primes may simultaneously be required to satisfy $p_j \equiv 1 \pmod{8 \cdot 3 \cdot 17}$.

Proof. Take the avoidance modulus to be the product of the odd primes dividing $3 \cdot 17 \cdot L$, with seven omitted. It is squarefree and prime to seven, as required by Theorem A8.1. At each of its prime factors, the two adjacent norms of i are both two. They are therefore units. At seven the fixed residue $1 + 3i$ instead gives

$$d_-(1 + 3i) = 9 \equiv 2 \pmod 7, \qquad d_+(1 + 3i) = 13 \equiv 6 \pmod 7.$$

Thus seven is avoided even when it divides L.

Write $\pi = 9 + 16a + 16bi$. Direct calculation gives

$$d_+(\pi) \equiv 4 \pmod 8, \qquad d_-(\pi) \equiv 64 \pmod{128}.$$

The exact powers of two in the statement follow. The exclusion of odd common factors supplied by Theorem A8.1 then gives every stated cross-sign gcd, including a position paired with itself. Finally the Gaussian norm is one modulo eight, and its residues at three and seventeen are the norm of i, namely one. These are precisely the required congruences on p_j. □

Corollary A41.2. *In the situation of Lemma A41.1, put*

$$U = \operatorname{lcm}_j(d_-(\pi_j)/4), \qquad V = \operatorname{lcm}_j(d_+(\pi_j)/4).$$

Then

$$\gcd(U, V) = 1, \qquad \gcd(U, L) = 16, \qquad \gcd(V, L) = 1.$$

For every integer h_* divisible by 64, the congruences

$$h \equiv 0 \pmod U, \qquad h \equiv 1 \pmod V, \qquad h \equiv h_* \pmod L$$

have an arithmetic progression of solutions.

Proof. Division by four makes every minus order have two-adic valuation four and every plus order odd. The cross-sign gcds make their least common multiples coprime. Lemma A41.1 excludes all odd factors of L from both least common multiples. The displayed gcds follow. The only nontrivial pairwise compatibility condition is $h_* \equiv 0 \pmod{16}$, which holds. The generalized Chinese remainder theorem now supplies the progression. □

The reduction orders at the two places of K above p_j are the unordered pair $d_-(\pi_j), d_+(\pi_j)$, by the source lemma `lem:v5-two-reductions`. Both reductions have full rational two-torsion. For a finite group of elliptic-curve points of order d, full two-torsion implies that the subgroup of doubles has order $d/4$. Every double is therefore killed by $d/4$. This justifies the divisions by four in Corollary A41.2; the source point P is itself a double.

Lemma A41.3. *The preceding statements also apply to any fixed finite list of rational contact coefficients, with a common square denominator for the slope and with arbitrarily deep fixed denominator divisibility.*

Proof. For coefficients $\gamma_j \in \mathbb{Q}$, choose a positive integer D_0 so that $D_0^2\gamma_j$ are integers, and remove repeated coefficients. Apply Lemma A41.1 with $c_j = D_0^2\gamma_j$ and put $v = D_0^2 w^2$. Then $u + c_j w^2 = u + \gamma_j v$ and $u/v > D_0^{-2} H_0$. Enlarge the prescribed lower bound before applying the lemma to obtain any desired slope bound. By choosing q divisible by suitable powers of a fixed finite set of primes, one obtains any prescribed lower bounds on their valuations in v. Enlarge Y to exclude any fixed finite list of primes. None of these choices changes the quantified form of Theorem A8.1. □

Lemma A41.4. *Let G be the closure of a cyclic subgroup generated by a point Q in a compact real Lie group, and let G^0 be its identity component. Suppose L is divisible by the order of G/G^0. Any arithmetic progression $h \equiv h_* \pmod{L}$, with an additional compatible finite collection of congruences, is dense after multiplication by Q in the component $h_*Q + G^0$. Arbitrarily large positive indices can be used.*

Proof. The compatible congruences give one coset modulo an integer L' divisible by L. A compact connected abelian Lie group is a torus, and multiplication by any nonzero integer is onto it. It follows that $[L']G = G^0$. Thus the two-sided progression has the stated closure. In a compact group, the closure of a positive cyclic orbit is also a group: positive return times approach the identity and consequently approximate the negative multiples. Removing a finite initial segment does not change the closure. This proves the positive-index assertion. □

For the construction, apply this lemma to the closure of (P, P^σ) in the product of the two real elliptic groups. The scalar point, the four integer root labels, the real compact neighborhoods, all contact coefficients, and the modulus L are fixed before the prime-pattern theorem is used. Corollary A41.2 then supplies the final shared index without disturbing those neighborhoods. No estimate uniform in the coefficients, q, L, or the number of contact positions is required.

A41.2 Pole realization with zero preparation

For $b \in \mathbb{Q}^\times$, put $B = 17b$. The zero-preparation predicate is

$$\begin{aligned} p^2 &= B(1-e)(1+3e), \\ h_0 &= e(B-b^2), \\ Bq^2 &= (s^2 - Bh_0)(s^2 + 3Bh_0), \end{aligned}$$

with rational e, p, q, s; h_0 is the displayed expression. No nonzero condition is imposed on these four witnesses.

Lemma 16.2 states exactly the following two implications. A solution excludes negative odd valuations of b outside $\{2, 3, 17\}$. Conversely a solution exists if the reduced denominator of b is a square and

$$\begin{aligned} &0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \\ &v_2(b) \geq 3, \qquad v_3(b) \geq 1, \qquad v_{17}(b) \geq 2. \end{aligned}$$

These are precisely the hypotheses verified for each of the four normalized pole inputs in the finite-algebra construction.

The special role of zero preparation can also be checked directly in the source argument. With preparation parameter zero, the auxiliary quantity there is

$$B(B-b^2)=B^2(1-b/17).$$

It is positive and is a square in the three specified completions, by the displayed bounds. If μ,θ are the positive squarefree representatives of B and this quantity, their supports are disjoint, $\mu\equiv 17 \pmod{24}$, $\theta\equiv 1 \pmod{24}$, and each is a local square at primes dividing the other. To see the last condition, an odd prime in the support of μ has positive odd valuation on b, so $1-b/17$ is a square unit; a prime in the support of θ outside $2,3,17$ must have $v_p(b)=0$ and $b\equiv 17 \pmod{p}$, making $B\equiv 17^2 \pmod{p}$. The conditions at $2,3,17$ were already imposed. The simultaneous-realization argument in Lemma 16.2 therefore applies without an additional hypothesis. Its deep restricted-converse and rank dependencies are retained in this use.

Lemma A41.5. *For $B\in\mathbb{Q}^\times$, the first conic is covered by the two formulas*

$$e=\frac{\tau^2+(4\varepsilon-1)B}{\tau^2+3B},\qquad p=\frac{-4B\tau(1-\varepsilon)}{\tau^2+3B},\qquad \varepsilon\in\{0,1\},\quad \tau^2+3B\neq 0.$$

Thus its parametrization imposes no additional restriction on the pole-realization theorem.

Proof. Substitution verifies the conic equation. If $e\neq 1$, take $\varepsilon=0$ and $\tau=p/(e-1)$. The conic equation gives $\tau^2+3B=4B/(1-e)\neq 0$ and recovers the displayed point. If $e=1$, then $p=0$, and $\varepsilon=1,\tau=0$ works. The case $e=-1/3,p=0$ is included in the first chart. □

The two structural coordinates attached to different pole inputs are independent. Once all four inputs satisfy the source hypotheses, choose each rational certificate separately and use Lemma A41.5. Zero second-square roots are included by the explicit finite support split in the finite-algebra construction. No common structural witness for distinct pole inputs is asserted or needed.

Role of this baseline. This section proves the intermediate 248-variable construction with its original shifted scalar index. It is not the final variable count. Section A43 subsequently replaces the shifted scalar argument and gives the fixed-input transformations to 238 variables. References below to arithmetic inputs refer to the internal sections just established, not external manuscript assumptions.

A42 The 248-variable arithmetic baseline

A42.1 The family and its arithmetic scope

All variable counts include the distinguished input X. For a fixed integer $N\geq 1$ let $A_N=149^{2N+1}$. The complete tables

`quadratic-family248-v17.json`
`polynomial-family248-v17.json`

specify 222 primitive integer quadratic polynomials and their sum of squares

$$P_N(X,Y_1,\dots,Y_{247})=\sum_{j=1}^{222}Q_{N,j}^2.$$

The parameter A_N is a coefficient, not a variable.

Theorem A42.1. *Using the internally proved arithmetic consequences stated below, there is a fixed integer $N_0 \geq 53$ such that every $N \geq N_0$ satisfies*

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^{247}\ P_N(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

For every $N \geq 1$ the polynomial P_N has degree exactly four, content one, and 3189 nonzero coordinate monomials. For $N \geq 53$,

$$H(P_N) = 149^{4N+2} + 1.$$

The construction specifies every member exactly. The proof does not determine a sufficient numerical value of N_0.

Set

$$K = \mathbb{Q}(\rho), \quad \rho^2 = 2, \quad D = (3-\rho)^2 = 11 - 6\rho, \qquad E : y^2 = x^3 - Dx,$$
$$P = 2(1+\rho, 2\rho) = \left(\frac{57 - 28\rho}{8}, \frac{-660 + 417\rho}{32}\right).$$

We use the following previously proved consequences, with their precise domains. The normalized group and index-height bound are established in the elliptic-model sections; Sections A41 and A40 give the simultaneous finite and prime preparation:

1. $2E(K) = \mathbb{Z}P$ and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n, together with the specified point charts and group domains.

2. The modular-realization lemma and contact-height theorem of Section A36 prove

$$h(u) \leq H M_5(u)^{7/2}, \qquad H = 2^{330789}. \tag{231}$$

   Here $M_5(u)$ is the odd contact radical outside 76176 for $\infty, 0, 1, 2401, -343/32$, and u avoids the four finite contacts. The same theorem also proves the sharper estimate

$$h(u) \leq C_3 M_5(u)^3 (1 + \log M_5(u)), \qquad C_3 = 2^{330770}. \tag{232}$$

   Both bounds are consequences of the modular input, rather than an additional *abc* assumption.

3. The pole obstruction outside $2, 3, 17$, and its simultaneous realization for positive inputs less than 17 with square reduced denominator and the stated local squareclasses and valuation bounds at those three primes. Its normalized quadratic equations are retained.

4. The simultaneous prime-pattern, conjugate reduction-order, finite congruence, and real approximation assertions used by the source. The weighted selection and fixed denominator dilation are justified from the proved counting estimates in Section A40.

The aggregate trace-height theorem used below is the proved consequence of the family height theorem described in Sections A38; its additive constant is ineffective. The contact amplification below is an elementary consequence of (231), not an additional arithmetic assumption. The finite coefficient calculation is separate from these arithmetic arguments.

A42.2 Two contact blocks and the comparison scales

Put

$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad \Theta_2 = \Theta \cup (2 - \Theta), \qquad F_2(t) = \prod_{\theta \in \Theta_2} (t - \theta).$$

The ten contacts are distinct, omit one, and are symmetric about one. Let Σ consist of the primes at most 76176 and the prime 2029147. Outside this set, all contact differences and all numbers $1 - \theta$ are units. Define $M_2^\circ(t)$ as the product of primes $p \notin \Sigma$ for which $v_p(t - \theta)$ is positive and odd for some $\theta \in \Theta_2$. Exact factorizations in the independent certificate verify this exceptional set. In particular

$$F_2(1) = -\frac{82429855236}{7878870169}, \qquad M_2^\circ(1) = 1.$$

Set $E_2 = 2029147$ and $H_2 = \log 7 + H E_2^{7/2}$. Then

$$h(t) \le H_2 M_2^\circ(t)^{7/4} \qquad (t \notin \Theta_2). \tag{233}$$

Indeed, apply (231) to $2 + 1/t$ and $2 + 1/(2 - t)$. The inverse projective matrices have maximum absolute row sums three and seven, respectively, so the inverse height change is at most $\log 7$. If M_+, M_- are the corresponding block radicals with Σ omitted, the original radicals are at most $E_2 M_+$ and $E_2 M_-$. Outside Σ the two radicals are coprime and their product is $M_2^\circ(t)$. Multiplying the two inequalities

$$h(t) - \log 7 \le H E_2^{7/2} M_\pm^{7/2}$$

when their left side is positive proves (233); if it is nonpositive, the conclusion is immediate.

The retained root points are

$$Q_1 = [n]P, \quad Q_2 = [a]P, \quad Q_3 = [t]P, \quad Q_4 = [b]P, \qquad B = \max(|n|, |a|, |t|, |b|),$$

with nonzero integer labels. Set

$$T = \frac{1}{4} \operatorname{Tr}_{K/\mathbb{Q}} \left(\sum_{i=1}^{4} x(Q_i) \right).$$

It is positive, and the aggregate trace theorem gives fixed constants $M_1 > 0$, $M_2 \ge 1$ such that

$$B^2 \le M_1 h(T) + M_2. \tag{234}$$

Use the two positive rational scales $a_* = 1, T$. No square condition on a scale is needed. Replacing the half-trace in the aggregate height theorem by its half changes its logarithmic height by at most $\log 2$; this is absorbed into M_2. For each scale put $t_* = a_* \lambda$ and $\delta = t_* - 1$. On the nonzero domain use the balanced contact and the local weight

$$R = \frac{F_2(t_*)}{\delta^{10}}, \qquad q = \frac{\delta^{10}}{F_2(t_*)} - 1 = R^{-1} - 1, \qquad \gamma = \frac{408}{409}. \tag{235}$$

The contact circuit stores $r = \delta^2$ and its powers r^2, r^3. Its actual quadratic normalization is

$$F_2(t_*)(1 + q) = r^2 r^3. \tag{236}$$

Thus $R^{-1} = 1 + q$ when $\delta \ne 0$, and no additional coordinate for this reciprocal is needed. No contact value can occur: $F_2(t_*) = 0$ would force $r = 0$ and $t_* = 1$, contradicting $F_2(1) \ne 0$. There

are two distinct exceptional strata. If $\delta = 0$, then $q = -1$ and $M_2^\circ(t_*) = 1$; no reciprocal is used. If $q = 0$, then $\delta \neq 0$ and $F_2(t_*) = \delta^{10}$, so again $M_2^\circ(t_*) = 1$. Indeed a consumed contact prime would have odd valuation on the left and valuation zero on the right, since δ is a unit there.

If both radicals divide a nonzero integer J to order k, then $h(T) \le h(\lambda) + h(T\lambda)$ and (233)–(234) imply

$$B^2 \le C|J|^{7/(4k)} + M_2, \qquad C = 2M_1H_2. \tag{237}$$

The two radicals need not be coprime for this implication.

A42.3 The scalar certificate and the point domains

Put $\beta = 51X^2 + 2504$, let $S = Q_2 = [a]P$, and write $C_S = S - P = [a-1]P$. The scalar normalizer is

$$\begin{aligned} \ell &= 149, \qquad A_N = \ell^{2N+1}, \\ r_s &= \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1), \\ R_s &= \frac{A_N r_s}{1+\ell^2 r_s}. \end{aligned} \tag{238}$$

All five factors of r_s are positive, since $\beta \ge 2504$, so $R_s > 0$. The retained square-abscissa chart for S is

$$x_S = h_S^2, \qquad j_S^2 = x_S^2 - D, \qquad S = (x_S, h_S j_S).$$

It has $h_S x_S j_S \neq 0$. The fixed-subtraction coordinate $g_S \in K$ satisfies

$$8(x_P x_S + D)g_S + (39 - 36\rho)h_S + (8 - 14\rho)j_S = 0.$$

The rational component of g_S has a retained inverse witness. At $S = P$ the coordinate g_S is zero, so this guard gives $a \neq 1$. All nonzero points of $\mathbb{Z}P$ are nontorsion. Thus C_S is a nonzero nontorsion point, and the usual fixed-subtraction formulas have no remaining zero denominator. Explicitly, define

$$\begin{aligned} N_S &= (-660 + 417\rho)h_S + (-228 + 112\rho)j_S, \\ D_S &= (-704 + 384\rho)h_S + (660 - 417\rho)j_S + (6888 - 5040\rho)g_S. \end{aligned}$$

Then $D_S \neq 0$ and $z(C_S) = N_S/D_S$, where $z = -x/y$. These identities are obtained from the chord-and-tangent formulas; they are the same fixed-subtraction identities independently checked in the exact function field by the supplied certificate.

The common point chart retains $\chi, f, S_H \in K$ with

$$\chi f = 1, \qquad S_H^2 = \chi - Df, \qquad H = (\chi, \chi S_H). \tag{239}$$

Write $H = [m]R_0 + T_0$, where $P = 2R_0$ and $T_0 \in E[2](K)$, and put $h = m/2$. Before any logarithm is used, the scalar ratio coordinate $Z = Z_0 + \rho Z_1$ is determined by

$$ZD_S = -N_S S_H.$$

When H is nontorsion this is $Z = z(C_S)/z(H)$. Set $F = \mathrm{Nm}_{K/\mathbb{Q}}(f) \neq 0$. The reduced scalar norm equations are

$$A = \frac{Z_0^2 + R_s Z_1^2}{R_s^4}, \qquad W_s = \frac{A^2 + R_s F^2}{R_s^4}, \qquad b_s = R_s(R_s + 9792)W_s^2. \tag{240}$$

The actual quadratic source stores $R_s^2, R_s^4, Z_1^2, F^2, A, W_s$ and a scaled square of W_s; thus these equations are quadratic after clearing. The normalized scalar pole has exactly the ordinary pole input b_s in (240). To verify its scaling literally, let M be the coefficient of the stored W_s-square coordinate in source row491, let K be the coefficient of $e\bar{b}$ in row499, and let L be the coefficient of $\bar{b}p^2$ in row494. The exact coefficient identities are

$$K = 82479248M, \qquad 3L = 17K.$$

The rational substitutions

$$b_s = -K\bar{b}, \quad e' = -e/K, \quad p' = p/K, \quad u' = u/K, \quad v' = -v, \quad w' = w/K$$

convert those rows to the retained normalized pole equations. The independent checker verifies these identities. Since $R_s > 0$ and $F \neq 0$, one has $W_s > 0$ and $b_s > 0$ at every rational source solution.

Lemma A42.2. *Let $p > 76176$ or $p = 149$, and suppose $\nu = v_p(R_s) > 0$ is odd. Then*

$$v_p(F) > 5\nu/4, \qquad v_p(Z_0) > 23\nu/8, \qquad v_p(Z_1) > 19\nu/8.$$

Once the point interpretations hold, these imply

$$v_p(a-1) > 19\nu/8.$$

Proof. The number $9792 = 2^6 \cdot 3^2 \cdot 17$ is a unit at p. The valuation of the nonzero pole input is $\nu + 2v_p(W_s)$, so the pole obstruction makes it positive and $v_p(W_s) > -\nu/2$. The two nonzero terms of a norm $u^2 + R_s v^2$ have opposite valuation parity. A zero term can be omitted. Hence the second norm in (240) gives

$$v_p(A) > 7\nu/4, \qquad v_p(F) > 5\nu/4.$$

The first norm gives the two asserted bounds on Z_0, Z_1.

At some place above p the value $f(H)$ has positive valuation, because the valuation of its norm is positive. Thus H is in the formal group there. The displayed bounds give $v(Z) > 19\nu/8 > 0$, so $v(z(C_S)) > 0$. On $y^2 = x^3 - Dx$, at a good odd prime a point with positive z-valuation is either formal or reduces to $(0,0)$. Indeed $z^2 = x/(x^2 - D)$, and D is a unit. Translation by $(0,0)$ sends z to $-z$ and exchanges these two cases. Choose $C' = C_S + \epsilon(0,0)$ to be formal. The integral identity

$$[2m]C' = [4(a-1)]H$$

kills both two-torsion terms. The formal logarithm is an isometry, so its ratio has valuation $v_p((a-1)/h) = v(Z)$. Since $h = m/2$ is p-integral, $v_p(a-1) \geq v(Z) > 19\nu/8$. □

The integer $\ell = 149$ is prime. For the five contact roots $0, 1, 2401, -343/32, 1/2$, the residues of $(\theta - 2504)/51$ are respectively $59, 97, 109, 108, 78$ modulo 149. Their 74th powers are all 148, so they are nonsquares. Trial division through 12 certifies primality. The preceding lemma applies at 149: the curve has good reduction there, 9792 is a unit, and the pole obstruction excludes only $2, 3, 17$. The generic contact cutoff is unnecessary at this separately verified prime. Thus r_s is an ℓ-adic unit when X is integral at ℓ. If $v_\ell(X) < 0$, then $v_\ell(r_s) = 10v_\ell(X)$. It follows that

$$v_\ell(R_s) = \begin{cases} 2N+1, & v_\ell(X) \geq 0, \\ 2N-1, & v_\ell(X) < 0. \end{cases} \tag{241}$$

This is always positive and odd. These finite facts are included in the independent exact certificate.

Lemma A42.2, using only its norm inequalities at this stage, gives $v_\ell(F) > 0$. Every finite allowed point in

$$E[2](K) \ \cup \ (P + E[2](K)) \ \cup \ (-P + E[2](K))$$

has reciprocal-abscissa norm an ℓ-adic unit. To check this, use the two-torsion translations and

$$\mathrm{Nm}(x_P) = 41^2/64, \quad \mathrm{Nm}(x_P - (3-\rho)) = 17^2/64, \quad \mathrm{Nm}(x_P + (3-\rho)) = 63^2/64.$$

The remaining torsion abscissas have norm seven. Thus none of these points can equal H. In particular $h \notin \{-1, 0, 1\}$, H and $H - P$ are nontorsion, and all required denominators and formal-logarithm interpretations are now justified. This exclusion precedes every use of a comparison logarithm.

Let $c = |a - 1| > 0$. Outside the fixed cutoff, every prime of $M_5(\beta)$ is a positive odd contact of r_s and R_s. The contact at infinity contributes none, since a negative valuation of $\beta = 51X^2 + 2504$ is even there. Lemma A42.2 therefore implies $M_5(\beta)^3 \mid c$. The prime $\ell = 149$ is below the defining cutoff of M_5, so it is automatically absent from $M_5(\beta)$. The forced-prime estimate is consequently independent of these radical factors. We obtain

$$\ell^{L_N} M_5(\beta)^3 \mid c, \qquad L_N = \left\lfloor \frac{19(2N-1)}{8} \right\rfloor + 1. \tag{242}$$

Together with (232), this gives

$$h(\beta) \le C_3 \ell^{-L_N} c \left(1 + \frac{\log c}{3}\right), \qquad c \ge \ell^{L_N}. \tag{243}$$

Indeed $M_5(\beta)^3 \le c/\ell^{L_N}$ and $\log M_5(\beta) \le (\log c)/3$.

A42.4 Plain parameter ratios and the local tests

For $f = 1/x$ and $z = -x/y$, extend their regular values at the identity by zero. The retained point relations are

$$U_0 = H, \quad V_0 = H - P, \quad U_i = [u_i]P, \quad V_i = U_i - Q_i \qquad (1 \le i \le 4).$$

The charts permit $U_i = O$ or $V_i = O$ for $i > 0$. The fixed subtraction for H stores $g, F_0 \in K$ with

$$8(x_P\chi + D)g + (39 - 36\rho) + (8 - 14\rho)S_H = 0, \qquad fF_0 = g^2.$$

Its affine numerator and denominator, after multiplying by the common unit $3 + 2\rho$, are

$$\begin{aligned} N_H &= (-312 - 69\rho) + (-236 - 120\rho)S_H, \\ D'_H &= (-576 - 256\rho) + (312 + 69\rho)S_H + (504 - 1344\rho)g. \end{aligned}$$

On the established domain, $F_0 = f(H - P)$ and $z(H - P) = N_H/D'_H$. The exact function-field certificate checks these identities against the addition law, including the exceptional point $-P+(0,0)$ already excluded above. For each i, retain the primary ratio L_i and an unconjugated alternate ratio $L'_{i,0}$ using

$$L'_{i,0} N_H = z(V_i) D'_H, \qquad L'_i = \sigma L'_{i,0}.$$

These are quadratic equations and mean precisely

$$L_i = \frac{z(U_i)}{z(H)}, \qquad L_i' = \sigma\Big(\frac{z(V_i)}{z(H-P)}\Big).$$

No logarithm polynomial or coordinate for $z(H-P)$ is retained. Put

$$D_H = f(H), \quad A_i = f(U_i), \quad B_i = \sigma f(V_i), \quad C_i = L_i - L_i', \quad I = L_1 - \beta,$$

$$G = L_3^2 - L_2(L_2-2)L_3L_4 + (L_2-2)^2L_4^2 - 1.$$

The required orders are two for D_H, four for C_2, C_3, C_4, G, one for C_1, two for I, and one for the nine A_i, B_i. The final input comparison uses order one.

A42.5 Five factors and extraction at one place

At a contact prime $p \notin \Sigma$, δ is a unit and $\nu = v_p(R) = v_p(F_2(t_*))$ is positive and odd, and $v_p(q) = -\nu$. This follows from the unique contact factor with positive odd valuation and does not require any condition on the valuation of the scale. All valuations of K are unramified and normalized by $v(p) = 1$. For a point on E, every finite value of $v(f)$ is even: if $v(x) < 0$ then $2v(y) = 3v(x)$, while if $v(x) > 0$ then $2v(y) = v(x)$; the remaining case is valuation zero. This also holds for conjugated reciprocal abscissas. Before excluding the exceptional H classes, the stored value for $f(H-P)$ still has even valuation: its literal equation is $f(H)F_0 = g^2$, and $f(H)$ has even valuation. Thus the parity argument and the preceding exclusion do not assume the validity of a division in the fixed subtraction chart.

Use

$$\begin{aligned}
T_1 &= 1 + q^7C_2^2 + q^2D_H^2 + q^4B_1^4,\\
T_2 &= 1 + q^7G^2 + q^2A_3^2 + q^4B_0^4,\\
T_3 &= 1 + q^7I^4 + qA_2^2 + q^2A_4^2 + q^4B_3^4,\\
T_4 &= 1 + q^7C_3^2 + q^2C_1^4 + q^4B_4^4,\\
T_5 &= 1 + q^7C_4^2 + q^2A_1^2 + q^4B_2^4, \qquad W = T_1T_2T_3T_4T_5.
\end{aligned}$$

Lemma A42.3. *At a contact prime, every factor is nonzero and has valuation at most zero. A factor of valuation zero is congruent to one in the residue field.*

Proof. First suppose every displayed test in a factor is nonzero. In each factor other than T_3, the three nonconstant valuations belong, respectively, to the odd integers, to $2 + 4\mathbb{Z}$, and to $4 + 8\mathbb{Z}$. For example T_1 has valuations

$$2v(C_2) - 7\nu, \qquad 2v(D_H) - 2\nu, \qquad 4v(B_1) - 4\nu.$$

The second has exact factor two because $v(D_H)$ is even and ν is odd. The third has exact factor four because $v(B_1)$ is even. The term $q^2C_1^4$ in T_4 has valuation $2(2v(C_1) - \nu)$, again twice an odd integer.

In T_3 the four nonconstant valuations are

$$4v(I) - 7\nu, \quad 2v(A_2) - \nu, \quad 2v(A_4) - 2\nu, \quad 4v(B_3) - 4\nu.$$

The first three have residues $\nu, -\nu, 2$ modulo four. These are pairwise distinct and nonzero because ν is odd and the A_i have even valuations. The fourth has residue four modulo eight; it therefore differs

from each of the first three and from zero. Thus, in every factor, all finite term valuations, including zero for the constant term, are distinct. Zero tests simply remove terms from this argument. The minimum is attained exactly once. This proves nonvanishing and the asserted bound. If the minimum is zero, every nonconstant term has positive valuation, so the residue is one. □

Proposition A42.4. *At each contact prime, the retained pole system and the cleared trace equations imply $\nu > 0$ and give all sixteen required test orders at one place above that prime.*

Proof. Write $W = w_0 + w_1\rho$. The literal cleared trace equations are

$$w_0^2 - 2w_0\tau - 2w_1^2 = 0, \qquad (1+q)\mathfrak{b} = \gamma^2\tau^2.$$

By Lemma A42.3, $W \neq 0$. The first equation therefore gives $w_0 \neq 0$ and $\tau = 1/\operatorname{Tr}(W^{-1}) \neq 0$; indeed $w_0 = 0$ would give $w_1 = 0$. Hence $\mathfrak{b} = \gamma^2 R/\operatorname{Tr}(W^{-1})^2$. The factor γ is a unit at every consumed contact prime. If $\ell = \max_{v|p} v(W) \leq 0$, the trace inequality gives $v_p(\mathfrak{b}) \leq \nu + 2\ell$. This valuation is odd; the retained pole obstruction forces it positive. Consequently $\nu > 0$ and $\ell > -\nu/2$. At a place attaining ℓ, the factor valuations are nonpositive and add to ℓ. Each factor valuation is therefore at least ℓ, and each of its term valuations is strictly greater than $-\nu/2$.

The resulting lower bounds, before integer rounding, are

test	strict lower bound	integer order
C_2, C_3, C_4, G	$13\nu/4$	4
I	$13\nu/8$	2
C_1	$3\nu/8$	1
D_H	$3\nu/4$	2
A_2	$\nu/4$	1
A_1, A_3, A_4	$3\nu/4$	1
B_0, B_1, B_2, B_3, B_4	$7\nu/8$	1

Here ν is a positive odd integer. The order two for D_H uses its even valuation. In fact each of the nine reciprocal-abscissa tests has order at least two; order one suffices for its subsequent formal-group use. Zero tests satisfy all the stated inequalities. □

A42.6 Integrality before the formal congruences

At a contact place selected by Proposition A42.4, the points H, U_i are formal in the primary completion, and $H - P, V_i$ are formal in the conjugate completion. The reciprocal-abscissa tests have positive even valuation, so every nonidentity point has formal depth at least one. In the parameter $z = -x/y$, write

$$\log_E(Q) = z(Q)\mathcal{A}(Q), \qquad \mathcal{A}(Q) \equiv 1 \pmod{p^{4v(z(Q))}}.$$

The value of the unit is taken to be one at the identity. This is the constant-term formal logarithm estimate proved in Sections A39.

The integral group identities

$$[m]U_i = [2u_i]H, \qquad [m-2]V_i = [2(u_i - n_i)](H - P)$$

kill the torsion in $H = [m]R_0 + T_0$. Their full logarithmic ratios are u_i/h and $(u_i - n_i)/(h-1)$, where $(n_1, n_2, n_3, n_4) = (n, a, t, b)$. At least one of $h, h-1$ is a p-adic unit, so at least one full ratio

is integral. The formal units preserve the valuation of its plain parameter ratio. The positive-order test C_i transfers integrality to the other plain ratio, and then to the other full ratio.

Only after this transfer may the relative errors of order four be used as absolute congruences. They give

$$L_i \equiv u_i/h \pmod{p^4}, \qquad L_i' \equiv (u_i - n_i)/(h-1) \pmod{p^4}.$$

Since $n_i = h(u_i/h) - (h-1)((u_i - n_i)/(h-1))$, the high cross tests give $L_i \equiv n_i \pmod{p^4}$ for $i > 1$. The first cross test gives $L_1 \equiv n \pmod{p}$, while I gives $L_1 \equiv \beta \pmod{p}$. These arguments include identity numerator points, half-integral h, and inert primes, where conjugation preserves the unique valuation.

A42.7 Global soundness for sufficiently large parameters

The scalar bounds (243) hold independently of the comparison logarithms. In particular $c \to \infty$ uniformly with N, and $B \geq c - 1$.

Let $\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1$. The local comparison shows that each fourth power of a comparison radical divides Δ. If $\Delta \neq 0$, then $|\Delta| \leq 14B^4$, and (237) gives

$$B^2 \leq C14^{7/16}B^{7/4} + M_2.$$

Since $7/4 < 2$, this bounds B by a fixed constant. Equation (243) gives $B \geq c - 1 \to \infty$ uniformly as $N \to \infty$, a contradiction. Thus $\Delta = 0$ eventually. A stronger form of the integral Lucas descent gives

$$\log B \geq (c-5)\log(c-2) \qquad (c \geq 8). \tag{244}$$

Here is the complete elementary argument. Put $A_0 = |a|$ and $w = \operatorname{sgn}(a)(a-2)b$. Then $t^2 - A_0tw + w^2 = 1$. For $\eta = (A_0 + \sqrt{A_0^2 - 4})/2$, every positive norm-one unit of $\mathbb{Z}[\eta]$ is a power of η. Indeed multiply a unit by a power of η to place it in $[1, \eta)$; its coefficient of η is $(v - v^{-1})/(\eta - \eta^{-1})$, an integer in $[0, 1)$, so it is zero and $v = 1$. Let $U_0 = 0$, $U_1 = 1$, and $U_{j+1} = A_0U_j - U_{j-1}$ for $j \geq 1$. The unit identities show that $|w| = U_j$ and $|t| \in \{U_{j-1}, U_{j+1}\}$ for some $j \geq 1$. Modulo $A_0 - 2$ one has $U_j \equiv j$; modulo $A_0 + 2$ one has $U_j \equiv (-1)^{j-1}j$. Since w is a nonzero multiple of $|a-2|$, it follows that $j \geq A_0 - 2$ if $a > 0$ and $j \geq A_0 + 2$ if $a < 0$. Induction gives $U_j \geq (A_0 - 1)^{j-1}$ for $j \geq 1$. Therefore $|t| \geq (A_0 - 1)^{A_0 - 4}$. Finally $A_0 \geq c - 1$, which proves (244).

If $n - \beta \neq 0$, let J be its nonzero reduced numerator. The local input comparisons make both radicals divide J, while $|J| \leq 2B\exp(h(\beta))$. For $B^2 \geq 2M_2$, (237) gives

$$\log B \leq 4\log(2C) + 7\log 2 + 7h(\beta) \leq C' + 7C_3\ell^{-L_N}c\left(1 + \frac{\log c}{3}\right).$$

Choose N sufficiently large that $7C_3\ell^{-L_N} \leq 1$. The right side is then at most $C' + c + (c\log c)/3$, which is strictly smaller than $(c-5)\log(c-2)$ for all sufficiently large c. Since $c \geq \ell^{L_N} \to \infty$ uniformly with N, this contradicts (244). Hence $n = \beta \in \mathbb{Z}$. If $X = u/v$ is reduced with $v > 0$, then $v^2 \mid 51$, and squarefreeness of 51 gives $v = 1$. All bounds depend only on fixed constants. Since L_N is increasing, a single N_0 works for every larger N. Its dependence on the aggregate additive constant is ineffective.

A42.8 Simultaneous completeness

Fix an integer input. The scalar construction can be made uniformly complete for all sufficiently large N, as follows. First choose a fixed real neighborhood of a nonzero reference point $[h_*]P$, with

$64 \mid h_*$, on which both real values of $|z(H)|$ are bounded below. Choose a fixed neighborhood of the identity for C_S on which its two z values are bounded. Then $|Z_0|, |Z_1|$ are bounded by a constant independent of the input and N. Also $|\operatorname{Nm}(f(H))| \le 1/7$ when $H \in \mathbb{Z}P$ is nonzero, by the positive-square abscissa bounds. For $R_s \ge 1$, equations (240) consequently give

$$W_s = O(R_s^{-3}), \qquad b_s = O(R_s^{-4}),$$

with fixed constants. Since $r_s \ge 1$, $R_s \ge A_N/(1+\ell^2) \to \infty$ uniformly in the input. Hence $0 < b_s < 17$ for all sufficiently large N on these neighborhoods.

The finite preparation uses no new arithmetic theorem. For integer input, r_s is a positive integer and

$$R_s(R_s + 9792) = \frac{A_N r_s(A_N r_s + 9792(1+\ell^2 r_s))}{(1+\ell^2 r_s)^2}.$$

The numerator is coprime to the displayed denominator, so its reduced denominator is square. At the three preparation primes the elementary input congruences give

$$v_2(R_s) \in \{2, 3\}, \qquad v_3(R_s) = 1, \qquad v_{17}(R_s) = 0.$$

In particular $r_s \equiv 15 \pmod{17}$ and $1 + 149^2 r_s \equiv 3 \pmod{17}$, so the normalizer denominator is also a unit at 17. Thus $R_s(R_s + 9792) = R_s^2(1 + 9792/R_s)$ is a local square at all three primes. Choose the shared index h_0 sufficiently divisible at the finite support of this rational number, and then choose the index $c = a - 1$ still more divisible. The points $H = [h_0]P$ and $C_S = [c]P$ are then formal at the required finite places, and all components of $f(H)$ and Z have the needed arbitrarily large valuations. Make W_s integral at every prime where the displayed numerator has odd valuation, and deep enough at $2, 3, 17$ to supply the pole bounds. At every other prime the valuation of b_s is even. These conditions give a square reduced denominator and all pole preparation hypotheses. The progression of Section A41 and the compact-real-group recurrence supplies the chosen real neighborhoods at the same time, together with the retained nonzero chart guards. It also permits c to be arbitrarily large and to satisfy the retained divisibility conditions for the Lucas construction.

After S is fixed, continuity preserves this finite scalar preparation and its guards for all h in one fixed congruence class modulo a sufficiently divisible integer L. Retain the source's integral Lucas construction, choose the four root labels, and fix both positive scales and compact neighborhoods for all comparison tests before selecting contact primes. The additional contact primes avoid the finite scalar preparation support. Hence these later congruences do not conflict with any scalar congruence.

Choose a fixed positive integer D_0 clearing all contact coefficients $D_0^2\theta/a_*$ and sufficiently divisible at the prepared primes, including Σ, and the prime divisors of the scales. In the prime pattern write

$$\lambda = \frac{u}{D_0^2 w^2}.$$

The zero contact makes u one of the selected primes, outside that finite set. Thus $v_p(\lambda) \le -2v_p(D_0)$ there. Make this upper bound low enough that $t_* = a_*\lambda$ has valuation below zero and all nonzero contacts. Then $v_p(\delta) = v_p(t_*)$, and

$$R = \prod_{\theta \in \Theta_2} \left(1 + \frac{1-\theta}{\delta}\right) \longrightarrow 1 \tag{245}$$

in every prepared completion. The weight $q = R^{-1} - 1$ tends to zero. Increasing the fixed denominator divisibility makes its valuation uniformly arbitrarily large. Thus all weighted test terms are small at every prepared place, and $\operatorname{Tr}(W^{-1})$ is arbitrarily close to two.

The factors in (245) are arbitrarily close to one. Requiring congruence to one modulo eight at 2, and modulo p at an odd prepared prime p, makes their product a local square. The pole input is $\gamma^2 R/\operatorname{Tr}(W^{-1})^2$, so it is also a square at $2, 3, 17$. The fixed factor has valuations

$$v_2(\gamma) = 3, \qquad v_3(\gamma) = v_{17}(\gamma) = 1.$$

Since the reciprocal trace is close to $1/2$, the pole input has valuations $4, 2, 2$ at those three primes. These satisfy all retained preparation inequalities.

This preparation is valid for any fixed nonzero rational scales. The prime-pattern input in Sections A40, A41 applies to arbitrary fixed distinct integer coefficients. After clearing $D_0^2\theta/a_*$ and removing repeated coefficients, it applies here without a square restriction on a_*. All denominator conditions are fixed before prime selection.

Select the contact primes with the inherited opposite-order separation. Take $H = [h]P$, $U_i = [hn_i]P$ and $V_i = [(h-1)n_i]P$. The common-index congruences preserve the scalar progression and make H, U_i formal at one place and $H - P, V_i$ formal at its conjugate. Only natural depth one is needed on either branch; the source's stronger primary preparation is also permitted. Both full logarithmic ratios are exactly n_i. Plain-parameter errors therefore have order at least four. All C_i, and I since $n_1 = \beta$, have valuation at least four, and the Lucas equation for the chosen labels gives $v(G) \geq 4$. Every reciprocal-abscissa test has valuation at least two. At the selected place $v_p(q) = -1$, so every nonconstant term of every packed factor has positive valuation. Thus $W \equiv 1$ there. At the opposite place, each factor has valuation at most zero, with residue one when its valuation is zero. The trace of W^{-1} is a unit with residue one or two. The comparison pole input has valuation one at every selected contact prime.

The exact prime-pattern identity is

$$R = \frac{a_*^{10} \prod_{\theta \in \Theta_2} (u - (\theta/a_*) D_0^2 w^2)}{(D_0^2 w^2)^{10} \delta^{10}}.$$

Every factor outside the selected prime numerators is a rational square, including a_*^{10} even when a_* is not itself a square. Thus at every prime outside those numerators the valuation is even. Multiplication by γ^2 and division by the squared trace preserve this parity. This argument includes primes dividing δ; no unit assertion about δ is made away from contacts. Equation (245) and the preparatory divisibility give the required local squareclasses and valuation bounds at $2, 3, 17$.

For the real preparation, write $r = \delta^2$ and let $\alpha_1, \ldots, \alpha_5 > 0$ be the five squared distances of the contact pairs from one. Then

$$R = \prod_{j=1}^{5} (1 - \alpha_j/r), \qquad q \sim \frac{\alpha_1 + \cdots + \alpha_5}{r} > 0 \quad \text{as } r \longrightarrow +\infty.$$

On the fixed compact real neighborhoods the tests remain bounded, so $W \to 1$ and $\operatorname{Tr}(W^{-1}) \to 2$. Consequently

$$\mathfrak{b} \longrightarrow \frac{\gamma^2}{4} = \frac{41616}{167281} \in (0, 17).$$

All factor values are positive at both real embeddings for sufficiently large λ, because $q > 0$ and all test powers are even. The simultaneous selection supplies the required real size after all fixed data have been chosen. The pole-realization input supplies its witnesses, and the exact power identities fill the quadratic circuit. This proves completeness for every integer input and all sufficiently large N.

A42.9 Exact quadratic specification

The local weights are specified by three scalar quadratic equations:

$$q_2 = q^2, \qquad q_4 = q_2^2, \qquad qq_7 = q_4^2.$$

At every contact $q \neq 0$, so $q_7 = q^7$. If $q = 0$, the contact radical is one; its possibly free q_7 value is not used for extraction. No inverse coordinate is required.

The four contact outputs are

$$r = (t_* - 1)^2, \qquad r_2 = r^2, \qquad r_3 = rr_2, \qquad F_2(t_*) = P_5(r) = \prod_{\theta \in \Theta} (r - (\theta - 1)^2).$$

Every integer from zero through five is a sum of two elements of $\{0, 1, 2, 3\}$, so the last expression is quadratic in the retained outputs. Its normalization is $F_2(t_*)(1 + q) = r_2 r_3$. The scalar normalizer is the quadratic equation $R_s + 22201 r_s R_s - A_N r_s = 0$. The scalar chain is $R_2 = R_s^2$, $R_4 = R_2^2$. The two scalar norm rows use R_4, as in (240).

The retained fourth powers of all five B_i are needed by the local factors; none is a logarithm correction. The three separately computed shared fourth powers are C_1^4, I^4, B_1^4, each with two rational coordinates. All other fourth-power coordinates and ordinary point equations are fixed by the literal coefficient tables. The exact count is

point, scalar, contact, ratio, and weight coordinates	210
three shared fourth powers	6
five factors and intermediate products for two detectors	32
total coordinates, including the input	248
point, scalar, contact, ratio, and weight quadratic rows	180
shared fourth-power rows	6
factor and product rows	36
total quadratic rows	222

The complete tables fix every coordinate, equation, coefficient, and normalization; the input is coordinate one. The producer clears denominators and primitive-normalizes each row before squaring. The independent checkers import no producer. The contact certificate recovers P_5 by Newton finite differences at six exact arguments, checks all five roots, and independently verifies $P_5((t - 1)^2) = F_2(t)$ as a degree-ten polynomial identity. It reconstructs every modified quadratic equation, compares the other equations with their pinned coefficient source, and checks the contact exceptional set and both height matrices. It also verifies the scalar pole scaling, the forced prime and its nonsquare tests, and the fourth-power extraction bounds. The balanced normalization, its two exceptional strata, the fixed factor γ, and the linear trace scale are verified directly. The projection proof uses these literal equations and the preceding arithmetic argument. It expands the entire sum of squares independently by unordered pairs of terms. The supplementary local certificate exhausts all odd residues of ν modulo eight and all possible valuation residues of every test. These complete residue classes verify the noncancellation assertion for all integer valuations, together with the exact extraction bounds and sufficient completeness depths. The infinite arithmetic arguments are the proofs above.

The expanded quartic has 3190 terms before powers of A_N belonging to the same coordinate monomial are collected, and 3189 monomials after collection. Its unique coefficient involving A_N^2 is $A_N^2 + 1$, at the square of the actual scalar contact-product coordinate. Every other coefficient is

a nonzero constant or a nonzero constant multiple of A_N. Consequently no support cancellation occurs for positive A_N. The largest constant coefficient C_0 has 1520 bits, the largest linear multiplier is 44402, and exact comparison gives

$$C_0 > 44402^2, \qquad 149^{210} + 1 < C_0 < 149^{214} + 1.$$

Thus $H(P_N) = \max(C_0, A_N^2 + 1)$ for every $N \geq 1$ and the asserted exact height holds for $N \geq 53$. This is a coefficient statement; it does not make the defining threshold effective.

A43 From the baseline to the 238-variable predecessor

A43.1 The arithmetic change: 248 to 245

Use the source field $K = \mathbb{Q}(\sqrt{2})$ and retain its elliptic curve, generator P, and all local conventions. The source scalar comparison uses $S - P$, where $S = [a]P$. Replace it by S itself. In the retained charts

$$S = (h_S^2, h_S j_S), \qquad H = (\chi, \chi S_H), \qquad z(S) = -h_S/j_S, \quad z(H) = -1/S_H,$$

so the new comparison is the two rational components of

$$Z j_S = h_S S_H.$$

The finite root chart excludes O and represents a nonzero point of the retained torsion-free subgroup; consequently $a \neq 0$ and $j_S \neq 0$. This eliminates the two shifted-point coordinates and their inverse guard: original 248-coordinate IDs 201,202,203, with original rows 135,136,177. Original rows 124,125 are replaced by the displayed two-component equation. The other 217 rows are unchanged up to compaction. There are now 245 coordinates and 219 residuals.

This step is not claimed to be a bijection between the old and new witnesses at fixed root tuple. Its soundness uses the unchanged two scalar norm tests. At a consumed odd prime with positive odd valuation ν of the scalar normalizer they give

$$v(F) > 5\nu/4, \qquad v(Z_0) > 23\nu/8, \qquad v(Z_1) > 19\nu/8.$$

The retained forced-149 exclusion is applied before taking logarithms. At a suitable place, H is formal and S is formal after the permitted two-torsion translation. Writing $H = [m]Q + T$ and $P = 2Q$, the relation

$$[2m]S' = [4a]H$$

kills the torsion. The same formal-logarithm isometry gives the required divisibility with the nonzero index $c = |a|$, and the retained height argument gives

$$h(\beta) \leq C_3\, 149^{-L_N} c(1 + \log(c)/3).$$

For sufficiently large c, the source's integral Lucas calculation in $|a|$ supplies the stronger lower bound $(c-4)\log(c-1)$ for $\log B$. Since $c \geq 149^{L_N}$ and $L_N \to \infty$, increasing the single ineffective N_0 ensures this size condition uniformly for every $N \geq N_0$.

For completeness, choose H with the retained finite depths and real bounds, then choose a positive sufficiently divisible a with $[a]P$ near O in the real compact group. Recurrence in the required finite-index subgroup gives this while retaining finite local conditions. The ratio is bounded near O. For every such $a \geq 3$, put $U_0 = 0, U_1 = 1, U_{j+1} = aU_j - U_{j-1}$. Choose positive j divisible by $a-2$, set $b = U_j/(a-2)$ and $t = U_{j+1}$. Then $U_j \equiv j \pmod{a-2}$ and

$$t^2 - a(a-2)tb + (a-2)^2 b^2 = 1.$$

The remaining contact selection is the retained source construction. In particular no condition $a \equiv 1$ is introduced.

A43.2 Root-product charts: 245 to 240

Put $d = 3-\sqrt{2}$ and $D = d^2$. For each of the four source root blocks the old field coordinates satisfy $x = h^2$, $j^2 = x^2 - D$ and

$$hI = jV + DVW + xQ, \qquad hK_1 = -jQ - xV - DQW,$$
$$VQ = xW, \qquad V^2 + Q^2 + 2jW + DW^2 = 1.$$

Here F_U and F_V are the retained reciprocal-abscissa-square tests of the associated points U, V: the source ratio equations give $I^2 = x(1 - DF_U)$ and $K_1^2 = x(1 - DF_V)$. The retained z_V is the plain formal coordinate $-x(V)/y(V)$, with the source zero convention at the identity. The primary comparison L satisfies $VS_H = IL$, so it represents the original primary formal-coordinate ratio wherever the denominator is nonzero, with the same cleared equation at zero. These are existing consumers, not newly imposed nonvanishing assumptions. Replace these eight field coordinates by seven A, B, C, E, i, k, u, imposing

$$A^2 + B^2 + 2CE = 0, \quad AC = d(i-k), \quad BE = d(i+k),$$
$$(A+B)^2 = 4Du, \quad F_U = u^2, \quad i^2 + DF_U = 1, \quad k^2 + DF_V = 1,$$
$$A - B + 2dkz_V = 0, \qquad (A+B)S_H = 2diL.$$

The last two equations replace the old reciprocal and primary ratio equations. Their number is unchanged. Take rational components with the explicit conventions of Section A43.6.

The forward map is

$$A = d(V+Q)/h, \quad B = d(V-Q)/h, \quad C = (x + j + DW)/h,$$
$$E = C - 2h, \quad i = I/h, \quad k = K_1/h, \quad u = (A+B)^2/(4D).$$

Here $h \neq 0$ follows from $j^2 = h^4 - D$ at a real embedding. The inverse is

$$h = (C-E)/2, \quad x = h^2, \quad j = (B^2 + C^2 - A^2 - E^2)/4,$$
$$V = h(A+B)/(2d), \quad Q = h(A-B)/(2d), \quad I = hi, \quad K_1 = hk, \quad W = (A^2 - B^2)/(4D).$$

For verification put $\alpha = (B^2 + C^2)/2$ and $\beta_1 = (A^2 + E^2)/2$. The new equations give $\alpha + \beta_1 = 2h^2$ and $\alpha\beta_1 = D$, using

$$(B^2 + C^2)(A^2 + E^2) = (AB - CE)^2 + (AC + BE)^2.$$

Thus $j^2 = h^4 - D$ and $h \neq 0$. Also $j \neq 0$, since $h^4 = d^2$ would force d to be a square in K, contrary to $\mathrm{Norm}(d) = 7$. Substitution proves both compositions and all retained equations. No division by $A \pm B, i, k, C, E$ is made, so their zero strata survive.

Four blocks save eight rational coordinates. Retaining the two components of j_S and adding one rational trace output $T = \frac{1}{4}\operatorname{Tr}_{K/\mathbb{Q}} \sum h_\ell^2$ gives $245 - 8 + 2 + 1 = 240$, with 214 residuals. The scalar ratio uses $h_S = (C_S - E_S)/2$ and the retained quadratic expression for j_S.

A43.3 Reciprocal scale: 240 to 239

The remaining scale equations are $T = \frac{1}{4}\operatorname{Tr}\sum h_\ell^2$ and $t_1 = \lambda T$. Replace λ by $\rho = 1/\lambda$, delete T, and impose $\rho t_1 = \frac{1}{4}\operatorname{Tr}\sum h_\ell^2$. The right side is positive, so $\rho \neq 0$ without a new guard.

For the exact source contact polynomial F, set $G(\rho) = \rho^{10} F(1/\rho)$. The cleared weight equation becomes

$$G(\rho)(1+q) = (1-\rho)^{10}.$$

Three existing power coordinates now represent ρ^2, ρ^4, ρ^5. Degrees zero through ten have quadratic representatives in $1, \rho, \rho^2, \rho^4, \rho^5$, so both displayed polynomials remain quadratic in the expanded coordinates. The exact coefficients are the pinned 239 IR, not an independently chosen contact polynomial. The old equation excludes $\lambda = 0$ since $F(0) = 0$. Multiplication or division by ρ^{10} proves the inverse maps. Values $\rho = 1$ and $q = 0$ are not discarded. The count is 239 coordinates and 213 residuals.

A43.4 Global scalar rescaling: 239 to 238

This section uses the 239 IR indices, before deleting coordinate 140. Write $R = x_{143}$; coordinates 139 and 140 represent R^2, R^4. For every other index impose $x_i = R^{w_i} y_{\mu(i)}$, where $\mu(i) = i$ for $i < 140$ and $\mu(i) = i - 1$ for $i > 140$. All unspecified weights are zero. The nonzero ordinary-coordinate weights are

$$\begin{aligned} w_i = -1: \quad & i \in [27, 32] \cup [55, 66] \cup [68, 71] \cup [184, 195] \\ & \cup [198, 205] \cup [212, 219] \cup [226, 233], \\ w_i = -2: \quad & i \in \{2, 3, 110, 111, 112, 130, 138, 145\}, \\ w_{113} = -4, \qquad & w_{128} = w_{129} = 1, \qquad w_{135} = 2. \end{aligned}$$

Intervals here include all integer indices. Retain $y_{139} = R^2$ with the explicit equation $y_{139} - y_{142}^2 = 0$. Delete old coordinate 140 and only its defining row 87. This equation must not be replaced by a tautology when powers are collected.

For each remaining row, the row-multiplier table below specifies the integer k for which its substituted old residual equals R^k times the new residual. Powers are represented using $1, R, R^2, R \cdot R^2, (R^2)^2$, subject to total degree two. The complete 212-row output is the specified final IR.

The zero-domain check is independent of the rescaled charts. Old rows 25,72–74,92 remain unchanged. If $z = X^2 \geq 0$, their scalar contact product is

$$\begin{aligned} r = {} & 260105411931107592 + 149961348584209227z \\ & + 11129085312521652z^2 + 328736168366319z^3 \\ & + 4385243879406z^4 + 22081616064z^5 > 0. \end{aligned}$$

The normalizer equation is $(1 + 22201r)R = A_N r$, so $R > 0$. The forward map divides by the indicated powers of R; the inverse multiplies by them and restores R^4. Every division is therefore legitimate on both solution sets. Together with the row identities this proves a fixed-input, fixed-parameter rational witness bijection. The count is 238 coordinates and 212 residuals.

A43.5 Sparse presentation of the 238-variable predecessor

The preceding section describes the raw 238-coordinate presentation. The predecessor presentation is the independently accepted composition in `composition238-007`. Apply its affine coordinate stages to the raw system: the Lucas P/E basis, four constant shifts, five non-Lucas shears, and the four root sum/difference bases. In each root block the old slots become

$$A = (X_1 + Y_1)/2, \quad B = (X_1 - Y_1)/2, \quad C = (J_1 + H_1)/2, \quad E = (J_1 - H_1)/2.$$

Here X_1 denotes a block coordinate, not the distinguished input. Every consumer is transformed, including the scalar and aggregate equations. The intermediate sequence also makes 19 invertible row additions, replaces the two root component rows by their sum and difference, and applies

recorded nonzero constant primitive normalizations. No variable-dependent divisor is introduced. These are applied in the exact order in `EXACT-MAPS.json`, rather than presumed to commute with the preceding R rescaling.

For each coordinate stage, that file states old coordinates as affine polynomials in new ones; unspecified coordinates are fixed. A raw solution is sent to the final system by applying the inverse coordinate stages in forward stage order. Recover the raw solution by applying the forward stages in reverse order. `INVERSE-MAPS.json` records these inverses. Row additions are undone by subtraction, sum/difference by half-sum/half-difference, and primitive normalization by its recorded nonzero rational multiplier. Thus simultaneous vanishing is equivalent in both directions with the input and A_N fixed. All 238 variables occur.

The exact final definitions are the 212 equations in Section A44; the coordinate and row transformations are specified below. The companion files are reproducibility aids only. All 212 explicit quadratic equations are listed in the appendix, in the IR order. Its table specifies every large integer coefficient. Squaring and collecting these final equations yields 2342 coordinate monomials (2343 parameter-separated terms). The sparse composition changes the polynomial's coefficients, but not its rational zero-set projection onto the input. Its independent review is identified by the prefix `07f3b18c`; the complete receipt is included in the delivery evidence.

A43.6 Field components and scalar gate meanings

Here is an explicit interpretation of the raw 239-coordinate dictionary, before the signed R rescaling and the subsequent sparse affine changes. Write its coordinates as x_i and put $\epsilon = \sqrt{2}$. The following formulas specify the component conventions; a suffix in a historical coordinate name is not itself a convention.

$$\begin{aligned}
j_S = x_2 + x_3\epsilon, \qquad \rho = x_4, \qquad X = x_1,\\
x(H) = x_{17} + 2x_{18}\epsilon,\\
f(H) = x_{19} + x_{20}\epsilon,\\
S_H = -1/z_H = x_{21} + x_{22}\epsilon,\\
f(V_0) = x_{23} + x_{24}\epsilon,\\
z_{V_n} = x_{23+2n} + x_{24+2n}\epsilon \quad (1 \le n \le 4),\\
L_n = x_{47+4n} + x_{48+4n}\epsilon,\\
M'_n = x_{49+4n} + x_{50+4n}\epsilon \quad (1 \le n \le 4),\\
Z = x_{128} + x_{129}\epsilon,\\
J_0 = x_{141} + x_{142}\epsilon.
\end{aligned}$$

Here J_0 is the retained reciprocal auxiliary, with $J_0^2 = f(H)f(V_0)$; its two cleared component equations are retained in the printed system. The raw pair M'_n is the unconjugated second ratio; the baseline notation is $L'_n = \overline{M'_n}$. These primes do not denote differentiation. With $D = (3-\epsilon)^2 = 11 - 6\epsilon$, the stored square components are

$$\begin{aligned}
f(V_0)^2 = x_{33} + 2x_{34}\epsilon, \qquad f(V_1)^2 = x_{37} + 2x_{38}\epsilon,\\
f(V_2)^2 = x_{39} + 2x_{40}\epsilon, \qquad f(V_3)^2 = x_{43} + 2x_{44}\epsilon, \qquad f(V_4)^2 = x_{47} + 2x_{48}\epsilon.
\end{aligned}$$

For $(a, b) = (33, 35), (39, 41), (43, 45), (47, 49)$, the relation is

$$D^2(x_a + 2x_{a+1}\epsilon)^2 = 193x_b - 132x_{b+1}\epsilon.$$

This follows by expanding $D^2 = 193 - 132\epsilon$, and records both exceptional multipliers. Likewise, for $n = 1, 2, 3, 4$,

$$f(U_n)^2 = x_{72+2n} + 2x_{73+2n}\epsilon.$$

In particular these are the F_U inputs to the root equations; the F_V inputs are the corresponding four squares of $V_1, \dots, V_4$ just displayed. The root variables A, B, C, E, i, k, u for the nth root block have ordinary components starting at $184 + 14(n-1)$: the successive pairs have offsets $0, 2, 4, 6, 8, 10, 12$. Thus $h = (C - E)/2$, the specified root expression for j, $f(U) = u$, and every root equation are literal equations in $\mathbf{Q}(\epsilon)$ with no further component factors.

The remaining special square conventions are

$$\begin{aligned}
(L_1 - \overline{M_1'})^2 &= x_{82} + 2x_{83}\epsilon, \\
(L_1 - (51X^2 + 2501))^2 &= x_{84} + 2x_{85}\epsilon, \\
f(H)^2 &= x_{86} + 2x_{87}\epsilon, \\
(L_n - \overline{M_n'})^2 &= x_{84+2n} + 2x_{85+2n}\epsilon \quad (2 \le n \le 4).
\end{aligned}$$

The Lucas products have ordinary components $L_2L_4 = x_{68} + x_{69}\epsilon$ and $L_2L_3 = x_{70} + x_{71}\epsilon$. The raw Lucas quantity is $F = G + 1 = x_{72} + 2x_{73}\epsilon$, where G is the baseline Lucas defect, and hence $F^2 = x_{94} + 4x_{95}\epsilon$; the defect square is recovered as $G^2 = F^2 - 2F + 1$. Put $\beta = 51X^2 + 2504$ and $I = L_1 - \beta$. Thus the square at 84 is $(I + 3)^2$, not I^2. The fourth-power pairs at $146, 148$ are ordinary components of $(L_1 - \overline{M_1'})^4$ and I^4, respectively; indeed $I^2 = (I+3)^2 - 6(I+3) + 9$. The pair at 150 is instead the ordinary pair of $\overline{f(V_1)^4}$, as is seen from $x_{150} = x_{37}^2 + 8x_{38}^2$ and $x_{151} = -4x_{37}x_{38}$. All detector-factor and detector-product pairs, and the two actual-W pairs at 103 and 118, use ordinary components. Their defining equations in the final system, transported backwards by the displayed maps, specify their signs. No imaginary multiplier is inferred from their names.

For completeness the scalar singletons that occur in the rescaling argument have the following literal meanings. Put $R = x_{143}$, $s = X^2 = x_{67}$ and $K = (2^2 \cdot 3 \cdot 5 \cdot 11 \cdot 109 \cdot 439)^{14}$. Then

$$\begin{aligned}
x_{125} &= (51s + 2501)^2/4, \\
1525940x_{126} &= 2386189056485 + 97317586470s + 992242485s^2 \\
&\quad + 15531948sx_{125} + 761674548x_{125} - 512x_{125}^2, \\
x_{127} &= -71677797244068 - 2921180304792s - 29762681184s^2 \\
&\quad - 429432240sx_{125} - 155645880sx_{126} \\
&\quad - 21080070840x_{125} - 7640381580x_{126}, \\
x_{139} &= R^2, \qquad x_{140} = R^4, \qquad x_{135} = x_{129}^2, \\
x_{136}R^4 &= x_{128}^2 + Rx_{135}, \\
x_{137} &= x_{86}^2 - 8x_{87}^2 = \mathrm{Norm}(f(H))^2, \\
x_{138}R^4 &= x_{136}^2 + Rx_{137}, \\
Kx_{130} &= x_{138}^2, \\
82479248x_{131} &= -x_{130}(R^2 + 9792R), \\
(22201x_{127} + 1)R &= A_N x_{127}.
\end{aligned}$$

These are polynomial identities even at a zero input; division by R is used only after the positivity argument. The other scalar square auxiliaries are the ordinary squares indicated by their defining rows. Thus the backward reconstruction from the final 212 equations consists of the explicit inverse

sparse affine maps, reinsertion of R^4 and the signed-power inverse map, followed by the component formulas above. It requires neither an external coordinate file nor an assumption that every stored pair has identical scaling.

A43.7 Index dictionary for the rescaling proof

The following raw 239-coordinate dictionary fixes the indices in the weight table. The field interpretation and every exceptional stored-component multiplier are specified in Section A43.6. The final sparse stage subsequently changes these positions by its explicit affine formulas.

Index	Coordinate label
1	`input`
2	`S.chart.j0`
3	`S.chart.j1`
4	`reciprocal.rho`
5	`detector0.normalized.scaled_e`
6	`detector0.normalized.s`
7	`detector0.normalized.p`
8	`detector0.factored.q`
9	`detector1.normalized.scaled_e`
10	`detector1.normalized.s`
11	`detector1.normalized.p`
12	`detector1.factored.q`
13	`normalized_scalar.scaled_e`
14	`normalized_scalar.s`
15	`normalized_scalar.p`
16	`normalized_scalar.factored.q`
17	`H.chart.x.0.quadratic`
18	`H.chart.x.1.quadratic`
19	`U0.f.0`
20	`U0.f.1`
21	`primary.negativeReciprocalZ0.0`
22	`primary.negativeReciprocalZ0.1`
23	`V0.f.0`
24	`V0.f.1`
25	`V1.z.0`
26	`V1.z.1`
27	`V2.z.0`
28	`V2.z.1`
29	`V3.z.0`
30	`V3.z.1`
31	`V4.z.0`
32	`V4.z.1`
33	`logV0.fSquare.0.quadratic`
34	`logV0.fSquare.1.quadratic`
35	`logV0.wPower2.0.quadratic`
36	`logV0.wPower2.1.quadratic`
37	`logV1.fSquare.0.quadratic`

```
logV1.fSquare.1.quadratic
logV2.fSquare.0.quadratic
logV2.fSquare.1.quadratic
logV2.wPower2.0.quadratic
logV2.wPower2.1.quadratic
logV3.fSquare.0.quadratic
logV3.fSquare.1.quadratic
logV3.wPower2.0.quadratic
logV3.wPower2.1.quadratic
logV4.fSquare.0.quadratic
logV4.fSquare.1.quadratic
logV4.wPower2.0.quadratic
logV4.wPower2.1.quadratic
L1.0
L1.1
LP1.0
LP1.1
L2.0
L2.1
LP2.0
LP2.1
L3.0
L3.1
LP3.0
LP3.1
L4.0
L4.1
LP4.0
LP4.1
inputSquare.quadratic
Lucas.C.0.quadratic
Lucas.C.1.quadratic
Lucas.AT.0.quadratic
Lucas.AT.1.quadratic
Lucas.F0.quadratic
Lucas.F1.quadratic
testSquare.1.0.0.quadratic
testSquare.1.0.1.quadratic
testSquare.1.1.0.quadratic
testSquare.1.1.1.quadratic
testSquare.1.2.0.quadratic
testSquare.1.2.1.quadratic
testSquare.1.3.0.quadratic
testSquare.1.3.1.quadratic
testSquare.12.0.0.quadratic
testSquare.12.0.1.quadratic
testSquare.12.1.0.quadratic
testSquare.12.1.1.quadratic
```

```
testSquare.40.0.0.quadratic
testSquare.40.0.1.quadratic
testSquare.40.1.0.quadratic
testSquare.40.1.1.quadratic
testSquare.40.2.0.quadratic
testSquare.40.2.1.quadratic
testSquare.40.3.0.quadratic
testSquare.40.3.1.quadratic
testSquare.40.4.0.quadratic
testSquare.40.4.1.quadratic
reciprocal.rhoSquared
reciprocal.rhoFourth
reciprocal.contactPolynomial
detector0.q1
reflected.detector0.weightPower7
detector0.E1.power2.quadratic
detector0.E1.power4.quadratic
detector0.actualW.0
detector0.actualW.1
detector0.normOver2w0
detector0.B
detector0.normalized.p2.quadratic
detector0.factored.v
detector0.factored.q2
detector1.t.quadratic
symmetric.detector1.deltaSquared
symmetric.detector1.deltaFourthPower
symmetric.detector1.contactPolynomial
detector1.q1
reflected.detector1.weightPower7
detector1.E1.power2.quadratic
detector1.E1.power4.quadratic
detector1.actualW.0
detector1.actualW.1
detector1.normOver2w0
detector1.B
detector1.normalized.p2.quadratic
detector1.factored.v
detector1.factored.q2
scalar.rPair1.quadratic
scalar.rFour.quadratic
scalar.contactProduct
scalar.zRatio.0
scalar.zRatio.1
scalar.W2.quadratic
scalar.b.quadratic
normalized_scalar.p2.quadratic
normalized_scalar.factored.v
```

```
normalized_scalar.factored.q2
scalar.ratio1_unscaled_square
reduced_scalar.ratio_node
reduced_scalar.H_norm_squared
reduced_scalar.root
scalar.r_power_2
scalar.r_power_4
V0.reciprocal.gOverH.0
V0.reciprocal.gOverH.1
scalar.forced_positive_parameter
reciprocal.rhoFifth
reflected.detector1.deltaSixthPower
amplified.C1.power4.0
amplified.C1.power4.1
residue.I.power4.0
residue.I.power4.1
residue.V1.power4.0
residue.V1.power4.1
amplified.det0.factor0.0
amplified.det0.factor0.1
amplified.det0.factor1.0
amplified.det0.factor1.1
amplified.det0.factor2.0
amplified.det0.factor2.1
amplified.det0.factor3.0
amplified.det0.factor3.1
amplified.det0.factor4.0
amplified.det0.factor4.1
amplified.det0.product1.0
amplified.det0.product1.1
amplified.det0.product2.0
amplified.det0.product2.1
amplified.det0.product3.0
amplified.det0.product3.1
amplified.det1.factor0.0
amplified.det1.factor0.1
amplified.det1.factor1.0
amplified.det1.factor1.1
amplified.det1.factor2.0
amplified.det1.factor2.1
amplified.det1.factor3.0
amplified.det1.factor3.1
amplified.det1.factor4.0
amplified.det1.factor4.1
amplified.det1.product1.0
amplified.det1.product1.1
amplified.det1.product2.0
amplified.det1.product2.1
```

```
amplified.det1.product3.0
amplified.det1.product3.1
rootProduct.1.A.0
rootProduct.1.A.1
rootProduct.1.B.0
rootProduct.1.B.1
rootProduct.1.C.0
rootProduct.1.C.1
rootProduct.1.E.0
rootProduct.1.E.1
rootProduct.1.i.0
rootProduct.1.i.1
rootProduct.1.k.0
rootProduct.1.k.1
rootProduct.1.u.0
rootProduct.1.u.1
rootProduct.2.A.0
rootProduct.2.A.1
rootProduct.2.B.0
rootProduct.2.B.1
rootProduct.2.C.0
rootProduct.2.C.1
rootProduct.2.E.0
rootProduct.2.E.1
rootProduct.2.i.0
rootProduct.2.i.1
rootProduct.2.k.0
rootProduct.2.k.1
rootProduct.2.u.0
rootProduct.2.u.1
rootProduct.3.A.0
rootProduct.3.A.1
rootProduct.3.B.0
rootProduct.3.B.1
rootProduct.3.C.0
rootProduct.3.C.1
rootProduct.3.E.0
rootProduct.3.E.1
rootProduct.3.i.0
rootProduct.3.i.1
rootProduct.3.k.0
rootProduct.3.k.1
rootProduct.3.u.0
rootProduct.3.u.1
rootProduct.4.A.0
rootProduct.4.A.1
rootProduct.4.B.0
rootProduct.4.B.1
```

```
rootProduct.4.C.0
rootProduct.4.C.1
rootProduct.4.E.0
rootProduct.4.E.1
rootProduct.4.i.0
rootProduct.4.i.1
rootProduct.4.k.0
rootProduct.4.k.1
rootProduct.4.u.0
rootProduct.4.u.1
```

Row multipliers for global rescaling. The omitted raw row is87. For every other raw row i, the substituted residual is R^{k_i} times new row j. Zero powers are omitted from this table; row order is unchanged apart from deletion87.

i	j	k_i
15	15	-1
16	16	-1
19	19	-1
20	20	-1
23	23	-1
24	24	-1
26	26	-2
27	27	-2
28	28	-2
29	29	-2
30	30	-2
31	31	-2
38	38	-2
39	39	-2
40	40	-2
41	41	-2
42	42	-2
43	43	-2
59	59	-4
60	60	-4
61	61	-4
62	62	-4
75	75	-4
76	76	-1
88	87	2
89	88	2
94	93	-4
137	136	-2
138	137	-2
139	138	-2
140	139	-2

141	140	-2
142	141	-2
143	142	-2
144	143	-2
147	146	-2
148	147	-2
149	148	-2
150	149	-2
151	150	-1
152	151	-1
153	152	-1
154	153	-1
155	154	-2
156	155	-2
157	156	-2
158	157	-2
159	158	-2
160	159	-2
161	160	-2
162	161	-2
169	168	-1
170	169	-1
171	170	-1
172	171	-1
173	172	-2
174	173	-2
175	174	-2
176	175	-2
177	176	-2
178	177	-2
179	178	-2
180	179	-2
187	186	-1
188	187	-1
189	188	-1
190	189	-1
191	190	-2
192	191	-2
193	192	-2
194	193	-2
195	194	-2
196	195	-2
197	196	-2
198	197	-2
205	204	-1
206	205	-1
207	206	-1
208	207	-1

209	208	-2
210	209	-2
211	210	-1
212	211	-1
213	212	-2

A43.8 Explicit ordered sparse transformations

At each coordinate stage z on the left denotes the old coordinates and y on the right the new coordinates; unlisted coordinates are fixed. After the stage rename y to z. Row indices below are one-based. Apply stages in order.

Stage 1. Lucas P/E basis.

$$z_{68} = (2)y_{63} + (1)y_{68}$$
$$z_{69} = (2)y_{64} + (1)y_{69}$$
$$z_{70} = (1)y_{68} + (1)y_{70}$$
$$z_{71} = (1)y_{69} + (1)y_{71}$$

Stage 2. $Q_{96} \leftarrow Q_{96} + (-36)Q_{34}$.

Stage 3. $Q_{11} \leftarrow Q_{11} + (-23/104)Q_{12}$.

Stage 4. $Q_{12} \leftarrow Q_{12} + (-2392/4879)Q_{11}$.

Stage 5. $Q_{15} \leftarrow Q_{15} + (-23/104)Q_{16}$.

Stage 6. $Q_{16} \leftarrow Q_{16} + (-2392/4879)Q_{15}$.

Stage 7. $Q_{19} \leftarrow Q_{19} + (-23/104)Q_{20}$.

Stage 8. $Q_{20} \leftarrow Q_{20} + (-2392/4879)Q_{19}$.

Stage 9. $Q_{23} \leftarrow Q_{23} + (-52/23)Q_{24}$.

Stage 10. $Q_{24} \leftarrow Q_{24} + (2392/4879)Q_{23}$.

Stage 11. $Q_{73} \leftarrow Q_{73} + (-381485)Q_{72}$.

Stage 12. $Q_{89} \leftarrow Q_{89} + (-56/57)Q_{90}$.

Stage 13. $Q_{90} \leftarrow Q_{90} + (-1596/1681)Q_{89}$.

Stage 14. $Q_{97} \leftarrow Q_{97} + (-72)Q_{35}$.

Stage 15. $Q_{9} \leftarrow Q_{9} + (-193/33)Q_{10}$.

Stage 16. $Q_{13} \leftarrow Q_{13} + (-193/33)Q_{14}$.

Stage 17. $Q_{17} \leftarrow Q_{17} + (-193/33)Q_{18}$.

Stage 18. $Q_{21} \leftarrow Q_{21} + (-193/33)Q_{22}$.

Stage 19. $Q_{48} \leftarrow Q_{48} + (3813373161796)Q_{49}$.

Stage 20. $Q_{61} \leftarrow Q_{61} + (3813373161796)Q_{62}$.

Stage 21. four constant shifts.

$$z_{140} = (-1672/18921) + (1)y_{140}$$
$$z_{67} = (-2501/51) + (1)y_{67}$$
$$z_{84} = (-9) + (1)y_{84}$$
$$z_{94} = (-1) + (1)y_{94}$$

Stage 22. non-Lucas shear.

$$z_{35} = (1)y_{35} + (1)y_{36}$$

Stage 23. non-Lucas shear.

$$z_{41} = (1)y_{41} + (1)y_{42}$$

Stage 24. non-Lucas shear.

$$z_{45} = (1)y_{45} + (1)y_{46}$$

Stage 25. non-Lucas shear.

$$z_{49} = (1)y_{49} + (1)y_{50}$$

Stage 26. non-Lucas shear.

$$z_{126} = (1)y_{125} + (1)y_{126}$$

Stage 27. four root sum/difference bases.

$$z_{183} = (1/2)y_{183} + (1/2)y_{185}$$
$$z_{185} = (1/2)y_{183} + (-1/2)y_{185}$$
$$z_{187} = (1/2)y_{187} + (1/2)y_{189}$$
$$z_{189} = (1/2)y_{187} + (-1/2)y_{189}$$
$$z_{184} = (1/2)y_{184} + (1/2)y_{186}$$
$$z_{186} = (1/2)y_{184} + (-1/2)y_{186}$$
$$z_{188} = (1/2)y_{188} + (1/2)y_{190}$$
$$z_{190} = (1/2)y_{188} + (-1/2)y_{190}$$
$$z_{197} = (1/2)y_{197} + (1/2)y_{199}$$
$$z_{199} = (1/2)y_{197} + (-1/2)y_{199}$$
$$z_{201} = (1/2)y_{201} + (1/2)y_{203}$$
$$z_{203} = (1/2)y_{201} + (-1/2)y_{203}$$
$$z_{198} = (1/2)y_{198} + (1/2)y_{200}$$
$$z_{200} = (1/2)y_{198} + (-1/2)y_{200}$$
$$z_{202} = (1/2)y_{202} + (1/2)y_{204}$$
$$z_{204} = (1/2)y_{202} + (-1/2)y_{204}$$
$$z_{211} = (1/2)y_{211} + (1/2)y_{213}$$
$$z_{213} = (1/2)y_{211} + (-1/2)y_{213}$$
$$z_{215} = (1/2)y_{215} + (1/2)y_{217}$$
$$z_{217} = (1/2)y_{215} + (-1/2)y_{217}$$
$$z_{212} = (1/2)y_{212} + (1/2)y_{214}$$
$$z_{214} = (1/2)y_{212} + (-1/2)y_{214}$$
$$z_{216} = (1/2)y_{216} + (1/2)y_{218}$$
$$z_{218} = (1/2)y_{216} + (-1/2)y_{218}$$
$$z_{225} = (1/2)y_{225} + (1/2)y_{227}$$
$$z_{227} = (1/2)y_{225} + (-1/2)y_{227}$$
$$z_{229} = (1/2)y_{229} + (1/2)y_{231}$$
$$z_{231} = (1/2)y_{229} + (-1/2)y_{231}$$
$$z_{226} = (1/2)y_{226} + (1/2)y_{228}$$
$$z_{228} = (1/2)y_{226} + (-1/2)y_{228}$$
$$z_{230} = (1/2)y_{230} + (1/2)y_{232}$$
$$z_{232} = (1/2)y_{230} + (-1/2)y_{232}$$

Stage 28. $(Q_{138}, Q_{140}) \leftarrow (Q_{138} + Q_{140}, Q_{138} - Q_{140})$ simultaneously.

Stage 29. $(Q_{139}, Q_{141}) \leftarrow (Q_{139} + Q_{141}, Q_{139} - Q_{141})$ simultaneously.

Stage 30. $(Q_{156}, Q_{158}) \leftarrow (Q_{156} + Q_{158}, Q_{156} - Q_{158})$ simultaneously.

Stage 31. $(Q_{157}, Q_{159}) \leftarrow (Q_{157} + Q_{159}, Q_{157} - Q_{159})$ simultaneously.

Stage 32. $(Q_{174}, Q_{176}) \leftarrow (Q_{174} + Q_{176}, Q_{174} - Q_{176})$ simultaneously.

Stage 33. $(Q_{175}, Q_{177}) \leftarrow (Q_{175} + Q_{177}, Q_{175} - Q_{177})$ simultaneously.

Stage 34. $(Q_{192}, Q_{194}) \leftarrow (Q_{192} + Q_{194}, Q_{192} - Q_{194})$ simultaneously.

Stage 35. $(Q_{193}, Q_{195}) \leftarrow (Q_{193} + Q_{195}, Q_{193} - Q_{195})$ simultaneously.

Primitive normalization. Finally multiply row i by the following factor f_i; unlisted factors are one. These nonzero constants do not change simultaneous vanishing.

i	f_i
5	358004241
6	18921
9	33
11	5512
12	258587/104
13	33
15	5512
16	258587/104
17	33
19	5512
20	258587/104
21	33
23	1219
24	258587/23
25	51
73	1/4
89	1078497
90	10602067/19
136	2
137	2
138	2
139	2
140	2
141	2
154	2
155	2
156	2
157	2
158	2
159	2
172	2
173	2
174	2

175	2
176	2
177	2
190	2
191	2
192	2
193	2
194	2
195	2
209	2

A43.9 Completion of the 238-variable predecessor

The soundness and completeness proof for the 248-coordinate baseline is established in Section A42. The scalar-S argument in Section A43 changes its nonzero integer index to $|a|$ and preserves both directions. The three subsequent coordinate changes and the displayed affine and row operations are rational witness bijections, with the only nonconstant inverse powers justified by $R > 0$. Their composition therefore proves the defining equivalence for this 238-variable predecessor. The further reductions proving Theorem 1.2 are given in Section A45. The same sufficiently large N_0 works for all larger N: the fixed aggregate bound first excludes the nonzero defect, and $149^{-L_N} \to 0$ then gives the Lucas contradiction uniformly. The explicit equations below specify the resulting family. Their integer square expansion has coefficient one on z_4^4, degree four, and the stated coefficient profile; this last step is finite integer arithmetic and is independently reproducible from the complete displayed system. In particular it does not make N_0 effective.

A44 The explicit 238-variable predecessor

Put $A_N = 149^{2N+1}$. The distinguished input is z_1. Define $Q_1, \dots, Q_{212}$ below and $P_N = \sum_{r=1}^{212} Q_r^2$.

$$\begin{aligned} Q_1 &= z_{17} - 11z_{19} + 12z_{20} \\ &\quad - z_{21}^2 - 2z_{22}^2 \end{aligned}$$

$$\begin{aligned} Q_2 &= 2z_{18} + 6z_{19} - 11z_{20} \\ &\quad - 2z_{21}z_{22} \end{aligned}$$

$$Q_3 = 1 - z_{17}z_{19} - 4z_{18}z_{20}$$

$$Q_4 = z_{17}z_{20} + 2z_{18}z_{19}$$

$$\begin{aligned} Q_5 = {} & -2795584 + 358004241z_{19}z_{23} + 716008482z_{20}z_{24} \\ & + 63271824z_{140} - 358004241z_{140}^2 - 716008482z_{141}^2 \end{aligned}$$

$$\begin{aligned} Q_6 = {} & 18921z_{19}z_{24} + 18921z_{20}z_{23} - 37842z_{140}z_{141} \\ & + 3344z_{141} \end{aligned}$$

$$Q_7 = z_{23}^2 + 2z_{24}^2 - z_{33}$$

$$Q_8 = z_{23}z_{24} - z_{34}$$

$$Q_9 = 2401z_{33}z_{34} - 6369z_{35}$$

$$Q_{10} = 33z_{33}^2 - 193z_{33}z_{34} + 264z_{34}^2 - 33z_{36}$$

$$\begin{aligned} Q_{11} = {} & 775761z_{21}z_{25} + 504136z_{21}z_{53} + 373756z_{21}z_{54} \\ & + 1551522z_{22}z_{26} + 373756z_{22}z_{53} + 1008272z_{22}z_{54} \\ & - 1542912z_{25} + 3027360z_{25}z_{140} - 8022504z_{25}z_{141} \\ & - 8022504z_{26}z_{140} + 6054720z_{26}z_{141} + 775761z_{53} \end{aligned}$$

$$\begin{aligned} Q_{12} = {} & 775761z_{21}z_{26} + 186878z_{21}z_{53} + 504136z_{21}z_{54} \\ & + 775761z_{22}z_{25} + 504136z_{22}z_{53} + 373756z_{22}z_{54} \\ & - 4011252z_{25}z_{140} + 3027360z_{25}z_{141} - 1542912z_{26} \\ & + 3027360z_{26}z_{140} - 8022504z_{26}z_{141} + 775761z_{54} \end{aligned}$$

$$Q_{13} = 2401z_{39}z_{40} - 6369z_{41}$$

$$Q_{14} = 33z_{39}^2 - 193z_{39}z_{40} + 264z_{40}^2 - 33z_{42}$$

$$\begin{aligned} Q_{15} = {} & 775761z_{21}z_{27} + 504136z_{21}z_{57} + 373756z_{21}z_{58} \\ & + 1551522z_{22}z_{28} + 373756z_{22}z_{57} + 1008272z_{22}z_{58} \\ & - 1542912z_{27} + 3027360z_{27}z_{140} - 8022504z_{27}z_{141} \\ & - 8022504z_{28}z_{140} + 6054720z_{28}z_{141} + 775761z_{57} \end{aligned}$$

$$\begin{aligned} Q_{16} = {} & 775761z_{21}z_{28} + 186878z_{21}z_{57} + 504136z_{21}z_{58} \\ & + 775761z_{22}z_{27} + 504136z_{22}z_{57} + 373756z_{22}z_{58} \\ & - 4011252z_{27}z_{140} + 3027360z_{27}z_{141} - 1542912z_{28} \\ & + 3027360z_{28}z_{140} - 8022504z_{28}z_{141} + 775761z_{58} \end{aligned}$$

$$Q_{17} = 2401z_{43}z_{44} - 6369z_{45}$$

$$Q_{18} = 33z_{43}^2 - 193z_{43}z_{44} + 264z_{44}^2 - 33z_{46}$$

$$\begin{aligned} Q_{19} = {} & 775761z_{21}z_{29} + 504136z_{21}z_{61} + 373756z_{21}z_{62} \\ & + 1551522z_{22}z_{30} + 373756z_{22}z_{61} + 1008272z_{22}z_{62} \\ & - 1542912z_{29} + 3027360z_{29}z_{140} - 8022504z_{29}z_{141} \\ & - 8022504z_{30}z_{140} + 6054720z_{30}z_{141} + 775761z_{61} \end{aligned}$$

$$\begin{aligned} Q_{20} = {} & 775761z_{21}z_{30} + 186878z_{21}z_{61} + 504136z_{21}z_{62} \\ & + 775761z_{22}z_{29} + 504136z_{22}z_{61} + 373756z_{22}z_{62} \\ & - 4011252z_{29}z_{140} + 3027360z_{29}z_{141} - 1542912z_{30} \\ & + 3027360z_{30}z_{140} - 8022504z_{30}z_{141} + 775761z_{62} \end{aligned}$$

$$Q_{21} = 2401z_{47}z_{48} - 6369z_{49}$$

$$Q_{22} = 33z_{47}^2 - 193z_{47}z_{48} + 264z_{48}^2 - 33z_{50}$$

$$
\begin{aligned}
Q_{23} = & -775761z_{21}z_{32} - 186878z_{21}z_{65} - 504136z_{21}z_{66} \\
& - 775761z_{22}z_{31} - 504136z_{22}z_{65} - 373756z_{22}z_{66} \\
& + 4011252z_{31}z_{140} - 3027360z_{31}z_{141} + 1542912z_{32} \\
& - 3027360z_{32}z_{140} + 8022504z_{32}z_{141} - 775761z_{66}
\end{aligned}
$$

$$
\begin{aligned}
Q_{24} = & 775761z_{21}z_{31} + 504136z_{21}z_{65} + 373756z_{21}z_{66} \\
& + 1551522z_{22}z_{32} + 373756z_{22}z_{65} + 1008272z_{22}z_{66} \\
& - 1542912z_{31} + 3027360z_{31}z_{140} - 8022504z_{31}z_{141} \\
& - 8022504z_{32}z_{140} + 6054720z_{32}z_{141} + 775761z_{65}
\end{aligned}
$$

$$Q_{25} = 2501 + 51z_1^2 - 51z_{67}$$

$$
\begin{aligned}
Q_{26} = & z_{55}z_{63} + 2z_{56}z_{64} - 2z_{63}z_{142} \\
& - z_{68}z_{142}
\end{aligned}
$$

$$
\begin{aligned}
Q_{27} = & z_{55}z_{64} + z_{56}z_{63} - 2z_{64}z_{142} \\
& - z_{69}z_{142}
\end{aligned}
$$

$$
\begin{aligned}
Q_{28} = & z_{55}z_{59} + 2z_{56}z_{60} - z_{68}z_{142} \\
& - z_{70}z_{142}
\end{aligned}
$$

$$
\begin{aligned}
Q_{29} = & z_{55}z_{60} + z_{56}z_{59} - z_{69}z_{142} \\
& - z_{71}z_{142}
\end{aligned}
$$

$$
\begin{aligned}
Q_{30} = & z_{59}^2 + 2z_{60}^2 - z_{68}z_{70} \\
& - 2z_{69}z_{71} - z_{72}z_{139}
\end{aligned}
$$

$$
\begin{aligned}
Q_{31} = & 2z_{59}z_{60} - z_{68}z_{71} - z_{69}z_{70} \\
& - 2z_{73}z_{139}
\end{aligned}
$$

$$
\begin{aligned}
Q_{32} = & z_{51}^2 - 2z_{51}z_{53} + 2z_{52}^2 \\
& + 4z_{52}z_{54} + z_{53}^2 + 2z_{54}^2 \\
& - z_{82}
\end{aligned}
$$

$$
\begin{aligned}
Q_{33} = & z_{51}z_{52} + z_{51}z_{54} - z_{52}z_{53} \\
& - z_{53}z_{54} - z_{83}
\end{aligned}
$$

$$
\begin{aligned}
Q_{34} = & 9 + z_{51}^2 - 102z_{51}z_{67} \\
& + 2z_{52}^2 + 2601z_{67}^2 - z_{84}
\end{aligned}
$$

$$Q_{35} = z_{51}z_{52} - 51z_{52}z_{67} - z_{85}$$

$$Q_{36} = z_{19}^2 + 2z_{20}^2 - z_{86}$$

$$Q_{37} = z_{19}z_{20} - z_{87}$$

$$
\begin{aligned}
Q_{38} = & z_{55}^2 - 2z_{55}z_{57} + 2z_{56}^2 \\
& + 4z_{56}z_{58} + z_{57}^2 + 2z_{58}^2 \\
& - z_{88}z_{139}
\end{aligned}
$$

$$Q_{39} = z_{55}z_{56} + z_{55}z_{58} - z_{56}z_{57} - z_{57}z_{58} - z_{89}z_{139}$$

$$Q_{40} = z_{59}^2 - 2z_{59}z_{61} + 2z_{60}^2 + 4z_{60}z_{62} + z_{61}^2 + 2z_{62}^2 - z_{90}z_{139}$$

$$Q_{41} = z_{59}z_{60} + z_{59}z_{62} - z_{60}z_{61} - z_{61}z_{62} - z_{91}z_{139}$$

$$Q_{42} = z_{63}^2 - 2z_{63}z_{65} + 2z_{64}^2 + 4z_{64}z_{66} + z_{65}^2 + 2z_{66}^2 - z_{92}z_{139}$$

$$Q_{43} = z_{63}z_{64} + z_{63}z_{66} - z_{64}z_{65} - z_{65}z_{66} - z_{93}z_{139}$$

$$Q_{44} = 1 + z_{72}^2 + 8z_{73}^2 - z_{94}$$

$$Q_{45} = z_{72}z_{73} - z_{95}$$

$$Q_{46} = z_4^2 - z_{96}$$

$$Q_{47} = z_{96}^2 - z_{97}$$

$$\begin{aligned} Q_{48} = {} & 7626746323592 - 76267463235920z_4 - 628059618995128z_4z_{96} \\ & + 732473028362323z_4z_{143} + 307309842082151z_{96} - 583791423311868z_{96}z_{143} \\ & + 718967979842546z_{97} + 162308064100032z_{97}^2 - 38129835686440z_{97}z_{143} \\ & - 7626746323592z_{98} - 3813373161796z_{98}z_{99} - 646146742577308z_{143} \\ & + 3813373161796z_{143}^2 \end{aligned}$$

$$Q_{49} = 1 - 10z_4 - 120z_4z_{96} + 210z_4z_{143} + 45z_{96} - 120z_{96}z_{143} + 210z_{97} + 45z_{97}^2 - 10z_{97}z_{143} - z_{98} - z_{98}z_{99} - 252z_{143} + z_{143}^2$$

$$Q_{50} = z_{99}^2 - z_{101}$$

$$Q_{51} = z_{101}^2 - z_{102}$$

$$Q_{52} = z_{103}^2 - 2z_{103}z_{105} - 2z_{104}^2$$

$$Q_{53} = 167281z_{99}z_{106} - 166464z_{105}^2 + 167281z_{106}$$

$$Q_{54} = z_7^2 - z_{107}$$

$$Q_{55} = 289 + 34z_5 - 3z_5^2 - 17z_{106}z_{107}$$

$$Q_{56} = 17z_5 - z_5z_{106} - 17z_6z_{108}$$

$$Q_{57} = z_8^2 - z_{109}$$

$$Q_{58} = z_6^2 + 34z_6z_{108} - 17z_{106}z_{109} - 867z_{108}^2$$

$$Q_{59} = z_{110}^2 - 2z_{110}z_{139} - z_{111}z_{139} + z_{139}^2$$

$$Q_{60} = z_{111}^2 - z_{112}z_{139}$$

$$Q_{61} = -141822723674536z_{111}z_{139} + 35893742479489z_{111}z_{144} + 192222345736537z_{112}z_{139} - 3813373161796z_{113}z_{114} + 39896049934224z_{139}^2 - 122376041313918z_{139}z_{144}$$

$$Q_{62} = z_{112}z_{144} - z_{113} - z_{113}z_{114}$$

$$Q_{63} = z_{114}^2 - z_{116}$$

$$Q_{64} = z_{116}^2 - z_{117}$$

$$Q_{65} = z_{118}^2 - 2z_{118}z_{120} - 2z_{119}^2$$

$$Q_{66} = 167281z_{114}z_{121} - 166464z_{120}^2 + 167281z_{121}$$

$$Q_{67} = z_{11}^2 - z_{122}$$

$$Q_{68} = 289 + 34z_9 - 3z_9^2 - 17z_{121}z_{122}$$

$$Q_{69} = 17z_9 - z_9z_{121} - 17z_{10}z_{123}$$

$$Q_{70} = z_{12}^2 - z_{124}$$

$$Q_{71} = z_{10}^2 + 34z_{10}z_{123} - 17z_{121}z_{124} - 867z_{123}^2$$

$$Q_{72} = 2601z_{67}^2 - 4z_{125}$$

$$Q_{73} = 3882987z_{67}z_{125} - 128z_{125}^2 - 381485z_{126}$$

$$Q_{74} = 31581660 + 2103220824z_{67} + 29762681184z_{67}^2 + 585078120z_{67}z_{125} + 155645880z_{67}z_{126} + 28680300z_{125} + 7629700z_{126} + z_{127}$$

$$Q_{75} = K_1z_{130}z_{139} - z_{138}^2$$

$$Q_{76} = 9792z_{130} + z_{130}z_{142} + 82479248z_{131}z_{142}$$

$$Q_{77} = z_{15}^2 - z_{132}$$

$$Q_{78} = K_7 - K_4z_{13} - z_{13}^2 + K_3z_{131}z_{132}$$

$$Q_{79} = 17z_{13} + K_2z_{13}z_{131} - 17z_{14}z_{133}$$

$$Q_{80} = z_{16}^2 - z_{134}$$

$$Q_{81} = z_{14}^2 - K_6 z_{14} z_{133} + K_5 z_{131} z_{134} - K_8 z_{133}^2$$

$$Q_{82} = z_{86}^2 - 8 z_{87}^2 - z_{137}$$

$$Q_{83} = z_{136}^2 + z_{137} z_{142} - z_{138} z_{139}$$

$$Q_{84} = z_{99} z_{100} - z_{102}^2$$

$$Q_{85} = z_{114} z_{115} - z_{117}^2$$

$$Q_{86} = z_{139} - z_{142}^2$$

$$Q_{87} = z_{129}^2 - z_{135}$$

$$Q_{88} = z_{128}^2 + z_{135} z_{142} - z_{136} z_{139}$$

$$Q_{89} = 24231 - 2810632 z_{17} + 31806201 z_{17} z_{140} + 127224804 z_{18} z_{141} - 6206088 z_{21} - 21721308 z_{22} + 44048088 z_{140} - 10293024 z_{141}$$

$$Q_{90} = 17709376 - 31806201 z_{17} z_{141} + 5621264 z_{18} - 63612402 z_{18} z_{140} + 10860654 z_{21} + 6206088 z_{22} + 5146512 z_{140} - 44048088 z_{141}$$

$$Q_{91} = 22201 z_{127} z_{142} + z_{142} - A_N z_{127}$$

$$Q_{92} = z_4 z_{97} - z_{143}$$

$$Q_{93} = z_{111} z_{112} - z_{139} z_{144}$$

$$Q_{94} = z_{82}^2 + 8 z_{83}^2 - z_{145}$$

$$Q_{95} = 4 z_{82} z_{83} - z_{146}$$

$$Q_{96} = -324 - 12 z_{51} z_{84} - 48 z_{52} z_{85} + 612 z_{67} z_{84} + 36 z_{84} + z_{84}^2 + 8 z_{85}^2 - z_{147}$$

$$Q_{97} = -24 z_{51} z_{85} - 12 z_{52} z_{84} + 1224 z_{67} z_{85} + 4 z_{84} z_{85} + 72 z_{85} - z_{148}$$

$$Q_{98} = z_{37}^2 + 8 z_{38}^2 - z_{149}$$

$$Q_{99} = 4 z_{37} z_{38} + z_{150}$$

$$Q_{100} = 1 + z_{86} z_{101} + z_{88} z_{100} + z_{102} z_{149} - z_{151}$$

$$Q_{101} = 2 z_{87} z_{101} + 2 z_{89} z_{100} + z_{102} z_{150} - z_{152}$$

$$Q_{102} = 2401 + 37249 z_{35} z_{102} + 2401 z_{36} z_{102} - 4802 z_{72} z_{100} + 2401 z_{78} z_{101} + 2401 z_{94} z_{100} - 2401 z_{153}$$

$$Q_{103} = 25476z_{35}z_{102} + 9604z_{73}z_{100} - 4802z_{79}z_{101} - 9604z_{95}z_{100} + 2401z_{154}$$

$$Q_{104} = 2401 + 37249z_{45}z_{102} + 2401z_{46}z_{102} + 2401z_{76}z_{99} + 2401z_{80}z_{101} + 2401z_{100}z_{147} - 2401z_{155}$$

$$Q_{105} = 25476z_{45}z_{102} - 4802z_{77}z_{99} - 4802z_{81}z_{101} - 2401z_{100}z_{148} + 2401z_{156}$$

$$Q_{106} = 2401 + 37249z_{49}z_{102} + 2401z_{50}z_{102} + 2401z_{90}z_{100} + 2401z_{101}z_{145} - 2401z_{157}$$

$$Q_{107} = 25476z_{49}z_{102} - 4802z_{91}z_{100} - 2401z_{101}z_{146} + 2401z_{158}$$

$$Q_{108} = 2401 + 37249z_{41}z_{102} + 2401z_{42}z_{102} + 2401z_{74}z_{101} + 2401z_{92}z_{100} - 2401z_{159}$$

$$Q_{109} = 25476z_{41}z_{102} - 4802z_{75}z_{101} - 4802z_{93}z_{100} + 2401z_{160}$$

$$Q_{110} = z_{151}z_{153} + 2z_{152}z_{154} - z_{161}$$

$$Q_{111} = z_{151}z_{154} + z_{152}z_{153} - z_{162}$$

$$Q_{112} = z_{155}z_{161} + 2z_{156}z_{162} - z_{163}$$

$$Q_{113} = z_{155}z_{162} + z_{156}z_{161} - z_{164}$$

$$Q_{114} = z_{157}z_{163} + 2z_{158}z_{164} - z_{165}$$

$$Q_{115} = z_{157}z_{164} + z_{158}z_{163} - z_{166}$$

$$Q_{116} = z_{103} - z_{159}z_{165} - 2z_{160}z_{166}$$

$$Q_{117} = z_{104} - z_{159}z_{166} - z_{160}z_{165}$$

$$Q_{118} = 1 + z_{86}z_{116} + z_{88}z_{115} + z_{117}z_{149} - z_{167}$$

$$Q_{119} = 2z_{87}z_{116} + 2z_{89}z_{115} + z_{117}z_{150} - z_{168}$$

$$Q_{120} = 2401 + 37249z_{35}z_{117} + 2401z_{36}z_{117} - 4802z_{72}z_{115} + 2401z_{78}z_{116} + 2401z_{94}z_{115} - 2401z_{169}$$

$$Q_{121} = 25476z_{35}z_{117} + 9604z_{73}z_{115} - 4802z_{79}z_{116} - 9604z_{95}z_{115} + 2401z_{170}$$

$$Q_{122} = 2401 + 37249z_{45}z_{117} + 2401z_{46}z_{117} + 2401z_{76}z_{114} + 2401z_{80}z_{116} + 2401z_{115}z_{147} - 2401z_{171}$$

$$Q_{123} = 25476z_{45}z_{117} - 4802z_{77}z_{114} - 4802z_{81}z_{116} - 2401z_{115}z_{148} + 2401z_{172}$$

$$Q_{124} = 2401 + 37249z_{49}z_{117} + 2401z_{50}z_{117} + 2401z_{90}z_{115} + 2401z_{116}z_{145} - 2401z_{173}$$

$$Q_{125} = 25476z_{49}z_{117} - 4802z_{91}z_{115} - 2401z_{116}z_{146} + 2401z_{174}$$

$$Q_{126} = 2401 + 37249z_{41}z_{117} + 2401z_{42}z_{117} + 2401z_{74}z_{116} + 2401z_{92}z_{115} - 2401z_{175}$$

$$Q_{127} = 25476z_{41}z_{117} - 4802z_{75}z_{116} - 4802z_{93}z_{115} + 2401z_{176}$$

$$Q_{128} = z_{167}z_{169} + 2z_{168}z_{170} - z_{177}$$

$$Q_{129} = z_{167}z_{170} + z_{168}z_{169} - z_{178}$$

$$Q_{130} = z_{171}z_{177} + 2z_{172}z_{178} - z_{179}$$

$$Q_{131} = z_{171}z_{178} + z_{172}z_{177} - z_{180}$$

$$Q_{132} = z_{173}z_{179} + 2z_{174}z_{180} - z_{181}$$

$$Q_{133} = z_{173}z_{180} + z_{174}z_{179} - z_{182}$$

$$Q_{134} = z_{118} - z_{175}z_{181} - 2z_{176}z_{182}$$

$$Q_{135} = z_{119} - z_{175}z_{182} - z_{176}z_{181}$$

$$Q_{136} = z_{183}^2 + 2z_{184}^2 + z_{185}^2 + 2z_{186}^2 + z_{187}^2 + 2z_{188}^2 - z_{189}^2 - 2z_{190}^2$$

$$Q_{137} = z_{183}z_{184} + z_{185}z_{186} + z_{187}z_{188} - z_{189}z_{190}$$

$$Q_{138} = -12z_{142}z_{191} + 8z_{142}z_{192} + z_{183}z_{187} + 2z_{184}z_{188} + z_{185}z_{189} + 2z_{186}z_{190}$$

$$Q_{139} = 4z_{142}z_{191} - 12z_{142}z_{192} + z_{183}z_{188} + z_{184}z_{187} + z_{185}z_{190} + z_{186}z_{189}$$

$$Q_{140} = 12z_{142}z_{193} - 8z_{142}z_{194} + z_{183}z_{189} + 2z_{184}z_{190} + z_{185}z_{187} + 2z_{186}z_{188}$$

$$Q_{141} = -4z_{142}z_{193} + 12z_{142}z_{194} + z_{183}z_{190} + z_{184}z_{189} + z_{185}z_{188} + z_{186}z_{187}$$

$$Q_{142} = -44z_{139}z_{195} + 48z_{139}z_{196} + z_{183}^2 + 2z_{184}^2$$

$$Q_{143} = 12z_{139}z_{195} - 22z_{139}z_{196} + z_{183}z_{184}$$

$$Q_{144} = z_{74} - z_{195}^2 - 2z_{196}^2$$

$$Q_{145} = z_{75} - z_{195}z_{196}$$

$$\begin{aligned} Q_{146} = {} & 11z_{74}z_{139} - 24z_{75}z_{139} - z_{139} \\ & + z_{191}^2 + 2z_{192}^2 \end{aligned}$$

$$Q_{147} = -3z_{74}z_{139} + 11z_{75}z_{139} + z_{191}z_{192}$$

$$\begin{aligned} Q_{148} = {} & 11z_{37}z_{139} - 24z_{38}z_{139} - z_{139} \\ & + z_{193}^2 + 2z_{194}^2 \end{aligned}$$

$$Q_{149} = -3z_{37}z_{139} + 11z_{38}z_{139} + z_{193}z_{194}$$

$$\begin{aligned} Q_{150} = {} & 6z_{25}z_{193} - 4z_{25}z_{194} - 4z_{26}z_{193} \\ & + 12z_{26}z_{194} + z_{185} \end{aligned}$$

$$\begin{aligned} Q_{151} = {} & -2z_{25}z_{193} + 6z_{25}z_{194} + 6z_{26}z_{193} \\ & - 4z_{26}z_{194} + z_{186} \end{aligned}$$

$$\begin{aligned} Q_{152} = {} & z_{21}z_{183} + 2z_{22}z_{184} - 6z_{51}z_{191} \\ & + 4z_{51}z_{192} + 4z_{52}z_{191} - 12z_{52}z_{192} \end{aligned}$$

$$\begin{aligned} Q_{153} = {} & z_{21}z_{184} + z_{22}z_{183} + 2z_{51}z_{191} \\ & - 6z_{51}z_{192} - 6z_{52}z_{191} + 4z_{52}z_{192} \end{aligned}$$

$$\begin{aligned} Q_{154} = {} & z_{197}^2 + 2z_{198}^2 + z_{199}^2 \\ & + 2z_{200}^2 + z_{201}^2 + 2z_{202}^2 \\ & - z_{203}^2 - 2z_{204}^2 \end{aligned}$$

$$\begin{aligned} Q_{155} = {} & z_{197}z_{198} + z_{199}z_{200} + z_{201}z_{202} \\ & - z_{203}z_{204} \end{aligned}$$

$$\begin{aligned} Q_{156} = {} & -12z_{139}z_{205} + 8z_{139}z_{206} + z_{197}z_{201} \\ & + 2z_{198}z_{202} + z_{199}z_{203} + 2z_{200}z_{204} \end{aligned}$$

$$\begin{aligned} Q_{157} = {} & 4z_{139}z_{205} - 12z_{139}z_{206} + z_{197}z_{202} \\ & + z_{198}z_{201} + z_{199}z_{204} + z_{200}z_{203} \end{aligned}$$

$$\begin{aligned} Q_{158} = {} & 12z_{139}z_{207} - 8z_{139}z_{208} + z_{197}z_{203} \\ & + 2z_{198}z_{204} + z_{199}z_{201} + 2z_{200}z_{202} \end{aligned}$$

$$\begin{aligned} Q_{159} = {} & -4z_{139}z_{207} + 12z_{139}z_{208} + z_{197}z_{204} \\ & + z_{198}z_{203} + z_{199}z_{202} + z_{200}z_{201} \end{aligned}$$

$$\begin{aligned} Q_{160} = {} & -44z_{139}z_{209} + 48z_{139}z_{210} + z_{197}^2 \\ & + 2z_{198}^2 \end{aligned}$$

$$Q_{161} = 12z_{139}z_{209} - 22z_{139}z_{210} + z_{197}z_{198}$$

$$Q_{162} = z_{76} - z_{209}^2 - 2z_{210}^2$$

$$Q_{163} = z_{77} - z_{209}z_{210}$$

$$Q_{164} = -1 + 11z_{76} - 24z_{77} + z_{205}^2 + 2z_{206}^2$$

$$Q_{165} = -3z_{76} + 11z_{77} + z_{205}z_{206}$$

$$Q_{166} = -1 + 11z_{39} - 24z_{40} + z_{207}^2 + 2z_{208}^2$$

$$Q_{167} = -3z_{39} + 11z_{40} + z_{207}z_{208}$$

$$Q_{168} = 6z_{27}z_{207} - 4z_{27}z_{208} - 4z_{28}z_{207} + 12z_{28}z_{208} + z_{199}$$

$$Q_{169} = -2z_{27}z_{207} + 6z_{27}z_{208} + 6z_{28}z_{207} - 4z_{28}z_{208} + z_{200}$$

$$Q_{170} = z_{21}z_{197} + 2z_{22}z_{198} - 6z_{55}z_{205} + 4z_{55}z_{206} + 4z_{56}z_{205} - 12z_{56}z_{206}$$

$$Q_{171} = z_{21}z_{198} + z_{22}z_{197} + 2z_{55}z_{205} - 6z_{55}z_{206} - 6z_{56}z_{205} + 4z_{56}z_{206}$$

$$Q_{172} = z_{211}^2 + 2z_{212}^2 + z_{213}^2 + 2z_{214}^2 + z_{215}^2 + 2z_{216}^2 - z_{217}^2 - 2z_{218}^2$$

$$Q_{173} = z_{211}z_{212} + z_{213}z_{214} + z_{215}z_{216} - z_{217}z_{218}$$

$$Q_{174} = -12z_{139}z_{219} + 8z_{139}z_{220} + z_{211}z_{215} + 2z_{212}z_{216} + z_{213}z_{217} + 2z_{214}z_{218}$$

$$Q_{175} = 4z_{139}z_{219} - 12z_{139}z_{220} + z_{211}z_{216} + z_{212}z_{215} + z_{213}z_{218} + z_{214}z_{217}$$

$$Q_{176} = 12z_{139}z_{221} - 8z_{139}z_{222} + z_{211}z_{217} + 2z_{212}z_{218} + z_{213}z_{215} + 2z_{214}z_{216}$$

$$Q_{177} = -4z_{139}z_{221} + 12z_{139}z_{222} + z_{211}z_{218} + z_{212}z_{217} + z_{213}z_{216} + z_{214}z_{215}$$

$$Q_{178} = -44z_{139}z_{223} + 48z_{139}z_{224} + z_{211}^2 + 2z_{212}^2$$

$$Q_{179} = 12z_{139}z_{223} - 22z_{139}z_{224} + z_{211}z_{212}$$

$$Q_{180} = z_{78} - z_{223}^2 - 2z_{224}^2$$

$$Q_{181} = z_{79} - z_{223}z_{224}$$

$$Q_{182} = -1 + 11z_{78} - 24z_{79} + z_{219}^2 + 2z_{220}^2$$

$$Q_{183} = -3z_{78} + 11z_{79} + z_{219}z_{220}$$

$$Q_{184} = -1 + 11z_{43} - 24z_{44} + z_{221}^2 + 2z_{222}^2$$

$$Q_{185} = -3z_{43} + 11z_{44} + z_{221}z_{222}$$

$$Q_{186} = 6z_{29}z_{221} - 4z_{29}z_{222} - 4z_{30}z_{221} + 12z_{30}z_{222} + z_{213}$$

$$Q_{187} = -2z_{29}z_{221} + 6z_{29}z_{222} + 6z_{30}z_{221} - 4z_{30}z_{222} + z_{214}$$

$$Q_{188} = z_{21}z_{211} + 2z_{22}z_{212} - 6z_{59}z_{219} + 4z_{59}z_{220} + 4z_{60}z_{219} - 12z_{60}z_{220}$$

$$Q_{189} = z_{21}z_{212} + z_{22}z_{211} + 2z_{59}z_{219} - 6z_{59}z_{220} - 6z_{60}z_{219} + 4z_{60}z_{220}$$

$$Q_{190} = z_{225}^2 + 2z_{226}^2 + z_{227}^2 + 2z_{228}^2 + z_{229}^2 + 2z_{230}^2 - z_{231}^2 - 2z_{232}^2$$

$$Q_{191} = z_{225}z_{226} + z_{227}z_{228} + z_{229}z_{230} - z_{231}z_{232}$$

$$Q_{192} = -12z_{139}z_{233} + 8z_{139}z_{234} + z_{225}z_{229} + 2z_{226}z_{230} + z_{227}z_{231} + 2z_{228}z_{232}$$

$$Q_{193} = 4z_{139}z_{233} - 12z_{139}z_{234} + z_{225}z_{230} + z_{226}z_{229} + z_{227}z_{232} + z_{228}z_{231}$$

$$Q_{194} = 12z_{139}z_{235} - 8z_{139}z_{236} + z_{225}z_{231} + 2z_{226}z_{232} + z_{227}z_{229} + 2z_{228}z_{230}$$

$$Q_{195} = -4z_{139}z_{235} + 12z_{139}z_{236} + z_{225}z_{232} + z_{226}z_{231} + z_{227}z_{230} + z_{228}z_{229}$$

$$Q_{196} = -44z_{139}z_{237} + 48z_{139}z_{238} + z_{225}^2 + 2z_{226}^2$$

$$Q_{197} = 12z_{139}z_{237} - 22z_{139}z_{238} + z_{225}z_{226}$$

$$Q_{198} = z_{80} - z_{237}^2 - 2z_{238}^2$$

$$Q_{199} = z_{81} - z_{237}z_{238}$$

$$Q_{200} = -1 + 11z_{80} - 24z_{81} + z_{233}^2 + 2z_{234}^2$$

$$Q_{201} = -3z_{80} + 11z_{81} + z_{233}z_{234}$$

$$
\begin{aligned}
Q_{202} = & -1 + 11z_{47} - 24z_{48} \\
& + z_{235}^2 + 2z_{236}^2
\end{aligned}
$$

$$
Q_{203} = -3z_{47} + 11z_{48} + z_{235}z_{236}
$$

$$
\begin{aligned}
Q_{204} = & 6z_{31}z_{235} - 4z_{31}z_{236} - 4z_{32}z_{235} \\
& + 12z_{32}z_{236} + z_{227}
\end{aligned}
$$

$$
\begin{aligned}
Q_{205} = & -2z_{31}z_{235} + 6z_{31}z_{236} + 6z_{32}z_{235} \\
& - 4z_{32}z_{236} + z_{228}
\end{aligned}
$$

$$
\begin{aligned}
Q_{206} = & z_{21}z_{225} + 2z_{22}z_{226} - 6z_{63}z_{233} \\
& + 4z_{63}z_{234} + 4z_{64}z_{233} - 12z_{64}z_{234}
\end{aligned}
$$

$$
\begin{aligned}
Q_{207} = & z_{21}z_{226} + z_{22}z_{225} + 2z_{63}z_{233} \\
& - 6z_{63}z_{234} - 6z_{64}z_{233} + 4z_{64}z_{234}
\end{aligned}
$$

$$
\begin{aligned}
Q_{208} = & 4z_2 + z_{197}z_{199} + 2z_{198}z_{200} \\
& - z_{201}z_{203} - 2z_{202}z_{204}
\end{aligned}
$$

$$
\begin{aligned}
Q_{209} = & 4z_3 + z_{197}z_{200} + z_{198}z_{199} \\
& - z_{201}z_{204} - z_{202}z_{203}
\end{aligned}
$$

$$
\begin{aligned}
Q_{210} = & 2z_2z_{128} + 4z_3z_{129} - z_{21}z_{203} \\
& - 2z_{22}z_{204}
\end{aligned}
$$

$$
\begin{aligned}
Q_{211} = & 2z_2z_{129} + 2z_3z_{128} - z_{21}z_{204} \\
& - z_{22}z_{203}
\end{aligned}
$$

$$
\begin{aligned}
Q_{212} = & 8z_4z_{110} - z_{189}^2 - 2z_{190}^2 \\
& - z_{203}^2 - 2z_{204}^2 - z_{217}^2 \\
& - 2z_{218}^2 - z_{231}^2 - 2z_{232}^2
\end{aligned}
$$

Large integer coefficients

The decimal digits on consecutive lines in each row are concatenated. Thus every K_j below denotes one fixed positive integer; it is not a variable or a parameter.

K_1 98194994952101164627950993228801351760409466991314003358996422433259542652766869107036815360000000000000

K_2 80990493410131000784375977023646274345820490095244015289194989569855772668241364832628280392076492800000000000000

K_3 45894612932407567111146386980066222129298277206382753305438274229182711786701067384893588888433459200000000000000

K_4 91789225864815134222292773960132444258596555441276550661087654845836542357340213476978717777686691840000000000000000

K_5 13768383879722270133343916094019866638789483316191482599163148226875481353601032021546807666653003776000000000000000000

K_6 27536767759444540266687832188039733277578966632382965198326296453750962707202064043093615333306007552000000000000000000

K_7 63189464886465357187317595452292806620184449792472930176268493164014380154729615417241036648948308361460984548118588982686734559746745916595796472604244988128011149397522493691010836323796918782341939200000000000000000000000000000

K_8 56870518397818821468585835070635259581660048132256371586416438476129421392566538755169329840534775253148860933067300844180611037720713249362168253438204893152100344577702443219097526914172269041077452800000000000000000000000000000000

A45 The 233-variable family: arithmetic and exact reductions

The 238-variable system above is retained as a predecessor, with its own coordinates and polynomial. This section constructs the final family of Theorem 1.2. Throughout this section the coefficient is $A_N = 149^{2N+1}$, with $N \in \mathbb{N}$. Statements about the defining property are for all sufficiently large N; statements about coefficient height for $N \geq 53$ have a different, purely finite meaning.

We first work before the optional sparse presentation of the 238-variable system. The preceding coordinate maps recover that raw presentation without changing the input. The first step below changes the scalar criterion and is proved arithmetically in both directions. The next two steps are rational witness bijections on the admitted family. Finally, invertible affine coordinate maps and constant row operations give the 207 displayed equations in Section A46.

A45.1 A three-residue scalar criterion: 238 to 237

Use the raw 238-coordinate indices of the global-rescaling construction. Write $R = x_{142} > 0$, $R_2 = x_{139} = R^2$, $u = x_{128}$, $v = x_{129}$, $D_1 = x_{135} = v^2$, and $w = x_{138}$. The physical formal-coordinate ratio is $Z = R(u + \sqrt{2}v) = Z_0 + \sqrt{2}Z_1$, and the physical scalar pole variable is $W = w/R^2$. Put $f = f(H)$ and $F = \mathrm{N}_{K/\mathbb{Q}}(f)$, where $K = \mathbb{Q}(\sqrt{2})$. The retained coordinates satisfy

$$f^2 = x_{86} + 2\sqrt{2}x_{87}, \qquad F^2 = x_{86}^2 - 8x_{87}^2.$$

Delete coordinate 137 and its defining row 82. Repurpose coordinate 136 as C. Replace raw row 88 and raw row 83, respectively, by

$$u^2 - RC = 0, \qquad C^2 + RD_1 + x_{86}^2 - 8x_{87}^2 - R_2 w = 0.$$

All other rows are retained. Compact the indices above the deleted coordinate by one; compact the row indices after row 82 by one. Thus there are 237 coordinates and 211 quadratic residuals. These instructions refer to the raw predecessor, not its affine sparse basis. In physical variables the new equations say

$$W = \frac{Z_0^4}{R^{10}} + \frac{Z_1^2}{R^5} + \frac{F^2}{R^4}, \qquad b = R(R + 9792)W^2. \tag{246}$$

The definitions of the positive normalizer, the pole certificate, and all non-scalar interfaces remain unchanged.

Lemma A45.1. *At each consumed scalar contact prime, with $\nu = v_p(R) > 0$ odd, the new criterion implies*

$$v_p(Z_0) > 19\nu/8, \qquad v_p(Z_1) > 9\nu/4, \qquad v_p(F) > 7\nu/4.$$

The convention $v_p(0) = +\infty$ is allowed for either component of Z.

Proof. The primes in question are 149 or primes greater than 76176, so 9792 is a unit and the curve has good unramified reduction. First, $W > 0$: $R > 0$ and $F \neq 0$, so the last summand in (246) is strictly positive. Thus $b \neq 0$. The unchanged pole obstruction rules out a negative odd valuation of b. Since $v_p(b) = \nu + 2v_p(W)$ is odd, it follows that $v_p(W) > -\nu/2$.

For the actual point on $y^2 = x^3 - Dx$, with $D = 11 - 6\sqrt{2}$, the finite valuation of x, hence that of $f = 1/x$, is even: if $v(x) > 0$, then $2v(y) = v(x)$; if $v(x) < 0$, then $2v(y) = 3v(x)$. If $v(x) = 0$ the conclusion is immediate. Consequently $v_p(F)$ is even, both at split primes (sum of the two even valuations) and at inert primes (residue degree two). The three summands of R^4W have valuations

$$4v_p(Z_0) - 6\nu \in 2 + 4\mathbb{Z}, \quad 2v_p(Z_1) - \nu \in 1 + 2\mathbb{Z}, \quad 2v_p(F) \in 4\mathbb{Z}.$$

Their finite valuations are distinct, so their minimum cannot cancel. Omit a zero summand. The last summand is nonzero. Since $v_p(R^4W) > 7\nu/2$, each finite valuation exceeds $7\nu/2$. Solving the three inequalities gives the assertion. □

The retained formal-group argument now applies with the same physical ratio Z. At a suitable place, H is formal, and the allowed two-torsion translation makes the comparison point formal. With $H = [m]Q + T$, $P = 2Q$, and comparison index $a \neq 0$, the identity $[2m]S' = [4a]H$ and the formal-logarithm isometry give $v_p(a) > 9\nu/4$. This uses no extra divisibility hypothesis on m. In particular every selected contact contributes exponent at least three. The forced normalizer still has $v_{149}(R) = 2N + 1$ or $2N - 1$. The bound on F excludes the same finite exceptional reference points before their ratios are interpreted. Writing $c_0 = |a|$ and

$$L'_N = \lfloor 9(2N - 1)/4 \rfloor + 1,$$

the unchanged contact and aggregate argument gives, for sufficiently large positive N,

$$149^{L'_N} M_5(\beta)^3 \mid c_0, \qquad h(\beta) \leq C_3\, 149^{-L'_N} c_0(1 + \log(c_0)/3).$$

Here M_5, β, and C_3 retain their meanings in the preceding height proof. All five-factor tests and both aggregate contacts are unchanged. The first rationality and Lucas-defect arguments therefore still apply. In the last contradiction replace L_N by L'_N: the upper bound

$$\log B \leq C' + 7C_3\, 149^{-L'_N} c_0(1 + \log(c_0)/3)$$

is incompatible with $\log B \geq (c_0 - 4)\log(c_0 - 1)$ for all sufficiently large N, uniformly in the input, because $c_0 \geq 149^{L'_N}$ and $L'_N \to \infty$. This proves soundness with one, still ineffective, threshold.

For completeness, retain the compact real neighborhoods for the reference and comparison points used in the scalar-S construction. On these neighborhoods Z_0, Z_1, F are uniformly bounded. The unchanged normalizer has $R \geq A_N/(1 + 149^2)$, so (246) gives $W = O(R^{-4})$ and $b = O(R^{-6})$. In particular $0 < b < 17$ for sufficiently large N, uniformly in the integer input. Its squareclass is still $R(R + 9792)$; its reduced denominator is a square. At the finite support of this squareclass, first choose H sufficiently formal and then the comparison index sufficiently divisible. This makes F and both components of Z arbitrarily deep. Thus the three terms of (246) can be made integral at every odd-valuation numerator prime and sufficiently deep at $2, 3, 17$, despite the fixed powers of R in their denominators. At all other primes b has even valuation. These are precisely the retained pole-realization hypotheses.

Compact-group recurrence meets these finite conditions and the real neighborhoods simultaneously. After fixing the comparison point, continuity preserves the conditions for H in a fixed progression. The later contact selection avoids the finite support already fixed; its ordering, nonzero Lucas labels, opposite-order conditions, and four pairs of root points are unchanged. Finally set $C = u^2/R$, $D_1 = v^2$, and $w = R^2W$, and fill the existing scaled pole witnesses. These are rational and satisfy the changed rows. The positive-normalizer proof also holds when the distinguished input is zero. This proves completeness for all integers with the same type of uniform, ineffective threshold. It does not assert a bijection of the old and new scalar witnesses.

A45.2 Reference identities and removal of two inverse coordinates

Now use raw 237-coordinate indices, and put

$$f = x_{19} + \sqrt{2}x_{20}, \quad S = x_{21} + \sqrt{2}x_{22}, \quad v = x_{23} + \sqrt{2}x_{24}, \quad g = x_{139} + \sqrt{2}x_{140}, \quad \chi = x_{17} + 2\sqrt{2}x_{18}.$$

Set

$$a = 57 - 28\sqrt{2}, \quad b_0 = 8 - 14\sqrt{2}, \quad c = 88 - 48\sqrt{2}, \quad d = 39 - 36\sqrt{2}, \quad A = a + cf.$$

The notation b_0 distinguishes this fixed field constant from the physical pole input b above; A is not the coefficient A_N. The old reference rows are

$$\chi f = 1, \qquad S^2 = \chi - Df, \qquad G := g^2 - fv = 0, \qquad (a\chi + c)g + b_0 S + d = 0.$$

Replace the inverse and reference rows by the two field equations

$$\mathcal{F} = Ag + b_0 fS + df = 0, \tag{247}$$

$$\mathcal{E} = (b_0 S - d)g - af + 8 = 0. \tag{248}$$

Retain $G = 0$. Delete coordinates 17 and 18, and raw rows 1–4 and 88–89. Insert the four rational component rows of (247)–(248); every other row is unchanged under $i \mapsto i$ for $i < 17$ and $i \mapsto i - 2$ for $i > 18$. The result has 235 coordinates and 209 rows.

Direct expansion in $\mathbb{Q}(\sqrt{2})$ gives

$$A(af - 8) = b_0^2 Df^2 + d^2 f - b_0^2, \tag{249}$$

$$\mathcal{F}(b_0 S - d) - A\mathcal{E} = b_0^2(fS^2 - 1 + Df^2). \tag{250}$$

For a new solution, the second identity proves the old curve equation and also proves $f \neq 0$, since its right-hand factor at $f = 0$ is -1. Define $\chi = 1/f$. Division of (247) by f recovers the old reference equation, and the recovered curve equation is $S^2 = \chi - Df$. This direction requires no division by A, S, g, or v.

For the forward direction, an old solution gives $\mathcal{F} = 0$ and $A\mathcal{E} = 0$. We must exclude $A = 0$, rather than discard that stratum without proof. Its only possible value is

$$f_* = -a/c = -291/392 - (17/196)\sqrt{2}, \qquad \mathrm{N}(f_*) = (41/56)^2.$$

This norm is a 149-adic unit. The scalar criterion of Section A45.1, at the forced positive odd contact, gives $v_{149}(\mathrm{N}\, f) > 7\nu/4 > 0$. Therefore $f \neq f_*$, and $\mathcal{E} = 0$. This proves both directions on the admitted A_N family in the range used for the defining theorem.

Writing $f = f_0 + \sqrt{2} f_1$, the inverse coordinate map is explicitly

$$x_{17} = \frac{f_0}{f_0^2 - 2f_1^2}, \qquad x_{18} = \frac{-f_1}{2(f_0^2 - 2f_1^2)}.$$

The denominator is nonzero because $f \neq 0$ in the field. Either rational component may vanish. Every retained coordinate is transported by the inverse index map. Thus these maps are inverse on the full solution sets under consideration, not just on prepared completeness witnesses.

A45.3 Keeping the square, eliminating the reference value: 235 to 233

In raw 235 coordinates, the retained fields are

$$f = x_{17} + \sqrt{2}x_{18}, \quad S = x_{19} + \sqrt{2}x_{20}, \quad v = x_{21} + \sqrt{2}x_{22}, \quad g = x_{137} + \sqrt{2}x_{138},$$

with $B = x_{84} + 2\sqrt{2}x_{85} = f^2$ and $w = x_{31} + 2\sqrt{2}x_{32} = v^2$. All occurrences of v are in the two component rows of $G = 0$ and the two component rows defining $w = v^2$. Put

$$T = 8 - af - 2dg.$$

Delete coordinates 21 and 22 and rows 5–8, and replace the four rows by the two rational components of

$$(a^2 + 2acf + c^2B)w - T^2 = 0. \tag{251}$$

This is quadratic in the retained coordinates: B is an existing coordinate pair, not an instruction to substitute a fresh square into the displayed residual. All 205 other rows are retained. Put the two new rows first; the old rows 1–4 follow, then the old rows 9–209. Compact coordinate indices above 22 by two. This specifies the corrected raw 233-coordinate, 207-row presentation.

Here is a two-sided verification, including the nonvanishing needed for the inverse. Define

$$Q = Av - T, \qquad R_1 = Av + b_0Sg + dg.$$

Then $R_1 = Q + \mathcal{E}$, and direct multiplication gives

$$\mathcal{F}g - AG = fR_1. \tag{252}$$

On an old solution, $f \neq 0$, $\mathcal{F} = \mathcal{E} = G = 0$, so $R_1 = Q = 0$. Together with $w = v^2$, this proves (251). Conversely, the retained $\mathcal{F} = \mathcal{E} = 0$ first recovers $fS^2 - 1 + Df^2 = 0$ by (250), hence $f \neq 0$. The unchanged scalar argument at 149 can now be applied to that actual reference point, before interpreting v or w. It excludes $A = 0$ by the same norm calculation as above. Set $v = T/A$. Since $B = f^2$, equation (251) gives $w = v^2$. Now $Q = 0$, hence $R_1 = 0$, and (252) gives $G = 0$. All deleted rows are restored.

For explicit rational formulas in the corrected raw 233 coordinates, write $f = f_0 + \sqrt{2}f_1$, $g = g_0 + \sqrt{2}g_1$, where $(f_0, f_1) = (x_{17}, x_{18})$ and $(g_0, g_1) = (x_{135}, x_{136})$. Then

$$\begin{aligned} A_0 &= 57 + 88f_0 - 96f_1, & A_1 &= -28 + 88f_1 - 48f_0, \\ T_0 &= 8 - 57f_0 + 56f_1 - 78g_0 + 144g_1, & T_1 &= 28f_0 - 57f_1 + 72g_0 - 78g_1, \\ v_0 &= (T_0A_0 - 2T_1A_1)/(A_0^2 - 2A_1^2), & v_1 &= (T_1A_0 - T_0A_1)/(A_0^2 - 2A_1^2). \end{aligned}$$

Neither component of A is assumed nonzero separately. Only its field norm is inverted. No division by T, g, S, w, or a component of f occurs. In particular the maps cover their zero strata. The input coordinate and A_N are fixed. The two inverse removals introduce no new arithmetic assumption and do not make the threshold effective.

A45.4 Exact sparse presentation and its inverse

All indices in this subsection refer to the corrected raw 233 system. The input is coordinate 1 and is fixed by every stage. A coordinate stage means substitute the displayed old coordinates as affine expressions in new coordinates, simultaneously within that stage; all unspecified coordinates are fixed. Stages are applied in the order printed, without commuting coordinate changes past row operations. A row operation acts on the current list of 207 residuals; its indices below are one-based. These operations preserve simultaneous vanishing over the rationals.

Stage 1. Lucas P/E basis.

$$x_{64} = 2y_{59} + y_{64}.$$

$$x_{65} = 2y_{60} + y_{65}.$$

$$x_{66} = y_{64} + y_{66}.$$

$$x_{67} = y_{65} + y_{67}.$$

Stage 2.

$$R_{91} \longleftarrow R_{91} + (-36)R_{32}.$$

Stage 3.

$$R_9 \longleftarrow R_9 + (\frac{-23}{104})R_{10}.$$

Stage 4.

$$R_{10} \longleftarrow R_{10} + (\frac{-2392}{4879})R_9.$$

Stage 5.

$$R_{13} \longleftarrow R_{13} + (\frac{-23}{104})R_{14}.$$

Stage 6.

$$R_{14} \longleftarrow R_{14} + (\frac{-2392}{4879})R_{13}.$$

Stage 7.

$$R_{17} \longleftarrow R_{17} + (\frac{-23}{104})R_{18}.$$

Stage 8.

$$R_{18} \longleftarrow R_{18} + (\frac{-2392}{4879})R_{17}.$$

Stage 9.

$$R_{21} \longleftarrow R_{21} + (\frac{-52}{23})R_{22}.$$

Stage 10.

$$R_{22} \longleftarrow R_{22} + (\frac{2392}{4879})R_{21}.$$

Stage 11.

$$R_{71} \longleftarrow R_{71} + (-381485)R_{70}.$$

Stage 12.

$$R_{92} \longleftarrow R_{92} + (-72)R_{33}.$$

Stage 13.

$$R_7 \longleftarrow R_7 + (\frac{-193}{33})R_8.$$

Stage 14.

$$R_{11} \longleftarrow R_{11} + (\frac{-193}{33})R_{12}.$$

Stage 15.

$$R_{15} \longleftarrow R_{15} + (\frac{-193}{33})R_{16}.$$

Stage 16.

$$R_{19} \longleftarrow R_{19} + (\frac{-193}{33})R_{20}.$$

Stage 17.

$$R_{46} \longleftarrow R_{46} + (3813373161796)R_{47}.$$

Stage 18.

$$R_{59} \longleftarrow R_{59} + (3813373161796)R_{60}.$$

Stage 19. four constant shifts.

$$x_{135} = -\frac{1672}{18921} + y_{135}.$$
$$x_{63} = -\frac{2501}{51} + y_{63}.$$
$$x_{80} = -9 + y_{80}.$$
$$x_{90} = -1 + y_{90}.$$

Stage 20. non-Lucas shear.

$$x_{31} = y_{31} + y_{32}.$$

Stage 21. non-Lucas shear.

$$x_{37} = y_{37} + y_{38}.$$

Stage 22. non-Lucas shear.

$$x_{41} = y_{41} + y_{42}.$$

Stage 23. non-Lucas shear.

$$x_{45} = y_{45} + y_{46}.$$

Stage 24. non-Lucas shear.

$$x_{122} = y_{121} + y_{122}.$$

Stage 25. four root sum/difference bases.

$$x_{178} = \frac{1}{2}y_{178} + \frac{1}{2}y_{180}.$$
$$x_{180} = \frac{1}{2}y_{178} - \frac{1}{2}y_{180}.$$
$$x_{182} = \frac{1}{2}y_{182} + \frac{1}{2}y_{184}.$$
$$x_{184} = \frac{1}{2}y_{182} - \frac{1}{2}y_{184}.$$
$$x_{179} = \frac{1}{2}y_{179} + \frac{1}{2}y_{181}.$$
$$x_{181} = \frac{1}{2}y_{179} - \frac{1}{2}y_{181}.$$

$$x_{183} = \frac{1}{2}y_{183} + \frac{1}{2}y_{185}.$$
$$x_{185} = \frac{1}{2}y_{183} - \frac{1}{2}y_{185}.$$
$$x_{192} = \frac{1}{2}y_{192} + \frac{1}{2}y_{194}.$$
$$x_{194} = \frac{1}{2}y_{192} - \frac{1}{2}y_{194}.$$
$$x_{196} = \frac{1}{2}y_{196} + \frac{1}{2}y_{198}.$$
$$x_{198} = \frac{1}{2}y_{196} - \frac{1}{2}y_{198}.$$
$$x_{193} = \frac{1}{2}y_{193} + \frac{1}{2}y_{195}.$$
$$x_{195} = \frac{1}{2}y_{193} - \frac{1}{2}y_{195}.$$
$$x_{197} = \frac{1}{2}y_{197} + \frac{1}{2}y_{199}.$$
$$x_{199} = \frac{1}{2}y_{197} - \frac{1}{2}y_{199}.$$
$$x_{206} = \frac{1}{2}y_{206} + \frac{1}{2}y_{208}.$$
$$x_{208} = \frac{1}{2}y_{206} - \frac{1}{2}y_{208}.$$
$$x_{210} = \frac{1}{2}y_{210} + \frac{1}{2}y_{212}.$$
$$x_{212} = \frac{1}{2}y_{210} - \frac{1}{2}y_{212}.$$
$$x_{207} = \frac{1}{2}y_{207} + \frac{1}{2}y_{209}.$$
$$x_{209} = \frac{1}{2}y_{207} - \frac{1}{2}y_{209}.$$
$$x_{211} = \frac{1}{2}y_{211} + \frac{1}{2}y_{213}.$$
$$x_{213} = \frac{1}{2}y_{211} - \frac{1}{2}y_{213}.$$
$$x_{220} = \frac{1}{2}y_{220} + \frac{1}{2}y_{222}.$$
$$x_{222} = \frac{1}{2}y_{220} - \frac{1}{2}y_{222}.$$
$$x_{224} = \frac{1}{2}y_{224} + \frac{1}{2}y_{226}.$$
$$x_{226} = \frac{1}{2}y_{224} - \frac{1}{2}y_{226}.$$
$$x_{221} = \frac{1}{2}y_{221} + \frac{1}{2}y_{223}.$$

$$x_{223} = \frac{1}{2}y_{221} - \frac{1}{2}y_{223}.$$
$$x_{225} = \frac{1}{2}y_{225} + \frac{1}{2}y_{227}.$$
$$x_{227} = \frac{1}{2}y_{225} - \frac{1}{2}y_{227}.$$

Stage 26.

$$(R_{133}, R_{135}) \longleftarrow (R_{133} + R_{135}, R_{133} - R_{135}).$$

Stage 27.

$$(R_{134}, R_{136}) \longleftarrow (R_{134} + R_{136}, R_{134} - R_{136}).$$

Stage 28.

$$(R_{151}, R_{153}) \longleftarrow (R_{151} + R_{153}, R_{151} - R_{153}).$$

Stage 29.

$$(R_{152}, R_{154}) \longleftarrow (R_{152} + R_{154}, R_{152} - R_{154}).$$

Stage 30.

$$(R_{169}, R_{171}) \longleftarrow (R_{169} + R_{171}, R_{169} - R_{171}).$$

Stage 31.

$$(R_{170}, R_{172}) \longleftarrow (R_{170} + R_{172}, R_{170} - R_{172}).$$

Stage 32.

$$(R_{187}, R_{189}) \longleftarrow (R_{187} + R_{189}, R_{187} - R_{189}).$$

Stage 33.

$$(R_{188}, R_{190}) \longleftarrow (R_{188} + R_{190}, R_{188} - R_{190}).$$

Stage 34. centered Lucas defect and square.

$$x_{68} = 1 + y_{68}.$$
$$x_{90} = 2 + 2y_{68} + y_{90}.$$
$$x_{91} = y_{69} + y_{91}.$$

After all stages, multiply each row by the following nonzero factor; unlisted factors are 1. This clears denominators and makes each row primitive. The sign convention is the one in the displayed equations.

Row	Multiplier

1	39778249
2	39778249
3	18921
4	18921
5	18921
6	18921
7	33
9	5512
10	$\frac{258587}{104}$
11	33
13	5512
14	$\frac{258587}{104}$
15	33
17	5512
18	$\frac{258587}{104}$
19	33
21	1219
22	$\frac{258587}{23}$
23	51
71	$\frac{1}{4}$
131	2
132	2
133	2
134	2
135	2
136	2
149	2
150	2
151	2
152	2
153	2
154	2
167	2
168	2
169	2
170	2
171	2
172	2
185	2
186	2
187	2
188	2
189	2
190	2
204	2

For completeness the inverse coordinate substitutions are printed next, in reverse stage order.

Here y denotes the new coordinate and x the old coordinate for that individual stage. They are simultaneous within a stage. Thus one obtains final coordinates from a raw witness by solving the forward stages in their original order; one recovers the raw witness from a final witness by substituting old-in-new formulas in reverse order. Row addition is inverted by subtraction, sum/difference by half-sum and half-difference, and row normalization by its reciprocal. These inverses introduce only fixed rational denominators.

Inverse coordinate stage 1.

$$y_{68} = -1 + x_{68}.$$

$$y_{90} = -2x_{68} + x_{90}.$$

$$y_{91} = -x_{69} + x_{91}.$$

Inverse coordinate stage 2.

$$y_{178} = x_{178} + x_{180}.$$

$$y_{179} = x_{179} + x_{181}.$$

$$y_{180} = x_{178} - x_{180}.$$

$$y_{181} = x_{179} - x_{181}.$$

$$y_{182} = x_{182} + x_{184}.$$

$$y_{183} = x_{183} + x_{185}.$$

$$y_{184} = x_{182} - x_{184}.$$

$$y_{185} = x_{183} - x_{185}.$$

$$y_{192} = x_{192} + x_{194}.$$

$$y_{193} = x_{193} + x_{195}.$$

$$y_{194} = x_{192} - x_{194}.$$

$$y_{195} = x_{193} - x_{195}.$$

$$y_{196} = x_{196} + x_{198}.$$

$$y_{197} = x_{197} + x_{199}.$$

$$y_{198} = x_{196} - x_{198}.$$

$$y_{199} = x_{197} - x_{199}.$$

$$y_{206} = x_{206} + x_{208}.$$

$$y_{207} = x_{207} + x_{209}.$$

$$y_{208} = x_{206} - x_{208}.$$

$$y_{209} = x_{207} - x_{209}.$$

$$y_{210} = x_{210} + x_{212}.$$

$$y_{211} = x_{211} + x_{213}.$$

$$y_{212} = x_{210} - x_{212}.$$

$$y_{213} = x_{211} - x_{213}.$$

$$y_{220} = x_{220} + x_{222}.$$

$$y_{221} = x_{221} + x_{223}.$$

$$y_{222} = x_{220} - x_{222}.$$

$$y_{223} = x_{221} - x_{223}.$$

$$y_{224} = x_{224} + x_{226}.$$

$$y_{225} = x_{225} + x_{227}.$$

$$y_{226} = x_{224} - x_{226}.$$

$$y_{227} = x_{225} - x_{227}.$$

Inverse coordinate stage 3.

$$y_{122} = -x_{121} + x_{122}.$$

Inverse coordinate stage 4.

$$y_{45} = x_{45} - x_{46}.$$

Inverse coordinate stage 5.

$$y_{41} = x_{41} - x_{42}.$$

Inverse coordinate stage 6.

$$y_{37} = x_{37} - x_{38}.$$

Inverse coordinate stage 7.

$$y_{31} = x_{31} - x_{32}.$$

Inverse coordinate stage 8.

$$y_{63} = \frac{2501}{51} + x_{63}.$$

$$y_{80} = 9 + x_{80}.$$

$$y_{90} = 1 + x_{90}.$$

$$y_{135} = \frac{1672}{18921} + x_{135}.$$

Inverse coordinate stage 9.

$$y_{64} = -2x_{59} + x_{64}.$$

$$y_{65} = -2x_{60} + x_{65}.$$

$$y_{66} = 2x_{59} - x_{64} + x_{66}.$$

$$y_{67} = 2x_{60} - x_{65} + x_{67}.$$

A45.5 Coordinate dictionary and index bridge

The dictionary below names the raw 233 coordinates before the preceding sparse stages. A name records its inherited role; after a coordinate stage its value is determined by that stage, not by reimposing its old definition. This convention is essential for shifted and sheared slots. The raw 238 to raw 237 map deletes 137; the raw 237 to raw 235 map deletes 17 and 18; the raw 235 to raw 233 map deletes 21 and 22. At each deletion all higher indices decrease by the number of deleted indices below them. The exceptional repurposed scalar slot is described in Section A45.1. These index maps together with the full predecessor dictionary specify every retained arithmetic variable.

Coordinate	Raw inherited role
1	`input`
2	`S.chart.j0.rescaled`
3	`S.chart.j1.rescaled`
4	`reciprocal.rho`
5	`detector0.normalized.scaled_e`
6	`detector0.normalized.s`
7	`detector0.normalized.p`
8	`detector0.factored.q`
9	`detector1.normalized.scaled_e`
10	`detector1.normalized.s`
11	`detector1.normalized.p`
12	`detector1.factored.q`
13	`normalized_scalar.scaled_e`
14	`normalized_scalar.s`
15	`normalized_scalar.p`
16	`normalized_scalar.factored.q`
17	`U0.f.0`
18	`U0.f.1`
19	`primary.negativeReciprocalZ0.0`
20	`primary.negativeReciprocalZ0.1`
21	`V1.z.0`
22	`V1.z.1`
23	`V2.z.0.rescaled`
24	`V2.z.1.rescaled`
25	`V3.z.0.rescaled`
26	`V3.z.1.rescaled`
27	`V4.z.0.rescaled`
28	`V4.z.1.rescaled`
29	`logV0.fSquare.0.quadratic`
30	`logV0.fSquare.1.quadratic`
31	`logV0.wPower2.0.quadratic`
32	`logV0.wPower2.1.quadratic`
33	`logV1.fSquare.0.quadratic`
34	`logV1.fSquare.1.quadratic`
35	`logV2.fSquare.0.quadratic`
36	`logV2.fSquare.1.quadratic`
37	`logV2.wPower2.0.quadratic`

```
logV2.wPower2.1.quadratic
logV3.fSquare.0.quadratic
logV3.fSquare.1.quadratic
logV3.wPower2.0.quadratic
logV3.wPower2.1.quadratic
logV4.fSquare.0.quadratic
logV4.fSquare.1.quadratic
logV4.wPower2.0.quadratic
logV4.wPower2.1.quadratic
L1.0
L1.1
LP1.0
LP1.1
L2.0.rescaled
L2.1.rescaled
LP2.0.rescaled
LP2.1.rescaled
L3.0.rescaled
L3.1.rescaled
LP3.0.rescaled
LP3.1.rescaled
L4.0.rescaled
L4.1.rescaled
LP4.0.rescaled
LP4.1.rescaled
inputSquare.quadratic
Lucas.C.0.quadratic.rescaled
Lucas.C.1.quadratic.rescaled
Lucas.AT.0.quadratic.rescaled
Lucas.AT.1.quadratic.rescaled
Lucas.F0.quadratic
Lucas.F1.quadratic
testSquare.1.0.0.quadratic
testSquare.1.0.1.quadratic
testSquare.1.1.0.quadratic
testSquare.1.1.1.quadratic
testSquare.1.2.0.quadratic
testSquare.1.2.1.quadratic
testSquare.1.3.0.quadratic
testSquare.1.3.1.quadratic
testSquare.12.0.0.quadratic
testSquare.12.0.1.quadratic
testSquare.12.1.0.quadratic
testSquare.12.1.1.quadratic
testSquare.40.0.0.quadratic
testSquare.40.0.1.quadratic
testSquare.40.1.0.quadratic
testSquare.40.1.1.quadratic
```

```
testSquare.40.2.0.quadratic
testSquare.40.2.1.quadratic
testSquare.40.3.0.quadratic
testSquare.40.3.1.quadratic
testSquare.40.4.0.quadratic
testSquare.40.4.1.quadratic
reciprocal.rhoSquared
reciprocal.rhoFourth
reciprocal.contactPolynomial
detector0.q1
reflected.detector0.weightPower7
detector0.E1.power2.quadratic
detector0.E1.power4.quadratic
detector0.actualW.0
detector0.actualW.1
detector0.normOver2w0
detector0.B
detector0.normalized.p2.quadratic
detector0.factored.v
detector0.factored.q2
detector1.t.quadratic.rescaled
symmetric.detector1.deltaSquared.rescaled
symmetric.detector1.deltaFourthPower.rescaled
symmetric.detector1.contactPolynomial.rescaled
detector1.q1
reflected.detector1.weightPower7
detector1.E1.power2.quadratic
detector1.E1.power4.quadratic
detector1.actualW.0
detector1.actualW.1
detector1.normOver2w0
detector1.B
detector1.normalized.p2.quadratic
detector1.factored.v
detector1.factored.q2
scalar.rPair1.quadratic
scalar.rFour.quadratic
scalar.contactProduct
scalar.zRatio.0.rescaled
scalar.zRatio.1.rescaled
scalar.W2.quadratic.rescaled
scalar.b.quadratic
normalized_scalar.p2.quadratic
normalized_scalar.factored.v
normalized_scalar.factored.q2
scalar.ratio1_unscaled_square.rescaled
parity_scalar.C_Z0square_over_R
reduced_scalar.root.rescaled
```

```
scalar.r_power_2
V0.reciprocal.gOverH.0
V0.reciprocal.gOverH.1
scalar.forced_positive_parameter
reciprocal.rhoFifth
reflected.detector1.deltaSixthPower.rescaled
amplified.C1.power4.0
amplified.C1.power4.1
residue.I.power4.0
residue.I.power4.1
residue.V1.power4.0
residue.V1.power4.1
amplified.det0.factor0.0
amplified.det0.factor0.1
amplified.det0.factor1.0
amplified.det0.factor1.1
amplified.det0.factor2.0
amplified.det0.factor2.1
amplified.det0.factor3.0
amplified.det0.factor3.1
amplified.det0.factor4.0
amplified.det0.factor4.1
amplified.det0.product1.0
amplified.det0.product1.1
amplified.det0.product2.0
amplified.det0.product2.1
amplified.det0.product3.0
amplified.det0.product3.1
amplified.det1.factor0.0
amplified.det1.factor0.1
amplified.det1.factor1.0
amplified.det1.factor1.1
amplified.det1.factor2.0
amplified.det1.factor2.1
amplified.det1.factor3.0
amplified.det1.factor3.1
amplified.det1.factor4.0
amplified.det1.factor4.1
amplified.det1.product1.0
amplified.det1.product1.1
amplified.det1.product2.0
amplified.det1.product2.1
amplified.det1.product3.0
amplified.det1.product3.1
rootProduct.1.A.0.rescaled
rootProduct.1.A.1.rescaled
rootProduct.1.B.0.rescaled
rootProduct.1.B.1.rescaled
```

```
rootProduct.1.C.0.rescaled
rootProduct.1.C.1.rescaled
rootProduct.1.E.0.rescaled
rootProduct.1.E.1.rescaled
rootProduct.1.i.0.rescaled
rootProduct.1.i.1.rescaled
rootProduct.1.k.0.rescaled
rootProduct.1.k.1.rescaled
rootProduct.1.u.0
rootProduct.1.u.1
rootProduct.2.A.0.rescaled
rootProduct.2.A.1.rescaled
rootProduct.2.B.0.rescaled
rootProduct.2.B.1.rescaled
rootProduct.2.C.0.rescaled
rootProduct.2.C.1.rescaled
rootProduct.2.E.0.rescaled
rootProduct.2.E.1.rescaled
rootProduct.2.i.0
rootProduct.2.i.1
rootProduct.2.k.0
rootProduct.2.k.1
rootProduct.2.u.0
rootProduct.2.u.1
rootProduct.3.A.0.rescaled
rootProduct.3.A.1.rescaled
rootProduct.3.B.0.rescaled
rootProduct.3.B.1.rescaled
rootProduct.3.C.0.rescaled
rootProduct.3.C.1.rescaled
rootProduct.3.E.0.rescaled
rootProduct.3.E.1.rescaled
rootProduct.3.i.0
rootProduct.3.i.1
rootProduct.3.k.0
rootProduct.3.k.1
rootProduct.3.u.0
rootProduct.3.u.1
rootProduct.4.A.0.rescaled
rootProduct.4.A.1.rescaled
rootProduct.4.B.0.rescaled
rootProduct.4.B.1.rescaled
rootProduct.4.C.0.rescaled
rootProduct.4.C.1.rescaled
rootProduct.4.E.0.rescaled
rootProduct.4.E.1.rescaled
rootProduct.4.i.0
rootProduct.4.i.1
```

```
230  rootProduct.4.k.0
231  rootProduct.4.k.1
232  rootProduct.4.u.0
233  rootProduct.4.u.1
```

A45.6 Completion of the 233-variable theorem

The full arithmetic proof of the 248-variable predecessor is retained in Section A42. The scalar-S change and root, reciprocal, and global-scaling changes prove the 238-variable predecessor as above. Section A45.1 proves both arithmetic directions for the new 237-variable scalar criterion, with one sufficiently large ineffective threshold uniform in the input. Sections A45.2 and A45.3 prove two-sided maps to 235 and 233 variables on this family. All exceptional denominators have been justified before their use. The constant affine and row maps of Section A45.4 then preserve rational solubility with the input fixed.

The final residuals have integer coefficients and degree at most two. Their sum of squares vanishes over $\mathbb{Q}$ exactly when each residual does. Consequently they prove the defining equivalence of Theorem 1.2. The coefficient calculation below proves quartic degree, content one, and the stated height. This finite calculation does not supply a numerical value of the arithmetic threshold N_0.

A46 The explicit 233-variable system

In this section only, $z_1, \dots, z_{233}$ are the final sparse coordinates, with distinguished input z_1. Put $A_N = 149^{2N+1}$ and $P_N = \sum_{r=1}^{207} Q_{r,N}^2$. We suppress the subscript N on individual residuals. Each K_j in this section is the fixed positive integer specified in the following table; these symbols are local to this section and do not refer to the predecessor's table.

$$\begin{aligned}
Q_1 = {} & -12042981952 + 95879821296 z_{17} - 191611825433 z_{17}^2 \\
& + 507888683232 z_{17} z_{18} + 612903260592 z_{17} z_{29} - 1654775158400 z_{17} z_{30} \\
& - 674479990044 z_{17} z_{135} + 1000502518848 z_{17} z_{136} - 124053139840 z_{18} \\
& - 383223650866 z_{18}^2 - 827387579200 z_{18} z_{29} + 2451613042368 z_{18} z_{30} \\
& + 1000502518848 z_{18} z_{135} - 1348959980088 z_{18} z_{136} + 191611825433 z_{29} \\
& + 491340931648 z_{29} z_{82} - 1344186590208 z_{29} z_{83} - 507888683232 z_{30} \\
& - 1344186590208 z_{30} z_{82} + 3930727453184 z_{30} z_{83} + 165304149024 z_{135} \\
& - 654431752548 z_{135}^2 + 1787157171072 z_{135} z_{136} - 249575558400 z_{136} \\
& - 1308863505096 z_{136}^2
\end{aligned}$$

$$
\begin{aligned}
Q_2 = {} & 3769142784 - 31013284960z_{17} + 63486085404z_{17}^2 \\
& - 191611825433z_{17}z_{18} - 206846894800z_{17}z_{29} + 612903260592z_{17}z_{30} \\
& + 250125629712z_{17}z_{135} - 337239995022z_{17}z_{136} + 47939910648z_{18} \\
& + 126972170808z_{18}^2 + 306451630296z_{18}z_{29} - 827387579200z_{18}z_{30} \\
& - 337239995022z_{18}z_{135} + 500251259424z_{18}z_{136} - 63486085404z_{29} \\
& - 168023323776z_{29}z_{82} + 491340931648z_{29}z_{83} + 191611825433z_{30} \\
& + 491340931648z_{30}z_{82} - 1344186590208z_{30}z_{83} - 62393889600z_{135} \\
& + 223394646384z_{135}^2 - 654431752548z_{135}z_{136} + 82652074512z_{136} \\
& + 446789292768z_{136}^2
\end{aligned}
$$

$$
\begin{aligned}
Q_3 = {} & -95304 + 590783z_{17} + 151368z_{17}z_{19} \\
& - 529788z_{17}z_{20} + 1665048z_{17}z_{135} - 1816416z_{17}z_{136} \\
& - 1201800z_{18} - 529788z_{18}z_{19} + 302736z_{18}z_{20} \\
& - 1816416z_{18}z_{135} + 3330096z_{18}z_{136} + 1078497z_{135} \\
& - 1059576z_{136}
\end{aligned}
$$

$$
\begin{aligned}
Q_4 = {} & 46816 - 600900z_{17} - 264894z_{17}z_{19} \\
& + 151368z_{17}z_{20} - 908208z_{17}z_{135} + 1665048z_{17}z_{136} \\
& + 590783z_{18} + 151368z_{18}z_{19} - 529788z_{18}z_{20} \\
& + 1665048z_{18}z_{135} - 1816416z_{18}z_{136} - 529788z_{135} \\
& + 1078497z_{136}
\end{aligned}
$$

$$
\begin{aligned}
Q_5 = {} & 216576 - 1078497z_{17} + 1059576z_{18} \\
& - 13376z_{19} + 151368z_{19}z_{135} - 529788z_{19}z_{136} \\
& + 46816z_{20} - 529788z_{20}z_{135} + 302736z_{20}z_{136} \\
& - 737919z_{135} + 1362312z_{136}
\end{aligned}
$$

$$
\begin{aligned}
Q_6 = {} & -60192 + 529788z_{17} - 1078497z_{18} \\
& + 23408z_{19} - 264894z_{19}z_{135} + 151368z_{19}z_{136} \\
& - 13376z_{20} + 151368z_{20}z_{135} - 529788z_{20}z_{136} \\
& + 681156z_{135} - 737919z_{136}
\end{aligned}
$$

$$
Q_7 = 2401z_{29}z_{30} - 6369z_{31}
$$

$$
\begin{aligned}
Q_8 = {} & 33z_{29}^2 - 193z_{29}z_{30} + 264z_{30}^2 \\
& - 33z_{32}
\end{aligned}
$$

$$
\begin{aligned}
Q_9 = {} & 775761z_{19}z_{21} + 504136z_{19}z_{49} + 373756z_{19}z_{50} \\
& + 1551522z_{20}z_{22} + 373756z_{20}z_{49} + 1008272z_{20}z_{50} \\
& - 1542912z_{21} + 3027360z_{21}z_{135} - 8022504z_{21}z_{136} \\
& - 8022504z_{22}z_{135} + 6054720z_{22}z_{136} + 775761z_{49}
\end{aligned}
$$

$$
\begin{aligned}
Q_{10} = {} & 775761z_{19}z_{22} + 186878z_{19}z_{49} + 504136z_{19}z_{50} \\
& + 775761z_{20}z_{21} + 504136z_{20}z_{49} + 373756z_{20}z_{50} \\
& - 4011252z_{21}z_{135} + 3027360z_{21}z_{136} - 1542912z_{22} \\
& + 3027360z_{22}z_{135} - 8022504z_{22}z_{136} + 775761z_{50}
\end{aligned}
$$

$$Q_{11} = 2401 z_{35} z_{36} - 6369 z_{37}$$

$$\begin{aligned} Q_{12} = {} & 33 z_{35}^2 - 193 z_{35} z_{36} + 264 z_{36}^2 \\ & - 33 z_{38} \end{aligned}$$

$$\begin{aligned} Q_{13} = {} & 775761 z_{19} z_{23} + 504136 z_{19} z_{53} + 373756 z_{19} z_{54} \\ & + 1551522 z_{20} z_{24} + 373756 z_{20} z_{53} + 1008272 z_{20} z_{54} \\ & - 1542912 z_{23} + 3027360 z_{23} z_{135} - 8022504 z_{23} z_{136} \\ & - 8022504 z_{24} z_{135} + 6054720 z_{24} z_{136} + 775761 z_{53} \end{aligned}$$

$$\begin{aligned} Q_{14} = {} & 775761 z_{19} z_{24} + 186878 z_{19} z_{53} + 504136 z_{19} z_{54} \\ & + 775761 z_{20} z_{23} + 504136 z_{20} z_{53} + 373756 z_{20} z_{54} \\ & - 4011252 z_{23} z_{135} + 3027360 z_{23} z_{136} - 1542912 z_{24} \\ & + 3027360 z_{24} z_{135} - 8022504 z_{24} z_{136} + 775761 z_{54} \end{aligned}$$

$$Q_{15} = 2401 z_{39} z_{40} - 6369 z_{41}$$

$$\begin{aligned} Q_{16} = {} & 33 z_{39}^2 - 193 z_{39} z_{40} + 264 z_{40}^2 \\ & - 33 z_{42} \end{aligned}$$

$$\begin{aligned} Q_{17} = {} & 775761 z_{19} z_{25} + 504136 z_{19} z_{57} + 373756 z_{19} z_{58} \\ & + 1551522 z_{20} z_{26} + 373756 z_{20} z_{57} + 1008272 z_{20} z_{58} \\ & - 1542912 z_{25} + 3027360 z_{25} z_{135} - 8022504 z_{25} z_{136} \\ & - 8022504 z_{26} z_{135} + 6054720 z_{26} z_{136} + 775761 z_{57} \end{aligned}$$

$$\begin{aligned} Q_{18} = {} & 775761 z_{19} z_{26} + 186878 z_{19} z_{57} + 504136 z_{19} z_{58} \\ & + 775761 z_{20} z_{25} + 504136 z_{20} z_{57} + 373756 z_{20} z_{58} \\ & - 4011252 z_{25} z_{135} + 3027360 z_{25} z_{136} - 1542912 z_{26} \\ & + 3027360 z_{26} z_{135} - 8022504 z_{26} z_{136} + 775761 z_{58} \end{aligned}$$

$$Q_{19} = 2401 z_{43} z_{44} - 6369 z_{45}$$

$$\begin{aligned} Q_{20} = {} & 33 z_{43}^2 - 193 z_{43} z_{44} + 264 z_{44}^2 \\ & - 33 z_{46} \end{aligned}$$

$$\begin{aligned} Q_{21} = {} & - 775761 z_{19} z_{28} - 186878 z_{19} z_{61} - 504136 z_{19} z_{62} \\ & - 775761 z_{20} z_{27} - 504136 z_{20} z_{61} - 373756 z_{20} z_{62} \\ & + 4011252 z_{27} z_{135} - 3027360 z_{27} z_{136} + 1542912 z_{28} \\ & - 3027360 z_{28} z_{135} + 8022504 z_{28} z_{136} - 775761 z_{62} \end{aligned}$$

$$\begin{aligned} Q_{22} = {} & 775761 z_{19} z_{27} + 504136 z_{19} z_{61} + 373756 z_{19} z_{62} \\ & + 1551522 z_{20} z_{28} + 373756 z_{20} z_{61} + 1008272 z_{20} z_{62} \\ & - 1542912 z_{27} + 3027360 z_{27} z_{135} - 8022504 z_{27} z_{136} \\ & - 8022504 z_{28} z_{135} + 6054720 z_{28} z_{136} + 775761 z_{61} \end{aligned}$$

$$Q_{23} = 2501 + 51 z_1^2 - 51 z_{63}$$

$$\begin{aligned} Q_{24} = {} & z_{51} z_{59} + 2 z_{52} z_{60} - 2 z_{59} z_{137} \\ & - z_{64} z_{137} \end{aligned}$$

$$Q_{25} = z_{51}z_{60} + z_{52}z_{59} - 2z_{60}z_{137} - z_{65}z_{137}$$

$$Q_{26} = z_{51}z_{55} + 2z_{52}z_{56} - z_{64}z_{137} - z_{66}z_{137}$$

$$Q_{27} = z_{51}z_{56} + z_{52}z_{55} - z_{65}z_{137} - z_{67}z_{137}$$

$$Q_{28} = z_{55}^2 + 2z_{56}^2 - z_{64}z_{66} - 2z_{65}z_{67} - z_{68}z_{134} - z_{134}$$

$$Q_{29} = 2z_{55}z_{56} - z_{64}z_{67} - z_{65}z_{66} - 2z_{69}z_{134}$$

$$Q_{30} = z_{47}^2 - 2z_{47}z_{49} + 2z_{48}^2 + 4z_{48}z_{50} + z_{49}^2 + 2z_{50}^2 - z_{78}$$

$$Q_{31} = z_{47}z_{48} + z_{47}z_{50} - z_{48}z_{49} - z_{49}z_{50} - z_{79}$$

$$Q_{32} = 9 + z_{47}^2 - 102z_{47}z_{63} + 2z_{48}^2 + 2601z_{63}^2 - z_{80}$$

$$Q_{33} = z_{47}z_{48} - 51z_{48}z_{63} - z_{81}$$

$$Q_{34} = z_{17}^2 + 2z_{18}^2 - z_{82}$$

$$Q_{35} = z_{17}z_{18} - z_{83}$$

$$Q_{36} = z_{51}^2 - 2z_{51}z_{53} + 2z_{52}^2 + 4z_{52}z_{54} + z_{53}^2 + 2z_{54}^2 - z_{84}z_{134}$$

$$Q_{37} = z_{51}z_{52} + z_{51}z_{54} - z_{52}z_{53} - z_{53}z_{54} - z_{85}z_{134}$$

$$Q_{38} = z_{55}^2 - 2z_{55}z_{57} + 2z_{56}^2 + 4z_{56}z_{58} + z_{57}^2 + 2z_{58}^2 - z_{86}z_{134}$$

$$Q_{39} = z_{55}z_{56} + z_{55}z_{58} - z_{56}z_{57} - z_{57}z_{58} - z_{87}z_{134}$$

$$Q_{40} = z_{59}^2 - 2z_{59}z_{61} + 2z_{60}^2 + 4z_{60}z_{62} + z_{61}^2 + 2z_{62}^2 - z_{88}z_{134}$$

$$Q_{41} = z_{59}z_{60} + z_{59}z_{62} - z_{60}z_{61} - z_{61}z_{62} - z_{89}z_{134}$$

$$Q_{42} = z_{68}^2 + 8z_{69}^2 - z_{90}$$

$$Q_{43} = z_{68}z_{69} - z_{91}$$

$$Q_{44} = z_4^2 - z_{92}$$

$$Q_{45} = z_{92}^2 - z_{93}$$

$$\begin{aligned} Q_{46} = {} & 7626746323592 - 76267463235920z_4 - 628059618995128z_4z_{92} \\ & + 732473028362323z_4z_{138} + 307309842082151z_{92} - 583791423311868z_{92}z_{138} \\ & + 718967979842546z_{93} + 162308064100032z_{93}^2 - 38129835686440z_{93}z_{138} \\ & - 7626746323592z_{94} - 3813373161796z_{94}z_{95} - 646146742577308z_{138} \\ & + 3813373161796z_{138}^2 \end{aligned}$$

$$\begin{aligned} Q_{47} = {} & 1 - 10z_4 - 120z_4z_{92} \\ & + 210z_4z_{138} + 45z_{92} - 120z_{92}z_{138} \\ & + 210z_{93} + 45z_{93}^2 - 10z_{93}z_{138} \\ & - z_{94} - z_{94}z_{95} - 252z_{138} \\ & + z_{138}^2 \end{aligned}$$

$$Q_{48} = z_{95}^2 - z_{97}$$

$$Q_{49} = z_{97}^2 - z_{98}$$

$$Q_{50} = z_{99}^2 - 2z_{99}z_{101} - 2z_{100}^2$$

$$Q_{51} = 167281z_{95}z_{102} - 166464z_{101}^2 + 167281z_{102}$$

$$Q_{52} = z_7^2 - z_{103}$$

$$\begin{aligned} Q_{53} = {} & 289 + 34z_5 - 3z_5^2 \\ & - 17z_{102}z_{103} \end{aligned}$$

$$Q_{54} = 17z_5 - z_5z_{102} - 17z_6z_{104}$$

$$Q_{55} = z_8^2 - z_{105}$$

$$\begin{aligned} Q_{56} = {} & z_6^2 + 34z_6z_{104} - 17z_{102}z_{105} \\ & - 867z_{104}^2 \end{aligned}$$

$$\begin{aligned} Q_{57} = {} & z_{106}^2 - 2z_{106}z_{134} - z_{107}z_{134} \\ & + z_{134}^2 \end{aligned}$$

$$Q_{58} = z_{107}^2 - z_{108}z_{134}$$

$$\begin{aligned} Q_{59} = {} & - 141822723674536z_{107}z_{134} + 35893742479489z_{107}z_{139} + 192222345736537z_{108}z_{134} \\ & - 3813373161796z_{109}z_{110} + 39896049934224z_{134}^2 - 122376041313918z_{134}z_{139} \end{aligned}$$

$$Q_{60} = z_{108}z_{139} - z_{109} - z_{109}z_{110}$$

$$Q_{61} = z_{110}^2 - z_{112}$$

$$Q_{62} = z_{112}^2 - z_{113}$$

$$Q_{63} = z_{114}^2 - 2z_{114}z_{116} - 2z_{115}^2$$

$$Q_{64} = 167281z_{110}z_{117} - 166464z_{116}^2 + 167281z_{117}$$

$$Q_{65} = z_{11}^2 - z_{118}$$

$$Q_{66} = 289 + 34z_9 - 3z_9^2 - 17z_{117}z_{118}$$

$$Q_{67} = 17z_9 - z_9z_{117} - 17z_{10}z_{119}$$

$$Q_{68} = z_{12}^2 - z_{120}$$

$$Q_{69} = z_{10}^2 + 34z_{10}z_{119} - 17z_{117}z_{120} - 867z_{119}^2$$

$$Q_{70} = 2601z_{63}^2 - 4z_{121}$$

$$Q_{71} = 3882987z_{63}z_{121} - 128z_{121}^2 - 381485z_{122}$$

$$Q_{72} = 31581660 + 2103220824z_{63} + 29762681184z_{63}^2 + 585078120z_{63}z_{121} + 155645880z_{63}z_{122} + 28680300z_{121} + 7629700z_{122} + z_{123}$$

$$Q_{73} = K_1z_{126}z_{134} - z_{133}^2$$

$$Q_{74} = 9792z_{126} + z_{126}z_{137} + 82479248z_{127}z_{137}$$

$$Q_{75} = z_{15}^2 - z_{128}$$

$$Q_{76} = K_7 - K_4z_{13} - z_{13}^2 + K_3z_{127}z_{128}$$

$$Q_{77} = 17z_{13} + K_2z_{13}z_{127} - 17z_{14}z_{129}$$

$$Q_{78} = z_{16}^2 - z_{130}$$

$$Q_{79} = z_{14}^2 - K_6z_{14}z_{129} + K_5z_{127}z_{130} - K_8z_{129}^2$$

$$Q_{80} = z_{82}^2 - 8z_{83}^2 + z_{131}z_{137} + z_{132}^2 - z_{133}z_{134}$$

$$Q_{81} = z_{95}z_{96} - z_{98}^2$$

$$Q_{82} = z_{110}z_{111} - z_{113}^2$$

$$Q_{83} = z_{134} - z_{137}^2$$

$$Q_{84} = z_{125}^2 - z_{131}$$

$$Q_{85} = z_{124}^2 - z_{132}z_{137}$$

$$Q_{86} = 22201z_{123}z_{137} + z_{137} - A_Nz_{123}$$

$$Q_{87} = z_4z_{93} - z_{138}$$

$$Q_{88} = z_{107}z_{108} - z_{134}z_{139}$$

$$Q_{89} = z_{78}^2 + 8z_{79}^2 - z_{140}$$

$$Q_{90} = 4z_{78}z_{79} - z_{141}$$

$$\begin{aligned} Q_{91} = & -324 - 12z_{47}z_{80} - 48z_{48}z_{81} \\ & + 612z_{63}z_{80} + 36z_{80} + z_{80}^2 \\ & + 8z_{81}^2 - z_{142} \end{aligned}$$

$$\begin{aligned} Q_{92} = & -24z_{47}z_{81} - 12z_{48}z_{80} + 1224z_{63}z_{81} \\ & + 4z_{80}z_{81} + 72z_{81} - z_{143} \end{aligned}$$

$$Q_{93} = z_{33}^2 + 8z_{34}^2 - z_{144}$$

$$Q_{94} = 4z_{33}z_{34} + z_{145}$$

$$\begin{aligned} Q_{95} = & 1 + z_{82}z_{97} + z_{84}z_{96} \\ & + z_{98}z_{144} - z_{146} \end{aligned}$$

$$\begin{aligned} Q_{96} = & 2z_{83}z_{97} + 2z_{85}z_{96} + z_{98}z_{145} \\ & - z_{147} \end{aligned}$$

$$\begin{aligned} Q_{97} = & 2401 + 37249z_{31}z_{98} + 2401z_{32}z_{98} \\ & + 2401z_{74}z_{97} + 2401z_{90}z_{96} - 2401z_{148} \end{aligned}$$

$$\begin{aligned} Q_{98} = & 25476z_{31}z_{98} - 4802z_{75}z_{97} - 9604z_{91}z_{96} \\ & + 2401z_{149} \end{aligned}$$

$$\begin{aligned} Q_{99} = & 2401 + 37249z_{41}z_{98} + 2401z_{42}z_{98} \\ & + 2401z_{72}z_{95} + 2401z_{76}z_{97} + 2401z_{96}z_{142} \\ & - 2401z_{150} \end{aligned}$$

$$\begin{aligned} Q_{100} = & 25476z_{41}z_{98} - 4802z_{73}z_{95} - 4802z_{77}z_{97} \\ & - 2401z_{96}z_{143} + 2401z_{151} \end{aligned}$$

$$\begin{aligned} Q_{101} = & 2401 + 37249z_{45}z_{98} + 2401z_{46}z_{98} \\ & + 2401z_{86}z_{96} + 2401z_{97}z_{140} - 2401z_{152} \end{aligned}$$

$$\begin{aligned} Q_{102} = & 25476z_{45}z_{98} - 4802z_{87}z_{96} - 2401z_{97}z_{141} \\ & + 2401z_{153} \end{aligned}$$

$$\begin{aligned} Q_{103} = & 2401 + 37249z_{37}z_{98} + 2401z_{38}z_{98} \\ & + 2401z_{70}z_{97} + 2401z_{88}z_{96} - 2401z_{154} \end{aligned}$$

$$\begin{aligned} Q_{104} = & 25476z_{37}z_{98} - 4802z_{71}z_{97} - 4802z_{89}z_{96} \\ & + 2401z_{155} \end{aligned}$$

$$Q_{105} = z_{146}z_{148} + 2z_{147}z_{149} - z_{156}$$

$$Q_{106} = z_{146}z_{149} + z_{147}z_{148} - z_{157}$$

$$Q_{107} = z_{150}z_{156} + 2z_{151}z_{157} - z_{158}$$

$$Q_{108} = z_{150}z_{157} + z_{151}z_{156} - z_{159}$$

$$Q_{109} = z_{152}z_{158} + 2z_{153}z_{159} - z_{160}$$

$$Q_{110} = z_{152}z_{159} + z_{153}z_{158} - z_{161}$$

$$Q_{111} = z_{99} - z_{154}z_{160} - 2z_{155}z_{161}$$

$$Q_{112} = z_{100} - z_{154}z_{161} - z_{155}z_{160}$$

$$Q_{113} = 1 + z_{82}z_{112} + z_{84}z_{111} + z_{113}z_{144} - z_{162}$$

$$Q_{114} = 2z_{83}z_{112} + 2z_{85}z_{111} + z_{113}z_{145} - z_{163}$$

$$Q_{115} = 2401 + 37249z_{31}z_{113} + 2401z_{32}z_{113} + 2401z_{74}z_{112} + 2401z_{90}z_{111} - 2401z_{164}$$

$$Q_{116} = 25476z_{31}z_{113} - 4802z_{75}z_{112} - 9604z_{91}z_{111} + 2401z_{165}$$

$$Q_{117} = 2401 + 37249z_{41}z_{113} + 2401z_{42}z_{113} + 2401z_{72}z_{110} + 2401z_{76}z_{112} + 2401z_{111}z_{142} - 2401z_{166}$$

$$Q_{118} = 25476z_{41}z_{113} - 4802z_{73}z_{110} - 4802z_{77}z_{112} - 2401z_{111}z_{143} + 2401z_{167}$$

$$Q_{119} = 2401 + 37249z_{45}z_{113} + 2401z_{46}z_{113} + 2401z_{86}z_{111} + 2401z_{112}z_{140} - 2401z_{168}$$

$$Q_{120} = 25476z_{45}z_{113} - 4802z_{87}z_{111} - 2401z_{112}z_{141} + 2401z_{169}$$

$$Q_{121} = 2401 + 37249z_{37}z_{113} + 2401z_{38}z_{113} + 2401z_{70}z_{112} + 2401z_{88}z_{111} - 2401z_{170}$$

$$Q_{122} = 25476z_{37}z_{113} - 4802z_{71}z_{112} - 4802z_{89}z_{111} + 2401z_{171}$$

$$Q_{123} = z_{162}z_{164} + 2z_{163}z_{165} - z_{172}$$

$$Q_{124} = z_{162}z_{165} + z_{163}z_{164} - z_{173}$$

$$Q_{125} = z_{166}z_{172} + 2z_{167}z_{173} - z_{174}$$

$$Q_{126} = z_{166}z_{173} + z_{167}z_{172} - z_{175}$$

$$Q_{127} = z_{168}z_{174} + 2z_{169}z_{175} - z_{176}$$

$$Q_{128} = z_{168}z_{175} + z_{169}z_{174} - z_{177}$$

$$Q_{129} = z_{114} - z_{170}z_{176} - 2z_{171}z_{177}$$

$$Q_{130} = z_{115} - z_{170}z_{177} - z_{171}z_{176}$$

$$Q_{131} = z_{178}^2 + 2z_{179}^2 + z_{180}^2 + 2z_{181}^2 + z_{182}^2 + 2z_{183}^2 - z_{184}^2 - 2z_{185}^2$$

$$Q_{132} = z_{178}z_{179} + z_{180}z_{181} + z_{182}z_{183} - z_{184}z_{185}$$

$$Q_{133} = -12z_{137}z_{186} + 8z_{137}z_{187} + z_{178}z_{182} + 2z_{179}z_{183} + z_{180}z_{184} + 2z_{181}z_{185}$$

$$Q_{134} = 4z_{137}z_{186} - 12z_{137}z_{187} + z_{178}z_{183} + z_{179}z_{182} + z_{180}z_{185} + z_{181}z_{184}$$

$$Q_{135} = 12z_{137}z_{188} - 8z_{137}z_{189} + z_{178}z_{184} + 2z_{179}z_{185} + z_{180}z_{182} + 2z_{181}z_{183}$$

$$Q_{136} = -4z_{137}z_{188} + 12z_{137}z_{189} + z_{178}z_{185} + z_{179}z_{184} + z_{180}z_{183} + z_{181}z_{182}$$

$$Q_{137} = -44z_{134}z_{190} + 48z_{134}z_{191} + z_{178}^2 + 2z_{179}^2$$

$$Q_{138} = 12z_{134}z_{190} - 22z_{134}z_{191} + z_{178}z_{179}$$

$$Q_{139} = z_{70} - z_{190}^2 - 2z_{191}^2$$

$$Q_{140} = z_{71} - z_{190}z_{191}$$

$$Q_{141} = 11z_{70}z_{134} - 24z_{71}z_{134} - z_{134} + z_{186}^2 + 2z_{187}^2$$

$$Q_{142} = -3z_{70}z_{134} + 11z_{71}z_{134} + z_{186}z_{187}$$

$$Q_{143} = 11z_{33}z_{134} - 24z_{34}z_{134} - z_{134} + z_{188}^2 + 2z_{189}^2$$

$$Q_{144} = -3z_{33}z_{134} + 11z_{34}z_{134} + z_{188}z_{189}$$

$$Q_{145} = 6z_{21}z_{188} - 4z_{21}z_{189} - 4z_{22}z_{188} + 12z_{22}z_{189} + z_{180}$$

$$Q_{146} = -2z_{21}z_{188} + 6z_{21}z_{189} + 6z_{22}z_{188} - 4z_{22}z_{189} + z_{181}$$

$$Q_{147} = z_{19}z_{178} + 2z_{20}z_{179} - 6z_{47}z_{186} + 4z_{47}z_{187} + 4z_{48}z_{186} - 12z_{48}z_{187}$$

$$Q_{148} = z_{19}z_{179} + z_{20}z_{178} + 2z_{47}z_{186} - 6z_{47}z_{187} - 6z_{48}z_{186} + 4z_{48}z_{187}$$

$$Q_{149} = z_{192}^2 + 2z_{193}^2 + z_{194}^2 + 2z_{195}^2 + z_{196}^2 + 2z_{197}^2 - z_{198}^2 - 2z_{199}^2$$

$$Q_{150} = z_{192}z_{193} + z_{194}z_{195} + z_{196}z_{197} - z_{198}z_{199}$$

$$Q_{151} = -12z_{134}z_{200} + 8z_{134}z_{201} + z_{192}z_{196} + 2z_{193}z_{197} + z_{194}z_{198} + 2z_{195}z_{199}$$

$$Q_{152} = 4z_{134}z_{200} - 12z_{134}z_{201} + z_{192}z_{197} + z_{193}z_{196} + z_{194}z_{199} + z_{195}z_{198}$$

$$Q_{153} = 12z_{134}z_{202} - 8z_{134}z_{203} + z_{192}z_{198} + 2z_{193}z_{199} + z_{194}z_{196} + 2z_{195}z_{197}$$

$$Q_{154} = -4z_{134}z_{202} + 12z_{134}z_{203} + z_{192}z_{199} + z_{193}z_{198} + z_{194}z_{197} + z_{195}z_{196}$$

$$Q_{155} = -44z_{134}z_{204} + 48z_{134}z_{205} + z_{192}^2 + 2z_{193}^2$$

$$Q_{156} = 12z_{134}z_{204} - 22z_{134}z_{205} + z_{192}z_{193}$$

$$Q_{157} = z_{72} - z_{204}^2 - 2z_{205}^2$$

$$Q_{158} = z_{73} - z_{204}z_{205}$$

$$Q_{159} = -1 + 11z_{72} - 24z_{73} + z_{200}^2 + 2z_{201}^2$$

$$Q_{160} = -3z_{72} + 11z_{73} + z_{200}z_{201}$$

$$Q_{161} = -1 + 11z_{35} - 24z_{36} + z_{202}^2 + 2z_{203}^2$$

$$Q_{162} = -3z_{35} + 11z_{36} + z_{202}z_{203}$$

$$Q_{163} = 6z_{23}z_{202} - 4z_{23}z_{203} - 4z_{24}z_{202} + 12z_{24}z_{203} + z_{194}$$

$$Q_{164} = -2z_{23}z_{202} + 6z_{23}z_{203} + 6z_{24}z_{202} - 4z_{24}z_{203} + z_{195}$$

$$Q_{165} = z_{19}z_{192} + 2z_{20}z_{193} - 6z_{51}z_{200} + 4z_{51}z_{201} + 4z_{52}z_{200} - 12z_{52}z_{201}$$

$$Q_{166} = z_{19}z_{193} + z_{20}z_{192} + 2z_{51}z_{200} - 6z_{51}z_{201} - 6z_{52}z_{200} + 4z_{52}z_{201}$$

$$Q_{167} = z_{206}^2 + 2z_{207}^2 + z_{208}^2 + 2z_{209}^2 + z_{210}^2 + 2z_{211}^2 - z_{212}^2 - 2z_{213}^2$$

$$Q_{168} = z_{206}z_{207} + z_{208}z_{209} + z_{210}z_{211} - z_{212}z_{213}$$

$$Q_{169} = -12z_{134}z_{214} + 8z_{134}z_{215} + z_{206}z_{210} + 2z_{207}z_{211} + z_{208}z_{212} + 2z_{209}z_{213}$$

$$Q_{170} = 4z_{134}z_{214} - 12z_{134}z_{215} + z_{206}z_{211} + z_{207}z_{210} + z_{208}z_{213} + z_{209}z_{212}$$

$$Q_{171} = 12z_{134}z_{216} - 8z_{134}z_{217} + z_{206}z_{212} + 2z_{207}z_{213} + z_{208}z_{210} + 2z_{209}z_{211}$$

$$Q_{172} = -4z_{134}z_{216} + 12z_{134}z_{217} + z_{206}z_{213} + z_{207}z_{212} + z_{208}z_{211} + z_{209}z_{210}$$

$$Q_{173} = -44z_{134}z_{218} + 48z_{134}z_{219} + z_{206}^2 + 2z_{207}^2$$

$$Q_{174} = 12z_{134}z_{218} - 22z_{134}z_{219} + z_{206}z_{207}$$

$$Q_{175} = z_{74} - z_{218}^2 - 2z_{219}^2$$

$$Q_{176} = z_{75} - z_{218}z_{219}$$

$$Q_{177} = -1 + 11z_{74} - 24z_{75} + z_{214}^2 + 2z_{215}^2$$

$$Q_{178} = -3z_{74} + 11z_{75} + z_{214}z_{215}$$

$$Q_{179} = -1 + 11z_{39} - 24z_{40} + z_{216}^2 + 2z_{217}^2$$

$$Q_{180} = -3z_{39} + 11z_{40} + z_{216}z_{217}$$

$$Q_{181} = 6z_{25}z_{216} - 4z_{25}z_{217} - 4z_{26}z_{216} + 12z_{26}z_{217} + z_{208}$$

$$Q_{182} = -2z_{25}z_{216} + 6z_{25}z_{217} + 6z_{26}z_{216} - 4z_{26}z_{217} + z_{209}$$

$$Q_{183} = z_{19}z_{206} + 2z_{20}z_{207} - 6z_{55}z_{214} + 4z_{55}z_{215} + 4z_{56}z_{214} - 12z_{56}z_{215}$$

$$Q_{184} = z_{19}z_{207} + z_{20}z_{206} + 2z_{55}z_{214} - 6z_{55}z_{215} - 6z_{56}z_{214} + 4z_{56}z_{215}$$

$$Q_{185} = z_{220}^2 + 2z_{221}^2 + z_{222}^2 + 2z_{223}^2 + z_{224}^2 + 2z_{225}^2 - z_{226}^2 - 2z_{227}^2$$

$$Q_{186} = z_{220}z_{221} + z_{222}z_{223} + z_{224}z_{225} - z_{226}z_{227}$$

$$Q_{187} = -12z_{134}z_{228} + 8z_{134}z_{229} + z_{220}z_{224} + 2z_{221}z_{225} + z_{222}z_{226} + 2z_{223}z_{227}$$

$$Q_{188} = 4z_{134}z_{228} - 12z_{134}z_{229} + z_{220}z_{225} + z_{221}z_{224} + z_{222}z_{227} + z_{223}z_{226}$$

$$Q_{189} = 12z_{134}z_{230} - 8z_{134}z_{231} + z_{220}z_{226} + 2z_{221}z_{227} + z_{222}z_{224} + 2z_{223}z_{225}$$

$$Q_{190} = -4z_{134}z_{230} + 12z_{134}z_{231} + z_{220}z_{227} + z_{221}z_{226} + z_{222}z_{225} + z_{223}z_{224}$$

$$Q_{191} = -44z_{134}z_{232} + 48z_{134}z_{233} + z_{220}^2 + 2z_{221}^2$$

$$Q_{192} = 12z_{134}z_{232} - 22z_{134}z_{233} + z_{220}z_{221}$$

$$Q_{193} = z_{76} - z_{232}^2 - 2z_{233}^2$$

$$Q_{194} = z_{77} - z_{232}z_{233}$$

$$Q_{195} = -1 + 11z_{76} - 24z_{77} + z_{228}^2 + 2z_{229}^2$$

$$Q_{196} = -3z_{76} + 11z_{77} + z_{228}z_{229}$$

$$Q_{197} = -1 + 11z_{43} - 24z_{44} + z_{230}^2 + 2z_{231}^2$$

$$Q_{198} = -3z_{43} + 11z_{44} + z_{230}z_{231}$$

$$Q_{199} = 6z_{27}z_{230} - 4z_{27}z_{231} - 4z_{28}z_{230} + 12z_{28}z_{231} + z_{222}$$

$$Q_{200} = -2z_{27}z_{230} + 6z_{27}z_{231} + 6z_{28}z_{230} - 4z_{28}z_{231} + z_{223}$$

$$Q_{201} = z_{19}z_{220} + 2z_{20}z_{221} - 6z_{59}z_{228} + 4z_{59}z_{229} + 4z_{60}z_{228} - 12z_{60}z_{229}$$

$$Q_{202} = z_{19}z_{221} + z_{20}z_{220} + 2z_{59}z_{228} - 6z_{59}z_{229} - 6z_{60}z_{228} + 4z_{60}z_{229}$$

$$Q_{203} = 4z_2 + z_{192}z_{194} + 2z_{193}z_{195} - z_{196}z_{198} - 2z_{197}z_{199}$$

$$Q_{204} = 4z_3 + z_{192}z_{195} + z_{193}z_{194} - z_{196}z_{199} - z_{197}z_{198}$$

$$Q_{205} = 2z_2z_{124} + 4z_3z_{125} - z_{19}z_{198} - 2z_{20}z_{199}$$

$$Q_{206} = 2z_2z_{125} + 2z_3z_{124} - z_{19}z_{199} - z_{20}z_{198}$$

$$Q_{207} = 8z_4z_{106} - z_{184}^2 - 2z_{185}^2 - z_{198}^2 - 2z_{199}^2 - z_{212}^2 - 2z_{213}^2 - z_{226}^2 - 2z_{227}^2$$

A46.1 Fixed integer constants for the final system

The following decimal strings are concatenated across line breaks.

Constant	Exact positive integer
K_1	98194994952101164627950993228801351760 40946699131400335899642243325954265276 68691070368153600000000000000
K_2	80990493410131000784375977023646274345 82049009524401528919498956985577266824 13648326282803920764928000000000000000
K_3	45894612932407567111146386980066222129 29827772063827533054382742291827117867 01067384893588888433459200000000000000000
K_4	91789225864815134222292773960132444258 59655544127655066108765484583654235734 02134769787177776866918400000000000000000
K_5	13768383879722270133343916094019866638 78948331619148259916314822687548135360 10320215468076666530037760000000000000000 0
K_6	27536767759444540266687832188039733277 57896663238296519832629645375096270720 20640430936153333060075520000000000000000 0
K_7	63189464886465357187317594522928066201 84449792472930176268493164014380154729 61541724103664894830836146098454811858 89826867345597467459165957964726042449 88128011149397522493691010836323796918 78234193920000000000000000000000000000000
K_8	56870518397818821468585835070635259581 66004813225637158641643847612942139256 65387551693298405347752531488609330673 00844180611037720713249362168253438204 89315210034457770244321909752691417226 90410774528000000000000000000000000000000 0

A46.2 Exact coefficient profile and its scope

Expanding these 207 integer squares and collecting coordinate monomials gives 2514 nonzero monomials (2515 terms if different powers of A_N are kept separate). The coefficient of z_4^4 is 1, so the degree is four and the content is one. All 233 coordinates occur. The only parameter-dependent residual is

$$Q_{86} = 22201 z_{123} z_{137} + z_{137} - A_N z_{123}.$$

Thus every parameter-dependent coefficient can be checked directly from its square; all other contributions are constant. The largest absolute coefficient independent of A_N is the integer C_* below, and the largest absolute coefficient of A_N is 44402.

```
32342558628366490523958090505137577127207666788272010 51
41212144475297826411641161871190307260931507206679020 69
11463330366381934462871195934931740025259644388552756 58
97256790971300704192409599390436731181452655668356998 50
60230695491071517703974860972512777670857670513959154 12
34706518778051221304687530464670525936269863999881933 07
18197810416613557658448307987213918747824284943564306 3
52016352853622784000000000000000000000000000000000000 00
000000000000000000
```

The coefficient of z_{123}^2 is $A_N^2 + 1$. The remaining parameter-dependent coefficients have absolute value at most $44402A_N$. Direct integer comparison gives $C_* < 149^{214} + 1$. For $N \geq 53$, $A_N \geq 149^{107} > 44402$; hence both C_* and $44402A_N$ are at most $A_N^2 + 1$. Therefore the exact height is

$$H(P_N) = A_N^2 + 1 = 149^{4N+2} + 1 \qquad (N \geq 53).$$

The displayed equations, constants and finite coefficient calculation specify the family without choosing a numerical defining member. In particular $N = 53$ is a coefficient threshold only. The assertion that P_N defines $\mathbb{Z}$ requires $N \geq N_0$, with N_0 still ineffective.

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Version 15 provenance

The final 233-variable system uses the three-residue 237-variable scalar criterion, the two-reference-identity reduction to 235 variables, and the corrected reference-square elimination to 233 variables. The full sparse presentation has 207 quadratic residuals and 2514 monomials. Its exact source IR has SHA256

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These reductions and finite transformations were developed and internally reviewed by separate Astra agents. The present manuscript integrates those arguments; its companion audit records source versions, equation generation, and subsequent review separately. Internal model review is not independent human peer review or formal verification. The earlier complete development follows with its provenance preserved.

The 238-variable predecessor: provenance

The 238-variable development begins with José Brox's 9 November 2026 version-17 sources, pinned to repository commit `d25e0aa6046662d9a878755aca8ed742821d3e61`. Its height, aggregate and local-construction arguments have been incorporated into this manuscript with internal references. Astra agents authored and independently reviewed the scalar-S replacement, root-product charts, reciprocal scale, global scalar rescaling, sparse transformations, exact expansions and source matching, under Elliot Glazer's direction. These are internal model reviews, not independent human peer review or formal verification. The companion dependency map preserves exact source hashes and transformation provenance. Earlier provenance follows unchanged.

AI usage statement

The Version 9 odd-product and asymmetric-depth improvements were developed by Gabriel Conant with ChatGPT and incorporated with additional source, convention, and finite-certificate checks. This manuscript was developed primarily by OpenAI Codex agents (Astra 6 and Sol 6.1) and Anthropic Claude agents (Opus 5.5), with auditing from Fable 5.1. Their contributions included mathematical exploration, construction and analysis of the polynomial, proof drafting, literature searches, computational checks, internal review, and drafting and revision of the exposition. Reviews by multiple AI agents formed part of this process; they do not constitute independent human peer review.